Raytheon Aircraft Company

Beechcraft

Super King Air® B200 & B200C

(Serials BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-1438, BB-1440 thru BB-1443; and BL-37 thru BL-138)

Pilot's Operating Handbook and FAA Approved Airplane Flight Manual

FAA Approved in the Normal Category based on 14 CFR Part 23. This document must be carried in the airplane at all times, and be kept within reach of the pilot during all flight operations. This handbook includes the material required to be furnished to the pilot by 14 CFR Part 23.

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: _____

for John Tigue
Raytheon Aircraft Company
DOA-230339-CE

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For Training Purposes Only

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NOTE

Where Beech Aircraft Corporation is referred to in this publication, it will be taken to read Raytheon Aircraft Company.

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Raytheon Aircraft Company

Member of GAMA
General Aviation Manufacturers Association

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LIST OF EFFECTIVE PAGES

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Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual

P/N 101-590010-147
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- Revision C8 . . . . . April, 2004
- Revision C7 . . . . . February, 2004
- Revision C6 . . . . . January, 2002
- Revision C5 . . . . . May, 2000
- Revision C4 . . . . . April, 1997
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- Revision C2 . . . . . Apil, 1996
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Hawker Beechcraft Corporation

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Beech Super King Air® B200 & B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual

P/N 101-590010-147C

May, 2011

Temporary Changes to this manual must be in the airplane for all flight operations.

PART NUMBER	SUBJECT	DATE
101-590010-147CTC1	Provides Emergency Procedures for Windshield Electrical Fault	April 8, 1997
101-590010-147CTC2	In-Flight Selection of Beta Range	July 21, 1997
101-590010-147CTC3	Structural Limitations	May, 2011

NOTE: This page should be filed in the front of the manual immediately in front of the *Log Of Revisions* page(s). This page replaces any *Log Of Temporary Changes* page dated prior to the date of this Log.

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Super King Air® B200 & B200C

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Revision C9 - August, 2004

PAGE	DESCRIPTION
Title Page	New
List of Effective Pages	New
Log of Revisions	New
2-13	Revised "Other Limitations"

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For Training Purposes Only

Raytheon Aircraft Company

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Super King Air® B200 & B200C

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Revision C8 - April, 2004

PAGE	DESCRIPTION
Title Page	New
List of Effective Pages	New
Log of Revisions	New
3A-5	Revised "Flaps Up Landing"
4-16	Revised "Normal Landing" and "Maximum Reverse Thrust Landing"

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Revision C7 - February, 2004

PAGE	DESCRIPTION
Title Page	New
List of Effective Pages	New
Log of Revisions	New
2-1 and 2-2	Revised Table of Contents
2-6	Deleted "Minimum Temperature Limits" and Shifted Data
2-7	Revised "Anti-Icing Additive" and "Fuel Biocide Additive"
4-21	Revised "Blending Anti-Icing Additive To Fuel"

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For Training Purposes Only

Raytheon Aircraft

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P/N 101-590010-147
Revision C6 - January, 2002

PAGE	DESCRIPTION
Title Page	Updated
List of Effective Pages	New
Log of Revisions	New
2-2	Revised Table of Contents
2-13 thru 2-14A	Revised "Other Limitations", Deleted "Fuselage Side Windows", "Windshield" and Added "Cracked or Shattered Windshield" and "Crack In Any Side Window (Cockpit or Cabin)" and Shifted Material
3A-1	Revised Table of Contents
3A-10	Shifted Data
3A-11 and 3A-12	Deleted "Cracked Windshield" and Added "Cracked or Shattered Windshield" and "Crack In Any Side Window (Cockpit or Cabin)" and Shifted Material

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Beech Super King Air® B200 & B200C

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"C5" Revision ..... May, 2000

Page	Description
Title Page	Updated
List Of Effective Pages	New
Log Of Revisions (C5)	New
1-1 and 1-2	Revised Table of Contents
1-3	Revised "Important Notice"
1-4 thru 1-4B	Revised "Revising the Handbook" and Shifted Data
2-1 and 2-2	Revised Table of Contents
2-12 and 2-13	Added "Approved Airplane Deicing/Anti-icing Fluids"
2-14	Revised "Limitations When Encountering Severe Icing Conditions"
3A-1	Revised Table of Contents
3A-11	Revised "Severe Icing Conditions"
4-1 and 4-2	Revised Table of Contents
4-22 thru 4-22B	Revised "Cold Weather Procedures (Snow, Slush, and Ice)" and Shifted Data
8-1 and 8-2	Revised Table of Contents
8-3 and 8-4	Revised "Publications", "Airplane Inspection Periods" and Shifted Data
8-5 and 8-6	Revised "External Power" and Shifted Data
8-7 thru 8-9	Revised "Shock Struts", "Oil System" and Shifted Data
8-10	Revised "Fuel System" and "Instrument Vacuum Air"
8-11	Revised "Cabin Air Filters"
8-12 thru 8-14	Revised "Deicing and Anti-icing of Airplane on the Ground" and Shifted Data
8-15	Revised "Consumable Materials"
8-16	Revised "Lamp Replacement Guide"
8-17 thru 8-24	Deleted Pages
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Beech Super King Air® B200/B200C

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P/N: 101-590010-147C

Log of Revisions

"C4" Revision ..... April, 1997

Page	Description
Title Page	Updated
Logo Page	Updated
Page C (C4)	New
3-4	Shifted Material
3-5	Revised "Environmental System Smoke or Fumes"
3-6	Removed Duplicate "Glide" Step
3-7	Removed Duplicate "Emergency Descent" Procedure
6-6 and 6-25	Revised "Loading Data Cargo Configuration"

C4

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BEECHCRAFT® SUPER KING AIR® B200/B200C

(SERIALS BB-734, BB-793, BB-829, BB-854 THRU BB-870, BB-874 THRU BB-891, BB-894, BB-896 THRU BB-911, BB-913 THRU BB-1438, BB-1440 THRU BB-1443; AND BL-37 THRU BL-138)

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"C3" Revision . . . . . July, 1996

Page	Description
Title Page	Updated
Page A (C3)	New
2-2	Revised Table of Contents
2-14	Added Data (ADDITIONAL ICING LIMITATIONS REQUIRED BY FAA AD 96-09-13)
3A-1	Revised Table of Contents
3A-10	Deleted, Shifted Data
3A-11	Added Data (ICING PROCEDURES REQUIRED BY FAA AD 96-09-13)
4-23	Revised Data (ICING FLIGHT)

C3

For Training Purposes Only

BEECHCRAFT® SUPER KING AIR® B200/B200C

(SERIALS BB-734, BB-793, BB-829, BB-854 THRU BB-870, BB-874 THRU BB-891, BB-894, BB-896 THRU BB-911, BB-913 THRU BB-1438, BB-1440 THRU BB-1443, AND BL-37 THRU BL-138)

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"C2" Revision . . . . . April, 1996

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Title Page	Updated
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1 of 2	New
2 of 2	New
1-10	Revised Data (MAXIMUM CERTIFICATED WEIGHTS)
2-1, 2-2	Revised Table of Contents
2-11	Revised Data (WEIGHT LIMITS)
2-12	Revised Data (ICING LIMITATIONS)
2-13	Revised Data (STRUCTURAL LIMITATIONS)
2-14 thru 2-14B	Shifted Data
2-15	Deleted Placard
2-16	Shifted Data
3-3	Revised Procedure (EMERGENCY AIRSPEEDS)
3-4	Revised Procedure (ENGINE FAILURE DURING GROUND ROLL)
3-6	Revised Procedure (CABIN OR CARGO UNLOCKED)
3A-7	Revised Procedure (LANDING GEAR MANUAL EXTENSION HYDRAULIC SYSTEM)
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"C2" Revision

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3A-8	Revised Procedure (LANDING GEAR MANUAL EXTENSION MECHANICAL SYSTEM)
4-1	Revised Table of Contents
4-27	Revised Procedures (PRACTICE LANDING GEAR MANUAL EXTENSION MECHANICAL SYSTEM, PRACTICE LANDING GEAR MANUAL EXTENSION HYDRAULIC SYSTEM, LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION HYDRAULIC SYSTEM)
4-28, 4-29	Shifted Data
6-5	Revised Data (DIMENSIONAL AND LOADING DATA)
6-6	Revised Data (LOADING DATA)
6-24	Revised Data (DIMENSIONAL AND LOADING DATA)
6-25	Revised Data (LOADING DATA)
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"C1" Revision . . . . . September, 1995

Page	Description
Title Page	Updated
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2-1, 2-2	Revised Table of Contents
2-10	Shifted Data
2-11	Revised Data (WEIGHT LIMITS)
2-12	Revised Data (ICING LIMITATIONS)
2-13, 2-14	Shifted Data
3A-1	Revised Table of Contents
3A-9	Revised Heading (ICE PROTECTION SYSTEMS)
3A-10	Added Procedure (OPERATIONS OUTSIDE THE FAR 25, APPENDIX C ICING ENVELOPE)
3A-11	Revised (CAUTION)
4-23	Revised Data (ICING FLIGHT)
4-24	Revised Data (IN FLIGHT)

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BEECHCRAFT® SUPER KING AIR® B200/B200C

SERIALS BB-734, BB-793, BB-829, BB-854 THRU BB-870, BB-874 THRU BB-891, BB-894, BB-896 THRU BB-911, BB-913 AND AFTER, BL-37 AND AFTER

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL

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INTRODUCTION

The format and contents of this Pilot's Operating Handbook and FAA Approved Airplane Flight Manual conform to GAMA (General Aviation Manufacturers Association) Handbook Specification Number 1 as revised: September 1, 1984. Use of this specification by all manufacturers will provide the pilot with the same type of data in the same place in all handbooks.

Attention is called to Section X (SAFETY INFORMATION). BEECHCRAFT feels that it is highly important to have Safety Information in condensed form in the hands of the pilots. The Safety Information should be read and studied. Periodic review will serve as a reminder of good piloting techniques.

WARNING

Use only genuine BEECHCRAFT or BEECHCRAFT approved parts obtained from BEECHCRAFT approved sources, in connection with the maintenance and repair of Beech airplanes.

Genuine BEECHCRAFT parts are produced and inspected under rigorous procedures to ensure airworthiness and suitability for use in Beech airplane applications. Parts purchased from sources other than BEECHCRAFT, even though outwardly identical in appearance, may not have had the required tests and inspections performed, may be different in fabrication techniques and materials, and may be dangerous when installed in an airplane.

Salvaged airplane parts, reworked parts obtained from non-BEECHCRAFT approved sources, or parts, components, or structural assemblies, the service history of which is unknown or cannot be authenticated, may have been subjected to unacceptable stresses or temperatures or have other hidden damage, not discernible through routine visual or usual nondestructive testing techniques. This may render the part, component, or structural assembly, even though originally manufactured by BEECHCRAFT, unsuitable and unsafe for airplane use.

BEECHCRAFT expressly disclaims any responsibility for malfunctions, failures, damage or injury caused by use of non-BEECHCRAFT approved parts.

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SUPER KING AIR B200/B200C
PILOT'S OPERATING HANDBOOK
AND
FAA APPROVED AIRPLANE FLIGHT MANUAL

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SECTION III.....EMERGENCY PROCEDURES
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SECTION V.....PERFORMANCE
SECTION VI.....WEIGHT AND BALANCE/EQUIPMENT LIST
SECTION VII.....SYSTEMS DESCRIPTION
SECTION VIII.....HANDLING, SERVICING AND MAINTENANCE
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SECTION X.....SAFETY INFORMATION

For Training Purposes Only

SECTION I
GENERAL
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IMPORTANT NOTICE

This handbook should be read carefully by the owner and the operator in order to become familiar with the operation of the airplane. Suggestions and recommendations have been made within it to aid in obtaining maximum performance without sacrificing economy. Be familiar with, and operate the airplane in accordance with, *the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* and/or placards which are located in the airplane.

As a further reminder, the owner and the operator should also be familiar with the Federal Aviation Regulations applicable to the operation and maintenance of the airplane, and FAR Part 91, General Operating and Flight Rules. Further, the airplane must be operated and maintained in accordance with FAA Airworthiness Directives which may be issued against it.

The Federal Aviation Regulations place the responsibility for the maintenance of this airplane on the owner and the operator, who should ensure that all maintenance is done by qualified mechanics in conformity with all airworthiness requirements established for this airplane.

All limits, procedures, safety practices, time limits, servicing, and maintenance requirements contained in this handbook are considered mandatory for continued airworthiness and to maintain the airplane in a condition equal to that of its original manufacture.

Raytheon Aircraft Authorized Outlets can provide recommended modification, service, and operating procedures issued by both the FAA and Raytheon Aircraft Company, which are designed to get maximum utility and safety from the airplane.

USE OF THE HANDBOOK

WARNINGS, CAUTIONS, AND NOTES

The following definitions apply to (WARNINGS), (CAUTIONS), and (NOTES) found throughout the handbook:

WARNING

Operating procedures, techniques, etc., which could result in personal injury or loss of life if not carefully followed.

CAUTION

Operating procedures, techniques, etc., which could result in damage to equipment if not carefully followed.

NOTE

An operating procedure, technique, etc., which is considered essential to emphasize.

REVISING THE HANDBOOK

The *Pilot's Operating Handbook* is designed to facilitate maintaining the documents necessary for the safe and efficient operation of the airplane. The handbook has been prepared in loose leaf form for ease in maintenance. It incorporates quick-reference tabs imprinted with the title of each section.

NOTE

In an effort to provide as complete coverage as possible, applicable to any configuration of the airplane, some optional equipment has been included in the scope of the handbook. However, due to the variety of airplane appointments and arrangements available, optional equipment described or depicted herein may not be designated as such in every case.

Immediately following the Title Page is a List Of Effective Pages. A complete listing of all pages is presented along with the current status of the material contained; i.e. Original, Reissued or Revised. Also, in the lower right corner of the blocked portion is a box containing a capital letter which denotes reissue of the manual. It will be advanced one letter, alphabetically, per reissue. A reissue of the manual or the revision of any portion will be received with a new List Of Effective Pages to replace the previous one. Reference to the List Of Effective Page(s) enables the user to determine the current issue, revision, or reissue in effect for each page in the handbook, (except for the Supplements Section).

When the handbook is originally issued, and each time it is revised or reissued, a new Log Of Revisions page is provided, immediately following the List Of Effective Pages. All Log Of Revisions pages must be retained until the handbook is reissued. A capital letter in the lower right corner of the Log Of Revisions page designates the Original Issue ("A") or reissue ("B", "C", etc.) covered by the Log page. If a number follows the letter, it designates the sequential revision (1st, 2nd, 3rd, etc.,) to the Original Issue or reissue covered by the Log Of Revisions page. Reference to the Log Of Revisions page(s) enables the user to determine the current issue, revision, or reissue in effect for each page in the handbook (except for the Supplements Section), and provides a record of changes made since the Original Issue or the latest reissue.

That portion of the text or an illustration which has been revised by the addition of, or a change in, information is denoted by a solid revision bar located adjacent to the area of change and placed along the outside margin of a page.

REVISION SERVICE

The following publications will be provided, at no charge, to the registered owner/operator of this airplane:

1. Reissues and revisions of Pilot's Operating Manual and FAA Approved Airplane Flight Manual.
2. Original issues and revisions of FAA Approved Airplane Flight Manual Supplements.
3. Original issues and revisions of Raytheon Aircraft Service Bulletins.

The above publications will be provided to the registered owner/operator at the address listed on the FAA Aircraft Registration Branch List or the Raytheon Aircraft Domestic/International Owners Notification List. Further, the owner/operator will receive only those publications pertaining to the registered airplane serial number. For detailed information on how to obtain "Revision Service" applicable to this manual or other Raytheon Aircraft Service Publications, consult any Raytheon Aircraft authorized outlet, or refer to the latest revision of Raytheon Aircraft Service Bulletin No. 2001.

Raytheon Aircraft Corporation expressly reserves the right to supersede, cancel, and/or declare obsolete, without prior notice, any part, part number, kit, or publication referenced in this manual.

The owner/operator should always refer to all supplements, whether STC Supplements or Raytheon Aircraft Supplements, for possible placards, limitations, emergency, abnormal, normal, and other operational procedures for proper operation of the airplane with optional equipment installed.

WARNING

It shall be the responsibility of the owner/operator to ensure that the latest revisions of publications referenced in this manual are utilized during operation, servicing, and maintenance of the airplane.

AIRPLANE FLIGHT MANUAL SUPPLEMENTS REVISION RECORD

Section IX contains the FAA Approved Airplane Flight Manual Supplements headed by a Log of Supplements page. On the Log page is a listing of the FAA Approved Supplemental Equipment available for installation on the airplane. When new supplements are received or existing supplements are revised, a new Log page will replace the previous one, since it contains a listing of all previous approvals, plus the new approval. The supplemental material will be added to the Section in accordance with the sequence specified on the Log page.

NOTE

Upon receipt of a new or revised supplement, compare the existing Log of Supplements in the handbook with the corresponding applicable Log page accompanying the new or revised supplement. It may occur that the Log page already in the handbook is dated later than the Log page accompanying the new or revised supplement. In any case, retain the Log page having the later date in the folio at the bottom-left corner of the page, and discard the older Log page.

SUPPLIER-ISSUED STC SUPPLEMENTS

When a new airplane is delivered from the factory, the handbook delivered with it contains either an STC (Supplemental Type Certificate) Supplement or a Raytheon Aircraft Flight Manual Supplement for every installed item requiring a supplement. If a new handbook for operation of the airplane is obtained at a later date, it is the responsibility of the owner/operator to ensure that all required STC Supplements (as well as weight and balance and other pertinent data) are transferred into the new handbook.

For Training Purposes Only

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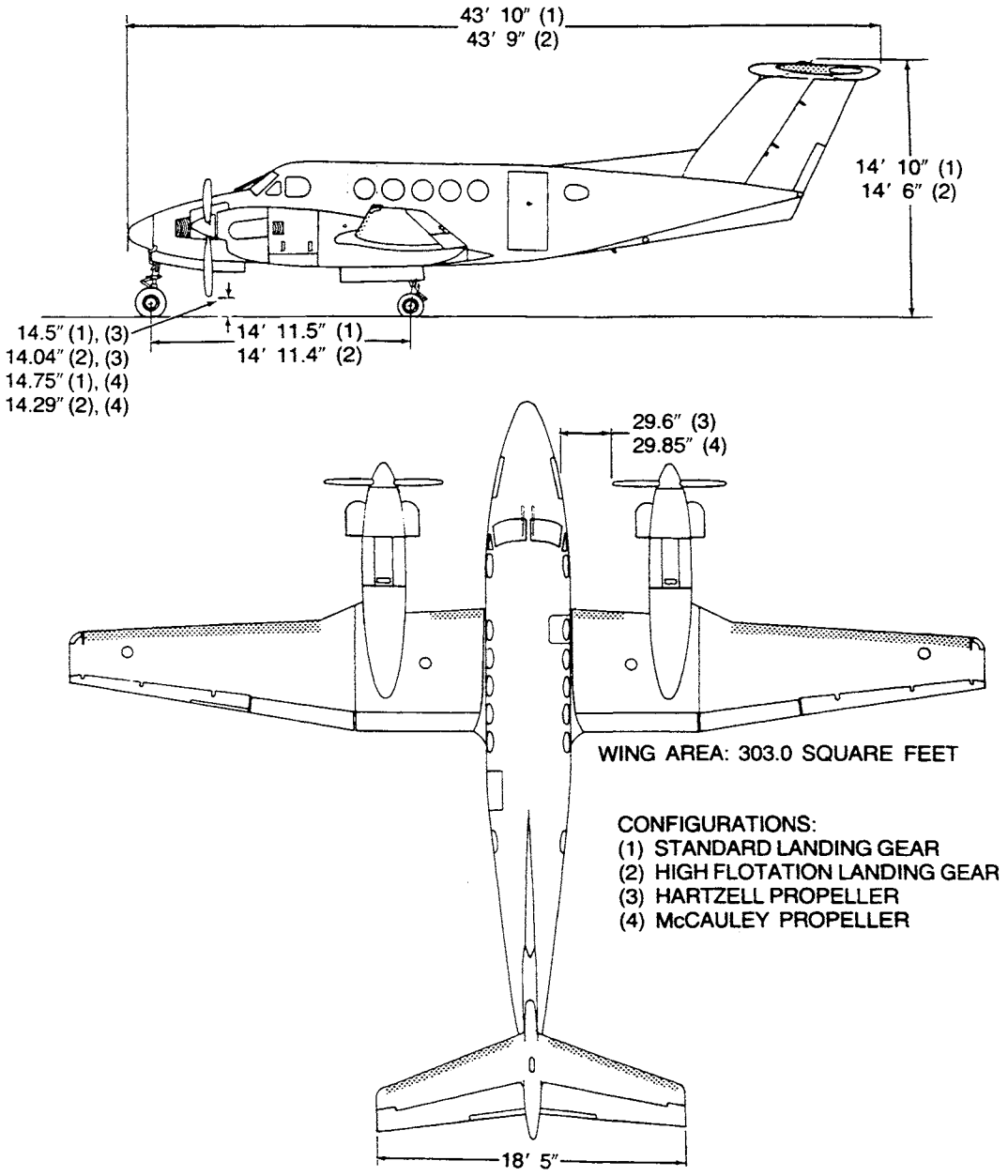
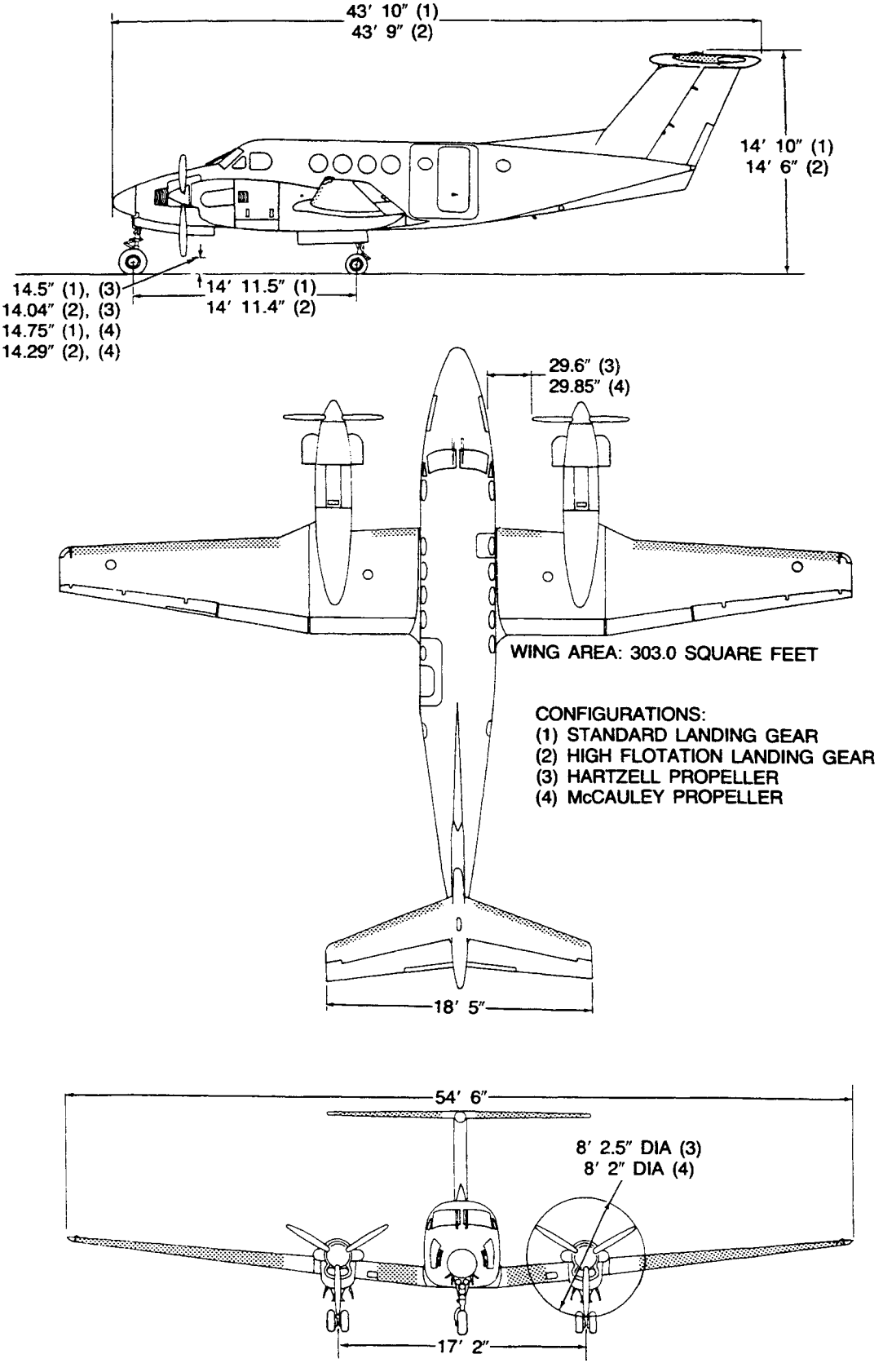


B200 THREE VIEW

200-607-1
001



B200C THREE VIEW

200-607-8
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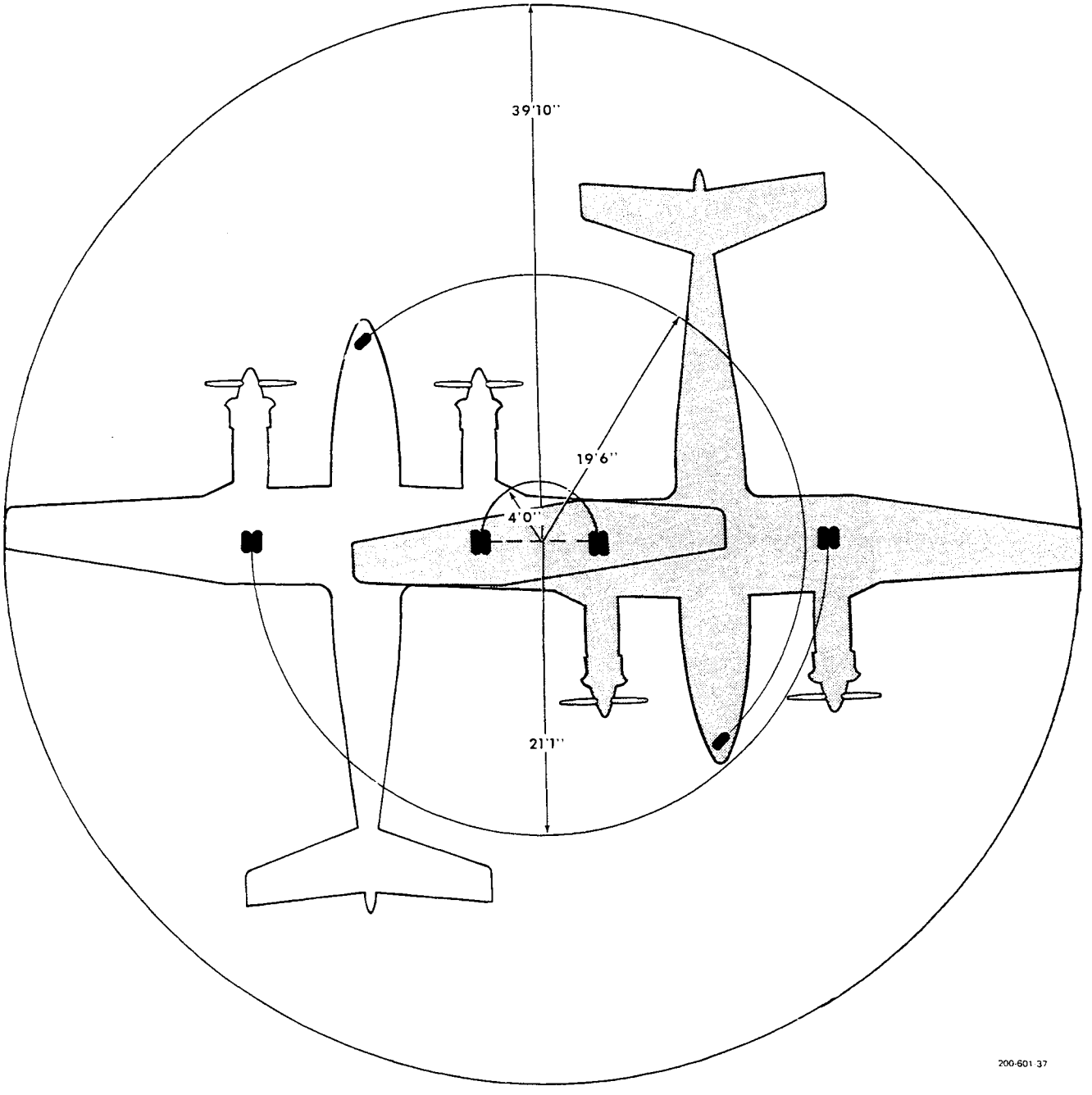
GROUND TURNING CLEARANCE



Radius for Inside Gear	4 feet	0 inches
Radius for Nose Wheel	19 feet	6 inches
Radius for Outside Gear	21 feet	1 inch
Radius for Wing Tip	39 feet	10 inches

TURNING RADII ARE PREDICATED ON THE USE OF PARTIAL BRAKING ACTION AND DIFFERENTIAL POWER

DESCRIPTIVE DATA

ENGINES

NUMBER OF ENGINES

Two

ENGINE MANUFACTURER

Pratt & Whitney Canada (Longueuil, Quebec, Canada)

ENGINE MODEL NUMBER

PT6A-42

ENGINE TYPE

Turbo-propeller Engine

NUMBER OF DRIVE SHAFTS

Two

1 Compressor (Gas Generator) Shaft

1 Power Turbine Shaft

COMPRESSOR STAGES AND TYPES

3 Axial-flow Stages

1 Centrifugal-flow Stage

COMBUSTION CHAMBER TYPE

Annular

TURBINE STAGES AND TYPES

COMPRESSOR (GAS GENERATOR) TURBINE

Single-stage Axial-flow Reaction Turbine

POWER TURBINE

Two-stage Axial-flow Reaction Turbine

ENGINE SHAFT-HORSEPOWER RATING

850 SHP

COMPRESSOR (GAS GENERATOR) SHAFT ROTATIONAL SPEED (N_1) LIMITS

Take-off, Maximum Continuous, Cruise Climb, Maximum Cruise, Normal Cruise Power: 101.5% N_1 (38,100 rpm)

PROPELLER ROTATIONAL SPEED (N_2) LIMITS

Take-off, Maximum Continuous, Cruise Climb, Maximum Cruise, Normal Cruise Power: 2000 rpm

PROPELLERS

NUMBER OF PROPELLERS

Two

PROPELLER MANUFACTURER

Hartzell Propeller, Inc. (Piqua, Ohio)
McCauley Propeller (Vandalia, Ohio)

NUMBER OF BLADES

Three

PROPELLER DIAMETER

Hartzell Propeller - 98.5 inches

McCauley Propeller - 98.0 inches

PROPELLER TYPE

Constant-speed, Full-feathering, Reversing, Counter-weighted, Hydraulically Actuated

PITCH RANGE (30-INCH STATION)

Hartzell Propeller: Feathered +90°, Reverse -9°

McCauley Propeller: Feathered+ 86.8°, Reverse -10°

FUEL

APPROVED ENGINE FUELS

COMMERCIAL GRADES

Jet A, Jet A-1, Jet B

MILITARY GRADES

JP-4, JP-5, JP-8

EMERGENCY ENGINE FUELS (SEE LIMITATIONS SECTION FOR LIMITATIONS)

COMMERCIAL AVIATION GASOLINE GRADES

80 Red (Formerly 80/87)
91/98
100LL Blue*
100 Green (Formerly 100/130)
115/145 Purple

**In some countries, this fuel is colored Green and designated "100L."*

MILITARY AVIATION GASOLINE GRADES

80/87 Red
100/130 Green
115/145 Purple

USABLE FUEL

Main Fuel System	386 gallons
Auxiliary Fuel System	158 gallons
Maximum Usable Fuel Quantity	544 gallons

APPROVED FUEL ADDITIVES

Anti-ice additive conforming to Specification MIL-I-27686 or equivalent, is permissible. Use not less than 20 fl oz of additive per 260 gallons of fuel or more than 20 fl oz additive per 104 gallons of fuel.

For Training Purposes Only

Fuel biocide "BIOBOR JF" in concentrations of 135 ppm or 270 ppm. Refer to the *Super King Air 200 Series Maintenance Manual and Pratt & Whitney Service Bulletin No. 1244* for concentrations to use and for procedures for adding BIOBOR JF to the airplane fuel.

ENGINE OIL

SPECIFICATION

Any oil specified by brand name in the latest revision of *Pratt & Whitney Service Bulletin Number 3001.* Refer to "CONSUMABLE MATERIALS" in the HANDLING, SERVICING & MAINTENANCE Section.

OIL CAPACITY (EACH ENGINE)

Total ..... 14.2 U.S. quarts
Undrainable ..... 1.5 U.S. quarts
Refill Capacity ..... 12.7 U.S. quarts
Operating Range ..... Full to 4.0 U.S. Quarts low

MAXIMUM CERTIFICATED WEIGHTS

Maximum Ramp Weight ..... 12,590 pounds
Maximum Take-off Weight ..... 12,500 pounds
Maximum Landing Weight ..... 12,500 pounds
Maximum Zero Fuel Weight ..... 11,000 pounds
Maximum Weight in Baggage Compartment:

BB-1052, BB-1091 and after, BL-58 and after, and prior airplanes with Beech Kit #101-5068-1 installed:

When Equipped with Fold-up Seats ..... 510 pounds
When Not Equipped with Fold-up Seats ..... 550 pounds

or;

Prior to BB-1091, except BB-1052, and prior to BL-58 without Beech Kit #101-5068-1 installed:

When Equipped with Fold-up Seats ..... 370 pounds
When Not Equipped with Fold-up Seats ..... 410 pounds

CABIN AND ENTRY DIMENSIONS

Cabin Width (Maximum) ..... 54 inches
Cabin Length (Maximum between pressure bulkheads) ..... 22 feet
Cabin Height (Maximum) ..... 57 inches
Airstair Entrance Door Width (Minimum) (B200) ..... 26.75 inches
Airstair Entrance Door Height (Minimum) (B200) ..... 51.5 inches
Airstair Entrance Door Width (Minimum) (B200C) ..... 20.2 inches
Airstair Entrance Door Height (Minimum) (B200C) ..... 46 inches
Cargo Door Width (Minimum) (B200C) ..... 49 inches
Cargo Door Height (Minimum) (B200C) ..... 52 inches
Pressure Vessel Volume ..... 393 cubic feet
Potential Cargo-area Volume ..... 253 cubic feet

SPECIFIC LOADINGS

WING LOADING: 41.3 pounds per square foot

POWER LOADING: 7.4 pounds per shaft horsepower

SYMBOLS, ABBREVIATIONS, AND TERMINOLOGY

The following glossary is applicable within this handbook.

GENERAL AIRSPEED TERMINOLOGY

CAS	*Calibrated Airspeed* is the indicated airspeed of an airplane corrected for position and instrument error. Calibrated airspeed is equal to true airspeed in standard atmosphere at sea level.
GS	*Ground Speed* is the speed of an airplane relative to the ground.
IAS	*Indicated Airspeed* is the speed of an airplane as shown on the airspeed indicator when corrected for instrument error. IAS values published in this handbook assume zero instrument error.
KCAS	*Calibrated Airspeed* expressed in knots.
KIAS	*Indicated Airspeed* expressed in knots.
M	*Mach Number* is the ratio of true airspeed to the speed of sound.
TAS	*True Airspeed* is the airspeed of an airplane relative to undisturbed air, which is the CAS corrected for altitude, temperature, and compressibility.
V_1	*Take-off Decision Speed.*
V_2	*Take-off Safety Speed*
V_A	*Maneuvering Speed* is the maximum speed at which application of full available aerodynamic control will not overstress the airplane.
V_F	*Design Flap Speed* is the highest speed permissible at which wing flaps may be actuated.
V_{FE}	*Maximum Flap Extended Speed* is the highest speed permissible with wing flaps in a prescribed extended position.
V_{LE}	*Maximum Landing Gear Extended Speed* is the maximum speed at which an airplane can be safely flown with the landing gear extended.
V_{LO}	*Maximum Landing Gear Operating Speed* is the maximum speed at which the landing gear can be safely extended or retracted.
V_{MCA}	*Air Minimum Control Speed* is the minimum flight speed at which the airplane is directionally controllable as determined in accordance with Federal Aviation Regulations. The airplane certification conditions include one engine becoming inoperative and windmilling, a 5-degree bank towards the operative engine, take-off power on operative engine, landing gear up, flaps in the take-off position, and most rearward C.G. For some conditions of weight and altitude, stall can be encountered at speeds above V_{MCA} as established by the certification procedure described above, in which event stall speed must be regarded as the limit of effective directional control.
V_{MCG}	*Ground Minimum Control Speed*
V_{MO}/M_{MO}	*Maximum Operating Limit Speed* is the speed limit that may not be deliberately exceeded in normal flight operations. V is expressed in knots and M in Mach Number.
V_R	*Rotation Speed*
V_S	*Stalling Speed* or the minimum steady flight speed at which the airplane is controllable.
V_{SO}	*Stalling Speed* or the minimum steady flight speed at which the airplane is controllable in the *landing configuration.*
V_{SSE}	*Intentional One-Engine-Inoperative Speed* is a speed above both V_{MCA} and stall speed, selected to provide a margin of lateral and directional control when one engine is suddenly rendered inoperative. Intentional failing of one engine below this speed is not recommended.

For Training Purposes Only

V_X	*Best Angle-of-Climb Speed* is the airspeed which delivers the greatest gain of altitude in the shortest possible horizontal distance with gear and flaps up.
V_{XSE}	*One-Engine-Inoperative Best Angle-of-Climb Speed* is the airspeed which delivers the greatest gain in altitude in the shortest possible horizontal distance with one engine inoperative.
V_Y	*Best Rate-of-Climb Speed* is the airspeed which delivers the greatest gain in altitude in the shortest possible time with gear and flaps up.
V_{YSE}	*One-Engine-Inoperative Best Rate-of-Climb Speed* is the airspeed which delivers the greatest gain in altitude in the shortest possible time with one engine inoperative.

METEOROLOGICAL TERMINOLOGY

Altimeter Setting	Barometric Pressure corrected to sea level.
Indicated Pressure Altitude	The number actually read from an altimeter when the barometric subscale has been set to 29.92 inches of mercury (1013.2 millibars).
IOAT	*Indicated Outside Air Temperature* is the temperature value read from an indicator.
ISA	International Standard Atmosphere in which: (1) The air is a dry perfect gas; (2) The temperature at sea level is 15° Celsius (59° Fahrenheit); (3) The pressure at sea level is 29.92 inches of mercury (1013.2 millibars); (4) The temperature gradient from sea level to the altitude at which the temperature is -56.5° Centigrade (-69.7° Fahrenheit) is -0.00198° Centigrade (0.003566° Fahrenheit) per foot and zero above that altitude.
OAT	*Outside Air Temperature* is the free air static temperature obtained either from the temperature indicator (IOAT) adjusted for compressibility effects, or from ground meteorological sources.
Pressure Altitude	Altitude measured from standard sea-level pressure (29.92 in. Hg) by a pressure (barometric) altimeter. It is the indicated pressure altitude corrected for position and instrument error. In this handbook, altimeter instrument errors are assumed to be zero. Position errors may be obtained from the Altimeter Correction graphs.
Station Pressure	Actual atmospheric pressure at field elevation.
Temperature Compressibility Effects	An error in the indication of temperature caused by airflow over the temperature probe. The error varies, depending on altitude and airspeed.
Wind	The wind velocities recorded as variables on the charts of this handbook are to be understood as the headwind or tailwind components of the reported winds.

POWER TERMINOLOGY

Beta Range	The region of the Power Lever control which is aft of the Idle Stop and forward of reversing range where blade pitch angle can be changed without a change of gas generator rpm.
Cruise Climb	Is the maximum power approved for normal climb. These powers are torque or temperature (ITT) limited.
High Idle	Obtained by placing the Condition Lever in the High Idle Position. This limits the power operation to a minimum of 70% of N_1 rpm.
Low Idle	Obtained by placing the Condition Lever in the Low Idle position. This limits the power operation to a minimum of 56% of N_1 rpm.
Maximum Continuous Power	Is the highest power rating not limited by time. Use of this rating is intended for emergency situations at the discretion of the pilot.

Maximum Cruise Power — Is the highest power rating for cruise and is not time limited.

Reverse — Reverse thrust is obtained by lifting the Power Levers and moving them aft of the Beta range.

SHP — *Shaft Horsepower.*

Take-off Power — Is the maximum power rating and is limited to a maximum of 5 minutes operation. use of this rating should be limited to normal take-off operations and emergency situations.

CONTROL AND INSTRUMENT TERMINOLOGY

Condition Lever (Fuel Shut-off Lever) — The fuel shut-off lever actuates a valve in the fuel control unit which controls the flow of fuel at the fuel control outlet and regulates the idle range from Low to High Idle.

ITT (Interstage Turbine Temperature) — Eight probes wired in parallel indicate the temperature between the compressor and power turbines.

N_1 Tachometer (Gas Generator RPM) — The tachometer registers the rpm of the gas generator with 100% representing a gas generator speed of 37,500 rpm.

Power Lever (Gas Generator N_1 RPM) — This lever serves to modulate engine power from full reverse thrust to take-off. The position for idle represents the lowest recommended level of power for flight operation.

Propeller Control Lever (N_2 RPM) — This lever is used to control the rpm setting of the propeller governor. Movement of the lever results in an increase or decrease in propeller rpm. Propeller feathering is the result of lever movement beyond the detents at the low rpm (high pitch) end of the lever travel.

Propeller Governor — The propeller governor senses changes in rpm and hydraulically changes propeller blade angle to compensate for the changes in rpm. Constant propeller rpm is thereby maintained at the selected rpm setting.

Torquemeter — The torquemeter system indicates the shaft output torque. Differential pressure from a mechanism within the reduction gear case causes a bellows and servo system to indicate torque on a meter. Instrument readout is in foot-pounds.

GRAPH AND TABULAR TERMINOLOGY

Accelerate-Go — The distance to accelerate to *Take-off Decision Speed* (V_1), experience an engine failure, continue accelerating to lift-off, then climb and accelerate in order to achieve *Take-off Safety Speed* (V_2) at 35 feet above the runway.

Accelerate-Stop — The distance to accelerate to *Take-off Decision Speed* (V_1) and bring the airplane to a stop.

AGL — *Above Ground Level*

Best Angle of Climb — The best angle-of-climb speed is the airspeed which delivers the greatest gain of altitude in the shortest possible horizontal distance with gear and flaps up.

Best Rate of Climb — The best rate-of-climb speed is the airspeed which delivers the greatest gain of altitude in the shortest possible time with gear and flaps up.

Clearway — A clearway is an area beyond the airport runway not less than 500 feet wide, centrally located about the extended centerline of the runway, and under the control of the airport authorities. The clearway is expessed in terms of a clear plane, extending from the end of the runway with an upward slope not exceeding 1.25 percent, above which no object nor any terrain protrudes. However, threshold lights may protrude above the plane if their height above the end of the runway is 26 inches or less and if they are located to each side of the runway.

For Training Purposes Only

Climb Gradient — The ratio of the change in height during a portion of a climb, to the horizontal distance traversed in the same time interval.

Demonstrated Crosswind — The maximum 90-degree crosswind component for which adequate control of the airplane during takeoff and landing was actually demonstrated during certification. The value shown is not limiting.

MEA — *Minimum Enroute Altitude*

Net Gradient of Climb — The gradient of climb with the flaps in the take-off position, and the landing gear retracted. "Net" indicates that the actual gradients of climb have been reduced by .8% to allow for turbulence and pilot technique. The Net Gradient of Climb graphs are constructed so that the value(s) obtained using the airport pressure altitude and outside air temperature will be the average gradient from 35 ft above the runway up to 1500 ft above the runway.

Route Segment — A part of a route. Each end of that part is identified by:
(1) a geographic location; or
(2) a point at which a definite radio fix can be established.

Take-off Flight Path — The minimum gradient of climb required to clear obstacles in excess of 35 feet, measured horizontally from reference zero and vertically at the altitude above the runway. Reference zero is the point where the airplane has reached 35 feet above the runway as determined from the Accelerate-Go graphs.

WEIGHT AND BALANCE TERMINOLOGY

Approved Loading Envelope — Those combinations of airplane weight and center of gravity which define the limits beyond which, loading is not approved.

Arm — The distance from the center of gravity of an object to a line about which moments are to be computed.

Basic Empty Weight — The weight of an empty airplane including full engine oil and unusable fuel. This equals *empty weight* plus the weight of unusable fuel, and the weight of all the engine oil required to fill the lines and tanks. *Basic empty weight* is the basic configuration from which loading data is determined.

Center of Gravity — A point at which the weight of an object may be considered concentrated for weight and balance purposes.

CG Limits — The extreme *center of gravity* locations within which the airplane must be operated at a given weight.

Datum — A vertical plane perpendicular to the airplane longitudinal axis from which fore and aft (usually aft) measurements are made for weight and balance purposes.

Empty Weight — The weight of an empty airplane before any oil or fuel has been added. This includes all permanently installed equipment, fixed ballast, full hydraulic fluid, full chemical toilet fluid, and all other operating fluids full, except that the engines, tanks, and lines do not contain any engine oil or fuel.

Engine Oil — That portion of the engine oil which can be drained from the engine.

Jack Point — Points on the airplane identified by the manufacturer as suitable for supporting the airplane for weighing or other purposes.

Landing Weight — The weight of the airplane at landing touchdown.

Leveling Points — Those points which are used during the weighing process to level the airplane.

Maximum Weight — The largest weight allowed by design, structural, performance or other limitations.

Moment — A measure of the rotational tendency of a weight, about a specified line, mathematically equal to the product of the weight and the arm.

Payload	Weight of occupants, cargo and baggage.
Ramp Weight	The airplane weight at engine start assuming all loading is completed.
Station	The longitudinal distance from some point to the zero datum or zero fuselage station.
Take-off Weight	The weight of the airplane at lift-off from the runway.
Tare	The weight which may be indicated by a set of scales before any load is applied.
Unusable Fuel	The fuel remaining after consumption of usable fuel.
Usable Fuel	That portion of the total fuel which is available for consumption as determined in accordance with applicable regulatory standards.
Useful Load	The difference between the airplane ramp weight and the basic empty weight.
Zero Fuel Weight	The airplane ramp weight minus the weight of usable fuel on board.

For Training Purposes Only

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Section 2
Limitations

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Section 2
Limitations

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AIRSPEED LIMITATIONS

The limitations included in this section have been approved by the Federal Aviation Administration, and they must be observed in the operation of the BEECHCRAFT Super King Air B200/B200C.

AIRSPEED LIMITATIONS			
SPEED	**KCAS**	**KIAS**	**REMARKS**
Maneuvering Speed VA (12,500 pounds)	182	181	Do not make full or abrupt control movements above this speed.
Maximum Flap Extension Extended Speed VFE Approach Position - 40% Full Down Position - 100%	 200 155	 200 157	 Do not extend flaps or operate with flaps in prescribed position above these speeds.
Maximum Landing Gear Operating Speed VLO Extension Retraction	 182 164	 181 163	 Do not extend or retract landing gear above the speeds given.
Maximum Landing Gear Extended Speed VLE	182	181	Do not exceed this speed with landing gear extended.
Air Minimum Control Speed VMCA	91	86	This is the lowest airspeed at which the airplane is directionally controllable when one engine suddenly becomes inoperative and the other engine is at take-off power. (See definition in Section I.)
Maximum Operating Speed VMO MMO	260 .52 Mach	259	Do not exceed this airspeed or Mach Number in any operation.

AIRSPEED INDICATOR MARKINGS

MARKING*	KCAS VALUE OR RANGE	KIAS VALUE OR RANGE	SIGNIFICANCE
Red Line	91	86	Air Minimum Control Speed (V_{MCA})
White Arc	80 to 155	75 to 157	Full-flap Operating Range
Wide White Arc	80 to 102	75 to 99	Lower Limit is the Stalling Speed (V_{SO}) at maximum weight with Full Flaps (100%) and idle power.
Narrow White Arc	102 to 155	99 to 157	Lower Limit is the Stalling Speed (V_S) at maximum weight with Flaps Up (0%) and idle power. Upper limit is the maximum speed permissible with flaps extended beyond Approach (more than 40%).
White Triangle	200	200	Maximum Speed with Approach (40%) Flaps.
Blue Line	122	121	One-Engine-Inoperative Best Rate-of-Climb speed
Red & White Hash-Marked Pointer	260 KCAS (259 KIAS) or value equal to .52 Mach, whichever is lower.		Maximum Speed for any operation.

BT04953

* The airspeed indicator is marked in IAS values.

POWER PLANT LIMITATIONS

NUMBER OF ENGINES

Two

ENGINE MANUFACTURER

Pratt & Whitney Canada (Longueuil, Quebec, Canada)

ENGINE MODEL NUMBER

PT6A-42

POWER LEVERS

Do not lift power levers in flight.

ENGINE OPERATING LIMITS

The following limitations shall be observed. Each column presents limitations. The limits presented do not necessarily occur simultaneously. Refer to the Pratt & Whitney Engine Maintenance Manual for specific actions required if limits are exceeded.

SUPER KING AIR B200/B200C

TEMPORARY CHANGE
TO THE
PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL

P/N 101-590010-147CTC2

PUBLICATION AFFECTED:	Super King Air B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual P/N 101-590010-147C, Reissued December, 1994 or Subsequent
AIRPLANE SERIAL NUMBERS AFFECTED:	BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-1438, BB-1440 thru BB-1443; and BL-37 thru BL-138
DESCRIPTION OF CHANGE:	Provides a Limitation prohibiting the selection of Beta Range in flight.
FILING INSTRUCTIONS:	Insert this Temporary Change in the B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual following page 2-4 (LIMITATIONS Section), and retain until rescinded or replaced.

LIMITATIONS

POWER PLANT LIMITATIONS

POWER LEVERS

Do not lift the power levers in flight. Lifting the power levers in flight, or moving the power levers in flight below the flight idle position, could result in a nose-down pitch and a descent rate leading to aircraft damage and injury to personnel.

APPROVED BY:
A. C. Jackson
Raytheon Aircraft Company
DOA CE-2

For Training Purposes Only

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OPERATING CONDITION	SHP	TORQUE FT-LBS (1)	MAXIMUM OBSERVED ITT°C	GAS GENERATOR RPM N_1 RPM	GAS GENERATOR RPM N_1 %	PROP RPM N_2	OIL PRESS PSI(2)	OIL TEMP °C (3) (4)
STARTING	---	---	1000 (5)	---	---	---	---	-40(min)
LOW IDLE	---	---	750 (6)	21,000	56(min)	---	60(min)	-40 to 99
HIGH IDLE	---	---	---	---	(7)	---	---	-40 to 99
TAKEOFF AND MAX CONT	850	2230	800	38,100	101.5	2000	100 to 135	0 to 99
MAX CRUISE	850	2230 (8)	800	38,100	101.5	2000	100 to 135	0 to 99
CRUISE CLIMB AND REC (NORMAL) CRUISE	850	2230 (8)	770	38,100	101.5	2000	100 to 135	0 to 99
MAX REVERSE (9)	800	---	750	---	88	1900	100 TO 135	0 to 99
TRANSIENT	---	2750 (5)	850	38,500(10)	102.6(10)	2200(5)	---	0 TO 104 (11)

BT04952

FOOTNOTES:

(1) Torque limit applies within range of 1600 - 2000 propeller rpm (N_2). Below 1600 propeller rpm, torque is limited to 1100 ft-lbs.

(2)When gas generator speeds are above 27,000 rpm (72% N_1) and oil temperatures are between 60°C and 71°C, normal oil pressures are:

100 to 135 psi below 21,000 feet; 85 to 135 psi at 21,000 feet and above.

During extremely cold starts, oil pressure may reach 200 psi. Oil pressure between 60 and 85 psi is undesirable; it should be tolerated only for the completion of the flight, and then only at a reduced power setting not exceeding 1100 ft-lbs torque. Oil pressure below 60 psi is unsafe; it requires that either the engine be shut down, or that a landing be made at the nearest suitable airport, using the minimum power required to sustain flight. Fluctuations of plus or minus 10 psi are acceptable.

(3) A minimum oil temperature of 55°C is recommended for fuel heater operation at take-off power.

(4) Oil temperature limits are -40°C and 99°C. However, temperatures of up to 104°C are permitted for a maximum time of 10 minutes.

(5) These values are time limited to 5 seconds.

(6) High ITT at ground idle may be corrected by reducing accessory load and/or increasing N_1 rpm.

(7) At approximately 70% N_1.

(8) Cruise torque values vary with altitude and temperature.

(9) This operation is time limited to 1 minute.

(10) These values are time limited to 10 seconds.

(11) Values above 99°C are time limited to 10 minutes.

EXTERNAL POWER LIMITS

External power carts will be set to 28.0 - 28.4 volts and be capable of generating a minimum of 1000 amps momentarily and 300 amps continuously.

GENERATOR LIMITS

Maximum sustained generator load is limited as follows:

In Flight:

Sea Level to 31,000 feet altitude 100%

Above 31,000 feet altitude 88%

For Training Purposes Only

Ground Operation 85%

During ground operation, also observe the following limitations:

GENERATOR LOAD	MINIMUM GAS GENERATOR RPM - N_1	
	WITHOUT AIR CONDITIONING	* WITH AIR CONDITIONING
0 - 75%	56%	62%
75% - 80%	60%	62%
80% - 85%	65%	65%

** Right engine only.*

STARTER LIMITS

Use of the starter is limited to 40 seconds ON, 60 seconds OFF, 40 seconds ON, 60 seconds OFF, 40 seconds ON, then 30 minutes OFF.

FUEL LIMITS

APPROVED ENGINE FUELS

COMMERCIAL GRADES

Jet A, Jet A-1, Jet B

MILITARY GRADES

JP-4, JP-5, JP-8

EMERGENCY ENGINE FUELS

COMMERCIAL AVIATION GASOLINE GRADES

80 Red (Formerly 80/87)
91/98
100LL Blue**
100 Green (Formerly 100/130)
115/145 Purple

**In some countries, this fuel is colored Green and designated "100L".

MILITARY AVIATION GASOLINE GRADES

80/87 Red
100/130 Green
115/145 Purple

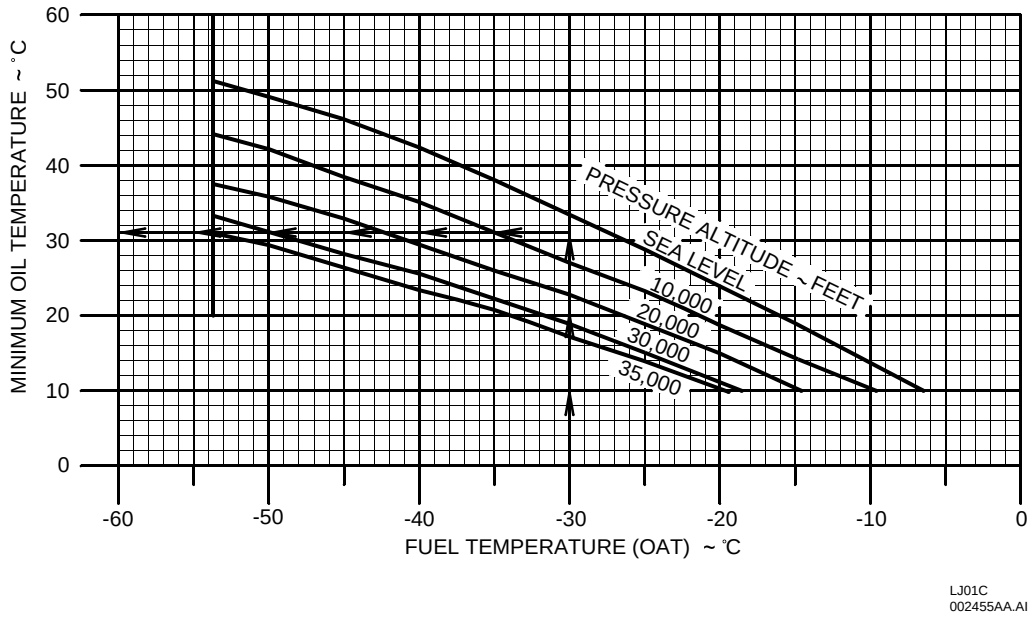
LIMITATIONS ON THE USE OF AVIATION GASOLINE

1. Operation is limited to 150 hours between engine overhauls.
2. Operation is limited to 20,000 feet pressure altitude (FL 200) or below if either standby pump is inoperative.
3. Crossfeed capability is required for climbs above 20,000 feet pressure altitude (FL 200).
4. Operation above 31,000 feet (FL 310) is prohibited.

APPROVED FUEL ADDITIVES

ANTI-ICING ADDITIVE

Engine oil is used to heat the fuel on entering the fuel control. Since no temperature measurement is available for the fuel at this point, it must be assumed to be the same as the OAT. The graph below is used to determine the minimum oil temperature required to maintain the fuel temperature above the freezing point of water, and thus prevent ice accumulations in the fuel control unit. Enter the graph at the known or forecast OAT and determine the minimum oil temperature required for each phase of flight. If the anticipated actual oil temperature is not equal to, or above this minimum temperature, anti-icing additive conforming to MIL-I-27686 or MIL-I-85470 must be added to the fuel. Refer to Section IV, NORMAL PROCEDURES, for blending procedures.



CAUTION

Prior to refueling, check with the fuel supplier to determine whether or not anti-icing additive has already been added to the fuel. If anti-icing additive is required, it must be properly blended with the fuel to avoid deterioration of the fuel cell sealant. The additive concentration shall be a minimum of 0.10% and a maximum of 0.15% by volume. To assure proper concentration by volume of fuel on board, blend only enough additive for the unblended fuel.

FUEL BIOCIDE ADDITIVE

Fuel biocide-fungicide BIOBOR JF in concentrations of 135 ppm or 270 ppm may be used in the fuel. BIOBOR JF may be used as the only fuel additive, or it may be used with the anti-icing additive conforming to MIL-I-27686 or MIL-I-85470 specification. Used together, the additives have no detrimental effect on the fuel system components.

Refer to the Super King Air 200 Series Maintenance Manual and to the latest revision of Pratt and Whitney Canada Engine Service Bulletin No. 3044 for concentrations to use and for procedures, recommendations and limitations pertaining to the use of biocidal/fungicidal additives in turbine fuels.

FUEL MANAGEMENT

USABLE FUEL (GALLONS X 6.7 = POUNDS)

Total Usable Fuel Quantity . . . . . 544 gallons (3645 pounds)
- Each Main Fuel Tank System . . . . . 193 gallons (1293 pounds)
- Each Auxiliary Fuel Tank. . . . . 79 gallons (529 pounds)

FUEL IMBALANCE

Maximum allowable fuel imbalance between wing fuel systems is 1000 pounds.

FUEL CROSSFEED

Crossfeeding of fuel is permitted only when one engine is inoperative.

WARNING

The airplane is approved for takeoff with one standby boost pump inoperative, but in such a case, crossfeed of fuel will not be available from the side of the inoperative standby boost pump.

FUEL GAGES IN THE YELLOW ARC

Do not take off if fuel quantity gages indicate in yellow arc or indicate less than 265 pounds of fuel in each main tank system.

AUXILIARY FUEL

Do not put any fuel into the auxiliary tanks unless the main tanks are full.

OPERATING WITH LOW FUEL PRESSURE

Operation of either engine with its corresponding fuel pressure annunciator (L FUEL PRESS or R FUEL PRESS) illuminated is limited to 10 hours before overhaul or replacement of the engine-driven fuel pump. Windmilling time need not be charged against this time limit.

OIL SPECIFICATION

Any oil specified by brand name in the latest revision of *Pratt & Whitney Service Bulletin Number 3001* is approved for use in the PT6A-42 engine.

NUMBER OF PROPELLERS

2

PROPELLER MANUFACTURER

Hartzell Propeller, Inc. (Piqua, Ohio)

McCauley Propeller (Vandalia, Ohio)

PROPELLER HUB AND BLADE MODEL NUMBERS

	HUBS	BLADES
Hartzell Propeller	Prior to BB-829, BL-37:	Prior to BB-829, BL-37:
	HC-B3TN-3G or HC-B3TN-3N	T10178B-3R
	BB-829 and After: BL-37 and After:	BB-829 and After: BL-37 and After:
	HC-B3TN-3G or HC-B3TN-3N	T10178K-3R
	or	
McCauley Propeller	HUBS	BLADES
	BB-1193 and After: BL-124 and After:	BB-1193 and After: BL-124 and After:
	3GFR34C702- ☒	☒ -100LA

NOTE

The letter appearing in the place of the ☒ represents minor variations in the propeller hub or blades. They do not affect eligibility or interchangeability.

Hartzell and McCauley propellers may be interchanged (in matched pairs only) on airplane serials BB-1193 and after, and BL-124 and after.

PROPELLER DIAMETER

Hartzell Propeller: 98.5 inches only

McCauley Propeller: 98.0 inches

PROPELLER BLADE ANGLES AT 30-INCH STATION

Hartzell Propeller: Feathered + 90°; Reverse - 9°

McCauley Propeller: Feathered + 86.8°; Reverse - 10°

PROPELLER ROTATIONAL SPEED LIMITS

Transients not exceeding 5 seconds ..... 2200 rpm
Reverse ..... 1900 rpm
All other conditions ..... 2000 rpm

PROPELLER ROTATIONAL OVERSPEED LIMITS

The maximum propeller overspeed limit is 2200 rpm and is time-limited to five seconds. Sustained propeller overspeeds faster than 2000 rpm indicate failure of the primary governor. Flight may be continued at propeller overspeeds up to 2080 rpm, provided torque is limited to 1800 foot-pounds. Sustained propeller overspeeds faster than 2080 rpm indicate failure of both the primary governor and the secondary governor, and such overspeeds are unapproved.

POWERPLANT INSTRUMENT MARKINGS

POWER PLANT INSTRUMENT MARKINGS			
INSTRUMENT	RED LINE MINIMUM LIMIT	GREEN ARC NORMAL OPERATING	RED LINE MAXIMUM LIMIT
INTERSTAGE TURBINE TEMPERATURE*	---	400°C to 800°C	800°C
TORQUE METER	---	400 ft-lbs to 2230 ft-lbs	2230 ft-lbs
PROPELLER TACHOMETER (N_2)	---	1600 rpm to 2000 rpm	2000 rpm
GAS GENERATOR TACHOMETER (N_1)	---	---	101.5%
OIL TEMPERATURE	---	10°C to 99°C	99°C
OIL PRESSURE**	60 psi	100 psi to 135 psi	200 psi

** Starting Limit (Dashed Red Line): 1000°C.*

*** A dual-band yellow/green arc extends from 85 to 100 psi, indicating the extended range of normal oil pressures for operation at, or above, 21,000 feet.*

MISCELLANEOUS INSTRUMENT MARKINGS

FUEL QUANTITY INDICATORS

Yellow Arc (No-Takeoff Range) ..... 0 to 265 pounds

CABIN DIFFERENTIAL PRESSURE GAGE

Green Arc (Approved Operating Range) ..... 0 to 6.6 psi
Red Arc (Unapproved Operating Range) ..... 6.6 psi to end of scale

PNEUMATIC GAGE

Green Arc (Normal Operating Range) ..... 12 to 20 psi
Red Line (Maximum Operating Limit) ..... 20 psi

GYRO SUCTION GAGE

Narrow Green Arc (Normal from 35,000 to 15,000 feet) ..... 2.8 to 4.3 in. Hg
Wide Green Arc (Normal from 15,000 feet to Sea Level) ..... 4.3 to 5.9 in. Hg
35K marked on face of gage at ..... 3.0 in. Hg
15K marked on face of gage at ..... 4.3 in. Hg

For Training Purposes Only

PROPELLER DEICE AMMETER

Green Arc (Normal Operating Range) ..... 14 to 18 amperes

WEIGHT LIMITS

Maximum Ramp Weight ..... 12,590 pounds
Maximum Take-off Weight:

All Except FAR Part 135 Operations ..... 12,500 pounds

FAR Part 135 Operations ..... As Limited by MAXIMUM ENROUTE WEIGHT Graph in Section V, PERFORMANCE

Maximum Landing Weight ..... 12,500 pounds
Maximum Zero Fuel Weight ..... 11,000 pounds
Maximum Weight in Baggage Compartment:

BB-1052, BB-1091 and after, BL-58 and after, and prior airplanes with Beech Kit #101-5068-1 installed:

When Equipped with Fold-up Seats ..... 510 pounds

When Not Equipped with Fold-up Seats ..... 550 pounds

or;

Prior to BB-1091, except BB-1052, and prior to BL-58 without Beech Kit #101-5068-1 installed:

When Equipped With Fold-up Seats ..... 370 pounds

When Not Equipped with Fold-up Seats ..... 410 pounds

CENTER OF GRAVITY LIMITS

AFT LIMIT

196.4 inches aft of datum at all weights

FORWARD LIMITS

185.0 inches aft of datum at 12,500 pounds, with straight line variation to 181.0 inches aft of datum at 11,279 pounds. 181.0 inches aft of datum at 11,279 pounds or less.

DATUM

The reference datum is located 83.5 inches forward of the center of the front jack point.

MEAN AERODYNAMIC CHORD (MAC)

The leading edge of the MAC is 171.23 inches aft of the datum.

The MAC length is 70.41 inches.

MANEUVER LIMITS

The BEECHCRAFT Super King Air B200 and B200C are Normal Category Airplanes. Acrobatic maneuvers, including spins, are prohibited.

FLIGHT LOAD FACTOR LIMITS

FLAPS UP	FLAPS DOWN
3.17 positive g's	2.00 positive g's
1.27 negative g's	0.00 g

For Training Purposes Only

MINIMUM FLIGHT CREW

One pilot

MAXIMUM OPERATING PRESSURE-ALTITUDE LIMITS

Normal Operation ..... 35,000 feet
Operation with Yaw Damp System Inoperative ..... 17,000 feet

MAXIMUM OUTSIDE AIR TEMPERATURE LIMITS

Sea Level to 25,000 feet pressure altitude ..... ISA+37°C
Above 25,000 feet pressure altitude ..... ISA+31°C

CABIN PRESSURIZATION LIMIT

Maximum Cabin Pressure Differential ..... 6.6 psi

MAXIMUM OCCUPANCY LIMIT

Including Crew ..... Fifteen (15)

SYSTEMS AND EQUIPMENT LIMITS

LANDING GEAR CYCLE LIMITS

Landing gear cycles (1 up - 1 down) are limited to one every 5 minutes for a total of 6 cycles followed by a 15 minute cool-down period.

AFT-FACING SEATS

Only aft-facing seats (placarded as such on the leg crossmember) are authorized in the aft-facing position. The seatback of each occupied aft-facing seat must be in the fully raised position for takeoff and landing. The headrest should be positioned properly for the occupant.

ICING LIMITATIONS

Minimum Ambient Temperature for Operation of Deicing Boots ..... -40°C
Minimum Airspeed for Sustained Icing Flight ..... 140 Knots

Sustained flight in icing conditions with flaps extended is prohibited except for approach and landings.

ICE VANES, LEFT and RIGHT, shall be extended for operations in ambient temperatures of +5°C or below when flight free of visible moisture cannot be assured.

ICE VANES, LEFT and RIGHT, shall be retracted for all takeoff and flight operations in ambient temperatures of above +15°C.

Once the manual override system is activated (i.e., anytime the ICE VANE EMERGENCY MANUAL EXTENSION handle has been pulled out), do not attempt to operate the ice vanes electrically until the override assembly inside the engine cowling has been properly reset on the ground. Even after the manual extension handle has been pushed back in, the manual override system is still engaged.

APPROVED AIRPLANE DEICING/ANTI-ICING FLUIDS

SAE AMS 1424 Type I
ISO 11075 Type I
SAE AMS 1428 Type II
ISO 11078 Type II

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Temporary Change to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual

P/N 101-590010-147CTC3

PUBLICATION AFFECTED:	Super King Air B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual P/N 101-590010-147C, Reissued December, 1994 or Subsequent
AIRPLANE SERIAL NUMBERS AFFECTED:	BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-1438, BB-1440 thru BB-1443; and BL-37 thru BL-138
DESCRIPTION OF CHANGE:	Replace Structural Limitations with reference to the *Super King Air 200 Series Airworthiness Limitations Manual*.
FILING INSTRUCTIONS:	Insert this Temporary Change into the B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual following page 2-12 (LIMITATIONS Section), and retain until rescinded or replaced.

LIMITATIONS

STRUCTURAL LIMITATIONS

Refer to the *Super King Air 200 Series Airworthiness Limitations Manual* for structural limitations.

FAA Approved by: Victoria E. Goff for

Randolph Shields, ODA Lead administrator
Hawker Beechcraft Corporation
ODA-230339-CE

SAE AMS 1428 Type IV. Only the following Type IV Fluids are approved:

Clariant Safewing MP IV 1957

Clariant Safewing MP IV 2001

UCAR ULTRA+ (Approved for use down to -15°C)

Octagon Max Flight Type IV

OTHER LIMITATIONS

STRUCTURAL LIMITATIONS (PRIOR TO S/N BB-1193 AND BL-73, EXCEPT BB-1158, BB-1167)

Cabin Door Forward and Aft Side Latches (or bayonets) (4) Safelife (B200 only) . . . . . 5,000 cycles

Cabin Door Upper Latch Hooks (2) and Attaching Hardware (B200 only) . . . . . 10,000 cycles

Cargo Door Cam-lock Actuator Cable Safelife (B200C only) . . . . . 8,000 cycles

Wing and Associated Structure Fatigue Safelife . . . . . 30,000 hours

Horizontal and Vertical Stabilizer . . . . . 45,000 hours

Windshield Frame Screws . . . . . 10,000 cycles

All Wing Attach Bolts, Nuts and Barrel Nut Assemblies:

Inconel Components . . . . . Replace every 15 calendar years of installed bolt and nut time

Refer to the Raytheon Aircraft Company Structural Inspection and Repair Manual and the Super King Air 200 Series Maintenance Manual for inspection and replacement procedures.

The above lives are based on airplane utilization, operation, and maintenance in the category of service for which the airplane was originally designed; specifically 95% or more time is accumulated from pressurized executive or corporate transportation wherein the majority of cruise is above 10,000 ft altitude (AGL) and flight duration is more than one hour. Should the airplane be used for missions other than that intended by design, notify the Customer Support Department of Raytheon Aircraft Company for a more appropriate life.

STRUCTURAL LIMITATIONS (S/N BB-1158, BB-1167, BB-1193 AND AFTER, BL-73 AND AFTER)

Refer to Chapter Four of the Super King Air 200 Series Maintenance Manual for structural limitations.

CRACKED OR SHATTERED WINDSHIELD

The following limitations apply when continued flight is required with a cracked outer or inner ply of the windshield.

1. Continued flight with a cracked windshield is limited to 25 flight hours.
2. Windshields which have a shattered inner ply will have numerous cracks which will obstruct forward vision and produce small particles or flakes of glass that can break free of the windshield and interfere with the crew's vision. These windshields must be replaced prior to the next flight unless a special flight permit is obtained from the local FAA Flight Standards District Office.
3. Crack(s) must not impair visibility.
4. Crack(s) must not interfere with the use of windshield wipers for flights requiring the use of the wipers.
5. Windshield anti-ice must be operational for flights in icing conditions.
6. The following placard must be installed in clear view of the pilot:

MAXIMUM AIRPLANE ALTITUDE IS LIMITED TO
25,000 FEET. CABIN ΔP MUST BE MAINTAINED
BETWEEN 2.0 AND 4.6 PSI DURING FLIGHT.

Windshields that have cracks in both the inner and outer plies must be replaced prior to the next flight unless a special flight permit is obtained from the local FAA Flight Standards District Office.

For Training Purposes Only

CRACK IN ANY SIDE WINDOW (COCKPIT OR CABIN)

The following limitations apply when continued flight is required with a cracked outer or inner ply in any side window. These limitations do not apply to minor compression-type chips (Clamshell) which may occur on the milled edge of cockpit side windows. Refer to the *Beech Super King Air 200 Series Maintenance Manual* for the disposition of such chips.

1. Continued flight is limited to 25 flight hours.
2. Flights must be conducted with the cabin depressurized. The following placard must be installed in clear view of the pilot:

PRESSURIZED FLIGHT IS PROHIBITED DUE TO A
CRACKED SIDE WINDOW. CONDUCT FLIGHT WITH THE
CABIN PRESSURE SWITCH IN THE DUMP POSITION.

CARGO LIMITATIONS (ALL SERIALS)

1. All cargo shall be properly secured by an FAA-approved cargo restraint system.
2. Cargo must be arranged to permit free access to all exits and emergency exits.

LIMITATIONS WHEN ENCOUNTERING SEVERE ICING CONDITIONS (Required By FAA AD 96-09-13)

WARNING

Severe icing may result from environmental conditions outside of those for which the airplane is certificated. Flight in freezing rain, freezing drizzle, or mixed icing conditions (supercooled liquid water and ice crystals) may result in ice build-up on protected surfaces exceeding the capability of the ice protection system, or may result in ice forming aft of the protected surfaces. This ice may not be shed using the ice protection systems, and may seriously degrade the performance and controllability of the airplane.

1. During flight, severe icing conditions that exceed those for which the airplane is certificated shall be determined by the following visual cues. If one or more of these visual cues exists, immediately request priority handling from Air Traffic Control to facilitate a route or an altitude change to exit the icing conditions.
   a. Unusually extensive ice accreted on the airframe in areas not normally observed to collect ice.
   b. Accumulation of ice on the upper surface of the wing aft of the protected area.
   c. Accumulation of ice on the propeller spinner farther aft than normally observed.
2. Since the autopilot may mask tactile cues that indicate adverse changes in handling characteristics, use of the autopilot is prohibited when any of the visual cues specified above exist, or when unusual lateral trim requirements or autopilot trim warnings are encountered while the airplane is in icing conditions.
3. All icing detection lights must be operative prior to flight into icing conditions at night. [NOTE: This supersedes any relief provided by the Master Minimum Equipment List (MMEL).]

For Training Purposes Only

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PLACARDS

On Overhead Panels in Pilot's Compartment:

OPERATION LIMITATIONS

THIS AIRPLANE MUST BE OPERATED AS A NORMAL CATEGORY AIRPLANE IN COMPLIANCE WITH THE OPERATING LIMITATIONS STATED IN THE FORM OF PLACARDS MARKINGS AND MANUALS.
NO ACROBATIC MANEUVERS INCLUDING SPINS ARE APPROVED.
THIS AIRPLANE APPROVED FOR VFR, IFR, & DAY & NIGHT OPERATION & IN ICING CONDITIONS.

CAUTION

STALL WARNING IS INOPERATIVE WHEN MASTER SWITCH IS OFF
STANDBY COMPASS IS ERRATIC WHEN WINDSHIELD ANTI-ICE AND/OR AIR-CONDITIONING IS ON.

C9200601 C

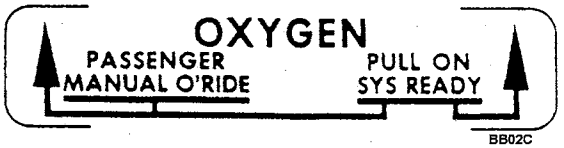
Aft of Overhead Light Control Panel:

AIRSPEEDS (IAS)

MAX GEAR EXTENSION	181 KNOTS
MAX GEAR RETRACT	163 KNOTS
MAX GEAR EXTENDED	181 KNOTS
MAX APPROACH FLAP	200 KNOTS
MAX FULL DOWN FLAP	157 KNOTS
MAX MANEUVERING	181 KNOTS

C9200602 C

Aft of Overhead Light Control Panel:



BB02C
991200AA

On Pilot's Window Sill:

TURN OFF STROBE LIGHTS WHEN TAXIING IN VICINITY OF OTHER AIRCRAFT OR DURING FLIGHT THROUGH FOG OR CLOUDS. STD POSITION LIGHTS TO BE TURNED ON FOR ALL NIGHT OPERATIONS.

C9200604 C

On Pilot's & Copilot's Window Sills:

WARNING

DO NOT SMOKE WHILE OXYGEN IS IN USE.
WITH DILUTER-DEMAND CREW MASK
FLOW STOPS WHEN BREATHING STOPS.

C9200605 C

On Curved Pedestal Adjacent to Power Levers:

CAUTION
REVERSE
ONLY WITH
ENGINES
RUNNING
C9200607 C

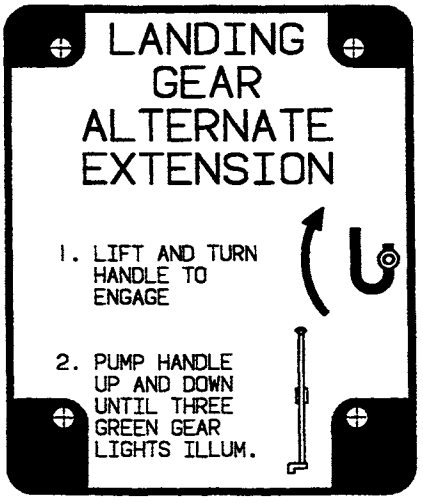
On Pedestal Near Cabin Pressurization Controller:

WARNING DE-PRESSURIZE CABIN
BEFORE LANDING
C9200609 C

Below Latch on Forward Side of Forward and Aft Partitions (B200):
Below Latch on Forward Side of Forward Partition and on Aft Side of Aft Partition (B200C):

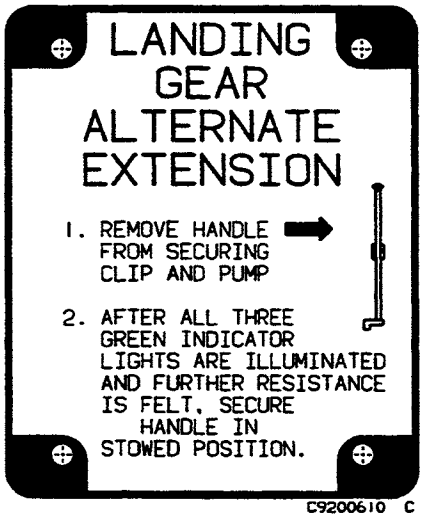
KICK HERE
FOR EMERGENCY EVACUATION
C9200611 C

On Floor Aft of Pedestal:
(On airplanes prior to BL-73 and BB-1193 (except BB-1158 and BB-1167) not incorporating Beech Kit P/N 101-8018-1):

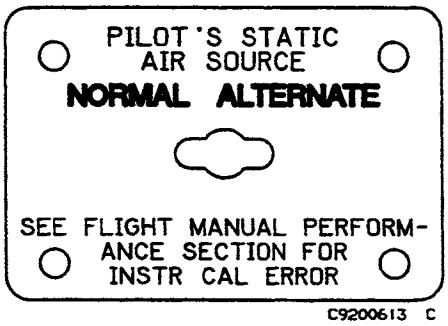


C958802C1339 C

On Floor Aft of Pedestal:
(On airplanes BB-1158, BB-1167, BB-1193 and after, BL-73 and after, and those prior airplanes incorporating Beech Kit P/N 101-8018-1):

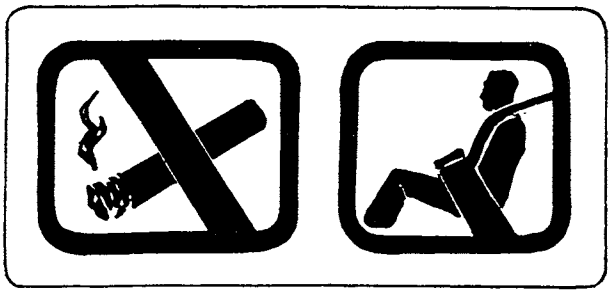


Below Right Circuit Breaker Panel:

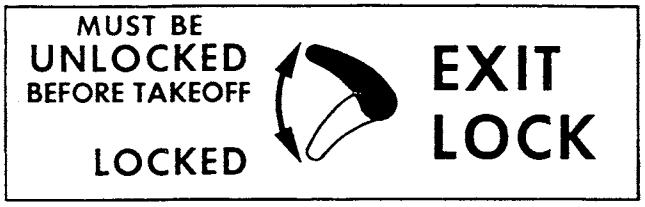


Center of Cabin on Passenger Servicing Unit (B200C):

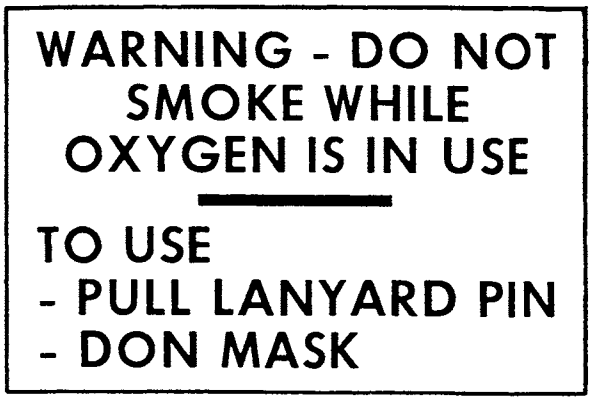
On Cabin Side of Forward and Aft Partitions (B200):



Below Emergency Exit Handle on Exit Lock Lever (Visible only when Lever is in Locked Position) (B200):

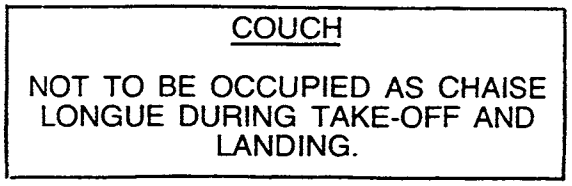


On Inside Surface of Each Oxygen Access Door:

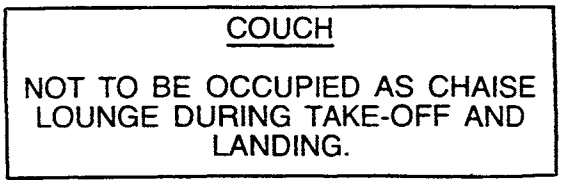


On Emergency Exit Handle:

On Window Frame Escutcheon Adjacent to Front Seats (airplanes equipped with a chaise lounge):

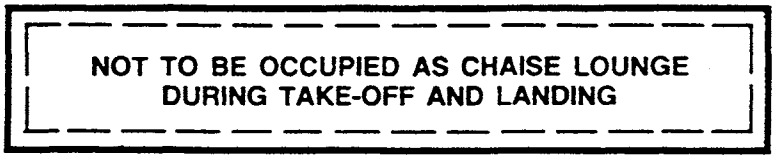


or:



or:

Stitched to the Aft End of the Couch Cushion (airplanes equipped with a chaise lounge):



On Inside Bottom of Couch Drawer (airplanes equipped with a two-placed couch):

COUCH LIMITATIONS
TOTAL WEIGHT OF OCCUPANTS
NOT TO EXCEED 340 LBS
MAX WT OF DRAWER CONTENTS
30 LBS PER DRAWER

Or:

COUCH LIMITATIONS
MAX WT OF DRAWER CONTENTS
30 LBS PER DRAWER

On Shoulder Harness Assembly For All Chairs:

SHOULDER HARNESS MUST BE WORN DURING TAKE-OFF AND LANDING WITH SEAT IN OUTBD POSITION, SEAT BACK UPRIGHT AND HEADREST FULLY EXTENDED

On Cupholders and Tables When Installed:

STOW BEFORE TAKE-OFF AND LANDING

On Trim Above Side Facing Seat: (Forward of Baggage Area)

FASTEN SEAT BELT DURING TAKEOFF AND LANDING

On Window Frame Adjacent to Lateral-Tracking Seats When Installed:

SEAT MUST BE LOCATED IN OUTBOARD POSITION FOR TAKEOFF AND LANDING

On Outside Surface of First Aid Oxygen Access Door:

FIRST AID OXYGEN ——— PULL

On Inside Surface of First Aid Oxygen Access Door:

WARNING - DO NOT SMOKE WHILE OXYGEN IS IN USE

TO USE
- TURN VALVE ON
- DON MASK

NOTE: CREW SYS MUST BE ON

On Aft Compartment Sidewall Adjacent to Fold-Up Seats:

FASTEN SHOULDER HARNESS AND SEAT BELT DURING TAKE OFF AND LANDING

On Center of Aft Bulkhead (When Not Equipped with Fold-up Seats): BB-1052, BB-1091, and after, BL-58 and after, and prior airplanes with Beech Kit No. 101-5068-1 or -5 installed:

COMPARTMENT CAPACITY
550 LBS
THIS INCLUDES BAGGAGE AND/OR EQUIPMENT AND/OR OPTIONAL FURNISHINGS
SEE WEIGHT AND BALANCE SECTION OF FLIGHT MANUAL FOR LOADING INSTRUCTIONS

Or:

Prior to BB-1052, BB-1091 and BL-58:

COMPARTMENT CAPACITY
410 LBS
THIS INCLUDES BAGGAGE AND/OR EQUIPMENT AND/OR OPTIONAL FURNISHINGS
SEE WEIGHT AND BALANCE SECTION OF FLIGHT MANUAL FOR LOADING INSTRUCTIONS

On Center of Aft Bulkhead (When Equipped with Fold-up Seats): BB-1052, BB-1091 and after, BL-58 and after, and prior airplanes with Beech Kit No. 101-5068-3 installed:

COMPARTMENT CAPACITY
TOTAL COMPARTMENT CAPACITY (INCLUDING BAGGAGE AND/OR PASSENGER) NOT TO EXCEED 510 POUNDS
SEE WEIGHT AND BALANCE SECTION OF FLIGHT MANUAL FOR LOADING INSTRUCTIONS

Or:

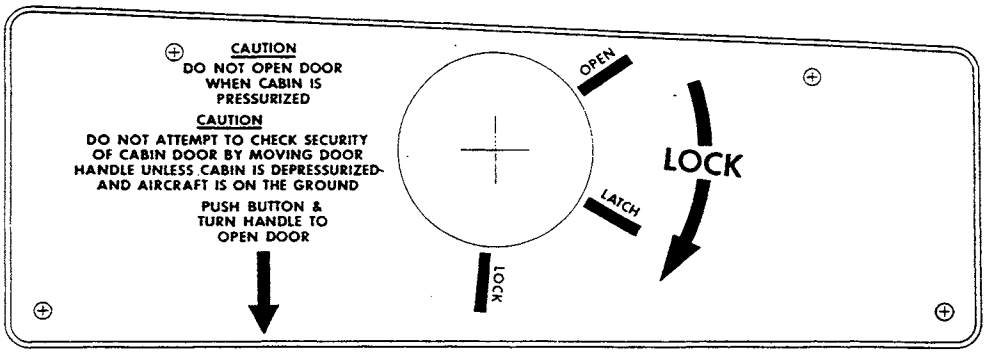
Prior to BB-1052, BB-1091 and BL-58:

COMPARTMENT CAPACITY
TOTAL COMPARTMENT CAPACITY (INCLUDING BAGGAGE AND/OR PASSENGER) NOT TO EXCEED 370 POUNDS
SEE WEIGHT AND BALANCE SECTION OF FLIGHT MANUAL FOR LOADING INSTRUCTIONS

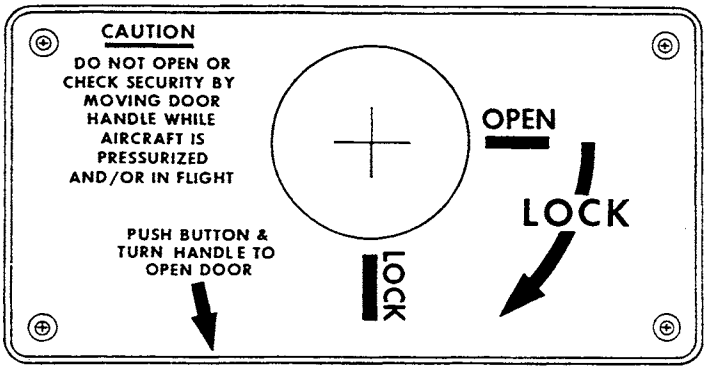
Above Coat Rod: (Aft Compartment):

CLOTHING ONLY

Inside Airstair Door Behind Handle (B200):

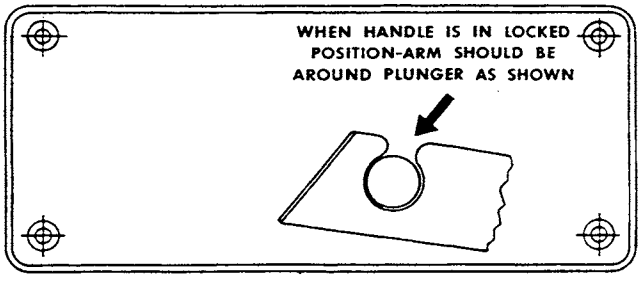


Inside Airstair Door Behind Handle (B200C):



Inside Airstair Door On Folding Step (B200 and B200C):

Inside Airstair Door Between Folding Steps (B200 and B200C):



On Escutcheon Near Upper Access Cover of Cargo Door (B200C):

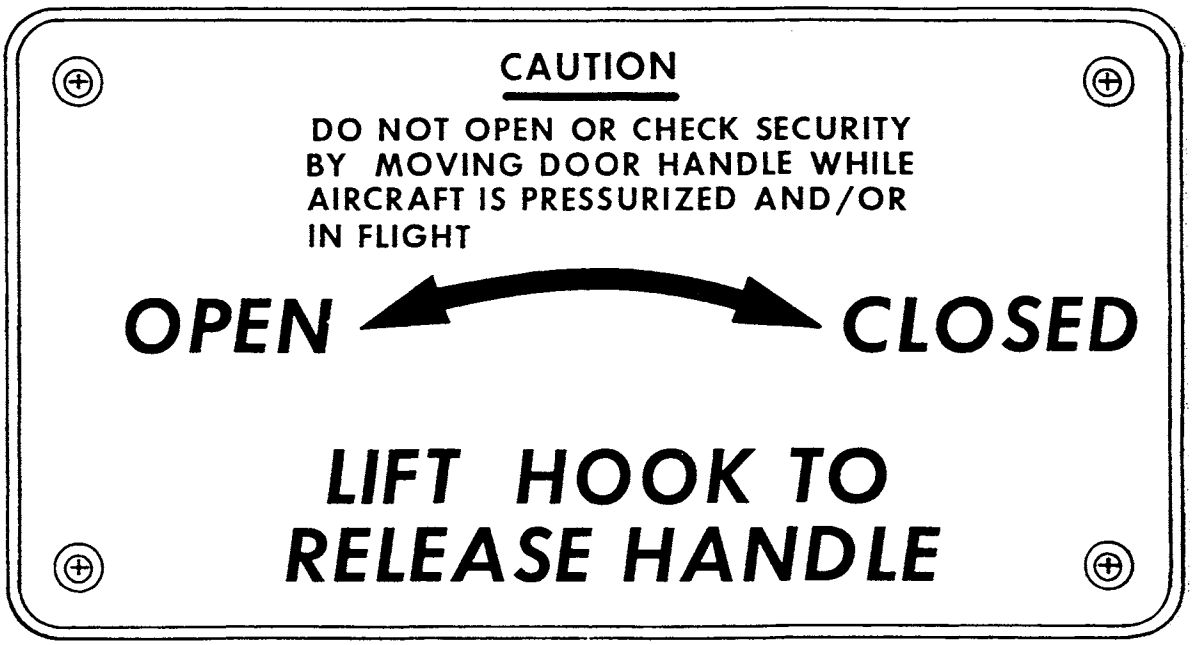
On Upper Access Cover of Cargo Door (B200C):

On Upper Access Cover of Cargo Door: (B200C)

On Cargo Door Upper Latch Handle (B200C):

On Lower Access Cover of Cargo Door: (B200C)

Inside Lower Access Cover of Cargo Door: (B200C)



KINDS OF OPERATIONS

The BEECHCRAFT Super King Air B200 and B200C are approved for the following types of operations when the required equipment is installed and operational as defined within the KINDS OF OPERATIONS EQUIPMENT LIST.

1. VFR Day
2. VFR Night
3. IFR Day
4. IFR Night
5. Know Icing Conditions

KINDS OF OPERATIONS EQUIPMENT LIST

This airplane may be operated in day or night VFR, IFR, or icing conditions when the appropriate equipment is installed and operable.

The following equipment list identifies the systems and equipment upon which type certification for each kind of operation was predicated. The systems and items of equipment listed must be installed and operable unless:

1. The airplane is operated in accordance with a current Minimum Equipment List (MEL) issued by the FAA.

   or;

2. An alternate procedure is provided in the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual for the inoperative state of the listed equipment.

Numbers on the Kinds of Operations Equipment List refer to quantities required to be operative for a specified condition.

NOTE

The following systems and equipment list does not include all equipment required by the FAR Parts 91 and 135 Operating Requirements. It also does not include components obviously required for the airplane to be airworthy, such as wings, empenage, engine, etc.

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING CONDITIONS	
ELECTRICAL POWER						
1. AC Volts/Frequency Meter	1	1	1	1	1	
2. Battery	1	1	1	1	1	
3. Battery Charge Monitor System and Annunciator	1	1	1	1	1	
4. DC Generator	2	2	2	2	2	
5. DC GEN Annunciator	2	2	2	2	2	
6. DC Load Meter	2	2	2	2	2	
7. Inverter	2	2	2	2	2	
8. INVERTER Annunciator	1	1	1	1	1	
ENGINE INDICATIONS						
1. ITT Indicator	2	2	2	2	2	
2. Tachometer (Gas Generator)	2	2	2	2	2	
3. Tachometer (Propeller)	2	2	2	2	2	
4. Torque Indicator	2	2	2	2	2	
ENGINE OIL						
1. Chip Detector System Including Annunciators	2	2	2	2	2	
2. Oil Pressure Indicator	2	2	2	2	2	
3. Oil Temperature Indicator	2	2	2	2	2	
4. OIL PRESS Annunciator	2	2	2	2	2	
ENVIRONMENTAL						
1. BL AIR FAIL Annunciators	2	2	2	2	2	
2. ALT WARN Annunciator (Cabin)	1	1	1	1	1	
3. Cabin Rate of Climb Indicator	1	1	1	1	1	
4. Differential Pressure/Cabin Altitude Indicator	1	1	1	1	1	
5. DUCT OVERTEMP Annunciator	1	1	1	1	1	
6. Outflow Valve	1	1	1	1	1	
7. Pressurization Controller	1	1	1	1	1	
8. Safety Valve	1	1	1	1	1	
9. Bleed Air Shutoff Valve	2	2	2	2	2	
FIRE PROTECTION						
1. Engine Fire Detector System And Annunciators	2	2	2	2	2	
FLIGHT CONTROLS						
1. Flap Position Indicator	1	1	1	1	1	
2. Flap System	1	1	1	1	1	
3. Stall Warning Horn	1	1	1	1	1	
4. Trim Tab Position Indicator (Rudder, Aileron, Elevator)	3	3	3	3	3	
5. Yaw Damp System	1	1	1	1	1	

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING CONDITIONS	
FUEL						
1. Engine Driven Boost Pump	2	2	2	2	2	
2. Fuel Crossfeed System Including Annunciator	1	1	1	1	1	
3. Standby Fuel Boost Pump	2	2	2	2	2	
4. FUEL PRESS Annunciator	2	2	2	2	2	
5. Fuel Quantity Indicating System Including Annunciators	2	2	2	2	2	
6. Firewall Fuel Shutoff System Including Annunciators	2	2	2	2	2	
7. Jet Transfer Pump	2	2	2	2	2	
8. Motive Flow Valve	2	2	2	2	2	
9. Fuel Flow Indicator	2	2	2	2	2	
ICE AND RAIN PROTECTION						
1. Alternate Static Air Source	0	0	1	1	1	
2. Engine Auto Ignition and Annunciators	2	2	2	2	2	
3. Engine Anti-Ice System and Annunciators	2	2	2	2	2	
4. Heated Fuel Vent	0	0	2	2	2	
5. Heated Windshield (Left)	0	0	0	0	1	
6. Pitot Heat	0	0	2	2	2	
7. Pneumatic Pressure Indicator	0	0	1	1	1	
8. Propeller Deicer System	0	0	0	0	1	
9. Stall Warning Heat (Lift Transducer and Mounting Plate)	0	0	0	0	1	
10. Surface Deicer System	0	0	0	0	1	
11. Wing Ice Light (Left)	0	0	0	0	1	
LANDING GEAR						
1. Landing Gear Position Indicator Lights	3	3	3	3	3	
2. Landing Gear Handle Light	1	1	1	1	1	
3. Landing Gear Aural Warning	1	1	1	1	1	
4. * Landing Gear Motor and Gearbox	1	1	1	1	1	
5. Alternate Landing Gear Extension System	1	1	1	1	1	
6. ** Hydraulic Fluid Low Annunciator	1	1	1	1	1	
7. ** Landing Gear Hydraulic Power Pack	1	1	1	1	1	
LIGHTS						
1. Cockpit and Instrument Lighting System	0	1	0	1	0	
2. CABIN DOOR Annunciator	1	1	1	1	1	
3. Landing Lights	0	1	0	1	0	
4. Position Lights	0	3	0	3	0	
5. Anticollision Lights System	0	1	0	1	0	

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING CONDITIONS	
NAVIGATION INSTRUMENTS						
1. Airspeed Indicator (Left)	1	1	1	1	1	
2. Altimeter (Left)	1	1	1	1	1	
3. Magnetic Compass	1	1	1	1	1	
4. Outside Air Temperature	1	1	1	1	1	
OXYGEN						
1. Oxygen System	1	1	1	1	1	
PROPELLER						
1. Prop Reversing System Including Annunciators	2	2	2	2	2	
2. Prop Governor Test Switch	1	1	1	1	1	
3. Prop Overspeed Governor	2	2	2	2	2	
4. Propeller Primary Low-Pitch Stop	2	2	2	2	2	
VACUUM SYSTEM						
1. Instrument Air System	0	1	1	1	1	
2. Gyro Suction Indicator	1	1	1	1	1	

BT04759

** On airplanes prior to BL-73 and BB-1193 (except BB-1158 and BB-1167) not incorporating Beech Kit P/N 101-8018-1*

*** On airplanes BB-1158, BB-1167, BB-1193 and after, BL-73 and after, and those prior airplanes incorporating Beech Kit P/N 101-8018-1*

NOTE

The above Kinds of Operations Equipment List does not include all specific flight instrument and communications/navigation equipment required by FAR Parts 91 and 135 Operating Requirements.

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SECTION III
EMERGENCY PROCEDURES
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All airspeeds quoted in this section are indicated airspeeds (IAS) and assume zero instrument error.

NOTE

Immediate action procedures are delineated by bold type with the remaining procedures following.

EMERGENCY AIRSPEEDS (12,500 LBS)

One-Engine-Inoperative Best Angle-of-Climb (V_{XSE}) ..... 115 kts
One-Engine-Inoperative Best Rate-of-Climb (V_{YSE}) ..... 121 kts
Air Minimum Control Speed (V_{MCA}) ..... 86 kts
One-Engine-Inoperative Enroute Climb ..... 121 kts
Emergency Descent ..... 181 kts
Maximum Range Glide ..... 135 kts

ENGINE FAILURE

NOTE

To obtain best performance with one engine inoperative, the airplane must be banked 3° to 5° into the operating engine while maintaining a constant heading.

EMERGENCY ENGINE SHUTDOWN

- *ENGINE TORQUE INCREASE - UNSCHEDULED (Ground or Flight) (Not Responsive to Power Lever Movement)*

- *ENGINE FIRE IN FLIGHT*

- *ENGINE FAILURE IN FLIGHT*

Affected Engine:

1. **Condition Lever - FUEL CUT OFF**
2. **Propeller Lever - FEATHER**
3. **Firewall Shutoff Valve - CLOSED**
4. **Fire Extinguisher (if installed) - ACTUATE (if required)**
5. Auto Ignition - OFF
6. Generator - OFF
7. Prop Sync - OFF
8. Electrical Load - MONITOR

ENGINE FIRE ON GROUND

Affected Engine:

1. **Condition Lever - FUEL CUT OFF**
2. **Firewall Shutoff Valve - CLOSED**
3. Ignition and Engine Start - STARTER ONLY
4. Fire Extinguisher (if installed) - ACTUATE (if required)

ENGINE FAILURE DURING GROUND ROLL

1. **Power Levers - IDLE**
2. **Brakes - AS REQUIRED**

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3. **Operative Engine - MAXIMUM REVERSE**

WARNING

Extreme care must be exercised when using single-engine reversing on surfaces with reduced traction.

If Insufficient Runway Remains for Stopping:

4. **Condition Levers - FUEL CUT OFF**
5. **Firewall Shutoff Valves - CLOSED**
6. **Master Switch - OFF (Gang bar down)**

ENGINE FAILURE AFTER LIFT-OFF (If Conditions Preclude an Immediate Landing)

1. **Power - MAXIMUM ALLOWABLE**
2. **Airspeed - MAINTAIN (take-off speed or above)**
3. **Landing Gear - UP**

NOTE

If the autofeather system (if installed) is being used, do not retard the failed engine power lever until the autofeather system has completely stopped propeller rotation. To do so will deactivate the autofeather circuit and prevent automatic feathering.

4. **Propeller Lever (inoperative engine) - FEATHER (or verify FEATHER if autofeather is installed)**
5. **Airspeed - V_{YSE} (after obstacle clearance altitude is reached)**
6. Flaps - UP
7. Clean-up (inoperative engine):
   a. Condition Lever - FUEL CUT OFF
   b. Propeller Lever - FEATHER
   c. Firewall Shutoff Valve - CLOSED
   d. Auto Ignition - OFF
   e. Autofeather (if installed) - OFF
   f. Generator - OFF
8. Electrical Load - MONITOR

ENGINE FAILURE IN FLIGHT BELOW AIR MINIMUM CONTROL SPEED (V_{MCA})

1. **Power - Reduce as required to maintain directional control.**
2. **Nose - Lower to accelerate above V_{MCA}.**
3. Power (operative engine) - AS REQUIRED
4. Failed Engine - SECURE (See EMERGENCY ENGINE SHUTDOWN)

ENGINE FLAMEOUT (2nd Engine)

1. **Power Lever - IDLE**
2. **Propeller Lever - DO NOT FEATHER**
3. **Condition Lever - FUEL CUT OFF**
4. **Conduct Air Start Procedures.**

NOTE

The propeller will not unfeather without engine operating.

SUPER KING AIR B200/B200C

TEMPORARY PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL CHANGE

P/N 101-590010-147CTC1

PUBLICATION AFFECTED:	Super King Air B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual P/N 101-590010-147C, Reissued December, 1994 or Subsequent
AIRPLANE SERIAL NUMBERS AFFECTED:	BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-1438, BB-1440 thru BB-1443; and BL-37 thru BL-138
DESCRIPTION OF CHANGE:	Provides an Emergency Procedure for Windshield Electrical Fault
FILING INSTRUCTIONS:	Insert this Temporary Change in the B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual following page 3-4 (EMERGENCY PROCEDURES Section), and retain until rescinded or replaced.

TEMPORARY EMERGENCY PROCEDURES

SMOKE AND FUME ELIMINATION

WINDSHIELD ELECTRICAL FAULT

The smell of an electrical overheat, or observing smoke and/or fire at the lower inboard corner of either windshield, adjacent to the center post, may indicate an overheat condition in the electrical power terminal for the normal heat mode of the pilot's or copilot's windshield heat.

1. WSHLD ANTI-ICE Switches - OFF

If Smoke and/or Fire Does Not Cease:

2. Conduct ELECTRICAL SMOKE OR FIRE procedure.

If Smoke and/or Fire Ceases:

3. Continue flight with Windshield Anti-ice OFF if possible.

If Windshield Anti-ice is Required:

4. If the source of the smell, smoke, or fire can be isolated to the pilot's or copilot's windshield, the opposite windshield (without the overheat condition) may be operated in the NORMAL or HI windshield heat mode.

APPROVED BY: _____
A. C. Jackson
Raytheon Aircraft Company
DOA CE-2

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FUEL SYSTEM

FUEL PRESSURE LOW (L or R FUEL PRESS Annunciator)

1. **Standby Pump (failed side) - ON**
2. FUEL PRESS Annunciator - EXTINGUISHED

SMOKE AND FUME ELIMINATION

Attempt to identify the source of smoke or fumes. Smoke associated with electrical failures is usually gray or tan in color, and irritating to the nose and eyes. Smoke produced by environmental system failures is generally white in color, and much less irritating to the nose and eyes. If smoke is prevalent in the cabin, cabin oxygen masks should not be intentionally deployed. If masks are automatically deployed due to an increase in cabin altitude, passengers should be instructed not to use them unless the cabin altitude exceeds 15,000 feet.

ELECTRICAL SMOKE OR FIRE

1. **Oxygen**
   a. **Oxygen System Ready - PULL ON (Verify)**
   b. **Crew (Diluter Demand Masks) - DON MASKS (100% position)**
   c. **Mic Selector - OXYGEN MASK**
   d. **Audio Speaker - ON**
2. Cabin Temp Mode - OFF
3. Vent Blower - AUTO
4. Aft Blower (if installed) - OFF
5. Avionics Master - OFF
6. Nonessential Electrical Equipment - OFF

If Fire or Smoke Ceases:

7. Individually restore avionics and equipment previously turned off.
8. Isolate defective equipment.

WARNING

Dissipation of smoke is not sufficient evidence that a fire has been extinguished. If it cannot be visually confirmed that no fire exists, land at the nearest suitable airport.

If Smoke Persists or if Extinguishing of Fire is Not Confirmed:

9. Cabin Pressure - DUMP
10. Land at the nearest suitable airport.

NOTE

Opening a storm window (after depressurizing) will facilitate smoke and fume removal.

ENVIRONMENTAL SYSTEM SMOKE OR FUMES

1. **Oxygen**
   a. **Oxygen System Ready - PULL ON (Verify)**
   b. **Crew (Diluter Demand Masks) - DON MASKS (100% position)**
   c. **Mic Selector - OXYGEN MASK**
   d. **Audio Speaker - ON**

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2. Cabin Temp Mode - OFF
3. Vent Blower - HI
4. Left Bleed Air Valve - ENVIR OFF

If Smoke Decreases:

5. Continue operation with left bleed air off.

If Smoke Does Not Decrease:

6. Left Bleed Air Valve - OPEN
7. Right Bleed Air Valve - ENVIR OFF
8. If smoke decreases, continue operation with right bleed air off.

NOTE

Each bleed air valve must remain closed long enough to allow time for smoke purging to positively identify the smoke source.

CABIN OR CARGO UNLOCKED (CABIN DOOR Annunciator)

WARNING

Do not attempt to check the security of the airstair or cargo door in flight. Remain as far from the door as possible with seatbelts securely fastened.

If the CABIN DOOR Annunciator Illuminates, or If an Unlatched Airstair/Cargo Door is Suspected:

1. **All Occupants - SEATED WITH SEAT BELTS SECURELY FASTENED**
2. Cabin Sign - NO SMOKE & FSB
3. Cabin Differential Pressure - REDUCE TO LOWEST VALUE PRACTICAL (zero preferred) by descending and/or selecting higher cabin altitude setting.
4. Oxygen - AS REQUIRED
5. Land at nearest suitable airport.

EMERGENCY DESCENT

1. **Oxygen - CREW REQUIRED (passengers as required)**
   a. **Oxygen System Ready - PULL ON (verify)**
   b. **Crew (Diluter Demand Masks) - DON MASKS**
   c. **Mic Selector - OXYGEN MASK**
   d. **Audio Speaker - ON**
   e. **Passenger Manual Drop Out - PULL ON**
   f. Passengers - PULL LANYARD PIN, DON MASK
2. **Power Levers - IDLE**
3. **Propeller Levers - FULL FORWARD**
4. **Flaps - APPROACH**
5. **Landing Gear - DN**
6. **Airspeed - 181 KNOTS MAXIMUM**

GLIDE

1. **Landing Gear - UP**
2. **Flaps - UP**

WARNING

Determine that procedures for re-starting first and second failed engines are ineffective before feathering second engine propeller.

3. **Propellers - FEATHERED**
4. **Airspeed - 135 KNOTS**
5. Glide Ratio - 2.0 nm for each 1000 feet of altitude. Decrease by 0.2 nm for each 10 knots of headwind.

ELECTRICAL

INVERTER INOPERATIVE (INVERTER Annunciator)

- **Select the other inverter.**

FLIGHT CONTROLS

UNSCHEDULED ELECTRIC ELEVATOR TRIM

1. **Airplane Attitude - MAINTAIN (using elevator control)**
2. **Control Wheel Disconnect Switch - DEPRESS FULLY (2nd level, ELEC TRIM OFF annunciator - ILLUMINATED)**

NOTE

Autopilot will disengage when the disconnect switch is depressed.

3. Manually retrim airplane.
4. Elevator Trim - OFF

CAUTION

DO NOT reactivate electric trim system until cause of malfunction has been determined.

UNSCHEDULED RUDDER BOOST ACTIVATION

Rudder boost operation without a large variation of power between the engines indicates a failure of the system.

1. **Directional Control - MAINTAIN USING RUDDER PEDALS**
2. **Rudder Boost - OFF**

If Condition Persists:

3. Rudder Boost Circuit Breaker - PULL
4. Either Bleed Air Valve - INSTR & ENVIR OFF
5. Rudder Trim - AS REQUIRED
6. Perform normal landing.

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ENVIRONMENTAL SYSTEMS

USE OF OXYGEN

WARNING

The following table sets forth the average time of useful consciousness (TUC) (time from onset of hypoxia until loss of effective performance) at various altitudes.

Cabin Pressure Altitude	TUC
35,000 feet	1/2 - 1 minute
30,000 feet	1 - 2 minutes
25,000 feet	3 to 5 minutes
22,000 feet	5 to 10 minutes
12 - 18,000 feet	30 minutes or more

1. **Oxygen System Ready - PULL ON (verify)**
2. **Crew (Diluter Demand Masks) - DON MASKS**
3. **Mic Selector - OXYGEN MASK**
4. **Audio Speaker - ON**
5. **Passenger Manual Drop Out - PULL ON**
6. Passengers - PULL LANYARD PIN, DON MASK
7. Oxygen Duration - CONFIRM (See OXYGEN SYSTEM in Section IV, NORMAL PROCEDURES for duration tables)
8. First Aid Oxygen - AS REQUIRED
   a. Oxygen Compartment - PULL OPEN
   b. ON/OFF Valve - ON
   c. Mask - DON

PRESSURIZATION LOSS (ALT WARN Annunciator)

1. **Oxygen**
   a. **Oxygen System Ready - PULL ON (verify)**
   b. **Crew - DON MASK**
   c. **Mic Selector - OXYGEN MASK**
   d. **Audio Speaker - ON**
   e. **Passenger Manual Drop Out - PULL ON**
   f. Passengers - PULL LANYARD PIN, DON MASK
2. **Descend as required.**

WARNING

Adequate oxygen pressure is not provided to the passengers for sustained flight at cabin altitudes above 34,000 feet. The highest recommended cabin altitude for sustained flight is 25,000 feet.

3. Range - DETERMINE FOR FINAL CRUISE ALTITUDE
4. Oxygen Duration - CONFIRM (See OXYGEN SYSTEM in Section IV, NORMAL PROCEDURES for duration tables)

AUTO-DEPLOYMENT OXYGEN SYSTEM FAILURE (ALT WARN Annunciator Illuminated, PASS OXY ON Annunciator Not Illuminated)

1. **Passenger Manual Drop Out - PULL ON**
2. First Aid Mask (if required) - DEPLOY MANUALLY

To Isolate Oxygen Supply to the Crew and First Aid Mask:

3. Oxygen Control Circuit Breaker - PULL
4. Passenger Manual Drop Out - PUSH OFF

HIGH DIFFERENTIAL PRESSURE (Cabin Differential Pressure Exceeds 6.6 PSI)

1. **Bleed Air Valves - ENVIR OFF**
2. **Oxygen (Crew and Passengers) - AS REQUIRED**
3. **Descend - AS REQUIRED**

BLEED AIR LINE FAILURE (L or R BL AIR FAIL Annunciator)

Warning annunciators should be monitored during engine start procedure. Either engine will extinguish both annunciators upon starting.

Illumination of a warning annunciator in flight indicates a possible rupture of a bleed air line aft of the engine firewall.

1. **Bleed Air Valve (affected engine) - INSTR & ENVIR OFF position**
2. Engine Instruments - MONITOR

NOTE

The bleed air warning annunciator will not extinguish after closing the Bleed Air Valve.

EMERGENCY EXIT

- Emergency Exit Handle - PULL (This is a plug-type door and opens into the cabin)

CAUTION

B200:
The outside handle may be locked from the inside with the EXIT LOCK lever. The inside EXIT-PULL handle will unlatch the door regardless of the position of the EXIT LOCK lever. Before flight, make certain the lock lever is in the unlocked position.
B200C:
The outside handle may be locked from the inside with a key. The inside handle will unlatch the door, regardless of the position of the key lock, by overriding the locking mechanism. Before flight, make certain the door is unlocked.

SPINS

If a Spin is Entered Inadvertently:

1. **Control Column - FULL FORWARD**
2. **Full Rudder - OPPOSITE DIRECTION OF SPIN**
3. **Power Levers - IDLE**

4. **Controls - NEUTRALIZE WHEN ROTATION STOPS**
5. **Execute a smooth pull out.**

NOTE

Federal Aviation Administration Regulations do not require spin demonstration of airplanes of this weight; therefore no spin tests have been conducted. The recovery technique is based on the best available information.

SECTION IIIA
ABNORMAL PROCEDURES

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All airspeeds quoted in this section are *indicated airspeeds (IAS)* and assume zero instrument error.

AIR START

WARNING

Airstart using the STARTER ASSIST procedures may momentarily cause the loss of attitude display on Electronic Flight Instrument System (EFIS) equipped airplanes, and lead to premature system failures. IF FLIGHT CONDITIONS DO NOT PERMIT THE TEMPORARY LOSS OF ATTITUDE REFERENCE, CONDUCT AIRSTART USING THE NO STARTER ASSIST PROCEDURES.

CAUTION

The pilot should determine the reason for engine failure before attempting an air start. Do not attempt an air start if N_1 indicates zero.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an overtemperature condition.

STARTER ASSIST

1. Cabin Temp Mode - OFF
2. Vent Blower - AUTO
3. Aft Blower (if installed) - OFF
4. Radar - STANDBY or OFF
5. Windshield Heat - OFF
6. Power Lever - IDLE
7. Propeller Lever (inoperative engine) - LOW RPM
8. Condition Lever - FUEL CUT OFF
9. Firewall Shutoff Valve - OPEN
10. Generator (inoperative engine) - OFF

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

11. Ignition and Engine Start - ON, IGNITION ON annunciator - ILLUMINATED)
12. Condition Lever - LOW IDLE
13. ITT and N_1 - MONITOR (1000°C MAXIMUM)
14. Ignition and Engine Start - OFF (N_1 above 50%)
15. Propeller Lever - AS REQUIRED
16. Power Lever - AS REQUIRED
17. Generator - ON
18. Auto Ignition - ARM
19. Prop Sync - ON
20. Cabin Temp Mode - AUTO

21. Aft Blower (if installed) - AS REQUIRED
22. Radar - AS REQUIRED
23. Windshield Heat - AS REQUIRED

NO STARTER ASSIST (Windmilling Engine and Propeller)

1. Power Lever - IDLE
2. Propeller Lever - FULL FORWARD
3. Condition Lever - FUEL CUT OFF
4. Engine Ice Vane (inoperative engine) - RETRACTED
5. Firewall Shutoff Valve - OPEN
6. Generator (inoperative engine) - OFF
7. Airspeed - 140 KNOTS MINIMUM
8. Altitude - BELOW 20,000 FEET
9. Auto Ignition - ARM (IGNITION ON annunciator - ILLUMINATED)
10. Condition Lever - LOW IDLE
11. ITT and N_1 - MONITOR (1000°C MAXIMUM)
12. Power - AS REQUIRED (after ITT has peaked)
13. Generator - ON
14. Prop Sync - ON

LANDING

FLAPS UP LANDING

Refer to Section V, PERFORMANCE, for Flaps Up Landing Distance and Approach Speed.

1. Approach Speed - CONFIRM
2. Autofeather (if installed) - ARM
3. Pressurization - CHECK
4. Cabin Sign - NO SMOKE & FSB
5. Flaps - UP

CAUTION

Do not silence the landing gear warning horn, since the flap actuated portion of the landing gear warning system will not be actuated during a flaps up landing.

6. Landing Gear - DN
7. Lights - AS REQUIRED

NOTE

Under low visibility conditions, landing and taxi lights should be left off due to light reflections.

8. Radar - AS REQUIRED
9. Surface Deice - CYCLE (as required)

NOTE

If crosswind landing is anticipated, determine Crosswind Component from Section V, PERFORMANCE. Immediately prior to touchdown, lower upwind wing and align the fuselage with the runway. During rollout, hold aileron control into the wind and maintain directional control with rudder and brakes. Use propeller reverse as desired.

10. Approach Speed - ESTABLISHED
11. Yaw Damp - OFF
12. Propeller Levers - FULL FORWARD
13. Power Levers - IDLE

After Touchdown:

14. Power Levers - LIFT AND SELECT REVERSE
15. Brakes - AS REQUIRED

ONE-ENGINE-INOPERATIVE APPROACH AND LANDING

1. Approach Speed - CONFIRM
2. Fuel Balance - CHECK
3. Pressurization - CHECK
4. Cabin Sign - NO SMOKE & FSB

When It is Certain that the Field Can Be Reached:

5. Flaps - APPROACH
6. Landing Gear - DN
7. Propeller Lever - FULL FORWARD
8. Airspeed - 10 KNOTS ABOVE NORMAL LANDING APPROACH SPEED
9. Interior and Exterior Lights - AS REQUIRED
10. Radar - AS REQUIRED
11. Surface Deice - CYCLE (as required)

When It is Certain There is No Possibility of a Go-Around:

12. Flaps - DN
13. Airspeed - NORMAL LANDING APPROACH SPEED
14. Perform normal landing.

NOTE

Single-engine reverse thrust may be used with caution after touchdown on smooth, dry, paved surfaces.

ONE-ENGINE-INOPERATIVE GO-AROUND

1. Power - MAXIMUM ALLOWABLE
2. Landing Gear - UP
3. Flaps - UP
4. Airspeed - INCREASE TO BLUE LINE, 121 KNOTS

SYSTEMS

ENGINE OIL SYSTEM

LOW OIL PRESSURE INDICATION

Oil pressure values between 60 and 85 psi are undesirable; they should be tolerated only for the completion of the flight, and then only at a reduced power setting not exceeding 1100 foot-pounds torque. Oil pressure values below 60 psi are unsafe; they require that either the engine be shut down or that a landing be made at the nearest suitable airport, using the minimum power required to sustain flight.

For Training Purposes Only

CHIP DETECT (L or R CHIP DETECT Annunciator)

Illumination of a CHIP DETECT annunciator indicates possible metal contamination in the engine oil supply. Illumination of a CHIP DETECT annunciator is not in itself cause for an engine to be shut down. Engine parameters should be monitored for abnormal indications. If parameters are abnormal, a precautionary shutdown may be made at the pilot's discretion. After illumination of a CHIP DETECT annunciator, cause of the malfunction should be determined and corrected prior to the next flight.

FUEL SYSTEM

CROSSFEED (One-Engine-Inoperative Operation)

1. Crossfeed - LEFT OR RIGHT, AS REQUIRED (FUEL CROSSFEED annunciator - ILLUMINATED)
2. Standby Pumps - OFF
3. Auxiliary Tank Transfer - AUTO
4. Fuel Balance - MONITOR

If Fuel is Required from the Inoperative Engine's Auxiliary Fuel Tank and the Reason for Shutdown was Not an Engine Fire or Fuel Leak:

5. Firewall Shutoff Valve (inoperative engine) - OPEN (FUEL PRESS annunciator - EXTINGUISHED)
6. No Transfer Light (inoperative engine) - EXTINGUISHED IN 30 TO 50 SECONDS.

TO DISCONTINUE CROSSFEED

- Crossfeed Flow Switch - OFF (centered)

AUXILIARY FUEL TRANSFER FAILURE (NO TRANSFER Light)

1. Auxiliary Tank Transfer - OVERRIDE
2. No Transfer Light - EXTINGUISHED (If light does not extinguish, auxiliary fuel may not be available)
3. Auxiliary Fuel Quantity - MONITOR
4. Auxiliary Tank Transfer - AUTO (when auxiliary fuel tank is empty)

ELECTRICAL SYSTEM

GENERATOR INOPERATIVE (L or R DC GEN Annunciator)

1. Loadmeter - VERIFY GENERATOR IS OFF (0% LOAD)
2. Generator - RESET, THEN ON

If generator will not reset:

3. Generator - OFF
4. Loadmeter - DO NOT EXCEED 100% (88% Above 31,000 feet)

BATTERY CHARGE RATE (BATTERY CHARGE Annunciator)

Ground Operations:

The BATTERY CHARGE annunciator will illuminate after an engine start. Do not take off with the annunciator illuminated unless a decreasing battery charge current is confirmed. See Nickel-Cadmium Battery Check in Section IV, NORMAL PROCEDURES.

In Flight:

In-flight illumination of the BATTERY CHARGE annunciator indicates a possible battery malfunction.

1. Battery Switch - OFF
2. BATTERY CHARGE Annunciator Extinguished - CONTINUE TO DESTINATION
3. BATTERY CHARGE Annunciator Still Illuminated - LAND AT NEAREST SUITABLE AIRPORT

For Training Purposes Only

EXCESSIVE LOADMETER INDICATION (over 100%)

1. Battery - OFF (monitor loadmeter)

If Loadmeter Still Indicates Above 100%:

2. Nonessential Electrical Equipment - OFF

If Loadmeter Indicates 100% or Below:

3. Battery - ON

CIRCUIT BREAKER TRIPPED

1. Nonessential Circuit - DO NOT RESET IN FLIGHT
2. Essential Circuit:
   a. Circuit Breaker - PUSH TO RESET
   b. If Circuit Breaker Trips Again - DO NOT RESET

BUS FEEDER CIRCUIT BREAKER TRIPPED (Fuel Panel Bus Feeders and Right Circuit Breaker Panel Bus Feeders)

- A short is indicated, do not reset in flight.

NOTE

The items that may be inoperative can be determined from the electrical system schematic in Section VII, SYSTEMS DESCRIPTION.

AVIONICS MASTER POWER SWITCH FAILURE

If the Avionics Master Pwr Switch Fails to Operate in the ON Position:

- Avionics Master Circuit Breaker - PULL

NOTE

Turning on the Avionics Master Pwr switch removes power that holds the avionics relay open. If the switch fails to the OFF position, pulling the Avionics Master circuit breaker will remove power to the relay and should restore power to the avionics buses.

LANDING GEAR SYSTEM

HYDRAULIC FLUID LOW (HYD FLUID LOW Annunciator)

If HYD FLUID LOW annunciator illuminates during flight, attempt to extend the landing gear normally upon reaching destination. If the landing gear fails to extend, follow LANDING GEAR MANUAL EXTENSION procedures.

LANDING GEAR MANUAL EXTENSION (HYDRAULIC SYSTEM)

On airplanes BB-1158, BB-1167, BB-1193 and after, BL-73 and after, and those airplanes incorporating Beech Kit P/N 101-8018-1

If the landing gear fails to extend after placing the Landing Gear Control down, perform the following:

1. Landing Gear Relay Circuit Breaker (pilot's subpanel) - PULL
2. Landing Gear Control - DN
3. Alternate Extension Handle - PUMP UP AND DOWN UNTIL THE THREE GREEN GEAR-DOWN ANNUNCIATORS ARE ILLUMINATED. WHILE PUMPING, DO NOT LOWER HANDLE TO THE LEVEL OF THE SECURING CLIP DURING THE DOWN STROKE AS THIS WILL RESULT IN THE LOSS OF PRESSURE.

If all three green gear-down annunciators are illuminated:

4. Alternate Extension Handle - STOW

For Training Purposes Only

5. Landing Gear Controls - DO NOT ACTIVATE (The Landing Gear Control and the Landing Gear Relay Circuit Breaker must not be activated. The landing gear should be considered UNSAFE until the system is cycled and checked with the airplane on jacks.)

If one or more green gear-down annunciators do not illuminate for any reason and a decision is made to land in this condition:

6. Alternate Extension Handle - CONTINUE PUMPING UNTIL MAXIMUM RESISTANCE IS FELT.
7. Alternate Extension Handle - DO NOT LOWER. LEAVE AT THE TOP OF THE UP STROKE.

Prior to Landing:

8. Alternate Extension Handle - PUMP UNTIL MAXIMUM RESISTANCE IS FELT. DO NOT STOW.

After Landing:

9. Alternate Extension Handle - CONTINUE PUMPING, WHEN CONDITIONS PERMIT, TO MAINTAIN HYDRAULIC PRESSURE UNTIL THE GEAR CAN BE MECHANICALLY SECURED. DO NOT STOW HANDLE. DO NOT ACTIVATE THE LANDING GEAR CONTROL OR THE LANDING GEAR RELAY CIRCUIT BREAKER. THE LANDING GEAR SHOULD BE CONSIDERED UNLOCKED UNTIL THE SYSTEM IS CYCLED AND CHECKED WITH THE AIRPLANE ON JACKS.

LANDING GEAR MANUAL EXTENSION (MECHANICAL SYSTEM)

On airplanes prior to BL-73 and BB-1193 (except BB-1158 and BB-1167) not incorporating Beech Kit P/N 101-8018-1

If the landing gear fails to extend after placing the Landing Gear Control down, perform the following:

1. Airspeed - ESTABLISH 130 KNOTS
2. Landing Gear Relay Circuit Breaker (pilot's subpanel) - PULL
3. Landing Gear Control - DN
4. Alternate Engage Handle - LIFT AND TURN CLOCKWISE TO THE STOP TO ENGAGE.
5. Alternate Extension Handle - PUMP UP AND DOWN UNTIL THE THREE GREEN GEAR-DOWN ANNUNCIATORS ARE ILLUMINATED. ADDITIONAL PUMPING WHEN ALL THREE ANNUNCIATORS ARE ILLUMINATED COULD DAMAGE THE DRIVE MECHANISM AND PREVENT SUBSEQUENT ELECTRICAL GEAR RETRACTION.

If all three green gear-down annunciators are illuminated:

6. Alternate Extension Handle - DO NOT STOW (Proceed to step 8.)

If one or more green gear-down annunciators do not illuminate for any reason and a decision is made to land in this condition:

7. Alternate Extension Handle - CONTINUE PUMPING UNTIL MAXIMUM RESISTANCE IS FELT, EVEN THOUGH THIS MAY DAMAGE THE DRIVE MECHANISM.
8. Landing Gear Controls - DO NOT ACTIVATE (The Landing Gear Control and the Landing Gear Relay Circuit Breaker must not be activated. The landing gear should be considered UNSAFE until the system is cycled and checked with the airplane on jacks.)

For Training Purposes Only

ENVIRONMENTAL SYSTEMS

DUCT OVERTEMPERATURE (DUCT OVERTEMP ANNUNCIATOR)

1. Vent Blower - HIGH
2. Cabin and Cockpit Air - PUSH IN (to increase airflow to cabin)

If Condition Persists:

3. Cabin Temp Mode - MAN HEAT
4. Manual Temp - DECR (for 60 seconds)

If Condition Persists, the Right Bypass Valve May Be Inoperative, Preventing Both Valves from Moving to the Colder Position:

5. Left Bleed Air Valve - ENVIR OFF

If the DUCT OVERTEMP Annunciator Does Not Extinguish After 2 Minutes:

6. Oxygen - AS REQUIRED
7. Right Bleed Air Valve - ENVIR OFF
8. Descend as required.

ICE PROTECTION SYSTEMS

ELECTROTHERMAL PROPELLER DEICE (AUTO SYSTEM)

Abnormal Readings on Deice Ammeter: (Normal Operation: 14 to 18 amps)

1. Zero Amps:
   a. Prop Deice - CHECK AUTO
   b. If OFF, reposition to AUTO after 30 seconds.
   c. If in AUTO position with zero amps reading, system is inoperative: position the switch to OFF.
   d. Use manual backup system. (No deice ammeter indication - monitor loadmeter)
2. Below 14 amps:
   a. Continue operation.
   b. If propeller imbalance occurs, increase rpm briefly to aid in ice removal.
3. Over 18 amps:
   a. If the Auto Prop Deice circuit breaker switch does not trip, continue operation.
   b. If propeller imbalance occurs, increase rpm briefly to aid in ice removal.
   c. If the Auto Prop Deice circuit breaker switch trips, use the manual system. Monitor loadmeter for excessive current drain.
   d. If the Prop Deice Control circuit breaker or the Left or Right Prop Deice circuit breaker trips, avoid icing conditions.

ELECTROTHERMAL PROPELLER DEICE (MANUAL SYSTEM)

On Airplanes Prior to BB-829:

1. To use manual system, hold switch in OUTER position for approximately 30 seconds, then in INNER position for approximately 30 seconds.
2. Monitor manual system current requirement using the airplane's loadmeters when the switch is in OUTER or INNER. A small needle deflection (approximately 5%) indicating the system is functioning.

Airplanes BB-829 and After, BL-37 and After:

3. To use manual system, hold manual propeller deice switch in MANUAL position for approximately 90 seconds, or until ice is dislodged from blades.
4. Monitor manual system current requirement with the airplane's loadmeters when the manual deice switch is in the MANUAL position. A small needle deflection (approximately 5%) indicates the system is functioning.

ENGINE ICE VANE FAILURE (L or R ICE VANE Annunciator)

1. Ice Vane Control Circuit Breaker - PULL
2. Airspeed - 140 - 160 KIAS
3. Manual Extension Handle - PULL OUT (ICE VANE EXT Annunciator Illuminated)
4. Airspeed - RESUME

If ICE VANE EXT Annunciator Does Not Illuminate:

5. Exit icing conditions.
6. Manual Extension Handle - PUSH IN (to retract vanes when required)

CAUTION

Do not activate ice vanes electrically once the manual system has been used until the override linkage has been reset after landing.

NOTE

The ICE VANE fail annunciator will be illuminated any time the position of the ice vane does not match the corresponding switch position. The switch may be repositioned to match the vane position without damaging the linkage as long as the Ice Vane Control circuit breaker is out.

STATIC AIR SYSTEM

PILOT'S ALTERNATE STATIC AIR SOURCE

THE PILOT'S ALTERNATE STATIC AIR SOURCE SHOULD BE USED FOR CONDITIONS WHERE THE NORMAL STATIC SOURCE HAS BEEN OBSTRUCTED. When the airplane has been exposed to moisture and/or icing conditions (especially on the ground), the possibility of obstructed static ports should be considered. Partial obstructions will result in the rate of climb indication being sluggish during a climb or descent. Verification of suspected obstruction is possible by switching to the alternate system and noting a sudden sustained change in rate of climb. This may be accompanied by abnormal indicated airspeed and altitude changes beyond normal calibrated differences.

Whenever any obstruction exists in the Normal Static Air System, or when the Alternate Static Air System is desired for use:

1. Pilot's Static Air Source (right side panel) - ALTERNATE
2. For Airspeed Calibration and Altimeter Correction, refer to Section V, PERFORMANCE.

NOTE

Be certain the static air valve is in the NORMAL position when the alternate system is not needed.

CRACKED OR SHATTERED WINDSHIELD

The following procedure should be used when one or more cracks occur in the inner or outer ply of the windshield. The procedure is also applicable if the windshield shatters. This usually occurs in the inner ply and is characterized by a multitude of cracks which will likely obstruct the crew members vision and may produce small particles or flakes of glass that can break free of the windshield.

1. Altitude - MAINTAIN 25,000 FT OR LESS, IF POSSIBLE
2. Pressurization Controller - RESET
    a. Cruise and Descent - MAINTAIN A CABIN DIFFERENTIAL PRESSURE OF 2.0 TO 4.6 PSI (A cabin differential pressure of 4.6 psi will produce approximately a 10,500-foot cabin altitude at an airplane altitude of 25,000 feet)
    b. Before Landing - DEPRESSURIZE CABIN PRIOR TO TOUCHDOWN
3. Other In-flight Considerations
    a. Visibility through a shattered windshield may be sufficiently reduced to dictate flying the airplane from the opposite side of the cockpit.
    b. Precautions should be taken to prevent particles or flakes of glass from a shattered inner ply of the windshield from interfering with the crew's vision.
    c. A cracked outer windshield ply may damage operating windshield wipers.
    d. Windshield heat may be inoperative in the area of the crack(s).
    e. The structural integrity of the windshield will be maintained.
4. Postflight Considerations - SEE SECTION II, LIMITATIONS

CRACK IN ANY SIDE WINDOW (COCKPIT OR CABIN)

1. Altitude - DESCEND IF REQUIRED
    a. Descend to an altitude not requiring oxygen, if possible, or
    b. Descend to at least 25,000 feet if passengers are on board.
2. Crew and Passengers - DON OXYGEN MASKS, IF REQUIRED
3. Pressurization Controller - RESET AS REQUIRED TO DEPRESSURIZE CABIN
4. Cabin Pressure Switch - DUMP
5. Postflight Considerations - SEE SECTION II, LIMITATIONS

SEVERE ICING CONDITIONS (Required By FAA AD 96-09-13)

THE FOLLOWING WEATHER CONDITIONS MAY BE CONDUCIVE TO SEVERE IN-FLIGHT ICING:

- Visible rain at temperatures below 0°C ambient air temperature.
- Droplets that splash or splatter on impact at temperatures below 0°C ambient air temperature.

PROCEDURES FOR EXITING THE SEVERE ICING ENVIRONMENT:

These procedures are applicable to all flight phases from takeoff to landing. Monitor the ambient air temperature. While severe icing may form at temperatures as cold as -18°C, increased vigilance is warranted at temperatures around freezing with visible moisture present. If the visual cues specified in Section II, LIMITATIONS for identifying severe icing conditions are observed, accomplish the following:

1. Immediately request priority handling from Air Traffic Control to facilitate a route or an altitude change to exit the severe icing conditions in order to avoid extended exposure to flight conditions more severe than those for which the airplane has been certificated.
2. Avoid abrupt and excessive maneuvering that may exacerbate control difficulties.
3. Do not engage the autopilot.
4. If the autopilot is engaged, hold the control wheel firmly and disengage the autopilot.
5. If an unusual roll response or uncommanded roll control movement is observed, reduce the angle-of-attack.
6. Do not extend flaps during extended operation in icing conditions. Operation with flaps extended can result in a reduced wing angle-of-attack with the possibility of ice forming on the upper surface further aft on the wing than normal, possibly aft of the protected area.
7. If the flaps are extended, do not retract them until the airframe is clear of ice.
8. Report these weather conditions to Air Traffic Control.

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NORMAL PROCEDURES
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All airspeeds quoted in this section are indicated airspeeds (IAS) and assume zero instrument error.

AIRSPEEDS FOR SAFE OPERATION (12,500 LBS)

Maximum Demonstrated Crosswind Component ..... 25 Knots
Takeoff (Flaps Up)
- Rotation ..... 95 Knots
- 50-ft Speed ..... 121 Knots

Takeoff (Flaps Approach)
- Rotation ..... 94 Knots
- 50-ft Speed ..... 106 Knots

Two-Engine Best Angle-of-Climb (V_X) ..... 100 Knots
Two-Engine Best Rate-of-Climb (V_Y) ..... 125 Knots
Cruise Climb:
- Sea Level to 10,000 feet ..... 160 Knots
- 10,000 to 20,000 feet ..... 140 Knots
- 20,000 to 25,000 feet ..... 130 Knots
- 25,000 to 35,000 feet ..... 120 Knots

Maximum Airspeed for Effective Windshield Anti-icing ..... 226 Knots
Maneuvering Speed (V_A) ..... 181 Knots
Turbulent Air Penetration ..... 170 Knots

CAUTION

For turbulent air penetration, use an airspeed of 170 knots. Avoid over-action on power levers. Turn off autopilot altitude hold. Keep wings level, maintain attitude and avoid use of trim. Do not chase airspeed and altitude. Penetration should be at an altitude which provides adequate maneuvering margins when severe turbulence is encountered.

Landing Approach:
- Flaps Down ..... 103 Knots

Balked Landing Climb ..... 100 Knots
Intentional One-Engine-Inoperative Speed (V_{SSE}) ..... 104 Knots
Air Minimum Control Speed (V_{MCA}) ..... 86 Knots

PROCEDURES BY FLIGHT PHASE

NOTE

Refer to all applicable Beech Supplements and STC Supplements for flight phase procedures for optional equipment installed in the airplane.

PREFLIGHT INSPECTION

After the first flight of each day, the Preflight Inspection may be omitted except for items marked with a "+". Fuel tank caps and engine oil quantity and filler cap need not be checked unless system(s) were serviced. External inspections with flaps down may be conducted at intervals deemed appropriate by the pilot.

CABIN/COCKPIT

1. Monogram Electric Toilet (if installed) - KNIFE VALVE OPEN
+ 2. Baggage - SECURE
3. Emergency Exit - SECURE AND UNLOCKED
4. Trim Tabs - SET TO "0" UNITS

For Training Purposes Only

CAUTION

The elevator trim system must not be forced past the limits which are indicated on the elevator trim indicator scale, either manually, electrically, or by action of the autopilot.

5. Condition Levers - FUEL CUT OFF
6. Landing Gear Control - DN
7. Control Locks - REMOVE
8. Parking Brake - SET
9. Ignition and Engine Start - ENSURE OFF
10. Battery - ON, CHECK 23 VOLT MINIMUM
11. Fuel Quantity - CHECK
12. Flap Control (if desired) - APPROACH, THEN DOWN (check indicator)
13. Battery - OFF
14. Oxygen System Preflight Inspection - COMPLETE (See OTHER NORMAL PROCEDURES.)

LEFT WING AND NACELLE

1. Door Seal - CHECK
2. Flaps - CHECK
3. Oil Breather Vent - CLEAR
+ 4. Brake Lines, Brake Wear, Brake Deice Lines (if installed) - CHECK
5. Fire Extinguisher (if installed) - CHECK PRESSURE
6. Inverter Cooling Louver - CLEAR
7. Aileron and Tab - CHECK
8. Flush Outboard Wing Fuel Tank Sump - DRAIN
9. Static Wicks (4) - CHECK
10. Navigation, Recognition, & Strobe Lights - CHECK
+ 11. Main Fuel Tank Cap - SECURE
12. Stall Warning Vane - CHECK
+ 13. Tiedown - REMOVED
14. Outboard Deice Boot and Stall Strip - CHECK
15. Ice Light - CHECK
16. Heated Fuel Vent - CLEAR
17. Ram Scoop Fuel Vent - CLEAR
18. Gravity Line Drain - DRAIN
19. Inverter Cooling Louvers - CLEAR
20. Wing Leading Edge Tank Sump - DRAIN
+ 21. Landing Gear (Doors, Wheel Well, Strut, Tires, Brakes) - CHECK
+ 22. Chock - REMOVE
23. Standby Pump Drain - DRAIN
24. Fuel Filter Strainer Sump - DRAIN
+ 25. Engine Oil - CHECK QUANTITY, CAP SECURE
26. Engine Compartment Door (Outbd) - SECURE, BLEED VALVE EXHAUST CLEAR
27. Exhaust Stack (Outbd) - CHECK FOR CRACKS
28. Top Cowling Locks (Outbd) - SECURE
29. Nacelle Cooling Ram Air Inlets - CLEAR

+ 30. Propeller - CHECK FOR NICKS, DEICE BOOT SECURE
31. Engine Intake - CLEAR
32. Top Cowling Locks (Inbd) - SECURE
33. Exhaust Stack (Inbd) - CHECK FOR CRACKS
34. Generator Cooling Inlet - CLEAR
35. Engine Compartment Door (Inbd) - SECURE, BLEED VALVE EXHAUST CLEAR
36. Heat Exchanger Inlet - CLEAR
+ 37. Auxiliary Fuel Tank Cap - SECURE
38. Hydraulic Landing Gear Service Door (if installed) - SECURE
39. Inboard Deice Boot - CHECK
40. Heat Exchanger Outlet - CLEAR
41. Hydraulic Landing Gear Vent Lines (if installed) - CLEAR
42. Auxiliary Fuel Tank Sump - DRAIN
43. Lower Antennas and Beacon - CHECK

NOSE SECTION

1. OAT Probe (Lower Fuselage) - CHECK
2. Avionics Panel - SECURE
3. Condenser Blower Outlet - CLEAR
+ 4. Nose Gear (Doors, Strut, Nose Gear Steering Stop Block, Tire, Wheel Well) - CHECK
+ 5. Chock - REMOVE
6. Landing and Taxi Lights - CHECK
7. Pitot Masts - COVERS REMOVED, CLEAR
8. Radome - CHECK
9. Windshield Wipers - CHECK
10. Ram Air Inlet - CLEAR
11. Avionics Panel - SECURE

RIGHT WING AND NACELLE

1. Auxiliary Fuel Tank Sump - DRAIN
2. Battery Drain - CLEAR
3. Battery Air Inlet - CLEAR, VALVE FREE
4. Heat Exchanger Outlet - CLEAR
5. Inboard Deice Boot - CHECK
6. Battery Exhaust - CLEAR
+ 7. Auxiliary Fuel Tank Cap - SECURE
8. Heat Exchanger Inlet - CLEAR
+ 9. Engine Oil - CHECK QUANTITY, CAP SECURE
10. Engine Compartment Door (Inbd) - SECURE, BLEED VALVE EXHAUST CLEAR
11. Exhaust Stack (Inbd) - CHECK FOR CRACKS
12. Top Cowling Locks (Inbd) - SECURE
13. Nacelle Cooling Ram Air Inlets - CLEAR
+ 14. Propeller - CHECK FOR NICKS, DEICE BOOT SECURE
15. Engine Intake - CLEAR
16. Top Cowling Locks (Outbd) - SECURE
17. Exhaust Stack (Outbd) - CHECK FOR CRACKS

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18. Generator Cooling Inlet - CLEAR
19. Engine Compartment Door (Outbd) - SECURE, BLEED VALVE EXHAUST CLEAR
20. Fuel Filter Strainer Sump - DRAIN
21. Standby Pump Drain - DRAIN
+ 22. Landing Gear (Doors, Wheel Well, Strut, Tires, Brakes) - CHECK
23. Fire Extinguisher (if installed) - CHECK PRESSURE
+ 24. Chock - REMOVE
25. Heated Fuel Vent - CLEAR
26. Ram Scoop Fuel Vent - CLEAR
27. Gravity Line Drain - DRAIN
28. Inverter Cooling Louvers - CLEAR
29. Wing Leading Edge Tank Sump - DRAIN
+ 30. External Power Door - CLOSED
31. Ice Light - CHECK
32. Outboard Deice Boot and Stall Strip - CHECK
+ 33. Tiedown - REMOVE
34. Flush Outboard Wing Fuel Tank Sump - DRAIN
+ 35. Main Fuel Tank Cap - SECURE
36. Navigation, Recognition, and Strobe Lights - CHECK
37. Static Wicks (4) - CHECK
38. Aileron and Bendable Tab - CHECK
39. Flaps - CHECK
40. Inverter Cooling Louver - CLEAR
+ 41. Brake Lines, Brake Wear, Brake Deice Lines (if installed) - CHECK
42. Oil Breather Vent - CLEAR

RIGHT AFT FUSELAGE

1. Lower Antennas - CHECK
2. Ventral Fin Drain Holes - CLEAR
3. Lower Aft Cabin Access Door - SECURE
+ 4. Tiedown - REMOVED
5. Oxygen Service Access Door - SECURE
6. Static Ports - CLEAR
7. ELT - ARMED
8. Cabin Air Exhaust - CLEAR
9. Access Panel - SECURE

TAIL SECTION

1. Ventral Fin and Static Wick (1) - CHECK
2. VOR antennas (Right and Left) - CHECK
3. Rudder, Rudder Tab, Stinger and Static Wicks (4) - CHECK
4. Horizontal Stabilizer, Boots and Static Wick (Right and Left) - CHECK
5. Elevator, Tab, and Static Wicks (3 Each Side) - CHECK
   Verify Tabs are in "0" (Neutral) Position.

NOTE

The elevator trim tab "0" (neutral) position is determined by observing that the trailing edge of the elevator trim tab aligns with the trailing edge of the elevator when the elevator is resting against the down stops.

6. Position Light, Tail Floodlights (Left and Right If Installed) - CHECK

LEFT AFT FUSELAGE

1. Access Panel - SECURE
2. Static Ports - CLEAR
3. Drains (Aft Compartment, Oxygen Discharge, Relief Tube) - CLEAR

BEFORE ENGINE STARTING

NOTE

Items marked with an "*" may be omitted at pilot's discretion after the first flight of each day.

* 1. Cabin Door (B200) or Cabin Door/Cargo Door (B200C) Circuitry Check - COMPLETE (See OTHER NORMAL PROCEDURES.)

2. Cabin Door (B200) or Cabin Door/Cargo Door (B200C) - LOCKED (See OTHER NORMAL PROCEDURES.)

WARNING

A crew member must close and lock the door.

* 3. Monogram Electric Toilet (if installed) - CONFIRM KNIFE VALVE OPEN

4. Load and Baggage - CONFIRM SECURE
5. Weight and C.G. - CHECKED
6. Seats - POSITIONED (SEATBACKS UPRIGHT, LATERAL-TRACKING SEATS - OUTBOARD POSITION)
7. Emergency Exit - CONFIRM SECURE AND UNLOCKED
8. Passenger Briefing - COMPLETE
9. Control Locks - CONFIRM REMOVED
10. Seats and Rudder Pedals - ADJUSTED
11. Seatbelts and Shoulder Harnesses - FASTENED
12. Oxygen System Ready - PULL ON

* 13. Oxygen System Preflight Inspection - CONFIRM COMPLETE (See OTHER NORMAL PROCEDURES.)

14. Fuel Panel Circuit Breakers - IN
15. Pilot's Instrument Panel - CHECK
    a. Compass Control - SLAVED
    b. Prop Sync Switch (if installed) TYPE I - OFF; TYPE II - ON

* 16. Pilot's Clock (Control Wheel) - CHECK AND SET

17. Pilot's Subpanel - CHECK
    a. Mic Selector Switch - NORMAL
    b. Parking Brake - CONFIRM SET
    c. Engine Ice Vane Switches - EXTEND

NOTE

The engine ice vanes should be extended for all ground operations to minimize ingestion of ground debris. Turn engine anti-ice off, when required, to maintain oil temperature within limits.

d. Pilot Air Control - AS REQUIRED
e. Defrost Air Control - AS REQUIRED
f. Landing Gear Control - DN
g. Landing Gear Relay Circuit Breaker - IN
h. All Other Switches - OFF

18. Avionics Panel Switches - AS REQUIRED
    - Radar - OFF OR STANDBY
19. Power Console - CHECK
    a. Power Levers - IDLE, FRICTION SET
    b. Propeller Levers - FULL FORWARD, FRICTION SET
    c. Condition Levers - FUEL CUT OFF, FRICTION SET
    d. Elevator, Aileron, and Rudder Trim Controls - SET
    e. Landing Gear Alternate Extension Handle - STOWED
20. Pedestal - CHECK
    a. EFIS Power Switches (if installed) - OFF
    b. EFIS Reversionary Switches (if installed) - NORMAL
    c. Cabin Pressure Switch - PRESS
    d. Rudder Boost Switch - ON
    e. Elevator Trim Switch - ON
    f. Pressurization Controller - SET
21. Copilot's Instrument Panel - CHECK
    - Compass Control - SLAVED

* 22. Copilot's Clock (Control Wheel) - CHECK AND SET

23. Copilot's Subpanel - CHECK
    a. Cabin Sign - NO SMOKE & FSB
    b. Vent Blower Switch - AUTO
    c. Bleed Air Valve Switches - OPEN OR ENVIR OFF
    d. Cabin Temp Mode Control - OFF
    e. Cabin/Cockpit Air Control - AS REQUIRED
    f. Copilot Air Control - AS REQUIRED
    g. Mic Selector Switch - NORMAL
    h. Oxygen Pressure - CONFIRM
    i. All Other Switches - OFF
24. Copilot's Circuit Breaker Panel - CHECK

* 25. Pilot's Static Air Source - NORMAL

* 26. Fire Extinguisher (Under Copilot's Chair) - CHECK

27. Fuel Panel - CHECK
    a. Battery - ON (L & R FUEL PRESS, L & R ICE VANE EXT annunciators - ILLUMINATED)
    * b. Firewall Shutoff Valves - CLOSE
    * c. Standby Pumps - ON (Listen For Operation, L & R FUEL PRESS annunciators - ILLUMINATED)
    * d. Firewall Shutoff Valves - OPEN (L & R FUEL PRESS annunciators - EXTINGUISHED)

* e. Standby Pumps - OFF (L & R FUEL PRESS annunciators - ILLUMINATED)
* f. Crossfeed - ALTERNATELY LEFT AND RIGHT (FUEL CROSSFEED annunciator - ILLUMINATED, L & R FUEL PRESS annunciators - EXTINGUISHED)
* g. Crossfeed - OFF
* h. Auxiliary Tank Transfer - AUTO
* i. No Transfer Lights - PRESS TO TEST (if extinguished)
  j. Fuel Quantity (Main and Auxiliary) - CHECK

* 28. Landing Gear Handle Lights - PRESS TO TEST
* 29. Hydraulic Fluid Sensor (if installed) - TEST (HYD FLUID LOW annunciator ILLUMINATED)
  30. Beacon - ON
  31. DC Volt/Loadmeters - PRESS TO CHECK VOLTAGE (23 volt minimum)
* 32. Annunciators - TEST
* 33. Stall Warning - TEST
* 34. Fire Detectors and Fire Extinguishers (if installed) - TEST

ENGINE STARTING (BATTERY)

1. Right Ignition and Engine Start - ON (R FUEL PRESS annunciator - EXTINGUISHED)
2. Right Condition Lever - LOW IDLE (at 12% N_1 or above)
3. ITT and N_1 - MONITOR (1000°C maximum)

CAUTION

If no ITT rise is observed within 10 seconds after moving the Condition Lever to LOW IDLE, move the Condition Lever to CUT OFF. Allow 60 seconds for fuel to drain and starter to cool, then follow ENGINE CLEARING procedures.

4. Right Oil Pressure - CHECK
5. Right Condition Lever - HIGH IDLE
6. Right Ignition and Engine Start - OFF (at 50% N_1 or above)
7. Right Generator - RESET, (HOLD for 1 sec) then ON (charge battery until loadmeter reads approximately 50% or less)

NOTE

In order to turn the generator ON, the generator control switch must be held in the spring-loaded RESET position for a minimum of one second, then released to ON.

NOTE

The BATTERY CHARGE annunciator will illuminate approximately 6 seconds after the generator is on the line. If the annunciator does not extinguish within 5 minutes, refer to NICKEL-CADMIUM BATTERY CHECK procedure (see OTHER NORMAL PROCEDURES).

8. Right Generator - OFF
9. Left Ignition and Engine Start - ON (L FUEL PRESS annunciator - EXTINGUISHED)
10. As Left N_1 rpm accelerates thru 12%:
    a. Left Condition Lever - LOW IDLE
    b. Right Generator - ON
11. ITT and N_1 - MONITOR (1000°C maximum)
12. Left Oil Pressure - CHECK

13. Left Ignition and Engine Start - OFF (at 50% N_1 or above)
14. D.C. Volt/Loadmeters - PRESS TO CHECK VOLTAGE (27.5-29.0 volts)
15. Left Generator - RESET, (HOLD for 1 sec) THEN ON
16. Right Condition Lever - REDUCE TO LOW IDLE or as required for air conditioner.

ENGINE STARTING (EXTERNAL POWER)

CAUTION

NEVER CONNECT AN EXTERNAL POWER SOURCE TO THE AIRPLANE UNLESS A BATTERY INDICATING A CHARGE OF AT LEAST 20 VOLTS IS IN THE AIRPLANE. If the battery voltage is less than 20 volts, the battery must be recharged, or replaced with a battery indicating at least 20 volts, before connecting external power. Only use an external power source fitted with an AN-type plug.

NOTE

When an external power source is used, it must be set to 28.0 to 28.4 volts and be capable of producing 1000 amperes momentarily and 300 amps continuously. The battery should be On to absorb transients present in some auxiliary power units. An EXT PWR annunciator is provided to alert the crew when an external DC power plug is connected to the airplane.

1. Avionics Master Switch - CONFIRM OFF
2. Left and Right Generator Switches - CONFIRM OFF
3. Battery - ON
4. External Power Source - TURN OFF, then CONNECT TO AIRPLANE
5. External Power Source - TURN ON
6. Voltmeter - 28.0 TO 28.4 VOLTS

NOTE

If the battery is partially discharged, the BATTERY CHARGE annunciator will illuminate approximately 6 seconds after the external power is on the line. If the annunciator does not extinguish within 5 minutes, refer to the NICKEL-CADMIUM BATTERY CHECK procedure (see OTHER NORMAL PROCEDURES).

7. Propeller Levers - FEATHER
8. Right Ignition and Engine Start - ON (R FUEL PRESS Annunciator - EXTINGUISHED)
9. Right Condition Lever - LOW IDLE (at 12% N_1 or above)
10. ITT and N_1 - MONITOR (1000°C maximum)

CAUTION

If no ITT rise is observed within 10 seconds after moving the Condition Lever to LOW IDLE, move the Condition Lever to FUEL CUT OFF, Allow 60 seconds for fuel to drain and starter to cool, then follow ENGINE CLEARING procedures.

11. Right Oil Pressure - CHECK
12. Right Ignition and Engine Start - OFF (at 50% N_1 or above)
13. Left Ignition and Engine Start - ON (L FUEL PRESS Annunciator - EXTINGUISHED)
14. Left Condition Lever - LOW IDLE (at 12% N_1 or above)

15. ITT and N_1 - MONITOR (1000°C maximum)
16. Left Oil Pressure - CHECK
17. Left Ignition and Engine Start - OFF (at 50% N_1 or above)
18. External Power Source - TURN OFF, DISCONNECT, SECURE DOOR
19. Left and Right Generators - RESET, (HOLD for 1 sec) THEN ON
20. Propeller Levers - FULL FORWARD

ENGINE CLEARING

The following procedure is used to clear an engine at any it is deemed necessary to remove internally trapped fuel and vapor, or if there is evidence of a fire within the engine. Air passing through the engine serves to purge fuel, vapor, or fire from the combustion section, gas generator turbine, power turbines and exhaust system.

1. Condition Lever - FUEL CUT OFF
2. Ignition and Engine Start - STARTER ONLY (for a maximum of 40 seconds)

CAUTION

Do not exceed the starter time limits; see Section II, LIMITATIONS.

3. Ignition and Engine Start - OFF

BEFORE TAXI

NOTE

Items marked with an "*" may be omitted at pilot's discretion after the first flight of the day.

NOTE

If excessive ITT's occur during any one of the following conditions, adjust the condition levers to a higher N_1 speed.
- When high generator loads are required.
- During operations at high ambient air temperatures.
- During operations at high field elevations.

If excessive ITT's are encountered, particularly if accompanied by a decreasing N_1, the associated generator should be turned off prior to attempting to accelerate the engine. If the right ITT is high, also turn off the air conditioner by selecting the CABIN TEMP MODE switch to OFF.

1. Inverters CHECK (Volts/Freq = 105-120 V/380-420Hz)
   a. Either Inverter - SELECT (Check Volts, Freq, INVERTER annunciator - EXTINGUISHED)
   b. Other Inverter - SELECT (Check Volts, Freq, INVERTER annunciator - EXTINGUISHED)
2. Inverter - SELECT NO. 1 OR NO. 2
3. Loadmeters - PARALLEL WITHIN 10%
4. Avionics Master - ON
5. EFIS Power (if installed) - ON
6. External Lights - AS REQUIRED
7. Cabin Lights - AS REQUIRED
8. Bleed Air Valves - OPEN OR ENVIR OFF - AS REQUIRED
9. Cabin Temp Mode - AUTO

For Training Purposes Only

10. Cabin Temperature - AS REQUIRED
11. Instruments - CHECK (EFIS brightness at minimum required, if installed)
* 12. Brake Deice (if installed) - CHECK
    a. Bleed Air Valves - OPEN
    b. Brake Deice - ON (BRAKE DEICE ON annunciator - ILLUMINATED)
    c. Condition Levers - HIGH IDLE (if Brake Deice is required)
    d. Brake Deice - OFF (BRAKE DEICE ON annunciator - EXTINGUISHED)
    e. Condition Levers - LOW IDLE

CAUTION

Do not leave brake deice on longer than required to check function of annunciators at ambient temperatures above 15°C.

NOTE

Brake deice control valves may become inoperative if valves are not cycled periodically. One cycle of the valves is required daily regardless of the weather conditions.

13. Flight Controls - CHECK PROPER DIRECTION AND FREEDOM OF MOVEMENT
14. Flaps - CHECK AND SET
15. Brakes - RELEASED AND CHECKED

CAUTION

Never taxi with a flat tire or flat shock strut. During taxi operations, particular attention should be given to propeller tip clearance. Extreme caution is required when operating on unimproved or irregular surfaces or when high winds exist. If operations produce a propeller rpm over 1600, retard propeller levers to the detent to limit rpm to 1600 to help reduce the possibility of ingestion of ground debris.

BEFORE TAKEOFF (Runup)

NOTE

Items marked with an "*" may be omitted at pilot's discretion after the first flight of the day.

1. Avionics and Radar - CHECK
2. EFIS (if installed) - TEST
3. Pressurization - CHECK AND SET
* a. Cabin Altitude Selector Knob - ADJUST SO THAT "CABIN ALT" DIAL INDICATES AN ALTITUDE 1000 FEET BELOW FIELD PRESSURE ALTITUDE
* b. Rate Control Selector Knob - SET INDEX AT 12-O'CLOCK POSITION
* c. Cabin Pressurization Switch - HOLD IN TEST
* d. Cabin Climb Indicator - CHECK FOR RATE OF DESCENT INDICATION
* e. Cabin Pressurization Switch - RELEASE TO PRESS WHEN PRESSURIZING IS CONFIRMED
  f. Cabin Altitude Selector Knob - ADJUST SO THAT INNER SCALE (ACFT ALT) INDICATES PLANNED CRUISE ALTITUDE PLUS 1000 FEET (If this setting does not result in an outer scale (CABIN ALT) indication of at least 500 feet above take-off field pressure altitude, adjust as required.)

* 4. Autopilot - CHECK

* 5. Elevator Trim - CHECK

a. Elevator Trim - ON

b. Pilot's and Copilot's Switches - CHECK OPERATION

c. Trim Quick Disconnect - CHECK (ELEC TRIM OFF annunciator - ILLUMINATED)

d. Elevator Trim - OFF, THEN ON (ELEC TRIM OFF annunciator - EXTINGUISHED)

WARNING

Operation of the electric trim system should occur only by movements of pairs of switches. Any movement of the elevator trim wheel while depressing only one switch denotes a system malfunction. The elevator trim control switch must then be turned OFF and flight conducted only by manual operation of the trim wheel.

6. Trim Tabs - CONFIRM SET

* 7. Primary Governors, Overspeed Governors and Rudder Boost - TEST

a. Rudder Boost - ON

b. Prop Governor Test Switch - HOLD TO PROP GOV TEST

c. Power Levers (Individually) - INCREASE UNTIL PROP IS STABILIZED AT 1830 TO 1910 RPM.

d. Propeller Lever - RETARD TO DETENT, THEN FULL FORWARD (to check primary governor)

e. Power Lever - CONTINUE TO INCREASE UNTIL RUDDER MOVEMENT IS NOTED. (Observe ITT and Torque Limits)

f. Power Lever - IDLE

g. Repeat Steps c - f

h. Prop Governor Test - RELEASE

* 8. Autofeather (if installed) - CHECK

a. Power Levers - APPROXIMATELY 500 FT-LBS TORQUE

b. Autofeather Switch - HOLD TO TEST (L & R AUTOFEATHER annunciators - ILLUMINATED)

c. Power Levers - RETARD INDIVIDUALLY:

1) At Approximately 410 ft-lbs - OPPOSITE ANNUNCIATOR EXTINGUISHED

2) At Approximately 260 ft-lbs - BOTH ANNUNCIATORS EXTINGUISHED (Propeller Starts to Feather)

NOTE

Autofeather annunciator will cycle on and off with each fluctuation of torque as the propeller feathers.

d. Power Levers - BOTH RETARDED (L & R AUTOFEATHER annunciators - EXTINGUISHED, neither propeller feathers)

9. Autofeather (if installed) - ARM

10. Propeller Feathering (Manual) - CHECK

* 11. Vacuum and Pneumatic Pressure - CHECK

a. Left Bleed Air Valve - INSTR & ENVIR OFF

1) L BL AIR OFF annunciator - ILLUMINATED

2) Gyro Suction Gage and Pneumatic Pressure Gage in Green Arc

b. Right Bleed Air Valve - INSTR & ENVIR OFF

1) R BL AIR OFF, L & R BLEED FAIL annunciators - ILLUMINATED

2) Gyro Suction Gage and Pneumatic Pressure Gage - Zero

c. Left Bleed Air Valve - OPEN

1) L BL AIR OFF, L & R BLEED FAIL annunciators - EXTINGUISHED

2) Gyro Suction Gage and Pneumatic Pressure Gage - IN GREEN ARC

d. Right Bleed Air Valve - OPEN (R BL AIR OFF annunciator - EXTINGUISHED)

12. Engine Ice Vanes - CHECK (see OTHER NORMAL PROCEDURES)
13. Systems For Icing Flight - CHECK, IF REQUIRED (see OTHER NORMAL PROCEDURES)
14. Fuel Quantity, Flight and Engine Instruments - CHECK

BEFORE TAKEOFF (Final Items)

1. Propeller Levers - CONFIRM FULL FORWARD
2. Flaps - CONFIRM SET
3. Trim - CONFIRM SET
4. Brake Deice (if installed) - OFF
5. Left and Right Bleed Air - OPEN
6. Aft Blower (if installed) - AS REQUIRED
7. Annunciators - EXTINGUISHED OR CONSIDERED
8. Ice Protection - ON, IF REQUIRED
   a. Auto Ignition - ON (L and R IGNITION ON annunciators - ILLUMINATED)
   b. Engine Ice Vanes - CONFIRM EXTENDED (L & R ENG ANTI-ICE annunciators - ILLUMINATED)
   c. Windshield Anti-ice - NORMAL/HI
   d. Prop Heat - AUTO
   e. Left and Right Fuel Vent Heat - ON
   f. Stall Warning Heat - ON
   g. Left and Right Pitot Heat - ON

CAUTION

Prolonged use of stall warning and pitot heat on the ground will damage the heating elements.

9. Generator Load - CHECK
10. Engine Ice Vanes (if not required) - CONFIRM RETRACTED
11. External Lights - AS REQUIRED
12. Transponder - ON
13. V_1, V_R, V_2, and Minimum Takeoff Power - CONFIRM

TAKEOFF

1. Brakes - HOLD
2. Power - SET (ensure minimum takeoff power is available)
3. Autofeather Annunciators (if installed) - ILLUMINATED
4. Brakes - RELEASED

NOTE

Increasing airspeed will cause torque and ITT to increase.

5. Landing Gear - UP (when positive climb established)

6. Flaps - UP (at blue line, 121 knots minimum)

CLIMB

1. Yaw Damp - ON
2. Climb Power - SET
3. Propellers - 1900 RPM
4. Prop Sync (if installed) - ON
5. Autofeather (if installed) - OFF
6. Engine Instruments - MONITOR
7. Cabin Sign - AS REQUIRED
8. Pressurization - CHECK
9. Aft Blower (if installed) - AS REQUIRED
10. Lights - AS REQUIRED

CRUISE

WARNING

Do not lift power levers in flight.

1. Cruise Power - SET PER CRUISE POWER TABLES OR GRAPHS
2. Engine Instruments - MONITOR
3. Auxiliary Fuel Gages - MONITOR (Ensure fuel is being transferred from the auxiliary tanks.)
4. Pressurization - MONITOR (Reset if cruise altitude changes by 1000 feet or more)

DESCENT

1. Pressurization - SET
   a. Cabin Altitude Selector Knob - SET PER PRESSURIZATION CONTROLLER SETTING FOR LANDING GRAPH, OR SO THAT "CABIN ALT" DIAL INDICATES LANDING FIELD PRESSURE ALTITUDE PLUS 500 FEET.
   b. Rate Control Selector Knob - AS DESIRED
2. Altimeter - SET
3. Cabin Sign - AS REQUIRED
4. Windshield Anti-ice - AS REQUIRED (NORMAL or HI well before descent into warm, moist air, to aid in defogging.)
5. Autofeather (if installed) - ARM
6. Fuel Balance - CHECK
7. Power - AS REQUIRED

NOTE

Approximately 75% N_1 is required to maintain the pressurization schedule during descent.

BEFORE LANDING

CAUTION

Propeller operation in the range of 1750-1850 rpm shold be avoided as it may cause ILS interference.

For Training Purposes Only

1. Approach Speed - CONFIRM
2. Pressurization - CHECK
3. Cabin Sign - NO SMOKE & FSB
4. Flaps - APPROACH
5. Landing Gear - DN
6. Lights - AS REQUIRED

NOTE

Under low visibility conditions, landing and taxi lights should be left off due to light reflections.

7. Prop Sync (TYPE I) - OFF
8. Radar - AS REQUIRED
9. Surface Deice - CYCLE (as required)

NOTE

If crosswind landing is anticipated, determine crosswind component from Section V, PERFORMANCE. Immediately prior to touchdown, lower upwind wing and align the fuselage with the runway by use of rudder. During rollout, hold aileron control into the wind and maintain directional control with rudder and brakes. Use propeller reverse as desired.

NORMAL LANDING

1. Flaps - DOWN
2. Airspeed - NORMAL LANDING APPROACH SPEED
3. Yaw Damp - OFF
4. Power Levers - IDLE
5. Propeller Levers - FULL FORWARD

CAUTION

To ensure constant reversing characteristics, the propeller levers must be in the high rpm position.

After Touchdown:

6. Power Levers - BETA RANGE OR REVERSE (AS REQUIRED)
7. Brakes - AS REQUIRED

MAXIMUM REVERSE THRUST LANDING

1. Flaps - DOWN
2. Airspeed - NORMAL LANDING APPROACH SPEED
3. Yaw Damp - OFF
4. Condition Levers - HIGH IDLE
5. Propeller Levers - HIGH RPM

After Touchdown:

6. Power Levers - LIFT AND REVERSE

7. Brakes - AS REQUIRED
8. Condition Levers - LOW IDLE

CAUTION

If possible, propellers should be moved out of reverse at approximately 40 knots to minimize blade erosion. Care must be exercised when reversing on runways with loose sand, dust, or snow on the surface. Flying gravel will damage propeller blades, and dust or snow may impair the pilot's visibility.

BALKED LANDING

1. Power - MAXIMUM ALLOWABLE
2. Airspeed - 100 KNOTS (when clear of obstacles, establish normal climb)
3. Flaps - UP
4. Landing Gear - UP

AFTER LANDING

1. Auto Ignition - OFF
2. Engine Ice Vanes - EXTEND

CAUTION

The engine ice vanes should be extended for all ground operations to minimize ingestion of ground debris. Turn engine anti-ice off, when required, to maintain oil temperature within limits.

3. Lights - AS REQUIRED
4. Ice Protection - OFF
5. Transponder - STANDBY
6. Radar - STANDBY or OFF
7. Trim Tabs - SET
8. Flaps - UP

SHUTDOWN AND SECURING

1. Parking Brake - SET
2. Standby Boost Pumps and Crossfeed - OFF
3. EFIS Power (if installed) - OFF
4. Avionics Master - OFF
5. Inverter - OFF
6. Autofeather (if installed) - OFF
7. Lights - OFF
8. Vent Blower - AUTO
9. Cabin Temp Mode - OFF
10. Aft Blower (if installed) - OFF
11. Radiant Heat (if installed) - OFF
12. Battery - CHARGED
13. ITT - STABILIZED AT MINIMUM TEMPERATURE FOR ONE MINUTE

14. Condition Levers - FUEL CUT OFF

CAUTION

Monitor ITT during shutdown. If sustained combustion is observed, proceed immediately to the engine clearing procedure. During shutdown, ensure that the compressors decelerate freely. Do not close the firewall fuel valves for normal engine shutdown.

15. Propeller Levers - FEATHER
16. DC Volt/Loadmeters - CHECK VOLTAGE (No voltage indicates current limiter is out)
17. Overhead Panel Switches - OFF
18. Oxygen System Ready - PUSH OFF
19. Battery and Generator Switches - OFF (below 15% N_1)

NOTE

N_1 decreasing below 15% indicates the starter relay is not engaged.

20. Control Locks - INSTALL
21. Emergency Exit - LOCKED
22. Electric Toilet (if installed) - SERVICE AS REQUIRED
23. Tiedowns and Chocks - AS REQUIRED
24. External Covers - AS REQUIRED
25. Propeller Locks - AS REQUIRED
26. Oil Quantity Check - AS REQUIRED

OTHER NORMAL PROCEDURES

OXYGEN SYSTEM

PREFLIGHT INSPECTION

1. Passenger Manual Drop Out - PUSH OFF
2. Oxygen System Ready - PULL ON
3. Crew Diluter Demand Masks - DON MASK, CHECK FIT AND OPERATION, AND STOW

WARNING

Beard and mustaches should be carefully trimmed so that they will not interfere with the proper sealing of an oxygen mask. The fit of the oxygen mask around the beard or mustache should be checked on the ground for proper sealing. Studies conducted by the military and the FAA conclude that oxygen masks do not seal over beards and mustaches. Hats and "earmuff" type headsets must be removed prior to donning crew oxygen masks. Headsets and eyeglasses worn by crew members may interfere with quick-donning capabilities.

4. Oxygen Duration - DETERMINE

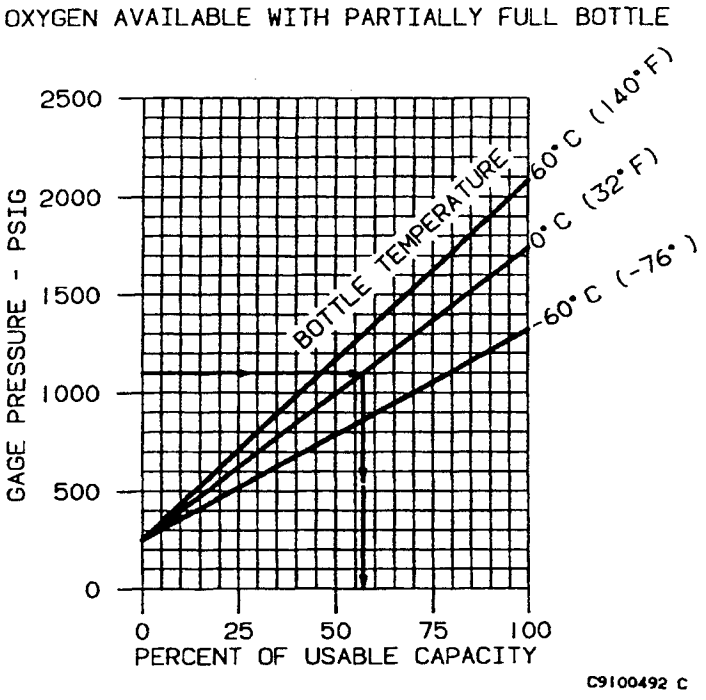
NOTE

A bottle of 1850 psi at 15°C is fully charged (100% capacity). Read duration directly from the table.

a. Read the oxygen pressure from the gage.

b. Read the IOAT.

c. Determine the percent of usable capacity from the following graph (eg., 1100 psi at 0°C = 57%).



d. Compute the oxygen duration in minutes from the table by multiplying the full bottle duration by the percent of usable capacity. eg.,

1) Pilot and copilot with masks set at 100% plus 6 passengers = 10 people using oxygen.

NOTE

Pilot and copilot are each counted as 2 people with diluter demand masks set at the 100% or NORMAL mode.

2) Cylinder volume = 115 cubic feet.

3) Duration with full bottle = 73 minutes

4) Duration with 57% capacity = .57 x 73 = 41 minutes.

OXYGEN DURATION WITH FULL BOTTLE (100% CAPACITY)

Stated Cylinder Size (Cu Ft)	†NUMBER OF PEOPLE USING																
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	†16	†17
	DURATION IN MINUTES																
22	144	72	48	36	28	24	20	18	16	14	13	12	11	10	*	*	*
49 or 50	317	158	105	79	63	52	45	39	35	31	28	26	24	22	21	19	18
66	422	211	140	105	84	70	60	52	47	42	38	35	32	30	28	26	24
76 or 77	488	244	162	122	97	81	69	61	54	48	44	40	37	34	32	30	28
115	732	366	244	183	146	122	104	91	81	73	66	61	56	52	48	45	43

**Will not meet oxygen requirements*

† For oxygen duration computations, count each diluter-demand crew mask in use as 2 (e.g., with 4 passengers and a crew of 2, enter the table at 8 people using).

HEATING/COOLING SYSTEM

1. Bleed Air Valves - OPEN (ENVIR OFF for more efficient cooling on the ground.)
2. Cabin Temperature Mode - AUTO
3. Vent Blower - AUTO
4. Radiant Heat or Aft Blower (if installed) - AS REQUIRED (The radiant heat system should be used in conjunction with manual temp control mode only.)
5. Temperature Control - AS REQUIRED
6. Cabin Air Control - AS REQUIRED (to divert cabin air flow to the cockpit)

NOTE

With the Cabin Temp Mode switch in AUTO, MAN HEAT or MAN COOL, the ventilation blower operates in the low position. For increased air circulation, turn the Vent Blower Switch to HIGH. For maximum cooling, the ventilation blower should be in the HIGH mode and the aft blower (if installed) should be selected ON. With air conditioner on, maintain at least 60% N_1 speed on the right engine. If below N_1 minimum speed, the AIR COND N_1 LOW annunciator will illuminate, and the air conditioner clutch will disengage. For maximum heating, the ventilation blowers should be selected to HIGH and the aft blower should be selected OFF.

RADIANT HEAT (IF INSTALLED)

Overhead radiant heat, operated by a switch in the environmental group, can be used in conjunction with an auxiliary power unit to warm the cabin prior to engine starting, and may be used for supplemental heat in flight, if required.

DEFROSTER AIR

1. Windshield Defroster Air Control (right side of pilot's control column) - ON (pull)

2. Pilot, Copilot, and Cabin Air Controls - OFF (if increased defroster air flow is required)

BLENDING ANTI-ICING ADDITIVE TO FUEL

The following procedure must be observed when blending anti-icing additive with fuel.

1. The additive must conform to specification MIL-I-27686 or MIL-I-85470.
2. The concentration of the additive must be a minimum of 0.10% and a maximum of 0.15% by volume. Thus, a 20 oz. can of additive is sufficient for 105 to 155 gallons of fuel.
3. When blending the additive as the airplane is being refueled, use the following precautions:
   a. Refuel at a rate of 30 to 45 gallons per minute. A rate of less than 30 GPM may be used when topping off the tanks.
   b. Start additive after fuel flow starts, and stop before fuel flow stops.
   c. Ensure additive is directed into the flowing fuel stream.
   d. Do not allow concentrated additive to contact coated interior of fuel cells or airplane surfaces.

ADDING BIOCIDE TO FUEL

Refer to the Super King Air 200 Series Maintenance Manual and to the latest revision of Pratt & Whitney Canada Engine Service Bulletin No. 3044 for concentrations to use, and for procedures, recommendations, and limitations pertaining to the use of biocidal/fungicidal additives in turbine fuels.

USE OF JET B, JP-4, AND AVIATION GASOLINE

Fuel quantity indicators will not indicate correctly when using Jet B, JP-4, or aviation gasoline due to the differences in their density and dielectric constants. The indicated fuel quantity should be corrected using the factors show below. These factors have been generalized in some cases for simplicity, thus results should be considered as approximations.

When using Jet B or JP-4 multiply the indicated fuel quantity by .96 to obtain the corrected fuel quantity.

When using aviation gasoline, multiply the indicated fuel quantity by .94 to obtain the corrected fuel quantity.

NICKEL-CADMIUM BATTERY CHECK (GROUND OPERATION ONLY)

Illumination of the BATTERY CHARGE annunciators indicates an above-normal charge current. Following an engine start, the battery recharge current is very high and causes the illumination of the BATTERY CHARGE anunciator. It should normally extinguish within five minutes. If it does not, or if it should reappear, the battery charge current should be monitored using the procedure shown below until it decreases to a level to extinguish the BATTERY CHARGE annunciator. Check the battery charge current every 90 seconds until the charge current decreases sufficiently to extinguish the annunciator. No decrease in charging current between checks indicates an unsatisfactory condition and the battery should be removed and checked by a qualified nickel-cadmium battery shop. Do not take off with annunciator illuminated unless a decreasing battery charge current is confirmed.

1. Either Generator - OFF
2. Voltmeter - ENSURE INDICATION OF 28 VOLTS
3. Battery Switch - OFF MOMENTARILY, NOTING DECREASE IN LOADMETER.

If Decrease in Loadmeter Exceeds 2.5%:

4. Battery - CONTINUE TO CHARGE, REPEATING STEP 3 EVERY 90 SECONDS.

5. Battery Charge Annunciator - EXTINGUISHED WHEN DECREASE IN LOADMETER IS LESS THAN 2.5%.

COLD WEATHER PROCEDURES (SNOW, SLUSH, AND ICE)

PREFLIGHT INSPECTION

Verify that the tires are not frozen to the ramp, and that the brakes are free of ice contamination. Deicing or anti-icing solutions may be used on the tires and brakes if they are frozen. Solutions which contain a lubricant, such as oil, must not be used as they will decrease the effectiveness of the brakes.

In addition to the normal exterior preflight inspection, special attention should be given all vents, openings, static ports, control surfaces, hinge points, and the wing, tail, and fuselage surfaces for accumulations of ice or snow. Removal of these accumulations is necessary prior to takeoff. Airfoil contours may be altered by the ice and snow to the extent that their lift qualities will be seriously impaired. Ice and snow on the fuselage can increase drag and weight. Frost that may form on the wing fuel tank bottom skins need not be removed prior to flight. Frost that may accumulate on other portions of the wing, the tail surfaces, or on any control surface, must be removed prior to flight.

Inspect the propeller blades and hubs for ice and snow. Unless engine inlet covers have been installed during snow or icing conditions, the propellers should be turned by hand in the direction of normal rotation to make sure they are free to rotate prior to starting engines.

The removal of frozen deposits by chipping or scraping is not recommended. A soft brush, squeegee, or mop may be used to clear snow that is not adhering to the surfaces. If use of deicing/anti-icing fluids are required to produce a clean airplane, special attention must be given to ensure that the pitot masts, static ports, fuel vents, cockpit windows and the area forward of the cockpit windows are free of the deicing/anti-icing solution. Both wings and both stabilizers must receive the same complete treatment. The type and concentration of deicing/anti-icing solution being applied and the rate of precipitation will affect the length of time the treatment will be effective. Refer to Chapter 12 of the *Beech Super King Air 200 Series Maintenance Manual* and Section VIII of this manual for additional information on deicing and anti-icing of airplanes on the ground. See Section II, Limitations, for a list of approved fluids.

Complete the normal preflight procedures, including a check of the flight controls for complete freedom of movement.

After engine start, exercise the propellers through low- and high-pitch and into reverse range to flush any congealed oil through the system.

If the optional brake deicing system is installed, turn it on prior to taxi if brakes require deicing.

TAXIING

Taxiing through deep snow or slush should be avoided when possible. Snow and slush can be forced into brake assemblies which may cause the brakes to freeze during a prolonged hold on the ground or during the subsequent flight. Keep flaps retracted during taxiing to avoid throwing snow or slush into flap mechanisms and to minimize damage to flap surfaces.

Glaze ice can be difficult to see. Therefore, taxi slowly and allow more clearance from objects when maneuvering the airplane.

BEFORE TAKEOFF

After completion of the normal Before Takeoff checklist, verify that the airplane is still free of frozen contaminants.

Ensure the runway is free from hazards such as snow drifts, glazed ice, and ruts.

WARNING

Ice, frost, or snow on top of deicing/anti-icing solutions must be considered as adhering to the airplane. Takeoff should not be attempted.

If visible moisture will be encountered during the takeoff, engine ice vanes must be extended to reduce the possibility of ice being ingested into the engine air inlet.

TAKEOFF

Allow additional take-off distance when snow or slush is on the runway. Extra cycling of the landing gear when above 500 feet AGL may help clear any contamination from the gear system.

When using FAA Approved SAE Type II or Type IV deicing/anti-icing fluids in the concentrated form, the control column force required to rotate for takeoff may temporarily increase approximately 20 pounds. The cruise, descent, approach and landing phases of flight are not affected by the use of these fluids.

LANDING

If it is possible that the brakes may be restricted by ice accumulations from previous ground or in-flight icing conditions, turn the brake deicing system (if installed) on during the descent.

Braking and steering are less effective on slick runways. Also, hydroplaning may occur under wet runway conditions at higher speeds. Use the rudder to maintain directional control until the tires make solid contact with the runway surface.

Selecting reverse thrust can effectively reduce stopping distances on slick runways; however, reverse thrust may cause snow or moisture to be thrown forward, temporarily reducing visibility.

SHUTDOWN AND SECURING

Avoid setting the parking brake, if possible. This will help reduce the possibility of freezing the brakes. Proper chocking can be used to prevent the airplane from rolling.

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ICING FLIGHT

This airplane is approved for flight in icing conditions as defined in FAR 25, Appendix C. These conditions do not include, nor were tests conducted in, all conditions that may be encountered (e.g., freezing rain, freezing drizzle, mixed conditions, or conditions defined as severe). Some icing conditions not defined in FAR 25 have the potential of producing hazardous ice accumulations, which: 1) exceed the capabilities of the airplane's ice protection equipment; and/or 2) create unacceptable airplane performance. Flight into icing conditions which lie outside the FAR-defined conditions is not prohibited; however, pilots must be prepared to divert the flight promptly if hazardous ice accumulations occur.

Refer to Section II for limitations relating to icing flight, and Section IIIA for abnormal procedures associated with icing equipment malfunctions and procedures required by FAA AD 96-09-13.

WARNING

Due to distortion of the wing airfoil, ice accumulations on the leading edges can cause a significant loss in rate of climb and in speed performance, as well as increases in stall speed. Even after cycling deicing boots, the ice accumulation remaining on the boots and unprotected areas of the airplane can cause large performance losses. For the same reason, the aural stall warning system may not be accurate and should not be relied upon. Maintain a comfortable margin of airspeed above the normal stall airspeed. In order to minimize ice accumulation on unprotected surfaces of the wing, maintain a minimum of 140 knots during operations in sustained icing conditions. In the event of windshield icing, reduce airspeed to 226 knots or below. Prior to a landing approach, cycle the deicing boots to shed any accumulated ice.

BEFORE TAKEOFF (RUNUP)

1. Auto Ignition - CHECK
   a. Power Levers - IDLE
   b. Auto Ignition - ARM (L & R IGNITION ON annunciators - ILLUMINATED)
   c. Power Levers - ADVANCE ABOVE 360-460 FT-LBS TORQUE (L & R IGNITION ON annunciators - EXTINGUISHED)
   d. Power Levers - IDLE (L & R IGNITION ON annunciators - ILLUMINATED)
   e. Auto Ignition - OFF (L & R IGNITION ON annunciators - EXTINGUISHED)
2. Engine Ice Vanes - CHECK (system initially on)
   a. Engine Ice Vanes - RETRACT (L & R ICE VANE EXT annunciators - EXTINGUISHED)
   b. Engine Ice Vanes (if required for takeoff) - EXTEND (L & R ICE VANE EXT annunciators - ILLUMINATED)
3. Windshield Anti-ice - CHECK
   a. Windshield Anti-ice - HI (Observe increase on left and right loadmeters.)
   b. Windshield Anti-ice - OFF, THEN NORMAL (Observe increase on left and right loadmeters.)
   c. Windshield Anti-ice - OFF
4. Electrothermal Propeller Deice - CHECK

CAUTION

Do not operate propeller deice when the propellers are static.

   a. Automatic Prop Deice - AUTO
   b. Deice Ammeter - 14 to 18 amps

For Training Purposes Only

- Monitor for 90 seconds to ensure automatic timer operation.

On Airplanes Prior to BB-829:

2) Manual Propeller Deice Switch - MOMENTARILY HOLD IN "INNER" POSITION, THEN "OUTER" (Small loadmeter deflection on both meters in each position indicates the manual system is operating).

On Airplanes BB-829 and After, BL-37 and After:

c. Manual Propeller Deice Switch - MOMENTARILY HOLD IN "MANUAL" POSITION (Small loadmeter deflection on both meters indicates manual system is operating).

NOTE

Use of current for the manual (backup) system is not registered on the propeller deice ammeter; however, it will be indicated as part of the airplane's load on the loadmeters (small needle deflection) when the system is switched on.

d. Automatic Propeller Deice Switch - OFF

5. Surface Deice System - CHECK
   a. Condition Levers - HIGH IDLE
   b. Pneumatic Pressure - CHECK
   c. Surface Deice Switch - SINGLE AND RELEASE
      1) Pneumatic Pressure - Will Decrease Momentarily
      2) Check boots visually, where possible, for inflation and hold down.
      3) Inflation time is 6 seconds for wings, followed by 4 seconds for horizontal stabilizer.
   d. Surface Deice Switch - MANUAL AND HOLD
      1) Pneumatic Pressure - Will Decrease Momentarily
      2) Check boots visually, where possible, for inflation.
   e. Surface Deice Switch - RELEASE
      - Check boots visually, where possible for hold down.
   f. Condition Levers - LOW IDLE

WARNING

Do not cycle boots during takeoff.

6. Stall Warning and Pitot Heat - CHECK

CAUTION

Prolonged use of stall warning and pitot heat on the ground will damage the heating elements.

IN FLIGHT

1. Engine Ice Protection

Before visible moisture is encountered at +5°C and below, or;

At night when freedom from visible moisture is not assured at +5°C and below (Operation of strobe lights will sometimes show ice crystals not normally visible).

a. Engine Ice Vanes - EXTEND (L & R ICE VANE EXT annunciators - ILLUMINATED)
b. Check proper operation by noting torque drop.

WARNING

If in doubt, extend the vanes. Engine icing can occur even though no surface icing is present. If freedom from visible moisture can not be assured, engine ice protection should be activated. Visible moisture is moisture of any form; clouds, ice crystals, snow, rain, sleet, hail or any combination of these. Operation of strobe lights will sometimes show ice crystals not normallly visible.

NOTE

If either engine's ice vane does not attain the selected position within 15 seconds, the appropriate L or R ICE VANE annunciator will illuminate. See ENGINE ICE VANE FAILURE in Section IIIA, ABNORMAL PROCEDURES.

2. Auto Ignition
   - Auto Ignition - ARM

NOTE

Auto Ignition must be ARMED for icing flight, precipitation, and operation during turbulence. To prevent prolonged operation of the ignitors with the system ARMED, do not reduce power levers below 500 ft-lbs torque.

3. Electrothermal Propeller Deice
   a. Automatic Prop Deice - AUTO (The system may be operated continuously in flight, and will function automatically until the switch is turned off.)
   b. Relieve propeller imbalance due to ice by increasing rpm briefly and returning to the desired setting. Repeat as necessary.

CAUTION

If the deice ammeter does not indicate 14 to 18 amperes, or the automatic timer fails to switch, refer to Section IIIA, ABNORMAL PROCEDURES.

4. Surface Deice

WARNING

All components of the surface deice system must be monitored during icing flight to ensure the system is functioning normally. These components include:
Pneumatic Pressure Gage. The gage should indicate 12.0 - 20.0 psi before boots are activated. The pressure will momentarily decrease when the boots are activated.
Gyro Suction Gage. The gage should indicate in the area of the green arc corresponding to the airplane altitude. The vacuum will momentarily decrease when the boots are activated.
Pneumatic Boots. Visually monitor the boots, where possible, to ensure ice is being removed.

CAUTION

Operation of the surface deice system in ambient temperatures below -40°C can cause permanent damage to the deice boots.

NOTE

Either engine will supply sufficient pneumatic pressure for deice operation.

When Ice Accumulates 1/2 to 1 inch:

a. Surface Deice Switch - SINGLE AND RELEASE

b. Repeat as required.

If Single Position of the Surface Deice Switch Fails:

c. Surface Deice Switch - MANUAL AND HOLD FOR A MINIMUM OF 6 SECONDS, THEN RELEASE.

d. Repeat as required.

5. Windshield Anti-ice - NORMAL/HI

NOTE

Maximum airspeed for effective windshield anti-icing is 226 Knots.

6. Left and Right Fuel Vent Heat - ON
7. Left and Right Pitot Heat - ON
8. Stall Warning Heat - ON
9. Ice Lights - AS REQUIRED
10. Alternate Static Air Source - Refer to Section IIIA, ABNORMAL PROCEDURES.

BEFORE LANDING

1. Surface Deice Switch - SINGLE AND RELEASE
2. Approach Speeds (Flaps Up or Flaps Down) - INCREASE IF RESIDUAL ICE REMAINS
3. Landing Distances - WILL INCREASE IF APPROACH SPEED IS INCREASED

NOTE

Prior to the landing approach, cycle the wing deice boots to shed as much residual ice as possible, regardless of the amount of ice remaining on the boots. Stall speeds can be expected to increase if ice is not shed from the deice boots.

PRACTICE DEMONSTRATION OF V_{MCA}

V_{MCA} demonstration may be required for multi-engine pilot certification. The following procedure shall be used at a safe altitude of at least 5000 feet above the ground in clear air only.

WARNING

IN-FLIGHT ENGINE CUTS BELOW V_{SSE} SPEED OF 104 KNOTS ARE PROHIBITED.

1. Landing Gear - UP
2. Flaps - UP
3. Airspeed - ABOVE 104 KNOTS (V_{SSE})
4. Propeller Levers - HIGH RPM
5. Power Lever (simulated inoperative engine) - IDLE
6. Power Lever (other engine) - MAXIMUM ALLOWABLE
7. Airspeed - Reduce approximately 1 knot per second until either V_{MCA} or stall warning is obtained.

CAUTION

Use rudder to maintain directional control (heading) and aileron to maintain 5° bank towards the operative engine (lateral attitude). At the first sign of either V_{MCA} or stall warning (which may be evidenced by: inability to maintain heading or lateral attitude, aerodynamic stall buffet, or stall warning horn sound) immediately initiate recovery: reduce power to idle on the operative engine and immediately lower the nose to regain V_{SSE}.

PRACTICE LANDING GEAR MANUAL EXTENSION (MECHANICAL SYSTEM)

On airplanes prior to BL-73 and BB-1193 (except BB-1158 and BB-1167) not incorporating Beech Kit P/N 101-8018-1

1. Airspeed - ESTABLISH 130 KNOTS
2. Landing Gear Relay Circuit Breaker (pilot's subpanel) - PULL
3. Landing Gear Control - DN
4. Alternate Engage Handle - LIFT AND TURN CLOCKWISE TO THE STOP TO ENGAGE
5. Alternate Extension Handle - PUMP UP AND DOWN UNTIL THE THREE GREEN GEAR-DOWN ANNUNCIATORS ARE ILLUMINATED. ADDITIONAL PUMPING WHEN ALL THREE ANNUNCIATORS ARE ILLUMINATED COULD DAMAGE THE DRIVE MECHANISM AND PREVENT SUBSEQUENT ELECTRICAL GEAR RETRACTION.

LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION (MECHANICAL SYSTEM)

On airplanes prior to BL-73 and BB-1193 (except BB-1158 and BB-1167) not incorporating Beech Kit P/N 101-8018-1

After a practice manual extension of the landing gear, the gear may be retracted electrically as follows:

1. Alternate Engage Handle - ROTATE COUNTERCLOCKWISE AND PUSH DOWN
2. Alternate Extension Handle - STOW
3. Landing Gear Control Circuit Breaker (pilot's subpanel) - PUSH IN
4. Landing Gear - UP

PRACTICE LANDING GEAR MANUAL EXTENSION (HYDRAULIC SYSTEM)

On airplanes BB-1158, BB-1167, BB-1193 and after, BL-73 and after, and those prior airplanes incorporating Beech Kit P/N 101-8018-1

1. Airspeed - BELOW 181 KNOTS
2. Landing Gear Relay Circuit Breaker (pilot's subpanel) - PULL
3. Landing Gear Control - DN
4. Alternate Extension Handle - PUMP UP AND DOWN UNTIL THE THREE GREEN GEAR-DOWN ANNUNCIATORS ARE ILLUMINATED.
5. Alternate Extension Handle - STOW

LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION (HYDRAULIC SYSTEM)

On airplanes BB-1158, BB-1167, BB-1193 and after, BL-73 and after, and those prior airplanes incorporating Beech Kit P/N 101-8018-1

After a practice manual extension, the gear may be hydraulically retracted as follows:

1. Alternate Extension Handle - CONFIRM STOWED
2. Landing Gear Relay Control Circuit Breaker (pilot's subpanel) - PUSH IN
3. Landing Gear - UP

For Training Purposes Only

SIMULATING ONE-ENGINE-INOPERATIVE (ZERO THRUST)

When establishing zero thrust operation, use the power setting listed below. By using this power setting to establish zero thrust, one avoids the inherent delays of restarting a shut-down engine and preserves almost instant power to counter any attendant hazard.

1. Propeller - 1600 RPM
2. Power Lever - SET 120 ft-lbs torque

NOTE

This setting will approximate Zero Thrust at low altitudes using recommended One-Engine-Inoperative Climb speeds.

NOISE CHARACTERISTICS

Approach to and departure from an airport should be made so as to avoid prolonged flight at low altitude near noise-sensitive areas. Avoidance of noise-sensitive areas, if practical, is preferable to overflight at relatively low altitudes.

For VFR operations over outdoor assemblies of persons, recreational and park areas, and other noise-sensitive areas, pilot's should make every effort to fly not less than 2000 feet above the surface, weather permitting, even though flight at lower level may be consistent with the provisions of government regulations.

NOTE

The preceding recommended procedures do not apply where they would conflict with Air Traffic Control clearances or instructions, or where, in the pilot's judgement, an altitude less than 2000 feet is necessary to adequately exercise his duty to see and avoid other airplanes.

The flyover noise level established in compliance with FAR 36 is:

79.2 dB(A) for Hartzell 3-blade propellers.

75.4 dB(A) for McCauley 3-blade propellers.

No determination has been made by the Federal Aviation Administration that the noise level of this airplane is or should be acceptable or unacceptable for operation at, into, or out of any airport.

CABIN DOOR ANNUNCIATOR CIRCUITRY CHECK (B200)

The following test shall be performed prior to the first flight of the day.

1. Perform the following annunciator circuitry check:
   a. Battery - ON
   b. With door open and mechanism in locked position, ensure CABIN DOOR annunciator is ILLUMINATED.
   c. With door closed and latched, but not locked, ensure the CABIN DOOR annunciator remains ILLUMINATED.
   d. With the door closed and locked, ensure that the CABIN DOOR annunciator is EXTINGUISHED.
   e. Battery - OFF
2. Ensure that the door is closed and locked using the following procedure:
   a. Ensure that the door handle will not move out of the locked position without depressing the release button.
   b. Lift the top door step and ensure that the red safety arm is around the plunger.
   c. Ensure that the green index mark on each of the 4 locking bolts aligns with the black pointer in the observation port.

CABIN/CARGO DOOR ANNUNCIATOR CIRCUITRY CHECK (B200C)

The following test checks both the cargo door and cabin door annunciator circuitry and shall be performed prior to the first flight of the day.

1. Perform the following annunciator circuitry check:
   a. Battery - OFF
   b. Cargo Door - CLOSED AND LOCKED
   c. With the cabin door closed, but not locked, ensure that the CABIN DOOR annunciator is ILLUMINATED.
   d. With the cabin door open, ensure that the CABIN DOOR annunciator is EXTINGUISHED.
   e. Battery - ON
   f. With the cabin door open, ensure that the CABIN DOOR annunciator is ILLUMINATED.
   g. With the cabin door closed and locked, ensure that the CABIN DOOR annunciator is EXTINGUISHED.
   h. Battery - OFF
2. Ensure the cargo door is closed and locked using the following procedure:
   a. Check upper handle position - CLOSED AND LOCKED (Open access panel on upper left side of door and attempt to open cargo door latches without releasing safety lock.)
   b. Ensure that the orange index mark on each of the four rotary cam locks aligns with the notch in the plate on the door frame.
   c. Check lower pin lock handle position - CLOSED AND LOCKED (Open access panel on lower forward area of door and attempt to rotate the handle without lifting the orange lock hook.)
   d. Ensure that the orange stripe on the latch pin linkage is aligned with the orange pointer. (Observe through window at lower aft corner.)
3. Ensure the cabin door is closed and locked using the following procedure:
   a. Ensure that the door handle will not move out of the locked position without depressing the release button.
   b. Lift the door step and ensure that the red safety arm is around the plunger.
   c. Ensure that the orange index mark on each of the six rotary cam locks aligns with the notch in the plate on the door frame.

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SECTION V
PERFORMANCE
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INTRODUCTION TO PERFORMANCE AND FLIGHT PLANNING

The graphs and tables in this Section present performance information for takeoff, climb, landing, and flight planning at various parameters of weight, power, altitude, and temperature. All FAA-approved performance information is included within this Section. Examples have been presented on all performance graphs.

The following example presents calculations for flight time, block speed, and fuel required for a proposed flight from Denver to Reno using the conditions listed below:

CONDITIONS

At Stapleton International (DEN):

Outside Air Temperature ..... 28°C (82°F)

Field Elevation ..... 5333 feet[1]

Altimeter Setting ..... 29.82 in. Hg

Wind ..... 330° at 10 knots

Runway 35L Length ..... 11,500 feet[1]

[1] Source: Jeppesen Approach Chart, July 11-80

Route of Trip[2]:

DEN - J116 - EKR - J173 - SLC - J154 - BAM - J32 - RNO

Route Segment Data:[2]

ROUTE SEGMENT	MAGNETIC COURSE	MAGNETIC VARIATION	DISTANCE NM	WIND AT FL 260 DIR/KNOTS	OAT AT FL 260 °C	OAT AT [3]18,000 FT °C	ALTIMETER SETTING IN. HG
DEN - EKR	265°	13°E	143[4]	350°/40	-10	-6	29.82
EKR - SLC	270°	15°E	192	350°/40	-10	-7	29.82
SLC - BVL	249°	16°E	81	340°/35	-20	0	29.75
BVL - BAM	250°	16°E	145	340°/35	-20	0	29.75
BAM - RNO	227°	17°	146[4]	290°/45	-20	-4	29.60

BT01883

[2]Source: Jeppesen High Altitude Enroute Chart US (HI) 2, Oct 24-80.

[3]To illustrate the MEA requirement, the floor of the Jet Airways (18,000 ft MSL) was used in this example.

[4]Includes distance between airport and VORTAC, per Jeppesen Airport Directory, 1978.

At Cannon International (RNO):

Outside Air Temperature ..... 32°C (90°F)

Field Elevation ..... 4412 feet[5]

Altimeter Setting ..... 29.60 in. Hg

Wind ..... 270° at 5 knots

Runway 25 Length ..... 6101 feet[5]

[5] Jeppesen Approach Chart, May 30-80.

To determine the pressure altitude at origin and destination airports, add 1000 feet to field elevation for each 1.00 in. Hg that the reported altimeter setting value is below 29.92 in. Hg, and subtract 1000 feet for each 1.00 in. Hg above 29.92 in. Hg. First, find the difference between 29.92 in. Hg and the reported altimeter setting. Then multiply the answer by 1000 to find the difference in feet between field elevation and pressure altitude.

Pressure Altitude at DEN:

29.92 in. Hg - 29.82 in. Hg = 0.10

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0.10 X 1000 feet = 100 feet

The pressure altitude at DEN is 100 feet above field elevation.

Pressure Altitude at DEN = 5333 + 100 = 5433 feet.

Pressure Altitude at RNO:

29.92 in. Hg - 29.60 in. Hg = 0.32

0.32 X 1000 feet = 320 feet

The pressure altitude at RNO is 320 feet above field elevation.

Pressure Altitude at RNO = 4412 + 320 = 4732 feet.

PERFORMANCE EXAMPLE

TAKE-OFF WEIGHT

Maximum take-off weight limit (from LIMITATIONS Section) = 12,500 pounds.

A summary of graphs provided in this handbook to restrict take-off weight is presented below:

1. MAXIMUM TAKE-OFF WEIGHT PERMITTED BY ENROUTE CLIMB REQUIREMENT (No Restriction)
2. TAKE-OFF WEIGHT TO ACHIEVE POSITIVE ONE-ENGINE-INOPERATIVE CLIMB AT LIFT-OFF (separate graphs for Flaps UP and Flaps APPROACH) (Optional)
3. MAXIMUM ENROUTE WEIGHT (FAR 135 OPERATIONS) (Required for FAR Part 135 Operations)
4. TAKE-OFF WEIGHT TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS (separate graphs for Flaps UP and Flaps APPROACH) (Optional)

MAXIMUM TAKE-OFF WEIGHT PERMITTED BY ENROUTE CLIMB REQUIREMENT

There is no weight restriction to meet Enroute Climb Requirements.

Maximum allowable take-off weight = 12,500 pounds.

TAKE-OFF WEIGHT TO ACHIEVE POSITIVE ONE-ENGINE-INOPERATIVE CLIMB AT LIFT-OFF

Enter the graphs at 5433 feet and 28°C to determine the maximum weight at which the accelerate-go procedures should be attempted.

Maximum Accelerate-Go Weight (Flaps UP).....12,500 pounds
Maximum Accelerate-Go Weight (Flaps APPROACH) .....12,450 pounds

MAXIMUM ENROUTE WEIGHT (FAR 135 OPERATIONS)

To determine the maximum take-off weight, the weight of the fuel used to reach the MEA is added to the maximum enroute weight.

From the TIME, FUEL, AND DISTANCE TO CLIMB Graph, the time, fuel, and distance required to climb from 5433 feet (28°C) to 18,000 feet (-6°C) are:

Time.....7 minutes
Fuel.....103 pounds
Distance.....22 nautical miles

Enter the MAXIMUM ENROUTE WEIGHT (FAR 135 OPERATIONS) graph at the conditions for the worst route segment of the trip, i.e., 0°C, 18,000 feet, and 29.75 in. Hg:

Maximum Enroute Weight to Meet FAR 135 Requirements.....12,550 pounds

Since this weight is greater than the Maximum Take-off Weight Limitation of 12,500 pounds, there is no limitation to meet FAR 135 Enroute Weight Requirements. Take-off Weight must not exceed 12,500 pounds.

TAKE-OFF WEIGHT TO MEET FAR 25 TAKE-OFF AND CLIMB CRITERIA

The following information has been presented to provide the option of limiting weight to obtain the performance specifications of FAR 25 during the critical take-off and initial climb flight segments. Their use is not mandatory and full compliance with other regulations applicable to FAR 25 is not implied.

The criteria for limiting weight involves the selection from the Take-off Weight graphs of the most adverse conditions of:

1. One-engine-inoperative climb
2. Field length to accelerate-stop
3. Field length to accelerate-go
4. The take-off flight path required to clear known obstacles beyond the runway

The performance graphs associated with the above conditions are:

TAKE-OFF WEIGHT - TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS Graphs

ACCELERATE-STOP Graphs

ACCELERATE-GO Graphs

NET GRADIENT OF CLIMB Graphs

TAKE-OFF FLIGHT PATH Graph

(Reference FAR 25.109, 25.111, 25.115, and 25.121)

The performance presented using this criteria is predicated on the autofeather system being armed and operable. The Ground Minimum Control Speed (V_{MCG}) has been determined to be 84 knots. At this speed, control within 25 feet of the runway center line is possible.

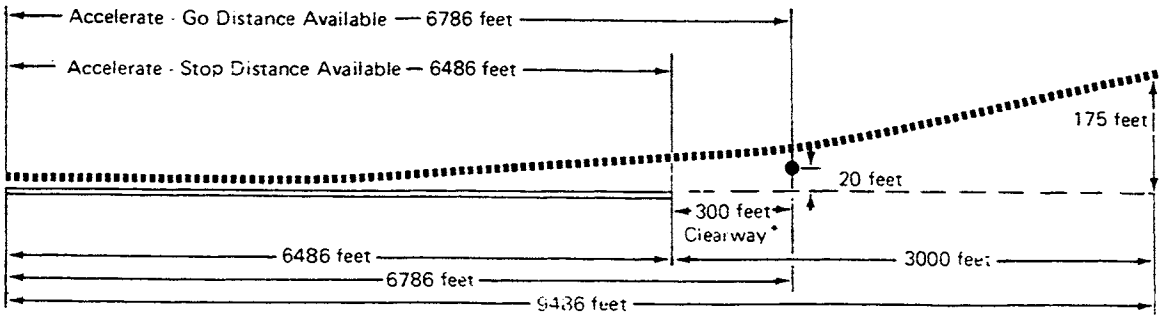
The following example illustrates the procedures required to obtain a take-off weight value using the CONDITIONS specified below, and illustrated by the TAKE-OFF FLIGHT PATH Diagram.

CONDITIONS (These conditions do not pertain to any particular airport location):

Outside Air Temperature ..... 28°C
Field Elevation ..... 5330 feet
Altimeter Setting ..... 29.82 in. Hg
Surface Wind ..... 140° at 11 knots
Runway 17 Length ..... 6486 feet
Pressure Altitude (29.92 - 29.82 = 0.10 in. Hg) 5330 + 100 ..... 5430 feet
Headwind Component (From WIND COMPONENTS Graph) ..... 9.5 knots
Obstacles (Height above runway):

20-foot-high wires 300 feet beyond end of runway

175-foot-high ridge 3000 feet beyond end of runway



** The maximum allowable clearway for this runway is 1621 feet. (25% of 6486 feet)*

TAKE-OFF FLIGHT PATH

1. From the TAKE-OFF WEIGHT - TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS Graphs, determine that the maximum take-off weights to meet FAR 25 climb requirements are 12,500 pounds with flaps at UP, and 12,300 pounds with flaps at APPROACH.
2. From the ACCELERATE-STOP Graphs, using 12,500 pounds with flaps at UP, and 12,300 pounds with flaps at APPROACH, determine that the resulting distances are less than the runway length. Therefore, accelerate-stop is not a limiting factor.
3. From the ACCELERATE-GO Graphs, using 12,500 pounds with flaps at UP, and 12,300 pounds with flaps at APPROACH, determine that the resulting distances are greater than the available accelerate-go distance of 6786 feet.(Refer to the TAKE-OFF FLIGHT PATH Diagram.)
4. Using the ACCELERATE-GO Graphs, enter in reverse using the 6786 feet value and determine the weight for which this accelerate-go distance is possible.
   a. Take-off weight of 10,650 pounds with flaps at UP
   b. Take-off weight of 11,700 pounds with flaps at APPROACH
5. Enter the TAKE-OFF FLIGHT PATH Graph to determine the minimum gradient of climb which will result in a flight path clear of the 175-foot-high ridge 3000 feet beyond the end of the runway. (Refer to the TAKE-OFF FLIGHT PATH Diagram above.)
   a. Ridge = 175 feet AGL
   b. From Reference Zero (9486 - 6786) = 2700 feet
   c. Minimum Gradient of Climb = 5.2%
6. Enter the NET GRADIENT OF CLIMB - FLAPS UP Graph at 10,650 pounds, and the NET GRADIENT OF CLIMB - FLAPS APPROACH Graph at 11,700 pounds.
   - Both resulting gradients are less than 5.2%.
7. Enter in reverse the NET GRADIENT OF CLIMB Graphs using the 5.2% net gradient of climb value to determine the weights for which a climb at this value is possible.
   a. Flaps UP = 9900 pounds
   b. Flaps APPROACH = 9250 pounds
8. Using the weight of 9900 pounds with the flaps at UP, and 9250 pounds with the flaps at APPROACH, the accelerate-go distance will be shortened. This in turn will decrease the minimum gradient of climb value required to clear the ridge. The allowable take-off weights to meet these requirements are between 10,650 pounds and 9900 pounds withe flaps at UP, or between 11,700 pounds and 9250 pounds with the flaps at APPROACH. Exact weight can be determined by an iterative process of assuming new weight halfway between these weights and using the procures outlined in Steps 3, 5, 6, 7, and 8. Determine new weights for first iteration as follows:
   a. Flaps UP:
      10,650 - 9900 = 750 ÷ 2 =375 + 9900 = 10,275 pounds
   b. Flaps APPROACH:
      11,700 - 9250 = 2450 ÷ 2 =1225 + 9250 = 10,475 pounds
9. (Step 3 procedures): From the ACCELERATE-GO Graphs, using 10,275 pounds for flaps at UP, and 10,475 pounds for flaps at APPROACH, the resulting distances are within the available accelerate-go distance of 6786 feet.
   a. 6100 feet with flaps at UP
   b. 5050 feet with flaps at APPROACH
10. (Step 5 procedure): Enter the TAKE-OFF FLIGHT PATH Graph and adjust the horizontal distance from Reference Zero and determine minimum gradient of climb.
    a. Flaps UP - From Reference Zero (9486 - 6100) = 3386 feet
       Minimum Gradient of Climb = 4.1%
    b. Flaps APPROACH - From Reference Zero (9486 - 5050) = 4436 feet
       Minimum Gradient of Climb = 3.2%
11. (Step 6 procedures): Enter the NET GRADIENT OF CLIMB - FLAPS UP Graph at 10,275 pounds for a 4.7% net gradient of climb. Enter the NET GRADIENT OF CLIMB - FLAPS APPROACH Graph at 10,475 pounds for a 3.5% net gradient of climb.

a. Since these results are greater than the minimum value, the take-off weights of 10,275 pounds with flaps at UP or 10,475 pounds with flaps at APPROACH may be used.

b. If an exact value is required, complete the next step and repeat Steps 3, 5, 6, 7, and 8.

12. (Step 7 procedures): Enter in reverse the NET GRADIENT OF CLIMB - FLAPS UP Graph using minimum gradient of climb of 4.1% for a take-off weight of 10,825 pounds, and the NET GRADIENT OF CLIMB - FLAPS APPROACH Graph using minimum gradient of climb of 3.2% for a take-off weight of 10,550 pounds.

13. (Step 8 procedures): Use the weights of Step 8 and Step 12 to obtain a new assumed weight.

    a. Flaps UP
    10,825 - 10,275 = 550 ÷ 2 = 275 + 10,275 = 10,550 pounds

    b. Flaps APPROACH
    10,725 - 10,475 = 250 ÷ 2 = 125 + 10,475 = 10,600 pounds

Use these assumed weights for second iteration.

14. After several additional iterations, the exact weights which will satisfy all the given conditions are:

    a. Flaps UP = 10,470 pounds with a 4.5% Net Gradient of Climb.

    b. Flaps APPROACH = 10,625 pounds with a 3.3% Net Gradient of Climb.

The fuel quantity required for start and taxi can be added to these weights.

TAKE-OFF DISTANCE

Enter the TAKE-OFF DISTANCE Graphs at 28°C, 5433 feet pressure altitude, 12,500 pounds, and 9.5 knots headwind component and obtain the following results:

	FLAPS 0%	FLAPS 40%
Ground Roll	2870 feet	2700 feet
Total Distance Over 50-foot Obstacle	4900 feet	3750 feet
Take-off Speed:		
At Rotation	95 knots	94 knots
At 50 Feet	121 knots	106 knots

BT01972

FLIGHT PLANNING

The following calculations provide information for flight planning at various parameters of weight, power, altitude, and temperature. Graphs and tables are included for: TIME, FUEL, AND DISTANCE TO CLIMB; TIME, FUEL, AND DISTANCE TO DESCEND; NORMAL CRUISE POWER AT 1700 RPM; MAXIMUM CRUISE POWER AT 1700 RPM; NORMAL CRUISE POWER AT 1800 RPM; MAXIMUM CRUISE POWER AT 1800 RPM; MAXIMUM RANGE POWER AT 1700 RPM; and HOLDING TIME.

Calculations for flight time, block speed, and fuel requirements for a proposed flight are detailed below.

Enter the ISA CONVERSION Graph at the conditions indicated:

DEN

Pressure Altitude ..... 5433 feet

OAT ..... 28°C

ISA Condition ..... ISA +23°C

DEN - SLC

Pressure Altitude ..... 26,000 feet

OAT ..... -10°C

ISA Condition ..... ISA +27°C

SLC - RNO

Pressure Altitude ..... 26,000 feet

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OAT .....-20°C

ISA Condition.....ISA +17°C

RNO

Pressure Altitude .....4732 feet

OAT .....32°C

ISA Condition.....ISA +27°C

Enter the TIME, FUEL, AND DISTANCE TO CLIMB graph at 28°C, to 5433 feet, and to 12,500 pounds and enter at -10°C, to 26,000 feet and to 12,500 pounds and read:

Time to Climb .....21 - 3 = 18 min

Fuel Used to Climb.....250 - 47 = 203 pounds

Distance Traveled .....62 - 8 = 54 nautical miles

Enter the TIME, FUEL, AND DISTANCE TO DESCEND graph at 26,000 feet, and enter again at 4732 feet, and read:

Time to Descend .....17 - 3 = 14 min

Fuel Used to Descend.....164 - 36 = 128 pounds

Descent Distance .....82 - 13 = 69 nautical miles

The estimated average cruise weight is approximately 11,600 pounds.

Enter the tables for NORMAL CRUISE POWER at 1800 RPM for ISA +10°C, ISA +20°C, and for ISA +30°C, and read the cruise speeds for 26,000 feet at 12,000 pounds and 11,000 pounds.

CRUISE TRUE AIRSPEEDS AT FL 260

12,000 POUNDS			11,000 POUNDS		
ISA + 10°C	ISA + 20°C	ISA + 30°C	ISA + 10°C	ISA + 20°C	ISA + 30°C
277	275	272	281	279	277

BT01886

Interpolate between these speeds for ISA + 27°C and ISA + 17°C at 11,600 pounds.

Cruise True Airspeed (ISA + 27°C) .....275 knots

Cruise True Airspeed (ISA + 17°C) .....278 knots

Enter the NORMAL CRUISE POWER at 1800 RPM graph at 26,000 feet and read the recommended torque settings for ISA +27°C (-3°C IOAT) and ISA + 17°C (-13°C IOAT):

ISA + 27°C (-3°C IOAT) .....1450 ft-lbs torque per engine

ISA + 17°C (-13°C IOAT) .....1515 ft-lbs torque per engine

Enter the *FUEL FLOW AT NORMAL CRUISE POWER at 1800 RPM Graph at 26,000 feet and read the fuel flow for ISA +27°C (-3°C IOAT)and ISA +17°C (-13°C IOAT):

ISA + 27°C (-3°C IOAT) Fuel Flow Per Engine .....265 lbs/hr

Total Fuel Flow .....530 lbs/hr

ISA + 17°C (-13°C IOAT) Fuel Flow Per Engine .....276 lbs/hr

Total Fuel Flow .....552 lbs/hr

BT01969

*NOTE: For flight planning, enter these graphs at the forecasted ISA condition; for enroute power settings and fuel flow, enter the graphs at the actual IOAT.

Time and Fuel used were calculated at Normal Cruise Power at 1800 RPM as follows:

$$\text{Time} = \frac{\text{Distance}}{\text{Ground Speed}}$$

Fuel Used = (Time) (Total Fuel Flow)

BT01970

Results are as follows:

ROUTE	DISTANCE	ESTIMATED GROUND SPEED	TIME AT CRUISE ALTITUDE	FUEL USED FOR CRUISE
	NM	KNOTS	HRS : MIN	LBS
DEN - EKR	89*	260	:21	181
EKR - SLC	192	256	:45	398
SLC - BVL	81	267	:18	167
BVL - BAM	145	266	:33	301
BAM - RNO	77*	245	:19	173

BT01884

*Distance required to climb or descend has been subtracted from segment distance.

TIME - FUEL - DISTANCE

ITEM	TIME HRS : MINS	FUEL POUNDS	DISTANCE NM
Start, Runup, Taxi, and Take-Off acceleration	00 : 00	90	0
Climb	00 : 18	203	54
Cruise	2 : 16	1220	584
Descent	00 : 14	128	69
Total	2 : 48	1641	707

BT01971

Block Speed: 707 NM ÷ 2 hours, 48 minutes = 253 knots

RESERVE FUEL

Reserve Fuel (45 minute at Maximum Range Power): Assume weight at end of cruise to be 11,000 pounds. Enter the tables for MAXIMUM RANGE POWER at 1700 RPM for ISA +10°C and ISA +20°C, and read the fuel flow for 26,000 feet at 11,000 pounds:

ISA +10°C.....380 lbs/hr Total Fuel Flow
ISA +20°C.....390 lbs/hr Total Fuel Flow

Interpolate to find fuel flow at ISA +17°C:

Total fuel flow for reserve = 380 + 7 = 387 lbs/hr
Reserve Fuel = 45 minutes X 387 lbs/hr = 291 lbs

TOTAL FUEL REQUIREMENT

1641 lbs + 291 lbs = 1932 lbs

ZERO FUEL WEIGHT LIMITATION

For this example, the following conditions were assumed:

Ramp Weight.....12,590 pounds
Weight of Usable Fuel Onboard.....1932 pounds

Zero Fuel Weight = Ramp Weight - Weight of Usable Fuel Onboard

Zero Fuel Weight = 12,590 - 1932 = 10,658 pounds

Maximum Zero Fuel Weight (from LIMITATIONS Section) .....= 11,000 pounds

Maximum Zero Fuel Weight Limitation has not been exceeded.

Anytime the Zero Fuel Weight exceeds the Maximum Zero Fuel Weight Limit by X amount, at least X pounds of payload must be off-loaded. If desired, additional fuel may then be added until the maximum ramp weight limitation of 12,590 pounds is again reached.

LANDING INFORMATION

The estimated Landing Weight is determined by subtracting the fuel required for the trip from the Ramp Weight:

Ramp Weight .....12,590 pounds

Fuel Required for Total Trip ..... -1641 lbs

Landing Weight (12,590 - 1641) .....10,949 pounds

NOTE: For the remainder of this example, a landing weight of 10,937 pounds has been assumed.

Enter the NORMAL LANDING DISTANCE WITHOUT PROPELLER REVERSING - FLAPS DOWN Graph at 32°C, 4732 feet, 10,937 pounds, and 4.7 knots headwind component:

Ground Roll .....1850 feet

Total Over 50-foot Obstacle .....3000 feet

Approach Speed .....99 knots

Enter the CLIMB - BALKED LANDING Graph at 32°C, 4732 feet, and 10,937 pounds

Rate of Climb .....1450 feet per minute

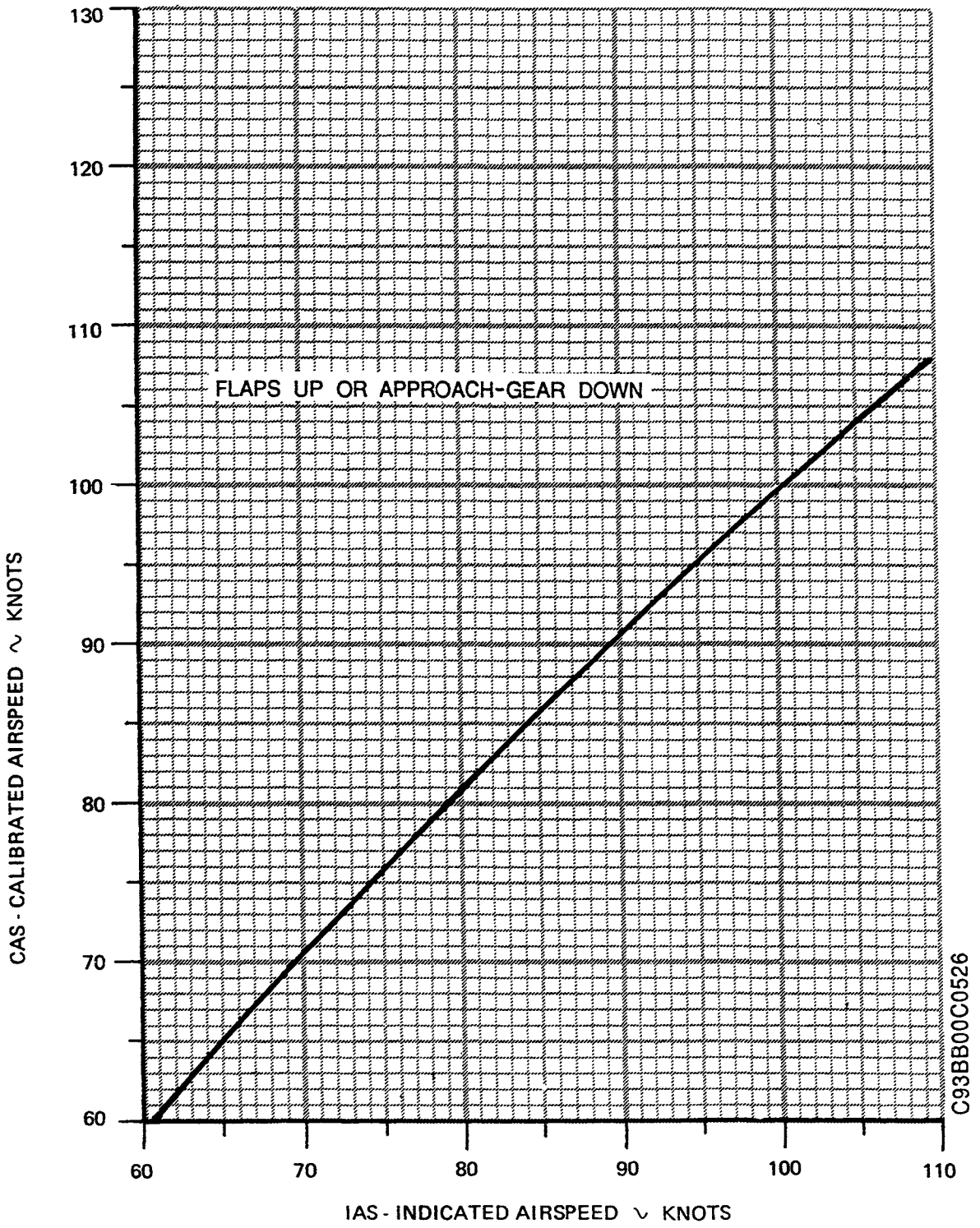
Climb Gradient.....12.8%

COMMENTS PERTINENT TO THE USE OF PERFORMANCE GRAPHS

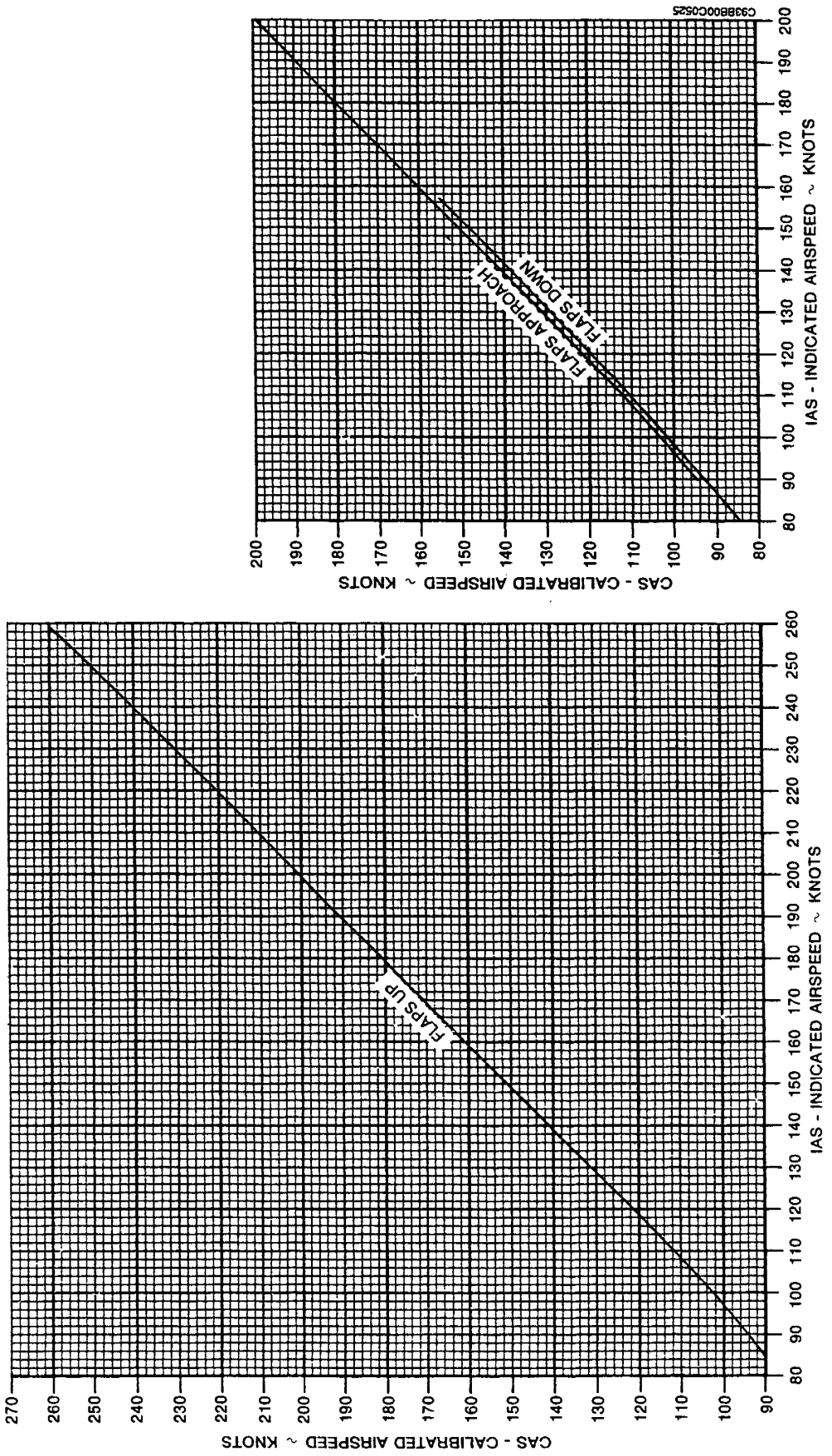
1. In addition to presenting the answer for a particular set of conditions, the example on the graph also presents the order in which the various scales on the graph should be used. For instance, if the first item in the example is OAT, then enter the graph at the known OAT.
2. The reference lines indicate where to begin following the guidelines. Always project to the reference line first, then follow the guidelines to the next known item by maintaining the same PROPORTIONAL DISTANCE between the guideline above and guideline below the projected line. For instance, if the projected line intersects the reference line in the ratio of 3UP down/ 7UP up between the guidelines, then maintain this same 3UP/7UP relationship between the guidelines all the way to the next known item or answer.
3. The associated conditions define the specific conditions from which performance parameters have been determined. They are not intended to be used as instructions; however, performance values determined from charts can only be achieved if the specific conditions exist.
4. Indicated airspeeds (IAS) were obained using the Airspeed Calibration - Normal System graph.
5. The full amount of usable fuel is available for all approved flight conditions.
6. Notes have been provided on various graphs and tables to approximate performance with ice vanes extended. The effect will vary, depending upon airspeed, temperature, altitude, and ambient conditions. At lower altitudes, where operation on the torque limit is possible, the effect of ice vane extension will be less, depending upon how much power can be recovered after the ice vanes have been extended.

AIRSPEED CALIBRATION – NORMAL SYSTEM

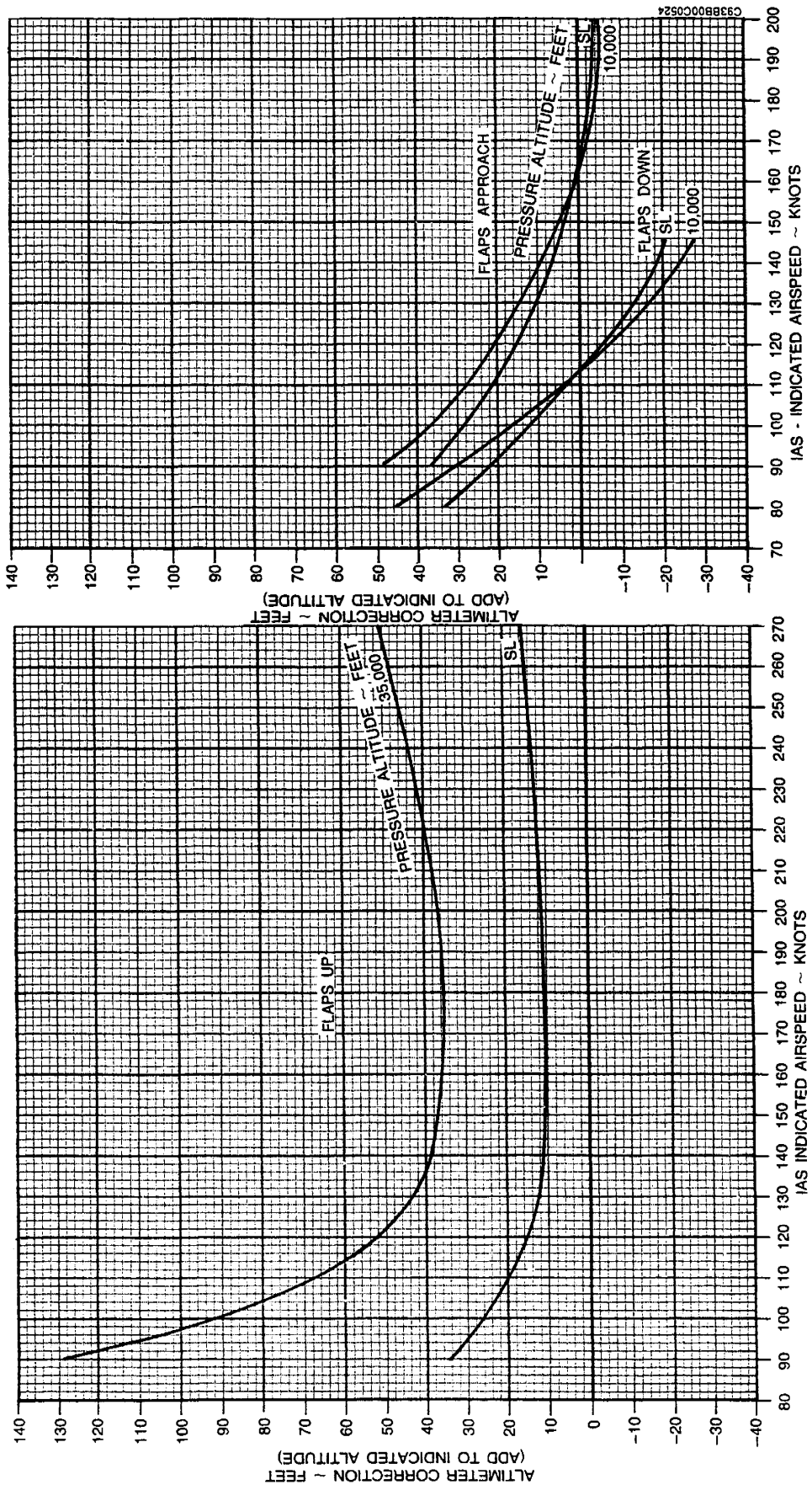
TAKE-OFF GROUND ROLL



AIRSPEED CALIBRATION – NORMAL SYSTEM

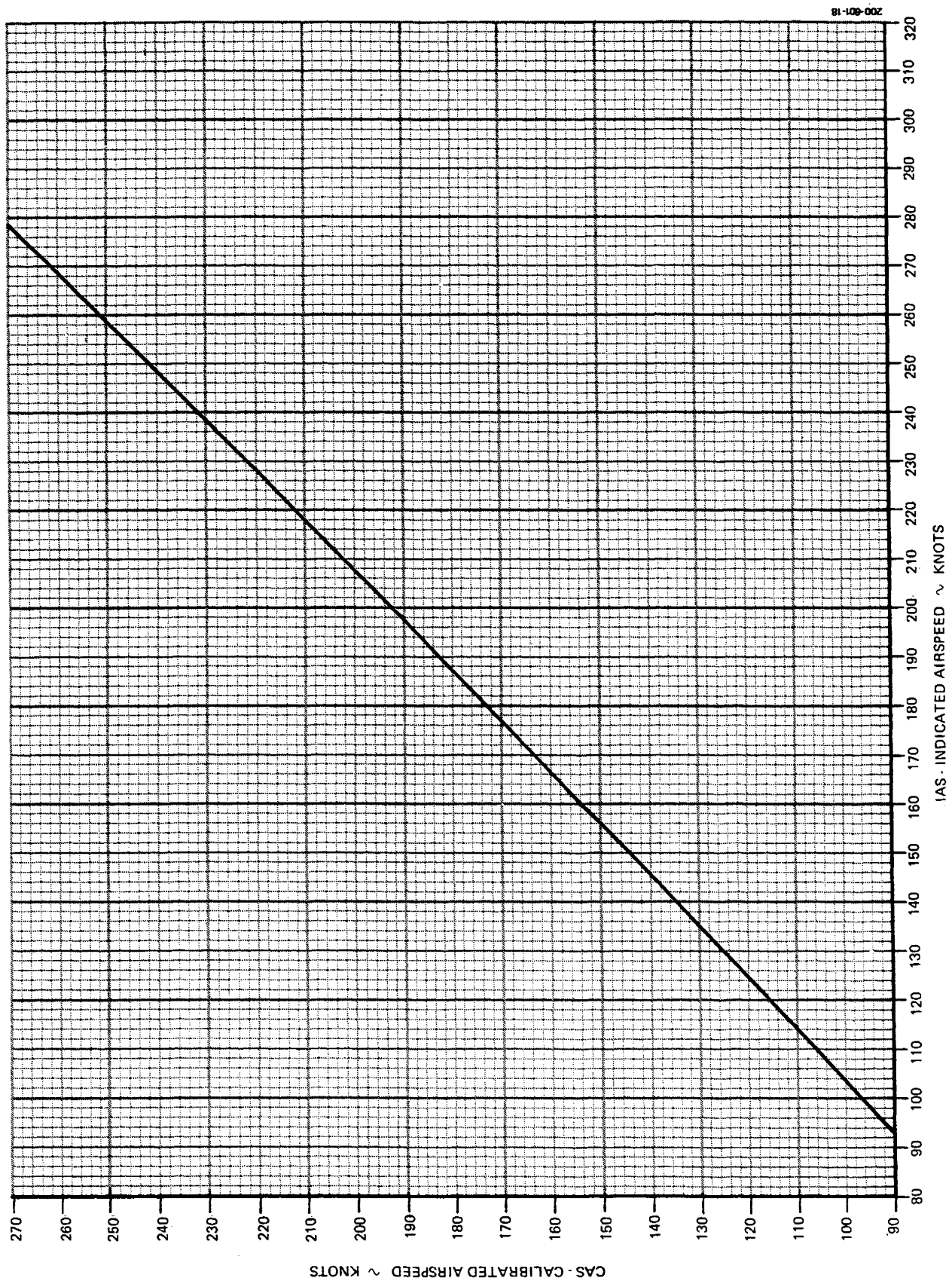


ALTIMETER CORRECTION – NORMAL SYSTEM



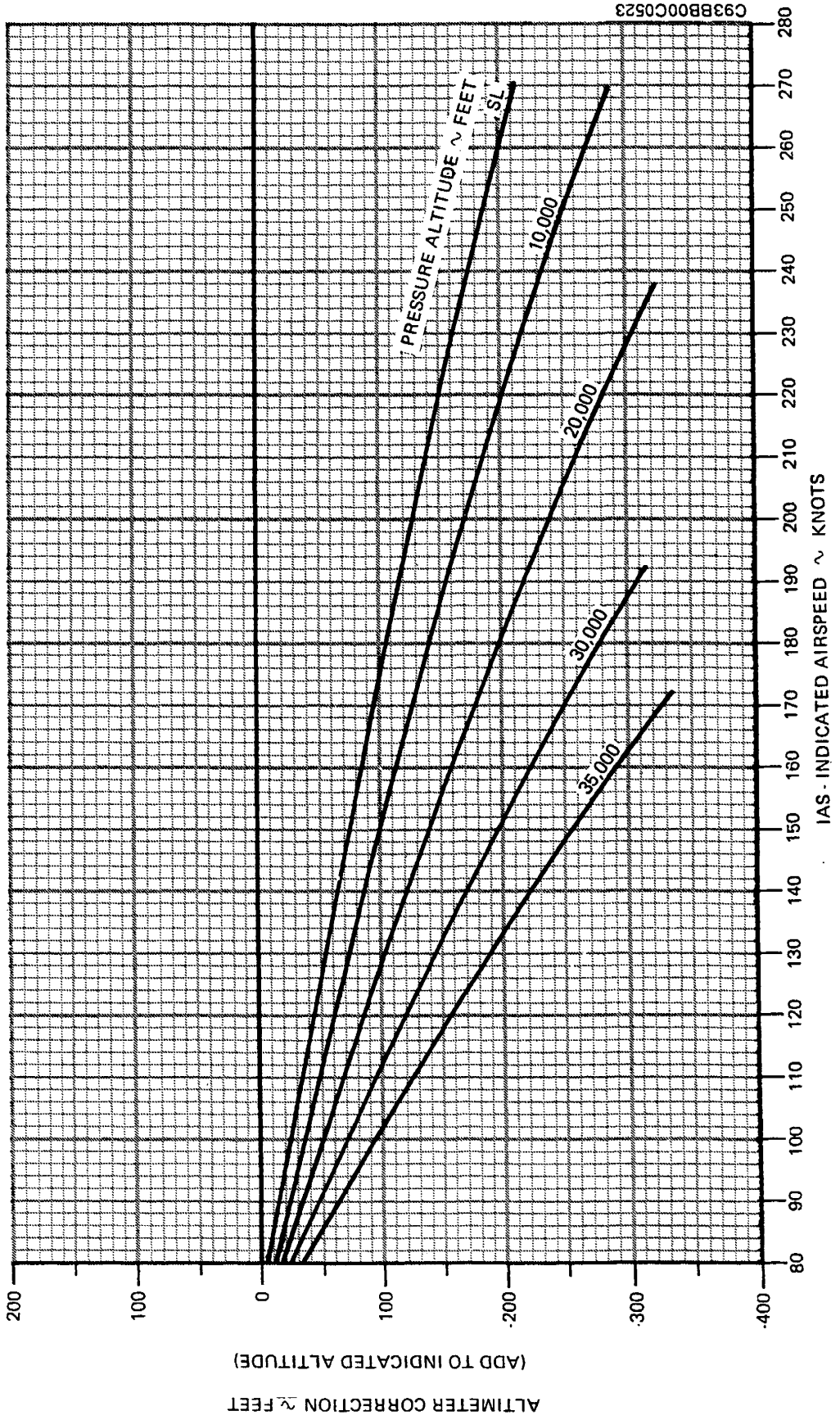
AIRSPEED CALIBRATION — ALTERNATE SYSTEM

APPLICABLE FOR ALL FLAP POSITIONS



ALTIMETER CORRECTION – ALTERNATE SYSTEM

APPLICABLE FOR ALL FLAP POSITIONS



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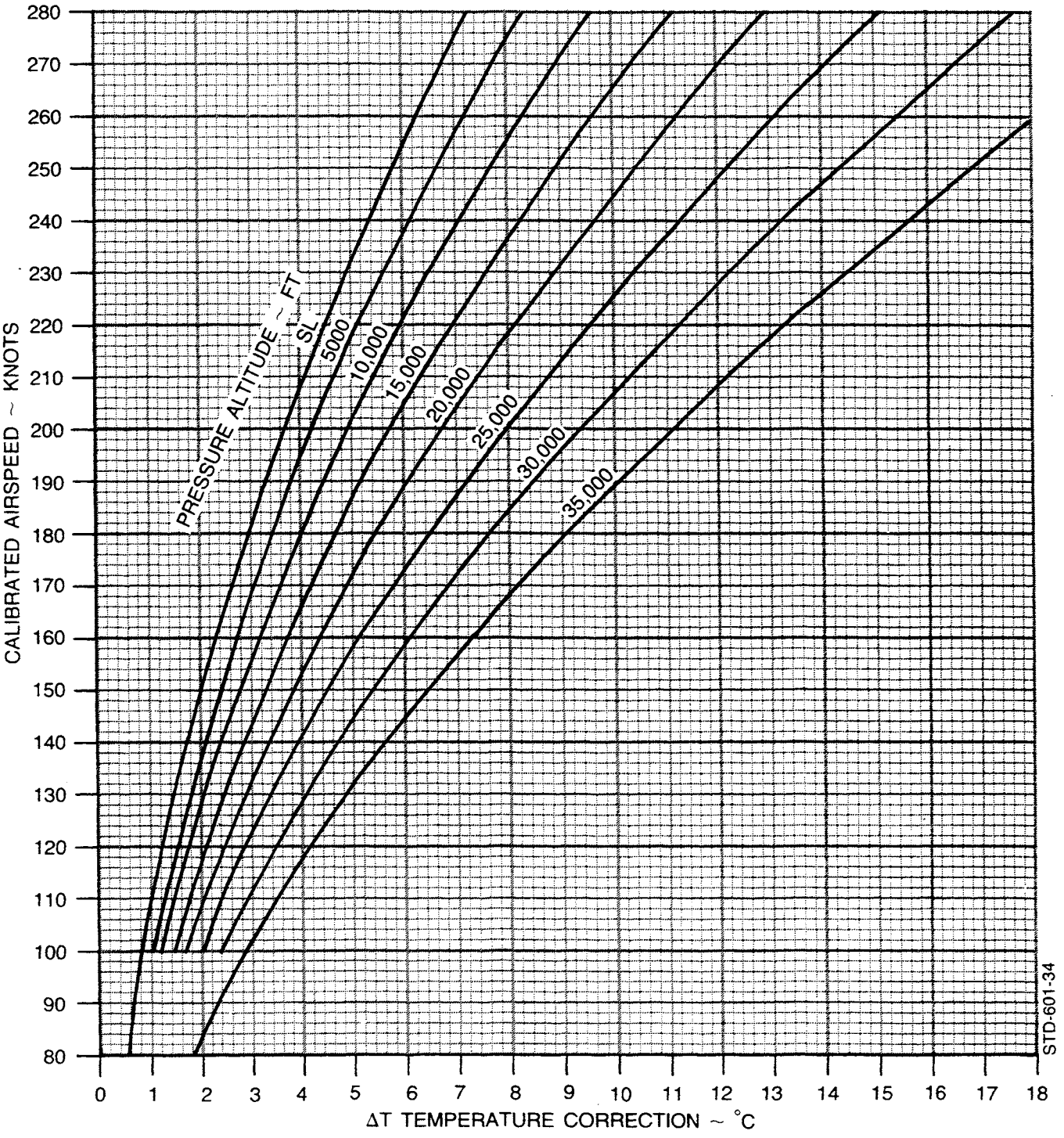
INDICATED OUTSIDE AIR TEMPERATURE CORRECTION

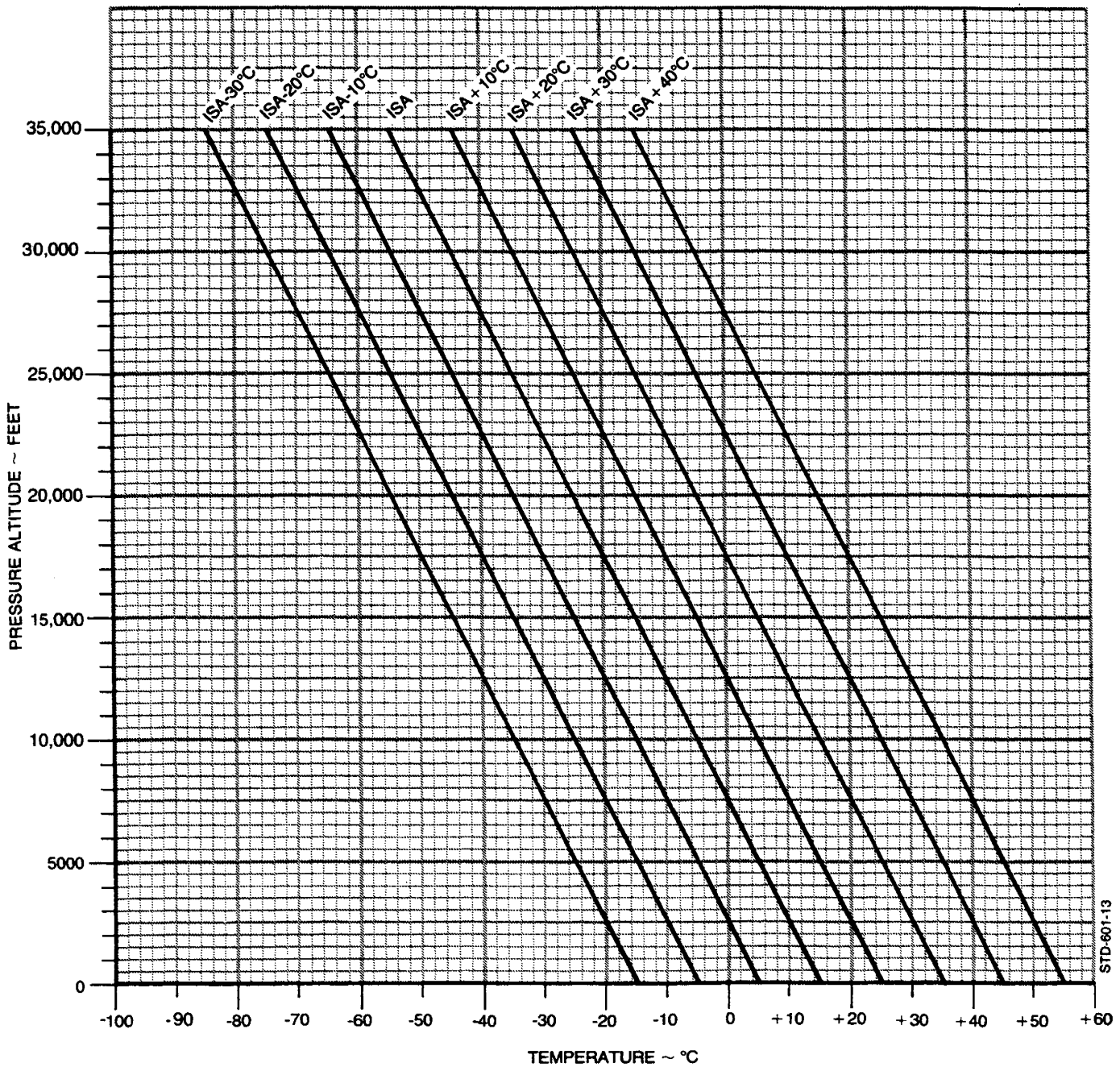
STANDARD DAY (ISA)

NOTE: SUBTRACT ΔT FROM INDICATED (GAGE) OAT TO OBTAIN TRUE OAT. (ΔT ASSUMES A RECOVERY FACTOR OF 0.7)



ISA CONVERSION

PRESSURE ALTITUDE vs OUTSIDE AIR TEMPERATURE



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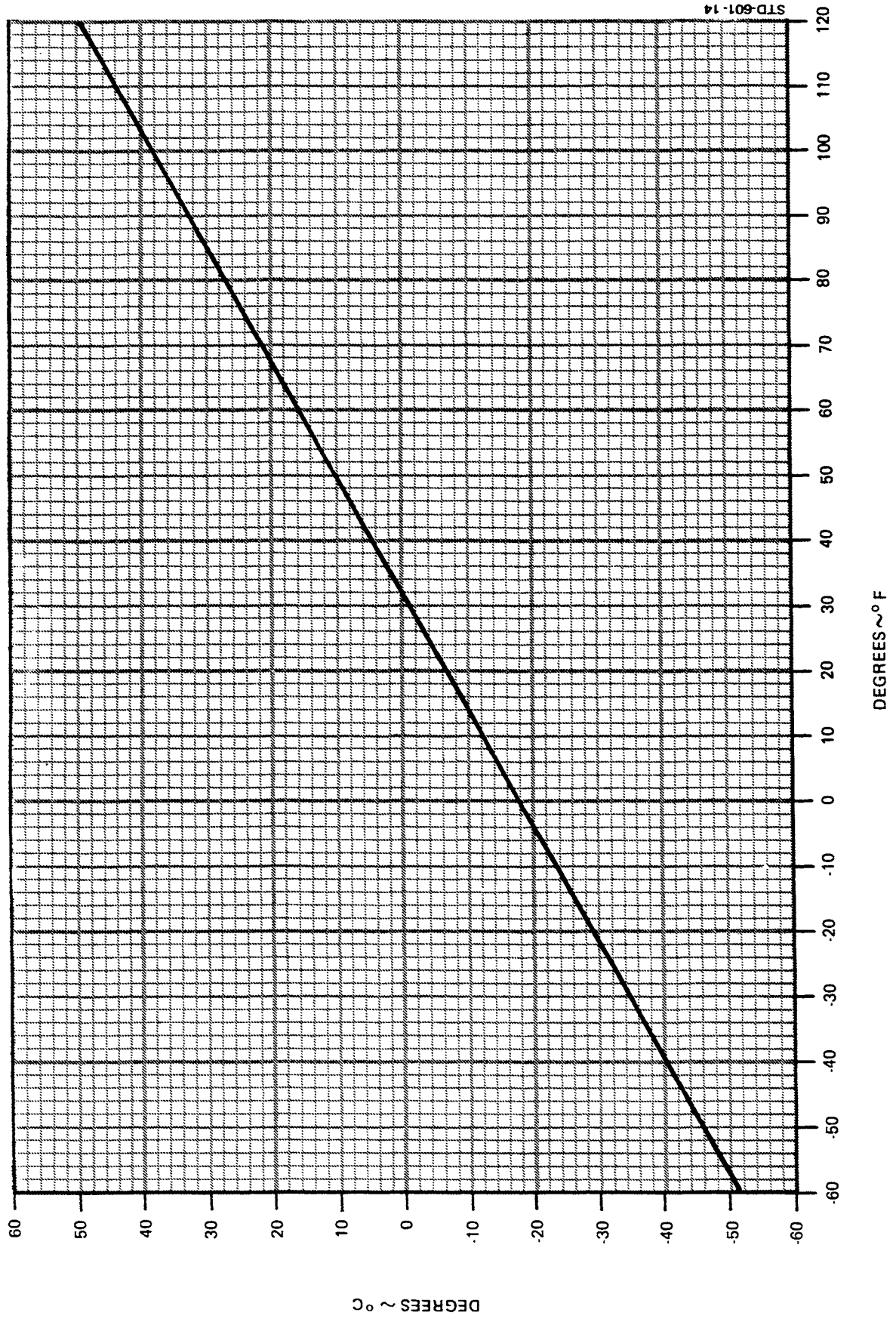
FAHRENHEIT TO CELSIUS TEMPERATURE CONVERSION



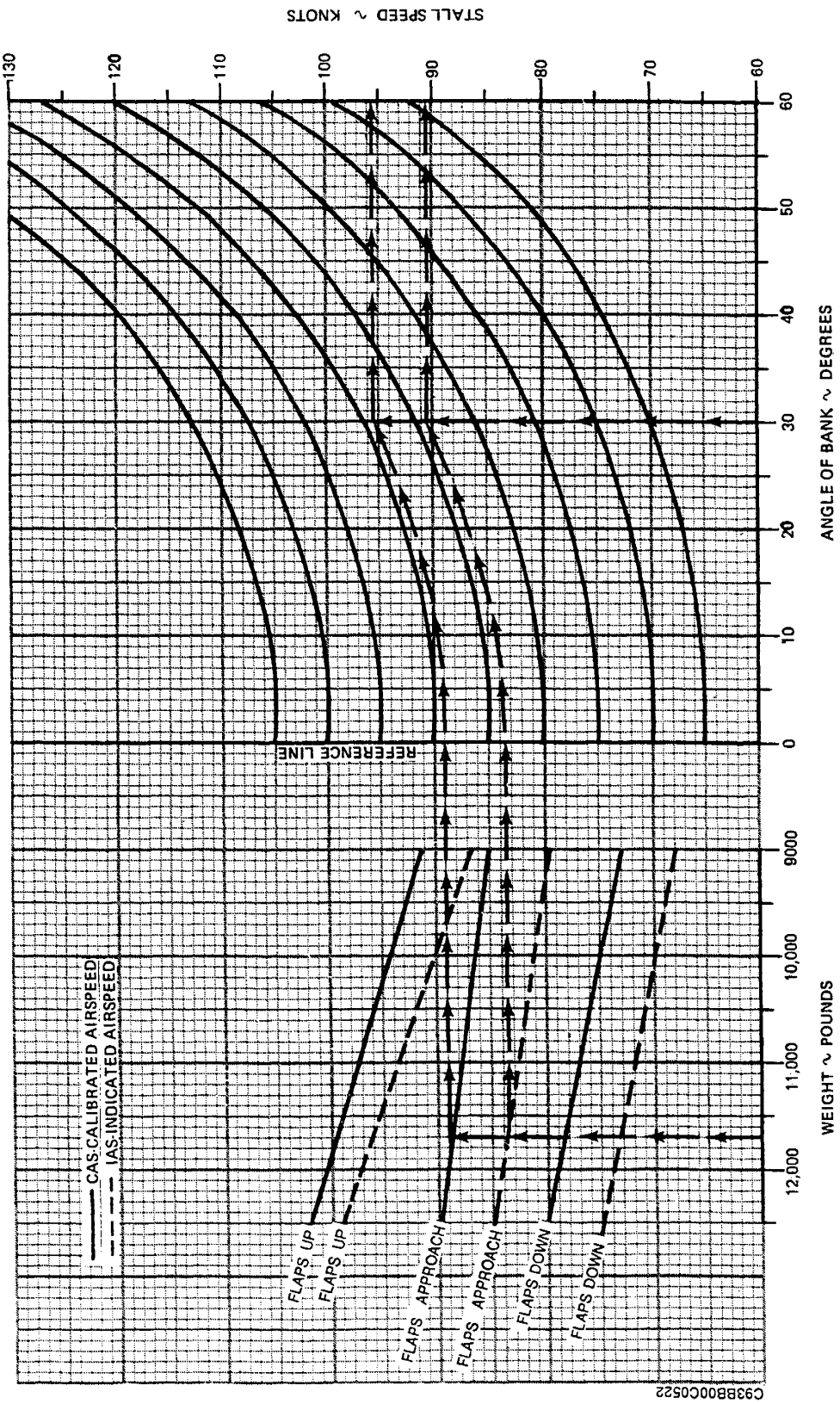
STALL SPEEDS – POWER IDLE

NOTES:

1. ALTITUDE LOSS EXPERIENCED WHILE CONDUCTING STALLS IN ACCORDANCE WITH FAR 23.201 WAS 800 FEET.
2. MAXIMUM NOSE DOWN PITCH ATTITUDE AND ALTITUDE LOSS DURING RECOVERY FROM ONE-ENGINE-INOPERATIVE STALLS PER FAR 23.205 ARE APPROXIMATELY 8° AND 300 FEET RESPECTIVELY.
3. A NORMAL STALL RECOVERY TECHNIQUE MAY BE USED. THE BEST PROCEDURE IS A BRISK FORWARD WHEEL MOVEMENT TO A NOSE DOWN ATTITUDE, LEVEL THE AIRPLANE AFTER AIRSPEED HAS INCREASED APPROXIMATELY 25 KNOTS ABOVE STALL.
4. LANDING GEAR POSITION HAS NO EFFECT ON STALL SPEED.

EXAMPLE:

WEIGHT	11,700 LBS
FLAPS	APPROACH
ANGLE OF BANK	30 DEG.
STALL SPEED	96 KTS CAS 90 KTS IAS



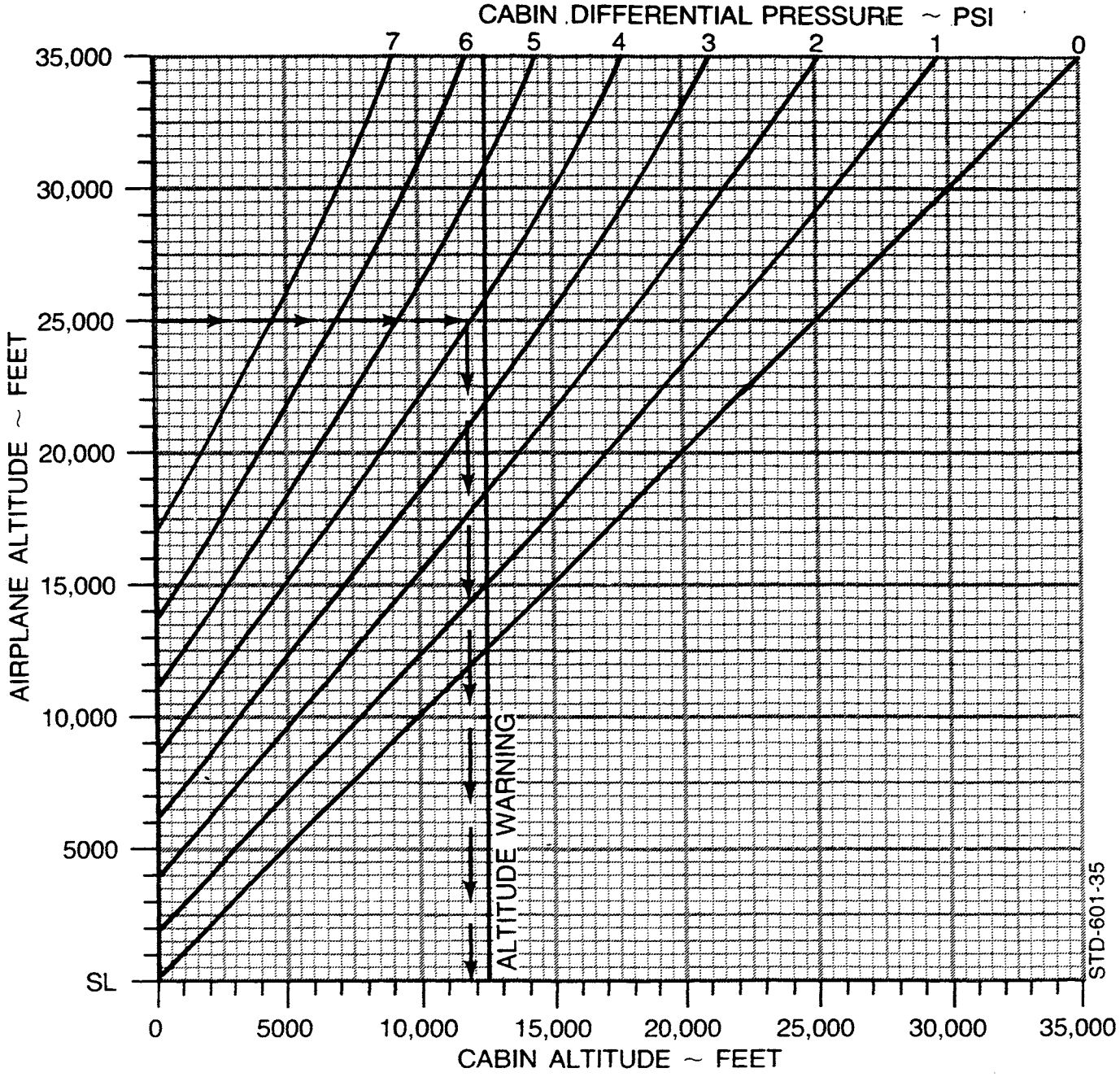
For Training Purposes Only

CABIN ALTITUDE FOR VARIOUS AIRPLANE ALTITUDES

EXAMPLE:

AIRPLANE ALTITUDE ..... 25,000 FT
CABIN DIFFERENTIAL PRESSURE..... 4.0 PSI

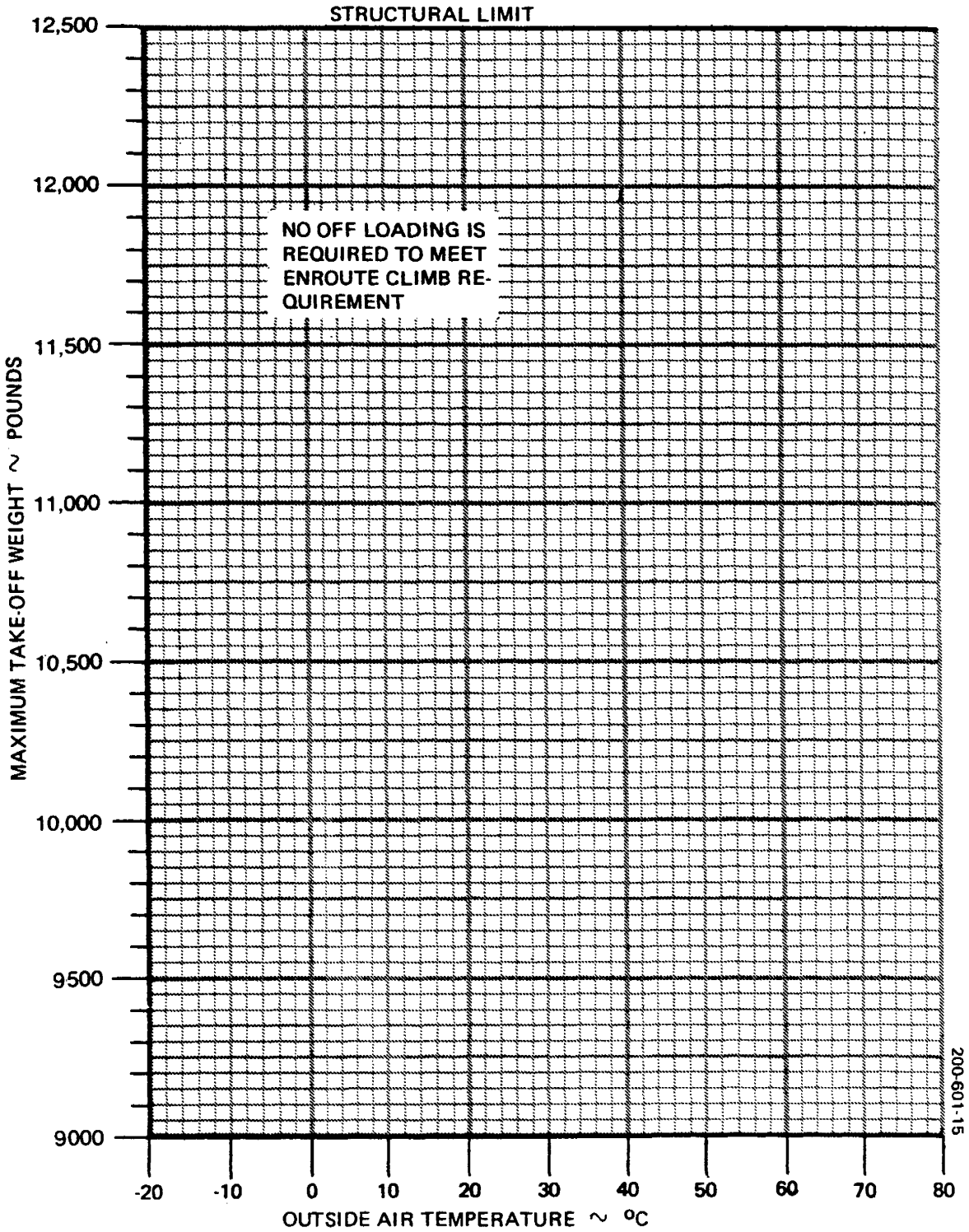
CABIN ALTITUDE..... 11,800 FT



MAXIMUM TAKE-OFF WEIGHT

PERMITTED BY ENROUTE CLIMB REQUIREMENT

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, NO OFF LOADING IS REQUIRED.



TAKE-OFF WEIGHT—FLAPS UP

TO ACHIEVE POSITIVE ONE-ENGINE-INOPERATIVE RATE-OF-CLIMB AT LIFT-OFF

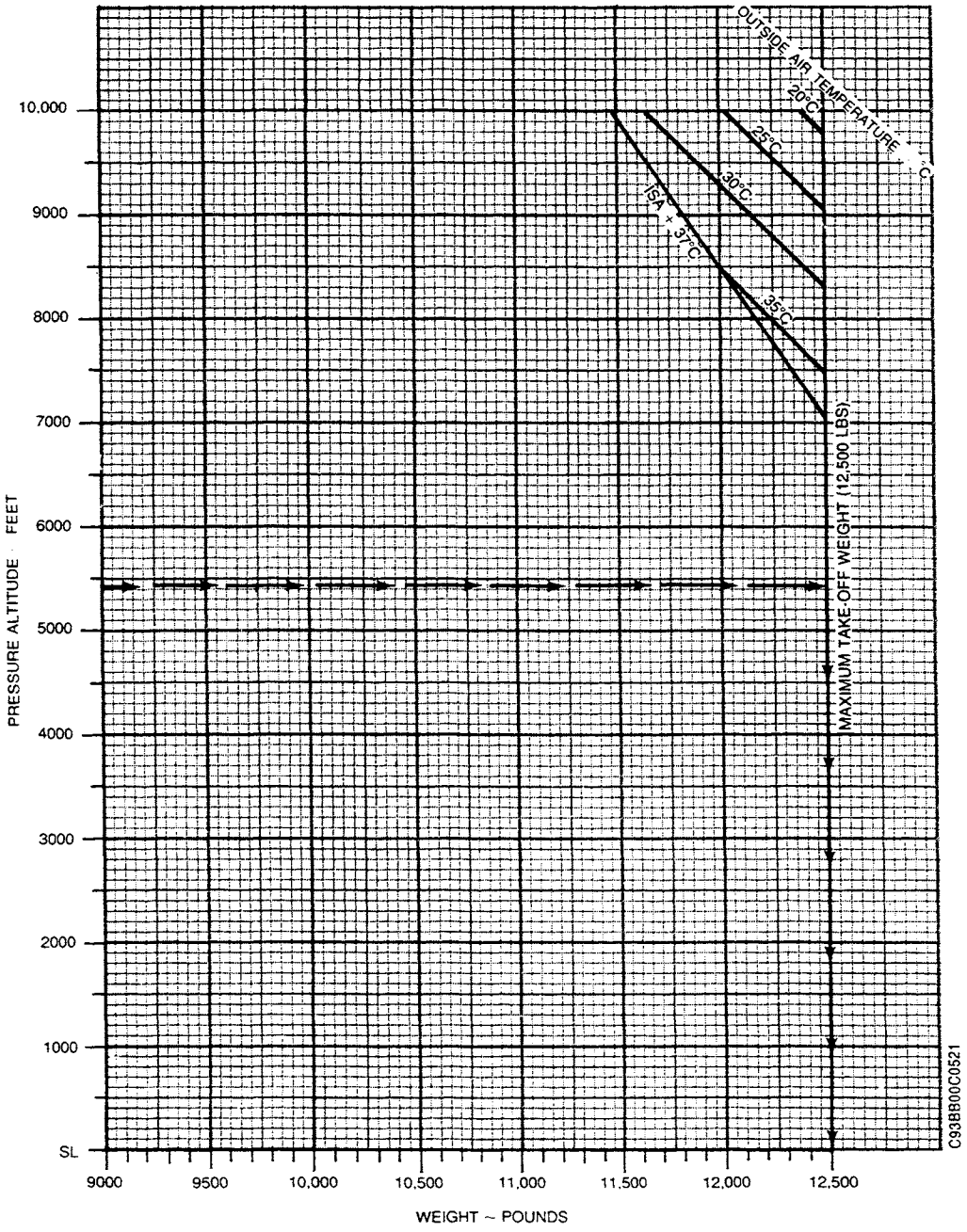
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	UP
LANDING GEAR	DOWN
INOPERATIVE PROPELLER	FEATHERED

EXAMPLE:

PRESSURE ALTITUDE	5433 FT
OAT	28°C
TAKE-OFF WEIGHT	12,500 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, NO OFF-LOADING IS REQUIRED.



TAKE-OFF WEIGHT - FLAPS APPROACH

TO ACHIEVE POSITIVE ONE-ENGINE INOPERATIVE RATE OF CLIMB AT LIFT-OFF

ASSOCIATED CONDITIONS:

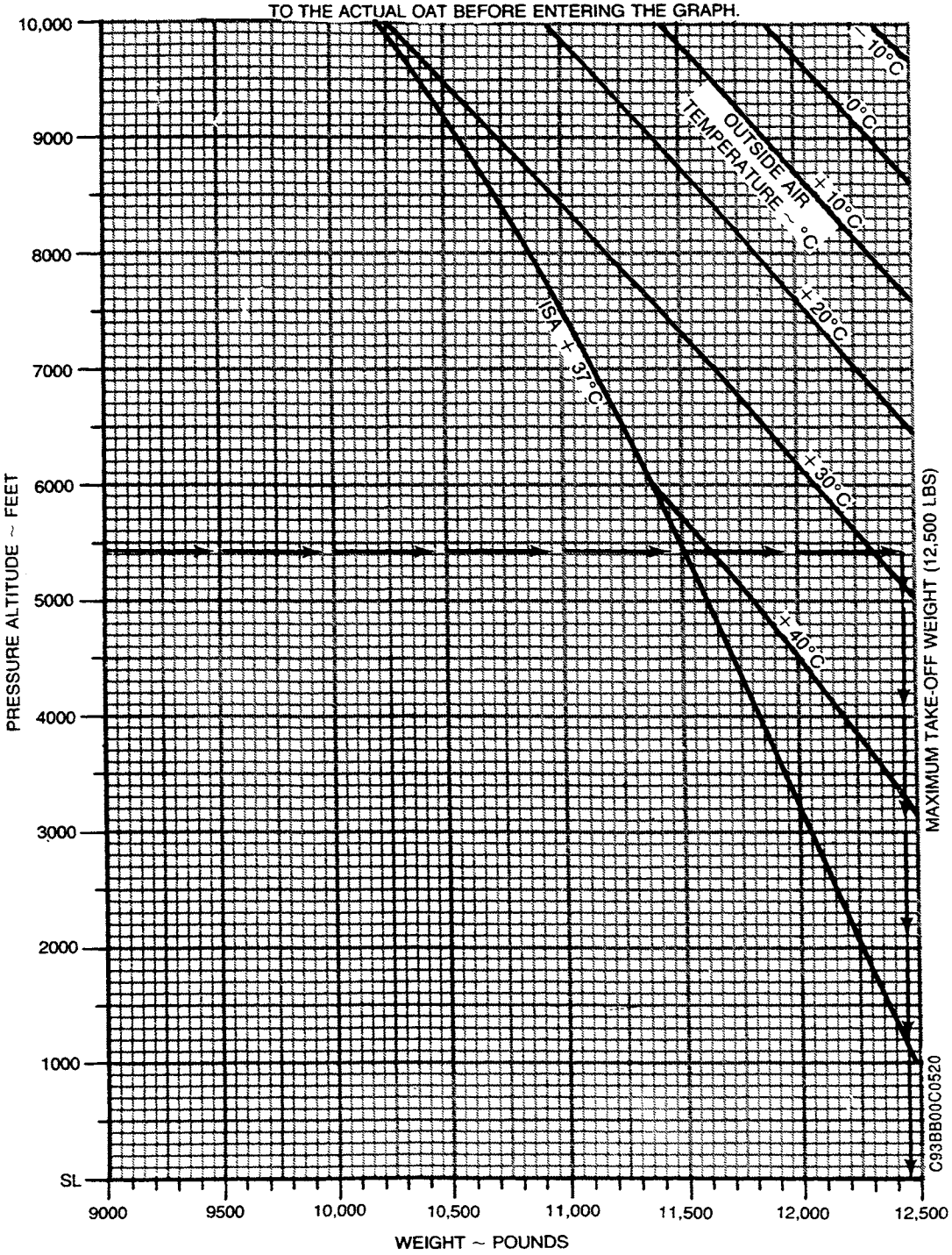
POWER .....TAKE-OFF
FLAPS.....APPROACH
LANDING GEAR....DOWN
INOPERATIVE
PROPELLER ... FEATHERED

EXAMPLE:

PRESSURE ALTITUDE...5433 FT
OAT.....28°C

TAKE-OFF WEIGHT.....12,450 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.



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MAXIMUM ENROUTE WEIGHT
[FAR PART 135 – OPERATIONS]

ASSOCIATED CONDITIONS:

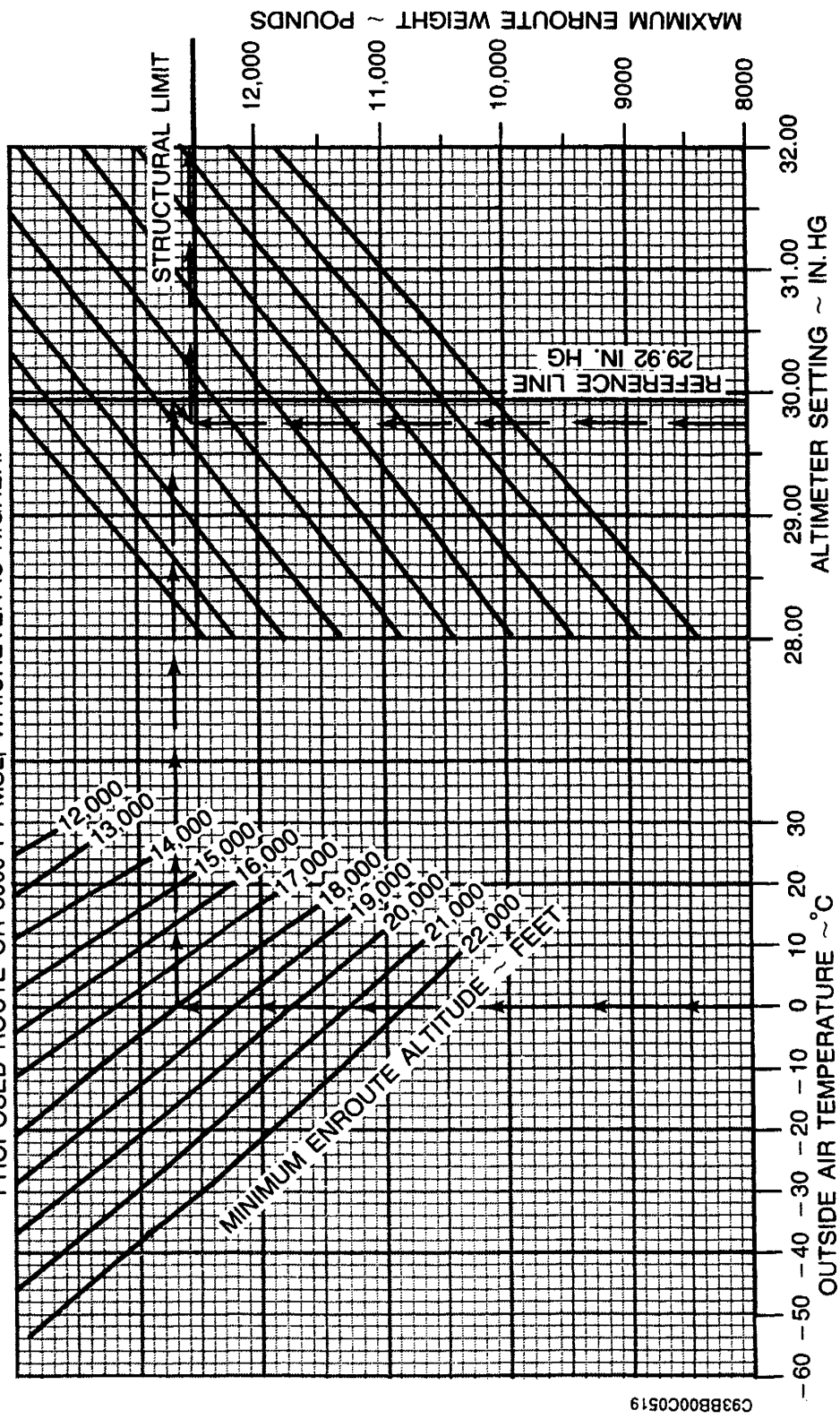
POWER ..... MAX CONTINUOUS
FLAPS ..... UP
LANDING GEAR ..... UP
INOPERATIVE PROPELLER ... FEATHERED

EXAMPLE:

OAT AT MEA ..... 0°C
MINIMUM ENROUTE ALTITUDE ... 18,000 FT
ALTIMETER SETTING ..... 29.75 IN. HG

MAXIMUM ALLOWABLE WEIGHT .. 12,500 LBS

NOTE: PER FAR 135, OPERATIONS OVER THE TOP OR IN IFR CONDITIONS REQUIRE THAT THE AIRPLANE BE CAPABLE OF CLIMBING 50 FT/MIN AT THE MEA'S OF THE PROPOSED ROUTE OR 5000 FT MSL, WHICHEVER IS HIGHER.



TAKE-OFF WEIGHT — FLAPS UP

TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS

ASSOCIATED CONDITIONS:

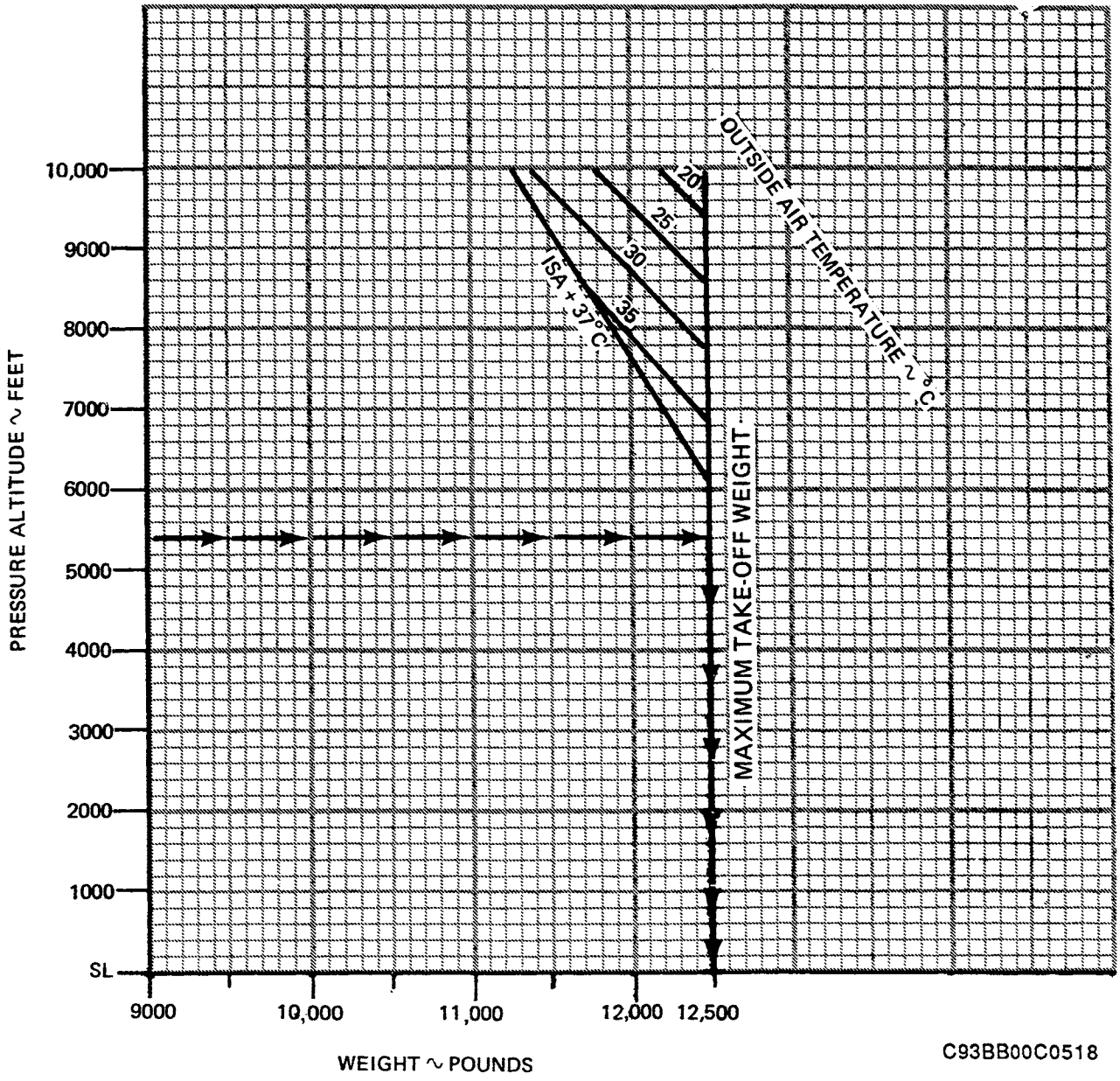
AIRPLANE . . . . . AIRBORNE
POWER. . . . . TAKE-OFF
FLAPS. . . . . UP
INOPERATIVE
PROPELLER. . . FEATHERED

EXAMPLE:

PRESSURE ALTITUDE. . . . . 5430 FT
OAT. . . . . 28°C

TAKE-OFF WEIGHT. . . . . 12,500 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, NO OFF LOADING IS REQUIRED.



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TAKE-OFF WEIGHT - FLAPS APPROACH

TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS

ASSOCIATED CONDITIONS:

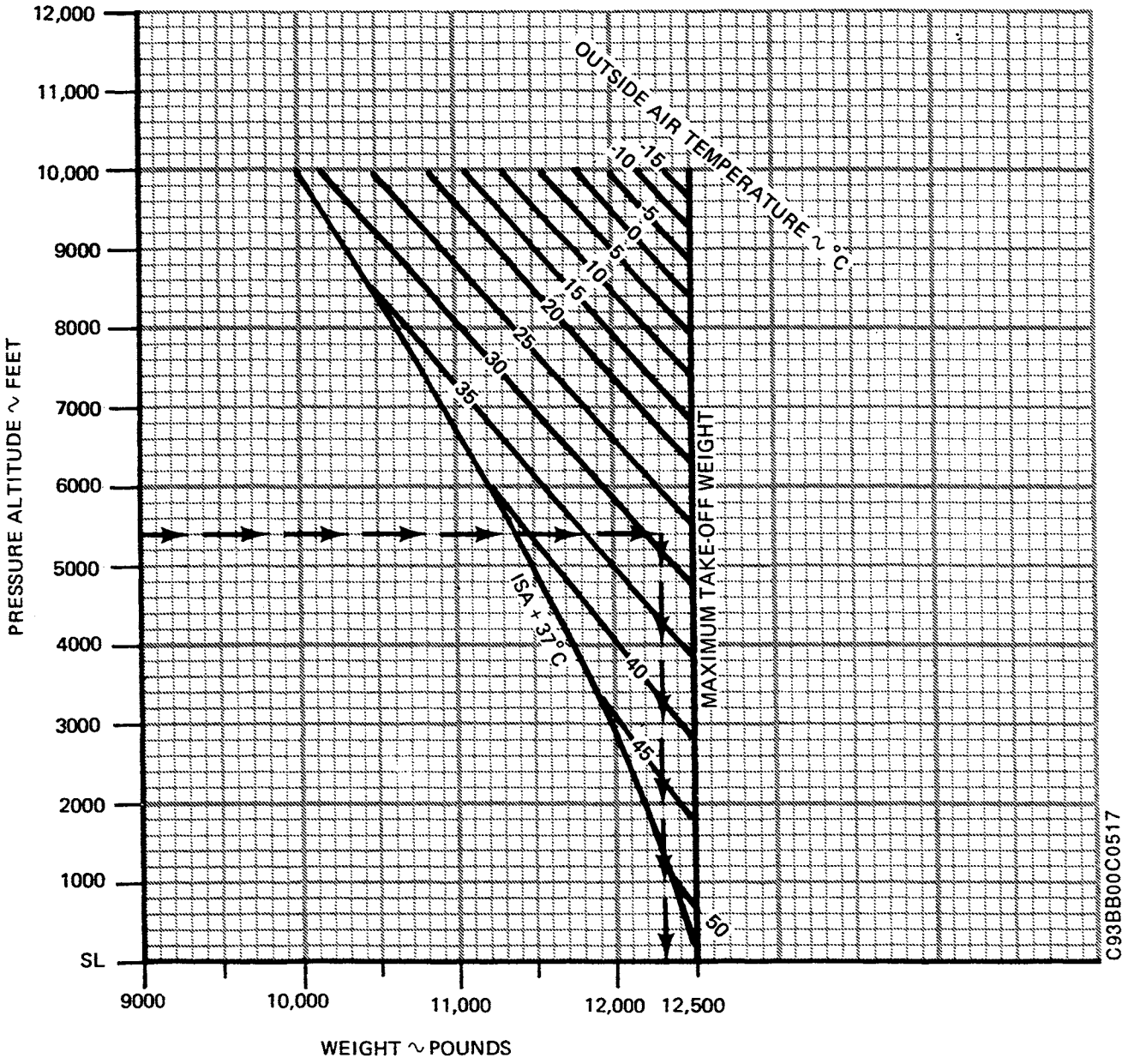
AIRPLANE..... AIRBORNE
POWER..... TAKE-OFF
FLAPS..... APPROACH
INOPERATIVE
PROPELLER... FEATHERED

EXAMPLE:

PRESSURE ALTITUDE..... 5430 FT
OAT.....28°C

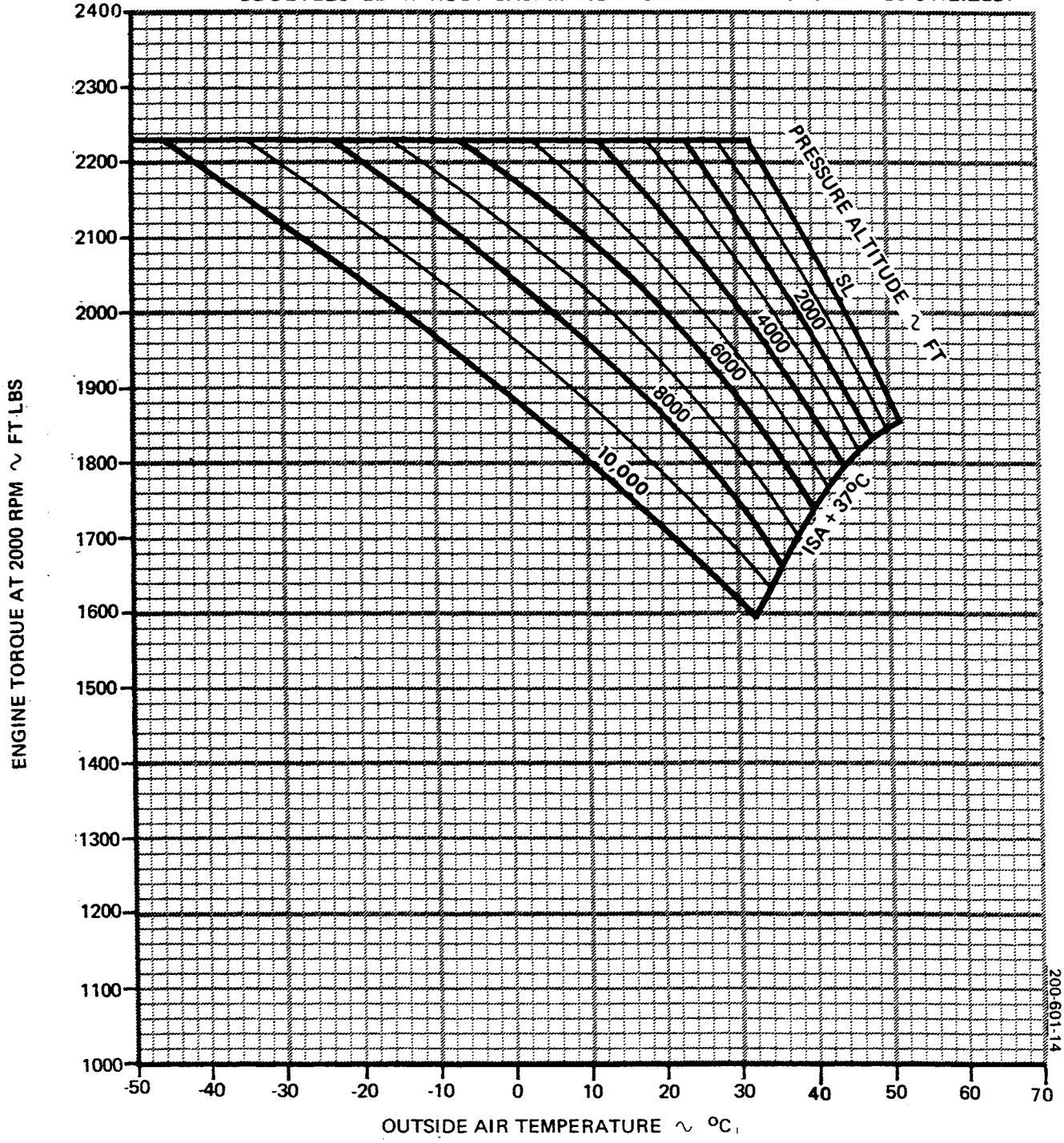
TAKE-OFF WEIGHT.....12,300 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10° C TO ACTUAL OAT BEFORE ENTERING ON GRAPH.



MINIMUM TAKE-OFF POWER AT 2000 RPM
WITH ICE VANES RETRACTED
(65 KNOTS)

NOTES: 1. TORQUE INCREASES APPROXIMATELY 20 FT-LBS FROM ZERO TO 65 KNOTS.
2. THE POWER (TORQUE) INDICATED IS THE MINIMUM VALUE FOR WHICH TAKE-OFF PERFORMANCE IN THIS SECTION CAN BE OBTAINED. EXCESS POWER WHICH CAN BE DEVELOPED WITHOUT EXCEEDING ENGINE LIMITATIONS MAY BE UTILIZED.



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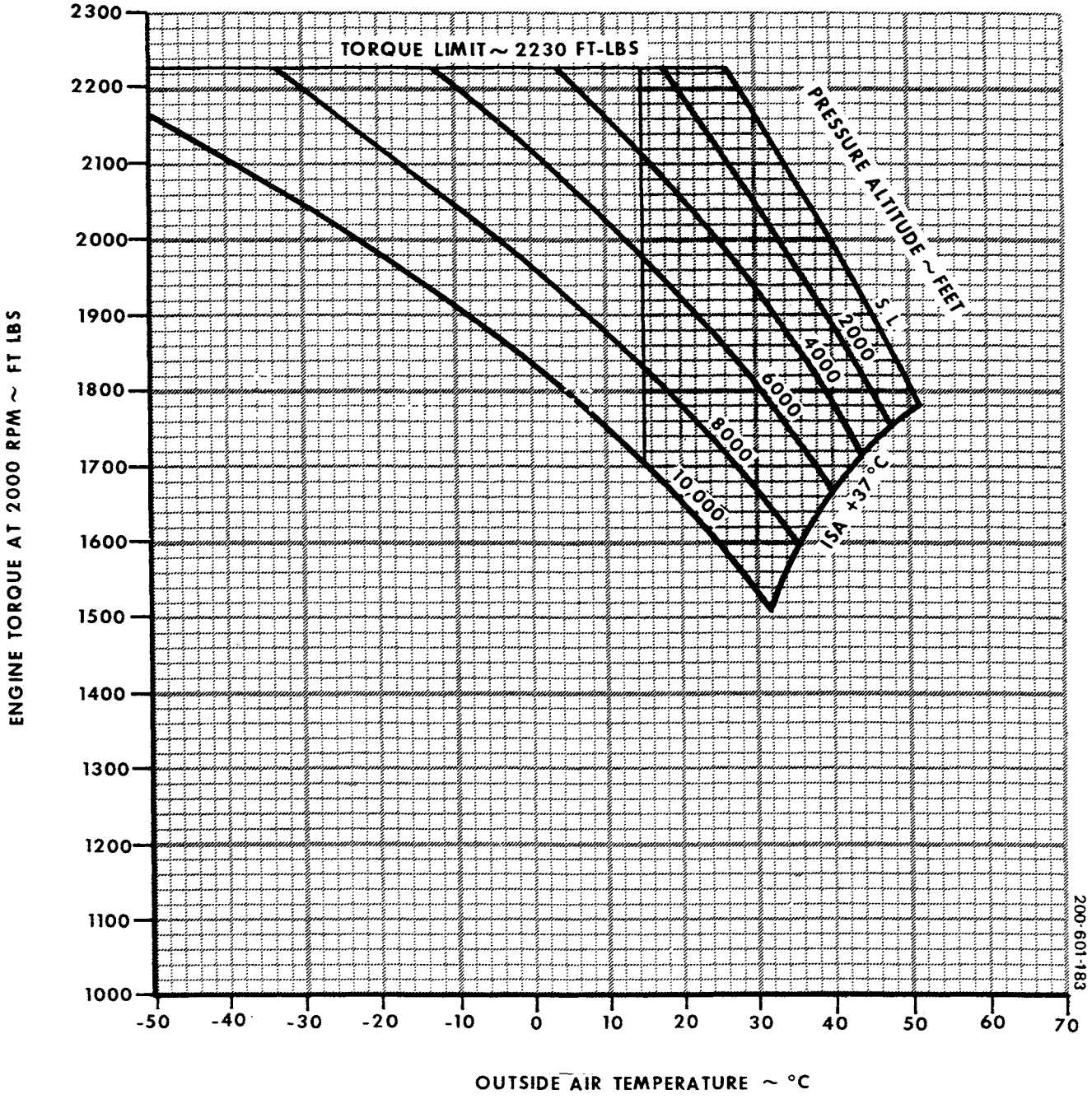
MINIMUM TAKE-OFF POWER
WITH ICE VANES EXTENDED
(65 KNOTS)

NOTES:

1. TORQUE INCREASES APPROXIMATELY 20 FT-LBS FROM ZERO TO 65 KNOTS.
2. THE POWER (TORQUE) INDICATED IS THE MINIMUM VALUE FOR WHICH TAKE-OFF PERFORMANCE WITH EXTENDED ICE VANES CAN BE OBTAINED. EXCESS POWER WHICH CAN BE DEVELOPED WITHOUT EXCEEDING ENGINE LIMITATIONS MAY BE UTILIZED.
3. TAKE-OFF WITH VANES EXTENDED IS PROHIBITED IN SHADED AREA.



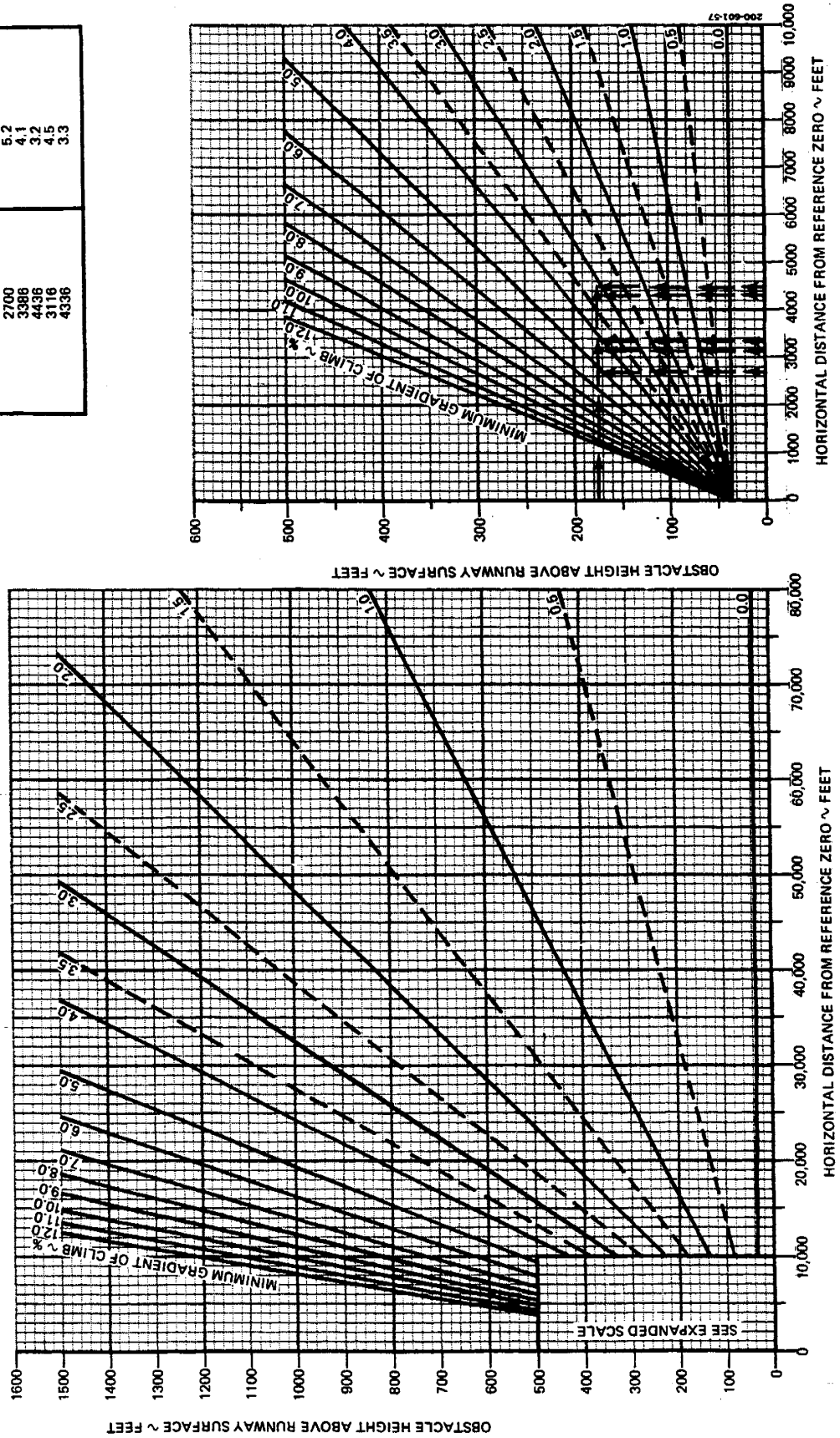
TAKE – OFF FLIGHT PATH

REFERENCE ZERO: THE POINT AT THE END OF THE TAKE-OFF RUN AT WHICH THE AIRPLANE IS 35 FEET ABOVE THE RUNWAY SURFACE.

EXAMPLE:

OBSTACLE HEIGHT . . . . . 175 FEET

HORIZONTAL DISTANCE FROM REFERENCE ZERO ~ FEET	MINIMUM GRADIENT OF CLIMB ~ %
2700	5.2
3386	4.1
4436	3.2
3116	4.5
4336	3.3



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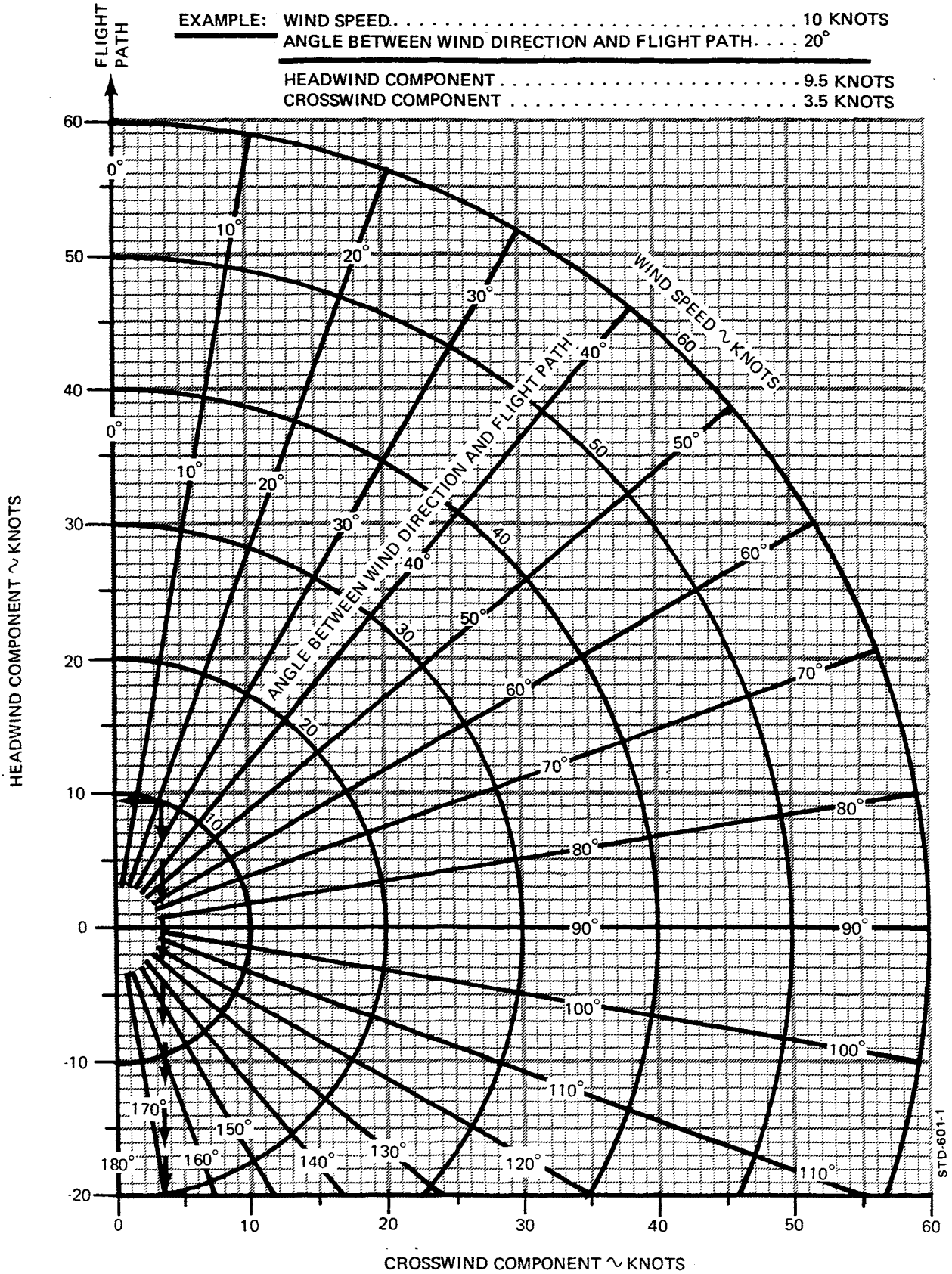
WIND COMPONENTS



TAKE-OFF DISTANCE – FLAPS UP

ASSOCIATED CONDITIONS:

POWER.....TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS.....UP
LANDING GEAR....RETRACT AFTER LIFT-OFF
RUNWAY.....PAVED, LEVEL, DRY SURFACE

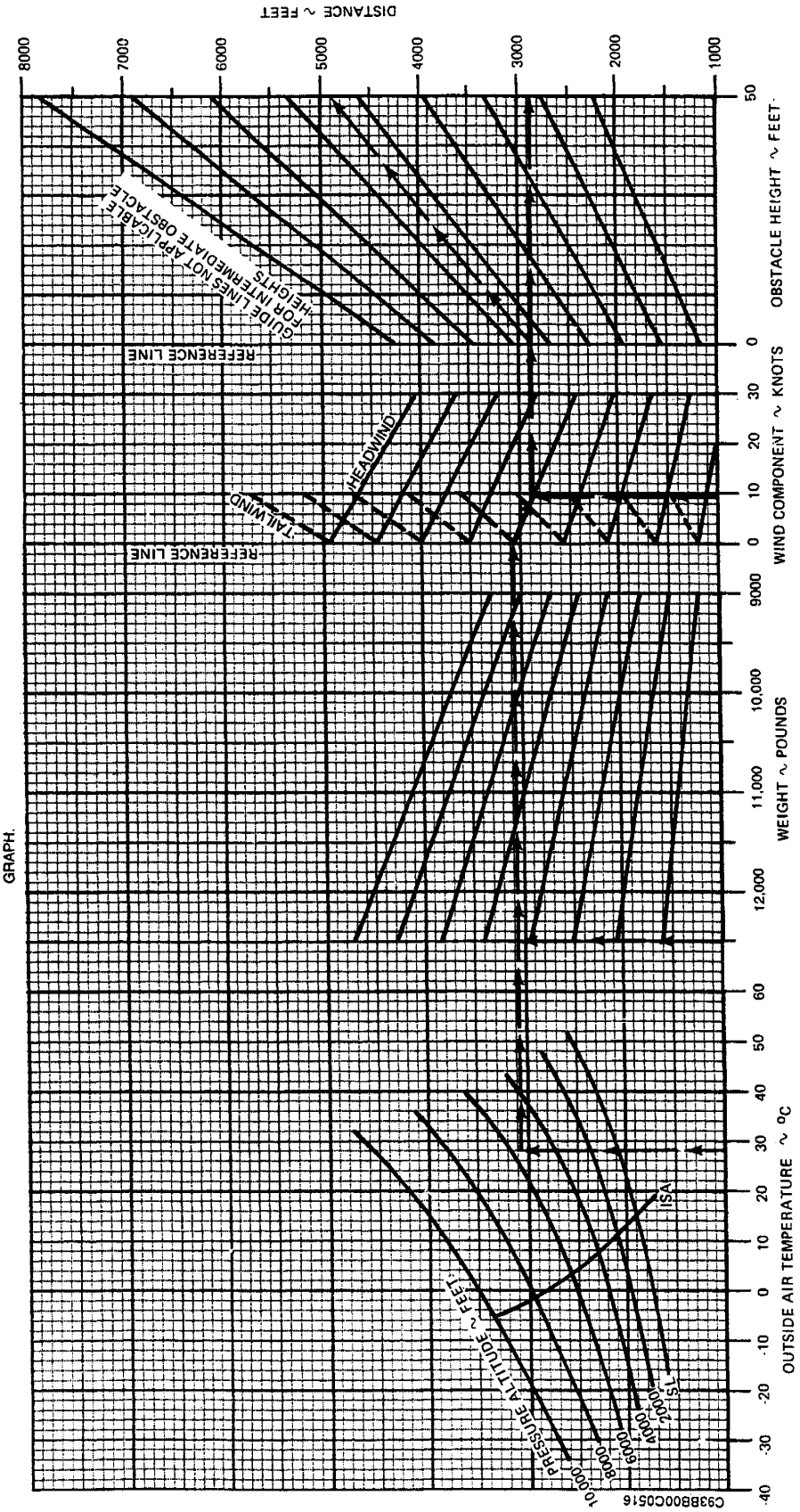
WEIGHT ~ POUNDS	TAKE-OFF SPEED ~ KNOTS	
	ROTATION	50 FT
12,500	95	121
12,000	95	119
11,000	95	115
10,000	95	111
9000	95	108

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10° C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.

EXAMPLE:

OAT.....28°C
PRESSURE ALTITUDE.....5433 FT
TAKE-OFF WEIGHT.....12,500 LBS
HEADWIND COMPONENT.....9.5 KTS

GROUND ROLL.....2870 FT
TOTAL DISTANCE OVER 50 FT OBSTACLE.....4900 FT
TAKE-OFF SPEED AT ROTATION.....95 KTS
AT 50 FT.....121 KTS



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ACCELERATE - STOP - FLAPS UP

WEIGHT ~ POUNDS	V_1 ~ KNOTS
12,500	95
12,000	95
11,000	95
10,000	95
9000	95

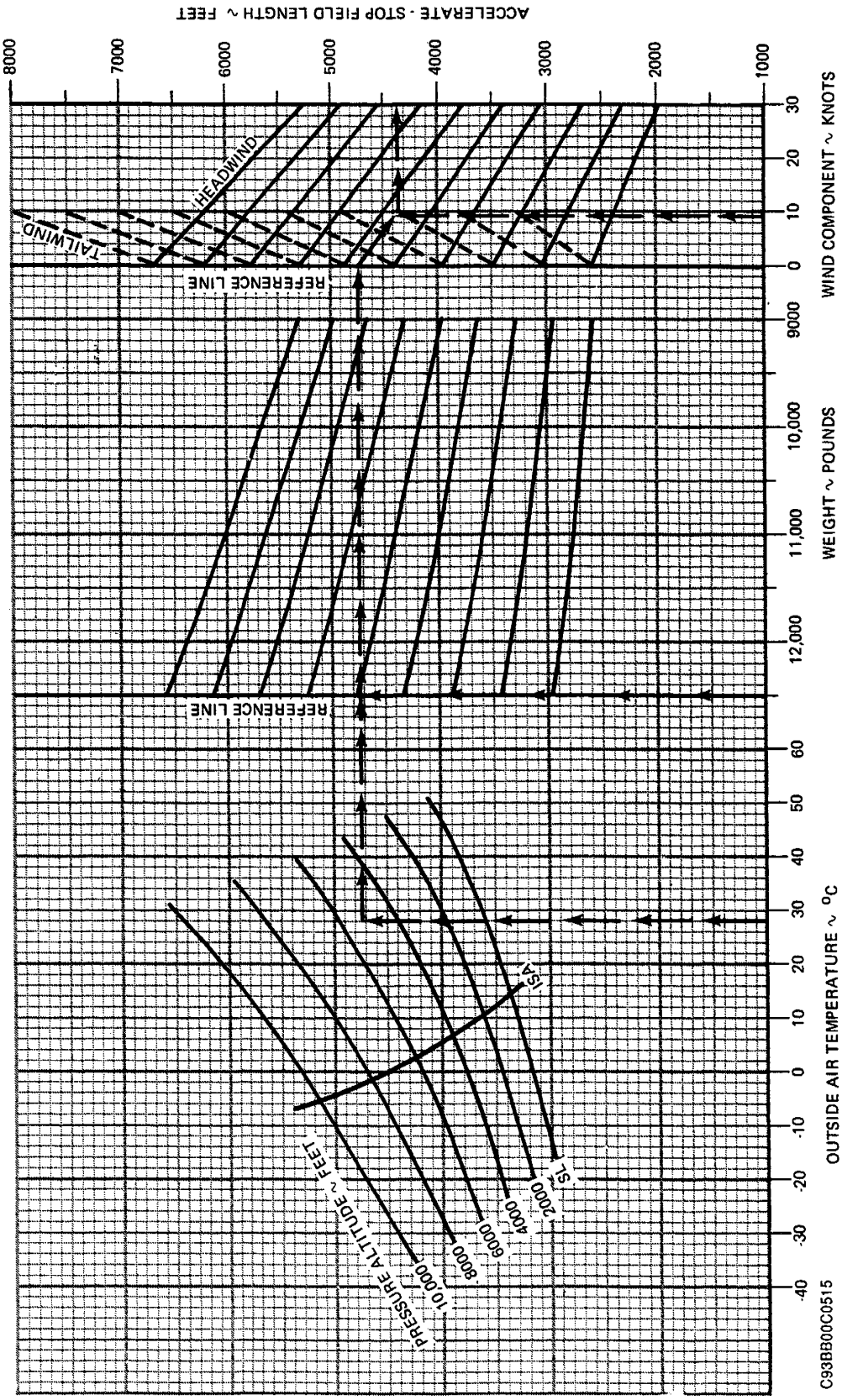
ASSOCIATED CONDITIONS:

POWER . . . . . 1. TAKE–OFF POWER SET BEFORE BRAKE RELEASE
2. BOTH ENGINES IDLE AT V_1 SPEED AND REVERSE OPERATING ENGINE
FLAPS . . . . . UP
AUTOFEATHER . ARMED
BRAKING . . . . MAXIMUM
RUNWAY . . . . PAVED, LEVEL, DRY SURFACE

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, PERFORMANCE IS NOT AFFECTED.

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
WEIGHT	12,500 LBS
HEADWIND COMPONENT	9.5 KTS
FIELD LENGTH	4375 FT
V_1	95 KTS



For Training Purposes Only

ACCELERATE - GO - FLAPS UP

ASSOCIATED CONDITION:

POWER . . . . . TAKE-OFF POWER SET
BEFORE BRAKE RELEASE
FLAPS . . . . . UP
AUTOFEATHER . ARMED
LANDING GEAR . RETRACT AFTER LIFT-OFF
RUNWAY . . . . PAVED, LEVEL, DRY SURFACE
OBSTACLE HEIGHT . . . . . 35 FEET

WEIGHT ~ POUNDS	SPEED ~ KNOTS V_R	SPEED ~ KNOTS V_2
12,500	95	121
12,000	95	119
11,000	95	115
10,000	95	111
9000	95	108

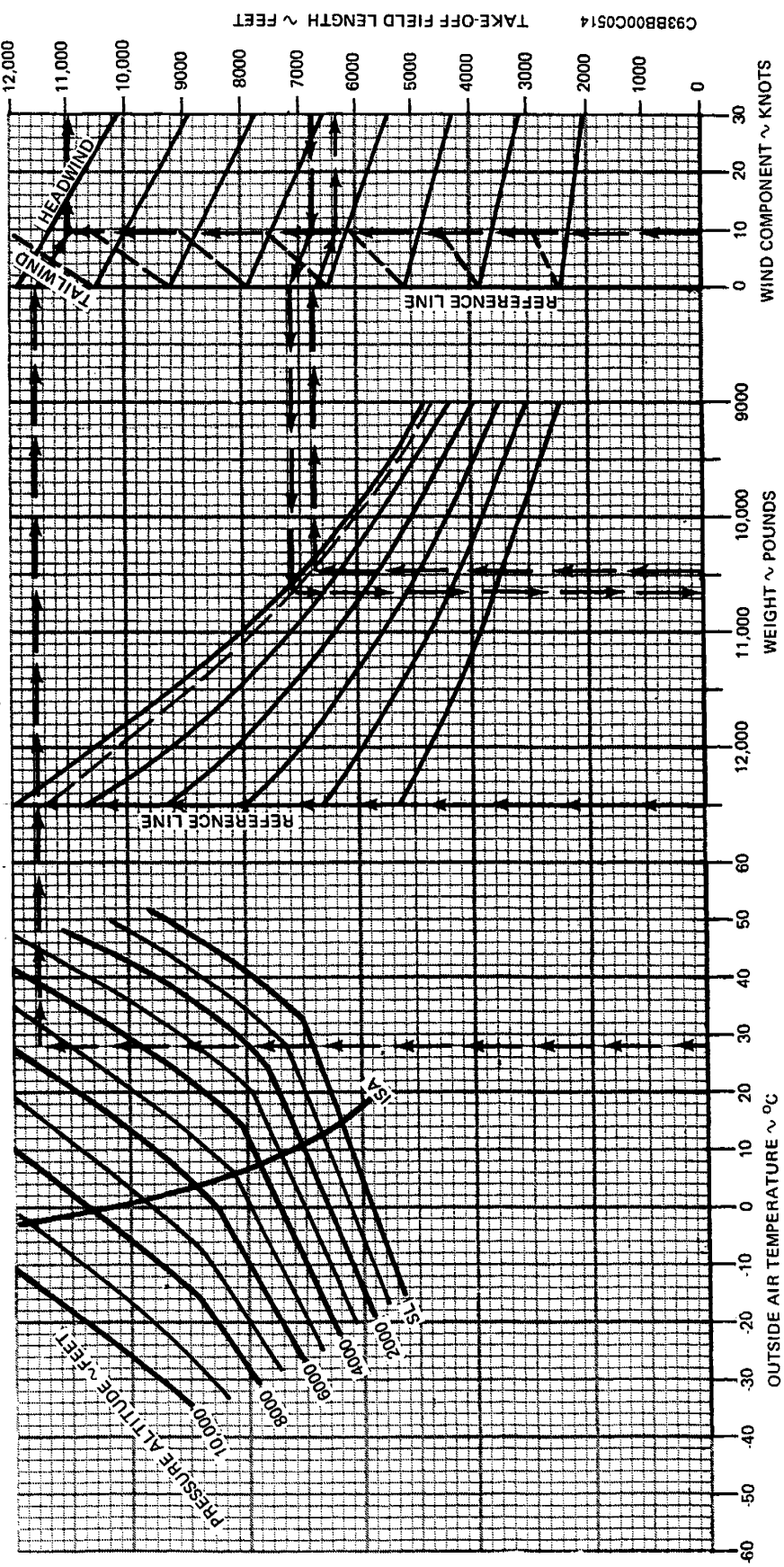
NOTE:
1. AIR DISTANCE IS 50% OF TAKE-OFF FIELD LENGTH
2. V_1 (ENGINE FAILURE SPEED) EQUALS V_R (ROTATION SPEED).
3. USABLE CLEAR WAY CANNOT EXCEED 25% OF THE RUNWAY LENGTH.
4. FOR OPERATION WITH ICE VANES EXTENDED, ADD 6°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.

EXAMPLE:

OAT . . . . . 28°C
PRESSURE ALTITUDE . . . . . 5430 FT
HEADWIND COMPONENT . . . . . 9.5 KTS

TAKE-OFF WEIGHT ~ POUNDS	TAKE-OFF FIELD LENGTH ~ FEET
12,500	10,950
10,650	6786
10,470	6370

SPEEDS (10,470 POUNDS) V_R . . . . . 95 KTS
V_2 . . . . . 113 KTS



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NET GRADIENT OF CLIMB - FLAPS UP

ASSOCIATED CONDITIONS:

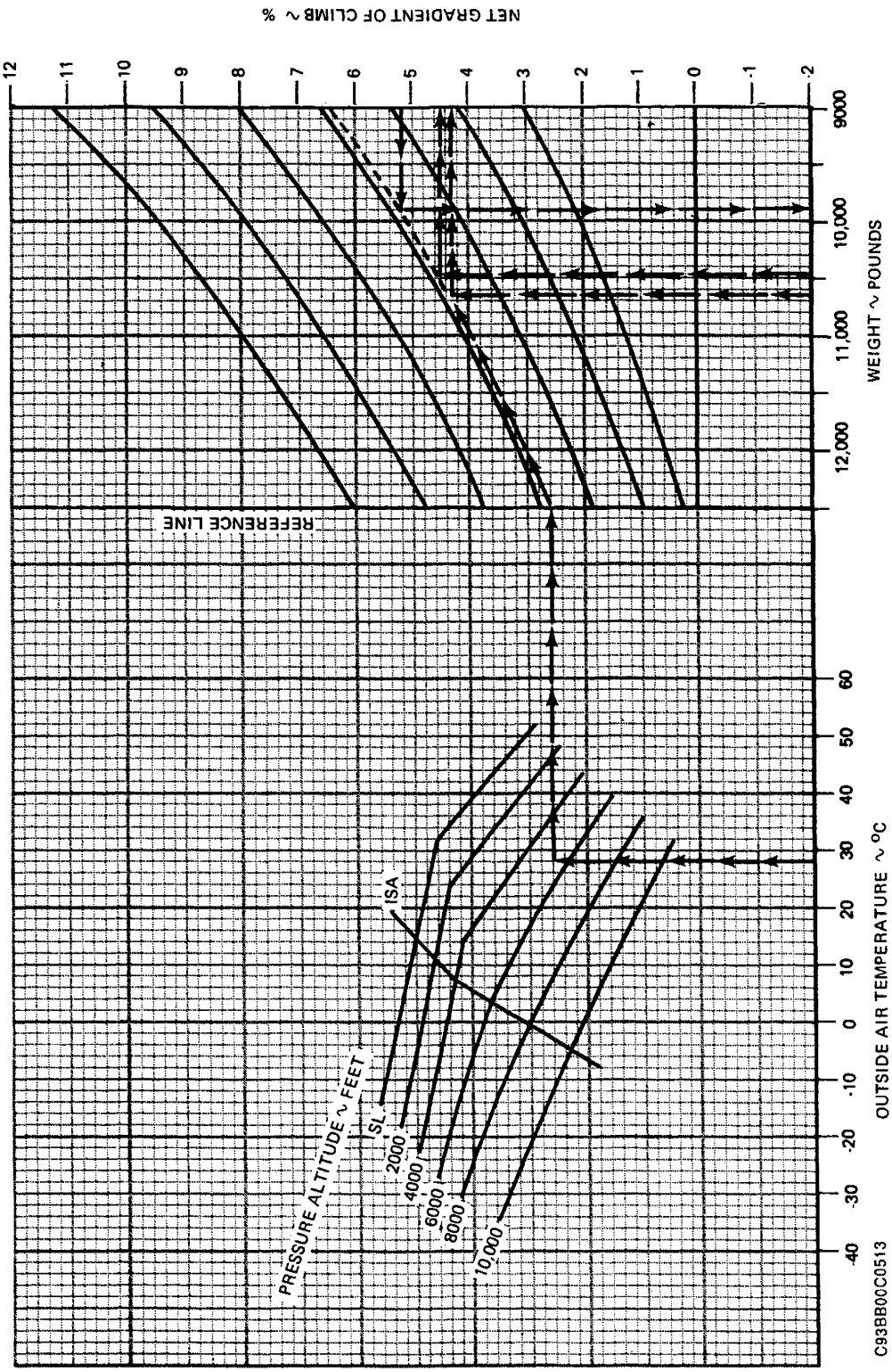
POWER	TAKE-OFF
FLAPS	UP
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

WEIGHT ~POUNDS	CLIMB SPEED ~ KNOTS
12,500	121
12,000	119
11,000	115
10,000	111
9000	108

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.

EXAMPLE:
OAT ..... 28°C
PRESSURE ALTITUDE ..... 5430 FT
CLIMB SPEED @ 10,470 LBS ..... 113 KTS

WEIGHT ~POUNDS	NET GRADIENT OF CLIMB ~ %
10,650	4.3
9900	5.2
10,470	4.5



TAKE-OFF DISTANCE – FLAPS APPROACH

ASSOCIATED CONDITIONS:

POWER.....TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS.....APPROACH
LANDING GEAR..RETRACT AFTER LIFT-OFF
RUNWAY.....PAVED, LEVEL, DRY SURFACE

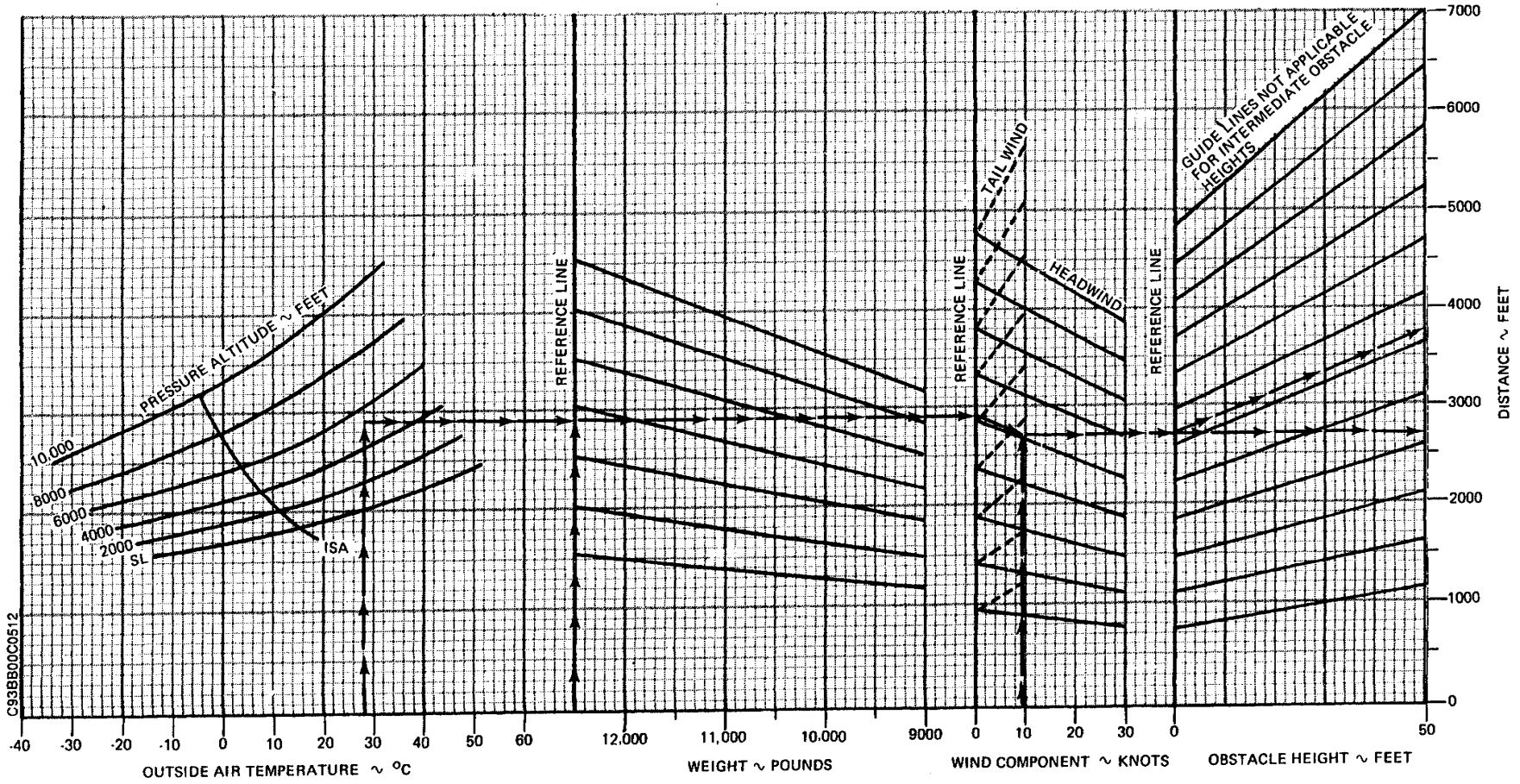
WEIGHT ∿ POUNDS	TAKE-OFF SPEED ∿ KNOTS	
	ROTATION	50 FT
12,500	94	106
12,000	94	105
11,000	94	103
10,000	94	101
9000	94	99

EXAMPLE:

OAT.....28°C
PRESSURE ALTITUDE.....5433 FT
TAKE-OFF WEIGHT.....12,500 LBS
HEADWIND COMPONENT.....9.5 KTS

GROUND ROLL.....2700 FT
TOTAL DISTANCE OVER 50 FT OBSTACLE.....3750 FT
TAKE-OFF SPEED AT ROTATION....94 KTS
AT 50 FT.....106 KTS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10° C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.



ACCELERATE - STOP - FLAPS APPROACH

ASSOCIATED CONDITIONS:

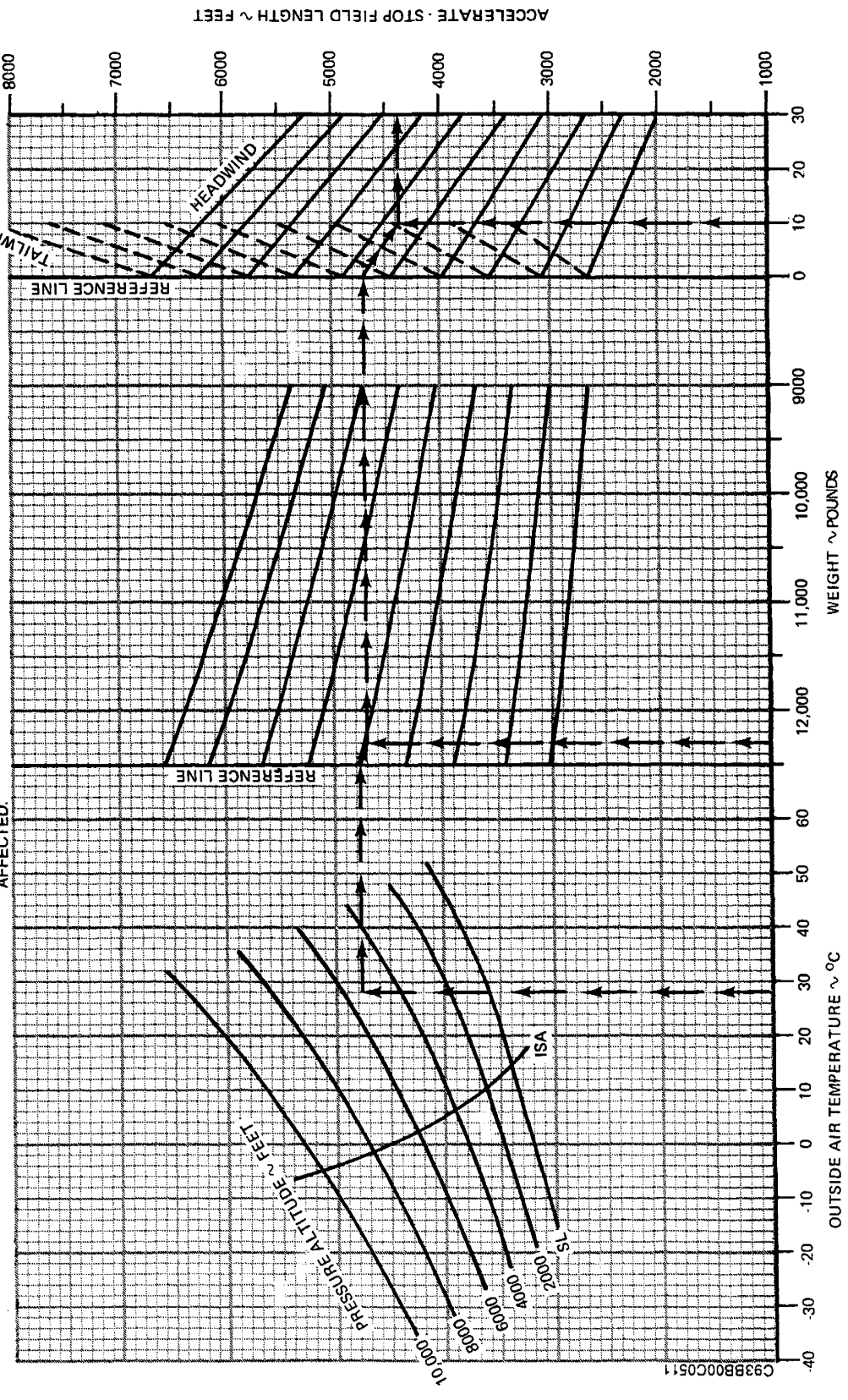
POWER . . . . . 1. TAKE-OFF POWER SET BEFORE BRAKE RELEASE
2. BOTH ENGINES IDLE AT V1
SPEED AND REVERSE OPERATING ENGINE
FLAPS . . . . . APPROACH
AUTOFEATHER . ARMED
BRAKING . . . . MAXIMUM
RUNWAY . . . . PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	V1 ~ KNOTS
12,500	94
12,000	94
11,000	94
10,000	94
9000	94

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, PERFORMANCE IS NOT AFFECTED.

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
WEIGHT	12,300 LBS
HEADWIND COMPONENT	9.5 KTS
FIELD LENGTH	4370 FT
V1	94 KTS



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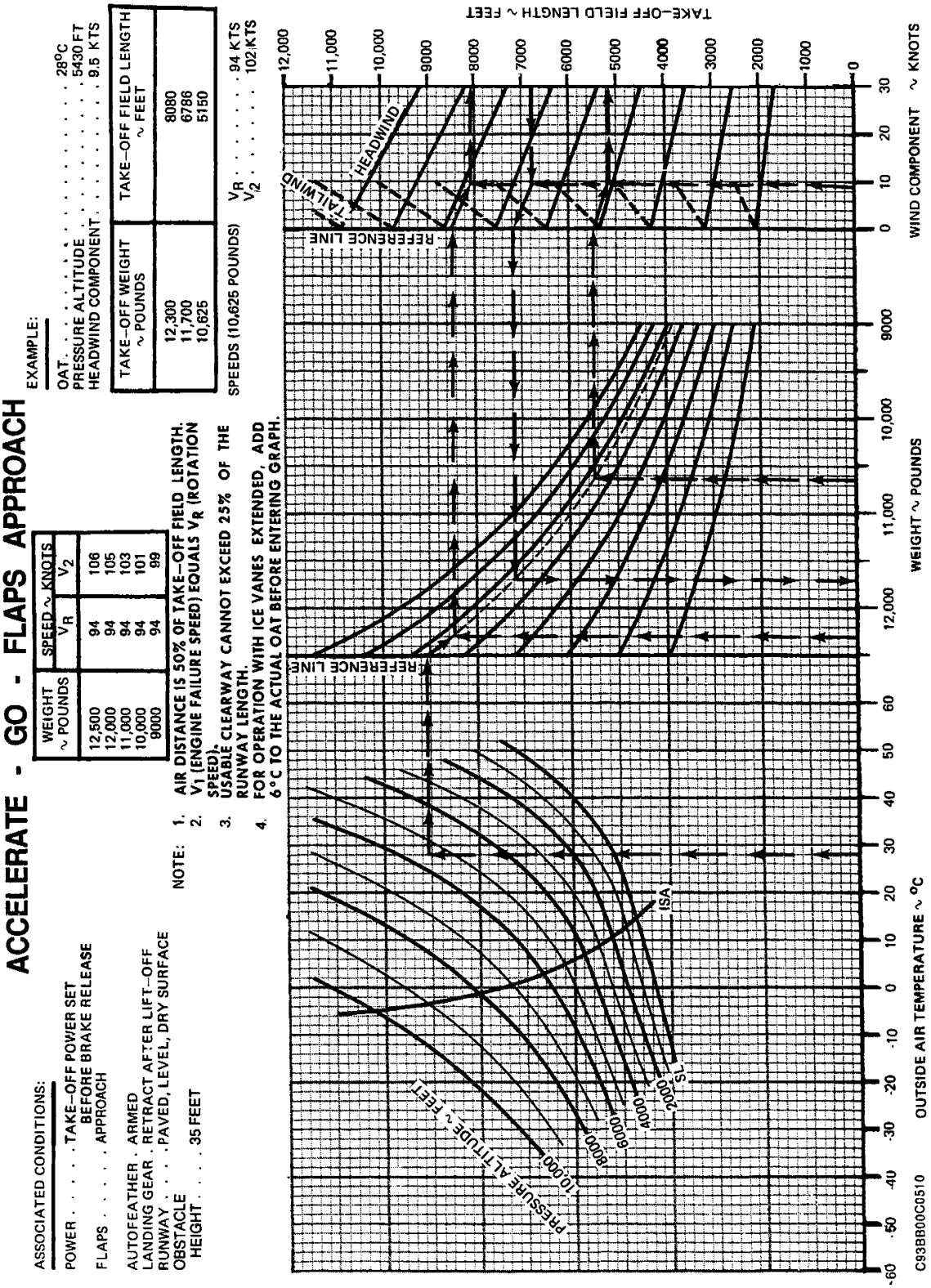

ACCELERATE - GO - FLAPS APPROACH
ASSOCIATED CONDITIONS:
POWER . . . . . TAKE-OFF POWER SET
BEFORE BRAKE RELEASE
FLAPS . . . . . APPROACH
AUTOFEATHER . ARMED
LANDING GEAR . RETRACT AFTER LIFT-OFF
RUNWAY . . . . PAVED, LEVEL, DRY SURFACE
OBSTACLE HEIGHT . . . 35 FEET
NOTE:
1. AIR DISTANCE IS 50% OF TAKE-OFF FIELD LENGTH.
2. V_1 (ENGINE FAILURE SPEED) EQUALS V_R (ROTATION SPEED).
3. USABLE CLEARWAY CANNOT EXCEED 25% OF THE RUNWAY LENGTH.
4. FOR OPERATION WITH ICE VANES EXTENDED, ADD 6°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.
WEIGHT ~ POUNDS
SPEED ~ KNOTS
V_R
V_2
12,500 94 106
12,000 94 105
11,000 94 103
10,000 94 101
9000 94 99
EXAMPLE:
OAT . . . . . 28°C
PRESSURE ALTITUDE . . . . . 5430 FT
HEADWIND COMPONENT . . . . . 9.5 KTS
TAKE-OFF WEIGHT ~ POUNDS
TAKE-OFF FIELD LENGTH ~ FEET
12,300 8080
11,700 6786
10,625 5150
SPEEDS (10,625 POUNDS)
V_R . . . . . 94 KTS
V_2 . . . . . 102 KTS
REFERENCE LINE
REFERENCE LINE
TAILWIND
HEADWIND
PRESSURE ALTITUDE ~ FEET
S.L.
2000
4000
6000
8000
10,000
ISA
OUTSIDE AIR TEMPERATURE ~ °C
WEIGHT ~ POUNDS
WIND COMPONENT ~ KNOTS
TAKE-OFF FIELD LENGTH ~ FEET
C93BB00C0510

NET GRADIENT OF CLIMB - FLAPS APPROACH

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	106
12,000	105
11,000	103
10,000	101
9000	99

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10° TO THE ACTUAL OAT BEFORE ENTERING GRAPH.

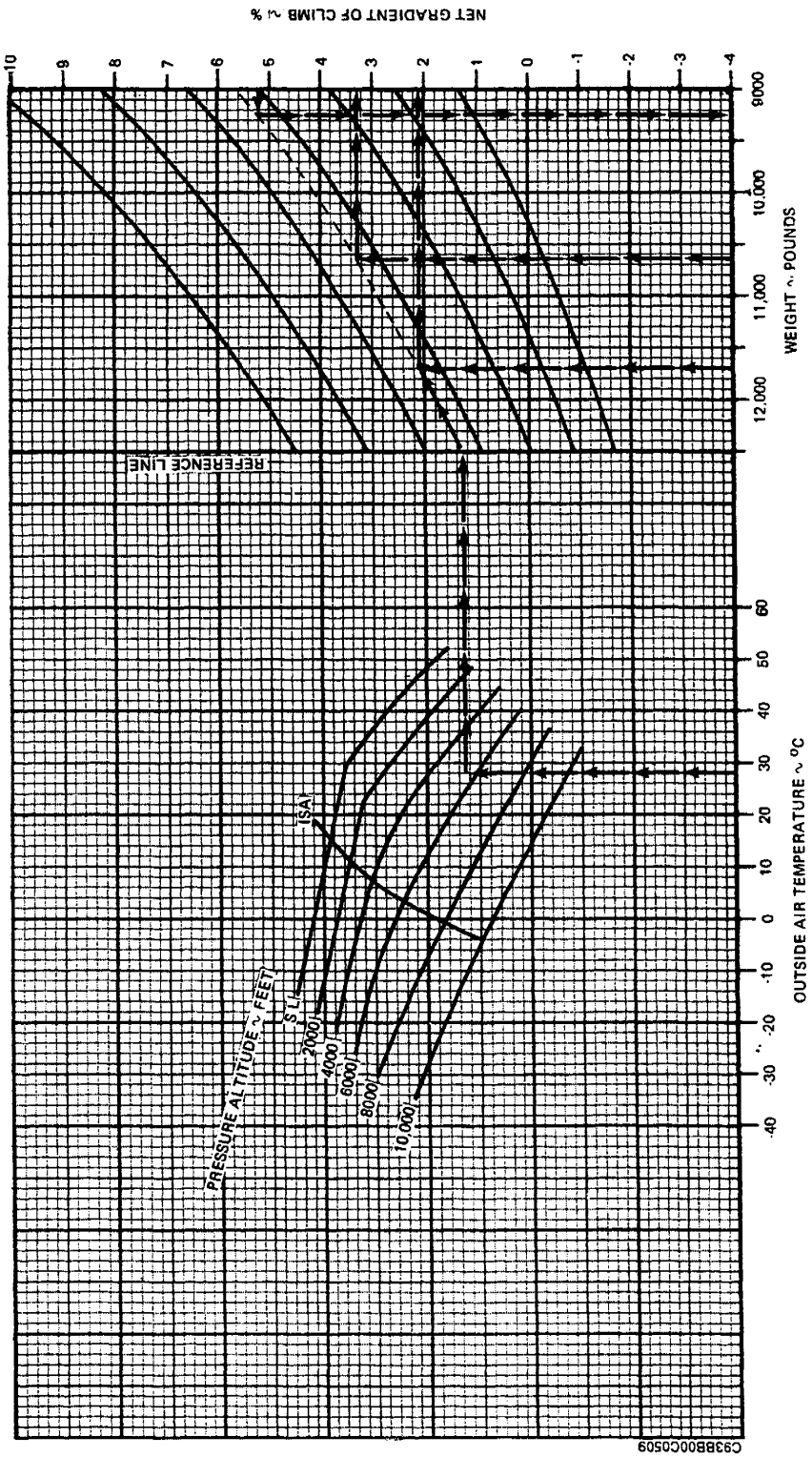
ASSOCIATED CONDITIONS:

POWER . . . . . TAKE-OFF
FLAPS . . . . . APPROACH
LANDING GEAR . UP
INOPERATIVE PROPELLER . FEATHERED

EXAMPLE:

OAT . . . . . 28°C
PRESSURE ALTITUDE . . 5430 FT
CLIMB SPEED @ 10,825 LBS . . . . 102 KTS

WEIGHT ~ POUNDS	NET GRADIENT OF CLIMB ~ %
11,700	2.1
9250	5.2
10,825	3.3



CLIMB – TWO ENGINES – FLAPS UP

ASSOCIATED CONDITIONS:

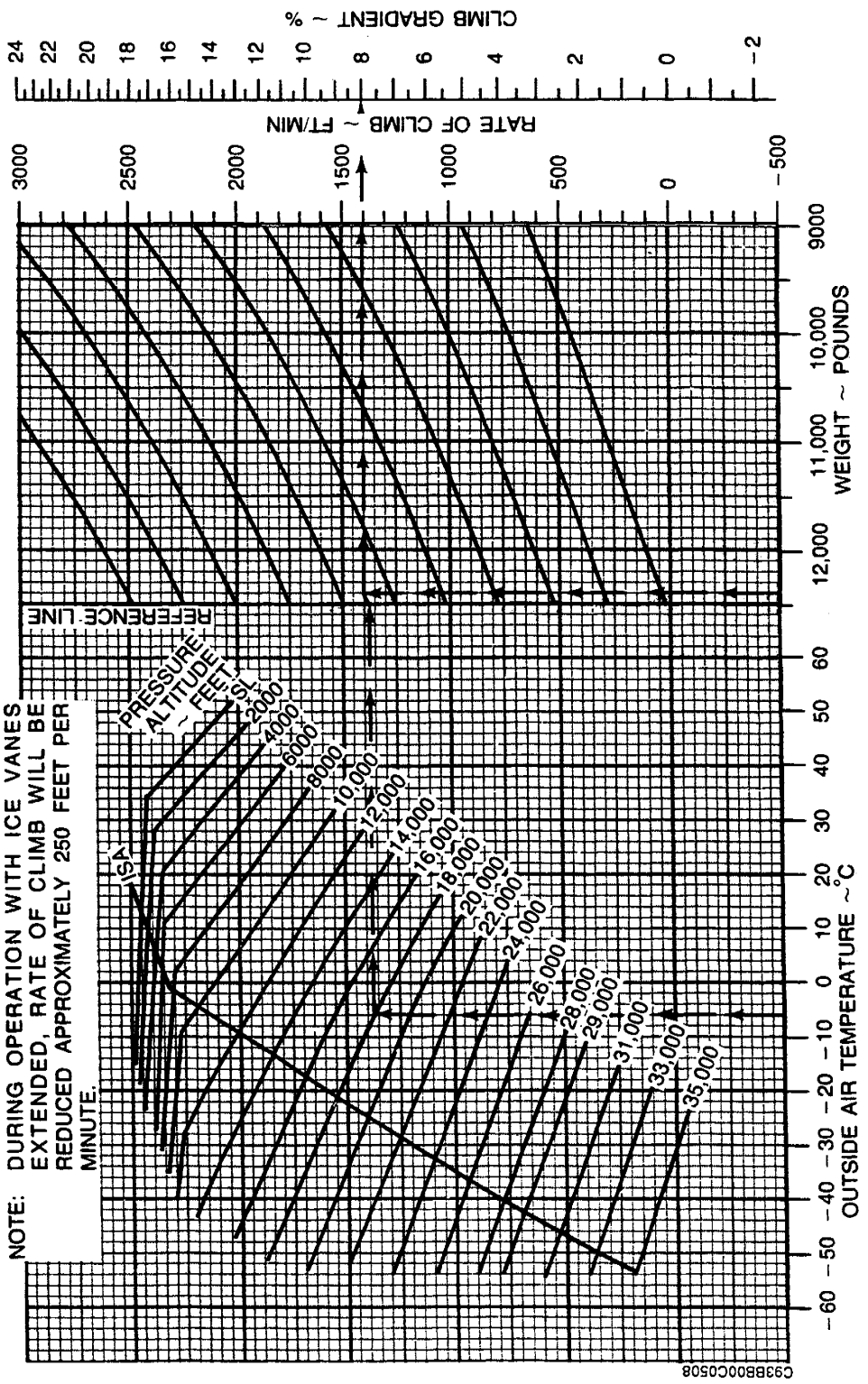
POWER	MAX CONTINUOUS
FLAPS	UP
LANDING GEAR	UP

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	125
12,000	124
11,000	121
10,000	118
9000	115

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,405 LBS
RATE OF CLIMB	1400 FT/MIN
CLIMB GRADIENT	8%
CLIMB SPEED	125 KTS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, RATE OF CLIMB WILL BE REDUCED APPROXIMATELY 250 FEET PER MINUTE.



For Training Purposes Only

CLIMB – TWO ENGINES – FLAPS APPROACH

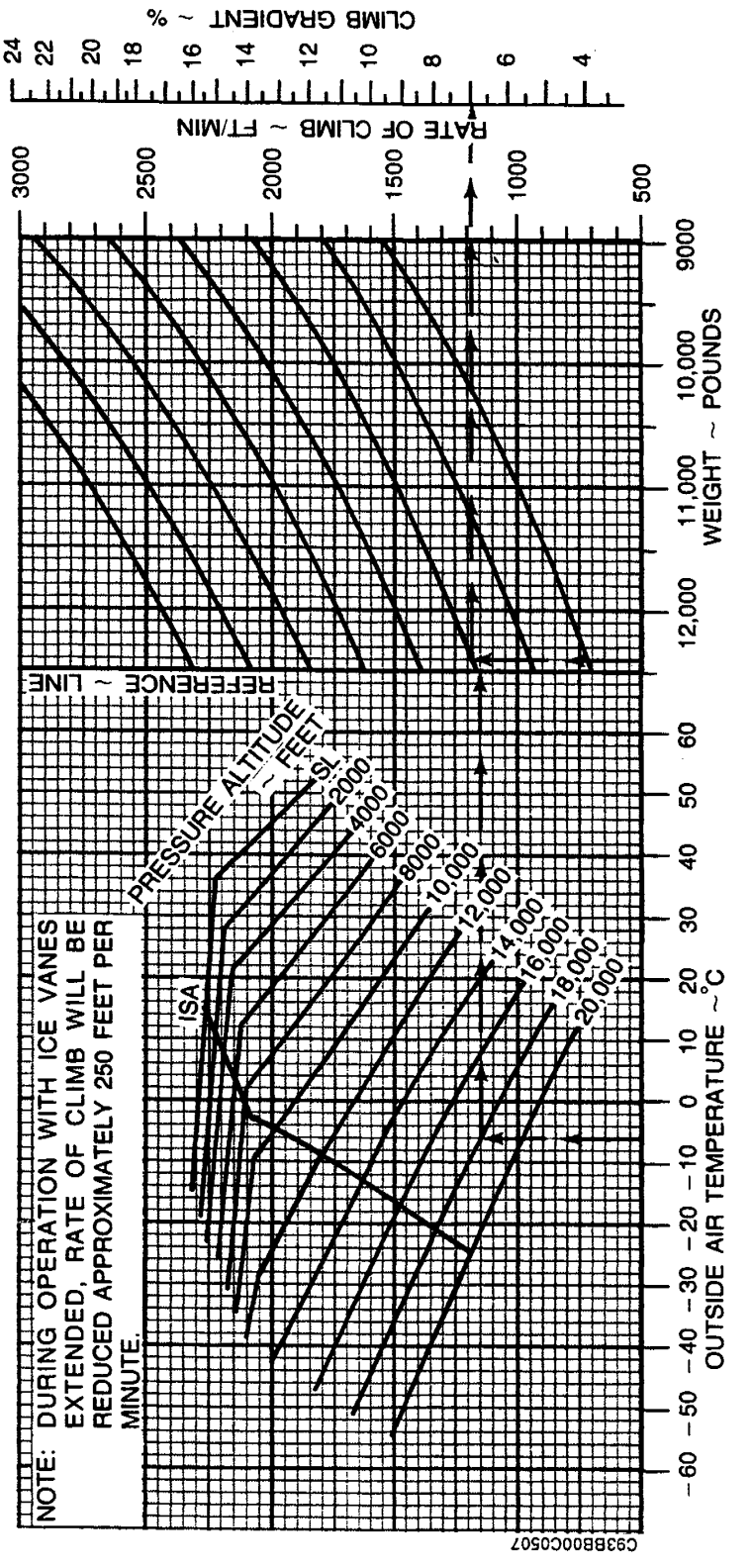
ASSOCIATED CONDITIONS:

POWER ..... MAX CONTINUOUS
FLAPS ..... APPROACH
LANDING GEAR ..... UP

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	125
12,000	124
11,000	121
10,000	118
9000	115

EXAMPLE:

OAT	–6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,405 LBS
RATE OF CLIMB	1190 FT/MIN
CLIMB GRADIENT	6.9%
CLIMB SPEED	125 KTS



CLIMB – ONE ENGINE INOPERATIVE

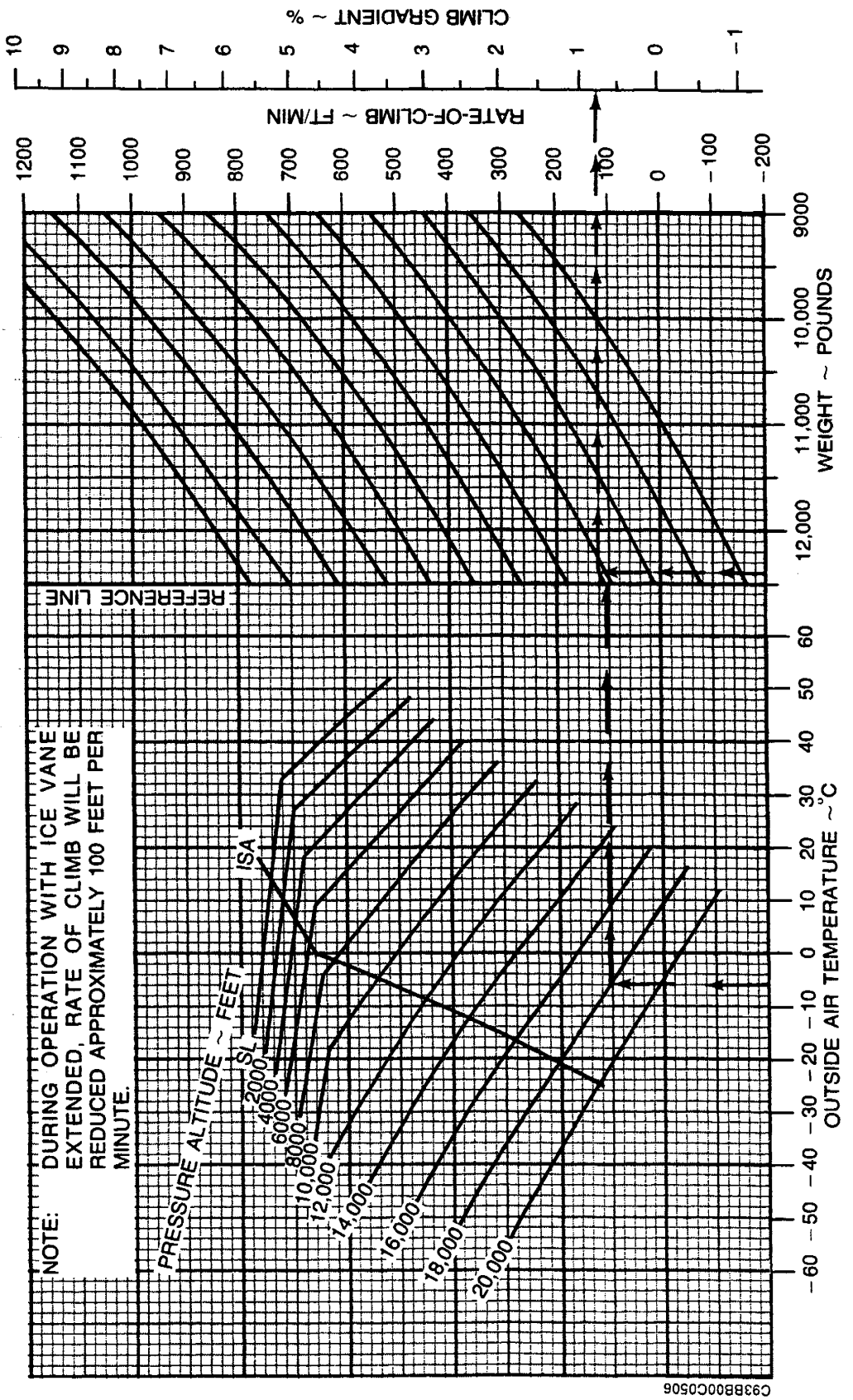
ASSOCIATED CONDITIONS:

POWER	MAX CONTINUOUS
FLAPS	UP
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	121
12,000	119
11,000	117
10,000	114
9000	111

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,405 LBS
RATE OF CLIMB	120 FT/MIN
CLIMB GRADIENT	0.8%
CLIMB SPEED	121 KTS



For Training Purposes Only

SERVICE CEILING – ONE ENGINE INOPERATIVE

ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED
FLAPS	UP

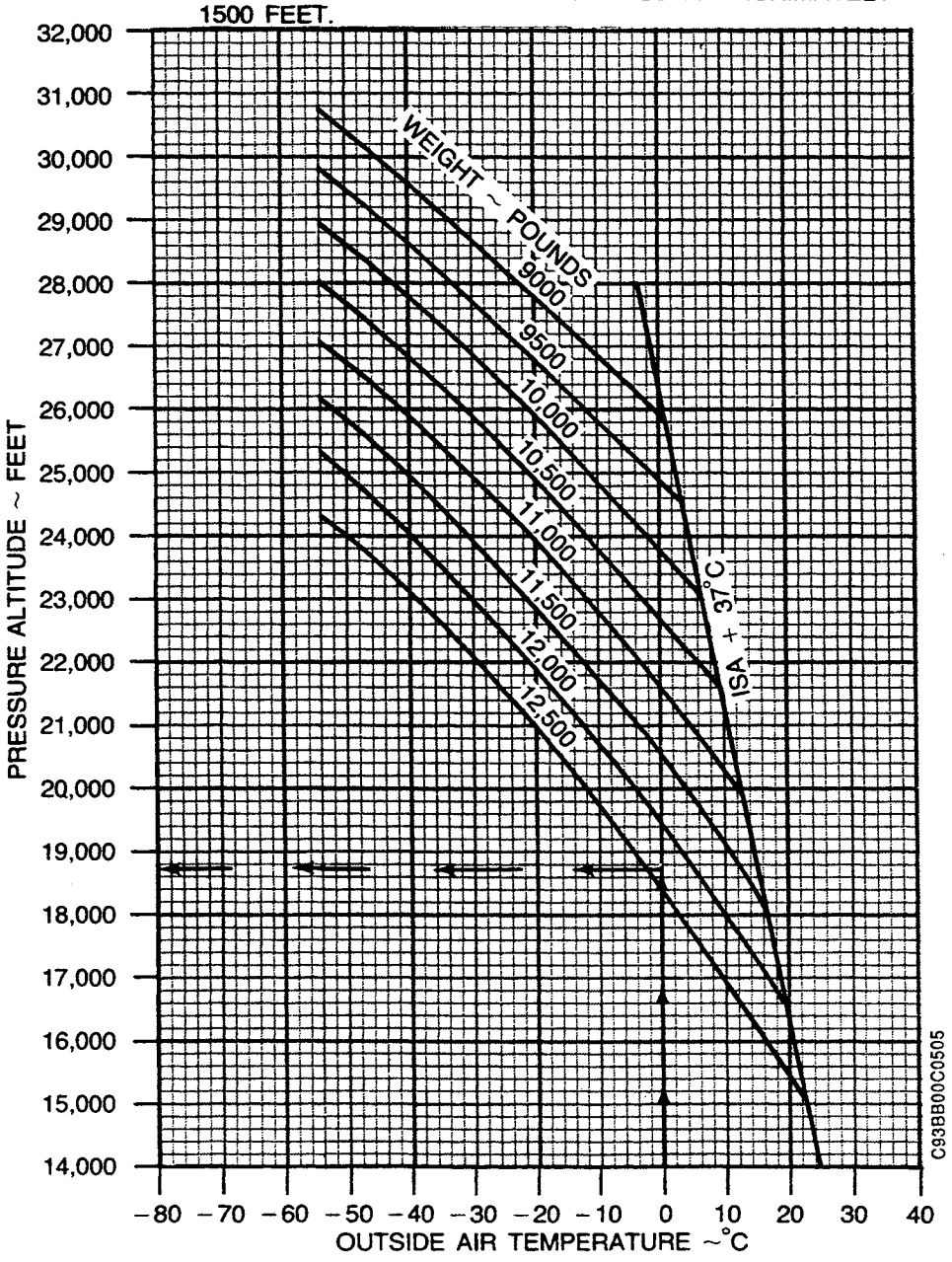
NOTE: SERVICE CEILING IS THE MAXIMUM PRESSURE ALTITUDE AT WHICH THE AIRPLANE IS CAPABLE OF CLIMBING 50 FT/MINUTE WITH ONE PROPELLER FEATHERED.

EXAMPLE:

OAT AT MEA	0°C
WEIGHT	12,393 LBS
ROUTE SEGMENT MEA	18,000 FT
SERVICE CEILING	18,730 FT

NOTE: SERVICE CEILING IS ABOVE ENROUTE MEA

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, SERVICE CEILING WILL BE LOWERED APPROXIMATELY 1500 FEET.



TIME, FUEL, AND DISTANCE TO CLIMB

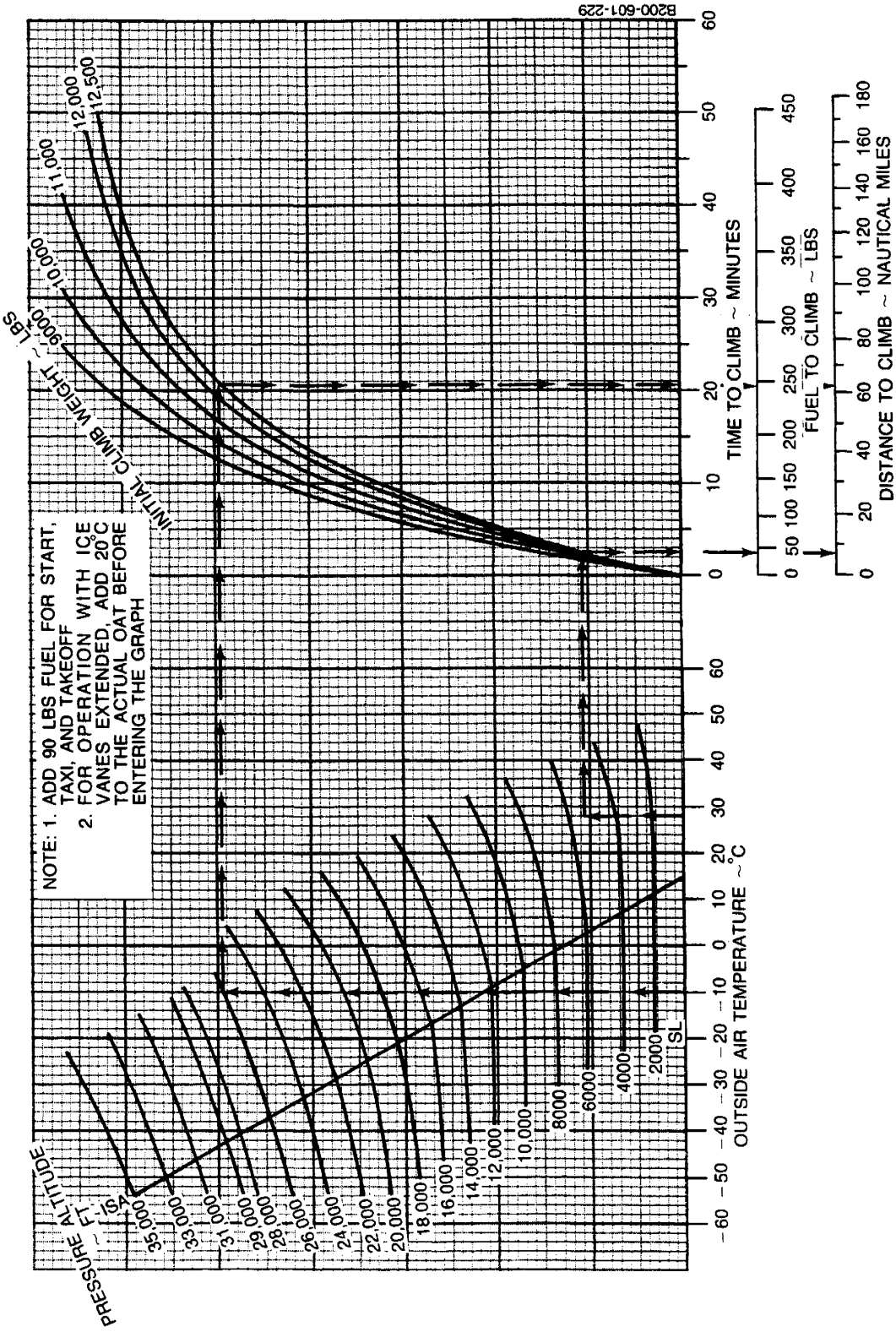
ASSOCIATED CONDITIONS:

PROPELLER SPEED	1900 RPM
ITT	770°C
OR TORQUE	2230 FT-LBS

ALTITUDE ~ FEET	CLIMB SPEED ~ KTS
SL TO 10,000	160
10,000 TO 20,000	140
20,000 TO 25,000	130
25,000 TO 35,000	120

EXAMPLE:

OAT AT TAKEOFF	28°C
OAT AT CRUISE	−10°C
AIRPORT PRESSURE ALTITUDE	5433 FT
CRUISE ALTITUDE	26,000 FT
INITIAL CLIMB WEIGHT	12,500 LBS
TIME TO CLIMB (21-3)	18 MIN
FUEL TO CLIMB (250-47)	203 LBS
DISTANCE TO CLIMB (62-8)	54 NM



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NORMAL CRUISE POWER

1700 RPM

ISA −30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	463	926	243	231	244	232	245	233
2000	-14	-19	2230	452	904	241	235	242	236	243	237
4000	-18	-23	2230	439	878	239	240	240	241	240	242
6000	-21	-27	2230	428	856	237	244	237	245	238	246
8000	-25	-31	2230	417	834	234	249	235	250	236	251
10,000	-29	-35	2230	406	812	232	254	233	255	234	256
12,000	-33	-39	2230	396	792	230	259	231	260	231	261
14,000	-36	-43	2230	389	778	227	264	228	265	229	266
16,000	-40	-47	2230	384	768	225	269	226	270	227	271
18,000	-44	-51	2230	379	758	223	274	224	276	225	277
20,000	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1700 RPM

ISA −20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	466	932	242	234	242	235	243	236
2000	-4	-9	2230	453	906	239	238	240	239	241	240
4000	-7	-13	2230	440	880	237	243	238	244	239	245
6000	-11	-17	2230	428	856	235	248	236	248	237	249
8000	-15	-21	2230	418	836	232	252	233	253	234	254
10,000	-19	-25	2230	408	816	231	257	231	258	232	259
12,000	-22	-29	2230	398	796	228	262	229	263	230	264
14,000	-26	-33	2230	391	782	225	267	226	269	227	270
16,000	-30	-37	2230	385	770	223	273	224	274	225	275
18,000	-33	-41	2230	381	762	220	278	221	279	223	280
20,000	-37	-45	2230	378	756	218	283	219	284	220	286
22,000	-41	-49	2155	364	728	212	284	213	287	215	288
24,000	-44	-53	2029	341	682	204	283	206	286	208	288
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1700 RPM

ISA −10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	469	938	240	237	241	238	241	238
2000	6	1	2230	456	912	238	241	239	242	239	243
4000	3	-3	2230	443	886	235	246	236	247	237	248
6000	-1	-7	2230	431	862	233	250	234	251	235	252
8000	-5	-11	2230	419	838	231	255	232	256	232	257
10,000	-9	-15	2230	408	816	228	260	229	261	230	262
12,000	-12	-19	2230	399	798	226	265	227	266	228	268
14,000	-16	-23	2230	392	784	224	271	225	272	226	273
16,000	-20	-27	2230	387	774	221	276	222	277	223	278
18,000	-23	-31	2230	383	766	218	281	219	282	221	284
20,000	-27	-35	2155	368	736	212	282	214	284	215	286
22,000	-31	-39	2055	350	700	206	283	208	285	209	287
24,000	-35	-43	1933	328	656	198	282	200	284	202	287
26,000	-39	-47	1800	305	610	190	279	192	282	195	285
28,000	-43	-51	1662	281	562	181	275	184	279	186	283
29,000	-45	-52	1595	270	540	176	273	179	277	182	281
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1700 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	473	946	238	239	239	240	240	241
2000	17	11	2230	459	918	236	244	237	245	238	246
4000	13	7	2230	446	892	234	249	235	250	236	250
6000	9	3	2230	433	866	231	253	232	254	233	255
8000	5	-1	2230	421	842	229	258	230	259	231	260
10,000	2	-5	2230	410	820	227	263	228	264	229	265
12,000	-2	-9	2230	400	800	224	268	225	270	226	271
14,000	-6	-13	2230	393	786	222	274	223	275	224	276
16,000	-9	-17	2230	388	776	219	279	220	280	221	282
18,000	-13	-21	2152	373	746	213	280	215	282	216	284
20,000	-17	-25	2057	355	710	207	281	208	283	210	285
22,000	-21	-29	1963	337	674	200	281	202	284	204	286
24,000	-25	-33	1842	316	632	192	280	195	283	197	285
26,000	-29	-37	1712	293	586	184	277	186	280	189	283
28,000	-33	-41	1579	270	540	175	272	178	277	180	281
29,000	-36	-42	1514	259	518	170	269	173	274	176	279
31,000	-40	-46	1391	239	478	160	263	164	270	167	275
33,000	-44	-50	1274	220	440	149	254	154	263	158	270
35,000	-49	-54	1153	201	402	133	237	143	254	150	263

For Training Purposes Only

NORMAL CRUISE POWER

1700 RPM

ISA +10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	30	25	2230	475	950	237	242	238	243	239	244
2000	27	21	2230	462	924	235	247	236	248	236	248
4000	23	17	2230	449	898	232	251	233	252	234	253
6000	19	13	2230	436	872	230	256	231	257	232	258
8000	15	9	2230	424	848	227	261	228	262	229	263
10,000	12	5	2230	412	824	225	266	226	267	227	269
12,000	8	1	2230	402	804	223	271	224	273	225	274
14,000	4	-3	2230	394	788	220	277	221	278	222	279
16,000	1	-7	2172	379	758	215	279	216	281	217	282
18,000	-3	-11	2064	358	716	208	279	210	281	211	283
20,000	-7	-15	1975	341	682	202	280	204	282	205	284
22,000	-11	-19	1873	325	650	195	279	197	282	199	284
24,000	-15	-23	1756	304	608	187	277	189	281	191	284
26,000	-19	-27	1630	281	562	178	274	181	278	183	281
28,000	-24	-31	1500	259	518	169	269	172	274	175	278
29,000	-26	-32	1438	248	496	164	264	167	271	270	276
31,000	-30	-36	1319	229	458	153	258	158	266	262	272
33,000	-34	-40	1205	210	420	141	247	148	258	253	267
35,000	-39	-44	1087	192	384	---	---	135	246	242	259

For Training Purposes Only

NORMAL CRUISE POWER

1700 RPM

ISA +20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	478	956	235	244	236	245	237	246
2000	37	31	2230	465	930	233	249	234	250	235	251
4000	33	27	2230	451	902	231	254	232	255	232	256
6000	29	23	2230	439	878	228	259	229	260	230	261
8000	26	19	2230	427	854	226	264	227	265	228	266
10,000	22	15	2230	415	830	223	269	224	270	226	272
12,000	18	11	2229	404	808	221	274	222	276	223	277
14,000	14	7	2135	382	764	214	275	216	277	217	278
16,000	10	3	2041	362	724	208	276	210	278	212	279
18,000	7	-1	1947	342	682	202	276	203	278	205	280
20,000	3	-5	1882	328	656	196	277	198	280	200	282
22,000	-1	-9	1802	313	616	190	278	192	281	194	284
24,000	-5	-13	1690	293	586	182	276	185	279	187	283
26,000	-9	-17	1560	271	542	173	271	176	276	178	280
28,000	-14	-21	1425	249	498	163	265	166	271	170	276
29,000	-16	-22	1364	238	476	157	261	162	268	165	273
31,000	-20	-26	1249	219	438	146	252	152	261	156	268
33,000	-25	-30	1139	202	404	131	234	141	253	147	262
35,000	---	---	---	---	---	---	---	---	---	---	---

For Training Purposes Only

NORMAL CRUISE POWER

1700 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	481	962	234	247	235	248	236	249
2000	47	41	2230	468	936	232	252	233	253	233	254
4000	43	37	2230	454	908	229	256	230	257	231	258
6000	39	33	2230	441	882	227	261	228	263	229	264
8000	36	29	2230	428	856	224	267	225	268	226	269
10,000	32	25	2187	410	820	220	270	221	271	222	273
12,000	28	21	2076	384	768	213	270	215	272	216	273
14,000	24	17	1989	363	726	207	271	209	273	210	274
16,000	20	13	1906	344	688	201	271	203	274	204	276
18,000	16	9	1828	326	652	195	272	197	275	199	277
20,000	12	5	1777	314	628	190	274	192	277	194	279
22,000	9	1	1719	302	604	185	275	187	279	189	282
24,000	5	-3	1608	281	562	176	273	179	277	182	280
26,000	0	-7	1489	260	520	167	268	171	273	174	278
28,000	-4	-11	1369	239	478	157	262	162	268	165	274
29,000	-6	-12	1310	229	458	152	257	157	265	161	271
31,000	-10	-16	1195	211	422	139	245	147	258	151	266
33,000	-15	-20	1078	193	286	---	---	134	245	141	258
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1700 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	483	966	233	249	234	250	235	251
2000	54	48	2230	470	940	231	253	232	254	232	255
4000	50	44	2230	456	912	228	258	229	259	230	260
6000	47	40	2230	442	884	226	263	227	264	228	265
8000	43	36	2160	420	840	220	265	222	267	223	268
10,000	39	32	2080	397	794	215	267	216	268	218	270
12,000	35	28	1975	370	740	208	267	210	268	211	270
14,000	31	24	1893	351	702	202	267	204	270	206	272
16,000	27	20	1815	333	666	197	268	198	271	200	273
18,000	23	16	1740	315	630	191	269	192	272	194	274
20,000	19	12	1696	304	608	185	271	188	274	190	277
22,000	15	8	1645	292	584	180	272	183	276	185	279
24,000	11	4	1546	273	546	172	270	175	274	178	278
26,000	7	0	1435	252	504	163	265	167	271	170	276
28,000	3	-4	1319	232	464	153	258	158	266	161	272
29,000	1	-5	1261	223	446	147	253	153	262	157	269
31,000	-4	-9	1151	205	410	133	238	142	254	148	263
33,000	-9	-13	1039	187	374	---	---	128	238	138	255
35,000	---	---	---	---	---	---	---	---	---	---	---

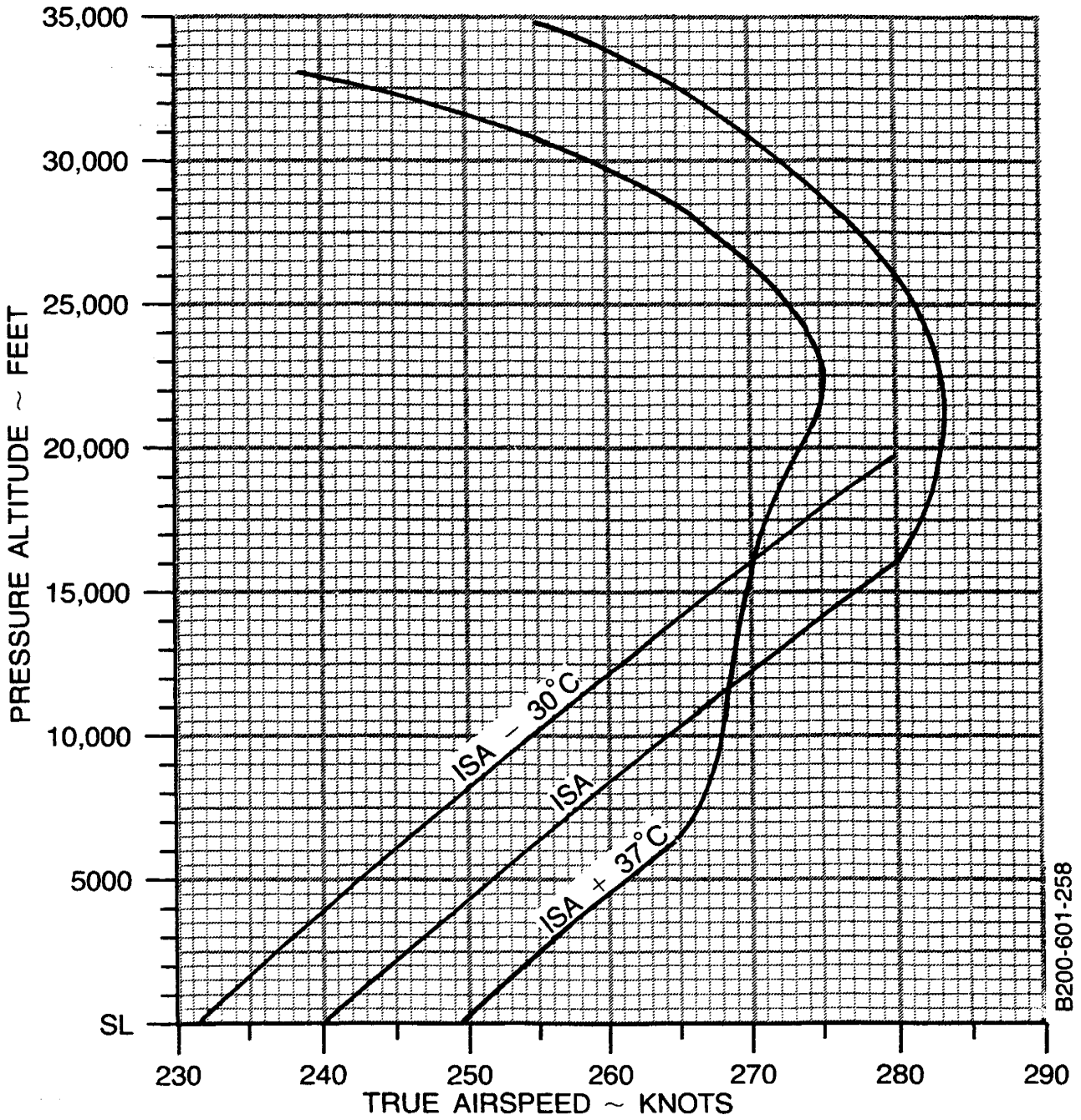
For Training Purposes Only

NORMAL CRUISE SPEEDS

1700 RPM

WEIGHT 11,000 LBS

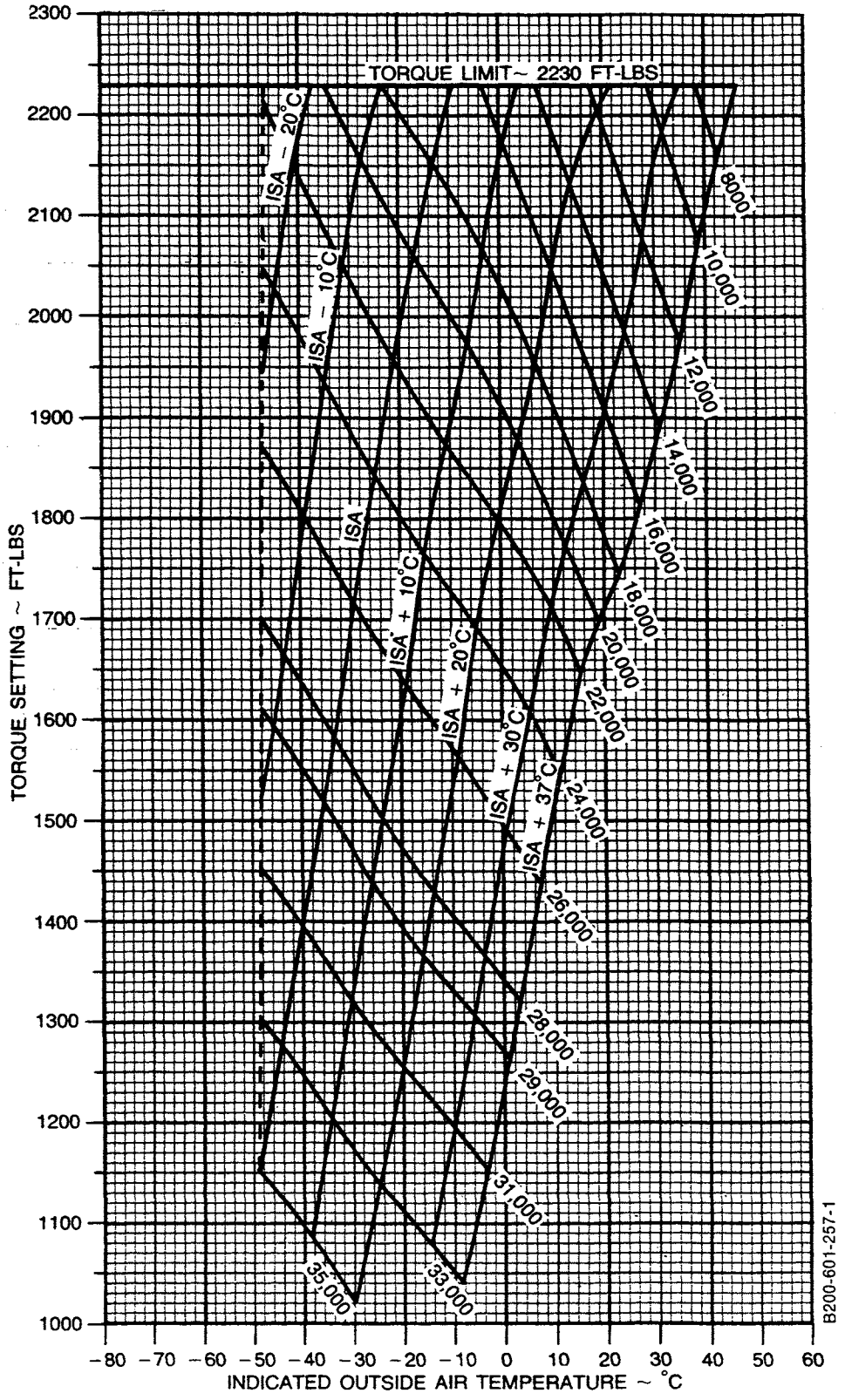
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS.



NORMAL CRUISE POWER

1700 RPM

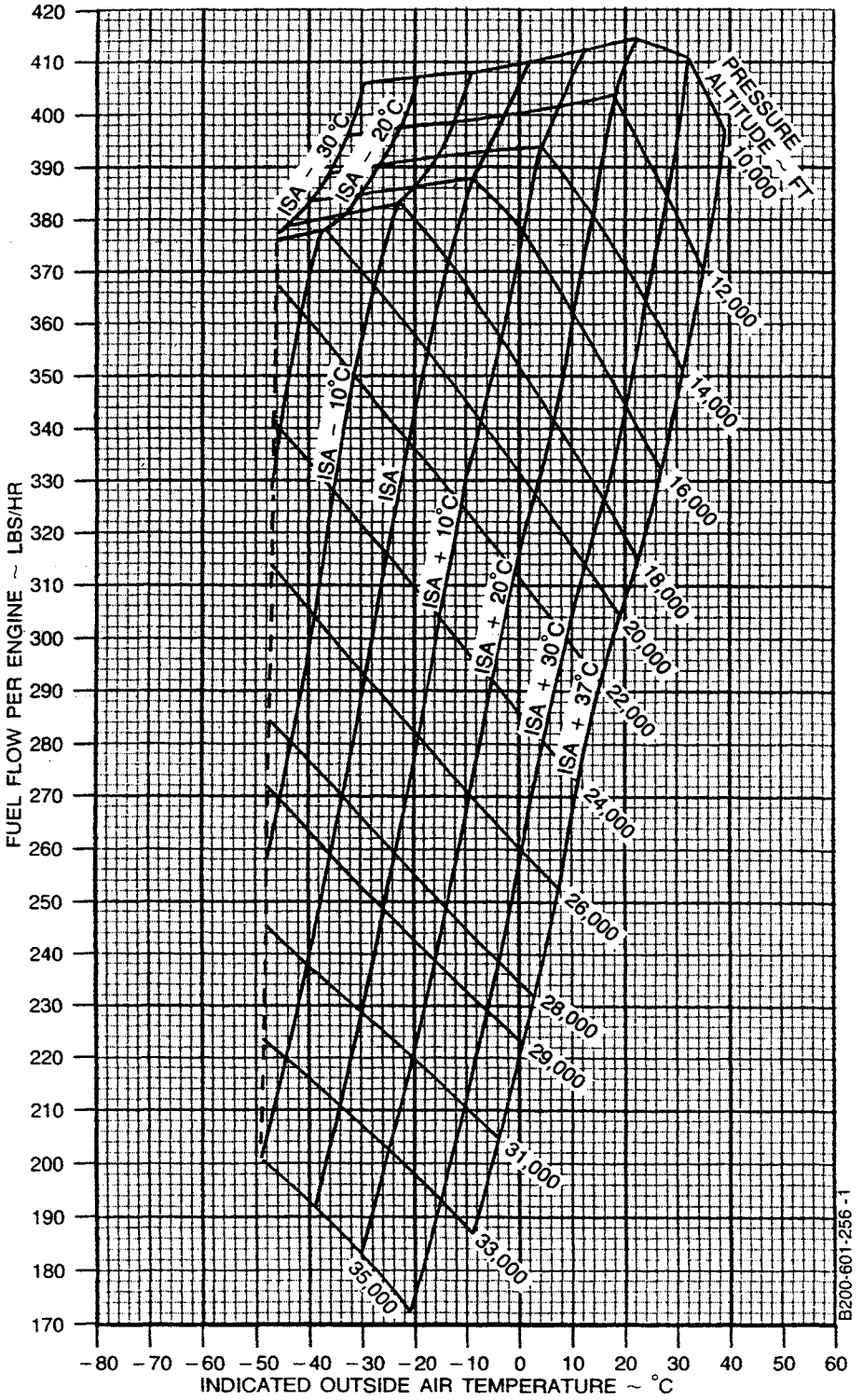
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%.



FUEL FLOW AT NORMAL CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10%.



RANGE PROFILE – NORMAL CRUISE POWER

1700 RPM

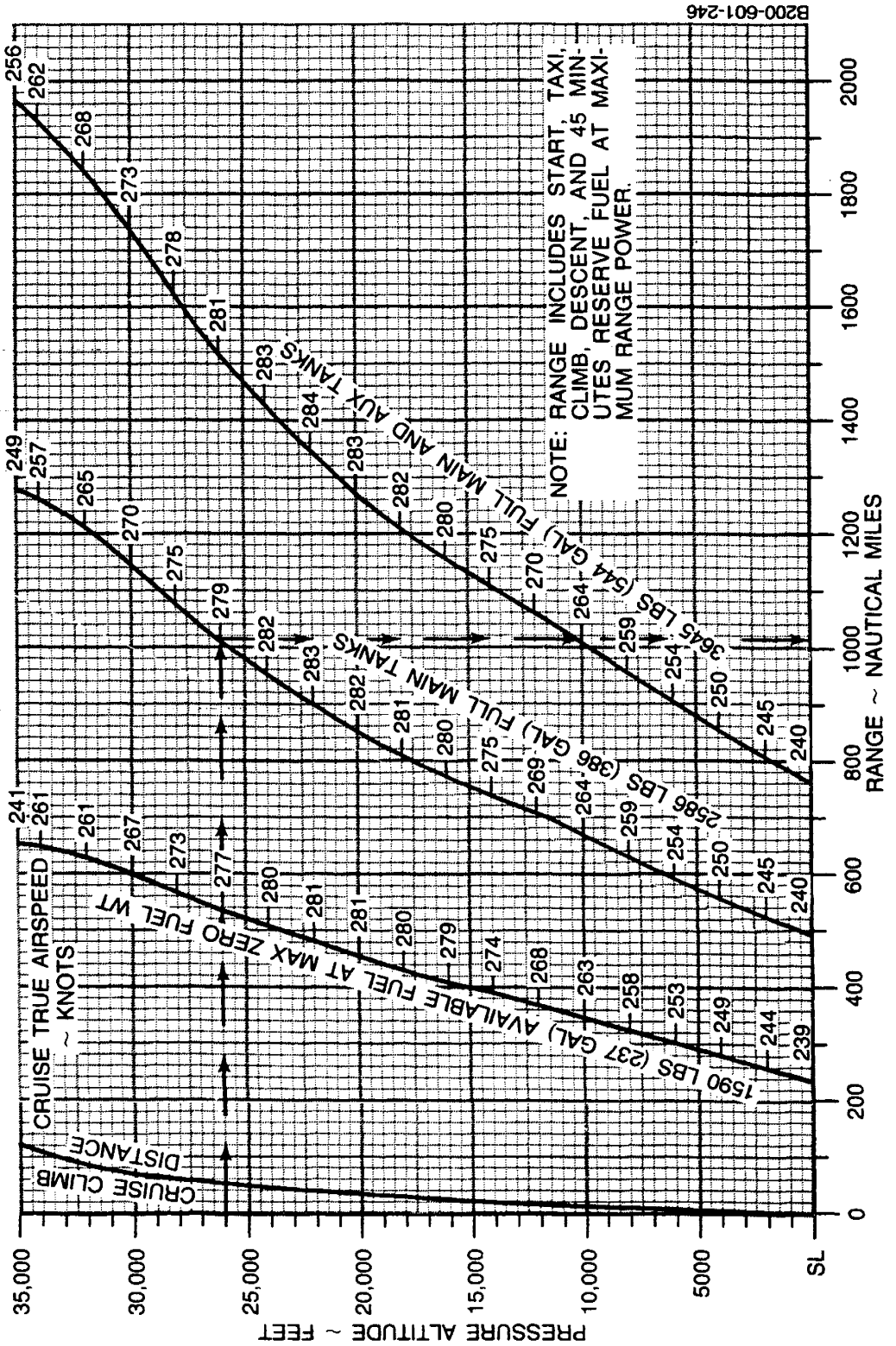
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	12,590 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL LOADING	2586 LBS
RANGE	1018 NM



For Training Purposes Only

MAXIMUM CRUISE POWER

1700 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	463	926	243	231	244	232	245	233
2000	-14	-19	2230	452	904	241	235	242	236	243	237
4000	-18	-23	2230	439	878	239	240	240	241	240	242
6000	-21	-27	2230	428	856	237	244	237	245	238	246
8000	-25	-31	2230	417	834	234	249	235	250	236	251
10,000	-29	-35	2230	406	812	232	254	233	255	234	256
12,000	-33	-39	2230	396	792	230	259	231	260	231	261
14,000	-36	-43	2230	389	778	227	264	228	265	229	266
16,000	-40	-47	2230	384	768	225	269	226	270	227	271
18,000	-44	-51	2230	379	758	223	274	224	276	225	277
20,000	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1700 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED – KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	- 5	2230	466	932	242	234	242	235	243	236
2000	-4	-9	2230	453	906	239	238	240	239	241	240
4000	-7	-13	2230	440	880	237	243	238	244	239	245
6000	-11	-17	2230	428	856	235	248	236	248	237	249
8000	-15	-21	2230	418	836	232	252	233	253	234	254
10,000	-19	-25	2230	408	816	230	257	231	258	232	259
12,000	-22	-29	2230	398	796	228	262	229	263	230	264
14,000	-26	-33	2230	391	782	225	267	226	268	227	270
16,000	-30	-37	2230	385	770	223	273	224	274	225	275
18,000	-33	-41	2230	381	762	220	278	221	279	223	280
20,000	-37	-45	2230	378	756	218	283	219	284	220	286
22,000	-41	-49	2230	375	750	215	288	216	290	217	292
24,000	-45	-53	2107	353	706	207	288	209	290	211	292
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1700 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED – KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	469	938	240	237	241	238	241	238
2000	6	1	2230	456	912	238	241	239	242	239	243
4000	3	-3	2230	443	886	235	246	236	247	237	248
6000	-1	-7	2230	431	862	233	250	234	251	235	252
8000	-5	-11	2230	419	838	231	255	232	256	232	257
10,000	-9	-15	2230	408	816	228	260	229	261	230	262
12,000	-12	-19	2230	399	798	226	265	227	266	228	268
14,000	-16	-23	2230	392	784	224	271	225	272	226	273
16,000	-20	-27	2230	387	774	221	276	222	277	223	278
18,000	-23	-31	2230	383	766	218	281	219	282	221	284
20,000	-27	-35	2230	380	760	215	286	217	288	218	289
22,000	-31	-39	2141	364	728	209	287	211	289	212	291
24,000	-35	-43	2018	342	684	202	286	204	289	205	291
26,000	-39	-47	1881	317	634	193	284	196	287	198	290
28,000	-43	-51	1734	292	584	184	280	187	284	189	287
29,000	-45	-52	1661	260	560	179	277	182	282	184	285
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1700 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	473	946	238	239	239	240	240	241
2000	17	11	2230	459	918	236	244	237	245	238	246
4000	13	7	2230	446	892	234	249	235	250	236	250
6000	9	3	2230	433	866	231	253	232	254	233	255
8000	5	-1	2230	421	842	229	258	230	259	231	260
10,000	2	-5	2230	410	820	227	263	228	264	229	265
12,000	-2	-9	2230	400	800	224	268	225	270	226	271
14,000	-6	-13	2230	393	786	222	274	223	275	224	276
16,000	-9	-17	2230	388	776	219	279	220	280	221	282
18,000	-13	-21	2230	385	770	216	284	217	286	219	287
20,000	-17	-25	2143	368	736	210	285	212	287	213	289
22,000	-21	-29	2043	350	700	203	285	205	288	207	290
24,000	-25	-33	1920	328	656	196	284	198	287	200	290
26,000	-29	-37	1789	305	610	187	281	190	285	192	288
28,000	-33	-41	1654	281	562	178	277	181	282	184	285
29,000	-35	-42	1588	270	540	173	275	177	280	179	284
31,000	-40	-46	1460	249	498	164	270	167	275	171	280
33,000	-44	-50	1334	229	458	153	261	158	269	162	275
35,000	-48	-54	1204	209	418	139	246	146	260	152	268

For Training Purposes Only

MAXIMUM CRUISE POWER

1700 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	30	25	2230	475	950	237	242	238	243	239	244
2000	27	21	2230	462	924	235	247	235	248	236	248
4000	23	17	2230	449	898	232	251	233	252	234	253
6000	19	13	2230	436	872	230	256	231	257	232	258
8000	16	9	2230	424	848	227	261	228	262	229	263
10,000	12	5	2230	412	824	225	266	226	267	227	269
12,000	8	1	2230	402	804	223	271	224	273	225	274
14,000	4	-3	2230	394	788	220	277	221	278	222	279
16,000	1	-7	2230	388	776	217	282	218	283	219	285
18,000	-3	-11	2146	371	742	211	283	213	285	214	287
20,000	-7	-15	2057	354	708	205	284	207	286	208	288
22,000	-11	-19	1950	337	674	198	283	200	286	201	288
24,000	-15	-23	1830	315	630	190	282	192	285	194	288
26,000	-19	-27	1703	292	584	181	279	184	283	186	286
28,000	-23	-31	1572	270	540	172	274	175	279	178	283
29,000	-25	-32	1507	259	518	167	271	171	277	174	281
31,000	-30	-36	1386	239	478	157	264	162	272	165	277
33,000	-34	-40	1271	221	442	146	255	152	265	156	272
35,000	-39	-44	1151	202	404	129	235	140	255	146	265

For Training Purposes Only

MAXIMUM CRUISE POWER

1700 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	478	956	235	244	236	245	237	246
2000	37	31	2230	465	930	233	249	234	250	235	251
4000	33	27	2230	452	904	231	254	232	255	232	256
6000	29	23	2230	439	878	228	259	229	260	230	261
8000	26	19	2230	427	854	226	264	227	265	228	266
10,000	22	15	2230	415	830	223	269	224	270	226	272
12,000	18	11	2230	404	808	221	274	222	276	223	277
14,000	14	7	2195	391	782	217	278	218	279	219	281
16,000	11	3	2115	373	746	211	279	212	281	214	283
18,000	7	-1	2034	355	710	205	280	207	282	208	284
20,000	3	-5	1954	340	680	199	281	201	284	203	286
22,000	-1	-9	1874	325	650	193	282	195	285	197	288
24,000	-5	-13	1761	304	608	185	280	188	284	190	287
26,000	-9	-17	1629	281	562	176	276	179	281	182	284
28,000	-13	-21	1493	259	518	166	270	170	276	173	280
29,000	-16	-22	1431	249	498	161	267	165	273	168	278
31,000	-20	-26	1314	229	458	150	259	155	267	160	274
33,000	-24	-30	1201	211	422	138	246	145	260	150	268
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1700 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	481	962	234	247	235	248	236	249
2000	47	41	2230	468	936	232	252	233	253	233	254
4000	43	37	2230	454	908	229	256	230	257	231	258
6000	39	33	2230	441	882	227	261	228	263	229	264
8000	36	29	2230	429	858	224	267	225	268	226	269
10,000	32	25	2223	416	832	221	271	223	273	224	274
12,000	28	21	2108	388	776	215	271	216	273	217	275
14,000	24	17	2038	371	742	209	273	211	275	212	277
16,000	20	13	1971	354	708	204	275	205	277	207	279
18,000	17	9	1907	338	676	198	276	200	279	202	281
20,000	13	5	1853	325	650	193	278	195	281	197	284
22,000	9	1	1782	311	622	187	279	190	282	192	285
24,000	5	-3	1672	291	582	179	277	182	281	184	284
26,000	1	-7	1555	270	540	171	273	174	278	177	282
28,000	-4	-11	1435	249	498	161	267	165	274	168	279
29,000	-6	-12	1376	240	480	156	264	160	271	164	277
31,000	-10	-16	1259	221	442	145	254	150	264	155	272
33,000	-15	-20	1142	203	406	126	231	139	254	145	264
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1700 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	484	968	233	249	234	250	235	251
2000	54	48	2230	470	940	231	253	232	254	232	255
4000	50	44	2230	456	912	228	258	229	259	230	259
6000	47	40	2230	442	884	226	263	227	264	228	265
8000	43	36	2189	424	848	222	266	223	268	224	269
10,000	39	32	2106	400	800	216	268	217	269	219	271
12,000	35	28	1997	374	748	209	268	211	270	212	271
14,000	31	24	1936	357	714	204	270	206	272	207	274
16,000	27	20	1874	341	682	199	272	201	274	202	276
18,000	23	16	1813	326	652	193	273	195	276	197	278
20,000	20	12	1767	314	628	189	275	191	278	193	281
22,000	16	8	1710	302	604	183	276	185	280	187	283
24,000	12	4	1609	283	566	175	274	178	279	181	282
26,000	8	0	1498	262	524	167	270	170	276	173	280
28,000	3	-4	1380	242	484	157	264	161	271	164	277
29,000	1	-5	1323	232	464	151	260	156	268	160	274
31,000	-3	-9	1212	214	428	139	248	146	260	151	269
33,000	-8	-13	1102	197	394	---	---	134	249	141	261
35,000	---	---	---	---	---	---	---	---	---	---	---

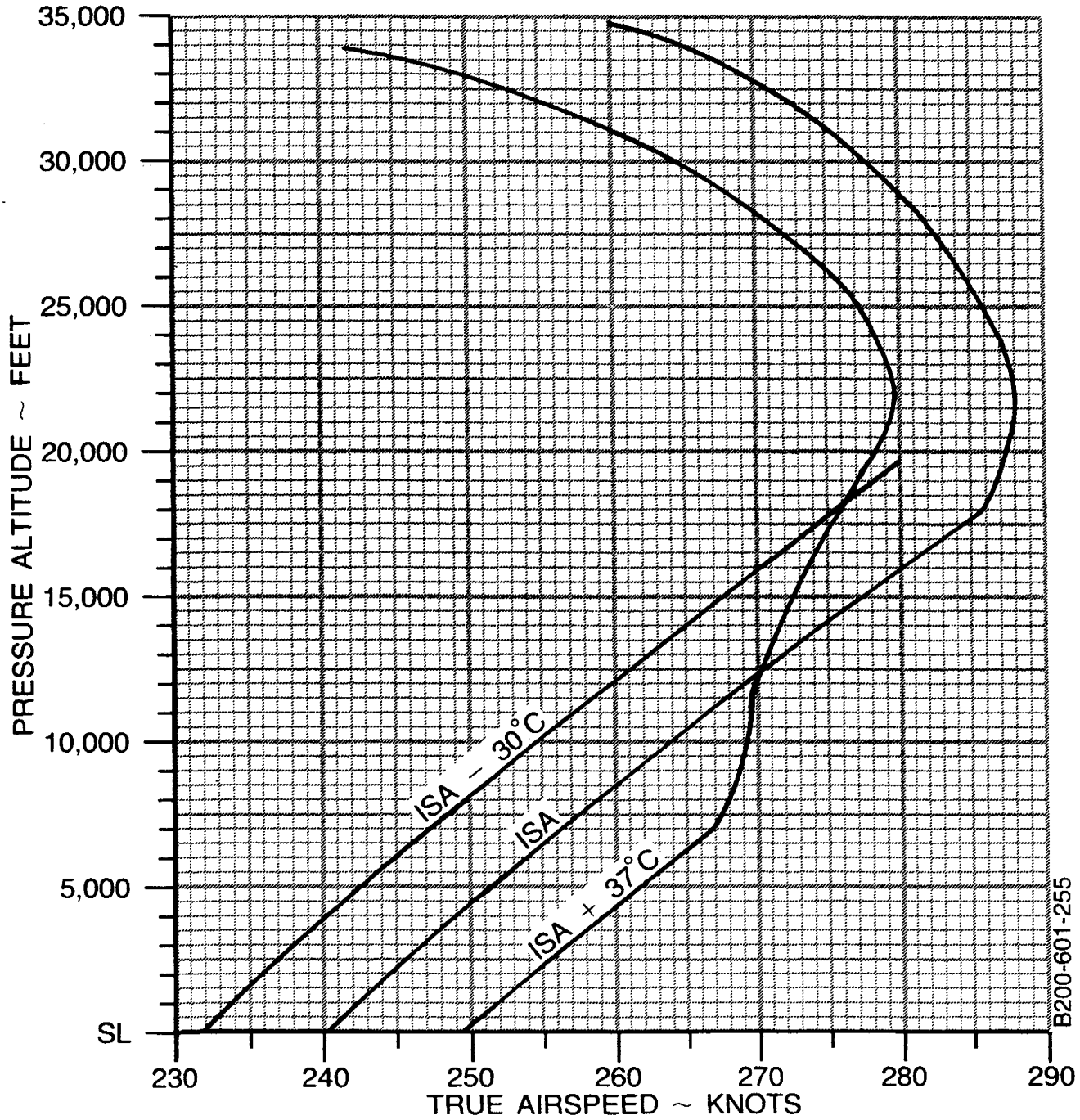
For Training Purposes Only

MAXIMUM CRUISE SPEED

1700 RPM

WEIGHT 11,000 LBS

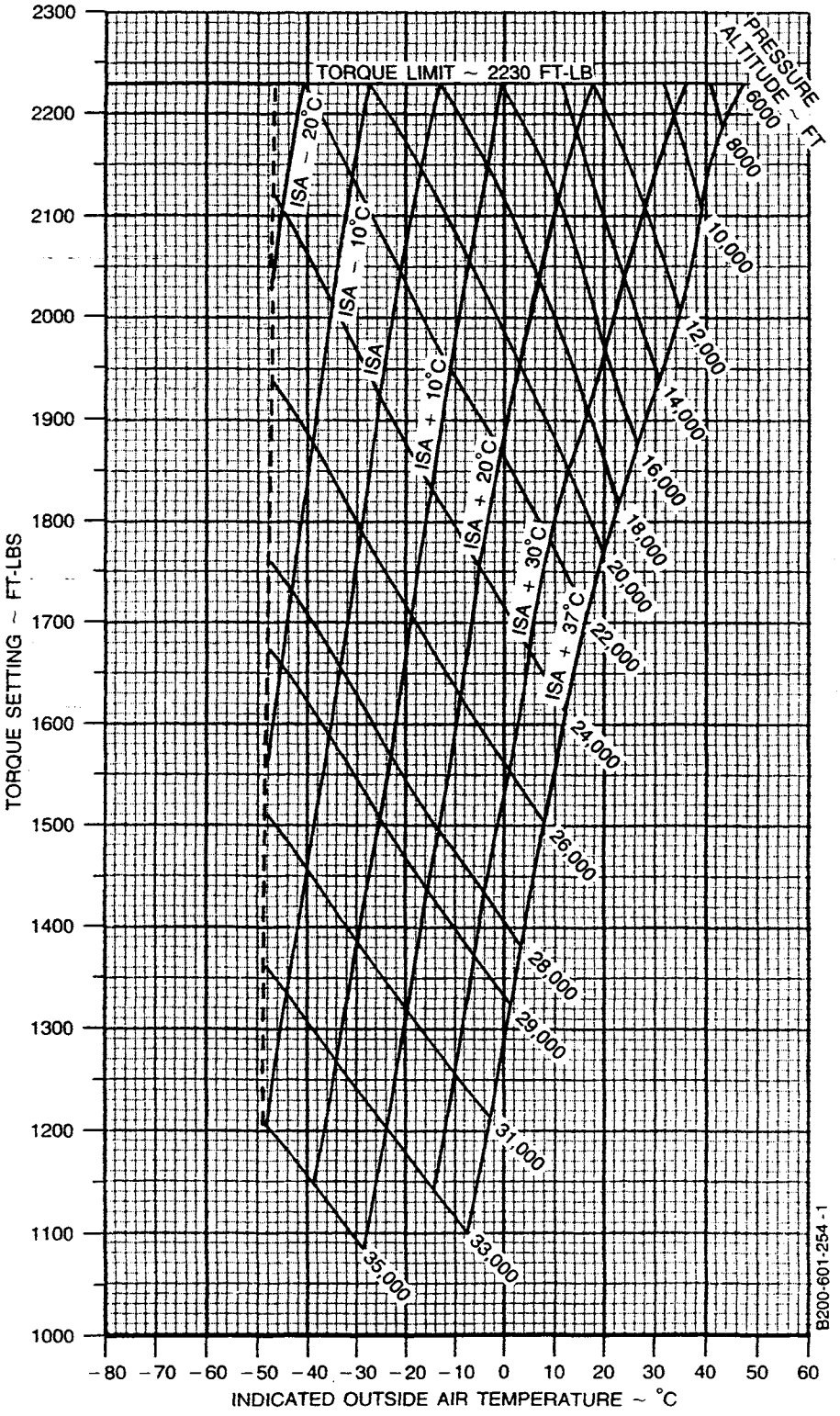
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS.



MAXIMUM CRUISE POWER

1700 RPM

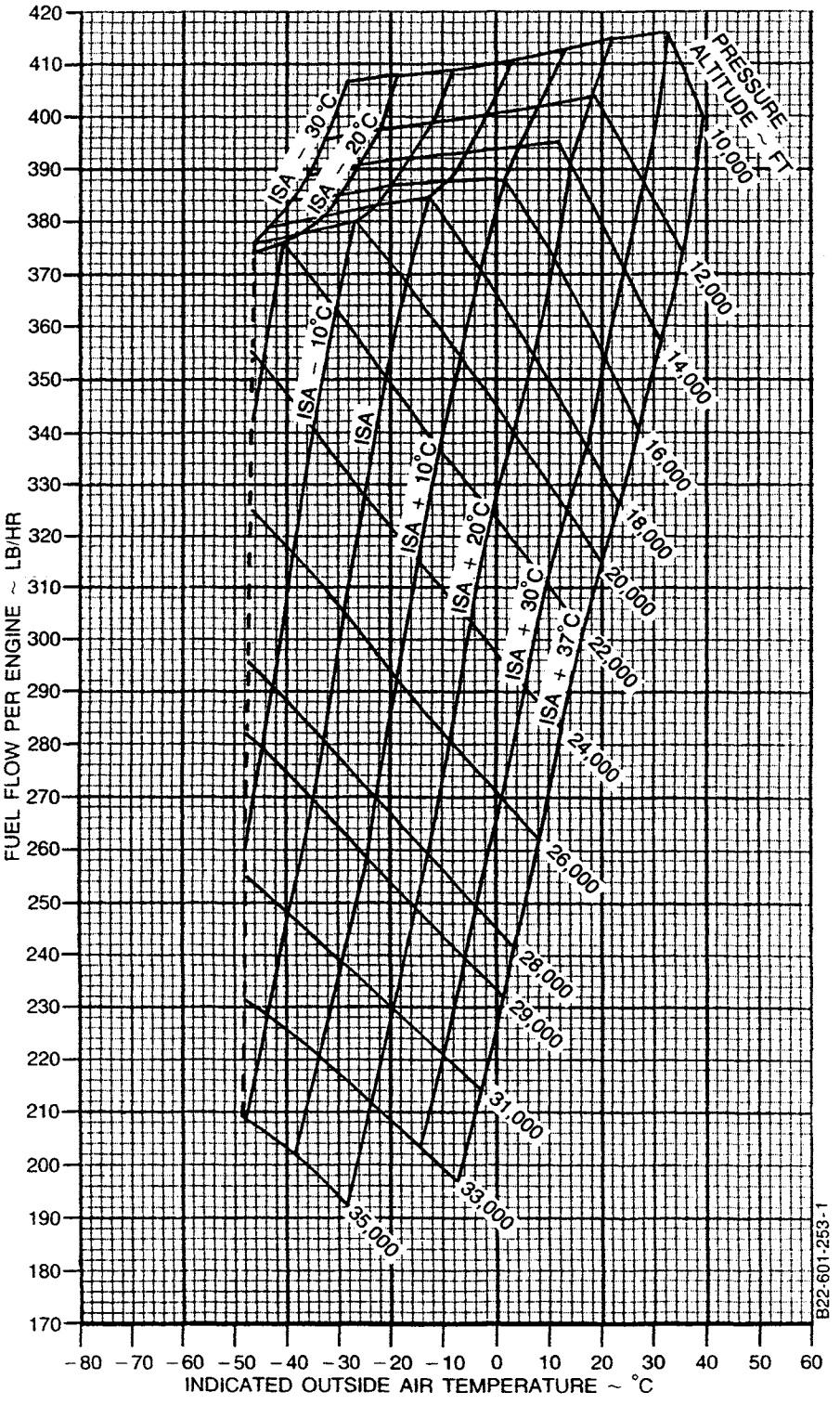
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%.



FUEL FLOW AT MAXIMUM CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10%.



RANGE PROFILE – MAXIMUM CRUISE POWER

1700 RPM

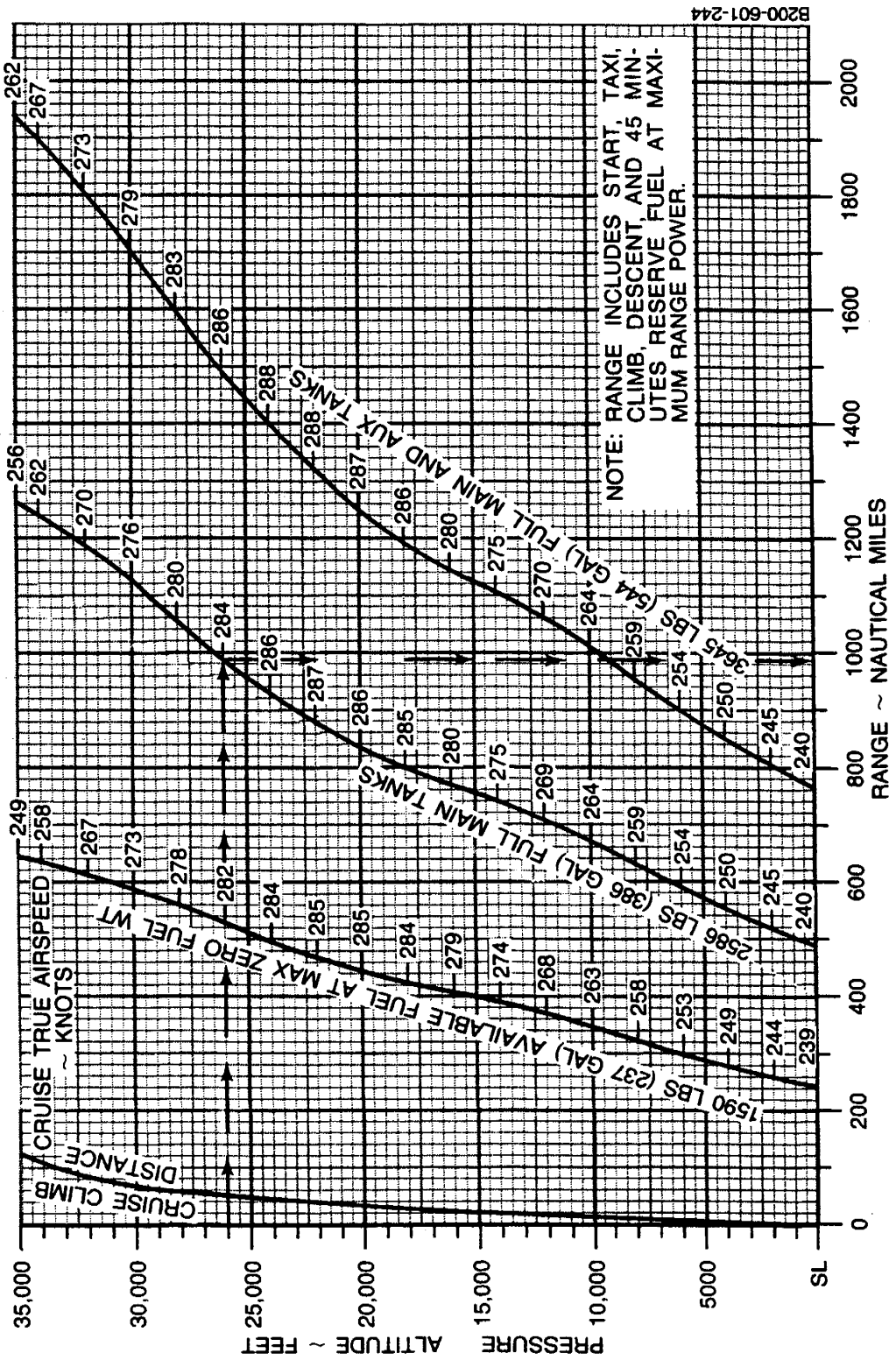
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	12,590 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL LOADING	2586 LBS
RANGE	988 NM



For Training Purposes Only

NORMAL CRUISE POWER

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	483	976	249	236	249	237	250	238
2000	-14	-19	2230	471	942	246	241	247	241	248	242
4000	-17	-23	2230	459	918	244	245	245	246	246	246
6000	-21	-27	2230	447	894	242	250	243	251	243	251
8000	-25	-31	2230	437	874	239	254	240	255	241	256
10,000	-29	-35	2230	426	852	237	259	238	260	239	261
12,000	-32	-39	2230	416	832	235	265	236	266	237	267
14,000	-36	-43	2230	409	818	233	270	233	271	234	272
16,000	-40	-47	2230	403	806	230	275	231	276	232	277
18,000	-43	-51	2230	398	796	228	281	229	282	230	283
20,000	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1800 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	486	972	247	239	248	240	248	241
2000	-3	-9	2230	473	946	245	244	245	244	246	245
4000	-7	-13	2230	460	920	242	248	243	249	244	250
6000	-11	-17	2230	448	896	240	253	241	254	242	255
8000	-15	-21	2230	438	876	238	258	238	259	239	260
10,000	-18	-25	2230	428	856	235	263	236	264	237	265
12,000	-22	-29	2230	417	834	233	268	234	269	235	270
14,000	-26	-33	2230	410	820	231	273	232	274	232	275
16,000	-29	-37	2230	404	808	228	279	229	280	230	281
18,000	-33	-41	2230	399	798	226	284	227	286	228	287
20,000	-37	-45	2153	383	766	220	286	222	288	223	290
22,000	-41	-49	2054	364	728	214	287	215	289	217	291
24,000	-45	-53	1936	342	684	207	286	208	289	210	291
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	488	976	245	242	246	243	247	243
2000	7	1	2230	475	950	243	246	244	247	244	248
4000	3	-3	2230	462	924	240	251	241	252	242	253
6000	-1	-7	2230	450	900	238	256	239	257	240	258
8000	-5	-11	2230	438	876	236	261	237	262	238	263
10,000	-8	-15	2230	427	854	234	266	234	267	235	268
12,000	-12	-19	2230	419	838	231	271	232	272	233	273
14,000	-16	-23	2230	412	824	229	277	230	278	231	279
16,000	-19	-27	2230	406	812	226	282	227	284	228	285
18,000	-23	-31	2157	389	778	221	284	222	286	223	287
20,000	-27	-35	2051	368	736	214	285	216	287	217	288
22,000	-31	-39	1959	350	700	208	286	210	288	211	290
24,000	-35	-43	1845	329	658	201	285	202	287	204	290
26,000	-39	-47	1720	306	612	192	282	194	286	196	288
28,000	-43	-51	1588	282	564	183	279	186	282	188	286
29,000	-45	-52	1523	270	540	178	276	181	280	184	284
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1800 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	21	15	2230	491	982	243	245	244	245	245	246
2000	17	11	2230	478	956	241	249	242	250	243	251
4000	13	7	2230	463	926	239	254	240	255	240	256
6000	9	3	2230	450	900	236	259	237	260	238	261
8000	6	-1	2230	439	878	234	264	235	265	236	266
10,000	2	-5	2230	429	858	232	269	233	270	234	271
12,000	-2	-9	2230	418	836	229	274	230	276	231	277
14,000	-5	-13	2230	411	822	227	279	228	281	229	282
16,000	-9	-17	2177	397	794	222	283	224	285	225	286
18,000	-13	-21	2048	373	746	215	282	216	284	217	286
20,000	-17	-25	1959	355	710	209	283	210	285	212	287
22,000	-21	-29	1871	338	676	202	284	204	286	206	288
24,000	-25	-33	1758	316	632	195	283	197	285	199	288
26,000	-29	-37	1635	293	586	187	280	189	283	191	286
28,000	-33	-41	1508	270	540	177	276	180	280	182	284
29,000	-35	-42	1447	259	518	172	273	175	278	178	282
31,000	-40	-46	1329	239	478	162	267	166	273	169	278
33,000	-44	-50	1218	220	440	151	258	156	267	160	274
35,000	-49	-54	1100	201	402	135	241	145	257	151	267

For Training Purposes Only

NORMAL CRUISE POWER

1800 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	31	25	2230	494	988	242	247	243	248	244	249
2000	27	21	2230	480	960	240	252	240	253	241	254
4000	23	17	2230	467	934	237	257	238	258	239	259
6000	19	13	2230	454	908	235	262	236	263	237	264
8000	16	9	2230	442	884	233	267	233	268	234	269
10,000	12	5	2230	431	862	230	272	231	273	232	274
12,000	8	1	2230	420	840	228	278	229	279	230	280
14,000	5	-3	2167	402	804	223	280	224	282	225	283
16,000	1	-7	2065	380	760	216	281	218	282	219	284
18,000	-3	-11	1964	359	718	210	281	211	283	212	285
20,000	-7	-15	1881	342	684	204	282	205	284	207	286
22,000	-11	-19	1787	325	650	197	282	199	285	200	287
24,000	-15	-23	1677	304	608	189	280	191	284	193	286
26,000	-19	-27	1557	282	564	180	277	183	281	185	284
28,000	-23	-31	1433	259	518	171	272	174	277	177	281
29,000	-26	-32	1373	249	498	166	269	169	275	172	279
31,000	-30	-36	1260	229	458	155	262	160	269	164	275
33,000	-34	-40	1152	211	422	143	250	150	262	155	270
35,000	-39	-44	1037	192	384	---	---	137	249	144	262

NORMAL CRUISE POWER

1800 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	497	994	240	250	241	251	242	251
2000	37	31	2230	483	966	238	255	239	255	240	256
4000	33	27	2230	469	938	236	259	237	261	238	261
6000	30	23	2230	456	912	233	265	234	265	235	267
8000	26	19	2230	444	888	231	270	232	271	233	272
10,000	22	15	2225	432	864	228	275	229	276	230	277
12,000	18	11	2114	404	808	221	275	223	276	224	278
14,000	14	7	2027	383	766	215	276	217	278	218	279
16,000	11	3	1940	362	724	209	277	211	279	212	281
18,000	7	-1	1852	342	684	203	278	205	280	206	282
20,000	3	-5	1793	329	658	198	280	200	282	201	285
22,000	-1	-9	1719	314	628	192	281	194	284	196	286
24,000	-5	-13	1614	294	588	184	279	187	283	189	286
26,000	-9	-17	1490	271	542	175	275	178	279	181	283
28,000	-14	-21	1362	249	498	165	268	168	274	171	279
29,000	-16	-22	1304	239	478	160	265	164	271	167	276
31,000	-20	-26	1194	220	440	148	255	154	265	158	272
33,000	-25	-30	1087	202	404	133	238	143	255	149	266
35,000	-30	-34	973	183	366	---	---	128	237	137	255

For Training Purposes Only

NORMAL CRUISE POWER

1800 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	499	998	239	252	240	253	241	254
2000	47	41	2230	485	970	237	257	238	258	238	259
4000	43	37	2230	471	942	234	262	235	263	236	264
6000	40	33	2230	458	916	232	267	233	268	234	269
8000	36	29	2153	434	868	226	269	227	270	228	271
10,000	32	25	2071	411	822	221	270	222	272	223	273
12,000	28	21	1968	384	768	214	270	215	272	217	274
14,000	24	17	1886	364	728	208	272	210	273	211	275
16,000	20	13	1809	344	688	202	273	204	275	205	277
18,000	16	9	1737	326	652	196	274	198	276	200	278
20,000	13	5	1691	314	628	192	276	194	279	195	281
22,000	9	1	1640	302	604	187	278	189	281	191	284
24,000	5	-3	1535	282	564	179	276	181	280	184	283
26,000	1	-7	1422	261	522	170	272	173	277	176	281
28,000	-4	-11	1308	240	480	160	266	164	272	167	277
29,000	-6	-12	1252	230	460	154	261	159	269	163	275
31,000	-10	-16	1142	212	424	141	248	149	261	153	269
33,000	-15	-20	1030	193	386	---	---	136	248	143	261
35,000	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1800 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	500	1000	238	254	239	255	240	256
2000	54	48	2219	485	970	235	258	236	259	237	260
4000	50	44	2176	465	930	231	262	232	263	233	264
6000	47	40	2117	444	888	226	264	227	265	228	266
8000	43	36	2043	420	840	221	266	222	267	223	268
10,000	39	32	1970	397	794	215	267	217	269	218	270
12,000	35	28	1871	371	742	209	267	210	269	212	271
14,000	31	24	1795	351	702	203	268	205	270	206	272
16,000	27	20	1722	333	666	197	269	199	272	201	274
18,000	23	16	1653	315	630	191	270	193	273	195	275
20,000	19	12	1614	304	608	187	273	189	276	191	279
22,000	16	8	1568	292	584	182	275	184	278	186	281
24,000	12	4	1475	273	546	174	273	177	277	180	281
26,000	7	0	1371	253	506	166	269	169	274	172	279
28,000	3	-4	1260	233	466	156	262	160	269	163	275
29,000	1	-5	1205	223	446	149	256	155	266	159	272
31,000	-4	-9	1099	205	410	135	241	144	257	150	266
33,000	-8	-13	992	188	376	---	---	130	242	139	258
35,000	---	---	---	---	---	---	---	---	---	---	---

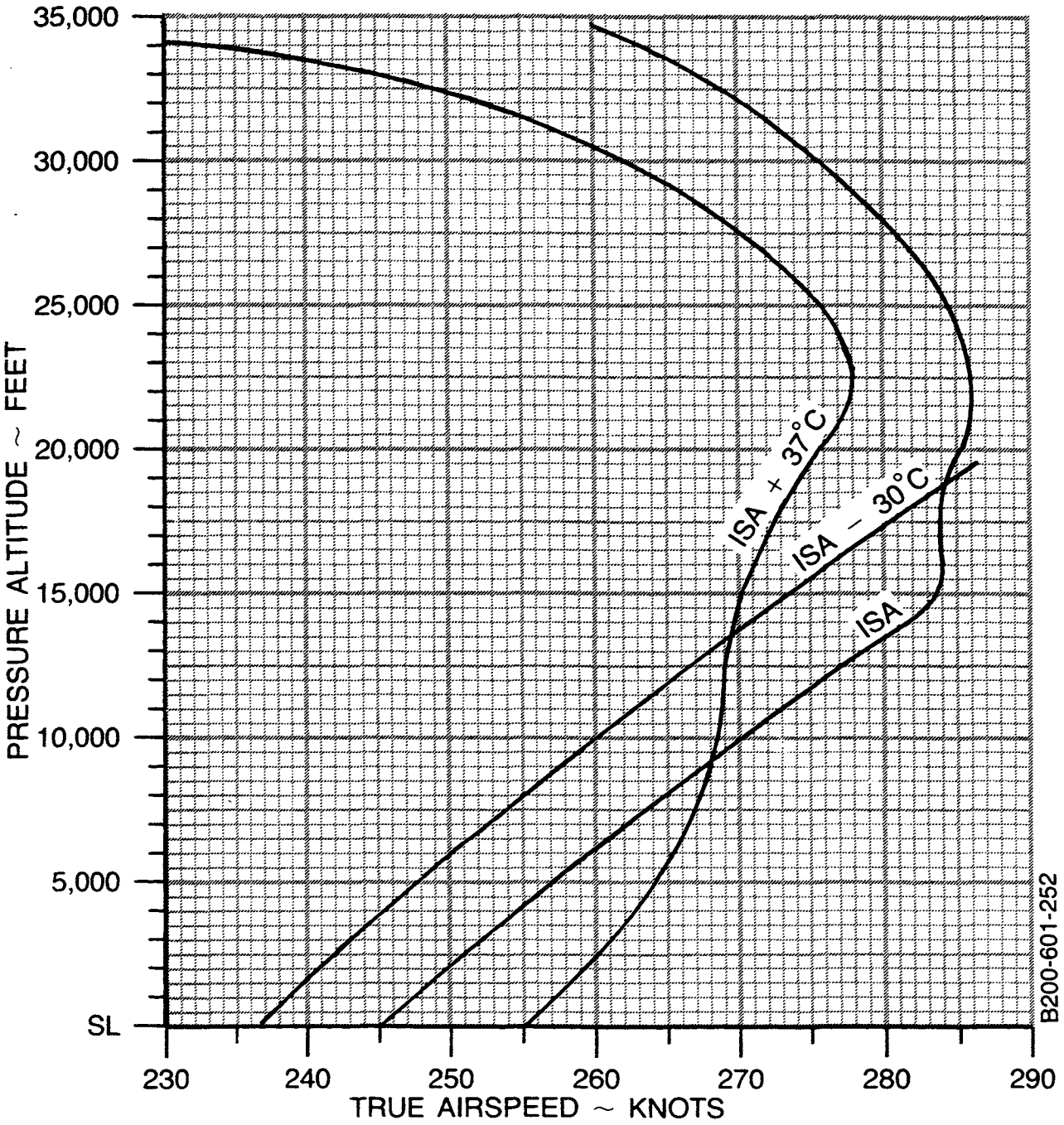
For Training Purposes Only

NORMAL CRUISE SPEEDS

1800 RPM

WEIGHT 11,000 LBS

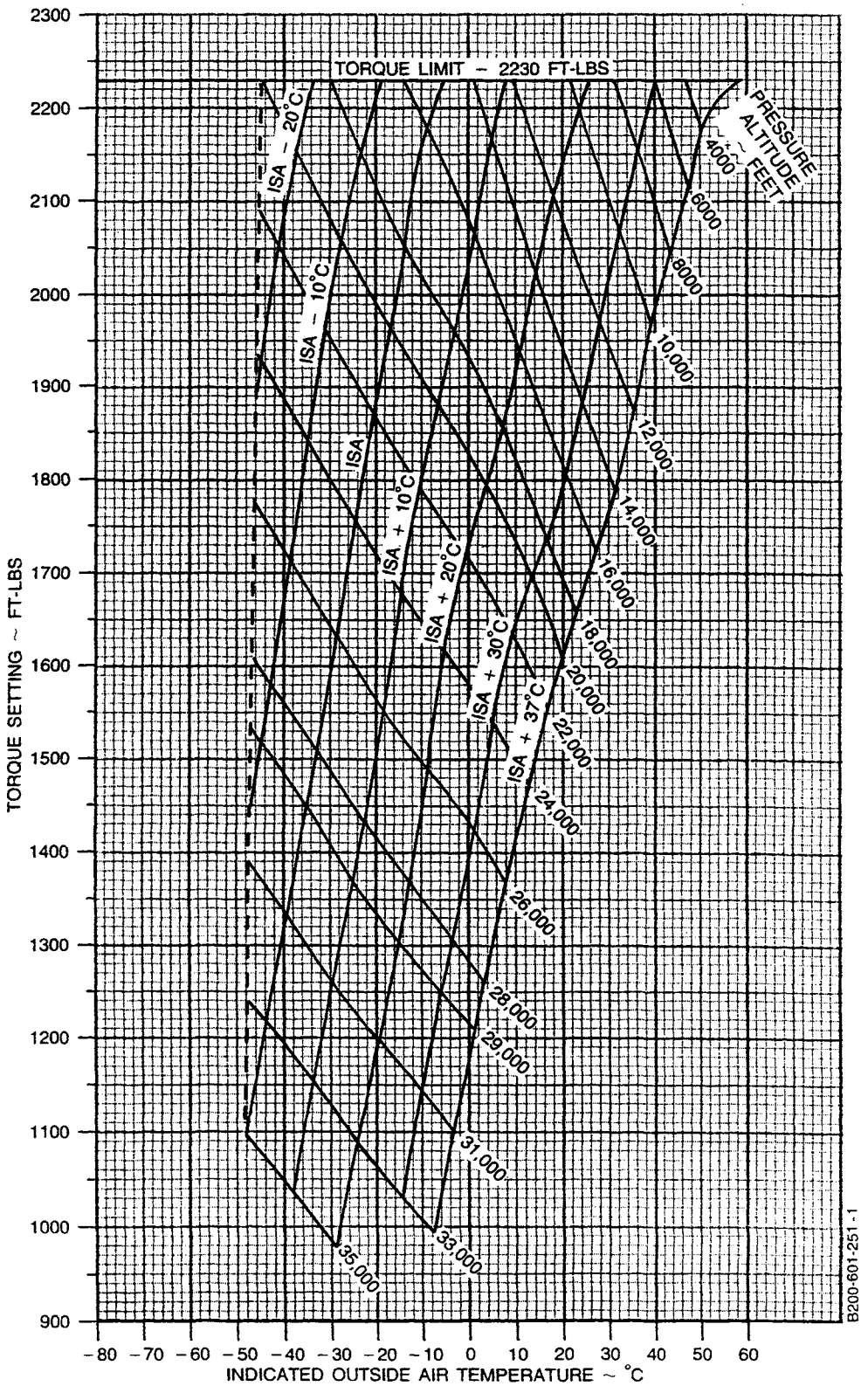
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS.



NORMAL CRUISE POWER

1800 RPM

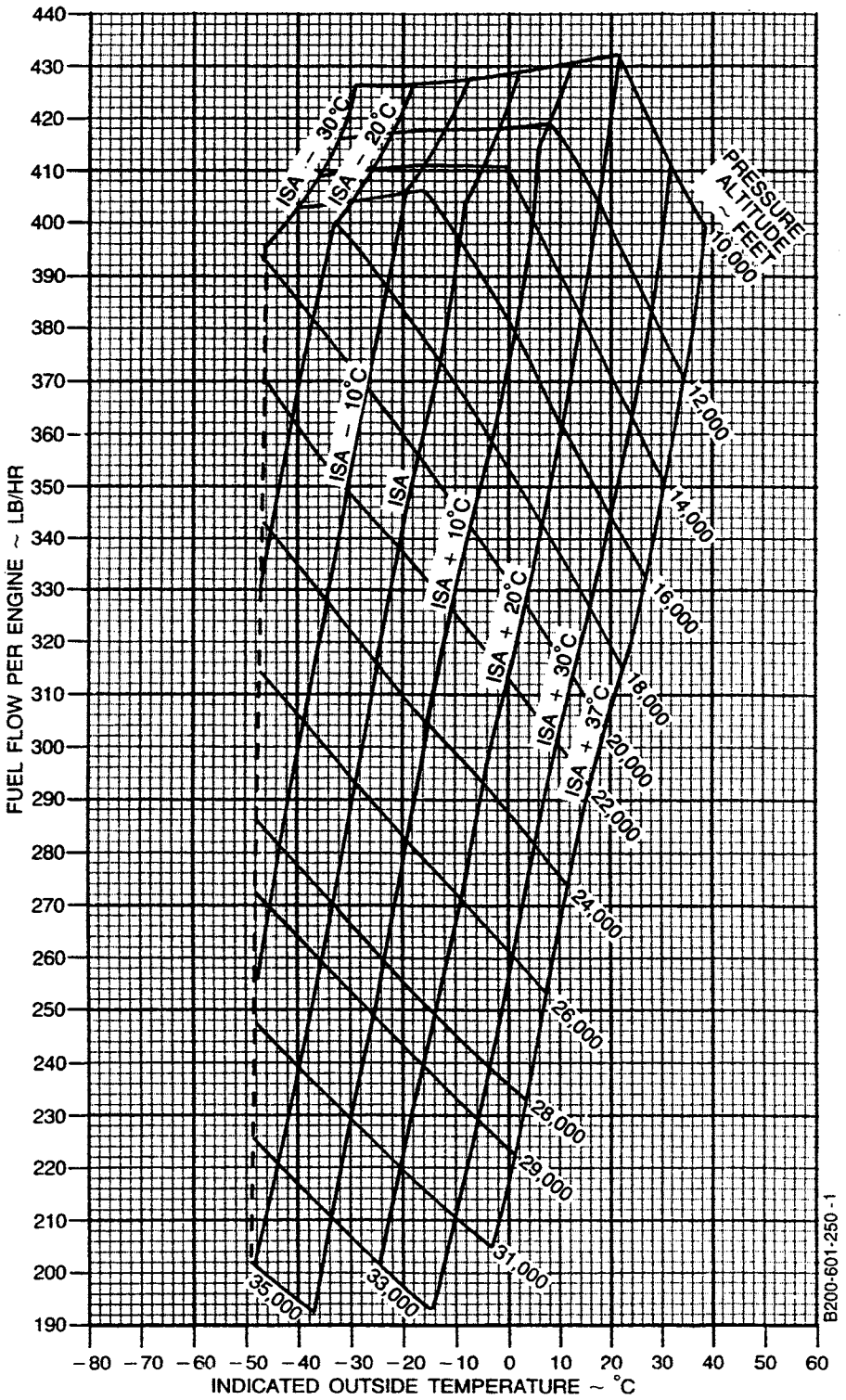
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%.



FUEL FLOW AT NORMAL CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10%.



RANGE PROFILE – NORMAL CRUISE POWER

1800 RPM

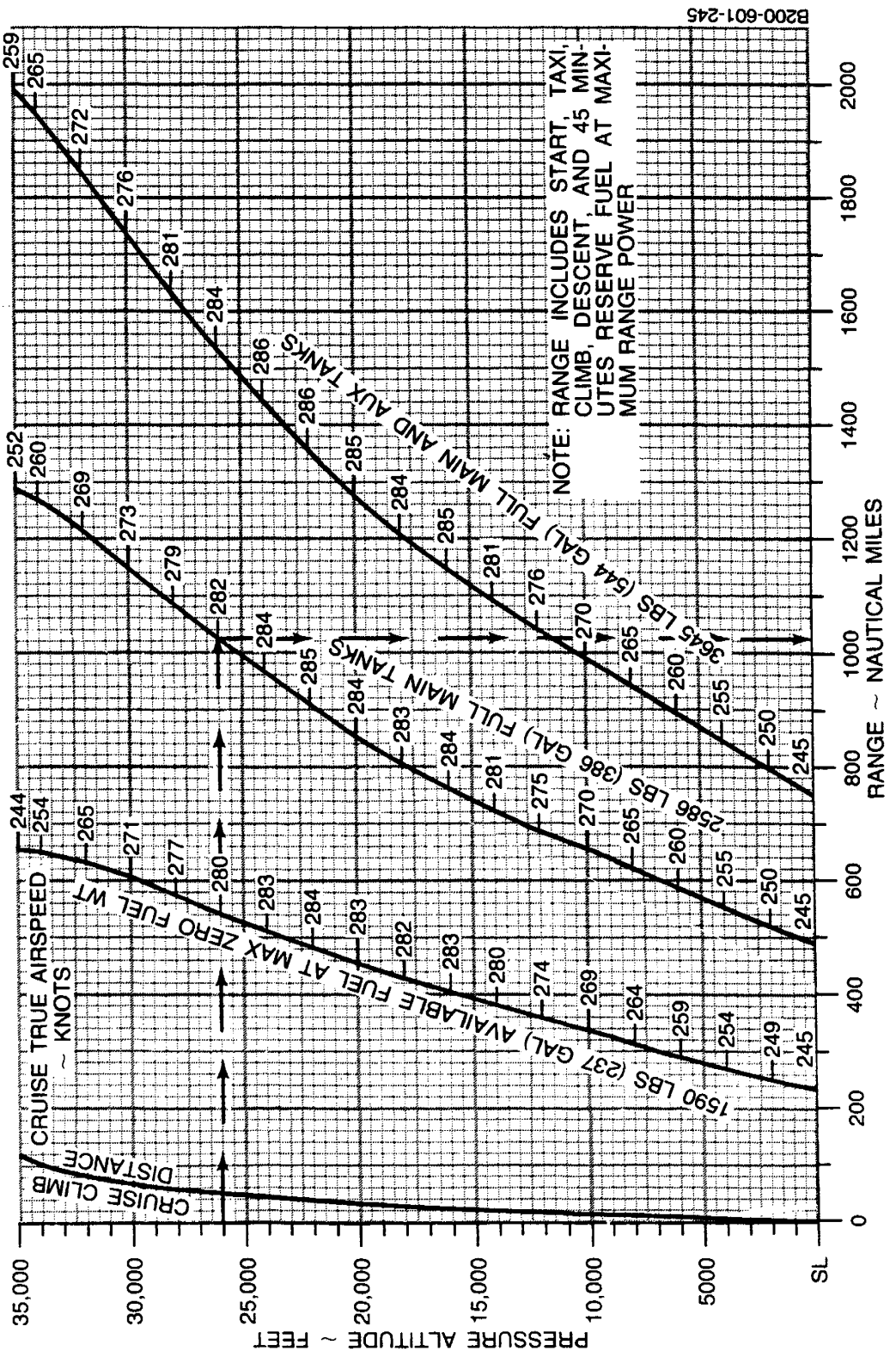
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	12,590 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL LOADING	2586 LBS
RANGE	1025 NM



For Training Purposes Only

MAXIMUM CRUISE POWER

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED – KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	483	966	250	236	249	237	250	238
2000	-14	-19	2230	471	942	246	241	247	241	248	242
4000	-17	-23	2230	459	918	245	245	245	246	246	247
6000	-21	-27	2230	447	894	242	250	243	251	243	251
8000	-25	-31	2230	437	884	239	254	240	255	241	256
10,000	-29	-35	2230	426	852	237	259	238	260	239	261
12,000	-32	-39	2230	416	832	235	265	236	266	237	267
14,000	-36	-43	2230	409	818	233	270	233	271	234	272
16,000	-40	-47	2230	403	806	230	275	231	276	232	277
18,000	-43	-51	2230	398	796	228	281	229	282	230	283
20,000	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1800 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	486	972	247	239	248	240	248	241
2000	-4	-9	2230	473	946	245	244	245	244	246	245
4000	-7	-13	2230	460	920	242	248	243	249	244	250
6000	-11	-17	2230	448	896	240	253	241	254	242	255
8000	-15	-21	2230	438	876	238	258	238	259	239	260
10,000	-18	-25	2230	428	856	235	263	236	264	237	265
12,000	-22	-29	2230	417	834	233	268	234	269	235	270
14,000	-26	-33	2230	410	820	231	273	232	274	232	275
16,000	-30	-37	2230	404	808	228	279	229	280	230	281
18,000	-33	-41	2230	400	800	226	284	227	286	228	287
20,000	-37	-45	2230	396	792	223	290	225	292	226	293
22,000	-41	-49	2146	379	758	217	292	219	293	220	295
24,000	-45	-53	2011	354	708	209	290	211	293	213	295
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	488	976	245	242	246	243	247	243
2000	7	-1	2230	475	950	243	246	244	247	244	248
4000	3	-3	2230	462	924	240	251	241	252	242	253
6000	-1	-7	2230	450	900	238	256	239	257	240	258
8000	-5	-11	2230	438	876	236	261	237	262	238	263
10,000	-8	-15	2230	427	854	234	266	234	267	235	268
12,000	-12	-19	2230	419	838	231	271	232	272	233	273
14,000	-16	-23	2230	412	824	229	277	230	278	231	279
16,000	-19	-27	2230	406	812	226	282	227	284	228	285
18,000	-23	-31	2230	401	802	224	288	225	290	226	291
20,000	-27	-35	2143	383	766	218	289	219	291	220	293
22,000	-31	-39	2042	364	728	211	290	213	292	214	294
24,000	-35	-43	1926	342	682	204	289	206	292	207	294
26,000	-39	-47	1796	318	636	196	287	198	290	199	293
28,000	-43	-51	1657	293	586	186	283	189	287	191	290
29,000	-45	-52	1587	280	560	182	281	184	285	187	288
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1800 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	21	15	2230	491	982	243	245	244	245	245	246
2000	17	11	2230	478	956	241	249	242	250	243	251
4000	13	7	2230	463	926	239	254	240	255	240	256
6000	9	3	2230	450	900	236	259	237	260	238	261
8000	6	-1	2230	439	878	234	264	235	265	236	266
10,000	2	-5	2230	429	858	232	269	233	270	234	271
12,000	-2	-9	2230	418	836	229	274	230	276	231	277
14,000	-5	-13	2230	411	822	227	280	228	281	229	282
16,000	-9	-17	2230	406	812	225	286	226	287	227	288
18,000	-13	-21	2140	388	776	218	287	220	289	221	290
20,000	-17	-25	2044	368	736	212	288	214	290	215	292
22,000	-21	-29	1949	350	700	206	288	207	291	209	292
24,000	-25	-33	1833	329	658	198	287	200	290	202	292
26,000	-29	-37	1708	305	610	190	285	192	288	194	291
28,000	-33	-41	1580	282	564	181	281	183	285	186	288
29,000	-35	-42	1518	271	542	176	279	179	283	181	287
31,000	-39	-46	1396	250	500	166	273	170	279	173	284
33,000	-44	-50	1277	230	460	156	266	160	273	164	279
35,000	-48	-54	1153	209	418	141	251	149	264	154	273

For Training Purposes Only

MAXIMUM CRUISE POWER

1800 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	31	25	2230	494	988	242	247	243	248	244	249
2000	27	21	2230	480	960	240	252	240	253	241	254
4000	23	17	2230	467	934	237	257	238	258	239	259
6000	19	13	2230	454	908	235	262	236	263	237	264
8000	16	9	2230	442	882	233	267	233	268	234	269
10,000	12	5	2230	431	862	230	272	231	273	232	274
12,000	8	1	2230	420	840	228	278	229	279	230	280
14,000	5	-3	2230	412	824	225	283	226	285	227	286
16,000	1	-7	2142	392	784	219	285	221	286	222	288
18,000	-3	-11	2045	372	744	213	286	214	287	216	289
20,000	-7	-15	1962	355	710	207	287	209	289	210	291
22,000	-11	-19	1860	337	674	200	286	202	289	203	291
24,000	-15	-23	1747	316	632	192	285	194	288	196	291
26,000	-19	-27	1626	293	586	184	282	186	286	188	289
28,000	-23	-31	1501	270	540	175	278	178	282	180	286
29,000	-25	-32	1440	260	520	170	275	173	280	176	284
31,000	-29	-36	1326	240	480	160	269	164	276	167	281
33,000	-34	-40	1216	221	442	149	260	155	270	159	277
35,000	-38	-44	1101	202	404	133	241	143	259	149	270

MAXIMUM CRUISE POWER

1800 RPM

IAS + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	497	994	240	250	241	250	242	251
2000	37	31	2230	483	966	238	255	239	255	240	256
4000	33	27	2230	469	938	236	259	237	260	238	261
6000	30	23	2230	457	914	233	265	234	266	235	267
8000	26	19	2230	445	890	231	270	232	271	233	272
10,000	22	15	2230	433	866	228	275	230	276	230	277
12,000	18	11	2157	411	822	223	277	224	279	225	280
14,000	15	7	2084	391	782	218	279	219	281	220	282
16,000	11	3	2011	373	746	212	281	214	283	215	285
18,000	7	-1	1936	355	710	207	283	208	285	210	287
20,000	3	-5	1862	340	680	201	284	203	287	204	289
22,000	-1	-9	1785	325	650	195	285	197	288	199	290
24,000	-5	-13	1678	305	610	187	284	190	287	192	290
26,000	-9	-17	1552	282	564	179	280	181	284	184	287
28,000	-13	-21	1422	260	520	169	274	172	279	175	284
29,000	-15	-22	1368	249	498	164	271	167	277	169	282
31,000	-20	-26	1257	230	460	153	264	158	272	162	278
33,000	-24	-30	1150	212	424	140	251	148	264	153	272
35,000	-29	-34	1036	193	386	121	226	135	250	142	264

For Training Purposes Only

MAXIMUM CRUISE POWER

1800 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	499	998	239	252	240	253	241	254
2000	47	41	2230	485	970	237	257	238	258	238	259
4000	43	37	2230	471	942	234	262	235	263	236	264
6000	40	33	2230	458	916	232	267	233	268	234	269
8000	36	29	2192	440	880	228	271	229	272	230	273
10,000	32	25	2107	416	832	222	272	223	274	224	275
12,000	28	21	1999	389	778	215	272	217	274	218	275
14,000	24	17	1935	371	742	210	274	212	276	213	278
16,000	20	13	1874	354	708	205	276	207	278	208	280
18,000	17	9	1815	338	676	200	278	202	281	203	283
20,000	13	5	1767	326	652	195	281	197	284	199	286
22,000	9	1	1700	312	624	190	282	192	285	193	288
24,000	5	-3	1597	292	584	182	281	184	284	186	287
26,000	1	-7	1486	271	542	173	277	176	282	179	285
28,000	-3	-11	1371	250	500	164	272	167	278	170	282
29,000	-6	-12	1315	240	480	159	268	163	275	166	280
31,000	-10	-16	1205	221	442	147	259	153	269	157	276
33,000	-14	-20	1092	203	406	131	239	141	258	148	269
35,000	-19	-24	968	184	368	115	220	124	236	135	257

MAXIMUM CRUISE POWER

1800 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	501	1002	238	254	239	255	240	256
2000	54	48	2230	487	974	236	259	237	260	237	261
4000	51	44	2210	470	940	232	263	233	264	234	265
6000	47	40	2148	448	896	228	265	229	267	230	268
8000	43	36	2072	424	848	222	267	223	268	224	270
10,000	39	32	1995	400	800	216	268	218	270	219	271
12,000	35	28	1893	374	748	210	268	212	270	213	272
14,000	31	24	1836	357	714	205	271	206	273	208	275
16,000	27	20	1780	341	682	200	273	202	275	203	277
18,000	23	16	1725	326	652	195	275	197	277	198	280
20,000	20	12	1684	315	630	190	278	192	281	194	283
22,000	16	8	1631	302	604	185	280	187	283	189	286
24,000	12	4	1536	283	566	178	278	180	282	183	285
26,000	8	0	1431	263	526	169	274	172	279	175	283
28,000	4	-4	1320	242	484	160	268	163	275	167	280
29,000	2	-5	1265	233	466	154	264	159	272	162	278
31,000	-3	-9	1160	215	430	142	253	149	265	153	273
33,000	-8	-13	1054	197	394	122	226:7	137	253	144	266
35,000	---	---	---	---	---	---	---	---	---	---	---

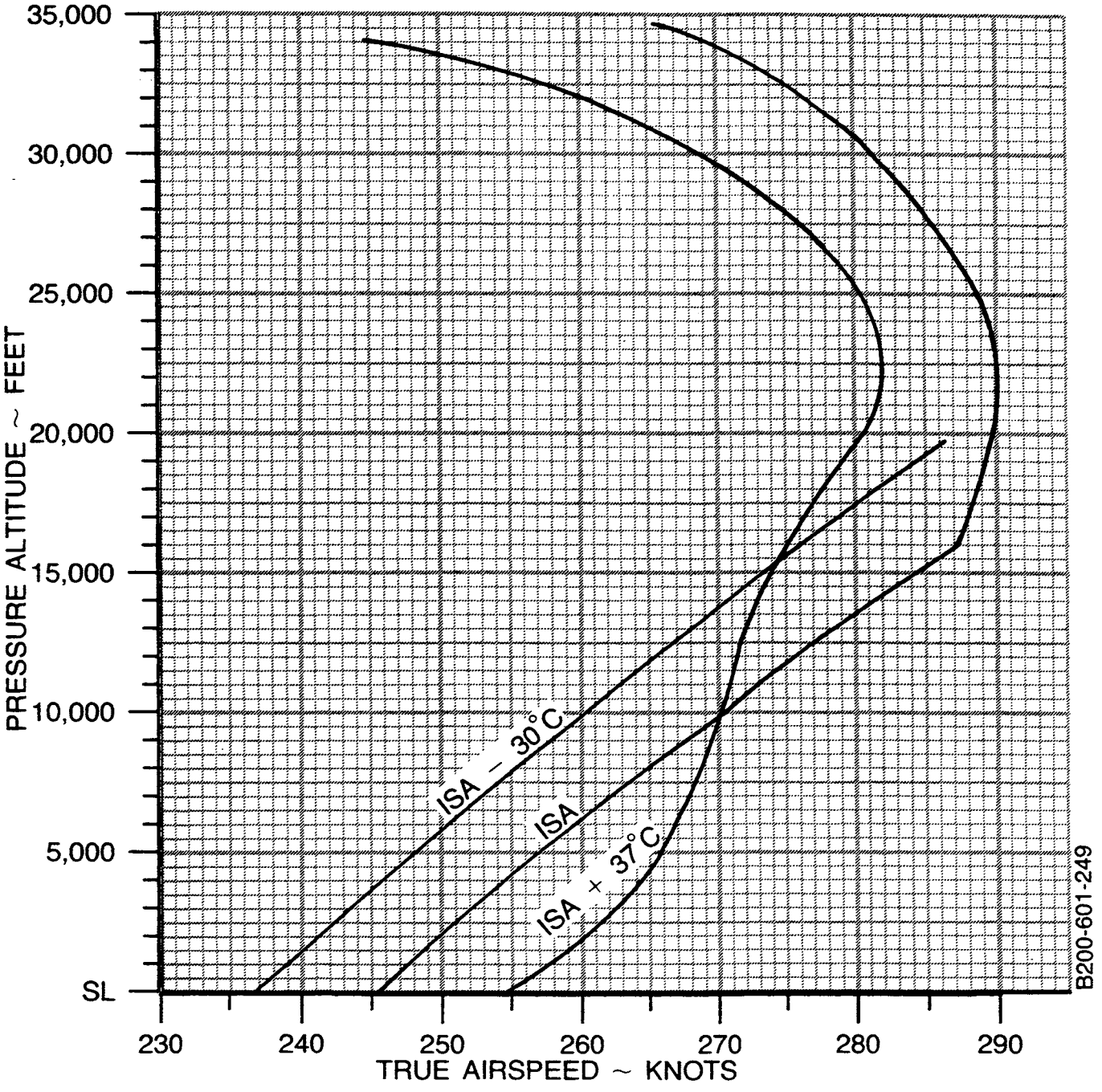
For Training Purposes Only

MAXIMUM CRUISE SPEEDS

1800 RPM

WEIGHT 11,000 LBS

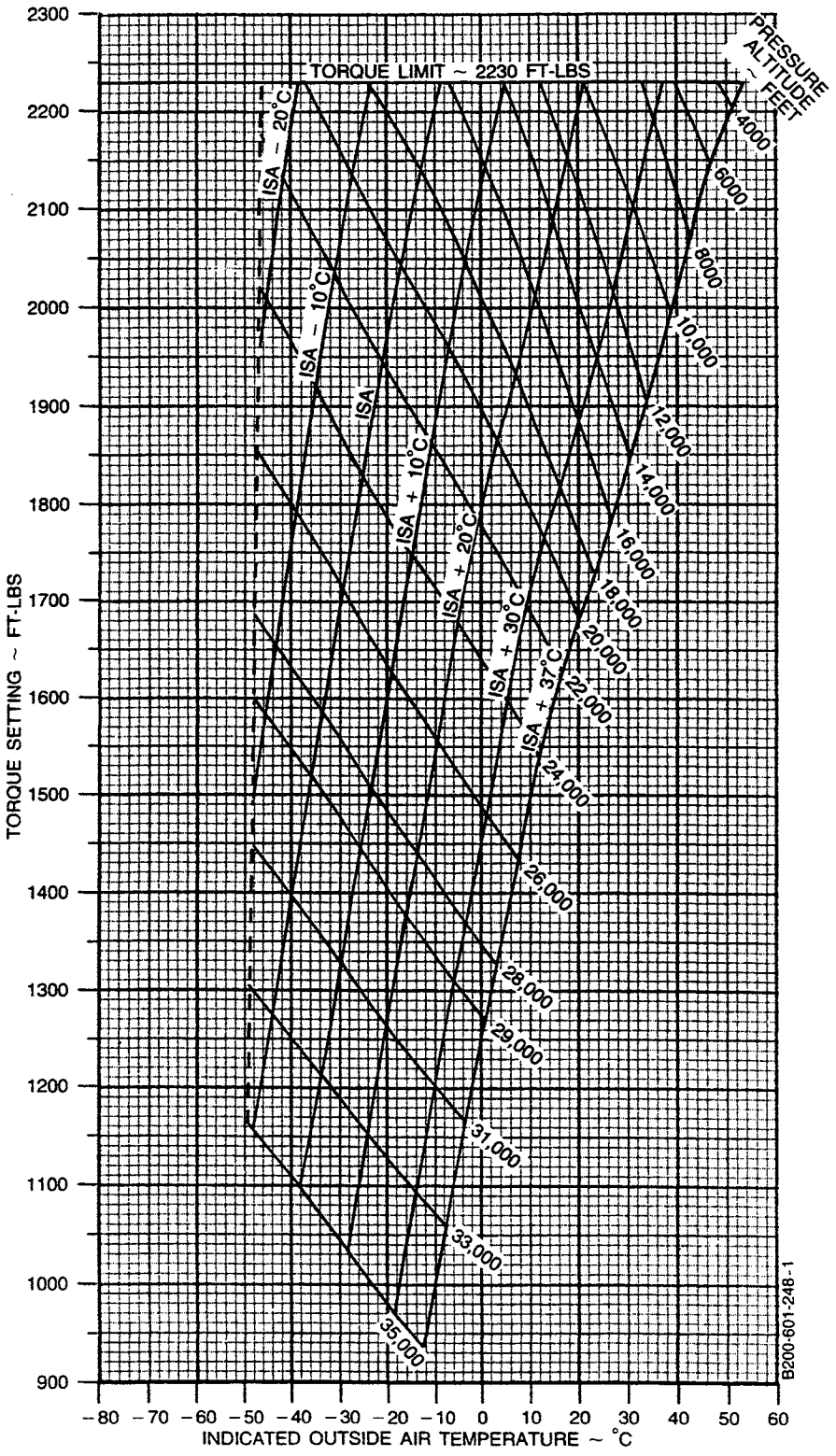
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS.



MAXIMUM CRUISE POWER

1800 RPM

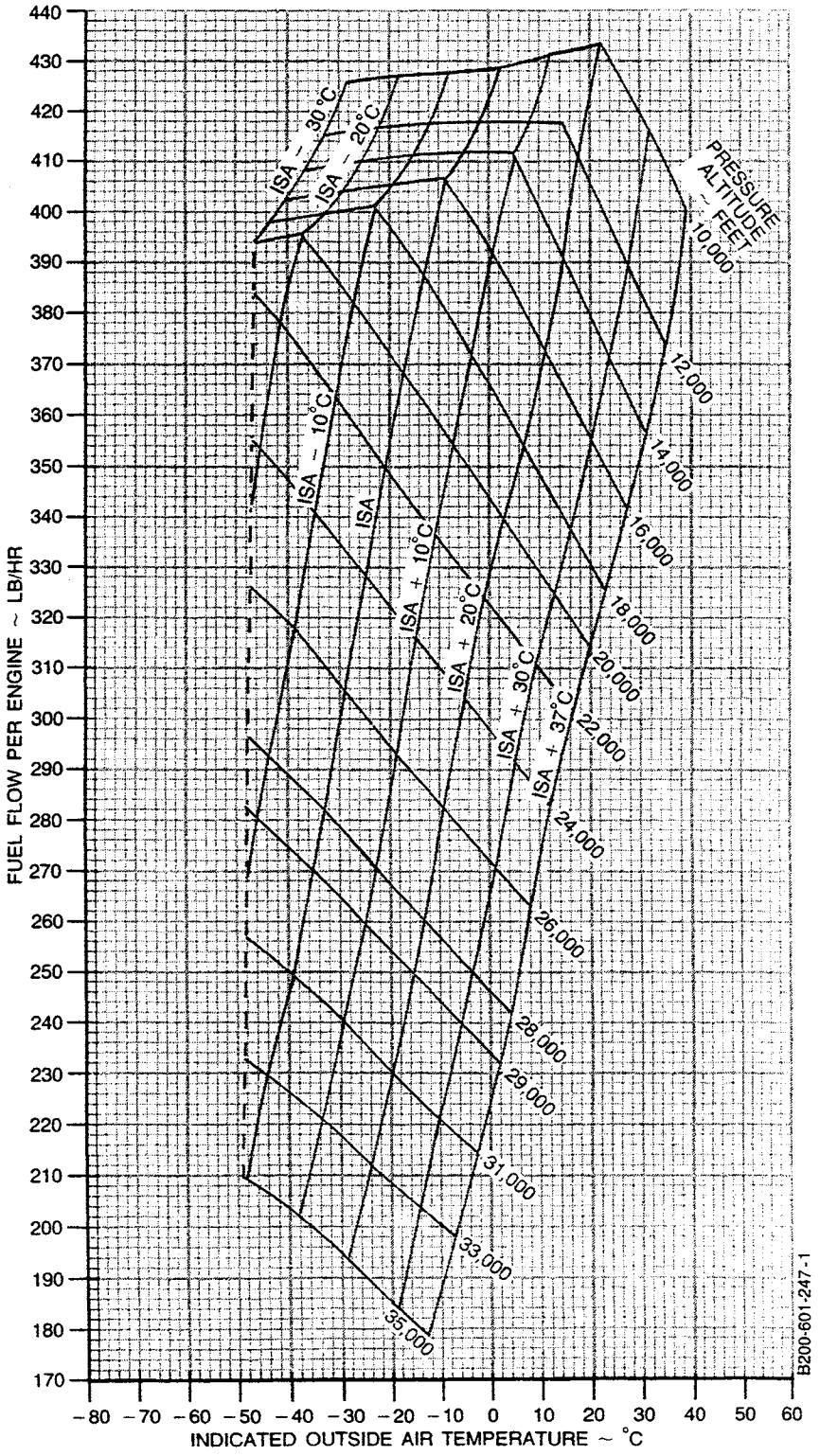
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%.



FUEL FLOW AT MAXIMUM CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10%.



RANGE PROFILE – MAXIMUM CRUISE POWER

1800 RPM

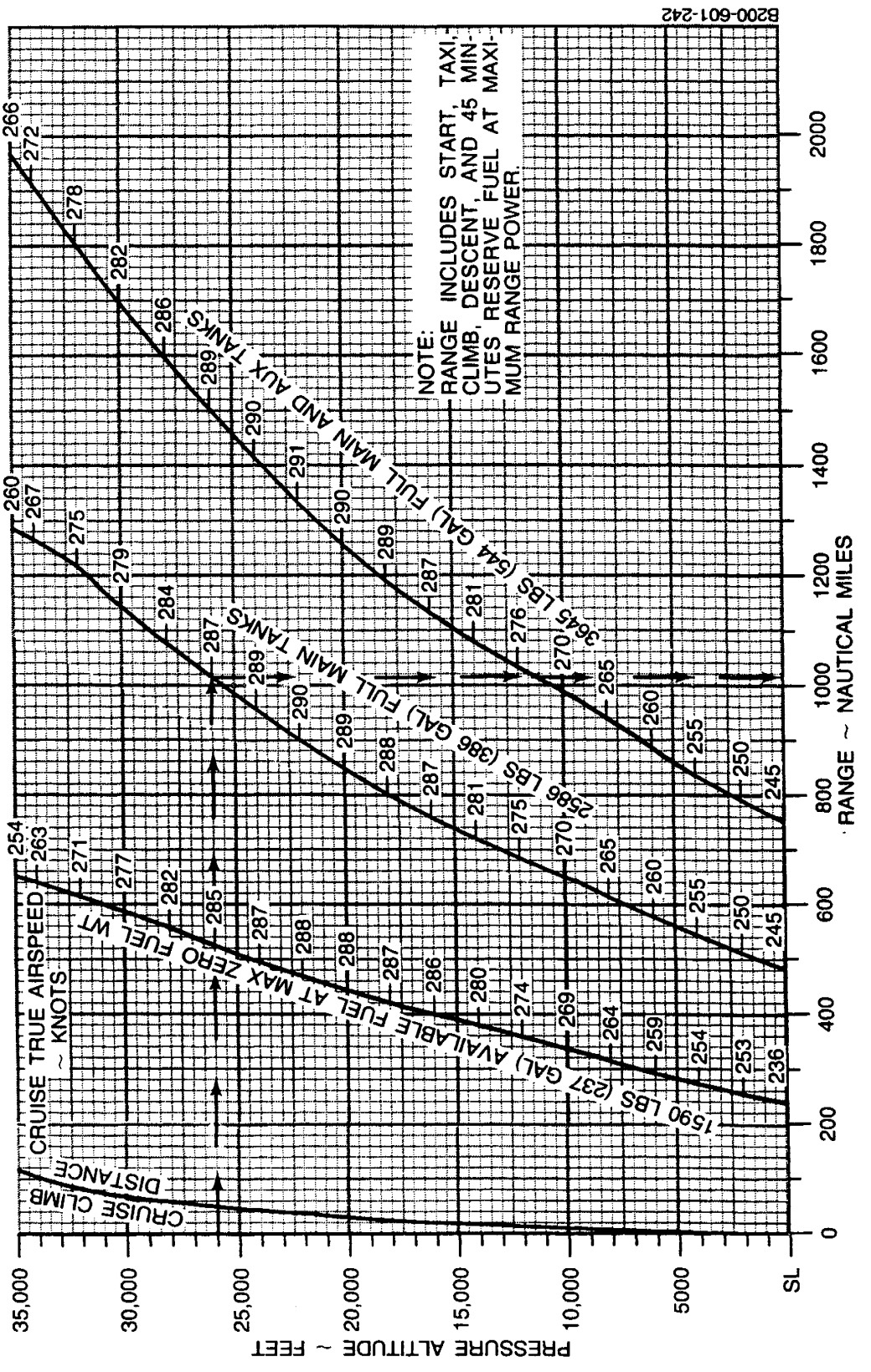
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	12,590 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL LOADING	2586 LBS
RANGE	1015 NM



For Training Purposes Only

MAXIMUM RANGE POWER

1700 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	-11	-15	1626	390	780	205	1561	383	766	203	1498	375	750	201
2000	-15	-19	1469	359	718	201	1406	351	702	199	1347	344	688	197
4000	-19	-23	1345	330	660	197	1280	322	644	195	1215	314	628	193
6000	-23	-27	1252	304	608	195	1184	296	592	193	1111	287	574	190
8000	-27	-31	1184	283	566	194	1107	273	546	191	1028	263	526	187
10,000	-31	-35	1127	262	524	193	1044	252	504	190	957	241	482	186
12,000	-35	-39	1096	246	496	194	1007	235	470	190	913	223	446	185
14,000	-39	-43	1057	233	466	195	962	221	442	190	858	208	416	184
16,000	-43	-47	1028	221	442	195	931	208	416	191	827	195	390	185
18,000	-47	-51	995	208	416	196	893	195	390	190	786	181	362	184
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---

For Training Purposes Only

MAXIMUM RANGE POWER

1700 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	-1	-5	1638	393	786	209	1603	389	778	208	1567	384	768	207
2000	-5	-9	1572	371	742	209	1513	364	728	207	1445	356	712	205
4000	-9	-13	1444	342	684	206	1370	334	668	203	1302	325	650	200
6000	-13	-17	1348	317	634	204	1272	308	616	200	1206	300	600	198
8000	-17	-21	1266	295	590	202	1188	285	570	199	1120	276	552	196
10,000	-21	-25	1182	271	542	200	1103	261	522	196	1033	252	504	193
12,000	-25	-29	1155	256	512	201	1069	245	490	197	991	236	472	194
14,000	-29	-33	1124	243	486	202	1031	231	462	198	947	220	440	193
16,000	-33	-37	1092	230	460	203	994	218	436	198	899	206	412	193
18,000	-37	-41	1068	219	438	204	967	206	412	199	870	194	388	193
20,000	-41	-45	1034	208	416	204	925	194	388	198	821	181	362	192
22,000	-45	-49	1020	200	400	206	902	184	368	199	790	169	338	192
24,000	-48	-53	1033	196	392	210	913	180	360	202	797	164	328	195
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

1700 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	9	5	1530	383	766	205	1509	381	762	206	1487	378	746	206
2000	5	1	1500	365	730	208	1477	362	724	208	1444	358	716	205
4000	1	-3	1473	347	694	210	1429	342	684	209	1381	336	672	208
6000	-3	-7	1411	326	652	210	1347	318	636	208	1275	310	620	205
8000	-7	-11	1337	304	608	209	1257	294	588	206	1181	285	570	202
10,000	-11	-15	1256	281	562	207	1173	270	540	204	1097	261	522	200
12,000	-15	-19	1222	266	532	209	1129	255	510	204	1045	244	488	200
14,000	-19	-23	1183	252	504	209	1083	239	478	204	995	228	456	200
16,000	-23	-27	1160	240	480	211	1054	227	454	205	958	215	430	200
18,000	-27	-31	1138	229	458	213	1024	215	430	206	919	202	404	200
20,000	-31	-35	1110	219	438	214	991	204	408	206	888	191	382	200
22,000	-35	-39	1089	211	422	215	966	195	390	207	857	180	360	200
24,000	-38	-43	1097	206	412	219	974	190	380	211	859	174	348	204
26,000	-42	-47	1100	201	402	223	986	186	372	216	866	170	340	208
28,000	-46	-51	1089	196	392	224	1003	184	368	221	881	167	334	213
29,000	-48	-52	1097	196	392	227	1002	182	364	223	889	166	332	215
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---

For Training Purposes Only

MAXIMUM RANGE POWER

1700 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	19	15	1514	385	770	207	1437	376	752	204	1382	369	738	202
2000	15	11	1419	359	718	205	1367	352	704	204	1341	349	698	204
4000	11	7	1387	340	680	207	1359	336	672	207	1329	332	664	207
6000	7	3	1361	322	644	209	1328	317	634	209	1289	313	626	209
8000	3	-1	1323	304	608	211	1278	298	596	210	1233	292	584	209
10,000	-1	-5	1283	285	570	212	1235	279	558	211	1165	271	542	208
12,000	-5	-9	1249	269	538	213	1192	262	524	212	1113	253	506	208
14,000	-9	-13	1225	257	514	215	1150	247	494	212	1057	236	472	207
16,000	-13	-17	1200	245	490	217	1109	234	468	213	1004	221	442	206
18,000	-16	-21	1177	235	470	219	1082	223	446	214	968	209	418	207
20,000	-20	-25	1169	227	454	222	1063	214	428	216	942	198	396	208
22,000	-24	-29	1148	219	438	223	1034	204	408	216	916	189	378	209
24,000	-28	-33	1118	209	418	223	1033	198	396	220	916	183	366	212
26,000	-32	-37	1094	201	402	224	1024	191	382	223	919	177	354	216
28,000	-36	-41	1125	202	404	230	1007	185	370	224	928	174	348	221
29,000	-38	-42	1131	202	404	233	1012	184	368	226	923	172	344	222
31,000	-42	-46	1163	205	410	241	1029	184	368	231	913	168	336	224
33,000	-45	-50	1258	218	436	254	1064	188	376	240	922	167	334	228
35,000	-49	-54	1128	198	396	237	1087	191	382	246	959	171	342	238

MAXIMUM RANGE POWER

1700 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	29	25	1603	398	796	214	1529	389	778	211	1454	380	760	208
2000	25	21	1501	371	742	212	1413	360	720	209	1332	350	700	205
4000	21	17	1387	342	684	209	1306	332	664	206	1252	326	652	204
6000	17	13	1313	319	638	208	1261	312	624	207	1222	307	614	206
8000	13	9	1264	299	598	209	1219	293	586	208	1186	289	578	208
10,000	9	5	1232	281	562	210	1187	275	550	209	1150	271	542	209
12,000	5	1	1211	267	534	212	1164	261	522	212	1123	255	510	211
14,000	1	-3	1189	254	508	214	1139	248	596	214	1091	242	484	213
16,000	-2	-7	1178	244	488	217	1122	237	474	216	1062	229	458	214
18,000	-6	-11	1159	233	466	219	1105	226	452	218	1027	216	432	215
20,000	-10	-15	1144	224	448	221	1089	217	434	221	986	204	408	215
22,000	-14	-19	1120	217	434	222	1075	210	420	223	976	197	394	218
24,000	-18	-23	1100	208	416	223	1035	199	398	222	969	190	380	221
26,000	-22	-27	1129	207	414	230	1005	190	380	222	948	182	364	222
28,000	-26	-31	1150	207	414	235	1027	189	378	228	922	174	348	222
29,000	-28	-32	1170	209	418	240	1039	189	378	231	921	173	346	224
31,000	-31	-36	1188	210	420	246	1064	191	382	238	935	172	344	229
33,000	-35	-40	---	---	---	---	1082	192	384	244	955	172	344	235
35,000	-39	-44	---	---	---	---	---	---	---	---	968	173	346	241

MAXIMUM RANGE POWER

1700 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	39	35	1552	394	788	213	1502	387	774	212	1446	380	760	210
2000	35	31	1511	374	748	215	1449	366	732	213	1385	358	716	211
4000	31	27	1465	354	708	216	1385	343	686	213	1299	333	666	209
6000	27	23	1418	334	668	218	1317	321	642	213	1225	310	620	209
8000	23	19	1343	311	622	217	1236	298	596	211	1142	286	572	207
10,000	19	15	1266	287	574	215	1160	274	548	209	1093	266	532	207
12,000	15	11	1259	274	548	218	1137	259	518	211	1063	250	500	208
14,000	12	7	1231	261	522	220	1119	247	494	214	1052	239	478	211
16,000	8	3	1210	249	498	222	1196	235	470	216	1035	227	454	214
18,000	4	-1	1190	239	478	224	1087	226	452	219	1024	217	434	217
20,000	0	-5	1163	228	456	225	1074	217	434	221	1014	209	418	220
22,000	-4	-9	1138	219	438	226	1051	207	414	222	997	200	400	222
24,000	-8	-13	1152	215	430	231	1030	199	398	224	967	190	380	223
26,000	-12	-17	1160	212	424	235	1040	195	390	228	931	180	360	222
28,000	-15	-21	1209	216	432	245	1057	194	388	234	935	177	354	226
29,000	-17	-22	1194	213	426	245	1076	196	392	238	945	177	354	229
31,000	-21	-26	---	---	---	---	1090	195	390	244	962	176	352	235
33,000	-25	-30	---	---	---	---	1096	195	390	247	979	177	354	257
35,000	-29	-34	---	---	---	---	---	---	---	---	998	179	358	262

MAXIMUM RANGE POWER

1700 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	49	45	1540	395	790	214	1473	387	774	212	1411	379	758	210
2000	45	41	1466	371	742	214	1406	363	726	212	1349	356	712	211
4000	41	37	1429	352	704	216	1374	345	690	215	1321	338	676	213
6000	37	33	1398	334	668	218	1340	326	652	217	1266	317	634	214
8000	34	29	1355	314	628	219	1295	307	614	218	1204	295	590	214
10,000	30	25	1308	295	590	220	1245	286	572	218	1152	275	550	214
12,000	26	21	1272	278	556	221	1205	269	538	219	1106	257	514	214
14,000	22	17	1245	265	530	223	1176	256	512	221	1065	242	484	215
16,000	18	13	1224	253	506	225	1154	244	488	223	1041	230	460	217
18,000	14	9	1207	242	484	227	1126	232	464	224	1012	217	434	218
20,000	10	5	1202	235	470	231	1097	221	442	225	995	208	416	220
22,000	6	1	1188	227	454	233	1079	212	424	227	974	199	398	222
24,000	2	-3	1162	218	436	234	1081	206	412	231	963	191	382	224
26,000	-1	-7	1201	218	436	242	1065	199	398	233	965	185	370	228
28,000	-5	-11	1202	215	430	246	1089	198	396	240	967	181	362	232
29,000	-7	-12	1201	214	428	247	1100	198	396	243	971	180	360	234
31,000	-11	-16	---	---	---	---	1106	198	396	247	992	180	360	241
33,000	-15	-20	---	---	---	---	---	---	---	---	1000	180	360	246
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

1700 RPM

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			12,000 POUNDS				11,000 POUNDS				10,000 POUNDS			
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS	FT-LB	LB/HR	LB/HR	KNOTS
SL	56	52	1586	402	804	219	1486	390	780	214	1428	383	766	213
2000	52	48	1493	376	752	217	1406	365	730	214	1341	357	714	212
4000	48	44	1423	353	706	217	1350	343	686	214	1291	336	672	213
6000	44	40	1369	331	662	218	1309	324	648	216	1251	316	632	214
8000	41	36	1329	313	626	219	1273	305	610	217	1212	297	594	216
10,000	37	32	1289	294	588	220	1235	287	574	219	1177	279	558	217
12,000	33	28	1253	277	554	221	1203	270	540	220	1145	263	526	219
14,000	29	24	1220	263	526	222	1170	256	512	222	1105	248	496	220
16,000	25	20	1203	252	504	225	1146	244	488	224	1081	236	472	222
18,000	21	16	1189	242	484	227	1126	233	466	226	1048	223	446	223
20,000	17	12	1187	234	468	230	1117	225	450	229	1015	212	424	223
22,000	13	8	1181	227	454	233	1100	216	432	231	991	202	404	225
24,000	9	4	1190	223	446	238	1081	207	414	233	993	195	390	229
26,000	6	0	1221	222	444	246	1071	201	402	235	985	188	376	232
28,000	2	-4	1204	216	432	247	1119	203	406	245	966	182	364	233
29,000	0	-5	---	---	---	---	1107	200	400	245	987	183	366	238
31,000	-4	-9	---	---	---	---	1106	198	396	248	1003	182	364	244
33,000	-8	-13	---	---	---	---	---	---	---	---	1003	181	362	247
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---	---

RANGE PROFILE – MAXIMUM RANGE POWER

1700 RPM

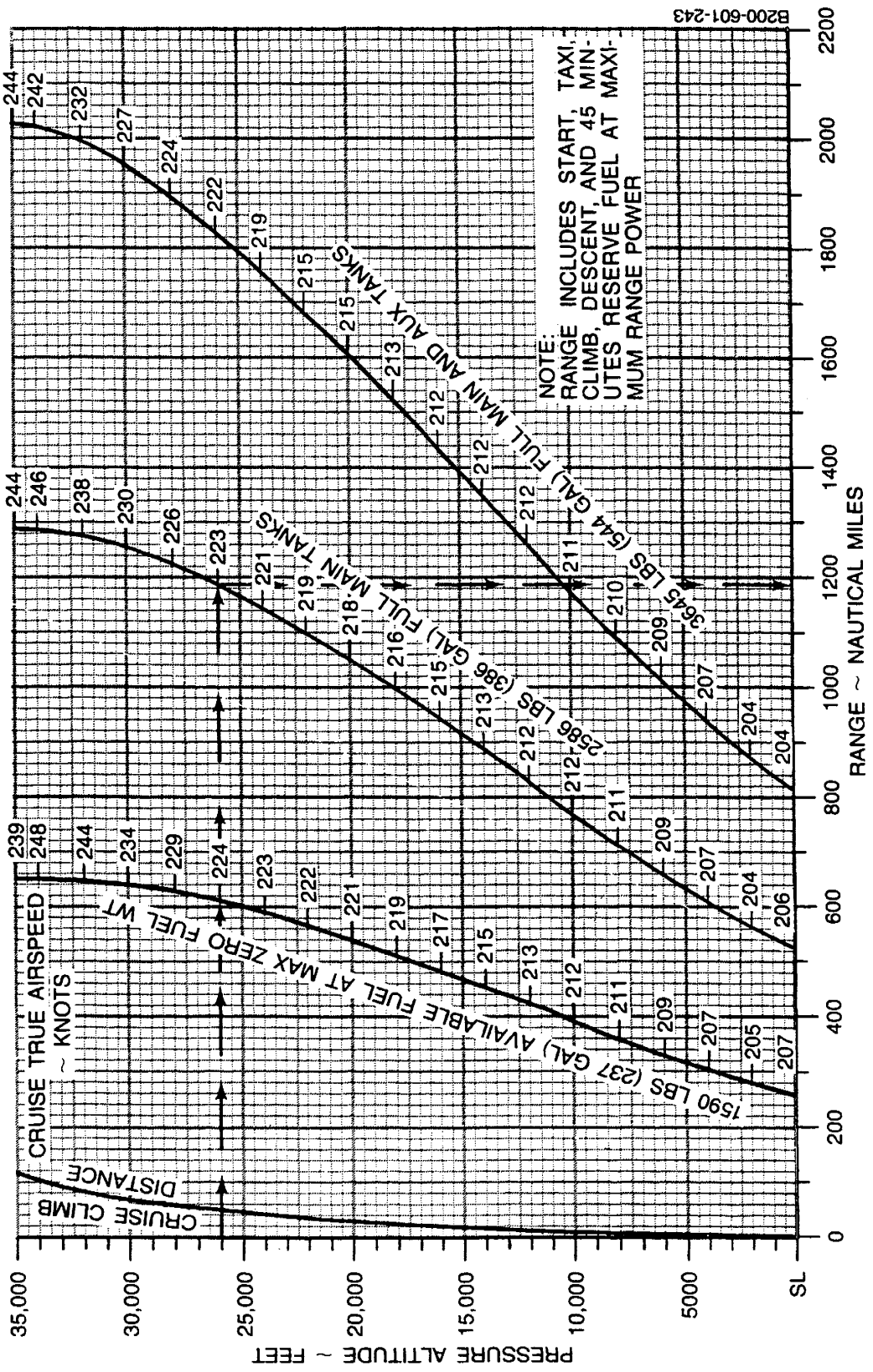
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	12,590 BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL LOADING	2586 LBS
RANGE	1188 NM



For Training Purposes Only

RANGE PROFILE – FULL MAIN AND AUX TANKS

STANDARD DAY
ZERO WIND

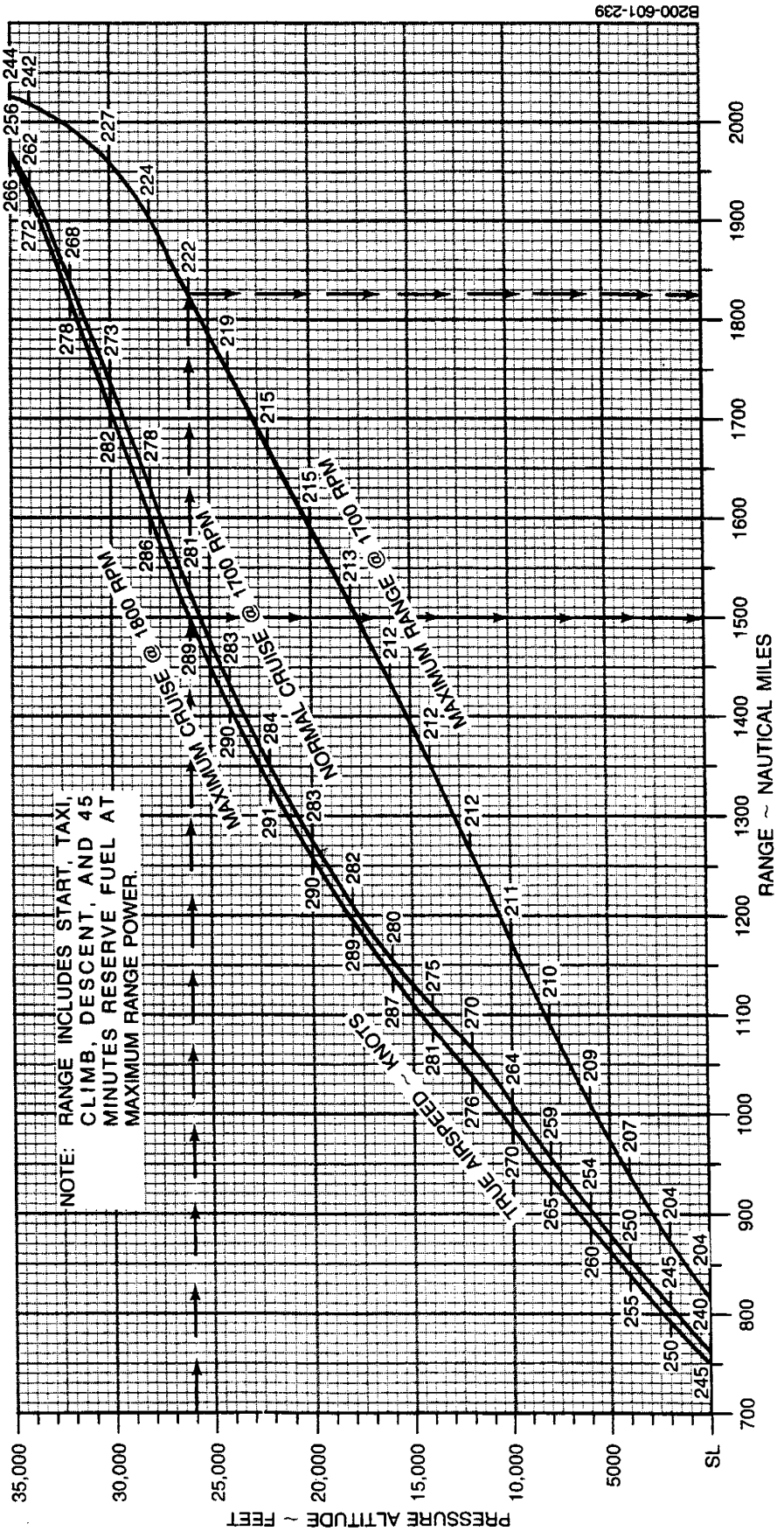
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

EXAMPLE:

PRESSURE ALTITUDE ..... 26,000 FT

RANGE @
MAX CRUISE POWER ..... 1502 NM
RANGE @
MAX RANGE POWER ..... 1828 NM



For Training Purposes Only

ENDURANCE PROFILE – FULL MAIN AND AUX TANKS

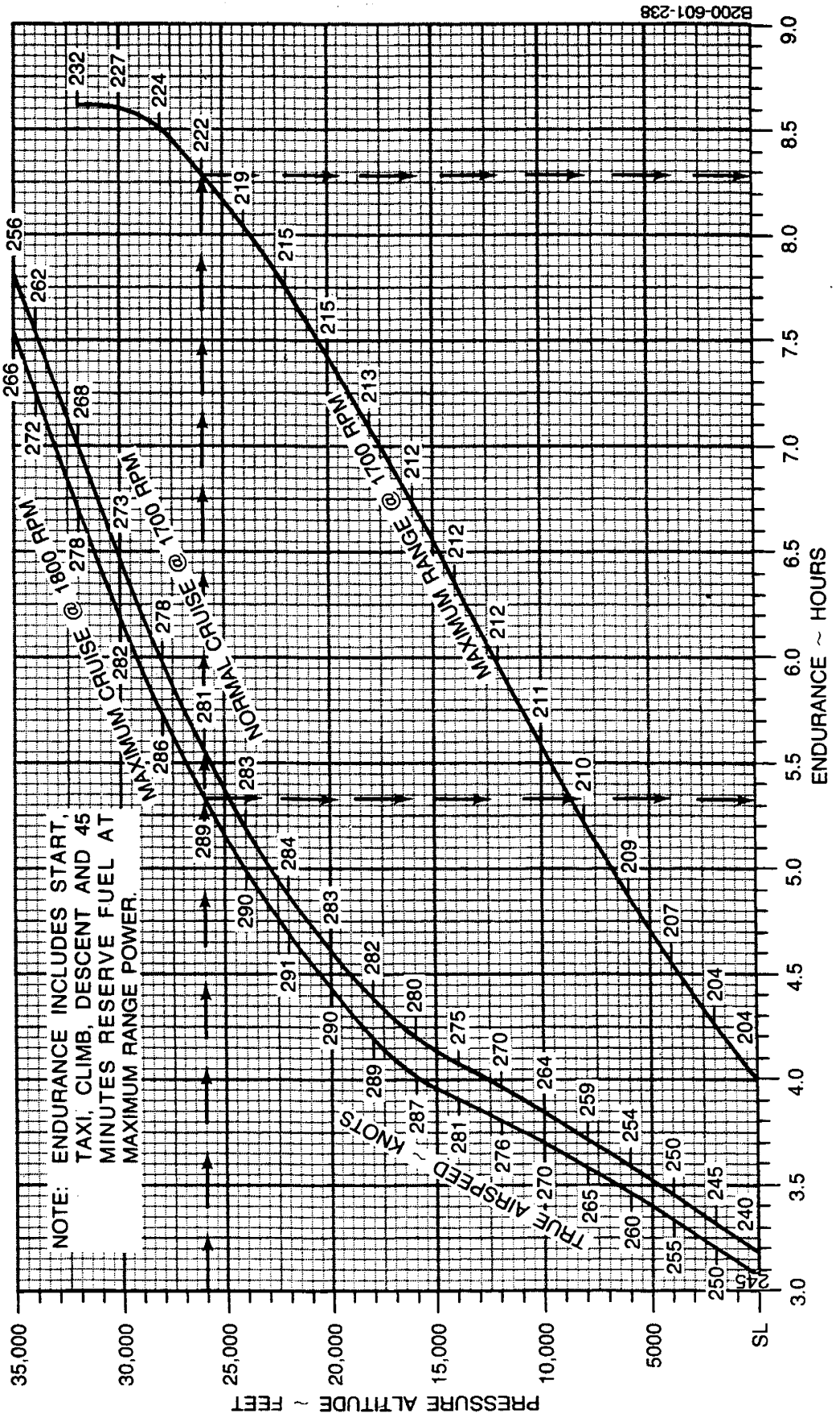
STANDARD DAY

ASSOCIATED CONDITIONS:

WEIGHT	12,590 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
ENDURANCE @ MAX CRUISE POWER	5.32 HRS
ENDURANCE @ MAX RANGE POWER	8.28 HRS



For Training Purposes Only

RANGE PROFILE – FULL MAIN TANKS

STANDARD DAY
ZERO WIND

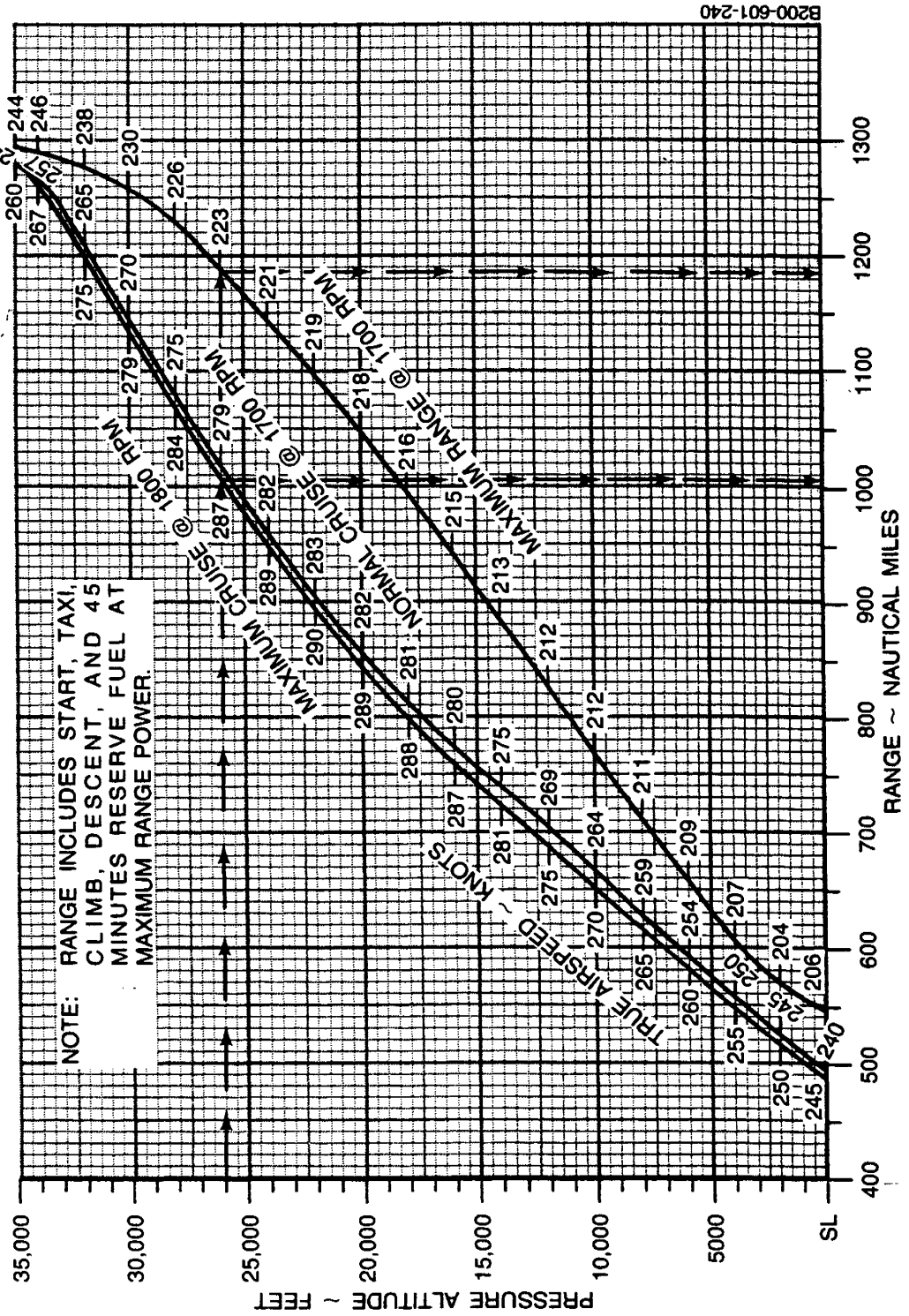
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

EXAMPLE:

PRESSURE ALTITUDE ..... 26,000 FT

RANGE @ MAX CRUISE POWER ..... 1004 NM
RANGE @ MAX RANGE POWER ..... 1185 NM



For Training Purposes Only

ENDURANCE PROFILE – FULL MAIN TANKS

STANDARD DAY

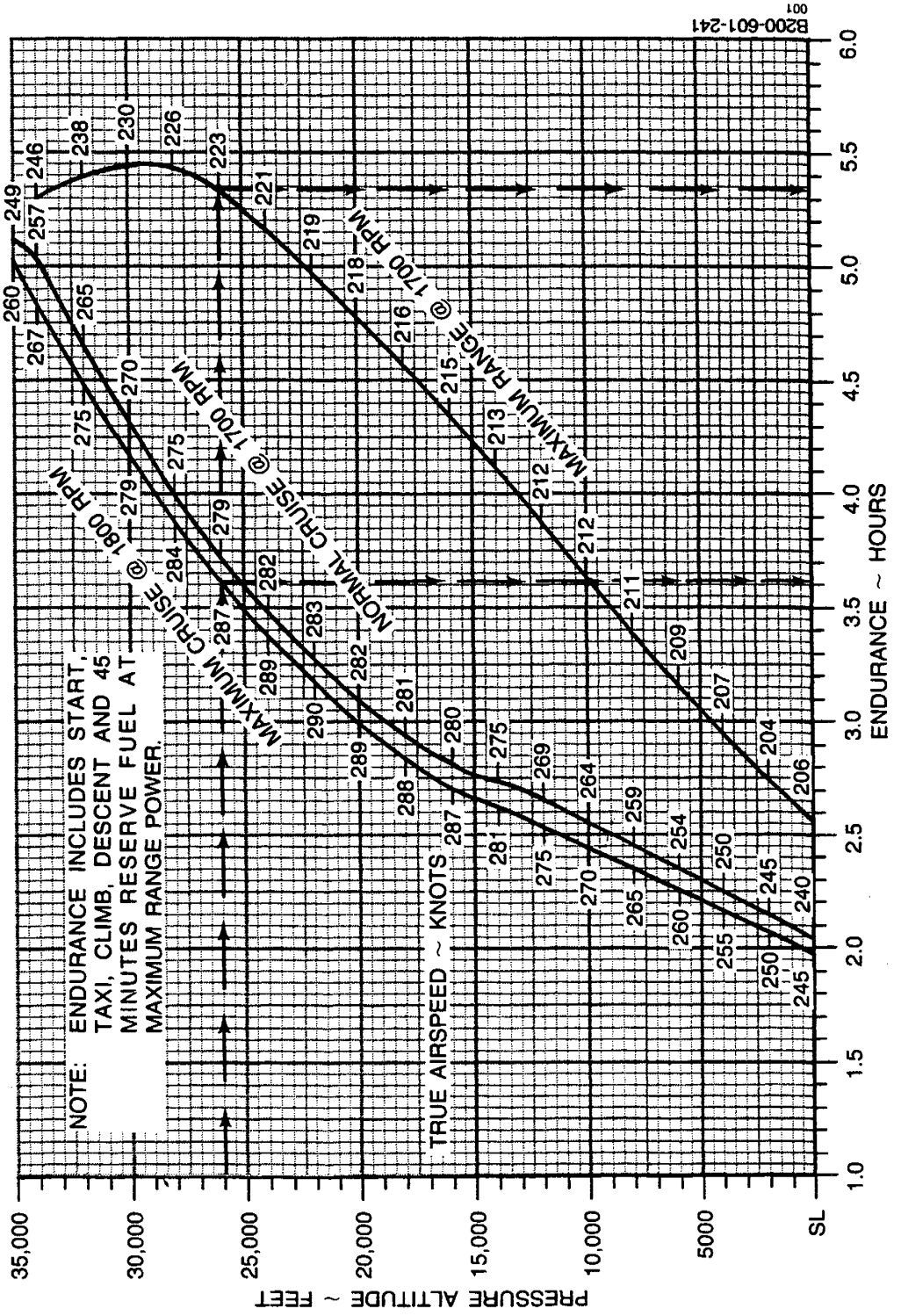
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

EXAMPLE:

PRESSURE ALTITUDE ..... 26,000 FT

ENDURANCE @ MAX CRUISE POWER ..... 3.61 HR
ENDURANCE @ MAX RANGE POWER ..... 5.33 HR



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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER
1900 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED – KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	-12	-15	2230	517	517	191	181	192	183	194	185
2000	-16	-19	2230	505	505	188	184	190	186	192	188
4000	-20	-23	2230	492	492	186	187	188	189	190	191
6000	-23	-27	2230	479	479	184	190	186	193	188	194
8000	-27	-31	2230	467	467	181	193	184	196	186	198
10,000	-31	-35	2230	454	454	179	197	181	199	183	201
12,000	-35	-39	2230	445	445	177	200	179	202	181	205
14,000	-39	-43	2230	439	439	174	203	176	206	179	208
16,000	-43	-47	2230	434	434	171	206	174	209	176	212
18,000	-47	-51	2155	416	416	165	205	169	209	171	213
20,000	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	-2	-5	2230	519	519	189	183	191	185	193	187
2000	-6	-9	2230	505	505	187	186	189	188	190	190
4000	-10	-13	2230	492	492	184	189	187	191	188	193
6000	-13	-17	2230	479	479	182	192	184	195	186	197
8000	-17	-21	2230	468	468	180	196	182	198	184	200
10,000	-21	-25	2230	456	456	177	199	180	201	182	204
12,000	-25	-29	2230	446	446	174	202	177	205	179	207
14,000	-29	-33	2230	440	440	172	205	174	208	177	211
16,000	-33	-37	2168	424	424	166	205	170	209	172	212
18,000	-37	-41	2058	401	401	159	202	163	207	166	211
20,000	-41	-45	1957	380	380	152	199	157	205	160	210
22,000	-45	-49	1849	358	358	143	194	149	202	154	208
24,000	-49	-53	1699	329	329	130	182	139	195	145	203
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

For Training Purposes Only

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE			TORQUE	FUEL FLOW	TOTAL	AIRSPEED ~ KNOTS					
ALTITUDE	IOAT	OAT	PER ENGINE	PER ENGINE	FUEL FLOW	@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	8	5	2230	522	522	188	185	189	187	191	189
2000	4	1	2230	508	508	185	188	187	190	189	192
4000	1	-3	2230	494	494	183	191	185	194	187	195
6000	-3	-7	2230	480	480	180	194	183	197	185	199
8000	-7	-11	2230	468	468	178	198	180	200	182	202
10,000	-11	-15	2230	456	456	175	201	178	203	180	206
12,000	-15	-19	2230	446	446	173	204	175	207	177	209
14,000	-19	-23	2166	430	430	167	203	170	207	173	211
16,000	-23	-27	2061	407	407	160	202	164	206	167	210
18,000	-27	-31	1959	385	385	153	199	158	204	161	209
20,000	-31	-35	1869	366	366	145	195	151	203	155	208
22,000	-35	-39	1766	345	345	136	189	143	199	149	206
24,000	-39	-43	1631	318	318	117	168	133	191	140	201
26,000	-44	-47	1490	291	291	---	---	117	174	130	193
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	18	15	2230	525	525	186	187	188	189	190	191
2000	14	11	2230	511	511	184	190	186	192	188	194
4000	11	7	2230	497	497	181	193	183	195	185	198
6000	7	3	2230	483	483	179	196	181	199	183	201
8000	3	-1	2230	470	470	176	199	179	202	181	204
10,000	-1	-5	2230	457	457	174	202	176	205	178	208
12,000	-5	-9	2175	437	437	169	203	172	206	174	209
14,000	-9	-13	2074	413	413	162	201	166	205	168	209
16,000	-13	-17	1970	391	391	155	198	159	204	162	208
18,000	-17	-21	1863	370	370	147	194	152	201	156	207
20,000	-21	-25	1778	352	352	138	190	145	199	150	205
22,000	-25	-29	1683	333	333	126	180	137	194	143	203
24,000	-29	-33	1551	306	306	---	---	125	183	134	197
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

For Training Purposes Only

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	28	25	2230	527	527	185	189	187	191	188	193
2000	25	21	2230	513	513	182	192	184	194	186	196
4000	21	17	2230	499	499	180	195	182	197	184	200
6000	17	13	2230	485	485	177	198	180	201	182	203
8000	13	9	2229	471	471	174	201	177	204	179	206
10,000	9	5	2127	442	442	168	200	171	203	174	206
12,000	5	1	2030	415	415	162	198	165	202	168	206
14,000	1	-3	1958	396	396	155	197	160	202	163	206
16,000	-3	-7	1878	376	376	149	195	154	201	157	206
18,000	-7	-11	1791	356	356	141	190	147	199	152	205
20,000	-11	-15	1712	339	339	132	184	140	196	145	203
22,000	-15	-19	1600	319	319	---	---	130	188	138	199
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	38	35	2230	530	530	183	190	185	193	187	194
2000	35	31	2230	515	515	181	193	183	196	185	198
4000	31	27	2230	500	500	178	197	180	199	183	201
6000	27	23	2160	475	475	173	197	176	200	178	203
8000	23	19	2064	447	447	167	196	170	199	173	202
10,000	19	15	1968	419	419	160	194	164	198	167	202
12,000	15	11	1883	395	395	154	192	158	197	161	201
14,000	11	7	1825	377	377	148	191	153	197	157	202
16,000	7	3	1766	360	360	141	188	147	196	152	202
18,000	3	-1	1706	344	344	134	184	141	194	146	202
20,000	-1	-5	1632	327	327	121	173	134	190	140	200
22,000	-6	-9	1538	307	307	---	---	123	181	133	196
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	48	45	2158	521	521	179	189	181	192	184	193
2000	44	41	2107	499	499	175	190	177	193	180	195
4000	41	37	2065	478	478	171	192	174	195	176	197
6000	37	33	2001	454	454	166	192	169	195	171	198
8000	33	29	1914	426	426	159	190	163	194	166	198
10,000	29	25	1826	398	398	153	188	157	193	160	197
12,000	25	21	1752	376	376	146	186	151	192	155	197
14,000	21	17	1695	359	359	139	183	145	191	150	197
16,000	17	13	1645	343	343	132	179	140	190	145	197
18,000	13	9	1602	329	329	122	172	134	188	140	197
20,000	9	5	1554	316	316	---	---	126	184	135	196
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

IAS + 37°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 11,000 POUNDS.

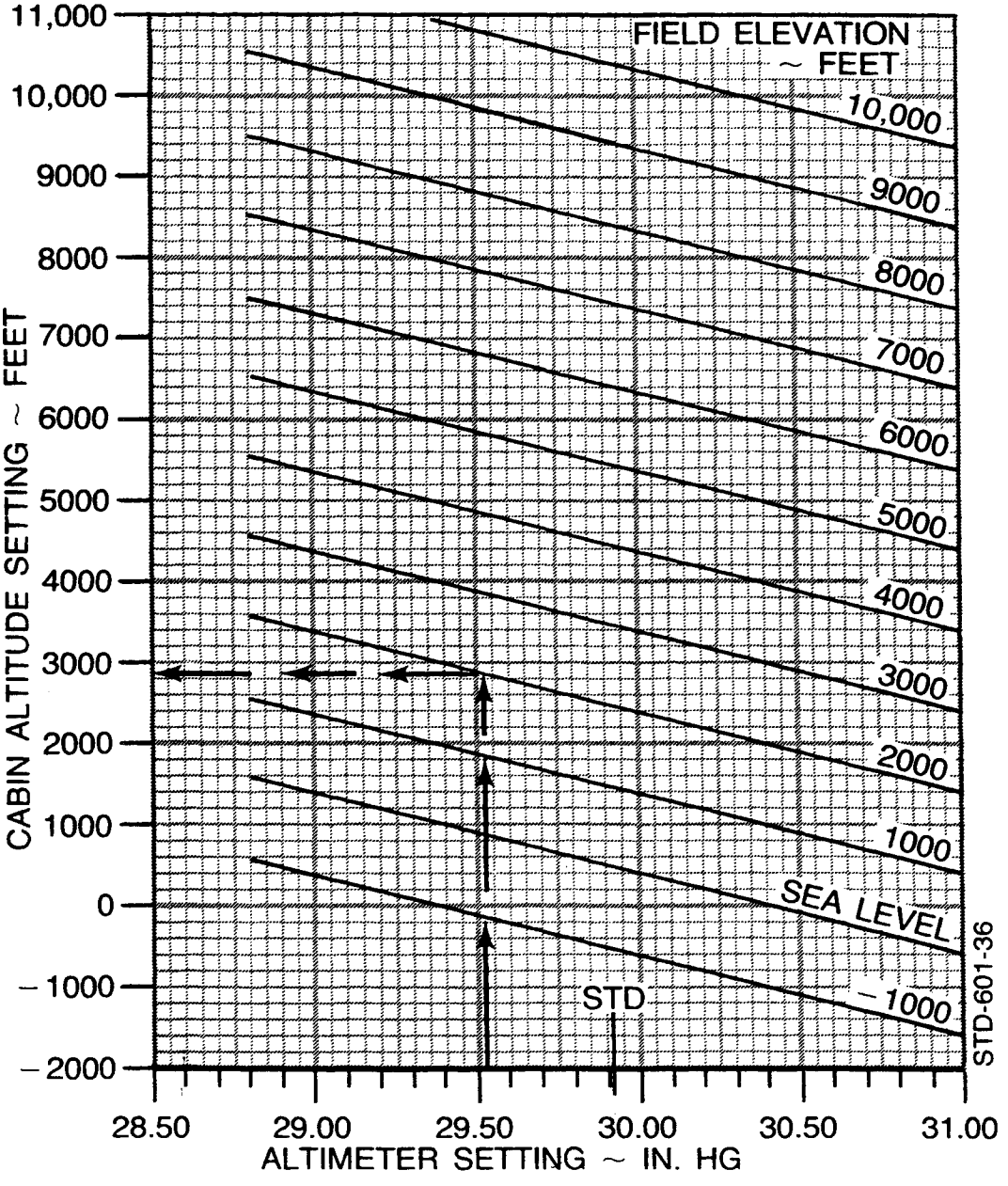
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS @ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
SL	55	52	2019	503	503	173	185	176	188	178	190
2000	51	48	1975	481	481	169	186	172	189	174	192
4000	47	44	1942	461	461	165	187	168	191	171	194
6000	44	40	1886	438	438	160	187	163	191	166	195
8000	40	36	1807	412	412	154	185	158	190	161	194
10,000	36	32	1728	385	385	147	183	152	189	156	194
12,000	32	28	1663	364	364	140	180	146	188	150	193
14,000	28	24	1612	347	347	133	177	140	187	145	193
16,000	23	20	1565	332	332	124	170	134	185	140	193
18,000	19	16	1523	318	318	---	---	127	181	135	192
20,000	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---

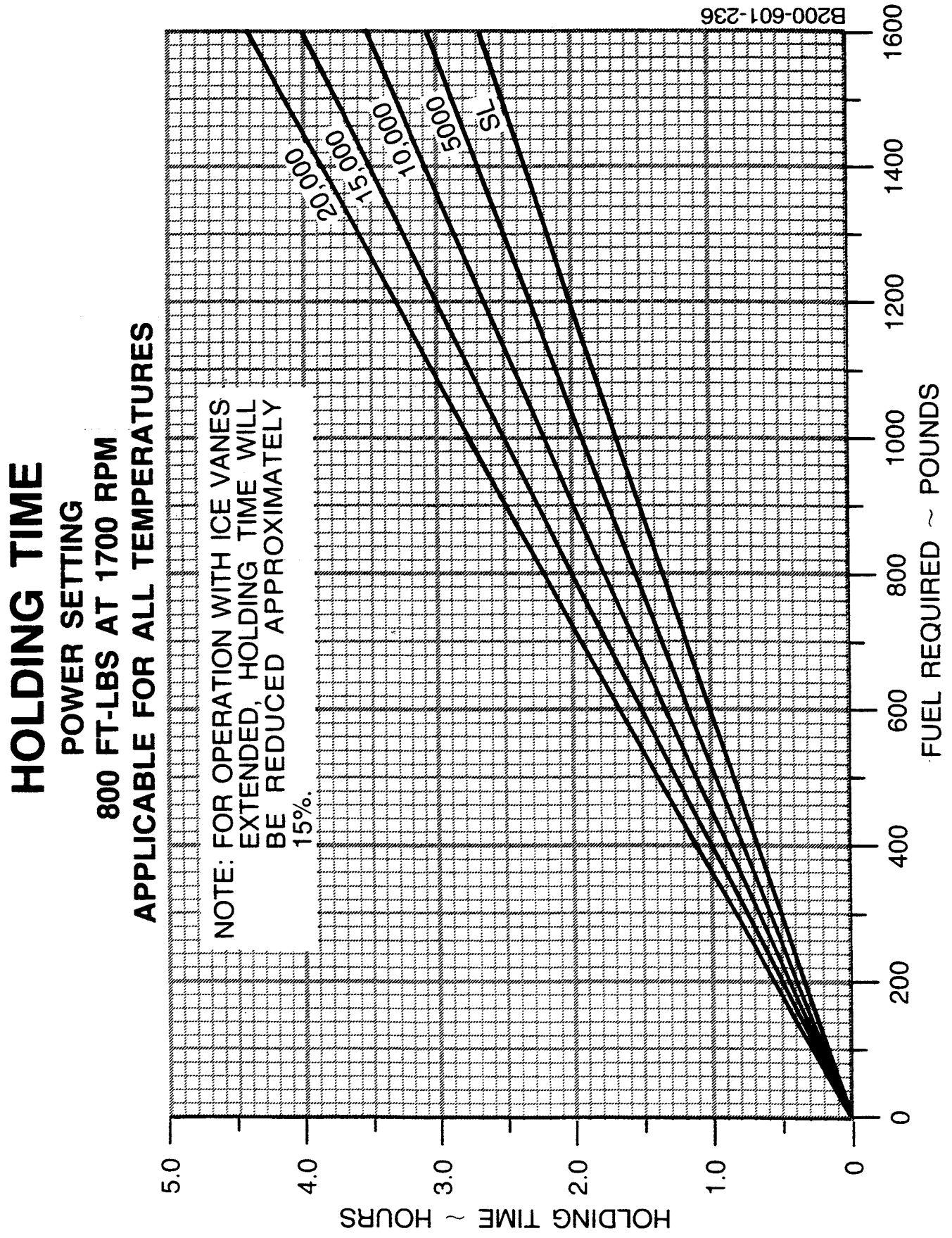
PRESSURIZATION CONTROLLER SETTING FOR LANDING

EXAMPLE

ALTIMETER SETTING ..... 29.52 IN. HG
LANDING FIELD ELEVATION ... 2000 FT

CABIN ALTITUDE SETTING .... 2885 FT





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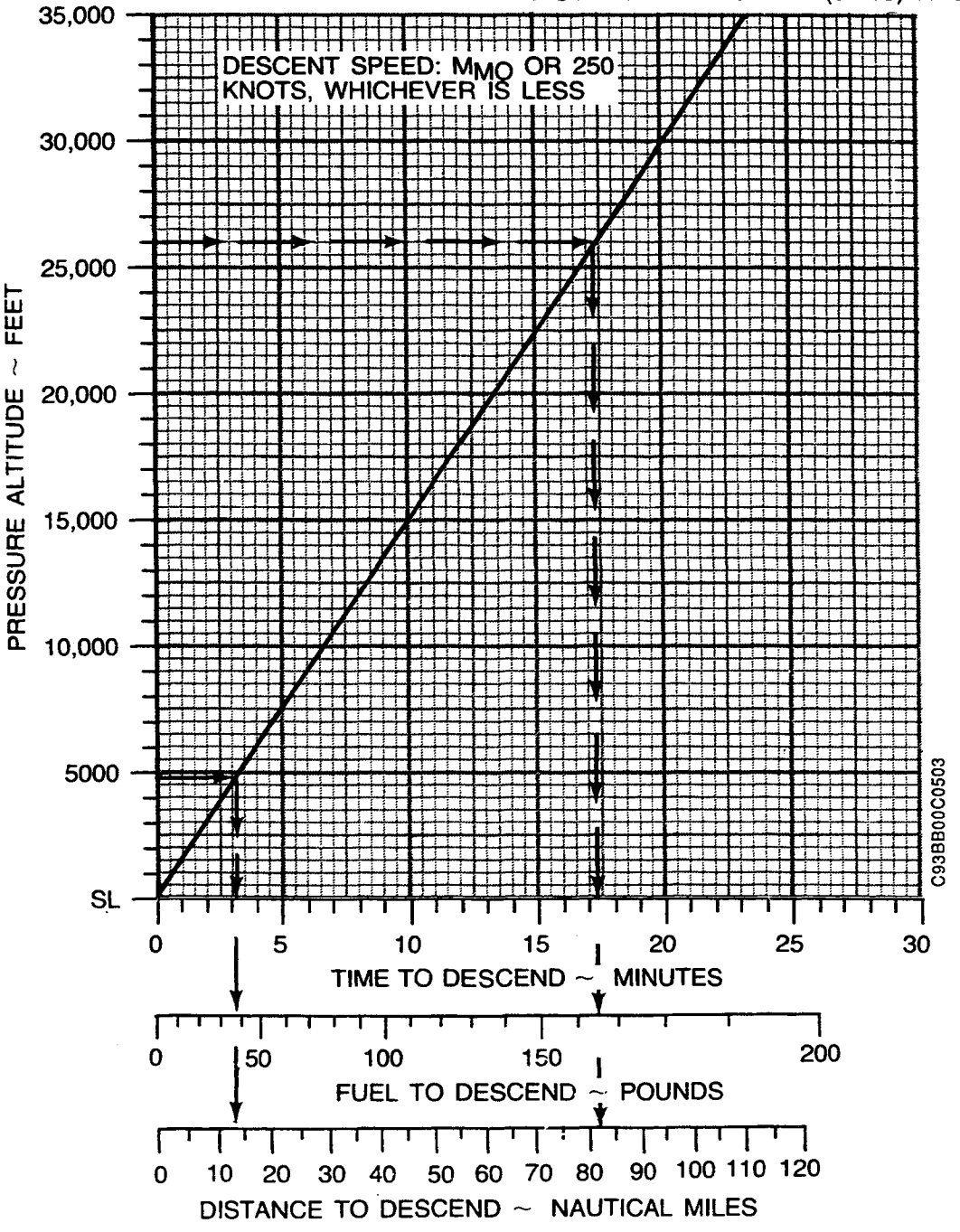
TIME, FUEL AND DISTANCE TO DESCEND

ASSOCIATED CONDITIONS:

POWER	AS REQUIRED TO DESCEND AT 1500 FT/MIN
GEAR	UP
FLAPS	UP

EXAMPLE:

INITIAL ALTITUDE	26,000 FT
FINAL ALTITUDE	4732 FT
TIME TO DESCENT (17-3)	14 MIN
FUEL TO DESCEND (164-36)	128 LBS
DISTANCE TO DESCEND (82-13)	69 NM



CLIMB – BALKED LANDING

CLIMB SPEED 100 KNOTS (ALL WEIGHTS)

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10° C TO ACTUAL OAT BEFORE ENTERING GRAPH.

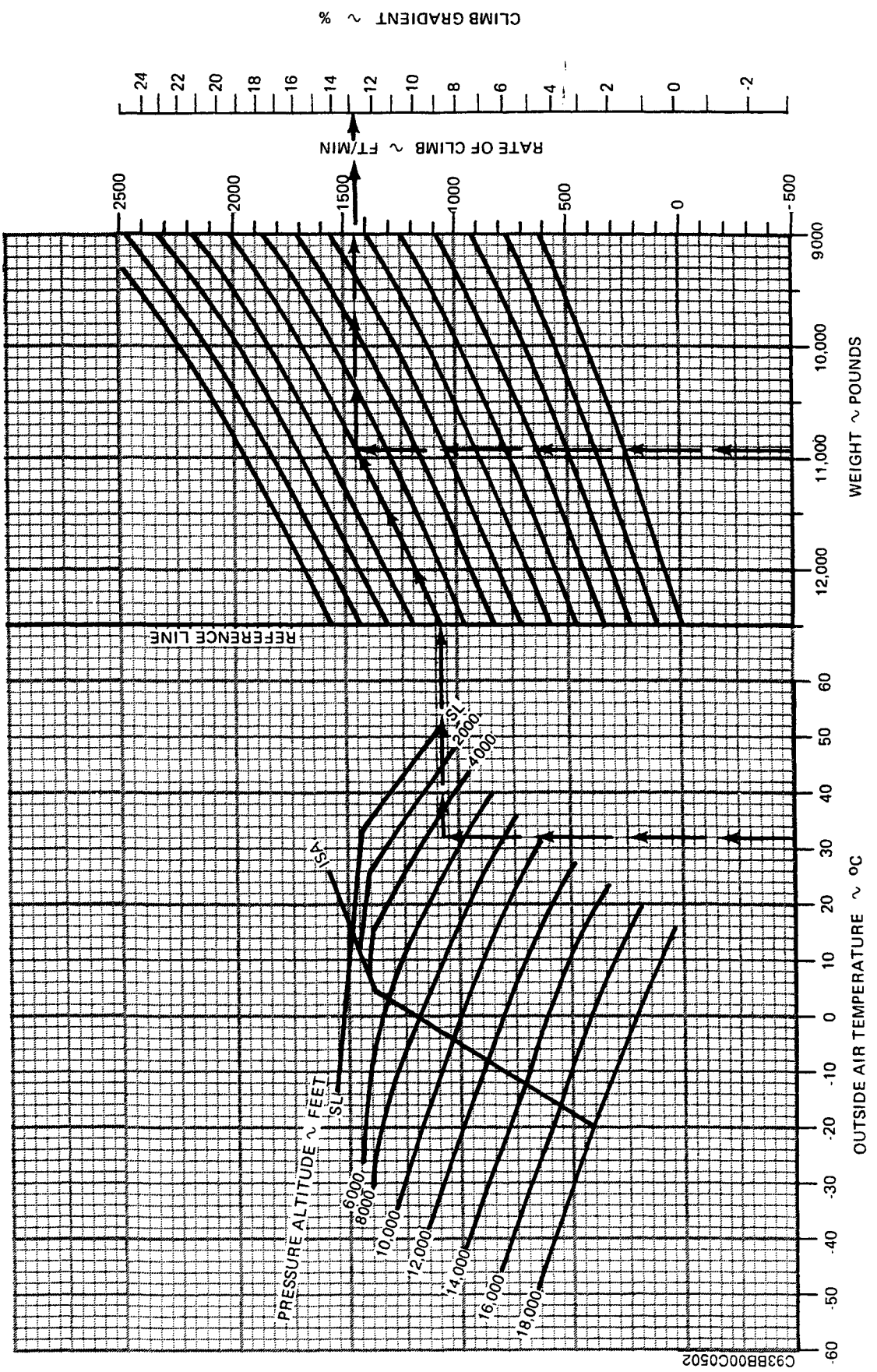
EXAMPLE:

OAT . . . . . 32°C
PRESSURE ALTITUDE . 4732 FT
WEIGHT . . . . . 10,937 LBS

RATE-OF-CLIMB . . . . 1450 FT/MIN
CLIMB GRADIENT . . . . 12.8 %

ASSOCIATED CONDITIONS:

POWER . . . . . TAKE-OFF
FLAPS . . . . . DOWN
LANDING GEAR . . DOWN



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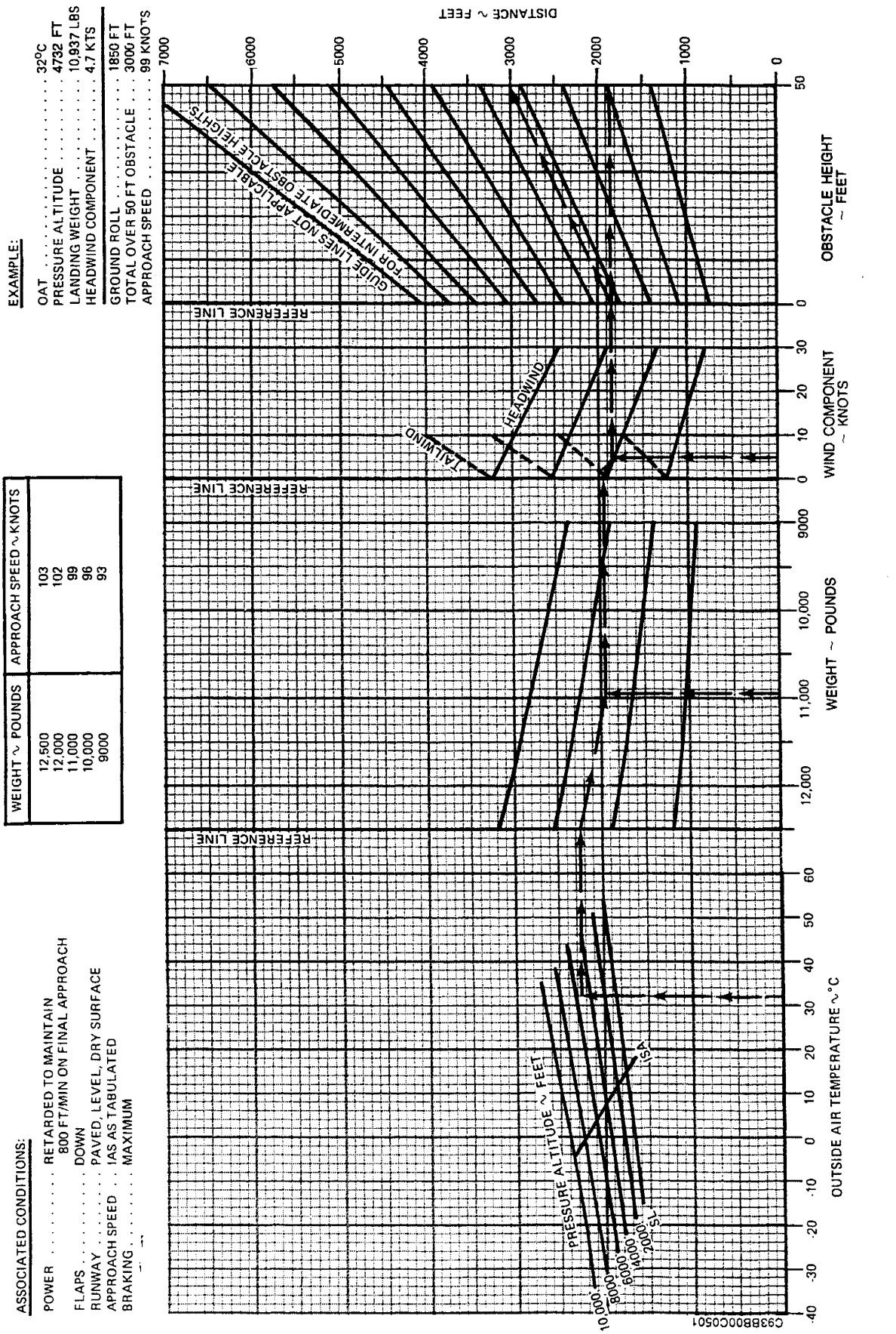
NORMAL LANDING DISTANCE WITHOUT PROPELLER REVERSING - FLAPS DOWN



EXAMPLE:

OAT	32°C
PRESSURE ALTITUDE	4732 FT
LANDING WEIGHT	10,937 LBS
HEADWIND COMPONENT	4.7 KTS
GROUND ROLL	1850 FT
TOTAL OVER 50 FT OBSTACLE	3000 FT
APPROACH SPEED	99 KNOTS

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
12,500	103
12,000	102
11,000	99
10,000	96
9000	93

ASSOCIATED CONDITIONS:

POWER	RETARDED TO MAINTAIN 800 FT/MIN ON FINAL APPROACH
FLAPS	DOWN
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS AS TABULATED
BRAKING	MAXIMUM

LANDING DISTANCE WITHOUT PROPELLER REVERSING

FLAPS UP

ASSOCIATED CONDITIONS

POWER . . . . . RETARDED TO MAINTAIN 900 FT/MIN ON FINAL APPROACH
PROPELLER CONTROLS . . . FULL FORWARD
FLAPS . . . . . UP
RUNWAY . . . . . PAVED, LEVEL, DRY SURFACE
APPROACH SPEED . . . . . IAS AS TABULATED
BRAKING . . . . . MAXIMUM

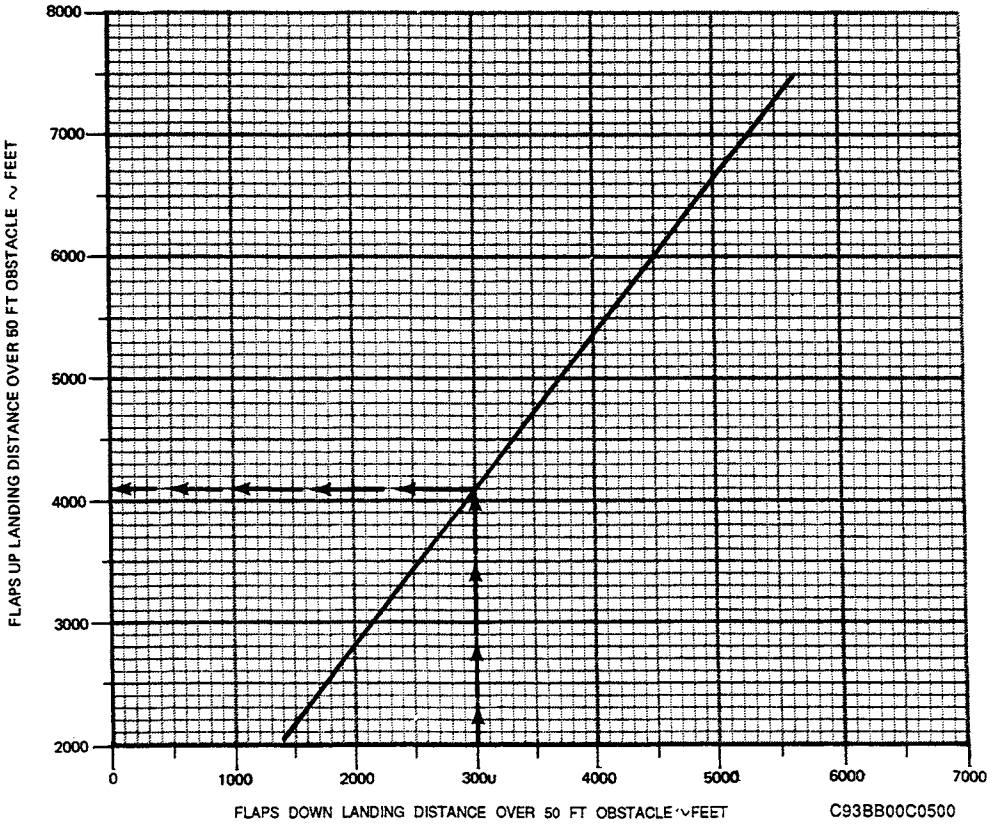
WEIGHT POUNDS	APPROACH SPEED ∿ KNOTS
12,500	132
12,000	130
11,000	126
10,000	122
9000	117

EXAMPLE:

FLAPS DOWN LANDING DISTANCE OVER 50 FT OBSTACLE . . . . 3000 FT
LANDING WEIGHT . . . . . 10,937 LBS

FLAPS UP LANDING DISTANCE OVER 50 FT OBSTACLE . . . . 4100 FT
APPROACH SPEED . . . . . 126 KNOTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN IS NORMAL PROCEDURE. USE THIS GRAPH WHEN IT IS NECESSARY TO LAND WITH FLAPS UP.
2. TO DETERMINE FLAPS-UP LANDING DISTANCE, READ FROM THE **LANDING DISTANCE WITHOUT PROPELLER REVERSING -FLAPS DOWN** GRAPH THE LANDING DISTANCE APPROPRIATE TO OAT, ALTITUDE, WEIGHT, WIND, AND 50-FT OBSTACLE. THEN ENTER THIS GRAPH WITH THE DERIVED VALUE AND READ THE FLAPS-UP LANDING DISTANCE.



LANDING DISTANCE WITHOUT PROPELLER REVERSING - FLAPS UP

For Training Purposes Only

LANDING DISTANCE WITH PROPELLER REVERSING - FLAPS DOWN

ASSOCIATED CONDITIONS:

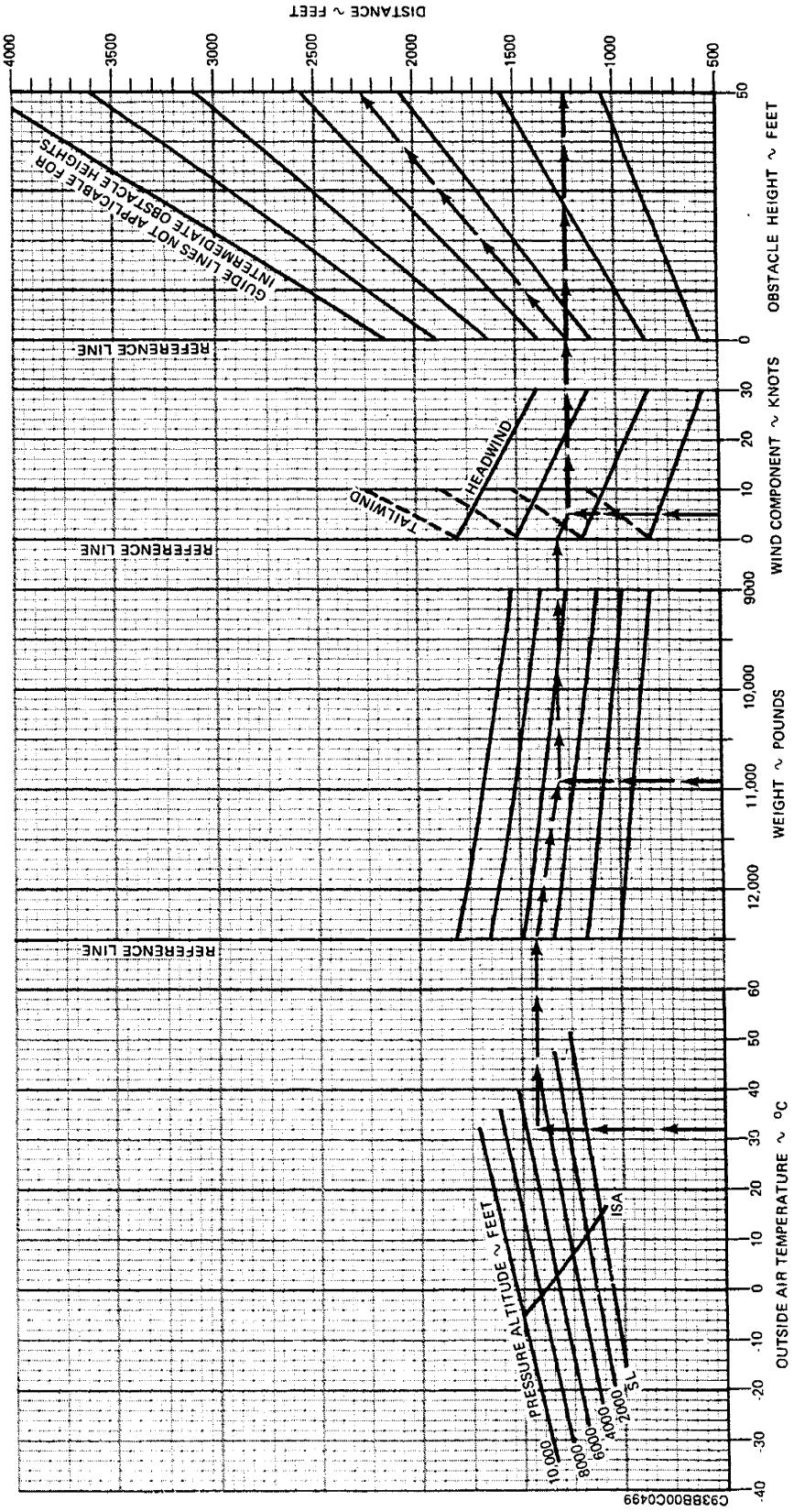
POWER ..... RETARD TO MAINTAIN 1000 FT/MIN ON FINAL APPROACH
FLAPS ..... DOWN
RUNWAY ..... PAVED, LEVEL, DRY SURFACE
APPROACH SPEED ..... IAS AS TABULATED
BRAKING ..... MAXIMUM
CONDITION LEVERS ..... HIGH IDLE
PROPELLER CONTROLS ..... FULL FORWARD
POWER LEVERS ..... MAXIMUM REVERSE AFTER TOUCHDOWN UNTIL FULLY STOPPED

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
12,500	103
12,000	102
11,000	99
10,000	96
9000	93

EXAMPLE:

OAT ..... 32°C
PRESSURE ALTITUDE ..... 4732 FT
LANDING WEIGHT ..... 10,937 POUNDS
HEADWIND COMPONENT ..... 4.7 KNOTS

GROUND ROLL ..... 1240 FEET
TOTAL OVER 50 FOOT OBSTACLE ..... 2260 FEET
APPROACH SPEED ..... 99 KNOTS



LANDING DISTANCE WITH PROPELLER REVERSING

FLAPS UP

ASSOCIATED CONDITIONS:

POWER. . . . . RETARD TO MAINTAIN 1000 FT/MIN ON FINAL APPROACH
FLAPS. . . . . UP
RUNWAY. . . . . PAVED, LEVEL, DRY SURFACE
APPROACH SPEED . . . . . IAS AS TABULATED
BRAKING. . . . . MAXIMUM
CONDITION LEVERS. . . HIGH IDLE
PROPELLER CONTROLS. FULL FORWARD
POWER LEVERS. . . . . MAXIMUM REVERSE AFTER TOUCHDOWN

WEIGHT ∿ POUNDS	APPROACH SPEED ∿ KNOTS
12,500	132
12,000	130
11,000	126
10,000	122
9000	117

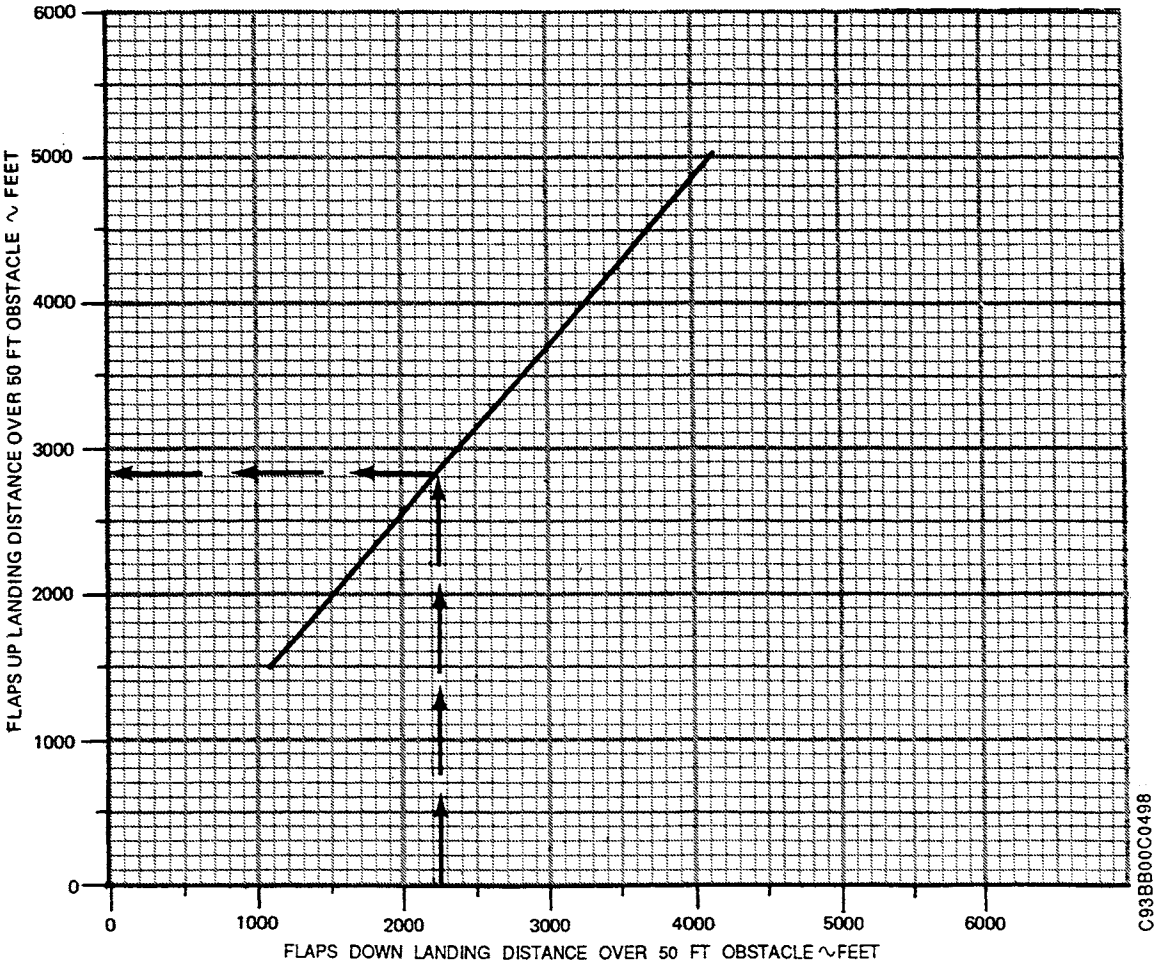
EXAMPLE:

FLAPS DOWN LANDING
DISTANCE OVER 50
FOOT OBSTACLE. . . . . 2260 FEET

LANDING WEIGHT. . . . . 10,937 POUNDS

FLAPS UP LANDING
DISTANCE OVER 50
FOOT OBSTACLE. . . . . 2820 FEET
APPROACH SPEED . . . . . 126 KNOTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN IS NORMAL PROCEDURE. USE THIS GRAPH WHEN IT IS NECESSARY TO LAND WITH FLAPS UP.
2. TO DETERMINE FLAPS-UP LANDING DISTANCE, READ FROM THE **LANDING DISTANCE WITH PROPELLER REVERSING – FLAPS DOWN** GRAPH THE LANDING DISTANCE APPROPRIATE TO OAT, ALTITUDE, WEIGHT, WIND, AND 50-FT OBSTACLE. THEN ENTER THIS GRAPH WITH THE DERIVED VALUE AND READ THE FLAPS-UP LANDING DISTANCE.



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SECTION VI
WEIGHT & BALANCE/EQUIP LIST
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B200

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Equipment List .....Prepared on an Individual Airplane Basis

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WEIGHING INSTRUCTIONS

Periodic weighing of the airplane may be required to keep the Basic Empty Weight current. Frequency of weighing is to be determined by the operator. All changes to the airplane affecting weight and/or balance are the responsibility of the airplane operator.

1. Airplane may be weighed on wheels or jacks points. Three jack points are provided: one on the nose section of the fuselage at station 83.5, and one on each wing center section rear spar at station 225.5. Wheel reaction locations should be measured as described in paragraph 6 below.
2. Fuel should be drained preparatory to weighing. Tanks are drained from the regular drain ports with the airplane in static ground attitude. When tanks are drained, 10.5 pounds of unusable fuel remains in the airplane at an arm of 188.0 inches. The remainder of the unusable fuel to be added to a drained system is 33.5 pounds at station 164.0.
3. Engine oil must be at the full level in each tank. Total engine oil aboard when both tanks are full is 62 pounds at an arm of 131.0 inches.
4. To determine airplane configuration at time of weighing, installed equipment is checked against the airplane equipment list or superseding forms. All equipment must be in its proper place during weighing.
5. The airplane is placed on the scales in a level attitude. Leveling screws are located on the fuselage entrance door frame. Leveling is accomplished with a plumb bob. Jack pad leveling may require the nose gear shock to be secured in the static position to prevent its extension. Wheel weighings can be leveled by varying the amount of air in the shocks and tires.
6. Measurement of the reaction arms for a wheel weighing is made using the nose jacking point for a reference. Using a steel measuring tape, measurements are taken with the airplane level on the scales from a reference (a plumb bob hung from the center of the nose jacking point) to the axle center line of the nose gear and then from the nose gear axle center line to the main wheel axle center line. The main wheel axles center line is best located by stretching a string across from one main wheel to the other. All measurements are to be taken with the tape level with the hangar floor and parallel to the fuselage center line. The locations of the wheel reactions will be approximately at an arm of 209 inches for main wheels and 30 inches for the nose wheel.
7. The Basic Empty Weight and Moment are determined from the scale readings. Items weighed which are not part of the empty airplane are subtracted, i.e., usable fuel. Unusable fuel and engine oil are added if not already in the airplane.
8. Weighing should always be made in a enclosed area which is free from air currents. The scales used should be properly calibrated and certified in accordance with the Bureau of Standards.

NOTE

The certificated maximum weight (this is 12,590 lbs ramp weight for a normal category airplane) may not be exceeded with:

a. Full fuel and minimum crew (one pilot, 170 lbs.) and/or

b. Each seat occupied and enough fuel for one-half hour operation at maximum continuous power.

For Training Purposes Only

BASIC EMPTY WEIGHT AND BALANCE

DATE: _____

SERIAL NO: _____

REGISTRATION NO: _____

PREPARED BY:

STRUT POSITION -	NOSE	MAIN
EXTENDED	29.4	208.5
COMPRESSED	30.8	210.5

JACK POINT LOCATION	
FORWARD	83.5
AFT	225.5

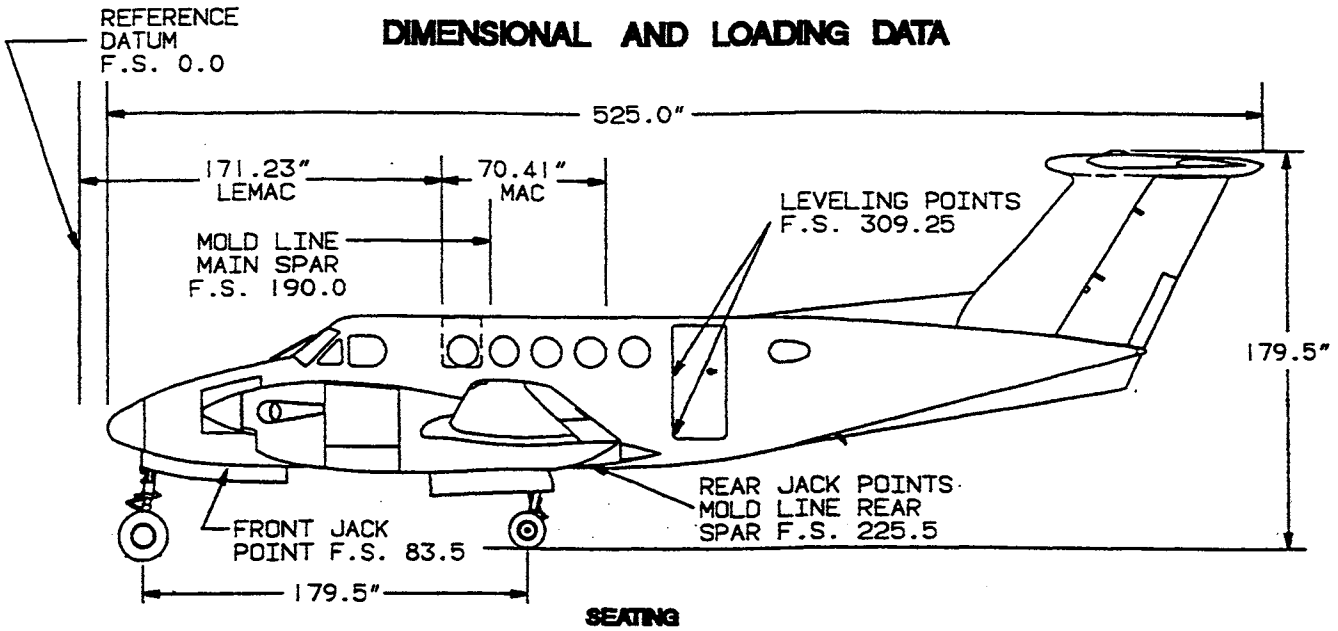
REACTION WHEEL - JACK POINTS	SCALE READING	TARE	NET WEIGHT	STATION OR ARM	MOMENT
LEFT MAIN					
RIGHT MAIN					
SUB TOTAL					
NOSE					
TOTAL (AS WEIGHED)					

SPACE BELOW PROVIDED FOR ADDITIONS AND SUBTRACTIONS TO AS WEIGHED CONDITION

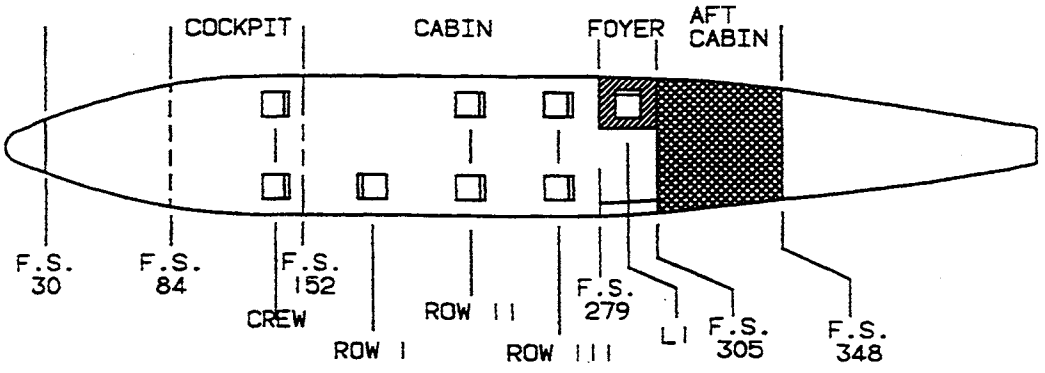
EMPTY WEIGHT			
ENGINE OIL UNUSABLE FUEL	62 44	131 170	8122 7480
BASIC EMPTY WEIGHT			

B200-601-217

DIMENSIONAL AND LOADING DATA



SEATING



OCCUPANT	CENTROID
CREW	F.S. 129
ROW I	F.S. 176
ROW II	F.S. 215
ROW III	F.S. 259
LI	F.S. 293

CONFIGURATION	BAGGAGE COMPARTMENT	BAGGAGE CAPACITY	CENTROID
SIDE PASSENGER SEAT AND TOILET	* FOYER -	150	F.S. 293
	AFT CABIN -	**	F.S. 325

NOTE: FOYER IS NOT EQUIPPED FOR LOOSE BAGGAGE; CLOTHING ON HANGERS MAY BE HUNG FROM THE ROD PROVIDED.

* LOADING DATA FOR STANDARD CONFIGURATIONS ONLY.

** 410 LBS FOR SERIALS PRIOR TO BB-1091, EXCEPT BB-1052 NOT EQUIPPED WITH BEECH KIT P/N 101-5068; 550 LBS FOR SERIALS BB-1052, BB-1091 AND AFTER, AND PRIOR AIRPLANES EQUIPPED WITH BEECH KIT P/N 101-5068.

C93B806C0904 C

LOADING DATA
CARGO CONFIGURATION

NOTES:

1. ALL CARGO IN SECTIONS A, B, C & D MUST BE SUPPORTED ON AND SECURED TO THE SEAT TRACKS BY AN FAA APPROVED SYSTEM.

2. CONCENTRATED CARGO LOADS IN SECTIONS A, B, C & D MUST NOT EXCEED 200 LBS. PER SQUARE FOOT & MUST BE SUPPORTED ON THE SEAT RAILS.

3. CARGO IN SECTION E IS TO BE SECURED BY BEECH FURNISHED BAGGAGE NET, WEBBING, OR STRAPS.

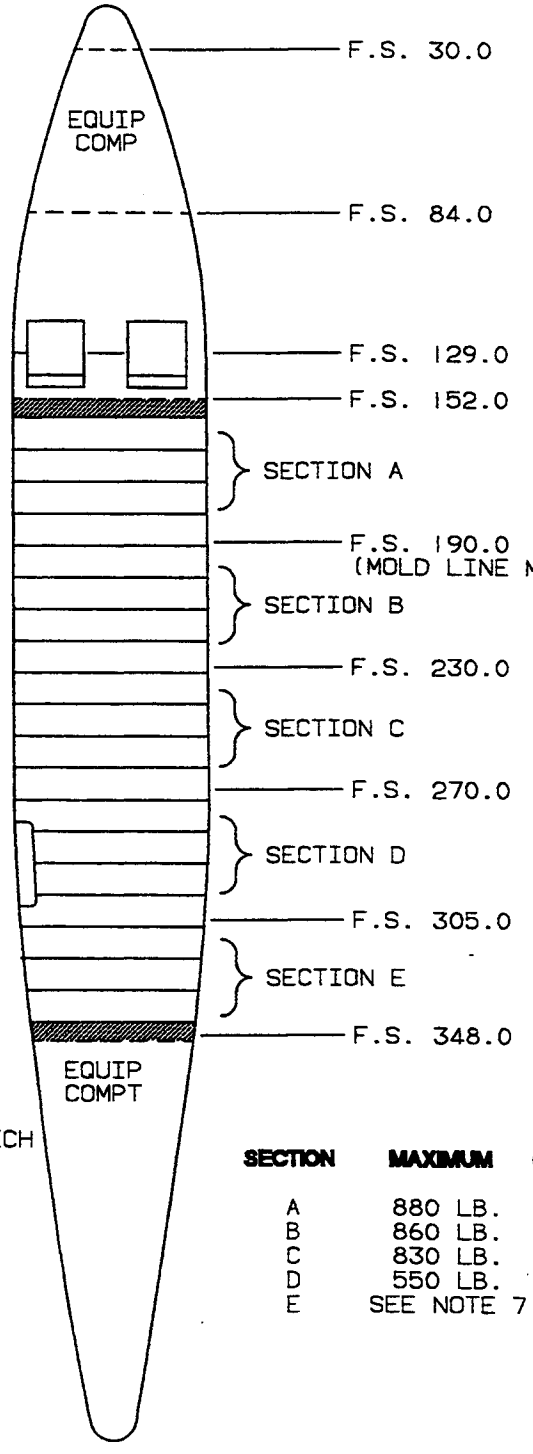
4. FOOTMAN LOOPS IN SECTION E ARE TO BE USED TO SECURE CARGO BAGGAGE IN THAT AREA ONLY.

5. CONCENTRATED FLOOR LOADINGS OF CARGO OR BAGGAGE IN SECTION E MUST NOT EXCEED 100 LBS. PER SQUARE FOOT.

6. ANY EXCEPTIONS TO THE ABOVE PROCEDURES WILL REQUIRE APPROVAL BY A LOCAL FAA OFFICE.

7. 410 LBS FOR SERIALS PRIOR TO BB-1091, EXCEPT BB-1052, NOT EQUIPPED WITH BEECH KIT P/N 101-5068; 550 LBS FOR SERIALS BB-1052, BB-1091 AND AFTER AND PRIOR AIRPLANES EQUIPPED WITH BEECH KIT P/N 101-5068.

CARGO TIEDOWN PROVISIONS ARE NOT PROVIDED.

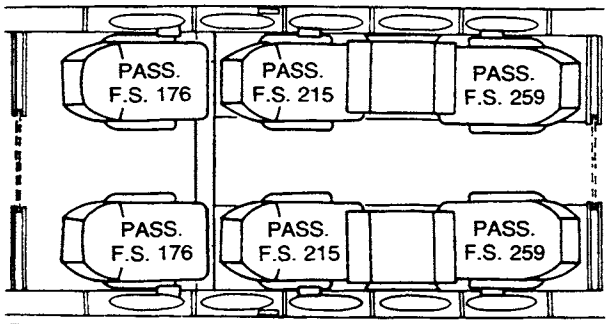


SECTION	MAXIMUM	CENTROID ARM
A	880 LB.	F.S. 171
B	860 LB.	F.S. 210
C	830 LB.	F.S. 250
D	550 LB.	F.S. 288
E	SEE NOTE 7	F.S. 325

BB06C960418 C

LOADING DATA (CARGO CONFIGURATION)

CABIN ARRANGEMENT DIAGRAM



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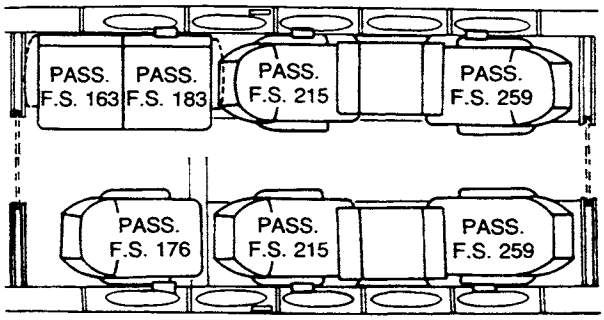

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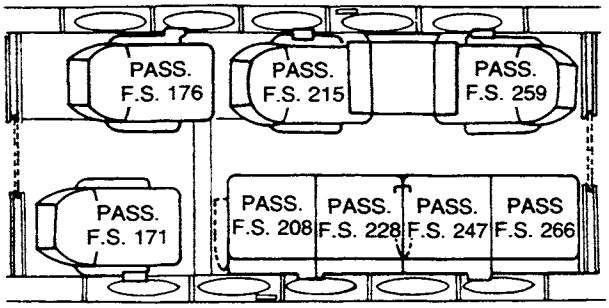

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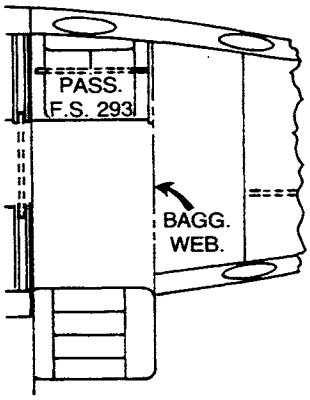
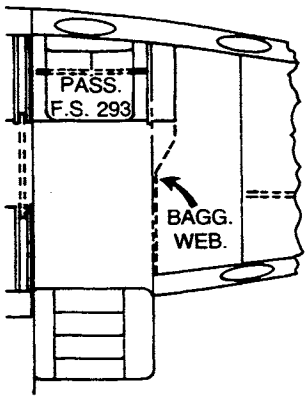
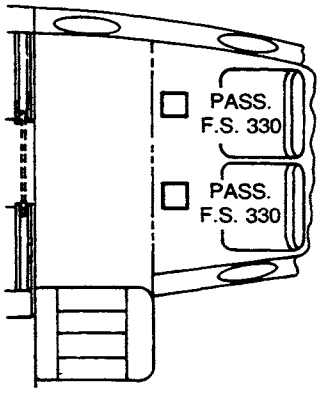


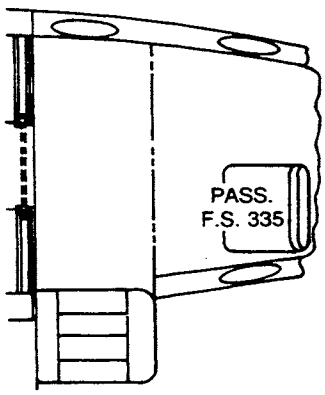
☐ SIDE FACING TOILET



☐ SIDE FACING TOILET WITH VANITY



☐ AFT FOLD UP SEATS



☐ FRONT FACING TOILET

NOTES:

The cabin seating may be arranged in different combinations. The diagrams marked ☒ above represent the seating arrangement established for this airplane prior to delivery. The passenger locations shown on the designated diagram are averages. Additional data for modified arrangements are noted. No diagrams are included for Hi-density versions.

200-603-29

USEFUL LOAD WEIGHTS AND MOMENTS

OCCUPANTS

USE COLUMNS MARKED X	CREW	CABIN CHAIRS				LAVATORY SEAT	
	F.S. 129	F.S. 171	F.S. 176	F.S. 215	F.S. 259	F.S. 293	F.S. 335
WEIGHT	MOMENT/100						
80	103	137	141	172	207	234	268
90	116	154	158	194	233	264	302
100	129	171	176	215	259	293	335
110	142	188	194	237	285	322	369
120	155	205	211	258	311	352	402
130	168	222	229	280	337	381	436
140	181	239	246	301	363	410	469
150	194	257	264	323	389	440	503
160	206	274	282	344	414	469	536
170	219	291	299	366	440	498	570
180	232	308	317	387	466	527	603
190	245	325	334	409	492	557	637
200	258	342	352	430	518	586	670
210	271	359	370	452	544	615	704
220	284	376	387	473	570	645	737
230	297	393	405	495	596	674	771
240	310	410	422	516	622	703	804

OCCUPANTS

USE COLUMNS MARKED X	TWO PLACE COUCH FORWARD POSITION		FOUR PLACE COUCH AFT POSITION				AFT FOLD UP SEATS
	F.S. 163	F.S. 183	F.S. 208	F.S. 228	F.S. 247	F.S. 266	F.S. 330
WEIGHT	MOMENT/100						
80	130	146	166	182	198	213	264
90	147	165	187	205	222	239	297
100	163	183	208	228	247	266	330
110	179	201	229	251	272	293	363
120	196	220	250	274	296	319	396
130	212	238	270	296	321	346	429
140	228	256	291	319	346	372	462
150	245	275	312	342	371	399	495
160	261	293	333	365	395	426	528
170	277	311	354	388	420	452	561
180	293	329	374	410	445	479	594
190	310	348	395	433	469	505	627
200	326	366	416	456	494	532	660
210	342	384	437	479	519	559	693
220	359	403	458	502	543	585	726
230	375	421	478	524	568	612	759
240	391	439	499	547	593	638	792
250	408	458	520	570	618	665	825

USEFUL LOAD WEIGHTS AND MOMENTS

BAGGAGE

WEIGHT	(Clothing on Hangers) FOYER F.S. 292	AFT CABIN F.S. 325
	MOMENT/100	
10	29	33
20	58	65
30	88	98
40	117	130
50	146	163
60	175	195
70	204	228
80	234	260
90	263	293
100	292	325
200		650
300		975
370		1203
400		1300
500		1625
510		1658
550		1788

*SEE CABIN ARRANGEMENT DIAGRAM

CABINET CONTENTS

USE COLUMNS MARKED X ↓	CHART CASES F.S. 148	FORWARD CABINET F.S. 158	MIDDLE CABINET F.S. 196	AFT CABINET F.S. 272	COUCH DRAWERS F.S. 173		
WEIGHT	MOMENT/100						
10	15	16	20	27	17		
20	30	32	39	54	35		
30	44	47	59	82	52		
40	59	63	78	109			
50	74	79					
60		95					
70		111					
80		126					
90		142					
100		158					

NOTE: Weight and Moment/100 of Cabinet Contents must be included in all loading computations.

USEFUL LOAD WEIGHTS AND MOMENTS

CARGO

WEIGHT	COMPARTMENT *				
	A	B	C	D	E
	F.S. 152-190	*F.S. 190-230*	*F.S. 230-270*	*F.S. 270-305*	*F.S. 305-348*
	CENTROID				
	F.S. 171	*F.S. 210*	*F.S. 250*	*F.S. 288*	*F.S. 325*
	MOMENT/100				
10	17	21	25	29	33
20	34	42	50	58	65
30	51	63	75	86	98
40	68	84	100	115	130
50	86	105	125	144	163
60	103	126	150	173	195
70	120	147	175	202	228
80	137	168	200	230	260
90	154	189	225	259	293
100	171	210	250	288	325
200	342	420	500	576	650
300	513	630	750	864	975
400	684	840	1000	1152	1300
500	855	1050	1250	1440	1625
510	872	1071	1276	1469	1658
550	940	1155	1375	1584	1788
600	1026	1260	1500		
700	1197	1470	1750		
800	1368	1680	2000		
830	1419	1743	2075		
860	1471	1806			
880	1505				

NOTE: All cargo must be supported by the seat tracks in a uniform distribution and tied down to the tracks by an FAA approved method.

*Refer to LOADING DATA CARGO CONFIGURATION.

USEFUL LOAD WEIGHTS AND MOMENTS

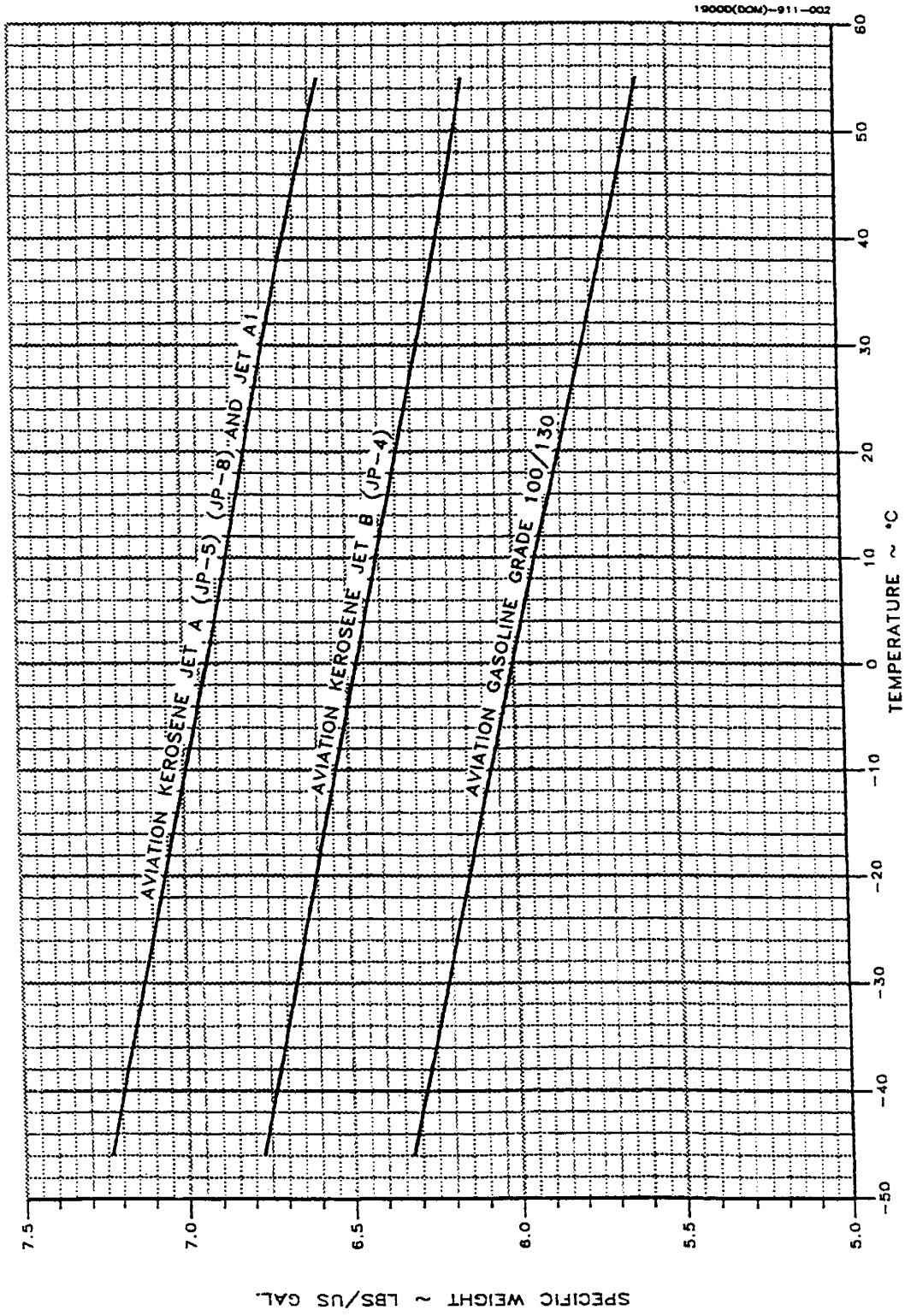
USABLE FUEL

GALLONS	6.4 LB/GAL		6.5 LB/GAL		6.6 LB/GAL		6.7 LB/GAL	
	WEIGHT	MOMENT 100	WEIGHT	MOMENT 100	WEIGHT	MOMENT 100	WEIGHT	MOMENT 100
10	64	99	65	100	66	102	67	103
20	128	197	130	200	132	203	134	206
30	192	305	195	310	198	314	201	319
40	256	423	260	430	264	436	268	443
50	320	542	325	550	330	559	335	567
60	384	662	390	672	396	683	402	693
70	448	782	455	794	462	807	469	819
80	512	904	520	918	528	932	536	946
90	576	1023	585	1039	594	1055	603	1071
100	640	1142	650	1160	660	1178	670	1196
110	704	1260	715	1280	726	1300	737	1319
120	768	1379	780	1400	792	1422	804	1443
130	832	1496	845	1519	858	1543	871	1566
140	896	1615	910	1640	924	1665	938	1690
150	960	1734	975	1761	990	1788	1005	1815
160	1024	1852	1040	1881	1056	1910	1072	1939
170	1088	1971	1105	2002	1122	2033	1139	2064
180	1152	2090	1170	2122	1188	2155	1206	2188
190	1216	2209	1235	2244	1254	2279	1273	2313
200	1280	2328	1300	2365	1320	2401	1340	2437
210	1344	2447	1365	2486	1386	2524	1407	2562
220	1408	2567	1430	2607	1452	2647	1474	2687
230	1472	2686	1495	2728	1518	2770	1541	2812
240	1536	2806	1560	2850	1584	2894	1608	2938
250	1600	2926	1625	2971	1650	3017	1675	3063
260	1664	3045	1690	3093	1716	3140	1742	3188
270	1728	3164	1755	3213	1782	3263	1809	3312
280	1792	3283	1820	3334	1848	3386	1876	3437
290	1856	3402	1885	3455	1914	3508	1943	3562
300	1920	3521	1950	3576	1980	3631	2010	3686
310	1984	3641	2015	3698	2046	3754	2077	3811
320	2048	3760	2080	3819	2112	3878	2144	3936
330	2112	3880	2145	3940	2178	4001	2211	4062
340	2176	3999	2210	4062	2244	4124	2278	4187
350	2240	4119	2275	4184	2310	4248	2345	4312
360	2304	4244	2340	4310	2376	4377	2412	4443
370	2368	4365	2405	4434	2442	4502	2479	4570
380	2432	4489	2470	4560	2508	4630	2546	4700
386	2470	4562	2509	4634	2548	4706	2586	4776
400	2560	4741	2600	4815	2640	4889	2680	4963
410	2624	4869	2665	4945	2706	5021	2747	5097
420	2688	4997	2730	5075	2772	5153	2814	5231
430	2752	5126	2795	5206	2838	5286	2881	5366
440	2816	5255	2860	5337	2904	5419	2948	5501
450	2880	5386	2925	5470	2970	5554	3015	5638
460	2944	5514	2990	5600	3036	5686	3082	5773
470	3008	5645	3055	5733	3102	5821	3149	5909
480	3072	5775	3120	5866	3168	5956	3216	6046
490	3136	5907	3185	5999	3234	6091	3283	6184
500	3200	6040	3250	6134	3300	6229	3350	6323
510	3264	6172	3315	6269	3366	6365	3417	6462
520	3328	6307	3380	6405	3432	6504	3484	6602
530	3392	6441	3445	6542	3498	6643	3551	6743
540	3456	6573	3510	6676	3564	6779	3618	6881
544	3482	6626	3536	6729	3590	6832	3645	6936

DENSITY VARIATION OF AVIATION FUEL

BASED ON AVERAGE SPECIFIC GRAVITY

FUEL	AVERAGE SPECIFIC GRAVITY
JET A (JP-5) (JP-8) AND JET A1	0.819 AT 15°C
JET B (JP-4)	0.764 AT 15°C
AV GAS GRADE 100/130	0.706 AT 15°C



LOADING INSTRUCTIONS

It is the responsibility of the airplane operator to ensure that the airplane is properly loaded. At the time of delivery, Beech Aircraft Corporation provides the necessary weight and balance data to compute individual loadings. All subsequent changes in airplane weight and balance are the responsibility of the airplane owner and/or operator.

The basic empty weight and moment of the airplane at the time of delivery are shown on the Basic Empty Weight and Balance form. Useful load items which may be loaded into the airplane are shown on the Useful Load Weight and Moment tables. The minimum and maximum moments approved by the FAA are shown on the Moment Limits vs. Weight graph or table. These moments correspond to the forward and aft center of gravity flight limits (landing gear down) for a particular weight. All moments are divided by 100 to simplify computations.

CARGO LOADING

The method of loading cargo, its placement in the airplane and the method of restraint should each be determined before starting the actual loading.

For loads that are evenly distributed in a given section, the useful Load Table under the heading of Cargo Compartment should be used. For any load that cannot be located at the centroid of a section or that extends over more than one section, it will be necessary to determine its own CG and its location in the airplane. Determine the CG arm (Fuselage Station) by measuring in inches, from a known location in the cabin to the CG of the load. Determine the "moment" for the load by multiplying the weight by the CG arm (Fuselage Station). This resultant should be divided by 100 to be compatible with other loading data.

COMPUTING PROCEDURE

1. Record the basic empty weight and moment from the Basic Empty Weight and Balance form (or from the latest superseding forms). The moment must be divided by 100 to correspond to Useful Load Moments.
2. Record the weight and corresponding moment of each item to be carried. These values are found on the Useful Load Weight and Moment tables.
3. Total the weight column and moment column. The total weight without usable fuel must not exceed the Maximum Zero Fuel Weight limitation of 11,000 pounds. All weight in excess of this limitation must be fuel. The auxiliary tanks may be used only when the main tanks are completely filled. The total take-off weight must not exceed the maximum allowable take-off weight and the total moment must be within the minimum and maximum moments shown on the Moment Limits Vs. Weight table or graph.
4. Using the page titled Useful Load Weights and Moments - Usable Fuel, determine the weight and corresponding moment of fuel to be used by subtracting the amount on board on landing from the amount on board at takeoff.
5. For landing configuration weight and balance, subtract the weight and moment of fuel to be used from the take-off weight and moment. The landing moment must be within the minimum and maximum moments shown on Weight and Moment Limits table for that weight. If the total moment is less than the minimum moment allowed, useful load items must be shifted aft, or forward load items reduced. If the total moment is greater than the maximum moment allowed, useful load items must be shifted forward, or aft load items reduced. If the quantity or location of load items is changed, the calculations must be revised and the moments rechecked.
6. Loadings may be made on the Weight and Balance Loading Form.

Beechcraft SUPER KING AIR B200/B200C

WEIGHT AND BALANCE LOADING FORM

SERIAL	REGISTRATION NO:	DATE:

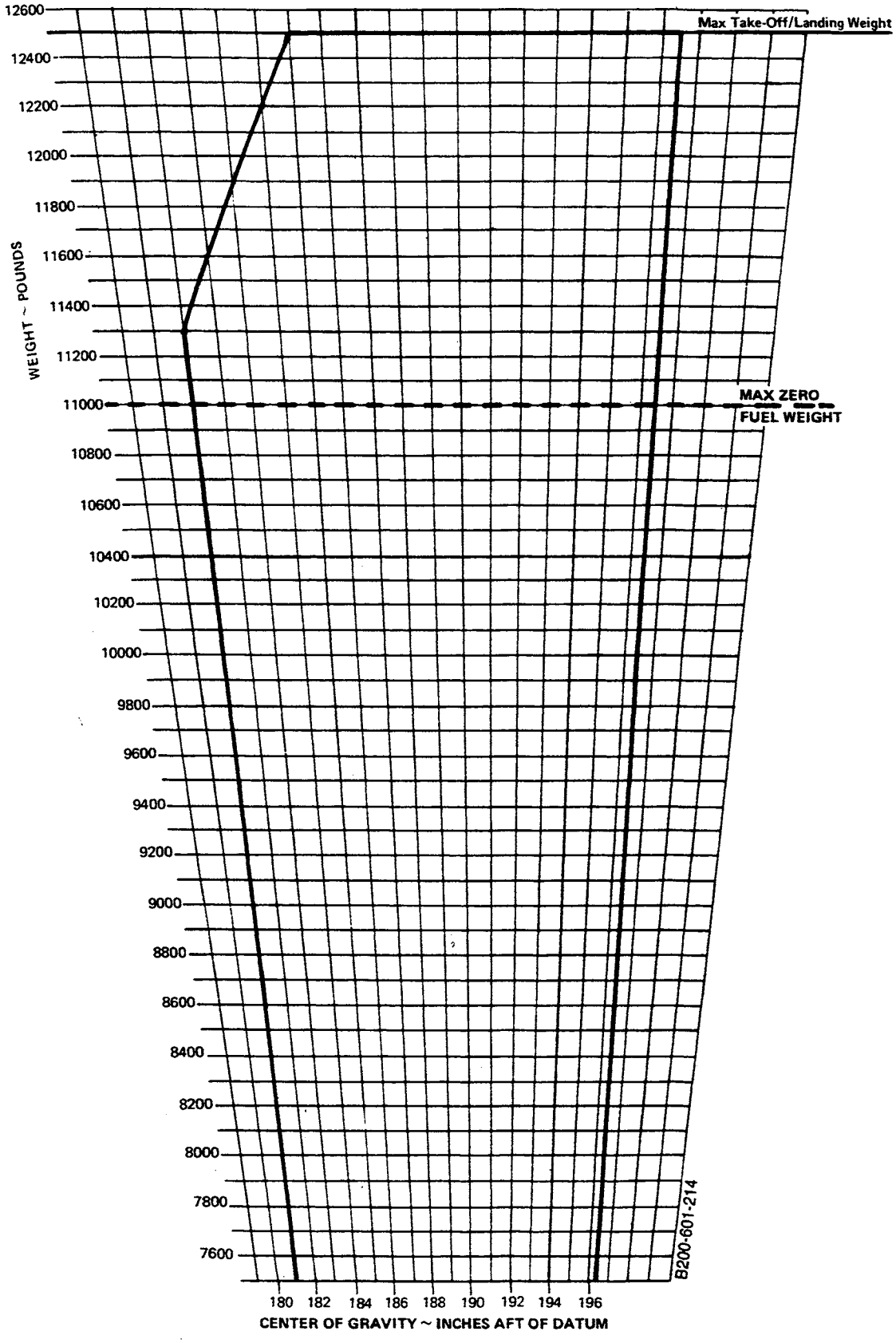
PASSENGERS OR CARGO		
ITEM LOCATION (ROW, F.S.,ETC)	WEIGHT	MOM/100
TOTAL PASSENGERS OR CARGO		

REF	ITEM	WEIGHT	MOM/100
1.	BASIC EMPTY COND.		
2.	PILOT		
3.	PASSENGERS OR CARGO		
4.	BAGGAGE		
5.	CABINET CONTENTS		
6.	**SUB TOTAL** ZERO FUEL CONDITION. DO NOT EXCEED 11,000 LBS		
7.	FUEL LOADING		
8.	**SUB TOTAL** RAMP CONDITION		
9.	*LESS FUEL FOR START, TAXI, AND TAKE OFF		
10.	**SUB TOTAL** TAKE OFF CONDITION		
11.	LESS FUEL TO DESTINATION		
12.	**LANDING CONDITION**		

*Fuel for start, taxi and take-off is normally 90 lbs at an average moment/100 of 177.

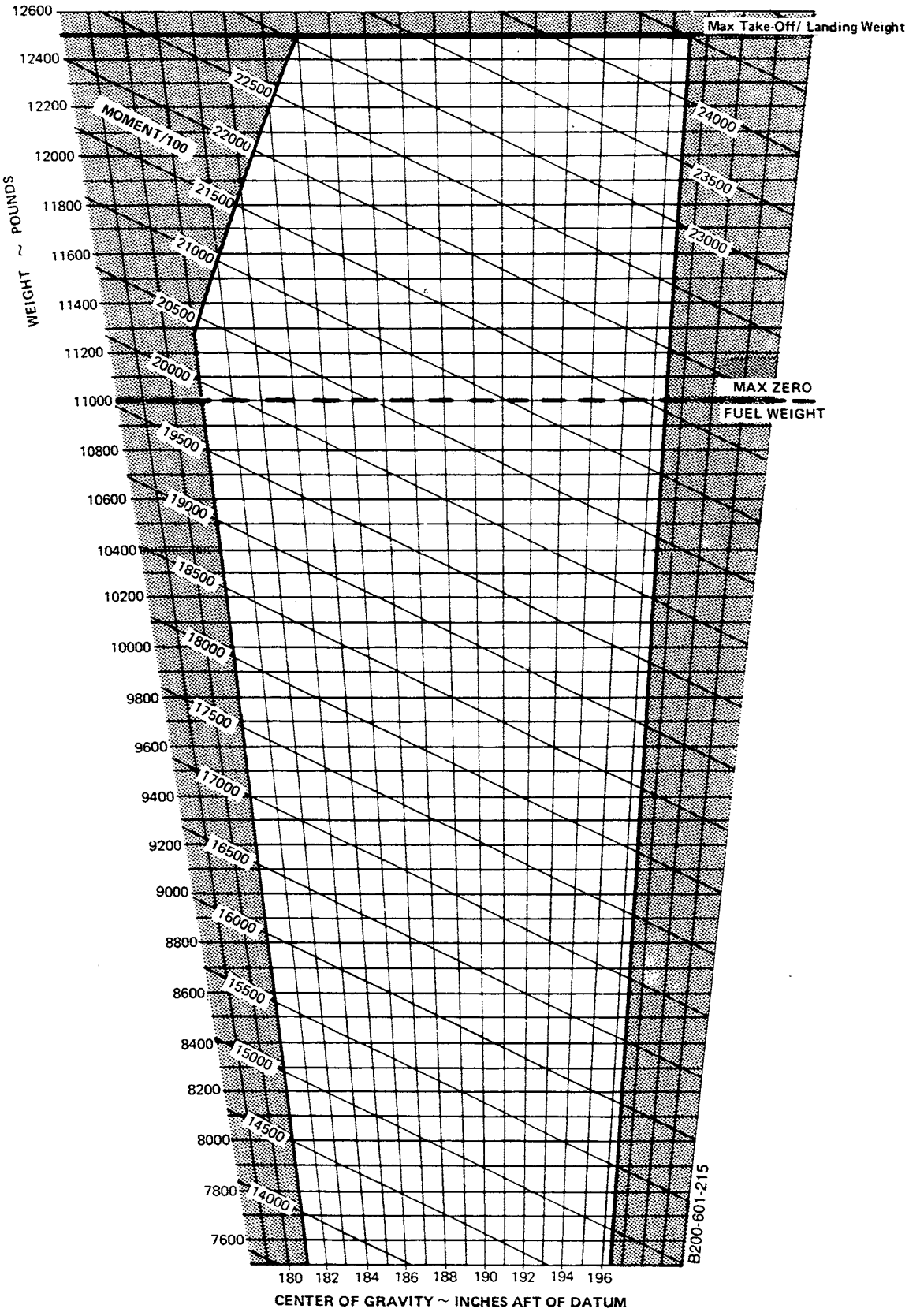
B200-601-219

WEIGHT AND BALANCE DIAGRAM



For Training Purposes Only

MOMENT LIMITS vs WEIGHT



MOMENT LIMITS VS. WEIGHT

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
7200	13032	14141
7250	13122	14239
7300	13213	14337
7350	13304	14435
7400	13394	14534
7450	13484	14632
7500	13575	14730
7550	13666	14828
7600	13756	14926
7650	13846	15025
7700	13937	15123
7750	14028	15221
7800	14118	15319
7850	14208	15417
7900	14299	15516
7950	14390	15614
8000	14480	15712
8050	14570	15810
8100	14661	15908
8150	14752	16007
8200	14842	16105
8250	14932	16203
8300	15023	16301
8350	15114	16399
8400	15204	16498
8450	15294	16596
8500	15385	16694
8550	15476	16792
8600	15566	16890
8650	15656	16989
8700	15747	17087
8750	15838	17185
8800	15928	17283
8850	16018	17381
8900	16109	17480
8950	16200	17578
9000	16290	17676
9050	16380	17774
9100	16471	17872
9150	16562	17971
9200	16652	18069
9250	16742	18167
9300	16833	18265
9350	16924	18363
9400	17014	18462
9450	17104	18560
9500	17195	18658
9550	17286	18756
9600	17376	18854
9650	17466	18953
9700	17557	19051
9750	17648	19149
9800	17738	19247
9850	17828	19345

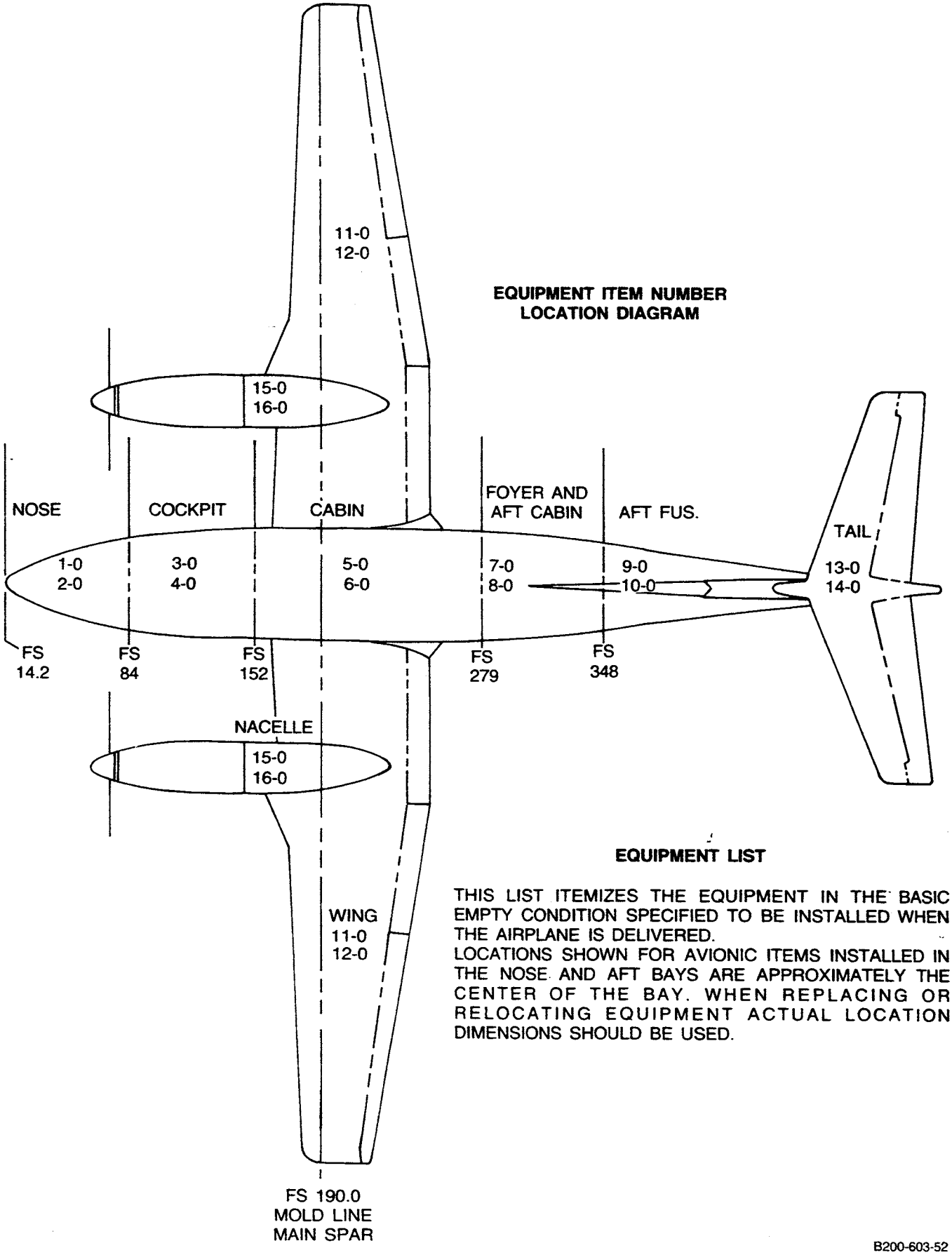
MAX ZERO FUEL WEIGHT

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
9900	17919	19444
9950	18010	19542
10000	18100	19640
10050	18190	19738
10100	18281	19836
10150	18372	19935
10200	18462	20033
10250	18552	20131
10300	18643	20229
10350	18734	20327
10400	18824	20426
10450	18914	20524
10500	19005	20622
10550	19096	20720
10600	19186	20818
10650	19276	20917
10700	19367	21015
10750	19458	21113
10800	19548	21211
10850	19638	21309
10900	19729	21408
10950	19820	21506
11000	19910	21604
11050	20000	21702
11100	20091	21800
11150	20182	21899
11200	20272	21997
11250	20362	22095
11300	20461	22193
11350	20570	22291
11400	20679	22390
11450	20789	22488
11500	20898	22586
11550	21008	22684
11600	21118	22782
11650	21228	22881
11700	21338	22979
11750	21449	23077
11800	21559	23175
11850	21670	23273
11900	21781	23372
11950	21892	23470
12000	22003	23568
12050	22115	23666
12100	22226	23764
12150	22338	23863
12200	22450	23961
12250	22562	24059
12300	22674	24157
12350	22787	24255
12400	22899	24354
12450	23012	24452
12500	23125	24550

CENTER OF GRAVITY LIMITS (LANDING GEAR DOWN)

WEIGHT CONDITION	FORWARD CG LIMIT	AFT CG LIMIT
12,500 LBS (MAX. TAKE-OFF OR LANDING)	185.0	196.4
11,279 LBS OR LESS	181.0	196.4

EQUIPMENT ITEM NUMBER LOCATION DIAGRAM



EQUIPMENT LIST

THIS LIST ITEMIZES THE EQUIPMENT IN THE BASIC EMPTY CONDITION SPECIFIED TO BE INSTALLED WHEN THE AIRPLANE IS DELIVERED.
LOCATIONS SHOWN FOR AVIONIC ITEMS INSTALLED IN THE NOSE AND AFT BAYS ARE APPROXIMATELY THE CENTER OF THE BAY. WHEN REPLACING OR RELOCATING EQUIPMENT ACTUAL LOCATION DIMENSIONS SHOULD BE USED.

B200-603-52

SECTION VI
WEIGHT & BALANCE/EQUIP LIST
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B200C

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Equipment List ..... Prepared on an Individual Airplane Basis

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WEIGHING INSTRUCTIONS

Periodic weighing of the airplane may be required to keep the Basic Empty Weight current. Frequency of weighing is to be determined by the operator. All changes to the airplane affecting weight and/or balance are the responsibility of the airplane operator.

1. Airplane may be weighed on wheels or jacks points. Three jack points are provided: one on the nose section of the fuselage at station 83.5, and one on each wing center section rear spar at station 225.5. Wheel reaction locations should be measured as described in paragraph 6 below.
2. Fuel should be drained preparatory to weighing. Tanks are drained from the regular drain ports with the airplane in static ground attitude. When tanks are drained, 10.5 pounds of unusable fuel remains in the airplane at an arm of 188.0 inches. The remainder of the unusable fuel to be added to a drained system is 33.5 pounds at station 164.0.
3. Engine oil must be at the full level in each tank. Total engine oil aboard when both tanks are full is 62 pounds at an arm of 131.0 inches.
4. To determine airplane configuration at time of weighing, installed equipment is checked against the airplane equipment list or superseding forms. All equipment must be in its proper place during weighing.
5. The airplane is placed on the scales in a level attitude. Leveling screws are located on the fuselage entrance door frame. Leveling is accomplished with a plumb bob. Jack pad leveling may require the nose gear shock to be secured in the static position to prevent its extension. Wheel weighings can be leveled by varying the amount of air in the shocks and tires.
6. Measurement of the reaction arms for a wheel weighing is made using the nose jacking point for a reference. Using a steel measuring tape, measurements are taken with the airplane level on the scales from a reference (a plumb bob hung from the center of the nose jacking point) to the axle center line of the nose gear and then from the nose gear axle center line to the main wheel axle center line. The main wheel axles center line is best located by stretching a string across from one main wheel to the other. All measurements are to be taken with the tape level with the hangar floor and parallel to the fuselage center line. The locations of the wheel reactions will be approximately at an arm of 209 inches for main wheels and 30 inches for the nose wheel.
7. The Basic Empty Weight and Moment are determined from the scale readings. Items weighed which are not part of the empty airplane are subtracted, i.e., usable fuel. Unusable fuel and engine oil are added if not already in the airplane.
8. Weighing should always be made in a enclosed area which is free from air currents. The scales used should be properly calibrated and certified in accordance with the Bureau of Standards.

NOTE

The certificated maximum weight (this is 12,590 lbs ramp weight for a normal category airplane) may not be exceeded with:

a. Full fuel and minimum crew (one pilot, 170 lbs.) and/or

b. Each seat occupied and enough fuel for one-half hour operation at maximum continuous power.

BASIC EMPTY WEIGHT AND BALANCE

DATE: _____

SERIAL NO: _____

REGISTRATION NO: _____

PREPARED BY: _____

STRUT POSITION -	NOSE	MAIN
EXTENDED	29.4	208.5
COMPRESSED	30.8	210.5

JACK POINT LOCATION	
FORWARD	83.5
AFT	225.5

REACTION WHEEL - JACK POINTS	SCALE READING	TARE	NET WEIGHT	STATION OR ARM	MOMENT
LEFT MAIN					
RIGHT MAIN					
SUB TOTAL					
NOSE					
TOTAL (AS WEIGHED)					

SPACE BELOW PROVIDED FOR ADDITIONS AND SUBTRACTIONS TO AS WEIGHED CONDITION

EMPTY WEIGHT			
ENGINE OIL UNUSABLE FUEL	62 44	131 170	8122 7480
BASIC EMPTY WEIGHT			

B200 601-217

Change from Passenger to Cargo Configuration

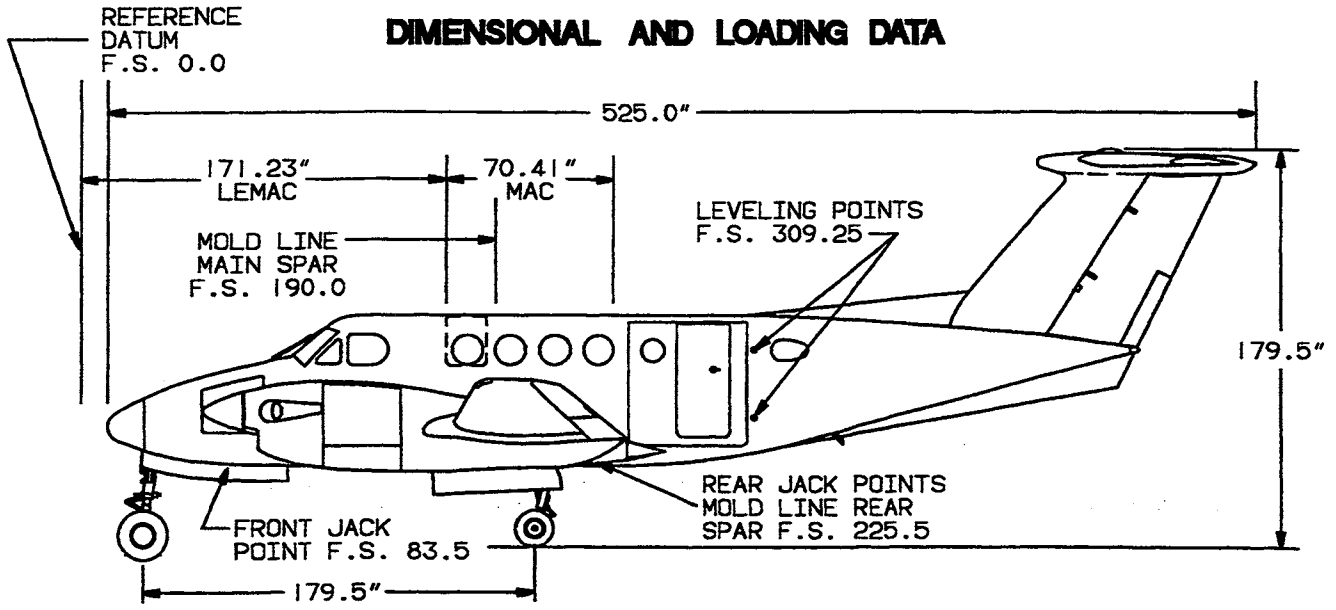
Date _____ Ser. No. _____ Reg. No. _____

Item	Weight	Arm	Moment
Basic Empty Wt - Pass Config. (Pg 25)			
Basic Empty Wt - Cargo Config.			

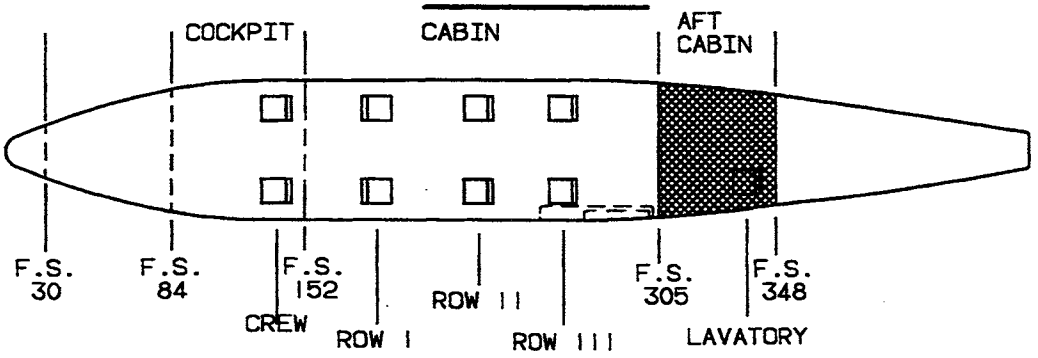
Basic Empty Wt - Pass Config.			
Basic Empty Wt - Cargo Config.			

For Training Purposes Only

DIMENSIONAL AND LOADING DATA



STANDARD SEATING



OCCUPANT	CENTROID
CREW	F.S. 129
ROW I	F.S. 171
ROW II	F.S. 222
ROW III	F.S. 258
LAV PASS	F.S. 335

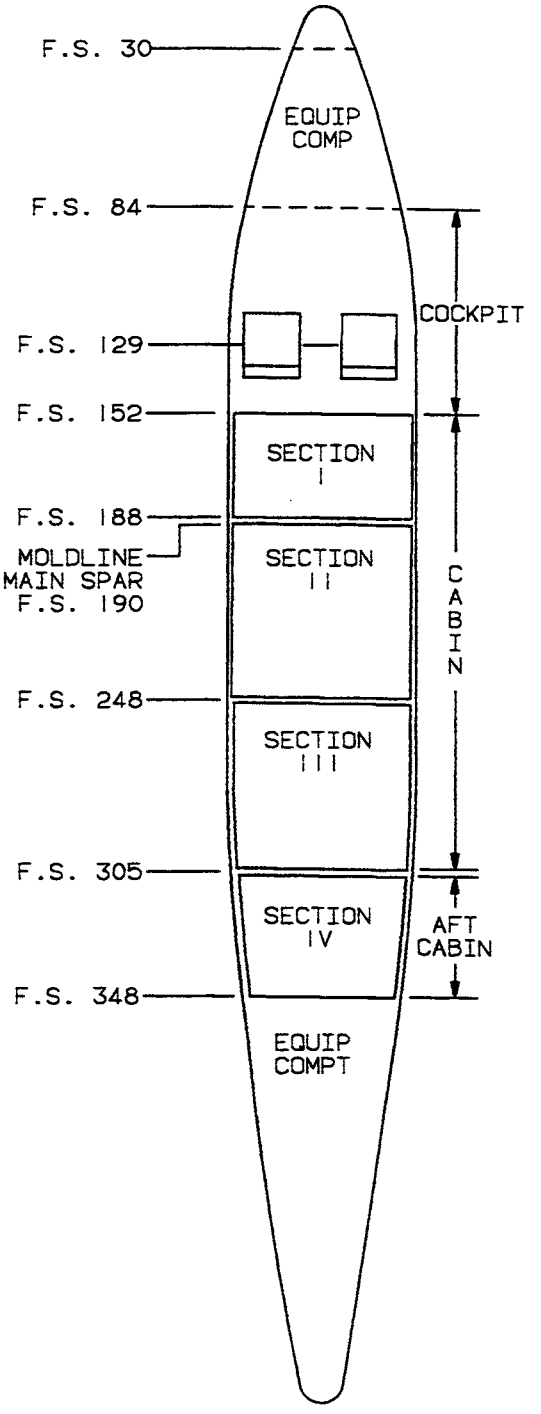
CONFIGURATION	BAGGAGE COMPARTMENT	BAGGAGE CAPACITY	CENTROID
AFT PASSENGER SEAT AND TOILET	AFT CREW - ▒▒▒▒	*	F.S. 325

NOTE *

MAXIMUM STRUCTURAL CAPACITY OF AFT CABIN IS 410 LBS FOR SERIALS PRIOR TO BL-58 NOT EQUIPPED WITH BEECH KIT P/N 101-5068; 550 LBS FOR SERIALS BL-58 AND AFTER, AND PRIOR AIRPLANES EQUIPPED WITH BEECH KIT P/N 101-5068. BAGGAGE CAPACITY IS AMOUNT REMAINING AFTER SUBTRACTING WEIGHT OF THE TOILET AND OTHER EQUIPMENT INSTALLED (SEE EQUIPMENT LIST) AND LAVATORY PASSENGER WHEN CARRIED.

C968806C1382 C

LOADING DATA
CARGO CONFIGURATION



SECTION	MAXIMUM STRUCTURAL CAPACITY	CENTROID ARM
I	800 LBS	F.S. 170
II	1300 LBS	F.S. 218
III	1370 LBS	F.S. 276
IV	SEE NOTE 7	F.S. 325

NOTES:

1. ALL CARGO IN SECTIONS I, II, & III MUST BE SUPPORTED ON AND SECURED TO THE SEAT TRACKS BY AN FAA APPROVED SYSTEM.

2. CONCENTRATED CARGO LOADS IN SECTIONS I, II & III MUST NOT EXCEED 200 LBS. PER SQUARE FOOT & MUST BE SUPPORTED ON THE SEAT RAILS.

3. CARGO IN SECTION IV IS TO BE SECURED BY BEECH FURNISHED BAGGAGE NET, WEBBING, OR STRAPS.

4. FOOTMAN LOOPS IN SECTION IV ARE TO BE USED TO SECURE CARGO/BAGGAGE IN THAT AREA ONLY.

5. CONCENTRATED FLOOR LOADINGS OF CARGO OR BAGGAGE IN SECTION IV MUST NOT EXCEED 100 LBS. PER SQUARE FOOR.

6. ANY EXCEPTIONS TO THE ABOVE PROCEDURES WILL REQUIRE APPROVAL BY A LOCAL FAA OFFICE.

7. 410 LBS FOR SERIALS PRIOR TO BL-58 NOT EQUIPPED WITH BEECH KIT P/N 101-5068; 550 LBS FOR SERIALS BL-58 AND AFTER, AND PRIOR AIRPLANES EQUIPPED WITH BEECH KIT P/N 101-5068.

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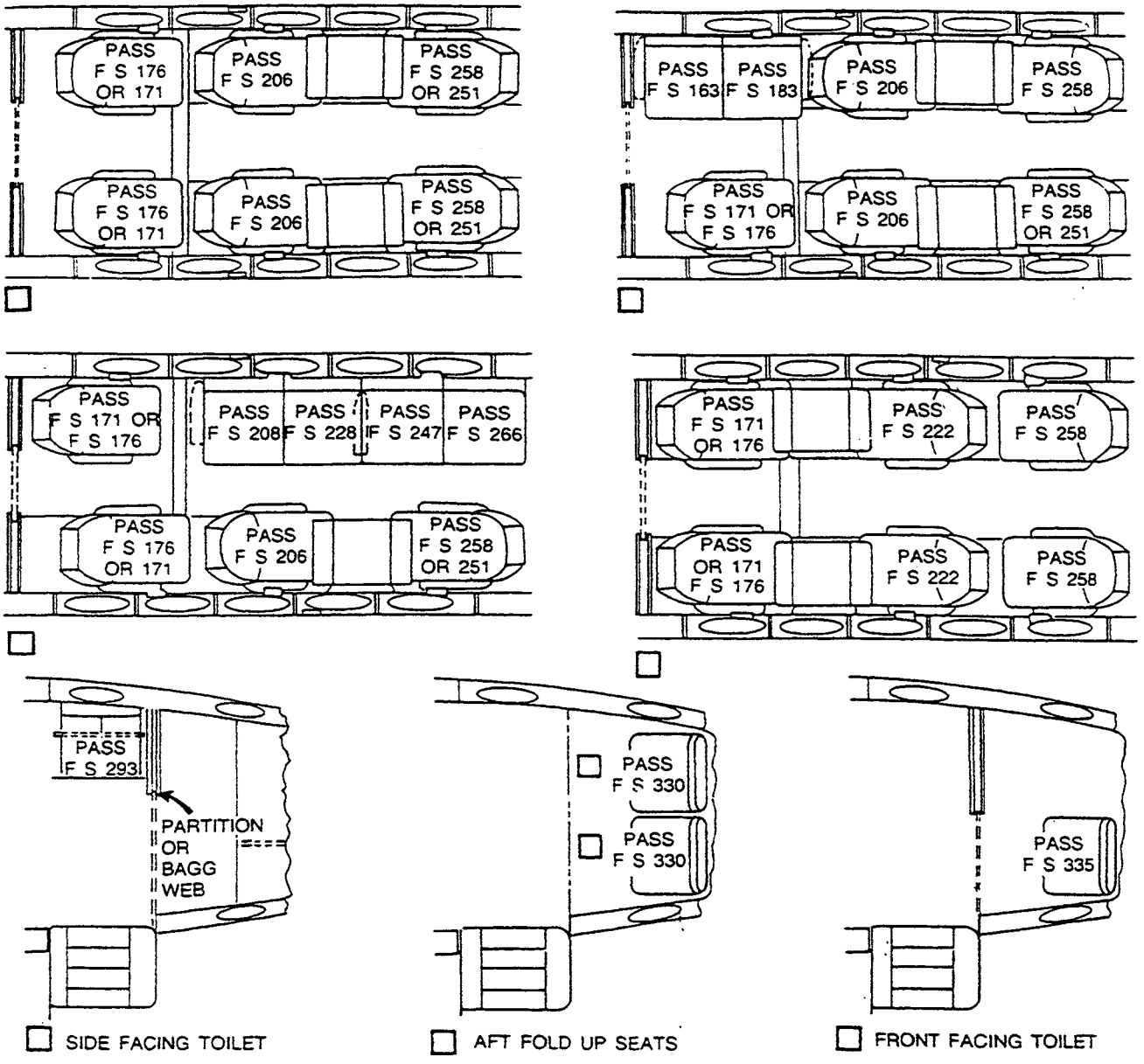
LOADING DATA (CARGO CONFIGURATION)

For Training Purposes Only

CABIN ARRANGEMENT DIAGRAM



NOTES:

The cabin seating may be arranged in different combinations. The diagrams marked ☒ above represent the seating arrangement established for this airplane prior to delivery. The passenger locations shown on the designated diagram are averages. Additional data for modified arrangements is noted. No diagrams are included for Hi-density versions.

B200-603-48

USEFUL LOAD WEIGHTS AND MOMENTS

OCCUPANTS

USE COLUMNS MARKED X ⬍	CREW	CABIN CHAIRS						LAVATORY
	F.S. 129	F.S. 171	F.S. 176	F.S. 206	F.S. 222	F.S. 251	F.S. 258	F.S. 335
WEIGHT	MOMENT/100							
80	103	137	141	165	178	201	206	268
90	116	154	158	185	200	226	232	302
100	129	171	176	206	222	251	258	335
110	142	188	194	227	244	276	284	369
120	155	205	211	247	266	301	310	402
130	168	222	229	268	289	326	335	436
140	181	239	246	288	311	351	361	469
150	194	257	264	309	333	377	387	503
160	206	274	282	330	355	402	413	536
170	219	291	299	350	377	427	439	570
180	232	308	317	371	400	452	464	603
190	245	325	334	391	422	477	490	637
200	258	342	352	412	444	502	516	670
210	271	359	370	433	466	527	542	704
220	284	376	387	453	488	552	568	737
230	297	393	405	474	511	577	593	771
240	310	410	422	494	533	602	619	804

OCCUPANTS

USE COLUMNS MARKED X ⬍	TWO PLACE COUCH FORWARD POSITION		FOUR PLACE COUCH AFT POSITION				AFT FOLD UP SEATS
	F.S. 163	F.S. 183	F.S. 208	F.S. 228	F.S. 247	F.S. 266	F.S. 330
WEIGHT	MOMENT/100						
80	130	146	166	182	198	213	264
90	147	165	187	205	222	239	297
100	163	183	208	228	247	266	330
110	179	201	229	251	272	293	363
120	196	220	250	274	296	319	396
130	212	238	270	296	321	346	429
140	228	256	291	319	346	372	462
150	245	275	312	342	371	399	495
160	261	293	333	365	395	426	528
170	277	311	354	388	420	452	561
180	293	329	374	410	445	479	594
190	310	348	395	433	469	505	627
200	326	366	416	456	494	532	660
210	342	384	437	479	519	559	693
220	359	403	458	502	543	585	726
230	375	421	478	524	568	612	759
240	391	439	499	547	593	638	792
250	408	458	520	570	618	665	825

USEFUL LOAD WEIGHTS AND MOMENTS

BAGGAGE

AFT CABIN	F.S. 325
WEIGHT	MOMENT/100
10	33
20	65
30	98
40	130
50	163
60	195
70	228
80	260
90	293
100	325
200	650
300	975
400	1300
500	1625
510	1658
550	1788

*SEE CABIN ARRANGEMENT DIAGRAM

CABINET CONTENTS

USE COLUMNS MARKED X→	CHART CASES	FORWARD CABINET	MIDDLE CABINET	AFT CABINET	COUCH DRAWERS		
	F.S. 148	F.S. 158	F.S. 196	F.S. 272	F.S. 173		
WEIGHT	MOMENT/100						
10	15	16	20	27	17		
20	30	32	39	54	35		
30	44	47	59	82	52		
40	59	63	78	109			
50	74	79					
60		95					
70		111					
80		126					
90		142					
100		158					

NOTE: Weight and Moment/100 of Cabinet Contents must be included in all loading computations.

USEFUL LOAD WEIGHTS AND MOMENTS

CARGO COMPARTMENT

WEIGHT	CABIN			AFT CABIN
	Section I F.S. 152-188	*Section II F.S. 188-248*	*Section III F.S. 248-305*	*Section IV F.S. 305-348*
	CENTROID			
	F.S. 170	*F.S. 218*	*F.S. 276*	*F.S. 325*
	MOMENT/100			
10	17	22	28	33
20	34	44	55	65
30	51	65	83	98
40	68	87	110	130
50	85	109	138	163
60	102	131	166	195
70	119	153	193	228
80	136	174	221	260
90	153	196	248	293
100	170	218	276	325
200	340	436	552	650
300	510	654	828	975
400	680	872	1104	1300
500	850	1090	1380	1625
510	867	1112	1408	1658
550	935	1199	1518	1788
600	1020	1308	1656	
700	1190	1526	1932	
800	1360	1744	2208	
900		1962	2484	
1000		2180	2760	
1100		2398	3036	
1200		2616	3312	
1300		2834	3588	
1370			3781	

NOTE: All cargo in Sections I, II & III must be supported by the seat tracks in a uniform distribution and tied down to the tracks by an FAA approved method.

*Refer to LOADING DATA CARGO CONFIGURATION

USEFUL LOAD WEIGHTS AND MOMENTS

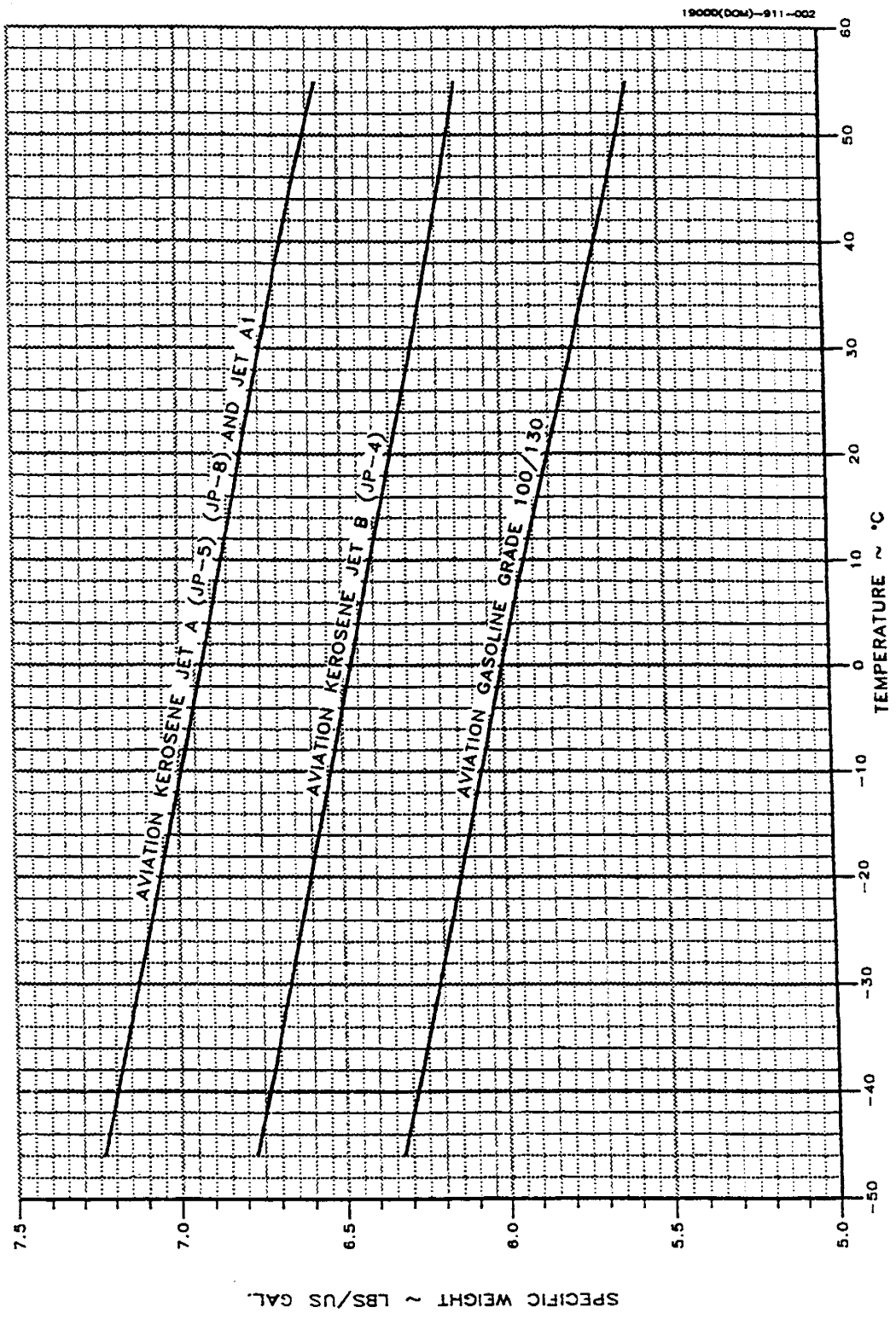
USABLE FUEL

GALLONS	6.4 LB/GAL		6.5 LB/GAL		6.6 LB/GAL		6.7 LB/GAL	
	WEIGHT	MOMENT 100	WEIGHT	MOMENT 100	WEIGHT	MOMENT 100	WEIGHT	MOMENT 100
10	64	99	65	100	66	102	67	103
20	128	197	130	200	132	203	134	206
30	192	305	195	310	198	314	201	319
40	256	423	260	430	264	436	268	443
50	320	542	325	550	330	559	335	567
60	384	662	390	672	396	683	402	693
70	448	782	455	794	462	807	469	819
80	512	904	520	918	528	932	536	946
90	576	1023	585	1039	594	1055	603	1071
100	640	1142	650	1160	660	1178	670	1196
110	704	1260	715	1280	726	1300	737	1319
120	768	1379	780	1400	792	1422	804	1443
130	832	1496	845	1519	858	1543	871	1566
140	896	1615	910	1640	924	1665	938	1690
150	960	1734	975	1761	990	1788	1005	1815
160	1024	1852	1040	1881	1056	1910	1072	1939
170	1088	1971	1105	2002	1122	2033	1139	2064
180	1152	2090	1170	2122	1188	2155	1206	2188
190	1216	2209	1235	2244	1254	2279	1273	2313
200	1280	2328	1300	2365	1320	2401	1340	2437
210	1344	2447	1365	2486	1386	2524	1407	2562
220	1408	2567	1430	2607	1452	2647	1474	2687
230	1472	2686	1495	2728	1518	2770	1541	2812
240	1536	2806	1560	2850	1584	2894	1608	2938
250	1600	2926	1625	2971	1650	3017	1675	3063
260	1664	3045	1690	3093	1716	3140	1742	3188
270	1728	3164	1755	3213	1782	3263	1809	3312
280	1792	3283	1820	3334	1848	3386	1876	3437
290	1856	3402	1885	3455	1914	3508	1943	3562
300	1920	3521	1950	3576	1980	3631	2010	3686
310	1984	3641	2015	3698	2046	3754	2077	3811
320	2048	3760	2080	3819	2112	3878	2144	3936
330	2112	3880	2145	3940	2178	4001	2211	4062
340	2176	3999	2210	4062	2244	4124	2278	4187
350	2240	4119	2275	4184	2310	4248	2345	4312
360	2304	4244	2340	4310	2376	4377	2412	4443
370	2368	4365	2405	4434	2442	4502	2479	4570
380	2432	4489	2470	4560	2508	4630	2546	4700
386	2470	4562	2509	4634	2548	4706	2586	4776
400	2560	4741	2600	4815	2640	4889	2680	4963
410	2624	4869	2665	4945	2706	5021	2747	5097
420	2688	4997	2730	5075	2772	5153	2814	5231
430	2752	5126	2795	5206	2838	5286	2881	5366
440	2816	5255	2860	5337	2904	5419	2948	5501
450	2880	5386	2925	5470	2970	5554	3015	5638
460	2944	5514	2990	5600	3036	5686	3082	5773
470	3008	5645	3055	5733	3102	5821	3149	5909
480	3072	5775	3120	5866	3168	5956	3216	6046
490	3136	5907	3185	5999	3234	6091	3283	6184
500	3200	6040	3250	6134	3300	6229	3350	6323
510	3264	6172	3315	6269	3366	6365	3417	6462
520	3328	6307	3380	6405	3432	6504	3484	6602
530	3392	6441	3445	6542	3498	6643	3551	6743
540	3456	6573	3510	6676	3564	6779	3618	6881
544	3482	6626	3536	6729	3590	6832	3645	6936

DENSITY VARIATION OF AVIATION FUEL

BASED ON AVERAGE SPECIFIC GRAVITY

FUEL	AVERAGE SPECIFIC GRAVITY
JET A (JP-5) (JP-8) AND JET A1	0.819 AT 15°C
JET B (JP-4)	0.764 AT 15°C
AV GAS GRADE 100/130	0.706 AT 15°C



LOADING INSTRUCTIONS

It is the responsibility of the airplane operator to ensure that the airplane is properly loaded. At the time of delivery, Beech Aircraft Corporation provides the necessary weight and balance data to compute individual loadings. All subsequent changes in airplane weight and balance are the responsibility of the airplane owner and/or operator.

The basic empty weight and moment of the airplane at the time of delivery are shown on the Basic Empty Weight and Balance form. Useful load items which may be loaded into the airplane are shown on the Useful Load Weight and Moment tables. The minimum and maximum moments approved by the FAA are shown on the Moment Limits vs. Weight graph or table. These moments correspond to the forward and aft center of gravity flight limits (landing gear down) for a particular weight. All moments are divided by 100 to simplify computations.

CARGO LOADING

The method of loading cargo, its placement in the airplane and the method of restraint should each be determined before starting the actual loading.

For loads that are evenly distributed in a given section, the useful Load Table under the heading of Cargo Compartment should be used. For any load that cannot be located at the centroid of a section or that extends over more than one section, it will be necessary to determine its own CG and its location in the airplane. Determine the CG arm (Fuselage Station) by measuring in inches, from a known location in the cabin to the CG of the load. Determine the "moment" for the load by multiplying the weight by the CG arm (Fuselage Station). This resultant should be divided by 100 to be compatible with other loading data.

COMPUTING PROCEDURE

1. Record the basic empty weight and moment from the Basic Empty Weight and Balance form (or from the latest superseding forms). The moment must be divided by 100 to correspond to Useful Load Moments.
2. Record the weight and corresponding moment of each item to be carried. These values are found on the Useful Load Weight and Moment tables.
3. Total the weight column and moment column. The total weight without usable fuel must not exceed the Maximum Zero Fuel Weight limitation of 11,000 pounds. All weight in excess of this limitation must be fuel. The auxiliary tanks may be used only when the main tanks are completely filled. The total take-off weight must not exceed the maximum allowable take-off weight and the total moment must be within the minimum and maximum moments shown on the Moment Limits Vs. Weight table or graph.
4. Using the page titled Useful Load Weights and Moments - Usable Fuel, determine the weight and corresponding moment of fuel to be used by subtracting the amount on board on landing from the amount on board at takeoff.
5. For landing configuration weight and balance, subtract the weight and moment of fuel to be used from the take-off weight and moment. The landing moment must be within the minimum and maximum moments shown on Weight and Moment Limits table for that weight. If the total moment is less than the minimum moment allowed, useful load items must be shifted aft, or forward load items reduced. If the total moment is greater than the maximum moment allowed, useful load items must be shifted forward, or aft load items reduced. If the quantity or location of load items is changed, the calculations must be revised and the moments rechecked.
6. Loadings may be made on the Weight and Balance Loading Form.

For Training Purposes Only

Beechcraft SUPER KING AIR B200/B200C

WEIGHT AND BALANCE LOADING FORM

SERIAL	REGISTRATION NO:	DATE:

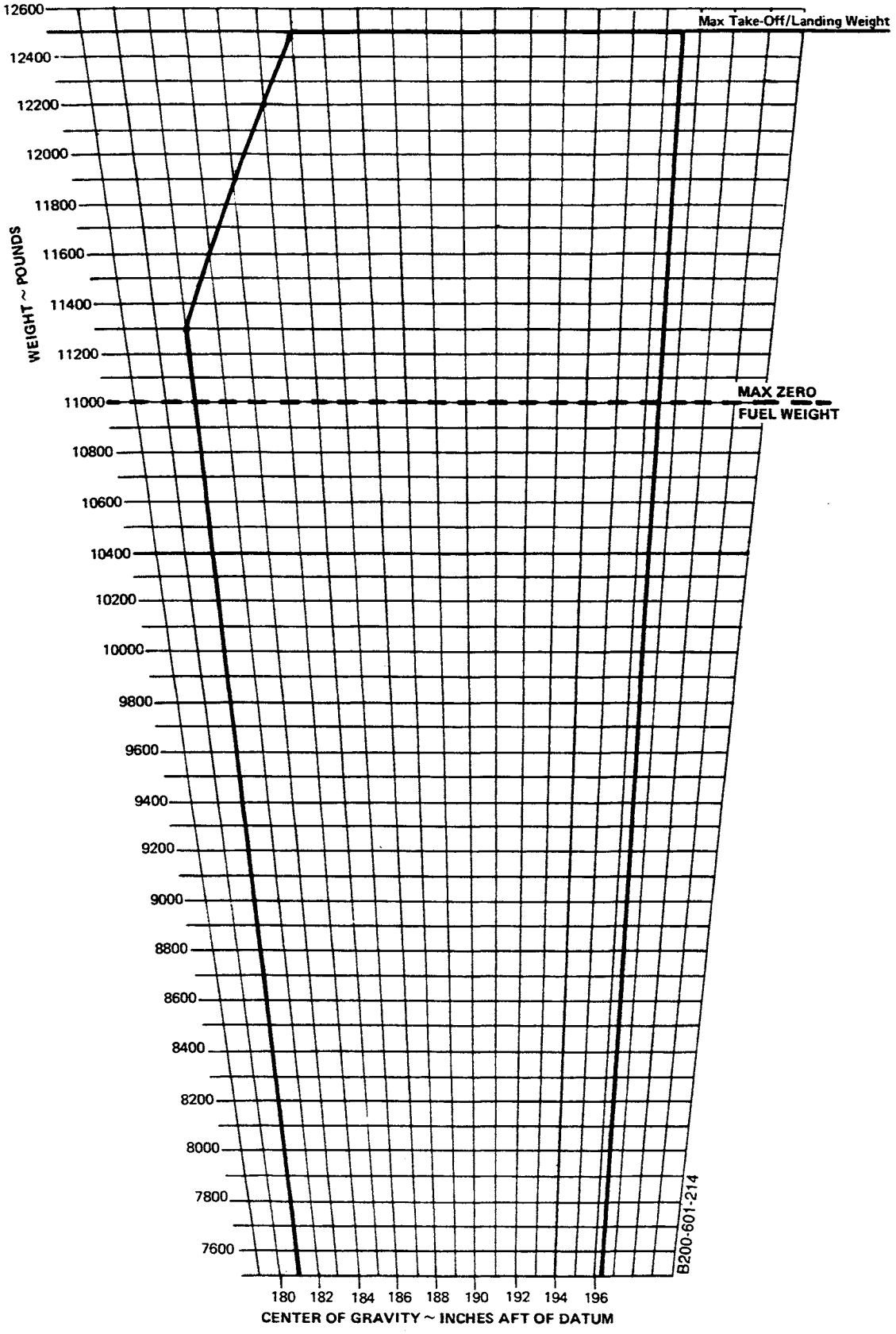
PASSENGERS OR CARGO		
ITEM LOCATION (ROW, F.S.,ETC)	WEIGHT	MOM/100
TOTAL PASSENGERS OR CARGO		

REF	ITEM	WEIGHT	MOM/100
1.	BASIC EMPTY COND.		
2.	PILOT		
3.	PASSENGERS OR CARGO		
4.	BAGGAGE		
5.	CABINET CONTENTS		
6.	**SUB TOTAL** ZERO FUEL CONDITION. DO NOT EXCEED 11,000 LBS		
7.	FUEL LOADING		
8.	**SUB TOTAL** RAMP CONDITION		
9.	*LESS FUEL FOR START, TAXI, AND TAKE OFF		
10.	**SUB TOTAL** TAKE OFF CONDITION		
11.	LESS FUEL TO DESTINATION		
12.	**LANDING CONDITION**		

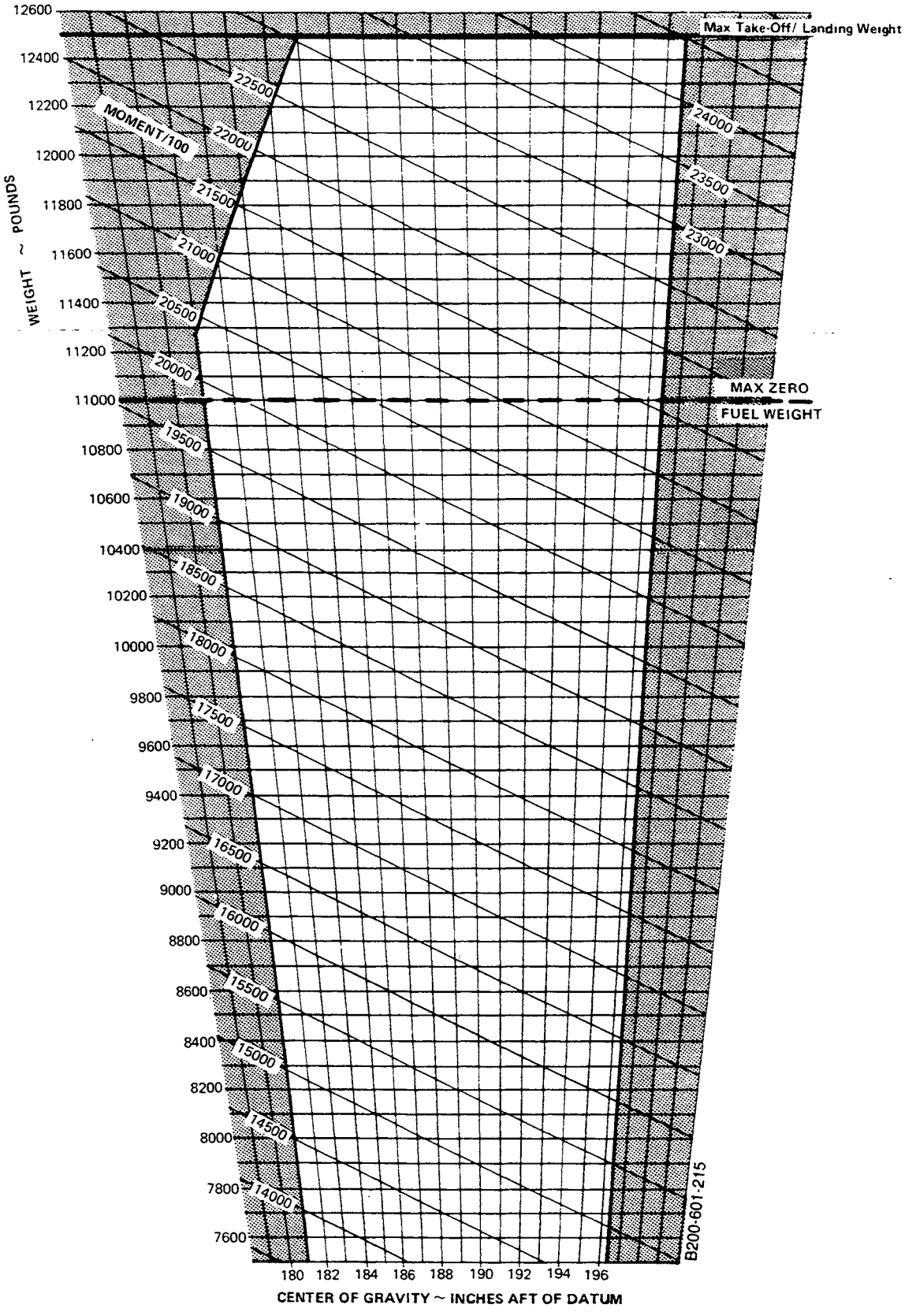
*Fuel for start, taxi and take-off is normally 90 lbs at an average moment/100 of 177.

B200-601-219

WEIGHT AND BALANCE DIAGRAM



MOMENT LIMITS vs WEIGHT



MOMENT LIMITS VS. WEIGHT

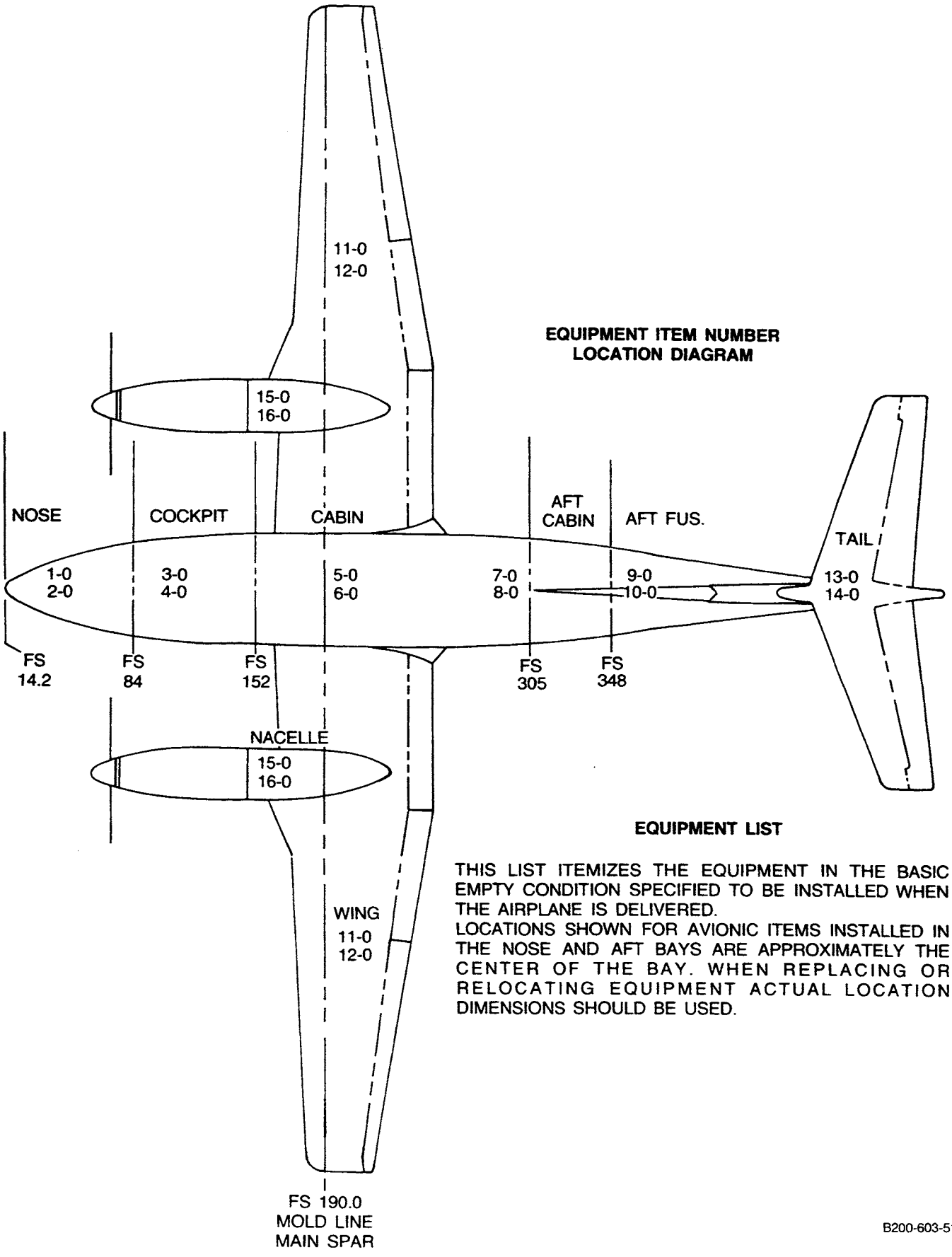
WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
7200	13032	14141
7250	13122	14239
7300	13213	14337
7350	13304	14435
7400	13394	14534
7450	13484	14632
7500	13575	14730
7550	13666	14828
7600	13756	14926
7650	13846	15025
7700	13937	15123
7750	14028	15221
7800	14118	15319
7850	14208	15417
7900	14299	15516
7950	14390	15614
8000	14480	15712
8050	14570	15810
8100	14661	15908
8150	14752	16007
8200	14842	16105
8250	14932	16203
8300	15023	16301
8350	15114	16399
8400	15204	16498
8450	15294	16596
8500	15385	16694
8550	15476	16792
8600	15566	16890
8650	15656	16989
8700	15747	17087
8750	15838	17185
8800	15928	17283
8850	16018	17381
8900	16109	17480
8950	16200	17578
9000	16290	17676
9050	16380	17774
9100	16471	17872
9150	16562	17971
9200	16652	18069
9250	16742	18167
9300	16833	18265
9350	16924	18363
9400	17014	18462
9450	17104	18560
9500	17195	18658
9550	17286	18756
9600	17376	18854
9650	17466	18953
9700	17557	19051
9750	17648	19149
9800	17738	19247
9850	17828	19345

MAX ZERO FUEL WEIGHT

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
9900	17919	19444
9950	18010	19542
10000	18100	19640
10050	18190	19738
10100	18281	19836
10150	18372	19935
10200	18462	20033
10250	18552	20131
10300	18643	20229
10350	18734	20327
10400	18824	20426
10450	18914	20524
10500	19005	20622
10550	19096	20720
10600	19186	20818
10650	19276	20917
10700	19367	21015
10750	19458	21113
10800	19548	21211
10850	19638	21309
10900	19729	21408
10950	19820	21506
11000	19910	21604
11050	20000	21702
11100	20091	21800
11150	20182	21899
11200	20272	21997
11250	20362	22095
11300	20461	22193
11350	20570	22291
11400	20679	22390
11450	20789	22488
11500	20898	22586
11550	21008	22684
11600	21118	22782
11650	21228	22881
11700	21338	22979
11750	21449	23077
11800	21559	23175
11850	21670	23273
11900	21781	23372
11950	21892	23470
12000	22003	23568
12050	22115	23666
12100	22226	23764
12150	22338	23863
12200	22450	23961
12250	22562	24059
12300	22674	24157
12350	22787	24255
12400	22899	24354
12450	23012	24452
12500	23125	24550

CENTER OF GRAVITY LIMITS (LANDING GEAR DOWN)

WEIGHT CONDITION	FORWARD CG LIMIT	AFT CG LIMIT
12,500 LBS (MAX. TAKE-OFF OR LANDING)	185.0	196.4
11,279 LBS OR LESS	181.0	196.4



EQUIPMENT ITEM NUMBER LOCATION DIAGRAM

EQUIPMENT LIST

THIS LIST ITEMIZES THE EQUIPMENT IN THE BASIC EMPTY CONDITION SPECIFIED TO BE INSTALLED WHEN THE AIRPLANE IS DELIVERED.
LOCATIONS SHOWN FOR AVIONIC ITEMS INSTALLED IN THE NOSE AND AFT BAYS ARE APPROXIMATELY THE CENTER OF THE BAY. WHEN REPLACING OR RELOCATING EQUIPMENT ACTUAL LOCATION DIMENSIONS SHOULD BE USED.

B200-603-51

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AIRFRAME

STRUCTURE

The BEECHCRAFT Super King Air B200/B200C is an all-metal, low-wing monoplane. It has fully cantilevered wings, and a T-tail empennage.

SEATING ARRANGEMENTS

The pilot and copilot seats are mounted in a separate forward compartment. Various configurations of passenger chairs and two- or four-place couch installations may be installed on the continuous tracks mounted on the cabin floor. One or two fold-up seats may be installed in the aft cabin area. The toilet is also equipped for use as a seat. Seating for up to 15 persons, including crew, is available. For additional information, refer to the "Cabin Arrangement Diagram" in Section VI, WEIGHT AND BALANCE/ EQUIPMENT LIST.

FLIGHT CONTROLS

CONTROL SURFACES

The airplane is equipped with conventional ailerons and rudder. It utilizes a T-tail horizontal stabilizer and elevator, mounted at the extreme top of the vertical stabilizer.

OPERATING MECHANISMS

The airplane is equipped with conventional dual controls for the pilot and copilot. The ailerons and elevators are operated by conventional control wheels interconnected by a T-bar. The rudder pedals are interconnected by linkage below the floor. These systems are connected to the control surfaces through push-rod and cable-and-bellcrank systems. Rudder, elevator, and aileron trim are adjustable with controls mounted on the center pedestal. A position indicator for each of the trim tabs is integrated with its respective control.

MANUAL ELEVATOR TRIM

Manual control of the elevator trim is accomplished with a handwheel located on the left side of the pedestal. It is a conventional trim wheel which is rolled forward for nose-down trim, and aft for nose-up trim.

ELECTRIC ELEVATOR TRIM

The electric elevator-trim system, if installed, is controlled by an ELEV TRIM - ON - OFF switch located on the pedestal, a dual-element thumb switch on each control wheel, a trim-disconnect switch on each control wheel, and a PITCH TRIM circuit breaker in the FLIGHT group on the right side panel. The ELEV TRIM switch must be ON for the system to operate. Both elements of either dual-element thumb switch must be simultaneously moved forward to achieve nose-down trim, aft for nose-up trim; when released, they return to the center (OFF) position. Any activation of the trim system by the copilot's thumb switch can be overridden by the pilot's thumb switch. A before take-off check of both dual element thumb switches should be made by moving each of the four switch elements individually. No one switch element should activate the system; moving the two switch elements on either the pilot's or copilot's control wheel in opposite directions should not activate the system - only the simultaneous movement of a pair of switch elements in the same direction should activate the electric elevator-trim system.

A bi-level, push-button, momentary-on, trim-disconnect switch is located inboard of the dual-element thumb switch on the outboard grip of each control wheel. The electric elevator-trim system can be disconnected by depressing either of these switches. If an autopilot is installed, depressing either trim-disconnect switch to the first of the two levels disconnects the autopilot and the yaw damp system; depressing the switch to the second level disconnects the autopilot, the yaw damp system, and the electric elevator-trim system. If an autopilot is not installed, depressing the switch to the first level does not do anything, since the yaw damp system is controlled by a separate YAW DAMP switch on the pedestal; depressing the switch to the second level disconnects the electric elevator-trim system. A green annnciator on the caution/advisory annunciator panel, placarded ELEC TRIM OFF, alerts the pilot whenever the system has has been disabled with a trim-disconnect switch and the ELEV TRIM switch is ON. The system can be reset by cycling the ELEV TRIM switch on the pedestal from ON to OFF, then back to ON again. The manual-trim control wheel can be used to change the trim anytime, whether or not the electric trim system is in the operative mode.

RUDDER BOOST

A rudder boost system is provided to aid the pilot in maintaining directional control in the event of an engine failure or a large variation of power between the engines. Incorporated into the rudder cable system are two pneumatic rudder-boosting servos that actuate the cables to provide rudder pressure to help compensate for asymmetrical thrust.

During operation, a differential pressure valve accepts bleed air pressure from each engine. When the pressure varies between the bleed air systems, the shuttle in the differential pressure valve moves toward the low pressure side. As the pressure difference reaches a preset tolerance, a switch on the low pressure side closes, activating the rudder boost system. The system is designed only to help compensate for asymmetrical thrust. Appropriate trimming is to be accomplished by the pilot. Moving either or both of the bleed air valve switches on the copilot's subpanel to the INSTR & ENVIR OFF position will disengage the rudder boost system.

The system is controlled by a toggle switch, placarded RUDDER BOOST - ON - OFF, located on the pedestal below the rudder trim wheel. The switch is to be turned ON before flight. A preflight check of the system can be performed during the run-up by retarding the power on one engine to idle and advancing power on the opposite engine until the power difference between the engines is great enough to close the switch that activates the rudder boost system. Movement of the appropriate rudder pedal (left engine idling, right rudder pedal moves forward) will be noted when the switch closes,

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indicating the system is functioning properly for low engine power on that side. Repeat the check with opposite power settings to check for movement of the opposite rudder pedal.

INSTRUMENT PANEL

The floating instrument panel design allows the flight instruments to be arranged in a group directly in front of the pilot and the copilot. Complete pilot and copilot flight instrumentation is installed, including dual navigation systems, two course indicators, dual gyro horizons, and dual turn and slip indicators.

The operation and use of the instruments, lights, switches, and controls located on the instrument panel is explained under the systems descriptions relating to the subject items.

ANNUNCIATOR SYSTEM

The annunciator system consists of a warning annunciator panel (with red readout) centrally located in the glareshield, and a caution/advisory annunciator panel (caution - yellow; advisory - green) located on the center subpanel. Two red MASTER WARNING flashers located in the glareshield (one in front of the pilot and one in front of the copilot) are a part of the system, as are two yellow MASTER CAUTION flashers (located just inboard of the MASTER WARNING flashers), and a PRESS TO TEST button located immediately to the right of the warning annunciator panel.

The annunciators are of the word-readout type. Whenever a fault condition covered by the annunciator system occurs, a signal is generated and the appropriate annunciator is illuminated.

If the fault requires the immediate attention and reaction of the pilot, the appropriate red warning annunciator in the warning annunciator panel illuminates and both MASTER WARNING flashers begin flashing. Any illuminated lens in the warning annunciator panel will remain on until the fault is corrected. However, the MASTER WARNING flashers can be extinguished by depressing the face of either MASTER WARNING flasher, even if the fault is not corrected. In such a case, the MASTER WARNING flashers will again be activated if an additional warning annunciator illuminates. When a warning fault is corrected, the affected warning annunciator will extinguish, but the MASTER WARNING flashers will continue flashing until one of them is depressed.

Whenever an annunciator-covered fault occurs that requires the pilot's attention but not his immediate reaction, the appropriate yellow caution annunciator in the caution/advisory panel illuminates, and both MASTER CAUTION flashers begin flashing. The flashing MASTER CAUTION lights can be extinguished by pressing the face of either of the flashing lights to reset the circuit. Subsequently, when any caution annunciator illuminates, the MASTER CAUTION flashers will be activated again. An illuminated caution annunciator on the caution/advisory annunciator panel will remain on until the fault condition is corrected, at which time it will extinguish. The MASTER CAUTION flashers will continue flashing until one of them is depressed.

The caution/advisory annunciator panel also contains the green advisory annunciators. There are no master flashers associated with these annunciators, since they are only advisory in nature, indicating functional situations which do not demand the immediate attention or reaction of the pilot. An advisory annunciator can be extinguished only by correcting the condition indicated on the illuminated lens.

The warning annunciators, caution annunciators, advisory annunciators and yellow MASTER CAUTION flashers feature both a "bright" and a "dim" mode of illumination intensity. The "dim" mode wil be selected automatically whenever all of the following conditions are met: a generator is on the line; the OVERHEAD FLOOD LIGHTS are OFF; the PILOT FLIGHT LIGHTS are ON; and the ambient light level in the cockpit (as sensed by a photoelectric cell located in the overhead light control panel) is below a preset value. Unless all of these conditions are met, the "bright" mode will be selected automatically. On later airplanes, and earlier airplanes with modified annunciator circuitry, The MASTER WARNING flasher also features both a "bright" and "dim" mode of illumination.

The lamps in the annunciator system should be tested before every flight, and anytime the integrity of a lamp is in question. Depressing the PRESS TO TEST button, located to the right of the warning annunciator panel in the glareshield, illuminates all the annunciator lights, MASTER WARNING flashers, and MASTER CAUTION flashers. Any lamp that fails to illuminate when tested should be replaced (refer to LAMP REPLACEMENT GUIDE in Section VIII, HANDLING, SERVICING AND MAINTENANCE).

WARNING PANEL ILLUSTRATION

L ENG FIRE	INVERTER	CABIN DOOR	ALT WARN	R ENG FIRE
L FUEL PRESS	———	———	———	R FUEL PRESS
*L OIL PRESS	*L GEN OVHT	*A/P TRIM FAIL	*R GEN OVHT	*R OIL PRESS
L CHIP DETECT	L BL AIR FAIL	*A/P DISC	R BL AIR FAIL	R CHIP DETECT

** Optional Equipment*

WARNING PANEL DESCRIPTION

NOMENCLATURE	COLOR	CAUSE FOR ILLUMINATION
L ENG FIRE	*Red*	*Fire in left engine compartment*
INVERTER	*Red*	*The inverter selected is inoperative*
CABIN DOOR	*Red*	*Cabin/cargo door open or not secure*
ALT WARN	*Red*	*Cabin altitude exceeds 12,500 feet*
R ENG FIRE	*Red*	*Fire in right engine compartment*
L FUEL PRESS	*Red*	*Fuel pressure failure on left side*
R FUEL PRESS	*Red*	*Fuel pressure failure on right side*
**L OIL PRESS*	*Red*	*Low oil pressure left engine*
**L GEN OVHT*	*Red*	*Left generator temperature too high*
**A/P TRIM FAIL*	*Red*	*Improper trim or no trim from autopilot trim command*
**R GEN OVHT*	*Red*	*Right generator temperature too high*
**R OIL PRESS*	*Red*	*Low oil pressure right engine*
L CHIP DETECT	*Red*	*Contamination in left engine oil is detected*
L BL AIR FAIL	*Red*	*Melted or failed plastic left bleed air failure warning line*
**A/P DISC*	*Red*	*Autopilot is disconnected*
R BL AIR FAIL	*Red*	*Melted or failed plastic right bleed air failure warning line*
R CHIP DETECT	*Red*	*Contamination in right engine oil is detected*

** Optional Equipment*

CAUTION/ADVISORY PANEL ILLUSTRATION

L DC GEN	††HYD FLUID LOW	†*PROP SYNC ON	RVS NOT READY	———	R DC GEN
———	———	———	DUCT OVERTEMP	———	———
L ICE VANE	———	BATTERY CHARGE	EXT PWR	———	R ICE VANE
*L AUTOFEATHER	———	*ELEC TRIM OFF	AIR COND N_1 LOW	———	*R AUTOFEATHER
L ICE VANE EXT	*BRAKE DEICE ON	LDG/TAXI LIGHT	PASS OXY ON	———	R ICE VANE EXT
L IGNITION ON	L BL AIR OFF	———	FUEL CROSSFEED	R BL AIR OFF	R IGNITION ON

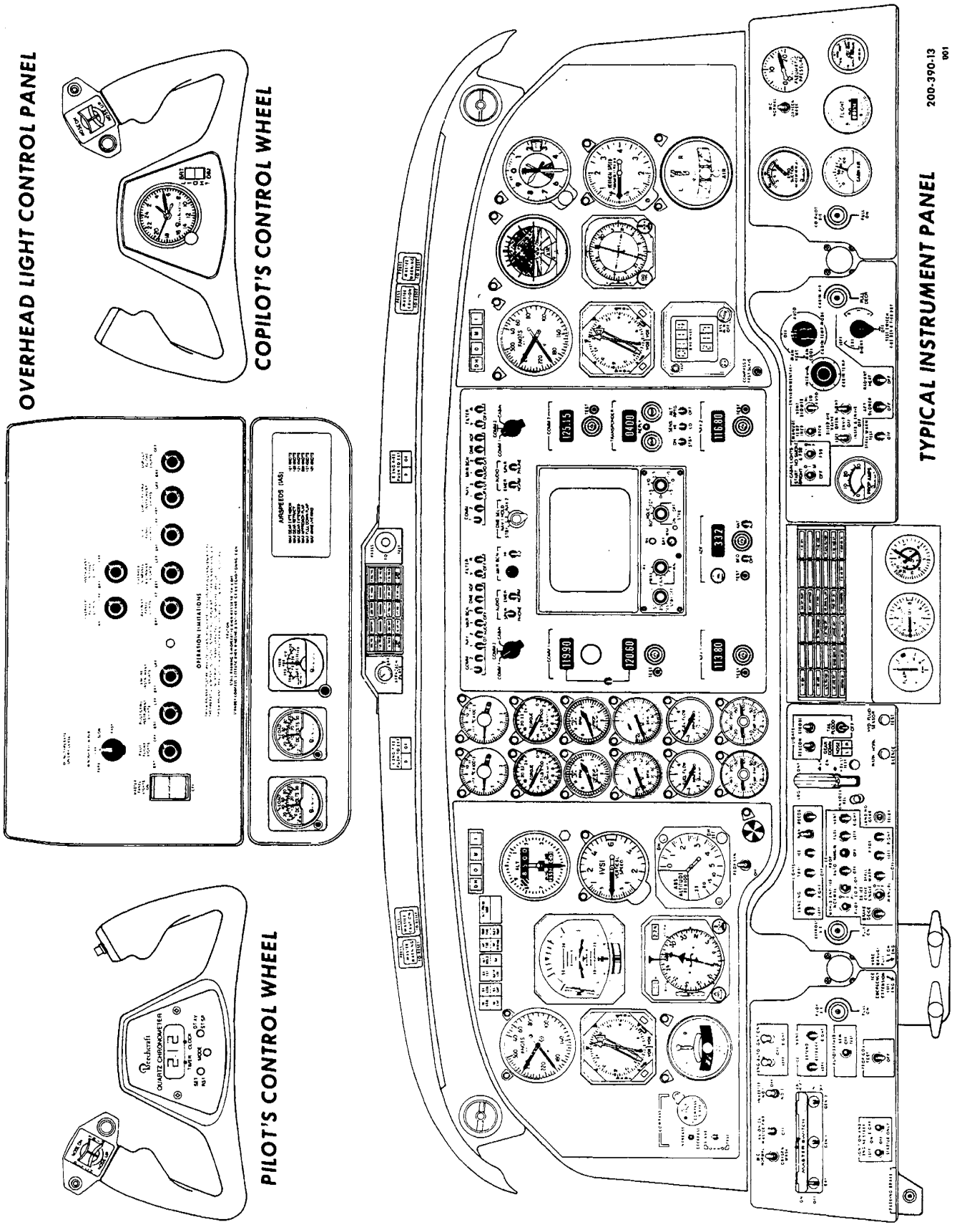
CAUTION/ADVISORY PANEL DESCRIPTION

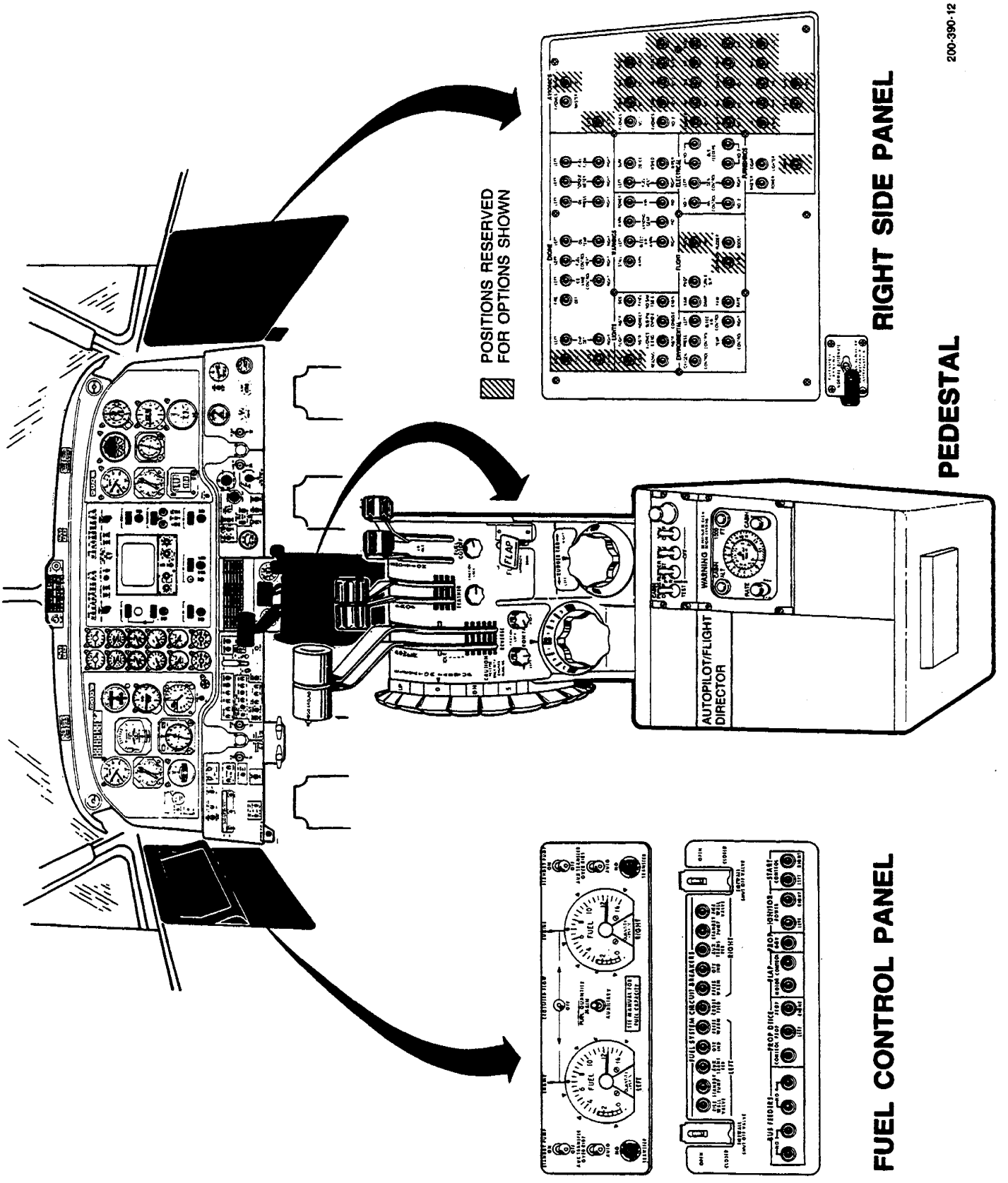
NOMENCLATURE	COLOR	CAUSE FOR ILLUMINATION
L DC GEN	*Yellow*	*Left generator off the line*
††HYD FLUID LOW	*Yellow*	*Hydraulic fluid in the landing gear system is low*
†PROP SYNC ON	*Yellow*	*Synchrophaser "ON" with landing gear extended*
RVS NOT READY	*Yellow*	*Propeller levers are not in the high rpm, low pitch position with landing gear extended*
R DC GEN	*Yellow*	*Right generator off the line*
DUCT OVERTEMP	*Yellow*	*Duct air too hot*
L ICE VANE	*Yellow*	*Left ice vane malfunction. Ice vane has not attained proper position*
BATTERY CHARGE	*Yellow*	*Excessive charge rate on battery*
EXT PWR	*Yellow*	*External power connector is plugged in*
R ICE VANE	*Yellow*	*Right ice vane malfunction. Ice vane has not attained proper position*
**L AUTOFEATHER*	*Green*	*Autofeather armed with power levers advanced above 90% N_1*
**ELEC TRIM OFF*	*Green*	*Electric trim de-energized by a trim disconnect switch on the control wheel with the system power switch on the pedestal turned on.*
AIR COND N_1 LOW	*Green*	*Right engine rpm too low for air conditioning load*
**R AUTOFEATHER*	*Green*	*Autofeather armed with power levers advanced above 90% N_1*
L ICE VANE EXT	*Green*	*Ice vane extended*
**BRAKE DEICE ON*	*Green*	*Brake deice system in operation*
LDG/TAXI LIGHT	*Green*	*Landing or taxi lights on with landing gear up*

**Optional Equipment*
†Not required when type II synchrophaser is used
†† On airplanes BB-1158, BB-1167, BB-1193 and after, BL-73 and after, and those prior airplanes incorporating Beech Kit P/N 101-8018-1

CAUTION/ADVISORY PANEL DESCRIPTION (Continued)

PASS OXY ON	*Green*	*Passenger oxygen system charged*
R ICE VANE EXT	*Green*	*Ice vane extended*
L IGNITION ON	*Green*	*Left starter/ignition switch is in the engine/ignition mode or left auto ignition system is armed and left engine torque is below approximately 410 ft-lbs*
L BL AIR OFF	*Green*	*Left environmental bleed air valve closed*
FUEL CROSSFEED	*Green*	*Crossfeed valve is open*
R BL AIR OFF	*Green*	*Right environmental bleed air valve is closed*
R IGNITION ON	*Green*	*Right starter/ignition switch is in the engine/ignition mode or right auto ignition system is armed and right engine torque is below approximately 410 ft-lbs*


OVERHEAD LIGHT CONTROL PANEL
COPILOT'S CONTROL WHEEL
PILOT'S CONTROL WHEEL
TYPICAL INSTRUMENT PANEL
200-390-13
001



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GROUND CONTROL

Direct linkage from the rudder pedals allows for nose wheel steering. When the rudder control is augmented by a main wheel brake, the nose wheel deflection can be considerably increased.

The minimum wing-tip turning radius, using partial braking action and differential engine power, is 39 feet 10 inches.

FLAPS

Two flaps are installed on each wing. Power is delivered from an electric motor to a gearbox mounted on the forward side of the rear spar. The gearbox drives four flexible driveshafts which are connected to jackscrews, one of which operates each flap. The motor incorporates a dynamic braking system, through the use of two sets of motor windings. This feature helps prevent overtravel of the flaps. A safety mechanism is provided to disconnect power to the electric flap motor in the event of a malfunction which would cause any flap to be three to six degrees out of phase with the other flaps.

The flaps are operated by a sliding switch handle on the pedestal just below the condition levers. Flap travel, from 0% (full up) to 100% (full down) is registered on an electric indicator on top of the pedestal. A side detent provides for quick selection of the APPROACH position (40% flaps). From the UP position to the APPROACH position, the flaps cannot be stopped in an intermediate position. Between APPROACH and DOWN, the flaps can be stopped anywhere by moving the handle to the DOWN position until the flaps reach the desired position, then moving the flap-switch handle back to APPROACH. The flaps can be raised to any position between DOWN and APPROACH by raising the handle to UP until the desired setting is reached, then returning the handle to APPROACH. Selecting the APPROACH position will stop flap travel anytime the flaps are deflected more than 40%.

The flap-motor power circuit is protected by a 20-ampere flap-motor circuit breaker placarded FLAP MOTOR, located on the left circuit breaker panel below the fuel control panel. A 5-ampere circuit breaker for the control circuit (placarded FLAP CONTROL) is also located on this panel.

Lowering the flaps will produce these results:

ATTITUDE - Nose UP

AIRSPEED - Reduced

STALL SPEED - Lowered

TRIM - Nose-Down Adjustment Required to Maintain Attitude

LANDING GEAR

MECHANICAL LANDING GEAR

A 28-volt motor and gear box, located on the forward side of the center-section main spar, extends and retracts the landing gear. The landing gear motor is controlled by the handle placarded LDG GEAR CONTROL - UP - DN on the pilot's right subpanel. The landing gear control handle must be pulled out of a detent before it can be moved from either the UP or the DN position. The motor incorporates a dynamic braking system controlled with "up" and "down" limit switches, which in conjunction with the landing gear locking mechanism prevents overtravel of the landing gear.

Torque shafts drive main gear actuators, and duplex chains drive the nose gear actuator. A spring-loaded friction-type overload clutch in the gearbox prevents damage to the structure and to the torque shafts in the event of a malfunction. A 150 amp limiter, located on the landing gear panel forward of the main spar under the center floorboard, protects the system from electrical overload.

The Beech air-oil type shock struts are filled with compressed air and hydraulic fluid. Linkage from the rudder pedals permits nose wheel steering when the nose gear is down. One spring-loaded link in the system absorbs some of the force applied to any of the interconnected rudder pedals until the nose wheel is rolling, at which time the resisting force is less and more pedal motion results in more nose wheel deflection. Since motion of the pedals is transmitted via cables and linkage to the rudder, rudder deflection occurs when force is applied to any of the rudder pedals. With the nose landing gear retracted, some of the force applied to any of the rudder pedals is absorbed by the spring-loaded link in the steering system so that there is no motion at the nose wheel, but rudder deflection still occurs. The nose wheel is self centering upon retraction.

When force on the rudder pedal is augmented by a main wheel braking action, the nose wheel deflection can be considerably increased.

A safety switch on the right main gear torque knee opens the control circuit when the strut is compressed. The safety switch also activates a solenoid-operated down-lock hook on the landing gear control handle located on the pilot's right subpanel. This mechanism prevents the landing gear control handle from being raised when the airplane is on the ground. The hook automatically unlocks when the airplane leaves the ground. In the event of a malfunction of the down-lock solenoid, the down lock can be released by pressing downward on the red down-lock release button. The release button is located just left of the landing gear control handle. The landing gear control handle should never be moved out of the DN detent while the airplane is on the ground; if it is, the landing gear warning horn will sound intermittently and the red gear-in-transit lights in the landing gear control handle will illuminate (provided the MASTER SWITCH is ON), warning the pilot to return the handle to the DN position.

Visual indication of landing gear position is provided by individual green GEAR DOWN annunciators placarded NOSE - L - R on the pilot's right subpanel. The annunciators may be checked in flight by pressing the annunciator. Two red, parallel-wired indicator lights located in the control handle illuminate to show that the gear is in transit or not locked. They also illuminate when the landing gear warning horn is actuated. Absence of illumination indicates that the gear is up and locked or down and locked. The red control handle

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lights may be checked by pressing the HDL LT TEST button located to the right of the landing gear control handle.

LANDING GEAR WARNING SYSTEM (MECHANICAL SYSTEM)

The landing gear warning system is provided to warn the pilot that the landing gear is not down and locked during specific flight regimes. Various warning modes result, depending upon the position of the flaps.

With the flaps in the UP or APPROACH position and either or both power levers retarded below approximately 80% N_1, the warning horn will sound intermittently and the landing gear control handle lights will illuminate. The horn can be silenced by pressing the WARN HORN silence button adjacent to the landing gear control handle; the lights in the landing gear control handle cannot be cancelled. The landing gear warning system will be rearmed if the power lever(s) are advanced sufficiently.

With the flaps beyond the APPROACH position, the warning horn and landing gear control handle lights will be activated regardless of the power settings, and neither can be cancelled.

MANUAL LANDING GEAR EXTENSION (MECHANICAL SYSTEM)

Manual landing gear extension is provided through a separate, chain-drive system. To engage the system, pull the LDG GEAR RELAY circuit breaker, located to the left of the landing gear control handle on the pilot's right subpanel, and ensure that the landing gear control handle is in the DN position. Pull up on the alternate engage handle (located on the floor) and turn it clockwise until it stops. This will electrically disconnect the motor from the system and lock the alternate drive system to the gear box. With the alternate drive locked in, the chain is driven by a continuous-action ratchet, which is activated by pumping the alternate extension handle located adjacent to the alternate engage handle. Stop pumping when all three green gear-down annunciators are illuminated. Further movement of the handle could damage the drive mechanism and prevent subsequent electrical gear retraction. Refer to LANDING GEAR MANUAL EXTENSION (MECHANICAL SYSTEM) in Section IIIA, ABNORMAL PROCEDURES. If any of the following conditions exist, it is likely that an unsafe gear indication is due to an unsafe gear and is not a false indication.

1. The inoperative gear down annunciator illuminates when tested.
2. The red light in the handle is illuminated.
3. The gear warning horn sounds when one or both power levers are retarded below a preset N_1.

After a practice manual extension of the landing gear, the gear may be retracted electrically. Refer to LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION (MECHANICAL SYSTEM) in Section IV, NORMAL PROCEDURES.

HYDRAULIC LANDING GEAR

The retractable tricycle landing gear is electrically controlled and hydraulically actuated. The system utilizes folding braces, called drag legs, that lock in place when the gear is fully extended. The nose gear actuator incorporates an internal mechanical down-lock to hold the gear in the fully extended position. The main gear incorporate mechanical locks on the drag leg and no locks on the actuators. The landing gear is held in the up-lock position by hydraulic pressure.

Hydraulic pressue to the system is supplied by a hydraulic power pack. A hydraulic reservoir located in the left center wing section provides hydraulic fluid to the power pack. The reservoir incorporates a dip stick to provide a visual check of fluid level.

Electrically actuated control valves control the flow of hydraulic fluid to the individual gear actuators. The control valves receive electrical power through the landing gear control handle.

Accidental retraction of the landing gear is prevented through safety switches located on the main landing gears.

LANDING GEAR EXTENSION AND RETRACTION (HYDRAULIC SYSTEM)

The nose and main landing gear assemblies are extended and retracted by a hydraulic power pack in conjunction with hydraulic actuators. The hydraulic power pack is located in the left center section, just forward of the main spar. One hydraulic actuator is located at each landing gear. The power pack consists of: a hydraulic pump, a 28VDC motor, a two section fluid reservoir, filter screens, gear selector valve, two solenoids, a fluid level sensor, and an up-lock pressure switch. For manual extension the system has a hand-lever-operated pump located on the floor between the crew seats. Hydraulic lines, one for normal extension, and one for retraction, routed from the power pack, and one for manual extension from the hand pump, are routed to the nose and main gear actuators. The normal extension lines and the manual extension lines are connected to the upper end of each hydraulic actuator. The hydraulic lines for retraction are fitted to the lower ends of the actuators. Hydraulic fluid under pressure generated by the power pack pump and contained in the accumulator acts on the piston faces of the actuators which are attached to folding drag braces resulting in the extension or retraction of the landing gear.

An internal mechanical lock in the nose gear actuator and the over-center action of the nose gear drag leg assembly lock the nose gear in the down position. Notched hook, lock link and lock link guide attachments fitted to each main gear upper drag leg provide positive down-lock action for the main gear.

Electrical overload to the system is prevented through the use of a 60 ampere circuit breaker located below the flooring near the hydraulic power pack.

The landing gear hydraulic power pack motor is controlled by the use of the landing gear control handle placarded LDG GEAR CONTROL - UP - DN located on the pilot's subpanel. The control handle must be pulled out of a detent before it can be moved from either the UP or DN position.

Safety switches, called squat switches, on the main gear torque knees open the control circuit when the strut is compressed. The squat switches must close to actuate a solenoid which moves a down-lock hook on the landing gear

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control handle to the released position. This mechanism prevents the landing gear control handle from being placed in the UP position when the airplane is on the ground. The hook automatically disengages when the airplane leaves the ground, and can be overridden by pressing down on the red down-lock release button located to the left of the landing gear control handle.

In flight, as the landing gear moves to the full down position, the down lock switches are actuated and interrupt current to the pump motor. When the red gear in-transit lights in the landing gear control handle extinguish and the three green GEAR DOWN annunciators illuminate, the landing gear is in the fully extended position.

Two gear select solenoids located on the valve body of the pump are energized through positioning of the landing gear control switch handle either to the UP or DN position. Once energized, the gear select valve is actuated, allowing hydraulic fluid to flow to the actuators.

Hydraulic system pressure performs the up-lock function, holding the landing gear in the retracted position. When the hydraulic pressure reaches 2775 ±55 psi, the up-lock pressure switch will cause the landing gear relay to open and interrupt the current to the pump motor. The same pressure switch will cause the pump to actuate, should the hydraulic pressure drop to approximately 2400 psi.

A caution annunciator, placarded HYD FLUID LOW, in the caution/advisory annunciator panel will illuminate (yellow) whenever the hydraulic fluid level in the hydraulic power pack is low. The annunciator is tested by pressing the HYD FLUID SENSOR TEST button located on the pilot's subpanel.

The landing gear control handle should never be moved out of the DN detent while the airplane is on the ground. If it is, the landing gear warning horn will sound intermittently, and the red gear-in-transit lights in the landing gear control handle will illuminate (provided the MASTER SWITCH is ON), warning the pilot to return the handle to the DN position.

Landing gear position is indicated by an assembly of three green annunciators. When illuminated, the annunciators indicate that the particular gear is down. Absence of illumination indicates that the gear is up.

Two red parallel-wired indicator lights, located in the landing gear control handle, illuminate to show that the gear is in transit or unlocked. The red lights in the handle also illuminate when the landing gear warning horn is actuated.

The red lights may be checked by pressing the HDL LT TEST button located adjacent to the landing gear control handle.

LANDING GEAR WARNING SYSTEM (HYDRAULIC SYSTEM)

The landing gear warning system is provided to warn the pilot that the landing gear is not down during specific flight regimes. Various warning modes result, depending upon the position of the flaps.

With the flaps in the UP or APPROACH position and either or both power levers retarded below approximately 80% N_1, the warning horn will sound intermittently and the landing gear control handle lights will illuminate. The horn can be silenced by pressing the WARN HORN silence button adjacent to the landing gear control handle; the lights in the landing gear control handle cannot be cancelled. The landing gear warning system will be rearmed if the power lever(s) are advanced sufficiently.

With the flaps beyond APPROACH position, the warning horn and landing gear switch handle lights will be activated regardless of the power settings, and neither can be cancelled.

MANUAL LANDING GEAR EXTENSION (HYDRAULIC SYSTEM)

An alternate extension handle, placarded LANDING GEAR ALTERNATE EXTENSION, is located on the floor on the pilot's side of the pedestal. To engage the system, pull the LANDING GEAR RELAY circuit breaker, located to the left of the landing gear control handle on the pilot's right subpanel, and ensure that the landing gear control handle is in the DN position. Remove the alternate extension handle from the securing clip and pump up and down. While pumping, do not lower the handle below the level of the securing clip during the down stroke as this will allow accumulated hydraulic pressure to bleed off. Continue the pumping action until the three green gear-down annunciators are illuminated, then stow the handle in the securing clip. If one or more gear down annunciators do not illuminate, the alternate handle must not be stowed. Instead, leave it at the top of the up stroke. Continue to pump the handle when conditions permit until the gear is mechanically secured after landing. Refer to LANDING GEAR MANUAL EXTENSION in Section IIIA, ABNORMAL PROCEDURES. If any of the following conditions exist, it is likely that an unsafe gear indication is due to an unsafe gear and is not a false indication.

1. The inoperative gear down annunciator illuminates when tested.
2. The red light in the handle is illuminated.
3. The gear warning horn sounds when one or both power levers are retarded below a preset N_1.

After a practice manual extension of the landing gear, the gear may be retracted hydraulically. Refer to LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION in Section IV, NORMAL PROCEDURES.

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BRAKE SYSTEM

The dual hydraulic brakes are operated by depressing the toe portion of either the pilot's or copilot's rudder pedals. The series system plumbing enables braking by either pilot or copilot.

Dual parking-brake valves are installed adjacent to the rudder pedals between the master cylinders of the pilot's rudder pedals and the wheel brakes. A control for the valves, placarded PARKING BRAKE, is located below the pilot's left subpanel. After the pilot's brake pedals have been depressed to build up pressure in the brake lines, both valves can be closed simultaneously by pulling out the parking brake handle. This retains the pressure in the brake lines. The parking brake is released by depressing the pedals briefly to equalize the pressure on both sides of the valve, then pushing in the parking brake handle to open the valve.

CAUTION

The parking brake should be left off and wheel chocks installed if the airplane is to be left unattended. Changes in the ambient temperature can cause the brakes to release or to exert excessive pressures.

TIRES

The airplane is normally equipped with dual 18x5.5 Type VII, 8-ply-rated, tubeless, rim-inflated tires on each main gear. For increased service life, 10-ply-rated tires of the same size may be installed.

Optionally, the airplane may be equipped with dual 22x6.75-10, 8-ply-rated, tubeless tires on each main gear. These tires provide higher flotation, and permit operation at approximately 2/3 the inflation pressure required for the standard 18x5.5 tires.

The nose gear is equipped with a 22x6.75-10, 8-ply-rated, tubeless tire.

BAGGAGE COMPARTMENT

The entire aft-cabin area (which is aft of the foyer) may be utilized as a baggage compartment. A nylon web is provided for the restraining of loose items. See "Dimensional and Loading Data" and "Cabin Arrangement Diagrams" in Section VI, WEIGHT AND BALANCE/EQUIPMENT LIST.

WARNING

Unless authorized by applicable Department of Transportation Regulations, do not carry hazardous material anywhere in the airplane.

Do not carry children in the baggage compartment unless secured in a seat.

CAUTION

Baggage and other objects should be secured by webs in order to prevent shifting in turbulent air.

Items stowed in the aft-cabin area are accessible in flight. The aft-cabin area can be closed off from the foyer by pulling the optional baggage compartment curtain across the opening and securing it with the snap fasteners provided. Alternately, a latching compartment door may be installed. The door is unlatched by rotating the latch handle clockwise, and latched by rotating the handle counterclockwise.

SEATS, SEATBELTS, AND SHOULDER HARNESSES

SEATS

COCKPIT

The pilot and copilot seats are adjustable fore and aft, as well as vertically. When the release lever under the front inboard corner of the seat is lifted, the seat can be moved forward or aft as required. When the release lever under the front outboard corner of the seat is lifted and no weight is on the seat, the seat will rise in half-inch increments to its highest position. When weight is on the seat and the lever is lifted, the seat will slowly move downward in half-inch increments until the lever is released, or until the seat reaches its lowest point of vertical travel. The armrests pivot at the aft end and can be raised to facilitate entry to and egress from the seats.

CABIN

Various configurations of passenger chairs and 2- or 4-place couches may be installed on the continuous tracks which are mounted on the cabin floor. All passenger chairs are placarded either FRONT FACING ONLY or FRONT OR AFT FACING on the horizontal leg cross brace. Only chairs placarded FRONT OR AFT FACING may be installed facing aft. All aft-facing chairs (and all forward-facing chairs that are equipped with shoulder harnesses) are equipped with adjustable headrests.

WARNING

Before takeoff and landing, the headrest should be adjusted as required to provide support for the head and neck when the passenger leans against the seatback.

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Some passenger chairs can be moved fore and aft, to suit legroom requirements of different passengers, by lifting a horizontal release lever that extends laterally under the front of adjustable seats. ("Front" is the direction opposite the seatback, regardless of whether the chair faces fore or aft.)

The seatbacks can be adjusted to any angle from fully upright to fully reclining, by depressing the release lever located on the side of the seat at the front inboard corner. When the lever is depressed and the passenger leans against the seatback, the seatback will slowly recline until the lever is released, or until the fully reclining position is attained. When no weight is placed against the seatback and the lever is depressed, the seatback will rise until the lever is released, or until the fully upright position is reached. The seatbacks of all occupied seats must be upright for takeoff and landing.

The passenger-chair seatback can also be folded flat over the seat cushion, after releasing the lock lever located on the side of the seat at the back inboard corner.

The optional lateral-tracking passenger chairs incorporate a flat, rectangular release lever underneath the front inboard corner of the seats. When this lever is lifted, the chairs can be adjusted fore and aft, as well as laterally. The seatback adjustments are the same as those on the standard passenger chairs. When occupied, these seats must be in the outboard position (i.e., against the cabin wall) for takeoff and landing.

Inboard armrests on passenger chairs - and both armrests on couches and lateral-tracking chairs - can be folded flush with the top of the seat cushions to facilitate entry to and egress from the seat. The armrests can be lowered by lifting the flat, rectangular release plate located under the front end of the armrest, then moving the armrest toward the front of the seat and downward. The armrest can be raised by pulling the armrest upward and toward the seatback until it locks into place.

The couches are not adjustable.

FOYER

Hinged seat-cushion halves mounted on top of the toilet form an extra passenger seat when the toilet is not is use.

AFT-CABIN AREA

One or two optional folding seats may be installed in the aft-cabin area. They are mounted on the cabin sidewall and swing inboard when unfolded. A latch mechanism on the leg locks the seats in place when they are unfolded. When this seating is not needed, the seat(s) may be folded against the cabin sidewall and held in place with retaining straps.

SEATBELTS

Every seat in the airplane is equipped with a seatbelt. The seatbelt can be lengthened by turning the male half of the buckle at a right angle to the belt, then pulling the male half in the direction away from the anchored end of the belt. The buckle is locked by sliding the male half into the female half of the buckle. The belt is then tightened by pulling the short end of the belt through the male half of the buckle until a snug fit is obtained. The buckle is released by lifting the large, hinged release lever on the female buckle half and pulling the male half of the buckle free. All occupants must wear seat belts during takeoff and landing.

SHOULDER HARNESSES

COCKPIT

The shoulder harness installations for the pilot and copilot seats consist of two straps each. Each strap is routed from the lower aft area of the seat, up the seatback, and through a retaining loop on top of the seatback. One strap is worn over each shoulder. Each strap terminates in a slotted bayonet-blade fastener which is aligned with one edge of the strap. When the two bayonet blades are placed together, the shoulder harness straps can be secured by sliding the male half of the seatbelt buckle through the bayonet slots and into the female half of the seatbelt buckle.

The shoulder harness straps proceed from inertia reels built into the crew chairs. Spring loading at the inertia reels keeps the shoulder harnesses snug, but allows the pilot and copilot all the freedom of movement normally required in flight. However, the inertia reels incorporate a locking device that will secure the harness straps in the event of sudden forward movement.

CABIN

The shoulder harness on passenger chairs consists of a single strap. It is routed through the top of the seatback and terminates in a triangular metal fastener. The strap is worn diagonally. It runs from the outboard shoulder to the inboard hip area, where it is secured by hooking the metal fastener around the securing stud on the male half of the seatbelt buckle.

The shoulder harness strap coils and uncoils from an inertia reel built into the passenger chair. Spring loading at the inertia reel keeps the shoulder harness strap snug, but allows considerable freedom of movement. However, the inertia reel incorporates a locking device that will secure the harness strap in the event of sudden forward movement. If the seat is equipped with a shoulder harness, it must be worn during takeoff and landing.

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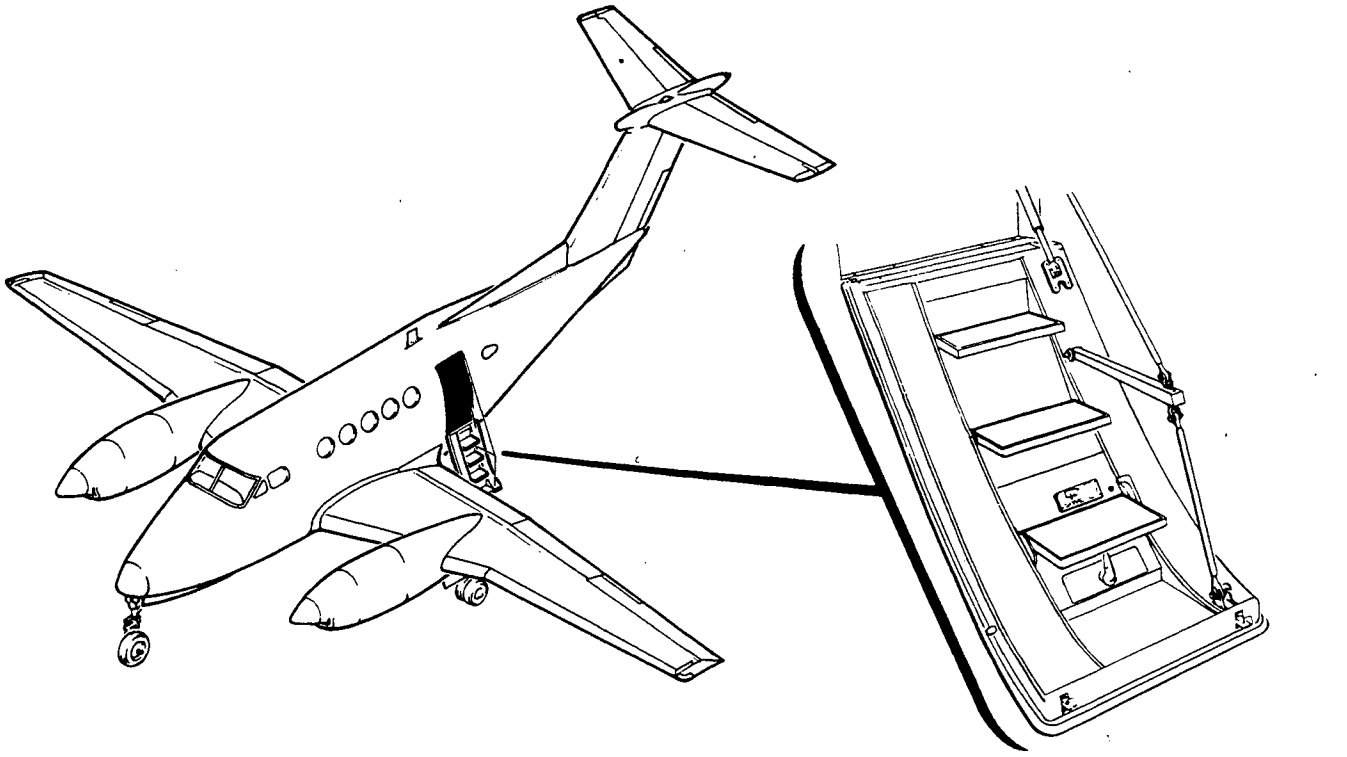
WARNING

Ensure the seatback is in the fully upright position and that the headrest is properly adjusted whenever the shoulder harness is used.

AFT-CABIN AREA

The shoulder harness for the aft-cabin area fold-up chairs is of a double-strap configuration. The middle portion of the strap is secured by a metal slip ring which is anchored to the aft pressure bulkhead. The two ends (which actually function as two separate straps) extend downward toward the seatbelt-buckle area. One end of the shoulder harness strap terminates in a slotted bayonet-blade fastener. The other end is attached to the upper edge of the shoulder harness adjuster. A short adjusting strap, which is also equipped with a slotted bayonet blade fastener, extends upward from the area of the seatbelt buckle and slides through the lower portion of the shoulder harness adjuster. A small, flexible adjusting tab is also attached to the lower edge of the adjuster.

One shoulder harness strap is worn over each shoulder. When the two bayonet blades are placed together, the shoulder harness straps can be secured by sliding the male half of the seatbelt buckle through the bayonet slots and into the female half of the seatbelt buckle. The shoulder harness strap can be lengthened by grasping the tab on the adjuster and pulling upward. The strap can be tightened by grasping the loose end of the adjusting strap and pulling it through the adjuster until the shoulder harness is snug.



B200-104-54

AIRSTAIR DOOR

DOORS, WINDOWS, AND EXITS

AIRSTAIR ENTRANCE DOOR (B200)

The airstair entrance door (cabin door) is hinged at the bottom. It swings outward and downward when opened. A stairway is built onto the outboard side of the door. Two of the stairsteps fold flat against the door when the door is closed. A hydraulic damper ensures that the door will swing down slowly when it opens. While the door is open, it is supported by a plastic-encased cable, which also serves as a handrail. Additionally, this cable is utilized when closing the door from inside the airplane. An inflatable rubber seal is installed around the perimeter of the door, and seats against the door frame as the door is closed. When weight is off the landing gear, engine bleed air supplies pressure to inflate the door seal, which provides a positive pressure-vessel seal around the door. The outside door handle can be locked with a key, for security of the airplane on the ground.

CAUTION

Only one person should be on the airstair door stairway at any one time.

The door locking mechanism is operated by rotating either the outside or the inside door handle. The handles are linked together, so they move together. Two latch bolts at each side of the door, and two latch hooks at the top of the door, lock into the door frame to secure the airstair door.

Whether unlocking the door from the outside or the inside, the release button adjacent to the door handle must be held depressed before the handle can be rotated (counterclockwise from inside the airplane, clockwise from outside) to unlock the door. Consequently, unlocking the door is a two-hand operation requiring deliberate action. The release button acts as a safety device to help prevent accidental opening of the door. As an additional safety measure, a differential-pressure-sensitive diaphragm is incorporated into the release-button mechanism. The outboard side of the diaphragm is open to atmoshperic pressure, the inboard side to cabin air pressure. As the cabin-to-atmospheric pressure differential increases, it becomes increasingly difficult to depress the release button, because the diaphragm moves inboard when either the outboard or inside release button is depressed.

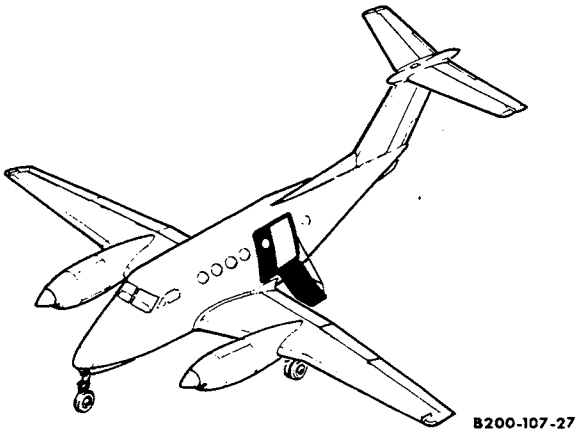
WARNING

Never attempt to unlock or even check the security of the door in flight.

If the CABIN DOOR annunciator illuminates in flight, or if the pilot has any reason whatever to suspect that the door may not be securely locked, the cabin should be depressurized (after first considering altitude), and all occupants instructed to remain seated with their seatbelts fastened. After the airplane has made a full-stop landing and the cabin has been depressurized, a crew member should check the security of the cabin door.

To close the door from outside the airplane, lift up the free end of the airstair door and push it up against the door frame as far as possible. Then grasp the handle with one hand and rotate it clockwise as far as it will go. The door will then move into the closed position. Then rotate the handle counterclockwise as far as it will go. The release button should pop out, and the handle should be pointing aft. Check the security of the airstair door by attempting to rotate the handle clockwise without depressing the release button; the handle should not move.

To close the door from inside the airplane, grasp the handrail cable and pull the airstair door up against the door frame. Then grasp the handle with one hand and rotate it counterclockwise as far as it will go, continuing to pull inward on the door. The door will then move into the closed position. Then turn the handle clockwise as far as it will go. The release button should pop out, and the handle should be pointing down. Check the security of the door by attempting to rotate the handle counterclockwise without depressing the release button; the handle should not move. Next, lift the folded stairstep that is just below the door handle. A placard adjacent to the round observation window advises the observer that the safety lock arm should be in position around the diaphragm shaft (plunger) when the handle is in the locked position. The placard also presents a diagram showing how the arm and shaft should be positioned. A red push-button switch near the window turns on a lamp inside the door, which illuminates the area observable through the window. If the arm is properly positioned around the shaft, proceed to check the indication in each of the visual inspection ports, one of which is located near each corner of the door. The green stripe painted on the latch bolt should be aligned with the black pointer in the visual inspection port. If any condition specified in this door-locking procedure is not met, DO NOT TAKE OFF.



AIRSTAIR ENTRANCE DOOR (B200C)

The airstair door is built into the cargo door. It is hinged at the bottom, and swings downward when opened. It has a stairway built onto the inboard side. Two of the stairsteps fold flat against the door when the door is closed. When the door is opened, a self-storing platform automatically folds down over the door sill to protect the rubber door seal. A hydraulic damper ensures that the door will swing down slowly when it opens. While the door is open, it is supported by a plastic-encased cable, which also serves as a handrail. Additionally, this cable is utilized when closing the door from inside the airplane. An inflatable rubber seal is installed around the perimeter of the door, and seats against the door frame as the door is closed. When the cabin is pressurized, air seeps into the rubber seal through small holes in the outboard side of the seal. The higher the cabin differential pressure, the more the seal inflates. This is a passive-seal system with no mechanical connection to a bleed air source. The outside door handle can be locked with a key, for security of the airplane on the ground.

For Training Purposes Only

CAUTION

Only one person should be on the airstair door stairway at any one time.

The door locking mechanism is operated by rotating either the outside or the inside door handle, both of which move simultaneously. Three hollow, crescent latches on each side of the door rotate to capture or release latch posts mounted in the cargo door to secure the airstair door. When latched, the airstair door becomes an integral part of the cargo door.

Whether unlocking the door from the outside or the inside, the release button adjacent to the door handle must be held depressed before the handle can be rotated (counterclockwise from inside the airplane, clockwise from outside) to unlock the door. Consequently, unlocking the door is a two-hand operation requiring deliberate action. The release button acts as a safety device to help prevent accidental opening of the door. As an additional safety measure, a differential-pressure-sensitive diaphragm is incorporated into the release-button mechanism. The outboard side of the diaphragm is open to atmoshperic pressure, the inboard side to cabin air pressure. As the cabin-to-atmospheric pressure differential increases, it becomes increasingly difficult to depress the release button, because the diaphragm moves inboard when either the outboard or inside release button is depressed.

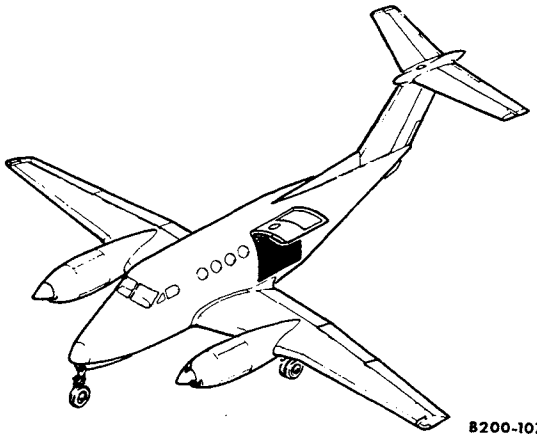
WARNING

Never attempt to unlock or even check the security of the door in flight.

If the CABIN DOOR annunciator illuminates in flight, or if the pilot has any reason whatever to suspect that the door may not be securely locked, the cabin should be depressurized (after first considering altitude), and all occupants instructed to remain seated with their seatbelts fastened. After the airplane has made a full-stop landing and the cabin has been depressurized, a crew member should check the security of the airstair door and the cargo door.

To close the door from outside the airplane, lift up the free end of the airstair door and push it up against the door frame as far as possible. Then grasp the handle with one hand and rotate it clockwise as far as it will go. The door will then move into the closed position. Then rotate the handle counterclockwise as far as it will go. The release button should pop out, and the handle should be pointing aft. Check the security of the door by attempting to rotate the handle clockwise without depressing the release button; the handle should not move.

To close the door from inside the airplane, grasp the handrail cable and pull the airstair door up against the door frame. Then grasp the handle with one hand and rotate it counterclockwise as far as it will go, continuing to pull inward on the door. The door will then move into the closed position. Then turn the handle clockwise as far as it will go. The release button should pop out, and the handle should be pointing down. Check the security of the door by attempting to rotate the handle counterclockwise without depressing the release button; the handle should not move. Next, lift the second folded airstep below the door handle. A placard adjacent to the round observation window advises the observer that the safety lock arm should be in position around the diaphragm shaft (plunger) when the handle is in the locked position. The placard also presents a diagram showing how the arm and shaft should be positioned. A red push-button switch near the window turns on a lamp inside the door, which illuminates the area observable through the window. If the arm is properly positioned around the shaft, proceed to check the orange stripe on each of the six rotary latches (three on each side of the airstair door) and ensure each is aligned with the notch in the plate on the door frame. Finally, turn the battery switch ON and check the warning annunciator panel in the cockpit; ensure that the red CABIN DOOR annunciator is extinguished. It will illuminate when the battery switch is ON and the airstair door is not closed and securely latched. With the BATtery switch OFF and the airstair door closed but not latched, the CABIN DOOR annunciator will illuminate. If any condition specified in this door-latching procedure is not met, DO NOT TAKE OFF.



CARGO DOOR (B200C)

A large, swing-up cargo door, hinged at the top, provides access for the loading of large items. The cargo door latch system is operated by two handles: one in the upper aft area of the door, and the other in the lower forward area of the door. Two separate access covers must be opened in order to operate the two handles. In order to move the upper aft handle out of the latched position, depress the black release button in the handle and rotate the yellow handle upward as far as it will go. This movement is transmitted via cables to two hollow, crescent latches on the forward side and two on the aft side of the cargo door. The latches rotate to release latch posts mounted in the cargo door frame.

In order to move the lower latch handle out of the CLOSED position (forward), lift the orange lock hook from the stud on the yellow latch handle, and rotate the handle aft as far as it will go. This movement is transmitted via linkage to four latch pins on the bottom of the cargo door. The pins move aft to disengage latch lugs mounted at the bottom of the cargo door frame.

CAUTION

After unlatching the bottom latch pins, close the forward latch handle access cover. If this cover is left open, it will rotate on its hinge until a portion of it extends below the bottom of the cargo door when the cargo door is opened. Then, when the cargo door is subsequently closed, the access cover will be broken.

To open the cargo door after it is unlatched, push out on the bottom of the door. After the cargo door is manually opened a few feet, gas springs take over and raise the door to the fully open position.

To close the cargo door, pull it down and inboard. The gas springs will resist the closing effort until the door is only open a few feet. Then, as the springs move over center, they begin applying a closing force to the door.

An inflatable rubber seal is installed around the perimeter of the cargo door, and seats against the door frame when closed. When the cabin is pressurized, air seeps into the rubber seal through small holes in the outboard side of the seal. The higher the cabin differential pressure, the more the seal inflates. This is a passive-seal system and has no mechanical connection to a bleed air source.

There are no latch handles on the outside of the cargo door, so it can be opened and closed from inside the airplane only.

To latch the cargo door after it is closed, rotate the lower forward latch handle forward until the orange lock hook engages the stud on the handle. Check the security of this handle by attempting to move it aft without raising the lock hook; it should not move. Close the access cover. Next, check the observation window at the lower aft corner of the cargo door. Ensure that the orange stripe on the latch pin linkage is aligned with the orange pointer in the observation window.

Next, rotate the upper aft latch handle down until the black release button pops up. Check the security of this handle by attempting to pull it out and up without depressing the release button; it should not move. Close the access cover. Then, insure that the orange stripe on each of the four rotary latches (two on each side of the cargo door) is aligned with the notch in the plate on the door frame. Finally, check the caution annunciator panel in the cockpit and ensure that the amber CABIN DOOR annunciator is extinguished. With the battery switch ON, it will be illuminated if either the airstair door or the cargo door is open. With the battery switch OFF, it will be illuminated only if the airstair door is closed but not securely latched. Perform the "Cabin/Cargo Door Circuitry Check" in the NORMAL PROCEDURES Section prior to the first flight of the day. If any condition specified in this door-latching procedures is not met, DO NOT TAKE OFF.

EMERGENCY EXIT

The emergency exit door is located on the right side of the fuselage at the forward end of the passenger compartment. From the inside, the door is released with a pull-down

handle, placarded EXIT-PULL. From the outside, the door is released with a flush-mounted, pull-out handle. The non-hinged, plug-type door removes completely from the frame into the cabin when the latches are released.

B200:

The door can be locked so that it cannot be removed or opened from the outside using the flush-mounted pull-out handle. The door is locked when the lock-lever (inside) is in the down or locked position. Locking the door is for security when the airplane is parked. The lock-lever should be in the up or unlocked position prior to flight, to allow removal of the door from the outside in the event of an emergency. Removal of the door from the inside is possible at all times using the EXIT-PULL handle, since this handle is not locked by the lock-lever. An exit lock placard is placed on the lock-lever so that it can be read when the lever is in the locked position.

B200C:

The door can be locked with a key from the inside, to prevent opening from the outside. The inside handle will unlatch the door, whether or not it is locked, by overriding the locking mechanism. The key lock should be unlocked prior to flight, to allow removal of the door from the outside in the event of an emergency. The keyhole is in the horizontal position when the door is locked. The key cannot be removed in this position.

A wiper-type disconnect for the air duct that supplies air to the eyeball outlet in the emergency exit door is located on the upper-aft edge of the door. As the door is removed, the duct is disconnected, since it is an integral part of the door.

Located on the lower-forward edge of the door is an electrical disconnect for the wiring that goes to the reading light and the fluorescent light in the emergency exit door. It will unplug as the door is being removed. Upon reinstalling the door, the electrical disconnect should be reconnected before moving the door into the closed position.

INTERIOR DOORS

Sliding doors are provided between the cockpit and cabin, and between the cabin and foyer. These doors provide privacy, and prevent the spilling of light from one compartment into another. The doors are closed by sliding the two partition-type door panels to the center of the aisle, where they are held together by a magnetic strip in the edge of each door.

CABIN WINDOWS

Each cabin window pane, which is composed of a sheet of polyvinyl butyral (PVB) laminated between two sheets of clear acrylic plastic, is stressed to withstand the cabin-to-ambient air pressure differential. It is then sealed into a window opening in the fuselage, and forms an integral part of the pressure vessel.

POLARIZED TYPE

Two dust panes are mounted inboard of the cabin window pane in each window frame. Each of these dust panes is composed of a film of polarizing material laminated between two sheets of acrylic plastic. The inboard dust pane rotates freely in the window frame and has a protruding thumb knob near the edge. Rotating the pane through an arc of 90° permits complete light regulation as desired. Rotation changes the relative alignment between the polarizing films, thus providing any degree of light transmission from full intensity to almost none.

WARNING

Do not look directly at the sun, even through polarized windows, because eye damage could result.

CAUTION

When the airplane is to be parked in areas exposed to intense sunlight, the polarized windows should be rotated to the clear position to prevent deterioration of the polarization coating. Sufficient ultraviolet protection is provided to prevent fading of the upholstery.

SHADE TYPE

A dust pane, which is a single sheet of tinted acrylic plastic, is mounted inboard of the cabin window pane in each window frame. An adjustable window shade is provided to control the amount of light admitted. The shade is adjusted by squeezing the two latch handles located on the lower center of the shade, then positioning the shade as desired. Detents in the shade tracks provide positive latching action at various positions.

SUNVISORS

OPERATING INSTRUCTIONS

TO OPERATE FROM STOWED POSITION:

Push straight back and pull down. Move along track to desired place and pivot out near windshield (or window), rotate knob clockwise to lock.

To change position:
Rotate knob counterclockwise to unlock, move to desired location and position, then relock knob by turning clockwise.

To stow:
Rotate knob counterclockwise to unlock, move along track to aft end, pivot up against headliner to allow catch to retain sunvisor assembly.

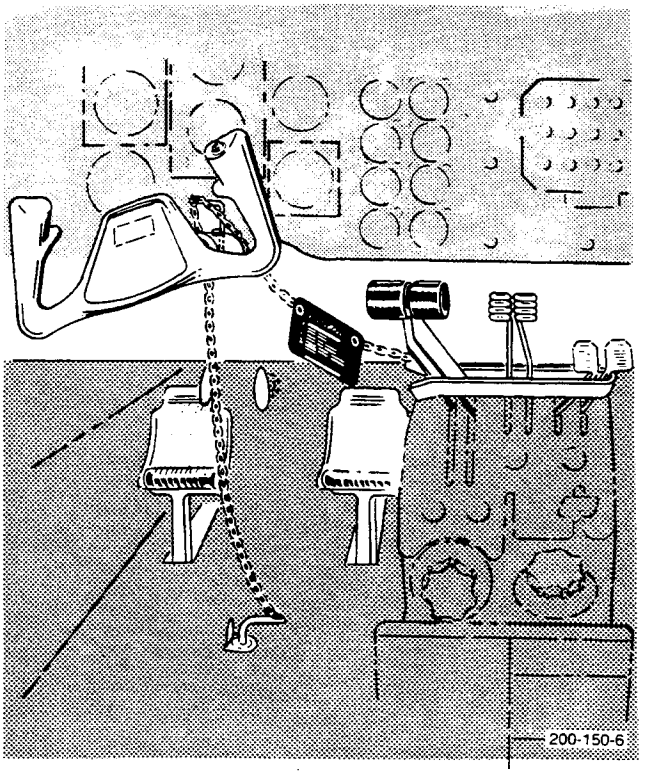
CONTROL LOCKS

The control locks consist of a U-shaped clamp and two pins, all connected by a chain. The pins lock the primary flight controls; the U-shaped clamp fits around the engine control

levers, serving to warn the pilot not to start the engines with the control locks installed. It is important that all the locks be installed and removed together, to preclude the possibility of attempting to taxi or fly the airplane with the engine control levers released, but with the pins still installed in the flight controls.

Install the control locks in the following sequence:

1. Position the U-clamp around the engine control levers.
2. Move the control column as necessary to align the holes, then insert the L-shaped pin that is attached to the middle of the chain (approx.). The control wheel position: full forward and rotated approximately 15° to the left.
3. Insert the L-shaped pin (attached to the end of the chain) through the hole provided in the floor aft of the rudder pedals. The rudder pedals must be centered to align the hole in the rudder bellcrank with the hole in the floor. The pin is then inserted until the flange is resting against the floor. This will prevent any rudder movement.



WARNING

Before starting engines, remove the control locks, reversing the above procedure.

CAUTION

Remove the control locks before towing the airplane. If towed with a tug while the rudder lock is installed, serious damage to the steering linkage can result.

ENGINES

The BEECHCRAFT Super King Air B200/B200C is powered by two Pratt & Whitney Canada PT6A-42 turbopropeller engines, each rated at 850 SHP. Each engine has a three-stage axial flow, single-stage centrifugal-flow compressor, which is driven by a single-stage reaction turbine. The power turbine - a two-stage reaction turbine counter-rotating with the compressor turbine - drives the output shaft. Both the compressor turbine and the power turbine are located in the approximate center of the engine, with their shafts extending in opposite directions. Being a reverse flow engine, the ram air supply enters the lower portion of the nacelle and is drawn in through the aft protective screens. The air is then routed into the compressor. After it is compressed, it is forced into the annular combustion chamber, and mixed with fuel that sprayed in through 14 nozzles mounted around the gas generator case. A capacitance discharge ignition unit and two spark igniter plugs are used to start combustion. After combustion, the exhaust passes through the compressor turbine and two stages of power turbine and is routed through two exhaust ports near the front of the engine. A pneumatic fuel control system schedules fuel flow to maintain the power set by the gas generator power lever. Propeller speed within the governing range remains constant at any selected propeller control lever position through the action of a propeller governor, except in the beta range where the maximum propeller speed is controlled by the pneumatic section of the propeller governor.

The accessory drive at the aft end of the engine provides power to drive the fuel pumps, fuel control, the oil pumps, the refrigerant compressor (right engine), the starter/generator, and the tachometer transmitter. At this point, the speed of the drive (N_1) is the true speed of the compressor side of the engine, 37,500 rpm (which corresponds to 100% N_1). Maximum continuous speed of the engine is 38,100 rpm, which equals 101.5% N_1, with a transient overspeed of 38,500 rpm, which equals 102.6% N_1.

The reduction gearbox forward of the power turbine provides gearing for the propeller and drives the propeller tachometer tramsmitter, the propeller overspeed governor, and the propeller governor. The turbine speed on the power side of the engine is 30,000 rpm. After reduction gearing the propeller rpm is 2000.

Engine torque at the propeller shaft is indicated by a torquemeter located inside the first stage reduction gear housing. The torquemeter is a hydromechanical torque measuring device. It consists of: a ring gear and case (helical splines between ring gear and case), torquemeter cylinder, torquemeter piston, valve plunger and spring, differential pressure sensor and servo transmitter combination, and servo indicator calibrated to indicate ft-lbs.

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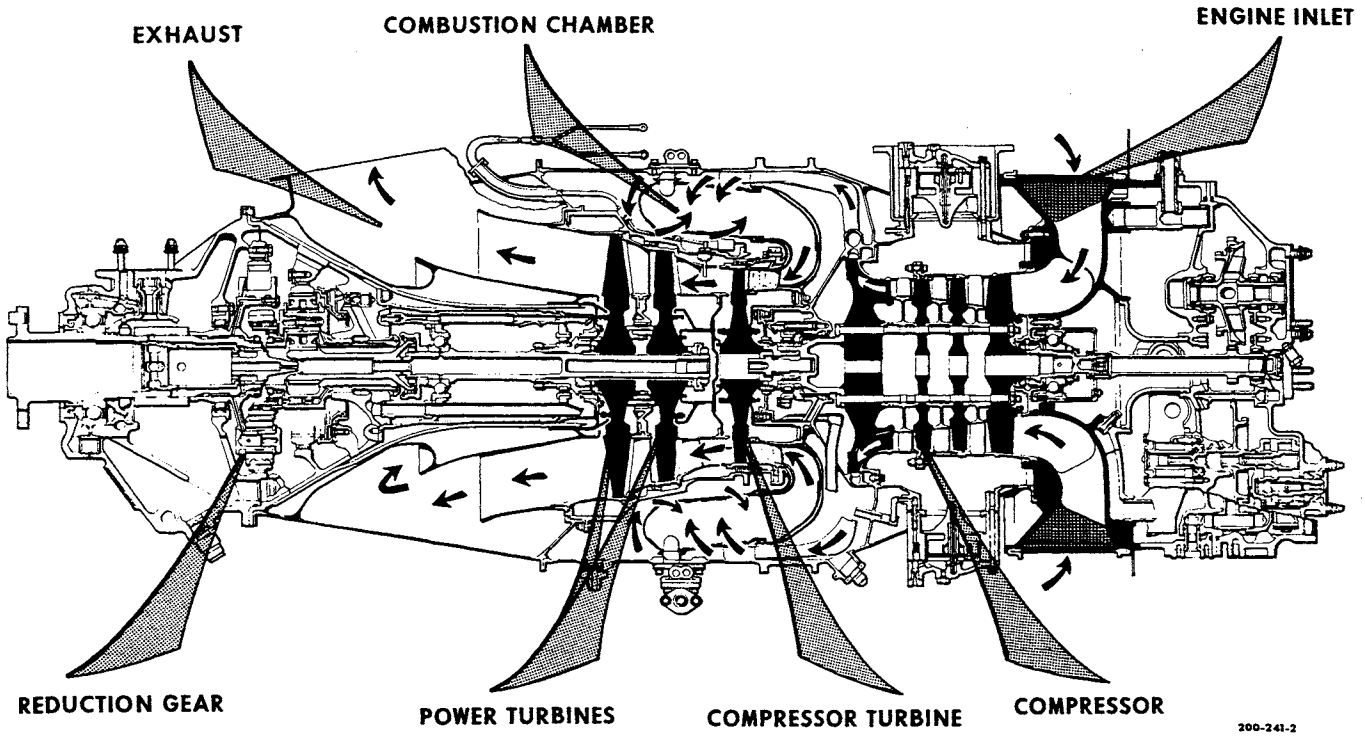
Torque at the power turbine shaft and the resisting torque at the propeller shaft gears is converted from rotary motion by the helical splines to a translating motion at the piston face. A change in torquemeter oil pressure results from the piston translation. The valve plunger and spring maintains oil pressure proportional to engine torque. The differential pressure sensor uses a bellows system to sense differences between torquemeter oil pressure and a reference pressure. Bellows movement drives the transmitter servo. The electric signal from the transmitter drives the servo motor in the torquemeter indicator. Torque is indicated by indicator needle position on a calibrated dial.



Deceleration on the ground is achieved by bringing the propeller blades through the Beta range into a reversing pitch by utilizing the pitch change mechanism. The power levers must be retarded below the IDLE position by raising them over a detent. Reversing power is avalilable in direct proportion to the retarding of the levers in the reversing range.

PROPULSION SYSTEM CONTROLS

The propulsion system is operated by three sets of controls; the power levers, propeller levers, and condition levers. The power levers serve to control engine power. The condition levers control the flow of fuel at the fuel control outlet and select fuel cutoff, low idle and high idle functions. The propeller levers are operated conventionally and control the constant speed propellers through the primary governor.

POWER LEVERS

The power levers provide control of engine power from idle through take-off power by operation of the gas generator (N_1) governor in the fuel control unit. Increasing N_1 rpm results in increased engine power.

PROPELLER LEVERS

Each propeller lever operates a speeder spring inside the primary governor to reposition the pilot valve, which results in an increase or decrease of propeller rpm. For propeller feathering, each propeller lever lifts the pilot valve to a position which causes complete dumping of high pressure oil, allowing the counterweights and feathering spring to change the pitch. Detents at the rear of lever travel prevent inadvertent movement into the feathering range. Operating range is 1600 to 2000 rpm.

CONDITION LEVERS

The condition levers have three positions; FUEL CUT-OFF, LOW IDLE and HIGH IDLE. Each lever controls the fuel cutoff function of the fuel control unit and limits idle speed at 56% N_1 for low idle, and 70% N_1 for high idle.

PROPELLER REVERSING

When the power levers are lifted over the IDLE detent, they control engine power through the Beta and reverse ranges.

CAUTION

Propeller reversing on unimproved surfaces should be accomplished carefully to prevent propeller erosion from reversed airflow and, in dusty conditions, to prevent obscuring the operator's vision.

Condition levers, when set at HIGH IDLE, keep the engines operating at 70% N_1 high idle speed for maximum reversing performance.

CAUTION

Power levers should not be moved into the reversing position when the engines are not running because the reversing system will be damaged.

FRICTION LOCKS

Four friction locks are located on the power quadrant of the pedestal. When they are rotated counterclockwise, the propulsion system control levers can be moved freely. As the friction locks are rotated clockwise, the control levers progressively become more resistant to movement, so that they will not creep out of position.

ENGINE INSTRUMENTATION

Engine instruments, located to the left of the center portion of the instrument panel, are grouped according to their function. At the top, the ITT (Interstage Turbine Temperature) indicators and torquemeters are used to set take-off power. Climb and cruise power are established with the torquemeters and propeller tachometers while observing ITT limits. Gas generator (N_1) operation is monitored by the gas generator tachometers. The lower grouping consists of the fuel flow indicators and the oil pressure/temperature indicators.

The ITT indicator gives an instantaneous reading of engine gas temperature between the compressor turbine and the power turbines.

The torquemeters give an indication in foot-pounds of the torque being applied to the propeller.

The propeller tachometer is read directly in revolutions per minute. The N_1 or gas generator tachometer is read in percent of rpm, based on a figure of 37,500 rpm at 100%. Maximum continuous gas generator speed is limited to 38,100 rpm or 104.5% N_1.

Proper observation and interpretation of these instruments provide an indication of engine performance and condition.

A propeller synchroscope, located to the left of the oil pressure/temperature indicators, operates to give an indication of synchronization of the propellers. If the right propeller is operating at a higher rpm than the left, the face of the synchroscope, a black and white cross pattern, spins in a clockwise rotation. Left, or counterclockwise, rotation indicates a higher rpm of the left propeller. This instrument aids the pilot in obtaining complete synchronization of propellers.

PROPELLER SYNCHROPHASER (OPTIONAL)

TYPE I SYSTEM

The Propeller Synchrophaser system automatically matches the RPM of the right propeller (slave propeller) to that of the left propeller (master propeller) and maintains the blades of one propeller at a predetermined relative position with the blades of the other propeller. To prevent the right propeller from losing excessive rpm if the left propeller is feathered while the synchrophaser is on, the synchrophaser has a limited adjustment range from the manual governor setting. Normal governor operation is unchanged but the synchrophaser will continuously monitor propeller rpm and reset the governor as required. A magnetic pickup mounted in each propeller overspeed governor and adjacent to each propeller deice brush block transmits electric pulses to a transistorized control box installed forward of the pedestal.

The control box converts any pulse rate differences into correction commands, which are transmitted to a stepping type actuator motor mounted on the right engine cowl forward support ring. The motor then trims the right propeller governor through a flexible shaft and trimmer assembly to exactly match the left propeller. The trimmer, installed between the governor control arm and the control cable, screws in or out to adjust the governor while leaving the control lever setting constant. A toggle switch installed adjacent to the synchroscope turns the system on. With the switch off, the actuator automatically runs to the center of its range of travel before stopping to assure normal function when used again. To operate the system, synchronize the propellers in the normal manner and turn the synchrophaser on. The system is designed for in-flight operations and is placarded to be off for takeoff and landing. Therefore, with the system on and the landing gear extended, the caution flashers and a yellow light on the caution/advisory annunciator panel, PROP SYNCH ON, will illuminate.

The right propeller rpm and phase will automatically be adjusted to correspond to the left. To change rpm, adjust both propeller controls at the same time. This will keep the right governor setting within the limiting range of the left propeller. If the synchrophaser is on but is unable to adjust the right propeller to match the left, the actuator has reached the end of its travel. To re-center, turn the switch off, synchronize the propellers manually, and turn the switch back on.

TYPE II SYSTEM

The Propeller Synchrophaser system is an electronic system certified for all operations including takeoff and landing. The system automatically matches the RPM of both propellers and positions them at a preset phase relationship in order to reduce cabin noise.

Before engaging the system, manually set the RPM of each engine to within 10 RPM of each other. When the prop sync

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switch is turned on, engagement will automatically occur when the relative phase angle of the propellers is within 30° of the preset angle. When the system engages, both propeller speeds are increased by one-half the holding range of the system. To maintain synchronization, the system increases the RPM of the slower propeller and simultaneously reduces the RPM of the faster propeller. The system will never reduce RPM below that selected by the propeller control lever.

To change RPM with the system on, adjust both propeller controls by the same amount. If the synchrophaser is on but does not maintain synchronization, the system has reached the end of its range. Increasing the setting of the slow propeller, or reducing the setting of the fast propeller, will bring the speeds within the limited synchrophaser range. If preferred, the synchrophaser switch may be turned off, the propellers re-synchronized manually, and the synchrophaser turned back on.

ENGINE LUBRICATION SYSTEM

Engine oil, contained in an integral tank between the engine air intake and the accessory case, cools as well as lubricates the engine. An oil radiator located inside the lower nacelle, keeps the engine oil temperature within the operating limits. A thermal element is used to regulate a bypass door which controls the volume of cooling air through the radiator. Engine oil also operates the propeller pitch change mechanism and the engine torquemeter system.

The lubrication system capacity per engine is 3.5 U.S. gallons. The oil tank capacity is 2.3 gallons with 5 quarts measured on the dipstick for adding purposes. Recommended oils and oil changing procedures are listed in the SERVICING Section.

MAGNETIC CHIP DETECTOR

A magnetic chip detector is installed in the bottom of each engine nose gearbox. This detector will activate a red annunciator, L CHIP DETECT or R CHIP DETECT, to alert the pilot of oil contamination indicating possible or pending engine failure.

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STARTING AND IGNITION SYSTEM

Each engine is started by a three-position switch located on the pilot's left subpanel placarded, IGNITION AND ENGINE START - LEFT - RIGHT - ON - OFF - STARTER ONLY. Each switch may be moved downward to the STARTER ONLY position to motor the engine for the purpose of clearing it of fuel without the ignition circuit on. The switch is spring loaded and will return to the center position when released. Moving the switch upward to the ON position activates both the starter and ignition, and the appropriate green IGNITION ON light on the annunciator panel will illuminate. When engine speed has accelerated through 50% N_1 or above on starting, the starter drive action is stopped by placing the switch in the center OFF position.

AUTO IGNITION

The auto ignition system provides automatic ignition to prevent engine loss due to combustion failure. This system is provided to ensure ignition during takeoff, landing, turbulence, and penetration of icing or precipitation conditions. Arming the system prior to takeoff and turning the system off after landing is required to assure the system being armed in the required conditions. To arm the system, move the required ENG AUTO IGNITION switches, located on the pilot's subpanel, from OFF to ARM. If for any reason the engine torque falls below approximately 410 foot-pounds, the igniter will automatically energize and the IGNITION ON light on the caution/advisory annunciator panel will illuminate. For extended ground operation, the system should be turned off to prolong the life of the igniter units.

INDUCTION AIR SYSTEM

The PT6A-42 is a reverse-airflow engine. The compressor wheels draw ambient air into the engine through the induction air inlet at the lower front of the engine nacelle. As airspeed increases, ram air pressure rises, compressing the air inside the induction air duct. The air then flows into an annular inlet-air chamber located at the aft end of the engine compartment. It then passes through a protective screen and into the primary compressor impeller, where it is further compressed. Then the air is forced through a stator ring and successively through the second and third axial-flow compressor stages. It is finally compressed in the centrifugal-flow compressor stage, then discharged into the turbine plenum assembly. Air from the plenum enters the annular combustion chamber through a series of holes in the aft end of the combustion chamber, and mixes with fuel that is sprayed into the combustion chamber through 14 nozzles mounted around the gas generator case. The air-fuel mixture burns inside the combustion chamber, then the hot gasses expand forward out of the chamber and pass through the compressor turbine stage, both stages of the power turbine, and out to the atmosphere through two exhaust ports located at the side of each nacelle, near the front.

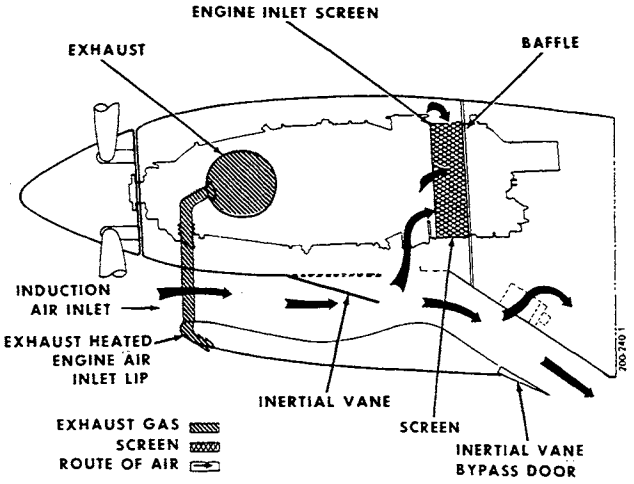
ICE PROTECTION

ENGINE AIR INLET

Engine exhaust heat is utilized for heating the engine air inlet lips. Hot exhaust is picked up by a scoop inside each engine exhaust stack and plumbed downward to connect into each side of the inlet lip. Exhaust flows through the inside of the lip downward to the bottom where it is plumbed out through the bottom of the nacelle. No shut-off or temperature indicator is necessary for this system.

ICE VANES (INERTIAL SEPARATOR SYSTEM)

An inertial separation system is built into each engine air inlet to prevent moisture particles from entering the engine inlet plenum under icing conditions. A movable vane and a bypass door are lowered into the airstream when operating in visible moisture at +5°C or colder, by energizing electrical actuators with the switches, placarded ICE VANE - EXTEND - RETRACT - LEFT & RIGHT, located on the pilot's subpanel. The vane deflects the ram airstream slightly downward to introduce a sudden turn in the airstream to the engine, causing the moisture particles to continue on undeflected, because of their greater momentum, and to be discharged overboard.



ENGINE ICE PROTECTION

While in the icing flight mode, the extended position of the vane and by-pass door is indicated by green annunciator lights, L ICE VANE EXT and R ICE VANE EXT.

In the non-ice-protection mode, the vane and by-pass door are retracted out of the airstream by placing the ice vane switches in the RETRACT position. The green annunciator lights will extinguish. Retraction should be accomplished at +15°C and above to assure adequate oil cooling. The vanes should be either extended or retracted; there are no intermediate positions.

If for any reason the vane does not attain the selected position within 15 seconds, a yellow L ICE VANE or R ICE VANE light illuminates on the caution/advisory panel. In this event, a mechanical backup system is provided, and is actuated by pulling the T-handles just below the pilot's subpanel placarded ICE VANE EMERGENCY - MANUAL EXTENSION - PULL - LEFT ENG - R ENG. Decrease airspeed to 160 knots or less to reduce forces for manual extension. Normal airspeed may then be resumed.

CAUTION

Once the manual override system has been engaged (i.e., anytime the manual ice vane T-handle has been pulled out), do not attempt to retract or extend the ice vanes electrically, even if the T-handle has been pushed back in, until the override linkage in the engine compartment has been properly reset on the ground. (See the maintenance manual for resetting procedure.)

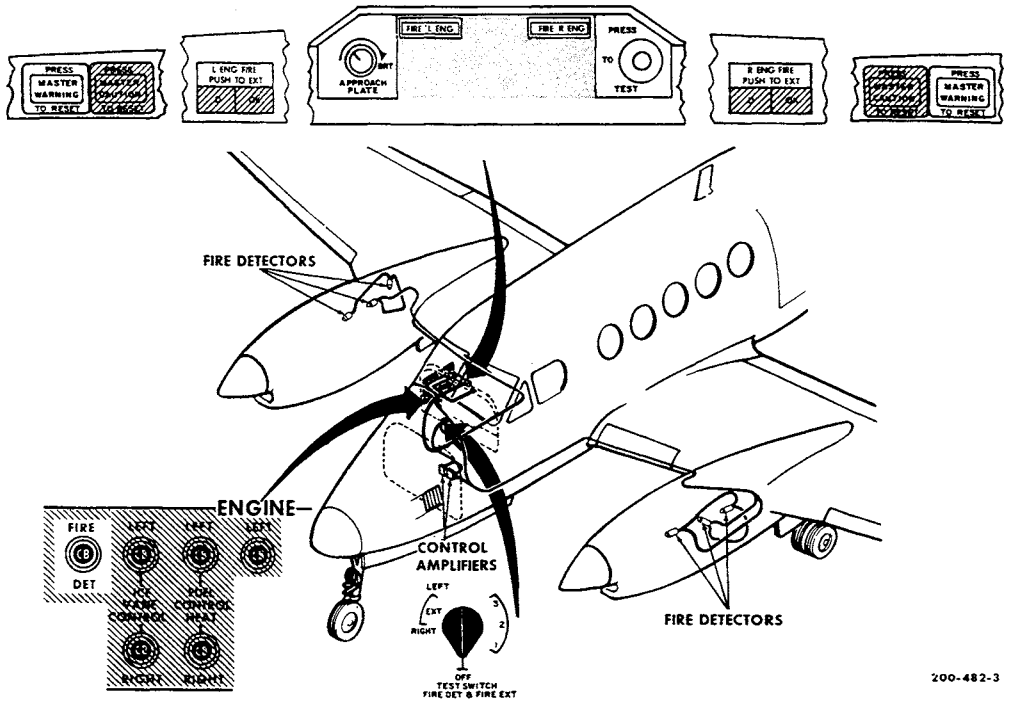
When the vane is successfully positioned with the manual system, the yellow annunciator lights will extinguish. The vane may also be retracted with the manual system. During manual system use, the electric motor switch position must match the manual handle position for a correct annunciator readout.

FUEL CONTROL

The engine fuel system consists of an engine-driven fuel pump, a fuel control unit (FCU), a flow divider and dump valve, dual fuel manifold, fourteen fuel nozzles, and two fuel drain valves. The fuel pump/fuel control unit assembly is mounted on the engine accessory case and is shaft-driven at a speed proportional to that of the compressor turbine. The fuel pump delivers fuel to the FCU. Engine power output is established by power lever position. The power lever is linked to the governor in the FCU which regulates fuel flow to the combustion section and thereby controls N_1 and power output. Increasing N_1 rpm results in increased engine power. System function depends upon the interaction of the fuel control unit governor and the propeller governor. The position of the fuel control unit metering valve is determined by differential pressures that vary proportionately with power required (as sensed by the fuel control unit) and propeller RPM.

The flow divider directs fuel from the metering valve to the primary and secondary fuel manifolds (or primary manifold only, depending on engine power requirements) and thence to the fuel nozzles. The flow divider also incorporates a dump valve that automatically drains residual fuel from both manifolds at engine shutdown. The fuel drain valves drain fuel from the combustion chamber at engine shutdown and at engine false starts. A fuel purge system has been installed in this case and on engine shutdown when fuel manifold pressure subsides, allows the dump poppett valve to open, the purge tank pressure forces fuel out of the engine fuel manifold lines through the nozzles and into the combustion chamber where the fuel is consumed. Constant fuel pressure is maintained by a fuel filter bypass valve and a pressure relief valve.

The oil-to-fuel heater mounted below the fuel pump on the accessory case is a heat exchanger that transfers heat from the engine lubricating oil to preheat the fuel. A fuel temperature-sensing oil bypass valve regulates the fuel temperature by either allowing oil to flow through the heater circuit, or bypass it to the engine oil tank.



FIRE DETECTION SYSTEM SCHEMATIC

FIRE DETECTION SYSTEM

The fire detection system is designed to provide immediate warning in the event of fire in either engine compartment. The system consists of the following: three photoconductive cells for each engine; a control amplifier for each engine; two red warning lights on the warning annunciator panel, one placarded FIRE L ENG, the other FIRE R ENG; a test switch on the copilot's left subpanel; and a circuit breaker placarded FIRE DET on the right side panel. The six photoconductive-cell flame detectors are sensitive to infrared radiation. They are positioned in each engine compartment so as to receive both direct and reflected rays, thus monitoring the entire compartment with only three photocells.

Conductivity through the photocell varies in direct proportion to the intensity of the infrared radiation striking the cell. As conductivity increases, the amount of current from the electrical system flowing through the flame detector increases proportionally. To prevent stray light rays from signaling a false alarm, a relay in the control amplifier closes only when the signal strength reaches a preset alarm level. When the relay closes, the appropriate left or right warning annunciators illuminate. When the fire has been extinguished, the cell output voltage drops below the alarm level and the relay in the control amplifier opens. No manual resetting is required to reactivate the fire detection system.

The test switch on the copilot's left subpanel, placarded TEST SWITCH - FIRE DET & FIRE EXT, has six positions: OFF - RIGHT EXT - LEFT EXT - 3 - 2 - 1 (If the optional engine-fire-extinguisher system is not installed, the RIGHT EXT and LEFT EXT positions on the left side of the test switch will not be installed.) the three test positions for the fire detector system are located on the right side of the switch (3 - 2 - 1). When the switch is rotated from OFF (down) to any one of these three positions, the output voltage of a corresponding flame detector in each engine is increased to a level sufficient to signal the amplifier that a fire is present. The following should illuminate: the red pilot and copilot MASTER WARNING flashers, and, if the optional engine-fire-extinguisher system is installed, the red lenses placarded L ENG FIRE - PUSH TO EXT and R ENGI FIRE - PUSH TO EXT on the fire-extinguisher activation switches. The system may be tested anytime, either on the ground or in flight. The TEST SWITCH should be placed in all three positions, in order to verify that the circuitry for all six fire detectors is functional. Illumination failure of all the fire detection system annunciators when the TEST SWITCH is in any one of the three flame-detector-test positions indicates a malfunction in one or both of the two detector circuits (one in each engine) being tested by that particular position of the TEST SWITCH.

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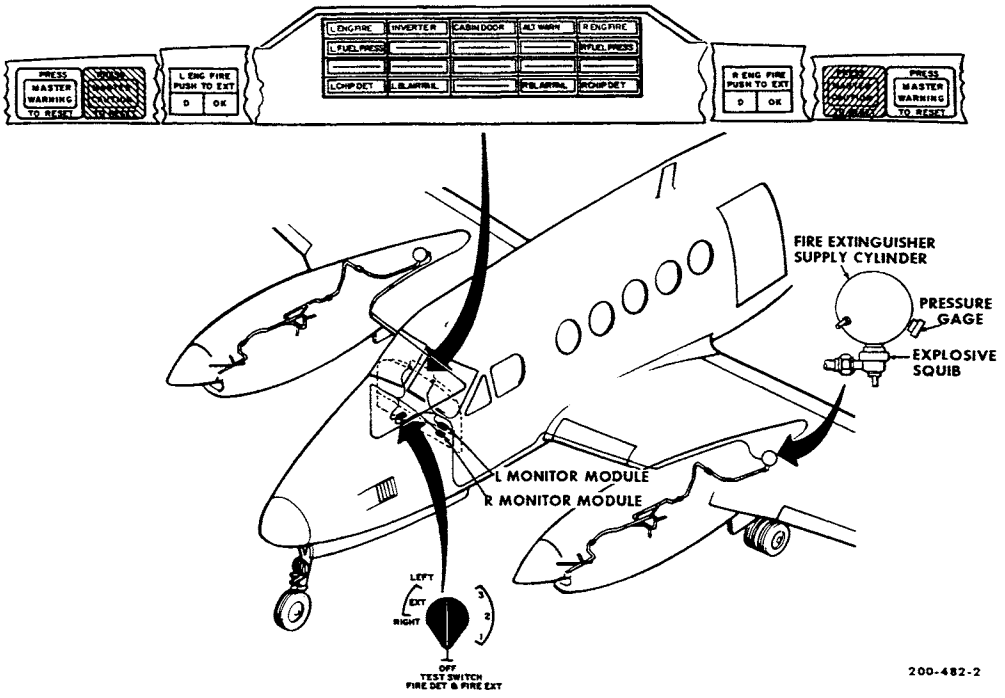

FIRE EXTINGUISHER SYSTEM SCHEMATIC

FIRE EXTINGUISHER SYSTEM

The optional engine-fire-extinguisher system incorporates a pyrotechnic cartridge inside the nacelle of each engine. When the activation valve is opened, the pressurized extinguishing agent is discharged through a plumbing network which terminates in strategically located spray nozzles.

The fire extinguisher control switches used to activate the system are located on the glareshield at each end of the warning annunciator panel. Their power is derived from the hot battery bus. Each push-to-activate switch incorporates three indicator lenses. The red lens, placarded L (or) R ENG FIRE - PUSH TO EXT, warns of the presence of fire in the engine. The amber lens, placarded D, indicates that the system has been discharged and the supply cylinder is empty. The green lens, placarded OK, is provided only for the test function, To discharge the cartridge, raise the safety-wired clear plastic cover and press the face of the lens. This is a one-shot system and will be completely expended upon activation. The amber D light will illuminate and remain illuminated, regardless of battery switch position, until the pyrotechnic cartridge has been replaced.

The fire-extinguisher-system test functions incorporated in the TEST SWITCH - FIRE DET & FIRE EXT test the circuitry of the fire extinguisher pyrotechnic cartridges. During preflight, the pilot should rotate the TEST SWITCH to each of the two positions (RIGHT EXT and LEFT EXT) and verify the illumination of the amber D light and the green OK light on each fire-extinguisher-activation switch on the glareshield.

A gage, calibrated in psi, is provided on each supply cylinder for determining the level of charge. The gages should be checked during preflight.

PROPELLER SYSTEM

DESCRIPTION

Each engine is equipped with a conventional three-blade, full-feathering, constant-speed, counter-weighted, reversing, variable-pitch propeller mounted on the output shaft of the reduction gearbox. The propeller pitch and speed are controlled by engine oil pressure, through single-action, engine-driven propeller governors. Centrifugal counterweights, assisted by a feathering spring, move the blades toward the low rpm (high pitch) position and into the feathered position. Governor boosted engine oil pressure moves the propeller to the high rpm (low pitch) hydraulic stop and reverse position. The propellers have no low rpm (high pitch) stops; this allows the blades to feather after engine shutdown.

Propeller tie-down boots are provided for use on the moored airplane to prevent windmilling at zero oil pressure.

PRIMARY LOW PITCH STOP

Low pitch propeller position is determined by the primary low pitch stop which is a mechanically actuated hydraulic stop. Beta and reverse blade angles are controlled by the power levers in the Beta and reverse range.

PROPELLER GOVERNORS

Two governors, a constant speed governor and an overspeed governor, control the propeller rpm. The constant speed governor, mounted on top of the gear reduction housing, controls the propeller through its entire range. The propeller control lever operates the propeller by means of this governor. If the constant speed governor should malfunction by requesting more than 2000 rpm, the overspeed governor cuts in at 2080 rpm and dumps oil from the propeller to keep the rpm from exceeding approximately 2080 rpm. A solenoid actuated by the PROP GOV TEST switch located on the pilot's subpanel, is provided for resetting the overspeed governor to approximately 1830 to 1910 rpm for test purposes.

If the propeller sticks or moves too slowly during a transient condition causing the propeller governor to act too slowly to prevent an overspeed condition, the power turbine governor, contained within the constant speed governor housing, acts as a fuel topping governor. When the propeller reaches 2120 rpm, the fuel topping governor limits the fuel flow to the gas generator, reducing N_1 rpm, which in turn prevents the propeller rpm from exceeding approximately 2200 rpm. During operation in the reverse range, the fuel topping governor is reset to approximately 95% propeller rpm before the propeller reaches a negative pitch angle. This ensures that the engine power is limited to maintain a propeller rpm somewhat less than that of the constant speed governor setting. The constant speed governor therefore will always sense an underspeed condition and direct oil pressure to the propeller servo piston to permit operation in Beta and reverse ranges.

AUTOFEATHER SYSTEM

The automatic feathering system provides a means of immediately dumping oil from the propeller servo to enable the feathering spring and counterweights to start the feathering action of the blades in the event of an engine failure. Although the system is armed by a switch on the pilot's subpanel, placarded AUTOFEATHER - ARM - OFF - TEST, the completion of the arming phase occurs when both power levers are advanced above 90% N_1 at which time both the right and left indicator lights on the caution/advisory annunciator panel indicate a fully armed system. The annunciator panel lights are green, placarded L AUTOFEATHER and R AUTOFEATHER. The system will remain inoperative as long as either power lever is retarded below 90% N_1 position. The system is designed for use only during takeoff and landing and should be turned off when establishing cruise climb. During takeoff or landing, if torquemeter oil pressure on either engine drops below a prescribed setting, the oil is dumped from the servo, the feathering spring starts the blades toward feather, and the autofeather system on the other engine is disarmed. Disarming of the autofeather portion of the operative engine is further indicated when the annunciator indicator light for that engine extinguishes.

FUEL SYSTEM

The fuel system consists of two separate systems connected by a valve-controlled crossfeed line. The fuel system for each engine is further divided into a main and auxiliary fuel system. The main system consists of a nacelle tank, two wing leading edge tanks, two box section bladder tanks, and an integral (wet cell) tank, all interconnected to flow into the nacelle tank by gravity. This system of tanks is filled from the filler located near the wing tip.

The auxiliary fuel system consists of a center section tank with its own filler opening, and an automatic fuel transfer system to transfer the fuel into the main fuel system.

When the auxiliary tanks are filled, they will be used first. During transfer of auxiliary fuel, which is automatically controlled, the nacelle tanks are maintained full. A swing check valve in the gravity feed line from the outboard wing prevents reverse fuel flow. Upon exhaustion of the auxiliary fuel, normal gravity transfer of the main wing fuel into the nacelle tanks will begin.

An anti-siphon valve is installed in each filler port which prevents loss of fuel or collapse of a fuel cell bladder in the event of improper securing or loss of the filler cap.

The two systems are vented through a recessed ram vent coupled to a protruding heated ram vent on the underside of the wing adjacent to the nacelle. One vent is recessed to prevent icing and the protruding vent, added as a backup, is heated to prevent icing.

All fuel is filtered with a firewall-mounted 20-micron filter. These filters incorporate an internal bypass which opens to permit uninterrupted fuel supply to the engine in the event of filter icing or blockage. In addition, a screen strainer is located at each tank outlet before the fuel reaches the boost and transfer pumps. The main engine driven fuel pump has an integral strainer to protect the pump.

A "differential pressure" fuel purge system is provided and is located in the aft compartment of each nacelle. The system purges the fuel that is left in the fuel manifolds at engine shutdown by forcing the fuel into the nozzles so that it is consumed in the combustion chamber.

FUEL PUMPS

The engine driven fuel pump (high pressure) is mounted on the accessory case in conjunction with the fuel control unit. Failure of this pump results in an immediate flameout. The primary boost pump (low pressure) is also engine driven and is mounted on a drive pad on the aft accessory section of the engine. This pump operates when the gas generator (N_1) is turning and provides sufficient fuel for start, takeoff, all flight conditions (except operation with hot aviation gasoline above 20,000 feet altitude) and operation with crossfeed.

In the event of a primary boost pump failure, the respective red FUEL PRESS light in the annunciator panel will illuminate. This light illuminates when pressure decreases below 10 ± 1 psi. The light will be extinguished by switching on the standby fuel pump on that side, thus increasing pressure above 11 ± 2 psi.

For Training Purposes Only

CAUTION

Engine operation with the fuel pressure light on is limited to 10 hours between overhaul, or replacement, of the engine driven fuel pump.

When using aviation gasoline during climbs above 20,000 feet, the first indication of insufficient fuel pressure will be an intermittent flicker of the FUEL PRESS lights. A wide fluctuation of the fuel flow indicator may also be noted. These conditions can be eliminated by turning on a standby pump.

An electrically driven standby boost pump (low pressure), located in the bottom of each nacelle tank, performs three functions; it is a backup pump for use in the event of a primary fuel boost pump failure, it is for use with hot aviation gasoline above 20,000 feet, and it is used during crossfeed operations. In the event of an inoperative standby pump, crossfeed can only be accomplished from the side of the operative pump.

SERIALS BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-1095, and BB-1097; BL-37 thru BL-57:

Electrical power to operate the standby boost pumps is controlled by lever lock toggle switches, placarded STANDBY PUMP - ON - OFF, located on the fuel control panel and is supplied power from two independent sources. One source of power for either the right or the left standby pump is provided through the number 3 or number 4 feeder bus and is protected by a 10-ampere circuit breaker located on the fuel control panel. This power is only available when the master switch is turned on. Another source of power comes directly from the battery through the hot battery bus and is protected by dual 5-ampere fuses located in the right wing center section. The fuse panel may be serviced through an access door on the bottom side of the wing outboard of the battery. This power source makes power available for the pumps at all times, regardless of the battery master switch position. These circuits are protected by diodes to prevent the failure of one circuit from disabling the other circuit. During shutdown, make certain both standby pump switches are off to prevent battery discharge.

SERIALS BB-1096, BB-1098 and After; BL-58 and After:

Electrical power to operate the standby boost pumps is controlled by lever lock toggle switches, placarded STANDBY PUMP - ON - OFF, located on the fuel control panel. It is supplied power from the number 3 or number 4 feeder bus, and is protected by a 10-ampere circuit breaker located on the fuel control panel. This power is only available when the master switch is turned on. These circuits are protected by diodes to prevent the failure of one circuit from disabling the other circuit.

AUXILIARY FUEL TRANSFER SYSTEM

The auxiliary tank fuel transfer system automatically transfers the fuel from the auxiliary tank to the nacelle tank without pilot action. Motive flow to a jet pump mounted in the auxiliary tank sump is obtained from the engine fuel plumbing system downstream from the engine driven boost pump and and routed through the transfer control motive flow valve. The motive flow valve is energized to the open position by the control system to transfer auxiliary fuel to the nacelle tank to be consumed by the engine during the initial portion of the flight. When an engine is started, pressure at the engine driven boost pump closes a pressure switch which, after a 30 to 50 second time delay to avoid depletion of fuel pressure during starting, energizes the motive flow valve. When the auxiliary fuel is depleted, a low level float switch de-energizes the motive flow valve after a 30 to 60 second time delay provided to prevent cycling of the motive flow valve due to sloshing fuel.

In the event of a failure of the motive flow valve or the associated control circuitry, the loss of motive flow pressure when there is still fuel remaining in the auxiliary fuel tank is sensed by a pressure switch and float switch, respectively, which illuminates a light placarded NO TRANSFER on the fuel control panel. During engine start, the pilot should note that the NO TRANSFER lights extinguish 30 to 50 seconds after engine start. A manual override is incorporated as a backup for the automatic transfer system. This is initiated by placing the AUX TRANSFER switch, located in the fuel control panel to the OVERRIDE position.

USE OF AVIATION GASOLINE

If aviation gasoline must be used as an emergency fuel, it will be necessary to determine how many hours the airplane is operated on gasoline. Since the gasoline is being mixed with the regular fuel, it is expedient to record the number of gallons of gasoline taken aboard for each engine. Each engine is permitted 150 hours of operation on aviation gasoline between overhauls. This means that if one engine has an average fuel consumption of 50 gallons per hour, for example, it is allowed 7500 gallons of aviation gasoline between overhauls. (Two engines; 15,000 gallons between overhauls.)

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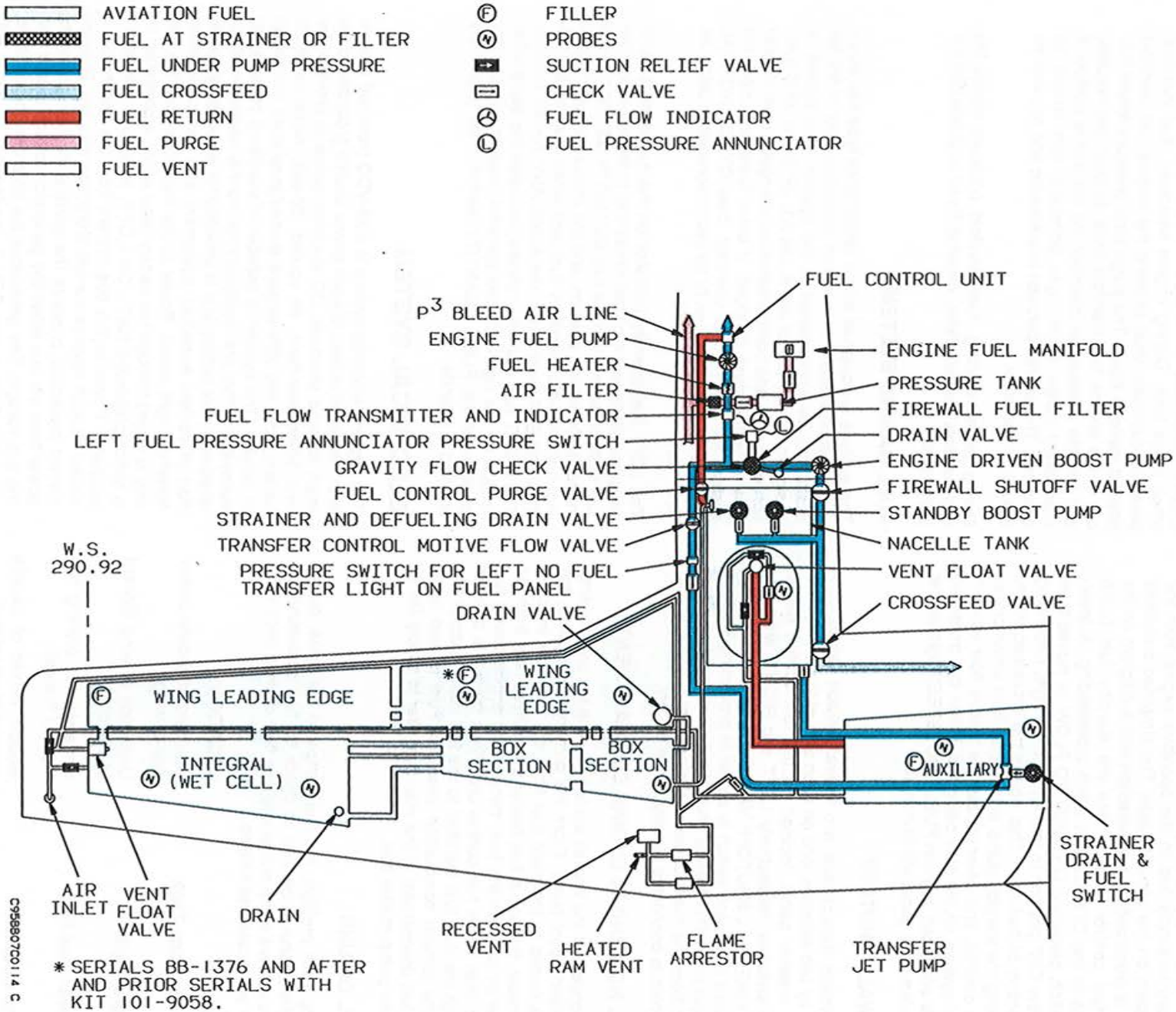

FUEL SYSTEM SCHEMATIC

For Training Purposes Only

CROSSFEED

During emergency single-engine operation, it may become necessary to supply fuel to the operative engine from the fuel system on the opposite side. The simplified crossfeed system is placarded for fuel selection with a diagram on the upper fuel control panel. Place the standby fuel pump switches in the OFF position when crossfeeding. A lever-lock switch, placarded CROSSFEED FLOW, is moved from the center OFF position to the left or to the right, depending on direction of fuel flow. This opens the crossfeed valve, energizing the standby pump on the side from which crossfeed is desired, and de-energizes the motive flow valve in the fuel system on the side being fed. When the crossfeed mode is energized, a green FUEL CROSSFEED light on the caution/advisory panel will illuminate.

FIREWALL SHUTOFF

The system incorporates two firewall shutoff valves controlled by two switches, one on each side of the fuel system circuit breaker panel, located on the fuel control panel. These switches, respectively LEFT and RIGHT, are placarded FIREWALL SHUTOFF VALVE - OPEN - CLOSED. A red guard over each switch is an aid in preventing inadvertent operation. The firewall shutoff valves receive electrical power from the main buses and also from the hot battery bus which is connected directly to the battery.

FUEL ROUTING IN ENGINE COMPARTMENT

Just forward of the firewall shutoff valve is the primary engine driven boost pump. From the primary boost pump, the fuel is routed to the main fuel filter, the fuel flow indicator transmitter, through a fuel heater that utilizes heat from the engine oil to warm the fuel, through the engine driven fuel pump, then to the fuel control unit. From there it is directed through the dual fuel manifold to the fuel outlet nozzles and into the annular combustion chamber. Fuel is also taken from just downstream of the main fuel filter to supply the jet transfer pump motive flow.

FUEL DRAINS

During each preflight, the fuel sumps on the tanks, pumps and filters should be bled to check for fuel contamination. There are five sump drains and one filter drain in each wing. They are located as follows:

DRAINS	LOCATION
Leading Edge Tank	Outboard of nacelle underside of wing
Integral Tank	Underside of wing forward of aileron
Firewall Fuel Filter	Underside of cowling forward of firewall
Sump Strainer	Bottom center of nacelle forward of wheel well
Gravity Feed Line	Aft of wheel well
Auxiliary Tank	At wing root just forward of the flap

FUEL PURGE SYSTEM

Engine compressor discharge air (P_3 air) pressurizes a small purge tank. During engine shutdown, fuel manifold pressure subsides, thus allowing the engine fuel manifold poppett valve to open. The purge tank pressure forces fuel out of the engine fuel manifold lines, through the nozzles, and into the combustion chamber. As the fuel is burned, a momentary surge in (N_1) gas generator rpm should be observed. The entire operation is automatic and requires no input from the crew.

During engine starting, fuel manifold pressure closes the fuel manifold poppett valve, allowing P_3 air to pressurize the purge tank.

FUEL GAGING SYSTEM

The airplane is equipped with a capacitance type fuel quantity indication system. A maximum indication error of 3% full scale may be encountered in the system. The system is designed for the use of Jet A, Jet A1, JP-5 and JP-8 aviation kerosene, and compensates for changes in fuel density due to temperature changes. If other fuels are used, the system will not indicate correctly. See OTHER NORMAL PROCEDURES in Section IV for instructions when using Jet B, JP-4, or aviation gasoline.

The LEFT fuel quantity indicator on the fuel control panel indicates the amount of fuel remaining in the left-side main fuel system tanks when the fuel QUANTITY SELECT switch in the MAIN (upper) position, and the amount of fuel remaining in the left-side auxiliary fuel tank when the fuel QUANTITY SELECT switch is in the AUXILIARY (lower) position. The RIGHT fuel quantity indicator indicates the same information for the right-side fuel systems, depending upon the position of the FUEL QUANTITY switch. The gages are marked in pounds.

ELECTRICAL SYSTEM

The airplane electrical system is a 28-VDC (nominal) system with the negative lead of each power source grounded to the main airplane structure. DC electrical power is provided by one 34-ampere-hour, air cooled, 20-cell, nickel-cadmium battery, and two 250-ampere starter/generators connected in parallel. The system is capable of supplying power to all subsystems that are necessary for normal operation of the airplane. A hot battery bus is provided for emergency operation of certain essential equipment and the cabin entry threshold light circuit. Power to the main bus from the battery is routed through the battery relay which is controlled by a switch placarded BAT - ON - OFF, located on the pilot's subpanel. Power to the bus system from the generators is routed through reverse-current-protection circuitry. Reverse current protection prevents the generators from absorbing power from the bus when the generator voltage is less than the bus voltage. The generators are controlled by switches, placarded GEN 1 and GEN 2, located on the pilot's subpanel.

NOTE

In order to turn the generator ON, the generator control switch must first be held upward in the spring-loaded RESET position for a minimum of one second, then released to the ON position.

Starter power to each individual starter/generator is provided from the main bus through a starter relay. The start cycle is controlled by a three-position switch for each engine, placarded IGNITION AND ENGINE START - ON - OFF - STARTER ONLY, on the pilot's subpanel. The starter/generator drives the compressor section of the engine through the accessory gearing. The starter/generator initially draws approximately 1100 amperes, then drops rapidly to about 300 amperes as the engine reaches 20% N_1.

Power is supplied from three sources: the battery, the right generator, and the left generator. The generator buses are interconnected by two 325-ampere current limiters. The entire bus system operates as a single bus, with power being supplied by the battery and both generators. There are four dual-fed sub-buses. Each sub-bus is supplied power from either generator main bus through a 60-amp limiter, a 70-amp diode, and a 50-amp circuit breaker. All electrical loads are divided among these buses except as noted on the accompanying Power Distribution Schematic. The equipment on the buses is arranged so that all items with duplicate functions (such as right and left landing lights) are connected to different buses. Among the loads on the generator buses are the number 1 and number 2 inverters. Through relay circuitry, the INVERTER selector switch activates the selected inverter, which provides 400-hertz, 115-volt, alternating current to the avionics equipment,and 400-hertz, 26 VAC to the torquemeters. The volt/frequency meter indicates the voltage and frequency of the alternating current being supplied to the avionics equipment.

The generators are controlled by individual voltage regulators which maintain a constant voltage during variations in engine speed and electrical load requirements. The generators are connected to the voltage regulating circuits by means of control switches located on the pilot's left subpanel. The voltage regulating circuit will automatically disable or enable a generator's output to the bus. The load on each generator is indicated by the respective left and right volt/loadmeter located in the overhead panel.

Overheating of the nickel-cadmium battery will cause the battery charge current to increase. Therefore, a yellow BATTERY CHARGE caution annunciator light is provided in the caution/advisory annunciator panel to alert the pilot of the possibility of battery overheating. A Battery Charge Current Detector will cause illumination of the yellow BATTERY CHARGE annunciator whenever the battery charge current is above 7 amps. Thus, the BATTERY CHARGE annunciator may occasionally illuminate for short intervals when heavy loads switch off. Following a battery-powered engine start, the battery recharge current is very high and causes illumination of the BATTERY CHARGE annunciator, thus providing an automatic self-test of the detector and the battery. As the battery approaches a full charge and the charge current decreases below 7 amps, the annunciator will extinguish. This will normally occur within a few minutes after an engine start, but may require a longer time if the battery has a low state of charge, low charge voltage per cell (20-cell battery), or low battery temperature. This system is designed for continuous monitoring of the battery condition.

Illumination of the BATTERY CHARGE annunciator in flight cautions the pilot that conditions may exist that may eventually damage the battery. The operator should check the battery charge current with the loadmeter. This is accomplished by turning off one generator and noting the load on the remaining generator. Turn off the battery and note the loadmeter change. If the change is greater than .025, the battery should be left off the bus and should be inspected after landing. If the annunciator remains on after the battery switch is moved to the OFF position, a malfunction is indicated in either the battery system or charge current detector, in which case the airplane should be landed as soon as practicable. The battery switch should be turned ON for landing in order to avoid electrical transients caused by power fluctuations.

EXTERNAL POWER

For ground operation, an external power socket, located under the right wing outboard of the nacelle, is provided for connecting an auxiliary power unit. A relay in the external power circuit will close only of the external source polarity is correct. The battery switch *must* be on before the external power relay will close and allow external power to enter the airplane electrical system. The battery will also tend to absorb voltage transients when operating avionics equipment and during engine starts. Otherwise, the transients might damage the many solid state components in the airplane.

For starting, an external power source capable of supplying up to 1000 amperes (300 amperes maximum continuous) should be used. A caution light on the caution/advisory annunciator panel, EXT PWR, is provided to alert the operator when an external DC power plug is connected to the airplane.

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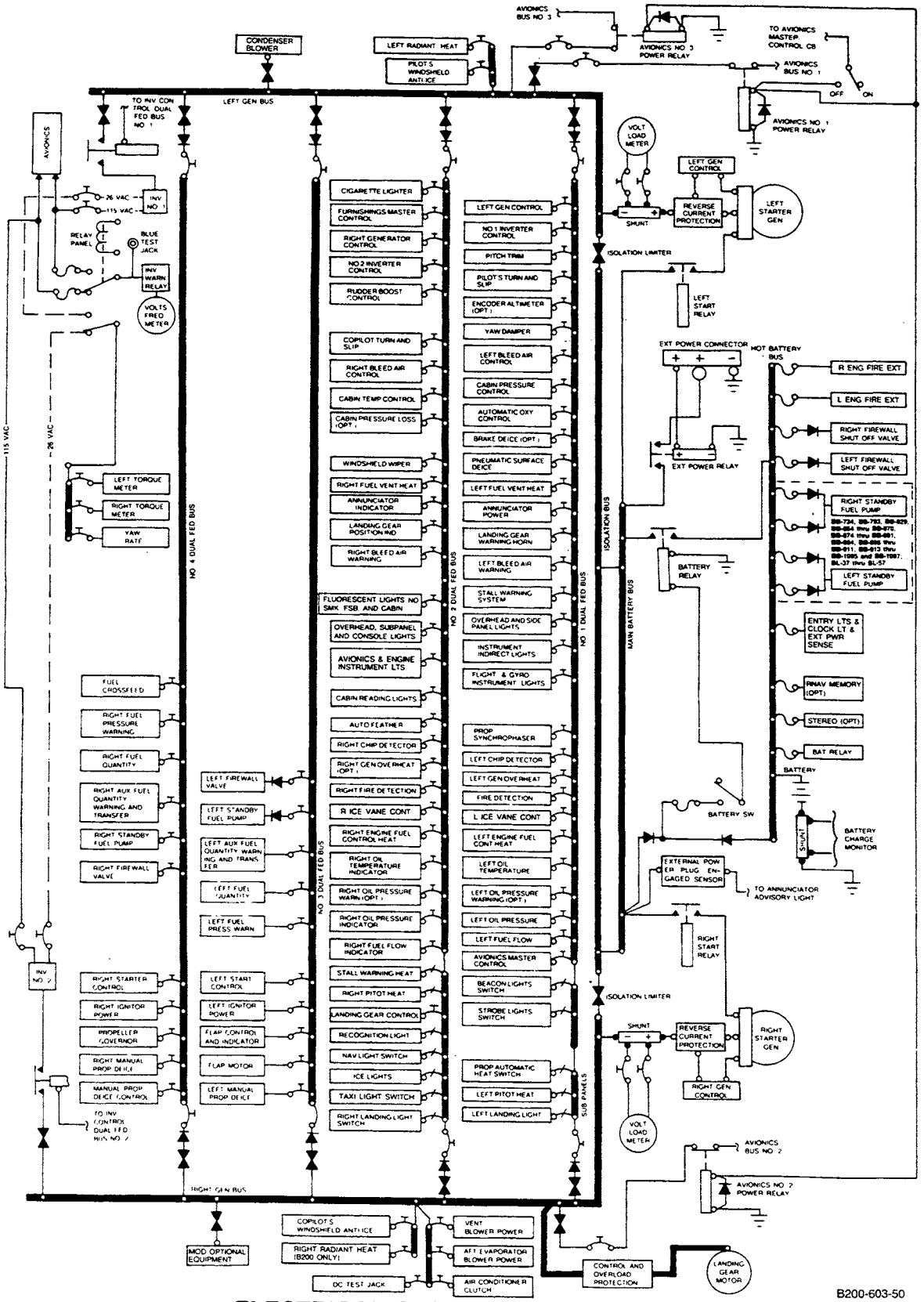

ELECTRICAL SYSTEM SCHEMATIC

LIGHTING SYSTEMS

COCKPIT

An overhead light control panel, easily accessible to both pilot and copilot, incorporates a functional arrangement of all lighting systems in the cockpit. Each light group has its own rheostat switch placarded BRT - OFF. The MASTER PANEL LIGHTS - ON - OFF switch controls the overhead light control panel lights,fuel control panel lights, engine instrument lights, radio panel lights, subpanel and console lights, pilot and copilot instrument lights, and gyro instrument lights. The instrument indirect lights in the glareshield and overhead map lights are individually controlled by separate rheostat switches.

The push-button FREE AIR TEMP switch, located on the left sidewall panel next to the gage, turns ON and OFF the lights near the outside air temperature gage.

CABIN

A three-position switch on the copilot's subpanel, placarded CABIN LIGHTS - START BRIGHT - DIM - OFF, controls the fluorescent cabin lights. The switch to the right of the interior light switch activates the cabin NO SMOKING/FASTEN SEAT BELT signs and accompanying chimes. This three-position switch is placarded CABIN LIGHTS - NO SMOKE & FSB - FSB - OFF.

The baggage-area light is controlled by a two-position switch just inside the airstair door aft of the door frame and is connected to the hot battery bus.

A threshold light is located forward of the airstair door at floor level, and an aisle light is located at floor level aft of the spar cover. A switch adjacent to the threshold light turns both these lights on and off. The switch also turns the exterior entry light on and off. When the airstair door is closed, all the lights controlled by the threshold light switch will extinguish.

When the master switch is on, the individual reading lights along the top of the cabin may be turned on or off by the passengers with a push-button switch adjacent to each light.

EXTERIOR

Switches for the landing lights, taxi lights, wing ice lights, navigation lights, recognition lights, rotating beacons, and wing-tip and tail strobe lights are located on the pilot's subpanel. They are appropriately placarded as to their function.

Tail floodlights, if installed, are incorporated into the horizontal stabilizers and are designed to illuminate both sides of the vertical stabilizer. A switch for these lights, placarded LIGHTS - TAIL FLOOD - OFF, is located on the pilot's subpanel.

A flush-mounted floodlight forward of the flaps in the bottom of the left wing may be installed. This entry light provides illumination of the area around the airstair door, to provide passenger convenience at night. It is controlled by the threshold light switch just inside the door on the forward door frame, and will extinguish automatically whenever the cabin door is closed.

ENVIRONMENTAL SYSTEM

The environmental system consists of the bleed air pressurization, heating and cooling systems, and their associated controls.

PRESSURIZATION SYSTEM

The pressurization system is designed to provide a normal working pressure differential of 6.5 ± .1 psi, which will provide cabin pressure altitudes of approximately; 2800 feet at an airplane altitude of 20,000 feet, 8600 feet at 31,000 feet, and 10,400 feet at 35,000 feet.

Bleed air from the compressor section of each engine is utilized to pressurize the pressure vessel. A flow control unit in the nacelle of each engine controls the pressure of the bleed air and mixes ambient air with it, in order to provide an air mixture suitable for the pressurization function. The mixture flows to the environmental bleed air shutoff valve, which is controlled by a switch placarded BLEED AIR VALVE - LEFT (or) RIGHT - OPEN - ENVIR OFF - INSTR & ENVIR OFF in the ENVIRONMENTAL controls group on the copilot's subpanel. When this switch is in either the ENVIR(onmental air) OFF or the INSTR(ument air) & ENVIR(onmental air) OFF position, the valve is closed. When it is in the OPEN position, the air mixture flows through the valve and to the air-to-air heat exchanger. Depending upon the position of the bypass valves, a greater or lesser volume of the air mixture will be routed through or around the heat exchanger. The temperature of the air flowing through the heat exchanger is lowered as heat is transferred to cooling fins, which are in turn cooled by ram airflow through the fins of the heat exchanger. The air leaving both (left and right) bypass valves, is then ducted into a single muffler, located under the right floorboard forward of the main spar, which helps ensure quiet operation of the environmental bleed air system. The air mixture is then ducted from the muffler into the mixing plenum, located under the copilot's floorboard.

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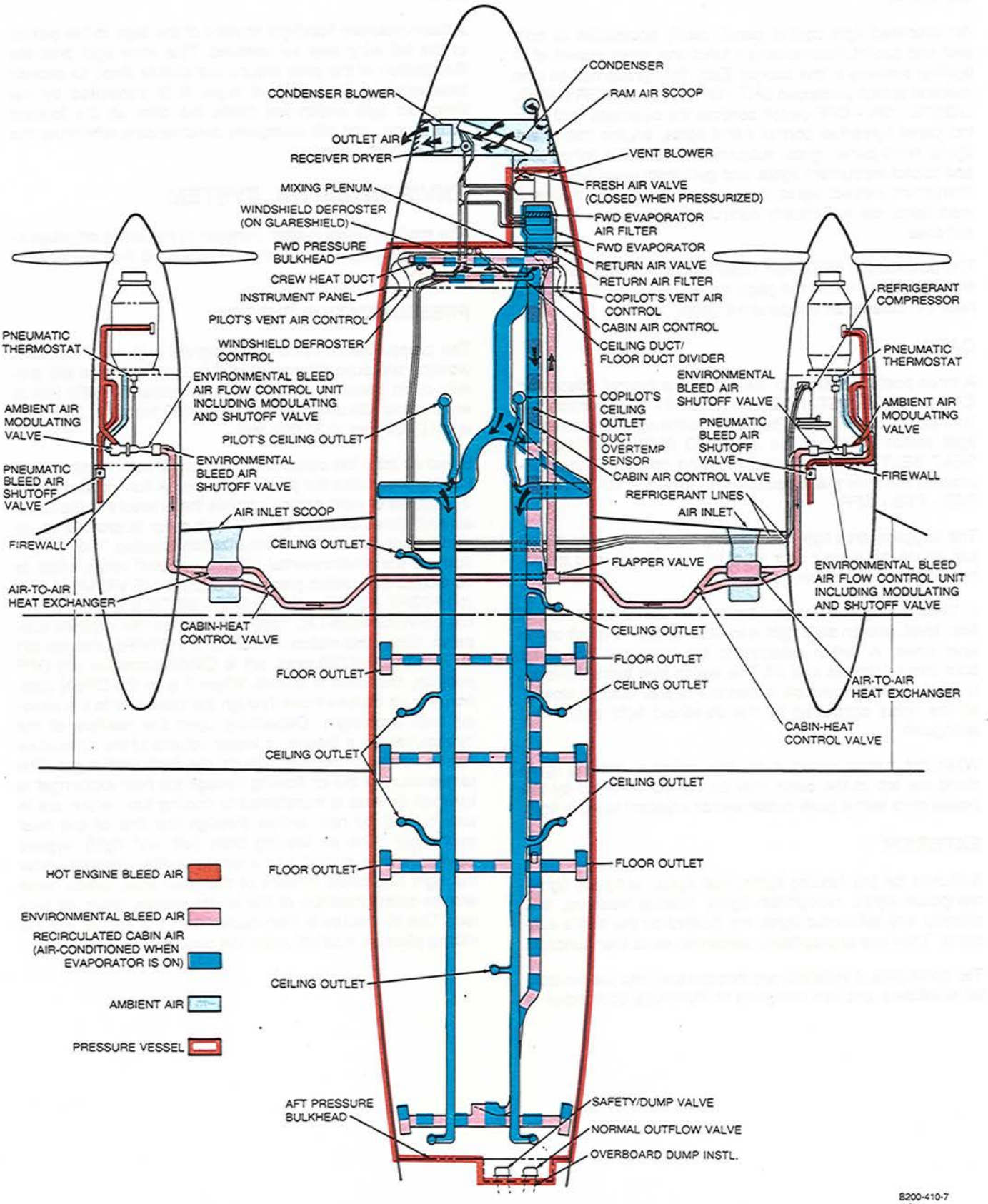

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B200 ENVIRONMENTAL SYSTEM SCHEMATIC

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A partition divides the mixing plenum into two sections. One section supplies the floor-outlet duct, and the other supplies the ceiling-outlet duct. Both sections receive recirculated cabin air from the forward vent blower. This air passes through the forward evaporator, so it will be cooled if the air conditioner is operating. Even in the event that the forward vent blower becomes inoperative, some air will still be circulated, due to a special nozzle in the discharge side of the mixing plenum.

The environmental bleed air duct is routed into the floor-duct section of the mixing plenum, then curves back to discharge the environmental bleed air toward the aft end of the floor-duct section of the mixing plenum. Forward of the discharge end of the environmental bleed air duct, warm air is tapped off and ducted up through the top of the mixing plenum and into the crew heat duct, which also receives recirculated cabin air from the mixing plenum. A valve on the forward side of the crew heat duct allows air to be tapped off for delivery to the windshield defroster when the DEFROST AIR knob on the pilot's left subpanel is pulled out. Eyeball outlets at the left and right sides of the glareshield are also supplied air from the defrost air tap.

The air from the environmental bleed air duct is mixed with recirculated cabin air (which may or may not be air conditioned) in the mixing plenum, then routed into the floor-outlet duct. This pressurized air is then introduced into the cabin through the floor registers. Finally, the air flows out of the pressure vessel through the outflow valve, located on the aft pressure bulkhead. A silencer on the outflow and safety/dump valves ensures quiet operation.

The mixture from both flow control units is delivered to the pressure vessel at a rate which can vary from about 8 to 16 pounds per minute, depending upon ambient temperature and pressure altitude. Pressure within the cabin and the rate of cabin-pressure changes are regulated by pneumatic modulation of the outflow valve, which controls the rate at which air can escape from the pressure vessel.

A vacuum-operated safety valve is mounted adjacent to the outflow valve on the aft pressure bulkhead. It is designed to serve three functions: to provide pressure relief in the event of malfunction of the normal outflow valve; to allow depressurization of the pressure vessel whenever the cabin pressure switch is moved into the DUMP position; and to keep the pressure vessel unpressurized while the airplane is on the ground with the left landing-gear safety switch compressed. A negative-pressure relief function is also incorporated into both the outflow and the safety valves. This prevents outside atmospheric pressure's exceeding cabin pressure by more than 0.1 psi during rapid descents, even if bleed air inflow ceases.

On airplanes BB-1284 and after, both the outflow valve and the safety valve are vented overboard, to preclude moisture build-up in the aft fuselage.

When the BLEED AIR VALVE switches on the copilot's subpanel are OPEN (up), the air mixture from the flow control units enters the pressure vessel. While the airplane is on the ground, a left landing gear safety-switch-activated solenoid valve in each flow control unit keeps the ambient-air intake port closed, allowing only bleed air to be delivered into the pressure vessel. At lift-off, the safety valve closes and the ambient air shutoff solenoid valve in the left flow control unit opens; approximately 6 seconds later, the solenoid in the right flow control unit opens. Consequently, by increasing the volume of airflow into the pressure vessel in stages, excessive pressure bumps during takeoff are avoided.

An adjustable cabin pressurization controller is mounted in the pedestal. It commands modulation of the outflow valve. A dual-scale indicator dial is mounted in the center of the pressurization controller. The outer scale (CABIN ALT) indicates the cabin pressure altitude which the pressurization controller is set to maintain. The inner scale (ACFT ALT) indicates the maximum ambient pressure altitude at which the airplane can fly without causing the cabin pressure altitude to climb above the value selected on the outer scale (CABIN ALT) of the dial. The indicated value on each scale is read opposite the index mark at the forward (top) position of the dial. Both scales rotate together when the cabin altitude selector knob, placarded CABIN ALT, is turned. The maximum cabin pressure altitude is selected by turning the cabin altitude selector knob until the desired setting on the CABIN ALT dial is aligned with the index mark. The maximum cabin altitude selected may be anywhere from -1000 to +15,000 feet MSL. The rate control selector knob is placarded RATE - MIN - MAX. The rate at which the cabin pressure altitude changes from the current value to the selected value is controlled by rotating the rate control selector knob. The rate of change selected may be from approximately 50-300 to approximately 2000 ±500 feet per minute.

The actual cabin pressure altitude is continuously indicated by the cabin altimeter, which is mounted in the right side of the panel that is located between the caution/advisory annunciator panel and the pedestal. Immediately to the left of the cabin altimeter is the cabin vertical speed (CABIN CLIMB) indicator, which continuously indicates the rate at which the cabin pressure altitude is changing.

The cabin pressure switch, located forward of the pressurization controller on the pedestal, is placarded CABIN PRESS - DUMP - PRESS - TEST. When this switch is in the DUMP (forward) position, the safety valve is held open, so that the cabin will depressurize and/or remain unpressurized. When it is in the PRESS (center) position, the safety valve is normally closed in flight, and the outflow valve is controlled by the pressurization controller, so that the cabin will pressurize. When the switch is held in the spring-loaded TEST (aft) position, the safety valve is held closed, bypassing the landing-gear safety switch, to facilitate testing of the pressurization system on the ground.

Prior to takeoff, the cabin altitude selector knob should be adjusted so that the ACFT ALT scale on the indicator dial indicates an altitude approximately 1000 feet above the planned cruise pressure altitude, and the CABIN ALT scale indicates an altitude at least 500 feet above the takeoff field pressure altitude. The rate control selector knob should be adjusted as desired; setting the index mark at the 12-o'clock position will provide the most comfortable cabin rate of climb. The cabin pressure switch should be checked to ensure that it is in the PRESS position. As the airplane

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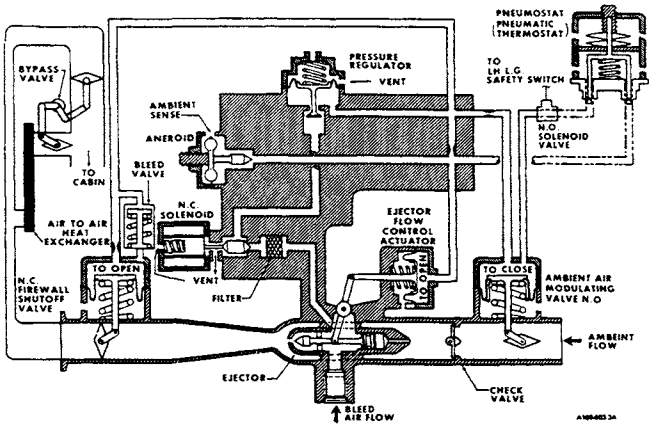
climbs, the cabin pressure altitude climbs at the selected rate of change until the cabin reaches the selected pressure altitude. The system then maintains cabin pressure altitude at the selected value. If the airplane climbs to an altitude higher than the value indexed on the ACFT ALT scale of the dial on the face of the controller, the cabin-to-ambient pressure differential will reach the pressure relief setting of the outflow valve and safety valve. Either or both valves will then override the cabin pressurization controller in order to limit the cabin-to-ambient pressure differential to 6.5 ± .1 psi. If the cabin pressure altitude should reach a value of 12,500 feet, a pressure-sensing switch mounted on the overhead light control panel will close. This causes the ALT WARN annunciator light to illuminate, warning the pilot of operation requiring the use of oxygen. If an auto-deployment oxygen system is installed, a pressure-sensing switch mounted on the cabin sidewall forward of the emergency exit also closes, signalling the passenger oxygen masks to drop out. During cruise operation, if the flight plan calls for an altitude change of 1000 feet or more, reselect the new altitude plus 1000 feet on the CABIN ALT dial.

During descent and in preparation for landing, the cabin altitude selector should be set to indicate a cabin altitude of approximately 500 feet above the landing field pressure altitude, and the rate control selector should be adjusted as required to provide a comfortable cabin-altitude rate of descent. The airplane rate of descent should be controlled so that the airplane altitude does not catch up with the cabin pressure altitude until the cabin pressure altitude reaches the selected value and stabilizes. Then, as the airplane descends to and reaches the cabin pressure altitude, the negative-pressure relief function modulates the outflow and the safety valve poppets toward the fully open position, thereby equalizing the pressure inside and outside the pressure vessel. As the airplane continues to descend below the pre-selected cabin pressure altitude, the cabin will be unpressurized and will follow the airplane rate of descent to touchdown.

FLOW CONTROL UNIT (THRU BB-1179, BL-69)

Each flow control unit consists of an ejector and an integral bleed air modulating valve, firewall shutoff valve, ambient air modulating valve, and a check valve that prevents the bleed air from escaping through the ambient air intake. The flow of bleed air through the flow control unit is controlled as a function of atmospheric pressure and temperature. Ambient air flow is controlled as a function of temperature only.



NOTE: Relative size of components intentionally exaggerated

LH LG - left hand Landing Gear

N.C. - normally closed

N.O. - normally open

BLEED AIR FLOW CONTROL UNIT (Thru BB-1179, BL-69)

When the BLEED AIR VALVE switches on the copilot's subpanel are OPEN, an electronic solenoid valve on each flow control unit opens to allow the bleed air into the unit. As the bleed air enters the flow control unit, it passes through a filter before going to the reference pressure regulator. The regulator will reduce the pressure to a constant value (18 to 20 psi). This reference pressure is then directed to the various components within the flow control unit that regulate the output to the cabin. One reference pressure line is routed to the firewall shutoff valve located downstream of the ejector. An orifice is placed in the line immediately before the shutoff valve to provide a controlled opening rate. At the same time, the reference pressure is directed to the ambient air modulating valve located upstream of the ejector. A pneumatic thermostat with a variable orifice is connected to the modulating valve. The pneumatic thermostat is located on the lower aft side of the fireseal forward of the firewall. The bimetallic sensing discs of the thermostat are inserted into the cowling intake. These discs sense ambient temperature and regulate the size of the thermostat orifices. Warm air will open the orifice; cold air will restrict it, until, at -30°F, the orifice will completely close. When the variable orifice is closed, the pressure buildup will cause the modulating valve to close off the ambient air source. An electric solenoid valve located in the line to the pneumatic thermostat is wired to the left landing gear safety switch. When the airplane is on the ground, the solenoid valve is closed, thereby directing the pressure to the modulating valve, causing it to shut off the ambient air source. The exclusion of ambient air allows faster cabin warmup during cold weather operation.

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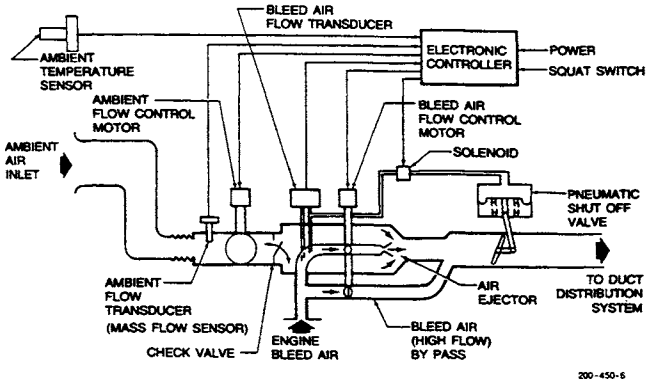
An electric circuit containing a time-delay relay is wired to the above-mentioned solenoid valves to allow the left valve to operate approximately 6 seconds before the right valve. This precludes the simultaneous opening of the shutoff valves, which would result in a sudden pressure surge into the cabin. A check valve, located downstream from the modulating valve, prevents the loss of bleed air through the ambient air intake. At the same time the above operations have been taking place in the control unit, reference pressure is directed to the ejector flow control actuator. This actuator is connected to another variable orifice of the pneumatic thermostat and avariable orifice controlled by an isobaric aneroid. The thermostat orifice is restricted by decreasing ambient temperature, and the isobaric aneroid orifice is restricted by decreasing ambient pressure. The restriction of either orifice will cause a pressure buildup on the ejector flow control actuator, permitting more bleed air to enter the ejector.

FLOW CONTROL UNIT (BB-1180 AND AFTER, BL-70 AND AFTER, AND EARLIER AIRPLANES IN COMPLIANCE WITH BEECH SERVICE BULLETIN 2002).

To meet cabin pressurization and heating requirements, bleed air is routed from the compressor section of each engine to electronic flow control valves mounted on the firewalls. The flow control unit controls the flow of ambient and bleed air. Each flow control unit consists of an ambient temperature sensor, an electronic controller and a pneumatic control valve. The control valve consists of an air ejector, bleed air flow modulating valve with bypass, bleed air valve switch, ambient air flow modulating valve, firewall shut-off valve, a check valve that prevents the bleed air from escaping through the ambient air intake, solenoid valve, bleed air flow transducer, and an ambient air flow transducer.

The ambient air flow modulating valve is actuated by a safety switch located on the left-hand landing gear. When weight is on the landing gear, the ambient air valve is closed off, preventing entry of contaminants into the environmental system during ground operations. After the engine is started and the flow control unit energized, the bleed air modulating valve will close and actuate the bleed air shaft microswitch. When the microswitch is actuated, the electronic controller signals the solenoid to open, this enables the pneumatic line to the firewall shutoff valve to be pressurized and to open the valve. The bleed air shaft continues to open until the desired bleed air flow rate to the cabin is attained. The correct flow rate is determined by the bleed air flow transducer, the ambient air temperature sensor, and the electronic controller.

As the airplane climbs to higher altitudes or enters a cooler environment, the ambient air flow is gradually reduced and the bleed air flow is increased to maintain cabin heat. At approximately 50°F ambient temperature, the ambient air valve is completely closed and the bleed air valve bypass section is opened to allow more bleed air flow around the air ejector.



BLEED AIR FLOW CONTROL UNIT (BB-1180 and After, BL-70 and After, and Earlier Airplanes in Compliance With Beech Service Bulletin 2002)

The bleed air mixture passes from the flow control valves to heat exchangers located in each side of the center section. The heat exchangers remove any excess heat from the bleed air mixture by passing it through an air cooled core. The cooling air from the heat exchanger core is ducted from the leading edge of the center section through the heat exchanger core and exhausted through the louvered plates on the lower side of the center section. Should full utilization of the bleed air be required for the cabin environment, bypass valves provide an alternate path around the heat exchangers. At the juncture of the LH and RH bleed air ducts is a check valve that assures continued cabin pressurization and heat, even though one engine is shut down. Forward of the check valve is the muffler which quiets the movement of air in the system. From the muffler, the bleed air mixture moves to the mixing plenum where it is mixed with recirculated cabin air. If additional cooling of the bleed air mixture is desired, cool air from the air conditioner may be injected into the mixing plenum.

On takeoff, the safety switch immediately reopens the LH ambient air valve. A time delay module integrated into the ambient air circuitry delays opening of the RH valve for approximately seven seconds after opening the LH valve. This delay prevents cabin pressurization surges.

Three-position toggles switches on the RH subpanel f the flow control valves. When these switches are in the up position (Bleed Air Valves OPEN), the flow control valves provide the required cabin heat and pressurization. With the

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switches in the center or neutral position (ENVIR OFF), the flow controls shut off all environmental bleed air. In this position the pneumatic bleed air, which tees off the bleed air lines forward of the flow control valves, is still in operation. When the switches are moved to the down or keying position (INST & ENVIR OFF), all bleed air to the cabin is shut off at the firewalls.

The rate of ambient air flow through the flow control valve is in direct proportion to the temperature of the air in which the airplane is operating. At high atmospheric temperatures, the amount of bleed air used is minimized by mixing ambient air inside the control valve through an air ejector. When atmospheric temperatures drop, the ambient air is reduced or shut off. An ambient air temperature sensor (thermistor), adjacent to the ambient air inlet, sends temperature signals to the electronic controller enabling the controller to determine the correct mix of ambient and bleed air for the cabin environment.

If bleed air pressure or temperature become excessive, the electronic controller will automatically shut off the environmental air at the affected valve.

UNPRESSURIZED VENTILATION

Fresh-air ventilation is provided from two sources. One source, which is available during both the pressurized and the unpressurized mode, is the bleed air heating system. This air mixes with recirculated cabin air and enters the cabin through the floor registers. The volume of air from the floor registers is regulated by using the CABIN AIR control knob located on the copilot's subpanel.

The second source of fresh air, which is available during the unpressurized mode only, is ambient air obtained (through a check valve) from the condensor section in the nose of the airplane. During pressurized operation, cabin pressure forces the check valve closed. During the unpressurized mode, a spring holds the check valve open, so that the forward blower can draw this air into the cabin. The ambient air then mixes with recirculated cabin air, goes through the forward blower, through the forward evaporator, (if it is operating, the air will be cooled), into the mixing plenum, into both the ceiling-outlet and the floor-outlet duct, and into the cabin through all the ceiling and floor outlets. Air ducted to each individual ceiling eyeball outlet can be directionally controlled by moving the eyeball in the socket. Volume is regulated by twisting the outlet to open or close the damper.

HEATING

When air is compressed, its temperature is increased. Therefore, the bleed air extracted from the compressor section of each engine for pressurization purposes is hot. This heat is utilized to warm the cabin.

When the left landing gear safety switch is in the on-the-ground position, the ambient air valve in each flow control unit is closed. Consequently, only bleed air is delivered to the environmental bleed air duct when the airplane is on the ground. In flight, the ambient air valve is open, and ambient air is mixed with the engine bleed air in the flow control unit. This environmental bleed air mixture is then routed into the cabin.

If the environmental bleed air mixture is too warm for cabin comfort, the bypass valve routes some or all of it through the air-to-air heat exchanger, located in the wing center section. The position of the damper in the cabin-heat control valve is determined by positioning of the controls in the ENVIRONMENTAL group on the copilot's subpanel. An air intake on the leading edge of the inboard wing brings ram air into the heat exchanger to cool the bleed air. After leaving the heat exchanger, the ram air is ducted overboard through louvers on the underside of the wing.

After the bleed air passes through or around the air-to-air heat exchanger, it is ducted to the mixing plenum. Some of the environmental bleed air is tapped off and delivered to the pilot/copilot heat duct, which is located below the instrument panel. An outlet at each end of this duct is provided to deliver warm air to the pilot and copilot. A mechanically controlled damper in each outlet permits the volume of airflow to be regulated. The pilot's damper is controlled by the PILOT AIR knob, located on the pilot's subpanel just below the outboard of the control column. The copilot's damper is controlled by the CO-PILOT AIR knob, located on the copilot's subpanel just below and outboard of the control column. The DEFROST AIR control knob is located on the pilot's subpanel just below and inboard of the control column. This knob controls a valve at the forward side of the pilot/copilot heat duct which admits air to two ducts that deliver the warm air to the defroster, located just below the windshields in the top of the glareshield. An air plenum built into the glareshield feeds air to "eyeball" outlets on the left and right sides. Defrost air is the air source for the "eyeball" outlets; thus the use of the DEFROST AIR control knob also controls air to the eyeball outlets.

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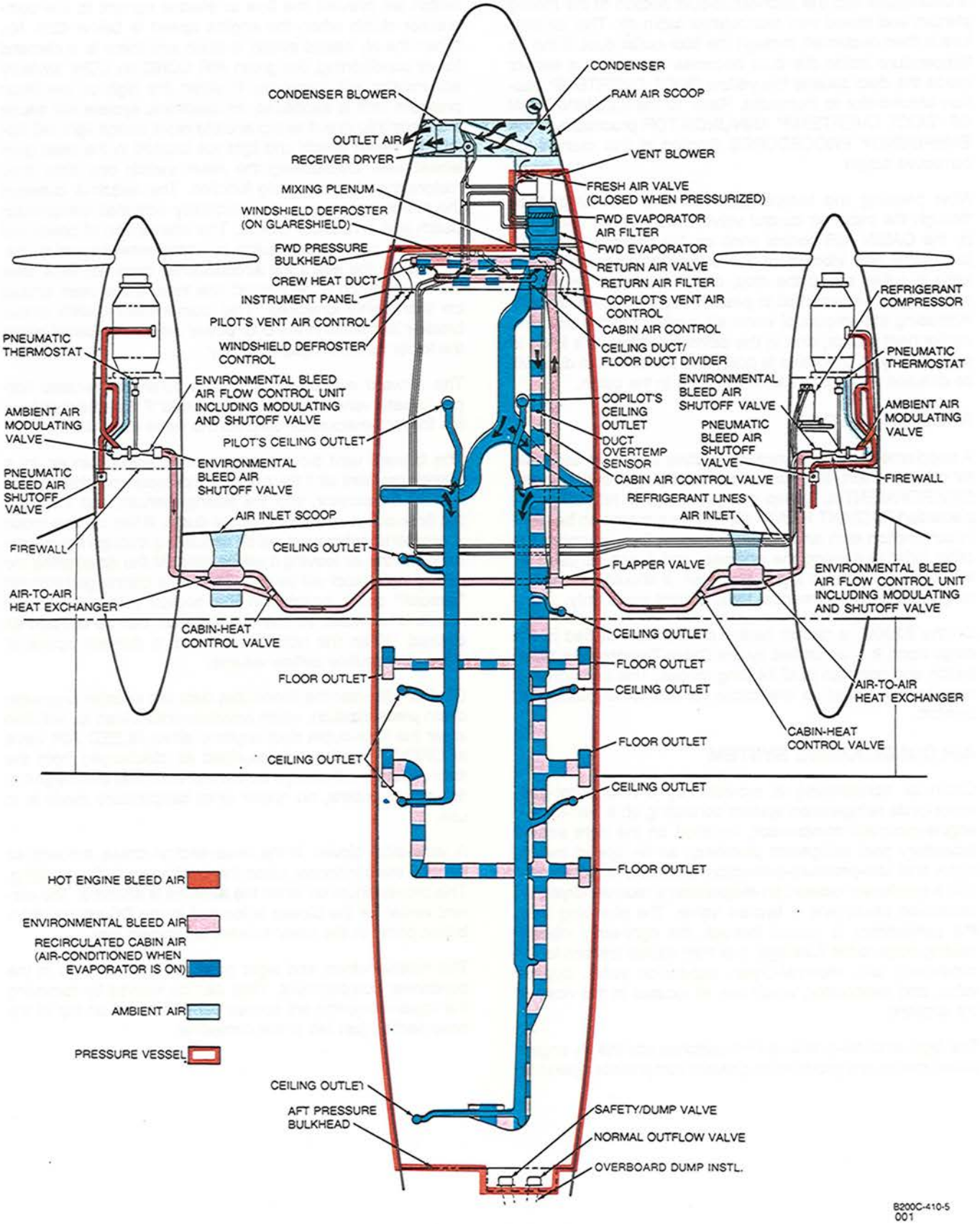

B200C ENVIRONMENTAL SYSTEM SCHEMATIC

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The remainder of the air in the environmental bleed air duct is discharged into the floor-outlet-duct section of the mixing plenum and mixed with recirculated cabin air. This air mixture is then ducted aft through the floor-outlet duct. If the air temperature inside the duct becomes excessive, a sensor inside the duct causes the yellow DUCT OVERTEMP caution annunciator to illuminate. Refer to the ILLUMINATION OF "DUCT OVERTEMP" ANNUNCIATOR procedure in the EMERGENCY PROCEDURES Section of this manual for corrective action.

After passing the temperature sensor, the air passes through the cabin air control valve. This valve is controlled by the CABIN AIR control knob on the copilot's subpanel, just below and inboard of the control column. When this knob is pulled out to the stop, only a minimum amount of warm air will be permitted to pass through the valve, thereby increasing the amount of warm air available to the pilot and copilot heat outlets, and to the defroster. When this knob is pushed fully in, the valve is open and the air in the duct will be directed to the floor-outlet registers in the cabin.

RADIANT HEATING

A supplemental electric radiant heating system is available for cabin comfort. It is turned on and off by a switch in the ENVIRONMENTAL group on the copilot's left subpanel placarded RADIANT HEAT - OFF. This system can be used in conjunction with an auxiliary power unit for warming the cabin prior to starting the engines, and it can be used as supplemental heat in flight. However, it should be used in conjunction with the manual temp control mode only.

On the B200C, a radiant heater element is installed in the cargo door. It is controlled by the Cabin Temperature Mode switch and operates in all heating modes. This unit provides supplemental heat to the cabin for additonal passenger comfort.

AIR CONDITIONING SYSTEM

Cabin air conditioning is provided by a refrigerant-gas-vapor-cycle refrigeration system consisting of: a belt-driven, engine-mounted compressor, installed on the right engine accessory pad; refrigerant plumbing; an N_1 speed switch; high- and low-pressure-protection switches; a condensor coil; a condensor blower; an evaporator; a receiver-dryer; an expansion valve; and a bypass valve. The plumbing from the compressor is routed through the right-wing inboard leading edge to the fuselage. It is then routed forward to the condensor coil, receiver-dryer, expansion valve, bypass valve, and evaporator, which are all located in the nose of the airplane.

The high- and low-pressure-limit switches and the N_1 engine speed switch are provided to prevent compressor operation outside of established limitation parameters. The N_1 speed switch will prevent the flow of electric current to the compressor clutch when the engine speed is below 62% N_1. When the N_1 speed switch is open and there is a demand for air conditioning, the green AIR COND N_1 LOW advisory annunciator will illuminate. If either the high or low freon pressure limit is exceeded, an electronic system will cause the electronic circuit to trip and the reset switch light will illuminate. Reset switch and light are located in the nose gear wheel well. Depressing the reset switch one time may restore the airconditioning function. The switch is between the power source and the electrically activated compressor clutch and condensor blower. The interruption of power will cause the compressor clutch and condenser blower to disengage. In the event the airconditioner does not work after the reset switch is depressed one time, the system should be thoroughly checked. The compressor-clutch circuit breaker is located in the D.C. power distribution panel inside the lower forward equipment bay.

The forward evaporator utilizes a solenoid-operated hot-gas-bypass valve to prevent icing. A 33°F thermal switch on the forward evaporator controls the valve solenoid.

The forward vent blower blows recirculated cabin air (plus outside ambient air if the cabin is unpressurized) through the forward evaporator, into the mixing plenum, and into both the floor-outlet and ceiling-outlet ducts. If the cooling mode is operating, refrigerant will be circulating through the evaporator and the air leaving it will be cool. All the air entering the ceiling-outlet duct will be cool. This air is discharged through "eyeball" outlet nozzles in the cockpit and cabin. Each nozzle is movable, so that the airstream can be directed as desired. When the nozzle is twisted, a damper opens or closes to regulate airflow volume.

Cool air will enter the floor-outlet duct, but in order to provide cabin pressurization, warm environmental bleed air will also enter the floor-outlet duct anytime either BLEED AIR valve is OPEN. Therefore, pressurized air discharged from the floor registers will always be warmer than that discharged at the ceiling outlets, no matter what temperature mode is in use.

A vane-axial blower in the nose section draws ambient air through the condensor when the cooling mode is operating. This blower shuts off when the airplane is airborne. The current limiter for the blower is located in the DC power distribution panel in the lower forward equipment bay.

The receiver-dryer and sight gage are located high in the condenser compartment. They can be viewed by removing the upper-compartment access panel, located on top of the nose section just left of the centerline.

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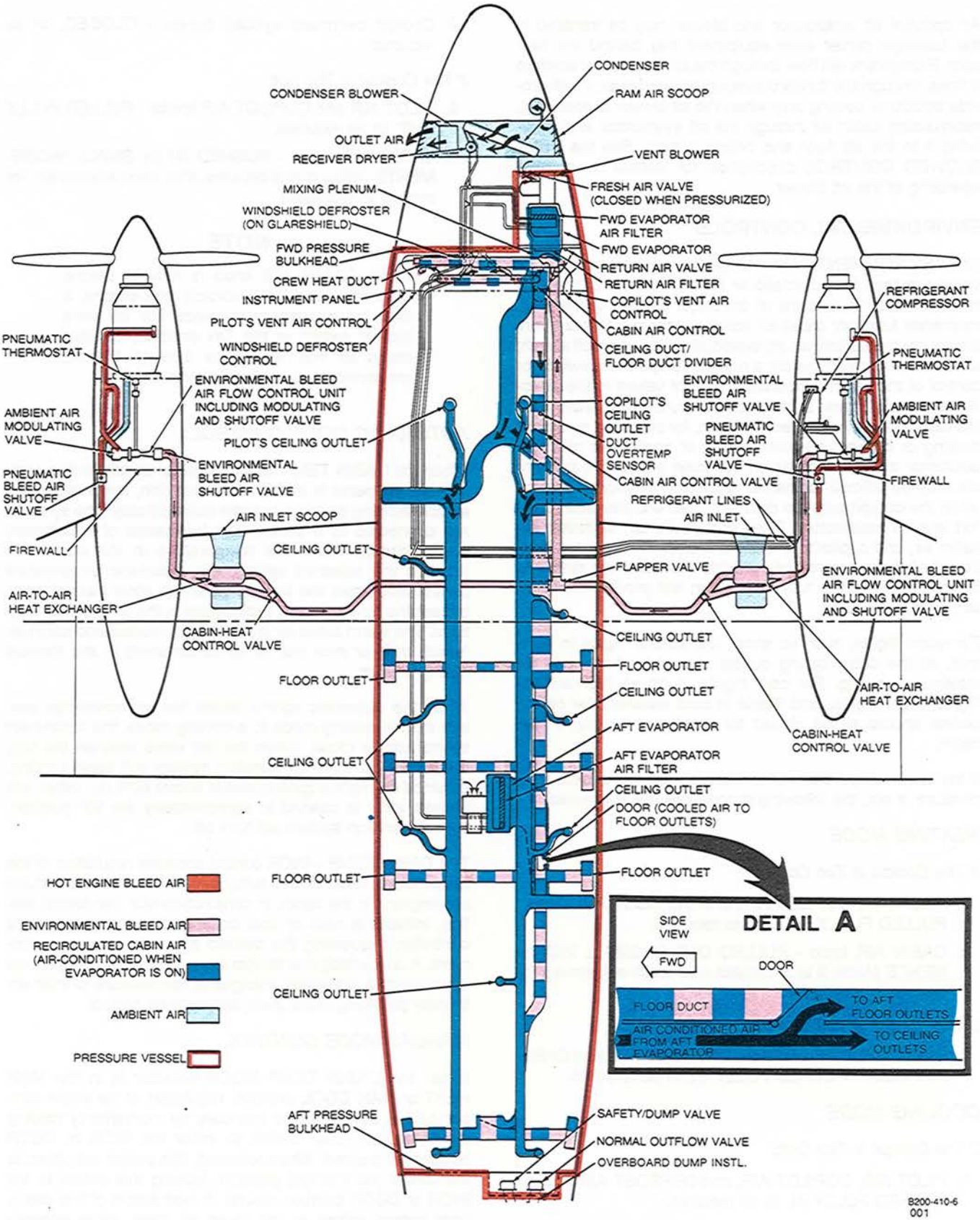

ENVIRONMENTAL SYSTEM SCHEMATIC
(Optional Aft Evaporator System Shown)

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An optional aft evaporator and blower may be installed in the fuselage center aisle equipment bay behind the rear spar. Refrigerant will flow through the aft evaporator anytime it flows through the forward evaporator. However, it will provide additional cooling only when the aft blower is operating, recirculating cabin air through the aft evaporator and delivering it to the aft floor and ceiling outlets. See the VENT BLOWER CONTROL description for details concerning operating of the aft blower.

ENVIRONMENTAL CONTROLS

The ENVIRONMENTAL control section on the copilot's subpanel provides for automatic or manual control of the system. This section contains all the major controls of the environmental function: bleed air valve switches; a forward vent blower control switch; an aft evaporator blower on/off switch; a radiant heat on/off switch; a manual temperature switch for control of the cabin temperature control valves in the air-to-air heat exchangers; a cabin-temperature-level control; and the cabin temp mode selector switch, for selecting automatic heating or cooling, manual heating or cooling, or off. Four additional manual controls on the main instrument subpanels may be utilized for partial regulation of cockpit comfort when the cockpit partition door is closed and the cabin comfort level is satisfactory. They are: pilot's air, defroster air, cabin air, and copilot's air control knobs. The fully out position of all these controls will provide the maximum airflow to the cockpit, and the fully in position will provide minimum airflow to the cockpit.

For warm flights, such as short, low-altitude flights in summer, all the cabin ceiling outlets should be fully open for maximum cooling. For cold flights, such as high-altitude flights, night flights, and flights in cold weather, the ceiling outlets should all be closed for maximum heating in the cabin.

If the cabin temperature is comfortable but the cockpit temperature is not, the following procedures are suggested:

HEATING MODE

If The Cockpit is Too Cold:

1. PILOT AIR, COPILOT AIR, and DEFROST AIR knobs - PULLED FULLY OUT, or as required.
2. CABIN AIR knob - PULLED OUT IN SMALL INCREMENTS (Allow 3 to 5 minutes after each adjustment for system to stabilize.)

If The Cockpit is Too Hot:

3. PILOT AIR, COPILOT AIR, DEFROST AIR, and CABIN AIR knobs - PUSHED FULLY IN, or as required.

COOLING MODE

If The Cockpit is Too Cold:

1. PILOT AIR, COPILOT AIR, and DEFROST AIR knobs - PUSHED FULLY IN, or as required.
2. Cockpit overhead eyeball outlets - CLOSED, or as required.

If The Cockpit is Too Hot:

3. PILOT AIR and COPILOT AIR knobs - PULLED FULLY OUT, or as required.
4. CABIN AIR knob - PUSHED IN IN SMALL INCREMENTS (Allow 3 to 5 minutes after each adjustment for system to stabilize.)

NOTE

If the CABIN AIR knob is fully in before obtaining satisfactory cockpit temperature, it may be necessary to place the aft vent blower switch in the ON position, so that cabin air will recirculate through the aft evaporator to provide additional cooling.

AUTOMATIC MODE CONTROL

When the CABIN TEMP MODE selector switch on the copilot's left subpanel is in the AUTO position, the heating and air conditioning systems operate automatically. The systems are connected to a control box by means of a balanced bridge circuit. When the temperature in the cabin has reached the selected setting, the automatic temperature control modulates the bypass valves to allow heated air to bypass the air-to-air heat exchangers in the wing center sections. The warm bleed air is mixed with recirculated cabin air (which may or may not be air-conditioned) in the forward mixing plenum.

When the automatic control drives the environmental system from a heating mode to a cooling mode, the cabin-heat control valves close. When the left valve reaches the fully closed position, the refrigeration system will begin cooling, provided the right engine speed is above 60% N_1. When the bypass valve is opened to approximately the 30° position, the refrigeration system will turn off.

The CABIN TEMP - INCR control provides regulation of the temperature level in the automatic mode. A temperature-sensing unit in the cabin, in conjunction with the control setting, initiates a heat or cool command to the temperature controller, requesting the desired pressure-vessel environment. A duct anticipator temperature probe (duct stat) allows the system to anticipate changes in temperature of inlet air, thereby providing more even temperature control.

MANUAL MODE CONTROL

When the CABIN TEMP MODE selector is in the MAN HEAT or MAN COOL position, regulation of the cabin temperature is accomplished manually by momentarily holding the MANUAL TEMP switch to either the INCR or DECR position as desired. When released, this switch will return to the center (no change) position. Moving this switch to the INCR or DECR position results in modulation of the cabin-heat control valves in the bleed air lines. Allow approximately 30 seconds per valve (1 minute total time) for the valves to move to the fully open or fully closed position. Only

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one valve at a time moves. Movement of these valves varies the amount of bleed air routed through the air-to-air heat exchanger. Consequently, the temperature of the incoming bleed air will vary. This bleed air mixes with recirculated cabin air (which will be air-conditioned if the refrigeration system is operating) in the mixing plenum, and is then ducted to the floor registers. As a result, the cabin temperature will vary according to the position of the cabin-heat control valves, whether or not the air conditioner is operating.

When the CABIN TEMP MODE selector is in the MAN COOL position, the air conditioner system will operate, provided the speed of the right engine is above 60% N_1.

NOTE

The air conditioner compressor will not operate unless the cabin-heat control valves are closed. To ensure that the valves are closed, hold the MANUAL TEMPerature switch in the DECRease position for one minute.

BLEED AIR CONTROL

Bleed air entering the cabin is controlled by the switches placarded BLEED AIR VALVE - LEFT - OPEN - RIGHT - ENVIR OFF - INSTR & ENVIR OFF. When the switch is in the OPEN position, the environmental flow control unit and the pneumatic instrument air valve are open. When the switch is in the ENVIR OFF position, the environmental flow control unit is closed and the pneumatic instrument air valve is open; in the INSTR & ENVIR OFF position, both are closed. For maximum cooling on the ground, turn the bleed air valve switches to the ENVIR OFF position.

VENT BLOWER CONTROL

The forward vent blower is controlled by a switch in the ENVIRONMENTAL group placarded VENT BLOWER - HI - LO - AUTO. When this switch is in the AUTO position, the forward vent blower will operate at low speed if the CABIN TEMP MODE selector switch is in any position other than OFF (i.e., MANual COOL, MANual HEAT, or AUTOmatic).

When the VENT BLOWER switch is in the AUTO position and the CABIN TEMP MODE selector switch is in the OFF position, the blower will not operate. Anytime the VENT BLOWER switch is in the LO position, the forward vent blower will operate at low speed, even if the CABIN TEMP MODE selector switch is OFF. Anytime the VENT BLOWER switch is in the HI position, the forward vent blower will operate at high speed, regardless of the position of the CABIN TEMP MODE selector switch (i.e., MAN COOL, MAN HEAT, OFF, or AUTO).

If the optional aft evaporator unit is installed in the air conditioning system, an aft blower is also installed under the floor.

The aft blower draws in cabin air, blows it across the aft evaporator, and to the aft floor and ceiling outlets. This blower operates at high speed only. It is controlled by a switch placarded AFT BLOWER - OFF, located in the ENVIRONMENTAL group on the copilot's subpanel. The blower is independent of any other control. The aft blower is intended for use only when maximum cabin cooling (air conditioning) is desired. If the blower should be turned on during a heating mode of operation, the door between the aft-blower duct and the warm-air (floor-outlet) duct will open. This will stop the flow of heated air to the aft floor registers, and deliver recirculated cabin air (which is not cooled, since refrigerant is not flowing through the aft evaporator) to the aft floor registers and ceiling outlets.

Both blower circuit breakers are located in the DC power distribution panel in the lower forward equipment bay.

OXYGEN SYSTEM

The Super King Air B200/B200C oxygen system is based on an adequate oxygen flow for a pressure altitude of 35,000 feet. The masks and Oxygen Duration Chart, Section IV, NORMAL PROCEDURES, are based on a flow rate of 3.9 Liters Per Minute (LPM-NTPD). The only exception is the diluter-demand crew mask. For oxygen duration computation, each diluter-demand mask being used is counted as two masks at 3.9 LPM-NTPD. At cabin altitudes above 20,000 feet, SELECT 100% MODE.

A push/pull handle (PULL ON - SYStem READY), located aft of the overhead light control panel, is used in conjunction with the automatically deployed passenger oxygen system. This handle operates a cable which opens and closes the shut-off valve located at the oxygen supply bottle in the aft, unpressurized area of the fuselage. When this handle is pushed in, no oxygen supply is available anywhere in the airplane. It should be pulled out prior to engine starting to ensure that oxygen will be immediately available anytime it is needed. When this handle is pulled out, the primary oxygen supply line is charged with oxygen, provided the oxygen supply bottle is not empty (Check the oxygen supply pressure gage on the right subpanel and verify that sufficient oxygen is available for the flight). The primary oxygen supply line delivers oxygen to the two crew oxygen outlets in the cockpit, to the first aid oxygen outlet in the toilet area, and to the passenger oxygen system shutoff valve.

The crew is provided with diluter-demand, quick-donning oxygen masks. These masks hang on the aft cockpit partition behind and outboard of the pilot and copilot seats. They are held in the armed position by spring-tension clips, and can be donned immediately with one hand. The diluter-demand crew masks deliver oxygen to the user only upon inhalation. Consequently, there is no loss of oxygen when the masks are plugged in and the PULL - ON - SYStem READY handle is pulled out, even though oxygen is immediately available upon demand.

A small lever on each diluter-demand oxygen mask permits the selection of two modes of operation: NORMAL and 100%. In the NORMAL position, air from the cockpit is mixed with the oxygen supplied through the mask. This reduces the rate of depletion of the oxygen supply, and it is

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more comfortable to use than 100% aviators breathing oxygen. However, in the event of smoke or fumes in the cockpit, the 100% position should be used to prevent the breathing of contaminated air. For this reason, the selector lever should be left in the 100% position when the masks are not in use.

Anytime the primary oxygen supply line is charged, oxygen can be obtained from the first aid oxygen mask located in the toilet area, by manually opening the overhead access door (placarded FIRST AID OXYGEN - PULL) and opening the ON-OFF valve inside the box. A placard (NOTE: CREW SYStem MUST BE ON) reminds the user that the PULL ON - SYStem READY handle in the cockpit must be pulled out before oxygen will flow from the first aid oxygen mask.

The passenger oxygen system is of the constant flow type. Anytime the cabin pressure altitude exceeds approximately 12,500 feet, a barometric-pressure switch automatically energizes a solenoid which opens the passenger oxygen system shut-off valve. The pilot can open the valve manually anytime by pulling out the PASSENGER MANUAL OveRIDE handle, located aft of the overhead light control panel. Once the passenger oxygen system shut-off valve has been opened (either automatically or manually), oxygen will flow into the passenger oxygen supply line, if the primary oxygen system line has been charged (i.e., if the oxygen supply bottle contains oxygen and the PULL ON - SYStem READY handle in the cockpit is pulled out). When oxygen flows into the passenger oxygen system supply line, a pressure-sensitive switch in the line closes a circuit to illuminate the green PASS OXYGEN ON annunciator on the cautionary/advisory annunciator panel. This switch will also cause the cabin lights (all fluorescent lights, the foyer light and the center baggage compartment light) to illuminate in the full bright mode, regardless of the position of the interior lights switch placarded CABIN LIGHTS - START BRIGHT - DIM - OFF located on the copilot's left subpanel.

The pressure of the oxygen in the passenger oxygen system supply line then automatically extends a plunger against each of the passenger oxygen mask dispenser doors, forcing the doors open. The oxygen masks then drop down about 9 inches below the dispensers. The lanyard valve pin at the top of the oxygen mask hose must be pulled out in order for oxygen to flow from the mask. The pin is connected to the oxygen mask via a flexible cord; when the oxygen mask is pulled down for use, the cord pulls the pin out of the lanyard valve. The lanyard valve pin must be manually reinserted into the valve in order to stop the flow of oxygen when the mask is no longer needed. The passenger oxygen can be shut off and the remaining oxygen isolated to the crew and first aid outlets by pulling the OXYGEN CONTROL circuit breaker in the ENVIRONMENTAL group on the right side panel, providing the PASSENGER MANUAL O'RIDE handle is pushed in to the OFF position.

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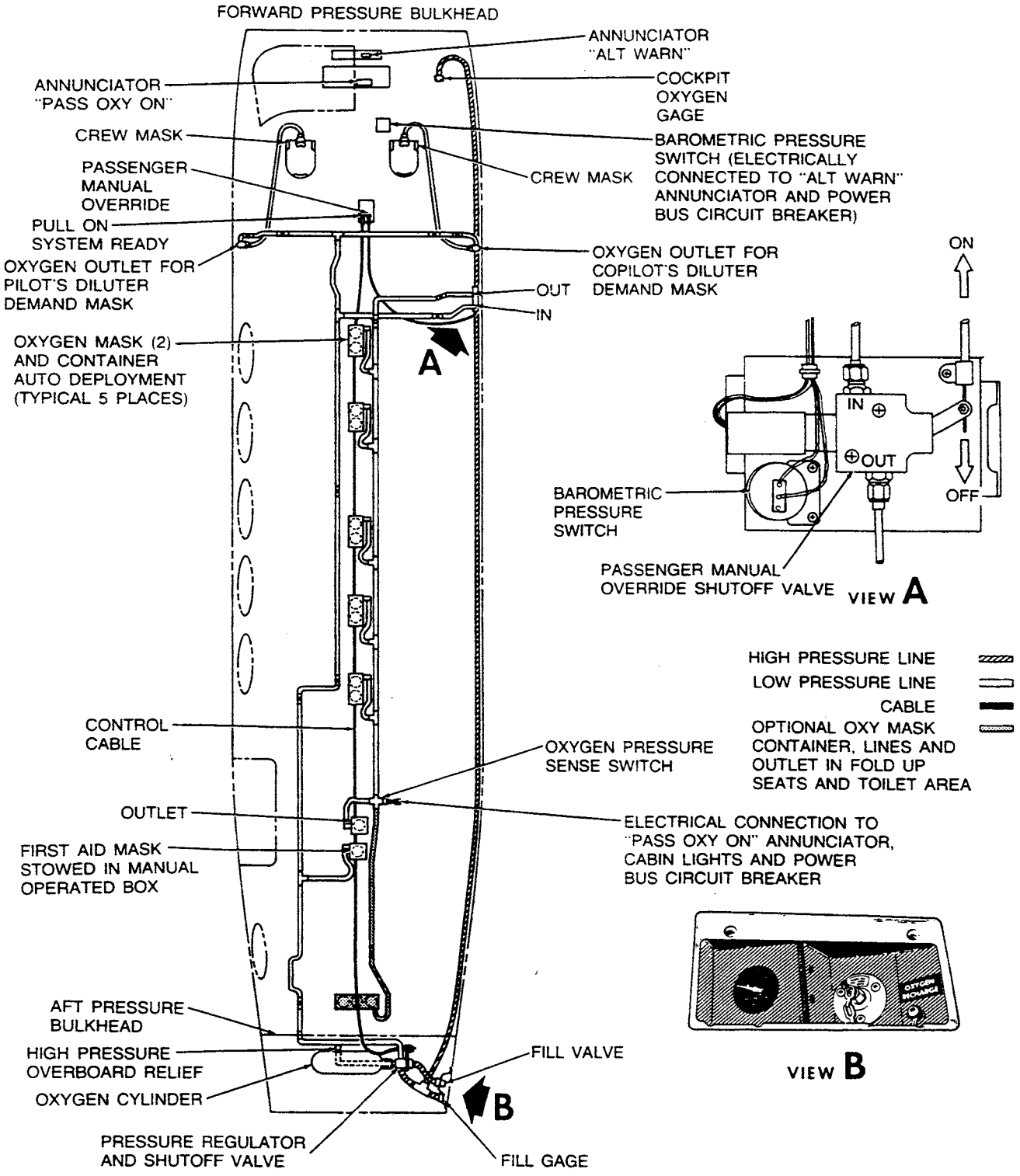
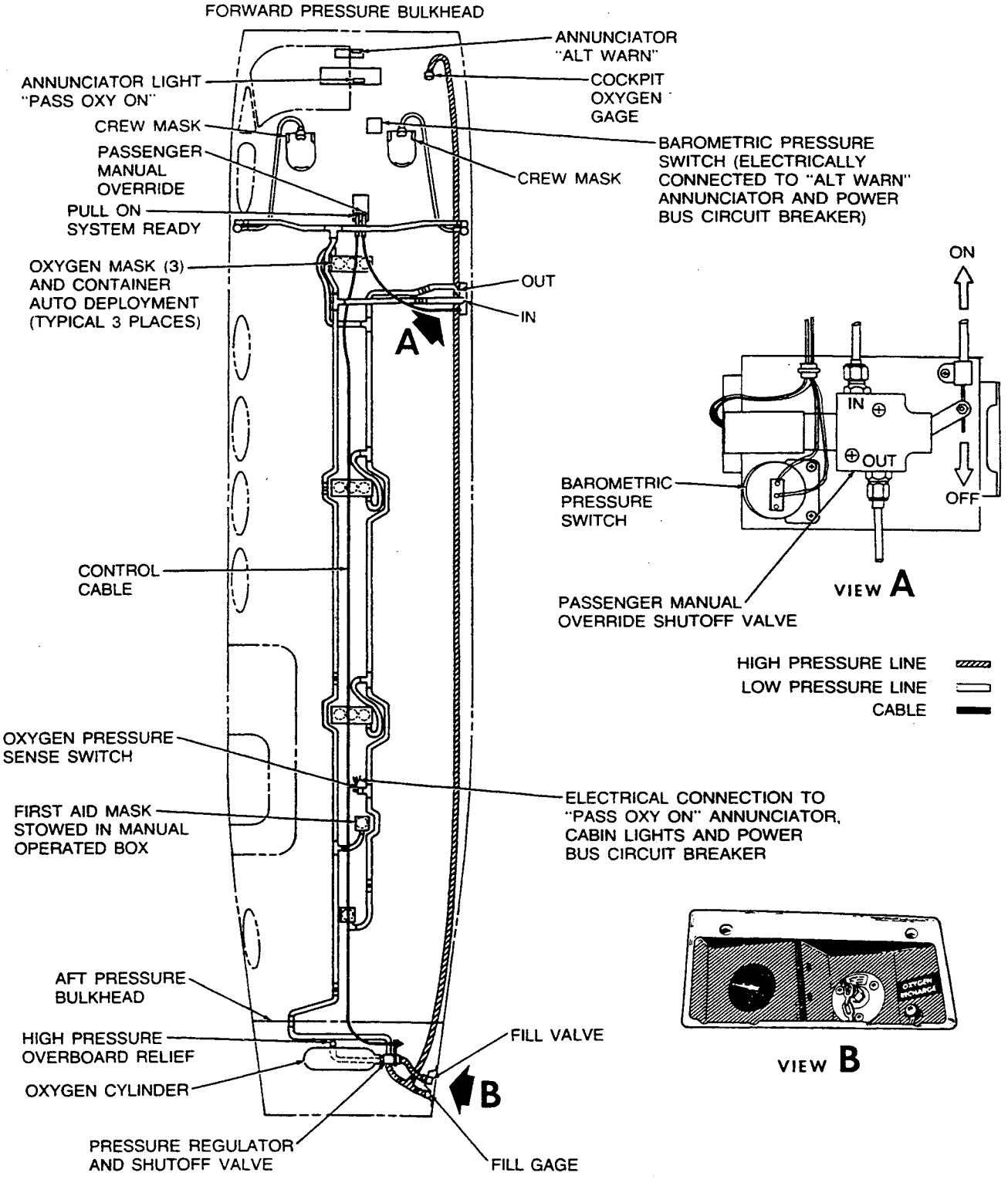


B200 OXYGEN SYSTEM SCHEMATIC



B200C OXYGEN SYSTEM SCHEMATIC

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PITOT AND STATIC SYSTEM

The pitot and static system provides a source of impact air and static air for operation of the flight instruments. A heated pitot mast is located to each side of the lower portion of the nose. Tubing from the left pitot mast is connected to the pilot's airspeed indicator, and tubing from the right pitot mast is connected to the copilot's airspeed indicator.

The normal static system provides two separate sources of static air - one for the pilot's flight instruments, and one for the copilot's. Each of the normal static air lines opens to the atmosphere through two static air ports - one on each side of the fuselage.

An alternate static air line is also provided for the pilot's flight instruments. In the event of a failure of the pilot's normal static air source (e.g., if ice accumulations should obstruct the static air ports), the alternate source should be selected by lifting the spring-clip retainer off the PILOT'S STATIC AIR SOURCE valve handle, located on the right side panel, and moving the handle aft to the ALTERNATE position. This will connect the alternate static air line to the pilot's flight instruments. The alternate line obtains static air just aft of the rear pressure bulkhead, from inside the unpressurized area of the fuselage.

The pilot's altimeter, vertical speed indicator, and airspeed indicator are connected to the pilot's static air source. When the system is switched to the pilot's alternate air source, the pilot's altimeter and vertical speed indicator are affected as well as the pilot's airspeed indicator. The copilot's airspeed indicator, altimeter, and vertical speed indicator are all on the copilot's static air source and cannot be switched to alternate source. See SCHEMATIC DIAGRAM of PITOT AND STATIC SYSTEM.

WARNING

The pilot's airspeed and altimeter indications change when the alternate static air source is in use. Refer to the Airspeed Calibration - Alternate System, and the Altimeter Correction - Alternate System graph in Section V, PERFORMANCE for operation when the alternate static air source is in use.

When the alternate static air source is not needed, ensure that the PILOT'S STATIC AIR SOURCE valve handle is held in the forward (NORMAL) position by the spring-clip retainer.

Three petcocks are provided to facilitate draining moisture from the static air lines. They are located behind an access cover below the circuit breakers on the right side panel. The drain valves should be opened to release any trapped moisture at each 150-hour inspection, and after exposure to visible moisture on the ground. They must be closed after draining.

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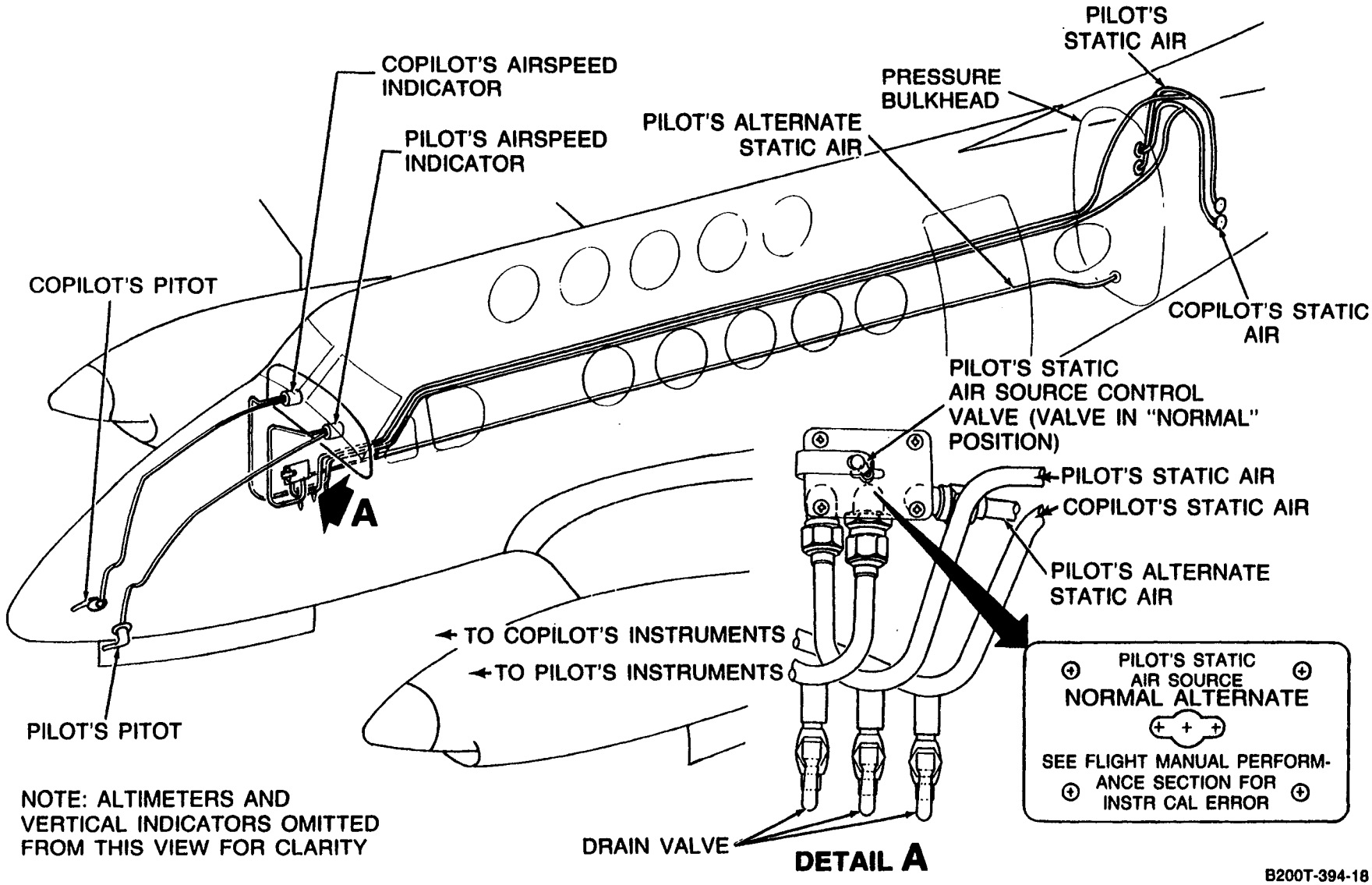

PITOT AND STATIC SYSTEM SCHEMATIC

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ENGINE BLEED AIR PNEUMATIC SYSTEM

High-pressure bleed air from each engine compressor, routed through the firewall shutoff valves and regulated at 18 psi, supplies pressure for surface deice system and vacuum source. Vacuum for the flight instruments is derived from a bleed air ejector. One engine can supply sufficient bleed air for all these systems.

During single-engine operation, a check valve in the bleed air line from each engine prevents flow back through the line on the side of the inoperative engine. A suction gage calibrated in inches of mercury, located on the copilot's subpanel, indicates instrument vacuum. To the right of the suction gage is a pneumatic pressure gage, calibrated in pounds per square inch, which indicates air pressure available to the deice distributor valve.

Refer to the Pneumatic Bleed Air and Surface Deice System Schematic.

BLEED AIR WARNING SYSTEM

The bleed air lines from the engines to the cabin are shielded with insulation to protect other components from heat. Heat is also dissipated in the air-to-air heat exchanger in the center wing section. The bleed air lines are accompanied in close proximity by plastic tubing from the engines to the cabin. One end of the tubing is plugged off; the other end is connected to a bleed air source in the cabin, to supply the line with pressure. Excessive heat on the plastic tubing caused by a ruptured bleed air line will cause the tubing to fail. Upon release of pressure in the tubing, a normally open switch in the line, located under the copilot's floor in the fuselage, will close, causing a circuit to be completed to the respective BL AIR FAIL light in the warning annunciator panel. When the indication of bleed air line failure becomes evident, the bleed air for that side should be turned off by placing the respective lever-lock BLEED AIR VALVE switch on the copilot's subpanel in the INSTR & ENVIR - OFF position.

AUTOMATIC DEVICES IN THE CONTROL SYSTEM

YAW DAMP

A yaw damp system is provided to aid the pilot in maintaining directional control, and to increase ride comfort. The system may be used at any altitude, and is required for flight above 17,000 feet. It should be deactivated for takeoff and landing.

If the system is equipped with an autopilot, the yaw damp system will be a part of the autopilot. Operating instructions for this system will be contained in the appropriate Airplane Flight Manual Supplement.

If an autopilot is not installed in the airplane, yaw damping is provided by an independent yaw damp system. The components include a yaw sensor, amplifier, and control valve. Regulated air pressure from the control valve is directed to the same pneumatic servos used for the rudder boost system. The system (on airplanes without an autopilot) is controlled by a YAW DAMP switch adjacent to the RUDDER BOOST switch on the pedestal. In the event the YAW DAMP switch is inadvertently left ON during takeoff or landing, the circuit for the yaw damping system will be interrupted by the left landing gear safety switch while the airplane is on the ground, rendering it inoperative.

STALL WARNING SYSTEM

The stall warning system consists of a transducer, a lift computer, a warning horn, and a test switch. Angle of attack is sensed by aerodynamic pressure on the lift transducer vane located on the left wing leading edge. When a stall is imminent, the output of the transducer activates a stall warning horn.

The system has preflight capability through the the use of a switch placarded STALL WARN - TEST - OFF on the copilot's left subpanel. Holding this switch in the TEST position activates the warning horn.

ICE PROTECTION SYSTEMS

WINDSHIELD HEAT

Windshield heat switches are located on the pilot's subpanel (inboard) and are placarded ICE - WSHLD ANTI-ICE - NORMAL - OFF - HI - PILOT - COPILOT.

Two levels of heat are provided. When the switches are in the NORMAL (up) position, heat is supplied to the major portion of the windshields. When they are in the HI (down) position, a higher level of heat is supplied to a smaller area of the windshields. Each switch must be lifted over a detent before it can be moved into the HI position. This lever-lock feature prevents inadvertent selection of the HI position when moving the switches from NORMAL to the OFF (center) position.

Controllers with temperature-sensing units provide for proper heat at the windshield surfaces. Five-ampere circuit breakers, located on a panel on the forward pressure bulkhead, protect the control circuits. The power circuit of each system is protected by a 50-ampere circuit breaker located in the power distribution panel under the floor forward of the main spar.

NOTE

Erratic operation of the magnetic compass may occur while windshield heat is being used.

PROPELLER ELECTRIC DEICE SYSTEM

PRIOR TO BB-829, AND PRIOR TO BL-37:

The propeller electric deice system includes: an electrically heated boot with two elements, (inner and outer) for each

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propeller blade, brush assemblies, slip rings, an ammeter, a timer for automatic operation, and a circuit for manual control for backup.

A 20-ampere circuit breaker switch on the pilot's subpanel, placarded PROP - AUTO - OFF, is provided to activate the automatic system. A deice ammeter on the right subpanel registers the amount of current (14 to 18 amperes) passing through the system being used. During AUTO operation, power to the timer will be cut off if the current rises above the circuit breaker switch rating. Current flows from the timer to the brush assembly and then to the slip rings installed on the spinner backing plate. The slip rings carry the current to the deice boots on the propeller blades. Heat from the boots reduces the grip of the ice, which is then thrown off by centrifugal force aided by the air blast over the propeller surfaces. Power to the two heating elements on each blade, the inner and outer element, is cycled by the timer in the following sequence: right propeller outer elements, right propeller inner elements, left propeller outer elements, and left propeller inner elements. Loss of one heating element on one side does not mean that the entire system will be inoperative.

Proper operation can be checked by noting the correct level of current usage on the ammeter. An intermittent flicker of the needle approximately every 30 seconds indicates switching to the next group of heating elements by the timer.

The manual prop deice system is provided as a backup to the automatic system. A control switch located on the left subpanel, placarded PROP - INNER - OUTER, controls the manual override relays. When the switch is in the OUTER position, the automatic timer is overridden and power is supplied to the outer elements of both propellers simultaneously. The switch is of the momentary type and must be held in position until the ice has been dislodged from the propeller surface. After deicing with the outer elements, the switch must be held in the INNER position to perform the same function for the inner elements of both propellers. The loadmeters will indicate approximately a .05 increase of load per meter when manual prop deice is operating. The prop deice ammeter will not indicate any load in the manual mode of operation.

BB-829 AND AFTER, BL-37 AND AFTER:

The propeller electric deice system includes: electrically heated deice boots, slip rings and brush block assemblies, a timer for automatic operation, an ammeter, three circuit breakers located on the fuel control panel for left and right propeller and control circuit protection, and two switches located on the pilot's subpanel for automatic or manual control of the system.

A circuit breaker switch located on the pilot's subpanel, placarded PROP - AUTO - OFF, is provided to activate the automatic system. Upon placing the switch to the AUTO position, the timer diverts power through the brush block and slip ring to all heating elements on one propeller. Subsequently, the timer then diverts power to all heating elements on the other propeller for the same length of time. This cycle will continue as long as the switch is in the AUTO position. The system utilizes a metal foil type single heating element energized by DC voltage. The timer switches every 90 seconds, resulting in a complete cycle in approximately 3 minutes.

A manual prop deice system is provided as a backup to the automatic system. A control switch located on the left subpanel, placarded PROP - MANUAL - OFF, controls the manual override relay. Upon placing the switch is in the MANUAL position, the automatic timer is overridden and power is then supplied to the heating elements of both propellers simultaneously. This switch is of the momentary type and must be held in position for aproximately 90 seconds to dislodge from the propeller surface. Repeat this procedure as required to avoid significant buildup of ice which will result in loss of performance, vibration, and impingement upon the fuselage. The prop deice ammeter will not indicate a load while the propeller deice system is being utilized in the manual mode. However, the loadmeters will indicate an approximate .05 increase of load per meter while the manual prop deice system is operating.

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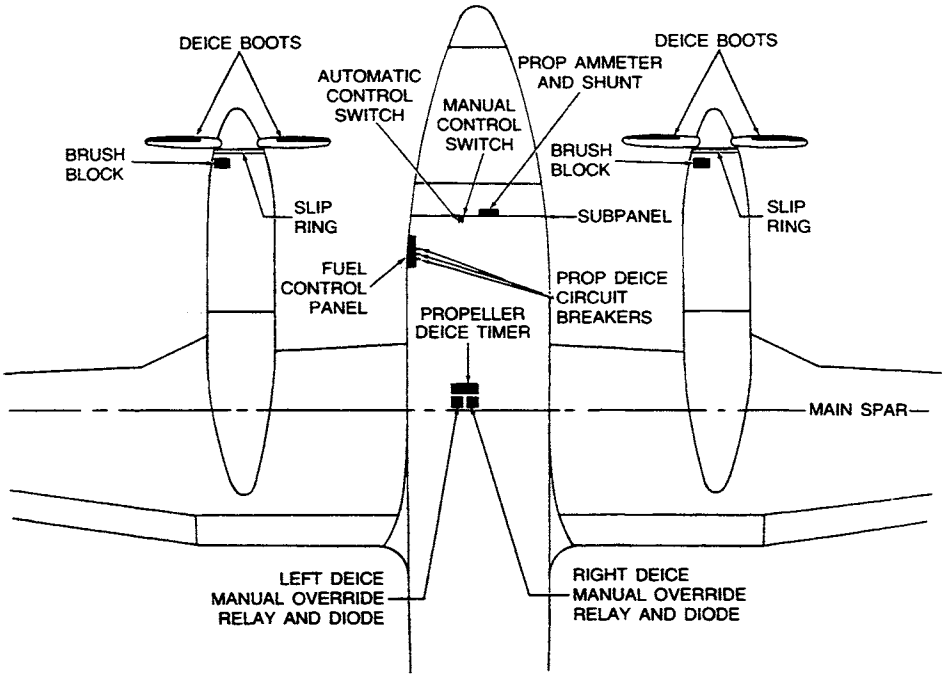


PROPELLER ELECTRIC DEICE SYSTEM SCHEMATIC

PITOT MAST HEAT

Heating elements are installed in the pitot masts located on the nose. Each heating element is controlled by an individual circuit breaker switch placarded PITOT - LEFT - RIGHT, located on the pilot's subpanel. It is not advisable to operate the pitot heat system on the ground except for testing or for short intervals of time to remove ice or snow from the mast.

SURFACE DEICE SYSTEM

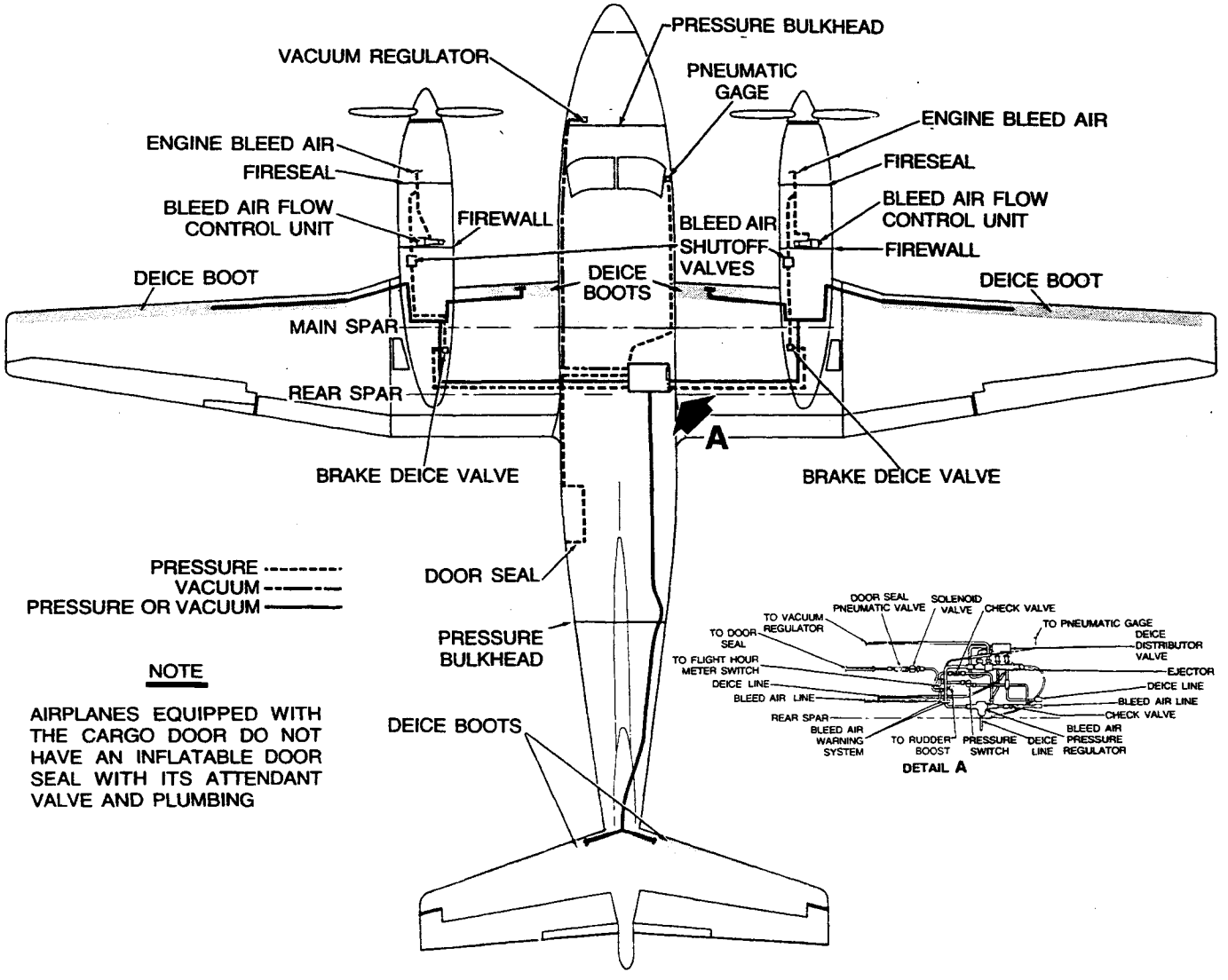
The surface deice system removes ice accumulations from the leading edges of the wings and horizontal stabilizers. Ice removal is accomplished by alternately inflating and deflating the deice boots. Pressure-regulated bleed air from the engines supplies pressure to inflate the boots. A venturi ejector, operated by bleed air, creates vacuum to deflate the boots and hold them down while not in use. To assure operation of the system in the event of failure of one engine, a check valve is incorporated in the bleed air line from each engine to prevent loss of pressure through the compressor of the inoperative engine. Inflation and deflation phases are controlled by a distributor valve.

CAUTION

Operation of the surface deice system in ambient temperatures below -40°C can cause permanent damage to the deice boots.

A three-position switch in the ICE group on the pilot's subpanel identified: DEICE CYCLE - SINGLE - OFF - MANUAL, controls the deicing operation. The switch is spring-loaded to return to the OFF position from SINGLE or MANUAL. When the SINGLE position is selected, the distributor valve opens to inflate the wing boots. After an inflation period of approximately 6 seconds, an electronic timer switches the distributor to deflate the wing boots, and a 4-second inflation begins in the horizontal stabilizer boots. When these boots have inflated and deflated, the cycle is complete.

When the switch is held in the MANUAL position, all the boots will inflate simultaneously and remain inflated until the switch is released. The switch will return to the OFF position when released. After the cycle, the boots will remain in the vacuum hold-down condition until again actuated by the switch.



PNEUMATIC BLEED AIR SYSTEM AND SURFACE DEICE SYSTEM SCHEMATIC

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For most effective deicing operation, allow at least 1/2 inch of ice to form before attempting ice removal. Very thin ice may crack and cling to the boots instead of shedding. Subsequent cyclings of the boots will then have a tendency to build up a shell of ice outside the contour of the leading edge, thus making ice removal efforts ineffective.

STALL WARNING VANE HEAT

The lift transducer is equipped with anti-icing capability on both the mounting plate and the vane. The heat is controlled by a switch in the ice group located on the pilot's right subpanel identified: STALL WARN. The level of heat is minimal for ground operation, but is automatically increased for flight operation through the left landing gear safety switch.

WARNING

The heating elements protect the lift transducer vane and face plate from ice. However, a buildup of ice on the wing may change or disrupt the airflow and prevent the system from accurately indicating an imminent stall. Remember that the stall speed increases whenever ice accumulates on any airplane.

FUEL HEAT

An oil-to-fuel heat exchanger, located on the engine accessory case, operates continuously and automatically to heat the fuel sufficiently to prevent ice from collecting in the fuel control unit when the procedure under "APPROVED FUEL ADDITIVE" located in LIMITATIONS Section is complied with.

Each pneumatic fuel control line is protected against ice by an electrically heated jacket. Power is supplied to each fuel control air line jacket heater by two switches actuated by moving the condition levers in the pedestal out of the fuel cutoff range. Fuel control heat is automatically turned on for all flight operations.

COMFORT FEATURES

TOILET

The side facing toilet is installed in the foyer and faces the airstair door. The foyer can be closed off from the cabin by sliding the two partition-type door panels to the center of the fuselage, where they are held closed by magnetic strips. A forward facing toilet, when installed, is located in the aft cargo area and is enclosed by the cargo partition. The toilet may be either the chemical type or the electrically flushing type. In either case, the two hinged lid half-sections must be raised to gain access to the toilet. A toilet tissue dispenser is contained in a slide-out compartment on the forward side of the toilet compartment.

CAUTION

If a Monogram electrically flushing toilet is installed, the sliding knife valve should be open at all times, except when actually servicing the unit. The cabinet below the toilet must be opened in order to gain access to the knife valve actuator handle.

RELIEF TUBES

A relief tube is contained in a special tilt-out compartment at the aft side of the toilet cabinet. A relief tube may also be installed in the cockpit and stowed under the pilot or copilot chair. The hose on the cockpit relief tube is of sufficient length to permit use by either pilot or copilot.

A valve lever is located on the side of the relief tube horn. This valve lever must be depressed at all times while the relief tube is in use. Each tube drains into the atmosphere through its own special drain port, which protrudes from the bottom of the fuselage. Each drain port atomizes the discharge to keep it away from the skin of the airplane.

NOTE

The relief tubes are for use during flight only.

CABIN FEATURES

FIRE EXTINGUISHERS

An optional portable fire extinguisher may be installed on the floor on the left side of the airplane forward of the airstair entrance door, just aft of the rearmost seat. Another one may also be installed underneath the copilot's seat.

WINDSHIELD WIPERS

The dual windshield wiper installation consists of an electric motor, arm and wiper assemblies, drive shafts, and converters, all located forward of the instrument panel. The system also includes a control switch, located in the upper left corner of the overhead light control panel, and a circuit breaker located on the right circuit breaker panel. The control knob, placarded WINDSHIELD WIPER - PARK - OFF - SLOW - FAST, controls the wipers. The wipers have two speeds, one for light and one for heavy precipitation. After the control is turned to PARK to bring the wiper arms to their most inboard position, spring-loading returns the control to the OFF position.

Windshield wipers may be used during either ground or flight operations.

CAUTION

Do not operate windshield wipers on dry glass.

CARGO RESTRAINT (B200C)

Beech Aircraft Corporation offers an FAA approved cargo restraint system as Kit No. 101-5040. Any other restraint system used in this airplane must be approved by the FAA. Such approval is the sole responsibility of the owner/ operator of the airplane.

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HANDLING, SERVICING & MAINTENANCE
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SECTION VIII

HANDLING, SERVICING & MAINTENANCE

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INTRODUCTION TO SERVICING

The purpose of this section is to outline to the Owner and Operator the requirements for maintaining the Super King Air B200 & B200C in a condition equal to that of its original manufacture. This information sets the time intervals at which the airplane should be taken to a Raytheon Aircraft Authorized Outlet for periodic servicing or preventive maintenance.

The Federal Aviation Regulations place the responsibility for the maintenance of this airplane on the Owner and the Operator, who should make certain that all maintenance is done by qualified mechanics in conformity with all airworthiness requirements established for this airplane.

All limits, procedures, safety practices, time limits, servicing and maintenance requirements contained in this handbook are considered mandatory.

Raytheon Aircraft Authorized Outlets can provide recommended modification, service, and operating procedures issued by both FAA and Beech Aircraft Corporation, which are designed to get maximum utility and safety from the airplane.

If a question arises concerning the care of the Super King Air B200 & B200C, it is important that the airplane serial number be included in any correspondence. The serial number appears on the Manufacturer's Identification Plaque, located either on the aft frame of the airstair door opening, or on the fuselage exterior, left side, aft of the airstair door.

WARNING

The Beech Super King Air B200 & B200C is a pressurized airplane. Drilling, modification, or any type of work which creates a break in the pressure vessel is considered the responsibility of the owner or facility performing the work. Obtaining approval of the work is, therefore, their responsibility.

PUBLICATIONS

The following publications for the Super King Air B200 & B200C are available through Raytheon Aircraft Authorized Outlets:

1. Pilot's Operating Handbook and FAA Approved Airplane Flight Manual
2. Pilot's Check List
3. Maintenance Manual
4. Component Maintenance Manual (Includes Supplier Data)
5. Structural Inspection and Repair Manual
6. Components Maintenance Manual
7. Wiring Diagram Manual
8. Parts Catalog
9. Service Instructions and Service Bulletins

The following publications will be provided, at no charge, to the registered owner/operator of this airplane:

- Reissues and revisions of Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.
- Original issues and revisions of FAA Approved Airplane Flight Manual Supplements.
- Original issues and revisions of Raytheon Aircraft Service Bulletins.

The above publications will be provided to the registered owner/operator at the address listed on the FAA Aircraft Registration Branch List or the Raytheon Aircraft Domestic/International Owners Notification List. Further, the owner/operator will receive only those publications pertaining to the registered airplane serial number. For detailed information on how to obtain "Revision Service" applicable to this handbook or other Raytheon Aircraft Service Publications, consult any Raytheon Aircraft authorized outlet, or refer to the latest revision of Raytheon Aircraft Service Bulletin No. 2001.

AIRPLANE INSPECTION PERIODS

Refer to the following for required inspections:

1. Structural Inspection and Repair Manual
2. Maintenance Manual

NOTE

The FAA may require other inspections by issuance of Airworthiness Directives applicable to the airplane, engines, propellers, and components. It is the responsibility of the owner/operator to ensure that all applicable Airworthiness Directives are complied with, and when repetitive inspections are required, to prevent inadvertent noncompliance with subsequent inspection requirements. It is also the responsibility of the owner/operator to ensure that all FAA-required inspections and most Raytheon-recommended inspections are accomplished by properly certificated mechanics at properly certificated agencies (both meeting FAR 91 and FAR 43 requirements). Consult a Raytheon Aircraft Authorized Outlet for assistance in determining and complying with these requirements.

SPECIAL CONDITIONS CAUTIONARY NOTICE

Airplanes operated for Air Taxi or other than normal operations and airplanes operated in humid tropics or cold and damp climates, etc., may need more frequent inspections for wear, corrosion, and/or lack of lubrication. In these areas,

periodic inspections should be performed until the operator can set his own inspection periods based on experience. The required periods do not constitute a guarantee that the item will reach the period without malfunction, as the aforementioned factors cannot be controlled by the manufacturer.

PREVENTIVE MAINTENANCE THAT MAY BE ACCOMPLISHED BY A CERTIFICATED PILOT

A certificated pilot may perform limited maintenance. Refer to FAR Part 43 for the items which may be accomplished. To ensure that proper procedures are followed, obtain a *Beech Super King Air 200 Series Maintenance Manual* prior to performing preventive maintenance. All other maintenance must be performed by properly certificated personnel. Contact a Raytheon Aircraft Authorized Outlet for further information.

NOTE

Pilots operating airplanes of other than U. S. registry should refer to the regulations of the country of registry for information on preventive maintenance that may be performed by a pilot.

ALTERATIONS OR REPAIRS TO AIRPLANE

The FAA should be contacted prior to any alterations on the airplane, to ensure that the airworthiness of the airplane is not violated.

NOTE

Alterations or repairs to the airplane must be accomplished by properly licensed personnel.

GROUND HANDLING

TOWING

The tow bar connects to the upper torque knee fitting of the nose strut. The airplane is steered with the tow bar when moving the airplane by hand, or an optional tow bar is available for towing the airplane with a tug. Although the tug will control the steering of the airplane, someone should be positioned in the pilot's seat to operate the brakes in case of an emergency.

CAUTION

Always ensure that the control locks are removed before towing the airplane. Serious damage to the steering linkage can result if the airplane is towed while the control locks are installed. Do not tow the airplane with a flat shock strut.

The nose gear strut has turn limit warning marks to warn the tug driver when turning limits of the gear will be exceeded. Damage will occur to the nose gear and linkage if the turn limit is exceeded. The maximum nose wheel turn angle is 48° left and right. When ground handling the airplane, do not use the propellers or control surfaces as hand holds to push or move the airplane.

CAUTION

Do not exert force on the propellers or control surfaces. Do not place weight on the stabilizers to raise the nose wheel. When towing, limit turns to prevent damage to the nose gear. Do not tow the airplane backward using the tail tie down ring as an attach point.

PARKING

The parking brake may be set by pulling outward on the parking brake control, located on the extreme left side, below the pilot's subpanel, and depressing the toe portion of the pilot's rudder pedals. The parking control closes dual valves in the brake lines that trap the hydraulic pressure applied to the brakes and prevents pressure loss through the master cylinders. To release the parking brake, depress the pilot's brake pedals to equalize the pressure on both sides of the parking brake valves and push the parking brake control fully in.

NOTE

Avoid setting the parking brake when the brakes are hot from severe usage, or when moisture conditions and freezing temperatures could form ice locks.

CAUTION

The parking brake should be left off and wheel chocks installed while the airplane is unattended. Changes in ambient temperature can cause the brakes to release or to exert excessive pressures.

TIE-DOWN

Three mooring eyes are provided: one underneath each wing, and one in the ventral fin. To moor the airplane, chock the wheels fore and aft, install the control locks, and tie the airplane down at all three points. If extreme weather is anticipated, it is advisable to nose the airplane into the wind before tying it down. Install engine inlet and exhaust covers, propeller tie-down boots (one blade down), and pitot mast covers when mooring the airplane.

NOTE

Unrestrained propellers are apt to windmill. Prolonged windmilling at zero oil pressures can result in bearing damage. Windmilling propellers are a SAFETY HAZARD.

JACKING AND LEVELING

The Beech Super King Air B200 & B200C is provided with three jacking points to raise the airplane for servicing. The forward point is on the left side of the nose wheel well opening near the aft end of the nose wheel doors. The main gear points are on the rear spar just inboard of the nacelle fairing. All three points are easily identified by the placarding JACK PAD adjacent to the jack points. The areas around the jack pads are unobstructed to facilitate the use of jacks. All adapters extend 0.7 inch or more below the structure surface.

Leveling screws are located on the fuselage entrance door frame. Leveling is accomplished with a plumb bob. Jack pad leveling may require the nose-gear shock strut to be secured in the static position to prevent its extension. Wheel weighings can be leveled by varying the amounts of air in the shocks and tires.

PROLONGED OUT-OF-SERVICE CARE

Refer to the *Beech Super King Air 200 Series Maintenance Manual.*

ENGINE CARE IN SALTY ENVIRONMENTS

When the airplane is operated in a salty atmosphere (such as near the sea) or off airstrips treated with salt:

1. Wash engine exterior as soon as possible with clean water.
2. Start engine and run at ground idle for a minimum of 10 minutes to remove moisture and salt residue.
3. Spray rust preventive material on fuel control assembly, controls linkage assembly, and any exposed metal parts.
4. Inspect the entire gearcase for corrosion and spray with rust preventive material at one-week intervals. Pay particular attention to the areas around studs and inserts.

SERVICING

EXTERNAL POWER

The airplane is equipped with an external power receptacle, located just outboard of the right engine in the lower side of the wing center section. The receptacle will accept a standard AN-type plug. The airplane electrical system is automatically protected from reverse polarity (i.e., positive ground) by a diode network. A amber EXT PWR caution light on the caution/advisory annunciator panel will flash when the external power plug is engaged.

External power can be used to operate all the airplane electrical equipment (this includes avionics checkouts) during ground operations without the engines running, and it can be used to start the engines. The external power unit should be capable of producing 1000 amperes for 5 seconds, 500 amperes for 2 minutes and 300 amperes continuously. A maximum continuous load of 350 amperes will damage the external power relay and power cables of the airplane.

CAUTION

Any current in excess of 1000 amperes may overtorque the drive shaft of the starter-generator or produce heat sufficeint to shorten the life of the unit.

The following precautions must be observed when using an external power source:

1. AVIONICS MASTER PWR Switch (pilot's left subpanel) - OFF
2. GENerator 1 and GENerator 2 Switches - OFF
3. BATtery Switch - ON
4. Volt/Loadmeter (overhead panel) - DEPRESS SWITCH on face of either meter, and read battery voltage.

CAUTION

NEVER CONNECT AN EXTERNAL POWER SOURCE TO THE AIRPLANE UNLESS A BATTERY INDICATING A CHARGE OF AT LEAST 20 VOLTS IS IN THE AIRPLANE. If the battery voltage is less than 20 volts, the battery must be recharged, or replaced with a battery indicating at least 20 volts, before connecting external power.

5. Auxiliary Power Unit Output Voltage - SET AT 28.0 - 28.4 VOLTS
6. Auxiliary Power Unit - TURN OFF before connecting to airplane.

CAUTION

Only use an external power source fitted with an AN-type plug.

7. External Power Source Plug - PLUG INTO AIRPLANE RECEPTACLE
8. Auxiliary Power Unit - TURN ON
9. Volt/Loadmeter (overhead panel) - DEPRESS SWITCH on face of either meter, and read voltage. (If external power is properly connected, the value will be greater than it was when reading battery voltage only.)

BATTERY

Servicing the 24-volt, 20-cell, air-cooled, nickel-cadmium battery is normally limited to checking the electrolyte level, cleaning the battery box and associated components, and equalizing the cells. For detailed servicing of the battery, refer to the *Beech Super King Air 200 Series Maintenance Manual.*

CAUTION

The electrolyte in the nickel-cadmium battery is an alkali solution. Use equipment reserved for nickel-cadmium batteries only. Even minute traces of acid can damage a nickel-cadmium battery.

Add only distilled water when liquid level is low. The battery electrolyte level is related to the amount of electrical charge stored in the battery. When the charge is low, the electrolyte will appear to be low, therefore the distilled water should only be added when the battery is fully charged.

BATTERY OVERHEAT FACTORS

Battery overheating can be caused or accelerated by the following factors:

1. Frequent engine starts and excessive engine cranking.
2. Airplane generator bus voltage too high.
3. Improper charging.
4. Unnecessary use of the airplane battery to run auxiliary equipment such as lights, avionics equipment, ventilation systems, etc., during ground operations.
5. Loose cell-to-cell connectors (links).
6. Ground operations using power units with voltage settings higher than the recommended airplane bus voltage, or power units with poor regulation.

MAINTENANCE PRACTICES TO PREVENT BATTERY OVERHEATING

1. Service the battery at the interval recommended in the maintenance manual; however, more frequent servicing may be necessary, depending upon the type of operation to which the airplane is subjected.
2. The voltage regulators should be checked periodically for proper calibration, thereby reducing the possibility of overcharging and concurrent rise in battery temperature.
3. Keep battery loads to a minimum during extended ground operation.
4. Reduce the probability of localized heating of the cells by checking torque values of the cell-to-cell connectors.
5. Keep battery clean.
6. When charging the battery with an auxiliary power unit, observe the following:
   a. Provide adequate ventilation for the battery compartment.
   b. The auxiliary power unit voltage should not exceed the specified airplane voltage. Make certain the unit is well regulated and that its ammeters are accurate.
   c. Open the battery cavity during charging, to allow visual monitoring and increased ventilation.

TIRES

The airplane is equipped with dual tires on the main gear, and a single tire on the nose gear. The standard configuration features 18x5.5, 8-ply-rated (or optionally, 10-ply-rated) tubeless tires on the main gear, and a 22x6.75-10, 8-ply-rated tire is installed on the nose gear.

Airplanes equipped with the optional high flotation landing gear are equipped with 22x6.75-10, 8-ply-rated tubeless tires on the main gear and on the nose gear.

CAUTION

Tires that have picked up a film of fuel, hydraulic fluid, or oil should be washed down as soon as possible, in order to prevent deterioration of the rubber.

Maintaining proper tire inflation pressures will help prolong tire service life. Check tires frequently to maintain pressures within recommended limits, and maintain equal pressures on both tires of each dual-wheel installation. Proper inflation pressures will help avoid damage from landing shocks, contact with sharp stones and ruts, and will minimize tread wear. When inflating the tires, inspect them for cuts, cracks, breaks, and tread wear.

NOTE

This procedure is recommended only for those airplanes equipped with rim-inflated tubeless tires. The leak rate of tires to be sealed by this method should not exceed 5% over a 24-hour period.

Refer to the *Beech Super King Air 200 Series Maintenance Manual* for more detailed inspection and repair procedures.

Inflate the standard main wheel tires (18x5.5) to 96 ±2 psi. Inflate the optional high flotation main wheel tires (22x6.75-10) to 62 ±2 psi. Both the standard and high flotation configuration nose wheel tires should be inflated to between 55 and 60 psi.

NOTE

While Raytheon Aircraft Company cannot recommend the use of recapped tires, tires retreaded by an FAA-approved repair station with a specialized service-limited rating for TSO-C62c may be used.

SHOCK STRUTS

Servicing the shock struts is normally part of each 200-hour phase inspection procedure. If it becomes necessary to service the shock struts due to the leakage of either the hydraulic oil or the air, the following procedure should be followed:

WARNING

Release the air pressure entirely before removing the valve adapter.

NEVER FILL SHOCK STRUTS WITH OXYGEN.

NOSE GEAR STRUT

1. Release all of the air from the strut by depressing the core of the air valve on top of the strut.
2. Remove the air valve and wipe clean. With the strut fully compressed, the end of the filler neck on the air valve should touch the oil. If the oil is below this level, add approved hydraulic fluid. Reinstall and safety the air valve.
3. With the airplane empty except for full fuel and oil, inflate the nose gear until the inner cylinder is extended approximately 3 to 3.5 inches.

MAIN GEAR STRUT

1. Release all of the oil from the strut through the air valve, and remove the core from the valve.
2. Fully compress the strut and attach a small hose over the air valve and immerse the other end of the hose in hydraulic fluid. Slowly extending the strut will create a partial vacuum, thereby drawing the oil into the cylinder. Cycling the strut slightly as it is extended will expel any trapped air. Slowly return the strut to the fully compressed position; this will force the excess oil back into the container, and the strut will be properly filled with oil.
3. With the airplane empty except for full fuel and oil, inflate the strut until the inner cylinder is extended approximately 3.93 to 4.19 inches. If the optional high flotation gear is installed, inflate the strut until the inner cylinder is extended approximately 5.56 to 5.93 inches.

BRAKE SYSTEM

Brake servicing is limited to maintaining adequate fluid in the reservoir mounted on the bulkhead in the upper left corner of the nose avionics compartment. A dipstick is provided as part of the reservoir lid to measure the fluid level. When the fluid is low, add sufficient quantity of approved hydraulic fluid to raise the level to the full mark on the dipstick.

Brake assemblies are equipped with automatic adjusters to assure a positive clearance between disc and lining when the brakes are not applied.

Each wheel cylinder (except those airplanes equipped with optional brake deice) is provided with a means of conveniently checking brake wear. The distance between the piston housing and the lining carrier will increase with lining wear. When the distance exceeds 0.250 inch (as indicated by the accompanying illustration) the brakes should be replaced. This check should be accomplished with brake pressure applied. For more detail on servicing of the wheels and brakes, and airplanes equipped with brake deice, refer to the *Beech Super King Air 200 Series Maintenance Manual.*

OIL SYSTEM

Servicing the engine oil system primarily involves maintaining the engine oil at the proper level, inspecting and cleaning or replacing the filter element, and changing the oil as conditions require. See the *Beech Super King Air 200 Series Maintenance Manual* and Pratt and Whitney Service Bulletin 3001 for oil system servicing procedures and requirements.

CAUTION

Do not mix different brands of oil when adding oil between oil changes. Different brands or types of oil may be incompatible because of the difference in their chemical structures.

The oil tank is provided with an oil filler neck and quantity dipstick cap which protrude through the accessory gearcase at the eleven-o'clock position. The dipstick is marked in U.S. quarts and indicates the last five quarts required to bring the system up full. Access to the dipstick cap is gained through an access door on the aft engine cowl. Service the oil system with oil as specified in Pratt and Whitney SB 3001. Do not mix different oil brands together. Total oil tank capacity is 2.3 U. S. gallons. When a dry engine is first serviced, it will require approximately 5 quarts in addition to tank capacity to fill the lines and cooler, giving a total system capacity of 14 U.S. quarts (3.50 U.S. gallons). The engine will trap approximately 1.5 quarts which cannot be drained; therefore, when performing an oil change, refill the system with 12 quarts and add additional oil based on the dipstick reading. While the airplane is standing idle, engine oil could possibly seep into the scavenge pump reservoir, causing a low dipstick reading.

Anytime an engine has been shut down for 12 hours or more, or if the oil has just been changed, run the engine for at least two minutes before checking the oil level.

NOTE

The normal oil level is at the one quart mark. Overfilling may cause a discharge of oil through the breather until a satisfactory level is reached.

CAUTION

Spilled oil should be removed immediately to prevent possible tire contamination or damage.

CHANGING THE ENGINE OIL

CAUTION

When changing to a different brand of oil, completely drain the airplane oil system. Remove the oil filter and immerse it in the brand of oil to be used. Reinstall the oil filter and drain plugs. Fill the system to the proper level, and ground run the engine for 20 minutes to thoroughly circulate the new brand of oil throughout the system. Completely drain the airplane oil system and again remove the oil filter and immerse it in the new brand of oil. Refill the airplane oil system. This will prevent chemical interaction between it and the new brand.

Refer to the *Beech Super King Air 200 Series Maintenance Manual* for detailed procedures for changing the engine oil.

OIL FILTER SERVICING

The engine oil filter is located under the square coverplate at the three-o'clock position of the compressor inlet case and just behind the aft fire seal. The filter element should be replaced according to the time periods specified in the *Beech Super King Air 200 Series Maintenance Manual.*

CAUTION

This filter element is not cleanable and must be replaced if it has been subjected to ultrasonic cleaning, or heavy contamination from the engine oil system.

Light contaminants collected on the external protective screen may be removed by hand flushing the element with Varsol. The filter element may be removed, inspected, and replaced in accordance with the *Beech Super King Air 200 Series Maintenance Manual.*

FUEL SYSTEM

FUEL HANDLING PRACTICES

All hydrocarbon fuels contain some dissolved, suspended water. The quantity of water contained in the fuel depends on temperature and the type of fuel. Kerosene, with its higher aromatic content, tends to absorb and suspend more water than aviation gasoline. Along with the water, it will suspend rust, lint, and other foreign materials longer. Given sufficient time, these suspended contaminants will settle to the bottom of the tank. However, the settling time for kerosene is five times that of aviation gasoline. Due to this fact, jet fuels require good fuel handling practices to assure that the airplane is serviced with clean fuel. If recommended ground procedures are carefully followed, solid contaminants will settle and free water can be reduced to 30 parts per million (ppm), a value that is currently accepted by the major airlines. Since most suspended matter can be removed from the fuel by sufficient settling time and proper filtration, it is not a major problem. Dissolved water has been found to be the major fuel contamination problem. Its effects are multiplied in airplanes operating primarily in humid regions and warm climates.

Dissolved water cannot be filtered from the fuel with micronic type filters, but can be released by lowering the fuel temperature, such as will occur in flight. For example, a kerosene fuel may contain 65 ppm (8 fl oz per 1000 gallons) of dissolved water at 80°F. When the fuel temperature is lowered to 15°F, only about 25 ppm will remain in solution. The difference of 40 ppm will have been released as supercooled water droplets which need only a piece of solid contaminant or an impact shock to convert them to ice crystals. Tests indicate that these water droplets will not settle since the specific gravity of ice is approximately equal to that of kerosene. The 40 ppm of suspended water seems like a very small quantity, but when added to suspended water in the fuel at the time of delivery, is sufficient to ice a filter. While the critical fuel temperature range is from 0°F to -20°F, which produces severe system icing, water droplets can freeze at any temperature below 32°F.

Water in jet fuel also creates an environment favorable to the growth of microbiological sludge in the settlement areas of the fuel cells. This sludge, plus other contaminants in the fuel, can cause corrosion of metal parts in the fuel system as well as clogging of the fuel filters.

Since fuel temperature and settling time affect total water content and foreign matter suspension, contamination can be minimized by keeping equipment clean. Use adequate filtration equipment and careful water drainage procedures, store the fuel in the coolest areas possible, and allow adequate settling time. Underground storage is recommended for fuels. Filtering the fuel each time it is transferred will minimize the quantity of suspended contaminants carried by the fuel.

The primary means of contamination control by the owner/operator is careful handling. This applies not only to fuel supply, but to keeping the airplane system clean. The following is a list of steps that may be taken to prevent and recognize contamination problems.

1. Know your supplier. It is impractical to assume that fuel free of contaminants will always be available, but it is feasible to exercise precaution and be watchful for signs of fuel contamination.
2. Assure, as much as possible, that the fuel obtained has been properly stored, filtered as it is pumped to the truck, and filtered again as it is pumped from the truck to the airplane.
3. Perform filter inspections to determine if sludge is present.
4. Periodically flush the fuel tanks and systems. The frequency of flushing should be determined by the climate and the presence of sludge.
5. Use only clean fuel servicing equipment.
6. After refueling, allow a three hour settle period, whenever possible, then drain a small amount of fuel from each drain.

CAUTION

Fuel spills on tires have a deteriorating effect and the tires should be cleaned promptly.

FILLING THE TANKS

When filling the airplane fuel tanks, always observe the following:

1. Make sure the airplane is statically grounded to the servicing unit and that the airplane and servicing unit are both grounded to ground.
2. The main filler cap is located on top of the outboard wing section, and the auxiliary filler cap is located on top of the center wing section. Do not rest fuel nozzle in tank filler neck, because this may damage the filler neck.

WARNING

Do not fill auxiliary tanks unless main tanks are full.

3. Allow a three-hour settle period whenever possible, then drain a small amount of fuel into a container from each drain point. Check fuel at each drain point for contamination.

FUEL GRADES AND TYPES

Aviation Kerosene Grades Jet A, Jet A-1, Jet B, JP-4, JP-5, and JP-8 may be mixed in any ratio. Aviation Gasoline Grades 80 (80/87), 100 (100/130), 100LL, and 115/145 are emergency fuels and may be mixed with the recommended fuels in any ratio; however, use of the lowest octane rating available is recommended. Operation on Aviation Gasoline shall be limited to 150 hours per engine during each time-between-overhaul (TBO) period. Refer to LIMITATIONS Section for additional limitations on the use of Aviation Gasoline.

CAUTION

Do not allow the fuel cells to dry out and crack. At a later servicing, the cracks would allow fuel to diffuse through the walls of the fuel cell. If any fuel cell is to remain empty for an extended interval, ensure that the fuel cells last contained jet fuel; if the fuel cells last contained aviation gasoline, coat them with oil.

DRAINING THE MAIN FUEL SYSTEM

A rapid defueling adapter is installed in the nacelle tank sump strainer. A valve in the adapter opens as a standard AN-type fuel connector is screwed into the adapter. This facilitates rapid, complete draining of the main fuel system.

DRAINING THE AUXILIARY FUEL SYSTEM

The auxiliary fuel system can be drained by either transferring it to the main system with the transfer system, or by draining it into a container by removing the drain plug located in the bottom of the auxiliary tank sump strainer.

The auxiliary tank need not be drained to replace the standby fuel pump.

ENGINE FUEL FILTERS AND SCREENS

NOTE

Inspect fuel filters and screens any time fuel that is suspected of contamination has been used. In addition to such precautions, all fuel filters and the tank should be cleaned any time the submerged boost pump is removed.

CLEANING FIREWALL FUEL FILTERS AND ENGINE DRIVEN FUEL PUMP FILTER

Refer to the *Beech Super King Air 200 Series Maintenance Manual* for cleaning and replacement of the firewall fuel filters.

Refer to the Pratt and Whitney Engine Maintenance Manual for the PT6A-41/42 engine for proper servicing procedures for the engine driven fuel pump screens and filters.

P3 AIR FILTER

An air filter is installed in the P3 air line between the gas generator case and the fuel control unit. The filter is located on the lower right side of the engine immediately aft of the rear fireseal mount ring. Inspect and/or replace in accordance with the *Beech Super King Air 200 Series Maintenance Manual.*

INSTRUMENT VACUUM AIR

Vacuum for the flight instruments is obtained by operating an ejector with bleed air from the engines. During operation, the ejector draws air in through the instrument filter and the gyros. A vacuum relief regulator valve regulates instrument pressure.

The instrument filter, located at the top of the avionics compartment, is of prime importance and should be replaced in accordance with the *Beech Super King Air 200 Series Maintenance Manual*, or more often if conditions warrant (smoky, dusty conditions).

The vacuum relief regulator valve, located on the forward pressure bulkhead in the bottom of the avionics compartment, is protected by a foam sponge type filter which should be cleaned in accordance with the *Beech Super King Air 200 Series Maintenance Manual.* If vacuum pressure rises above a normal reading, clean the filter and recheck vacuum pressure before attempting to adjust the valve.

SERVICING THE OXYGEN SYSTEM

OXYGEN COMPONENTS

Oxygen for unpressurized, high-altitude flight is supplied by a cylinder located in the compartment immediately aft of the aft pressure bulkhead. A 22-, 49-, 66-, 76-, or 115-cubic-foot cylinder may be installed. The oxygen system is serviced by a filler valve accessible by removing an access plate on the right side of the aft fuselage. The system has two pressure gages, one located on the right subpanel in the crew compartment for in-flight use, and one adjacent to the filler valve for checking system pressure during filling.

Refer to OXYGEN SYSTEM SCHEMATIC in Section VII, SYSTEMS DESCRIPTION.

A shutoff valve and regulator, located on the cylinder, controls the flow of oxygen to the crew and passenger outlets. The shutoff valve is actuated by a push-pull type control located aft of the overhead light control in the cockpit. The regulator is a constant-flow type which supplies low pressure oxygen through system plumbing to the outlets.

OXYGEN SYSTEM PURGING

Offensive odors may be removed from the oxygen system by purging. The system should also be purged anytime system pressure drops below 50 psi, or a line in the system is opened. Purging is accomplished simply by connecting a recharging cart into the system and permitting oxygen to flow through the lines and outlets until any offensive odors have been carried away. The following precautions should be observed when purging or servicing the oxygen system:

1. Avoid any operation that could create sparks. Keep all burning cigarettes or fire away from the vicinity of the airplane when the outlets are in use.
2. Inspect the filler connection for cleanliness before attaching it to the filler valve.
3. Make sure that your hands, tools, and clothing are clean, particularly of grease or oil stains, for these contaminants are extremely dangerous in the vicinity of oxygen.
4. As a further precaution against fire, open and close all oxygen valves slowly during filling.

FILLING THE OXYGEN SYSTEM

When filling the oxygen system, only use Aviator's Breathing Oxygen, MIL-O-27210.

WARNING

DO NOT USE MEDICAL or INDUSTRIAL OXYGEN. It contains moisture which can cause the oxygen valve to freeze.

Fill the oxygen system slowly by adjusting the recharging rate with the pressure regulating valve on the servicing cart, because the oxygen, under high pressure, will cause excessive heating of the filler valve. Fill the cylinder (22-cubic-foot cylinder installation) to a pressure of 1800 ± 50 psi at a temperature of 70°F. This pressure may be increased an additional 3.5 psi for each degree of increase in temperature; similarly, for each degree of drop in temperature, reduce the pressure for the cylinder by 3.5 psi. The oxygen system, after filling, will need to cool and stabilize for a short period before an accurate reading on the gage can be obtained. The 49-, 64-, 76-, or 115-cubic-foot cylinders may be charged to a pressure of 1850 ± 50 psi at a temperature of 60°F. When the system is properly charged, disconnect the filler hose from the filler valve and replace the protective cap on the filler valve.

OXYGEN CYLINDER RETESTING

Oxygen cylinders used in the airplane are of two types. Light weight cylinders, stamped "3HT" on the plate on the side, must be hydrostatically tested every three years and the test date stamped on the cylinder. This bottle has a service life of 4,380 pressurizations or 24 years, whichever occurs first, and then must be discarded. Regular weight cylinders, stamped "3A" or "3AA", must be hydrostatically tested every five years and stamped with the retest date. Service life on these cylinders is not limited.

AIR CONDITIONER

If an extended period of time occurs during which the air conditioning system is not operated, moisture may condense and settle in the system low spots, resulting in corrosion of the refrigerant lines. Also, the system seals may dry out, shrink, and crack, due to the lack of lubrication. In order to protect the integrity of the system, the air conditioner should be operated at least 10 minutes every month.

CAUTION

Do not attempt to operate the air conditioner when the ambient temperature is below 10°C (50°F). If for several weeks it is impossible to obtain an ambient temperature of at least 10°C (50°F), the recommended monthly interval for operating the air conditioner may be extended somewhat.

For air conditioner system servicing information, refer to *Beech Super King Air 200 Series Maintenance Manual.*

WARNING

Refrigerant and oil are under pressure within the refrigeration system. Injury to personnel or damage to the system could occur if the maintenance is not performed properly. The refrigerant system should be serviced only by qualified air conditioner technicians.

CABIN AIR FILTERS

A flexible, fiberglass-type air filter covers the coils of the forward air conditioner evaporator. When an aft evaporator is installed, another flexible filter is used at the aft evaporator coils. A foam-rubber type recirculated-air filter is also installed over the return-air valve, at floor level forward of the copilot's rudder pedals. All these filters should be inspected in accordance with the *Beech Super King Air 200 Series Maintenance Manual.*

FORWARD EVAPORATOR FILTER REPLACEMENT

1. Remove the access door in the nose-wheel-well keel under the refrigerant plumbing.
2. Pull the filter down and out of the retaining springs on the evaporator coil.
3. Fold the new filter to insert it through the access doors. The filter must be carefully inserted between the coil assembly and the refrigerant plumbing under the retaining springs.
4. Replace the access doors.

AFT EVAPORATOR FILTER REPLACEMENT (WHEN OPT. EVAPORATOR INSTALLED)

1. Remove the carpet and floor panel behind the rear spar, and remove the cover of the evaporator plenum.
2. Remove the old filter from behind the retaining springs on the evaporator coil.
3. Insert the new filter between the retaining springs and the evaporator coil.
4. Replace the plenum cover, floor panel, and carpet.

RETURN AIR FILTER REPLACEMENT

1. Reach forward of the copilot's rudder pedals to the pressure bulkhead at floor level.
2. Pull out the two (left and right) flexible filters.
3. Insert one new filter into the left half of the channel on top of the filter support grill, and one into the right half.
4. Tamp the filters into place by hand and smooth them out.

DEICING AND ANTI-ICING OF AIRPLANES ON THE GROUND

Deicing is the removal of ice, frost, and snow from the airplane's exterior after it has formed. Anti-icing is a means of keeping the surface clear of subsequent accumulations of ice, snow and frost.

Snow and ice on an airplane will seriously affect its performance. Even formation of a smooth covering of ice on the wing will change the contour of the wing, producing an increase in drag and a reduction in effective lift coefficient. Frost or frozen snow may present an even greater hazard since the surface texture is rough and will seriously disrupt the smooth flow of air across the wing.

SNOW REMOVAL

The best way to remove snow is to brush it off with a squeegee, soft brush, or mop. Exercise care so as not to damage any components that may be attached to the outside of the airplane, such as antennas, vents, stall warning devices, etc. Remove loose snow from the airplane before heating the airplane interior; otherwise, at low temperatures, the snow may melt and refreeze to build up a considerable depth of ice. Never attempt to chip or break frozen snow from the airplane. If the airplane has been hangared and snow is falling, coat the airplane surfaces with an anti-icing solution; snow falling on the warm surface will have a tendency to melt, then refreeze.

After snow has been removed from the airplane, inspect the airplane for evidence of residual snow, particularly in the area of control surface gaps and in the hinge areas. Carefully inspect the static ports for evidence of obstruction. Check the exterior of the airplane for damage to external components that may have occurred during the snow removal operations.

Control surfaces should be moved to ascertain that they have full and free movement. The landing gear mechanism, doors, wheel wells, uplocks and microswitches should be checked for ice deposits that may impair function.

When the airplane is hangared to melt snow, any melted snow may freeze again if the airplane is subsequently moved into subzero temperatures. Any measures taken to remove frozen deposits while the airplane is on the ground must also prevent the possibility of refreezing of the liquid.

Following snow removal, should freezing precipitation continue, the airplane surface should be treated for anti-icing.

FROST REMOVAL

Heavy frost that cannot be removed by wiping with a gloved hand or soft towel must be removed by placing the airplane in a warm hangar or by the application of a deicing fluid.

After removal of all frost from the airplane exterior, check all external components for damage that may have occurred during frost removal.

ICE REMOVAL

Moderate or heavy ice and residual snow deposits should be removed with a deicing fluid. No attempt should be made to remove ice deposits or break an ice bond by force.

After completing the deicing process, the airplane should be inspected to ensure that its condition is satisfactory for flight. All external surfaces should be examined for residual ice or snow, particularly in the vicinity of control surface gaps and hinges. Static ports should be carefully inspected for any signs of obstruction.

Control surfaces should be moved to ascertain that they have full and free movement. The landing gear mechanism, doors, wheel wells, uplocks and microswitches should be checked for ice deposits that may impair function.

When the airplane is hangared to melt ice, any melted ice may freeze again if the airplane is subsequently moved into subzero temperatures. Any measures taken to remove frozen deposits while the airplane is on the ground must also prevent the possible refreezing of the liquid.

Following ice removal, should freezing precipitation continue, the airplane surface should be treated for anti-icing.

DEICING AND ANTI-ICING FLUID APPLICATION

Airplane deicing fluids may be used diluted or undiluted according to manufacturers recommendations for deicing. For anti-icing purposes, the fluids should always be used undiluted. Deicing fluids may be applied either heated or unheated. Refer to Section II, LIMITATIONS, for a listing of approved airplane deicing/anti-icing fluids.

NOTE

Type II and Type IV deicing fluids should only be applied at low pressure by trained personnel with proper equipment.

If a sprayer is not available, deicing fluid may be brushed or painted onto the airplane's surface.

MISCELLANEOUS MAINTENANCE

CLEANING

EXTERIOR PAINTED SURFACES

CAUTION

Polyester urethane undergoes a curing process for a period of 30 days after application. Wash uncured painted surfaces with a mild non-detergent soap (MILD detergents can be used on urethane finishes) and cold or lukewarm water only. Use soft cloths, keeping them free of dirt and grime. Any rubbing of the surface should be done gently and held to a minimum to avoid damaging the paint film. Rinse thoroughly with clear water. Stubborn oil or soot deposits may be removed with automotive tar removers.

Prior to cleaning, cover the wheels, making certain the brake discs are covered. Attach the pitot cover securely, and plug or mask off all other openings. Be particularly careful to mask off all static air buttons before washing or waxing. Use special care to avoid removing lubricant from lubricated areas.

Hand washing may be accomplished by flushing away loose dirt with clean water, then washing with a mild soap and water, using soft cleaning cloths or a chamois. Avoid harsh, abrasive or alkaline soaps or detergents which could cause corrosion or scratches. Thorough clear-water rinsing prevents buildup of cleaning agent residue, which can dull the paint's appearance. To remove oily residue or exhaust soot, use a cloth dampened with an automotive tar remover. Wax or polish the affected area if necessary.

WARNING

Do not expose elevator, rudder, and aileron trim tab hinge lines and their pushrod systems to the direct stream or spray of high-pressure soap-and-water washing equipment. Fluid dispensed at high pressure could remove the protective lubricant, allowing moisture from heavy or prolonged rain to collect at hinge lines, and then to freeze at low temperatures. After high-pressure or hand washing, and at each periodic inspection, lubricate trim tab hinge lines and trim tab pushrod end fittings (Brayco 300 per Federal Specification VV-L-800 preferred).

When using high-pressure washing equipment, keep the spray or stream clear of wheel bearings, propeller hub bearings, etc., and openings such as pitot tubes, static air buttons, and battery and avionics equipment cooling ducts which should be securely covered or masked off. Avoid directing high-pressure sprays toward the fuselage, wings, and empennage from the rear, where moisture and chemicals might more easily enter the structure, causing corrosion damage to structural members and moving parts.

CAUTION

When cleaning wheel well areas with solvent, especially if high-pressure equipment is used, exercise care to avoid washing away grease from landing gear components. After washing the wheel well areas with solvent, lubricate all lubrication points, or premature wear may result.

During the curing period, do not make prolonged flights in heavy rain or sleet, and avoid all operating conditions which might cause abrasion or premature finish deterioration.

CAUTION

Do not apply wax, polish, rubbing compound, or abrasive cleaner to any uncured painted surface. Use of such items can permanently damage the surface finish. Also, waxes and polishes seal the paint from the air and prevent curing.

Waxing of polyester urethane finishes, although not required, is permitted; however, never use abrasive cleaner type waxes, polishes, or rubbing compounds, as these products cause eventual deterioration of the characteristic urethane gloss.

For waxing, select a high quality automotive or aircraft waxing product. Do not use a wax containing silicones, as silicone polishes are difficult to remove from surfaces. A buildup of wax on any exterior paint finish will yellow with age; therefore, wax should be removed periodically. Generally, alphatic naphtha is adequate and safe for this purpose.

NOTE

Before returning the airplane to service, remove all maskings and coverings, and relubricate as necessary.

WINDOWS AND WINDSHIELDS

WINDOWS

The plastic windows should be kept clean and waxed. To prevent scratches, wash the windows carefully with plenty of mild soap and water, using the palm of the hand to dislodge dirt and mud. Flood the surface with clean water to rinse away dirt and soap. After rinsing, dry the windows with a clean, moist chamois. Rubbing the surface of the plastic with a dry cloth should be avoided, as it builds up an electrostatic charge on the surface, which attracts dust particles.

If oil or grease is present on the surface of the plastic, remove it with a cloth moistened with kerosene, aliphatic naptha, or hexene, then rinse the surface with clear water. Never use gasoline, benzine, alcohol, acetone, carbon tetrachloride, fire-extinguisher or anti-ice fluid, lacquer thinner, or glass cleaner. These materials will soften the plastic and may cause it to craze.

If it is desired to use a commercial cleaner to clean the plastic windows, use only cleaners that are approved by Raytheon Aircraft Company and follow the directions on the container. It will not be necessary to apply wax to windows after use of the above commercial cleaners, as these cleaners contain wax, as well as cleaning agents.

After thoroughly cleaning, wax the surface with a good grade of commercial wax that does not have an acrylic base. The wax will fill in minor scratches and help prevent further scratching. Apply a thin, even coat of wax and bring it to a high polish by rubbing lightly with a clean, dry, soft, flannel cloth. Do not use a power buffer; the heat generated by the buffing pad may soften the plastic.

POLARIZED CABIN WINDOWS

The polarized cabin windows consist of two plastic window panels installed with the polarized surfaces facing each other in a sealed assembly. To clean the interior exposed surface of the window requires only careful application of the practices for cleaning plastic windows. If it should become necessary to clean the inner surface of the sealed assembly and the inside of the pressure glass, the sealed assembly may be removed by removing the escucheon, four screws, and the sealed assembly. Clean the interior windows and reinstall the sealed assembly and escucheon.

WINDSHIELDS

Glass windshields with antistatic coating should be cleaned as follows:

1. Wash excessive dirt and other substances from the glass with clean water.
2. Clean the windshield with mild soap and water or a 50/50 solution of isopropyl alcohol and water. Wipe the glass surface in a straight rubbing motion with a soft cloth or sponge. Never use any abrasive materials or any strong acids or bases to clean the glass.
3. Rinse the glass thoroughly and dry, but do not apply wax.

SURFACE DEICE BOOT CLEANING

The deice boots are made of soft, flexible stock, which may be damaged if fuel hoses are dragged over the surface of the boots or if ladders and platforms are rested against them. Keep deice boots free of oil, fuel, paint remover, solvents, and other injurious substances. Deice boots should be cleaned regularly with a mild soap and water solution. The temperature of the solution should not exceed 180°F.

ENGINE

Clean the engine with neutral solvent. Spray or brush the fluid over the engine, then wash off with water and allow to dry.

CAUTION

Do not use solutions which may attack rubber or plastic. Protect engine switches, controls and seals; fluid applied at high pressure can unseat seals, resulting in contamination of the sealed systems.

INTERIOR CARE

To remove dust and loose dirt from the upholstery, headliner, and carpet, clean the interior regularly with a vacuum cleaner.

Blot up any spilled liquid promptly with cleansing tissue or rags. Do not pat the spot. Press the blotting material firmly and hold it for several seconds. Continue blotting until no more liquid is taken up. Scratch off sticky materials with a dull knife, then spot clean the area.

Oily spots may be cleaned with household spot removers, used sparingly. Before using any solvent, read the instructions on the container and test it on an obscure place on the fabric to be cleaned. Never saturate the fabric with a volatile solvent; it may damage the padding and backing materials.

Soiled upholstery and carpet may be cleaned with foam-type detergent, used according to the manufacturer's instructions. To minimize wetting the fabric, keep the foam as dry as possible and remove it with a vacuum cleaner.

CAUTION

The colors of many leathers may only be accomplished by surface dye processing. The color may be rubbed off by continuously dragging hard or coarse material across the leather. While working in the cabin, use protective covers on the leather upholstery. Use only mild detergent with a soft cloth to clean soiled leather.

The plastic trim, instrument panel, and control knobs need only be wiped with a damp cloth. Oil and grease on the control wheel and control knobs can be removed with a cloth moistened with kerosene. Volatile solvents, such as those mentioned in the article on care of plastic windows, should never by used, since they soften and craze the plastic.

ICE VANES (INERTIAL SEPARATOR SYSTEM)

RESETTING OVERRIDE ASSEMBLY FOR NORMAL OPERATION

Once the manual extension handle has been pulled out, do not use the electric actuator system on the affected side again until the override is disengaged as follows:

1. Open the forward cowling.
2. Push up on the latch where it protrudes into the slot in the override assembly until the clevis bolt on the side of the manual override arm is free to travel forward in the slot.
3. If necessary, center the clevis bolt on the end of the actuator arm in the semicircular slot in the override assembly. This will align the clevis bolt with the pin in the override assembly. When in this position, the actuator arm clevis engages the cam slot and is properly locked in place for the extent of travel required to operate the inertial vane.

CAUTION

Failure to properly position the end of the actuator arm in the semicircular slot of the override assembly will prevent full travel of the actuator and could result in damage to the electrical actuator.

CONSUMABLE MATERIALS

Refer to the *Beech King Air 200 Series Maintenance Manual* for consumables (type and brand name) approved for use in the Beech Super King Air B200 & B200C.

LAMP REPLACEMENT GUIDE

EXTERIOR

Entry Light (under Left Wing)	MS24513-4174
Wing Ice Light	A7079B-24
Landing Lights	4596
Rotating Beacons (Upper/Lower)	A7079B-24 / A7079B-25
Strobe Light Flash Tubes	55-0221-1
Empennage Floodlight	DS0079-BJ
Tail Navigation Light	A-2064-1683
Taxi Light	4587
Wing Navigation Light	A7512-24
Wing-tip Recognition Light	1982

PASSENGER COMPARTMENT

Aft Dome and Baggage Compartment Light	303
Aisle Light	MS25231-313
Spar Cover LIght	1495
Cabin Sign Light	D158-100-4
Cabin Door Observer and Overhead Lights	1864
Cabin Table Light	1309
Flood Light	MS35478-1683
Fluorescent Light Tube	5108WW
O.A.T. Light	MS25237-327
Reading and Chair Light	1864
Step Light	313
Threshold Light	313
Window Fluorescent light	PW 20B or 6900026

FLIGHT COMPARTMENT

All Edge-lighted Placards and Panels	D158-100-5
Cabin Door Lock Light	1864
Fuel Quantity Indicator Light	267
Glareshield Light	1864
Instrument Indirect Lights (Red/White)	1864R / 1864
Map Overhead Light (Red/White)	1309 / 1495
Map Overhead Floodlight	303
Lights for all other Instruments, Indicators, Annunciators, and Switches	327

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SECTION IX

SUPPLEMENTS

NOTE

When a new airplane is delivered from the factory, the handbook delivered with it contains either an STC (Supplemental Type Certificate) Supplement or a Beech Flight Manual Supplement for every installed item requiring a supplement. If a new handbook for operation of the airplane is obtained at a later date, it is the responsibility of the owner/operator to ensure that all required Flight Manual Supplements and STC Supplements (as well as weight and balance and other pertinent data) are transferred into the new handbook.

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Hawker Beechcraft Corporation

LOG OF SUPPLEMENTS

Super King Air® B200/B200C Landplanes

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual

P/N 101-590010-147C

June, 2010

FAA Supplement must be in the airplane for all flight operations when subject equipment is installed.

PART NUMBER	SUBJECT	REV NO.	DATE
131536	Beech Cargo Restraint System Kit No. 101-5040-1 P	1	January, 1981
90-590010-81	King KNS-81 Integrated Navigation System	3	September, 1985
90-590010-85	Ground Communications Electric Power Bus	7	September, 1994
101-590010-23	Woodward Electronic Propeller Synchrophaser (Type 1 System)	3	January, 1981
101-590010-29	Collins AP-105/FD-108Y/Z or AP-105/FD-109Y/Z Automatic Flight Control System	7	January, 1981
101-590010-35	King KFC-300 Automatic Flight Control System	4	December, 1982
101-590010-55	King KNC-610 Area Navigation System	8	April, 1984
*101-590010-65	Pilot's Operating Handbook Supplement for Aeroplanes Registered in Australia	G	April 21, 1982
101-590010-69	King KNR-665 or KNR-665A Area Navigation System	3	January, 1981
101-590010-71	Collins ANS-31/31A Area Navigation System or Collins NCS-31/31A Navigation Control System	8	December, 1983
101-590010-87	Camera Well Installation	2	January, 1981
101-590010-93	Sperry SPZ-200A/STARS IV D or SPZ-200A/SPI-80/81 or SPZ-200A/SPI-400/500 Series Automatic Flight Control System	7	May, 1982
101-590010-97	Bendix RNS 3500 Area Navigation System	4	December, 1983
101-590010-105	AirData AD611/D Area Navigation/Vertical Navigation System	6	December, 1983
101-590010-107	Brake Deice System	5	August, 1993
101-590010-109	Collins LRN-70 (ONTRAC IIIA)/LRN-80 (ONTRAC III) VLF/OMEGA Navigation System	7	December, 1983
*101-590010-113	King KFC-300 Automatic Flight Control System Modified for United Kingdom Registered Aircraft		Jan 10, 1977
101-590010-125	Flight With Cabin Entrance Door Removed	3	January, 1981
101-590010-137	Global GNS-500A VLF/OMEGA Navigation System, Series 2	5	December, 1983
101-590010-139	Lear-Siegler Model 23069-016 300-Amp Starter/Generator	2	December, 1984

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Log Of Supplements (Cont'd)
P/N 101-590010-147C
June, 2010

PART NUMBER	SUBJECT	REV NO.	DATE
101-590010-145	Collins FCS-80 Automatic Flight Control System, Category 1, and Collins EFIS-85 (EFIS-85A-1a or EFIS-85A-2a) Electronic Flight Instrument System	4	January, 1984
101-590010-165	Canadian Marconi CMA-734 (-003 with Mod 7) VLF/OMEGA Navigation System	1	May, 1981
101-590010-169	Airplanes Equipped With A Secondary Encoding Altimeter	3	January, 1994
101-590010-171	Collins LRN-85 VLF/OMEGA Navigation System (Dual System Optional)		February, 1981
*101-590010-173	High Flotation Landing Gear	3	April, 1999
101-590010-175	Foster AirData RNAV-612 Area Navigation System	1	December, 1983
101-590010-177	Global GNS-500A VLF/OMEGA Navigation System Series 3A, 3B, 3C, or 4 with Optional Vandling Navigation Data Bank	4	July, 1987
*101-590010-179	Auxiliary Electric Power Bus (Serial BB-882 only)		June, 1981
*101-590010-181	Auxiliary Electric Power Bus (Serials BB-900, BB-916 and BB-927 only)		July, 1981
*101-590010-185	Pilot's Operating Handbook Supplement for Aeroplanes Registered in the United Kingdom	C	August, 1988
101-590010-187	Auxiliary DC Bus	1	August, 1993
101-590010-193	Collins ANS-31C Area Navigation System	3	September, 1984
*101-590010-195	Auxiliary Electric Power Bus (Serials BB-1063, BB-1066 and BB-1077 only)	1	July, 1982
*101-590010-197	Auxiliary Electric Power Bus (Serials BB-1074 and BB-1075 only)		June, 1982
*101-590010-199	Delco Carousel IV E Inertial Navigation System (Serial BL-53 only)		August, 1982
*101-590010-201	Auxiliary DC Bus (Serial BB-1089 only)		August, 1982
*101-590010-203	Teledyne TDL-424 Loran Navigator (Serial BL-53 only)		August, 1982
101-590010-205	Sperry SPZ-4000 Automatic Flight Control System, Category I, and Sperry EDZ-600/ EDZ-800/ EDZ-601/ EDZ-801/ EDZ-603/ EDZ-803 Electronic Flight Instrument System (EFIS)	5	November, 1988
101-590010-211	Collins FMS-90 Flight Management System		March, 1983
101-590010-213	Collins FCS-65 Automatic Flight Control System, Category I, with Collins EFIS-85 (EFIS-85A-1a or EFIS-85-2a) Electronic Flight Instrument System or Collins EHIS-74 Electronic Horizontal Situation Indicator or Collins Electro-Mechanical Instruments.	7	April 26, 1995
*101-590010-219	Hydraulic Landing Gear (Serials BB-1158 and BB-1167 only)	1	March, 1984
*101-590010-221	McCauley Four-bladed Propeller System (BL-73 thru BL-112)		March, 1984
101-590010-225	King KNS-660 Flight Management System	2	September, 1986
101-590010-227	Foster LNS616A Long Range NAV System	3	April, 1985
101-590010-229	Lear-Siegler Model 23085-001 300-Amp Starter/Generator	1	August, 1993
*101-590010-231	Auxiliary Power Unit, Kit No. 101-5079	1	March, 1986
*101-590010-233	Beech Modified U.S. Forest Service Surveillance Airplane (Serial BL-124 only)		Sept, 1985

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PART NUMBER	SUBJECT	REV NO.	DATE
*101-590010-235	Beech Modified Chinese Dual Camera Well Aerial Surveillance Airplanes (Serials BB-1204 and BB-1205 only) and Chinese Flight Inspection Airplane (Serial BB-1206 only)	2	October, 1990
*101-590010-241	Royal Hong Kong Auxiliary Air Force Multi-Mission Surveillance Airplane, Serials BL-128 and BL-130	A5	June, 1993
101-590010-243	Sperry MLZ-910 Microwave Landing Receiver System		February, 1988
101-590010-245	King KNS-660 Flight Management System with King KLN-670 Global Positioning System		January, 1988
*101-590010-247	High Density Configuration (HDC) Airplanes Equipped with External Baggage Pod and Certified in accordance with FAR 135 Appendix A	2	August, 1990
101-590010-249	King Single or Dual KFC 400 Automatic Flight Control System, Category I, with Bendix EFS-10 Electronic Flight Instrument System (3-Tube EFIS or 4-Tube EFIS) or King Electro-Mechanical Instruments and optional King KNS 660 Flight Management System	4	February, 1995
101-590010-253	Foster LNS616B Long Range NAV System	1	November, 1991
*101-590010-257	Thailand Flight Inspection Airplane (S/N BB-1315 Only) and Thailand Aerial Surveillance Airplanes (BB-1436, BB-1441 & BB-1443 Only)	A2	April, 1992
*101-590010-259	Sperry SPZ-4000 Automatic Flight Control System, Category I, and Sperry EDZ-605 Electronic Flight Instrument System (EFIS) when modified in accordance with Sperry Service Bulletin 21-1989-183 dated 11-10-89 (Serial BB-1322 only)	1	February, 1990
101-590010-263	Collins FCS-65 Automatic Flight Control System, Category I, and Collins EFIS-85B(14) Electronic Flight Instrument System	4	January, 1994
*101-590010-265	Auxiliary Power Unit, Kit No. 101-5120		August, 1991
*101-590010-269	Japan Air System Trainer	3	November, 1993
101-590010-271	Collins FCS-65 Automatic Flight Control System, Category I, and Collins EFIS-85B(14) (5-Tube) Electronic Flight Instrument System (EFIS)	3	January, 1994
101-590010-313	Honeywell SPZ-4000 Automatic Flight Control System, Category I, and Honeywell EDZ-605 Electronic Flight Instrument System (EFIS) (S/N BB-1425 and after), (S/N BL-138 and after)		April, 1992
*101-590010-317	Turkish Ministry of Defense Aerial Surveillance Airplane (BB-1434 Only)		September, 1992
101-590010-353	Continuous Loop Fire and Overheat Detection System		June, 1995
101-590010-373	ARTEX ELT 110-4-002 With Remote Cockpit Switch (Raytheon Aircraft Kit P/N 101-3210 or 101-3214)		December, 1998
101-590010-391	Pitot Heat Annunciators		August, 2000
*101-590010-399	Bendix/King CAS 66A TCAS I System – C-12F (BL-77 and BL-78)		March, 2000
*101-590010-401	Fairchild A200S Solid-State Cockpit Voice Recorder – C-12F (BL-77 and BL-78)		March, 2000
*101-590010-403	Fairchild F1000 Series Solid-State Flight Data Recorder (SSFDR) – C-12F (BL-77 and BL-78)		March, 2000
*101-590010-411	Airplanes Registered In France		June, 2002
*101-590010-417	Wing Fatigue Limitations (BB-950 Only)		July, 2001

Log Of Supplements (Cont'd)
P/N 101-590010-147C
June, 2010

PART NUMBER	SUBJECT	REV NO.	DATE
*101-590010-441	Wing Fatigue Limitations (BB-1125 and BB-1189)		September, 2003
*101-590010-465	Wing Fatigue Limitations (BB-968, BB-1008, BB-1054, BB-1087, and BB-1178)		September, 2005
*101-50010-505	Wing Fatigue Limitations (BB-944 and BB-1048 Only)		June, 2010
101-590037-23	Decca Doppler 72 Navigation System (TANS Computer, Type 9447C)	1	January, 1981

NOTE: Supplements applicable to equipment other than that installed may, at the discretion of the owner/operator, be removed from the manual.

** Supplements marked with an asterisk will not be supplied with handbooks sold through Authorized Hawker Beechcraft Outlets due to their limited applicability. If a document is required for your airplane, please order the document through normal channels.*

BEECHCRAFT SUPER KING AIR
200C, 200CT, B200C, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
BEECH CARGO RESTRAINT SYSTEM KIT No. 101-5040-1 P

GENERAL

The information in this supplement is FAA Approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane is flown with the Beech Cargo Restraint System in use.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Beechcraft Drawing No. 101-5040, provided with Kit No. 101-5040-1P, contains all limitations required for use of the Beech Cargo Restraint System. Kit Drawing No. 101-5040 must be attached to this supplement any time the Beech Cargo Restraint System is used.

EMERGENCY PROCEDURES - No change

NORMAL PROCEDURES

Beech Drawing No. 101-5040, provided with Kit No. 101-5040-1P shall, when accompanied by this page become an integral part of this supplement and constitute the NORMAL PROCEDURES Section thereof.

PERFORMANCE - No change

WEIGHT AND BALANCE - No change

SYSTEMS DESCRIPTION - No change

HANDLING, SERVICING AND MAINTENANCE - No change

Approved: Donald H Pete

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: January, 1981
P/N 131536

For Training Purposes Only

BEECHCRAFT KING AIR C90 (LJ-668 thru LJ-1062, except LJ-670), C90A (LJ-1063 and after) E90, F90 SUPER KING AIR B200, B200C, B200T, and B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the KING KNS-81 INTEGRATED NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the King KNS-81 Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of this manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

For airplanes so equipped, the Pilot's Guide for the TACAN-compatible KNS-81 Integrated Navigation System, KPN 006-8369-00 5K dated 11/82 or later revision, is available from: King Radio Corporation, 400 N. Rogers Road, Olathe, KS 66062. It is the responsibility of the owner/operator to maintain the above pilot's guide in current status.

LIMITATIONS

1. The Area Navigation mode may not be used as a primary system under IFR conditions except on approved approach procedures, approved airways, and random area navigation routes when approved by Air Traffic Control.

2. The Area Navigation and VOR-PAR modes can only be used with colocated facilities (VOR and DME signals originate from the same geographical location).

3. VOR or VOR-PAR modes may be selected only when flying directly to or from a VORTAC facility. TAC mode may be selected only when flying to or from a VORTAC or TAC-only facility.

4. An area navigation installation located on the right instrument panel may be used for primary navigation only if a qualified pilot occupies the right seat.

5. In addition to this supplement, operation of the TACAN-compatible KNS-81 system must be in conformity with King Radio Pilot's Guide for the TACAN-Compatible KNS-81 Integrated Navigation System, KPN 006-8369-00 5K, dated 11/82, or later revision.

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude and angle of bank.

1. If NAV flag appears while in the Area Navigation mode, use CHK button to check for validity of raw DME and VOR data.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

FAA Approved
Revised: September, 1985
P/N 90-590010-81

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3. If NAV flag appears and/or DME information is lost during an approach, execute published missed approach and utilize another approved facility.

NORMAL PROCEDURES

For airplanes equipped with the optional TACAN-compatible KNS-81, refer to the Pilot's Guide (item 5 of Limitations) for procedures related to the TACAN mode of operation.

PREFLIGHT

AREA NAVIGATION FUNCTIONAL TEST

The following procedure applies only to airports equipped with, or in range of, a colocated VOR/DME STATION.

1. Place the KNS-81 in VOR mode.

2. Find and record the angle from the VOR station by centering the course deviation needle with the TO/FROM flag giving a "FROM" indication.

3. Program a waypoint radial angle equal to the OBS value determined in Step 2.

4. Program a waypoint distance equal to the indicated DME value.

5. Place the KNS-81 in RNAV.

The KNS-81 is operating properly if the distance to waypoint is 0 + 1.0 NM and the course deviation needle is within a dot of being centered.

PROGRAMMING

Pertinent information (waypoint number, station frequency, waypoint radial, and waypoint distance) can be entered into the memory. Programming may be completed prior to takeoff or during the flight. Any combination of navigational facilities (RNAV waypoint, VOR/DME, ILS) may be loaded into the computer, however, it is desirable that each facility be numbered and loaded in the sequence it is to be used.

RNAV WAYPOINTS

1. Turn the system on by rotating the ON/OFF switch clockwise.

2. Put waypoint 1 in the WPT window by turning the WPT knob. Turn the knob in either direction to get "1".

3. Select the waypoint 1 frequency using the data input controls which are the two concentric knobs on the right.

4. Select the waypoint 1 radial by depressing the DATA button. This will move the > <(caret) from FRQ to RAD. Select the new radial with the data input controls.

5. Select the waypoint 1 distance by again depressing the DATA button. This will move the > <from RAD to DST. Select the new distance with the data input controls.

6. This completes the programming for the first waypoint. Follow these procedures for all selected waypoints.

CONVENTIONAL VOR

1. The programming technique for conventional navigation directly toward or away from a VOR facility without a colocated DME is similar to that for RNAV waypoints. Putting the waypoint number and frequency

into the memory is accomplished in the same manner. The RAD and DST displays will display dashes during VOR and VOR-PAR operation.

ILS APPROACH (Front course and Back course)

1. Programming and ILS approach is accomplished in the same manner as programming conventional VOR.

MISSED APPROACH

1. If the published missed approach utilizes an RNAV waypoint or VOR facility, it may be entered into the memory any time prior to the approach. This is accomplished in the same manner set forth in CONVENTIONAL VOR and RNAV WAYPOINTS in this section.

INFLIGHT

1. Preset waypoints may be recalled from memory and put into active use as required. Turn the WPT knob as required to select the desired waypoint. The preset waypoint number, frequency, radial and distance will appear in their respective displays. The WPT display will blink to indicate that the waypoint displayed is other than the active waypoint.

2. Verify that the data is correct.

NOTE

Revisions to the waypoint data can be programmed at this time by entering the new waypoint parameters.

3. When return to the active waypoint is desired, press the RTN button. The active waypoint along with its data will be displayed.

4. When navigation to the displayed (blinking WPT) waypoint is desired, press the USE button. The WPT display will cease blinking and the displayed waypoint becomes the active waypoint.

5. The raw VOR & DME data can be checked at any time by pressing the CHK button. The radial from the VOR will be displayed above RAD and the DME distance will be displayed above DST.

RNAV OPERATION

If the system is receiving valid signals from a colocated VOR-DME facility, it will supply linear deviation information to the Horizontal Situation Indicator (or Course Deviation Indicator). Enroute (RNAV) sensitivity, available by turning the MODE selector knob until RNAV is displayed, provides a constant course width of ± 5 NM full scale.

Approach (RNAV-APR) sensitivity, available by turning the MODE selector knob until RNAV-APR is displayed, provides a constant course width of ± 1¼ NM full scale. Approach sensitivity should be selected just prior to final approach course interception. Time and distance to the waypoint, and computed groundspeed are displayed on the DME display.

CONVENTIONAL VOR OPERATION

VOR or VOR-PAR modes are selected by turning the MODE selector knob until VOR or VOR-PAR is displayed. In VOR mode the remote DME is automatically tuned when the KNS-81 is selected as the tuning source. Upon lock-on, distance, groundspeed and time to the VORTAC station will be displayed on the DME display. The HSI (CDI) will display conventional angular crosstrack deviation from the selected course (±10° full scale). In VOR-PAR mode, operation is identical to VOR except the HSI (CDI) will display crosstrack deviation of ±5 NM full scale from the selected course. Course width will be constant irrespective of distance from the VORTAC. The VOR-PAR mode is not available in the TACAN-compatible KNS-81.

Anytime the RAD button is engaged, the radial from the waypoint/station will be displayed on the DME

knots display along with an "F" on the DME time to station display.

NOTE

The RAD switch is not the momentary type, therefore, the switch must be pressed again for the normal DME information to be displayed.

ILS OPERATION

Whenever an ILS Frequency is put "IN USE" the mode display will remain the same (either VOR, VOR-PAR, RNAV, RNAV-APR displayed) but the RAD & DST displays will be blanked. Absence of the LOC/GS functions is annunciated by the NAV and GS flags in the HSI (CDI). Only angular deviation is provided in the ILS Mode.

RNAV APPROACH

The RNAV Approach (RNAV-APR) mode may be used for runway location (by placing a waypoint at the approach end of the runway) during an approach to an airport. Turn the MODE selector knob to select RNAV-APR. In RNAV-APR the deviation needle on the HSI (CDI) will display crosstrack deviation of ±1¼ NM full scale. All other aspects of the RNAV-APR mode are identical to the RNAV mode.

RANGE MONITOR (OPTIONAL)

The range monitor feature provides for the separation of the RNAV computed range to a waypoint from the steering guidance of the pilot's horizontal situation indicator. Pressing the SPLIT switch in the cluster of nav function switches will result in the RNAV computer (NAV-2) distance and bearing being presented on the pilot's horizontal situation indicator. NAV-1 steering will be retained. This is indicated by the background illumination of the DME-2, HSI-1 and RMI-2 legends on the SPLIT switch.

On an ILS approach, for example, it is desirable to know distance to the outer marker and then to the runway threshold. By selecting SPLIT and setting the appropriate NAV-2 frequency and waypoint parameters into the system, the distance to the desired fix will be continuously displayed while ILS steering guidance on the horizontal situation indicator will be conventional. The result is the ability to fly a localizer or full ILS steering situation while retaining RNAV computed distance to a selected fix.

CAUTION

It is imperative the NAV-2 switch be selected during RNAV operation or the NAV-1 or NAV-2 switch be selected during conventional VOR/ILS operation. If the SPLIT switch is activated, the range display will be based on the NAV-2 frequency and waypoint parameters, and the pilot's horizontal situation indicator will display conventional VOR steering based on the selected NAV-1 frequency.

PERFORMANCE - No change

WEIGHT AND BALANCE - No change

SYSTEM DESCRIPTION

The King KNS-81 is an integrated navigation system combining a 200 channel VOR/Localizer receiver, a 40 channel glideslope receiver and a digital RNAV computer with a capability of preselection and storage of 9, or on later models 10, VOR/LOC frequencies and equivalent sets of RNAV waypoint parameters. A DME System must be used in conjunction with the KNS-81.

The KNS-81 can be operated in any one of three basic modes: VOR, RNAV, or ILS. To change from one

mode to another the rotary MODE selector knob on the left side of the panel is rotated, except that the ILS Mode is entered automatically whenever an ILS frequency is channeled as the ACTIVE frequency. The display will annunciate the mode by lighting a message beside the WPT display, except in the ILS mode in which case the RAD & DST displays are blanked to denote the ILS mode. In addition to the standard VOR & RNAV enroute (RNAV) modes, the KNS-81 has a constant course width or parallel VOR mode (VOR-PAR) and an RNAV approach mode (RNAV-APR). The same rotary MODE selector knob is used to place the unit in either of these secondary modes. The VOR-PAR mode is not available in the TACAN-compatible KNS-81.

All waypoint information (station frequency, waypoint distance and waypoint radial) is entered with the increment/decrement rotary switch on the right side of the panel and displayed in their respective displays. The small knob affects the least significant digits while the large knob changes the most significant digits. The tenth's position of waypoint radial and distance can be changed by pulling the small knob to the out position. The type of data being selected is indicated by the illuminated carets (> <) located by either FRQ, RAD or DST. Frequency, radial or distance information for a waypoint can be selected sequentially by pressing the DATA push button. The increment/decrement switch changes only the information being displayed with the carets.

The KNS-81 can store frequency, radial and distance information for up to ten waypoints. The waypoint number of the data being displayed is located above the message WPT. This waypoint number is changed by rotating the WPT selector knob (small center knob) on the left side of the panel. If the waypoint in use is different from the displayed Waypoint (WPT blinking), pressing the USE button will cause the displayed WPT to become the waypoint in use.

DISPLAYS

1. FRQ, RAD, DST Display
   a. FRQ Display

      Displays frequency from 108.00 to 117.95 MHz in increments of .05 MHz. Least significant digit displays only zero or five.

   b. RAD Display

      Displays ground station radial on which waypoint is located from 0.0 to 359.9 degrees.

   c. DST Display

      Displays the offset distance of the waypoint from the ground station over a range of 0.0 to 199.9 NM.

2. VOR, PAR, RNAV, RNAV-APR Displays

   System mode lights

3. WPT Display

   Displays waypoint number of data being displayed.

4. Carets (> <) Display

   Indicates which waypoint data (FRQ, RAD or DST) the increment/decrement rotary switch will change.

5. DME Indicator (Remote)

   Displays NM to/from the waypoint/station, KT ground speed and MIN time to the waypoint/station. Also, the waypoint radial is displayed whenever the KNS-81 RAD button is pressed.

6. RMI Display (Optional)

   Displays the bearing to the waypoint/station.

CONTROLS

1. WPT/MODE Control

   Dual concentric knobs.

   a. The outer knob selects the MODE of unit operation. Turning the knob clockwise causes the mode to sequence thru VOR, VOR-PAR, RNAV, RNAV-APR and then back to the VOR mode.

   b. The center knob selects the WPT to be displayed. Turning the knob causes the displayed waypoint to increment by one thru the waypoint sequence of 1, 2, 8, 9, 1, or on later models 0, 1, ... 8, 9, 0.

2. USE Button

   Momentary pushbutton which, when pressed, causes the active waypoint to take on the same value as the displayed waypoint.

3. RTN Button

   Momentary pushbutton which, when pressed, causes the active waypoint to return to the display.

4. RAD Button

   Push-on, push-off button which, when pushed on, causes the radial from the waypoint and "F" to be displayed on the remote DME display.

5. CHK Button

   Momentary pushbutton which, when pressed, causes the raw data from the NAV Receiver and DME to be displayed. The radial from the VOR Ground Station will be displayed on the RAD display and the distance from the station will be displayed on the DST display. There is no effect on any other data output.

6. DATA Button

   Momentary pushbutton which, when pressed, causes the caret (> <) display to change from FRQ tc RAD to DST and back to FRQ.

7. ON/OFF/Pull ID Control

   Rotary switch/potentiometer which, when turned clockwise, applies power to the KNS-81 anc increases NAV audio level. The switch may be pulled out to hear VOR ident.

8. DATA INPUT Control

   Dual concentric knobs with the center knob having an "in" and "out" position.

   a. Frequency Data

      The outer knob varies the 1 MHz and 10 MHz digits and the center knob varies the frequency in .05 MHz increments which carry to/from the .1 MHz digit regardless of whether the switch is in its "in" or "out" position.

   b. Radial Data

      The outer knob varies the 10 degree digit with a carryover occuring from the tens to hundreds position. The center knob in the "in" position varies the 1 degree digit and in the "out" position varies the 0.1 degree digit.

   c. Distance Data

      The outer knob varies the 10 NM digit with a carryover occuring from the tens to hundreds place.

The center knob in the "in" position varies the 1 NM digit and in the "out" position varies the 0.1 NM digit.

HANDLING, SERVICE AND MAINTENANCE - No change

Approved:

Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

For Training Purposes Only

Beechcraft®

SUPER KING AIR® B200, B200C, B200T, B200CT
KING AIR® C90 (LJ-668 thru LJ-1062, except LJ-670), C90A (LJ-1063 and after), E90, F90 and B100 LANDPLANES

Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement for the GROUND COMMUNICATIONS ELECTRIC POWER BUS

This Supplement is applicable to the following manual(s):

P/N 90-590010-61, P/N 90-590010-87, P/N 90-590012-3, P/N 90-590024-5, P/N 90-590024-23, P/N 90-590024-35, P/N 90-590024-43, P/N 90-590024-61, P/N 90-590024-69, P/N 100-590038-1, P/N 101-590010-147, P/N 101-590010-327, P/N 101-590037-29, P/N 101-590037-57, P/N 109-590010-3 and P/N 109-590010-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Ground Communications Electric Power Bus in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Operation of the Ground Communications Electric Power Bus is restricted to ground operation only. No authorization is extended to operate this system in any flight configuration.

The Ground Communications Electric Power Bus must not be used when refueling the airplane.

EMERGENCY PROCEDURES

No change.

ABNORMAL PROCEDURES

No change.

NORMAL PROCEDURES

SYSTEM ACTIVATION:

1. Battery and Generator Switches - OFF
2. GND COMM PWR Switch - Press ON (GND COMM PWR annunciator illuminates.)

NOTE

Activation of this system by any means other than the prescribed procedure is not approved.

SYSTEM DEACTIVATION:

1. GND COMM PWR Switch - Press OFF (GND COMM PWR annunciator extinguishes)

CAUTION

Ensure the ground communications power switch is off before leaving the airplane so a drain on the battery will not occur.

PERFORMANCE

No change.

WEIGHT & BALANCE

No change.

SYSTEMS DESCRIPTION

The Ground Communications Electric Power Bus provides electric power directly from the battery (auxiliary battery, if installed) when selected by the pilot. Control of the system consists of a push on/push off solenoid-held annunciator switch located on the instrument panel. Circuit protection is provided by the GND COMM circuit breaker. Audio is provided in the headphones, with speaker audio optional. Subsequent activation of the battery switch (or appropriate generator switch with engine running) will result in automatic disconnection of the ground communications bus from the comm system; however, the normal method for deactivation of the system is accomplished by pressing the GND COMM PWR switch.

HANDLING, SERVICING & MAINTENANCE

No change.

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT B200, B200C, B200T, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

WOODWARD ELECTRONIC PROPELLER SYNCHROPHASER

(TYPE I SYSTEM)

The information in this document is FAA Approved material which, together with the basic Airplane Flight Manual is applicable and must be attached to the basic manual when the airplane is modified by the installation of the Woodward Electronic Propeller Synchrophaser in accordance with STC SA250CE.

The information in this document supersedes the basic manual only where covered in the items contained herein. For Limitations, Procedures, and Performance not contained in the supplement, consult the basic Airplane Flight Manual.

LIMITATIONS

When the *Type I System* is installed the following placard must be mounted on or near the synchrophaser control switch:

PROP SYN

+

OFF FOR T.O. LDG

NORMAL PROCEDURES

1. Synchronize the engines manually.
2. Position control switch to ON position.
3. If a change in rpm setting is desired, move both master (left) and slave propeller governor control levers together.
4. If synchronization is not maintained with the switch ON, indicating the actuator has reached the end of its travel, turn switch OFF and repeat procedures above. With the switch in the OFF position, the actuator is returned to the center of its travel.

PERFORMANCE

No change in the airplane performance results from the installation of the synchrophaser.

FUNCTIONAL TEST

The rpm range of the synchrophaser may be checked in cruise by slowly moving only the master propeller control toward both high and low rpm until propellers are no longer synchronized.

Note the range of rpm over which the slave engine remains synchronized with the master engine. This is the limited range provided for safety and is the maximum speed adjustment range beyond which the slave engine cannot be adjusted by the synchrophaser.

Approved: Donald H. Peters

for W. H. Schultz
Beech Aircraft Corporation
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BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
COLLINS AP-105/FD-108Y/Z AUTOMATIC FLIGHT CONTROL SYSTEM
or the
COLLINS AP-105/FD-109Y/Z AUTOMATIC FLIGHT CONTROL SYSTEM
Category I

GENERAL

The information in this supplement is FAA-approved material and must be attached to the *FAA Approved Airplane Flight Manual* when the airplane has been modified by installation of the Collins AP-105/FD-108Y/Z or AP-105/FD-109Y/Z Automatic Flight Control System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic *FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
2. Do not use autopilot under 200 feet above terrain.
3. Do not use autopilot or yaw damper during takeoff or landing.
4. Pilot must be seated at the controls with the seat belt fastened during autopilot operations.
5. Do not use propeller in the range of 1750-1850 rpm during coupled ILS approaches.
6. Do not extend landing gear above 15,000 feet with autopilot engaged.
7. Do not use CAA version of autopilot on the Super King Air 200T, 200CT, B200T, B200CT when the optional wing-tip fuel tanks are installed. The following placard must be mounted on the instrument panel or glareshield in pilot's view:

AUTOPILOT NOT
APPROVED FOR USE
WITH TIP TANKS

NOTE: The CAA version of autopilot has 522-2900-027 Autopilot Computer and 777-1406-027 Yaw Damper Computer.

EMERGENCY PROCEDURES

The autopilot can be disengaged by any of the following methods:

1. Press the AP/YD disconnect switch on the pilot's or copilot's control wheel.

2. Move the engage lever to the DISENGAGE position.

3. Engage the go-around mode

4. Pull out (off) the AP power circuit breaker.

5. Turn off the airplane master switch.

6. Turn off the avionics master switch.

The following conditions will cause the autopilot to disengage automatically:

1. Any interruption or failure of power.

2. Vertical gyro failure indication.

3. Flight control system power or circuit failure.

In the event of an engine failure:

1. Disengage the autopilot, retrim the airplane, and re-engage the autopilot. Maintain 120 KIAS for single engine approach speed until landing is assured.

Maximum altitude losses during malfunction tests were:

OPERATION CONFIGURATION	ALTITUDE LOSS
Climb	200 feet
Cruise	400 feet
Maneuvering	120 feet
Descent	430 feet
Approach/ILS Coupled	80 feet
Single Engine Approach/ILS Coupled	90 feet

NORMAL PROCEDURES

NOTE

The switches on the mode selector are the push-on, push-off type that activate solenoid held switches. When engaged, a green bar appears at the edge of the button. For operation at night the switches are illuminated by the OVERHEAD SUBPANEL AND CONSOLE light control. When the switch is on, the green bar will illuminate with an ON placarding visible.

The autopilot incorporates its own annunciator panel located just above the flight director display on the instrument panel. The modes and indications given on the annunciator panel are placarded on the face of the plastic lenses and illuminate when the respective conditions are indicated. Depending upon the type of flight director installation, dimming of the annunciator panel is provided for by a switch located adjacent to the annunciator panel; or by the airplane's navigation lights switch.

PREFLIGHT CHECK

The preflight check assures the pilot that the safety and failure warning features of the system are operating properly.

1. Turn on airplane power, an inverter and the avionics master switch. Check that the vertical gyro has erected (GYRO flag out of view), and that the gyrostabilized magnetic compass is slaved. Tune the navigation receiver to a local VOR frequency. Set the heading marker under the lubber line, and select HDG mode.

NOTE

The pressure of air flow that normally opposes movement of control surfaces is absent during any preflight check. It is possible to get a hardover control surface deflection if an autopilot command is allowed to remain active for any appreciable time. If it is desired to check operation of the TURN or PITCH controls, move them only as required to check control operation, and then return them to the center position.

2. Engage the autopilot. Check that the controls resist movement. Move the heading marker to either side of the lubber line. Observe that the command bars indicate a bank toward the new heading and the control wheel responds in the appropriate direction.

3. Press the VG FAST ERECT button. Observe that the GYRO and COMPUTER flags appear and the autopilot disengages. Release VG FAST ERECT button.

   FD-109Y/Z Only

   Press the TEST button on the flight director indicator. Observe that the GYRO and CMPTR flags appear and the attitude display indicates 20 degrees right bank and 10 degrees pitch up. Release the TEST button.

4. Pull the control wheel aft to mid-travel and engage the autopilot. Push forward on the control wheel and hold. The trim wheel should move in the nose-up direction, and after a few seconds, the ELEV TRIM annunciator should illuminate. Continue to hold the control wheel and press the control wheel trim switches to the NOSE DN position. Observe that the TRIM FAIL annunciator illuminates. Pull the control wheel aft and hold. The trim wheel should move to the nose-down direction, and again the ELEV TRIM annunciator should illuminate. Press the control wheel trim switches to the NOSE UP position and observe the TRIM FAIL annunciator illuminates.

5. Press the AP/YD disconnect button on the control wheel. Observe that the engage levers move to the DISENGAGE position and that the flight controls operate freely.

6. Set the course arrow to a VOR radial approximately 15 degrees either side of the course from the airplane position to the VOR station. Select NAV/LOC mode. Observe that the N/L ARM annunicator illuminates and the command bars still follow the heading marker for bank commands.

7. Move the course arrow to center the course deviation bar. Observe that the course arrow indicates the proper radial to the VOR station and that the to-from arrow points to the head of the course arrow.

8. Move the heading marker so that the command bars indicate a bank, select any lateral mode (HDG, NAV/LOC, APPR) on the mode selector and engage the autopilot. Press the GA button on the left power lever and observe the GA annunciator illuminates, the autopilot disengages and the command bars move to indicate a wings-level, pitch-up attitude.

ENGAGING AUTOPILOT

1. Place the ENGAGE-DISENGAGE switch lever on the autopilot controller in the ENGAGE position.

NOTE

The autopilot and flight director are coupled when both units are engaged. If the optional AP CPLD pushbutton switch is installed, then the autopilot is either coupled or uncoupled from the flight director. When coupled, the autopilot accepts guidance commands from the flight director. When uncoupled, the autopilot accepts pitch and roll commands from the vertical command and turn controls as selected by the pilot. The autopilot may be uncoupled by depressing the AP CPLD push button on the mode selector to the OFF position.

2. The autopilot may be engaged in any reasonable attitude and in either the coupled or uncoupled mode. The autopilot will smoothly acquire the command attitude. When uncoupled, the autopilot will maintain the bank (heading, if bank is less than 2 degrees) and pitch attitude at the time of engagement.

MANEUVERING

1. To change flight functions, press the desired mode button on the mode selector. The green bar will appear at the edge of the button and the autopilot annunciator lights on the instrument panel will illuminate, indicating the respective modes in operation.

2. With the AP CPLD mode switch off, use the TURN knob for turns.

3. In any function except "after glideslope capture", use the autopilot pitch control for climbing and descending. Movement of the pitch control determines a pitch rate that is proportional to knob displacement.

4. With the optional altitude preselect feature, the flight director gives commands to climb/descend to and level off at the altitude selected on the altitude alerter. A rate of climb/descent toward the selected altitude must be established manually or with the autopilot. Movement of the altitude alerter selector knob during the level-off maneuver results in desengagement of the mode. If a new altitude is selected while tracking a previously selected altitude, a command at a rate of approximately 700 FPM toward the newly selected altitude is given. If a different rate is desired, the altitude preselect mode must be disengaged, the desired rate established, and the mode reengaged.

5. If any vertical mode button (except altitude preselect before the selected altitude is captured) has been selected, it will automatically release when the AP pitch control is rotated.

6. When the HDG mode is selected, the autopilot will command the airplane to turn and maintain the heading set on the heading marker.

YAW DAMPER OPERATION

1. The rudder channel of the autopilot may be selected separately for yaw damping by moving the YD lever on the autopilot controller to the ENGAGE position.

2. To disengage the yaw damper, press the disconnect button on the pilot's or copilot's control wheel to the first detent or manually place the YD lever to the DISENGAGE position.

3. Refer to EMERGENCY PROCEDURES for other means of disconnecting the yaw damper.

DISCONNECTING AUTOPILOT

1. Press the release button on the outboard horn of either control wheel to the first detent or manually place the AP lever to the disengage position to disengage the autopilot for transition to manual control.

NOTE

After assuming manual control, fly the airplane using the same Course Indicator and Approach Horizon used to monitor autopilot operation prior to assuming manual control.

VOR FLYING

1. Tune NAV receiver to the appropriate frequency.

2. Set the desired course to or from the station on the pilot's Course Indicator by turning the course knob.

3. Set the desired beam intercept heading with the HDG knob. The intercept angle with respect to the radio beam may be any angle of 90 degrees or less. Depress the HDG button on the mode selector and establish the intercept heading.

4. Depress the NAV/LOC button on the mode selector. The system is then armed to capture the beam as indicated by the N/L ARM annunciator light on the instrument panel. At the point of capture the N/L CAP annunciator light will illuminate, indicating that the system has captured the selected course. Correction for proper tracking of the radial is automatically provided.

NOTE

Except as described below, do not select a different VOR frequency or course once a course and intercept have been programmed or capture achieved. To select a different course or VOR frequency, return the HDG mode, select the course and/or frequency and then return to the NAV/LOC mode.

5. Radio course may be changed over a VOR station when operating in NAV/LOC mode as long as the course change is not more than 10 degrees. If the course change is more than 10 degrees, HDG mode should be selected to establish a new intercept and then NAV/LOC mode reselected to set up a new capture.

6. When crossing a VOR station in the NAV/LOC mode, the flight director automatically switches to a dead reckoning (DR) mode until the airplane is positioned to receive reliable VOR signals. While in the DR mode, the flight director maintains the heading required to track over the station. On airplanes S/N BB-88, BB-202, BB-204 and after, BT-3 and after, BL-1 and after and BN-1 and after this mode is annunciated by flashing of the N/L CAP light.

7. The APPR mode should be used in lieu of the NAV/LOC mode for flying a VOR approach.

AUTOMATIC APPROACH - FRONT COURSE

NOTE

The localizer and glideslope are captured automatically and independently of each other on an ILS front course approach. Either one may capture before the other. The localizer is always captured from a selected heading, but the glideslope may be captured from any of the vertical modes and from above or below the glideslope.

1. To intercept the localizer beam, turn the NAV receiver to the correct ILS frequency. Set the course selector to the inbound runway heading and set the heading marker to the desired intercept angle. Program DH if radio altimeter is installed.

2. Press the APPR button on the mode selector. The N/L ARM and GS ARM annunciator lights will illuminate indicating the system is armed for localizer and glideslope capture. As the airplane approaches the localizer beam, the N/L CAP annunciator light will illuminate, indicating the system has captured the localizer course. At the point of glideslope intercept, the G/S CAP annunciator light will illuminate and all vertical modes preselected will be cleared, indicating the system is in glideslope operation.

3. The DH lights on the pilot's and copilot's instrument panels will illuminate when the airplane reaches the decision height previously selected on the radio altimeter, if installed.

4. FD-109Y/Z Only
   At approximately 200 feet above the ground, and if a radio altimeter has been installed and is operating, the runway symbol on the flight director will begin to rise toward the airplane symbol, while continuing to show localizer deviation.

5. Go-around mode may be activated by pressing the GA button on the left power lever, and may be actuated from any lateral mode (HDG, NAV/LOC, APPR) with the following results:

   a. Illuminates the GA light on the autopilot annunciator panel.

   b. Disengages the autopilot.

   c. Gives command presentation for wing level 7° nose up climb attitude.

NOTE

The heading marker may be preset to the go-around heading after the localizer is captured. After go-around airspeed and power settings are established, the autopilot may be engaged in the HDG mode. Select PITCH SYNC or any desired vertical mode (except ALT SEL) to clear the go-around mode, If PITCH SYNC is selected before HDG, command bars go out of view. To track outbound on the ILS course, use the NAV/LOC mode to prevent false glide-slope capture.

6. To assume manual control of the airplane for landing, press the disengage button on the control wheel.

BACK COURSE APPROACH

As in a front course approach, the localizer is captured automatically. The airplane should be maneuvered into the approach area by setting the heading marker and functioning in the HDG mode.

1. Tune the NAV receiver to localizer frequency.

2. Set course selector to front course inbound localizer bearing.

3. Set heading marker for desired intercept heading. (Intercept angle must be within 75° of the back course inbound localizer bearing).

4. Select APPR on the mode selector. When the airplane heading is within 75° of the back course inbound localizer bearing, the system will automatcally switch to back course operation and the BACK LOC annunciator light will illuminate.

5. Use the PITCH control on the autopilot controller to establish and maintain the desired rate of descent.

NOTE

The HDG mode should be used if intercept is within one mile of the runway due to the large radio deviations encountered when flying over the localizer transmitter.

6. The DH lights on the pilot's and copilot's instrument panels will illuminate when the airplane reaches the decision height previously selected on the radio altimeter, if installed.

7. If minimum altitude is attained before visual contact is achieved, arrest the rate of descent using the pitch control. The ALT HOLD mode may then be used to hold altitude until time to a missed approach has elapsed.

8. FD-109Y/Z Only
   The rising runway symbol on the flight director indicator operates as in Front Course Approach.

9. Go-around mode may be activated by pressing the GA button on the left power lever, with the results as specified in the AUTOMATIC APPROACH-FRONT COURSE PROCEDURE.

SPECIAL NOTES

1. The V bars on the flight director indicator will disappear to the top of the instrument when the optional AP CPLD button on the autopilot controller is pressed to the off position and all lateral modes are cleared.

2. When the autopilot engage lever is in the DISENGAGED position, the system may be used as a manual flight director system by selecting the desired mode of operation on the mode selector.

3. To synchronize the vertical command to airplane attitude, depress the PITCH SYNC button on the pilot's control wheel. When the autopilot is engaged, the vertical command remains synchronized after the first actuation of the PITCH SYNC switch.

4. Altitude hold information is displayed on the V-bars in flight director function by pushing the ALT button on the mode selector.

5. After selection of APPR mode, test functions for Nav, Marker Beacons and Radio Altimeter are disarmed to prevent operation.

6. The APPR mode should be used for flying all approaches (Front Course, Back Course, VOR and RNAV). Fly all approaches at 120 kts with 40% flap.

7. The autopilot may disengage during manual extension of the landing gear. Reengage if required after gear extension.

8. FD-108Z/FD-109Z Only
   To maintain a desired indicated airspeed, press the IAS button on the mode selector. To maintain a desired climb or descent, press the VS button on the mode selector.

PERFORMANCE - No Change

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
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BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

KING KFC 300 AUTOMATIC FLIGHT CONTROL SYSTEM (RNAV AND VNAV OPTIONAL) CATEGORY I

The information in this supplement is FAA-approved material and must be attached to the *FAA Approved Airplane Flight Manual* when the airplane has been modified by installation of the King KFC 300 Automatic Flight Control System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic *FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. Do not use autopilot on the Super King Air 200T, 200CT, B200T and B200CT when the optional wing-tip fuel tanks are installed.
2. Do not use autopilot at gross weights in excess of 12,500 lbs.
3. During autopilot operations, pilot must be seated at the controls with seat belt fastened.
4. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
5. Do not use autopilot under 200 feet above terrain.
6. Do not use autopilot or yaw damper during takeoff or landing.
7. Do not use propeller in the range of 1750-1850 rpm during coupled ILS approach.
8. The Vertical Navigation (VNAV) function does not adversely affect any airplane system. Flight operations must not be predicated on its use as the primary source of vertical guidance.
9. Reengagement of autopilot following use of the control-wheel-steering switch (placarded PITCH SYNC & CWS) shall be made at a rate of climb or descent not to exceed 500 feet per minute, if altitude hold is engaged.

NORMAL PROCEDURES

NOTE

The autopilot incorporates its own annunciator panel located just above the flight director display on the instrument panel. The modes and indications given on the annunciator panel are placarded on the face of the plastic lenses and illuminate when the respective conditions are indicated. Located on the left side of the autopilot annunciator panel is a light sensitive dimmer switch to control light intensity for night operation. The switches on the mode selector are push-on, push-off type. When engaged, the corresponding flight director annunciator light illuminates. For operation at night, the OVERHEAD SUBPANEL AND CONSOLE light control for the backlighting of the switches is located on the overhead console.

PREFLIGHT

Neither the autopilot nor the flight director may be turned on if the attitude flag is visible on the flight command indicator or if the gyros are not up to operating speed.

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1. Check that all circuit breakers for the autopilot are in.

2. Turn the battery, inverter, and avionics switches on and engage the flight director to bring the command bars into view.

3. Engage the autopilot and yaw damper. (The autopilot will not engage when the flight director is inoperative.) Check that the system can be overpowered in all three axes.

4. Press and hold the preflight test button located on the lower left corner of the flight command indicator. This will activate a self test cycle provided to preflight the autopilot and flight director system. Note the following sequence for the test cycle:

   a. All autopilot and flight director warning and mode lights should illuminate and remain lighted until the test has been completed.

   b. A simulated climbing right turn of 10° pitch up and 10° right roll will appear on the attitude display.

   c. The command bars will remain centered with the airplane reference symbol until the flight director, autopilot computer, and servos check valid.

   d. All three servo actuator monitors will trip.

   e. The autopilot and yaw damper will disengage to demonstrate proper computer monitor operation.

   f. The command bars will come into exact alignment with the original horizon display after the flight director computer, autopilot computer, and servos check valid.

5. Release the preflight test button and check to see that warning and mode lights extinguish. A warning light illuminated after the test button is released indicates malfunction for that mode.

6. Check for proper trim action as follows:

   a. Position the command bars approximately 5° above the airplane reference symbol with the vertical trim switch (on mode controller).

   b. Engage the autopilot and yaw damper. Trim should run in nose up direction 3 to 4 seconds after autopilot engagement.

   c. Depress and hold the TRIM TEST switch (on pedestal). The trim should cease operation and the A/P TRIM FAIL light should illuminate in 15 to 20 seconds.

   d. Release and depress the TRIM TEST switch. Actuate the control wheel trim switches to the nose down direction. The trim should run in the nose down direction and the A/P TRIM FAIL light should illuminate immediately.

   e. Repeat the procedures with the command bars positioned approximately 5° below the aircraft symbol. Trim operation should be reversed from the previous as specified.

7. Disengage the autopilot by depressing the AP/YD DISC switch on the pilot's control wheel.

INFLIGHT

ENGAGING THE AUTOPILOT

1. Engage the flight director by depressing the FLT DIR switch on the mode controller or by depressing the PITCH SYNC & CWS switch on the pilot's control wheel. The existing pitch attitude and heading information will be retained on the flight command indicator command bars as they are brought into view.

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2. Engage the autopilot. The autopilot action is always in response to, and consistent with, flight director commands. When engaged by the solenoid-held toggle switch on the mode controller, the autopilot will respond to any operating mode through a fader circuit, which allows engagement into an unsatisfied flight director command without an abrupt control transient.

3. Depressing the control wheel steering (PITCH SYNC & CWS) button located on the horn of the pilot's control wheel, allows the pilot to momentarily revert to manual control in pitch and roll (yaw damper stays engaged), while retaining his previous mode program, and conveniently resuming that profile upon disengagement.

4. The autopilot, together with the yaw damper, provides three-axis rate stabilization, automatic turn coordination, and automatic elevator trim as well as automatic response to all flight director modes.

HEADING CONTROL OPERATION

The flight control system is electrically connected to the directional gyro for heading hold information whenever the system is in basic flight director mode. Heading hold is automatically disengaged when an incompatible lateral mode is engaged or when the control wheel PITCH SYNC & CWS switch is depressed and held.

Pressing the HDG SEL mode button automatically causes the airplane to execute a pre-selected heading change as set on the pictorial navigation indicator with either the HDG control knob on the indicator or the HDG SEL spring loaded knob on the mode controller. Heading changes using HDG SEL mode will bank the airplane two degrees for every degree of heading change selected up to a maximum bank angle of 25 degrees.

VERTICAL CONTROL OPERATION

1. Vertical Trim:

   Operation of the vertical trim switch (on mode controller), provides a convenient means of adjusting the reference parameter of all the vertical modes except glideslope and vertical navigation. This permits the pilot to change his vertical reference without disengaging and re-engaging modes.

2. Altitude Hold:

   The altitude hold mode may be engaged by pressing the ALT mode switch on the mode controller. The airplane will maintain the pressure altitude existing at the time the switch is depressed. Altitude hold may be engaged at any rate-of-climb or descent. For best performance, engagement should be made after establishing stabilized operation in any other vertical mode. Altitude hold is automatically disengaged when any other vertical mode is selected. The vertical trim switch may be used to trim the referenced altitude up or down at approximately 500 fpm.

3. Altitude Select:

   This mode allows the pilot to select, arm, and upon approaching the pre-set altitude, obtain an automatic visual pitch command to capture and hold the preselected altitude. Prior to selecting the function, the pilot must set the desired altitude (by means of rotary control knobs) into the selected altitude readout of the vertical navigation computer. The ALT ARM button on the vertical navigation computer may be depressed any time during climb or descent, to arm the altitude capture circuitry and the ALT ARM annunciator will illuminate. As the airplane approaches the selected altitude, the ALT ARM annunciator will extinguish, and as the airplane passes through the selected altitude, the altitude hold will automatically engage and the ALT HOLD annunciator light will illuminate.

4. Indicated Airspeed Hold:

   Engaging the indicated airspeed hold mode will introduce a computed, visually-displayed pitch command to maintain the reference airspeed. The mode is utilized by maneuvering the airplane, and setting engine power, to attain the desired speed in climb, descent, or level flight and then depressing the IAS button. The reference airspeed may be adjusted at a rate of approximately one knot per second

by operation of the vertical trim switch on the mode controller.

5. Speed Profile:

   Engaging the speed profile mode will introduce a visually displayed pitch command on the flight command indicator which varies the indicated climb or descent speed as a function of altitude. During climb, airspeed is decreased at the rate of 1.75 knots per 1000 feet. The proper initial airspeed must be set up by the pilot and correct power settings maintained before depressing the SPD PRF button. After engagement, airspeed reference may be trimmed using the vertical trim switch.

6. Vertical Navigation:

   The vertical navigation computer provides a computed pitch command, which is displayed on the flight command indicator, to capture and maintain a vertical track angle in ascent or descent to a selected waypoint or VORTAC facility.

7. Altitude Alerting:

   Two altitude alert lights, one on the vertical navigation computer and one on the servoed altimeter, provide altitude alerting with the vertical navigation computer. When the airplane climbs, or descends, to within 1000 feet of the selected altitude, the alert lights illuminate and remain illuminated through a 500 foot altitude warning band. At 500 feet from the selected altitude, the lights extinguish. When the airplane reaches the selected altitude, a two-second aural tone indicates the desired flight altitude has been achieved. The two-second tone is also heard when the lights first illuminate upon penetration of a warning band.

   An MDA toggle switch on the vertical navigation computer allows the pilot to activate the MDA annunciator and warning horn for warning when the minumum descent altitude is reached.

8. Go-around:

   Engagement of the go-around mode using the go-around button on the left power lever will introduce a wings level 7°, nose-up display on the flight command indicator command bars. Operation of go-around cancels all other vertical modes and also disengages the autopilot, if the autopilot has been engaged. The autopilot may be re-engaged in the go-around mode. The go-around mode may be used as a take-off pitch reference, if desired, by engaging go-around mode on the runway. Momentary operation of the trim switch disengages the go-around mode.

FLYING RADIO FACILITIES

VOR PROCEDURES

1. Tune the NAV receiver to the appropriate frequency. (RNAV mode selector in VOR-DME.)

2. Set the desired course to or from the station on the pilot's course indicator.

3. Set the desired intercept heading. (Heading hold or HDG SEL may be used)

   NOTE

   The intercept angle with respect to the VOR radial selected may be any angle up to 90°.

4. Arm the navigation mode by depressing the NAV switch on the mode controller. The NAV ARM light on the flight director annunciator panel illuminates indicating that the system is armed to capture the selected radial. At the point of capture, the NAV ARM light on the annunciator extinguishes and the NAV CPLD annunciator light illuminates, indicating the system has captured the selected course.

5. The selected track may be changed while in the tracking mode, by setting a new course using the

COURSE knob on the pictorial navigation indicator. Course changes made at less than 4° per second with the COURSE knob are acquired without leaving the tracking mode, however if the COURSE knob is moved at a rate exceeding 4° per second, a pre-programmed intercept angle of 45° is automatically engaged without having to return to the heading mode. When over the navigational facility, the course selection should be made at the change from "to" to "from", for best results.

AREA NAVIGATION ENROUTE

1. Tune the NAV and DME receivers supplying information to the area navigation computer to the radio facility (VORTAC) being used. The signal must be valid.

2. Set the area navigation bearing and distance to establish the desired waypoint.

3. Set the area navigation mode switch to RNAV position.

4. Set the desired course using the COURSE knob on the pilot's pictorial navigation indicator.

5. Set the desired intercept heading (heading hold or HDG SEL may be used).

NOTE

The intercept angle, relative to the RNAV radial, may be any angle of 90° or less.

6. Arm the navigation mode by depressing the NAV switch on the mode controller. The NAV ARM light on the flight director annunciator panel illuminates indicating that the system is armed to capture the selected radial. At the point of capture, the NAV ARM light on the annunciator will extinguish and the NAV CPLD annunciator light illuminates, indicating the system has captured the selected course.

VERTICAL NAVIGATION (VNAV)

Vertical navigation provides a computed pitch command, displayed on the flight command indicator, to capture and maintain a vertical track angle in ascent or descent to an RNAV waypoint or VORTAC facility. The following prerequisites must be fulfilled prior to flight director/autopilot coupling to the vertical navigation system:

1. Tune the NAV and DME receivers supplying information to the RNAV computer to the radio facility (VORTAC) being used. The signal must be valid.

2. The desired course to a selected waypoint or VORTAC facility must be set on the pictorial navigation indicator and a "TO" flag must be in view.

3. The RNAV computer mode switch must be placed in the RNAV or APPR position.

4. The selected climb or descent track angle must be less than 55 degrees.

Programming the vertical navigation computer:

1. Preset the desired altitude in the selected altitude window.

2. Set the altitude of the VORTAC facility being used, using the VTAC ALT - 1000 FT tab.

3. If altitude acquisition is desired prior to reaching the selected waypoint or VORTAC facility, program the mileage offset (0 to 30 miles) using the DIST BIAS MILES knob. (Bias is the distance short of the selected waypoint.)

If the NAV receiver, RNAV computer, servo altimeter and DME are valid, the vertical track angle will be indicated on the right display scale of the flight command indicator in degrees of angle, to a maximum of ± 5°. As the airplane flies toward the waypoint or VORTAC facility at a constant altitude, the displayed

vertical track angle will slowly increase. When the vertical track angle has reached a value desired by the pilot, the pilot must manually couple VNAV by depressing the VNAV CPLD button on the VNAV computer. The vertical track angle displayed upon engagement becomes the reference flight path angle and the display pointer then becomes a deviation display above or below the selected flight path. The maximum scale deflection in the VNAV coupled mode is ± 250 feet. Selection of the VNAV CPLD mode automatically activates ALT ARM to capture the selected altitude.

4. **On appropriately equipped airplanes the HDG SEL mode may be used with VNAV coupled to deviate from the selected course momentarily, for radar vectors, traffic, etc. HDG SEL turns of more than 90° are not recommended.**

NOTE

VNAV accuracy to make good the selected VNAV waypoint cannot be guaranteed with tracking offsets of more than 2 nautical miles on the selected RNAV course.

APPROACH

1. VOR

VOR approaches may be made by coupling VOR in the approach mode. This gives proper responses for a close in non-precision approach.

2. ILS FRONT COURSE

a. Tune the NAV receiver to the correct ILS frequency, set the course selector to the inbound runway heading, set the heading bug to the desired intercept angle, and set the decision height on the radio altimeter.

NOTE

With both NAV receivers tuned to the same ILS facility, if the number 2 NAV receiver deviates more than 35 millivolts from the number 1 NAV receiver on either localizer or glideslope, the appropriate APPR CPLD or GS CPLD annunciator will flash indicating monitor limits have been exceeded.

Localizer and glideslope are captured automatically on front course. The localizer must be captured before glideslope capture is possible.

If the airplane heading is within 90° of the back course heading, the REV LOC annunciator will illuminate.

b. Engage HDG mode and arm the APPR mode. The APPR ARM annunciator will illuminate, indicating the system is armed to capture the localizer beam. As the airplane nears the beam, the APPR CPLD annunciator will illuminate and the system will intercept the localizer. At the point of glide path intercept, the GS CPLD annunciator will illuminate and all vertical modes will be disengaged, indicating the system is locked on to the glide slope.

c. The decision height light on the flight command indicator will illuminate when the aircraft reaches the decision height previously selected by the pilot on the radio altimeter.

d. To assume manual control of the airplane for landing, depress the autopilot disengage switch on the pilot's control wheel.

e. Disengage the autopilot at no less than 200 feet above the ground prior to manually landing the aircraft.

f. Go-around mode may be selected by pressing the go-around button on the left power lever any time the pilot needs to execute a missed approach. The autopilot will be disengaged and the flight

command indicator will command a 7° nose-up wings level attitude.

3. ILS BACK COURSE

a. Tune the NAV receiver to the correct ILS frequency, set the course selector to the inbound front course runway heading, set the heading indicator to establish the desired intercept angle and set decision height on the radio altimeter.

NOTE

With both NAV receivers tuned to the same ILS facility, if the number 2 NAV receiver deviates more than 35 millivolts from the number 1 NAV receiver on localizer, the appropriate APPR CPLD annunciator will flash indicating monitor limits have been exceeded.

b. Engage HDG mode and arm the APPR mode. The APPR ARM annunciator will illuminate, indicating the system is armed to capture the localizer beam. As the aircraft nears the beam, the APPR CPLD annunciator will illuminate and the system will intercept the localizer. If the airplane heading is within 90° of the back course heading, the REV LOC annunciator will light.

c. Indicated airspeed hold or pitch attitude hold may be used to establish a descent while on reverse localizer.

d. Disengage the autopilot at no less than 200 feet above the ground prior to manually landing the aircraft.

e. Go around operation is the same as for front course operation.

4. RNAV APPROACH

a. Tune the NAV receiver and DME to the appropriate VORTAC frequency.

b. Set RNAV bearing and distance as given on the navigation charts for RNAV approaches. Set RNAV to APPR mode when within ten miles of the selected waypoint.

c. Set vertical navigation to give minimum descent altitude and bias as desired. Set the MDA switch to the MDA WARN position.

d. Set the required course on the pictorial navigation indicator and establish an intercept angle to the inbound radial. Arm the approach mode.

e. After RNAV approach is coupled, observe the vertical navigation deviation on the flight command indicator and depress the VNAV CPLD button when desired descent angle is displayed.

f. When the MDA annunciator on the flight director indicator illuminates, a go-around should be executed unless the pilot has the field in sight.

DISENGAGING THE AUTOPILOT

1. Monitor the controls prior to disengagement. Under normal operating conditions the automatic pitch trim will have the aircraft properly trimmed in the pitch axis at the pitch attitude existing when the system is disengaged.

2. Disengage the system by pressing the pilot's or copilot quick disconnect switch, pilot/copilot trim

switches, or returning the autopilot engage switch to OFF. The flight director may be turned off, then on, which will also disengage the autopilot.

SPECIAL NOTES

1. The V-bars on the flight director indicator will disappear to the top of the instrument when no flight director modes are engaged.
2. The V-bars must be in view before the autopilot can be engaged.
3. When the autopilot is not engaged, the system may be used as a manual flight director system.

EMERGENCY PROCEDURES

The autopilot can be disengaged by any of the following methods:

1. Press the A/P disconnect switch on the pilot's or copilot's control wheel.
2. Move the on-off switch to the off position.
3. Engage the go-around mode. (Yaw damper will remain engaged.)
4. Pull the flight director/autopilot circuit breaker out (off).
5. Turn off the aircraft master switch.
6. Turn off the avionics master switch.
7. Any interruption or failure of power.
8. Operate main trim switches UP or DN. (Yaw damper will remain engaged.)

The following conditions will cause the autopilot to disengage automatically:

1. Vertical gyro failure indication.
2. Flight control system power or circuit failure.

The following will cause a servo to disengage:

1. Rapidly overpowering any servo will cause disengagement of only that servo through operation of the servo monitor. The servo may be re-engaged by turning off the autopilot and waiting for the monitor light to extinguish before re-engaging the autopilot.
2. A hardover failure in any of the primary servos will result in only that servo being automatically disengaged.

In the event of an engine failure:

1. Disengage the autopilot, retrim the aircraft, and re-engage the autopilot. Maintain at least 120 knots for single-engine approach.

Maximum altitude losses during malfunction tests were:

CONFIGURATION	ALTITUDE LOSS
Climb	Negative loss (40 ft gain)
Cruise	170 ft
Maneuvering	50 ft
Descent	420 ft
Approach/ILS Coupled	50 ft
Single Engine Approach/ILS Coupled	80 ft

Approved: Donald H Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT KING AIR C90 (LJ-502 thru LJ-1062), C90A (LJ-1063 and after), E90, F90, A100, B100, and SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T & B200CT LANDPLANES PILOT'S OPERATING HANDBOOK and FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the KING KNC-610 AREA NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the King KNC-610 Area Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. This system may not be used as a primary system under IFR conditions except on approved approach procedures, approved area navigation airways, and random area navigation routes when approved by Air Traffic Control.

2. This system can only be used with co-located facilities. (VOR and DME signals originate from the same geographical location).

3. An area navigation installation located on the right instrument panel may be used for primary navigation only if a qualified pilot occupies the right seat.

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude, and angle of bank.

1. VOR or Distance flag appears while in RNAV mode:
   a. Selected Frequency - CHECK FOR CORRECT FREQUENCY
   b. VOR or Distance flag intermittent or lost - UTILIZE OTHER NAV EQUIPMENT AS REQUIRED
2. VOR or Distance flag appears while in APPR mode:
   a. If flag appears while on an approach, execute published missed approach and utilize other approved facility.

NORMAL PROCEDURES

1. VHF NAV - ON
2. DME - ON
3. Mode Selector - SELECT VOR/DME RNAV, or APPR (whichever is appropriate)

NOTE

Assure Mode Selector switch is in VOR/DME position for best Flight Director performance in an ILS approach mode.

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P/N 101-590010-55

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4. NAV Frequency - SET
5. DME Frequency - SET
6. Waypoint Bearing - SET WAYPOINT RADIAL FROM VORTAC
7. Waypoint Distance - SET WAYPOINT DISTANCE FROM VORTAC
8. OBS Control - DESIRED MAGNETIC COURSE
9. Self-Test - PRESS BUTTON (must have VOR reception)

PERFORMANCE - No change

WEIGHT AND BALANCE - No change

SYSTEMS DESCRIPTION

The RNAV function of the King KNC-610 system performs a vector computation that results in a visual display of the magnetic bearing and distance to or from a selected waypoint. The computer, in effect, moves the selected reference facility (VORTAC or colocated VOR/DME facility) to a different location called a waypoint. The waypoint, which is expressed in terms of nautical miles along a selected radial from the VORTAC, is programmed into the system by the pilot.

Steering guidance is presented as a left/right display on the Horizontal Situation Indicator (HSI). The display format differs from the conventional VOR course deviation of ± 10 degrees called "angular course deviation". Rather, course deviation is presented in nautical miles from the course centerline. This feature, referred to as "linear course deviation", provides for a constant course width irrespective of the distance to the waypoint. Two levels of sensitivity are available for area navigation. The enroute sensitivity, available when the Mode Selector switch is positioned to RNAV, provides a constant course width of ± 5 nautical miles. Approach sensitivity, available when the Mode Selector switch is positioned to APPR, provides a constant course width of ± 1.25 nautical miles. Approach sensitivity should be used when within ten nautical miles of the terminal waypoint.

The Range Monitor feature (optional) provides for the separation of the RNAV computed range to a waypoint from the steering guidance of the pilot's Horizontal Situation Indicator. Selecting the Range Monitor switch to the RANGE MONITOR position will connect the RNAV computer to the NAV 2 receiver. The pilot's Horizontal Situation Indicator will be retained on the NAV 1 receiver.

On an ILS approach, for example, it is desirable to know distance to the outer marker and then to the runway threshold. By selecting RANGE MONITOR and setting the appropriate NAV 2 frequency and waypoint parameters into the system, the distance to the desired fix will be continuously displayed while ILS steering guidance on the Horizontal Situation Indicator will be conventional. The result is the ability to fly a localizer or full ILS steering situation while retaining RNAV computed distance to a selected fix.

CAUTION

It is imperative the Range Monitor switch be placed in the NORMAL position during RNAV operation. If left in the RANGE MONITOR position, the range display will be based on the NAV 2 frequency and waypoint parameters, and the pilot's Horizontal Situation Indicator will display conventional VOR steering based on the selected NAV 1 frequency.

HANDLING, SERVICING AND MAINTENANCE - No change

Approved: Donald W. Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: April, 1984
P/N 101-590010-55

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BEECHCRAFT LANDPLANE
MODELS 200/200C/B200/B200C
PILOT'S OPERATING HANDBOOK SUPPLEMENT
FOR
AEROPLANES REGISTERED IN AUSTRALIA

GENERAL

This document is to be attached to the Pilot's Operating Handbook for aeroplanes on the Australian Register. The contents are in addition to, or override, the contents of the Pilot's Operating Handbook and FAA Approved Flight Manual. Conversion charts for feet-metres, pounds-kilograms, inches-millimetres, and U.S. gallons-liters are shown.

DEPARTMENT OF TRANSPORT

(D.O.T.)

P/N 101-590010-65
AMENDMENT G: 21 APRIL, 1982

For Training Purposes Only

APPROVED
FLIGHT MANUAL
FOR
BEECH MODELS 200/200C/B200/B200C

DEPARTMENT OF TRANSPORT

P/N 101-590010-65
AMENDMENT G: 21 APRIL, 1982

For Training Purposes Only

FLIGHT MANUAL APPROVAL

Nationality and Registration Marks	
Aeroplane Serial Number	VH –
Manufacturer	BEECH AIRCRAFT CORPORATION
Designation of Aeroplane	Beech Model
Certification Category or Categories	Normal

This Flight Manual has been approved by the Secretary to the Department of Transport and is the Flight Manual referred to in Certificate of Airworthiness Number.

Date: 25th June 1981

for Secretary, D.O.T.

Any person finding this Manual should return it to the nearest Regional Office of the Department of Transport, Air Transport Group.

INTRODUCTION

This Flight Manual applies only to the particular aeroplane identified by registration marking and serial number on the cover page. It is the responsibility of the pilot in command to be familiar with the contents of this manual and to comply with all directions contained herein relating to the operation of the aeroplane.

Unless exempt under the provisions of Air Navigation Regulation 113-(3), it is required that the Flight Manual be carried in the aeroplane on all flights.

This Flight Manual shall be read in conjunction with the Pilot's Operating Handbook. Where conflict may arise between these documents, the data contained in the Australian Flight Manual shall take precedence over that in the Pilot's Operating Handbook.

The Federal Aviation Administration operational regulations are not applicable in Australia, and where reference is made in the Pilot's Operating Handbook to those regulations, the pilot should refer to the appropriate Australian requirements.

Amendments will be issued by the Secretary as necessary and will take the form of replacement pages with changes to the text indicated by a vertical line in the margin together with the amendment number. It is the owner's responsibility to incorporate in this manual all such amendments and to enter the date of incorporation and his signature on the appropriate amendment record sheet. No other entries or endorsements may be made to this Flight Manual.

The aeroplane has been certificated on the basis of the equipment fitted at the time of certification. Changes in equipment are subject to approval by the Secretary.

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GENERAL AMENDMENT RECORD SHEET

Amendment	Paragraph(s) Affected	Signature	Date of Incorporation
A	3.9 Pages 23, 24, 24A		July 15, 1975
B	2.7		Sep. 30, 1977
C	1.2, 2.1, 3.9		Oct. 25, 1977
D	Introduction (i), (ii), (iii), (iv), (vii), 3.9, 4.4, 37A, 37I, 39.		Feb. 14, 1978
E	(i) Pages 37A, 37J		Oct. 23, 1979
F	ALL (REISSUE OF MANUAL)		June 24, 1981
G	(pages affected) Cover Sheet, Title Page iii, 2A, 2N, 2O, 2P		April 21, 1982

Incorporation of a General Amendment must be certified by inserting the date of incorporation and signature in the appropriate columns. All amendments must be embodied consecutively. This page will be reissued with each General Amendment, and previous copies should be retained in the Flight Manual to serve as a record of amendments issued. Superseded Flight Manual pages should be removed and destroyed.

Date: 21 April 1982

for Secretary, D.O.T.

PARTICULAR AMENDMENT RECORD SHEET

Amendment	Paragraph(s) Affected	Signature	Date of Incorporation

Incorporation of a Particular Amendment must be certified by inserting the date of incorporation and signature in the appropriate columns. All amendments must be embodied consecutively. This page will be reissued with each Particular Amendment, and previous copies should be retained in the Flight Manual to serve as a record of amendments issued. Superseded Flight Manual pages should be removed and destroyed.

Date: _____

_____
for Secretary, D.O.T.

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(iv) Particular Amendment Record Sheet
(v) List of Contents
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P/N 101-590010-147 B200/B200C

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SECTION 2 - LIMITATIONS

2.1 APPROVED FUEL GRADES AND ADDITIVES

The use of wide cut type fuel is restricted. Prior approval must be obtained from the Secretary, Department of Transport. Refer A.N.O. 108.46.

2.2 ALTITUDE LIMITATIONS

Maximum Operating Altitude.....35,000 feet

Note: Maximum operating altitude of 35,000 feet subject to compliance with oxygen system requirements of A.N.O. 108.26.

2.3 REQUIRED EQUIPMENT FOR VARIOUS CONDITIONS OF FLIGHT.

Information given in P.O.H. Section II, Required Equipment for Various Conditions, is not applicable. Refer to relevant sections of A.N.O. Part 20 and Section 5 of this supplement.

For other information related to aeroplane limitations refer to the Pilots Operating Handbook.

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SECTION 4 - PERFORMANCE

4.1 PERFORMANCE CHARTS

The charts in this Section contain data establishing weight limitations for takeoff and landing in accordance with Air Navigation Orders Section 20.7.4.

Extrapolation outside the boundaries of the performance charts is not permitted. When the outside air temperature and/or pressure height is below the lowest range scheduled on the charts, the aeroplane performance shall be assumed to be no better than that appropriate to this lowest range. The performance information is not valid when the outside air temperature and/or pressure height exceed the maximum values for which this information is scheduled.

4.2 TAKE-OFF

The weight of the aeroplane for takeoff shall not exceed the lesser of:

(i) the maximum takeoff weight specified in Section 2 of this Manual; and
(ii) the weight for takeoff determined from the take-off weight chart of this Section.

The take-off chart is based on factored take-off distances from rest to a height of 50 feet with the engines operating at take-off power. The surface corrections on the chart are based on standard factors related to strips with a firm surface. Soft ground and unusually long and/or wet grass will increase the take-off distance over that scheduled and the pitot should therefore ensure that adequate strip length is available to cover these conditions.

The technique used in establishing the take-off distance is such that the aeroplane is held on or close to the ground until the appropriate take-off safety speed is approached, and the climb away then commenced so that this speed is achieved and maintained at or before the 50 foot height point.

The required 6 percent climb gradient for two engine operation and the required 1 percent climb gradient for single engine operation can be achieved at the maximum take-off weight specified in Section 2 of this manual.

4.3 SUPPLEMENTARY TAKE-OFF WEIGHT CHART

The Supplementary Take-off Weight Chart is based on factored take-off distances from rest to a height of 50 feet with the engines operating at take-off power. It may be used in lieu of the basic Take-off Weight Chart only for Airfield Pressure Heights of 3000 feet or less.

The technique used and take-off safety speeds remain the same.

4.4 LANDING

The weight of the aeroplane for landing shall not exceed the lesser of:

(i) the maximum landing weight specified in Section 2 of this Manual; and
(ii) the weight for landing determined from the landing weight chart of this Section.

The landing weight chart is based on factored landing distances from a height of 50 feet to stop, and is applicable to sealed, short dry grass, earth or gravel strips. Wet and/or slippery surfaces will increase the landing distances over that scheduled and the pilot should therefore ensure that adequate strip length is available to cover these conditions.

The technique used in establishing the landing distance is such that the aeroplane approaches at the given approach speed, with power as required to maintain 1000 ft/min rate of descent through the 50 foot height point at the strip threshold. After passing the 50 foot height point, touchdown is effected as soon as possible. After touchdown, maximum wheel braking and maximum available propeller beta range is used to bring the aeroplane to a stop.

The required 3.2 percent climb gradient during a baulked landing can be achieved at all altitudes at the maximum landing weight specified in Section 2 of this Manual.

For Training Purposes Only

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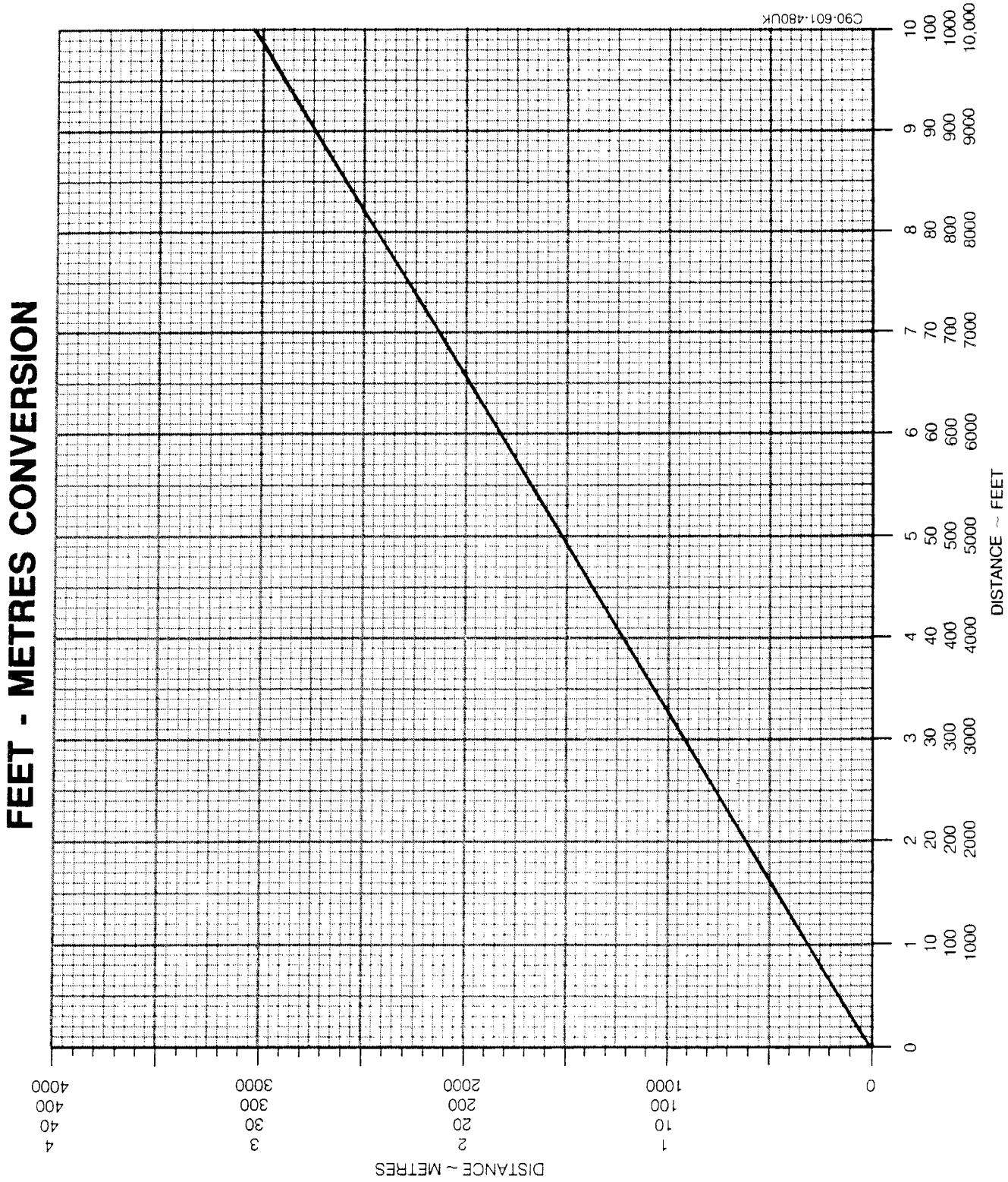
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C90-601-480UK
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DISTANCE ~ METRES
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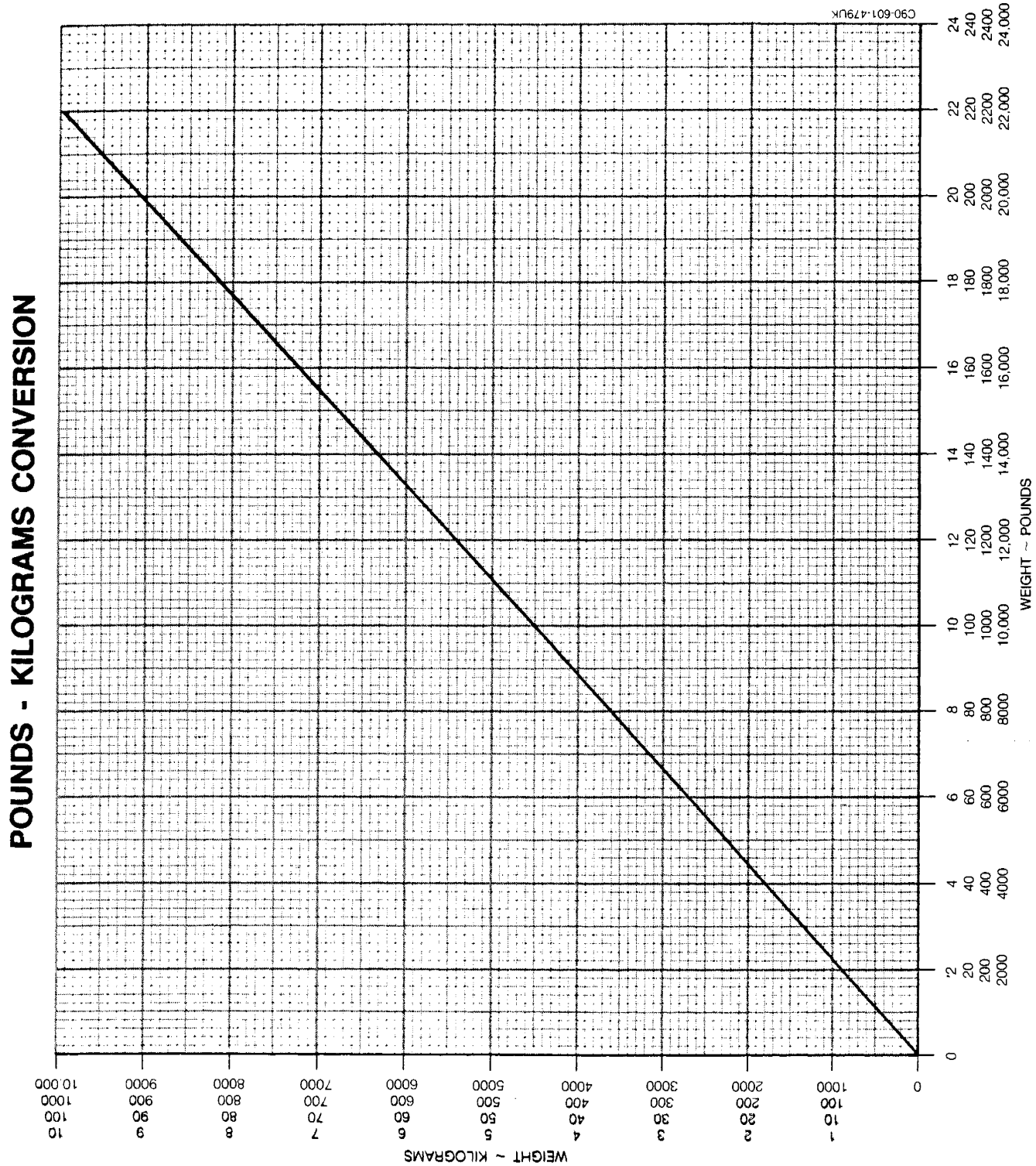
0 100 200 300 400
0 10 20 30 40
0 1 2 3 4


POUNDS - KILOGRAMS CONVERSION
C90-601-479UK
WEIGHT ~ POUNDS
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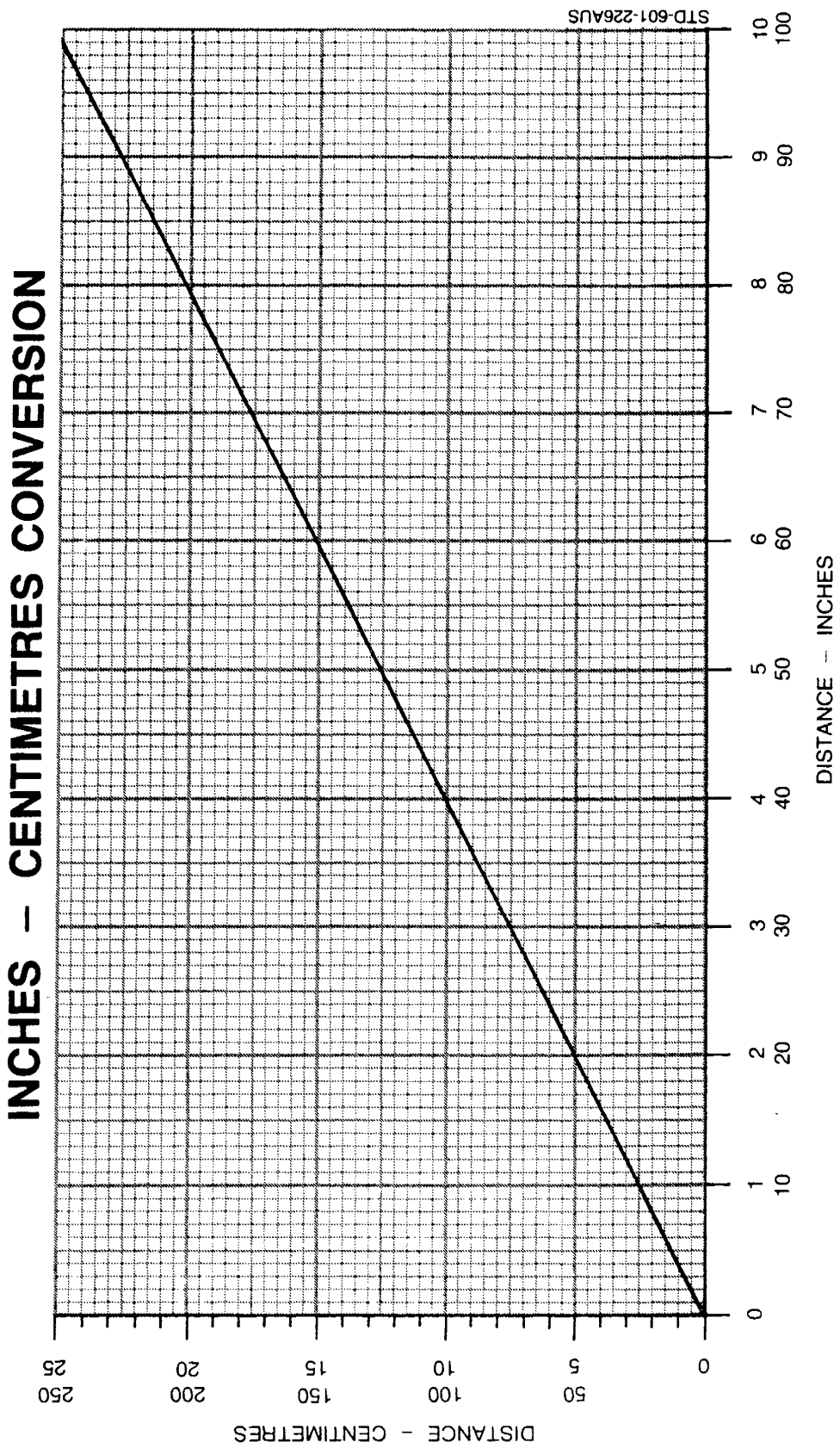
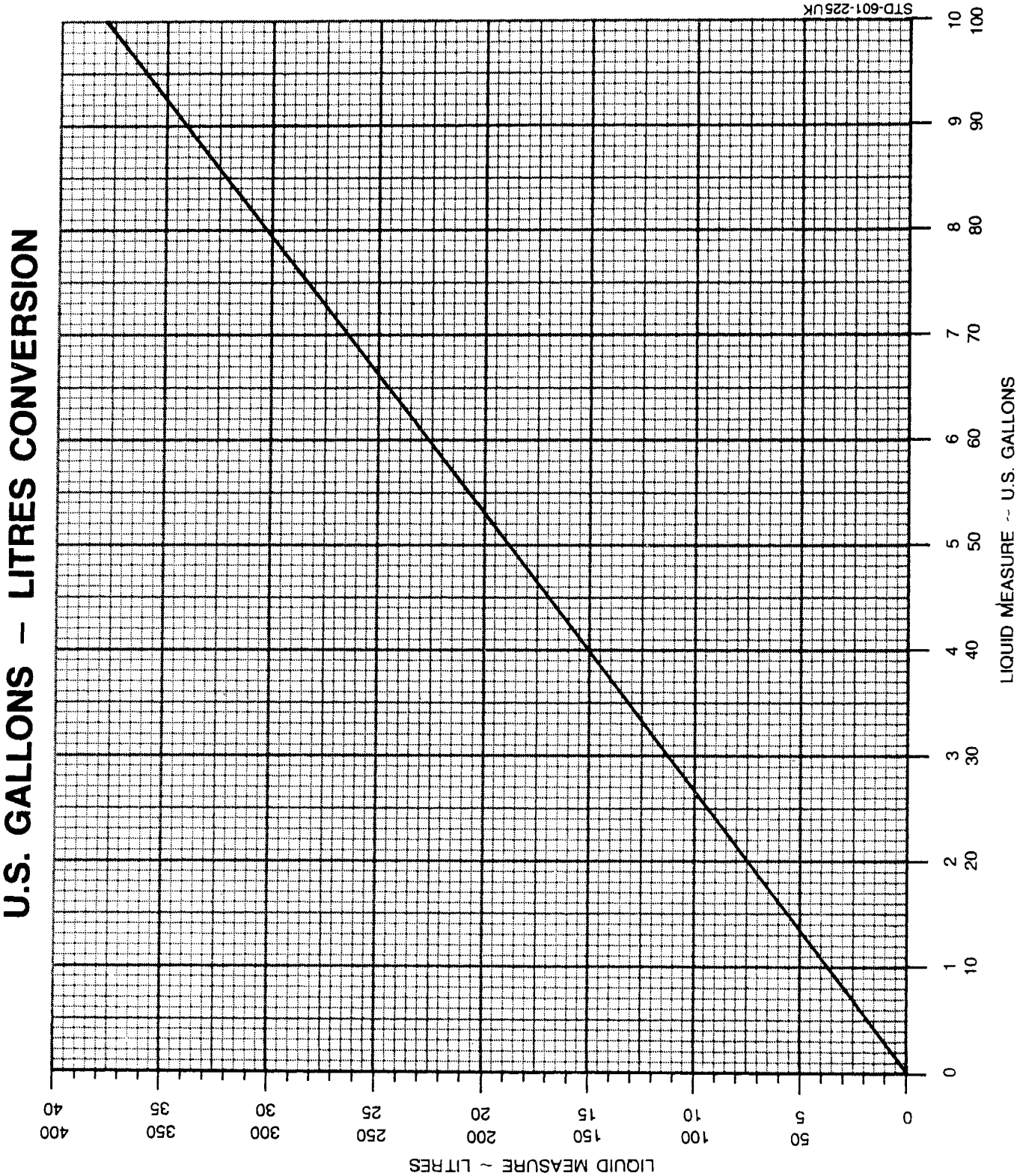
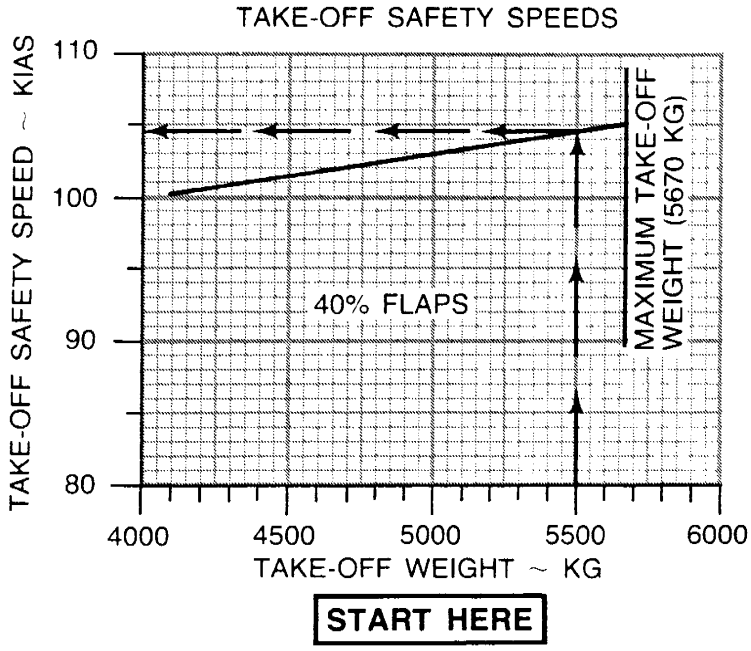
WEIGHT ~ KILOGRAMS
0 1000 100 10 1 2000 200 20 2 3000 300 30 3 4000 400 40 4 5000 500 50 5 6000 600 60 6 7000 700 70 7 8000 800 80 8 9000 900 90 9 10,000 1000 100 10

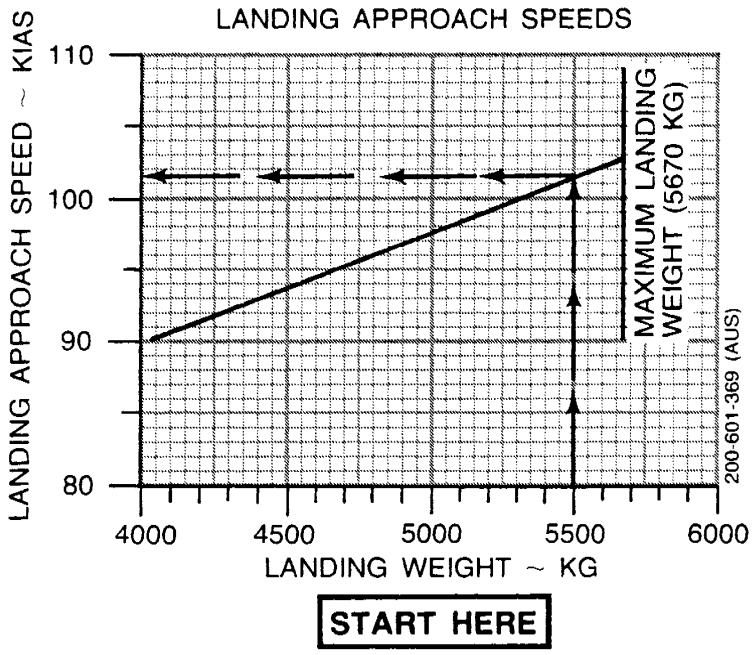

INCHES – CENTIMETRES CONVERSION
STD-601-226AUS
DISTANCE – INCHES
0
1
2
3
4
5
6
7
8
9
10
10
20
30
40
50
60
70
80
90
100
DISTANCE – CENTIMETRES
0
5
10
15
20
25
50
100
150
200
250


U.S. GALLONS – LITRES CONVERSION
STD-601-225UK
LIQUID MEASURE ~ U.S. GALLONS
0 1 2 3 4 5 6 7 8 9 10
0 10 20 30 40 50 60 70 80 90 100
LIQUID MEASURE ~ LITRES
0 5 10 15 20 25 30 35 40
0 50 100 150 200 250 300 350 400

TAKE-OFF SAFETY SPEEDS

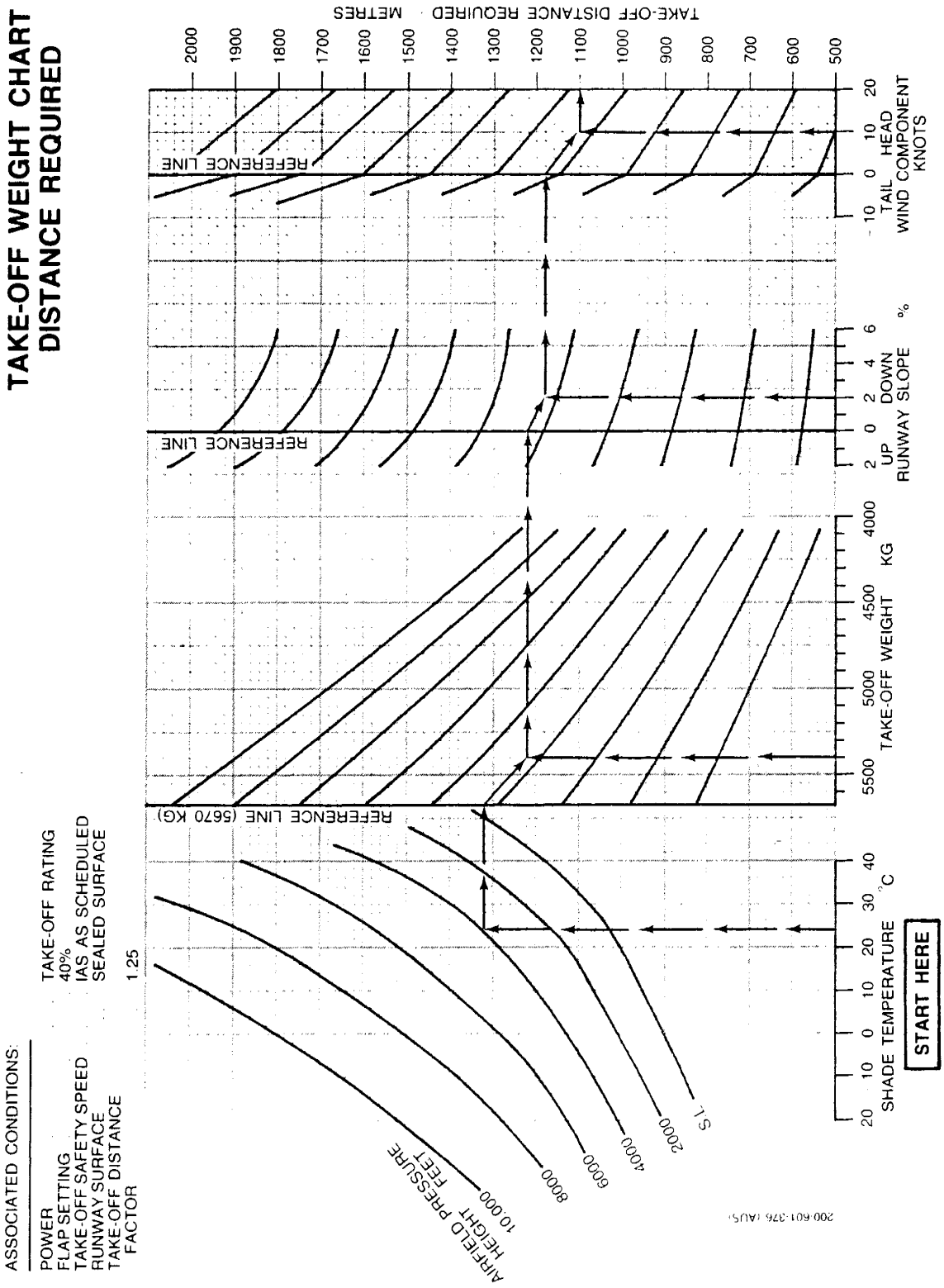


LANDING APPROACH SPEEDS



For Training Purposes Only

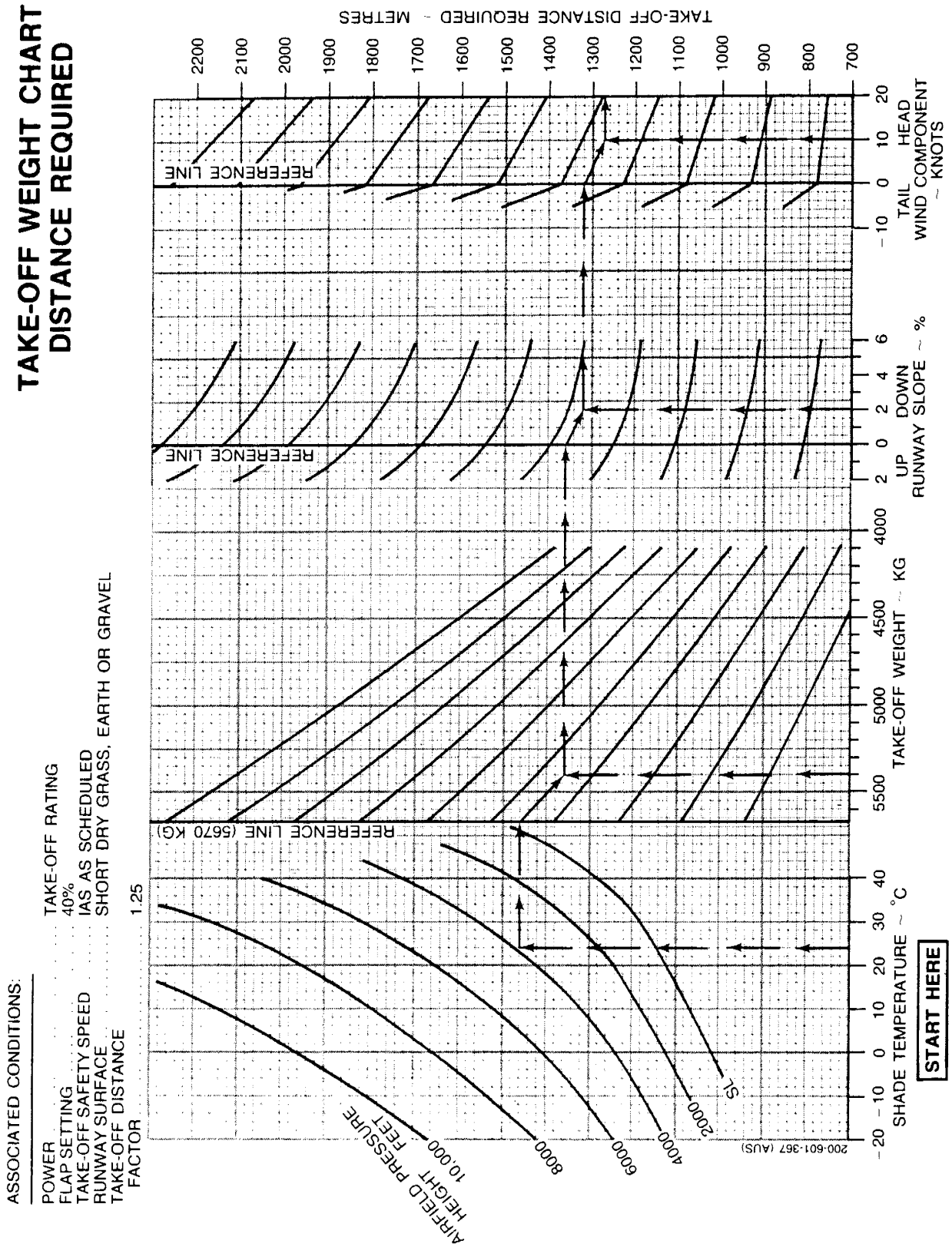
TAKE-OFF WEIGHT CHART
DISTANCE REQUIRED



ASSOCIATED CONDITIONS:

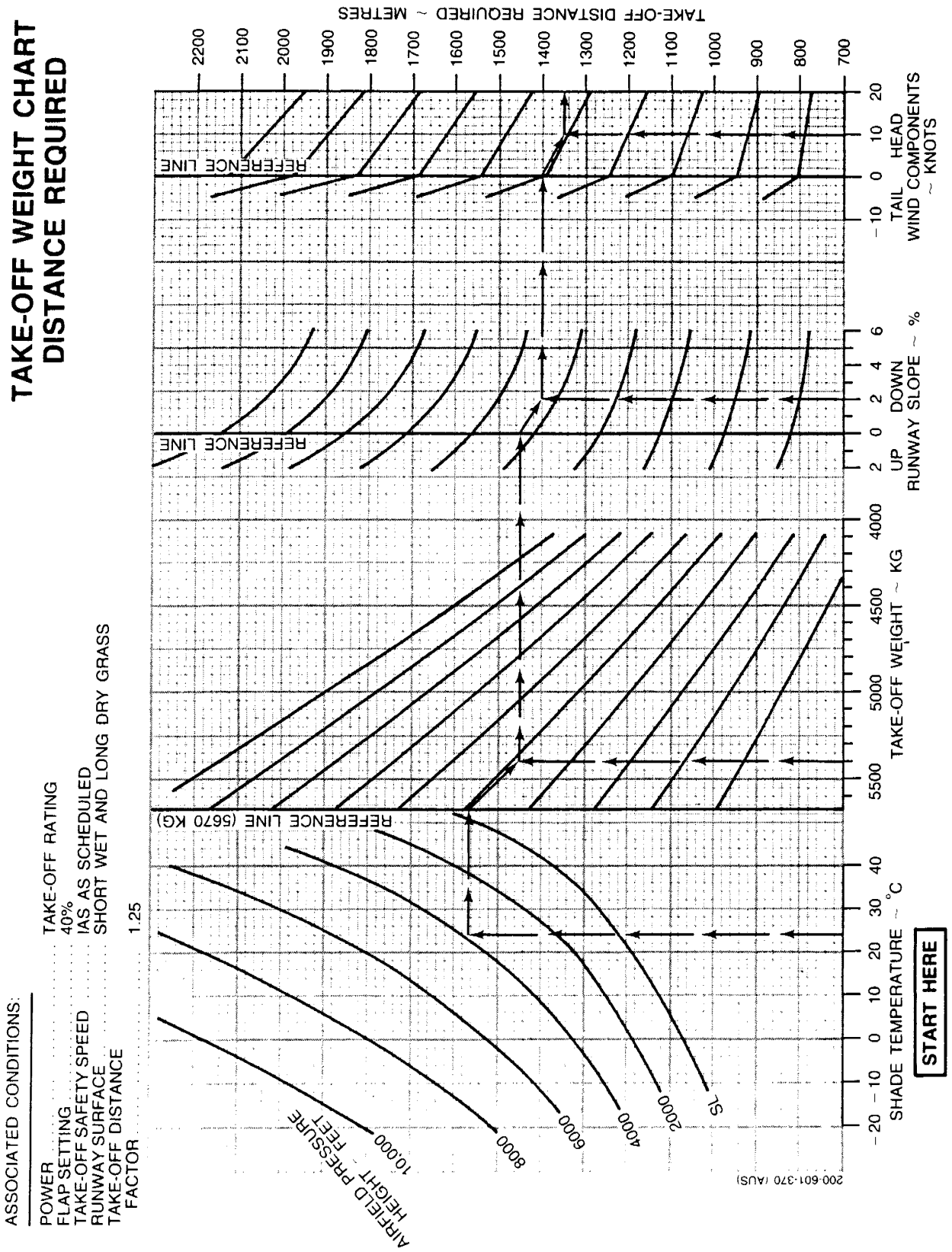
POWER	TAKE-OFF RATING
FLAP SETTING	40%
TAKE-OFF SAFETY SPEED	IAS AS SCHEDULED
RUNWAY SURFACE	SEALED SURFACE
TAKE-OFF DISTANCE FACTOR	1.25

TAKE-OFF WEIGHT CHART
DISTANCE REQUIRED



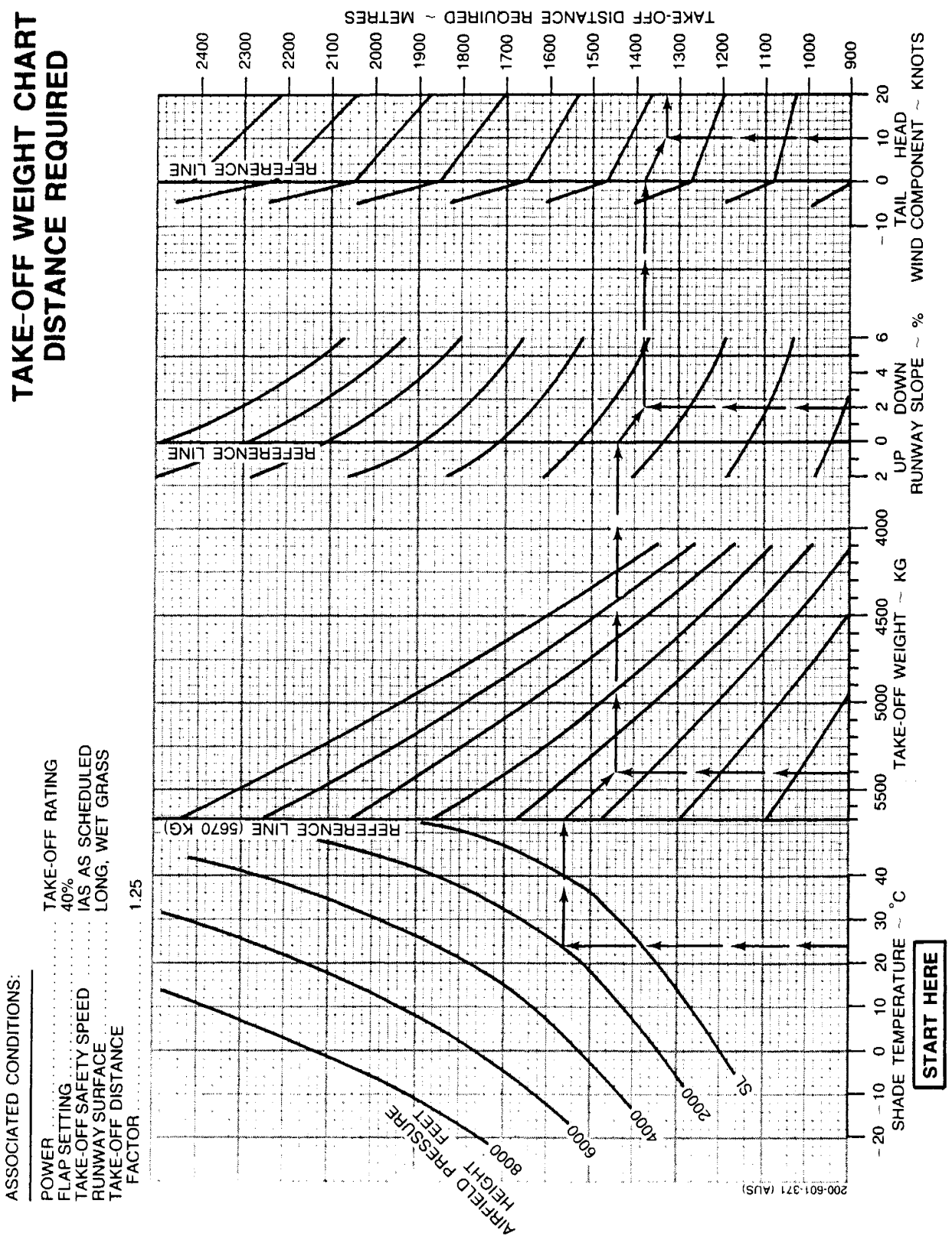
For Training Purposes Only

TAKE-OFF WEIGHT CHART
DISTANCE REQUIRED



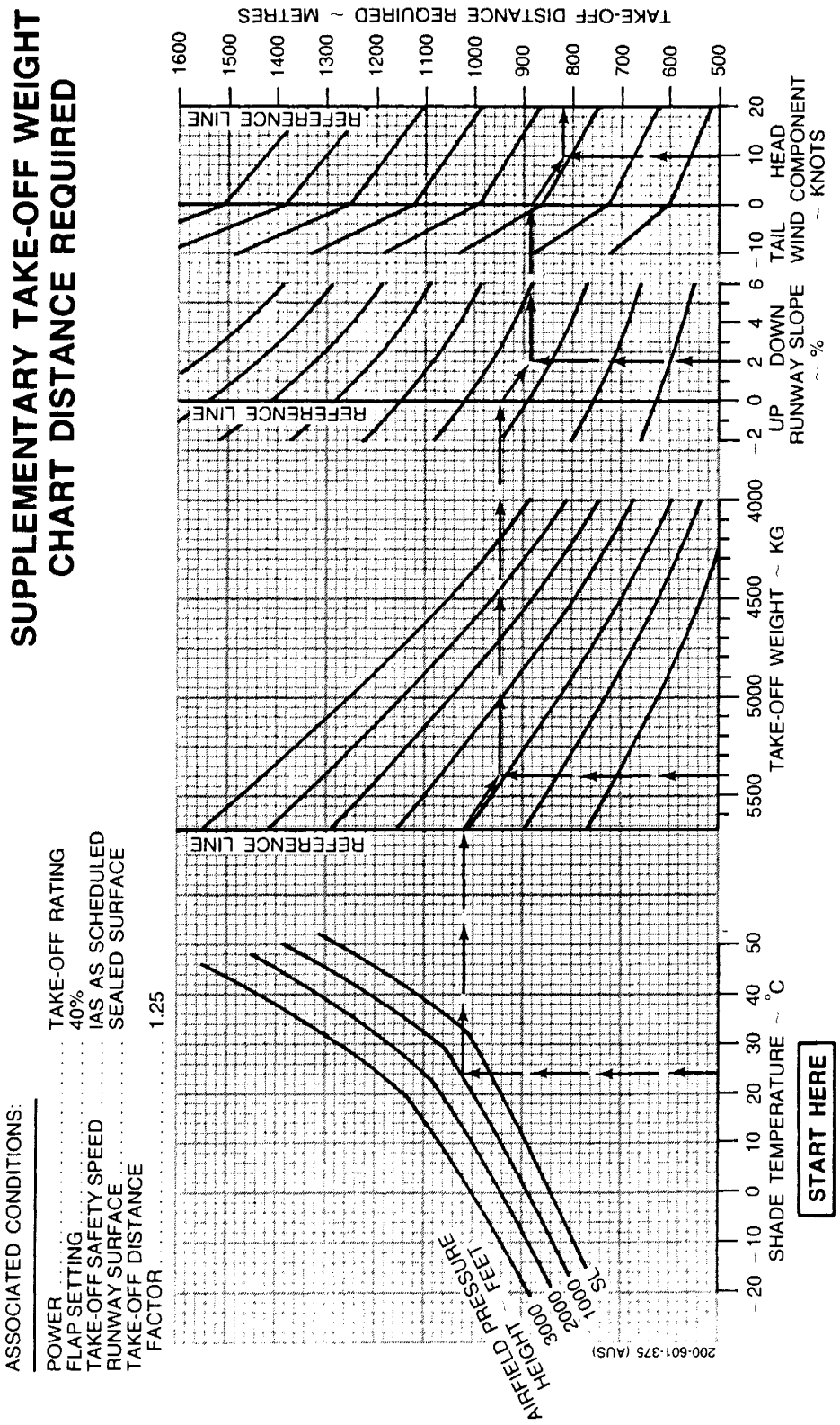
For Training Purposes Only

TAKE-OFF WEIGHT CHART
DISTANCE REQUIRED



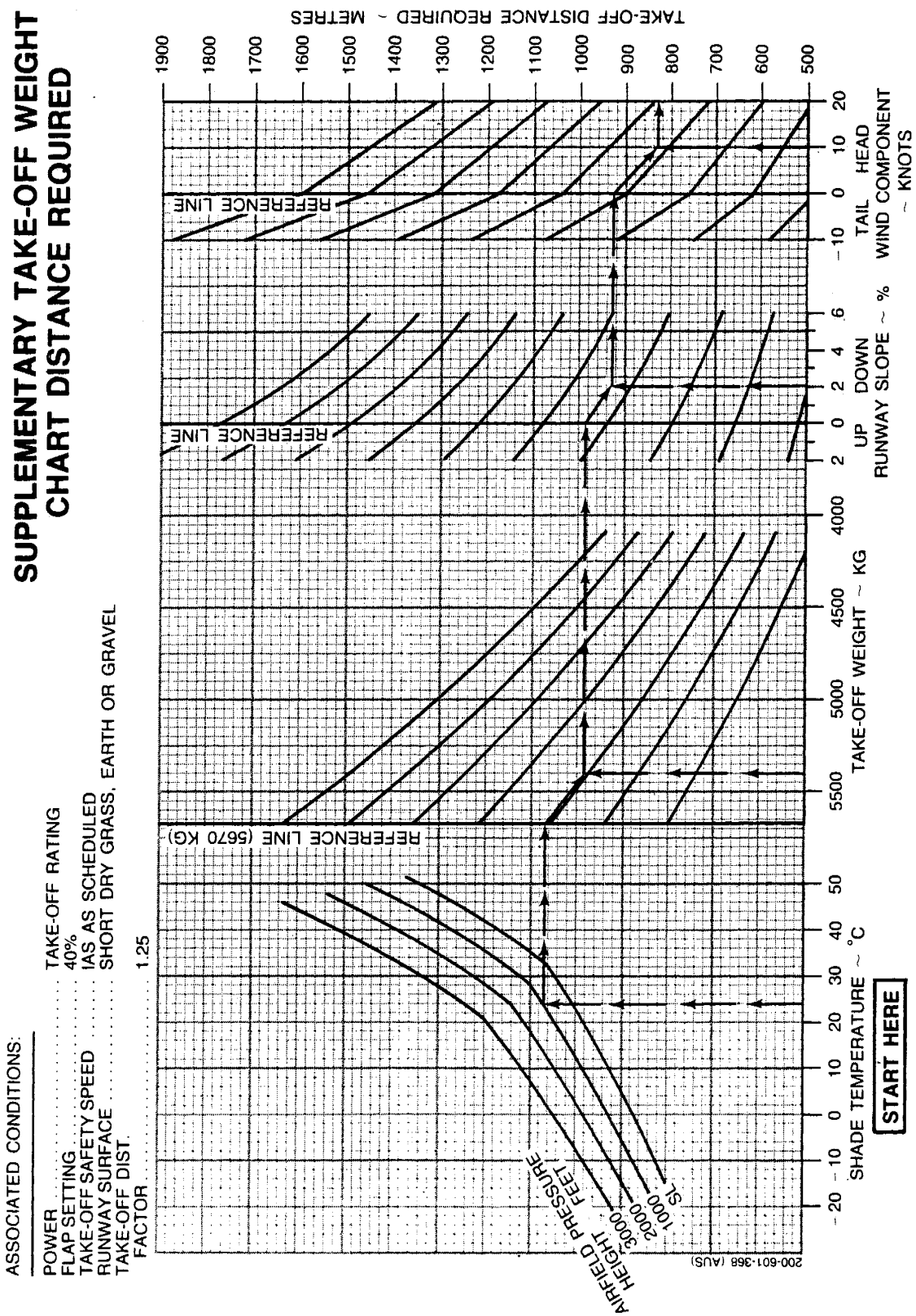
For Training Purposes Only

SUPPLEMENTARY TAKE-OFF WEIGHT CHART DISTANCE REQUIRED



For Training Purposes Only

SUPPLEMENTARY TAKE-OFF WEIGHT CHART DISTANCE REQUIRED



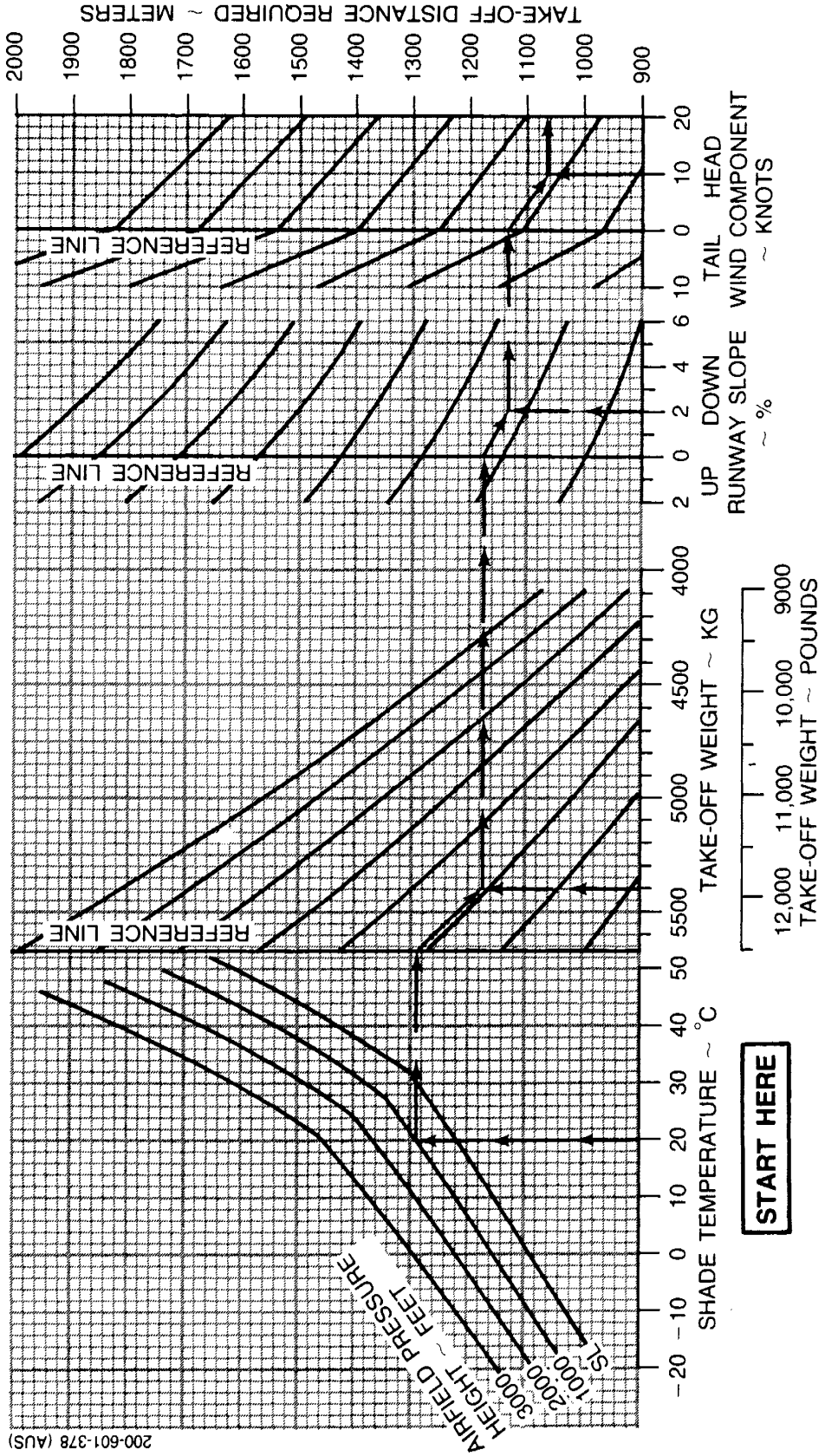
For Training Purposes Only

TAKE-OFF DISTANCE GRAPH - FLAPS 0% SEALED SURFACE

ASSOCIATED CONDITIONS:

POWER ..... TAKE-OFF POWER 2000 RPM
FLAP SETTING 0%
TAKE-OFF SAFETY SPEED ..... KIAS AS SCHEDULED
RUNWAY SURFACE ..... SEALED HARD SURFACE
TAKE-OFF DISTANCE FACTOR 1.25

WEIGHT ~ KILOGRAMS	TAKE-OFF SAFETY SPEED ~ KNOTS
5670	121
5400	119
5000	115
4500	111
4100	108

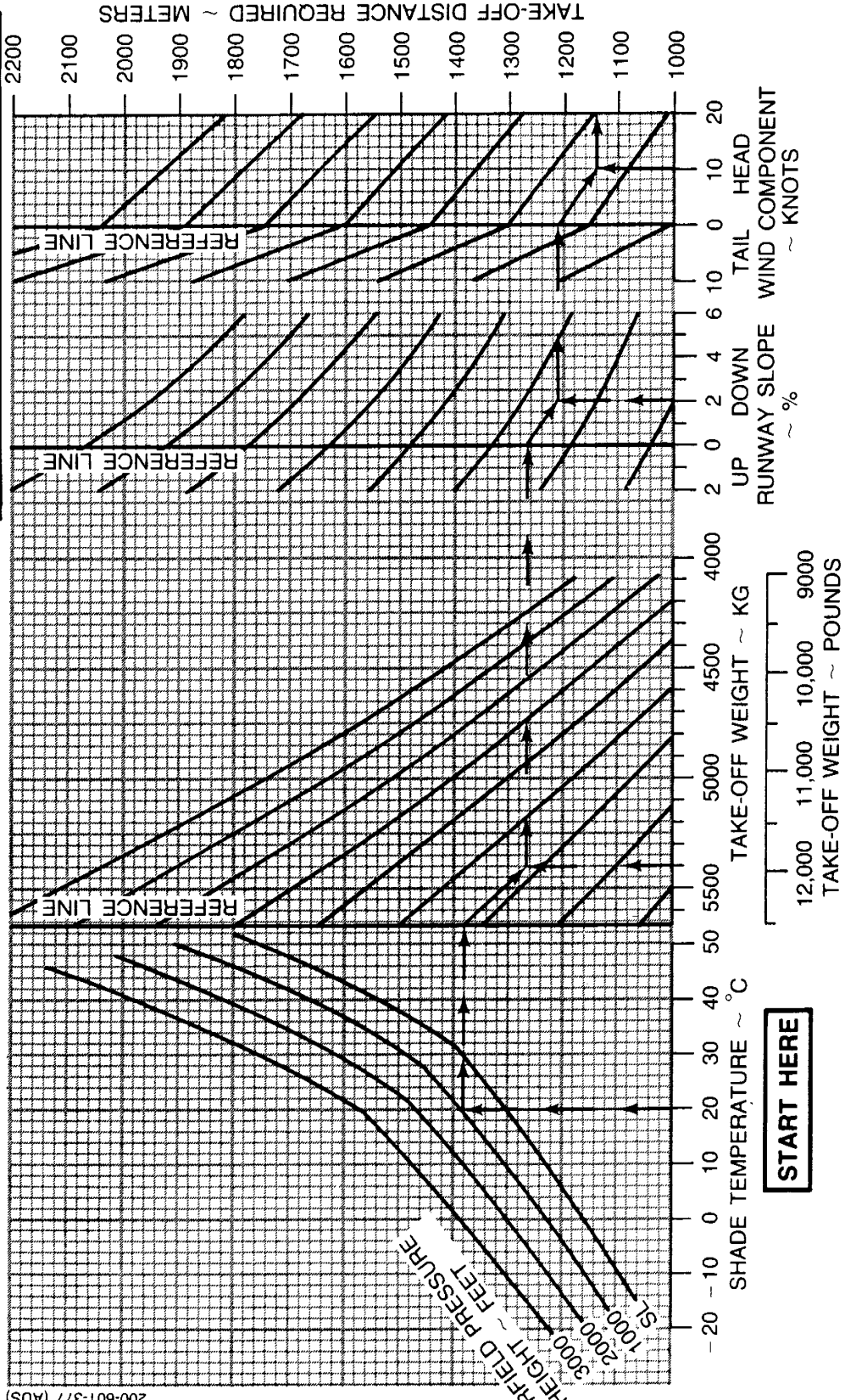


TAKE-OFF DISTANCE GRAPH - FLAPS 0% DRY GRASS

ASSOCIATED CONDITIONS:

POWER ..... TAKE-OFF POWER 2000 RPM
FLAP SETTING 0%
TAKE-OFF SAFETY SPEED ..... KIAS AS SCHEDULED
RUNWAY SURFACE ..... SHORT DRY GRASS, EARTH OR GRAVEL
TAKE-OFF DISTANCE FACTOR 1.25

WEIGHT ~ KILOGRAMS	TAKE-OFF SAFETY SPEED ~ KNOTS
5670	121
5400	119
5000	115
4500	111
4100	108



LANDING WEIGHT CHART

LANDING WEIGHT CHART TO BE PROVIDED BY HAWKER PACIFIC

SECTION 5 - INSTRUMENT AND EQUIPMENT INSTALLATIONS

5.1 MANDATORY INSTRUMENTS AND INDICATORS

The aeroplane shall not be operated unless, in addition to the minimum flight and navigation instruments required by Air Navigation Orders Section 20.18, the following indicators and instruments are also installed:

(a) Position indicators

- (i) Trim position indicators
- (ii) Flap position indicators
- (iii) Landing gear position indicator

(b) Power Plant Instruments and Indicators

- (i) Fuel quantity indicator for each tank (auxiliary and main)
- (ii) Fuel pressure indicator for each engine
- (iii) Oil pressure indicator for each engine
- (iv) Oil temperature indicator for each engine
- (v) Propeller tachometer for each engine
- (vi) Interstage turbine temperature indicator (ITT)
- (vii) Gas generator tachometer
- (viii) Torquemeter

(c) Other Airworthiness Instruments or Indicators

- (i) Stall warning horn
- (ii) Gear up warning horn
- (iii) Two electrical loadmeters

For Training Purposes Only

SECTION 6 - LOADING DATA

6.1 GENERAL

This Section contains basic weight and centre of gravity information necessary to ensure correct loading of the aeroplane and comprises Aeroplane Weight and Loading System pages. Both of these documents, separately approved by the Secretary of the Department of Transport or an aircraft weight control design signatory, are to be carried in the Flight Manual at all times.

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6.2 AEROPLANE WEIGHT

Aeroplane Type: .....

Registration Marking: VH- .....

Issue	Date	Date of Expiry

Aeroplane Weight and Centre of Gravity Data:

Item	Weight kg	Arm mm aft of datum	Index Unit kg mm	Cabin Configuration

NOTE: The above weight(s) include .....

Approval Stamp

6.3 LOADING SYSTEM

Aeroplane Type: .....

Registration Marking: VH-.....

Issue	Date	Date of Expiry

Approval Stamp

For Training Purposes Only

SECTION 7 - RADIO SYSTEMS AND OPERATIONAL LIMITATIONS

7.1 GENERAL

The radio communication and radio navigation systems in the aeroplane are approved for the types of operation and maximum operating altitudes shown in the Radio Systems Approval table of this Section. Approval of a radio system for a particular type of operation is signified in this table by inclusion of the maximum operating altitude of the equipment (expressed in thousands of feet) under the appropriate headings.

Before the aeroplane may engage in the types of operation for which the radio systems are approved, the instruments, radio systems and equipment which are required, in accordance with the appropriate Sections of Air Navigation Orders Part 20, for the operation to be performed shall be installed and airworthy to the standards required for that operation.

NOTE: Reported unserviceability of instruments, radio systems and equipment will be indicated on the Maintenance Release.

7.2 RADIO SYSTEMS APPROVAL

System	Equipment Type	Type of Operation		
		I.F.R.	Limited I.F.R.	V.F.R. and Night V.M.C.

SECTION 8 - SUPPLEMENTS

8.1 GENERAL

Flight Manual Supplements covering the special operations for which this aeroplane is approved are listed below.

The operations listed shall be conducted in accordance with the limitations and instructions contained in the appropriate Supplements included in this Manual.

SUPPLEMENT TITLE

I.F.R. and/or Limited I.F.R. Operations
Night V.M.C. Operations
* Automatic Flight Control System Operation
* Transport Category Operation

SUPPLEMENTAL CRUISE PERFORMANCE (Model 200/200C)

*Provided by Hawker Pacific

For Training Purposes Only

Flight Manual Supplement

I.F.R. AND/OR LIMITED I.F.R. OPERATIONS

This aeroplane is eligible for operation under the Instrument Flight Rules and/or Limited Instrument Flight Rules provided:

(a) The radio communication and radio navigation systems installed in the aeroplane are approved for the particular type of operation - see Flight Manual Supplement "Radio Systems - Operational Limitations", and

(b) The instruments, radio systems and equipment which are airworthy to Instrument Flight Rules standards are appropriate, in accordance with Air Navigation Orders Part 20, to the operation to be performed.

NOTE: Reported unserviceability of instruments, radio systems and equipment will be indicated on the Maintenance Release. Air Navigation Orders Part 20 require the placarding or removal of unserviceable instruments, radio systems and equipment.

For Training Purposes Only

Flight Manual Supplement

NIGHT V.M.C. OPERATIONS

This aeroplane is eligible for Night V.M.C. operations provided that the radio communication and radio navigation systems installed in the aeroplane are approved for this type of operation - see Flight Manual Supplement "Radio Systems - Operational Limitations".

NOTE: Reported unserviceability of instruments, radio systems and equipment will be indicated on the Maintenance Release. Air Navigation Orders Part 20 require the placarding or removal of unserviceable instruments, radio systems and equipment.

For Training Purposes Only

SUPPLEMENTAL

CRUISE PERFORMANCE

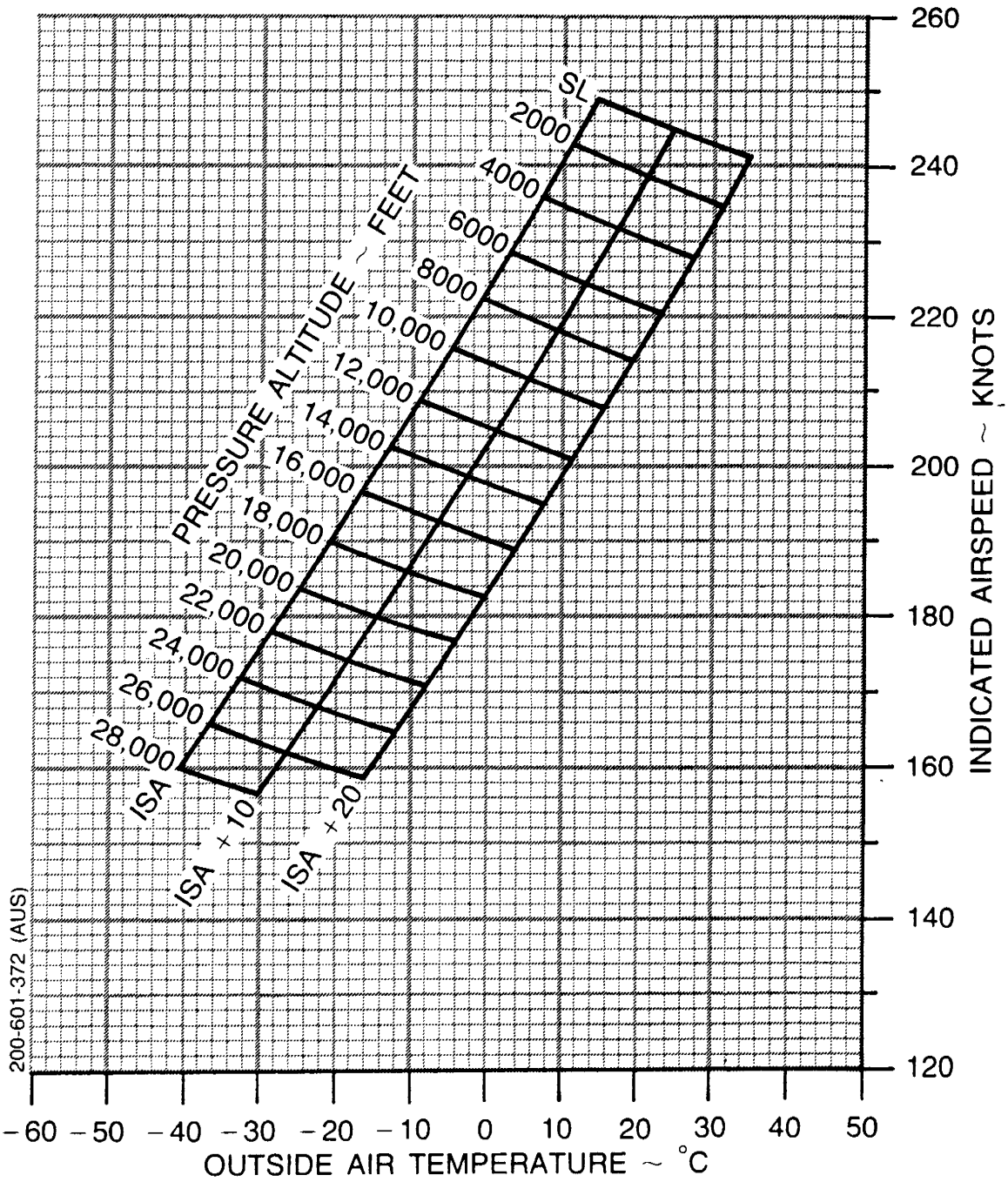
MODEL 200/200C

For Training Purposes Only

INDICATED CRUISE AIRSPEED

ASSOCIATED CONDITIONS:

CRUISE SPEED ..... 250 KTAS
PROP RPM ..... 1700 RPM
ICE VANES ..... RETRACTED

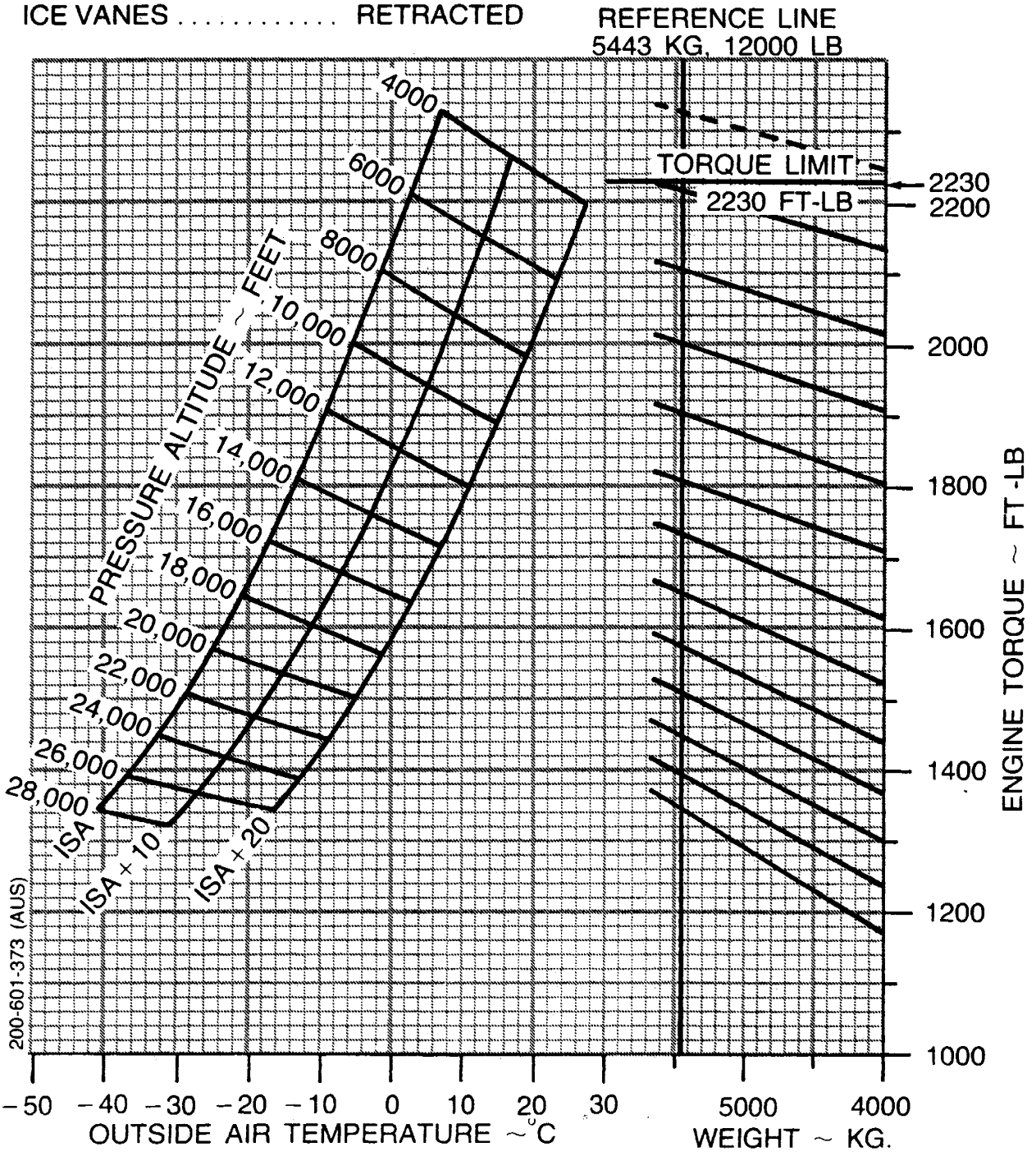


For Training Purposes Only

CRUISE TORQUE SETTING

ASSOCIATED CONDITIONS:

CRUISE SPEED ..... 250 KTAS
PROP. SPEED ..... 1700 RPM
ICE VANES ..... RETRACTED

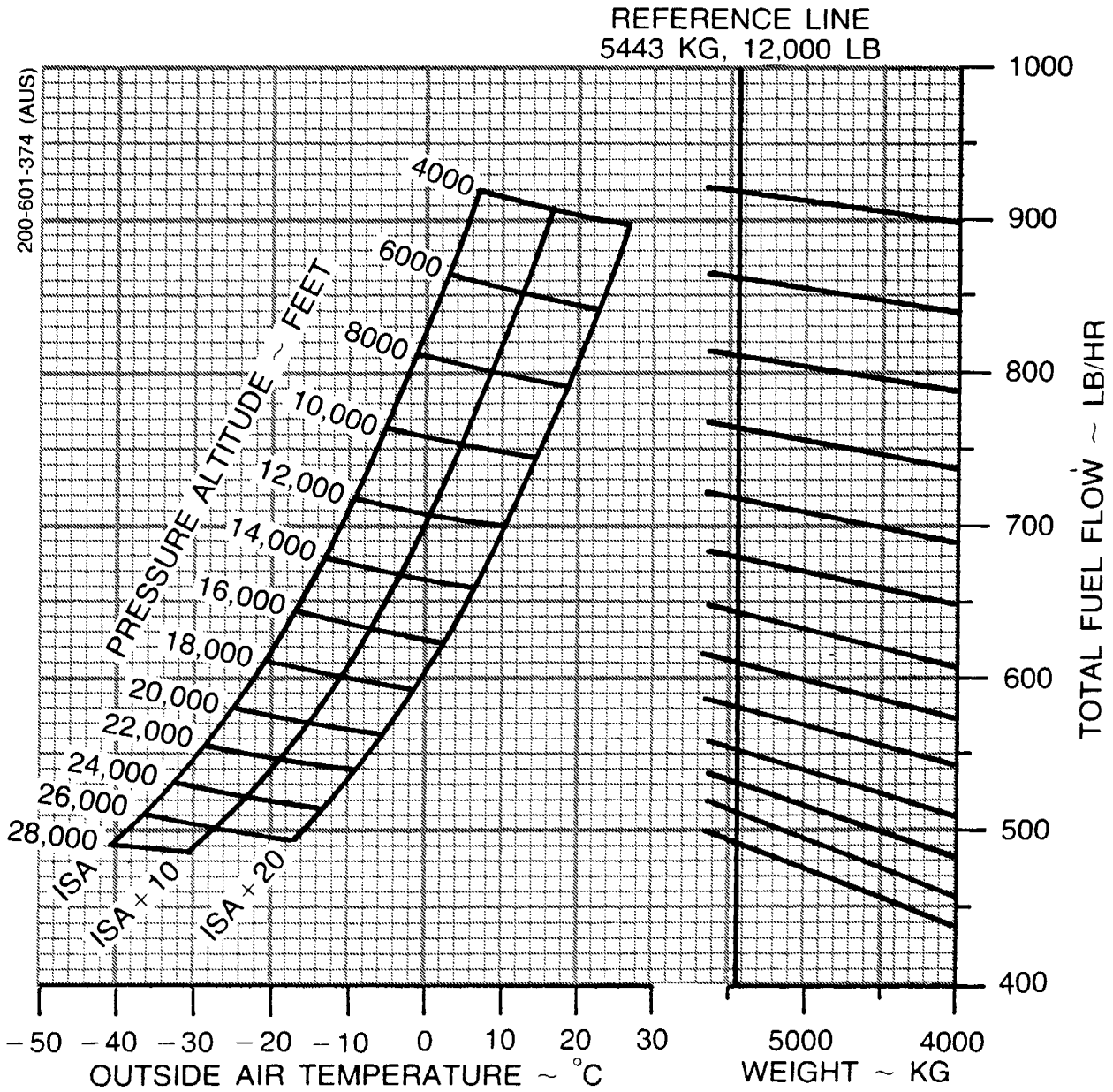


For Training Purposes Only

FUEL FLOW SETTING

ASSOCIATED CONDITIONS:

CRUISE SPEED ..... 250 KTAS
PROP. SPEED ..... 1700 RPM
ICE VANES ..... RETRACTED



For Training Purposes Only

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T & B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the KING KNR-665 OR KNR-665A AREA NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* when the airplane has been modified by installation of the King KNR-665 or -665A Area Navigation System in accordance with Beech-Approved data.

The information in this supplement supersedes or adds to the basic *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. The Area Navigation Function may not be used as a primary system under IFR conditions except on approved approach procedures, approved area navigation airways, and random area navigation routes when approved by Air Traffic Control.

2. The Area Navigation Function can only be used with co-located facilities (VOR and DME signals originate from same geographical location).

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude, and angle of bank.

1. If NAV flag appears while in the ENROUTE mode, check for correct frequency.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

3. If NAV flap appears during an approach while in the APPR mode, execute published missed approach and utilize another approved facility.

NORMAL PROCEDURES

The King KNR-665 or -665A Area Navigational System is a push-button tuned, navigational unit with a ten waypoint memory capacity. Included is the capability to select the VOR/DME, localizer and glideslope frequencies, electronically "move" the VOR to a phantom location called a Waypoint, and set the navigation course in the Flight Director.

The KNR-665 or -665A functions in three modes. In the VOR mode, the unit operates as a conventional VOR Converter with a course deviation scale factor of ± 10 degrees presented on the Pictorial Navigation Indicator. This mode is also utilized for localizer/glideslope approaches with a conventional display. Two

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P/N 101-590010-69

For Training Purposes Only

Area Navigation modes are available. They are designated ENROUTE and APPR for use in enroute and terminal/approach navigation. For Area Navigation, the course deviation is presented in nautical miles on the Pictorial Navigation Indicator rather than in degrees as with the VOR mode. This is referred to as "constant course width". The ENROUTE mode provides a constant course width of ± 5 nautical miles (1 nautical mile per 1 dot deviation). APPR mode has a constant course width of ± 1.25 nautical miles (1/4 nautical mile per 1 dot deviation) and should be used when within ten miles of the terminal waypoint.

CONTROL FUNCTIONS

The KNR-665 or -665A Area Navigation System is programmed and operated from a panel mounted control unit. Information such as station frequency, course, waypoint radial and waypoint distance is entered into memory from the keyboard on the control unit. During the flight, the desired waypoints are recalled from memory and the modes of operation are selected from the control unit.

1. Mode Switch:
   This three position switch selects conventional VOR/DME operation designated "VOR", enroute Area Navigation designated "ENROUTE", and terminal/approach Area Navigation designated "APPR".

2. 0 thru 9 Keys:
   Each depression of one of these keys enters one digit into the FREQ/KEYBOARD window.

3. FREQ/KEYBOARD Window:

   a. Displays the Waypoint facility frequency when display is constant.

   b. Serves as a "scratch pad" to confirm the input of the keyboard when display is flashing.

4. COURSE Window:

   Displays the selected course in degrees.

5. WPT Radial Window:

   Displays the VOR radial on which the Waypoint is placed.

6. WPT DISTANCE Window:

   Displays the distance along the selected VOR radial on which the Waypoint is placed.

7. Load Keys:

   Load keys are located to the right of the FREQ/KEYBOARD, COURSE, WPT RADIAL and WPT DISTANCE windows. The load keys cause data from the keyboard to be loaded into the respective windows.

8. WPT-CRS-DSPY Window:

   Annunciates the waypoint and course being displayed.

9. WPT-CRS-IN USE Window:

   Annunciates the waypoint and course in use.

10. → Transfer Key:

    Puts the displayed waypoint into use.

11. CRS 1 Key:

    Selects Course 1 (Inbound course).

For Training Purposes Only

12. CRS 2 Key:

    Selects Course 2 (Outbound course).

13. AUTO CRS Key:

    Computes and enters the direct course from present position to the facility. VOR (In VOR mode) or waypoint (In ENROUTE or APPR mode).

14. KYBD CLR Key:

    The Keyboard Clear Key clears the "scratch pad" (FREQ/KEYBOARD window when flashing).

15. NAV TEST Key:

    Initiates an automatic, three-part, sequential self-test. Active in ENROUTE mode only.

PREFLIGHT

The preflight check consists of a sequential test of the entire RNAV system, including a test of the computation accuracy of the computer and all displays. The RNAV system will not test with the autopilot engaged or with the navigation receiver tuned to an ILS frequency.

1. Set mode selector switch to ENROUTE

2. Press and hold NAV TEST key for approximately 15 seconds to initiate three-part self-test.

    Part 1. All lamp segments are illuminated to numeral "8" (except the extreme left digit of FREQ/KEYBOARD window which illuminates to numeral "1").

    Part 2. The airplane is placed over the VOR, the waypoint is located 30.0 miles on the 90.0° radial, the selected course is 30°. The FREQ/KEYBOARD, DSPY, and IN USE windows are extinguished. The Pictorial Navigation Indicator course needle will rotate to 30°.

    Part 3. The course required to fly to the waypoint is computed (90°) and entered into the COURSE window. The Pictorial Navigation Indicator course needle will rotate to 90°.

Failure to satisfy the preflight test requirements indicates an inoperative RNAV computer. Set the mode selector to VOR and use navigational units in conventional VOR/DME operation.

PROGRAMMING

Pertinent information (frequency, course, waypoint radial, waypoint distance, waypoint number) for up to ten waypoints is entered into memory from the panel mounted keyboard unit. Programming the computer may be completed prior to take-off or during the flight. Any combination of navigational facilities (RNAV waypoint, VOR/DME, ILS) may be loaded into the computer; however, it is desirable that each facility be numbered and loaded in the sequence it is to be used.

RNAV WAYPOINTS

1. Select the first waypoint and the inbound course by depressing keyboard number "1" and CRS 1 pushbuttons. These numbers will appear in the top center DSPY window over WPT and CRS.

NOTE

If the navigational facility is at or near the departure point and the first route segment is outbound from that facility, depress keyboard number "1" and CRS 2 pushbuttons.

2. Select the VORTAC frequency by depressing the keyboard buttons in the number sequence. A total of five digits must be entered to complete the frequency input (i.e., frequency 113.8 entered as 113.80). The frequency will appear flashing in the FREQ/KEYBOARD window. Upon confirming the proper frequency has been entered on the keyboard, it is stored into memory by depressing the load key adjacent to the FREQ/KEYBOARD window. The flashing presentation will become steady which confirms frequency storage.

3. Select inbound course to the waypoint on the keyboard. The frequency number in the FREQ/KEYBOARD window will be replaced by the flashing course numbers. Confirm accuracy of course numbers and store into memory with the load key adjacent to the COURSE window. The inbound course number in the FREQ/KEYBOARD window will transfer to the COURSE window and the VORTAC frequency will reappear in the FREQ/KEYBOARD window.

4. Select outbound course from the waypoint by depressing CRS 2 pushbutton. The number 2 will appear in the DSPY window over CRS and adjacent to WPT number 1. Select and load the outbound course value using the same procedure as the inbound course.

5. Select the waypoint radial on the keyboard and enter into memory (after checking) by depressing the load key adjacent to the WPT RADIAL window.

6. Select the waypoint distance following the same procedure as with the selection of the waypoint radial. Enter the value into memory by depressing the load key adjacent to the WPT DISTANCE window.

NOTE

WPT RADIAL and WPT DISTANCE are decimal readouts. Program these values to the nearest tenth unit.

7. This completes the programming for the first waypoint and inbound/outbound courses. Follow these procedures for all selected waypoints up to a maximum of ten.

NOTE

If an error is noted while the value in the FREQ/KEYBOARD window is still flashing, depress the KYBD CLR button and select the correct value on the keyboard. This will not affect any information already stored in the memory. If the error is noted after the value has been loaded, select the proper value on the keyboard, confirm its accuracy in the flashing FREQ/KEYBOARD window, and reload the value into the appropriate window.

CONVENTIONAL VOR

The programming technique for conventional navigation directly toward or away from a VOR facility is similar to that for RNAV waypoints. Inputing the waypoint number, course number, frequency and course values into the memory is accomplished in the same manner. Since the station is not to be electronically "moved" to a new location (waypoint), no values are programmed into the WPT RADIAL and WPT DISTANCE windows.

ILS APPROACH (Front Course and Back Course)

Programming an ILS approach is accomplished in the same manner as programming conventional VOR. However, it is essential that only the inbound front course localizer bearing be entered into the COURSE window for both front course and back course approaches. This will assure that the Pictorial Navigation Indicator and autopilot maintain the proper left/right logic.

MISSED APPROACH

If the published missed approach utilizes an RNAV waypoint or VOR facility, it may be entered into memory

anytime prior to the approach. It is recommended that WPT "O" (keyboard numeral 0) be reserved for this operation. Any other waypoint storage (1 thru 9) could be used; however, habitual use of WPT "O" eliminates the possibility of error that could be experienced when selecting an intermediate digit during this critical flight phase.

ENROUTE OPERATION

Prior to take-off, select ENROUTE on the mode switch. Flight in this mode is recommended even if navigating directly toward or away from a VORTAC facility. The ENROUTE mode provides the advantages of "constant course width" and smoothes the received signals to improve autopilot operation. An exception to this procedure would be caused by the lack of a DME signal co-located with the VOR facility. In this case, the VOR mode would be selected.

1. Place WPT 1 / CRS 1 in the DSPY window by depressing keyboard number "1" and CRS 1 pushbuttons. This calls up waypoint 1 information from the memory bank and displays that information in the appropriate windows for checking.

NOTE

At this point, changes to a waypoint parameter may be made by replacing the original numbers with a new entry without affecting the other parameters that are in memory.

2. Depress ⟶ (transfer key) to place the displayed information into active use. The waypoint numbers WPT 1 / CRS 1 will appear in the IN USE window. This action also automatically tunes the VOR/DME receivers to the appropriate frequencies and causes the course needle on the Pictorial Navigation Indicator to be driven to the displayed course.

3. As the waypoint is approached, recall the outbound course by depressing the CRS 2 pushbutton. This places WPT 1 / CRS 2 in the DSPY window. The waypoint information and outbound course are displayed for checking. However, navigation continues on the inbound course. The IN USE window will flash WPT 1 / CRS 1 to advise the waypoint and course currently displayed are not in use. Transition to the outbound course is accomplished by depressing the ⟶ (transfer key). The displayed waypoint parameters will be placed in use and WPT 1 / CRS 2 will appear steady in the IN USE window.

4. Follow these procedures for subsequent waypoints.

AUTOCOURSE OPERATION

The "autocourse" function allows for navigation from the airplane's present location direct to an IN USE waypoint or VORTAC.

1. DIRECT TO WAYPOINT

   Depress AUTO CRS pushbutton. The direct course will be computed and displayed in the COURSE window. The course needle on the Pictorial Navigation Indicator will be driven to the displayed course.

2. DIRECT TO VORTAC

   Select VOR on mode switch. Depress AUTO CRS pushbutton. The direct course will be computed and displayed as above. (To obtain the advantages of "constant course width", load 0 nautical miles into the WPT DISTANCE window and return the mode switch to ENROUTE.)

Approved: Donald H. Peter

For W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: January, 1981
P/N 101-590010-69

For Training Purposes Only

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT AND KING AIR C90 (LJ-668 thru LJ-1062, except LJ-670), C90A (LJ-1063 and after), E90, F90, A100 AND B100 LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

COLLINS ANS-31 OR ANS-31A AREA NAVIGATION SYSTEM OR COLLINS NCS-31 OR NCS-31A NAVIGATION CONTROL SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the *FAA Approved Airplane Flight Manual* when the airplane has been modified by installation of the Collins ANS-31 Area Navigation System or the Collins NCS-31 Navigation Control System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic *FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

Except as noted, all references to the ANS-31 or NCS-31 systems also apply respectively to the ANS-31A or NCS-31A systems.

LIMITATIONS

1. The Area Navigation mode may not be used as a primary system under IFR conditions except on approved approach procedures, approved airways, and random area navigation routes when approved by Air Traffic Control.

2. The Area Navigation mode can only be used with colocated facilities. (VOR and DME signals originate from same geographical location.)

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude and angle of bank.

1. If NAV flag appears while in the enroute mode, check for correct frequency.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

3. If NAV flag appears during an approach, execute published missed approach and utilize another approved facility.

NORMAL PROCEDURES

The Collins ANS-31/NCS-31 Systems are push-button operated navigation computers with ten waypoint memory capacities. They contain a numerical keyboard for data entry and digital displays for data readout. Included is the capability to tune the VOR/DME, localizer and glideslope receivers, and electronically "move" the VOR to a phantom location called a waypoint. A waypoint is a convenient navigational position either at or within reception range of a selected VOR/DME station. The position of the waypoint is a function of its bearing and distance from the station.

In addition to the navigation function, the NCS-31 system can provide for the frequency control of two VHF communications radios, two ADF radios, two ATC transponders, and a second VOR and DME radio. Refer to the appropriate Collins manual for specific operating instructions of this feature.

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For Training Purposes Only

The ANS-31/NCS-31 systems operate in three fundamental modes: VOR, localizer/glideslope, and Area Navigation. In the VOR mode, the units operate as conventional VOR converters with an "angular course deviation" scale factor of ± 10 degrees presented on the Horizontal Situation Indicator. The localizer/glideslope mode presents data in a conventional display with an "angular course deviation" scale factor appropriate to the specific approach facility.

CAUTION

Always operate the ANS-31/NCS-31 systems in the VOR mode when the system is used during takeoff.

For Area Navigation, course deviation is presented in nautical miles on the Horizontal Situation Indicator rather than in degrees as with the VOR mode. This feature, referred to as "linear course deviation", provides for a constant course width irrespective of the distance to the waypoint. Two levels of sensitivity are available in the Area Navigation mode. They are designated ENROUTE and APPROACH for use in enroute and terminal/approach navigation. The ENROUTE sensitivity, available when the flight control system is not in the approach mode, provides a constant course width of ± 10 nautical miles. APPROACH sensitivity, available when the flight control system is in the approach mode, provides a constant course width of ± 2 nautical miles. APPROACH sensitivity should be used when within ten nautical miles of the terminal waypoint.

DISPLAYS AND CONTROLS

The ANS-31/NCS-31 systems are programmed and operated from a panel mounted control unit. Information such as waypoint number, station frequency, station elevation, waypoint bearing, and waypoint distance are entered into memory from the keyboard on the control unit. During the flight, the desired waypoints are recalled from memory and the modes of operation are selected on the control unit.

1. WPT Window:

   Identifies the waypoint defined by the displayed data. The letter "P" precedes the waypoint number when the displayed waypoint/frequency/code data is inactive/preset. The "P" blinks when the displayed data is on the scratch pad only.

2. FREQ Window:

   Displays the programmed VOR/DME/localizer frequency (108.00 through 117.95).

3. EL 100' Window:

   a. Displays the programmed VOR/DME station elevation in hundreds of feet.

   b. Displays the VOR-only mode of operation (VOR).

   c. Displays the localizer mode of operation (LOC).

4. BRG Window:

   a. Displays the waypoint bearing from the VOR/DME station (000.0° through 359.9°).

   b. Displays the localizer bearing in the LOC mode (000° through 359°).

   c. Displays the held VOR/DME frequency when DME HOLD is in use.

5. DIST Window:

   a. Displays the waypoint distance from the VOR/DME station (000.0 through 249.9 nautical miles).

   b. Annunciates DME HOLD is in use (dh).

6. TEST Button:

   Momentary push button to initiate ANS-31/NCS-31 self-test.

7. Data Keyboard:

   Ten digital (0 through 9) momentary keys for entry of numerical data.

8. CLR Key:

   Momentary key to clear scratch pad for correction of entry errors or revision of stored data.

9. PRE Key:

   Momentary key to store data displayed on scratch pad into memory.

10. WPT Key:

    Momentary key to display active waypoint data on the scratch pad. When used in conjunction with a digital key, the WPT key recalls the desired waypoint data from memory to the scratch pad.

11. USE Key:

    Momentary key to transfer displayed data on the scratch pad to the navigation computer and the VOR/DME/localizer receivers. Annunciates active data by blanking the display for 1/2 second before displaying active data.

NOTE

NAV/DME TEST and DME HOLD keys are located on the panel mounted control unit (ANS-31) or the mode select unit (NCS-31). The momentary NAV/DME TEST key only serves to initiate the VOR and DME receivers self-test. The particular system under test should be monitored for proper test indications. These tests will not affect the ANS-31/NCS-31 operation provided the key is not depressed for more than 8 seconds in the enroute mode, or more than 1 second in the approach mode. The DME HOLD key is an on-off pushbutton to hold the DME frequency.

A second panel-mounted component, the remote readout unit, displays the active navigation information in use. Distance or time to the waypoint, waypoint number, computed ground speed, waypoint passage alert, and various navigational modes are displayed on this unit.

1. MILES/MIN Window:

   Displays either the distance or time to or from the waypoint as selected by the MILES/MIN toggle switch.

2. WPT Window:

   a. Displays the active waypoint number in use.

   b. Displays the active VOR/DME frequency when DME HOLD is in use.

3. KTS Window:

   a. Displays the computed ground speed.

   b. Annunciates the VOR-only mode of operation is in use (VOR).

c. Annunciates the localizer mode of operation is in use (LOC).

d. Annunciates computer is in dead reckoning mode of operation (d-r).

e. Annunciates DME HOLD is in use (dh).

4. ALERT Light:

a. Indicates approach of waypoint (steady light).

b. Indicates crossing the TO/FROM line (flashing light).

c. Indicates recovery of valid VOR/DME signal after a prolonged loss in the dead reckoning, enroute mode (flashing light).

d. **(ANS-31A/NCS-31A only):** When depressed, displays bearing to waypoing (RNAV mode) or bearing to station (VOR mode) in place of distance or time to waypoint.

MEMORY FUNCTION

CAUTION

Memory function is intended for a maximum use period of three hours with airplane's main electrical system off. Memory should be turned off if down time is to exceed three hours.

The status of the ANS-31/NCS-31 computer memory is annunciated on the upper scratch pad display of the control unit when avionic power is applied. If the memory has been erased, as would normally be the case when avionic power was last removed, the letters "POC" appear in the WPT and EL 100′ display windows.

Actuation of the remote memory switch prior to removing avionic power will hold the programmed data in the computer memory. Subsequent reapplication of avionic power will confirm the program has been saved by showing the letters "POC" only in the WPT display window. The memory save function permits the computer to be programmed in advance of the flight and held in storage without the airplane's main electrical system activated. A small light adjacent to the memory switch indicates the memory function has been selected.

PREFLIGHT

SELF-TEST

This abbreviated self-test prescribes a procedure to check the ANS-31/NCS-31 prior to flight. Complete self-test procedures for maintenance checks are available in the system maintenance section (523-0765313/523-0765291) of the Collins ANS-31/NCS-31 Navigation Control System Instruction Manual (523-0765309/523-0765286). Abbreviated self-test procedures without fault isolation are available in the Collins ANS-31/NCS-31 Self-Test Guide (523-0765453/523-0765454).

1. Press the TEST button once on the control unit. The figure "8" will appear in all positions of the upper scratch pad.

2. Press the TEST button the second time. The figure "8" will extinguish from all positions and the figure "2" will appear in the WPT display. All other positions will be blank.

3. Press the TEST button the third time.

a. Press the USE button, the numeral "1" key, and the numeral "8" key. The figure "8" will again appear in all positions of the upper scratch pad.

b. Press the USE button, the numeral "2" key, and the numeral "8" key. The figure "8" will appear in all positions of the lower scratch pad.

c. Press the USE button, the numeral "3" key, and the numeral "8" key. The figure "8" will appear in all positions (except position 5) of the remote readout unit.

4. Press the CLR key to exit the self-test program.

CAUTION

Engaging the ANS-31/NCS-31 self-test in flight may disrupt navigation calculations.

AREA NAVIGATION FUNCTIONAL TEST

The following procedure applies only to airports equipped with, or in range of, a colocated VOR/DME station.

1. Place the MILES/MIN switch on the remote readout unit in the MILES position.
2. Press WPT key.
3. Press any number key.
4. Enter the local VOR/DME station frequency and elevation.
5. Press USE key.
6. Adjust the course control knob on the Horizontal Situation Indicator to center the deviation bar.
7. The course arrow on the Horizontal Situation Indicator will point to the local station and the remote readout unit will display the distance.

PROGRAMMING

Pertinent information (waypoint number, station frequency, station elevation, waypoint bearing, and waypoint distance) for up to ten waypoints is entered into memory from the control unit. Programming the computer may be completed prior to take-off or during the flight. Any combination of navigational facilities (RNAV waypoint, VOR/DME, ILS) may be loaded into the computer; however, it is desirable that each facility be numbered and loaded in the sequence it is to be used.

RNAV WAYPOINTS

1. Press the WPT key. One of two display conditions will occur on the control unit.
   a. The display will be blank indicating the absence of an active waypoint. This is a normal display when loading the initial waypoint parameters.
   b. Active waypoint data will appear on the display.
2. Select the first waypoint by pressing the keyboard number "1" key.
   a. If the waypoint has not been previously stored in the memory, only the letter "P" and the waypoint number "1" will appear on the display.
   b. If the selected waypoint has been preset, the letter "P", the waypoint number, and the waypoint parameters will appear on the display.
3. Select the VOR/DME frequency by pressing the keyboard number keys in the proper sequence. A total of five digits must be entered to complete the frequency input (i.e., frequency 113.8 entered as 113.80). Prior data is blanked when the first frequency digit is entered. The letter "P" will blink as long as data displayed is on the scratch pad only (not stored in memory.)

Entries beyond the allowable range of navigational frequencies (108.00 through 117.95 MHz in .05 MHz increments) are annunciated immediately by the letters "CLR" appearing on the right edge of the scratch pad. Further entries are inhibited until the CLR key is pressed to erase the false digit.

4. Select the VOR/DME station elevation in hundreds of feet by pressing the keyboard number keys in the appropriate sequence. Two digits must be entered. Use a leading zero for elevations less than 1000 feet.

5. Successively press the keyboard number keys to select the waypoint bearing (radial) and waypoint distance from the station. All four digits must be entered, using leading zeros as required. Bearing and distance entries are not required when the waypoint is colocated with the VOR/DME station site.

   Entries beyond the allowable range of values for bearing (000.0° through 359.9°) and distance (000.0 through 249.9 nautical miles) are annunciated immediately by the letters "CLR" appearing on the right edge of the scratch pad. Further entries are inhibited until the CLR key is pressed to erase the false digit.

NOTE

If an error is noted during the programming, corrections or revisions of data within the allowable range of values can be made by pressing the CLR key. Data is erased by fields (FREQ, EL 100', BRG, and DIST) in the reverse order of entry each time CLR is pressed. Enter the correct data. Values for fields of correct data that were erased must be reentered.

6. Press the PRE key to place the displayed data into memory. This action will cause the display to go blank.

7. This completes the programming for the first waypoint. Follow these procedures for all selected waypoints up to a maximum of ten.

CONVENTIONAL VOR

The programming technique for conventional navigation directly toward or away from a VOR facility without a colocated DME is similar to that for RNAV waypoints. Inputing the waypoint number and frequency into the memory is accomplished in the same manner. Since the station has no DME, it cannot be electronically "moved" to a new location (waypoint). Therefore, no values are programmed in the EL 100', BRG or DIST displays. Only angular deviation on the Horizontal Situation Display is available in this mode.

ILS APPROACH (Front Course and Back Course)

Programming an ILS approach is accomplished in the same manner as programming conventional VOR. The control unit decodes the frequency as it is entered. Upon detecting the frequency is in the ILS range, the letters "LOC" are annunciated immediately in the EL 100' display thereby inhibiting an elevation entry. Although not required for ILS operation, the localizer bearing (000° through 359°) may be programmed into the BRG display for convenient reference. Only angular deviation is provided in the ILS mode.

MISSED APPROACH

If the published missed approach utilizes an RNAV waypoint or VOR facility, it may be entered into memory any time prior to the approach. It is recommended that WPT "O" (keyboard numeral 0) be reserved for this operation. Any other waypoint storage (1 thru 9) could be used; however, habitual use of WPT "O" eliminates the possibility of error that could be experienced when selecting an intermediate digit during this critical flight phase.

INFLIGHT

Preset waypoints may be recalled from memory and put into active use as required.

1. Press the WPT key. If an active waypoint is displayed on the remote readout unit, the waypoint data will appear on the control unit display. Otherwise, the display will be blank.

2. Press the appropriate number key to select the desired waypoint. The preset waypoint data will replace any active waypoint data on the control unit display. The letter "P" is annunciated adjacent to the waypoint number to indicate that this is not the active waypoint. Information displayed on the remote readout unit, Horizontal Situation Indicator, and signals supplied to the flight control system will continue to reference the active waypoint and selected course.

3. Verify that the displayed data is correct.

NOTE

Revisions to the waypoint data can be programmed at this time by entering the new waypoint parameters. Entry of the first frequency digit blanks the remainder of the display.

4. When reference to the next waypoint is desired, press the USE key. The letter "P" is blanked to indicate that this is now the active waypoint. The Horizontal Situation Indicator NAV flag will momentarily come into view, the deviation signals supplied to the course deviation bar and flight control system will be zero, and the remote readout unit will be blanked until the NAV radios complete retuning the new active waypoint.

5. Select the desired course on the Horizontal Situation Indicator course arrow.

NOTE

Any waypoint may be used without being preset (PRE key) by entering the waypoint data in the normal manner and immediately pressing the USE key. The waypoint data will be put into active use and also stored into memory.

RNAV OPERATION

This is the normal mode of operation. If the VOR/DME radios are receiving valid signals from a colocated VOR/DME station, the ANS-31/NCS-31 computer will supply linear deviation information to the Horizontal Situation Indicator. The ENROUTE sensitivity, available when the flight control system is not in the approach mode, provides a constant course width of ±10 nautical miles. APPROACH sensitivity, available when the flight control system is in the approach mode, provides a constant course width of ±2 nautical miles. APPROACH sensitivity should be used when within ten nautical miles of the terminal waypoint.

Distance or time to the waypoint, waypoint number, and computed groundspeed are displayed on the remote readout unit. The ANS-31/NCS-31 computer combines inputs from the encoding altimeter with the VOR/DME station elevation to correct DME slant range error.

NOTE

The RNAV mode of operation requires the programming of station elevation to correct DME slant range error. Operation in this mode is recommended even if navigating directly toward or away from a VOR/DME facility. This provides the advantages of linear deviation and smooths the received signals to improve autopilot operation.

CONVENTIONAL VOR OPERATION

This is the mode of operation when either DME is not available or the DME is not colocated with the desired VOR facility. The VOR mode is annunciated by the letters "VOR" appearing on the control unit display in place of station elevation, and on the remote readout unit in place of ground speed. Raw DME distance will be displayed on the remote readout unit if a valid DME signal is received. However, slant range correction and computed ground speed will not be available. The ANS-31/NCS-31 computer supplies angular deviation information to the Horizontal Situation Indicator.

ILS OPERATION (Front Course and Back Course)

This is the mode of operation when the navigation receiver is tuned to a localizer frequency. The localizer mode is annunciated by the letters "LOC" appearing on the control unit display in place of station elevation, and on the remote readout unit in place of ground speed. Raw DME distance will be displayed on the remote readout unit if a valid DME signal is received. It is essential that only the inbound front course localizer bearing be set on the Horizontal Situation Indicator for both front course and back course approaches. This will assure the Flight Director display and autopilot maintain the proper left/right logic. Only angular deviation information is provided in the ILS mode.

DEAD RECKONING OPERATION

The ANS-31/NCS-31 will automatically enter the dead reckoning mode from either the enroute or approach RNAV mode whenever the VOR or DME signal is lost, or when passing over the VOR/DME station being used for navigation. Navigation calculations will continue using the ground speed and wind values available at the time the dead reckoning mode is entered. Changes in ground speed or wind velocity while in the dead reckoning mode will result in degradation of the accuracy of position estimates.

The loss of the VOR or DME signal for less than 9 seconds in the enroute mode or 1 second in the approach mode will not affect normal operation.

When operating in the enroute mode, loss of signal for more than 9 seconds forces the ANS-31/NCS-31 into dead reckoning. The dead reckoning mode is annunciated by displaying the letters "d-r" in place of ground speed on the remote readout unit. Recovery of the signal after 9 seconds but before 72 seconds returns the ANS-31/NCS-31 to the enroute mode and replaces the letters "d-r" with the normal ground speed display. If an invalid signal condition exceeds 72 seconds, the NAV flag on the Horizontal Situation Indicator will come into view and automatic reentry to the enroute mode will be inhibited.

Signal recovery after the NAV flag has been displayed is indicated by the ALERT light flashing. Normal operation may be regained by pressing the WPT key, the desired waypoint number key, and the USE key.

NOTE

A flashing ALERT light may also indicate crossing the TO/FROM line. This is verified by a zero distance or time to the waypoint displayed on the remote readout unit. Press the WPT key to extinguish the light.

When operating in the approach mode, loss of signal for more than 1 second forces the ANS-31/NCS-31 into dead reckoning. The letters "d-r" will again appear on the remote readout unit. Loss of signal in excess of 9 seconds causes the NAV flag on the Horizontal Situation Indicator to come into view. Recovery of the signal at any time returns the ANS-31/NCS-31 to the normal approach mode of operation and replaces the letters "d-r" with ground speed.

When operating under conventional VOR (including DME HOLD) conditions, the system will not enter dead reckoning in the event of an invalid signal. However, the NAV flag will be displayed and the annunciation "VOR" on the remote readout unit will be blanked.

WAYPOINT ALERT

Active waypoint approach is annunciated by an illuminated ALERT light on the remote readout unit when within 24 seconds flying time from the waypoint. This feature is available only in the enroute and approach RNAV modes of operation.

Crossing the TO/FROM line is indicated by a flashing ALERT light and reversal of the TO/FROM arrow on the Horizontal Situation Indicator. The ALERT light will automatically extinguish 24 seconds after crossing the TO/FROM line or it may be manually extinguished by pressing the WPT key.

DME HOLD OPERATION

The DME HOLD function inhibits changing the DME receiver frequency. Engaging DME HOLD and then selecting a new waypoint forces the ANS-31/NCS-31 into either a conventional VOR or LOC mode of operation according to the newly selected frequency.

If the waypoint to be selected is a conventional VOR or LOC waypoint, engage the DME HOLD as follows:

1. Press the DME HOLD key.
2. Select the new waypoint data on the scratch pad by pressing the WPT key and the appropriate waypoint number key.
3. Press the USE key once. The upper scratch pad of the control unit will display the letter "P", waypoint number, frequency and the letters "VOR" or "LOC". The lower scratch pad will display the active frequency on which the DME is to be held and the letters "dh" flashing on and off.
4. Verify the displayed data.
5. Press the USE key the second time. The NAV receiver will be tuned to the new waypoint frequency. The DME will remain tuned to the previously active frequency. The held DME frequency and the letters "dh" will be displayed steadily on the lower scratch pad, and also will appear on the remote readout unit in the place of waypoint number and computed ground speed. Raw DME distance to the held DME facility will be displayed on the remote readout unit.

NOTE

Only VOR radials may be flown with angular deviation provided.

Releasing the DME HOLD key will tune the DME receiver to the active NAV frequency. The lower scratch pad on the control unit will be cleared. Raw DME distance (if the signal is valid), waypoint number, and the letters "VOR" or "LOC" will appear on the remote readout unit.

CAUTION

The DME HOLD function should not be used when navigating between RNAV waypoints. These waypoints require valid signals from colocated VOR and DME facilities to establish their geographical positions. If the VOR and DME receivers are not tuned to a colocated facility, the DME HOLD function will cause raw DME distance to the held facility to be displayed on the remote readout unit and angular deviation to the VOR facility on the Horizontal Situation Indicator.

PERFORMANCE - No change

Approved: Donald H Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR 200 AND B200 LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the CAMERA WELL INSTALLATION

GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* when the airplane has been modified for special photographic purposes by installation of a forward camera well, an aft camera well, or both a forward and an aft camera well in accordance with Beech-approved data.

The information in this Supplement supersedes or adds to the basic *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the Handbook are advised always to refer to the Supplement for possibly superseding information and placarding applicable to operation of the airplane.

The camera well installation results in a modification of the bottom contour of the fuselage, in order to accommodate the optical glass camera ports and port covers or doors.

The airplane is eligible for certification in the Restricted Category for special photographic purposes when it is equipped with a Beech-installed camera well installation and an appropriate Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement is attached to the Handbook.

The airplane has been issued a Multiple Airworthiness Certificate permitting operation in the Restricted Category under FAR 21.187.

LIMITATIONS

OPERATION UNDER RESTRICTED CATEGORY AIRWORTHINESS CERTIFICATION

Operation shall be in accordance with the provisions and limitations of FAR 91.39 (Restricted Category Civil Aircraft; Operation Limitations) anytime the airplane is to be flown above 25,000 feet with the cabin pressurized and any internal camera well cover removed.

The airplane shall be identified "RESTRICTED", per FAR 45.23(b).

Anytime an internal camera well cover is removed, a camera shall be installed in the camera well for pressurized flight. The cabin shall be unpressurized while removing or exchanging camera in flight.

Occupancy shall be restricted to those persons specified in FAR 91.39(c). Personnel required for performance of photographic missions qualify.

Each time the airplane is operated in the Restricted Category, a post-flight inspection of the camera installation shall be made to assure camera security, and to assure that no chips or cracks are present in the optical glass, and a record of the inspection shall be made.

Maximum cabin pressurization shall be 6.1 p.s.i.

OPERATION UNDER NORMAL CATEGORY AIRWORTHINESS CERTIFICATION

The internal camera well cover(s) (identified by part number) shall be installed for all flight.

Maximum cabin pressurization shall be 6.1 p.s.i.

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EMERGENCY PROCEDURES

No Change

NORMAL PROCEDURES

PREFLIGHT INSPECTION

If the camera well installation is equipped with external sliding doors, the doors should be closed before starting the engines, in order to protect the optical glass from flying debris.

If the camera well installation is equipped with external bolt-on covers, they should be securely in place prior to a non-photographic mission, and removed prior to a photographic mission.

TAKEOFF, CLIMB, CRUISE, DESCENT, APPROACH, AND LANDING

On airplanes equipped with external sliding doors below the camera wells, the doors should remain closed except during the actual photographic session, in order to protect the optical quality of the glass against unnecessary exposure to possible degradation.

PERFORMANCE

Charts Affected	Forward Camera Well Only	Aft Camera Well Only; OR Dual (Both Forward and Aft) Camera Wells
Takeoff and Landing	No Change	No Change
Climb Performance	No Change	No Change
Cruise and Range Values	Decrease 2%	Decrease 4%

Approved: Donald H Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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BEECHCRAFT SUPER KING AIR 200, 200C, B200 & B200C LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

SPERRY SPZ-200A/STARS IV D AUTOMATIC FLIGHT CONTROL SYSTEM or SPERRY SPZ-200A/SPI-80/81 AUTOMATIC FLIGHT CONTROL SYSTEM or SPERRY SPZ-200A/SPI-400/500 SERIES AUTOMATIC FLIGHT CONTROL SYSTEM
CATEGORY I

GENERAL

The information in this supplement is FAA-Approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Sperry SPZ-200A/STARS IV D or SPI-80/81 or SPI-400/500 Series Automatic Flight Control System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. During autopilot operations, pilot must be seated at the controls with seat belt fastened.
2. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
3. Do not use autopilot under 200 feet above the terrain.
4. Do not use autopilot or yaw damper during takeoff or landing.
5. Do not use propeller in the range of 1750-1850 rpm during coupled ILS approach.
6. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
7. The accuracy of the VNAV system has not been shown to meet the requirements of advisory circular 90-45A and is therefore not to be used as the primary source of vertical guidance information.

EMERGENCY PROCEDURES

If the AP TRIM FAIL and MASTER WARNING annunciators illuminate while the autopilot is engaged, immediately disconnect while restraining the control wheel for a possible out of trim force.

IN THE EVENT OF AN AUTOPILOT MALFUNCTION, disengagement can be accomplished by one of the following:

1. Pressing (momentarily) to the first level the pilot's or copilot's AP/YD & TRIM DISC button on the control wheel.
2. Pressing (momentarily) the go-around button on the left power lever.

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3. Pressing the TEST button on the Autopilot Engage Controller.

4. Opening the autopilot circuit breaker.

5. If necessary, the autopilot may be overpowered by either pilot.

IN THE EVENT OF AN ENGINE FAILURE:

1. Disengage the autopilot, retrim the airplane and re-engage the autopilot. Maintain at least 120 knots for one-engine-inoperative approach.

MAXIMUM ALTITUDE LOSSES DURING MALFUNCTION TESTS WERE:

CONFIGURATION	ALTITUDE LOSS
Climb	250 feet
Cruise	410 feet
Maneuvering	100 feet
Descent	500 feet
Approach/ILS Coupled	80 feet
One-engine-inoperative Approach/ILS Coupled	50 feet

NORMAL PROCEDURES

FLIGHT DIRECTOR OPERATION

The pilot can select any of the following flight modes using the flight director mode selector pushbuttons which provide both the flight director and autopilot mode control and annunciation. All modes on the mode selector have push on - push off capability. When no mode has been selected, the command bars on the Attitude Display Indicator (ADI) are retracted from view.

1. Standby (SBY): In this mode, flight director is ready for operation of the other modes. The command bars on the Attitude Display Indicator are retracted from view. Depressing the SBY pushbutton tests all flight control system annunciators and resets the flight director mode.

2. Heading Select (HDG): This mode provides lateral commands on the Attitude Display Indicator command bar to acquire and maintain the heading displayed on the heading cursor on the Horizontal Situation Indicator. The desired heading is selected by turning the heading select knob.

3. Low Bank: This mode, used only with Heading Select, provides reduced bank-angle commands on the Attitude Display Indicator command bar. The pushbutton LOW BANK switch is located on the autopilot controller.

4. VOR/RNAV/Front Course Localizer (NAV): Selecting NAV with the radio tuned to a VOR or localizer frequency results in the NAV ARM light illuminating. The desired radial or inbound course is set on the Horizontal Situation Indicator course selector by turning the course knob. When NAV is selected with the airplane outside the edge of the beam, the HDG mode is automatically selected. The heading select cursor can be used to set the intercept angle to the beam. When the capture point is reached, the HDG mode drops out, the NAV mode goes from ARM to CAP, and the flight director command bars provide commands to track the VOR radial or localizer beam. Crosswind correction is automatically provided.

5. ILS Approach (APR): Selecting APR with the radio tuned to a localizer frequency will result in both the APR ARM and NAV ARM lights illuminating which indicate glideslope and localizer signals respectively are armed. The inbound course should be set on the Horizontal Situation Indicator course selector. The heading mode will be automatically selected if the airplane is outside the edge of the localizer beam. An appropriate intercept angle should be set with the heading select cursor. When the localizer is captured, the HDG mode drops out and the NAV mode goes from ARM to CAP. The APR mode goes from ARM to CAP upon intercepting the glideslope. Glideslope capture will not occur until after the localizer has been captured. If ALT, VS or IAS modes were selected, they will drop out at glideslope capture.

6. Back Course Localizer (BC): This mode provides lateral commands on the flight director to fly back course approaches or outbound on the front course. The inbound front course should be set on the Horizontal Situation Indicator course selector. Lateral beam sensor operation is as described in the NAV mode.

7. VOR/RNAV Approach (VOR APR): Selecting VOR APR with the radio tuned to a VOR frequency results in the VOR APR ARM and NAV ARM lights illuminating. Course intercept and capture operation is as described in the NAV mode. When the course capture is accomplished, the ARM lights will extinguish and the VOR APR CAP and NAV CAP lights will illuminate. This mode provides optimum gains for the VOR/RNAV approach.

8. Altitude Hold (ALT): This mode provides pitch commands on the flight director to maintain the engaged altitude. If the autopilot is engaged, selection of the ALT mode should be made at vertical speeds less than 2000 feet per minute.

9. Altitude Preselect (ALT SEL): This mode operates in conjunction with the Altitude Alert Controller on which the desired altitude is set. Pressing ALT SEL will illuminate the ARM annunciator indicating the flight director is armed to automatically capture the selected altitude. Pitch hold, IAS or VS modes may be used to fly to the selected altitude. When the difference between the airplane's altitude and the selected altitude is approximately one quarter of the vertical rate, the ARM and any selected vertical mode annunciators will extinguish and the CAP annunciator will illuminate. The flight director will command a programmed flare to the selected altitude. When the altitude is reached, the altitude hold mode automatically engages and the CAP annunciator will extinguish.

10. Vertical Speed (VS): The VS mode maintains the existing vertical speed at the time of selection through pitch commands on the flight director.

11. Airspeed Hold (IAS): The IAS mode is used to maintain a constant indicated airspeed by controlling pitch attitude. The IAS mode is selected when the airplane is at the desired airspeed. The flight director commands pitch attitude changes to maintain the selected airspeed.

12. Go-Around: This mode provides commands to the flight director when an approach is to be terminated. A fixed pitch-up, wings-level command of 7 degrees is presented. The GO-AROUND mode cancels all other modes and is selected by depressing the go-around switch on the left power lever. Selection of the GO-AROUND mode will disengage the autopilot; however, the yaw damper will be retained.

    After GO-AROUND is selected, any roll mode can be selected and will cancel the wings level roll command. The GO-AROUND mode is cancelled by either selecting another pitch mode, selecting TCS or engaging the autopilot.

13. Vertical Navigation (VNAV): This optional mode operates in conjunction with the optional VNAV computer/controller to compute and display pitch commands to capture and maintain a vertical track angle in ascent or descent to a selected waypoint or VORTAC facility.

AUTOPILOT FEATURES

1. Autopilot Controller: The autopilot controller provides the means of engaging the yaw damper and autopilot. The controller also contains a pitch wheel and turn knob for manual control of the autopilot.

2. Yaw Damper Engage: The yaw damper is engaged by depressing the Y/D ENGAGE button on the autopilot controller. The YAW DAMPER may be engaged independently of the autopilot to provide yaw stabilization.

3. Autopilot Engage: When the A/P ENGAGE button is pressed the yaw damper is engaged and, with no modes selected on the Flight Director mode selector, pitch attitude is held, roll attitude is brought to zero, and airplane heading is maintained.

NOTE

To engage the autopilot, the yaw damper must be operable.

4. Elevator Trim Annunciator: The elevator trim light illuminates when a sustained signal is being applied to the elevator servo. IF EITHER THE DN OR UP TRIM ANNUNCIATOR IS ILLUMINATED, THE AUTOPILOT SHOULD NOT BE ENGAGED.

5. Soft Ride Mode: Selection of the SOFT RIDE mode on autopilot controller reduces autopilot response in both roll and pitch axes. This mode should be used for turbulence penetration or any other time that softer response is desired.

6. Turn Knob: Rotation of the turn knob out of its detent results in a roll command proportional to, and in direction of, the turn knob rotation. If HDG, NAV, APR, BC, or VOR APR is on the flight director mode selector, rotation of the turn knob cancels the mode. These modes cannot be reselected and the autopilot cannot be engaged until the turn knob is in its detent.

7. Pitch Wheel: Rotation of the pitch wheel results in a change of pitch attitude proportional to the rotation of the pitch wheel and in the direction of wheel movement. If IAS, ALT, VS, ALT SEL CAP, or VNAV CAP is on the flight director mode selector, rotation of the pitch wheel cancels the mode.

8. Touch Control Steering: When the touch control steering (TCS) switch on the control wheel is depressed, the elevator and aileron axes of the autopilot will disengage, the AP ENGAGE annunciator will extinguish, and the pilot can manually control the airplane. When the TCS switch is released, the autopilot will re-engage and illuminate the AP ENGAGE annunciator. TCS can only be used with the turn knob in the center detent.

   If no mode has been selected on the flight director, the existing pitch attitude will be maintained when the TCS switch is released. If the airplane roll attitude is more than 6 degrees, the roll attitude will be maintained. If the roll attitude is less than 6 degrees, the airplane heading will be maintained.

   The TCS feature also allows the pilot to modify the commanded flight path from the flight director computer. When coupled to ALT, VS, IAS, or APR, touch control steering can be selected and the altitude, vertical speed, airspeed or position on the glideslope can be manually changed through pitch attitude or power changes. Upon release of the TCS switch, the new reference will be held or the autopilot will recouple to APR. If the autopilot is coupled to the roll mode, TCS allows maneuvering in roll while the switch is depressed. Upon release, the autopilot will couple to the previously selected lateral mode.

NOTE

If the ALT hold mode is engaged, vertical speed should be less than 2000 feet per minute prior to release of the TCS switch.

AUTOPILOT COUPLING TO FLIGHT DIRECTOR

The autopilot uses the Flight Director computer for autopilot commands. Whenever autopilot is engaged, it will fly the roll and/or pitch mode selected except for the GO-AROUND mode. When the autopilot is engaged and a roll mode is on the flight director, operation of the turn knob will cancel the selected roll mode. When the autopilot is engaged with a pitch mode on the flight director, operation of the pitch wheel will cancel the selected pitch mode. If a roll mode is selected prior to moving the pitch wheel, the command indicator will synchronize to the existing attitude.

The following modes on the flight director are also autopilot modes:

HDG - Heading Select
NAV - VOR, RNAV or Front Course Localizer Beam Tracking
APR - Front Course ILS Beam Tracking
BC - Back Course Localizer Beam Tracking
VOR APR - VOR or RNAV Approach
ALT - Altitude Hold
ALT SEL - Altitude Preselect
VS - Vertical Speed Hold

IAS - Airspeed Hold
VNAV - Vertical Navigation
LOW BANK

If no mode is selected, the autopilot will fly heading hold and pitch hold. When GO-AROUND mode is selected, the autopilot will disengage; however, the yaw damper will remain engaged. Re-engagement of the autopilot when the Flight Director is in the GO-AROUND mode results in a pitch attitude hold command for the autopilot. If a Flight Director mode is previously selected, the autopilot will couple to the selected mode.

AUTOPILOT OPERATION

PREFLIGHT

1. Elevator Trim Annunciator - CHECK (Observe that autopilot trim light on autopilot controller is not indicating UP or DN. A steady UP or DN light denotes automatic synchronization is not functioning and the autopilot should not be engaged.)

2. Turn Knob - IN CENTER DETENT POSITION.

3. Autopilot - TEST

   a. Control Wheel to mid-travel - DEPRESS AP ENGAGE ANNUNCIATOR SWITCH

   b. Control movement - CHECK (that the system can be overpowered by slowly moving the controls of all three axes.)

CAUTION

If autopilot disengages, do not use.

   c. Elevator Trim followup - CHECK

      (1) Hold control wheel forward of mid-travel, trim wheel will run nose up after approximately 2 seconds. Actuate control wheel trim switches to nose down. AP TRIM FAIL and MASTER WARNING annunciators will illuminate.

      (2) Hold control wheel aft of mid-travel. AP TRIM FAIL annunciator will extinguish and trim wheel will run nose down after approximately 2 seconds. Actuate control wheel trim switches to nose up. AP TRIM FAIL and MASTER WARNING annunciators will illuminate.

   d. AP/YD & TRIM DISC Button - DEPRESS THROUGH SECOND LEVEL (Autopilot will disengage and ELEC TRIM OFF annunciator will illuminate.

   e. Re-engage Autopilot.

   f. Autopilot TEST Button - DEPRESS (Autopilot will disengage and AP DISC and MASTER WARNING annunciators will illuminate.)

CAUTION

If autopilot does not disengage when the test button is depressed, it indicates autopilot torque monitors are not functioning properly. DO NOT USE AUTOPILOT IN FLIGHT UNTIL CORRECTIVE ACTION HAS BEEN TAKEN.

   g. Annunciators - CLEAR (AP DISC annunciator will extinguish by depressing the control wheel AP/YD & TRIM DISC button and the MASTER WARNING annunciator will extinguish by depressing its face.)

h. ELEV TAB CONTROL Switch - OFF, then ON (resets electric trim and ELEC TRIM OFF annunciator will extinguish.)

i. Elevator Trim - RESET AS REQUIRED.

IN-FLIGHT ENGAGEMENT OF AUTOPILOT

1. All autopilot and Flight Director Circuit Breakers - IN
2. Elevator Trim Indicator - CHECK (Observe that autopilot trim annunciators on autopilot controller are not illuminated.)
3. Turn Knob - IN CENTER DETENT POSITION
4. Autopilot Controller AP ENGAGE Annunciator Switch - DEPRESS

DISENGAGING THE AUTOPILOT

The autopilot may be disengaged by:

1. Actuation of the AP/YD & TRIM DISC buttons on either control wheel to the first level. (Copilot's button causes MASTER WARNING and AP DISC annunciators to illuminate.)

NOTE

The AP/YD & TRIM DISC button is a two-level switch. The first level disengages the autopilot while the second level disengages both autopilot and electric trim.

2. Pressing the TEST button on the Autopilot Engage Controller. (MASTER WARNING and AP DISC annunciator will illuminate.)
3. Pressing the go-around mode switch on the left power lever. (AP DISC annunciator will illuminate.)

NOTE

The AP DISC annunciator light is extinguished by depressing the pilot's AP/YD & TRIM DISC button. The MASTER WARNING annunciator is extinguished by depressing its face.

DISENGAGING THE YAW DAMPER

The yaw damper may be disengaged by actuation of the AP/YD & TRIM DISC button on either control wheel to the first level.

PREFLIGHT ALTITUDE ALERTING

The alerting sequence can be verified by the following procedure:

1. Set the altitude on the altitude alert controller for 1500 feet above the altitude on the pilot's altimeter.
2. Using the baro set on the pilot's altimeter, adjust altitude toward the alert controller reading. At 1000 feet from the desired altitude, the alert light on the altimeter will illuminate and the alert tone will sound.
3. Within 250 feet of selected altitude, the alet light will extinguish.
4. Then using the baro set to return to the original altitude, the alert light will illuminate and the tone will sound when the altimeter has deviated over 250 feet from that on the alert controller. At 1000 feet from the selected altitude, the alert light will extinguish.

VERTICAL NAVIGATION OPERATION (optional)

The VNAV computer/controller (VNCC) provides for both altitude alerting and vertical navigation.

Altitude Alerting: The desired altitude is selected by placing the VNCC selector switch to ALT and setting the display to the desired altitude using the SET knob. This altitude can be attained by using either the altitude preselect (ALT SEL) or vertical navigation (VNAV) modes of the flight director.

Vertical Navigation: To utilize the VNAV for capturing and maintaining a vertical track to the selected altitude, the VORTAC station elevation (STA EL), along track offset distance (TO/FR), and vertical angle (VANG) must be set after selecting the desired altitude.

STA EL: Station elevation is set in 100 foot increments by selecting the STA EL position and using the SET knob. When using RNAV in conjunction with VNAV, it is only necessary to clear the dashes from the display by setting any arbitrary STA EL such as zero (000) unless the RNAV system does not correct for slant range error.

TO/FR: The TO and FR settings, respectively represent the distance prior to or beyond a VORTAC or waypoint at which a desired altitude is to be reached. These distances are set in tenths of nautical miles up to a maximum of 35.0 miles on the -902 VNCC or 99.9 miles on the -906 VNCC.

VANG: If ALT and STA EL parameters have been set, selection of the VANG position results in the display of the vertical angle from present position to the selected altitude at the offset distance. The display can also be set to a desired vertical angle in tenths of a degree up to a maximum of 6 degrees using the set knob unless the Flight Director VNAV mode has been selected. This display will revert to computed angle if an attempt is made to set an angle less than the computed angle.

After the above values have been set, the Flight Director VNAV mode may be used to either capture the displayed vertical angle or if a desired angle has been set, to arm and capture the desired angle when it is reached. The Flight Director will then display commands to follow the vertical angle so as to level off at the selected altitude at the specified along track offset distance. The VNCC supplies a vertical deviation signal similar to a glideslope signal which is displayed on the course indicator. When the selected altitude is reached, the STA EL, TO/FR, and VANG settings revert to dashes to prevent erroneous VORTAC data from being used on subsequent VNAV uses. Furthermore, should the computed angle reach the preset maximum, the above displays will be cancelled in the same manner. This automatic cancellation will be preceded by flashing of the display for the last degree of computed vertical path angle.

Proper operation of the VNAV mode requires that the distance information to the VNAV corresponds with the VORTAC (or waypoint) utilized for navigation. Always assure this requirement is satisfied prior to engaging the VNAV mode and that the VNAV mode is disconnected prior to changing the navigation mode coupled to the flight director.

NOTE

DME (or waypoint) invalid renders the VNAV inoperative during the invalid condition, but does not disconnect the VNAV mode.

PERFORMANCE - No Change

Approved: Donald St. Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT, KING AIR E90, C90 (LJ-668, thru LJ-1062, EXCEPT LJ-670), AND C90A (LJ-1063 and after) LANDPLANES
PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

BENDIX RNS 3500 AREA NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the *FAA Approved Airplane Manual* when the airplane has been modified by installation of the Bendix RNS 3500 Area Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic *FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. The Area Navigation Function may not be used as a primary system under IFR conditions except on approved approach procedures, approved area navigation airways, and random area navigation routes when approved by Air Traffic Control.

2. The Area Navigation Function can only be used with co-located facilities. (VOR and DME signals originate from the same geographical location.)

3. The Area Navigation installation located on the right instrument panel may be used for primary navigation, only if qualified copilot occupies the right seat.

4. The Bendix RNS 3500 Area Navigation System must have a minimum electrical power supply of 23.0 volts to function properly and to provide accurate information.

5. Waypoints are limited to a maximum distance from the VOR/DME ground station of 199.9 nautical miles.

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude and angle of bank.

1. If NAV flag appears while in the enroute mode, check for correct frequency.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

3. If NAV flag appears during an approach while in the APR mode, execute published missed approach and utilize another approved facility.

NORMAL PROCEDURES

The RNS-3500 Area Navigation System, is an airborne system that provides a capability for navigating and flying an airplane on any desired course within the coverage of co-located VOR/DME (TACAN) ground stations. The RNS-3500 system has been designed to compliment the airplane NAV system.

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For Training Purposes Only

The RNS-3500 system is a station oriented two-dimensional navigation device whose primary position determining data sources are: (1) barometric encoded altitude, (2) VOR bearing, and (3) DME distance relative to a selected VORTAC or colocated VOR/DME station installed at some specified elevation above mean sea level.

When operated within the service volume of a colocated VOR/DME ground station, this system will provide position information relative to any desired or specified location defined as a waypoint. This way point is sometimes referred to as a phantom station. The position of the way point is arbitrary and is defined in terms of bearing and distance from the reference ground station. Area navigation is accomplished by continuously solving the horizontal navigation triangle. In addition, the RNS-3500 system contains circuitry that corrects for the slant range DME distance to provide the true ground distance to the VOR/DME station.

Correcting the slant range error places the navigation triangle in the horizontal plane which permits the RNS-3500 system to operate at all altitudes and distances within the service volume of any selected VOR/DME (VORTAC) facility.

The position information computed by the RNS-3500 system is presented in terms of bearing and distance to or from the waypoint. This information is provided in terms of deviation from a selected track through the waypoint in the horizontal plane determined by the OBS setting.

Linearized deviation signals are proportional to cross-track error in the horizontal plane at any distance from the waypoint and are developed for display on conventional Horizontal Situation/Course Deviation Indicators and for use by autopilot and flight director systems.

In addition to outputs for flight instruments and flight control systems, the RNS-3500 system provides outputs to drive the Radio Magnetic Indicator (RMI) and external distance indicators. The system also has the capability of providing for a pilot display of such items as ground speed, time, distance, and bearing, all of which are relative to the waypoint. Remote mode and alert annunciator drive capabilities are additional features of the RNS-3500 system.

CONTROL FUNCTIONS

MODE SWITCH:

Controls the mode of operation (off, VOR/LOC, RNAV, or APR) for the system.

(T) TEST PUSHBUTTON:

Used to initiate a comprehensive system test.

DISPLAY SWITCH:

Controls selection of readout display.

SBY:

The parameters for the standby waypoint are displayed and may be reprogrammed as desired.

ACT:

The parameters for the active waypoint are displayed.

BRG/DST:

Displays bearing and distance to the waypoint if the MODE selector is in the RNAV or APR position. Displays bearing and distance to the VOR/DME (VORTAC) station if the MODE selector is in the VOR/LOC position.

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KTS/TME:

Displays ground speed and time relative to the waypoint in the RNAV or APR mode or to the VOR/DME (VORTAC) station in the VOR/LOC mode.

V Pushbutton:

Depressing the V pushbutton reverts RMI and remote distance display readouts to NAV receiver and DME system to provide VOR/DME station orientation without interrupting RNAV guidance to HSI and autopilot.

ADDRESS PUSHBUTTONS:

SBY:

Depressing the standby (SBY) address pushbutton allows the SBY W/P number to be changed from 0 through 31. Use of this pushbutton is required only when displaying SBY W/P parameters and then only if another address pushbutton has been depressed; otherwise, the SBY W/P number is constantly addressed as indicated by the W/P legend being illuminated.

EL:

Depressing the station elevation (EL) address pushbutton when the DISPLAY selector is set to SBY, causes the EL legend to flash indicating that the station elevation of the SBY W/P can be altered.

BRG:

Depressing the bearing (BRG) address pushbutton when the DISPLAY selector is set to SBY, causes the BRG lamp to flash indicating that the bearing of the SBY W/P can be altered.

DST:

Depressing the distance (DST) address pushbutton when the DISPLAY selector is set to SBY, causes the DST lamp to flash indicating that the distance to the SBY W/P can be altered.

NOTE

After SBY, EL, BRG, or DST information has been inserted the associated legend will continue to flash until another address pushbutton is depressed or the position of the DISPLAY selector is changed.

PADDLE SWITCHES:

There are four paddle switches each of which is associated with a specific vertical column of 7-segment indicators. The left most paddle switch controls the left most digit in the EL, BRG, or DST display. The second paddle from the left controls the second column of digits, etc. The up/down action of the switch permits incrementing or decrementing of the digit in the addressed display window. Holding the paddle switch actuated in the up or down position slews the associated digit in an increasing or decreasing count direction.

NUMERIC DISPLAY:

Seven-segment numeric displays that illuminate to indicate W/P number, elevation, bearing, distance, speed, and time data.

SBY W/P:

Two digit display that indicates the standby waypoint number.

ACT W/P:

Two digit display that indicates the active waypoint number.

EL:

Single display that indicates waypoint station elevation to the nearest thousand feet.

BRG:

Three digit display that indicates bearing to waypoint or station.

DST:

Four digit display that indicates distance to waypoint or station.

KTS:

Three digit display that indicates ground speed in nm/hr.

TME:

Four digit display that indicates time to waypoint in minutes and tenths of minutes.

AMBIENT LIGHT SENSOR:

Intensity of all seven-segment numeric displays and logic controlled legends is automatically controlled by an ambient light sensor and its associated circuitry. The legends are W/P, EL, KTS, BRG, TME, and DST.

PANEL LIGHTING:

There are integral panel lamps that provide halo lighting for display address pushbuttons and back lighting for the MODE and DISPLAY controls.

The light intensity for these controls is set by the instrument panel light dimmer control.

PROGRAMMING

Pertinent information (elevation, bearing, distance and waypoint number) for up to thirty two waypoints is entered into memory from the panel mounted unit. Programming the computer may be completed prior to take-off or during flight. The computer memory is non-volatile and program data is not lost with a power loss. Any combination of navigational facilities (RNAV waypoint VOR/DME, ILS) may be loaded into the computer; however, it is desirable that each facility be numered and loaded in the sequence it is used.

Turn the MODE control to VOR/LOC. If the seven-segment numeric displays illuminate, power is being applied to the equipment. If displays are not illuminated, check the circuit breaker and input power.

Check that VOR, DME and barometric altimeter are on and operating. Tune VOR and DME to the same ground station or to the frequency of the portable VOR/DME test sets.

Set MODE to VOR/LOC, RNAV, or APR. Select SBY W/P number for which parameters are to be altered. Set DISPLAY to SBY.

Select values for way point parameters (elevation, bearing, and distance). Depress EL address pushbutton; insert station EL by actuating paddle switch. Depress BRG address pushbutton; insert station BRG by actuating paddle switches. Depress DST address pushbutton; insert station DST actuating paddle switches.

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PREFLIGHT OR SYSTEM TEST

This is a comprehensive system test of the RNS-3500 Area Navigation System. It tests associated airplane wiring and interface circuits, including the NAV, DME, ALT, HSI, and optional equipment such as RMI, remote distance display and annunciators.

Place autopilot in heading hold mode.

NOTE

The RNS-3500 provides a signal which causes some flight control systems to automatically revert to the heading hold mode whenever the test (T) is initiated.

Tune VOR and DME to the same frequency. Display the active W/P data by setting the display control to the active position, to show the parameters to be tested.

Press the test (T) pushbutton and hold for the duration of the test.

Lighting Test (2 seconds)

All displays will be lighted to 8's and all amber panel legends will be dashed (center lamp segment lighted). The HSI nav flag will be displayed, and the remote distance indicator will indicate a flagged condition.

Computed BRG/DST Test

For the remainder of the test (until the pushbutton is released) the active W/P bearing and distance parameters are computed and displayed. The computed values should agree with the active W/P bearing and distance within normal system tolerances. The bearing accuracy should be within ±2° and distance within ±1.5% or ±0.3 NMI. A W/P with a distance of 10 nm or more should be used. Accuracy will be degraded slightly when no nav signal is present. In the RNAV or APR mode the optional remote distance indicator and RMI will repeat the computed bearing and distance. The RMI (optional) will point to the active W/P bearing.

Left-Right, To-From Test

With the test button still depressed, turn the CRS knob to center the L-R needle with a "To" indication. The CRS should point to the active W/P bearing within 2° with a signal, or 10° without a signal. Turn the CRS knob to center the L-R needle with a "From" indication. The CRS should point to the reciprocal of the active W/P bearing within the above tolerances.

CONVENTIONAL VOR OR CO-LOCATION OF WAYPOINTS AND VORTACS

The constant course width feature of RNAV can be used when flying conventional VOR airways. Set the mode control the RNAV and the active bearing and distance parameters to zero. This method will locate the waypoint at the same location as the vortac. Linear needle displacement may be flown. During flight, the direction and true ground distance to the vortac will be continuously displayed on the control display unit. The Left/Right indicator will indicate the distance in nautical miles left or right of the airway.

APPROACH PROCEDURES

Flying the RNS-3500 system for an approach is similar to making a localizer approach. However, the system is using VOR and DME information and the MDA will be higher than when flying a precision approach. Insert the waypoint parameters from the approach chart. These parameters must be taken from an approved RNAV approach procedure. Place the mode selector in the APR mode. Select the inbound heading on the course indicator of the HSI. Activate each successive W/P by depressing the transfer switch as each W/P is passed. Similar to a localizer approach, the left/right needle movement will be more active than enroute because full needle deflection is equal to 1.25 nm when in the APR mode. If landing cannot be initiated upon reaching MDA, follow the missed approach procedure outlined on the approved plate, using additional waypoints as required.

IN–FLIGHT PROCEDURES

Flying the RNS-3500 system enroute corresponds to flying VOR airways, except flying is now to or from waypoints. The waypoint parameters move the VORTAC. Once this is accomplished, the HSI, OBS, remote DME indicator and RMI functions are the same as the pilot has been using for VOR and DME. Insert the waypoint parameters as explained in the programming procedures. Waypoint parameters may be taken from RNAV enroute charts, or plotted and measured on other charts. In either case, it is important to assure that altitude and distance from the VORTAC will result in a Satisfactory NAV/DME signal. Random waypoints may be used only during VFR conditions. Select the proper VORTAC frequency. Place mode selector in RNAV mode. Select desired course to first waypoint on the HSI. Pilot must select the next outbound course at each waypoint and the next inbound course and waypoint at each change-over point. Transfer to next active W/P; select the proper VOR/DME frequency. Select proper course on the HSI for next W/P at the change-over point, and continue to fly the left-right needle on the nav indicator.

PERFORMANCE - No change

Approved: Donald St Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR
200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT KING AIR A100, B100, F-90, E90 C90 (LJ-668 thru LJ-1062, EXCEPT LJ-670), and C90A (LJ-1063 AND AFTER) LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

AIRDATA AD611/D AREA NAVIGATION/VERTICAL NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the AirData AD611/D Area Navigation/Vertical Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The RNAV function of the AirData AD611/D system performs a vector computation that results in a visual display of the magnetic bearing and distance to or from a selected waypoint. The computer, in effect, moves the selected reference facility (VORTAC or colocated VOR/DME facility) to a different location called a waypoint. The waypoint, which is expressed in terms of nautical miles along a selected radial from the VORTAC, is programmed into the system on the Manual Waypoint Setter.

Steering guidance is presented as a left/right display on the Horizontal Situation Indicator(HSI). The display format differs from the conventional VOR course deviation of 10 degrees called "angular course deviation". Rather, course deviation is presented in nautical miles from the course centerline. This feature, referred to as "linear course deviation", provides for a constant course width irrespective of the distance to the waypoint. Two levels of sensitivity are available for area navigation. The enroute senstivity, available when the APPR pushbutton on the system's range indicator is not activated, provides a constant course width of 5 nautical miles. Approach sensitivity, available with the APPR pushbutton depressed, provides a constant course width of 1.25 nautical miles. Approach sensitivity should be used when within ten nautical miles of the terminal waypoint.

The Multi-Waypoint Memory System is an option to the basic AD611/D Area Navigation Computer System. This system consists of a Horizontal Display Unit (61HDU) and the Data Entry Unit (61DEU). These units may be used in conjunction with or in lieu of the Manual Waypoint Setter. The Multi-Waypoint System stores RADIAL/DISTANCE and TRACK/FREQUENCY information for up to 10 different waypoints. The memory does not erase when electrical power to the unit is turned off.

The VNAV function of the AirData AD611/D does not depend on interconnection with the encoding altimeter nor does it drive the vertical needle on the HSI. It does not depend on wind or groundspeed, but is based solely on vertical triangulation. Two versions of the VNAV controller are available, the 61CAC and the 541CAC. With the 61CAC installed, the VNAV displays as a set of numbers the MSL altitude the airplane "should be at" for a 3.0 degree (330 feet per nautical mile) approach slope to the runway waypoint. The pilot compares the computed "should be at" altitude with the standard altimeter in the cockpit and manually adjusts the airplane flight path as required. The display takes into account the MSL elevation of the waypoint, which is a value entered by the pilot. There is also provision for entering an MDA value appropriate to the approach conditions. The "count down" of the altitude display ceases at the MDA value and the screen blinks to indicate that further descent must be based on visual observations.

Airplanes equipped with the 541CAC operate the same as those equipped with the 61CAC except that a 1.5° (165 feet per nautical mile) slope is also available. Operation is the same as for the 3° slope except NO MDA FUNCTION IS PROVIDED. A 1.5° descent should NEVER be used to a waypoint placed at a runway elevation.

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For Training Purposes Only

LIMITATIONS

1. The area navigation system may not be used as a primary system under IFR conditions except on approved approach procedures, approved airways, and random area navigation routes when approved by Air Traffic Control.

2. This system can only be used with colocated facilities. (VOR and DME signals originate from same geographical location.)

3. An area navigation installation located on the right instrument panel may be used for primary navigation only if a qualified pilot occupies the right seat.

4. The vertical navigation system does not adversely affect any other airplane system. The computed vertical slope on those systems with the 61CAC controller is preprogrammed to a single value of 3.0 degrees. Systems with the optional 541CAC controller offer an additional 1.5 degree vertical slope. This mode MUST NOT be used for an approach descent. Pending publication of certification requirements, use of the 3.0 degree computed vertical slope to stabilize the flight path is permitted provided the maximum/minimum altitudes specified in the published procedures are observed.

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude and angle of bank.

1. If NAV flag appears while in the enroute mode, check for correct frequency.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

3. If NAV flag appears during an approach, execute published missed approach and utilize another approved facility.

NORMAL PROCEDURES

The AirData AD611/D system is programmed and operated from a Digital Range/Mode Control unit, one or more Waypoint Setter Units, and a Command Altitude Computer for VNAV display. Frequency selection and course display are provided by the standard navigation controls and HSI.

CONTROLS AND DISPLAYS

DIGITAL RANGE/MODE CONTROL UNIT (RNAV 61 DRM)

1. RNAV ON-OFF Pushswitch:

   Used to activate and deactivate the RNAV system. It is a push on/push off switch that is backlighted whenever it is in the ON state. When selected ON, it connects the RNAV computer to the HSI. When selected OFF, the HSI display presents conventional VOR/LOC information.

2. APPR Pushswitch:

   Used to activate or deactivate the RNAV approach mode of operation. This operation increases the sensitivity of the HSI presentation and is used when approaching a waypoint in an approach to landing. The switch is backlighted whenever it is switched ON.

3. Digital Display:

   Normally indicates the distance to the waypoint in nautical miles from present position. The airplane's

standard DME distance indicator will continue to display DME distance to the reference VORTAC.

4. BRG Pushbutton:

   Used to temporarily cause the digital display to indicate the magnetic bearing from the airplane to the selected waypoint. Valid VOR and DME signals must be received for this function.

5. TEST Pushbutton:

   Illuminates the three diagnostic annunciator lights to verify their operation. Temporarily causes the digital display to indicate the waypoint DISTANCE value entered on the active waypoint setter unit. Also, a reference bearing output is sent to the HSI which causes the left/right needle to center when the course selector is set to the RADIAL value entered on the active waypoint setter unit. Depressing both the TEST and BRG buttons simultaneously causes the waypoint RADIAL value entered on the active waypoint setter unit to appear on the digital display.

   These tests require at least 10 nautical miles to be set into the waypoint DISTANCE and reception of a valid VOR signal.

6. Diagnostic Lights:

   Each of the three fault annunciators will flash and the digital display will be blank under the specified conditions.

   DTW: Indicates that "distance to waypoint" computation cannot be made. This can be an excessive distance (over 199.9 N.M. to waypoint), excessive RADIAL setting (over 359.9°) or a computer malfunction.

   VOR: Indicates that computation quality of VOR signal has been lost.

   DME: Indicates a loss of DME signal.

WAYPOINT SETTER UNIT (RNAV 61 WPS)

1. RADIAL Thumbwheels:

   Set to indicate the bearing from the VOR to the waypoint. The DTW diagnostic annunciator will flash if a RADIAL entry exceeds 359.9 degrees or results in a distance-to-waypoint exceeding 199.9 nautical miles.

2. DISTANCE Thumbwheels:

   Set to indicate the distance from the VOR to the waypoint.

3. ACTIVATE Pushbutton:

   Depressing white pushbutton, located above the RADIAL thumbwheels, activates that waypoint setter unit, placing its RADIAL and DISTANCE information into the RNAV computer. In systems containing more than one waypoint setter unit, the number 1 unit is automatically activated when the RNAV ON-OFF switch is selected ON. Any other waypoint setter unit can then be activated by depressing the ACTIVATE pushbutton on the desired waypoint setter unit.

   Depressing the ACTIVATE pushbutton also performs a "fast update" function for the RNAV computer each time it is depressed. Fast update allows current VOR and DME information on airplane position into the computer without averaging out the errors in these signals. Fast update would be used after channeling a new frequency into the NAV equipment, after regaining DME lock-on, or after changing a thumbwheel setting on an active waypoint setter unit.

4. Waypoint Indicator Light:

   Yellow light, located above DISTANCE thumbwheels, illuminates whenever its waypoint setter unit is activated. These lights are numbered when more than one waypoint setter unit is installed.

HORIZONTAL DISPLAY UNIT (61 HDU)

1. RADIAL/DISTANCE Pushswitch:

   Depressing pushswitch causes RADIAL (upper line) and DISTANCE (lower line) information to be displayed from either the active waypoint or from a new waypoint being entered into the MEMORY.

2. TRACK/FREQUENCY Pushswitch:

   Depressing pushswitch causes prestored TRACK and FREQUENCY information for the selected waypoint to be displayed.

DATA ENTRY UNIT (61 DEU)

1. ACTIVE Pushbutton:

   This pushbutton selects the Memory Waypoint System for use when installed in conjunction with a Manual Waypoint Setter (61WPS). The pushbutton also engages the AD611/D computer "Fast Update".

2. MEMORY Pushbutton:

   Pressing pushbutton opens the MEMORY allowing data to be stored in the MEMORY. After data has been entered, pressing the pushbutton again closes the MEMORY and permanently stores the data.

3. Thumbwheel "SCRATCH PAD":

   Set to indicate information to be entered into MEMORY.

4. ENTER Pushbuttons:

   Pushbuttons are pressed after MEMORY is opened to enter "SCRATCH PAD" data into the waypoint MEMORY; Upper pushbuttons enter RADIAL or TRACK information and the lower pushbuttons enter DISTANCE or FREQUENCY information.

5. WAYPOINT Number Window:

   This window displays the number of the waypoint currently selected.

6. WAYPOINT Select Pushbuttons:

   These pushbuttons, located above and below the waypoint number window, are used to change the waypoint number selected to a larger number (upper pushbutton) or a smaller number (lower pushbutton).

COMMAND ALTITUDE COMPUTER (VNAV 61CAC)

1. Digital Display:

   Normally displays COMMAND ALTITUDE (altitude MSL that airplane currently "should be at" in order to achieve a 3.0°/330 feet per nautical mile approach descent angle to the runway waypoint). It also displays values set for waypoint altitude and MDA.

2. ON-STBY Switch:

   Turns the VNAV computer ON. W/P MSL and MDA MSL values may be set in either ON or STBY switch positions.

3. W/P-SET-MDA Switch:

   W/P position enters the value set by the W/P MSL knob into the VNAV computer. MDA position enters

the value set by the MDA MSL knob into the VNAV computer. It should be in the SET position for normal VNAV operation.

4. W/P MSL Knob:

   Sets the altitude MSL of the waypoint.

5. MDA MSL:

   Sets the selected minimum decision altitude.

COMMAND ALTITUDE COMPUTER (VNAV 541CAC)

1. Digital Display:

   Normally displays COMMAND ALTITUDE (altitude MSL that airplane currently "should be at" in order to achieve a 3°/330 feet per nautical mile approach descent angle to the runway waypoint or a 1.5°/165 feet per nautical mile cruise descent). It also displays values set for waypoint altitude and, in the 3° descent, MDA.

2. ON Lamp/Pushbutton:

   An illuminated push-on/push-off switch which illuminates when in the ON position. Turns the VNAV computer ON.

3. ALT-SET-MDA Switch:

   ALT position enters the value set by the rotary knob into the computer for desired altitude at the waypoint. MDA position enters the value set by the rotary knob for minimum decision altitude into the computer. It should be in the SET position for normal VNAV operation.

4. 1.5°/3° Lamp/Pushbuttons:

   When pushed these switches select either the 1.5° or 3° descent angle. The appropriate switch illuminates when that descent angle is selected.

5. Rotary Knob:

   Sets the altitude MSL of the waypoint or the minimum decision altitude, depending on the setting of the ALT - SET - MDA Switch.

PREFLIGHT (MANUAL WAYPOINT SELECTOR)

The preflight check is to test the computation accuracy of the computer and to assure the proper operation of the controls and displays. This procedure should be completed prior to programming for the intended flight.

1. Depress RNAV pushswitch to ON.
2. Set RADIAL thumbwheels to 000.0°.
3. Set DISTANCE thumbwheels to 25.0 NM.
4. Set NAV 1 receiver to a VOR or VORTAC within receiving range.
5. Press and hold TEST button. Adjust course control on HSI to produce centered needle with "TO" indication. Check that:

   a. Digital display indicates 25.0 ± 1 NM.

   b. The course setting is 000 ± 2 degrees.

6. Press and hold BRG and TEST buttons. Check that:

   a. Digital display indicates 0 ± 1 degree.

7. Release BRG and TEST buttons.

NOTE

If any of the preflight tests are not within the prescribed tolerances, the RNAV system will not meet the required standards of accuracy. Corrective adjustment or maintenance is required. This procedure does not test the DME.

PREFLIGHT (Multi-Waypoint Memory System)

To preflight check the RNAV system using the Multi-Waypoint Memory System prior to flight, the following procedure should be used.

1. Depress RNAV pushswitch to ON.
2. Press ACTIVE pushbutton on Data Entry Unit (providing unit is used in conjunction with one or more manual waypoint setters).
3. Firmly press MEMORY pushbutton to open MEMORY (Horizontal display unit readout will flash indicating MEMORY is open).
4. Set thumbwheel "SCRATCH PAD" to 000.00.
5. Press upper ENTER button to enter data into MEMORY.
6. Set thumbwheel "SCRATCH PAD" to 25.00.
7. Press lower ENTER pushbutton.
8. Set NAV 1 receiver to VOR or VORTAC within receiving range.
9. Press and hold TEST button. Adjust course control on HSI to produce centered needle with "TO" indication.

   a. Digital display indicates 25.0 ± 1 NM.

   b. Course setting is 000 ± 2 degrees.

10. Press and hold BRG and TEST buttons. Check that digital display indicates 0 ± 1 degree.
11. Release BRG and TEST buttons.

NOTE

If any of the preflight tests are not within the prescribed tolerances, the RNAV system will not meet the required standards of accuracy. Corrective adjustment or maintenance is required. This procedure does not test the DME.

PROGRAMMING

1. RNAV ON-OFF Pushswitch - ON (switch illuminated)

NOTE

The number 1 waypoint setter unit is automatically selected when the RNAV pushswitch is turned ON.

2. Waypoint Definition - Determine in terms of RADIAL and DISTANCE from a specific VORTAC.

NOTE

The maximum allowable RADIAL setting is 359.9 degrees. If a RADIAL of 360.0 degrees is desired, use a value of 000.0 degrees. The maximum allowable DISTANCE setting is 199.9 NM. The maximum allowable range from the airplane to the waypoint is also 199.9 NM. If any of these restrictions are exceeded, select a waypoint that is within these values.

3. Manual Waypoint Setter Units or Multi-Waypoint Memory System.

   a. Manual Waypoint Setters - Set RADIAL and DISTANCE thumbwheels.

   b. Multi-Waypoint Memory System - Enter data.

      1. Press MEMORY pushbutton to "open" MEMORY - Digital display will blink.

      2. Select category of data to be entered - (RADIAL/DISTANCE or TRACK/FREQUENCY) and press appropriate pushswitch.

      3. Select waypoint number.

      4. Set proper data on "SCRATCH PAD" thumbwheels.

      5. Enter data - If RADIAL or TRACK data, press upper ENTER pushbutton; If DISTANCE or FREQUENCY data, press lower ENTER pushbutton.

NOTE

RADIAL/DISTANCE information entered is used by the RNAV to compute course information. TRACK/FREQUENCY information entered is used as a reference only and is provided as a reminder of what information is pertinent to that waypoint. TRACK/FREQUENCY information is not used by the RNAV for computation of any kind.

      6. Check digital display for proper data entry.

      7. Press MEMORY pushbutton to close MEMORY, digital display will stop blinking.

NOTE

The MEMORY does not erase when power to the RNAV unit is turned off.

4. NAV Receivers (VOR and DME) - ON. Frequency set.

5. Digital Display - Check to insure that distance to waypoint value appears.

6. HSI Course Control - SET to desired magnetic course.

ENROUTE

Using the AirData AD611/D system enroute corresponds to flying VOR airways, except navigation is now to or from waypoints. The waypoint parameters (radial and distance) in effect "move" the VORTAC. Once this is accomplished, the horizontal situation indicator and AD611/D digital range indicator will provide guidance to the waypoint similar to conventional VOR/DME navigation. The only notable difference is that the course deviation needle on the HSI will maintain a constant sensitivity of ± 5 nautical miles irrespective of the distance to the waypoint. The range indicator will count down to approximately 0.2 nautical mile when, upon reaching the waypoint, the "TO" flag will change to "FROM".

When the next waypoint is required for navigation, depress the ACTIVATE pushbutton on the next waypoint setter unit in sequence, confirm the proper VORTAC frequency is set, and set the desired magnetic course on the horizontal situation indicator.

The next waypoint is selected on the Multi-Waypoint Memory System by pressing the appropriate waypoint select pushbutton until the desired waypoint number appears in the waypoint number window.

NOTE

If an ILS frequency is selected on NAV 1 while in an RNAV mode, the NAV flag will appear on the horizontal situation indicator and the VOR diagnostic light will flash. The RNAV must be selected OFF for ILS or conventional VOR operation (except for Approach Range Monitor operation).

Data may be entered into the MEMORY system or data already entered may be reviewed while the system is being used in flight. The MEMORY may be opened at any time to enter or review data. When this happens, the waypoint data currently being used is locked in to the computer; the RNAV continues to use this data for navigation while the MEMORY is open. Data may be entered at this time as previously described. After all desired data changes or reviews have been made, the MEMORY is closed. This causes the data displays to revert back to the currently active waypoint.

APPROACH

Using the AirData AD611/D system for an approach is similar to making a localizer approach. However, the system is using VOR and DME information and the MDA will be higher than when conducting a precision approach. Insert the waypoint parameters from the approach chart into the waypoint setter units. These parameters must be taken from an approved RNAV approach procedure for IFR operations. Activate the approach mode by depressing the APPR pushswitch. This will increase the horizontal situation indicator navigation sensitivity to a ± 1.25 nautical miles course width. For smoother operation, the computed distance to the waypoint should not exceed 30 nautical miles while in the approach mode.

Set the appropriate inbound course to each waypoint in turn and depress the ACTIVATE pushbutton on the appropriate waypoint setter unit to establish the next waypoint. If landing cannot be made upon reaching the MAP, follow the missed approach procedure outlined on the approved plate.

VERTICAL NAVIGATION (VNAV 61CAC)

The digital display screen of the Command Altitude Computer indicates the altitude the airplane "should be at" on a descent profile of 3.0° (330 feet per nautical mile) to the runway waypoint. The screen will count down as the airplane proceeds toward the runway waypoint and will count up as the airplane flies from the runway waypoint. The display will stop counting when the airplane should be at the MDA, at which point the display will flash the MDA value. The maximum altitude of the display is 9900 feet.

1. ON-STBY Switch - STBY.
2. SET Switch - W/P Position.
3. W/P MSL Knob - TURN until altitude MSL of runway waypoint shows in display.
4. SET Switch - MDA Position.

5. MDA MSL Knob - TURN until altitude MSL of MDA shows in display.

6. SET Switch - Center Position.

Immediately prior to reaching descent waypoint:

7. ON-STBY Switch - ON.

8. Command Altitude - READ from digital display.

9. MDA Reached - Display starts flashing and stops decreasing.

CAUTION

It is essential that the runway end waypoint setter unit be activated when the Command Altitude Computer is being used. DO NOT activate the VNAV while navigating to the Final Approach Fix waypoint.

VERTICAL NAVIGATION (VNAV 541CAC)

PROCEDURES FOR 3° DESCENT ANGLE

The digital display screen of the Command Altitude Computer (CAC) indicates the altitude the airplane "should be at" on a descent profile of 3.0° (330 feet per nautical mile) to the runway waypoint. The screen will count down as the airplane proceeds toward the runway waypoint and will count up as the airplane flies from the runway waypoint. The display will stop counting when the airplane should be at the MDA, at which point the display will flash the MDA value. The maximum altitude of the display is 9900 feet.

1. ALT-SET-MDA Switch - ALT Position.

2. Rotary Knob - TURN until altitude MSL of runway waypoint shows in display.

3. ALT-SET-MDA Switch - MDA Position.

4. Rotary Knob - TURN until altitude MSL of MDA shows in display.

5. ALT-SET-MDA Switch - SET Position.

Immediately prior to reaching descent waypoint:

6. ON Button - PRESS.

7. 3° Button - PRESS.

8. Command Altitude - READ from digital display.

9. MDA Reached - Display starts flashing and stops decreasing.

CAUTION

It is essential that the runway end waypoint setter unit be activated when the Command Altitude Computer is being used DO NOT activate the VNAV while navigating to the Final Approach Fix waypoint.

PROCEDURES FOR 1.5° DESCENT ANGLE

The digital display screen of the Command Altitude Computer indicates the altitude the airplane "should be

at" on a cruise descent profile of 1.5° (165 feet per nautical mile) to a selected altitude at a waypoint. The screen will count down as the airplane proceeds toward the waypoint. The display screen will commence flashing while counting down when the airplane is within 1000 feet of the set altitude. The 3° lamp also commences flashing at this time to assure the pilot is aware that the 3° slope must be selected to complete an approach.

1. Complete steps 1 thru 4 for 3° slope.
2. ALT-SET-MAD Switch - ALT Position.
3. ON Lamp/Pushbutton - ON.
4. 1.5° Lamp/Pushbutton - PUSH, 3° lamp extinguishes and 1.5° lamp illuminates.

Within 1000 feet of set altitude the CAC display and 3° lamp commence flashing. Countdown of altitude continues.

CAUTION

The 1.5° slope is NOT to be used as an approach descent.

5. 3° Lamp/Pushbutton - PUSH, 3° altitude data presented

or

6. ALT-SET-MDA Switch - SET Position, 3° altitude data presented.

APPROACH RANGE MONITOR

The Approach Range Monitor feature provides for the separation of the RNAV computed range to a waypoint from the steering guidance of the pilot's horizontal situation indicator. Selecting the Approach Range Monitor switch to the RANGE MONITOR position will connect the RNAV computer to the NAV 2 receiver. The pilot's horizontal situation indicator will be retained on the NAV 1 receiver.

On an ILS approach, for example, it is desirable to know distance to the outer marker and then to the runway threshold. By selecting RANGE MONITOR and setting the appropriate NAV 2 frequency and waypoint parameters in the waypoint setter unit, the distance to the desired fix will be continuously displayed while ILS steering guidance on the horizontal situation indicator will be conventional. The result is the ability to fly a localizer or full ILS steering situation while retaining RNAV computed distance to a selected fix.

CAUTION

It is imperative the Approach Range Monitor switch be placed in the NORMAL position during RNAV operations. If left in the RANGE MONITOR position, the range display will be based on the NAV 2 frequency and waypoint setter unit parameters, and the pilots horizontal situation indicator will display conventional VOR steering based on the selected NAV 1 frequency.

PERFORMANCE - No Change

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

Beechcraft®

Super King Air® 200, 200C, 200T, 200CT, B200, B200C, B200T & B200CT Landplanes

Pilot's Operating Handbook and

FAA Approved Airplane Flight Manual Supplement

for the

Brake Deice System

This Supplement is applicable to the following manual(s):

101-590010-127, 101-590010-147, 101-590037-3, 101-590037-29 & 101-590037-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane has been modified by installation of a Brake Deice System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. The brake deice system is not to be operated above 15°C ambient temperature.
2. The brake deice system is not to be operated longer than 10 minutes (one deice timer cycle) with the landing gear retracted. If operation does not automatically terminate 10 minutes after gear retraction, the system must be manually selected off.
3. Maintain 85% N_1 or higher during periods of simultaneous brake deice and wing boot operation. If inadequate pneumatic pressure is developed for proper wing boot inflation, select brake deice system off.
4. Both sources of instrument bleed air must be in operation. Select brake deice system off during single engine operation.

EMERGENCY PROCEDURES

RUDDER BOOST OPERATION

The rudder boost system may not operate when the brake deice system is in use. Consequently, increased rudder-pedal forces should be anticipated in the event of single-engine operation. Availability of the rudder boost system will be restored to normal when the brake deice system is turned off.

ILLUMINATION OF BLEED AIR FAIL ANNUNCIATOR

If either BLEED AIR FAIL light illuminates in flight, shut off the INSTR & ENVIR bleed air valves on the affected engine. Select brake deice system off.

NOTE

BLEED AIR FAIL lights may momentarily illuminate during simultaneous wing boot and brake deice operation at low N_1 speeds. If lights immediately extinguish, they may be disregarded.

NORMAL PROCEDURES

AFTER STARTING

If Brakes Require Deicing:

1. Bleed Air Valves - OPEN
2. Brake Deice - ON (check annunciator illuminated; check for operation)

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NOTE

Brake deice control valves may become inoperative if valves are not cycled periodically. One cycle of operation is required daily regardless of weather conditions.

3. Condition Levers - HIGH IDLE

NOTE

Once brakes have been deiced, the condition levers may be returned to LOW IDLE.

BEFORE LANDING

If it is possible that brakes may be restricted by ice accumulations from previous ground operation or inflight icing conditions:

1. Brake Deice - ON (check annunciator illuminated)

NOTE

If automatic timer has terminated brake deice operation after last retraction of the landing gear, the landing gear must be extended to obtain further operation of the system.

PERFORMANCE

Use of the brake deice system during certain ambient conditions may reduce available engine power. Consult the MINIMUM TAKE-OFF POWER chart in the FAA Performance Section of the POH/AFM to determine the minimum torque value permitted for take-off. If this value cannot be obtained, without exceeding engine limitations, the brake deice system must be selected off until after the take-off has been completed.

Use of the brake deice system in flight will result in an ITT rise of approximately 20°C. Observe ITT limitations when setting climb and cruise power.

WEIGHT AND BALANCE/EQUIPMENT LIST

No change.

SYSTEMS DESCRIPTION

High temperature engine compressor bleed air is directed onto the brake assemblies by a distributor manifold on each main landing gear. This high pressure air is supplied by the standard bleed air pneumatic system which also provides regulated pressure to the surface deice system and vacuum source. High temperature air from the pneumatic system is routed through a solenoid control valve in each main wheel well, through a flexible hose on the main gear strut, and to the distribution manifold around the brake assembly.

A switch on the pilot's subpanel, placarded BRAKE DEICE, controls the brake deice system. When this switch is activated, both solenoid control valves are opened and an indicator light, BRAKE DEICE ON, on the lower annunciator panel is illuminated to advise the system is in operation.

The brake deice system may be operated as required on a continuous basis with the landing gear extended provided the appropriate LIMITATIONS are observed. To avoid excessive wheel well temperatures with the landing gear retracted, a timer is incorporated to automatically terminate system operation approximately ten minutes after the landing gear is retracted. The system indicator light should be monitored and the control switch selected OFF when the light extinguishes or if brake deice operation has not automatically terminated within approximately ten minutes. The landing gear must be extended before the timer is reset and the system can be activated again.

HANDLING, SERVICING AND MAINTENANCE

No change.

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BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT, KING AIR B100, E90, F90, C90 (LJ-668 THRU LJ-1062, EXCEPT LJ-670) AND C90A (LJ-1063 AND AFTER) LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the COLLINS LRN-70 (ONTRAC IIIA) AND LRN-80 (ONTRAC III) VLF/OMEGA NAVIGATION SYSTEMS

GENERAL

The information in this supplement is FAA-approved material and must be attached to the *FAA Approved Airplane Flight Manual* when the airplane has been modified by installation of the Collins LRN-70 (ONTRAC III A) or LRN-80 (ONTRAC III) VLF/OMEGA Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic *FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

The LRN-70/LRN-80 meet the approved criteria for VLF/Omega navigation systems as established by AC 20-101A. It may be used as a means of VFR/IFR enroute navigation provided the following limitations are observed:

1. Only enroute operations in the conterminous United States and Alaska are approved.

2. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).

3. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.

4. VOR and DME equipment must be installed and operating during navigation on approved RNAV routes.

5. IFR flight is not approved based on VLF/OMEGA navigation into any area in which such operation could not be approved based on the installed equipment required by paragraph 3. This includes overwater flights beyond the range of approved ground navigation facilities.

6. The VLF/OMEGA position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

   a. Prior to compulsory reporting points during IFR operation when not under radar surveillance or control.

   b. At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.

   c. At hourly intervals during operation off of approved RNAV routes.

7. The VLF/OMEGA position information should be updated when a cross-check with other onboard approved navigation equipment reveals an error greater than 2 nautical miles along track or crosstrack.

8. Navigation shall not be predicated on the use of this system during periods of dead reckoning.

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9. Following a period of dead reckoning, the VLF/OMEGA position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment.

EMERGENCY PROCEDURES

An emergency battery pack is provided in the event the primary power source is lost. If the primary power is interrupted, the system will switch to the emergency battery pack until the primary power is restored, at which time the system will switch back to primary power. The system cannot be turned on using only the emergency battery. Battery power will only function with loss of primary power. It is important the system power switch be selected off at the conclusion of flight to avoid depleting the emergency battery. The system can operate from the emergency battery pack for approximately 30 seconds at 70°F. The battery pack will recharge in approximately ten hours when the system is on the primary power source.

In the event that the WARN light illuminates, or any time the navigational information is questioned based on information from other onboard navigation equipment, the other onboard navigational equipment must be used as the primary source of navigational information.

NORMAL PROCEDURES

The LRN-70/LRN-80 is a very low frequency (VLF) radio navigation system designed for area navigation utilizing existing VLF and OMEGA transmitting stations. Airplane position is determined by automatically making position fixes by time referencing radio signals from several stations. Generally, two stations can determine a position on the surface of the earth. From changes in latitutde/longitude position, the system derives distance to go, bearing, desired track, cross track error, ground speed and time to waypoint/destination. With an airplane true airspeed input, the system can compute the current wind direction and speed.

Navigation is computed along a Great Circle flight path to provide the shortest distance between positions. Course deviation is presented in nautical miles on the Pilot's Course Indicator rather than in degrees as with conventional VOR navigation. This feature provides for a constant course width of approximately ± 3 nautical miles regardless of the distance to the waypoint.

CONTROLS AND DISPLAYS

The system is programmed and operated from a pedestal mounted control/display unit (CDU). The control section is used to enter data in to the computer and select the computed information to be displayed. The display section provides navigational information to the pilot. It also displays the data to be entered into the computer.

1. Data Keyboard:
   Ten momentary keys for entry of data into the computer. Digit keys (0 through 9) enter numerical data. N (north), E (east), S (south), W (west) keys enter the sign of latitude/longitude information.

2. CLR Key:
   Clears the computer of data in the position selected by the FUNCTION selector.

3. CAL Key:
   Activates the calibration of the system to a known position.

4. ON/OFF Switch:
   Applies power to all system components.

5. WAYPOINT Selector:
   Ten position switch to select either the start position or one of nine waypoint/destination positions.

6. LAMP TEST Switch:
   Tests all lights on the display section, except BATT. All digits display the number "8".

7. DIM Switch:
   Adjusts background lighting for display section.

8. FUNCTION Selector:
   Twelve position switch to select the desired input or output mode of operation.

9. Numerical Display Registers:
   Left display is a five digit register to indicate latitude, bearing, desired track, wind direction, and time to selected waypoint. Right display is a six digit register to indicate longitude, distance to selected waypoint, cross track error, ground speed, magnetic variation, wind speed, DME, GMT, and date.

10. Function Indicators:
    Lamps to indicate S (south) and N (north) for latitude; E (east) and W (west) for longitude and magnetic variation.

11. DR (Dead Reckoning) Light:
    Illuminates when less than two VLF/OMEGA transmitting stations are being received or when only stations of unusable geometry are being received (see Enroute Operation). In the DR mode, the airplane's position is updated based on true airspeed, heading, and last known wind information. DR will remain illuminated until the system is recalibrated or the position is checked.

12. WARN light:
    Illuminates with a computer or system malfunction or improper pilot operation procedure. Navigational information should be considered unreliable.

13. BATT Light:
    Illuminates when the system is operating on the emergency battery pack.

14. SY Light:
    Illuminates to indicate the received OMEGA stations are being synchronized.

15. STD Light:
    Illuminates when the Frequency Standard is stabilizing. The computer is ready to navigate when the STD and SY lights extinguish.

16. Station Received Indicators:
    Illuminate to indicate those VLF and OMEGA stations which are being received.

NOTE

The LRN-70 is a condensed version of the LRN-80 intended primarily for operation in the United States. VLF station availability is limited to Maine, Washington, Maryland and England in this installation.

VLF STATIONS	*OMEGA STATIONS*
1. -MAIN	A. -NORWAY
2. -WASHINGTON	B. -LIBERIA
3. -AUSTRALIA	C. -HAWAII
4. -ENGLAND	D. -NORTH DAKOTA
5. -MARYLAND	E. -LA REUNION
6. -TBD	F. -ARGENTINA
7. -JAPAN	G. -TRINIDAD
8. -NORWAY	H. -JAPAN
9. -SPARE	

17. ENROUTE Light:
    Illuminates when the system is calibrated and navigating.

18. OFFSET Light:
Illuminates when the selected waypoint is offset by an input of radial and DME distance.

19. ALERT Light:
Illuminates approximately one minute from selected waypoint/destination.

20. L-R Lights:
Illuminate in conjunction with the FUNCTION selector XTK (crosstrack) position. Indicate airplane's position left or right of the desired track.

21. DECIMALS:
Each decimal point illuminates in the proper position appropriate to the data selected by the FUNCTION selector.

22. FD NAV/FD VLF Selector:
Lighted push switch which selects and indicates which course information is displayed on the pilot's course indicator (NAV or VLF). Selection of VLF provides for display of VLF course information.

NOTE

For the airplanes listed below, localizer information is always displayed when an ILS frequency is tuned on the selected NAV control. Models 200 (prior to BB-746 except BB-706), C90 (prior to LJ-625), E90 (prior to LW-344), F90 (prior to LA-82), B100 (prior to BE-104), 200C (prior to BL-14), 200T (prior to BT-21).

23. VLF MSG Light:
Illuminates to call attention to the CDU for one or more of the following messages: DR, WARN, BATT, or ALERT.

PROGRAMMING

Prior to take-off, the departure point latitude/longitude data is entered into the computer. This is referred to as a START position. Although the following procedures specify use of the START (S) position for system calibration, any waypoint position may be used for this purpose, provided that the position used is selected each time the START (S) position appears in the procedures. After calibration is complete, the waypoint position used for calibration may then be used as an additional enroute waypoint. In addition, the date and time of day are entered for the computer to compensate for diurnal shift while navigating. Coordinates (latitude/longitude) for up to nine waypoints (ten if system is calibrated) can be entered while the airplane is on the ground or after take-off.

1. Turn on airplane's avionics master-switch.
2. Set system master switch to ON.
3. Depress TEST switch. Verify all lights on display section are functional.
4. Set WAYPOINT switch to S (start).
5. Set FUNCTION switch to DATE.

NOTE

Verify that six zero's are shown on the right display register. This indicates the computer is ready to accept data. The STD and SY light should be illuminated.

6. Enter present month (one or two digits) day (two digits) and year (two digits) by depressing the keyboard buttons in the number sequence. The date entered must be the current Greenwich Mean Date.

DATA FORMAT: XX.XX.XX i.e., 12-29-75 or 1-01-76

NOTE

Leading zeros need not be entered, but all subsequent zeros must be entered.

7. Set FUNCTION switch to GMT. Enter current Greenwich Mean Time (within +5 minutes).

DATA FORMAT: XX.XX i.e., 22-06 or 02-27

8. Set FUNCTION switch to VAR. Enter existing variation to the nearest 0.1 degree. If variation is not known to a 0.1 degree, enter a zero for this digit.

DATA FORMAT: E/W XX-X i.e., E9.5 or W10.0

NOTE

Magnetic variation need not be entered unless it is desired that the bearing and desired track outputs be referenced to magnetic north. If no variation is entered, bearing and desired track information displayed will be referenced to true north and the N function indicator will illuminate in both functions. Variation *MUST* be entered for accurate WIND calculations.

9. Enter departure position into computer.

   a. Assure WAYPOINT switch is set to S (start).

   b. Set FUNCTION switch to LAT. Enter latitude (5 digits) of departure position, N (north) or S (south) must be entered prior to latitude digits.

DATA FORMAT: N/S-XX.XX.X i.e., N-33.40.2

   c. Set FUNCTION switch to LON. Enter longitude (6 digits) of departure position. E (east) or W (west) must be entered prior to longitude digits.

DATA FORMAT: E/W-XXX.XX i.e., W-117.48.0

10. Calibrate the system to the previously entered present position.

NOTE

Do not attempt to calibrate the system or calculate distances until the STD and SY lights have extinguished. This may require a period of up to 10 minutes in very low start up temperatures. This time may be used to enter desired waypoints.

If the airplane is located in or near a hangar, or near a noise source where good signals cannot be received, the system cannot synchronize to the OMEGA pattern. In these conditions, the SY light will remain illuminated until adequate signals are received and the system has synchronized.

a. Set the WAYPOINT selector to S (start).

b. Set the FUNCTION selector to POS (CAL key will flash).

c. Press the CAL key ONCE, verify that displayed data agrees with the actual start position (CAL key will continue to flash).

d. Press the CAL key again (it will stop flashing).

e. Note that the ENROUTE light illuminates within approximately 60 seconds, (DR will illuminate if insufficient navigation signals are received).

NOTE

After the ENROUTE light has illuminated, the system is navigating and sensing airplane movement. It is preferable to calibrate the system near the start of the take-off roll to minimize any errors which may be incurred while taxiing long distances while being exposed to extraneous ground interference.

WAYPOINT ENTRIES

Coordinates (latitude/longitude) for up to nine (ten after system calibration) waypoints can be entered while the airplane is on the ground or after take-off. Once entered, waypoints remain in the computer until new waypoints are entered. Waypoints may be changed any time during the flight by eliminating one previously entered. If entered waypoint data is incorrect, press CLR button and re-enter. All navigational information is automatically cleared when the ON/OFF switch is set to OFF. The waypoint data and last known position will be retained.

1. Set WAYPOINT switch to 1 and FUNCTION switch to LAT. Enter latitude of first waypoint. North or south must be entered prior to latitude digits.

DATA FORMAT: N/S-XX.XX.X i.e., N-39.56.0

2. WAYPOINT switch remains at 1. Set FUNCTION switch to LON. Enter longitude of first waypoint. East or west must be entered prior to longitude digits.

DATA FORMAT: E/W-XXX.XX.X i.e., W-86.03.0

3. Set WAYPOINT switch to 2 through 9. Enter coordinates of additional waypoints as required.

If the bearing (RAD) and distance (DME) from a known latitude/longitude waypoint are known, this information can be inserted into the computer to define a new waypoint OFFSET from the latitude/longitude waypoint. This is accomplished as follows:

1. Enter latitude and longitude of reference waypoint as described above.

2. Set FUNCTION selector to DME - enter distance to the offset position to the nearest 0.1 mi.

DATA FORMAT: XXX.X i.e., 30.0

3. Set FUNCTION selector to RAD/VAR

   a. Enter bearing from waypoint (RAD) to the nearest 0.1 degree leading zeros must be entered.

   b. Enter variation (VAR).

DATA FORMAT: XXX.X and E/WXX.X, i.e., 0.94.2 and E9.5

The OFFSET light will illuminate indicating the selected waypoint has been offset by the values of RAD and DME.

PREFLIGHT

The preflight check is to test the computation accuracy of the computer and to assure its capability for proper operation. This procedure should be completed prior to programming for the intended flight.

Coordinates for a calibrated start position and destination are programmed into the computer along with the current date and Greenwich Mean Time. Upon selecting the BRG/DIS enroute mode, a predetermined bearing and distance should be displayed.

1. Set WAYPOINT switch to S (start).
2. Set FUNCTION switch to DATE. Enter current date.
3. Set FUNCTION switch to GMT. Enter current Greenwich Mean Time.
4. Set FUNCTION switch to LAT. Enter: N-33.40.2.
5. Set FUNCTION switch to LON. Enter: W-117.48.0.
6. Calibrate the system (See Step 10 Above - PROGRAMMING).

NOTE

The test waypoint may be entered while waiting for the STD light to extinguish.

7. Set WAYPOINT switch to 1.
8. Set FUNCTION switch to LAT. Enter: N-44.38.9
9. Set FUNCTION switch to LON. Enter: W-67.16.9
10. Set FUNCTION switch to BRG/DIS (after system calibration). The numerical display registers should indicate a bearing of 058.6 degrees and a distance of 2406.6 nautical miles.

ENROUTE OPERATION

The desired waypoint/destination may be selected prior to take-off or when airborne. When the desired waypoint is selected, the computer determines the desired track (DTK), bearing (BRG), and distance (DIS) from the airplane's position when the waypoint is selected. If autopilot tracking is desired, set the displayed value of desired track on the pilot's course indicator and select FD VLF with the selector switch. The computed course information is now displayed on the pilot's Course Indicator. Autopilot tracking is accomplished by selecting the flight director's navigation mode and engaging the autopilot. While enroute, the following functions of navigational information are available. These functions are selected by placing the FUNCTION switch to the appropriate position and reading the display.

POS Function (Latitude/Longitude)

This position displays the current latitude and longitude in degrees, minutes, and tenths of minutes. Airplane position is updated every five seconds and is stable to approximately 2/10 minutes of latitude.

BRG/DIS Function (Bearing/Distance)

The bearing and distance to the waypoint as selected on the WAYPOINT switch will be displayed. Bearing is defined as the Great Circle angle from the airplane's current position to the waypoint as referenced to North (true or magnetic). The N light will be displayed if the bearing is defined to true North (i.e., no variation has been entered). If a magnetic course or route is desired, the magnetic variation for the area of flight should be entered and periodically updated in the VAR position of the FUNCTION switch. The N light will not be displayed if the bearing is magnetic. Bearing is calculated to 1/10 degree and distance to 1/10 nautical mile. The display is updated every four seconds.

DTK/XTK Function (Desired Track/Cross Track)

The desired track is defined as a Great Circle path from the airplane's position to the selected waypoint. This track is established when the waypoint is selected on the WAYPOINT switch. The DTK function displays to the nearest 1/10 degree the angle of the desired track flight path as referenced to North (true or magnetic). A magnetic desired track will be displayed provided the area magnetic variation has been entered.

Bearing (BRG) and desired track (DTK) will always agree when the airplane is centered on the prescribed Great Circle Course. If a deviation is made from the course, the bearing will update to provide the angle, as referenced to North, from the current position to the selected waypoint. The desired track angle will vary only as required to define the previously established Great Circle course.

NOTE

If it is desired to approach a waypoint on a predetermined track, this track may be inserted into the computer by (1) selecting DTK/XTK, (2) entering desired track to the nearest 0.1 degree. (If 0.1 degree is not known, enter a zero for this digit). The entered desired track will be referenced to the entered magnetic variation. This track should also be set on the pilot's course indicator.

Approximately one minute from the selected waypoint, the ALERT light will flash. The WAYPOINT selector should be set to the next waypoint, the desired track read in the DTK/XTK mode, and the Horizontal Situation Indicator course set to agree.

Crosstrack is defined as the lateral deviation from the desired track. The crosstrack error is displayed to the nearest 1/10 nautical mile. The L/R lights indicate whether the airplane is to the left or right of the desired track.

TIME/GS Function (Time/Ground speed)

Time to any of the programmed waypoints is available by setting the WAYPOINT switch to the desired waypoint. The computed time is presented in hours, minutes, and tenths of minutes and is updated every 4 seconds and assumes the airplane is on track and headed toward the waypoint.

The GS (ground speed) function is independent of the selected waypoint and indicates the true ground speed in knots. The ground speed display will remain valid even when not tracking to a waypoint. Ground speed is not accurate until navigation has been established for 3 minutes. If TAS and HDG inputs are supplied to the system, ground speed is updated every six seconds; otherwise, ground speed is updated every three to five minutes.

WIND FUNCTION

Wind direction, to the nearest 0.1 degree referenced to true north, is displayed in the left display register and wind speed, in knots, is displayed in the right register. If the computer does not have enough information to calculate the wind, the following flags will appear when the WIND position is selected:

1. The left display register will indicate four zeros if no VAR is entered.

2. The right display register will indicate.

   a. The digit 1 with no TAS input

   b. The digit 2 with no HDG input

   c. The digit 3 with no HDG and no TAS input.

The heading input comes from the pilot's compass system and can not be entered manually. True airspeed must be manually entered, unless the airplane is equipped with an Air Data System with a TAS output. TAS may be manually inserted into the computer by: (1) selecting the WIND position, (2) entering TAS in knots (three digits), and (3) selecting a different function, such as BRG/DIS, for several seconds and then returning to the WIND position. The displayed wind information will not be reliable for approximately 6 minutes after TAS is manually inserted. During this time, TIME and GS are also not reliable.

Unusable Station Geometry

If the airplane's position is such that the geometry of the received stations is not suitable for navigation, the system will go into the DR (dead reckoning) mode. This is because the geometry of the received stations makes for potentially large navigation errors. This condition usually happens in cases of only two station reception, but in very rare situations can happen with three stations. The airplane's position will be updated based on true airspeed, heading, and last known wind information. When the airplane flys out of the unusable geometry condition, the ENROUTE light will again illuminate. However, DR will remain illuminated, and navigation is not approved utilizing the LRN-70/LRN-80. To resume normal navigation and extinguish the DR light, the system's position must be recalibrated to a visual ground fix or to other approved navigation equipment.

INFLIGHT CALIBRATION

During flight, accumulated error can be reduced by calibrating over an entered waypoint or an offset waypoint.

1. Set the WAYPOINT selector to the upcoming waypoint or reference waypoint (if offset).

2. If the calibration point is the waypoint (no offset), assure that the DME position reads zero. If offset, enter the DME and RAD/VAR to define the offset.

3. Set the Function Selector to POS.

4. Prior to crossing the calibration point, press the CAL key once, the latitude/longitude of the waypoint will be displayed for verification and the CAL key will flash.

5. Directly at the reference point, press the CAL key a second time. The computer will calibrate on the coordinates of the selected waypoint (or waypoint offset). The enroute light will extinguish for up to 30 seconds during the calibration process and illuminate after calibration.

NOTE

To minimize DME slant range error, all inflight calibrations using a VORTAC as the reference waypoint and VOR/DME equipment to define the offset calibration point should be restricted to a distance greater than twice the airplane altitude and less than 60 NM from the VORTAC.

OMEGA RE-SYNCHRONIZATION PROCEDURE

The system will automatically synchronize to the OMEGA stations with system turn on. Re-synchronization is only to be done if incorrect synchronization is suspected.

Never re-synchronize for normal calibration (up-date) of the system. Once the system has synchronized and has not been switched off, the OMEGA receivers lock to the frequency standard and the system will remain synchronized.

If the system has mis-synchronized, do not attempt to re-synchronize until the amber STD light extinguishes.

1. Set FUNCTION switch to WIND
2. Set WAYPOINT switch to S
3. Press CAL key

NOTE

The SY light will illuminate for approximately two to six minutes during synchronization. The system must be calibrated after manual synchronization.

SYNCHRONIZATION BYPASS

When the system is synchronizing (SY light illuminated), it cannot be used for preflight navigation computations. If OMEGA reception is poor, such as in or near a hangar, the system may stay in the synchronization mode indefinitely. The synchronization bypass procedure permits preflight navigation computations prior to normal synchronization.

BYPASS PROCEDURE

1. Set FUNCTION switch to GMT
2. Enter all 8's on display
3. SY light will extinguish

RESYNCHRONIZATION PROCEDURE

1. Set FUNCTION switch to GMT
2. Press CLR key
3. Set FUNCTION switch to Wind
4. Set WAYPOINT switch to S
5. Press CAL key once.

Or

1. Turn system OFF; then back ON.

CAUTION

Prior to flight operations, the system must be synchronized to the OMEGA format and calibrated. Use of the LRN-70/LRN-80 as an additional means of navigation is not authorized while using fewer than two OMEGA stations.

GENERAL NOTES

1. An illuminated BATT light indicates the system is operating on the emergency battery pack and not on the primary power source. The system cannot be turned on without primary power first being applied. The emergency battery pack should not be used for ground check or warm-up.

2. To redefine a new desired track from the airplane's present position to the waypoint being used, set the WAYPOINT selector to another waypoint position for at least five seconds, then switch back to the original waypoint.

3. While enroute, the system calibration can be checked utilizing VOR/DME equipment. This can be accomplished by entering, as waypoints, known VORTAC stations (or fixes offset from known VORTAC stations) and comparing the system indications relative to these waypoints against the VOR and DME indications. Any fix whose latitude and longitude are known (visual, VOR, VOR/DME, ADF or combination thereof) may be utilized to check for system calibration by entering its location as a waypoint and comparing the system indications relative to the fix with those of other approved equipment.

Approved: Donald St Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT KING AIR 65-90, 65-A90, B90, C90, E90, 100, A100, B100, SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED FLIGHT MANUAL SUPPLEMENT

for

FLIGHT WITH CABIN ENTRANCE DOOR REMOVED

GENERAL

The information in this supplement is FAA Approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane is flown with the cabin entrance door removed.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below.

Beechcraft drawing number 100-4006, included in the applicable model kit, specifies required modifications which shall be accomplished prior to flight with the cabin entrance door removed.

LIMITATIONS

1. Pilot must use an approved headset at all times during flight (models 100, A100 and B100 only) Trim-in-motion warning is not audible without headset.

2. Maximum speed 208 knots CAS (205 knots IAS).

3. All occupants, except crew, must be properly equipped with parachutes when conducting operations with the cabin entrance door removed. Crew (pilot and copilot) must be at their stations secured by restraint straps, or equipped with parachutes if away from pilot or copilot seats.

4. Smoking is prohibited.

5. When flown for the purpose of dropping objects or equipment from the doorway, a suitable guard rail or equivalent safety device must be provided.

6. All loose articles must be tied down or stowed.

7. Parachutist's static lines shall be kept free of pilot's controls and control surfaces.

8. All flights with the cabin entrance door removed must be for the purpose of intentional egress of personnel, material or equipment.

9. The following placard must be in view of the pilot:

"FOR FLIGHT WITH CABIN DOOR REMOVED, SEE AIRCRAFT OPERATING LIMITATIONS WITH DOOR REMOVED."

10. Center of gravity limits:
    Aft limit: No change
    Forward limit: No change.

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EMERGENCY PROCEDURES - No change.

NORMAL PROCEDURES - No change.

PERFORMANCE

MAXIMUM TAKE-OFF WEIGHT: No change.

TAKE-OFF: No change.

CLIMB:

Models 100, A100, B100, 200, 200C, 200T, 200CT, B200, B200C, B200T, and B200CT: Subtract 40 ft/min rate of climb from climb graph (TWO ENGINES and ONE ENGINE INOPERATIVE)

Models 65-90, 65-A90, B90, C90, and E90: No change.

CRUISE

SPEEDS

Models 100, A100, B100, 200, 200C, 200T, 200CT, B200, B200C, B200T, and B200CT: Subtract 8 knots true airspeed from graph values.

Models 65-90, 65-A90, B90, C90 and E90: No change.

LANDING: No change.

WEIGHT AND BALANCE:

All airplane loadings must account for the items removed from the airplane. Weight & Center of Gravity data for the airstair entrance door are included in Beech drawing No. 100-4006. Consult the airplane Equipment List or superseding forms for weight and C.G. data on other items removed. For any items not listed above use actual weight and arm.

Approved: Donald H Peters

For W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
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For Training Purposes Only

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT AND KING AIR C90 (LJ-668 thru LJ-1062, EXCEPT LJ-670), C90A (LJ-1063 AND AFTER) AND E90 LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

GLOBAL GNS-500A VLF/OMEGA NAVIGATION SYSTEM SERIES 2

General

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Global GNS-500A, Series 2 VLF/OMEGA Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

LIMITATIONS

The GNS-500A Series 2 may be used as an additional means of enroute navigation during VFR/IFR conditions provided the following limitations are observed:

1. Only enroute operations in the conterminous United States and Alaska are approved.

2. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).

3. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.

4. VOR and DME equipment must be installed and operating during navigation on approved RNAV routes.

5. IFR flight is not approved based on VLF/OMEGA navigation into any area in which such operation could not be approved based on the installed equipment required by paragraph 3. This includes overwater flights beyond the range of approved ground navigation facilities.

6. The VLF/OMEGA position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

   a. Prior to compulsory reporting points during IFR operation when not under radar surveillance or control.

   b. At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.

   c. At hourly intervals during operation off of approved RNAV routes.

7. The VLF/OMEGA position information should be updated when a cross-check with other onboard approved navigation equipment reveals an error greater than 2 nautical miles along track or crosstrack.

8. Navigation shall not be predicated on the use of this system during periods of dead reckoning.

9. Following a period of dead reckoning, the VLF/OMEGA position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment.

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10. IFR flight is not approved based on VLF/OMEGA navigation when the VLF/OMEGA is operating in the search mode.

EMERGENCY PROCEDURES

A display Quality Factor over a value of "100" represents the possible failure of a system component.

1. Revert to the other installed navigation equipment appropriate to the available ground facilities.
2. Refer to the GNS-500A Series 2 Operator's Manual, Report No. 1054-2 for failure analysis.

NORMAL PROCEDURES

1. PREDEPARTURE

A. POWER ON

1. Mode Selector Switch - M
2. DIM knob - as required
3. Depress BACK key to test readouts and CDU annunciators

B. GMT and DATE ENTRY

1. Display Selector Switch - GMT/DATE
2. GMT/DATE - VERIFY

If data displayed is accurate:

3. Depress ENTER key twice

If update is required:

4. Key the GMT
5. Depress ENTER key once
6. Key the DATE (day/month/year)
7. Depress ENTER key once

C. DEPARTURE COORDINATES ENTRY

1. Display Selector Switch - POS
2. Position Coordinates - VERIFY

If data displayed is accurate:

3. Depress ENTER key once

If update is required:

4. Insert departure point latitude
5. Depress ENTER key once
6. Insert departure point longitude
7. Depress ENTER key once
8. Verify position display accurate
9. Depress flashing ENTER key

A. WAYPOINT ENTRY

To verify waypoint coordinates:

1. Display Selector Switch - WPT
2. Depress desired waypoint number key
3. Verify waypoint coordinates
4. Repeat Steps 2 and 3 for remaining waypoints.

To update or change waypoint coordinates:

NOTE

For waypoint 1 perform steps 1, 2, 3, and 7 only.

1. Display Selector Switch - WPT
2. Verify desired waypoint number
3. Depress WPT DEF key
4. Insert waypoint latitude
5. Depress ENTER key once
6. Insert waypoint longitude
7. Depress ENTER key once
8. Repeat Steps 2 through 7 for remaining waypoints

To erase stored waypoint coordinates:

1. Display Selector Switch - WPT
2. Depress desired waypoint number key
3. Depress WPT DEF key
4. Depress BACK key
5. Depress ENTER key once

B. MAGNETIC VARIATION ENTRY

NOTE

Magnetic variation is automatically computed between 70° North and 60° south latitudes. For all other areas refer to Manual Magnetic Variation Entry.

C. INITIAL LEG SELECTION

1. Depress LEG CHG key
2. Depress desired "FROM" waypoint reference number
3. Depress desired "TO" waypoint reference number
4. Verify new BRG and DIS reasonable

D. MANUAL TAS ENTRY

1. Display Selector Switch - Q/TAS
2. Key TAS in knots
3. Depress ENTER key

*D. RUNWAY LINE UP *(Departure point)*

1. Verify DR light flashing
2. Display Selector Switch - POS
3. Depress HOLD key
4. Depress ENTER key
5. Verify: VLF light steady - DR light off

2. ENROUTE OPERATION

A. MANUAL LEG CHANGE AT WAYPOINT

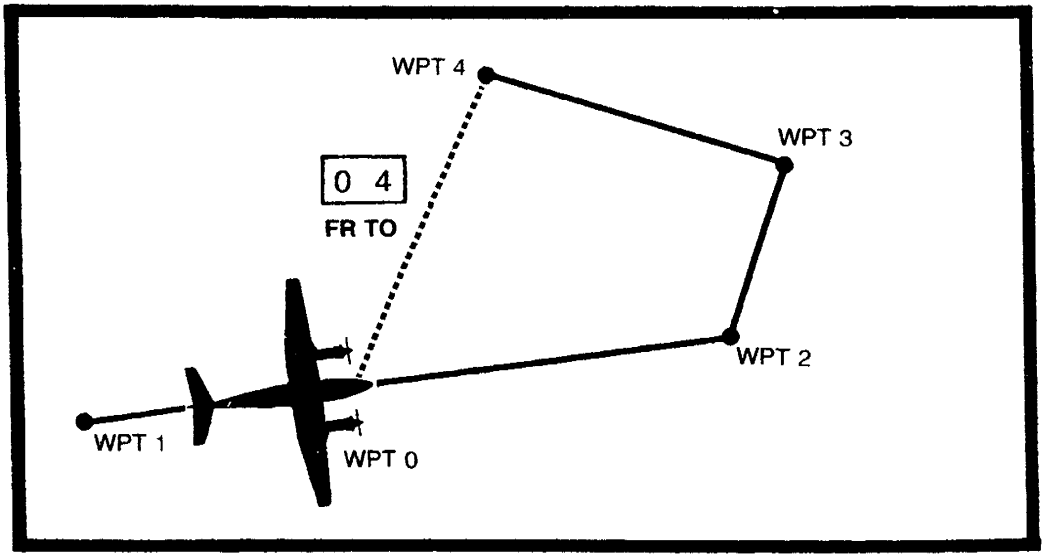
1. Mode Selector Switch - M
2. Depress LEG CHG key
3. Depress desired "FROM" and "TO" waypoint reference numbers
4. Verify new BRG and DIS reasonable

B. AUTOMATIC LEG CHANGE AT WAYPOINT

1. Mode Selector Switch - A
2. Verify proper leg changes
3. Verify new BRG and DIS reasonable

C. PRESENT POSITION DIRECT ANY WAYPOINT

1. Depress LEG CHG key
2. Depress "0" key
3. Depress desired "TO" waypoint reference number
4. Verify new BRG and DIS reasonable



D. WAYPOINT DEFINE/REDEFINE

1. Display Selector Switch - WPT
2. Depress waypoint reference number key
3. Depress WPT DEF key
4. Insert waypoint latitude
5. Depress ENTER key once
6. Insert waypoint longitude
7. Depress ENTER key

E. DEAD RECKONING (TO VLF) (See Limitations, Par. 6)

Whenever the system is operating in the VLF mode and the Quality Factor deteriorates to a value of 8 or greater, the system will automatically revert to the Dead Reckoning (DR) mode.

When the Quality Factor improves to a value of 7 or lower, the system will, after approximately 30 seconds, automatically revert to the VLF mode. The green VLF light will come on steady and the DR light will be flashing to inform the pilot that the system had been operating in the DR mode.

To Turn The Flashing DR Light Off:

1. Display Selector Switch - POS
2. Depress HOLD key
3. Depress ENTER key

F. POSITION (ACCURACY) CHECK AND UPDATE (See Limitations Par. 5 and 7)

1. Display Selector Switch - POS
2. Depress HOLD key over identifiable point
3. Check POS display against known coordinates
4. Insert known latitude if required
5. Depress ENTER key
6. Insert known longitude if required

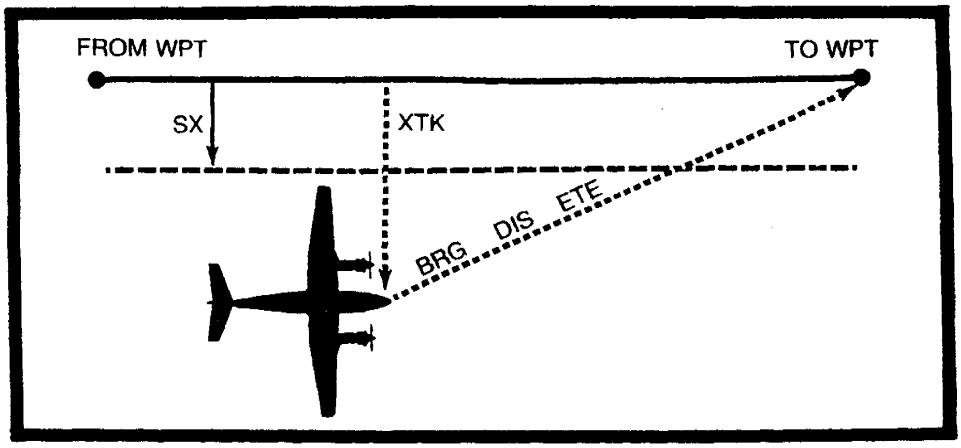
7. Depress ENTER key
8. Depress flashing ENTER key

NOTE

To update latitude only: Steps 4, 5, 7, 8
To update longitude only: Steps 6, 7, 5, 8

G. *PARALLEL COURSE*

1. Display Selector Switch - XTK/SX
2. Depress "R" or "L" key
3. Key offset distance (NM and tenths)
4. Depress ENTER key



H. *GNS-500A INITIALIZATION ENROUTE*

Five Minutes (Minimum) Prior to Reaching Known Point:

1. Mode Selector Switch - M
2. DIM knob - as required
3. Verify DATE and GMT
4. Enter coordinates of point
5. Verify flashing VLF light - OFF
6. Insert manual TAS (if required)

Within 10 NM of Known Point:

7. Verify DR light flashing
8. Display Selector Switch - POS
9. Depress HOLD key
10. Depress ENTER key
11. Verify VLF light ON steady

NOTE

VLF initiated automatically with AUTO TAS.

Over Known Point:

12. Depress HOLD key

Shortly After Passing the Point:

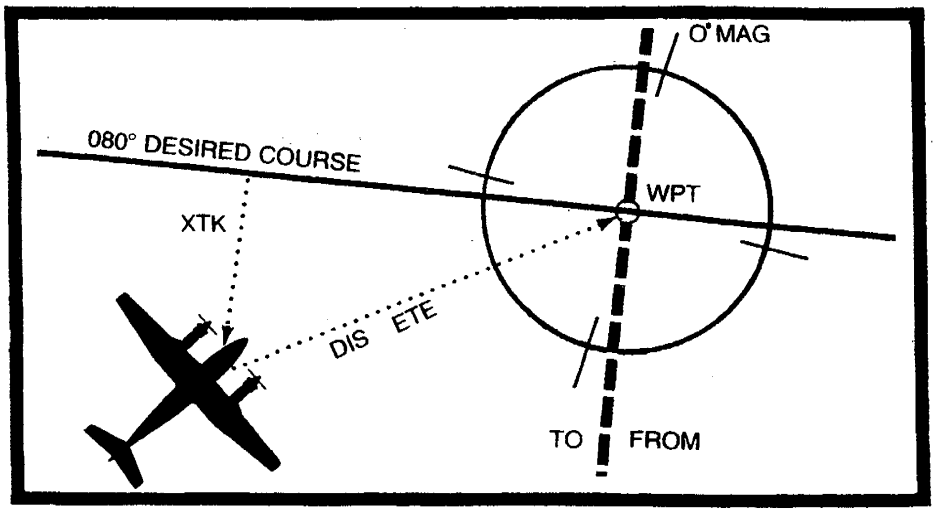
13. Re-insert proper coordinates of the point and ENTER
14. Verify: VLF light steady - DR light off

Enter the Following Information as Required:

Waypoint coordinates
Local magnetic variation (if required)
FROM/TO leg selection

I. PSEUDO-VORTAC

1. Mode Selector Switch - M
2. If necessary, defining coordinates of waypoint to be used (use Waypoint Define/Redefine procedure)
3. Depress LEG CHG key
4. Depress waypoint reference number key twice (may use "0" for present position pseudo VORTAC)
5. Display Selector Switch - BRG/VAR
6. Insert magnetic variation and depress ENTER key once
7. Key desired bearing
8. Depress ENTER key



J. MANUAL MAGNETIC VARIATION ENTRY

NOTE

This procedure required only for latitudes above 70 degrees North and below 60 degrees South.

1. Display Selector Switch - BRG/VAR
2. Insert Local Variation (E or W first)
3. Depress ENTER key

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To Return to Automatic Variation:

1. Display Selector Switch - BRG/VAR
2. Depress "E" or "W" key

K. RATE-AIDING MODE

With the X-2 computer program (noted in 3.B), the following applies:
In the event that VLF Communications station reception drops to less than 3 stations, the system automatically incorporates a rate-aiding function utilizing TAS, Heading and Magnetic Variation inputs. When reverting to this mode, the HSI NAV flag will appear if all the data is not entered. Refer to Paragraph 2.D, for manual entry of TAS data, if required.

3. COMPUTER ACCESS MODE

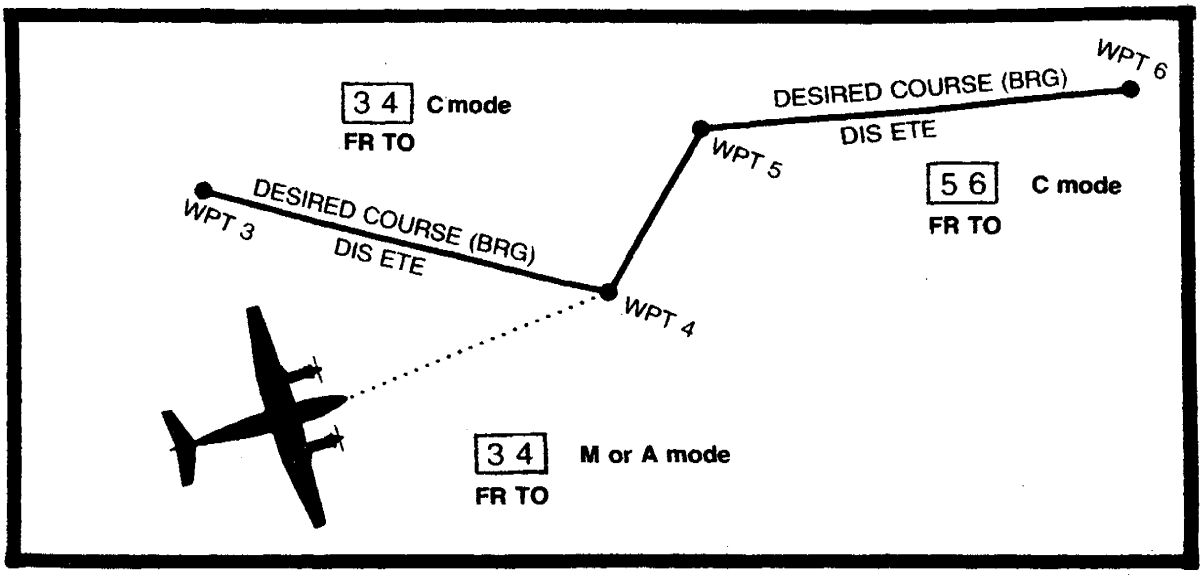
A. FLIGHT PLANNING

1. Waypoint entries must be made or have been made previously
2. With Mode Selector Switch - M or A enter planned groundspeed (Manual TAS Entry)

NOTE

Omit groundspeed entry if airborne.

3. Mode Selector Switch - C
4. Depress LEG CHG key
5. Key waypoint reference numbers defining desired leg
6. Use Display Selector Switch to select: DIS, ETE, Desired Course (BRG)
7. Repeat Steps 2 through 6 for remaining legs desired
8. Mode Selector Switch - M or A



B. COMPUTER PROGRAM IDENTIFICATION

1. Mode Selector Switch - C
2. Display Selector Switch - Q/TAS
3. Key the Code 1777 into the Left Data Display

4. Depress ENTER key
5. The number that appears in the Right Data Display represents the computer version and Comm slot tuning configuration shown.

COMPUTER PROGRAM	RIGHT DATA DISPLAY	COMMUNICATIONS STATION SLOT NUMBER						
		0	1	2	4	5	6	7
X-2	1	G	R	-SEE NOTE-			---	---

NOTE: For the X-2 program, receiver slots are tuned to receive the individual stations according to the following:

1. Slot 0 is always tuned to "G", Slot 1 to "R".
2. Slot 2 is not used. The system scans Stations A, J, L, M, S and W and selects the station according to the following:

   a. The most distant station is never used.

   b. Normally M is used only when A is not available. A and M are both tuned if airplane is within 1000 NM of either station.

C. STATION USAGE

This procedure allows the operator to determine which stations are being used for navigation and the order in which they were selected. Up to eight stations may be used simultaneously for navigation, with any additional stations being received held as backups.

1. Mode Selector Switch - C
2. Display Selector Switch - Q/TAS
3. Key the desired entry code into the left data display
4. Depress the ENTER key

The left data display will blank momentarily and when it reappears a number will also appear in the right data display. This number represents a particular station as shown in the table below.

ENTRY CODES		RIGHT DATA DISPLAY	
First station	- 1170	0-	No station
Second station	- 1171	1-A	Annapolis, Maryland
Third station	- 1172	2-G	Rugby, England
Fourth station	- 1173	3-J	Yosami, Japan
Fifth station	- 1174	4-L	Lualualei, Hawaii
Sixth station	- 1175	5-M	Cutler, Maine
Seventh station	- 1176	6-O	Helgeland, Norway
Eighth station	- 1177	7-P	Panama, Canal Zone
		10-S	Australia
		11-R	Anthorne, England
		13-W	Jim Creek, Washington
		30-	Norway Omega
		31-	Liberia Omega
		32-	Hawaii Omega
		33-	North Dakota Omega
		34-	La Reunion Omega
		35-	Argentina Omega
		36-	Trinidad Omega
		37-	Japan Omega

D. *DETERMINING COMM RECEIVER/SLOT LOCATION CORRELATION*

This procedure provides a means of determining which Comm station the computer has tuned each receiver "slot" to.

1. Mode Selector Switch - C
2. Display Selector Switch - Q/TAS
3. Key the code for the Comm slot desired. The digits keyed will appear in the left data display and the ENTER light will illuminate.
4. Depress the ENTER key

The left data display will blank momentarily and when it reappears a number will also appear in the right data display. This number represents the Comm station which that particular slot is tuned to as shown in the table below.

ENTRY CODES	RIGHT DATA DISPLAY	
Comm Station tuned in Slot 0-660	0-	No station
Comm Station tuned in Slot 1-661	1-A	Annapolis, Maryland
Comm Station tuned in Slot 2-662	2-G	Rugby, England
Comm Station tuned in Slot 4-664	3-J	Yosami, Japan
Comm Station tuned in Slot 5-665	4-L	Lualualei, Hawaii
Comm Station tuned in Slot 6-666	5-M	Cutler, Maine
Comm Station tuned in Slot 7-667	6-O	Helgeland, Norway
	7-P	Panama, Canal Zone
	10-S	Australia
	11-R	Anthorne, England
	13-W	Jim Creek, Washington

E. *INDIVIDUAL COMMUNICATIONS OMEGA STATION SIGNAL AMPLITUDE*

This procedure is designed to provide a means of checking the amplitude (or strength) of each station's signal individually.

1. Mode Selector Switch - C
2. Display Selector Switch - Q/TAS
3. Key the code for the Comm slot or OMEGA station desired. The digits keyed will appear in the left data display and the ENTER light will illuminate.
4. Depress the ENTER key

The left data display will blank momentarily and when it reappears a six-digit number will appear in the right data display.

COMM ENTRY CODES	OMEGA ENTRY CODES	
Station in Slot 0 - 560	Norway	-570
Station in Slot 1 - 561	Liberia	-571
Station in Slot 2 - 562	Hawaii	-572
Station in Slot 4 - 564	North Dakota	-573
Station in Slot 5 - 565	La Reunion	-574
Station in Slot 6 - 566	Argentina	-575
Station in Slot 7 - 567	Trinidad	-576
	Japan	-577

LEGEND (Right Data Display)	COMM	OMEGA
Minimum signal for navigational use:	14000	26000
Maximum amplitude:	77777	50000

If no OMEGA stations are received during flight, refer to the *GNS-500A Maintenance Manual, Report No. 1030,* for failure analysis.

4. CDU ANNUNCIATORS

A. "VLF" (Green)

1. Steadily illuminated: The "VLF" annunciator light is illuminated steadily whenever the GNS-500A is operating in the navigational mode with sufficient VLF/OMEGA signals to be used for navigation.

2. Flashing: Indicates that the GNS-500A is on and lacking either GMT/DATE or Position inputs.

3. OFF:

   a. If the "VLF" and amber "DR" annunciators are off, the navigation computer has failed.
   b. If the "VLF" annunciator is off while the amber "DR" annunciator is flashing, the system is ready to be placed into the navigational mode (see Paragraph 1 D.)

B. "DR" (Amber)

1. Steady illuminated: The "DR" annunciator light is illuminated steadily whenever the GNS-500A is not operating with sufficient VLF/OMEGA signals to be used for navigation.

NOTE

Operation near a ground power unit may cause electrical interference, preventing station aquisition and causing steady illumination of the DR light. This has no affect on data entry or computer functions.

2. Flashing:

   a. VLF annunciator off: Indicates that the system has sufficient inputs for navigation and is ready to be placed into the "VLF" navigation mode. If the system had not yet been in the VLF mode this indication will continue until the pilot manually selects the VLF mode. (See Runway Lineup, 1.D.) If the system had previously been in the VLF mode, this indication will continue for approximately 30 seconds and then automatically change to that of subparagraph b. below.

   b. VLF annunciator steadily illuminated: (With the DR annunciator flashing) indicates that the system is operating in the VLF mode but had been in the DR mode for a period of time. The Nav Warning Flag will retract when the VLF light comes on steady.

3. OFF:

   a. If the "VLF" and amber "DR" annunciators are off, the navigation computer has failed.

   b. The "DR" annunciator will be off during normal operations in the "VLF" navigational mode.

C. "ENTER" (White)

1. Steadily illuminated: The word ENTER will be steadily illuminated whenever the computer expects to receive some new information.

2. Flashing:

   a. The ENTER light will flash if the computer needs to have the information just entered verified by the pilot.

   b. Continued flashing will result whenever totally unreasonable information is being programmed, i.e., latitude over 90 or GMT over 2400.

D. WAYPOINT ALERT

Flashing digits in the FROM/TO display window indicate that the automatic leg change is impending.

5. NAVIGATION DISPLAYS

Once a navigation leg or pseudo-VORTAC mode has been selected, left and right course information will be displayed on the HSI when VLF is selected. A green FLT DIR VLF light will illuminate on the flight director annunciator panel when VLF information is being supplied to the HSI. Course information is provided for enroute navigation with a constant course width of ± 7½ nautical miles irrespective of the distance to the waypoint. On typical installations, the HSI DME readout will display raw DME distance to a tuned VORTAC, not distance to a VLF waypoint, unless indicated otherwise. On installations which include a VLF mode switch on the flight director selector, VLF distance will be displayed on the HSI when the VLF mode switch is selected.

NOTE

For the below listed airplanes the NAV 1 frequency selector must be placed in the VOR range. Frequency selection in the ILS range will automatically switch the flight director to the ILS mode. C90 (prior to LJ-925), E90 (prior to LW-342), 200 (prior to BB-746), 200C (prior to BL-14).

The GNS-500A drives the normal VOR/LOC warning flag when VLF information is displayed on the HSI. The warning flag will be in view anytime the VLF information is invalid (e.g., invalid leg selection or computer malfunction) or the system goes into the DR mode.

The course deviation indicator (CDI) displacement on the HSI is independent of the course selected (similar to a localizer). Therefore, to get relative bearing as well as displacement information, the current selected leg bearing must be set on the HSI course selector. In the MANUAL or AUTO modes, the BRG displayed on the GNS-500A display unit is always the bearing from the present position to the selected "TO" waypoint. Therefore, the selected leg bearing will be displayed only when the CDI is centered. The bearing of the selected leg will change in flight because the Great Circle route between selected waypoints will be displayed. Therefore, the course selector must be reset periodically to agree with the displayed bearing.

After a selected cross-track (SX) has been entered, the CDI will be centered when the airplane is positioned on the selected parallel leg. A green VLF SX light will illuminate on the flight director annunciator panel to indicate the displayed course guidance is with respect to a pilot-selected offset parallel leg.

6. AUTOPILOT OPERATION

Depending on the system installed, once the navigation leg on the pseudo-VORTAC mode has been selected, the flight director may be coupled to the GNS-500A by either:

1. placing the flight director switch in the FD VLF position and selecting the NAV mode, or
2. selecting the flight director VLF mode switch.

NOTE

To obtain reliable autopilot operation, the current selected leg bearing must be set on the HSI course selector.

If a course change of more than 30° is made at a waypoint, the new leg and course must be selected prior to waypoint passage. When operating in the AUTO mode and a course change of more than 30° is anticipated, MANUAL must be selected and the leg and course change made prior to waypoint

passage. To obtain improved autopilot performance during large course changes, select the flight director HDG and NAV ARM (or VLF ARM) modes, and establish a heading which will recapture the new course.

When VLF is operating in the surveillance mode "SM", the flight director will revert to the DR mode for grid turns. The heading mode "HDG" must be used for autopilot tracking of the VLF course signal.

7. MARINE SURVEILLANCE SEARCH PATTERNS

WARNING

GNS-500A operating modes are not FAA approved if either "SQR" or "LDR" is illuminated on the Multiple Annunciator Unit (MAU).

A. REQUIRED EQUIPMENT

1. Marine Surveillance PROM board Series 2.
2. Multiple Annunciator Unit (MAU).

B. INDENTIFYING A MARINE SURVEILLANCE SYSTEM

A GNS-500A Marine Surveillance System can be readily distinguished from the normal GNS-500A by the presence of the MAU. The operator may be assured that the correct program has been installed by the following procedure.

1. Mode Selector Switch - M
2. Display Selector Switch - GMT/DATE
3. Verify that GMT illuminates on the Multiple Annunciator Unit.
4. Rotate Display Selector Switch to verify operation of the MAU.

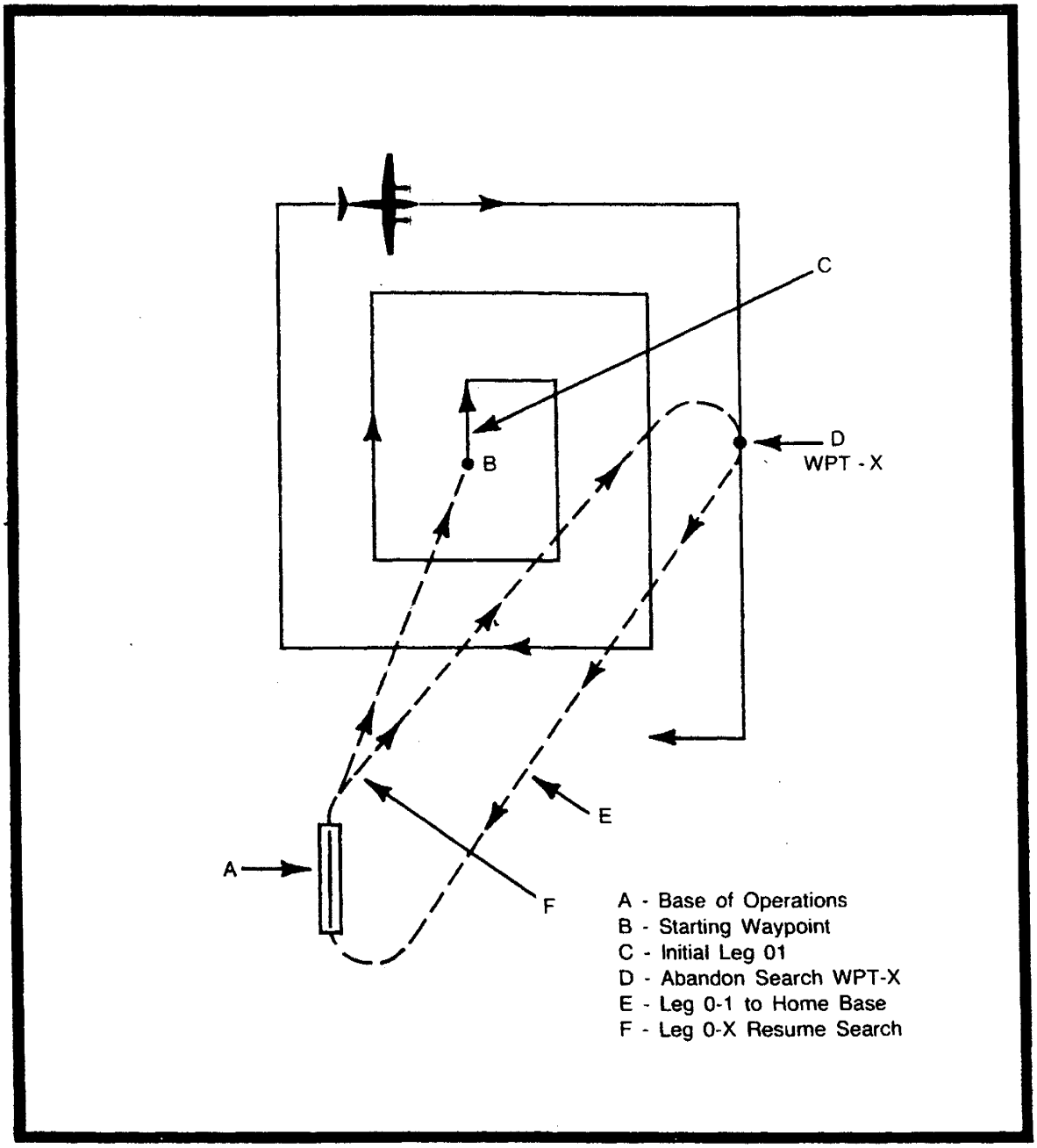
C. GNS-500A SEARCH PATTERN PROGRAMMING

Normal navigation functions are performed by the GNS-500A Marine Surveillance System when the Marine Search Mode is not active, thus allowing normal Compute Mode functions of flight planning and system self-checking. When entering into the Marine Surveillance Mode, Compute Mode "C" functions are restricted to the parameters of the "LDR" or "SQR" pattern selected. Normal navigation functions are annunciated by observing a dash in the Waypoint Display window after the selection of Mode "C". If 0-9 is displayed, a marine surveillance pattern has been programmed and may be active as indicated by the Multiple Annunciator Unit.

D. PROGRAMMING AN EXPANDING SQUARE SEARCH

1. Determine the starting latitude and longitude of the search pattern. Using normal waypoint entry procedures, enter latitude and longitude as the "Starting Waypoint" (Point B of the Expanding Square Search Pattern).

EXPANDING SQUARE SEARCH PATTERN



2. Select Mode "C" with Mode Selector Switch.

3. Depress WAYPOINT DEFINE key and enter the waypoint number assigned to the start of the search. Verify "SM" being illuminated on the MAU.

4. Select "BRG/VAR" with Display Selector Switch.

5. Re-enter the Magnetic Variation being displayed. Depress "E" or "W" followed by local variation in degrees. If operation with respect to True North is desired, depress "W" followed by the "O" key. The magnetic variation entered at this time will be used during the resultant search. Magnetic variation will automatically return upon termination of the Marine Surveillance operating mode.

6. Depress ENTER key.

7. Input the initial leg bearing.

8. Depress ENTER key which will estinguish.

9. Select "DIS/ETE" with Display Selector Switch.

10. Input initial leg length. The length of the initial leg in nautical miles and tenths will appear in the Left Data Display.

11. Depress ENTER key and the display will extinguish.

12. Select "XTK/SX" with Display Selector Switch.

13. Define direction of pattern rotation. Depress either "L" or "R" key, which will appear in the Right Data Display.

14. Depress ENTER key which will extinguish.

NOTE

This completes programming of an expanded square search pattern. Programming may be reaccomplished prior to flight or enroute and held in memory until required. Proceed in "M" or "A" mode to the search area using normal navigation procedures. Refer to the course line between "A" and "B" in the Expanding Square Search Pattern. As the approach is made to the search area, the airplane heading should be similar to that of the initial search leg. When 0.6 minutes from this location, complete the following steps to activate the search program.

1. Select Mode "C" with Mode Selector Switch.

NOTE

If computer access mode is used after programming of surveillance mode, the following is required to recall the surveillance mode.

   a. WPT/DEFINE - Depress

   b. Starting WPT number - Depress

   c. Verify "SM" is illuminated on MAU.

2. Depress LEG CHANGE key

3. Input "01", the initial search leg, (Leg C of the Expanding Square Search Pattern). Verify "SQR" is illuminated on MAU.

NOTE

There are 01 through 99 search legs available with GNS-500A search patterns.

4. Select Mode "A" with Mode Selector Switch and proceed with the search.

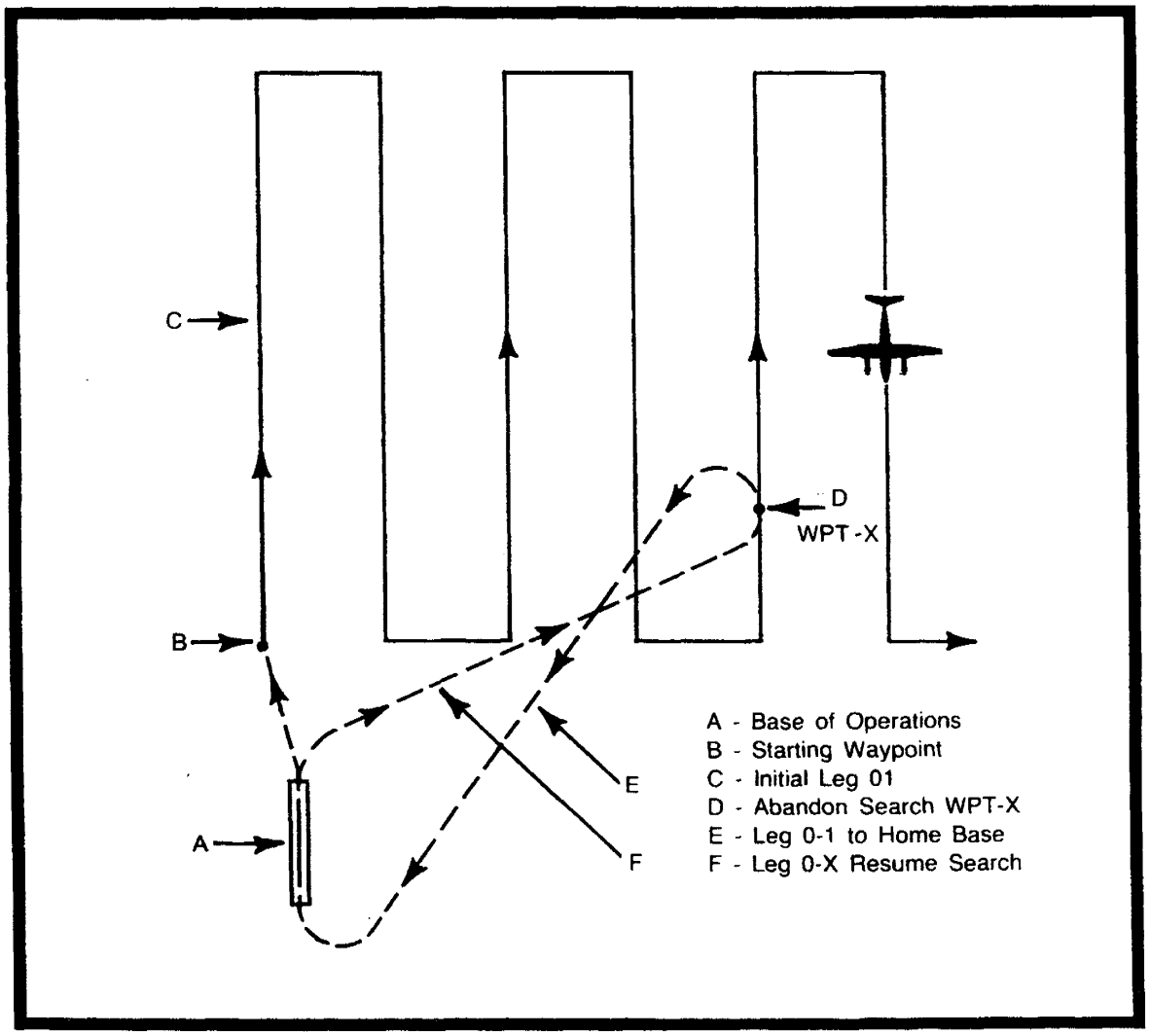
NOTE

The GNS-500A Maritime Surveillance Mode is now selected and active. Alway fly search patterns in Mode "A" to allow automatic sequencing of legs.

E. PROGRAMMING A LADDER SEARCH

1. Determine the starting latitude and longitude of the search pattern. Using normal waypoint entry procedures, enter latitude and longitude as the "Starting Waypoint" (Point B of the Ladder Search Pattern).

LADDER SEARCH PATTERN



2. Select Mode "C" with Mode Selector Switch.

3. Depress WAYPOINT DEFINE key and enter the waypoint number assigned to the start of the search. Verify "SM" being illuminated on MAU.

4. Select "BRG/VAR" with Display Selector Switch.

5. Re-enter the magnetic variation being displayed. Depress "E" or "W" followed by the local variation in degrees. If operation with respect to True North is desired, depress "W" followed by the "O" key. Magnetic variation entered at this time will be used during the resultant search. Automatic magnetic variation will return upon termination of the Marine Surveillance Operating mode.

6. Depress ENTER key.

7. Insert the initial leg bearing.

8. Depress ENTER key which will extinguish.

9. Select "DIS/ETE" with Display Selector Switch.

10. Insert initial leg length. The length of the initial leg in nautical miles and tenths will appear in the Left Data Display.

11. Depress ENTER key which will extinguish.

12. Select "XTK/SX" with Display Selector Switch.

13. Define direction of initial turn by keying "L" or "R" followed by the digits for spacing between legs. "L" or "R" followed by spacing between legs in nautical miles and tenths will appear in the Right Data Display.

14. Depress ENTER key which will extinguish.

NOTE

Programming of a ladder search is now completed. Programming may be accomplished prior to flight or enroute and held in memory until required. Proceed to the search area using normal navigation procedures. Refer to the course line between "A" and "B" in the LADDER SEARCH PATTERN. When 0.6 minutes from this location, complete the following step to activate the search program.

1. Select Mode "C" with Mode Selector Switch

NOTE

If computer access mode is used after programming of surveillance mode, the following is required to recall the surveillance mode.

a. WPT DEFINE - Depress

b. Starting WPT number - Depress

c. Verify "SM" is illuminated on MAU.

2. Depress LEG change key.

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3. Insert "01", the initial search leg. Reference "C" (LADDER SEARCH PATTERN). Verify "LDR" is illuminated on MAU.

NOTE

There are 01 through 99 search legs available with GNS-500A search patterns.

4. Select Mode "A" with mode selector switch and proceed with the search.

NOTE

The GNS-500A Maritime Surveillance Mode is now selected and active. Always fly search patterns in Mode "A" to allow automatic sequencing of legs.

F. TERMINATING THE EXPANDING SQUARE (SQR) OR LADDER SEARCH (LDR)

1. With mode selector switch set to "A", manual entry of any leg change instructions will immediately terminate the active search. This procedure should be used upon completion of search, low fuel, etc. For example, present position to home base might be programmed leg change 0-1. Refer to Leg E of the Expanding Square Search Pattern or of the Ladder Search Pattern. If an abandoned search is to be resumed from this point at a later date, or with another airplane, record the present position coordinates and airplane heading at this time. Verify "SQR" or "LDR" extinguished on MAU.

2. To return to standard navigation displays in Compute Mode: Set Mode Selector Switch to "C"; depress "WPT DEF" key; then, depress ENTER key. Normal flight planning displays are now available in Mode "C".

G. RESUMING THE SEARCH

Utilizing the previous recorded latitude and longitude identifying the point at which the search had been abandoned (Point D on either Expanding Square Search Pattern or Ladder Search Pattern), proceed from the base of operations to "WPT X" (Leg F of either "SQR" or "LDR").

NOTE

The previously recorded latitude and longitude must be defined as a WPT other than the starting point of the search.

1. Select Mode "C" 0.6 minutes from "WPT X."

NOTE

In the event Search Mode was cancelled from Compute Mode, or if another airplane is continuing the search, WPT DEF and insertion of original search starting waypoint may be necessary as if verifying the MAU annunciator light functions.

NOTE

It may be helpful to approach "WPT X" while flying the previously recorded airplane heading.

2. Depress "LEG CHG" key. The FROM/TO Display will display the last search leg flown and the ENTER key will illuminate.

3. Depress ENTER key. The two digit leg number identifying the last leg flown will be displayed in the FROM/TO window and, the ENTER light will extinguish. Depending upon the search pattern programmed, the MAU "SQR" or "LDR" annunciator will illuminate at this time.

4. Select Mode "A" with the Mode Selector Switch and resume search.

NOTE

This procedure allows resumption of search at any point on any leg.

H. MAU ANNUNCIATORS

1. "SM" (Orange) - SURVEILLANCE MODE OF OPERATION

2. "SQR" (Orange) - EXPANDING SQUARE SEARCH PATTERN

3. "LDR" (Orange) - LADDER SEARCH PATTERN

PERFORMANCE - No Change

SYSTEMS DESCRIPTION

The GNS-500A, Series 2 is an automatic navigation system which utilizes a digital computer in conjunction with Very Low Frequency (10-30 kHz) radio transmissions to continuously keep track of the airplane's position and to provide Great Circle navigation to pilot-selected waypoints or destinations.

The transmissions used by the system are provided by both Very Low Frequency U.S. Naval Communications stations and the stations comprising the OMEGA Navigation network. The computer automatically selects eight signals which are used simultaneously to optimize the navigational coverage in the geographical area of the flight.

The system is capable of VLF/OMEGA navigation using as few as two stations, provided the Rubidium Frequency Standard in the OEU is operational. If it is not, three stations are required.

Should the number or quality of incoming signals become unacceptable, the computer will automatically revert to Dead Reckoning (DR) navigation using the airplane's Heading and True Airspeed inputs together with the previously computed winds until such time as these factors improve to an acceptable level. The pilot can select for display the computer's assessment of its ability to function with the signals available, called Quality Factor or "Q". A Dead Reckoning "DR" annunciator light will indicate periods of unacceptable signal reception.

The computer is constantly processing all available inputs. While the displays of Present Position, Distance-to-go, and Cross-track, are updated once each updated once each second, the displays of Drift Angle, Track, Track Angle Deviation, Wind, Groundspeed, and Estimated Time Enroute are averaged over ten second periods.

The GNS-500A, Series 2 utilizes latitude and longitude information to extablish the desired track and to provide positional information. The equipment operator must determine the latitude and longitude of the airplane starting point or reference point and the lattitude and longitude of waypoints to be overflown enroute, as well as the final destination. This information can be derived from airport reference charts, area navigation charts or airport and navaid positional reference data, as found in the "Enroute Supplements" published by the Defense Mapping Agency Aero Centers.

The continuously operating clock and the non-volitile memory will provide GMT and date, last present position coordinates, the coordinates of the waypoints entered last and the magnetic variation of the present position as soon as the system is energized. Prior to departure these values must be checked and updated, if necessary, particularly if the system has been removed from the airplane.

The pilot will have available a digital display on the CDU of any pertinent navigational data required. Course guidance inputs for the HSI, autopilot, and flight director systems are continuously provided. Enroute, the pilot may select either automatic or manual leg changes at each waypoint. Should it be desired to deviate slightly form the planned course, the GNS-500A offers a variety of methods for the pilot to quickly establish the alternate course guidance he may require, such as present position direct to any waypoint, waypoint bypassing, offset parallel courses, and pseudo-VORTAC establishment at any waypoint location. Should a completely new course become necessary, the pilot need only define the new waypoints.

Approved Donald A Peters

for W.H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, AND B200CT LANDPLANE

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

LEAR-SIEGLER MODEL 23069 016 300-AMP STARTER/GENERATOR

GENERAL

The information in this supplement is FAA approved and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane is equipped with Lear-Siegler Model 23069-016 300-amp Starter/Generators (P/N 100-389014-5) in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the Manual are advised always to refer to this supplement for possibly superseding information and placards/markings applicable to operation of the airplane.

LIMITATIONS

The following is a list of minimum N_1 speeds for the designated generator loads and types of operation indicated.

TYPE OF OPERATION	MINIMUM N1 SPEED ~ % RPM	MAXIMUM GENERATOR LOAD ~ % OF LOAD
*GROUND (From S.L. to 5000 ft)	52% 55% 65% 70%	50% 66% 90% 100%
**FLIGHT	75%	100%

*Always observe engine ITT limits when operating at low N1 speeds. High ITT may be decreased by reducing accessory load and/or increasing N1 speed.
**This flight operation limit is for airspeeds of 116 KIAS and higher. Observe engine ITT limits.

EMERGENCY PROCEDURES - No Change

NORMAL PROCEDURES - No Change

PERFORMANCE - No Change

Approved: Donald H Petz

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: December, 1984
P/N 101-590010-139

For Training Purposes Only

BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT
B200, B200C; B200T, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND
FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the
COLLINS FCS-80 AUTOMATIC FLIGHT CONTROL SYSTEM
CATEGORY I
and
COLLINS EFIS-85 (EFIS-85A-1a or EFIS-85A-2a)
ELECTRONIC FLIGHT INSTRUMENT SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Collins FCS-80 Automatic Flight Control System with or without the Collins EFIS-85 Electronic Flight Instrument System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Collins Pilot's Guides for the Electronic Flight Instrument System (EFIS-85A-1a) No. 523-0772833, dated March 9, 1983 or later issue, (EFIS-85A-2a) No. 523-0773278, dated October 15, 1983 or later issue, and the Vertical Navigation Indicator (VNI-80A) No. 523-0772019, dated May 31, 1982 or later issue, are available from Collins General Aviation Division, 400 Collins Road NE, Cedar Rapids, Iowa 52498. It is the responsibility of the owner/operator to maintain the appropirate pilot's guides in current status.

LIMITATIONS

1. Do not use autopilot on the Super King Air 200T, 200CT, B200T, or B200CT when the optional wing-tip fuel tanks are installed.
2. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
3. Do not use autopilot under 200 feet above terrain.
4. Do not use autopilot or yaw damper during takeoff or landing.
5. Pilot must be seated at the controls with the seat belt fastened during autopilot operations.
6. Do not use propeller speeds in the range of 1750-1850 rpm during ILS approaches.
7. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
8. Disengage autopilot if TRIM FAIL annunciator illuminates.
9. Operation of the VNI-80A VNAV system must be in conformity with the Collins Pilot's Guide, No. 523-0772019, dated May 31, 1982 or later issue.
10. The VNAV system has not been shown to meet the accuracy requirements of Advisory Circular 90-45A and is therefore not to be used as the primary source of vertical guidance information.
11. Autopilot coupled VNAV descent angles exceeding 10 degrees are prohibited.

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12. Operation of the EFIS-85A-1a or EFIS-85A-2a system must be in conformity with the Collins Pilot's Guide, No. 523-0772833, dated March 9, 1983 or later issue or No. 523-0773278, dated October 15, 1983, or later issue, respectively.

13. Navigation and checklist data on the Multifunction Display (MFD-85) is provided only as an aid and is not approved as primary information.

14. IFR operation is prohibited in the CMPST (Composite) mode.

15. VOR/DME approaches are prohibited in the LINEAR ENABLED (linear deviation) mode on airplanes equipped with the EFIS-85A-1a system.

EMERGENCY PROCEDURES

The autopilot can be disengaged by any of the following methods:

1. Press the AP-YD/TRIM disconnect switch on the pilot's or copilot's control wheel.
2. Move the engage levers to the DISENGAGED position.
3. Select the go-around mode. (Yaw damper will remain on.)
4. Pull out (off) the AP POWER circuit breaker. (Yaw damper will remain on.)
5. Pull out (off) the AP ENGAGE circuit breaker. (Yaw damper will remain on.)
6. Pull out (off) the YAW DAMP circuit breaker.
7. Turn off the airplane MASTER SWITCH (Gang Bar - DOWN)
8. Turn off the AVIONICS MASTER PWR switch. (Yaw damper will remain on.)
9. Operate the electric trim.

The following conditions will cause the autopilot to disengage automatically:

1. Any interruption or failure of power.
2. Vertical gyro failure indication.
3. Automatic flight control system power or circuit failure.

In the event of an engine failure:

1. Disengage the autopilot, retrim the airplane, and re-engage the autopilot. Maintain at least 120 KIAS until landing is assured.

Maximum altitude losses during malfunction tests were:

OPERATION CONFIGURATION	ALTITUDE LOSS
Climb	160 feet
Cruise	220 feet
Maneuvering	90 feet
Descent	450 feet
Approach/ILS Coupled	50 feet
Single Engine Approach/ILS Coupled	80 feet

For the optional EFIS-85 system:

1. In the event of a Display Processor failure: Activate the AUX DRIVE switch.

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2. In the event of an EADI or EHSI display failure: Activation of the CMPST switch will provide a composite display on the remaining indicator.

3. In the event of a number 1 dual fed electrical bus failure: Activation of the AUX DRIVE and CMPST switches will provide attitude information (minimum) on the EHSI in the composite format. (Inverter #2 is required for AC power.)

NORMAL PROCEDURES

NOTE

The push buttons on the flight director mode selector are the push-on, push-off type. When engaged, amber "gull wings" appear above the selected button. For operation at night, the push buttons are illuminated by the OVERHEAD SUBPANEL & CONSOLE LIGHTS control.

The FCS-80 either incorporates its own annunciator panel or provides mode annunciation on the EADI should the optional EFIS-85 be installed. The annunciator panel is located just above the flight director display on the instrument panel. The modes and indications given on the annunciator panel are placarded on the face of the plastic lenses and illuminate when the respective conditions are indicated. Mode annunciation will be displayed on the EADI if the EFIS-85 is installed.

PREFLIGHT CHECK

AUTOPILOT

1. Pull the control wheel aft to mid-travel and engage the autopilot by moving the AP paddle switch forward to the ENGAGED position. Push forward on the control wheel and hold. The trim wheel should move in the nose-up direction. Pull the control wheel aft and hold. The trim wheel should move in the nose-down direction.

2. Move the pitch wheel (UP - DN) and the TURN knob. The control wheel should respond accordingly.

3. Set HSI/EHSI heading marker under lubber line, select HDG mode, and move the heading marker. The control wheel should respond accordingly.

4. Depress the autopilot-yaw damp/trim disconnect switch on the control wheel. Ensure that the AP paddle switch moves aft to the DISENGAGED position. If the amber AP DISC annunciator (or AP/L or AP/R EADI annunciator) begins flashing, recycle the AP-YD/TRIM disconnect switch to extinguish it.

5. Apply force to the primary controls. They should not resist movement.

NOTE

The position of the elevator trim tab should be checked after the autopilot is disengaged, to ensure that the trim servo did not move it from the desired position.

ELECTRIC ELEVATOR TRIM

1. Depress the ELEV TRIM push-button switch (on pedestal) to ON (ELEV TRIM illuminates on face of the push button).

2. Check operation of dual-element thumb switches as detailed in the Pilot's Operating Handbook except that if the pilot and copilot initiate trim in opposite directions, the trim will stop.

WARNING

Operation of the electric trim system should occur only by movement of pairs of switches. Any movement of the elevator trim wheel while actuating only one switch denotes a system malfunction. If a malfunction of the electric trim system is indicated, electric trim must be disengaged and trim changes made with manual trim ONLY.

DIGITAL AIR DATA SYSTEM (if installed)

1. Press and hold the cockpit-mounted ADC TEST switch for the Air Data Computer. Check for the following:
   A. Air data instrument flags come into view, and the instruments go into the loss-of-data mode.
   B. Altitude pointer goes to the 250-foot mark.
   C. Vertical speed pointer goes to 6000-ft/min rate of descent mark.
   D. Airspeed Indicator IAS and V_{MO} pointers go to zero.
2. Release the ADC TEST switch and check for the following:
   A. Data transmissions to the air data instruments are restored.
   B. All air data instrument flags go out of view.
3. Press and hold the TEST switch on the Vertical Speed Indicator, and check for the following:
   A. The warning flag comes into view.
   B. The pointer slews to the 6000-ft/min rate of climb mark.
4. Release the TEST switch, and check for the following:
   A. The flag goes out of view.
   B. The pointer returns to zero.
5. Press and hold the PUSH TEST switch on the Airspeed Indicator, and check for the following:
   A. The warning flag comes into view, and both pointers (IAS and V_{MO}) slew to 160 knots.
   B. After approximately 1 second, both pointers slew to zero.
6. Release the PUSH TEST switch and check for the following:
   A. The flag goes out of view.
   B. The barber pole (V_{MO} indicator) returns to 260 knots.

The following items apply only to installations incorporating the Collins Barometric Altimeter:

7. Press and hold the TEST switch on the Barometric Altimeter, and check for the following:
   A. The warning flag comes into view in approximately ½ second.

B. The altitude pointer goes to the 750-foot mark.

NOTE

The output from the altimeter to the transponder is inhibited during the self test.

8. Release the altimeter TEST switch, and check for the following:

   A. The flag goes out of view.

   B. The altitude pointer returns to the actual altitude.

9. Turn the BARO knob on the altimeter and observe that the baroset digits and altitude pointer respond accordingly.

10. Push BARO knob in to obtain baroset values in in. Hg. Set the barometric display to the correct local pressure value. (Pull BARO knob out if display of baroset values in millibars is desired.)

11. Press and hold the ALT ALERT annunciator/switch on the Altitude Preselector/Alerter, and check for the following:

   A. The ALT ALERT annunciator illuminates.

   B. The warning flag comes into view.

   C. Both remote ALT ALERT annunciators illuminate.

12. Release the ALT ALERT annunciator/switch, and check for the following:

   A. The ALT ALERT annunciator extinguishes.

   B. The warning flag goes out of view.

   C. Both remote ALT ALERT annunciator extinguish.

GENERAL OPERATION

ENGAGING AUTOPILOT

Hold the AP ENGAGED - DISENGAGED switch lever on the autopilot controller in the ENGAGED position momentarily. The yaw damp system may be engaged separately by moving the YD paddle switch to the ENGAGED position.

DISENGAGING AUTOPILOT

Press the control wheel AP-YD/TRIM disconnect switch for normal autopilot and/or YD disengagement.

OPERATING MODES

The autopilot has three modes of operation: Manual Mode, Autopilot Sync Mode, and Guidance Mode.

MANUAL MODE

When the autopilot is engaged (engage lever in the ENGAGED position) and no modes are selected on the

flight guidance panel or with the go-around button, the autopilot is in the manual mode. The autopilot accepts pitch and roll rate commands from the pitch wheel (UP - DN) and the TURN knob on the autopilot panel.

GUIDANCE MODE

When the autopilot is engaged and a lateral and/or vertical mode is selected, the autopilot is in the guidance mode and accepts steering commands from the flight guidance computer (or EFIS-85 Display Processor Unit, if installed).

AUTOPILOT SYNC MODE

Autopilot sync mode is controlled by the PITCH SYNC & CWS (control wheel steering) button on the control wheel. With the autopilot engaged and in either manual or guidance mode, depressing the PITCH SYNC & CWS button allows the pilot to maneuver the airplane, with servo assistance, without disengaging the autopilot. When the PITCH SYNC & CWS button is released, the autopilot maintains the attitude existing at the time the button is released if in manual mode, or follows commands received from the flight guidance computer if in guidance mode.

NOTE

The PITCH SYNC & CWS button simultaneously controls autopilot sync mode and the flight director system vertical sync mode.

FLIGHT DIRECTOR

The flight director system supplies steering commands for the pilot and the autopilot. When the autopilot is engaged, the pilot monitors autopilot performance on the attitude director indicator. When the autopilot is not engaged, the pilot flies the airplane manually in response to the attitude director indicator commands. When the EFIS-85 is installed, the flight director system supplies steering commands to the EFIS which in turn supplies these commands for the pilot and autopilot.

EFIS-85 SYSTEM (if installed)

The EFIS-85 Electronic Flight Instrument System replaces the conventional electro-mechanical Attitude Director (ADI) and Horizontal Situation (HSI) indicators. The CRT displays (EADI & EHSI) provide the necessary information for the pilot to respond to flight director and navigation commands in the same manner as with conventional instruments. The system also includes additional features which allow the pilot to select alternate formats as a navigation aid or in the event of specific failures. The operator is cautioned that the primary displays (attitude and directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate modes is described in the Collins EFIS-85 Pilot's Guide. The Owner/Operator of the airplane is responsible for obtaining and maintaining this manual.

1/2 BANK

In ½ BANK mode, the bank limit in HDG or NAV mode is reduced to one-half the normal value. The ½ BANK mode is interlocked so that selection of APPR mode or localizer capture clears the ½ BANK mode if selected, or prevents its being selected.

VERTICAL MODES

All of the air data hold modes (i.e., altitude, and the optional vertical speed and indicated airspeed) operate identically. Commands are generated to hold the reference value which exists at the time the desired mode is selected. Only one of the air data hold modes can be selected at any given time.

When the optional slewing bugs are installed on the airspeed and vertical speed indicators, selecting either the IAS HOLD mode or the VS HOLD mode causes the respective bug to immediately slew to the existing value. This becomes the reference value that the system will hold. The reference indicated airspeed or vertical speed can be changed while the IAS or VS HOLD mode is engaged by: 1) turning the respective PUSH TEST knob; or 2) manually flying the airplane to the new reference value with the PITCH SYNC & CWS button depressed.

WARNING

Do not slew the IAS bug beyond airspeed limits nor the VS bug beyond 6000 ft/min with the respective IAS HOLD or VS HOLD mode engaged.

TAKEOFF (FLIGHT DIRECTOR ONLY)

While rotating to climb attitude, press the PITCH SYNC & CWS button and establish climb airspeed. As soon as the desired climb attitude is established, release the PITCH SYNC & CWS button on the control wheel. The command bars then command flight to the pitch attitude that existed at the time the PITCH SYNC & CWS button was released. Selection of another vertical mode cancels pitch hold mode.

WARNING

Autopilot must be disengaged for takeoff.

NOTE

A lateral mode must be selected to bring the command bars into view and to obtain vertical sync.

CLIMB

Engage the autopilot. Make heading changes by moving the heading marker on the HSI (or EHSI). The flight control system maneuvers the airplane in a coordinated turn to the new heading. The course arrow can be set to the first VOR course. Adjust the COURSE knob until the desired course appears in the course display window on the HSI/EHSI. The head of the course arrow now indicates the direction of the selected course.

CRUISE ALTITUDE CAPTURE (WITH ALTITUDE PRESELECT)

Set desired altitude on Altitude Preselector. Engage ALT SEL mode. Using desired vertical mode, maneuver the airplane toward the desired altitude. As the airplane approaches the selected altitude, the ALT CAP annunciator (ALTS annunciator on EADI) will illuminate, and the VS or IAS button (if selected) will automatically disengage.

CRUISE ALTITUDE CAPTURE (WITHOUT ALTITUDE PRESELECT)

Prior to reaching cruise altitude, lower vertical speed to not more than 500 feet per minute. When at the cruise altitude, press the ALT button.

VOR NAVIGATION

To establish the airplane on a desired VOR course, perform the following:

1. With the autopilot engaged and HDG mode selected, maintain the lateral flight path by adjusting the heading marker as required. Any vertical mode may be selected to maintain the desired vertical flight path.

2. Tune the navigation receiver to the desired VOR and verify the station identifier.

3. Set the COURSE display on the course indicator to the desired course. The head of the course arrow now represents the desired VOR course.

4. Set the heading marker on the course indicator to the desired intercept heading, and allow the autopilot to establish the airplane on the intercept heading (intercept angle not to exceed 90 degrees).

5. With the airplane established on the intercept heading, press the NAV button on the mode selector. The NL ARM annunciator (white VOR annunciator on EADI) will illuminate, indicating that the system is armed for VOR capture.

With the above procedures completed, the airplane follows the selected heading to the point of beam capture. At VOR course capture, the NL CAP annunciator (green VOR annunciator on EADI) illuminates and the HDG and NL ARM/white VOR annunciators extinguish.

After capture, the system commands rollout on the VOR course, and provides automatic crosswind correction.

Normally, the lateral deviation bar on the HSI/ EHSI represents angular deviation from the selected course. However, if the optional linear deviation feature is in operation, the lateral deviation bar represents linear deviation. In that case, full-scale deflection (2 dots) represents an offset of 10 NM from the selected course in NAV mode, or 2 NM in APPR mode. To select the linear deviation feature, place the flight director deviation selector switch (FLT DIR DEV - ANGULAR - LINEAR ENABLED) on the pilot's instrument panel down into the LINEAR ENABLED position. If valid VOR and DME signals are being received and either NAV or APPR mode is selected on the flight guidance panel, the LIN DEV annunciator in the FCS-80 annunciator panel (LIN annunciator on EADI) will illuminate, to advise the pilot that the lateral deviation bar is indicating linear deviation.

NOTE

EFIS-85 equipped airplanes display linear deviation only with a full-scale offset of 10 NM. This presentation is displayed in both NAV and APPR modes. Consequently, VOR/DME approaches are prohibited in the LINEAR ENABLED (linear deviation) mode.

When passing over a VOR station, the autopilot computer may go into the Dead Reckoning submode, indicated by the flashing NL CAP annunciator (DR annunciator on EADI). In the Dead Reckoning submode, the bank angle goes to zero and the airplane maintains the average course value flown just before entering the signal cone. After passing the station, the system will resume tracking the outbound course when the computer switches out of Dead Reckoning submode, indicated by steady illumination of the NL CAP annunciator (extinguishing of DR annunciator on EADI).

Outbound course changes may be selected when passing the VOR station, provided the course change is not more than 60 degrees. Set the course arrow to the new outbound course at the time the to-from arrow changes from inbound to outbound. The autopilot will turn the airplane to attain the new selected course, and station passage will be as described above.

VOR APPROACH

To fly a typical VOR approach, track in to the station in NAV mode. The flight control system tracks the selected course to the station. The to-from pointer will reverse at station passage. Set the heading marker under the lubber line and select HDG mode. Then proceed as follows:

1. Track outbound by adjusting the heading marker to position the airplane on the desired track. Set the course arrow to the published inbound course at this time. To start the procedure turn, adjust the heading marker to the appropriate heading for the published procedure turn.

2. Continue adjusting the heading marker not more than 135 degrees at a time for the 180-degree portion of the procedure turn. If letdown is desired during the turn, use the pitch wheel on the Autopilot Panel to set the desired rate of descent. If altitude hold is desired, press the ALT button on the mode selector at the desired altitude.

3. Maneuver the airplane to the intercept angle (not to exceed 90 degrees) in the HDG mode. When established on the desired intercept course, select APPR mode. The NL ARM annunciator (white VOR annunciator on EADI) will illuminate. As the flight control system intercepts the selected inbound course, the HDG and NL ARM/white VOR annunciators will extinguish, and the NL CAP annunciator (green VOR annunciator on EADI) will illuminate.

FRONT COURSE APPROACH

On an ILS front course approach, the localizer and glideslope are automatically and independently captured. The localizer is always captured from a selected heading. The glideslope can be captured from any of the vertical modes and from above or below the glideslope. To fly a front course approach, perform the following:

1. Set the heading marker to current airplane heading. Select HDG and any vertical mode on the flight guidance panel.

2. Turn the navigation receiver to the localizer frequency and verify the station identifier. Set the course arrow to the published inbound course. The runway symbol at the bottom of the ADI/EADI appears when the ILS frequency is tuned and a valid localizer signal is being received. With HDG mode still selected, set the heading marker to the desired intercept heading (intercept angle not to exceed 75°) and allow the autopilot to establish the airplane on the intercept heading. With the airplane established on the intercept heading, press the APPR button on the flight guidance panel.

3. The HDG, NL ARM, and GS ARM annunciators (HDG, white LOC/GS annunciators on EADI) illuminate to indicate that the system is armed for ILS capture. As the airplane nears the localizer, the HDG and NL ARM annunciators (HDG and white LOC annunciators on EADI) extinguish and the NL CAP annunciator (green LOC annunciator on EADI) illuminates. The airplane makes a coordinated rollout on the localizer.

NOTE

The published missed approach heading may be set with the heading marker at this time.

4. Maneuver the airplane toward the glideslope using the pitch knob or any vertical mode. When the glideslope is captured, the GS ARM annunciator (white GS annunciator on EADI) extinguishes and the GS CAP annunciator (green GS annunciator on EADI) illuminates. Any selected vertical mode automatically releases at glideslope capture. Commands are generated to maintain the center of the localizer and glideslope.

5. If a radio altimeter is installed, the decision height (DH) annunciator illuminates when the preselected radio altitude is reached.

6. At decision height, disengage the autopilot for landing or select go-around mode, as required.

BACK-COURSE APPROACH

As in a front course approach, the localizer is captured automatically. Glideslope circuits are automatically inhibited during a back course approach. To perform a back course approach, perform the following:

1. Set heading marker to current airplane heading. Select HDG mode on the flight guidance panel. Any vertical mode may be selected.

2. Tune navigation receiver to localizer frequency and verify the station identifier. Set course arrow to the published inbound (front) course. Note that course arrow is always set to the front localizer course. With HDG mode still selected, set heading marker to the desired intercept heading (intercept angle not to exceed 75°) and allow the autopilot to establish the airplane on the intercept heading.

3. With the airplane established on the intercept heading, press the APPR button on the mode selector. The NL ARM, and BL annunciators (white LOC or B/C annunciator on EADI) illuminate to indicate that the system is armed for back localizer capture.

4. As the airplane nears the center of the localizer, the HDG and NL ARM annunciators (HDG and white LOC or B/C annunciators on EADI) extinguish and the NL CAP annunciator (green LOC or B/C annunciator on EADI) illuminates. The airplane makes a coordinated turn and rollout to track the center of the back course localizer.

5. If desired, set minimum altitude on the Altitude Preselector and select ALT SEL mode. Use the pitch wheel on the autopilot panel to establish the desired rate of descent. If minimum altitude is reached before visual contact is made and ALT SEL mode has not been used, select ALT hold to maintain minimum altitude until visual contact is made or until time to missed approach has elapsed. The autopilot must be disengaged for landing.

6. The decision height (DH) annunciator and go-around operation are the same as for a front course approach.

NOTE

It is recommended that the HDG mode be used if less than 1 mile from the runway, due to the large radio deviations encountered when flying near the localizer transmitter on a back course approach.

GO-AROUND

The runway heading or published missed approach heading may be set with the heading marker on the HSI/EHSI as soon as localizer capture occurs.

Execute a missed approach as follows:

1. Press the GA button. All other modes will cancel, the autopilot will disengage (yaw damp remains engaged), and steering commands will be provided for a wings-level, 7°-pitch-up attitude.

NOTE

Go-around mode can be selected anytime during the approach.

2. The 7°-nose-up pitch reference can be changed by depressing the PITCH SYNC & CWS button on the control wheel, flying the airplane to the desired pitch attitude, then releasing the PITCH SYNC & CWS button. Wings-level lateral commands continue to be presented.

3. After airplane cleanup, go-around power settings, and airspeed are established, select HDG mode on the flight guidance panel (cancels go-around mode). Commands are generated to maintain the desired heading selected with the heading marker on the HSI/EHSI, and to maintain the pitch attitude that existed at the time HDG mode was selected.

4. Engage the autopilot. Make heading changes by adjusting the heading marker. Use the pitch wheel on the autopilot panel to change pitch attitude.

TURBULENCE MODE

Pressing the TURB button softens the autopilot gains to provide proper control in turbulence. The TURB button is a push-on, push-off switch. An interlock disables TURB mode when APPR mode is selected. The "gull wings" above the button illuminate when the TURB mode is selected.

SPECIAL NOTES

1. With the autopilot disengaged, always maneuver the airplane to "fly" the ADI/EADI airplane symbol into the command bars to satisfy pitch and bank commands.

2. The pilot may preset the missed approach heading during the approach (after localizer capture occurs) so that missed approach steering commands will be available by selecting the HDG mode.

3. Course may be changed over a VOR station when operating in NAV mode as long as the course change is not more than 60 degrees. If the course change is more than 60 degrees, select a new intercept heading with the HSI/EHSI heading marker, select HDG, set the desired VOR course, then select NAV mode again. The system reverts back to HDG and VOR capture mode, and a new capture cycle will occur.

4. When flying inbound toward a VOR at normal cruise speeds, do not set up a VOR capture problem within approximately 15 miles of the VOR station. The system does not have time to solve the problem and stabilize on course before passing over the station.

5. Limit all VOR intercept angles to 90 degrees or less, and LOC front course and back course intercept angles to 75 degrees or less. When setting the capture angle, allow for speed and distance from station, so that the system can capture without excessive overshoots.

6. When tracking outbound on a VOR course (system in VOR capture mode), the pilot can retune the navigation receiver to the next VOR and track inbound on the proper course without selecting HDG mode. Retune the navigation receiver at the appropriate segment changeover point. The system will remain in VOR capture mode during the retuning. If the signal from the new station is reliable, the to-from pointer on the HSI/EHSI will change and indicate "to" the new station. If the signal is unreliable, the NAV flag on the HSI/EHSI will appear. In this case, HDG mode should be selected to maintain the flight path.

7. Closely monitor outbound tracking after VOR station passage to ensure that the system is maintaining the desired track, especially after course changes or during low-altitude operations. The NL CAP annunciator will flash anytime the system is in Dead Reckoning mode (DR annunciator on EADI).

8. Unless otherwise annunciated, distance displayed on the pilot's HSI/EHSI is raw DME distance.

9. The APPR mode should be used for flying all approaches (Front Course, Back Course, VOR, and RNAV). Fly the final segment of all autopilot-coupled approaches at or above 120 KIAS, flaps 40%, and landing gear extended.

10. Linear deviation is inhibited if a localizer frequency is tuned, regardless of the position of the selector switch.

11. If the autopilot automatically disengages, the flashing AP DISC annunciator (AP/L or AP/R annunciator on EADI) can be extinguished by depressing the AP-YD/TRIM disconnect switch on the control wheel.

PERFORMANCE - No Change

WEIGHT AND BALANCE/EQUIPMENT LIST - No Change

AUTOPILOT ANNUNCIATOR PANEL

ANNUNCIATOR	*CONDITIONS UNDER WHICH IT ILLUMINATES*	*COLOR*
NL ARM	After APPR or NAV mode is selected, after an ILS or VOR frequency is tuned, and before course capture.	Amber
NL CAP Steady	After APPR or NAV mode is selected and signal captured.	Green
Flashing	System is in dead reckoning.	Green
GS ARM	After APPR mode is selected, a localizer frequency is tuned, and a glideslope valid is received on a front course approach	Amber
GS CAP	After glideslope capture in APPR mode on a front course approach.	Green
ALT ARM	Selection of altitude preselect mode.	Amber
ALT CAP	Preselected altitude has been captured.	Green
HDG	Selection of HDG mode. Also illuminates when in NL ARM submode.	Green
GA	Selection of go-around mode.	Green
BL	After APPR or NAV mode is selected, a localizer frequency is tuned, and the difference between airplane heading and the course arrow exceeds approximately 105 degrees.	Green
TRIM FAIL A (or) B	Dual monitoring when elevator trim system fails or when trim runs away when autopilot is engaged.	Red
ALT HLD	Selection of altitude hold mode.	Green
AP DISC (flashing)	Autopilot is disengaged.	Amber
LIN DEV	Linear deviation is being displayed.	Green
IAS	Selection of IAS hold mode.	Green
VS	Selection of VS hold mode.	Green
EFIS-85A-1a	Refer to Collins Pilot's Guide No. 523-0772833 for description of EFIS-85 annunciators	
EFIS-85A-2a	Refer to Collins Pilot's Guide No. 523-0772833 for description of EFIS-85 annunciators.	

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For Training Purposes Only

WARNING FLAGS

The flight control system flags are positive monitoring. The proper operating signals must be present to keep the associated flags from view. Limited system operation is possible with some of the flags in view. The following paragraphs describe malfunctions that cause each flag to appear, and specify the information that remains reliable in each case. Check to determine whether or not the apparent malfunction can be corrected by resetting circuit breakers, recycling an associated receiver, or slaving or erecting a gyro. With an EFIS-85 installed, a flag condition results in removal of the data from the display and display of a red box with an abbreviation of the failed function inscribed. Pilot action remains the same.

HEADING

The HEADING flag indicates a failure of the compass system. All heading display and command information is unusable. For enroute operation, set the flight guidance panel mode select buttons to off to remove the command bars from view. For terminal operation, VOR/LOC deviation and glideslope information is correct.

GLIDESLOPE

The GS flags indicate a malfunction of the glideslope receiver or an unreliable glideslope signal when tuned to a localizer frequency. Vertical commands in approach mode are unusable. Command information in the heading or NAV mode is correct. Attitude, heading, and VOR/LOC position information is correct.

NAV

The NAV flag indicates either a malfunction of the navigation receiver or an unreliable radio signal, and warns that the lateral deviation displays and roll commands computed from VOR or localizer information are unusable.

COMPUTER

The computer flag monitors the active inputs to the flight director system. If the CMPTR flag appears, command information is unusable. Attitude, heading, VOR/LOC position, and glideslope information may still be reliable. Display of one or more additional flags indicates which information is unreliable.

GYRO

The GYRO flag indicates failure of the vertical reference gyro or attitude circuits. Attitude and command information is unusable. VOR/LOC position and heading information is still correct.

HANDLING, SERVICING AND MAINTENANCE

SERVICING SCHEDULE

1. If a Collins Altimeter is installed, one additional static-line drain is installed. It is located inside the nose avionics compartment in the pilot's static air line that provides static pressure to the digital air data computer. This drain should be opened to release any trapped moisture at each 100-hour inspection, and after exposure to visible moisture on the ground. It must be closed after draining.

2. If a Collins EFIS-85 is installed, fans are installed to aid in cooling of the Display Processor Unit (DPU-85), Multi-function Processor Unit (MPU-85), and the pilot CRT displays. At each 100-hour inspection and at any sign of abnormal equipment operation, verify these fans are operating properly.

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR
200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT AND KING AIR B100 LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the
CANADIAN MARCONI CMA-734 (-003 WITH MOD 7) VLF/OMEGA NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Canadian Marconi CMA-734 (-003 with Mod 7) VLF/Omega Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

The CMA-734 (-003 with Mod 7) meets the approval criteria for VLF/Omega navigation systems as established by AC 20-101A. It may be used as a means of VFR/IFR enroute navigation provided the following limitations are observed:

1. Only enroute operations within the conterminous United States and Alaska are approved.
2. IFR operations are limited to the Relative (normal) mode of navigation. IFR operations are not approved in the Absolute Omega (hyperbolic) navigation mode.
3. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys, between peaks in mountainous terrain, or below Minimum Enroute Altitude (MEA).
4. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.
5. VOR and DME equipment must be installed and operating during navigation on approved RNAV routes.
6. IFR flight is not approved based on VLF/Omega navigation into any area in which such operation could not be approved based on the installed equipment required by paragraph 4. This includes over-water flights beyond the range of approved ground navigation facilities.
7. The VLF/Omega position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

    a. Prior to compulsory reporting points during IFR operation when not under radar surveillance and control.
    b. At or prior to arrival at each enroute waypoint during operation on approved RNAV routes.
    c. At hourly intervals during operation on approved RNAV routes.

8. The VLF/Omega position information should be updated when a cross-check with other on-board approved navigation equipment reveals an error greater than 2 nautical miles along-track or cross-track.
9. Navigation shall not be predicated on the use of this system during extended periods of dead reckoning.
10. Following a period of dead reckoning, the VLF/Omega position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment.

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1. Power Interruption
   The only situation requiring immediate pilot action occurs when power to the system is restored after a significant period of primary power interruption. If the interruption exceeds 7 seconds, the system is automatically put in HOLD mode from the time power is restored. GMT and present position must be updated.

   a. Note incorrect displayed GMT to obtain time elapsed since power interruption.
   b. Enter correct GMT.
   c. Note displayed present position. Based on elapsed time (from "a") and best estimate of airplane track and ground speed, calculate present position at the time power was restored.
   d. Enter best estimated position as calculated in "c". The system will automatically update this position for airplane movement since power was restored.
   e. Provided acceptable Omega signals are available, the SYN annunciator should extinguish within 3 minutes.
   f. Operation should now be normal. However, position should be treated with caution until confirmed by a positive position check.

2. SYS Annunciator Illuminated
   In the event that the SYStem annunciator illuminates, no VLF/Omega displayed information can be relied upon. The other on-board navigation equipment must be used as the primary source of navigational information. Refer to Computer Access Mode in NORMAL PROCEDURES Section for failure analysis.

3. Compass Failure

   a. In the event of a compass failure, the system will continue to operate. However, the airplane rate of turn must not exceed 20 degrees per minute.
   b. By flying the CROSS TRACK and TRACK ANGLE ERROR at zero, the airplane will follow the DESIRED TRACK.

NORMAL PROCEDURES

The CMA-734 (-003 with Mod 7) is a fully automatic, computerized navigation system designed for point to point, area navigation. The system utilizes the Omega ground transmitter navigation network and the U.S. Navy Very Low Frequency (VLF) communication stations to provide continuous position and navigational information on a world-wide basis. In the unlikely event that adequate signals are not available from at least three stations, Omega and/or VLF, the system automatically goes into a Dead Reckoning (DR) mode of operation. As soon as adequate signals are again received, the system automatically reverts to its normal mode of operation.

Navigation is computed along a Great Circle flight path to provide the shortest distance between positions. Course deviation is presented in nautical miles on the pilot's Course Deviation Indicator (CDI) rather than in degrees as with conventional VOR navigation. Full-scale deflection represents a constant course width of approximately · 7½ nautical miles regardless of the distance to the waypoint.

After initialization, the CMA-734 (-003 with Mod 7) will automatically operate in the Relative (normal) mode of navigation. Relative navigation requires an accurate initial position input and provides the best overall accuracy for navigation. It is the only mode approved for IFR enroute operations. However, position accuracy will slowly degrade with time unless manually updated.

An Absolute Omega mode of navigation utilizing the principles of hyperbolic navigation with lane identification can be manually selected if the airplane position is not precisely known at initialization. Refer to Manual Lane Resolution in NORMAL PROCEDURES Section. This mode is not approved for IFR operations.

CONTROLS AND DISPLAYS

The CMA-734 (-003 with Mod 7) is programmed and operated from a pedestal mounted Control/Display Unit (CDU). The control section is used to enter data into the computer and select the computed information to be displayed. The display section, along with the pilot's Horizontal Situation Indicator (HSI), provides navigational information to the pilot. The display section also shows the data to be entered into the computer.

1. ON/OFF Switch:
   Applies power to CMA-734 (-003 with Mod 7) system. Detented in both positions and must be pulled out to unlock before changing.
2. Display Mode Selector Switch:
   Ten position switch to select the desired output or input data on the left and right numerical displays.
3. Dimmer Control (DIM):
   Adjusts illumination intensity of displays.
4. Waypoint Number Display (WPT):
   Indicates which waypoint coordinates are being displayed on the left and right displays.
5. From/To Waypoint Display (FR/TO):
   Displays FROM and TO waypoint numbers of leg being navigated.
6. Annunciators (L, R, N, S and L, R, E, W):
   Displays Left or Right, North or South, East or West, or are blanked, dependent upon the position of the Display Mode Selector Switch.
7. Left/Right Numerical Displays:
   Displays the data selected by the Display Mode Selector Switch.
8. Status Annunciators:
   When illuminated, indicate respectively, SYStem failure (Red warning lamp and SYS); Dead Reckoning mode of operation (DR); Position AMBiguity (AMB); Omega SYNchronization status (SYN); Relative (normal) mode of operation (VLF).
9. Data Keyboard:

   a. Provides 12 keys, of which 10 (0 through 9) are used for entry of numerical data into the computer. N (north), E (east), S (south), W (west) keys enter the sign of latitude/longitude information.
   b. Hold Key (HLD):
      Allows displayed present position information to be frozen (immobilized).
   c. Backspace Key (BK):
      Allows data entry to be backspaced one digit at a time if an error is made during entry. Also frees information frozen by Hold Key operation.

10. Waypoint Define Key (WPT DEF):
    Allows entry of waypoint coordinates.
11. Leg Change Key (LEG CHG):
    Allows manual initialization of leg change.
12. ENTER Indicator Light:
    Indicates system is in Data Entry Mode.
13. Enter Key (ENT):
    Transfers entered data into the computer.
14. Automatic/Manual Leg Change Switch (AUTO/MAN):
    Allows for automatic sequential selection of route legs, or manual selection of route legs.
15. FD NAV/FD VLF Selector:
    Lighted, panel mounted, push switch which selects and indicates which course information is displayed on the pilot's horizontal situation indicator (NAV or VLF/OMEGA).

NOTE

On airplane Serial Numbers BL-5, BB-343, BB-646, BB-676 and BB-689 only, localizer information is always displayed when an ILS frequency is tuned on the selected NAV control.

16. HSI NAV WARNING FLAG:
    Appears to call attention to the CDU for displayed information.

PROGRAMMING

Switching on the CMA-734 (-003 with Mod 7) automatically initializes a synchronization sequence which, under normal signal conditions, takes a maximum of three minutes. Confirmation that the synchronization process has finalized is indicated when the DR and SYN annunciators extinguish and the VLF annunciator illuminates. Synchronization is an independent function which allows the pilot to carry out other programming procedures while the synchronization sequence is in progress.

Prior to departure, Greenwich Mean Time (GMT), date, and the present position (latitude and longitude) are entered into the computer. The time of day and date are entered to compensate for diurnal shift while navigating. Coordinates (latitude and longitude) for up to nine waypoints can be entered while the airplane is on the ground or after takeoff.

After the pilot enters the data and initializes the first leg, the system automatically navigates from waypoint to waypoint (CDU switched to AUTO). Manual operation is also provided (CDU switched to MAN), which allows waypoints to be bypassed or sequence changed to allow for flight plan alterations.

Magnetic variation for the entire globe is programmed into the memory, and the value appropriate to present position is automatically selected. All angular displays are referenced to magnetic north.

POWER ON

1. Turn on airplane's Avionics Master Switch.
2. Pull the ON/OFF switch out to unlock. Set to ON.

NOTE

SYN and DR annunciators illuminate. SYN annunciator extinguishes within approximately three minutes. DR annunciator extinguishes and VLF annunciator illuminates when the system starts tracking three or more VLF/Omega stations. E/W and N/S annunciators flash continuously.

3. Set Display Mode Selector switch to TEST.
4. Verify that displays and annunciators are as follows:

    a. All numerical readouts show 8's.
    b. All annunciators are illuminated.
    c. All decimal points are illuminated.
    d. N, S, E, W annunciators and WPT and TO numerical readouts are flashing.

GMT AND DATE ENTRY

1. Set Display Mode Selector switch to GMT/D.
2. Insert Greenwich Mean Time via Data Keyboard to an accuracy of one minute. Display resolution is to 0.1 minutes. (e.g., 09:30.5). An entry must be made for the tenth minute display.
3. Press ENT key once. ENTER indicator remains illuminated.
4. Insert day (one or two digits), month (two digits), and year (two digits) via Data Keyboard (e.g., 29.12.79 or 01.01.80). The date entered must be the current Greenwich Mean Date.
5. Press ENT key once. ENTER indicator extinguishes, indicating the GMT and DATE have been accepted.

NOTE

DR illuminates for about 90 seconds after entry of GMT and date.

GMT will not advance until both GMT and date have been inserted and the ENTER indicator extinguishes.

In-flight update of GMT will cause the system to revert to DR mode momentarily. Re-check position after the DR annunciator extinguishes.

PRESENT POSITION ENTRY

1. Set Display Mode Selector switch to POS.
2. Insert present position latitude. N (north) or S (south) must be inserted prior to latitude digits (e.g., N 37.41.5).
3. Press ENT key once. ENTER indicator remains illuminated.
4. Insert present position longitude. E (east) or W (west) must be entered prior to longitude digits (e.g., W 97.13.2).
5. Press ENT key once. ENTER indicator extinguishes, indicating present position coordinates have been accepted.

NOTE

If ENTER indicator flashes, unreasonable data has been inserted. Turn Display Mode Selector switch to WPT; then back to POS. Repeat steps 2 through 5.

WAYPOINT ENTRY

Coordinates (latitude/longitude) for up to nine waypoints can be entered while the airplane is on the ground or in flight. Once entered, waypoints remain in the computer until new waypoints are entered. Coordinates of the first enroute waypoint are normally entered as Waypoint 1, with subsequent waypoints being entered sequentially. Waypoints may be changed any time by entering new coordinate data in any of the nine waypoint positions.

1. Set Display Mode Selector switch to WPT.
2. Press 1 on Data Keyboard. Number 1 appears in waypoint number display.
3. Press WPT DEF key. ENTER indicator illuminates.
4. Enter latitude of initial enroute waypoint.
5. Press ENT key once.
6. Enter longitude of initial enroute waypoint.
7. Press ENT key once. ENTER indicator extinguishes.
8. Repeat steps 2 through 7, selecting next waypoint number in step 2 for up to nine waypoints.
9. When all waypoints have been entered, display each waypoint in sequence by pressing the appropriate waypoint number on the Data Keyboard. Verify the latitude and longitude for each waypoint are correct.

RUNWAY LINE-UP (Departure Point)

1. Verify that SYS, DR, and SYN annunciators are extinguished, and that VLF annunciator is illuminated.
2. Verify that airplane compass is in the slaved mode.
3. Verify that latitude and longitude displayed in POS are correct. If incorrect, re-enter.

WARNING

Initial position must be correct. Any initial position error will be retained until updated.

NOTE

If a CDU annunciator is illuminated, the NAV flag on the pilot's HSI will also appear. This flag cannot be retracted except when the AMB annunciator is illuminated. In this case the NAV flag can be retracted by pressing the BK button.

INITIAL LEG SELECTION

1. Press LEG CHG key. ENTER indicator illuminates.
2. Press 0; then number of first selected waypoint.
3. Press ENT key once. ENTER indicator extinguishes.
4. FRom number and TO number windows show zero and selected waypoint number.
5. Set Display Mode Selector switch to WPT. Press 0 on Data Keyboard. Verify displayed latitude and longitude is that of airplane when ENT key was pressed in step 3.
6. Set Display Mode Selector switch to BRG/DIS. Verify displayed bearing and distance are reasonable.
7. Set FD NAV/FD VLF push switch to FD/VLF mode
8. Set Display Mode Selector switch to DSRTK/TK. Set HSI course indicator to displayed "Desired Track".

NOTE

If the NAV flag appears, it may be necessary to press the BK key to obtain a VLF/Omega display on the HSI.

The bearing and distance displayed refer to the Great Circle route from airplane present position to the selected waypoint. Waypoint zero will be the latitude and longitude of the airplane at the moment ENT key was pressed.

TOP OF CLIMB

The following procedure can be used to reduce the time required for the system to update the computed ground speed errors developed during the climb.

1. Set Display Mode Selector switch to GS/ETE.
2. Insert true airspeed or approximate ground speed. ENTER indicator illuminates.
3. Press ENT key once. ENTER indicator extinguishes.

The displayed ground speed and estimated time enroute will adjust for the headwind/tailwind conditions. The time required will depend on the extent of the adjustment.

ENROUTE OPERATION

1. MANUAL LEG CHANGE AT WAYPOINT

   a. Set mode selector switch to MAN.
   b. Press LEG CHG key. ENTER indicator illuminates.
   c. Press present waypoint number on Data Keyboard, then next selected waypoint number. FRom number and TO number windows show present and next waypoint numbers.

d. Press ENT key. ENTER indicator extinguishes.
e. Verify new BRG and DIS are reasonable.
f. Set displayed DSRTK on HSI course indicator.

NOTE

Waypoints may be selected in any sequence in this mode.

2. AUTOMATIC LEG CHANGE AT WAYPOINT

a. Set mode selector switch to AUTO.
b. Two minutes before reaching waypoint, the TO waypoint number display flashes and the VLF WPT annunciator illuminates. Before the airplane reaches the waypoint, the FR/TO waypoint display changes automatically to show the next leg waypoints.
c. Verify new BRG and DIS are reasonable.
d. Set displayed DSRTK on HSI course indicator.

NOTE

When in the automatic leg change mode, the system will switch to the next highest waypoint number; therefore, the bearing and distance will be from present position to whatever latitude and longitude are stored in that waypoint. When last desired waypoint appears in the TO window, switch to manual.

3. PRESENT POSITION DIRECT TO ANY WAYPOINT

a. Press LEG CHG key. ENTER indicator illuminates.
b. Press 0 on Data Keyboard, then number of desired TO waypoint.
c. Press ENT key once. ENTER indicator extinguishes.
d. Verify new BRG and DIS are reasonable.
e. Set displayed DSRTK on HSI course indicator.

NOTE

Waypoint 0 (present position) becomes a new stored waypoint and replaces those coordinates established in INITIAL LEG SELECTION, step 2.

4. BEARING AND DISTANCE DIRECT TO ANY WAYPOINT

a. Set Display Mode Selector switch to BRG/DIS.
b. Press 0 on Data Keyboard.
c. Press desired waypoint number on Data Keyboard.
d. Great Circle bearing/distance from present position to selected waypoint is displayed.
e. In a similar manner, bearing/distance between any two waypoints can be displayed.

NOTE

During this function, the ENTER indicator will illuminate and the normal navigation program continues uninterrupted. To return to normal navigation displays, rotate the Display Mode Selector switch to any other position.

DISPLAYED DATA

The following functions of navigational information displays are available. These functions are selected by placing the Display Mode Selector switch to the appropriate position.

1 GREENWICH MEAN TIME AND DATE (GMT/D)

a. Time, to the nearest tenth of a minute, is shown in the left display.
b Date, in the order day/month/year, is shown in the right display. Day must be the correct Greenwich date.

2. HEADING AND DRIFT ANGLE (HDG/DA)
Heading is shown in the left display; drift in the right. Appropriate annunciation (L or R) above the DA indicates drift left or right of the heading. If compass heading input fails, heading and drift angle displays go to 000/000.

3 DESIRED TRACK AND TRACK (DSRTK/TK)

a. Desired track is the computed Great Circle course, as referenced in degrees to magnetic North, from WPT FR to WPT TO and is shown in the left display. This value will update periodically as the airplane progresses along the Great Circle route. The pilot should maintain this displayed value on the HSI course indicator.
b Track is the actual track made good as referenced in degrees to magnetic North. It is shown in the right display.

4 CROSS TRACK DISTANCE AND TRACK ANGLE ERROR (XTK/TKE)

a. Cross track distance, to the nearest tenth of a nautical mile, is shown in the left display. Appropriate annunciation (L or R) above the XTK indicates whether the airplane's present position is to the left or right of the desired track (DSRTK).
b. Track angle error, to the nearest degree, is shown in the right display.

5. PRESENT POSITION (POS)
Airplane present position coordinates in latitude and longitude, to the nearest tenth of a minute are displayed.

6. WAYPOINT (WPT)
The latitude and longitude which have been stored for a given waypoint can be displayed by pressing the appropriate waypoint number on the Data Keyboard.

7. BEARING AND DISTANCE (BRG/DIS)

a. Bearing, shown in the left display, is the Great Circle angle from the present position to the next waypoint as referenced to magnetic North.
b. Distance along the Great Circle path to the next waypoint is shown in the right display.

NOTE

Bearing (BRG) and desired track (DSRTK) will always agree when the airplane is centered on the prescribed Great Circle course. If a deviation is made from the course, the bearing will update to provide the angle, as referenced from magnetic North, from the present position to the selected waypoint. The desired track angle will vary only as required to define the previously established Great Circle course.

8. GROUND SPEED AND ESTIMATED TIME ENROUTE (GS/ETE)

a. Ground speed is an estimated speed and is shown in knots in the left display.

b. Estimated Time Enroute is shown in the right display. The ETE is calculated at the current ground speed, using the distance from present position to a line perpendicular to the desired track through the TO waypoint.

9. WIND DIRECTION AND SPEED (WIND)

   a. With optional true airspeed (TAS) input provided to the system:

      (1) Wind direction is shown in the left display and is always referenced to True North.
      (2) Wind speed is shown in the right display.

   b. Without TAS input to the system: No wind information is available. Display should indicate 000/0.

10. TEST

    See COMPUTER ACCESS MODE, paragraphs 2 through 9.

POSITION CHECK AND UPDATE

During flight, accumulated errors to present position can be reduced by updating over a known reference point.

1. Set Display Mode Selector switch to POS.
2. Press HOLD (HLD) key when directly over the known reference point. N or S and E or W annunciators flash continuously.
3. Compare displayed coordinates with those of reference point.
4. If displayed position is satisfactory, press BK key to return system to normal mode of operation.
5. If update is required:

   a. Insert known latitude.
   b. Press ENT key.
   c. Insert known longitude.
   d. Press ENT key.

6. N or S and E or W annunciators stop flashing.

NOTE

While in HOLD, the system keeps track of airplane movement. Therefore, when BK key is pressed in step 4 or ENT key pressed in step 5d, the position is immediately updated to include all position movement since pressing HOLD key.

WARNING

Do not press HOLD key before arriving at known reference point as this will cause an error in the new position inserted.

DEAD RECKONING (DR) AND AMBIGUITY (AMB)

1. The system is constantly monitoring the accuracy of the position in latitude and longitude. It does this by evaluating the DR data, the predicted position and the VLF/Omega position which is resolved from all available stations. The signal-to-noise ratio and station geometry are evaluated. If signals are received from fewer than three stations, the DR annunciator illuminates and the VLF annunciator extinguishes. After approximately three minutes, the AMB annunciator will illuminate.

The DR annunciator means there are insufficient signals for VLF/Omega navigation. The AMB annunciation signifies that the position error as estimated by the system is greater than 4 NM.

If the system loses synchronization for any reason, the AMB and DR annunciators stay illuminated and the SYN annunciator also illuminates.

2. When the system synchronizes again, the SYN annunciator will extinguish, and the AMB and DR annunciators will remain illuminated.

3. When sufficient signals are available for VLF/Omega navigation, the DR annunciator will extinguish and the VLF annunciator will illuminate.

4. In the VLF mode, the AMB annunciator will not extinguish automatically. To extinguish the AMB annunciator, the indicated position must be verified and updated, or re-confirmed, as necessary. To re-confirm the position, set Display Mode Selector switch to POS and press ENT twice. To update position, refer to POSITION CHECK AND UPDATE.

   The AMB annunciator will not extinguish.

5. During Absolute Omega (hyperbolic) operation, the system monitors the position as in Paragraph 1. If the position error is estimated to exceed 4 nm, the AMB annunciator illuminates.

   The HSI warning flag is activated. When the operator updates position or verifies it to be correct, the VLF MESSAGE and HSI FLAG can be cancelled by pressing BK key.

6. When in full DR operation, the system uses the last VLF/Omega estimate of groundspeed and track.

MANUAL LANE RESOLUTION

On recovery from an extended period of DR, or if the AMB annunciator is on and no VOR or other means is available to update position, then a manual lane resolution can be forced by selecting the Absolute Omega (hyperbolic) mode (VLF annunciator extinguished), as detailed below:

TO FORCE ABSOLUTE OMEGA NAVIGATION TRIGGER FOR LANE RESOLUTION:

1. Turn Display Mode Selector switch to TEST. Press 660 on Keyboard. Press ENT twice. 660 should appear in left display. Press ENT once. Press 100 on Keyboard. Press ENT once. Verify that VLF annunciator is extinguished.
2. Press 310 on keyboard. Press ENT twice. 310 should appear in left display. Press ENT once. Press 1 on the keyboard. Press ENT once. The right display then modifies as follows:
   0 - LAR completed
   1 - LAR in progress
   2 - LAR armed, insufficient signals to trigger
   3 - LAR armed, inadequate geometry for trigger

TO RETURN TO RELATIVE NAVIGATION MODE:

3. Turn Display Mode Selector switch to TEST. Press 660 on Keyboard. Press ENT twice. 660 should appear in left display. Press ENT once. Verify 660 appears in left display. Verify 0 appears in WPT number display and that ENT is illuminated. Press 20 on Keyboard. Press ENT once.
4. Turn Display Mode Selector switch to any display mode except TEST and verify VLF annunciator is illuminated and all other annunciators are extinguished.
5. Treat position (POS) with caution until verified by some other approved navigation procedure.

COMPUTER ACCESS MODE

1. FLIGHT PLANNING

   a. Waypoint entries must have been previously made.

b. Turn Display Mode Selector switch to BRG/DIS.
c. Press desired FRom and TO waypoint number on data keyboard.
d. After several seconds the Great Circle bearing will be displayed in the left window, and the Great Circle distance will be displayed in the right window.
e. If 0 is selected as the FRom waypoint, then the data will be from the present airplane position to the selected waypoint.
f. If bearing and distance to Waypoint 0 from any waypoint is desired, press that waypoint, then 0. This 0 is not airplane present position, but that established in paragraph 5 of INITIAL LEG SECTION.

2. STATION USAGE

This procedure allows the operator to determine which VLF/Omega stations were used for the last navigation fix.

a. Display Mode Selector switch to TEST.
b. Press 737 on data keyboard.
c. Press ENT key twice. ENT annunciator extinguishes. 737 appears in left display window
d. Press WPT DEF key.
e. Press 0 on data keyboard.

Memory content appears on right display window.

This number is the numerical sum of the station identifiers.

OMEGA STATIONS			VLF STATIONS		
1	=	Norway	400	=	NWC (Australia)
2	=	Liberia	1000	=	NDT (Japan)
4	=	Hawaii	2000	=	GBR (England)
10	=	North Dakota	4000	=	NAA (Maine)
20	=	La Reunion	10,000	=	NPM (Hawaii)
40	=	Argentina	20,000	=	NSS (Maryland)
100	=	Australia	40,000	=	NLK (Washington State)
200	=	Japan			

Thus, the number 54356 indicates:

40,000	=	NLK
10,000	=	NPM
4000	=	NAA
200	=	Japan
100	=	Australia
40	=	Argentina
10	=	North Dakota
4	=	Hawaii
2	=	Liberia
54356		TOTAL

NOTE

As a temporary measure, reception of the Australia Omega station is disabled. See SERVICE NOTE later in this supplement for information regarding reception of this station.

f. Turning the Display Mode Selector switch to any other position restores the system to normal operation.
g. This function has no effect on the navigation in process.

3. INDIVIDUAL OMEGA STATION SIGNAL-TO-NOISE RATIO

This procedure provides a means of checking individual station signal strength.

a. Display Mode Selector switch to TEST.
b. Press 140 on data keyboard. 140 appears in left display window.
c Press ENT key twice.
d. Press WPT DEF key.
e. Press 3 on data keyboard. ENTER indicator extinguishes.

NOTE

Memory content appears in right display window. This number is the signal strength of Omega station Norway, based on increasing strength scale 0 to 100. Thirteen or above is accepted for navigation.

f. Press the HLD (hold) key once to increment memory location by one. Note that the number in the left display window is now 141. The right display window number is now the quality factor for Liberia.

140 = Norway
141 = Liberia
142 = Hawaii
143 = North Dakota
144 = Reunion
145 = Argentina
146 = Australia
147 = Japan

140 to 147 is the 10 kHz Omega Frequency
150 to 157 is the 13 kHz Omega Frequency
160 to 167 is the 11 kHz Omega Frequency (Not displayed in relative mode)

g. The left display number can be backed up in steps of one by pressing BK (back) key.

4. INDIVIDUAL VLF STATION SIGNAL-TO-NOISE RATIO

a. Follow the same procedure used for Omega signal strength in Paragraph 3, with the following exceptions.
b. In step b, press 170 on data keyboard.
c. In step f, read:

170 = Australia
171 = Japan
172 = England
173 = Maine
174 = Hawaii
175 = Maryland
176 = Washington State

5. DISPLAY OF SYSTEM FAILURE STATUS

Illumination of the red Failure Warning lamp and/or SYS annunciator, if all other displays are blank, indicates a catastrophic system failure. No operator action is possible.

If an input such as compass heading fails or if a failure occurs in the system itself, the SYS annunciator will illuminate. The operator can generally identify the failure in the following manner:

a. Display Mode Selector switch to TEST.

b. Press 670 on data keyboard.
c. Press ENTER key twice. 670 appears in left display window. ENTER indicator extinguishes.
d. The number in the right display window is the numerical sum of the system failure identification.

0 = normal
1 = analog-to-digital failure
2 = true-airspeed failure
4 = compass failure
20 = oscillator drift
100 = self-test receiver
400 = processor/memory failure

Thus, for example, the number 24 would consist of:

4 = compass failure
20 = oscillator drift

6. DISPLAY OF RECEIVER FAILURE

a. Same as in Paragraph 5, steps a through c, except substitute "666" in steps b and c.
b. The number in the right display window is the numerical sum of the receiver failure identification:

0 = normal
1 = 10 kHz receiver failure
2 = 13 kHz receiver failure
4 = 11 kHz receiver failure
10 = VLF receiver failure

7. DISPLAY OF SYNCHRO FAILURE

a. Same as in Paragraph 5, steps a through c, except substitute "671" in steps b and c.
b. The number in the right display window is the numerical sum of the synchro failure identification:

0 = normal
2 = true-airspeed discrete failure
4 = compass discrete failure
12 = true-airspeed synchro failure
24 = compass synchro failure

8. MAGNETIC VARIATION

The system automatically computes magnetic variation for the present latitude and longitude. This variation is applied to the magnetic compass input automatically.

To read the magnetic variation in use:

a. Display Mode Selector switch to TEST.
b. Press 1032 on data keyboard.
c. Press ENTER key twice. 1032 appears in left display window.
d. Press WPT DEF key.
e. Press 4 on data keyboard. ENTER indicator extinguishes.

The right display is the variation in degrees X 10. If the L and R annunciators are both on, the variation is WEST. If both L and R annunciators are off, the variation is EAST.

9. STATION REJECTION

Any VLF/Omega station can be manually rejected if required.

a. Display Mode Selector switch to TEST.
b. Press 657 on data keyboard.
c. Press ENTER key twice. 657 appears in left display window. ENTER indicator extinguishes.

If 0 appears in right display window, all stations are being used. Station can be rejected by means of the data keyboard.

d. Press ENTER key once.
e. Insert desired content from table, via data keyboard. For example, to reject Omega Stations Hawaii and Australia, press 104 on DATA keyboard.
f. Press ENT key. ENT indicator extinguishes.

OMEGA STATIONS			VLF STATIONS		
0	=	use all stations	400	=	reject NWC (Australia)
1	=	reject Norway	1000	=	reject NDT (Japan)
2	=	reject Liberia	2000	=	reject GBR (England)
4	=	reject Hawaii	4000	=	reject NAA (Maine)
10	=	reject North Dakota	10000	=	reject NPM (Hawaii)
20	=	reject Reunion	20000	=	reject NSS (Maryland)
40	=	reject Argentina	40000	=	reject NLK (Washington State)
100	=	reject Australia			
200	=	reject Japan			

10. AUTOMATIC STATION REJECTION

Inhibit provisions are included in the system to automatically diselect the Liberian station. This occurs as a function of airplane position and time of day. To inhibit this automatic feature:

a. Set display mode selector switch to TEST.
b. Press 661 on data keyboard.
c. Press ENTER key twice. 661 will appear in left display window.
d. Press ENTER key once. Verify ENTER light illuminates.
e. Press 1 on data keyboard.
f. Press ENTER key once. Verify ENTER light extinguishes.

CONTROL DISPLAY UNIT ANNUNCIATORS

1. SYS (System)

When illuminated, there is an internal failure, or one of the system inputs has failed. See COMPUTER ACCESS MODE, Paragraphs 5, 6 and 7.

2. DR (Dead Reckoning)

Whenever the VLF/Omega system does not receive sufficient signals for navigation, the DR annunciator will be on steady.

3. AMB (Ambiguity)

a. If the displayed position is estimated to be outside a circular area which exceeds a radius of four nautical miles, the AMB annunciator will be on steady.
b. If the AMB annunciator is flashing, there has been some loss of operator-inserted data due to a main power recycling. Restart system as in GMT AND DATE ENTRY and PRESENT POSITION ENTRY. Verify all waypoints are correct. AMB annunciator should extinguish.

4. DR and AMB

   If both DR and AMB are on steady, this indicates that the system has not received sufficient stations for some time. If the displayed latitude and longitude in POS can be substantiated with a visual ground fix or other approved navigation equipment, the ENT key should be pressed twice. AMB will extinguish. DR extinguishes when full operation is resumed.

5. SYN (Synchronization)

   When illuminated, this indicates synchronization to the Omega sequence is in progress. During this time, the DR annunciator will be illuminated.

6. VLF

   The VLF annunciator will illuminate when the system is in its primary (relative) navigation mode - VLF/Omega.

7. WAYPOINT ALERT

   Two minutes before reaching the active waypoint, the TO waypoint number flashes and continues to do so as airplane approaches active waypoint.

NAVIGATION DISPLAYS

Once a navigation leg has been selected, left and right course information will be displayed on the HSI when the flight director switch is selected to the FD VLF position. Course information is provided for enroute navigation with a constant course width of ± 7½ nautical miles irrespective of the distance to the waypoint. The TO/FROM flag on the HSI reads TO when the airplane has not passed a line at right angles to the desired track which passes through the active TO waypoint. It reads FROM when the airplane has passed that line. If the Automatic/Manual leg change switch is in the automatic mode, the flag will always indicate TO.

The CMA-734 (-003 with Mod 7) drives the normal VOR/LOC warning flag when VLF/Omega information is displayed on the HSI. The warning flag will be in view anytime the VLF/Omega information is invalid or suspect (e.g. system failure SYS; SYN; AMB annunciators). The HSI DME readout will display raw DME distance to the tuned VORTAC; not distance to a VLF/Omega waypoint, unless annunciated otherwise.

The course deviation indicator (CDI) displacement on the HSI is independent of the course selected (similar to a localizer). Therefore, to get relative bearing as well as displacement information, the current desired track (DSRTK) must be set on the HSI course indicator. The BRG displayed on the CMA-734 (-003 with Mod 7) display unit is always the bearing from the present position to the selected TO waypoint. Therefore, the selected leg bearing will be displayed only when the CDI is centered. The bearing of the selected leg will change in flight because the path established between waypoints is a Great Circle route. Therefore, the course indicator must be reset periodically to agree with the displayed desired track (DSRTK).

AUTOPILOT OPERATION

Once the navigation leg has been selected and the flight director switch is in the FD VLF position, the autopilot can be coupled to the CMA-734 (-003 with Mod 7) using the normal NAV mode.

NOTE

To obtain reliable autopilot operation, the displayed desired track (DSRTK) must be maintained on the HSI course indicator.

If a course change of more than 30° is made at a waypoint while operating in the VLF/Omega MANual

mode, the new leg and course indication should be selected and set prior to waypoint passage. When operating in the AUTO mode, the only required action is to set the new course.

PERFORMANCE - No change.

HANDLING, SERVICING, AND MAINTENANCE

The Australian station is currently under construction. Contact a representative of Canadian Marconi Company or Beech Aircraft Service to find out how to enable reception of Omega Station Australia when appropriate

Approved:

For W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: May, 1981
P/N 101-590010-165

For Training Purposes Only

Beechcraft®

SUPER KING AIR® 200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT & KING AIR® C90 (LJ-668 thru LJ-1062, except LJ-670), C90A (LJ-1063 and after), E90, F90, A100 & B100 LANDPLANES

Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement for AIRPLANES EQUIPPED WITH A SECONDARY ENCODING ALTIMETER

This Supplement is applicable to the following manual(s):

90-590010-61, 90-590010-87, 90-590012-3, 90-590024-5, 90-590024-23, 90-590024-35, 90-590024-69, 100-590032-1, 100-590038-1, 101-590010-127, 101-590010-147, 101-590010-327, 101-590037-3, 101-590037-29, 101-590037-57, 109-590010-3, 109-590010-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of a secondary encoding altimeter on the copilot's instrument panel in addition to the primary encoding altimeter installed on the pilot's instrument panel in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth in this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

No change.

EMERGENCY PROCEDURES

No change.

ABNORMAL PROCEDURES

No change.

NORMAL PROCEDURES

No change.

PERFORMANCE

No change.

WEIGHT AND BALANCE/EQUIPMENT LIST

No change.

SYSTEMS DESCRIPTION

Airplanes equipped with a secondary encoding altimeter on the copilot's side may be configured in either of the following ways.

1. Encoded altitude information from either altimeter is provided to both transponders using the Altimeter Select switch as follows:

   Altimeter Select Switch on No. 1 - Pilot's altimeter will provide encoded information to both transponders.

   Altimeter Select Switch on No. 2 - Copilot's altimeter will provide encoded information to both transponders.

2. Encoded altitude information from the pilot's altimeter is provided only to the No. 1 transponder. Encoded altitude information from the copilot's altimeter is provided only to the No. 2 transponder. Airplanes configured in this manner do not have an Altimeter Select Switch.

NOTE

Encoded altitude information is not dependent on the altimeter setting.

HANDLING, SERVICING, AND MAINTENANCE

No change.

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FAA Approved
Revised: January, 1994
P/N 101-590010-169

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BEECHCRAFT SUPER KING AIR 200, 200C, 200T 200CT, B200, B200C, B200T, B200CT & KING AIR B100 LANDPLANES PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

COLLINS LRN-85 VLF/OMEGA NAVIGATION SYSTEM (DUAL SYSTEM OPTIONAL)

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Collins LRN-85 VLF/OMEGA Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

The Collins LRN-85 meets the approved criteria for VLF/OMEGA navigation systems as established by AC 20-101A. It may be used as a means of VFR/IFR enroute navigation provided the following limitations are observed:

1. Only enroute operations in the conterminous United States and Alaska are approved.
2. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).
3. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.
4. VOR and DME equipment must be installed and operating during navigation on approved RNAV routes.
5. IFR flight is not approved based on VLF/OMEGA navigation into any area in which such operations could not be approved based on the installed equipment required by paragraph 3. This includes overwater flights beyond the range of approved ground navigation facilities.
6. The VLF/OMEGA position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

   a. Prior to compulsory reporting points during IFR operation when not under radar surveillance or control.
   b. At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.
   c. At hourly intervals during operation off of approved RNAV routes.

7. The VLF/OMEGA position information should be updated when a cross-check with other onboard approved navigation equipment reveals an error greater than 2 nautical miles along track or crosstrack.
8. Navigation shall not be predicated on the use of this system during periods of dead reckoning.
9. Following a period of dead reckoning, the VLF/OMEGA position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment.

EMERGENCY PROCEDURES

An emergency battery pack is included in the system. If the primary power is interrupted, the system will switch to the emergency battery pack until the primary power is restored, at which time the system will switch back to primary power. The system cannot be turned on using only the emergency battery. Battery power will only function with loss of primary power. It is important the system power switch be selected off

at the conclusion of flight to avoid depleting the emergency battery. The system can operate from the emergency battery pack for approximately 15 minutes at 70°F. The battery pack will recharge in approximately ten hours when the system is on the primary power source.

In the event that the WARN annunciator illuminates, and anytime the navigational information is questioned based on information from other onboard navigation equipment, the other onboard navigation equipment must be used as the primary source of navigation information.

NORMAL PROCEDURES

BEFORE TAKEOFF

Prior to departure, the operator verifies the Greenwich Mean Time (GMT) and date, enters present position, and if required, latitude and longitude for up to 99 waypoints. The waypoints may be entered by the operator or called from the computer memory of over 5000 NAVAIDS and airports. After the operator enters the data and initializes the first leg, the LRN-85 automatically navigates from waypoint to waypoint (AUTO/MAN Switch in AUTO position), or the operator may use the manual operation (AUTO/MAN Switch in MAN position) mode to bypass waypoints or change waypoints in order to allow for flight plan alterations. The LRN-85 may also be interfaced with the autopilot through the flight director system. The LRN-85 continuously performs a self-test routine. Annunicators advise the operator of the system's operational status.

SYSTEM START-UP

1. Airplane Master Switch - ON
2. Avionics Master Power Switch - ON

NOTE

The system cannot be turned on using the internal emergency battery only; airplane electrical power must be applied.

3. LRN-85 Mode Selector Switch - PULL and TURN to NAV position. (Knob must be pulled out before it can be turned.)
4. Verify that all displays illuminate for approximately 5 seconds.
5. Verify that all displays except the red "DR" annunciator extinguish after approximately 5 seconds. (The "DR" annunciator will remain illuminated until the airplane position is entered and the system has been initialized.)
6. Verify that the red warning annunciator ("WRN") is extinguished. If it remains illuminated, refer to the WARNING ("WRN") ANNUNCIATOR write-up and clear the warning before continuing.

GREENWICH MEAN TIME (GMT) VERIFICATION AND ENTRY

1. Set Mode Selector Switch to NAV.
2. Set Function Selector Switch to GMT/DATE.
3. Ensure that Alpha Register displays "GMT".
4. Verify GMT and DATE in Left and Right Registers.

If displayed data is correct:

5. Depress ENTER key.

If displayed data is not correct:

5. Enter current GMT to nearest minute. Data format: x-x-x-x (i.e., 12:30 is entered as 1-2-3-0).
6. Depress ENTER key once.
7. Enter Date in format of Month/Day/Year. Alpha register will display "MDY". Data Format: MM-DD-YY (i.e., 0-7-0-4-8-0)

NOTE

Do Not update GMT or DATE after position has been entered. If an incorrect GMT or DATE is entered, position must be re-initialized.

WAYPOINT SELECTION AND ENTRY USING LATITUDE AND LONGITUDE

To verify waypoint coordinates:

1. Turn FUNCTION selector to WPT
2. Press WPT key
3. Key waypoint number,

Data Format: XX (i.e., waypoint 55 as 5-5)

4. Press ENTER key
5. Verify coordinates of waypoint presented in display

If data displayed is accurate:

6. Press ENTER key to select next waypoint in numerical order. (Pressing BACK key will select previous waypoint)

To update or change waypoint coordinates:

7. Enter waypoint latitude
8. Depress ENTER key
9. Enter waypoint longitude
10. Depress ENTER key

Data format: N33.40.2,W97.13.2 (i.e., N-3-3-4-0-2-ENTER, W-9-7-1-3-2-ENTER)

NOTE

Latitudes and Longitudes may be entered in either order. Leading zeros need not be entered. Waypoints may be entered in any order. However, they should be entered in flight plan sequence if automatic leg switching is desired. The system will not automatically advance to a waypoint with coordinates N00.00.0, E000.00.0. These coordinates should be entered after the final waypoint in a route to prevent automatic advance to an erroneous waypoint.

WAYPOINT SELECTION AND ENTRY USING COORDINATES FROM COMPUTER MEMORY

Refer to ADVANCED PROCEDURES Section

SYSTEM INITIALIZATION

Close proximity to hangars, ground power carts, and terminal buildings can cause interference with VLF/Omega signals. For best results, the system should be initialized away from such noise sources.

1. Set function selector to POS.
2. Depress WPT key.
3. Key in start position waypoint number (start position must be one of the 99 waypoints.)
4. Depress ENTER key.
5. Verify start position coordinates in display.
6. Depress BACK/HOLD key.
7. Depress ENTER key a second time.

8. Verify both DR and NAV annunciators extinguished.
9. Verify NAV annunciator illuminates after 15 seconds.

NOTE

Do not attempt to initialize the system until the green NAV annunciator is on. The NAV annunciator will illuminate when the system is receiving sufficient signals to navigate. The red DR annunciator will remain on until position has been entered and the system is initialized.

Data Format: Initialize with waypoint no. 01: i.e., WPT-0-1-ENTER-ENTER

LEG SELECTION

1. Press LEG key.
2. Key in the FROM waypoint number.
3. Key in the TO waypoint number.
4. Press ENTER.

Data Format: To define leg between WPT 36 to WPT 64: LEG-3-6-6-4-ENTER

MAGNETIC/TRUE NORTH SELECTION

Place the MAG/TRUE Selector switch in the MAG or TRUE position as desired.

AUTOMATIC/MANUAL LEG CHANGE SELECTION

The AUTO/MAN selector switch determines whether legs are advanced manually by the operator or automatically by the computer. The difference between the two modes occurs when the airplane approaches the TO waypoint. With the switch set to AUTO, the system will automatically advance to the next leg 60 seconds prior to reaching the TO waypoint. The LEG key will begin to flash 1.5 minutes before the selected waypoint is reached to alert the pilot.

IN-FLIGHT PROCEDURES

FLIGHT DIRECTOR/AUTOPILOT INTERFACE

VLF/OMEGA navigation information may be presented on the flight director by use of a selector switch located on the pilot's instrument panel. When accomplished, all enroute modes of flight director/autopilot operation are available for use with the VLF/OMEGA system.

CAUTION

The VLF/OMEGA system MUST have a true airspeed input, either automatically from a TAS sensor or manually inserted, to ensure proper flight director/autopilot VLF/OMEGA course tracking.

DETERMINATION OF BEARING AND DISTANCE TO WAYPOINT

With the MODE selector set to NAV and the FUNCTION selector set to BRG/DIS, the Bearing to the TO waypoint from present position is displayed in the left register to the nearest degree. BRG is referenced to true or magnetic north, depending on the position of the MAG/TRUE switch. If the BRG is referenced to true north, the N annunciator will be on. The distance (DIS) between present position and the TO waypoint is displayed in the right register to tenths of a nautical mile.

DETERMINATION OF DESIRED TRACK, AND CROSSTRACK DISTANCE

With the MODE selector set to NAV and the FUNCTION selector set to DTK/XTK, the Desired Track (DTK) from the FROM waypoint through the TO waypoint is displayed in the left register to the nearest degree. DTK is referenced to true or magnetic north, depending on the position of the MAG/TRUE switch. If DTK is referenced to true north, the N annunciator will be on. Note that desired track is a slowly changing function of the remaining distance to the waypoint.

The crosstrack distance (XTK) between present position and the defined course is displayed in the right register to tenths of a nautical mile. The L and R lights indicate whether the airplane is left (L) or right (R) of track.

NOTES

1. If a parallel course set crosstrack (SX) has been entered, the E annunciator will illuminate.

2. DTK and TK are also output on hardware synchros. These outputs are either true or magnetic, depending on the state of an external program pin, usually controlled by a cockpit switch. The MAG/TRUE switch on the CDU controls only the CDU displays.

3. Normally, track legs are defined between waypoints, or from present position to a waypoint. An additional feature allows any desired track through any waypoint to be keyed in. XTK and autopilot outputs will then be referenced to the new desired course. The FROM waypoint display will read 00.

DETERMINATION OF ESTIMATED TIME ENROUTE, AND GROUND SPEED

With the MODE selector set to NAV and the FUNCTION selector set to ETE/GS, ground speed to the nearest knot is displayed in the right register. Ground speed is updated once per second but integrated over the previous two minutes.

The Estimated Time Enroute (ETE) to the nearest tenth of a minute is displayed in the left register. ETE is defined as the remaining along-track distance divided by the current ground speed.

Estimated Time of Arrival (ETA = ETE + GMT) may be accessed by pressing the ENTER key in the ETE/GS function position. In this mode, "ETA" will be displayed in the alpha register. Pressing ENTER again will resume the ETE display.

DETERMINATION OF WIND DATA

With the MODE selector set to NAV and the FUNCTION selector set to WIND, the wind direction to the nearest degree is displayed in the left register. Wind direction is referenced to true or magnetic north, depending on the position of the MAG/TRUE switch. If wind direction is referenced to true north, the N annunciator is on.

The wind velocity to the nearest knot is displayed in the right register.

Wind outputs are available only if true airspeed (TAS) and Heading (HDG) are supplied to the system. Both TAS and HDG may be keyed in if external interfaces are not available. If TAS and HDG are not available and have not been keyed in, the left register will read OFF.

NOTE

Wind outputs are updated every second, but are integrated over the previous two minutes.

DETERMINATION OF TRUE AIRSPEED AND DRIFT ANGLE

With the FUNCTION selector set to TAS/DA and the MODE selector set to NAV, True Airspeed (TAS) is displayed in the left register to the nearest knot. The TAS display repeats the input of an external TAS sensor. If external TAS is not available and has not been keyed in (see TAS Update below), OFF will be displayed in the left register.

Drift Angle (DA) to the nearest degree is displayed in the right register. Drift Angle is defined as the difference between the airplane's Track and Heading. The L and R annunciators indicate whether the Heading is left (L) or right (R) of Track. If external or keyed-in Heading is not available, OFF will be displayed in the right register.

Manual TAS Entry

If external TAS is not available, TAS may be keyed in manually. To ensure that WIND and DR calculations are accurate, manually entered TAS should be as accurate as possible. This may require frequent updating. To update TAS, perform the following:

1. Turn FUNCTION selector to TAS/DA
2. Key in TAS value in knots
3. Press ENTER key.

NOTE

Prior to climb-out or descent, an average TAS should be entered. If the external TAS sensor is valid, manual TAS entry is inhibited.

Manual TAS Clearing

To resume use of external TAS, manually entered TAS must be cleared as follows:

1. Turn FUNCTION switch to TAS/DA
2. Key in 999.
3. Press ENTER key.

LEG CHANGES AT WAYPOINT

Automatic Sequencing

With the AUTO/MAN switch in AUTO, the change to the next sequential leg is automatic. This occurs 60 seconds prior to reaching the TO waypoint. If the TO waypoint number is less than the FROM waypoint number, the next TO waypoint will be one less than the current TO waypoint. If the current FROM waypoint is 00 (present position), the next TO waypoint number will be determined by the prior leg.

NOTE

The system will not automatically advance to a waypoint with coordinates N00.00.0, E000.00.0. These coordinates should be entered after the final waypoint in a route to prevent automatic advance to an erroneous waypoint.

Manual Sequencing

With the AUTO/MAN switch in MAN, the change to the next leg may be performed manually as follows:

1. Turn FUNCTION selector to BRG/DIS
2. Press LEG key
3. Press ENTER key.

The next leg numbers in sequence will appear in the FROM/TO waypoint display.

EXAMPLE: 1. FROM/TO display currently reads 05/06

2. Turn FUNCTION selector to BRG/DIS.

3. Press LEG-ENTER.

4. FROM/TO display now reads 06/07.

NOTE

The LEG key will flash approximately 1.5 minutes prior to reaching the TO waypoint.

The flashing leg light may be cancelled by pressing LEG-0-ENTER.

LEG CHANGE (WAYPOINT BYPASS)

A new leg from present position to any waypoint may be entered as follows:

1. Turn FUNCTION selector to BRG/DIS
2. Press LEG key.
3. Key in number of desired TO waypoint.
4. Press ENTER key.

EXAMPLE: To establish a new leg to WPT 36

1. FROM/TO display reads 05/06 (old leg)

2. Turn FUNCTION selector to BRG/DIS

3. Press LEG-3-6-ENTER

4. FROM/TO display now reads 00/36 (new leg).

LEG CHANGE (DEFINE NEW LEG)

A new leg between any two waypoints may be entered as follows:

1. Turn FUNCTION selector to BRG/DIS
2. Press LEG key.
3. Key in numbers of the desired FROM and TO waypoints.
4. Press ENTER key.

NOTE

After all leg changes, verify that the new BRG/DIS is reasonable.

EXAMPLE: To establish a new leg between waypoints 07 and 14:

1. FROM/TO display reads 05/06 (old leg)

2. Turn FUNCTION switch to BRG/DIS

3. Press LEG-0-7-1-4-ENTER

4. FROM/TO display now reads 07/14 (new leg).

POSITION UPDATE

Any error in present position may be eliminated by updating the system over a known waypoint. To update the system, perform the following:

1. Turn FUNCTION selector to POS
2. Press WPT key.
3. Key in number of waypoint to be used for update.
4. Press ENTER key.
5. As the airplane flies directly over the checkpoint, press the HOLD (BACK/HOLD) key.

NOTE

The latitude/longitude displayed is the difference between the system's present position and the selected waypoint.

6. After one second, the left and right registers will display the latitude and longitude components of error in degrees/minutes/tenths format. This display will be frozen until the operator completes step 7.

To correct the error:

7. If the operator wishes to correct this error out of the system, press ENTER. After a delay of a few seconds, present position will be displayed with the error corrected.

To retain the error:

7. If the error is acceptable, simply rotate the FUNCTION selector to any other position and the HOLD mode is cancelled.

NAVIGATION STATUS ANNUNCIATION

"Nav" Annunciator

The NAV annunciator will illuminate when enough VLF and Omega signals are received for navigation. This should be within three minutes after system turn-on if the system is located where signals can be received. After the NAV annunciator comes on, the red DR (Dead Reckoning) annunciator will remain on until the system is initialized.

CAUTION

Do not initialize the system until the NAV light comes on.

"Dr" (Dead Reckoning) Annunciator

The red DR annunciator will illuminate when the system is turned on and remain on until the system has

been initialized. During the flight, if insufficient signals are being received for good navigation, the DR annunciator will illuminate and the NAV annunciator will extinguish. During periods of DR, the system will navigate on the basis of last known winds, current TAS and Heading, and any remaining VLF or Omega stations.

If the DR annunciator illuminates, perform the following:

1. Ensure that TAS and Heading inputs are correct.
2. When sufficient stations are again being received and the NAV annunciator is on, check indicated position for accuracy.
3. If the DR period was short and the current position is accurate, the DR annunciator may be turned off by turning the FUNCTION selector to POS and pressing the 0 (zero) key.
4. If the accuracy of the present position cannot be ascertained, the system must be initialized at a known position.
5. If a period of DR was longer than 30 seconds, the system will attempt to use Omega signals to re-lane. When signals are reacquired after a dead reckoning period of 30 seconds or longer, the green NAV annunciator will flash until Omega signals are sufficient for re-laning. Re-laning can correct position errors up to 200 nautical miles. After re-laning is complete, the NAV annunciator stays on continuously. The DR annunciator will also stay on until position is checked (see items 2, 3, and 4 above).

Warning ("WRN") Annunciator

The system constantly monitors itself for hardware, software, and receiver failures. If the system detects a failure, the red WRN annunciator will illuminate. If this happens, the operator should perform the following:

1. Set the MODE selector to NAV and rotate the FUNCTION selector through several function positions. If the system fails to respond to rotating the FUNCTION selector, switch power off for five seconds, then turn power back on. The system should reset normally and position should be checked. If the system will not reset, it is inoperative and service personnel should be contacted.
2. If the system responds normally to the FUNCTION selector, set the MODE selector to TEST and key the code 0-0-0-2-ENTER. Record the number displayed in the right register. The Warning Codes Table below indicates the appropriate user action. In all cases, the warning code should be recorded and transmitted to service personnel.
3. The WRN annunciator can be cleared by keying in the code 1-0-0-2-ENTER in the TEST mode.

WARNING CODES TABLE

CODE DISPLAYED	PROBLEM INDICATED	PROBLEM/ USER ACTION
0	No Warning	Warning Light should be OFF.
1	EPROM Memory Failure	Computer Memory has failed – system is unreliable – refer to service personnel.
2	RAM Memory Failure	Computer Memory has failed – system is unreliable – refer to service personnel.
3, 4, 5	CDU I/O Failure	CDU interface failure CDU/computer interface may be unreliable.

CODE DISPLAYED	PROBLEM INDICATED	PROBLEM/ USER ACTION
6	Program Failure	Computer program has gotten "lost". Navigation may be unreliable. Check system position.
7, 8, 9	Receiver Warning	Receiver computer/navigation computer interface has failed. Position data unreliable. Refer to service personnel.
A, B	Time Error	System clock has failed. GMT/date outputs may be incorrect as well as ground speed and wind. Position data may be degraded due to inaccurate diurnal shift.
C	Divergence	The VLF/Omega stations received disagree on system position by more than 5 miles. System should be updated over known position.
D	Flunks	At least two receivers diverge substantially from the system position. Suspect stations should be de-selected and the system reinitialized over a known position.
E	Mass Memory Failure	Mass Memory has failed – system is unreliable – refer to service personnel.
F	Static Memory	Static Memory failed to hold all information during power off period. Operator should check coordinates of all waypoints as well as GMT/date. (In LRN-85 systems, this warning will occur when either the CRU or AEU has been removed from the rack.)

"BATT" Annunciator

The battery annunciator indicates that the system is operating on the internal emergency battery. Under normal conditions, this battery will last approximately 15 minutes at 21°C (70°F). Immediate steps should be taken to restore external power.

SHUTDOWN AND SECURING

After landing, shut down the system as follows:

1. Check WRN annunciator:

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For Training Purposes Only

If WRN annunciator is illuminated:

– Perform the test procedure given under WARNING ("WRN") ANNUNCIATOR Section above.

If WRN annunciator is not illuminated:

2. Pull and turn the Mode Selector Switch counterclockwise to the OFF position.
3. Verify that all displays are blank.
4. Verify that the BATT annunciator is extinguished.

ADVANCED PROCEDURES

The LRN-85 has the capability to perform many additional navigation functions that are extensions to the basic navigation functions described. Also, the system allows many additional procedures that greatly reduce pilot workload. These functions and procedures are described in the paragraphs that follow.

STATION DESELECT/RESELECT

The LRN-85 computer automatically selects the best stations in a given area. Occasionally, NOTAMSs are published indicating that selected stations should not be used for a period of time. In addition, user experience sometimes indicates that navigation accuracy is improved if certain stations are not used in a given area. In these cases, the operator can command the computer not to use certain stations. This is accomplished by "deselecting" the station(s).

To Deselect an Omega or VLF station, perform the following:

1. Turn MODE selector to TEST.
2. Key in 60xx, where xx is the station code from the STATION CODES TABLE below.
3. Press ENTER. The three-letter identifier for the station will be displayed in the alpha register.

STATION CODES TABLE

STATION	CODE	STATION LIGHT	LRN NAME	CALL SIGN
Cutler ME	01	1	MNE	NAA
Jim Creek WA	02	2	WSH	NLK
North West Cape Aust	03	3	AUS	NWC
Rugby GB	04	4	GBR	GBR
Annapolis MD	05	5	MRY	NSS
Lualualei HA	06	6	HAW	NPM
Yosami Japan	07	7	JPN	NDT
Spare	08	8	---	---
Omega Norway	51	A	ONW	---
Omega Liberia	52	B	OLB	---
Omega Hawaii	53	C	OHA	---
Omega N. Dakota	54	D	OND	---
Omega La Reunion	55	E	ORE	---
Omega Argentina	56	F	OAR	---
Omega Trinidad	57	G	OTR	---
Omega Japan	58	H	OJP	---

To Display Stations Deselected By The Computer:

1. Turn MODE selector to TEST
2. Key in 1998
3. Press ENTER

The deselected stations will appear in the station present display.

NOTE

All stations are selected after system power is turned ON.

ALPHANUMERIC DATA ENTRY

In using several of the advanced procedures, it is necessary to enter alphanumeric data (letters or letters and number). To key in alphanumeric data, use the following procedure:

NOTE

Each input that is a letter must be preceded by pressing the ALPHA key. If the desired input is a number that is part of a letter/number combination, it too must be preceded by pressing the ALPHA key.

1. Press the ALPHA key
2. Press the desired letter key once for the first letter appearing on the key, twice for the second letter appearing on the key, three times for the third letter appearing on the key, and four times for the number appearing on the key.
3. Repeat steps 1 and 2 for each letter or number to be entered.
4. When the desired input is displayed correctly in the alpha display, press the ENTER key.
5. If an error is made when entering a letter, the BACK key may be used to clear that entry.

EXAMPLE: To enter LAX:

1. Press ALPHA key
2. Press the L key (key 4) three times
3. Press ALPHA key
4. Press the A key (key 1) once
5. Press ALPHA key
6. Press the X key (key 8) three times
7. Verify that LAX appears in the alpha display
8. Press ENTER Key

EXAMPLE: ALPHA-4-4-4-ALPHA-1-ALPHA-8-8-8-ENTER

(L) (A) (X)

Alphanumeric Waypoint Names (User-Defined)

The operator can give a three-character name to each waypoint, if desired. To enter the name, the Function Selector is set to the WPT position, and (if necessary) the waypoint coordinates are entered.

EXAMPLE: If the operator wants waypoint 12 to be named "LAX":

1. Mode Selector - NAV Position
2. Function Selector - WPT Position
3. Select waypoint 12 with the following keystrokes:
   WPT-1-2-ENTER
4. Assign the name LAX with the following keystrokes:
   ALPHA-4-4-4-ALPHA-1-ALPHA-8-8-8-ENTER
   (L) (A) (X)

EXAMPLE: If the operator wants waypoint 76 to be named "Y6A":

1. Mode Selector - NAV Position
2. Function Selector - WPT Position
3. Select waypoint 76 with the following keystrokes:
   WPT-7-6-ENTER
4. Assign the name "Y6A" with the following keystrokes:
   ALPHA-9-ALPHA-6-6-6-6-ALPHA-1-ENTER
   (Y) (6) (A)

NOTE

Duplicate names are allowed.

VHF NAVAID LOOK-UP

The LRN-85 has the coordinates of most of the world's VOR's, DME's and VORTAC's preprogrammed into the system's memory. These coordinates can be accessed by their three-letter identifier, and loaded into any numbered waypoint. To load the coordinates of a VORTAC (VORTAC here includes all VHF Navaids, VOR's and DME's), first select the desired waypoint. Set FUNCTION selector to WPT and key in WPT-WPT#-ENTER. Next, press the MEM key, then enter the three-letter international code for the station as described in ALPHANUMERIC DATA ENTRY. The coordinates of the desired station will be loaded into the selected waypoint's latitude and longitude. If no station was found with the entered identifier, zeros will be displayed. For example, to load the coordinates of the LAX VORTAC into waypoint 12, the operator does the following:

1. Turn FUNCTION selector to WPT
2. Key in WPT 12: WPT-1-2-ENTER
3. Press MEM key
4. Key in LAX: ALPHA-4-4-4-ALPHA-1-ALPHA-8-8-8-ENTER.
   (L) (A) (X)

The coordinates of the LAX VORTAC will be loaded into waypoint 12 and displayed for verification. The letters LAX will appear in the alpha register.

NOTE

While VHF Navaid identifiers are not duplicated within a country, there are occasional duplications in different countries. If the selected name is duplicated in different locations around the world, the coordinates of the closest Navaid will be displayed and the ENTER key will continue to flash, indicating the duplication. The operator should verify the displayed coordinates as approximately correct. If the operator wishes to select a Navaid other than the closest, press MEM-ENTER again. The coordinates of the next closest Navaid with the same identifier will be displayed. This process can be repeated until all coordinates having the same identifier are displayed. When the desired coordinates are found, press ENTER again to load them into the waypoint.

AIRPORT LOOK-UP

The LRN-85 has memory-stored coordinates of all the airports in the United States with runways over 5000 feet long, and many of the world's other major airports. The procedure for loading the coordinates of an airport is the same as loading a VHF Navaid, except the MEM key is pressed twice.

For example, to load the coordinates of Los Angeles International Airport (LAX) into waypoint 12, the operator performs the following:

1. Turn FUNCTION selector to WPT
2. Key in WPT 12: WPT-1-2-ENTER
3. Press MEM key
4. Press MEM key
5. Key in LAX: ALPHA-4-4-4-ALPHA-1-ALPHA-8-8-8-ENTER
   (L) (A) (X)

The coordinates of LAX airport will be loaded into waypoint 12 and displayed for verification. The letters LAX will appear in the alpha register and the ARPT annunciator will light, indicating an airport waypoint.

COMPUTE MODE

The distance, bearing, desired track and enroute time between any two waypoints may be calculated in the Compute mode (CMPT). First, set the Mode Selector to CMPT. To compute the BRG, DIS, DTK and ETE between any two waypoints, enter the waypoint numbers in the FROM and TO legs.

For example, to calculate the distance between waypoints 06 and 37:

1. Turn Mode Selector to CMPT, Function Selector to BRG/DIS.
2. Enter the leg: LEG-0-6-3-7-ENTER.
3. The bearing and distance between waypoints 06 and 37 are displayed in the upper registers.

The user may enter a test ground speed in the ETE/GS position and calculate ETE for various legs.

NOTES

1. Pressing LEG-ENTER will advance the computer to the next leg.

2. Use of the Compute mode does not affect position, navigation, or autopilot outputs in any fashion. When the NAV mode is reselected, all outputs will be referenced to the original leg.

3. The accumulated distance and accumulated time along a sequence of waypoints can be displayed in Extended Functions.

4. Waypoint coordinate entry, position entry, GMT and date entry are inhibited in the Compute mode. Dashes are displayed in these functions.

5. In the Compute mode, leading blanks in the CDU display are replaced by leading dashes to prevent inadvertent misinterpretation of the displayed data.

EXTENDED FUNCTIONS (XFCN)

By setting the FUNCTION selector to XFCN (Extended Functions), the user can access several other input/output functions not specified as CDU switch positions. These functions are:

1. Set Crosstrack (SX)
2. Heading (HDG)
3. Variation (VAR)
4. Track (TK)
5. Track Angle Error (TKE)
6. Accumulated Distance (ACD)
7. Cross Load (CRL) (optional with dual installation)

These functions are accessed by setting the FUNCTION selector to XFCN and repeatedly pressing the ENTER or BACK keys until the selected function name is displayed. The selected function code is displayed in the alpha register. Each of the following inputs/outputs can be accessed through XFCN.

SX - Set Crosstrack Parallel (Offset) Course

The system may be used to define a track that is right or left (up to 99.9 nautical miles) of the desired great circle course. To establish an offset navigation leg, perform the following:

1. Turn FUNCTION selector to XFCN (Extended Function)
2. Press ENTER key until SX appears in the alpha display
3. Press the L (1) key or R (3) key
4. Key in crosstrack to tenths of a nautical mile
5. Press ENTER
6. Return FUNCTION selector to desired position

The LRN-85 now outputs analog deviation relative to the offset navigation leg. The displayed crosstrack (XTK) will be referenced to the new course center line.

CAUTION

A selected parallel track should not be entered while the airplane is within 1.5 minutes of a waypoint.

Once a track is defined by a FROM/TO leg, the track can be modified by defining an offset. If the user has entered a non-zero Set Crosstrack (SX), the E (East) annunciator will be on in the DTK/XTK mode.

HDG - Heading

The airplane heading in whole degrees is displayed in the right register. Heading may be true or magnetic, depending on the position of the MAG/TRUE switch. If the external airplane sensor is not supplying heading information, OFF is displayed.

To manually enter a heading, simply key in the heading value (in true or magnetic as indicated by the MAG/TRUE switch).

NOTE

If the external heading input is valid, (flag out of view) the manual heading entry is inhibited.

VAR - Variation

The magnetic variation, to the nearest whole degree, is displayed in the right register. This variation is output from the computer's variation mode. (NOTE: Output only.)

TK - Track

The airplane's track over the ground to the nearest whole degree is displayed in the right register. TK may be true or magnetic, depending on the MAG/TRUE switch. (NOTE: Output only.)

TKE - Track Angle Error

The airplane's track angle error (DTK - TK) to the nearest whole degree is displayed in the right register. L indicates that the track is left of the DTK; R indicates that the track is right of the DTK. (NOTE: Output only.)

ACD - Accumulated Distance

This function is used in the Compute mode only to display the total distance along a course of waypoints. For example, to compute the distance from waypoint 10 to waypoint 16 along the course from 10 to 11, 11 to 12, 12 to 13, 13 to 14, 14 to 15 and 15 to 16, key in LEG-10-16-ENTER. The total distance from waypoint 10 to waypoint 16 will be displayed to tenths of a nautical mile in the right register.

Note that the TO waypoint number MUST be greater than the FROM waypoint number.

If waypoint 00 is selected as the FROM waypoint, the displayed distance will be the direct course from present position to the TO waypoint.

The estimated time enroute (ETE) for the accumulated distance is displayed in the left register. The accumulated distance is divided by the Compute mode ground speed. The resulting time along the legs is displayed to tenths of a minute in the left register.

CRL - Cross Load (optional with dual installation)

The lattitude, longitude, alpha identifier, radial and DME offset can be transferred between LRN-85 systems. Select the LRN-85 that is to be loaded with data from the second LRN-85. Place the function selector in "XFCN" position. The display will show as follows:

0 0 0 0

C R L "FR" "TO"

Key in the starting waypoint number 01-99 and the ending waypoint number 01-99. The cross load will begin with the starting waypoint and finish with the end waypoint. A typical display would appear as follows:

0 1 0 9

C R L "FR" "TO"

As the cross load takes place, the upper left two digits in the display will increment. The cross loading is finished when the left two digits indicate one more than the right two digits.

Offset Waypoints

In addition to entering the latitude/longitude coordinates for up to 99 waypoints, the DME (distance) and RAD (radial) can also be entered for each waypoint number. This DME and RAD input will define a new waypoint offset from the latitude/longitude entered. The waypoint coordinates, DME and RAD, will be held in the computer when the system Mode selector is set to OFF. Offset waypoints may be changed anytime during the flight by first changing the latitude/longitude, then the DME and RAD.

1. Set the Function selector to WPT
2. Select the desired waypoint
3. Enter waypoint latitude, e.g., N-3-9-5-6-0-ENTER
4. Enter waypoint longitude, e.g., W-8-6-0-3-0-ENTER
5. Set Function selector to RAD/DME
6. Enter the radial to tenth of a degree, e.g., 3-4-2-6-ENTER

NOTE

Tenths of a degree must be entered. If unknown, enter zero.

7. Enter the offset distance (DME) to tenths of a nautical mile, e.g., 3-3-0-ENTER.
8. The OFST annunciator will now be on.

NOTES

Radials are assumed to be magnetic if the MAG/TRUE switch is set to MAG at the time the offset waypoint is used in a FROM/TO leg. Radials are assumed to be true if the MAG/TRUE switch is set to TRUE.

The DME offset of any waypoint is cleared when new coordinates are entered.

WAYPOINT EDITING

WAYPOINT COPY

The present position or any one of the 99 waypoints may be moved to another waypoint. To copy one waypoint to another, set the FUNCTION selector to WPT. Select the desired destination waypoint using the WPT key. Press MEM followed by the number of the source waypoint followed by ENTER. The coordinates, alphas, and offset of the source waypoint are transferred to the destination waypoint. To move present position to any waypoint, use waypoint 0 as the source.

Example: To move waypoint 36 to waypoint 55, key the following:

Select WPT 55: WPT-5-5-ENTER
Move WPT 36: MEM-3-6-ENTER.

The information from source waypoint 36 has now been copied to waypoint 55. The original information is still retained in waypoint 36.

WAYPOINT SCAN

If a location has been programmed in any of the 99 waypoints, and assigned a one, two or three alphanumeric identifier, its waypoint number can be found without manually cycling thru all 99 waypoints.

For example, if waypoint 64 was programmed and given a name X2Y, by entering X2Y, the computer will scan all the waypoint locations and stop at waypoint 64 and display the information.

To scan waypoint locations:

a. Set the FUNCTION SELECTOR to WPT.
b. Select any waypoint.
c. Press WPT key.
d. Key in each character of the waypoint name by pressing the ALPHA key and then pressing the key for the desired character.
e. After the last alphanumeric character is entered, press the enter key.

Example: To find X2Y, press WPT, ALPHA-8-8-8. ALPHA-2-2-2-2, ALPHA-9, ENTER.

NOTE: The above procedure will work only if the APRT annunciator at that waypoint location is not illuminated.

WAYPOINT INSERT

To insert a new waypoint in the 99-waypoint sequence without replacing the old waypoint, select the waypoint to be inserted by pressing WPT-HOLD-WPT#-ENTER. The coordinates, alphas and offsets of the selected waypoint are moved to the next higher waypoint and each succeeding waypoint is advanced. Waypoint 99 is lost.

Example: To insert a new waypoint 36 between waypoint 35 and old waypoint 36, key WPT-HOLD-3-6-ENTER.

Zeros are displayed in the coordinates of waypoint 36. New coordinates for waypoint 36 can be entered in the normal fashion. Waypoint 37 now equals the old coordinates of waypoint 36.

TEST MODE

GENERAL

Various system performance characteristics may be examined and various commands given through the use of the TEST mode. Use of this mode for station deselection and warning analysis has been described earlier. This section describes how various other data can be accessed.

NOTE

For all TEST mode displays and commands, the MODE selector is set to TEST. The FUNCTION selector has no effect in this mode; therefore, it can be set to any position.

COMPOSITE STATION MAGNITUDES

The relative station magnitudes can be accessed in the TEST mode. Turn the MODE selector to TEST and key the code 0000-ENTER for VLF station magnitude and 0001-ENTER for Omega station magnitude. Each displayed digit represents the composite magnitude of a station.

		LEFT REGISTER	**RIGHT REGISTER**
VLF CODE	0000-ENTER	1 2 3 4	5 6 7 8
OMEGA CODE	0001-ENTER	A B C D	E F G H

A magnitude of 1 is the minimum usable signal. The maximum received signal is 9.

OTHER SYSTEM STATUS DISPLAYS

Various other system parameters can be examined in the test mode:

TEST CODE	DISPLAY
0002-ENTER	Warning Display: see WARNING CODES TABLE in NORMAL PROCEDURES.
0004-ENTER	Frequency Standard Status: OFFF = Not Stabilized, 0010 = Stabilized.
0007-ENTER	Autopilot roll in degrees. "F" as the leading character indicates that the roll is left. "0" indicates that the roll is right.
0012-ENTER	Battery Charger Status: FFFF indicates battery is charged; 0000 indicates battery is charging (right register).
0013-ENTER	Synchronization Status: FFFF indicates synchronization is complete (right register); 0000 indicates synchronization is incomplete (right register).
0014-ENTER	Clear Waypoint Memory: Entering this code sets all waypoint coordinates, alphas, and offsets to 0.
0016-ENTER	Navigation Program Number (PID): The left register displays the navigation software program number installed in the system. The right register displays the release date for the software (month/day/year).
0017-ENTER	Memory Program Number (MID): The left register displays the waypoint memory program number which is installed in the system. The right register displays the release date for the data (month/day/year).
0019-ENTER	Receiver Program Number (RCV): The left register displays the receiver software program number installed in the system.

SYSTEMS DESCRIPTIONS

The LRN-85 is a very-low-frequency (VLF) radio navigation system designed for area navigation utilizing existing VLF and OMEGA transmitting stations. Airplane position is determined by automatically making position fixes by time referencing radio signals from several stations. Generally, reception of signals from two stations is sufficient to determine a position. From changes in latitude/longitude position, the system derives distance to go, bearing, desired track, cross track error, ground speed, and time to waypoint/destination. With an airplane true airspeed input, the system can compute the current wind direction and speed.

Navigation is computed along a Great Circle flight path to provide the shortest distance between positions. Course deviation is presented in nautical miles on the Pilot's Course Indicator rather than in degrees as with conventional VOR navigation. This feature provides for a constant course width of approximately ±7.5 nautical miles regardless of the distance to the waypoint. The waypoints can be entered by the operator or called from the computer memory of over 5000 NAVAIDS and airports.

CONTROL/DISPLAY UNIT (CDU)

The system is programmed and operated from a pedestal-mounted Control/Display Unit (CDU). The

control section is used to enter data into the computer and select the computed information to be displayed. The display section provides navigation information to the pilot. It also displays the data to be entered into the computer.

DATA DISPLAYS

Left Register

The display window at the upper left of the CDU displays numeric data selected with the CDU Function Selector.

Right Register

The display window at the upper right of the CDU displays numeric data selected with the CDU Function Selector.

Alpha Register

The display window directly above the CDU Function Selector displays a 3-character identifier associated with a waypoint, abbreviation for an extended function, or a test code.

From/To Display

The display windows to the right of the Alpha Register display the selected FROM and TO waypoint numbers.

ANNUNCIATORS

NOMENCLATURE	MEANING
N	If angle or bearing is being displayed, it is referenced to True North. If latitude is being displayed, it is North Latitude.
S	Latitude being displayed is South Latitude.
W	If longitude is being displayed, it is West Longitude. If magnetic variation is being displayed, the variation is West.
E	If longitude is being displayed, it is East Longitude. If magnetic variation is being displayed, the variation is East.
OFST	The selected or TO waypoint has been OFFSET by an input radial and DME distance.
ARPT	The selected or TO waypoint called from computer memory is an AIRPORT.
L	The present crosstrack is LEFT of the desired track.
R	The present crosstrack is RIGHT of the desired track.
BATT	The system is operating on the emergency BATTERY.
DR	The system is in DEAD RECKONING mode.
NAV	The system is receiving signals from enough stations to NAVIGATE

NOMENCLATURE	MEANING
WRN	The WARNING annunciator indicates either a hardware malfunction or reception of unreliable navigation signals.
A	Omega Norway Station is being received.
B	Omega Liberia Station is being received.
C	Omega Hawaii Station is being received.
D	Omega North Dakota Station is being received.
E	Omega La Reunion Station is being received.
F	Omega Argentina Station is being received.
G	Omega Trinidad Station is being received.

NOTE

Omega Station G is scheduled to move from Trinidad to Australia in late 1980. Refer to HANDLING, SERVICING, AND MAINTENANCE Section for changeover procedure.

H	Omega Japan Station is being received.
1	VLF Cutler, Maine Station is being received.
2	VLF Jim Creek, Washington Station is being received.
3	VLF North West Cape, Australia Station is being received.
4	VLF Rugby, Great Britain Station is being received.
5	VLF Annapolis, Maryland Station is being received.
6	VLF Lualualei, Hawaii Station is being received.
7	VLF Yosami, Japan Station is being received.
8	(Spare annunciator; should not illuminate except during testing.)

KEYBOARD

DATA KEYS - Ten alphanumeric keys are provided for the insertion of data into the computer.

BACK/HOLD KEY - BACK eliminates previous keyboard entry to correct input data. HOLD is used to correct position error.

ENTER KEY - Transfers displayed data into the computer.

ALPHA KEY - Allows for entry of alphanumeric information into the computer.

WAYPOINT (WPT) KEY - Selects a waypoint for input/output operation or selection.

LEG KEY - Enables FROM/TO waypoint numbers to be entered for route definition. LEG key will flash approximately 1.5 minutes prior to reaching the TO waypoint.

MEMORY (MEM) KEY - Selects computer memory for waypoint data look-up.

CONTROL PANEL

MODE SELECTOR - this switch has four positions:

1. ON/OFF - Power is applied to the equipment by pulling, then rotating the switch clockwise from the OFF position. The switch must be pulled out to rotate it to the OFF position.
2. TEST - Various system performance characteristics may be examined in this mode.
3. CMPT - Allows use of the computer for distance/time calculations and course computations without disturbing navigation.
4. NAV - Normal navigation functions are displayed in this mode.

FUNCTION SELECTOR - Allows for entry and readout of navigation information provided by the LRN. This switch has 10 positions:

1. XFCN - Various Extended Function (XFCN) outputs are displayed in the left and right registers with the XFCN name appearing in the alpha register.
2. GMT/DATE - Current Greenwich Mean Time (GMT) in hours and minutes, is displayed in the left register, and the GMT date (month/day/year) is displayed in the right register.
3. RAD/DME - This function is used to offset the position of the selected waypoint. The input radial (RAD), to tenths of a degree, from a waypoint is displayed in the left register and the distance (DME), to tenths of a mile, is displayed in the right register.
4. WPT - Coordinates of the selected waypoint are displayed in the left and right registers.
5. POS - Position, when enroute, displays the airplanes present position in latitude and longitude in the left and right registers, respectively.
6. BRG/DIS - The bearing (BRG) from present position to the TO waypoint, referenced to either true or magnetic north, is displayed in the left register in degrees. The distance (DIS) from present position to the TO waypoint is displayed in the right register to tenths of a nautical mile.
7. DTK/XTK - Desired track (DTK), referenced to either true or magnetic north, is displayed in the left register in degrees. Crosstrack distance (XTK) is displayed in the right register to tenths of a nautical mile.
8. ETE/GS - Estimated time enroute (ETE) to the TO waypoint is displayed in the left register to tenths of a minute. Ground speed (GS) in knots is displayed in the right register.
9. WIND - Wind direction, referenced to either true or magnetic north, is displayed in the left register in degrees. Wind velocity in knots is displayed in the right register.
10. TAS/DA - True Airspeed (TAS) in knots is displayed in the left register. Airplane drift angle (DA) in degrees is displayed in the right register.

LT (Lamp Test) - When pressed, all displays and annunciators are illuminated for test purposes.

DIM (Dim Control) - Allows illumination intensity of displays to be adjusted as required.

NOTE

Keyboard backlighting is controlled by the OVERHEAD SUBPANEL & CONSOLE LIGHTS Switch on the overhead light control panel.

MAG/TRUE SWITCH - Selects magnetic or true north as reference for radial, bearing, desired track, track, heading, and wind.

AUTO/MAN SWITCH - Allows for automatic sequential or manual selection of route legs.

HANDLING, SERVICING, AND MAINTENANCE

AUSTRALIA/TRINIDAD OMEGA STATION "G" RELOCATION

Omega Station G, currently located in Trinidad, is scheduled to be moved to Australia in late 1981. The coordinates of both stations are contained in the LRN-85 memory. A hardwired jumper inside the CRU/RPU will have to be removed in order to change Omega Station G from Trinidad to Australia.

This change can also be temporarily affected from the CDU by setting the Mode Selector to TEST, and keying the following:

1-0-1-1-ENTER

1-0-1-2-ENTER (Alpha register displays "OTR")

This change must be re-entered everytime the system is turned on until the hardware jumper is removed by qualified service personnel.

Approved: Donald H Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

Raytheon Aircraft

Beech Super King Air® B200 & B200C Landplanes

PILOT'S OPERATING HANDBOOK
AND
FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for

High Flotation Landing Gear

This Supplement is applicable to the following Manual(s):

101-590010-147 and 101-590010-327

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: _____
A.C. Jackson
Raytheon Aircraft Company
DOA CE-2

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FAA Approved
Revised: April, 1999
P/N 101-590010-173

For Training Purposes Only

Raytheon Aircraft

Beech Super King Air® B200 & B200C Landplanes

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

P/N 101-590010-173

LOG OF REVISIONS

"A3" Revision ..... April, 1999

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Title Page 1 of 2	Updated
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A3

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BEECHCRAFT DUKE B60, KING AIR C90 (LJ-668 THRU LJ-1062, except LJ-670), C90A (LJ-1063 and after) E90, F90, A100, B100, SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the FOSTER AIRDATA RNAV 612 AREA NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA-Approved Airplane Flight Manual when the airplane has been modified by installation of the Foster AirData RNAV-612 Area Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA-Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. The Area Navigation mode may not be used as a primary system under IFR conditions except on approved approach procedures, approved airways, and random area navigation routes when approved by Air Traffic Control.

2. The Area Navigation mode can only be used with colocated facilities. (VOR and DME signals originate from the same geographical location.)

EMERGENCY PROCEDURES

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude and angle of bank.

1. If NAV flag appears while in the enroute mode, check for correct frequency.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

3. If NAV flag appears during an approach, execute published missed approach and utilize another approved facility.

NORMAL PROCEDURES

ANNUNCIATOR AND DIGITAL DISPLAY SELF-TEST

1. Rotate MODE SELECT knob to VOR/LOC.

2. Press Test Button "T" and hold. Digital display segments will read all 8's. Release Test Button.

3. Press each waypoint button (1 through 4, and AUTO WPT), and release. Each lamp will illuminate briefly.

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4. Rotate MODE SELECT knob through all positions. Corresponding annunciator lamps will illuminate in each position. In CRS mode, note position of OBS indication and compare with CRS digital display. Push Test Button "T" in CRS mode to display OBS setting to 0.1° resolution.

RNAV SELF-TEST

1. Select RNAV mode. Active waypoint button will be solidly lit.

2. Press Test Button "T" and hold. Active waypoint address will be displayed. BRG and RNG annunciators will be solidly lit.

3. Check for centered needles on proper BRG value with TO flag.

4. Release Test Button "T".

PROGRAMMING

Waypoints 1 through 4 and AUTO WPT function as standard programmable waypoints when manually programmed. AUTO WPT may be programmed for automatic computation of waypoint addresses on either VOR cardinal radials or the airplane's VOR/DME Present Position.

PROGRAMMABLE WAYPOINTS

1. Rotate MODE SELECT knob to RAD/DIST.

2. Press desired waypoint button. Button to be programmed will blink continuously, while waypoint button in use remains solidly lit. RAD annunciator blinks continuously, and DST annunciator is solidly lit.

3. Use DATA knobs to enter desired Radial in left display. Small knob changes "tenths" digit in 0.1° steps. Large knob changes "tens" digit in 10° steps. Overflow from any digit carries over to next digit to left. "Rate Sensing" provides accelerated display changes with rapid knob movements, to speed programming.

4. Press blinking waypoint button again. DST annunciator blinks continuously and RAD annunciator is solidly lit.

5. Use DATA knobs to enter desired Distance in right display. Small knob changes "tenths" digit in 0.1 NM steps. Large knob changes "tens" digit in 10 NM steps. Overflow and rate sensing same as (3).

6. Rotate MODE SELECT knob to FREQ/ELEV.

7. Push the blinking waypoint button again. The FREQ annunciator will commence blinking.

8. Program the NAV receiver to the correct frequency. Rotate the DATA knobs to program the VOR frequency. Display of the three digits left of the decimal point is controlled by the large knob and appears in the left display. The first digit right of the decimal point, and the left most digit in the right display are controlled by the small knob. The last digit displays 0 or 5 only, for 50kHz channel spacing.

9. Push blinking waypoint button again. The ELV annunciator will blink.

10. Program the VOR station MSL elevation (in thousands) in the right hand digit of the right display by rotating the large DATA knob. Waypoint address is now stored.

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NOTE

When reprogramming an active (in use) waypoint, waypoint lamp status will change from solidly lit to blinking. New waypoint information will immediately modify active waypoint address, HSI/CDI display, and autopilot control.

AUTOMATIC WAYPOINTS

1. Set HSI/CDI OBS to desired course between departure and destination points and tune NAV receiver to first reference VORTAC.

2. Rotate MODE SELECT knob to SET (AUTO WPT). AUTO WPT flashes. RAD and DST annunciators are solidly lit.

3. Rotate large DATA knob until desired Cardinal Radial (0°, 90°, 180°, 270°) appears in left display and waypoint distance appears in right display. Note that only Cardinal Radials can be selected.

   If the courseline does not intercept the selected cardinal radial within 299.9 NM of the VORTAC, the right display will show horizontal bars. In this case, the courseline, cardinal radial or VORTAC must be altered.

4. Rotate MODE SELECT knob to RNAV and activate Automatic Waypoint by pressing the AUTO WPT button. Displays will show bearing and Range to Waypoint. BRG and RNG annunciators will be solidly lit. If Auto WPT "DIST" is invalid, displays will be all bars and DST annunciator will blink.

   HSI/CDI steering and autopilot will respond to established courseline, as set by the OBS.

5. Program and activate each new automatic waypoint as the active waypoint is crossed, by tuning the NAV receiver to new VORTAC frequency ahead of the airplane along the established courseline, and repeating steps 2 through 4.

6. As sequential VORTAC stations with differing magnetic variations are selected, modify the OBS setting to maintain the desired track to the destination.

PRESENT POSITION WAYPOINTS

1. Establish airplane in RNAV MODE over desired geographic location.

2. Rotate MODE SELECT knob to RAD/DIST. Press AUTO WPT causing it to blink.

3. Press VOR/DME pushbutton to transfer into memory, the aircraft's Present Position waypoint (radial and distance) relative to selected VORTAC.

CROSSTRACK OFFSET

Crosstrack (XTK) is used to create an RNAV courseline which is parallel to and up to 20 NM either side of the courseline to a waypoint. In the RNAV mode, crosstrack may be programmed simultaneously for all waypoints in memory.

1. Rotate the MODE SELECT knob from RNAV position to CRS/XTK position. XTK annunciator and R annunciator are solidly lit.

2. The small DATA knob controls "tenths" and the large knob controls "units". Rotate knobs clockwise for right crosstrack offset and counterclockwise for left crosstrack offset. L annunciator is solidly lit for left crosstrack settings and R annunciator is off. Displayed crosstrack offset is now stored in memory. L or R and XTK annunciators will flash during entry of new data.

3. Activate pre-programmed XTK in each waypoint by placing the steering mode switch at top of RNAV612 in XTK position. L or R and XTK will be solidly lit when XTK mode is selected.

NOTE

XTK cannot be programmed if mode knob is turned to XTK from VOR/LOC position. XTK can only be programmed when mode knob is turned to XTK position from RNAV mode.

NAVIGATION

RNAV NAVIGATION PROCEDURES

The RNAV612 is in the RNAV mode at all times when the MODE SELECT knob is in the RNAV position, or when it is rotated from the RNAV position to CRS/XTK, RAD/DIST, FREQ/ELEV or SET (AUTO WPT). Digital bearing and distance to the waypoint are continuously displayed (except during programming, self-test or recall of other data), and the BRG and RNG annunciators are solidly lit.

The RNAV provides left/right steering, flag and to/from signals to the HSI/CDI. RMI steering and flag signals are optional. The steering method (Enroute, Approach, Crosstrack) is determined by the position of the steering selector switch.

WAYPOINT ADDRESS DISPLAY

Any active or inactive waypoint address (radial and distance) may be recalled and digitally displayed. The RAD and DST annunciators will be solidly lit during recall and display.

ACTIVE WAYPOINT

With MODE SELECT knob in RNAV position, press and hold the active waypoint button to display its address.

2. In the RNAV mode, with the MODE SELECT knob in RAD/DIST position, the active waypoint address is automatically displayed.

INACTIVE WAYPOINTS

1. With the MODE SELECT knob in the RAD/DIST position, inactive waypoint addresses may be displayed by pressing the desired waypoint pushbutton. Returning to the RNAV mode will restore the active waypoint BRG/RNG display.

2. With the MODE SELECT knob in the VOR/LOC or CRS/XTK position, press and hold any inactive waypoint button to display its address radial and distance in the left and right digital displays.

GROUNDSPEED/TIME-TO-WAYPOINT DISPLAY

In the RNAV mode, press and hold the KTS/MIN button. The left digital display will show groundspeed (knots) to the waypoint. The right digital display will show time (minutes) to the waypoint. The KTS and MIN annunciators will be solidly lit.

RNAV groundspeed computation will normally be valid after 15 miles of flight toward a waypoint. When orbiting a waypoint, groundspeed of 60 KTS or less will not be displayed.

PRESENT POSITION DISPLAY

PRESENT POSITION is commonly used to identify the exact airplane position relative to the VORTAC while flying computed RNAV routes.

With the MODE SELECT knob in any position except OFF, press and hold the VOR/DME button. The left digital display will show the radial from the VOR to the airplane position. If the station is a VORTAC, the right digital display will show airplane distance from the VORTAC along this radial. The VOR and RAD (VOR Radial) and DME annunciators will be solidly lit.

HSI/CDI OBS SETTING AND CROSSTRACK OFFSET DISPLAY

In the RNAV mode, rotate the MODE SELECT knob to the CRS/XTK position. The left digital display shows CDI/HSI OBS setting and the CRS annunciator will be solidly lit.

Rotation of the OBS knob will cause a corresponding change in the digital CRS display. This feature can be used to reduce pilot setting error when selecting a course.

NOTE

To obtain 0.1° course OBS resolution, push Test Button "T".

While in this mode, the right digital display shows any crosstrack offset programmed for the active waypoint, and the XTK and R or L annunciators will be solidly lit.

VOR/LOC OPERATION

The RNAV612 is in either the VOR or Localizer mode at all times when the MODE SELECT knob is in the VOR/LOC position, or when it is rotated from the VOR/LOC position to CRS/XTK.

When the RNAV612 is in the VOR/LOC mode and the NAV receiver is channeled to a VORTAC station, the digital displays show VOR bearing and DME distance TO the station. The VOR and DME annunciators will be solidly lit.

When the localizer receiver is used (Localizer mode), the left display will show dashes and annunciators will be off. The RNAV612 displays continuous digital distance to the terminal DME in the right display, and the DME annunciator will be solidly lit.

If there is no terminal DME, the right display will also show dashes and the DME annunciator will be off.

HSI/CDI OBS SETTING DISPLAY

In the VOR/LOC mode, rotate the MODE SELECT knob to CRS/XTK. The left digital display will show the CDI/HSI OBS setting and the CRS annunciator will be solidly lit. The right digital display will be blank, indicating XTK is neither operational nor programmable in the VOR/LOC mode.

NOTE

To obtain 0.1° CRS display resolution, push Test Button "T" while in CRS mode.

GROUNDSPEED/TIME-TO-VORTAC DISPLAY

In the VOR/LOC mode, press and hold the KTS/MIN button. The left digital display will show groundspeed

(knots) to the station. The right digital display will show time (minutes) to the VORTAC. The KTS and MIN annunciators will be solidly lit.

PRESENT POSITION DISPLAY

With the MODE SELECT knob in the VOR/LOC or CRS/XTK position, press and hold the VOR/DME button. The left digital display will show the radial from the VOR to the airplane position. If this station is a VORTAC, the right digital display will show the airplane distance from the VORTAC along this radial. The VOR and RAD (VOR Radial) and DME annunciators will be solidly lit.

RNAV STEERING

RNAV612 provides linear (constant course width) HSI/CDI steering information in the RNAV mode, with Enroute (ENR), Approach (APP) or Crosstrack (XTK) steering.

ENROUTE (ENR)

With the RNAV612 steering selector in the Enroute (ENR) position, full scale course width is a constant ±5 NM.

APPROACH (APR)

The RNAV612 steering selector may be moved to the Approach (APR) position for increased HSI/CDI steering resolution. Full scale course width is ±1.25 NM.

Approach mode steering sensitivity should be used only on RNAV instrument approaches inside the Final Approach Fix.

CROSSTRACK (XTK)

The RNAV612 may be programmed to establish a parallel courseline up to 20 NM either side of an RNAV course. When Crosstrack (XTK) steering has been selected, the HSI/CDI needle will center when the aircraft is on the programmed parallel offset course. As with enroute steering, full scale HSI/CDI needle deflection is ±5 NM.

VOR/LOC STEERING

When a VOR station has been selected in the VOR/LOC mode, HSI/CDI steering will be based on conventional ±10° full scale track angle deviation from the selected courseline. The HSI/CDI needle will center when the aircraft is on the courseline set into the OBS. When the localizer has been selected, HSI/CDI needle sensitivity will assume standard localizer course width characteristics.

DIGITAL STEERING

The RNAV612 has fully digital navigation steering capability in both the RNAV and VOR modes, including left/right steering and range to waypoint or DME distance to the VORTAC. The RNAV612 digital steering is a viable and precise method of navigation, and can also be employed as a backup to the HSI/CDI.

Digital steering follows the same rule as ADF steering. When inbound TO a waypoint or VOR, fly right to reduce the displayed bearing and left to increase it.

APPROACH RANGE MONITOR

The Approach Range Monitor feature provides for the separation of the RNAV computed range to a

waypoint from the steering guidance of the pilot's horizontal situation indicator. Selecting the Approach Range Monitor switch to the RANGE MONITOR position will connect the RNAV computer to the NAV 2 receiver. The pilot's horizontal situation indicator will be retained on the NAV 1 receiver.

On an ILS approach, for example, it is desirable to know distance to the outer marker and then to the runway threshold. By selecting RANGE MONITOR and setting the appropriate NAV 2 frequency and waypoint parameters, the distance to the desired fix will be continuously displayed while ILS steering guidance on the horizontal situation indicator will be conventional. The result is the ability to fly a localizer or full ILS steering situation while retaining RNAV computed distance to a selected fix.

CAUTION

It is imperative the Approach Range Monitor switch be placed in the NORMAL position during RNAV operations. If left in the RANGE MONITOR position, the range display will be based on the NAV 2 frequency, and the pilots horizontal situation indicator will display conventional VOR steering based on the selected NAV 1 frequency.

Approved: Donald H. Peters

For W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT SUPER KING AIR 200, 200C,
200T, 200CT, B200, B200C, B200T, B200CT
AND KING AIR F90 LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

GLOBAL GNS-500A VLF/OMEGA NAVIGATION SYSTEM
SERIES 3A, 3B, 3C, OR 4
WITH OPTIONAL VANDLING NAVIGATION DATA BANK

THIS SUPPLEMENT IS APPLICABLE TO
PILOT'S OPERATING HANDBOOK AND
FAA APPROVED AIRPLANE FLIGHT MANUAL
P/N's: 101-590010-127, 101-590037-3B,
101-590010-147, 101-590037-29, 109-590010-3,
AND 109-590010-57

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:

Donald H Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised:July,1987
P/N 101-590010-177

For Training Purposes Only

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Global GNS-500A Series 3A, 3B, 3C, or 4 VLF/OMEGA Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

The appropriate Global Navigation GNS-500A Operator's Manuals, Report 1080 for Series 3 installations, Report 1230 for Series 4 installations and Reports 1230 and 1250 for Series 4 installations with the optional Special Mission software package are available from Global Navigation Incorporated, 2144 Michelson Drive, Irvine, California 92715. The appropriate Vandling Corporation NDB-2 Operator's Manual, Report Number 10 is available from Vandling Corporation, 16270 Raymer Street, Van Nuys, California 91406. It is the responsibility of the owner/operator to maintain the appropriate operator's manuals in current status.

LIMITATIONS

The GNS-500A Series 3A, 3B, 3C, or 4 may be used as an additional means of enroute navigation during VFR/IFR conditions provided the following limitations are observed:

1. Operation must be in conformity with the Global Navigation GNS-500A Operator's Manual, Report 1080 dated September 1, 1980, or later issue for Series 3A, revised June 1, 1981 or later issue for Series 3B, revised October 1, 1982 or later issue for Series 3C, or Report 1230 dated April 1, 1986 or later issue for Series 4, and Report 1230 dated April 1, 1986 or later issue and Report 1250 dated February 1,1987 or later issue for Series 4 with optional Special Mission software package. Operation of the Optional Vandling Navigation Data Bank must be in conformity with the Vandling NDB-2 Operator's Manual, Report Number 10 dated January 19, 1981 or later issue for Series 3A or 3B, revised November 1, 1981 or later issue for Series 3C, or revised April 1, 1986 or later issue for Series 4.
2. Only enroute operations in the continuous United States and Alaska are approved.
3. The system is not to be used for navigation in terminal areas, during departures from or approaches to airports, into valleys or between peaks in mountainous terrain, or below Minimum Enroute Altitude (MEA).
4. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.
5. VOR and DME equipment must be installed and operating during navigation on approved RNAV routes.
6. IFR flight is not approved based on VLF/OMEGA navigation into any area in which such operation could not be approved based on the installed equipment required by paragraph 4. This includes overwater flights beyond the range of approved navigation facilities.
7. The VLF/OMEGA position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:
   a. Prior to compulsory reporting points during IFR operation when not under radar surveillance or control.
   b. At or prior to arrival at each enroute waypoint during operations on an approved RNAV route.

c. At hourly intervals during operation off of approved RNAV routes.

8. Update VLF/OMEGA position information when a cross-check with other onboard approved navigation equipment reveals an error greater than 2 nautical miles along track or crosstrack.
9. Navigation shall not be predicated on the use of this system during periods of dead reckoning.
10. Following a period of dead reckoning, the VLF/OMEGA position information must be verified and updated as required by a visual ground fix, or by using other approved navigation equipment.

EMERGENCY PROCEDURES

Refer to Global Navigation GNS-500A Operator's Manual, Report 1080 dated September 1, 1980 or later issue for Series 3A, revised June 1, 1981 or later issue for Series 3B, revised October 1, 1982 or later issue for Series 3C or Report 1230 dated April 1, 1986 or later issue for Series 4, and Report 1230 dated April 1,1986 or later issue and Report 1250 dated February 1,1987 or later issue for Series 4 with optional Special Mission software package.

NORMAL PROCEDURES

Refer to Global Navigation GNS-500A Operator's Manual, Report Number 1080 dated September 1, 1980 or later issue for Series 3A, revised June 1, 1981 or later issue for Series 3B, revised October 1, 1982 or later issue for Series 3C, or Report 1230 dated April 1, 1986 or later issue for Series 4, or Report 1230 dated April 1, 1986 or later issue and Report 1250 dated February 1,1987 or later issue for Series 4 with optional Special Mission software package.

Refer to Vandling Corporation NDB-2 Operator's Manual, Report Number 10, dated January 19, 1981 or later issue for Series 3A and 3B, revised November 1, 1981 or later issue for Series 3C or revised April 1, 1986 or later issue for Series 4, for operation of navigation data bank, if installed.

Computer Program Identification:

1. Turn system-ON
2. When initialization page automatically appears, note program version (3A, 3B, 3C, or 4A) on the bottom line. The display will read:

PROG 3A	XXXX/X
PROG 3B	XXXX/X
PROG 3C	XXXX/X
PROG 4A	XXXX/X

VLF/OMEGA AUTOMATIC FLIGHT CONTROL SYSTEM INTERFACE

When tracking a VLF/OMEGA course with AFCS, utilize the heading mode to accomplish course changes greater than 25°. The enroute navigation mode may then be re-engaged to capture and track the new course.

PERFORMANCE

No Change

WEIGHT & BALANCE

No Change

SYSTEMS DESCRIPTION

No Change

HANDLING, SERVICING, AND MAINTENANCE

No Change

BEECHCRAFT SUPER KING AIR
B200 LANDPLANE (BB-882 ONLY)
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
AUXILIARY ELECTRIC POWER BUS

GENERAL

The information in this supplement is FAA Approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary Electric Power Bus in accordance with Beech approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Equipment connected to this bus is by specific customer request and does not satisfy minimum equipment requirements. No authorization is extended to permit any flight operations to be predicated solely upon the use of this system. Any use is limited to discretionary action by the pilot.

EMERGENCY PROCEDURES - No change.

NORMAL PROCEDURES

1. AUX COMM PWR switch - ON (AUX COMM 1 PWR ON Annunciator will illuminate.)
2. Battery and Generator switches - OFF.

WARNING

The auxiliary electric power bus should be used only when conservation of electric power is paramount. Use of this system will render most lights, electrically driven flight instruments, avionics, anti-ice, and pressurization systems inoperative and will necessitate manual extension of the landing gear. Prior to turning Battery and Generator switches OFF, the pilot should consider existing flight conditions which may require the use of equipment not connected to the auxiliary power bus to assure safe flight. If safe flight cannot be assured using only the auxiliary bus, the Battery switch should be left on and nonessential electric loads turned OFF individually.

CAUTION

Ensure that the Auxiliary Comm Power switch is OFF before leaving the airplane so a drain on the battery will not occur.

PERFORMANCE - No change.

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SYSTEMS DESCRIPTION

The auxiliary electric power bus provides electric power from an auxiliary battery when selected by the pilot. Control of the system consists of one switch mounted in the master switch cluster on the pilot's subpanel and an annunciator light on the instrument panel. If auxiliary power is desired, the switch can be turned ON, supplying electrical power directly from the battery to that system connected to the bus. The system tied to the auxiliary electric power bus (listed below) can be readily identified by noting the placarding of the COMM 1 circuit breaker. Activation of this system does not satisfy the procedures outlined in the Pilot's Operating Handbook and if circumstances warrant, the user of this manual must refer to the Approved Handbook for appropriate action.

The following listed equipment is connected to the auxiliary bus:

(1) The Number 1 Communication system (emergency audio only).

Approved: Donald H Peters

For W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Issued: June, 1981
P/N 101-590010-179

For Training Purposes Only

BEECHCRAFT SUPER KING AIR
B200 LANDPLANES (BB-900, BB-916, BB-927 ONLY)

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the AUXILIARY ELECTRIC POWER BUS

GENERAL

The information in this supplement is FAA Approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary Electric Power Bus in accordance with Beech approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Equipment connected to this bus is by specific customer request and does not satisfy minimum equipment requirements. No authorization is extended to permit any flight operations to be predicated solely upon the use of this system. Any use is limited to discretionary action by the pilot.

EMERGENCY PROCEDURES - No change

NORMAL PROCEDURES

1. AUX PWR switch - ON (AUX PWR ON Annunciator will illuminate.)
2. Battery and Generator switches - OFF

WARNING

The auxiliary electric power bus should be used only when conservation of electric power is paramount. Use of this system will render most lights, electrically driven flight instruments, avionics, anti-ice, and pressurization systems inoperative and will necessitate manual extension of the landing gear. Prior to turning Battery and Generator switches OFF, the pilot should consider existing flight conditions which may require the use of equipment not connected to the auxiliary power bus to assure safe flight. If safe flight cannot be assured using only the auxiliary bus, the Battery switch should be left on and nonessential electric loads turned OFF individually.

CAUTION

Ensure that the Auxiliary Power switch is OFF before leaving the airplane so a drain on the battery will not occur.

PERFORMANCE - No change.

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SYSTEMS DESCRIPTION

The auxiliary electric power bus provides electric power directly from the battery when selected by the pilot. Control of the system consists of one switch mounted in the master switch cluster on the pilot's subpanel and an annunciator light on the instrument panel. If auxiliary power is desired, the switch can be turned ON, supplying electrical power directly from the battery to those systems connected to the bus. Systems tied to the auxiliary electric power bus (listed below) can be readily identified by noting the placarding of the applicable circuit breaker. Activation of this system does not satisfy the procedures outlined in the Pilot's Operating Handbook and if circumstances warrant, the user of this manual must refer to the Approved Handbook for appropriate action.

The following listed equipment is connected to the auxiliary bus:

(1) The Number 1 Communication System (emergency audio only)
(2) The ILS portion of the Number 1 Navigation System
(3) The Glareshield Flood Lights

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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BEECHCRAFT SUPER KING AIR 200, 200C, B200, AND B200C LANDPLANES

PILOT'S OPERATING HANDBOOK SUPPLEMENT FOR AEROPLANES REGISTERED IN THE UNITED KINGDOM

GENERAL

This document is to be attached to the Pilot's Operating Handbook for aeroplanes on the United Kingdom Register. The contents are in addition to, or override, the contents of the PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL.

PILOT'S OPERATING HANDBOOK
AND
FAA APPROVED AIRPLANE FLIGHT MANUAL (P.O.H.)

200, 200C	P/N 101-590010-127
B200, B200C	P/N 101-590010-147

INTRODUCTION

It is the responsibility of the pilot to be familiar with the contents of this supplement, including all the amendments and any relevant equipment supplements.

The current amendment state of this copy is given on the Amendment Record Sheet. Amendments to the text will be indicated by a marginal line together with the amendment number, unless the page is a graph, in which case the amendment number appears at the foot of the page. The date of approval is shown on each page.

Amendments are published by the manufacturer and approved by the Federal Aviation Administration.

A list of the approved equipment supplements which have been embodied in this copy is recorded in the Record of Equipment Supplements. Equipment supplements are published either by the manufacturer or by an organization producing a modification to the aeroplane, and approved by the Federal Aviation Administration and the Civil Aviation Authority.

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AMENDMENT RECORD SHEET

AMENDMENT NO. AND DATE OF APPROVAL	DESCRIPTION OF AMENDMENT	PAGES AFFECTED
Amendment "A" February, 1982	Add Altitude Limitation Information - Revised Date	3, 8, 49
Amendment "B" May, 1987	Revise Minimum Flight Crew Revise Maximum Operational Altitude Information Revise Structural Limitations Add Note to Flaps Up Landing Revise Weight Scale in Figure 7 Add Note to Associated Conditions	3, 8, 9, 12, 25, 38, 49
Amendment "C" August, 1988	Revise Number of Occupants - Aeroplane Serialization Revise Dual Generator Failure Revise Illustrated Example	3, 9, 11, 22, 49

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RECORD OF EQUIPMENT SUPPLEMENTS

The following approved equipment supplements have been embodied in this copy.

SUPPLEMENT NUMBER AND NAME OF PUBLISHER	DATE OF APPROVAL	DATE OF EMBODIMENT AND SIGNATURE
101-590010-113 Beech Aircraft Corporation King KFC-300 Automatic Flight Control System	January 10, 1977	

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DEFINITIONS

The following definitions are in addition to those shown in Section 1 of the Pilot's Operating Handbook.

GROSS PERFOR-MANCE:	The average performance which a fleet of aeroplanes can be expected to achieve if satisfactorily maintained and flown in accordance with the associated techniques described in the manual.
HEAVY ICING:	An accumulation of ice which continues to build up despite deicing procedures. It is sufficiently serious to cause marked alteration in speed, altitude, or track, and would seriously affect the safety of the flight.
HEIGHT:	The vertical distance between the lowest part of the aeroplane and the relevant datum.
LIGHT ICING:	An accumulation of ice which can be disposed of by operating deicing equipment, and which presents no serious hazard. Light icing will not cause alteration in speed, altitude or track.
MODERATE ICING:	An accumulation of ice in which deicing procedures provide marginal protection; the ice continues to accumulate, but not at a rate sufficiently serious to affect the safety of the flight unless it continues over an extended period of time.
NET PERFOR-MANCE:	Net performance is the "gross" performance, diminished by the amounts specified in British Civil Airworthiness Requirements to allow for various contingencies which cannot be directly accounted for operationally, e.g. the need to manoeuvre, unavoidable variations in piloting technique, temporary below average performance, etc. It is improbable that the net performance will not be achieved in operation, provided the aeroplane is flown in accordance with the recommended technique.
TAKEOFF SAFETY SPEED:	The speed used in the determination of take-off performance which has been selected to provide a safe margin above stall and V_{MCA}.
FINAL TAKE-OFF CLIMB SPEED:	The minimum speed for climb after take-off in the flaps retracted configuration.

LIMITATIONS

CERTIFICATION CATEGORY

This type of aeroplane is eligible for certification in the Transport Category (passenger). However, this particular aeroplane may be restricted to a particular use and to some other category, and this will be stated in the Certificate of Airworthiness.

WIND CONDITIONS

Maximum Demonstrated Crosswind is 25 knots. Taxiing has been satisfactorily demonstrated in 90° crosswinds up to 40 knots.

MAXIMUM TAKE-OFF AND LANDING WEIGHT

The maximum take-off weight is 12,500 pounds, and the maximum landing weight is 12,500 pounds. Per

BCAR, Chapter K2-4 for Group C aeroplanes, neither of these limitations is affected by aerodrome elevations up to 10,000 feet, or by ambient air temperatures up to ISA + 30°C.

MINIMUM FLIGHT CREW

One Pilot

TYPE OF OPERATION

Night flying is permited when the required equipment is installed, and when allowed by the Air Navigation Legislation. Flight in icing conditions has been demonstrated as satisfactory only when conducted under the limitations presented under ICING FLIGHT in this Section.

ICING FLIGHT

All flights under heavy icing conditions or extended flights under moderate icing conditions are prohibited. Flight into areas of light to moderate icing conditions should be undertaken, with caution, only when the essential equipment for extended flight under light icing conditions is installed and operable.

Flight under moderate icing conditions for short periods of time and extended flight under light icing conditions has been demostrated as satisfactory only when the essential equipment listed in Section II is installed and operable.

MAXIMUM OPERATIONAL ALTITUDE (YAW DAMPER SYSTEM INOPERATIVE)

Maximum operational altitude is 17,000 feet with yaw damper system inoperative. If the yaw damper becomes inoperative enroute above 17,000 feet, flight may be continued as planned to destination.

MAXIMUM OPERATIONAL ALTITUDE

Serials prior to BB-983 ..... 29,000 ft

Serials prior to BL-55 ..... 31,000 ft

Serials prior to BB-54 (except BB-38, BB-39, BB-42 and BB-44) if modified by kits 101-5007, 101-5008 and 101-5060 ..... 35,000 ft

Serials BB-38, BB-39, BB-42, BB-44, BB-54 thru BB-982 and BL-1 thru BL-54 if modified by Kit 101-5060 ..... 35,000 ft

Serials BB-983 and after, and BL-55 and after ..... 35,000 ft

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PRESSURIZATION SYSTEM (SAFELIFE CYCLES)

Maximum Fuselage Pressurization Cycles 10,000

STRUCTURAL LIMITATIONS

Wing Structure Safe Life
- B200-B200C ..... 12,000 hrs
- 200-200C ..... 16,000 hrs

Empennage Safe Life ..... 14,000 hrs

OXYGEN

All occupants must be breathing oxygen above 30,000 feet.

NUMBER OF OCCUPANTS

The total number of persons carried shall not exceed twelve, nor exceed the number of seats provided. For aeroplane serials BB-180 and after, the number of persons carried shall not exceed fifteen, nor exceed the number of seats provided. Children under the age of three, carried in the arms of passengers, are excluded from this count.

FUEL CAPACITY

Total of 453.0 IMP gallons (2059 Litres) usable, 321.4 IMP gallons (146.1 Litres) usable in two main systems of 160.7 IMP gallons (730.5 Litres) each, and 131.6 IMP gallons (598 Litres) usable in two auxiliary tanks of 65.8 IMP gallons (299 Liters) each.

AUTOPILOT

Minimum permissible altitudes with autopilot engaged in cruise are:

Collins AP-105	1000 ft above terrain
Collins APS-80	1000 ft above terrain
King KFC 300	1000 ft above terrain

When coupled to ILS glideslope and localizer, the minimums are:

Collins AP-105	200 ft
Collins APS-80	200 ft
King KFC 300	200 ft

In the event of an engine failure during cruise or an ILS approach, disengage the autopilot. Do not extend the landing gear above 15,000 feet with the Collins AP-105 autopilot engaged.

WINDSCREEN HEAT

The windscreen must be heated at least during take-off and landing.

EMERGENCY ENGINE SHUTDOWN

-ENGINE TORQUE INCREASE - UNSCHEDULED (Ground or Flight)
(Not Responsive to Power Lever Movement)
-ENGINE FIRE IN FLIGHT
-ENGINE FAILURE IN FLIGHT
-ILLUMINATION OF MAGNETIC CHIP DETECTOR ANNUNCIATOR

Affected Engine:

1. Power Lever - IDLE
2. Condition Lever - CUT OFF
3. Propeller - FEATHER
4. Fuel Firewall Valve - CLOSED
5. Fire Extinguisher - ACTUATE (if required)
6. Engine Auto Ignition - OFF
7. Generator - OFF
8. Propeller Synchrophaser - OFF
9. Electrical Load - MONITOR

EMERGENCY DESCENT

NOTE

The time required to accomplish an emergency descent from 35,000 feet to 15,000 feet pressure altitude will not exceed 4 minutes.

ELECTRICAL SYSTEM FAILURES

GENERATOR OVERHEAT (Overheat Detector - GEN LH or GEN RH Illuminated)

On Ground

1. Do Not Take-off
2. Reduce Electrical Load To a Minimum

NOTE

A generator overheat warning could be due to a combination of high ambient air temperature and heavy electrical load. Observe minimum gas generator N_1 limits.

In Flight

1. Generator (Affected Side) - OFF
2. Operating Generator - DO NOT EXCEED 1.0 LOAD

DUAL GENERATOR FAILURE

In the event that a dual generator failure occurs (neither generator switch will reset), turn both generator switches off and continue on battery only. Shed electrical loads as follows:

Circuit Breakers:

(1) Number 2 Bus Feeder Circuit Breakers (Right Panel) - OFF (PULL)
(2) Number 1 Avionics Circuit Breaker (Right Panel) - OFF (PULL)
(3) Number 3 Avionics Circuit Breaker (Right Panel) - OFF (PULL), If Installed
(4) Left Engine Fuel Control Heat Circuit Breaker (Right Panel) - OFF (PULL)
(5) Furnishings Bar Circuit Breaker (Right Panel) - OFF (PULL), If Installed
(6) Number 3 Bus Feeder Circuit Breakers (Left Panel) - OFF (PULL)
(7) Number 4 Bus Feeder Circuit Breakers (Left Panel) - OFF (PULL)

Switches:

(8) All switches off except:

- (a) Battery
- (b) Rotating Beacons
- (c) Left Pitot Heat
- (d) Avionics Master Switch
- (e) Number 2 VHF Communication
- (f) Left Bleed Air Valve OPEN
- (g) Indirect Instrument Lights
- (h) Pilot's Flight Instrument Lights

This will result in 30 minutes or more of battery duration. Attitude reference under these conditions is by an air-driven attitude indicator on the co-pilot's panel or, if installed, a standby attitude indicator. The landing gear motor, gear down lights, and wing flaps have been deactivated. Extend the landing gear manually. For landing gear down indication, retard throttles to assure that landing gear warning horn does not sound.

If icing conditions are encountered, use of propeller de-ice and windshield anti-ice systems will significantly reduce the battery duration and therefore should be minimized.

TOTAL AVIONICS FAILURE

(1) Avionics Master Power Switch - OFF; THEN ON;

If Power Not Restored:

(2) Avionics Master Control Circuit Breaker - OFF (PULL)

NOTE

Pulling OFF the avionics master control circuit breaker should restore avionics power if master switch has failed to the OFF position.

LOSS OF PRESSURIZATION

In the event of pressurization loss above 10,000 feet, USE OXYGEN AND DESCEND AS REQUIRED.

NOTES

1. In the event the PASS OXY ON light does not illuminate following illumination of the ALT WARN annunciator, pull PASSENGER MANUAL ORIDE valve to deploy passenger masks. First aid mask can only be deployed manually.
2. If oxygen quantity is insufficient to sustain both passengers and crew, the supply can be isolated to the crew and First Aid outlets by pulling the oxygen control circuit breaker located in the environmental section of the circuit breaker panel. PASSENGER MANUAL ORIDE must be in the OFF position.

NORMAL PROCEDURES

AIRSPEEDS FOR SAFE OPERATION

Air Minimum Control Speed (V_{MCA}) 40% Flaps ..... 85 KIAS

NOTE

The 40% flap V_{MCA} requires full rudder pedal travel and a 5° bank towards the operative engine.

STANDBY COMPASS ACCURACY

The accuracy of the standby compass as presented on the compass deviation card assumes the following.

a. Power - Any normal cruise power setting.
b. Windscreen heat - OFF.
c. Anti-ice equipment - OFF.
d. All radio equipment - ON.
e. Radar - on STANDBY
f. Other normal electrical equipment for flight - ON.

CAUTION

1. Standby compass is erratic when windscreen anti-ice is on.
2. Either the left or right generator off, or both generators off, may cause additional compass deviation.

SHORT FIELD APPROACH AND LANDING

For a short field two-engine landing manoeuvre to final approach, extend the landing gear and set flaps 40%. On final approach, select flaps 100% and decelerate so as to penetrate the 30-foot height point at the approach speed appropriate to the weight. After touchdown, lift and reverse power levers.

FLAPS UP LANDING:

1. Propeller Controls - FULL FORWARD
2. Approach Speed - 27 knots above flaps 100% landing approach speed

NOTE

Gross landing distances will not exceed values scheduled in Figure 14.

Use of the data in this section is mandatory only when required by an Air Navigation Order or Regulation; otherwise its use, although most desirable on the grounds of safety, is at the descretion of the pilot in command of the aeroplane.

COMPLIANCE WITH THE AIR NAVIGATION ORDER AND GENERAL REGULATIONS

1. This aeroplane is classified in Performance Group C.
2. For compliance with the regulations governing flight over water, the true airspeed is 250 knots.

PERFORMANCE BASES

The following performance is based on the Model 200/200C with high flotation gear.

The Model B200/B200C will have at least the same performance.

TAKE-OFF SAFETY SPEED

The variation of take-off safety speed with weight is shown in Figure 1.

FINAL TAKE-OFF CLIMB SPEED

The variation of final take-off climb speed with weight is shown in Figure 2.

APPROACH SPEED

The variation of landing approach speed with weight is shown in Figure 3.

ENROUTE CLIMB SPEEDS

The variations of enroute climb speeds with weight for both power units operating conditions and for one power unit inoperative condition is shown in Figure 4. The bank angle with one engine inoperative should not exceed five degrees.

TAKE-OFF SAFETY SPEED

FLAPS 40%

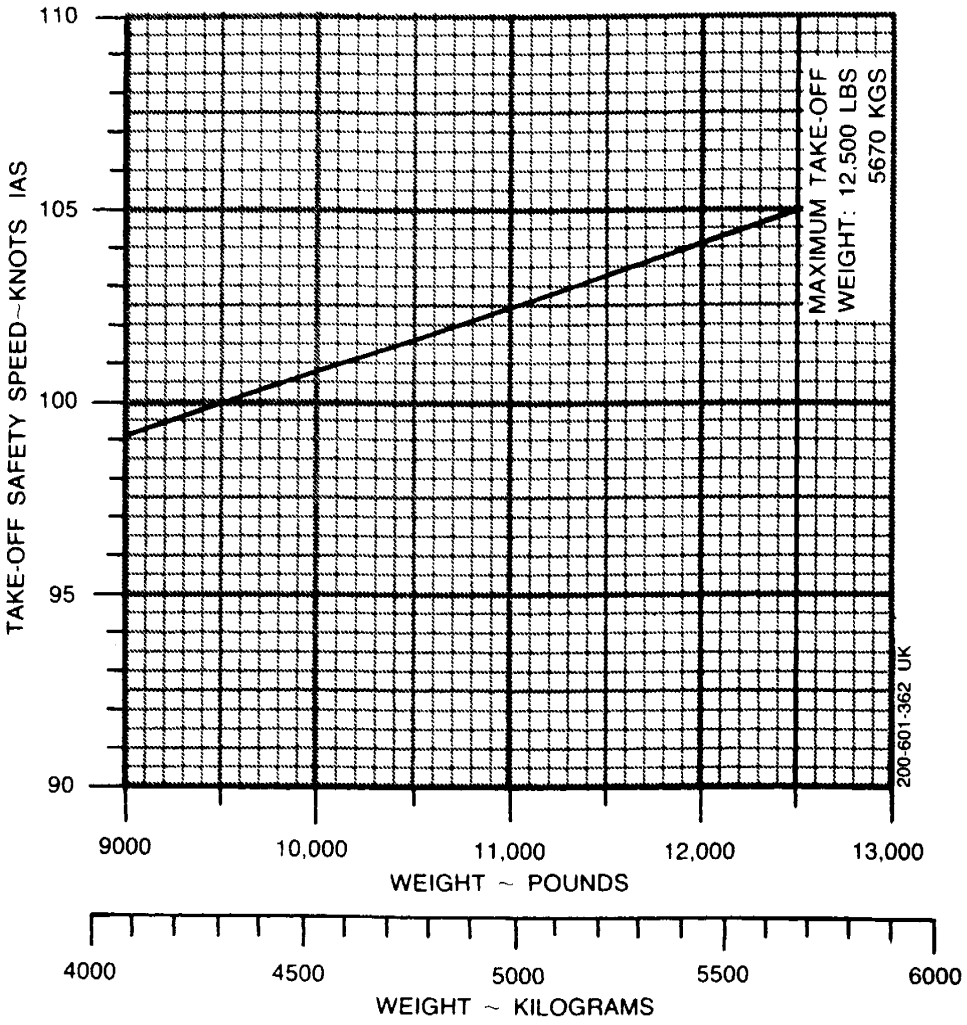

Figure 1

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FINAL TAKE-OFF CLIMB SPEED

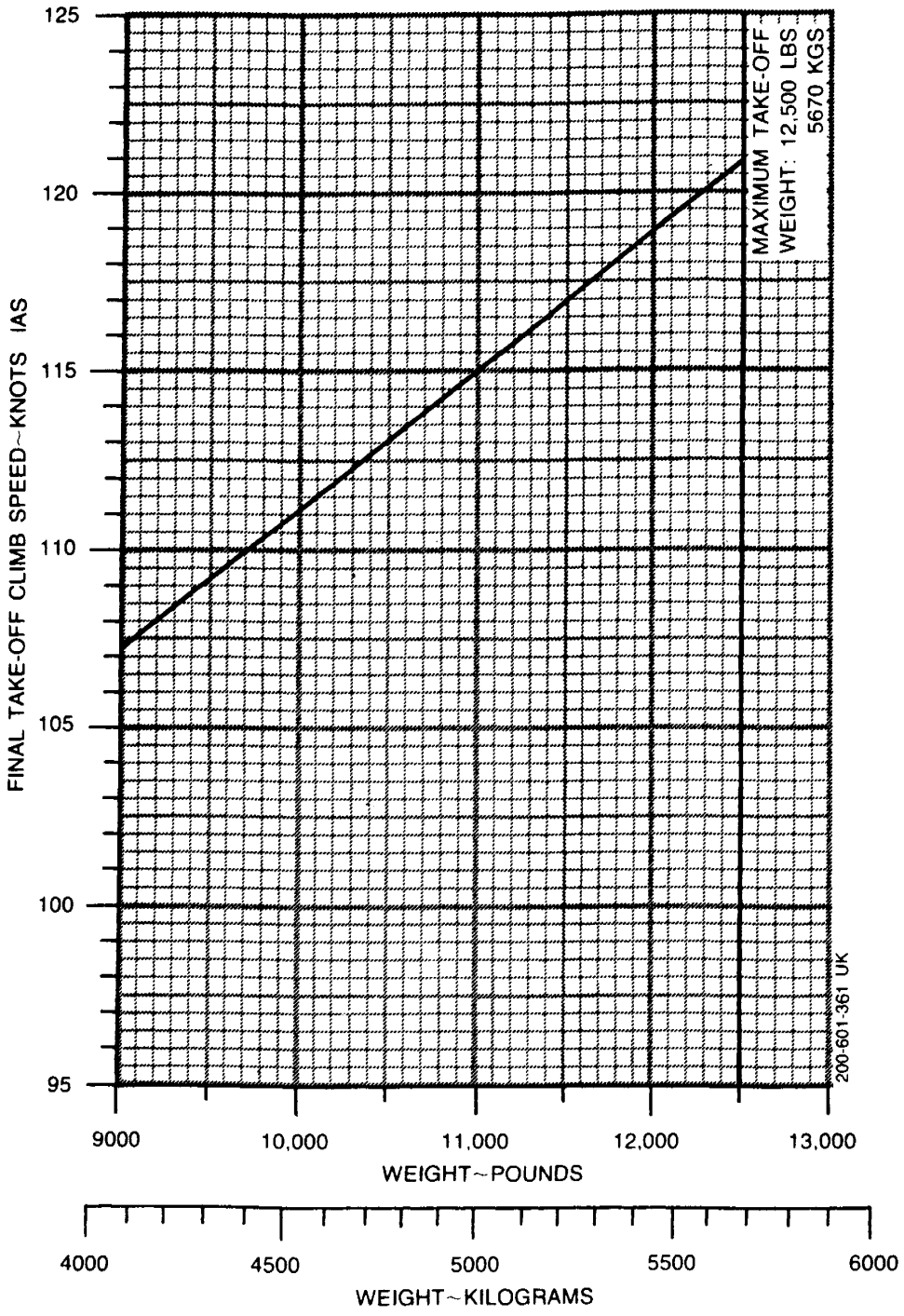

Figure 2

LANDING APPROACH SPEED

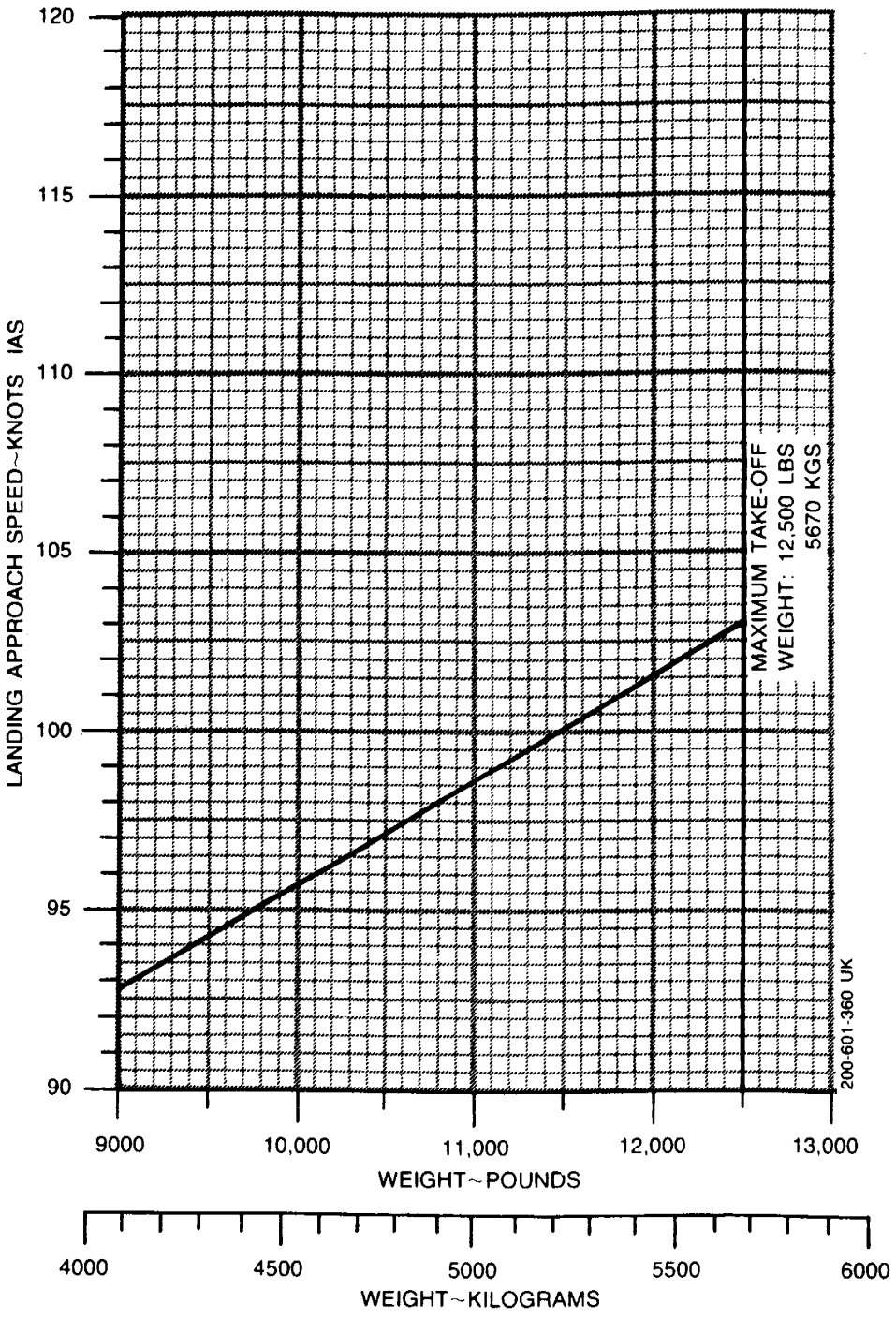

Figure 3

ENROUTE CLIMB SPEEDS

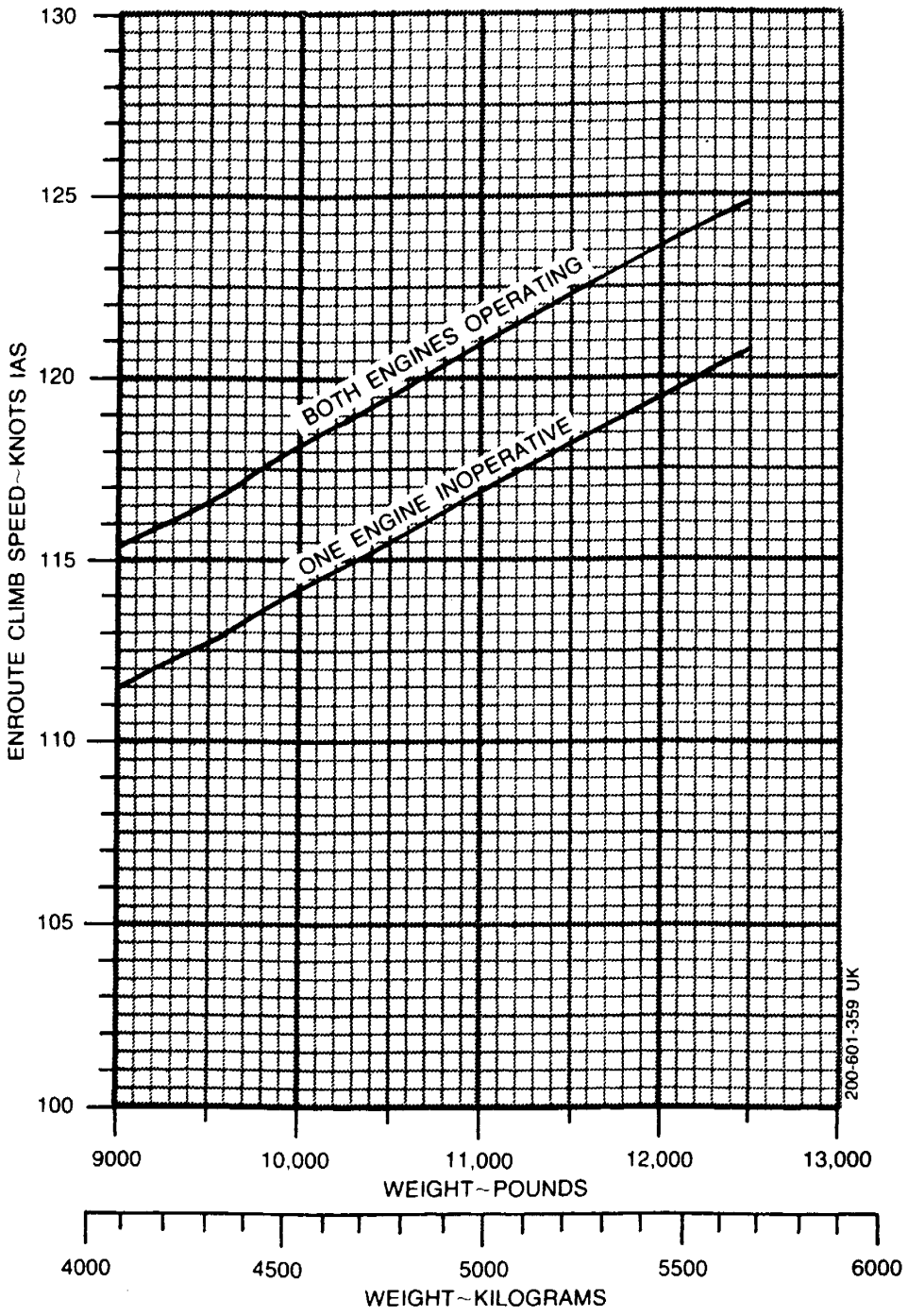

Figure 4

PERFORMANCE (Continued)

TAKE-OFF DISTANCE REQUIRED

The take-off distance required from rest to the 50-foot height point is shown in Figure 5 for varying air temperatures, aerodrome altitudes, weights, components of reported wind and uniform runway slopes.

ASSOCIATED CONDITIONS

Engines: Both operating with take-off power (see Engine Limitations), set prior to brake release.

Wing Flaps: Extended 40%.

Undercarriage: Selected "up" at 50 feet.

Technique: Initiate rotation above V_{MCA}, and then climb away so as to attain a screen height of 50 feet at the take-off safety speed.

Runway: Hard, dry surface (see Note 1).

The example given by the arrowed dashed lines shows that with an air temperature of 13°C (ISA + 10°C), an aerodrome altitude of 6000 feet, a weight of 10,000 pounds, (4536 kilograms) a component of reported headwind of 20 knots and a uniform downhill runway slope of 1.0 percent, the take-off distance required is 840 metres.

NOTES:

1. For operations from dry grass runways with freshly cut grass and firm subsoil, the distances for a hard dry surface runway should be increased by 14 percent.

2. The take-off distance required includes the British Civil Airworthiness Requirement Factor of 1.25. This means that distances obtained from Figure 5 may be equated directly to the take-off emergency distance available.

3. The wind correction grids are factored so that 50 percent of headwind components and 150 percent of tailwind components are obtained. Components of reported winds may, therefore, be used directly in the grids.

4. The take-off ground roll will not exceed 67 percent of the total take-off distance.

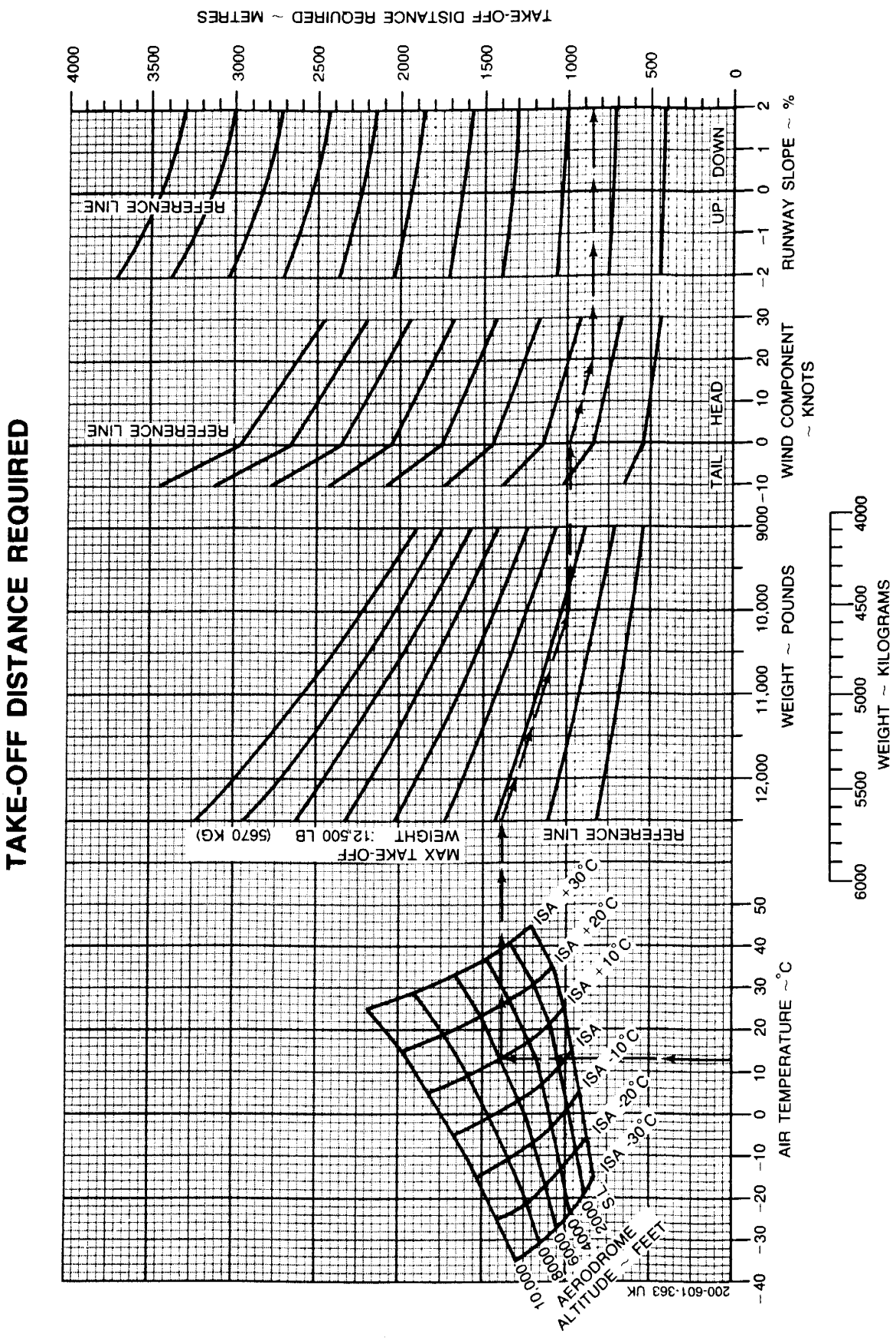
TAKE-OFF DISTANCE REQUIRED
TAKE-OFF DISTANCE REQUIRED ~ METRES
4000
3500
3000
2500
2000
1500
1000
500
0
REFERENCE LINE
REFERENCE LINE
MAX TAKE-OFF WEIGHT :12,500 LB (5670 KG)
REFERENCE LINE
UP DOWN
RUNWAY SLOPE ~ %
2
1
0
-1
-2
TAIL HEAD
WIND COMPONENT ~ KNOTS
30
20
10
0
-10
9000
10,000
11,000
12,000
WEIGHT ~ POUNDS
WEIGHT ~ KILOGRAMS
4000
4500
5000
5500
6000
ISA +30°C
ISA +20°C
ISA +10°C
ISA
ISA -10°C
ISA -20°C
ISA -30°C
50
40
30
20
10
0
-10
-20
-30
-40
AIR TEMPERATURE ~ °C
10,000
8000
6000
4000
2000
S.L.
AERODROME ALTITUDE ~ FEET
200-601-363 UK

Figure 5

NET TAKE-OFF FLIGHT PATH

INTRODUCTION

The information provided in this section makes possible the construction of a complete net take-off flight path to a height of 1500 feet above the aerodrome. It is assumed that both engines will be working until at least the 200-foot point is reached.

After this, the flight path may be constructed on the assumption that both engines continue to function for all or part of the climb to 1500 feet. If the aeroplane is operated in weather conditions such that obstacles cannot be located by visual observation before reaching the minimum safe altitude, compliance with the relevant operating regulations must be established by using the one-engine-inoperative net flight path data from the point at which loss of visual observation is expected to occur.

In circumstances where it is quite clear that all obstacles are less than 1500 feet above the aerodrome and it has been established that they can be cleared by the margins required by the operating regulations, there is no need to proceed further with the construction of the net take-off flight path. In the event of obstacles being above this height, obstacle clearance must be established by using the one-engine inoperative enroute performance, shown in Figure 13 after completion of the net take-off flight path.

The technique which is scheduled for establishing the net take-off flight path is not mandatory and does not have to be followed in the event of an engine failure. However, it has been chosen because it gives the best overall obstacle clearance, in conjunction with satisfactory cooling.

When making a normal take-off and climb away with both engines operating from an aerodrome where obstacle clearance with one power unit inoperative is critical, the pilot must ensure that the aeroplane exceeds by an adequate amount the net take-off flight path assumed in assessing the obstacle clearance.

A fully worked example of a straight path is given in the text and illustrated in Figure 6. Whilst the example given assumes a weight and so obtains a distance, it is also possible to obtain the optimum weight for a given distance. Thus, the take-off distances and flight paths may be calculated for a range of weights, with other conditions remaining constant, and knowing the distance of the obstruction from the start of take-off, the optimum weight is found graphically by plotting distance to (obstacle + clearance height), against weight.

PRESENTATION

The information required to construct a net take-off flight path is given in Figures 7 to 9. Should changes in heading become necessary, the radius of turn should be accounted for as shown in Figure 11, and the effect of changes in wind components should be allowed for as shown in Figure 10.

A number of terms relating to the net take-off flight path presentation are listed and defined below. They are included here and not in the definition section, since they may vary with different presentations and, therefore it must not be assumed that their meaning is generally accepted.

(i) Reference Zero - This is the zero to which the coordinates of the various points in the net take-off flight path are referred. It is defined by a vertical datum, which passes through the 50-foot height point at the end of take-off distance required, and a horizontal datum which is 50 feet below this point.

(ii) First Segment - Extends from the reference zero (i) to the completion of landing gear and wing flap retraction. (Assumes that landing gear retraction will be initiated at the 50-foot height point and wing flaps will be retracted while accelerating at the 200-foot height point.)

(iii) Second Segment - This extends from (ii) to the 1500-foot point, or the point at which the one-engine inoperative net take-off flight path is assumed to commence.

(iv) Third Segment - When applicable, this extends from the expected height at which location of obstacles by visual observation will no longer be possible and, therefore, where use of the one-engine-inoperative data becomes necessary.

ILLUSTRATED EXAMPLE

Examples of the use of the net take-off flight path charts are given below and illustrated in Figure 6. It is assumed that (A) one engine fails at 300 feet above reference zero and (B) engine failure does not occur.

In the examples, the air temperature is 13°C (ISA + 10°), the aerodrome altitude is 6000 feet, the weight is 10,000 pounds (4536 kilograms), and the component of reported headwind is 20 knots. The same examples have been shown in Figures 7 through 9 and values so obtained are quoted in the text on the facing page of each chart.

The position of Point A is established from the take-off distance required or available, which ever is the less, and, therefore, the distance of Point A to any obstruction can be determined. Calculation of the segments then proceeds as follows:

(A) First Segment (A.B.)
From Figure 7, the horizontal distance to landing gear and flaps retracted point = 620 metres

Second Segment (B.C.)
From Figure 8, net gradient of climb = 21.4 percent.
(Height at which engine failure is assumed to commence = 300 feet
Height increment = 300 − 200 = 100 feet

Horizontal distance = $\frac{100 \times 100}{21.4}$ = 467 feet = 142 metres

Total distance to Point C (300 feet) = 620 + 142 = 762 metres

Third Segment (C.D.)
From Figure 9
Net gradient of climb = 6.75 percent
Height increment = 1500 − 300 = 1200 feet

Horizontal distance = $\frac{1200 \times 100}{6.75}$ = 17,778 feet = 5419 metres

Total of 3 segments
Total distance to Point D (1500 feet) = 762 + 5419 = 6181 metres

(B) First Segment (A.B.)
From Figure 7, the horizontal distance to undercarriage and flaps fully retracted point = 620 metres

Second Segment (B.E.)
From Figure 8
Net gradient of climb = 21.4 percent
Height increment = 1500 − 200 = 1300 feet

Horizontal distance = $\frac{1300 \times 100}{21.4}$ = 6075 feet = 1852 metres

Total of 2 segments
Total distance to Point E (1500 feet) = 620 + 1852 = 2472 metres

NET TAKE-OFF FLIGHT PATH
EXAMPLE

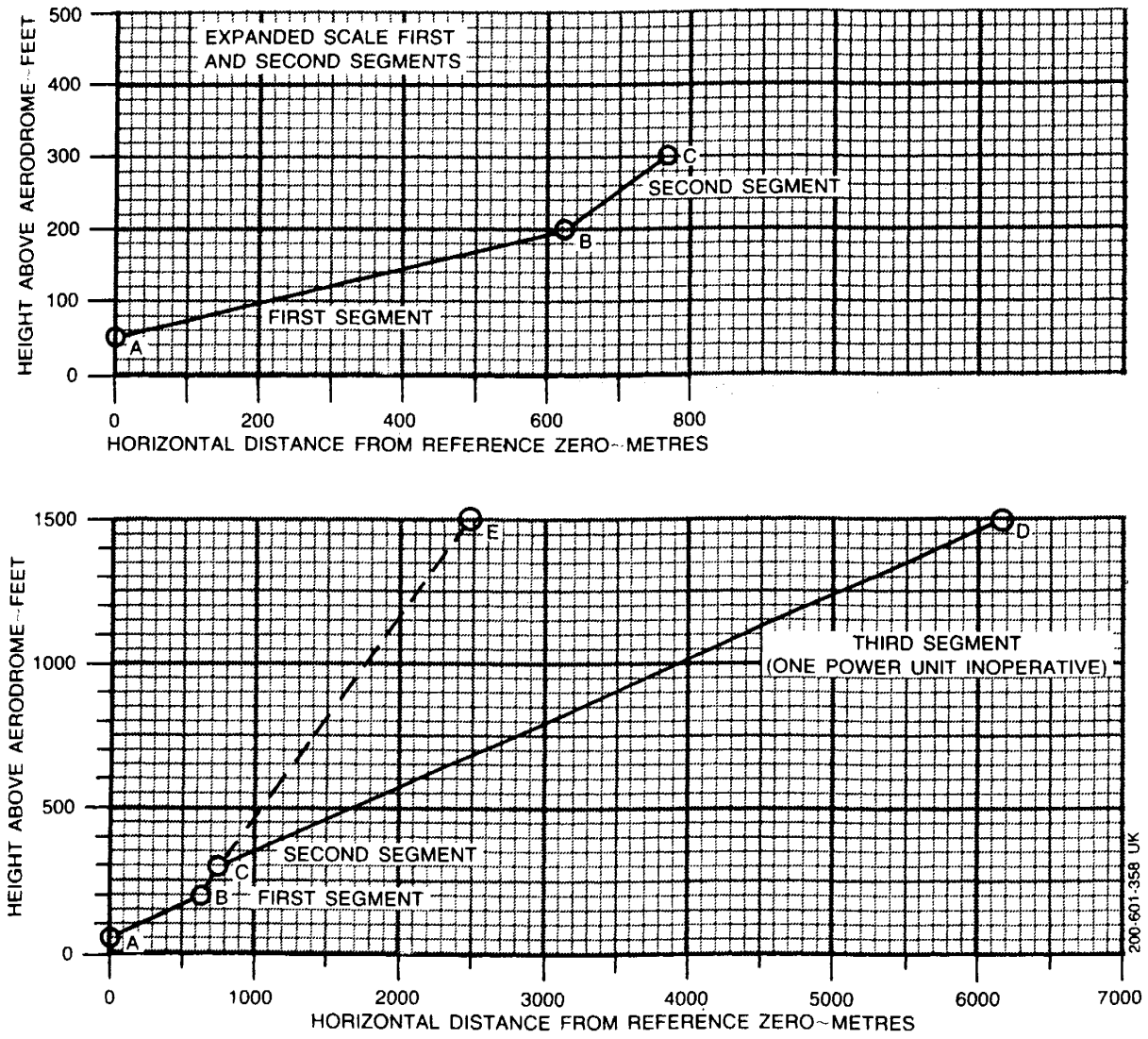

Figure 6

NET TAKE-OFF FLIGHT PATH I - FIRST SEGMENT

The horizontal distance travelled from reference zero to the completion of landing gear and wing flap retraction, for the attainment of final take-off climb speed, is shown in Figure 7 for varying air temperatures, aerodrome altitudes, weights and reported wind components.

ASSOCIATED CONDITIONS

Engines: Both operating at take-off power (see Engine Limitations).

Wing Flaps: Extended 40%, retract during acceleration at 200 feet.

Landing Gear: Selected "up" at 50 feet.

Airspeed: Take-off safety speed (see Figure 1) appropriate to the weight up to 200 feet height, then accelerate to final take-off climb speed (see Figure 2).

The example given by the arrowed dashed lines shows that with an air temperature of 13°C (ISA + 10°C) an altitude of 6000 feet, a weight of 10,000 pounds (4536 kilograms), with a component of reported headwind of 20 knots, the horizontal distance travelled is 620 metres.
NOTE:

1. The wind grids are factored so that 50 percent of headwind components and 150 percent of tailwind components are obtained. Components of reported winds may, therefore, be used directly in the grids.

2. The data given in Figure 7 has been derived from gross performance reduced by a margin of two percent climb gadient.

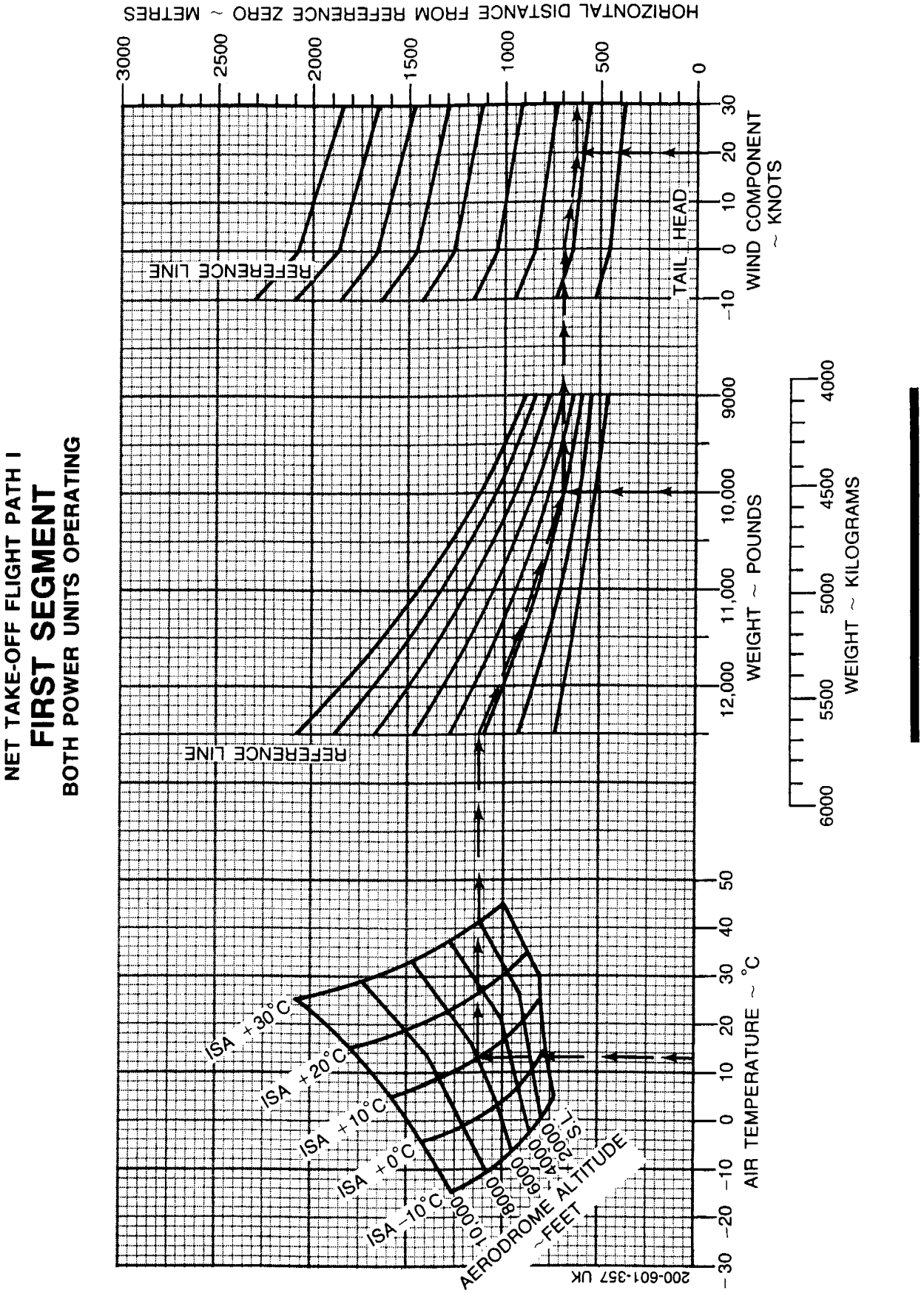
NET TAKE-OFF FLIGHT PATH I
FIRST SEGMENT
BOTH POWER UNITS OPERATING
HORIZONTAL DISTANCE FROM REFERENCE ZERO ~ METRES
3000
2500
2000
1500
1000
500
0
REFERENCE LINE
REFERENCE LINE
TAIL
HEAD
WIND COMPONENT ~ KNOTS
30
20
10
0
−10
WEIGHT ~ POUNDS
9000
10,000
11,000
12,000
WEIGHT ~ KILOGRAMS
4000
4500
5000
5500
6000
AIR TEMPERATURE ~ °C
50
40
30
20
10
0
−10
−20
−30
ISA +30°C
ISA +20°C
ISA +10°C
ISA +0°C
ISA −10°C
AERODROME ALTITUDE ~ FEET
S.L.
2000
4000
6000
8000
10000
200-601-357 UK

Figure 7

NET TAKE-OFF FLIGHT PATH II - SECOND SEGMENT

The net gradient of climb between the end of the First Segment and either the 1500-foot height point or the point at which the one power unit inoperative net take-off flight path is assumed to commence, is shown in Figure 8 for varying air temperatures, aerodrome altitudes, weights and reported wind components.

ASSOCIATED CONDITIONS

Engines: Both operating at take-off power (see Engine Limitations).

Wing Flaps: Retracted.

Landing Gear: Retracted.

Airspeed: Final take-off climb speed appropriate to the weight (see Figure 2).

The example given by the arrowed dashed lines shows that with an air temperature of 13°C (ISA + 10°C), an aerodrome altitude of 6000 feet, a weight of 10,000 pounds (4536 kilograms), and a component of reported headwind of 20 knots, the net gradient of climb is 21.4 percent, 71 ft/100 metre.

NOTE:

1. The wind grids are factored so that 50 percent of headwind components and 150 percent of tailwind components are obtained. Components of reported winds may, therefore, be used directly in the grids.

2. The data given in Figure 8 has been derived from gross performance reduced by a margin of two percent climb gradient.

NET TAKE-OFF FLIGHT PATH II
SECOND SEGMENT
BOTH POWER UNITS OPERATING

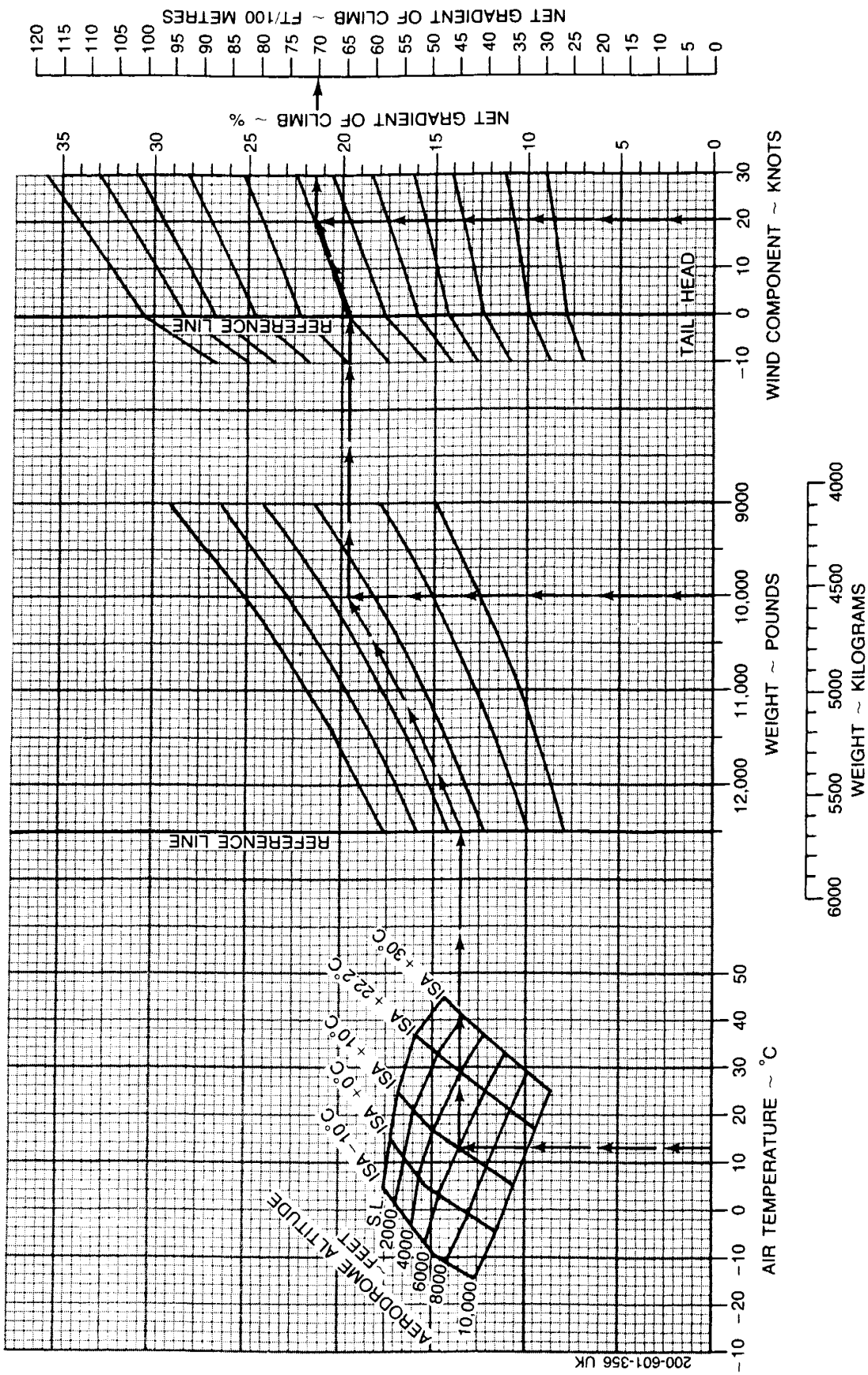

Figure 8

FAA Approved
August, 1981
P/N 101-590010-185

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NET TAKE-OFF FLIGHT PATH III - THIRD SEGMENT

The net gradient of climb between the point at which the one engine inoperative net take-off flight path is assumed to commence and the 1500-foot point is shown in Figure 9 for varying air temperatures, aerodrome altitudes, weights, and reported wind components.

ASSOCIATED CONDITIONS

Engines: Operative engine at take-off power (see Engine Limitations). The propeller of the inoperative engine is feathered.

Wing Flaps: Retracted.

Landing Gear: Retracted.

Airspeed: Final take-off climb speed appropriate to the weight (see Figure 2).

The example given by the arrowed dashed lines shows that with an air temperature of 13°C (ISA + 10°C), and aerodrome altitude of 6000 feet, a weight of 10,000 pounds (4536 kilograms), and with a component of reported headwind of 20 knots, the net gradient of climb is 6.75 percent, 22.1 ft/100 metre.

NOTE:

1. The wind grids are factored so that 50 percent of headwind components and 150 percent of tailwind components are obtained. Components of reported winds may, therefore, be used directly in the grids.

2. The data given in Figure 9 has been derived from gross performance reduced by a margin of 0.8 percent climb gradient.

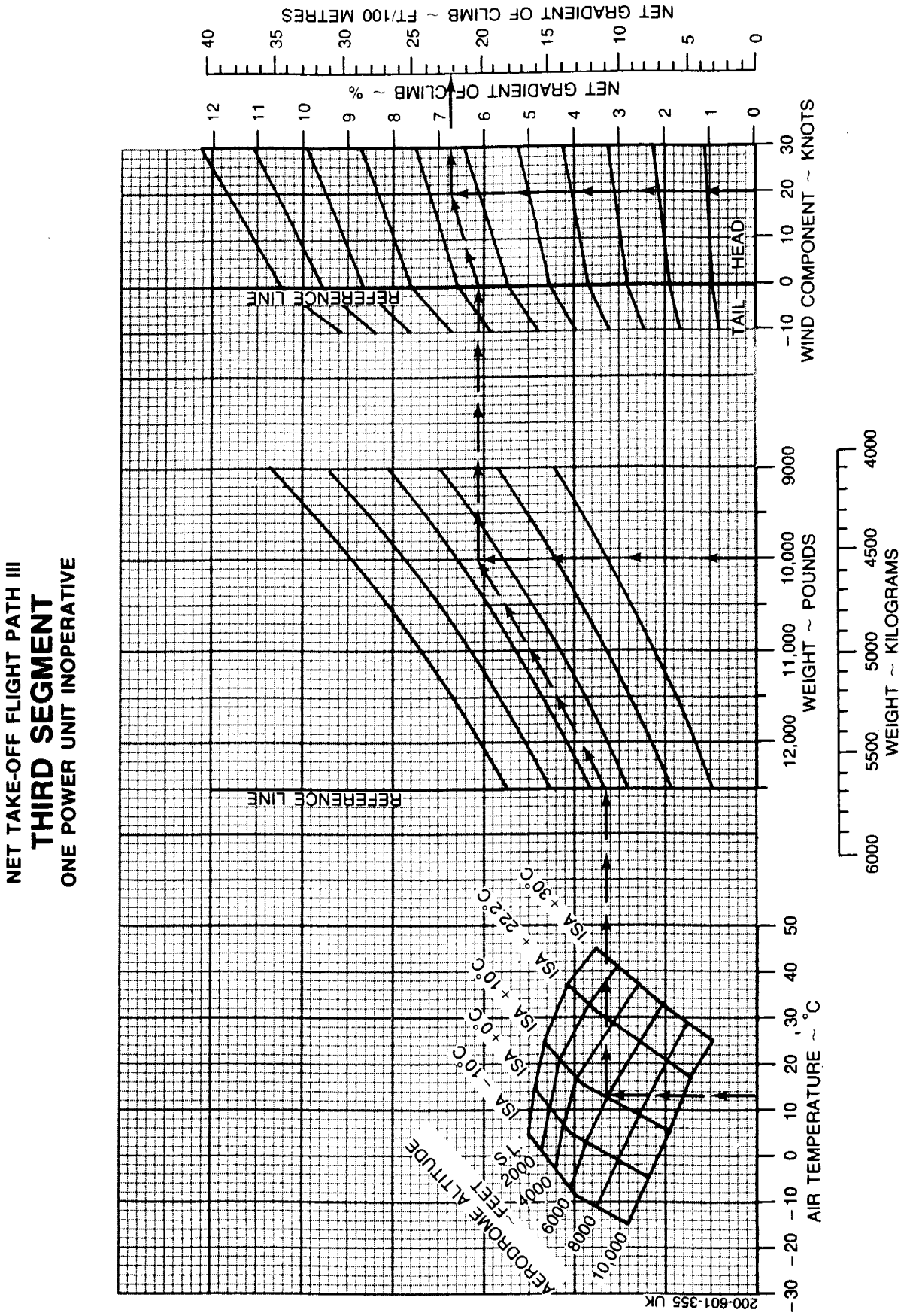

NET TAKE-OFF FLIGHT PATH III
THIRD SEGMENT
ONE POWER UNIT INOPERATIVE
NET GRADIENT OF CLIMB ~ FT/100 METRES
NET GRADIENT OF CLIMB ~ %
REFERENCE LINE
REFERENCE LINE
TAIL
HEAD
WIND COMPONENT ~ KNOTS
WEIGHT ~ POUNDS
WEIGHT ~ KILOGRAMS
AIR TEMPERATURE ~ °C
AERODROME ALTITUDE ~ FEET
ISA
ISA + 10°C
ISA - 10°C
ISA + 22°C
ISA + 30°C
200-601-355 UK

Figure 9

NET TAKE-OFF FLIGHT PATH - RADIUS OF A STEADY TURN

The radius of a steady rate of turn (180° per minute) in the second and third segments is shown in Figure 11 for varying air temperatures, aerodrome altitudes, and weights.

ASSOCIATED CONDITIONS

The associated conditions are specified for the appropriate segments.

The example given by the arrowed dashed lines shows that with an air temperature of 13°C (ISA + 10°C), an aerodrome altitude of 6000 feet, and a weight of 10,000 pounds, (4536 kilograms), the radius of turn is 1136 metres.

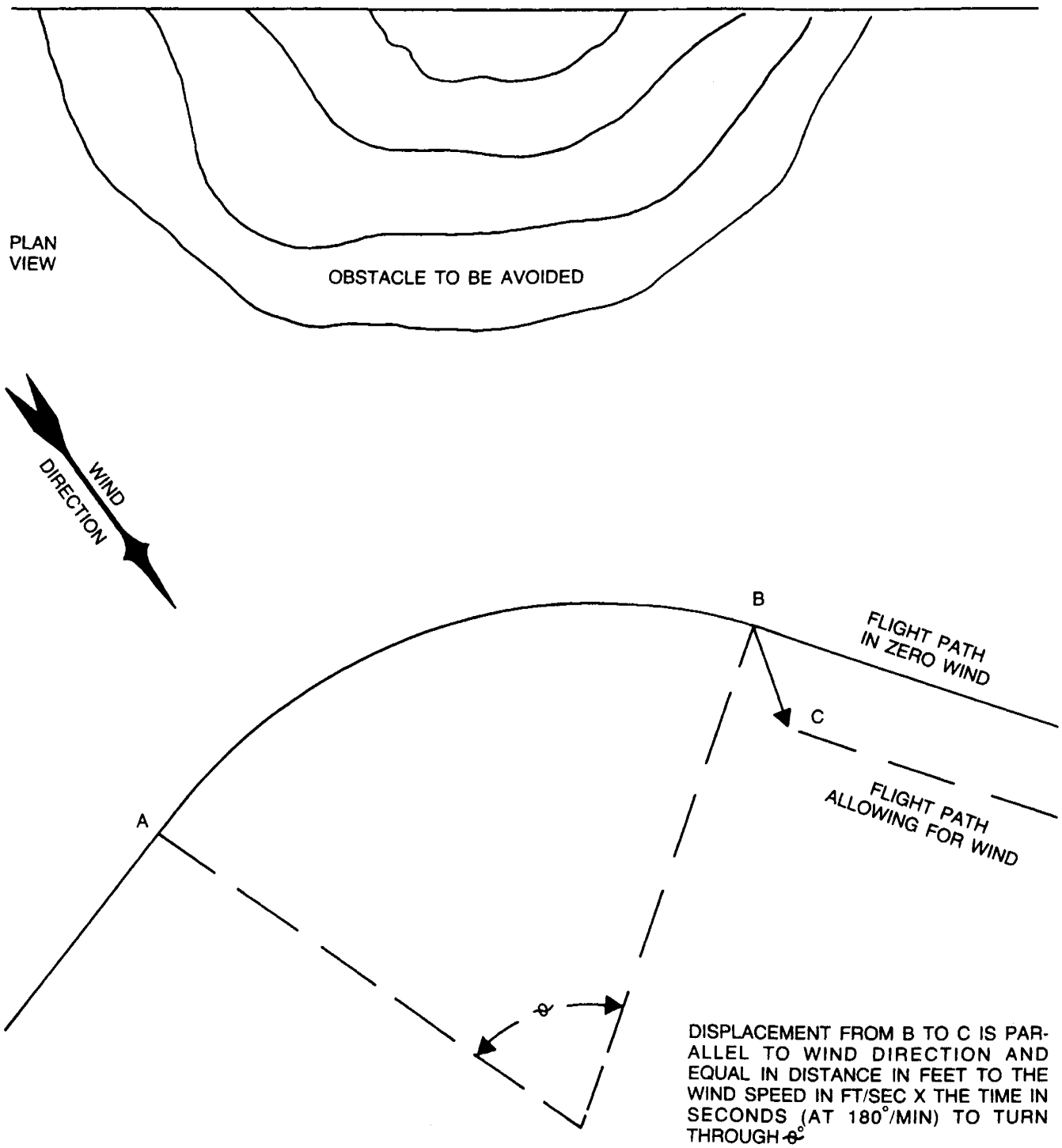
NOTE: The maximum height loss will not exceed 25 feet for a 90° change in heading.

ASSESSMENT OF A STEADY TURN

EXAMPLE: WINDSPEED = 20 KNOTS
Θ = 80°

20 KTS x 1.688 = 33.76 FT/SEC
180 DEG/MIN ÷ 60 = 3.0 DEG/SEC
80 DEG ÷ 3.0 = 26.67 SEC
DISTANCE BC = 33.76 x 26.67 = 900.38 FT (274 metres)

ASSESSMENT OF A STEADY TURN



STD-601-37UK

Figure 10

NET TAKE-OFF FLIGHT PATH
RADIUS OF A STEADY TURN

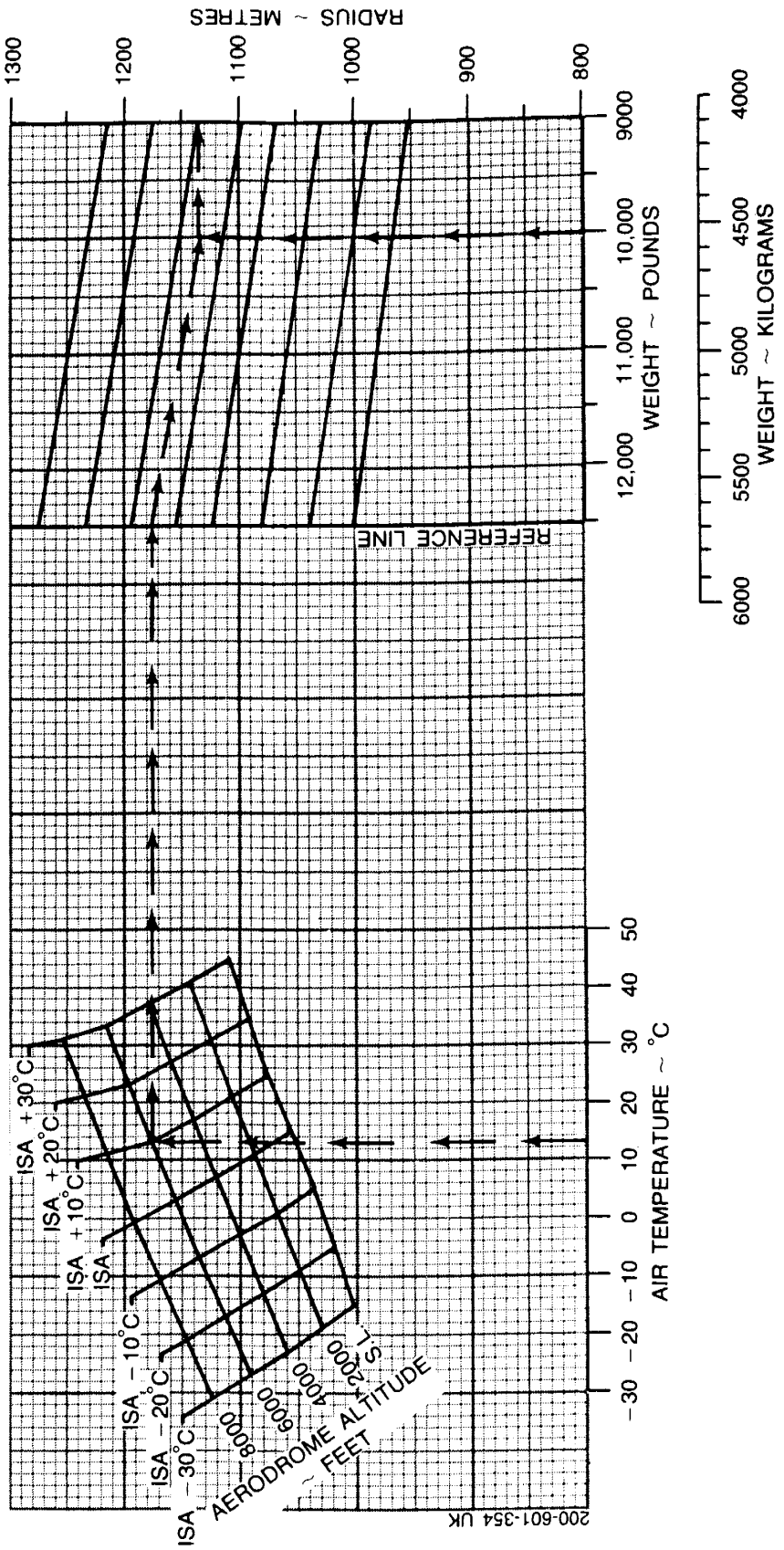


Figure 11

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EN ROUTE PERFORMANCE CEILING AND GROSS RATE OF CLIMB BOTH POWER UNITS OPERATING

The performance ceiling with both power units operating may be obtained from Figure 12 for varying weight, altitude and air temperature. The chart may also be used to determine the gross pressure rates of climb.

ASSOCIATED CONDITIONS

Engines: Both engines operating at maximum continuous power (see Engine Limitations).

Wing Flaps: Retracted.

Landing Gear: Retracted.

Airspeed: En route climb speed (see Figure 4).

The example (A) given by the arrowed dashed lines shows that for a weight of 11,500 pounds (5216 kilograms) in an atmosphere of ISA + 30°C the performance ceiling is 30,500 feet.

The example (B) given by the arrowed dashed lines shows that for a weight of 11,500 pounds (5216 kilograms), an altitude of 6000 feet and an atmosphere of ISA + 30°C, the gross rate-of-climb is 1890 ft/min.

NOTE:

The performance ceiling is the maximum altitude which may be assumed when establishing compliance with operating regulations dealing with en route flight. It does not prohibit flying at a higher altitude (although at some altitudes the operating regulations may require oxygen to be carried), but it is unlikely that the performance ceiling will be achieved unless maximum continuous power and the airspeed quoted in the Associated Conditions are used towards the end of the climb.

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EN ROUTE PERFORMANCE CEILING AND GROSS PRESSURE RATE-OF-CLIMB

BOTH POWER UNITS OPERATING

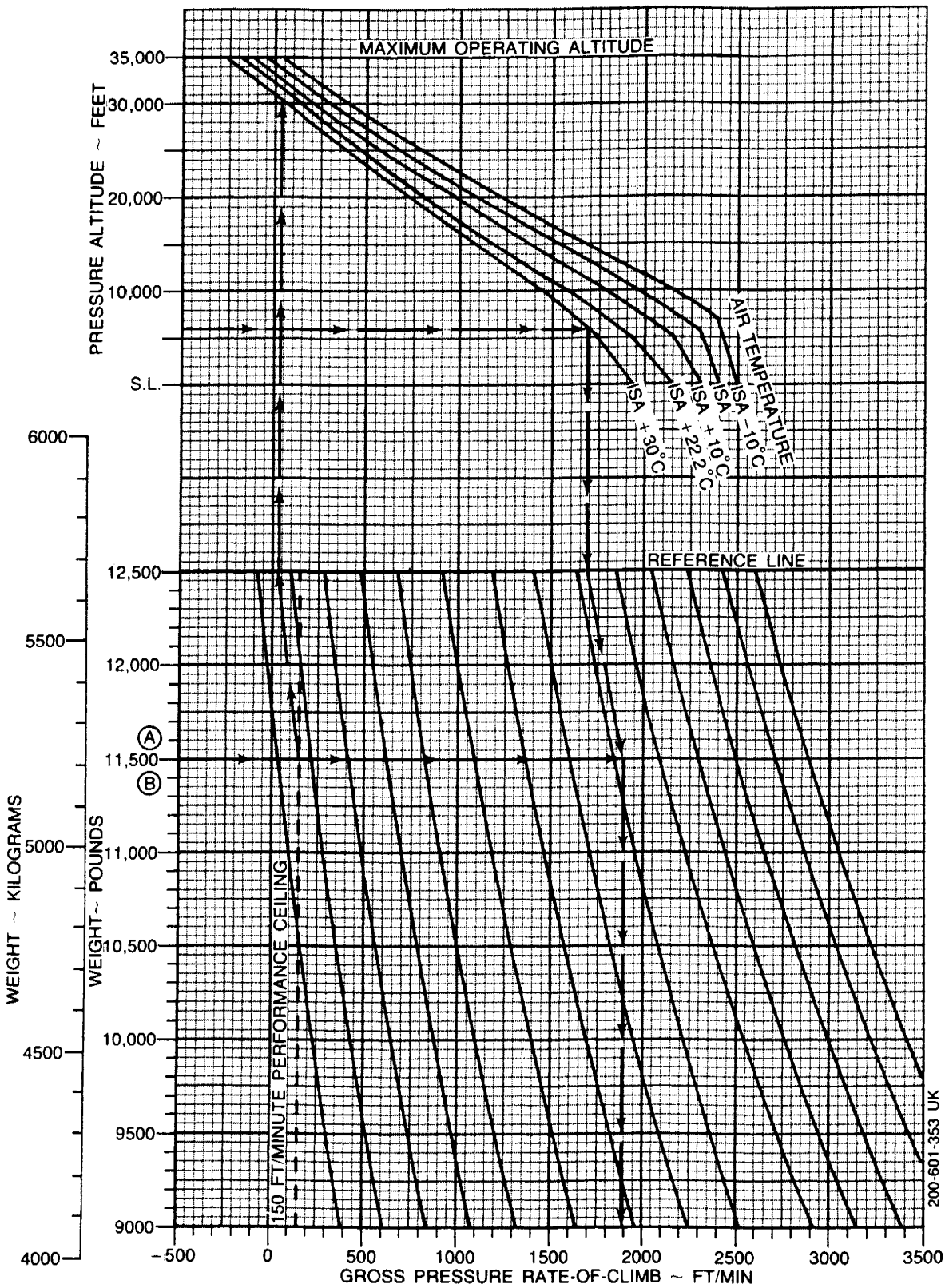

Figure 12

EN ROUTE NET GRADIENT OF CLIMB ONE POWER UNIT INOPERATIVE

The one power unit inoperative en route net gradient of climb is shown in Figure 13 for varying altitudes, air temperatures, and weights.

ASSOCIATED CONDITIONS

Engine: Operative engine at maximum continuous power (see Engine Limitations). The propeller of the inoperative engine is feathered.

Wing Flaps: Retracted.

Landing Gear: Retracted.

Airspeed: En route climb speed appropriate to the weight (see Figure 4).

The example given by the arrowed dashed lines shows that an altitude of 6000 feet, in an atmosphere of ISA + 10°C, and at a weight of 9500 pounds (4309 kilograms), the net gradient of climb is 6.45 percent, 21.2 ft/100 metre.

NOTES:

1. When constructing the flight path profile over a range of altitudes, the number of points plotted shall be sufficient to enable a smooth curve to be drawn.

2. The data given in Figure 13 has been derived from gross performance reduced by a margin of one percent climb gradient. For gross rates of climb, see Figure 16.

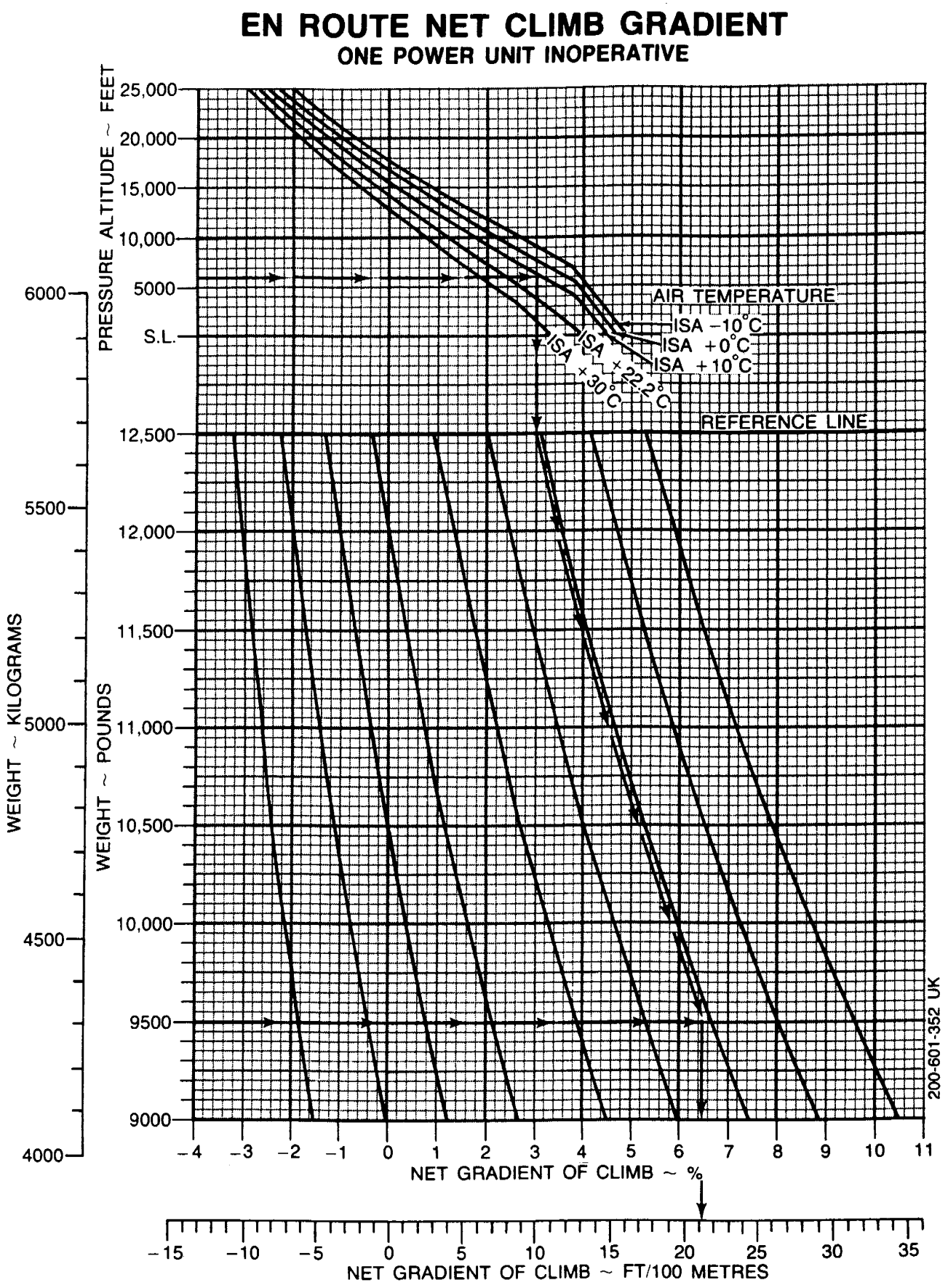

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Figure 13

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NORMAL LANDING DISTANCE REQUIRED

The landing distance required is shown in Figure 14 for varying air temperatures, aerodrome altitudes, weights, reported wind components, and uniform runway slopes.

ASSOCIATED CONDITIONS

Engines: Distances were measured using power as required to maintain 800 ft/min on final approach, and power levers reversed after touchdown.

Wing Flaps: Extended 100%.

Landing Gear: Extended.

Technique: Approach speed appropriate to weight, Figure 3.

Runway: Hard dry or short dry grass surface (see Note 2).

The example given by the arrowed dotted lines shows that at an air temperature of 13°C (ISA + 10°C), an aerodrome altitude of 6000 feet, at a weight of 10,000 pounds (4536 kilograms), a forecast headwind component of 20 knots, and an uphill runway slope of 1.0 percent, the landing distance required is 840 metres.

NOTES:

1. The wind grids are factored so that 50 percent of headwind components and 150 percent of tailwind components are obtained. Components of reported winds may, therefore, be used directly in the grids.

2. Dry grass runway surfaces assume freshly cut grass and firm subsoil.

3. The landing distance required includes the British Civil Airworthiness Requirement field length factor of 1.43. This means that distances obtained from Figure 14 may be equated directly to the landing distance available.

4. Gross landing distances for the abnormal case, flaps-up landing, will not exceed the factored distances for the flaps 100% landings scheduled in Figure 14.

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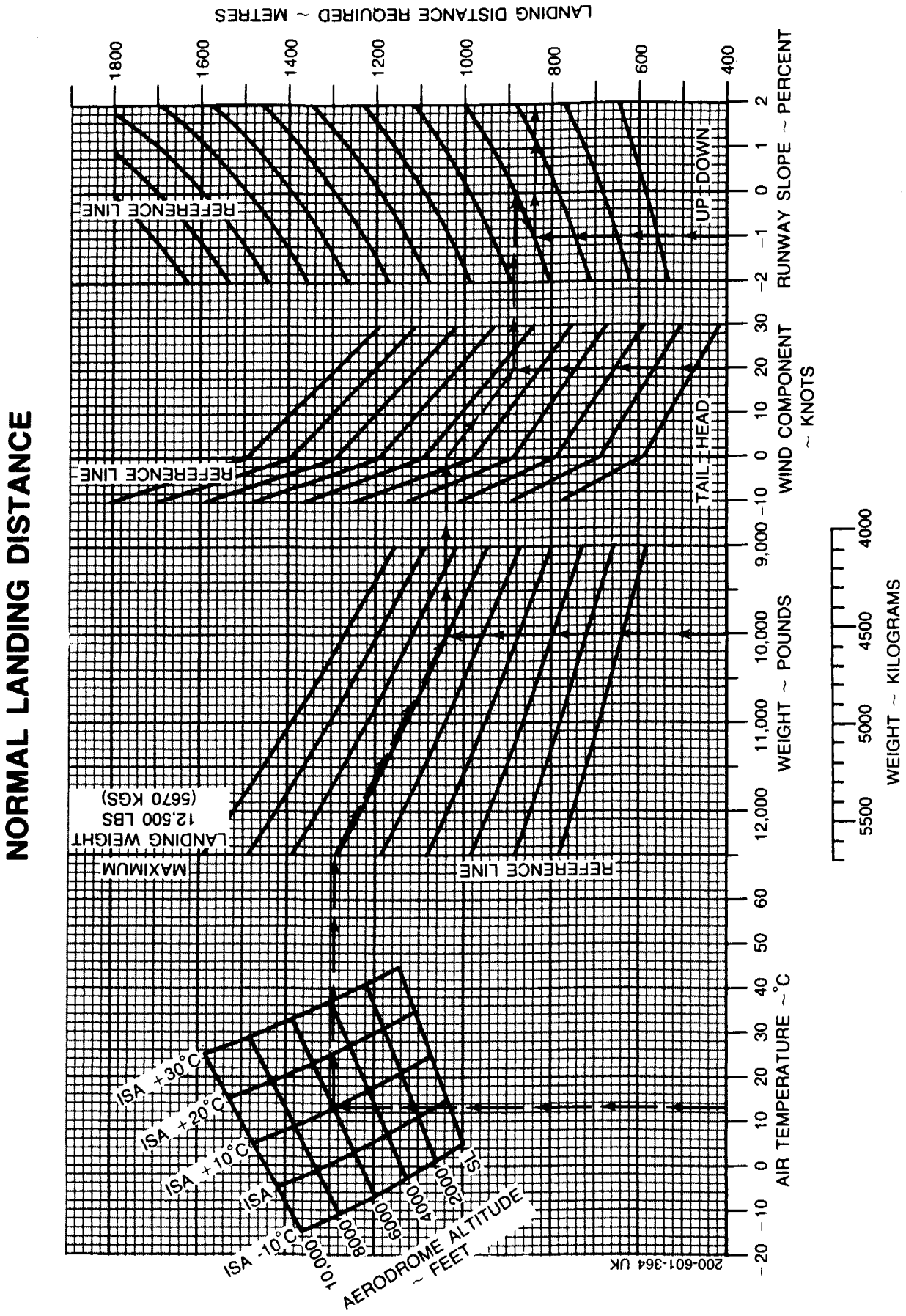
NORMAL LANDING DISTANCE
LANDING DISTANCE REQUIRED ~ METRES
1800
1600
1400
1200
1000
800
600
400
REFERENCE LINE
REFERENCE LINE
REFERENCE LINE
MAXIMUM
LANDING WEIGHT
12,500 LBS
(5670 KGS)
ISA +30°C
ISA +20°C
ISA +10°C
ISA
ISA -10°C
10,000
8000
6000
4000
2000
SL
AERODROME ALTITUDE
~ FEET
AIR TEMPERATURE ~ °C
−20 −10 0 10 20 30 40 50 60
WEIGHT ~ POUNDS
12,000 11,000 10,000 9,000
WEIGHT ~ KILOGRAMS
5500 5000 4500 4000
WIND COMPONENT
~ KNOTS
TAIL HEAD
−10 0 10 20 30
RUNWAY SLOPE ~ PERCENT
UP DOWN
−2 −1 0 1 2
200-601-364 UK

Figure 14

SHORT FIELD LANDING DISTANCE REQUIRED (ALL ENGINES OPERATING)

The landing distance required is shown in Figure 15 for varying air temperatures, aerodrome altitudes, weights, reported wind components, and uniform runway slopes.

ASSOCIATED CONDITIONS

Engines: Distances were measured using power as required to maintain 1000 ft/min on final approach, and power levers reversed after touchdown.

Wing Flaps: Extended 100%.

Landing Gear: Extended.

Technique: Approach speed appropriate to weight, Figure 3.

Runway: Hard dry or short dry grass surface (see Note 2).

The example given by the arrowed dotted lines shows that at an air temperature of 13°C (ISA + 10°C), an aerodrome altitude of 6000 feet, at a weight of 10,000 pounds (4536 kilograms), a forecast headwind component of 20 knots, and an uphill runway slope of 1.0 percent, the landing distance required is 610 meters.

NOTES:

1. The wind grids are factored so that 50 percent of headwind components and 150 percent of tailwind components are obtained. Components of reported winds may, therefore, be used directly in the grids.

2. Dry grass runway surfaces assume freshly cut grass and firm subsoil.

3. Operational techniques which are compatible with the distances obtained from Figure 15 are given in Normal Procedures; Short Field Approach and Landing.

4. The landing distance required includes the British Civil Airworthiness Requirement field length factor of 1.43. This means that distances obtained from Figure 15 may be equated directly to the landing distance available.

5. One engine inoperative or inoperative reverse system requires the use of "Normal Landing Distance Required" chart.

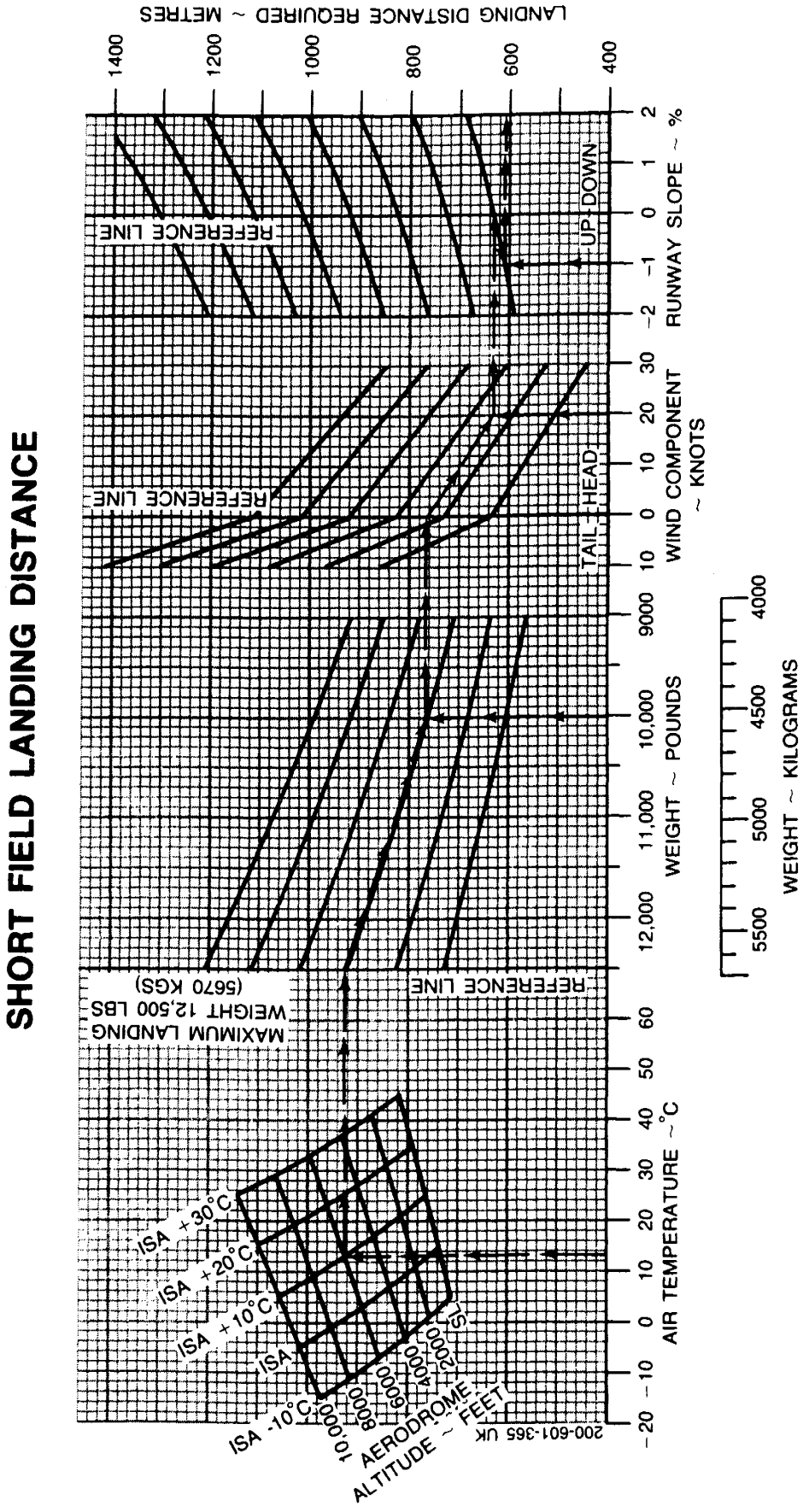
SHORT FIELD LANDING DISTANCE
LANDING DISTANCE REQUIRED ~ METRES
1400
1200
1000
800
600
400
REFERENCE LINE
REFERENCE LINE
UP DOWN
TAIL HEAD
WIND COMPONENT ~ KNOTS
RUNWAY SLOPE ~ %
2 1 0 −1 −2 30 20 10 0 10 9000 10,000 11,000 12,000
WEIGHT ~ POUNDS
5500 5000 4500 4000
WEIGHT ~ KILOGRAMS
REFERENCE LINE
(5670 KGS)
MAXIMUM LANDING
WEIGHT 12,500 LBS
60 50 40 30 20 10 0 −10 −20
AIR TEMPERATURE ~ °C
ISA +30°C
ISA +20°C
ISA +10°C
ISA
ISA -10°C
10,000
8000
6000
4000
2000
(SEA LEVEL)
AERODROME
ALTITUDE ~ FEET
200-601-365 UK

Figure 15

APPENDIX A

"GROSS" PERFORMANCE DATA

This appendix contains additional performance data which is given for general information and for Series and Renewal Certificate of Airworthiness flight test purposes only. It is not to be used for establishing compliance with the operating regulations.

The performance scheduled is "gross" data, that is to say, it is the expected performance of an average aircraft of its type, and no margins have been subtracted. The rates of climb are "pressure" rates and may be used directly compared with the performance measured by the aircraft's altimeter set to 1013.2 millibars.

Stall speeds have been scheduled for a center of gravity location of 185 inches (19.56% MAC) for use in airworthiness test flights.

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APPENDIX A

CLIMB AFTER TAKE-OFF - ONE POWER UNIT INOPERATIVE

The pressure rate of climb for an after take-off configuration, one power unit inoperative, is shown in Figure 16 for varying altitudes, air temperatures and weights.

ASSOCIATED CONDITIONS

Engine: Maximum take-off power (see Engine Limitations).

Propeller Left Engine: Feathered.

Wing Flaps: Retracted.

Landing Gear: Retracted.

Airspeed: Final take-off climb speed. (See Figure 2)

EXAMPLE:

To find the "altimeter" rate of climb knowing altitude, temperature, and weight: the example given by the dashed lines shows that, at an altitude of 6000 feet and a temperature of 13°C (ISA + 10°C), the altimeter (or pressure) rate of climb is 860 feet per minute at a weight of 10,000 pounds (4536 kilograms).

NOTE:

The true ("tape line") rate of climb, if required, may be obtained by multiplying the altimeter rate of climb by the ratio of the ambient temperature in °K (absolute) to the ISA temperature (also in °K) appropriate to the altitude.

$$\text{Temperature } ^\circ\text{K} = \text{Air Temperature } ^\circ\text{C} + 273.16^\circ$$

The rates-of-climb for climb after takeoff may be used also for enroute gross pressure rates-of-climb.

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CLIMB AFTER TAKE-OFF

LEFT POWER UNIT INOPERATIVE

GROSS PRESSURE RATE-OF-CLIMB

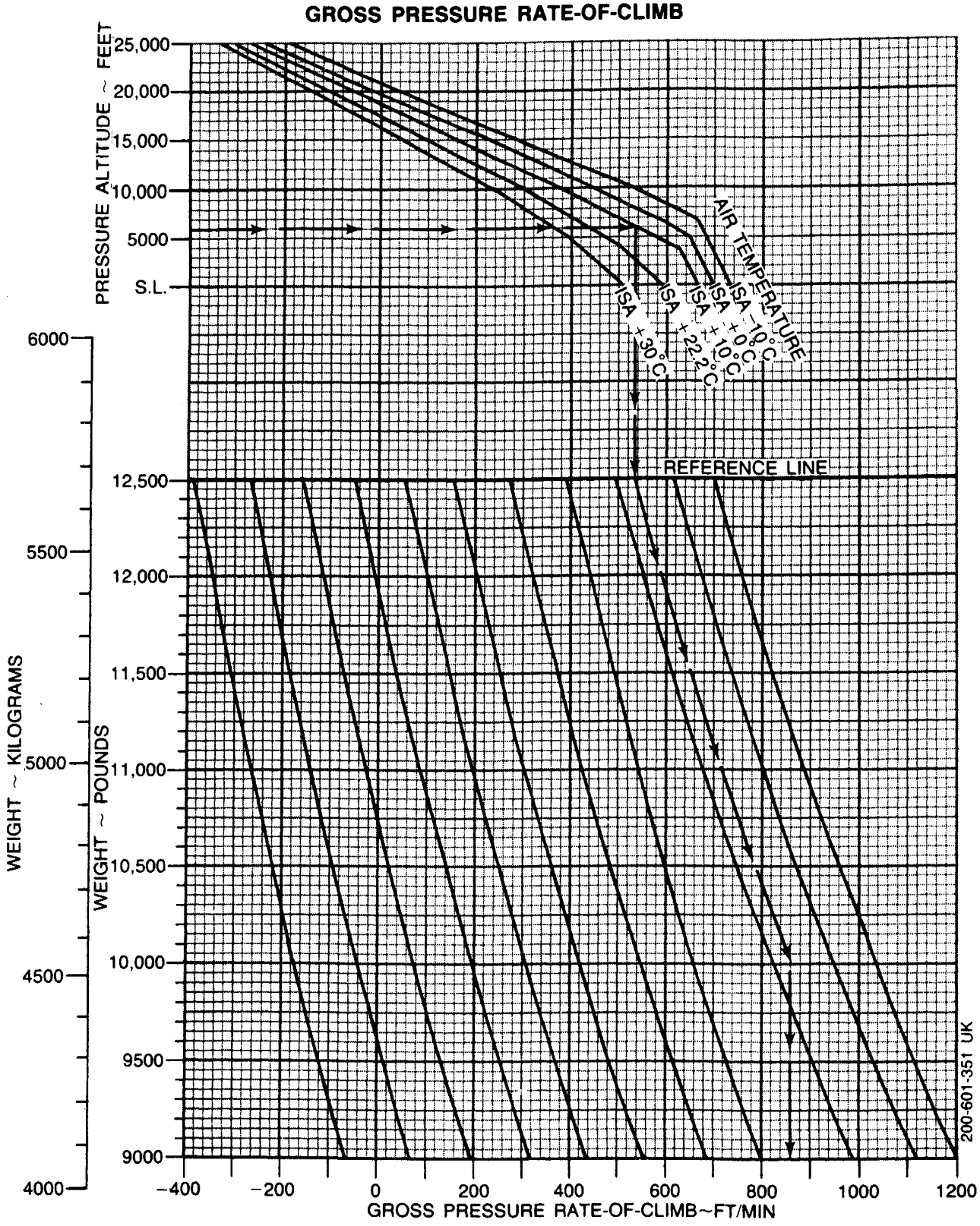
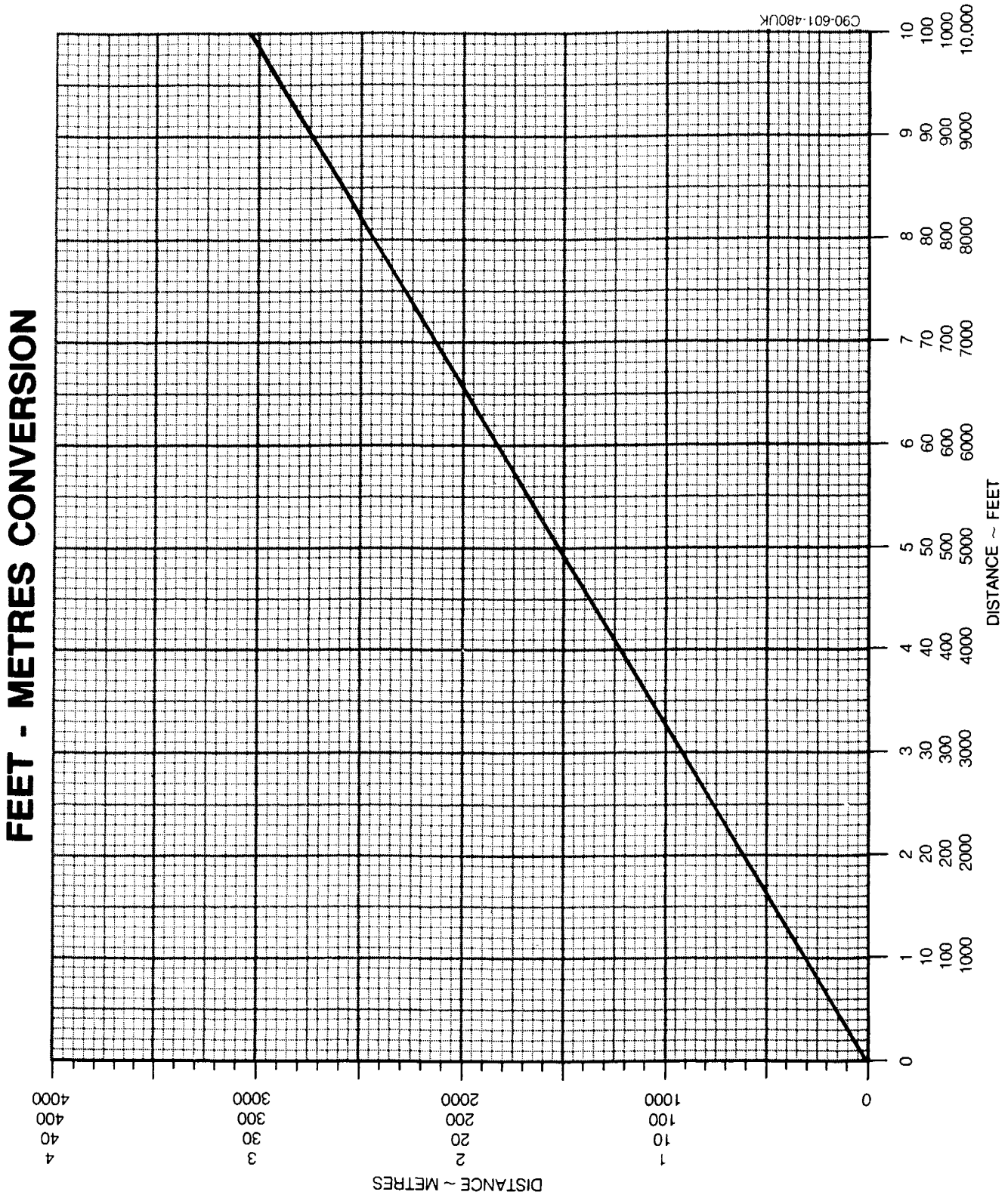


Figure 16

FEET - METRES CONVERSION



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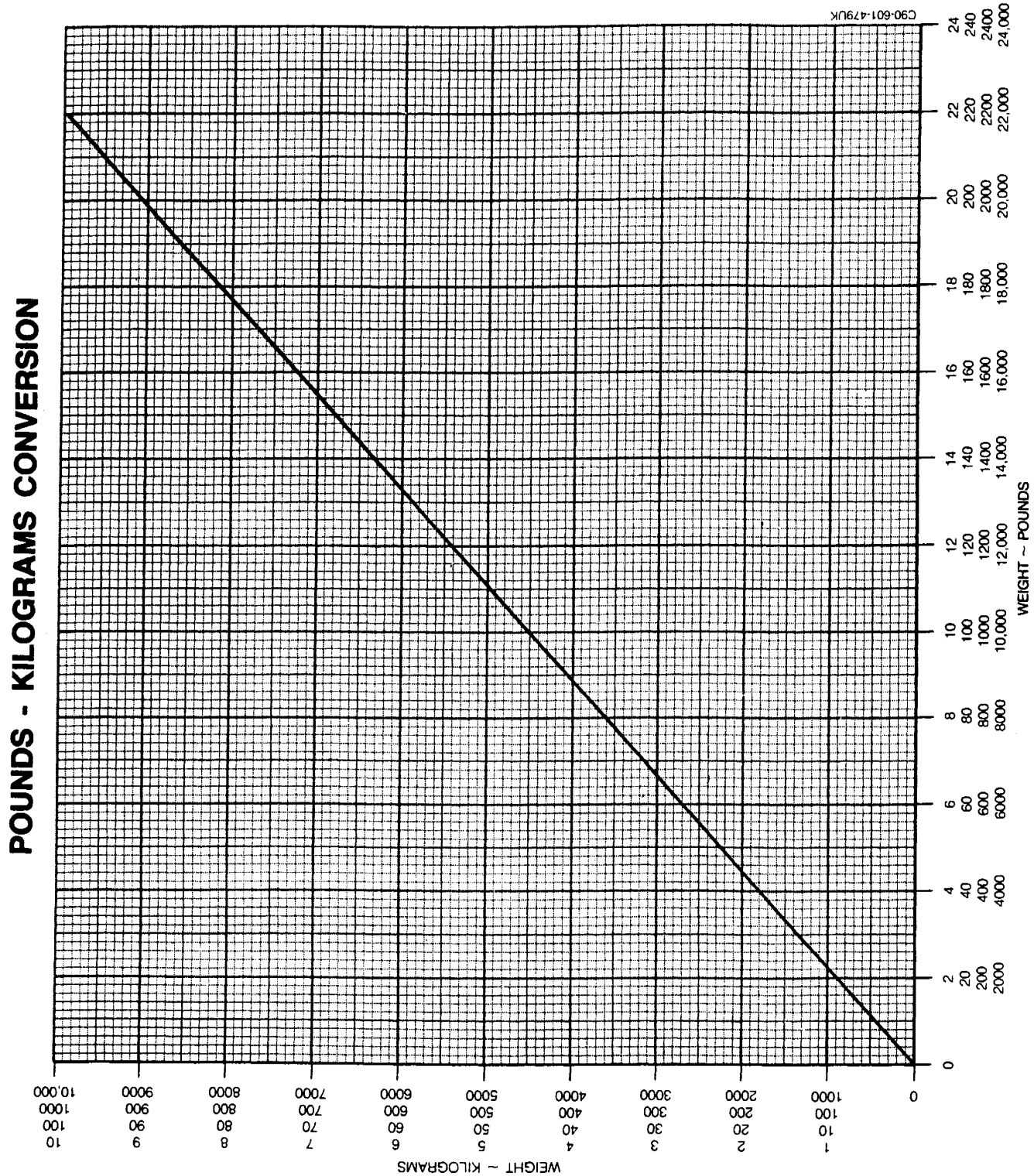
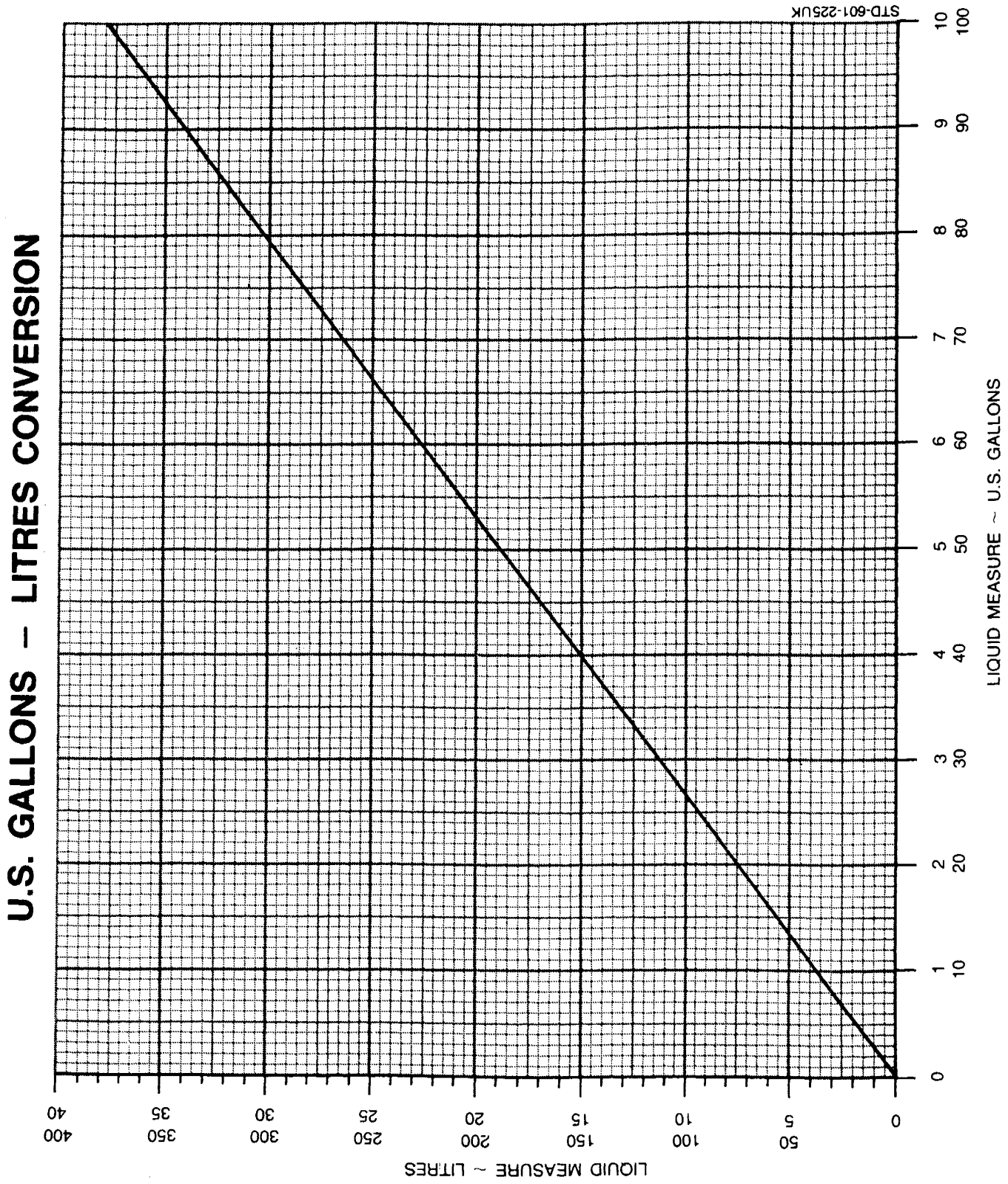

POUNDS - KILOGRAMS CONVERSION
C90-601-479UK
WEIGHT ~ POUNDS
0
2 20 200 2000
4 40 400 4000
6 60 600 6000
8 80 800 8000
10 100 1000 10,000
12 120 1200 12,000
14 140 1400 14,000
16 160 1600 16,000
18 180 1800 18,000
20 200 2000 20,000
22 220 2200 22,000
24 240 2400 24,000
WEIGHT ~ KILOGRAMS
0
1 10 100 1000
2 20 200 2000
3 30 300 3000
4 40 400 4000
5 50 500 5000
6 60 600 6000
7 70 700 7000
8 80 800 8000
9 90 900 9000
10 100 1000 10,000



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USABLE FUEL

Total of 453.0 (2059 Litres) IMP gallons usable. 321.4 (1461 Litres) IMP gallons usable in two main systems of 160.7 (730.5 Litres) IMP gallons each, and 131.6 (598 Litres) IMP gallons usable in two auxiliary tanks of 65.8 (299 Litres) IMP gallons each.

PRESSURIZATION SYSTEM

If the cabin pressure altitude should reach a value of 10,000 feet (± 500 feet), a pressure-sensing switch mounted on the forward pressure bulkhead will close. This causes the CABIN ALT caution light (amber) to illuminate, warning the pilot of approaching operation requiring the use of oxygen. A red CABIN ALT annunciator light illuminates at a cabin pressure altitude of 14,000 feet (+ 500 feet, −0 feet), indicating depressurization of the cabin.

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W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

Beechcraft®

Super King Air® B200, B200C, B200T & B200CT Landplanes

Pilot's Operating Handbook and

FAA Approved Airplane Flight Manual Supplement

for the

Auxiliary DC Bus

This Supplement is applicable to the following manual(s):

101-590010-147, 101-590037-29 & 101-590037-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual (POH/AFM) when the airplane has been modified by installation of the Auxiliary DC Bus in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

LIMITATIONS

Equipment connected to this bus is by specific customer request and does not satisfy minimum equipment requirements. No authorization is extended to permit any flight operations to be predicated soley upon the use of this system. Any use is limited to discretionary action by the pilot.

EMERGENCY PROCEDURES

No Change

ABNORMAL PROCEDURES

No Change

NORMAL PROCEDURES

1. AUX DC BUS Switch - ON (AUX BUS ON Annunciator will illuminate.)
2. Battery and Generator Switches - OFF

WARNING

The Auxiliary DC Bus should be used only when conservation of electric power is paramount. Use of this system will render most lights, electrically driven flight instruments, avionics, anti-ice, and pressurization systems inoperative and will necessitate manual extension of the landing gear. Prior to turning Battery and Generator Switches OFF, the pilot should consider existing flight conditions which may require the use of equipment not connected to the Auxiliary DC Bus to assure safe flight. If safe flight cannot be assured using only the auxiliary bus, the Battery Switch should be left ON and nonessential electric loads turned OFF individually.

CAUTION

Ensure that the AUX DC BUS Switch is OFF before leaving the airplane so a drain on the battery will not occur.

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PERFORMANCE

No Change

WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

The Auxiliary DC Bus provides electric power directly from the battery when selected by the pilot. Control of the system consists of one switch mounted in the master switch cluster on the pilot's subpanel and an annunciator light on the instrument panel. If auxiliary power is desired the switch can be turned ON, supplying electrical power directly from the battery to those systems connected to the bus. Systems tied to the Auxiliary DC Bus (listed below) can be readily identified by noting the placarding of the applicable circuit breaker. Activation of this system does not satisfy the procedures outlined in the POH/AFM and if circumstances warrant, the user of this manual must refer to the approved handbook for appropriate action.

The following listed equipment is connected to the auxiliary bus:

1. The Number 2 Communication System
2. The ILS portion of the Number 2 Navigation System
3. The Number 2 Compass System
4. The Number 2 Audio System
5. The Glareshield Flood Lights

HANDLING, SERVICING & MAINTENANCE

No Change

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BEECHCRAFT KING AIR C90A, F90 (LA-202, LA-205 and after)
AND
SUPER KING AIR B200, B200C, B200T and B200CT
LANDPLANES
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
COLLINS ANS-31C AREA NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Collins ANS-31C Area Navigation System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

The ANS-31C Quick Reference Guide, CPN 523-0771051-001117, dated 10/30/81 or later issue is available from Avionics Group, Rockwell International, Cedar Rapids, Iowa 52498. It is the responsibility of the owner/operator to maintain the above operator's manual in current status.

LIMITATIONS

1. Operation must be in conformity with the ANS-31C Quick Reference Guide CPN 523-0771051-001117, dated 10/30/81 or later issue.

2. The Area Navigation mode may not be used as a primary system under IFR conditions except on approved approach procedures, approved airways, and random area navigation routes when approved by Air Traffic Control.

3. The Area Navigation mode can only be used with colocated facilities. (VOR and DME signals originate from the same geographical location.)

4. An Area Navigation installation located on the right instrument panel may be used for primary navigation only if a qualified pilot occupies the right seat.

5. Operate the ANS-31C system in the VOR mode when the system is used during takeoff.

EMERGENCY PROCEDURES

Refer to ANS-31C Quick Reference Guide, CPN 523-0771051-001117, dated 10/30/81 or later issue.

NORMAL PROCEDURES

Refer to ANS-31C Quick Reference Guide, CPN 523-0771051-001117, dated 10/30/81 or later issue.

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RNAV PRGM ANNUNCIATOR/SWITCH (If installed)

Use of this switch allows the operator to program the RNAV in flight with minimum interruption of existing navigation commands.

Rotate the RNAV mode switch to ON then press face of RNAV PRGM annunciator/switch. Enter data into RNAV control display unit (CDU). When desired, the RNAV may be utilized by pressing the RNAV PRGM switch then pressing the CDU USE button.

NOTE

Navigation commands to the flight control system will be momentarily interrupted after first switch action and prior to second switch action for both operations.

PERFORMANCE - No Change

WEIGHT & BALANCE - No Change

Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: September, 1984
P/N 101-590010-193

For Training Purposes Only

BEECHCRAFT SUPER KING AIR
B200 LANDPLANES (BB-1063, BB-1066 AND BB-1077 ONLY)
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
AUXILIARY ELECTRIC POWER BUS

GENERAL

The Information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary Electric Power Bus in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Equipment connected to this bus is by specific customer request and does not satisfy minimum equipment requirements. No authorization is extended to permit any flight operations to be predicated soley upon the use of this system. Any use is limited to discretionary action by the pilot.

EMERGENCY PROCEDURES - No change.

NORMAL PROCEDURES

1. AUX PWR switch - ON (AUX PWR ON Annunciator will illuminate.)
2. Battery and Generator switches - OFF

WARNING

The auxiliary electric power bus should be used only when conservation of electric power is paramount. Use of this system will render most lights, electrically driven flight instruments, avionics, anti-ice, and pressurization systems inoperative and will necessitate manual extension of the landing gear. Prior to turning Battery and Generator switches OFF, the pilot should consider existing flight conditions which may require the use of equipment not connected to the auxiliary power bus to assure safe flight. If safe flight cannot be assured using only the auxiliary bus, the Battery switch should be left on and nonessential electric loads turned OFF individually.

CAUTION

Ensure that the Auxiliary Power switch is OFF before leaving the airplane so a drain on the battery will not occur.

PERFORMANCE - No change

FAA Approved
Revised: July, 1982
P/N 101-590010-195

For Training Purposes Only

SYSTEM DESCRIPTION

The auxiliary electric power bus provides electric power from an auxiliary battery when selected by the pilot. Control of the system consists of one switch mounted in the master switch cluster on the pilot's subpanel and an annunciator light on the instrument panel. If auxiliary power is desired, the switch can be turned ON, supplying electrical power directly from the battery to those systems connected to the bus. Systems tied to the auxiliary electric power bus (listed below) can be readily identified by noting the placarding of the applicable circuit breaker. Activation of this system does not satisfy the procedures outlined in the Pilot's Operating Handbook and if circumstances warrant, the user of this manual must refer to the Approved Handbook for appropriate action.

The auxiliary battery is maintained in a state of full charge by the airplane electric system during normal flight conditions. Illumination of the AUX TEST annunciator with the STB HORIZ PWR switch in the TEST position indicates an adequate voltage level. For maintenance testing, which should be performed periodically to assure proper operation, refer to J.E.T. Technical Publication P/N TP-336, revised February, 1980 or later. The manual may be obtained from: JET Electronic and Technology INC., 5353 52nd Street, Grand Rapids, Michigan 49508.

The following listed equipment is connected to the auxiliary bus:

(1) The Number 2 Communication system (Including Audio)
(2) The Number 2 Navigation system
(3) The Number 2 Transponder
(4) The Number 2 Encoding Atimeter
(5) The Glareshield Floodlights.

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Revised: July, 1982
P/N 101-590010-195

For Training Purposes Only

BEECHCRAFT SUPER KING AIR
B200 LANDPLANE (BB-1074 AND BB-1075 ONLY)
PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the AUXILIARY ELECTRIC POWER BUS

GENERAL

The Information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary Electric Power Bus in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Equipment connected to this bus is by specific customer request and does not satisfy minimum equipment requirements. No authorization is extended to permit any flight operations to be predicated solely upon the use of this system. Any use is limited to discretionary action by the pilot.

EMERGENCY PROCEDURES - No change.

NORMAL PROCEDURES

1. AUX PWR switch - ON (AUX PWR ON Annunciator will illuminate.)
2. Battery and Generator switches - OFF

WARNING

The auxiliary electric power bus should be used only when conservation of electric power is paramount. Use of this system will render most lights, electrically driven flight instruments, avionics, anti-ice, and pressurization systems inoperative and will necessitate manual extension of the landing gear. Prior to turning Battery and Generator switches OFF, the pilot should consider existing flight conditions which may require the use of equipment not connected to the auxiliary power bus to assure safe flight. If safe flight cannot be assured using only the auxiliary bus, the Battery switch should be left on and nonessential electric loads turned OFF individually.

CAUTION

Ensure that the Auxiliary Power switch is OFF before leaving the airplane so a drain on the battery will not occur.

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For Training Purposes Only

PERFORMANCE - No change

SYSTEM DESCRIPTION

The auxiliary electric power bus provides electric power from an auxiliary battery when selected by the pilot. Control of the system consists of one switch mounted in the master switch cluster on the pilot's subpanel and an annunciator light on the instrument panel. If auxiliary power is desired, the switch can be turned ON, supplying electrical power directly from the battery to those systems connected to the bus. Systems tied to the auxiliary electric power bus (listed below) can be readily identified by noting the placarding of the applicable circuit breaker. Activation of this system does not satisfy the procedures outlined in the Pilot's Operating Handbook and if circumstances warrant, the user of this manual must refer to the Approved Handbook for appropriate action.

The auxiliary battery is maintained in a state of full charge by the airplane electric system during normal flight conditions. Illumination of the AUX TEST annunciator with the STB HORIZ PWR switch in the TEST position indicates an adequate voltage level. For maintenance testing, which should be performed periodically to assure proper operation, refer to J.E.T. Technical Publication P/N TP-336, revised February, 1980 or later. The manual may be obtained from: JET Electronic and Technology INC., 5353 52nd Street, Grand Rapids, Michigan 49508.

The following listed equipment is connected to the auxiliary bus:

(1) The Number 2 Communication system (Including Audio)
(2) The Number 2 Navigation system
(3) Electrical Vibrator for the Number 2 Encoding Altimeter
(4) The Glareshield Floodlights.

Approved: Donald St Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Issued: June, 1982
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For Training Purposes Only

BEECHCRAFT SUPER KING AIR B200C LANDPLANE (BL-53 ONLY)
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
DELCO CAROUSEL IV E INERTIAL NAVIGATION SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Delco Electronics Carousel IV E Inertial Navigation system in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

The Delco Carousel IV E INS Operator's Manual, dated June 1, 1978 or later, is available from Delco Electronics, General Motors Corporation, Santa Barbara Operations, Santa Barbara, California. It is the responsibility of the owner/operator to maintain the appropriate operator's manual in current status.

LIMITATIONS

The Delco Carousel IV E INS may be used as an additional means of enroute navigation during VFR/IFR conditions provided the following limitations are observed:

1. Operation must be in conformity with the Delco Carousel IV E INS Operator's Manual, dated June 1, 1978 or later.

2. The INS is approved for enroute operation within conterminous United States and Alaska in accordance with the enroute criteria of AC 90-45A.

3. Prior to RNAV operation, the flight crew must verify satisfactory TACAN update accuracy.

4. The INS must not be coupled to the Flight Director or Autopilot during takeoff, approach or landing.

5. Automatic alignment and calibration of the INS can be performed only while the airplane is parked. Alignment degradation may occur if alignment is performed at latitudes greater than 80°.

EMERGENCY PROCEDURES

Refer to Delco Carousel IV E Operator's Manual, dated June 1, 1978 or later.

NORMAL PROCEDURES

Refer to Delco Carousel IV E Operator's Manual, dated June 1, 1978 or later.

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For Training Purposes Only

To accomplish a system update using the TACAN navigation receiver, complete the following steps:

INSERTION OF TACAN STATION DATA

1. WYPT - Set selector
2. "7" & "9" - Press simultaneously; now flashing in FROM TO window
3. THUMBWHEEL - Select number to represent TACAN station, NOT 0
4. LATITUDE - Load
5. INSERT/ADV - Press
6. LONGITUDE - Load
7. INSERT/ADV - Press (light goes out)
8. INSERT/ADV - Press (R.H. displays last previously inserted altitude)
9. "2" or "8" - Press (indicates next entry is altitude)
10. ALTITUDE - Load

NOTE

Round off to nearest thousand (e.g. 2000 is N2) Defaults to 0.

11. INSERT/ADV - Press (light goes out)
12. INSERT/ADV - Press (L.H. displays last previously inserted channel no.)
13. "2" or "8" - Press (indicates next entry is station channel number)
14. CHANNEL NO. - Load

NOTE

Channel # is not used by the INS, therefore load is not required, except for cross-reference when tuning the TACAN.

15. INSERT/ADV - Press (light goes out)
16. INSERT/ADV - Press (station lat-long reappear)
17. Repeat steps 3 thru 15 for each station
18. To return INS to normal operation momentarily set data selector to POS

AIDED TACAN OPERATION

1. Insure INS is in NAV mode
2. Tune TACAN to station desired for NAV UPDATE

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For Training Purposes Only

3. WYPT - Set selector
4. "7" & "9" - Press simultaneously
5. WAYPT CHG - Press
6. Select number corresponding to desired TACAN station
7. INSERT/ADV - Press (number should be flashing)
8. DIS/TIME - Set selector
9. Verify distance displayed on INS agrees with distance on DME
10. DSRTK/STS - Set selector
11. "4" - Press for TACAN position update
12. INSERT/ADV - Press (insert light goes out, display reverts to normal DSRTK/STS display)
13. Verify that TACAN UPDATE annunciator is illuminated

    With the data selector in DSRTK/STS, and the normal display selected, the second digit (relative performance indicator) will be decremented gradually during TACAN mixing. The rate at which it is decremented is a function of both time and TACAN station geometry. In single TACAN a "1" in this digit indicates complete updating has been accomplished. (This usually takes about ten minutes with good station geometry.)

14. To discontinue TACAN MIX:
    a. Select station "0" or
    b. Rotate selector to DSRTK/STS and press "5" INSERT/ADV

INS AUTOMATIC FLIGHT CONTROL SYSTEM INTERFACE

When tracking an INS course with the AFCS, utilize the heading mode to accomplish course changes greater than 25°. The enroute navigation mode may then be re-engaged to capture and track the new course.

Engagement of INS mode for Flight Director presents INS course and distance information on pilot's HSI.

PERFORMANCE

No Change

WEIGHT & BALANCE

No Change

Approved: Donald H. Carter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR
B200 LANDPLANES (BB-1089 ONLY)
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
AUXILIARY DC BUS

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary DC Bus in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

Equipment connected to this bus is by specific customer request and does not satisfy minimum equipment requirements. No authorization is extended to permit any flight operations to be predicated solely upon the use of this system. Any use is limited to discretionary action by the pilot.

EMERGENCY PROCEDURES - No change.

NORMAL PROCEDURES

1. AUX DC BUS Switch - ON (AUX BUS ON Annunciator will illuminate.)
2. Battery and Generator Switches - OFF

WARNING

The Auxiliary DC Bus should be used only when conservation of electric power is paramount. Use of this system will render most lights, electrically driven flight instruments, avionics, anti-ice, and pressurization systems inoperative and will necessitate manual extension of the landing gear. Prior to turning Battery and Generator Switches OFF, the pilot should consider existing flight conditions which may require the use of equipment not connected to the Auxiliary DC Bus to assure safe flight. If safe flight cannot be assured using only the auxiliary bus, the Battery Switch should be left on and nonessential electric loads turned OFF individually.

CAUTION

Ensure that the AUX DC BUS Switch is OFF before leaving the airplane so a drain on the battery will not occur.

PERFORMANCE - No change.

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For Training Purposes Only

SYSTEMS DESCRIPTION

The Auxiliary DC Bus provides electric power directly from the battery when selected by the pilot. Control of the system consists of one switch mounted in the master switch cluster on the pilot's subpanel and an annunciator light on the instrument panel. If auxiliary power is desired the switch can be turned ON, supplying electrical power directly from the battery to those systems connected to the bus. Systems tied to the Auxiliary DC Bus (listed below) can be readily identified by noting the placarding of the applicable circuit breaker. Activation of this system does not satisfy the procedures outlined in the Pilot's Operating Handbook and if circumstances warrant, the user of this manual must refer to the approved handbook for appropriate action.

The following listed equipment is connected to the auxiliary bus:

(1) The Number 2 Communication System
(2) The ILS portion of the Number 2 Navigation System
(3) The Number 2 Audio System
(4) The Glareshield Flood Lights

Approved: Donald H Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR B200C, (BL-53 ONLY)
LANDPLANE
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE MANUAL SUPPLEMENT
for the
TELEDYNE TDL-424 LORAN NAVIGATOR

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Teledyne TDL-424 Loran Navigator in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

The Teledyne TDL-424 Operator's Manual, dated February 15, 1978, or later issue is available from Teledyne Systems Company, Radio Navigation Systems, 19601 Nordhoff Street, Northridge, California 91324. It is the responsibility of the owner/operator to maintain the above operator's manual in current status.

LIMITATIONS

The TDL-424 Loran may be used as an additional means of enroute navigation during VFR conditions provided the following limitations are observed:

1. Operation must be in conformity with Teledyne TDL-424 Loran Navigator Operator's Manual, dated February 15, 1978 or later issue.

2. Only enroute operations in the conterminous United States and Alaska are approved.

3. The TDL-424 front panel (display and control head) must be placed in the center console.

EMERGENCY PROCEDURES

Refer to Teledyne TDL-424 Loran Navigator Operator's Manual, dated February 15, 1978, or later issued.

NORMAL PROCEDURES

Refer to Teledyne TDL-424 Loran Navigator Operator's Manual, dated February 15, 1978, or later issued.

LORAN AUTOMATIC FLIGHT CONTROL SYSTEM INTERFACE

Desired track information must be manually set on the HSI for each navigation leg. Course information only is presented on the HSI. DME information is unaffected by selecting LORAN on the Flight Director.

HSI presentation provides course width of ± 6 NM full scale in NAV ARM mode, all other modes are ± 0.5 NM.

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P/N 101-590010-203

For Training Purposes Only

PERFORMANCE - No change

WEIGHT & BALANCE - No change

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
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P/N 101-590010-203

For Training Purposes Only

BEECHCRAFT SUPER KING AIR

200, B200, B200C, B200T, and B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE

FLIGHT MANUAL SUPPLEMENT

for the

SPERRY SPZ-4000 AUTOMATIC FLIGHT CONTROL SYSTEM

CATEGORY 1

and

SPERRY EDZ-600/EDZ-800/EDZ-601/EDZ-801/EDZ-603/EDZ-803

ELECTRONIC FLIGHT INSTRUMENT SYSTEM (EFIS)

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK
AND FAA APPROVED AIRPLANE FLIGHT MANUALS
P/N 101-590010-127
P/N 101-590010-147
P/N 101-590037-29

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
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For Training Purposes Only

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Sperry SPZ-4000 Automatic Flight Control System, alone or in conjunction with the Sperry Electronic Flight Instrument System, in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

Sperry Pilot's Operating Handbooks for the Electronic Flight Instrument System are available from Sperry Flight Systems, P.O. Box 29000, Phoenix, Arizona 85038. It is the responsibility of the owner/operator to maintain the appropriate Pilot's Operating Handbook in current status.

The appropriate manual depending on system(s) installed is specified below:

— Single system (pilot only, EDZ-600 or EDZ-800): Handbook No. 62-0052-00-00 dated September, 1983 or later.
— Single system (pilot only EDZ-601 or EDZ-801): Handbook No. 62-0058-00-00 dated March, 1984 or later.
— Dual system (pilot and copilot, EDZ-600 or EDZ-800): Handbook No. 62-0036-00-03 dated September, 1983 or later.
— Dual system (pilot and copilot, EDZ-601 or EDZ-801): Handbook No. 62-0056-00-01 dated January, 1984 or later.
— Single and dual systems (EDZ-603 or EDZ-803) without the Multifunction Display: Handbook No. A62-0080-00-00, dated April, 1986 or later.
— Single and dual systems (EDZ-603 or EDZ-803) with the Multifunction Display: Handbook No. A62-0083-00-00, dated December, 1986 or later.

LIMITATIONS

1. During autopilot operations, pilot must be seated at the controls with seat belt fastened.
2. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
3. Do not use autopilot under 200 feet above the terrain. This includes takeoffs and landings.
4. Do not use autopilot or yaw damper during takeoff or landing.
5. Do not use propeller in the range of 1750-1850 rpm during coupled ILS approach.
6. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
7. The accuracy of the VNAV system has not been shown to meet the requirements of Advisory Circular 90-45A and is therefore not to be used as the primary source of vertical guidance information. VNAV is to be used only when proceeding directly toward or away from a VORTAC or RNAV waypoint.
8. The "T-bar" airspeed display (if installed) is not to be used as a primary source of airspeed information.
9. Operation of the EFIS must be in conformity with the applicable Sperry Pilot's Operating Handbook.

EMERGENCY PROCEDURES

AIR START

STARTER ASSIST

CAUTION

The pilot should determine the reason for engine failure before attempting an air start.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an over-temp condition.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. EFIS ON/OFF Switch(es) (if installed) - OFF (if conditions permit)
5. Windshield Heat - OFF
6. Power Lever - IDLE
7. Condition Lever - CUT-OFF
8. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

9. Ignition and Start Switch - ON (up), Check IGNITION Annunciator - ON
10. Condition Lever - LOW IDLE
11. Ignition and Start Switch - OFF (N_1 above 50%)
12. Propeller Lever - AS REQUIRED
13. Power Lever - AS REQUIRED
14. Generator - RESET, then ON
15. Engine Auto Ignition - ARM
16. EFIS ON/OFF Switch(es) (if installed) - ON
17. Electrical Equipment - AS REQUIRED

AUTOPILOT MALFUNCTION

If the AP TRIM FAIL and MASTER WARNING annunciators illuminate while the autopilot is engaged, immediately disconnect while restraining the control wheel for a possible out-of-trim force. Disengagement can be accomplished by any one of the following:

1. Press (momentarily) to the first level the pilot's or copilot's AP/YD & TRIM DISC button on the control wheel.
2. Opening the autopilot (AP CONTROL) circuit breaker.
3. If necessary, the autopilot may be overpowered by either pilot.

MAXIMUM ALTITUDE LOSSES DURING MALFUNCTION TESTS WERE:

CONFIGURATION	ALTLITUDE LOSS
Climb	130 ft
Cruise	310 ft
Maneuvering	140 ft
Descent	510 ft
Approach/ILS Coupled	90 ft
One-engine-inoperative Approach/ILS Coupled	80 ft

ENGINE FAILURE

— Disengage the autopilot, retrim the airplane and reengage the autopilot. Maintain at least 120 knots for one-engine-inoperative approach.

EFIS FAILURE

If the EFIS displays are missing or abnormal, refer to alternate instrument for usable data. The copilot instruments are approved as backup to the pilot instruments. EFIS reversionary modes may be selected as follows:

1. In the event of an Electronic Horizontal Situation Indicator (EHSI) failure:

   — Press the REV button on the Display Controller. The EADI will provide a composite attitude/navigation display.

   or

   — If a Multifunction Display is installed, select HSI on the Mode Controller. The MFD will display HSI information.

2. In the event of an Electronic Attitude-Director Indicator (EADI) failure, press the REV button twice. The EHSI will provide a composite attitude/navigation display.

3. In the event of primary symbol generator failure (both EADI and EHSI go blank or annunciates SG FAIL):

   a. If an auxilliary (backup) symbol generator is installed (two tube system), press the AUX DRIVE button on the pilot's instrument panel. The standby symbol generator will drive the EADI and EHSI.
   b. If a dual EFIS system is installed, press the failed side REV button three times. Cross side information will be displayed.
   c. If a Multifunction Display is installed, select the failed to SG position on the MFD controller. The Multifunction generator will drive the failed side EFIS and the MFD will blank.

4. Use flight director for instrument approaches in the composite mode.

NOTE

The composite format utilizes a course deviation indicator with automatic reverse course steering. Always fly toward the needle to intercept course centerline.

NORMAL PROCEDURES

The Automatic Flight Control System (AFCS) is an integrated Flight Director, Autopilot, and Yaw Damper System. Flight Director modes are selected using push button switches on the Flight Director Mode Selector. The Autopilot and Yaw Damper are engaged using push button switches on the Autopilot Controller. Whenever the autopilot is engaged, it will automatically couple to the selected flight director mode. The autopilot may also be controlled manually using the Pitch Wheel and Turn Knob on the controller.

BEFORE ENGINE STARTING

EFIS (if installed)

— EFIS ON/OFF Switch(es) (if installed) - OFF

AFTER STARTING, AND TAXIING

EFIS (if installed)

— EFIS ON/OFF Switch(es) (if installed) - ON

Altitude Alert

1. Altitude Alert Controller (or VNAV if installed) — Set to 1500 feet above altitude indicated on pilot's altimeter.
2. Pilot's baro set — ADJUST to increase altitude toward alerter setting.
3. VERIFY — alert light on pilot's altimeter illuminates and Aural Alert sounds at 1000 feet from alerter setting.
4. VERIFY — alert light is extinguished at 250 feet from alerter setting.
5. Pilot's baro set — ADJUST toward original setting.
6. VERIFY — alert light illuminates and horn sounds at 250 feet from alerter setting.
7. Altitude Alert Controller (or VNAV if installed) — SET to initial clearance altitude.
8. VERIFY — alert light extinguished.

Autopilot

Perform prior to each flight to verify proper operation of torque limiters, torque monitors, autopilot electric trim system, and failure annunciators.

1. Autopilot Controller TRIM UP, TRIM DN Annunciators — CHECK not illuminated. A steady illumination of TRIM UP or TRIM DN annunciator indicates that automatic synchronization is not functioning and autopilot should not be engaged.
2. Turn Knob — IN CENTER DETENT POSITION
3. Elevator Trim Control Switch — ON
4. Control Wheel — TO MID TRAVEL
5. Autopilot Controller AP Button — PRESS. AP ENGAGE and YD ENGAGE annunciators on autopilot controller will flash. Servo clutches will engage. FD flag on ADI/EADI in view.
6. Control Movement — CHECK when pushed ON.

    a. Rudder Pedals — OVERPOWER SLOWLY. YD ENGAGE annunciator stops flashing.
    b. Control Wheel — OVERPOWER SLOWLY in both pitch and roll axes. AP ENGAGE annunciator stops flashing. FD flag retracts.

WARNING

If autopilot or yaw damper disengages during overpower test, DO NOT USE.
If AP ENGAGE or YD ENGAGE annunciator continues to flash, DO NOT USE.

7. Elevator Trim Follow-up — CHECK

    a. Control Wheel — HOLD AFT OF MID TRAVEL. Trim wheel should run nose down after approximately 3 seconds. Trim down annunciator should illuminate after approximately 8 seconds.
    b. Control Wheel — HOLD FORWARD OF MID TRAVEL. Trim wheel should run nose up after approximately 3 seconds, TRIM UP annunciator should illuminate after approximatley 8 seconds, and AP TRIM FAIL annunciator and MASTER WARNING flasher should illuminate after approximately 15 seconds.

8. AP/YD & TRIM DISC Button — DEPRESS THROUGH SECOND LEVEL. Autopilot and yaw damper should disengage and ELECT TRIM OFF annunciator should illuminate. AP ENGAGE and YD ENGAGE annunciators on instrument panel should flash 5 times and autopilot off aural alert should sound for one second (or Voice Advisory System, if installed, should announce "AUTOPILOT").
9. MASTER WARNING Flasher — PRESS TO RESET.
10. Elevator Trim Control Switch — OFF, then ON (resets electric trim; ELECT TRIM OFF annunciator should extinguish)
11. Elevator Trim — SET AS REQUIRED.

NOTE

Autopilot Preflight Check is disabled during inflight operations.

EFIS (if installed)

1. SELF TEST

Press the "TEST" button on the Display Controller. Verify the following:

- Radar altimeter reads test value.
- Course Select, Heading Select, DH Set and ET are replaced with dashes.
- All flags in view.
- Command cue out of view.
- Optional Comparison Monitor annunciates COMPARE (EDZ-800) or CHECK (EDZ-600) on the EADI with ATT, HDG and ILS annunciated as installed.

NOTE

Localizer frequency must be tuned on both NAV receivers to annunciate ILS.

- For EDZ-600/EDZ-800
  If Flight Director Lamp test is good, "TEST PASS" will appear in upper right corner of EADI. "TEST FAIL" indicates a problem in the EFIS components or interconnection of the flight director computer.
- For EDZ-601/EDZ-801/EDZ-603/EDZ-803
  If Flight Director Lamp test is good. "PASS" will appear in left center of EADI. "FD FAIL" in the left center of the EADI indicates a problem in the EFIS components or interconnection of the flight director computer.
- Self-test in the composite mode causes the same results as previously noted. In addition (EDZ-601/EDZ-801/EDZ-603/EDZ-803) the digital heading readout is replaced with a red "FAIL" and the expanded localizer scale and pointer are removed.

NOTE

Self-test is inhibited during glideslope capture.

2. SWITCHING TEST
   Check each switch function on the Display Controller and Source Controller. Switching should operate as described under Item 17 of GENERAL OPERATION.

GENERAL OPERATION

All flight director modes (except Go-Around) are selected by push button switches on the Flight Director Mode Selector. Go-Around is selected by pressing the Go-Around button on the left engine power lever. Annunciation for the mode selected is provided on the mode selector and pilot's instrument panel.

The Electronic Flight Instrument System, if installed, replaces the conventional electro-mechanical Attitude-Director (ADI) and Horizontal Situation (HSI) indicators. The CRT displays (EADI & EHSI) provide the necessary information for the pilot to respond to flight director and navigation commands in the same manner as conventional instruments. The EDZ-600/EDZ-601/EDZ-603 utilizes standard (4.6" x 5") displays and the EDZ-800/EDZ-801/EDZ-803 utilizes large format (5" x 6") displays. The system includes additional features which allow the pilot to select alternate formats as navigation aids or in the event of specific failures. The operator is cautioned that the primary displays (attitude and directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate formats are described in the appropriate Sperry Pilot's Operating Handbooks. The owner/operator of the airplane is responsible for obtaining and maintaining the appropriate handbook.

1. Standby Lamp Test (STBY) Mode selector button — PRESS. All flight director modes are reset and all flight control system annunciators are illuminated. Flight director command cue is biased out of view and FD flag on ADI/EADI is in view as long as STBY is pressed. If STBY is pressed for more than 5 seconds:

   a. AP TRIM FAIL and MASTER WARNING will illuminate.
   b. Autopilot off aural alert will sound (or Voice Advisory System, if installed, will announce "AUTOPILOT").

2. Heading Select Mode (HDG)

   a. Heading Cursor — SET desired heading on HSI/EHSI.
   b. Mode Selector HDG button — PRESS. Flight director commands a turn to capture and hold selected heading.

3. VOR/RNAV Enroute Mode (NAV)

   a. Navigation Receiver — TUNE to desired VOR frequency.
   b. Course Selector — SET desired course on HSI/EHSI.
   c. Heading Cursor — SET desired intercept heading on HSI/EHSI.
   d. Mode Selector NAV button — PRESS. Mode selector annunciates (NAV) ARM (white NAV on EADI) and HDG mode is automatically selected if airplane is outside computed capture point.
   e. At the computed capture point — HDG mode is cancelled, mode selector annunciates (NAV) CAP (green NAV on EADI). Flight director commands a turn to capture and track selected course.

NOTE

At transition from armed to captured flight director modes, a white box appears for five seconds around the green capture mode annunciators on the EADI.

4. VOR/RNAV Approach Mode (NAV/APR) for optimized NAV mode tracking, for VOR/RNAV approaches.

   a. Navigation Receiver — TUNE to desired VOR frequency.
   b. Course Selector — SET desired course on HSI/EHSI.
   c. Heading Cursor — SET desired intercept heading on HSI/EHSI.
   d. Mode Selector APR button — PRESS. Mode selector annunciates (NAV) ARM and (APR) ARM. VOR APR is annunciated on the instrument panel (white VAPP on the EADI) and HDG mode is automatically selected if airplane is outside computed capture point.
   e. At computed capture point — HDG mode is cancelled and mode selector annunciates (NAV) CAP and (APR) CAP (green VAPP on EADI). Flight director commands a turn to capture and track selected course.

5. Localizer Approach (NAV)

   a. Navigation Receiver — TUNE to desired localizer frequency.
   b. Course Selector — SET inbound localizer course on HSI/EHSI.
   c. Heading Cursor — SET desired intercept heading on HSI/EHSI.
   d. Mode Selector NAV Button — PRESS. Mode selector annunciates (NAV) ARM (white LOC on EADI), and HDG mode is automatically selected if airplane is outside computed capture point.
   e. At computed capture point — HDG mode is cancelled. Mode selector annunciates (NAV) CAP (green LOC on EADI). Flight director commands a turn to capture and track localizer beam.
   f. Mode Selector — SELECT vertical flight director mode, as required.

6. Back Course Approach (BC)

   a. Navigation Receiver — TUNE to desired localizer frequency.
   b. Course Selector — SET inbound front course on HSI/EHSI.
   c. Heading Cursor — SET desired intercept heading on HSI/EHSI.
   d. Mode Selector BC Button — PRESS. Mode selector annunciates (BC) ARM (white BC on EADI) and HDG mode is automatically selected if airplane is outside computed capture point.
   e. At computed capture point HDG mode is cancelled. Mode Selector annunciates (BC) CAP (green BC on EADI). Flight director commands a turn to capture and track localizer back course.
   f. Mode Selector — SELECT vertical flight director mode, as required.

7. ILS Approach (APR)

   a. Navigation Receiver — TUNE to desired ILS frequency.
   b. Course Selector — SET inbound localizer course on HSI/EHSI.
   c. Heading Cursor — SET desired intercept heading HSI/EHSI.
   d. Mode Selector APR Button — PRESS. Mode selector annunciates (NAV) ARM and (APR) ARM (white LOC and GS on EADI) and HDG mode is automatically selected if airplane is outside computed localizer capture point.
   e. Mode Selector — SELECT vertical flight director mode, as required, to intercept glideslope.
   f. At computed localizer capture point — HDG mode is cancelled. Mode selector annunciates (NAV) CAP and (APR) ARM (green LOC on EADI). Flight director commands a turn to capture and track localizer beam.

g. At computed glideslope capture point — Flight Director Vertical modes are cancelled (except ALT SEL ARM). Mode selector annunciates (NAV) CAP and (APR) CAP (green LOC and GS on EADI). Flight director commands a pitch change to capture and track glideslope.

NOTE

Glideslope capture will not occur until after localizer has been captured. Pitch wheel is disabled while in glideslope (APR CAP) mode.

8. Pitch Hold/Sync Mode
With no vertical flight director modes selected:

   a. Heading Cursor — SET to desired heading.
   b. Mode Selector — SELECT lateral flight director mode as required. Flight director pitch command is to hold attitude existing when lateral mode was selected.

   To set/change command pitch attitude:

   a. TCS Switch — PRESS and HOLD
   b. Maneuver airplane to desired pitch attitude. Trim elevator as required.
   c. TCS Switch — RELEASE. Flight director command is to hold new pitch attitude.

9. Altitude Hold Mode (ALT)
To hold altitude:

   a. Maneuver airplane to desired altitude.
   b. Mode Selector ALT Button — PRESS.

   To change altitude hold reference:

   a. TCS Switch — PRESS and HOLD
   b. Maneuver airplane to new altitude.
   c. TCS Switch — RELEASE. Flight director command is to hold new altitude.

NOTE

Use ALT SEL mode for large altitude changes.

10. Altitude Alerting
The desired altitude is selected on the Altitude Alert controller (or the ALT PRE position on the VNAV controller, if installed) using the SET knob. Then altitude can be attained by using the altitude preselect mode of the flight director.

11. Altitude Preselect Mode (ALT SEL)

   a. Altitude Preselect (or VNAV) Controller — SET desired altitude (ALT PRE position on VNAV).
   b. Mode Selector ALT SEL Button — PRESS. Mode Selector annunciates (ALT SEL) ARM (white ASEL on EADI).
   c. Maneuver airplane to intercept selected altitude. (Use Pitch Hold, IAS, VS or GA mode, as required.)
   d. Altitude Capture Sequence — VERIFY:

      1) 1000 feet prior to selected altitude — alert light on pilot's altimeter illuminates and aural alert sounds for one second.
      2) 250 feet prior to selected altitude — alert light extinguishes.
      3) At computed altitude capture point — mode selector annunciates (ALT SEL) CAP (green ASEL on EADI). Other vertical flight director modes are cancelled. Flight director commands a flare to selected altitude.
      4) At selected altitude — Altitude Select Mode is cancelled, and Altitude Hold Mode is automatically selected (green ALT is annunciated on EADI).

NOTE

Changing selected altitude on Altitude Preselect (or VNAV) Controller during altitude capture (ALT SEL CAP) will automatically cancel ALT SEL CAP, and ALT SEL ARM will be selected.

ALT SEL mode is referenced to baro corrected altitude displayed on pilot's altimeter.

12. Airspeed Hold Mode (IAS)
To hold airspeed:

    a. Maneuver airplane to desired indicated airspeed.
    b. Mode Selector IAS Button — PRESS.

    To change airspeed hold reference:

    a. TCS Switch — PRESS and HOLD
    b. Maneuver airplane to new airspeed.
    c. TCS Switch — RELEASE. Flight director command is to hold new indicated airspeed.

13. Vertical Speed Hold Mode (VS)
Operation of VS mode is identical to IAS mode except that flight director command is to maintain vertical speed.

14. Go-Around Mode (GA)
Go-Around Button (on left power lever) — PRESS. All flight director modes are cancelled. Flight director commands a 7-degree-pitch-up, wings-level attitude, then Heading Hold is commanded when roll attitude becomes less than 6 degrees (green GA is annunciated on EADI).
To cancel heading hold:

    a. Heading Cursor — SET desired heading on HSI.
    b. Mode Selector — SELECT desired lateral mode.

    To cancel go-around mode:

    a. Mode Selector — SELECT vertical flight director mode (ALT, IAS, VS)
    OR;
    b. TCS Switch — PRESS
    OR;
    c. Autopilot — ENGAGE

15. Vertical Navigation Mode (VNAV) (Optional)
The VNAV mode operates in conjunction with an optional Dual Waypoint VNAV Controller.
Normal VNAV (Two independent waypoints) (See illustration).

    a. Navigation Receiver — TUNE to desired VORTAC frequency.
    b. DME Receiver — CHECK, not in HOLD.
    c. WPT Select Button — PRESS to select WPT 1 or WPT 2 as desired.
    d. Course Selector — SET DESIRED COURSE.
    e. VNAV DIsplay Selector Switch — SET to WPT ALT. SET desired waypoint altitude using SET knob.
    f. VNAV Display Selector Switch — SET to STA EL. SET VORTAC station elevation using SET knob.
    g. VNAV Display Selector Switch — SET to TO or FROM, as required. SET distance (alongtrack offset) that the waypoint is ahead of (TO) or beyond (FROM) the VORTAC.
    h. VNAV Display Selector Switch — SET to VANG. SELECT desired vertical angle for climb or descent by:

        1) OBSERVE angle displayed on VNAV Controller. Angle will increase as airplane approaches the waypoint.
        2) Mode Selector VNAV Button — PRESS when desired angle is displayed. Mode selector annunciates (VNAV) ARM (white VNAV on EADI) momentarily, then (VNAV) CAP (green VNAV on EADI). Displayed angle will stop increasing. HSI/EHSI glideslope pointer will display deviation from vertical angle. Flight director commands a flare to selected vertical angle.
        OR;
        1) SELECT desired vertical angle using SET knob.
        2) Mode Selector VNAV Button — PRESS. Mode selector annunciates (VNAV) ARM (white VNAV on EHSI). HSI/EHSI glideslope pointer displays deviation from vertical angle.
        3) When airplane reaches selected angle — Mode selector annunciates (VNAV) CAP (green VNAV on EHSI). Flight director commands a flare to selected vertical angle.

NOTE

VNAV angles are limited to 6 degrees and angle displayed in VANG position will flash for angles between 5 and 6 degrees.

Preselect altitude (ALT PRE) will be set equal to WPT ALT when VNAV mode is selected. Changes to ALT PRE are not allowed when VNAV mode is selected.

i. Approaching WPT ALT — Flight director commands a flare to WPT ALT.
j. At WPT ALT — VNAV mode is cancelled (WPT ALT, VANG set to dashes; TO, FROM set to zero) and Altitude Hold mode is automatically selected.

NOTE

Once the VNAV mode has been selected, no changes to the active WPT data are allowed. Data for the other (Alternate) WPT may be entered by pressing the WPT select button on the VNAV Controller. The number of the active WPT will remain illuminated (steady) and the number of the alternate WPT will flash.

VNAV Profile (Two waypoints linked together) (See illustration)

a. SET desired WPT data for both WPT 1 and WPT 2.
b. VNAV Controller PROFILE Button — PRESS, PROFILE button will illuminate.

NOTE

Profile mode is inhibited if:
1) Station elevation (STA EL) for WPT 1 and WPT 2 is not the same.
2) VANG for second WPT has been slewed ahead.
3) Airplane has passed computed flare point from first VANG to second VANG.

No changes will be allowed to WPT 1 or WPT 2 data while in the PROFILE mode.

Before PROFILE is selected, VANG for second WPT is computed angle between airplane and second WPT ALT. After PROFILE is selected, VANG for second WPT is computed angle between first WPT ALT and second WPT ALT.

To cancel PROFILE mode, press illuminated PROFILE button.

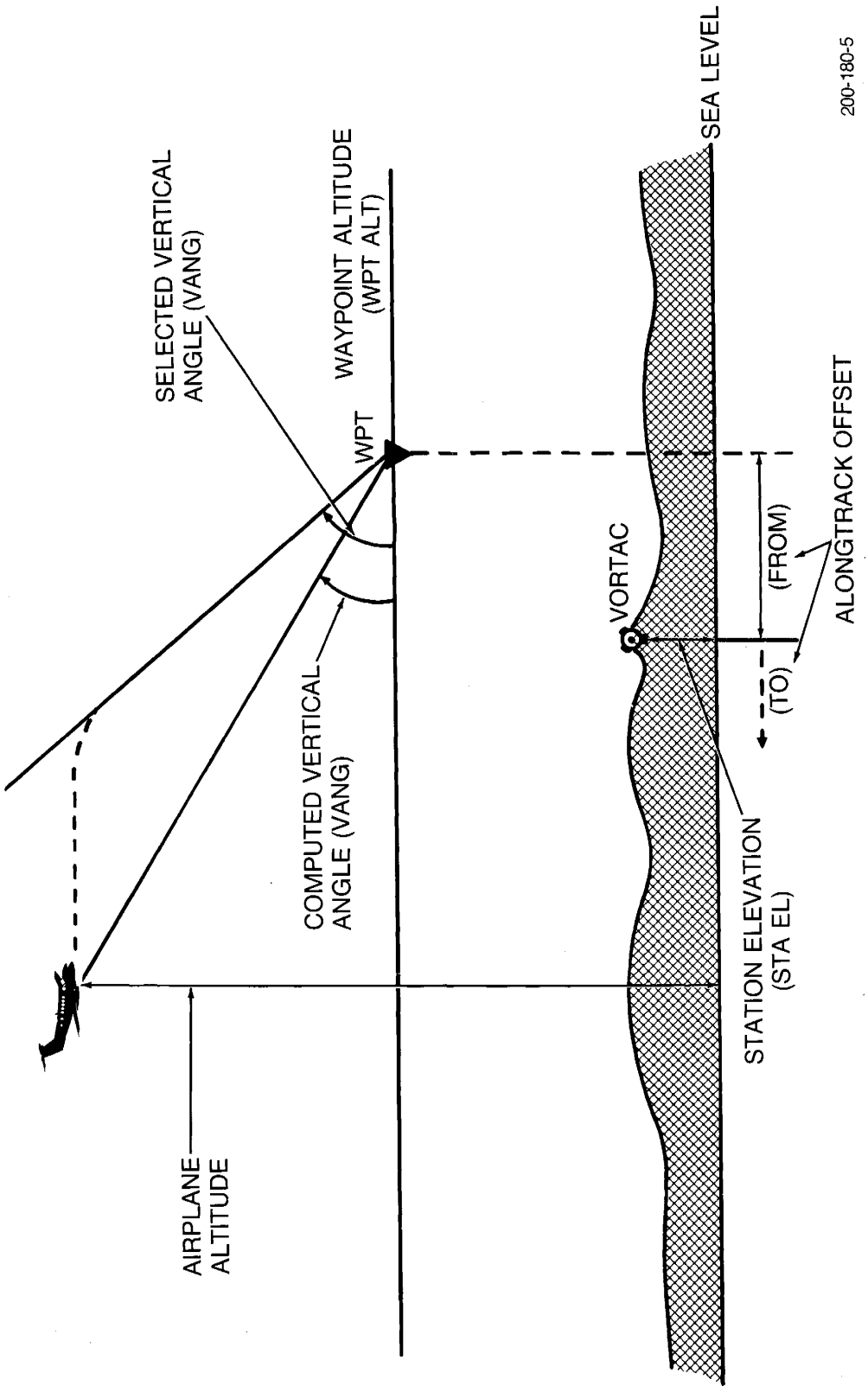
c. Mode Selector VNAV Button — PRESS. Mode selector annunciates (VNAV) ARM. HSI/EHSI glideslope pointer displays deviation from first vertical angle.
d. When airplane reaches VANG for first WPT — Mode Selector annunciates (VNAV) CAP. Flight director commands a flare to first VANG.
e. At computed flare point to second VANG — PROFILE button flashes then extinguishes; WPT data for first WPT is cancelled; mode selector VNAV annunciators extinguish momentarily, then annunciate (VNAV) ARM momentarily, then annunciate (VNAV) CAP. VNAV controller WPT button annunciates number of second WPT. Flight director commands a flare to second VANG. ALT PRE is set to WPT ALT for second WPT.
f. Approaching second WPT ALT — Flight director commands a flare to second WPT ALT.
g. At second WPT ALT — VNAV mode is cancelled and Altitude Hold is automatically selected.

16. Air Data Command Display (Optional)
To Command Airspeed Hold:

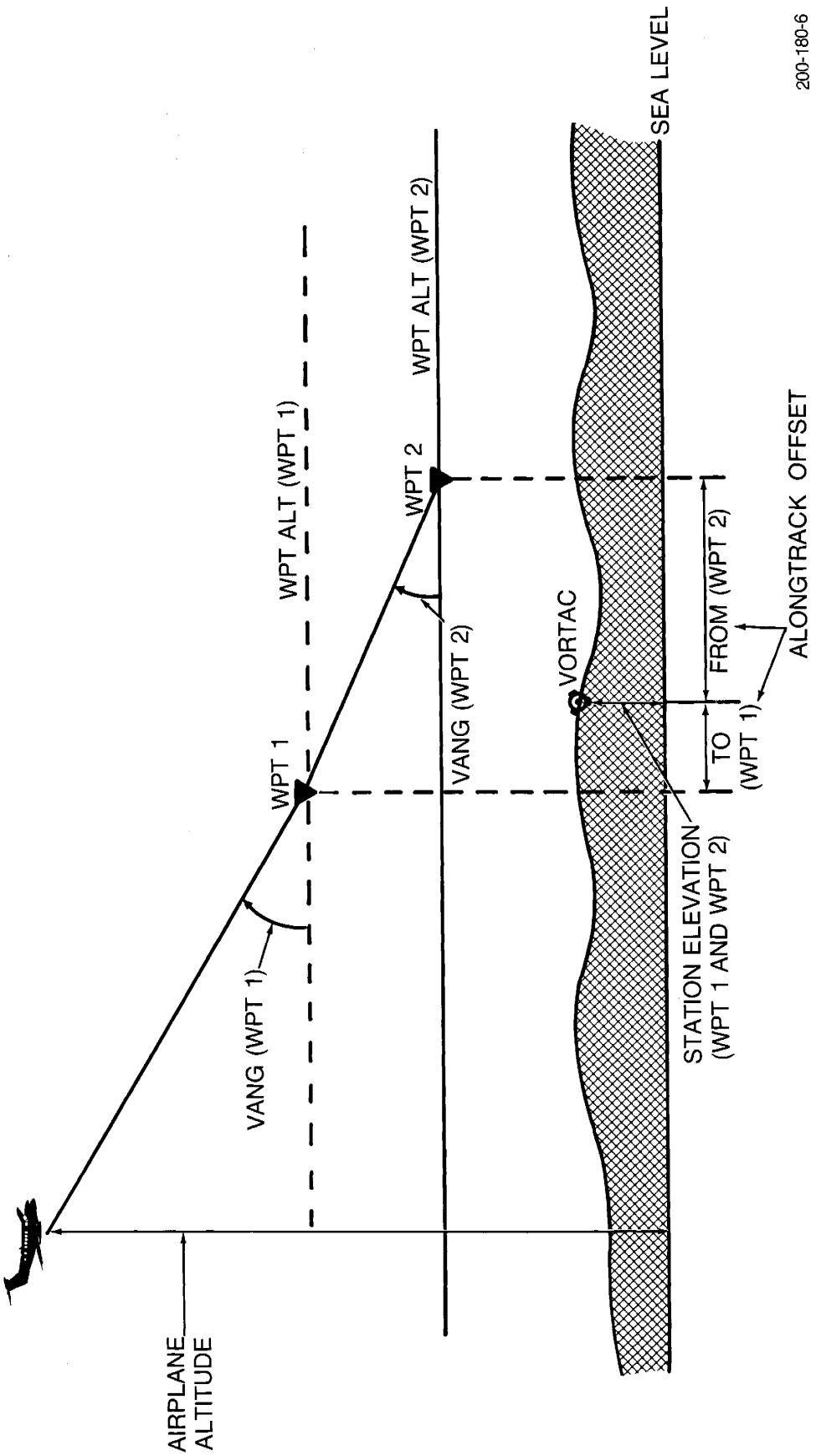
a. Mode Selector IAS Button — PRESS. Command display will indicate existing IAS in knots.
b. Autopilot Controller Pitch Wheel — ROTATE to increase or decrease airspeed shown on Command Display. Flight director will command pitch up, or down, to capture and track displayed airspeed.

To Command Vertical Speed Hold:

a. Mode Selector VS Button — PRESS. Command Display will indicate existing vertical speed in feet per minute.
b. Autopilot Controller Pitch Wheel — ROTATE to increase or decrease vertical speed shown on Command Display. Flight director will command pitch up, or down, to capture and track displayed vertical speed.


SEA LEVEL
200-180-5
SELECTED VERTICAL ANGLE (VANG)
WAYPOINT ALTITUDE (WPT ALT)
WPT
VORTAC
(FROM)
(TO)
ALONGTRACK OFFSET
COMPUTED VERTICAL ANGLE (VANG)
STATION ELEVATION (STA EL)
AIRPLANE ALTITUDE

NORMAL VNAV



VNAV - TWO WAYPOINTS LINKED TOGETHER

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17. EFIS Source Controller button definitions (if installed):
Single Installation (pilot's side only):

a. "NAV 1" Button — Selects NAV 1 as displayed navigation source.
b. "NAV 2" Button — Selects NAV 2 as displayed navigation source.
c. "HDG", "ATT" and Long Range NAV Button (if installed) — Select optional heading, attitude, and applicable long range navigation sources.

Dual Installation (pilot and copilot side):

a. "NAV" Button — Selects NAV 1 or NAV 2 as displayed navigation source with successive toggling of button.
b. "HDG" Button — Selects heading source, CMP 1 or CMP 2, with successive toggling of button.
c. "ATT" Button — Selects attitude (vertical gyro) source, VG1 or VG2, with successive toggling of button.
d. Long Range Navigation Buttons (if installed) — Select applicable long range navigation sources (INS, VLF, etc.)

18. EFIS Source Controller Initiated Flight Director Mode resets:
Pilot input to the EFIS through the SC-800 source controller initiates Flight Director and Autopilot mode changes as follows:

PILOT ACTION	FD/AP MODE CHANGE
Select a NAV course	All lateral modes reset except HDG
Select a HDG source	All lateral modes reset
Select an ATT source	Autopilot will disengage
Select an incompatible HDG and NAV source (such as true heading with VOR)	All lateral modes reset

NOTE

When selecting a NAV source, HDG will also reset if a flight director NAV mode is armed.

AUTOPILOT OPERATION

1. In-Flight Engagement of Autopilot

a. All AFCS and Autopilot Circuit Breakers — IN
b. Autopilot Controller TRIM UP and TRIM DN annunciators — CHECK. If either annunciator is illuminated, do not engage autopilot.
c. Turn Knob — IN CENTER DETENT POSITION.
d. Autopilot Controller AP Button — PRESS. Controller will annunciate AP Engage and YD ENGAGE.

2. Disengaging the Autopilot
The autopilot may be disengaged by:

a. Pressing the AP/YD & TRIM DISC button on either control wheel to the first level.

NOTE

The AP/YD & TRIM DISC button is a two-level switch. The first level disengages the autopilot and yaw damper, the second level disengages the electric trim.

b. Pressing the Go-Around button on the left power lever (Yaw Damper remains engaged).
c. Actuating the Trim Switches on either control wheel (YAW Damper remains engaged).
d. Pressing the AP button on the Autopilot Controller (Yaw Damper remains engaged).

NOTE

Upon autopilot disengagement, AP ENGAGE annunciator on the instrument panel will flash 5 times then extinguish, and the autopilot-off aural alert will sound for 1 second (or the Voice Advisory System, if installed, will announce "AUTOPILOT").

In addition, if the autopilot disenages automatically due to a detected failure, the AP DISC annunciator and the MASTER WARNING flasher will illuminate. The AP DISC annunciator is extinguished by pressing the AP/YD & TRIM DISC button on either control wheel. The MASTER WARNING flasher is reset by pressing its face.

3. Engaging the Yaw Damper
The Yaw Damper is automatically engaged whenever the Autopilot is engaged. The Yaw Damper may be engaged without engaging the Autopilot by pressing the YD button on the Autopilot Controller.

4. Disengaging the Yaw Damper
The Yaw Damper may be disengaged by:

   a. Pressing the AP/YD & TRIM DISC button on either control wheel to the first level.
   b. Pressing the YD button on the Autopilot Controller.

NOTE

Upon Yaw Damper disengagement, the YD ENGAGE annunciator on the instrument panel will flash 5 times, then extinguish.

5. Flight Director Coupling
When engaged, the Autopilot automatically couples to any flight director mode selected (except Go-Around).

6. Attitude Hold
When the Autopilot is engaged with no flight director modes selected, pitch attitude and heading are held. (If roll attitude is greater than 6 degrees, the Autopilot will command wings level, then heading hold).
To change Attitude

   a. TCS Switch — PRESS and HOLD
   b. Manually maneuver airplane to desired attitude
   c. TCS Switch — RELEASE. New pitch attitude and heading will be held. (If roll attitude is greater than 6 degrees, new roll attitude will be held.)

7. Turn Knob
To command a turn — ROTATE Turn Knob on Autopilot Controller out of detent in desired direction. Autopilot will command roll attitude proportional to amount of Turn Knob rotation.

NOTE

Rotating Turn Knob out of detent position with Autopilot engaged will cancel all lateral flight director modes.

8. Pitch Wheel
To change pitch attitude — ROTATE Pitch Wheel on Autopilot Controller in desired direction. Autopilot will command pitch attitude change proportional to amount of Pitch Wheel movement.

NOTE

Rotating the Pitch Wheel with the Autopilot engaged will cancel ALT, ALT SEL/CAP, IAS, VS, and VNAV CAP. (If optional Air Data Command Display is installed, IAS or VS will not be cancelled, but will be changed proportional to amount of wheel movement.)

9. Touch Control Steering (TCS)
   TCS allows the pilot to override autopilot modes (by disengaging pitch and roll servos) in two ways:

   a. In Attitude Hold, ALT, IAS, or VS mode, TCS synchronizes the hold references. When TCS is released, the new reference is held.
   b. In HDG, VOR, VOR APR, LOC, BC, ILS or VNAV mode, TCS allows corrective or supervisory override without mode disengagement. When TCS is released, the autopilot will recouple to the selected mode.

10. Soft Ride
    To soften autopilot response for turbulence penetration — PRESS Soft Ride Button on Autopilot Controller.

NOTE

Soft Ride is automatically cancelled if the autopilot is disengaged.

11. Bank Limit
    To limit bank angle in HDG mode — PRESS Bank Limit Button on Autopilot Controller. Autopilot and Flight Director commands are limited to 17 degrees.

NOTE

Low Bank Limit will be cancelled if HDG mode is cancelled, but will be reselected automatically if HDG is reselected. To cancel Low Bank Limit — PRESS Bank Limit Button when LOW is annunciated.

SHUTDOWN AND SECURING

— EFIS Switch(es) (if installed) - OFF before FUEL CUTOFF

PERFORMANCE

No change.

WEIGHT & BALANCE

No change.

SYSTEMS DESCRIPTION

No change.

HANDLING, SERVICING AND MAINTENANCE

The Sperry EADI and EHSI are equipped with internal temperature sensors. When the predetermined temperature limit is exceeded, a signal is generated to activate mechanical annunciators located on the Avionics Junction Box. A black color on the annunciator indicates normal operation. Yellow indicates an overtemperature condition. At each 100 hour inspection, and at any sign of abnormal EFIS operation, inspect the annunciators and verify operation of the EADI/EHSI and Symbol Generator cooling fans.

BEECHCRAFT SUPER KING AIR B200, B200C, B200T, and B200CT LANDPLANES PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

COLLINS FMS-90 FLIGHT MANAGEMENT SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Collins FMS-90 Flight Management System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Collins Pilot's Guide for the FMS-90 Flight Management System No. 523-0771373, dated July 31, 1981 or later issue, is available from Collins General Aviation Division, 400 Collins Road NE, Cedar Rapids, Iowa 52498. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status.

LIMITATIONS

The Collins FMS-90 Flight Management System meets the approval criteria for VOR/DME and VLF/Omega Area Navigation Systems as established by AC 90-45A and AC 20-101B. Operation must be in conformity with the Collins FMS-90 Flight Management System Pilot's Guide No. 523-0771373 dated July 31, 1981 or later issue. The FMS-90 may be used as a means of VFR/IFR navigation provided the following limitations are observed:

VOR/DME AREA NAVIGATION

1. The VOR/DME Area Navigation mode may not be used as a primary system under IFR conditions except on approved approach procedures, approved airways, and random area navigation routes when approved by Air Traffic Control.

2. The VOR/DME Area Navigation mode can only be used with colocated facilities. (VOR and DME signals originate from same geographical location.)

VLF/OMEGA AREA NAVIGATION

1. Only enroute operations in the conterminous United States and Alaska are approved.

2. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).

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3. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.

4. VOR and DME equipment must be installed and operating during navigation on approved RNAV routes.

5. IFR flight is not approved based on VLF/Omega navigation into any area in which such operations could not be approved based on the installed equipment required by paragraph 3. This includes overwater flights beyond the range of approved ground navigation facilities.

6. The VLF/Omega position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

   a. Prior to compulsory reporting points during IFR operations when not under radar surveillance or control.
   b. At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.
   c. At hourly intervals during operation off of approved RNAV routes.

7. The VLF/Omega position information should be updated when a cross-check with other onboard approved navigation equipment reveals an error greater than 2 nautical miles along-track or cross-track.

8. Navigation shall not be predicated on the use of this system during periods of dead reckoning.

9. Following a period of dead reckoning, the VLF/Omega position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment.

EMERGENCY PROCEDURES

Refer to the Collins FMS-90 Flight Management System Pilot's Guide No. 523-0771373 dated July 31, 1981 or later issue.

In the event that the FAIL or WRN (warning) annunciators illuminate, and anytime the navigational information is questioned based on information from other onboard navigation equipment, the other onboard navigation equipment must be used as the primary source of navigation information.

CAUTION

DME may unlock due to loss of signal with certain combinations of distance from station, altitude and angle of bank.

1. If NAV flag appears while enroute in the VOR/DME mode, check for correct frequency.

2. If VOR or DME equipment is intermittent or lost, utilize other navigation equipment as required.

3. If NAV flag appears during an approach, execute published missed approach and utilize another approved facility.

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NORMAL PROCEDURES

Refer to the Collins FMS-90 Flight Management System Pilot's Guide No. 523-0771373 dated July 31, 1981 or later issue.

NAVIGATION PROGRAM NUMBER (PID)

To ascertain the navigation program number (PID):

1. Position the Mode Selector to TEST. (The Fuction Selector has no effect in TEST mode and can be set to any position.)

2. Key in the code.

   a. 0005 for CRU program number
   b. 0006 for AEU program number
   c. 0007 for receiver program number

3. Press the ENTER key. (The left register displays the navigation software program number installed in the system. The right register displays the release date for the software. Registers read left- to-right year-month-day.)

CAUTION

Additional codes have been defined for fault diagnosis and repair. Entering codes not listed can interfere with navigation computations and result in significant navigation errors. These codes are erased whenever the FMS-90 is turned off.

PERFORMANCE

No change

WEIGHT AND BALANCE

No change

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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Issued: March, 1983
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For Training Purposes Only

Beechcraft®

Super King Air® B200, B200C, B200T & B200CT Landplanes

Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement

for the

Collins FCS-65 Automatic Flight Control System Category 1 with Collins EFIS-85 (EFIS-85A-1a or EFIS-85A-2a) Electronic Flight Instrument System or Collins EHSI-74 Electronic Horizontal Situation Indicator or Collins Electro-Mechanical Instruments

This Supplement is applicable to the following manual(s):

101-590010-147 and 101-590037-29

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane has been modified by installation of the Collins FCS-65 Automatic Flight Control System with Collins EFIS-85 Electronic Flight Instrument System or Collins EHSI-74 Electronic Horizontal Situation Indicator or Collins Electro-Mechanical Instruments in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Collins Pilot's Guides for the Electronic Flight Instrument System (EFIS-85A-1a) No. 523-0772833 dated March 9, 1983 or later issue, or (EFIS-85A-2a) No. 523-0773278 dated October 15, 1983 or later issue, the Vertical Navigation Indicator (VNI-80A) No. 523-0772019 dated May 31, 1982 or later issue, and the Collins Pilot's Guides for the Electronic Horizontal Situation Indicator (EHSI-74) No. 523-0772602 dated November 15, 1985 or later issue, and No. 523-0774211 dated February 24, 1993 or later issue, are available from Collins General Aviation Division, 400 Collins Road NE, Cedar Rapids, Iowa 52498. It is the responsibility of the owner/operator to maintain the appropriate pilot's guides in current status.

LIMITATIONS

1. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
2. Do not use autopilot below the following altitudes:
   a. Approach - 200 feet above ground level.
   b. Climb - 500 feet above ground level.
   c. All other operations - 1000 feet above ground level.
3. Do not use autopilot or yaw damper during takeoff or landing.
4. Pilot must be seated at the controls with the seat belt fastened during autopilot operations.
5. On serials prior to BT-35, BN-5, BL-139, and BB-1444, except BB-1439, do not use propeller speeds in the range of 1750-1850 rpm during ILS approaches.
6. On serials BT-35 & after, and BN-5 & after, do not use propeller speeds in the range of 1300-1400 rpm during ILS approaches.
7. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
8. Operation of the VNI-80A VNAV system must be in conformity with the Collins Pilot's Guide, No. 523-0772019, dated May 31, 1982 or later issue.
9. The VNAV system has not been shown to meet the accuracy requirements of Advisory Circular 90-45A and is therefore not to be used as the primary source of vertical guidance information.
10. Operation of the EFIS-85A-1a or EFIS-85A-2a system must be in conformity with the Collins Pilot's Guide, No. 523-0772833, dated March 9, 1983 or later issue or No. 523-0773278, dated October 15, 1983 or later issue, respectively.
11. Operation of the EHSI-74 Electronic HSI utilizing an HPU-74 (Collins P/N 622-6198-001) must be in conformity with the Collins Pilot's Guide, No. 523-0772602, dated November 15, 1985 or later issue.
12. Operation of the EHSI-74 Electronic HSI utilizing an HPU-74 (Collins P/N other than that referenced in 11., above) must be in conformity with the Collins Pilot's Guide, No. 523-0774211, dated February 24, 1993 or later issue.

13. Navigation and checklist data on the Multifunction Display (MFD-85) is provided only as an aid and is not approved as primary information.
14. IFR operation is prohibited in the CMPST (Composite) mode.
15. VOR/DME approaches are prohibited in the LINEAR ENABLED (linear deviation) mode on airplanes equipped with the EFIS-85 system.

EMERGENCY PROCEDURES

AIR START (FOR OPTIONAL EFIS-85 SYSTEM)

WARNING

Airstart using the STARTER ASSIST procedures may momentarily cause the loss of attitude display on Electronic Flight Instrument System (EFIS) equipped airplanes, and lead to premature system failures. IF FLIGHT CONDITIONS DO NOT PERMIT THE TEMPORARY LOSS OF ATTITUDE REFERENCE, CONDUCT AIRSTART USING THE NO STARTER ASSIST PROCEDURES.

CAUTION

The pilot should determine the reason for engine failure before attempting an air start. Do not attempt an air start if N_1 indicates zero.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an overtemperature condition.

STARTER ASSIST

NOTE

EFIS power should be turned off prior to a STARTER ASSIST airstart if conditions permit.

WARNING

Standby Attitude and Air Data displays must be monitored during STARTER ASSIST air starts.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF (if installed)
3. Radar - STANDBY or OFF
4. EFIS ON/OFF Switch(es) - OFF (if conditions permit)
5. Windshield Heat - OFF
6. Power Lever - IDLE
7. Condition Lever - CUT-OFF
8. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

9. Ignition and Start Switch - ON (up), Check IGNITION Annunciator - ON
10. Condition Lever - LOW IDLE
11. Ignition and Start Switch - OFF (N_1 above 50%)
12. Propeller Lever - AS REQUIRED
13. Power Lever - AS REQUIRED
14. Generator - ON
15. Engine Auto Ignition - ARM
16. EFIS ON/OFF Switch(es) - ON
17. Electrical Equipment - AS REQUIRED

EFIS-85 SYSTEM FAILURE

1. In the event of a Display Processor failure: Activate the AUX DRIVE switch.
2. In the event of an EADI or EHSI display failure: Activation of the CMPST switch will provide a composite display on the remaining indicator.
3. In the event of a number 1 dual fed electrical bus failure: Activation of the AUX DRIVE and CMPST switches will provide attitude information (minimum) on the EHSI in the composite format (inverter #2 is required for AC power).

EHSI-74 SYSTEM FAILURE

- Refer to copilot's HSI.

AUTOPILOT/ELEVATOR TRIM MALFUNCTIONS (FOR OPTIONAL EFIS-85 SYSTEM OR OPTIONAL EHSI-74 SYSTEM OR ELECTRO-MECHANICAL INSTRUMENTS)

1. Hold control wheel firmly. Be prepared to accept control wheel loads.
2. Fully depress red DISC TRIM/AP YD switch on pilot's or copilot's control wheel.
3. Recover - The maximum altitude losses observed during malfunction tests were:

Climb	450 ft
Cruise	440 ft
Maneuvering	150 ft
Descent	450 ft
Approach	100 ft

NOTE

An elevator trim malfunction will activate the master warning flasher and the red trim annunciator. The master warning flasher may be delayed for small trim requirements.

If necessary, the autopilot can also be disengaged by:

4. Pressing the AP ENG switch on the autopilot control head. (Yaw damper remains engaged.)
5. Activating the PITCH TRIM switch on the pilot's or copilot's control wheel. (Yaw damper remains engaged.)
6. Pressing the go around (GA) switch. (Yaw damper remains engaged.)
7. Pulling the autopilot circuit breaker.
8. Moving the avionics master switch to the OFF position.
9. Moving the gang bar to the OFF position.

CAUTION

If the autopilot is overpowered in pitch, the electric elevator trim will operate in the direction opposing the overpower force.

AUTOMATIC AUTOPILOT DISENGAGEMENT (FOR OPTIONAL EFIS-85 SYSTEM OR OPTIONAL EHSI-74 SYSTEM OR ELECTRO-MECHANICAL INSTRUMENTS)

The following conditions will cause the autopilot to disengage or servo torque to go to approximately zero torque:

DISENGAGE

1. Major degradation, interruption, or failure of electric power.
2. Autopilot internal monitors.
   a. System integrity checks.
   b. Loss of vertical gyro.
   c. Pitch attitudes in excess of 30°.
   d. Roll attitudes in excess of 45°.

ZERO TORQUE

The autopilot also monitors pitch and roll rate and normal acceleration. Pitch rates in excess of 4°/second, roll rates in excess of 10°/second, or accelerations in excess of 1.0 g cause autopilot servo torque to decrease to approximately zero torque until the rate or acceleration is below the above limits.

ENGINE FAILURE (FOR OPTIONAL EFIS-85 SYSTEM OR OPTIONAL EHSI-74 SYSTEM OR ELECTRO-MECHANICAL INSTRUMENTS)

1. Disengage autopilot.
2. Follow POH/AFM engine failure procedures.
3. Retrim.
4. Re-engage autopilot.
5. Maintain at least 120 KIAS until landing is assured.
6. Disengage autopilot for large power or airspeed variations.

NORMAL PROCEDURES

BEFORE TAKEOFF (RUNUP)

1. Autopilot - CHECK
   a. Position pitch trim to the take-off position, turn ELEV TRIM switch on and engage the autopilot. Check that YAW and AP annunciators are on.
   b. With control wheel in the forward position and the autopilot engaged, operate pilot then copilot pitch trim switches in both directions to ensure that the autopilot disengages. The yaw damper remains engaged.
   c. Center the control wheel and engage the autopilot. Apply forward pressure on control wheel. Note that pitch trim travels nose up and that the amber TRIM light comes on. Apply rearward pressure on control wheel. Note that pitch trim travels nose down and that the amber TRIM light comes on.
   d. Hold the control wheel and disengage the autopilot by depressing the DISC TRIM/AP YD button on the pilot's control column to the first level. Note that the YAW DIS and AP DIS annunciators illuminate. Further depress the DISC TRIM/AP YD button. Note that the ELEC TRIM OFF annunciator illuminates. Cycle the electric trim switch on the console, engage the autopilot, and repeat the check using the copilot's DISC TRIM/AP YD switch.
   e. Engage yaw damper. Note YAW annunciator is on. Check for additional resistance to movement of rudder pedals. Disengage yaw damper.
   f. Push the TEST button on the mode annunciator panel above the ADI to test the annunciators.

g. Reset pitch trim to the take-off position.

h. Move all primary flight controls through their full travel in both directions. Verify that controls move in proper direction and no restrictions to free movement are present.

2. Elevator Trim - CHECK

a. Verify the ELEV TRIM is ON.

b. Check operation of dual-element thumb switches as detailed in the POH/AFM.

WARNING

Operation of the electric trim system should occur only by movement of pairs of switches. Any movement of the elevator trim wheel while actuating only one switch denotes a system malfunction. If a malfunction of the electric trim system is indicated, electric trim must be disengaged and trim changes made with manual trim ONLY.

3. Digital Air Data System (if installed) - CHECK

a. Press and hold the cockpit-mounted ADC TEST switch for the Air Data Computer. Check for the following:

1) Air data instrument flags come into view, and the instruments go into the loss-of-data mode.

2) Altitude pointer goes to the 250-foot mark.

3) Vertical speed pointer goes to 6000-ft/min rate of descent mark.

4) Airspeed Indicator IAS and V_{MO} pointers go to zero.

b. Release the ADC TEST switch and check for the following:

1) Data transmissions to the air data instruments are restored.

2) All air data instrument flags go out of view.

c. Press and hold the TEST switch on the Vertical Speed Indicator, and check for the following:

1) The warning flag comes into view.

2) The pointer slews to the 6000-ft/min rate of climb mark.

d. Release the TEST switch, and check for the following:

1) The flag goes out of view.

2) The pointer returns to zero.

e. Press and hold the PUSH TEST switch on the Airspeed Indicator, and check for the following:

1) The warning flag comes into view, and both pointers (IAS and V_{MO}) slew to 160 knots.

2) After approximately 1 second, both pointers slew to zero.

f. Release the PUSH TEST switch and check for the following:

1) The flag goes out of view.

2) The barber pole (V_{MO} indicator) returns to 260 knots.

g. Press and hold the TEST switch on the Barometric Altimeter, and check for the following:

1) The warning flag comes into view in approximately 1/2 second.

2) The altitude pointer goes to the 750-foot mark.

NOTE

The output from the altimeter to the transponder is inhibited during the self test.

h. Release the altimeter TEST switch, and check for the following:

1) The flag goes out of view.

2) The altitude pointer returns to the actual altitude.

i. Turn the BARO knob on the altimeter and observe that the baroset digits and altitude pointer respond accordingly.

j. Push BARO knob in to obtain baroset values in in. Hg. Set the barometric display to the correct local pressure value. (Pull BARO knob out if display of baroset values in millibars is desired.)

k. Press and hold the ALT ALERT annunciator/switch on the Altitude Preselector/Alerter, and check for the following:

1) The ALT ALERT annunciator illuminates.

2) The warning flag comes into view.

3) Both remote ALT ALERT annunciators illuminate.

l. Release the ALT ALERT annunciator/switch, and check for the following:

1) The ALT ALERT annunciator extinguishes.

2) The warning flag goes out of view.

3) Both remote ALT ALERT annunciators extinguish.

IN FLIGHT OPERATION

AUTOPILOT OPERATION

To engage the autopilot, press the AP ENG switch on the autopilot control panel. If a red fault annunciator is on (AP, TRIM), it requires one operation of the AP ENG switch to clear the fault and another operation to engage the autopilot. If the fault does not clear, the autopilot will not engage.

An automatic pre-engagement test is performed each time the autopilot is engaged. This test requires approximately 2 seconds and is identified during the test by annunciation of red TRIM and AP.

The autopilot may be engaged in any reasonable attitude. When engaged with no lateral mode selected beyond 30 degrees bank, 15 degrees pitch up or 10 degrees pitch down, the autopilot will return the airplane to these limits. No steering information will be presented on the flight director until a lateral mode is selected.

To disengage the autopilot, press the AP ENG switch or operate the DISC TRIM/AP YD DISC switch on the pilot's or copilot's control wheel to first detent. Refer to EMERGENCY PROCEDURES for other means of disconnecting.

CONTROL WHEEL SYNCHRONIZATION (CWS)

Autopilot sync mode is controlled by the PITCH SYNC & CWS (control wheel steering) button on the control wheel. With the autopilot engaged and in either manual or guidance mode, depressing the PITCH SYNC & CWS button cancels vertical modes and allows the pilot to maneuver the airplane without disengaging the autopilot. When the PITCH SYNC & CWS button is released, the autopilot maintains the attitude existing at the time the button is released if in manual mode, or maintains the new pitch attitude and follows commands received from the autopilot computer if in a lateral mode. CWS mode does not cancel GS when in APPR mode.

VERTICAL TRIM SWITCH

This control switch is used for vertical control of the flight control system. Activating the switch for more than one second with a vertical hold mode selected causes the vertical mode to be cancelled (except GA, GS, and ALT SEL ARM) and the system to revert to basic pitch operation. Refer to Special Note 1.

1/2 BANK (1/2Φ)

In 1/2φ mode, the bank limit in HDG or NAV mode is reduced to approximately 1/2 the normal value. Capture of NAV or APPR mode clears the 1/2φ mode, if selected.

SOFT RIDE (SR)

SR mode is used to reduce the autopilot response to turbulent conditions. The pilot may use SR mode at his discretion. SR is cancelled by APPR mode and when engaging the autopilot.

YAW DAMPER OPERATION

The yaw damper is automatically engaged any time the autopilot is engaged.

The rudder channel of the autopilot may be selected separately for yaw damping by the YAW ENG switch on the autopilot panel.

To disengage the yaw damper, operate the DISC TRIM/AP YD disc switch, or press the YAW ENG switch. The YAW ENG switch is disabled if the autopilot is engaged.

ELECTRIC PITCH TRIM OPERATION

The autopilot computer provides manual electric trim operated by the split pitch trim switches located on the pilot's and copilot's control wheels. Electric trim operation is annunciated by the amber TRIM annunciator (when the trim speed is above the preset annunciation threshold).

MODE SELECTION

All modes with the exception of GA are selected by pushing the desired push-on/push-off button on the mode control panel. Annunciator lights on the mode control panel and mode annunciator panel indicate the selected mode of operation.

The flight director system supplies steering commands for the pilot and the autopilot. When the autopilot is engaged, the pilot monitors autopilot performance on the attitude director indicator. When the autopilot is not engaged, the pilot flies the airplane manually in response to the attitude director indicator commands. A lateral mode must be selected to bring the command bars into view for manual flight director operation.

The EFIS-85 Electronic Flight Instrument System may replace the conventional electro-mechanical Attitude Director (ADI) and Horizontal Situation (HSI) Indicators, or an EHSI-74 Electronic HSI may replace only the conventional electro-mechanical Horizontal Situation Indicator (HSI). The CRT displays (EADI and/or EHSI) provide the necessary information for the pilot to respond to flight director and navigation commands in the same manner as with conventional instruments. The system also includes additional features which allow the pilot to select alternate formats as a navigation aid or in the event of specific failures. The operator is cautioned that the primary displays (attitude and/or directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate modes is described in the Collins EFIS-85 Pilot's Guide or applicable Collins EHSI-74 Pilot's Guide. The Owner/Operator of the airplane is responsible for obtaining and maintaining these Pilot's Guides.

HEADING MODE (HDG)

Set desired heading and select HDG mode. The autopilot will turn the airplane to and maintain the selected heading. For proper operation, the HDG marker should not be displaced from the airplane heading by more than 150 degrees when the HDG mode is selected.

NAV MODE (NAV)

Set desired heading and course prior to selecting NAV mode. When NAV mode is selected, the HDG and NAV ARM annunciators will illuminate. The airplane will follow the selected heading until the centerline of the selected radio course is approached. HDG mode will then clear and the airplane will turn to track the beam centerline. The ARM annunciator will extinguish to indicate course capture and NAV will remain on. Crosswind correction, up to 30 degrees, is automatically computed after course capture.

CAUTION

Premature Glideslope and localizer capture may occur in the APPR ARM mode if selected when the airplane is far from the localizer centerline. Therefore, use the HDG mode to position the airplane near the localizer intercept before selecting the APPR ARM mode.

NOTE

1. A vertical mode, if selected, will automatically clear at glideslope capture.

2. All approaches must be made in APPR or B/C mode.

3. EFIS-85A-1a equipped airplanes display linear deviation with a full-scale offset of 10 NM. This presentation is displayed in both NAV and APPR modes. Consequently, VOR/DME approaches are prohibited in the LINEAR ENABLE (linear deviation) mode.

BACK COURSE MODE (B/C)

When the B/C mode is selected, localizer capture follows the same sequence as a front course approach. Correct lateral steering information is displayed on the flight director, but vertical information is displayed as a function of the vertical mode selected.

NOTE

The inbound (front) course must be selected on the course indicator.

ALTITUDE HOLD MODE (ALT)

ALT can be selected during all modes of operation except after glideslope capture in APPR mode. Deviations from the selected altitude are displayed on the flight director.

NOTE

The autopilot maintains the airplane at the selected altitude by changing the pitch attitude of the airplane. The pilot must maintain proper power settings to assure a safe airspeed.

INDICATED AIRSPEED MODE (IAS)

The IAS mode may be selected during all modes of operation except after glideslope capture in APPR mode. Deviations from the airspeed at the time the mode was selected are displayed on the flight director as pitch commands.

VERTICAL SPEED MODE (VS)

The VS mode may be selected during all modes of operation except after glideslope capture in APPR mode. Deviation in vertical speed provides commands to maintain the vertical speed present at the time of mode selection. Rate of ascent/descent should be established for 10 seconds before selecting VS mode.

ALTITUDE PRESELECT MODE (ALT SEL) (IF INSTALLED)

Dial in the preselected altitude on the Altitude Preselector/Alerter and press the ALT SEL mode on the mode selector. The ALT SEL annunciator will illuminate. Any vertical mode except altitude hold may be used in conjunction with ALT SEL.

An aural alert tone sounds and annunciator lights on the altitude panel and flight director illuminate when the airplane reaches 1000 ft from the preselected altitude. At 300 feet from the preselected altitude the altitude annunciator extinguishes. Thereafter, the alert tone and altitude annunciators will warn when the airplane deviates more than 300 feet from the preselected altitude. As the airplane reaches the preselected altitude the ALT SEL annunciator extinguishes and ALT illuminates.

The use of vertical trim switch or altitude preselect dial, when the airplane is within 1000 ft of the preselected altitude and before altitude capture, will cancel the ALT SEL mode.

DESCENT MODE (DSC)

When the DSC mode is selected, the autopilot begins a gradual descent, stabilizing at a rate-of-descent of 950 to 1350 feet per minute. The stabilized rate-of-descent will vary depending on altitude, with the lower rates-of-descent occurring at high altitude and the higher rates-of-descent occurring at lower altitudes. After the vertical speed has stabilized (DSC annunciator extinguished and VS annunciator illuminated), the pilot may vary the rate-of-descent by operation of the vertical trim switch.

On those airplanes equipped with the digital air data system, the altitude preselect mode is automatically activated when the DSC mode is selected.

NOTE

Desired altitude must be selected on the altitude alerter prior to selection of the DSC mode.

GO-AROUND MODE

The GA mode is activated by the GA switch. When the GA switch is pushed, while in any lateral mode, the autopilot will disengage and a 7° pitch up, wings level attitude will be displayed on the flight director. GA annunciator will illuminate and remain on until a lateral mode is reselected.

Operation of the PITCH SYNC & CWS switch will clear the GA mode and synchronize the vertical command to the airplane attitude. Re-engagement of the autopilot while in go-around clears the GA mode and synchronizes the autopilot commands to the airplane pitch attitude at the time of engagement and continues to maintain wings level.

Selection of a lateral mode will also clear the GA mode and synchronize the vertical commands to airplane pitch attitude at the time of mode selection.

SPECIAL NOTES

1. Vertical Trim Switch Operation
   a. In basic pitch mode, with the autopilot or a lateral mode selected, activating the vertical trim switch gives a 0.5° command followed by a continuous slew of 1.0°/second.
   b. In VS hold mode the synchronized vertical speed is changed in 200 ft/min increments with each momentary activation of the vertical trim switch.
   c. In IAS hold mode the synchronized airspeed is changed in one-knot increments with each momentary activation of the switch.
   d. In the altitude hold mode, the synchronized altitude is changed by 25 ft with each momentary activation of the switch.
2. Basic Attitude Mode - With no lateral mode selected, the steering display is biased out of view and the ADI is used as a standard artificial horizon. No computed steering information is displayed on the ADI.
3. Bank command limit is ±30° with turn knob, ±25° in lateral modes, ±15° in APPR (after GS capture), and ±10° in NAV track.
4. Vertical mode information is displayed by pushing the desired vertical mode on the mode select panel. If a vertical mode other than GS is selected during manual flight director operation, it will automatically clear when the autopilot is engaged.
5. To synchronize the vertical commands in flight director function to the airplane attitude, actuate the PITCH SYNC & CWS switch on the control wheel after any lateral mode has been selected.
6. When making any coupled approach, primary flight instruments must be monitored by the pilot. Presence of a VOR/LOC/GS flag must be considered as a warning of system or signal failure.
7. When flying inbound toward a VOR, do not set up a VOR capture problem within approximately 5 miles of the VOR. Depending on the speed of the airplane, the system may not have time to solve the problem and stabilize on course before passage over the station.
8. Course may be changed over a VOR station when operating in NAV mode, as long as the course change is not more than 30°. If the course change is more than 30°, the HDG mode should be selected to establish a new intercept and then NAV mode should be reslected to set up a new capture.
9. For optimum NAV mode operation, select intercept angle so the system maintains straight and level flight in the NAV mode a minimum of 30 seconds prior to capture.
10. When making a station passage in NAV mode, the computer may automatically switch into heading memory sub-mode to allow a smooth station passage. During this time, the dead reckoning (DR) annunciator will illuminate.
11. Maximum recommended capture angles are 90° or less in NAV mode and 60° or less in APPR mode.
12. Fly all coupled approaches at or above 120 KIAS, flaps APPROACH and landing gear extended.
13. Always check the mode select panel annunciator display to verify that proper switching has taken place.
14. Do not select any vertical modes until 3 minutes after applying power.
15. Optional Dual Flight Director Systems:
    a. Disengage the autopilot prior to transferring the autopilot to either flight director.
    b. Deselect 1/2ϕ mode prior to transferring the autopilot to either flight director.
16. The vertical gyro connected to the autopilot system is not transferred if optional VG switching feature is installed.

PERFORMANCE

No Change

WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

The Collins FCS-65 is an integrated flight control system which can be functionally divided into two general subsystems.

1. Flight Director System (FDS)
   The FDS consists of an attitude director indicator (ADI), a horizontal situation indicator (HSI) and associated components.

The ADI presents attitude data, raw radio information and steering commands. The HSI presents the navigation situation. EFIS ON/OFF SWITCH(ES) - Control power to the EFIS symbol generators, EADI and/or EHSI. The switches should be selected OFF during engine start to protect the EFIS system from low/transient voltages.

2. Autopilot System (APS)
   The APS consists of an autopilot computer, mode select panel, air data sensor, three primary servos, a vertical reference and a trim servo. The system also requires compass and navigation system inputs from the avionics system.

ANNUNCIATORS

Annunciation is displayed on the mode select panel and remote annunciator panel. Annunciation is color-keyed for better recognition of mode status with green for active condition, amber for arm condition, and red for warning condition. The following table shows the annunciations and their association with each other:

ANNUN	CONDITION TO ACTIVATE	COLOR ACTIVE	COLOR ARM	COLOR WARN
HDG	Shows heading mode selected. Also, will light automatically if NAV or APPR is selected but capture has not occurred.	Green		
NAV	Selection of navigation mode.	Green		
APPR	Selection of approach mode.	Green		
GS	Indicates glideslope mode.	Green		
AP	Indicates autopilot is engaged.	Green		
ALT	Selection of altitude hold mode.	Green		
IAS	Selection of IAS hold mode.	Green		
VS	Selection of vertical speed hold mode.	Green		
GA	Selection of go-around mode.	Green		
YAW	Indicates yaw damper is engaged.	Green		
DR	Computation is in Heading Memory.	Green		
ARM	Indicates an armed condition prior to capture.		Amber	
DSC	Indicates the selection of descent mode.	Green		
1/2ϕ	Indicates selection of half bank mode.	Green		
SR	Indicates selection of soft ride mode.	Green		
B/C	Indicates selection of back course mode.	Green		
DIS	Disengage annunciation for YAW and AP. Flashes for 5 seconds then steady indication when system is disengaged. Lights steady when PITCH SYNC button is depressed.		Amber	
ALT SEL	Shows that altitude preselect is armed for automatic capture.		Amber	
AP	Indicates first failure in dual safety functions - Pilot should take over and disengage.			Red
TRIM	Indicates Trim Servo failure in autopilot or manual electric operation.			Red
TRIM	Indicates Elevator Trim-in-Motion, both manual electric and autopilot operation.	Amber		*Amber

** Sustained Annunciation*

ANNUNCIATOR NOTES:

1. During annunciator test, CLM annunciator will illuminate even though the function is not installed.

2. When a mode annunciator is flashing it indicates that signals used for that mode are not usable.

HANDLING, SERVICING & MAINTENANCE

If a Collins altimeter is installed, one additional static-line drain is installed. It is located inside the nose avionics compartment in the pilot's static airline that provides static pressure to the digital air data computer. This drain should be opened to release any trapped moisture at each 150-hour inspection and after exposure to visible moisture on the ground. It must be closed after draining.

BEECHCRAFT SUPER KING AIR B200 LANDPLANE
(Serial BB-1158 only)
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

HYDRAULIC LANDING GEAR

GENERAL

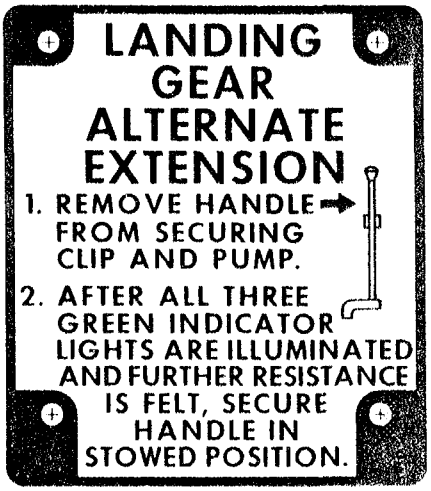
The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been equipped with Hydraulic Landing Gear, in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

PLACARDS

On Floor Aft of Pedestal:



EMERGENCY PROCEDURES

LANDING GEAR MANUAL EXTENSION

1. Airspeed - ESTABLISH 130 KNOTS
2. Landing Gear Relay Circuit Breaker (Pilot Subpanel) - PULL
3. Landing Gear Switch Handle - DOWN
4. Extension Lever - PUMP up and down until the three green GEAR DOWN indicator lights are illuminated.
5. Extension Lever - SECURE in clip (STOWED POSITION)

FAA Approved
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P/N 101-590010-219

For Training Purposes Only

WARNING

If for any reason the three green GEAR DOWN indicator lights do not illuminate (e.g. in case of an electrical failure) continue pumping until sufficient resistance is felt to ensure that the landing gear is down.

WARNING

After a manual (emergency) landing gear extension has been made, do not move any landing gear controls, or reset any switches or circuit breakers until the airplane is on jacks; since the failure may be in the gear-up circuit and the landing gear might retract on the ground.

LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION

After a practice manual extension the landing gear may be retracted as follows:

1. Extension Lever - STOW (Flush against floor under securing clip)
2. Landing Gear Relay Control Circuit Breaker (Pilot's Subpanel) - PUSH IN
3. Landing Gear Switch Handle - UP

NORMAL PROCEDURES

No Change

PERFORMANCE

No Change

WEIGHT AND BALANCE

No Change

SYSTEMS DESCRIPTION

ANNUNCIATOR SYSTEM

A caution annunciator placarded HYD FLUID LOW, in the caution/advisory annunciator panel will illuminate whenever the hydraulic fluid level in the hydraulic power pack is low.

LANDING GEAR

The retractable tricycle landing gear is electrically controlled and hydraulically actuated. The system utilizes folding braces, called drag legs, that lock in place when the gear is fully extended. The individual landing gear actuators incorporate internal/mechnical down-locks to hold the gear in the fully extended position. The landing gear is held in the up-lock position by hydraulic pressure.

Hydraulic pressure to the system is supplied by a hydraulic power pack. A hydraulic reservoir located in the left center wing section provides hydraulic fluid to the power pack. The reservoir incorporates a dip stick to provide a visual check of fluid level.

An electrically actuated control valve controls the flow of hydraulic fluid to the individual gear actuators. The control valve receives electrical power through the landing gear control switch.

Accidental retraction of the landing gear is prevented through safety switches located on the main landing gears.

HYDRAULIC EXTENSION AND RETRACTION SYSTEM

The nose and main landing gear assemblies are extended and retracted by a hydraulic power pack in conjuction with hydraulic actuators. The hydraulic power pack is located in the left center section, just forward of the main spar. One hydraulic actuator is located at each landing gear. The power pack consists of: a hydraulic pump, a 28VDC motor, a two section fluid reservoir, filter screens, gear selector valve and two solenoids, a fluid level sensor, and an up-lock pressure switch. For manual extension the system has a hand-lever-operated pump located on the floor between the crew seats. Three hydraulic lines, one for normal extension, and one for retraction, routed from the power pack, and one for emergency extension from the hand pump, are routed to the nose and main gear actuators. The normal extension lines and the manual extension lines are connected to the upper end of each hydraulic actuator. The hydraulic lines for retraction are fitted to the lower ends of the actuators. Hydraulic fluid under pressure generated by the power pack pump and contained in the accumulator acts on the piston faces of the actuators which are attached to folding drag braces resulting in the extension or retraction of the landing gear.

An internal mechanical lock in the nose gear actuator and the over-center action of the nose gear drag leg assembly lock the nose gear in the down position. Notched hook, lock link and lock link guide attachments fitted to each main gear upper drag leg provide positive down-lock action for the main gear.

Electrical overload to the system is prevented through the use of a 60 ampere circuit breaker located below the flooring near the hydraulic power pack.

The landing gear hydraulic power pack motor is controlled by the use of the landing gear switch handle placarded LDG GEAR CONT- UP-DN located on the pilot's subpanel. The switch handle must be pulled out of a detent before it can be moved from either the UP or DN position.

Safety switches, called squat switches, on the main gear torque knees open the control circuit when the strut is compressed. The squat switches must close to actuate a solenoid which moves a down-lock hook on the LDG GEAR CONT switch to the released position. This mechanism prevents the LDG GEAR CONT switch handle being placed in the UP position when the airplane is on the ground. The hook automatically disengages when the airplane leaves the ground, and can be overridden by pressing down on the red down-lock release button located to the left of the landing gear control switch handle.

In flight, as the landing gear moves to the full down position, the down-lock switches are actuated and interrupt current to the pump motor. When the red gear in-transit light in the landing gear control handle extinguishes and the three green GEAR DOWN indicators illuminate the landing gear is in the fully extended position.

Two gear select solenoids located on the valve body of the pump are energized through positioning of the landing gear control switch handle to either the UP or DN position. Once energized the gear select valve is actuated allowing hydraulic fluid to flow to the actuators.

Hydraulic system pressure performs the up-lock function, holding the landing gear in the retracted position. When the hydraulic pressure reaches approximately 2775 ±55 psi, the up-lock pressure switch will cause the landing gear relay to open and interrupt the current to the pump motor. The same pressure switch will cause the pump to actuate, should the hydraulic pressure drop to approximately 2400 psi.

A caution annunciator placarded HYD FLUID LOW, in the caution/ advisory annunciator panel will illuminate (yellow) whenever the hydraulic fluid level in the hydraulic power pack is low. The annunciator is tested by pressing in the HYD FLUID SENSOR TEST button located on the pilot's subpanel.

The LDG GEAR CONT switch handle should never be moved out of the DN detent while the airplane is on the ground. If it is, the landing gear warning horn will sound intermittently and the red gear-in-transit lights in the LDG GEAR CONT switch handle will illuminate (provided the MASTER SWITCH is ON), warning the pilot to return the handle to the DN position.

Landing gear position is indicated by an assembly of three individual lights. One light in each segment, when illuminated, makes the segment appear green and indicates that particular gear is DN. Absence of illumination indicates gear UP.

Two red parallel-wired indicator lights located in the LDG GEAR CONT switch handle illuminate, to show that the gear is in transit or unlocked. The red lights in the handle also illuminate when the landing gear warning horn is actuated.

The red control lights may be checked by pressing HDL LT TEST button located adjacent to the LDG GEAR CONT switch handle.

LANDING GEAR WARNING SYSTEM

The landing gear warning system is provided to warn the pilot that the landing gear is not down during specific flight regimes. Various warning modes result, depending upon the position of the flaps.

With the flaps in the UP or APPROACH position and either or both power levers retarded below a certain power level, the warning horn will sound intermittently and the LDG GEAR CONT switch handle lights will illuminate. The horn can be silenced by pressing the WARN HORN SILENCE button adjacent to the LDG GEAR CONT switch handle; the lights in the LDG GEAR CONT switch handle cannot be cancelled. The landing gear warning system will be rearmed if the power levers are advanced sufficiently.

With the FLAPS beyond APPROACH position, the warning horn and landing gear switch handle lights will be activiated regardless of the power settings, and neither can be cancelled.

MANUAL LANDING GEAR EXTENSION

A hand pump handle, placarded LANDING GEAR ALTERNATE EXTENSION, is located on the floor aft of the pedestal. The pump is located under the floor below the handle and is used when manual extension of the gear is required.

To engage the system pull the LANDING GEAR RELAY circuit breaker located adjacent to the LDG GEAR CONT switch handle on the pilot's subpanel and ensure that the LDG GEAR CONTROL switch handle is in the DN position. Remove the pump handle from the securing clip and PUMP the handle up and down until three green GEAR DOWN indicator lights illuminated and further resistance is felt. Place the handle in the full down position and secure in the retaining clip.

WARNING

If for any reason the three green GEAR DOWN indicators do not illuminate (e.g., in the case of an electrical system failure), continue pumping until sufficient resistance is felt to ensure that the gear is down.

WARNING

After a manual (emergency) landing gear extension has been made, do not move any landing gear controls, or reset any switches or circuit breakers until the airplane is on jacks, since the failure may be in the gear-up circuit and the landing gear might retract on the ground.

After a practice manual extension of the landing gear, the gear may be retracted hydraulically by pushing the LANDING GEAR RELAY circuit breaker in and moving the LDG GEAR CONT switch handle to the UP position.

HANDLING, SERVICING AND MAINTENANCE

Refer to the BEECHCRAFT Super King Air B200 series maintenance manual for lubrication and servicing.

Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

For Training Purposes Only

PILOT'S OPERATING HANDBOOK
AND
FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT
FOR THE
BEECHCRAFT SUPER KING AIR B200C(C12F)
(BL-73 THRU BL-112) LANDPLANES
EQUIPPED WITH THE
McCAULEY FOUR-BLADED PROPELLER SYSTEM
(MODEL 4HFR34C754/94LA-0)

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the McCauley Four-bladed Propeller System (Model 4HFR34C754/94LA-0) in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

NUMBER OF PROPELLERS

2

PROPELLER MANUFACTURER

McCauley Accessory Division, Vandalia, Ohio

PROPELLER BLADE AND HUB MODEL NUMBERS

HUB

4HFR34C754

BLADES

94LA-0

PROPELLER DIAMETER

94 inches (93 inches minimum repair diameter)

PROPELLER BLADE ANGLES AT 30-INCH STATION

Feathered: 87.5° ± .3°

Reverse: -10° ± .4°

PROPELLER ROTATIONAL SPEED LIMITS

Transients not exceeding 5 seconds	2200 rpm maximum
Reverse	1900 rpm maximum
All other conditions	2000 rpm maximum
Continuous operation on the ground	1150 rpm minimum

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CAUTION

Ground operation in feather at 600 rpm or less is approved. Propeller blade erosion may result from ground operation in feather.

PROPELLER ROTATIONAL OVERSPEED LIMITS

The maximum propeller overspeed limit is 2200 rpm and is time limited to five seconds. Sustained propeller overspeeds faster than 2000 rpm indicate failure of the primary governor. Sustained propeller overspeeds faster than 2080 rpm indicate failure of both the primary and secondary governors.

PROPELLER DEICE AMMETER

Green Arc (Normal Operating Range) ..... 18 to 24 amperes

EMERGENCY PROCEDURES

ELECTRICAL SMOKE OR FIRE

Airplanes equipped with McCauley four-bladed propellers are also equipped with non-opening cockpit side windows. Smoke evacuation procedures are identical to those of the basic airplane except that the storm windows are not used to facilitate smoke and fume removal.

NORMAL PROCEDURES

BEFORE TAKEOFF (RUNUP)

1. Ground Idle Low Pitch Stops - CHECK
   a. Condition Levers - HIGH IDLE
   b. Power Levers - IDLE (note propeller rpm)
   c. Prop Test Switch - HOLD TO "GND IDLE STOP" (note decrease in propeller rpm on left and right engines)
   d. Prop Test Switch - RELEASE (note propeller rpm increase to the value in Step "b")
   e. Condition Levers - LOW IDLE
2. Autofeather - CHECK
   a. Prop Test Switch - HOLD TO "GND IDLE STOP"
   b. Power Levers - APPROXIMATELY 500 FT-LBS TORQUE
   c. Autofeather Switch - HOLD TO TEST (both autofeather annunciators illuminated)
   d. Power Levers - RETARD INDIVIDUALLY:
      1) At Approximately 400 FT-LBS - OPPOSITE ANNUNCIATOR EXTINGUISHED
      2) At Approximately 260 FT-LBS - BOTH ANNUNCIATORS EXTINGUISHED

NOTE

Autofeather annunciators will cycle on and off with each fluctuation of torque as the propeller feathers.

e. Power Levers - BOTH RETARDED (both annunciators extinguished, neither propeller feathers)

f. Prop Test Switch - RELEASE

3. Autofeather Switch - ARM

BEFORE LANDING

NOTE

No limitations on propeller rpm exist during ILS approaches with the McCauley four-bladed propeller system installed.

NOISE CHARACTERISTICS

The flyover noise level established in compliance with FAR 36 is 76.75 dB(A).

PERFORMANCE

No Change.

SYSTEMS DESCRIPTION

The McCauley Model 4HFR34C754/94LA-0 is a four-bladed, full-feathering, constant-speed, counterweighted, variable-pitch, reversing propeller system. The system is controlled by engine oil pressure through single-acting, engine-driven governors. Operation of the system is identical to operation of the three-bladed propeller system with the exception of ground operation. Due to a limitation on minimum rotational speed for ground operation, an electrical solenoid is incorporated into the system to reset the propeller blades to a lower pitch (higher rpm) position whenever the airplane is on the ground. The solenoid is actuated into low pitch whenever the landing gear squat switch makes contact or the power levers are pulled aft of the flight idle detent. Operation of the solenoid may be verified during preflight by using the "GND IDLE STOP" test switch on the pilot's left subpanel, as discussed in NORMAL PROCEDURES.

HANDLING, SERVICING AND MAINTENANCE

No Change.

Approved: Donald H Peters

For W.H.Schultz
Beech Aircraft Corporation
DOA CE-2

For Training Purposes Only

BEECHCRAFT SUPER KING AIR B200, B200C, B200T, B200CT, and KING AIR C90A LANDPLANES PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the KING KNS-660 FLIGHT MANAGEMENT SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the King KNS-660 Flight Management System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The King Pilot's Guide for the KNS-660 Flight Management System, P/N 006-8394-00, dated August 2, 1984 or later, is available from King Radio Corp., 400 N. Rogers Road, Olathe, Kansas 66062. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status.

LIMITATIONS

The King KNS-660 Flight Management System meets the approval criteria for TACAN, VOR/DME and VLF/Omega Area Navigation Systems as established by Advisory Circular 90-45A and Advisory Circular 20-101B. Operation must be in conformity with the King KNS-660 Pilot's Guide, P/N 006-8394-00, dated August 2, 1984 or later.

The KNS-660 may be used as a means of VFR/IFR navigation provided the following limitations are observed:

If the data base has expired, IFR navigation is prohibited unless each waypoint and navaid is verified for accuracy by reference to current approved data.

Use of a TACAN sensor with VOR/TAC ground station is prohibited.

TACAN OR VOR/DME SENSOR

1. IFR RNAV approaches are approved for published FAA certified approaches.
2. The KNS-660 is approved as a two-dimensional VOR/DME or TACAN (not VOR/TAC) RNAV system for enroute, terminal area and approach operation in accordance with AC 90-45A and is not geographically limited.

VLF/OMEGA SENSOR

1. Only enroute operations in the conterminous United States and Alaska are approved for IFR.
2. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).

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3. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.

4. IFR flight is not approved based on VLF/Omega navigation into any area in which such operations could not be approved based on the installed equipment required by paragraph 3. This includes overwater flights beyond the range of approved ground navigation facilities.

5. Auto Update function must be verified or the VLF/Omega position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

   a. Prior to compulsory reporting points during IFR operations when not under radar surveillance or control.
   b. At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.
   c. Prior to and at hourly intervals during operation off of approved RNAV routes.

6. Navigation shall not be predicated on the use of this system during periods of dead reckoning or when a NAV flag is visible.

7. Following a period of dead reckoning, invalid operation (NAV flag visible), or operation in the Omega WARN mode, the VLF/Omega position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment. Position must also be updated if an accuracy check reveals a navigation error of greater than 2 NM.

VOR/DME AND VLF/OMEGA BLEND SENSOR

The KNS-660 is approved as a means of enroute navigation in accordance with AC 90-45A.

EMERGENCY PROCEDURES

1. If KNS-660 system information is intermittent or lost, utilize the remaining operational navigation equipment as required.

2. KNS-660 System Malfunction

   a. Press FMS/NAV CTL Switch - NAV CTL illuminated.
   b. CDU ON/OFF Switch - OFF
   c. KNS-660 Circuit Breaker - PULL
   d. VOR #1 - TUNE normally using the NAV 1 frequency control head. DME will be normally tuned by using the VOR receiver.
   e. Course Selector - MANUALLY SET.

NORMAL PROCEDURES

Normal operating procedures are outlined in the KNS-660 Pilot Guide, P/N 006-8394-00, dated August 2, 1984 or later revisions.

SYSTEM ANNUNCIATORS

1. Waypoint (WPT)

   Illuminates 90 seconds prior to each waypoint when operating in the OBS method of operation, when approaching the last waypoint in the flight plan or when going direct to a waypoint not in the flight plan.

   Illuminates 15 seconds prior to the automatic course change turn when operating in the AUTO/LEG method of operation.

2. Message (MSG)

   Identical to the message light on the Control Display Unit (CDU).

3. Dead Reckoning (DR)

   Illuminates when the system is dead reckoning.

4. X-Track (XTRK)

   Illuminates whenever a left or right cross track (parallel offset) is activated.

SYSTEM SWITCHES

1. Flight Management System (FMS)

   Activation of the FMS/NAV CTL switch (FMS illuminated) allows the KNS 660 navigation data to be displayed on the pilot's HSI and both RMI's. Active tuning of the VOR and DME sensors is now controlled by the KNS-660.

2. Navigation Control (NAV CTL)

   Activation of the FMS/NAV CTL switch (NAV CTL illuminated) allows basic NAV (VOR/LOC/GS) and DME data to be displayed on the pilot's HSI and VOR bearing on both RMI's. Active tuning of the VOR and DME sensors is now controlled by the pilot's NAV 1 frequency control head.

3. HSI Reference Data (OMEGA DIST/BRG)

   Activation of the HSI REF DATA switch (OMEGA DIST/BRG illuminated) allows bearing and distance to the active KNS-660 VLF/OMEGA waypoint to be displayed on the pilot's HSI bearing pointer and DME readout during operation in NAV CTL. VOR #1 drives the pilot's HSI course needle and both RMI's during this operation.

PILOT'S HSI

1. FMS Selected (FMS Illuminated and "blanked" active frequency display window in NAV 1 frequency control head).

   a. The KNS-660 provides automatic course needle drive while using the AUTO/LEG method of operation.

   b. The KNS-660 does not provide course needle drive while using the OBS method of operation. The pilot must set the selected course manually either via the course knob on the HSI or through the KNS-660 Control Display Unit.

   c. The deviation needle provides left/right steering information from the KNS-660.

   d. The DME display provides distance to the waypoint in nautical miles and the selection (via a pushbutton to the right of the DME display) of either groundspeed in knots or estimated time enroute in hours and minutes.

   e. The green RMI pointer provides bearing information derived from either the ADF or the KNS-660 depending upon whether ADF or NAV respectively are selected by the RMI selector.

2. NAV CTL Selected (NAV CTL Illuminated)

   a. The pilot must set the selected course manually via the course knob on the HSI.

   b. VOR/LOC/GS tuning is via the NAV 1 frequency control head.

c. VOR/LOC signals provide left/right steering information.

d. The DME display provides distance to the station in nautical miles and selection of either groundspeed in knots or estimated time enroute.

e. The green RMI pointer provides bearing information derived from either the ADF or the NAV 1 receiver depending upon whether ADF or NAV respectively are selected by the RMI selector.

PILOT'S AND COPILOT'S RMI

1. FMS Selected (FMS Illuminated and "blanked" active frequency display window in NAV 1 Frequency Control Head).

   The single bar (green) needle provides bearing information derived from either the ADF or the KNS-660 depending upon whether ADF or NAV respectively are selected on the RMI.

2. NAV CTL Selected (NAV CTL switch illuminated)

   The single bar (green) needle provides bearing information derived from either the ADF or the NAV 1 receiver depending upon whether ADF or NAV respectively are selected on the RMI.

AUTOPILOT/FLIGHT DIRECTOR COUPLING - No change

PERFORMANCE - No change

WEIGHT AND BALANCE - No change

SYSTEMS DESCRIPTION

The King KNS-660 is an integrated long range multi-sensor flight management system. It may be used to manage the entire range of navigational functions including trip planning, long range great circle navigation, and instrument approaches. The system consists of a cockpit mounted Control Display Unit (CDU) and a remote-mounted Navigation Computer. The KNS-660 serves as a computer processor/display for inputs from a choice of sensors such as VOR/DME, Omega/VLF, INS, TACAN or TAS.

The KNS-660 incorporates a King Data Base which must be updated every 28 days via diskette and a KDL-569 Data Loader. In addition, a self generated user data base may be loaded and maintained by the owner/operator through the CDU keyboard independent of the King Data Base. This data base consists of 175 storage locations accommodating any combination of ground stations and/or airports. If required, any number of these 175 user generated waypoints can be secured against cockpit alteration.

The KNS-660 will support 100 flight plans consisting of 25 alpha/numeric designated waypoints. These flight plans may be composed of any combination of waypoints taken from the data base or the user generated waypoints.

Dedicated special function keys are physically and visually separated from the alpha/numeric keys used in the generation and storage of flight plans and waypoints.

The KNS-660 provides for manual operation to selected waypoints or automatic operation providing uninterrupted navigation throughout a complete flight plan. Navigation sensors may be selected separately or blended within the computer.

On airplanes so equipped, the KNS-660 interfaces with the KRG-358 Radar Graphics Unit, providing a radar graphics presentation on the weather radar display. This includes such items as flight plan waypoints, selected course, and reference ground station.

Approved: Donald H. Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT SUPER KING AIR B200, B200C, B200T and B200CT
and
BEECHCRAFT KING AIR F90 (LA-202, LA-205 and after)
and C90A LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

FOSTER LNS616A LONG RANGE NAV SYSTEM

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Foster LNS616A Long Range NAV System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Foster Pilot's Guide for the LNS616A Long Range NAV System, P/N 002A0550, revised May, 1984 or later, is available from Foster Airdata Systems, Inc., 7020 Huntley Road, Columbus, Ohio 43229. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status.

LIMITATIONS

1. The Foster LNS616A Pilot's Guide, P/N 002A0550, revised May, 1984 or later, must be immediately available to the flight crew whenever navigation is predicated on the use of the LNS616A.
2. If the LNS616A data base has expired, IFR navigation is prohibited unless each REFerence is verified for accuracy against current approved data.
3. No flight operation shall be predicated on the LNS616A whenever a NAVigation flag is displayed in the pilot's HSI or whenever the system annunciates an unreliable condition.
4. The vertical guidance system (VNAV) is not approved for IFR.
5. The LNS616A must be checked for accuracy (reasonableness) prior to use as a means of navigation and following a period of operation with a NAV flag or aquisition of a new VORTAC or LORAN-C Group Repetition Interval (GRI).

VOR/DME ONLY (VO) MODE

1. The LNS616A is approved as a two-dimensional VOR/DME RNAV system in accordance with AC 90-45A for enroute terminal area and approach operation for IFR or VFR and is not geographically limited.
2. RNAV functions may be used only with co-located VOR/DME facilities.
3. IFR approaches must be conducted in the APProach mode in accordance with approved instrument approach procedures.
4. APProach mode is limited to approach operation with ground speeds of under 180 knots.

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LORAN ONLY (LO) OR LORAN/VOR/DME (LV) MODES (IF INSTALLED)

1. The LNS616A is approved as a two-dimensional LORAN-C navigation system in accordance with AC 20-121 for enroute and terminal area operations for IFR or VFR and is limited to the following LORAN-C coverage areas:

LORAN-C COVERAGE AREA	GRI
SOUTHEAST U.S. CHAIN	7980
GREAT LAKES CHAIN	8970
NORTHEAST U.S. CHAIN	9960
U.S. WEST COAST CHAIN	9940

Charts of the approved coverage areas are included in this supplement.

2. During LORAN-C operation, other navigation equipment required for the specific type of operation must be installed and operable.
3. The "LO" and "LV" modes are not approved for IFR approaches.

EMERGENCY PROCEDURES

1. If the NAV flag appears enroute, verify correct frequency and decoded station ident or utilize other navigation equipment as required.
2. If the NAV flag appears during approach, execute missed approach and utilize another approved navigation facility.

NORMAL PROCEDURES

Normal operating procedures are outlined in the Foster Pilot's Guide for the LNS616A, P/N 002A0550, revised May, 1984, or later.

SYSTEM ANNUNCIATORS

1. Waypoint (WPT)
   - Illuminates approximately three miles prior to waypoint passage

2. Parallel Track (PTK)
   - Illuminates whenever a parallel offset is established left or right of the desired track

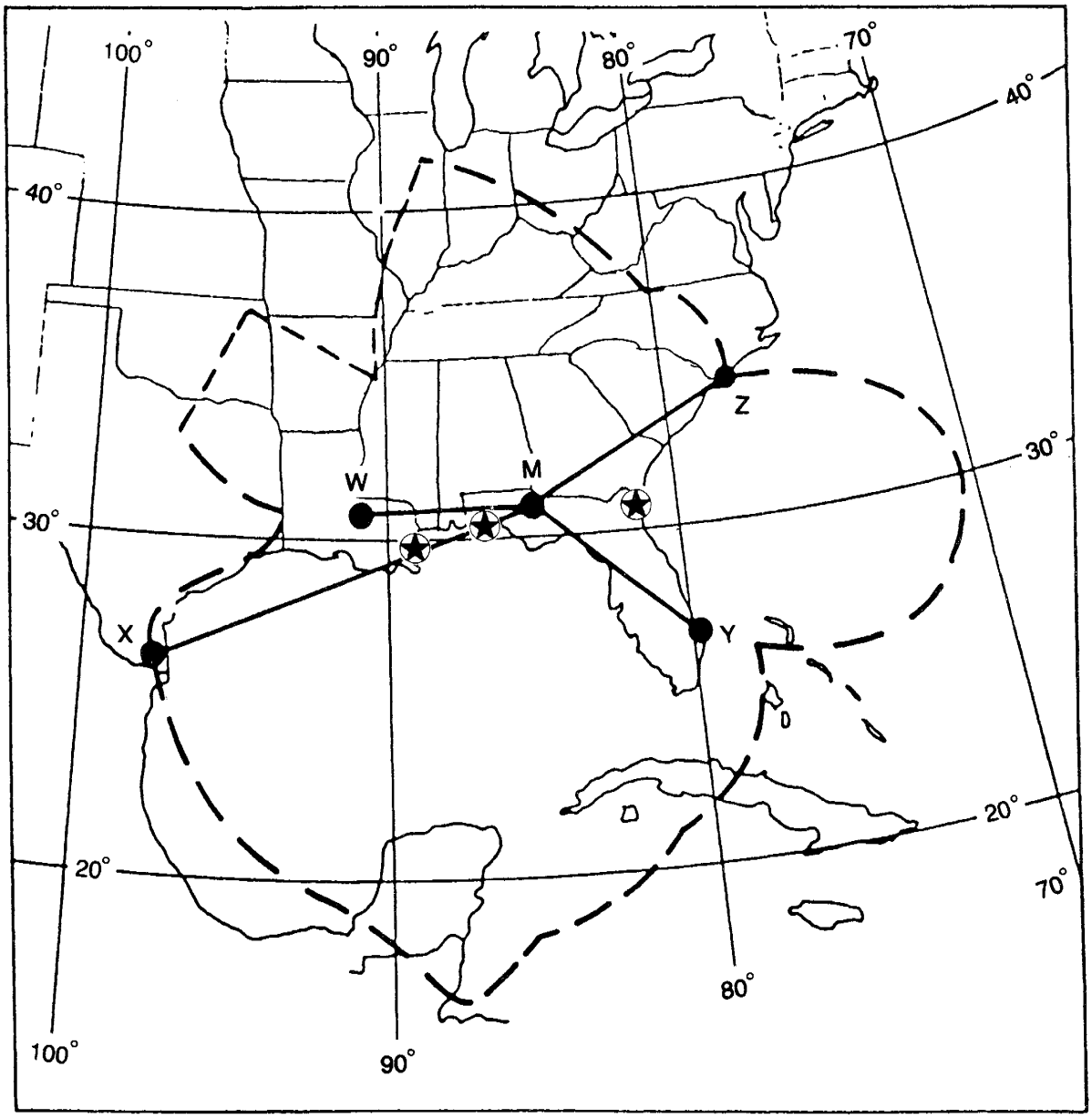
SYSTEM INTEGRATION

1. The LNS616A utilizes the #1 NAV control head to tune VOR/DME. LORAN-C is tuned by the LNS616A control head.
2. VOR/DME and RNAV steering information from the LNS616A is supplied to the pilot's HSI only. RMI indicators provide raw VOR information only.
3. The HSI will display RNAV information when the control head is in the "NAV", "TME/WND", "GRI", or "AUX" position. RNAV information will also be supplied in the "PRG" and "REF" mode when these modes are entered from the "NAV" mode. When "REF" and "PRG" are entered from the "VOR/LOC" mode, VOR or Localizer steering information will be displayed.
4. The HSI will display the following LNS616A system derived information:

   a. OBS Course Arrow
      - VOR mode selects course to VOR
      - NAV mode selects course to Waypoint

LORAN-C

SOUTHEAST U.S. CHAIN ——————— GRI 7980



Charts reproduced courtesy of the Department of Transportation

LEGEND:

- ● TRANSMITTING
- ⍟ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	MALONE
W	GRANGEVILLE
X	RAYMONDVILLE
Y	JUPITER
Z	CAROLINA BEACH

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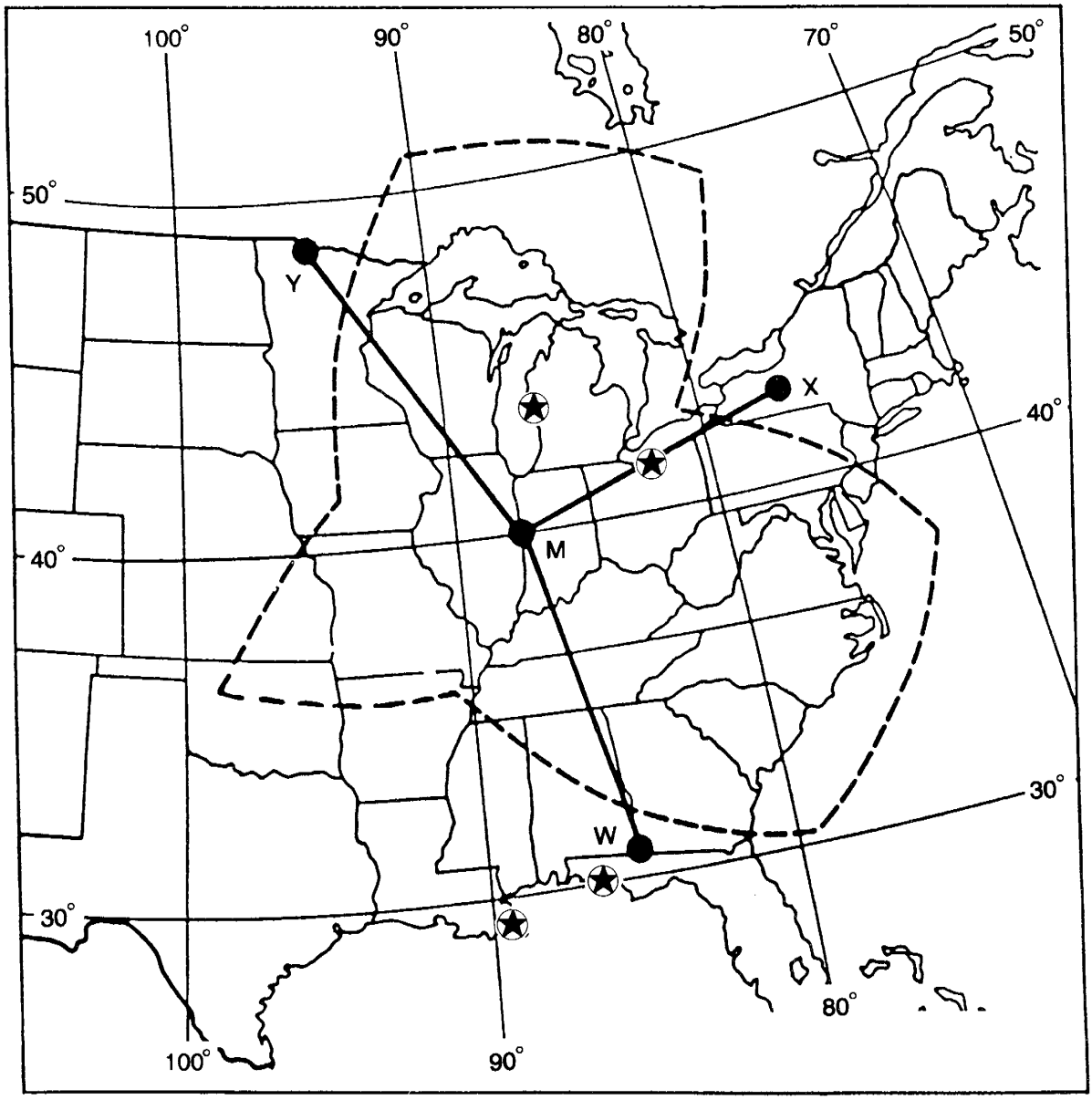
LORAN-C

GREAT LAKES CHAIN ——— GRI 8970



Charts reproduced courtesy of the Department of Transportation

LEGEND:

● TRANSMITTING

✪ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	DANA
W	MALONE
X	SENECA
Y	BAUDETTE

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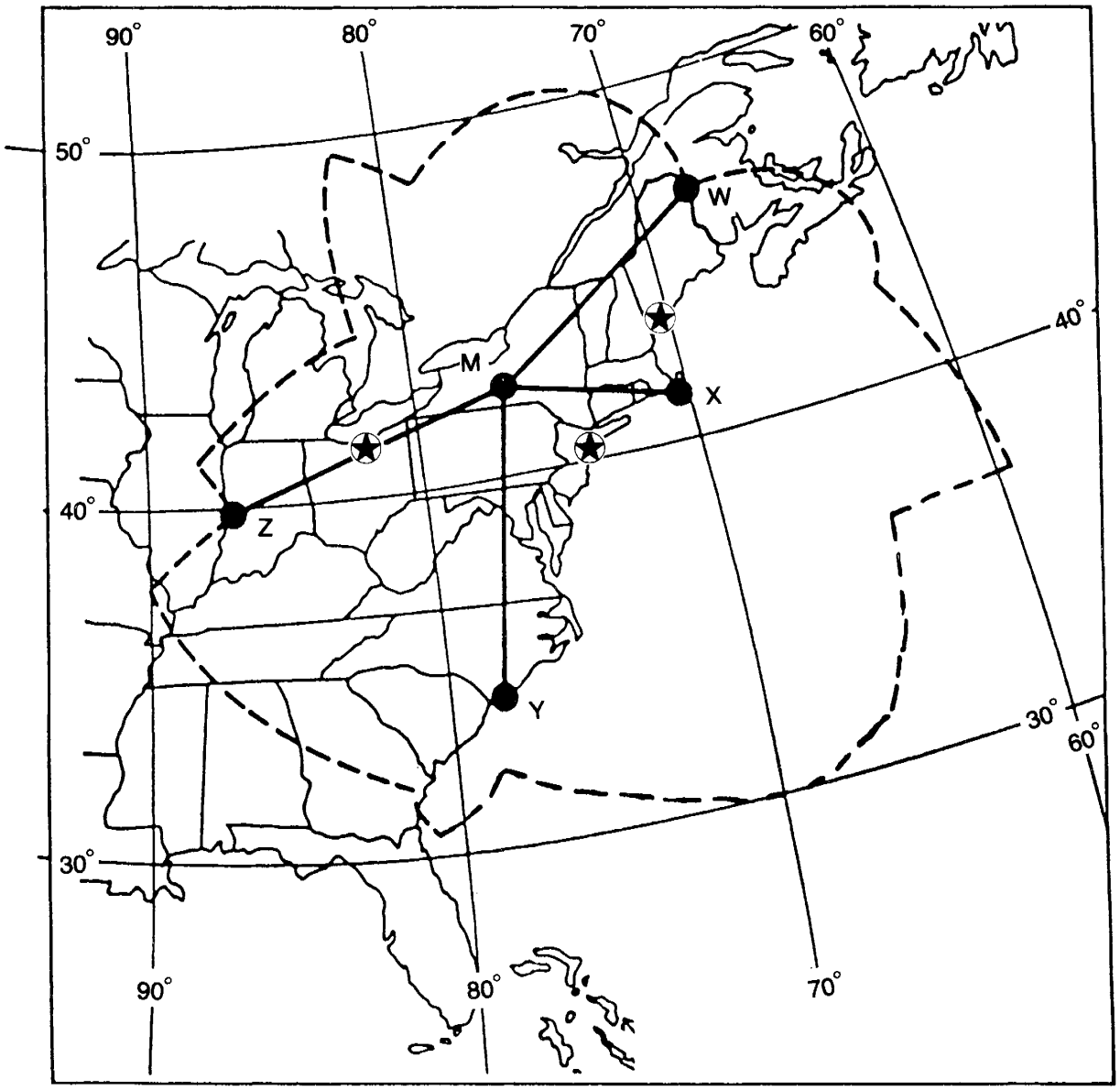
LORAN-C

NORTHEAST U.S. CHAIN ——— GRI 9960



Charts reproduced courtesy of the Department of Transportation

LEGEND:

● TRANSMITTING

✪ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	SENECA
W	CARIBOU
X	NANTUCKET
Y	CAROLINA BEACH
Z	DANA

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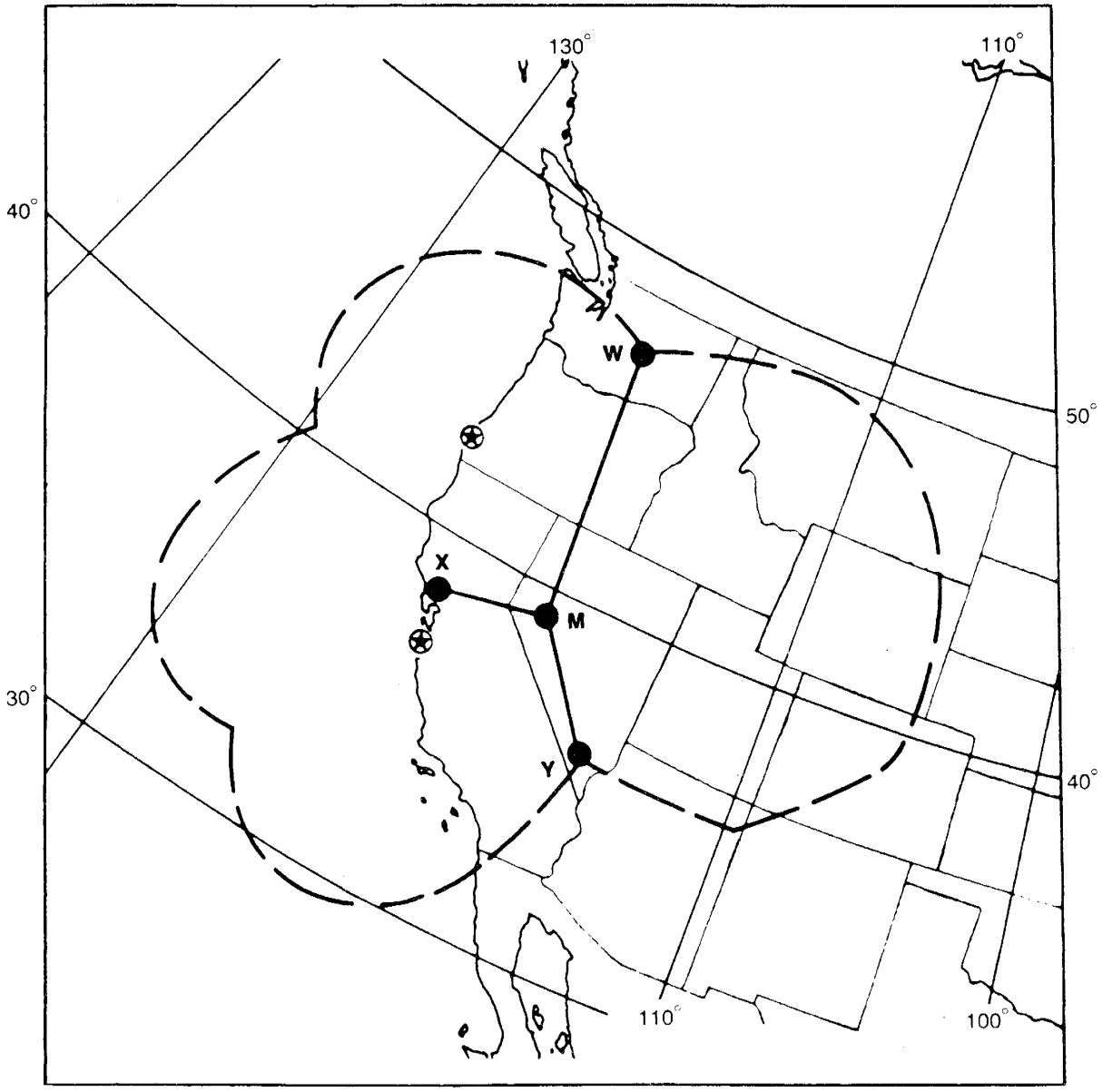
LORAN-C

U.S. WEST COAST CHAIN ——— GRI 9940



Charts reproduced courtesy of the Department of Transportation

LEGEND:

● TRANSMITTING

✪ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	FALLON
W	GEORGE
X	MIDDLETOWN
Y	SEARCHLIGHT

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b. CDI - Left or Right course information with full scale sensitivity as follows:
"VOR" Mode - 10 degrees
"NAV" Mode - a) APPROACH - 1.25 NM
b) ENROUTE - 5.0 NM

c. To/From Symbol
- The selected course will take the airplane To/From the VOR or waypoint.

d. NAV Flag
- LNS616A Navigation is unreliable.

5. When operating in "LEG" mode, the pilot must manually set the course indicator to agree with desired track (DTK) appearing on the NAV page of the LNS616A control head unless the installation includes an automatic slewing capability.

FAILURES AND WARNINGS

The LNS616A Long Range Navigation System annunciates failures or conditions in which LORAN-C navigation is unreliable in the following manner:

1. The LNS616A constantly monitors signal-to-noise ratio, and displays signal quality (MWXYZ) on a scale from 0 to 9 (best case is 9, worst case is 0) on the control head. Use only stations which display signal-to-noise ratings of 3 or greater.
2. Geometry quality (Q) is constantly monitored and displayed in "GRI" mode on the control head on a scale from 0 to 9 (best case is 9, worst case is 0). Low "Q" indicates marginal geometry when station ratings are acceptable.
3. Excessive range is indicated by a low signal-to-noise number and a low "Q" number. Use only stations which display signal-to-noise ratios of 3 or greater.
4. A blink condition is indicated by a signal-to-noise ratio of "0" for the appropriate station that the computer has detected a blink condition.
5. A steering failure will be indicated by a NAV flag on the HSI.

SYSTEM OPERATION WITHOUT LORAN-C INSTALLED

1. When LNS616A's LORAN-C receiver is not installed, "LO" mode causes the HSI to flag and navigation computations of LNS616A control head to cease functioning. The initial display after turn-on will also delete the LORAN-C software revision levels, "N 0.0 C 0.0" (bottom display line), when the receiver is not installed.
2. Without LNS616A LORAN-C receiver installed, "POS"pushbutton will only display the airplane's position relative to received VORTAC stations.
3. GRI mode is ineffective in the use of stored GRI data (PAGE ONE and PAGE TWO) unless the LNS616A remote LORAN-C receiver is installed. When the LNS616A LORAN-C receiver is not installed, GRI mode is operationally useful only to display the airplane's position in Latitude/Longitude when "POS" pushbutton is pressed.
4. LNS616A installations without LORAN-C receivers may use the "POS" pushbutton in all modes except "GRI" and "AUX" to display the decoded Morse code IDENT in the upper left of the display, and the airplane's radial and distance from the received VORTAC station. In "GRI" and "AUX" mode, pressing "POS" displays the airplane's Latitude/Longitude and during "REF" mode, pressing "POS" pushbutton programs the airplane's Latitude/Longitude into REF "PPP". If IDENT, VOR or DME signals are not available, pressing "POS" displays bars (---) in the associated display elements.
5. Messages relating to LORAN-C operation do not apply when LORAN-C receiver component of the LNS616A system is not installed.
6. DATA control of GRI pages may be implemented as described, but will not result in operational use without the LORAN-C receiver installed or operational in the LNS616A system.
7. When the LORAN-C receiver portion of LNS616A is not installed or is inoperative, the LAT/LNG page of "AUX" mode shows bars (---).

REFERENCE DATA BASE

It is the pilot's responsibility to ensure that current and accurate VORTAC reference data is being used for navigation and that the current data base cartridge is installed in the D1681 Database/Interface Unit. Effective date for the pre-programmed navigation references may be determined in two ways:

1. Upon turning on the LNS616A system, Software Revision Level and Database Effective Date will be displayed. The bottom line of the third display indicates chart effective date expressed in month, day and year (i.e., 05/23/84). Database is valid for 56 days following effective date.

2. Database Chart Effective date is also printed on the 61WDC Database Cartridge which is installed in the D1681 Database/Interface unit located in the nose avionics compartment.

AUTOPILOT/FLIGHT DIRECTOR COUPLING

No change.

PERFORMANCE

No change.

WEIGHT AND BALANCE

No change.

Approved: Donald H Peters

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

Beechcraft®

Super King Air® 200, 200C, 200T, 200CT, B200, B200C, B200T & B200CT Landplanes

Pilot's Operating Handbook and

FAA Approved Airplane Flight Manual Supplement

for the

Lear-Siegler Model 23085-001 300-Amp Starter/Generator

This Supplement is applicable to the following manual(s):

101-590010-127, 101-590010-147, 101-590037-3, 101-590037-29 & 101-590037-57

Airplane Serial Number: _____.

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane is equipped with Lear-Siegler Model 23085-001 300-amp Starter/Generators in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

STARTER LIMITS

For cranking and engine clearing, use of the starter is limited to 30 seconds ON, 3 minutes OFF, 30 seconds ON, 30 minutes OFF. For engine wash, use of the starter is limited to 30 seconds ON, 15 minutes OFF (soak), 30 seconds ON, 10 minutes OFF, 30 seconds ON, 10 minutes OFF, 30 seconds ON, 30 minutes OFF.

GENERATOR LIMITS

GENERATOR LOAD	MINIMUM GAS GENERATOR RPM - N_1		
	GROUND - AIR CONDITIONING OFF	GROUND - AIR CONDITIONING ON *	IN FLIGHT (ALL ALTITUDES)
0% - 75%	IDLE	62%	IDLE
75% - 100%	63%	68%	85%

* Right engine only

BT03853

EMERGENCY PROCEDURES

No Change

ABNORMAL PROCEDURES

No Change

NORMAL PROCEDURES

No Change

PERFORMANCE

No Change

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WEIGHT AND BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

No Change

HANDLING, SERVICING AND MAINTENANCE

No Change

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P/N 101-590010-229

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BEECHCRAFT SUPER KING AIR B200 AND B200C LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the AUXILIARY POWER UNIT KIT NO. 101-5079

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA-Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary Power Unit, BEECHCRAFT Kit No. 101-5079, in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

— The auxiliary power unit is not approved for in-flight operation.

EMERGENCY PROCEDURES

ILLUMINATION OF APU FIRE - PUSH ANNUNCIATOR/SWITCH

1. Press APU FIRE - PUSH annunciator/switch.
2. Move APU MASTER switch to the OFF position.
3. Evacuate the airplane.

NORMAL PROCEDURES

CAUTION

Prior to turning airplane Battery Master switch ON, ensure that the APU MASTER switch is in the OFF position and that the APU FAIL - ON - OFF and GEN FAIL - ON - OFF switches are disengaged.

Prior to Actuation of the Auxiliary Power Unit:

1. Move the Battery Master switch to the ON position.
2. APU MASTER Switch - ARM
3. FIRE TEST Button - DEPRESS (APU FIRE - PUSH annunciator will illuminate)

To Actuate the Auxiliary Power Unit:

1. Move the APU MASTER switch to the ARM position.
2. Press the APU FAIL - ON - OFF annunciator/switch. The segment placarded ON will illuminate and the APU will be activated.
3. Depress RPM Monitor Switch.
4. When the APU rpm stabilizes, press the GEN FAIL - ON - OFF annunciator/switch. The segment placarded ON will illuminate and the generator will come on line.

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5. APU digital indications on control head:

   a. RPM when RPM switch is depressed
   b. EGT when TEMP switch is depressed
   c. Electrical load in amps as a normal indication

CAUTION

APU generator must remain on line during APU operation.

To Actuate the Auxiliary Air Conditioner:

— Move the AIR COND - OFF - ON switch to the ON position.

To Deactivate the Auxiliary Air Conditioner:

— Move the AIR COND - OFF - ON switch to the OFF position.

To Deactivate the Auxiliary Power Unit:

1. Press the GEN FAIL - ON - OFF annunciator/switch. The segment placarded OFF will illuminate and the generator will go off line.
2. Press the APU FAIL - ON - OFF annunciator/switch. The segment placarded OFF will illuminate and the auxiliary power unit will be deactivated.
3. Move the APU MASTER switch to the OFF position.

The Auxiliary Power Unit may also be deactivated by:

1. Pressing the O/S TEST switch.
2. Pressing the FIRE TEST switch.
3. Pressing the APU FIRE - PUSH annunciator/switch.
4. Moving the APU MASTER switch to OFF.

CAUTION

If the APU is deactivated by other than normal procedures, ensure that the APU FAIL - ON - OFF switch is disengaged and the APU MASTER switch is turned OFF.

PERFORMANCE - No change

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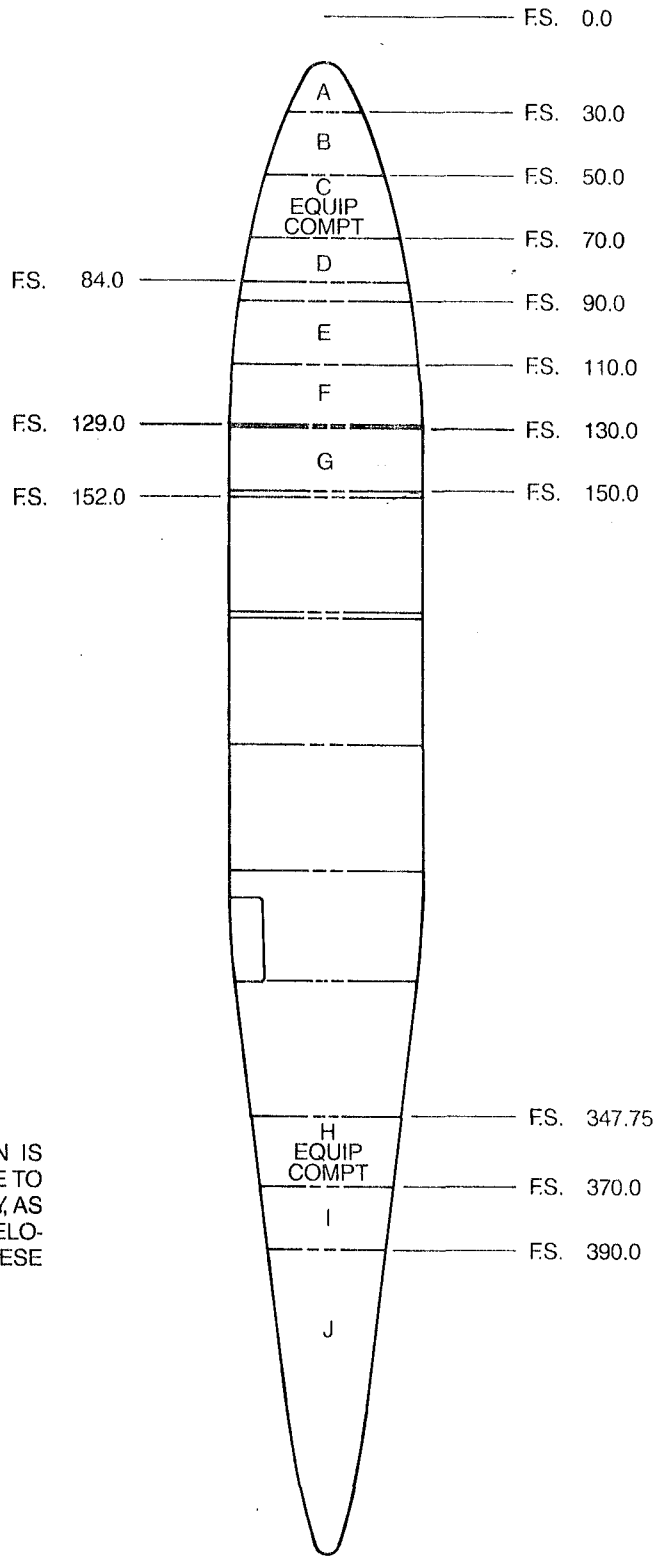
LOADING LIMITS - SUPER KING AIR B200

SECTION	MAXIMUM STRUCTURAL CAPACITY	CENTROID ARM
A	30 LB	F.S. 20
B	50 LB	F.S. 40
C	110 LB	F.S. 60
D	190 LB	F.S. 80
E	80 LB	F.S. 100
F	225 LB	F.S. 120
G	545 LB	F.S. 140
H	95 LB	F.S. 360
I	210 LB	F.S. 380
J	50 LB	F.S. 430

WARNING

WEIGHT OF THE AUXILIARY POWER UNIT INSTALLATION IS COUNTED AGAINST THE WEIGHT LIMITS LISTED ABOVE. DUE TO VARIATIONS IN ORIGINAL EQUIPMENT, IT MAY BE NECESSARY, AS PART OF THE AUXILIARY POWER UNIT INSTALLATION, TO RELOCATE OTHER COMPONENTS SO AS NOT TO EXCEED THESE LIMITS.



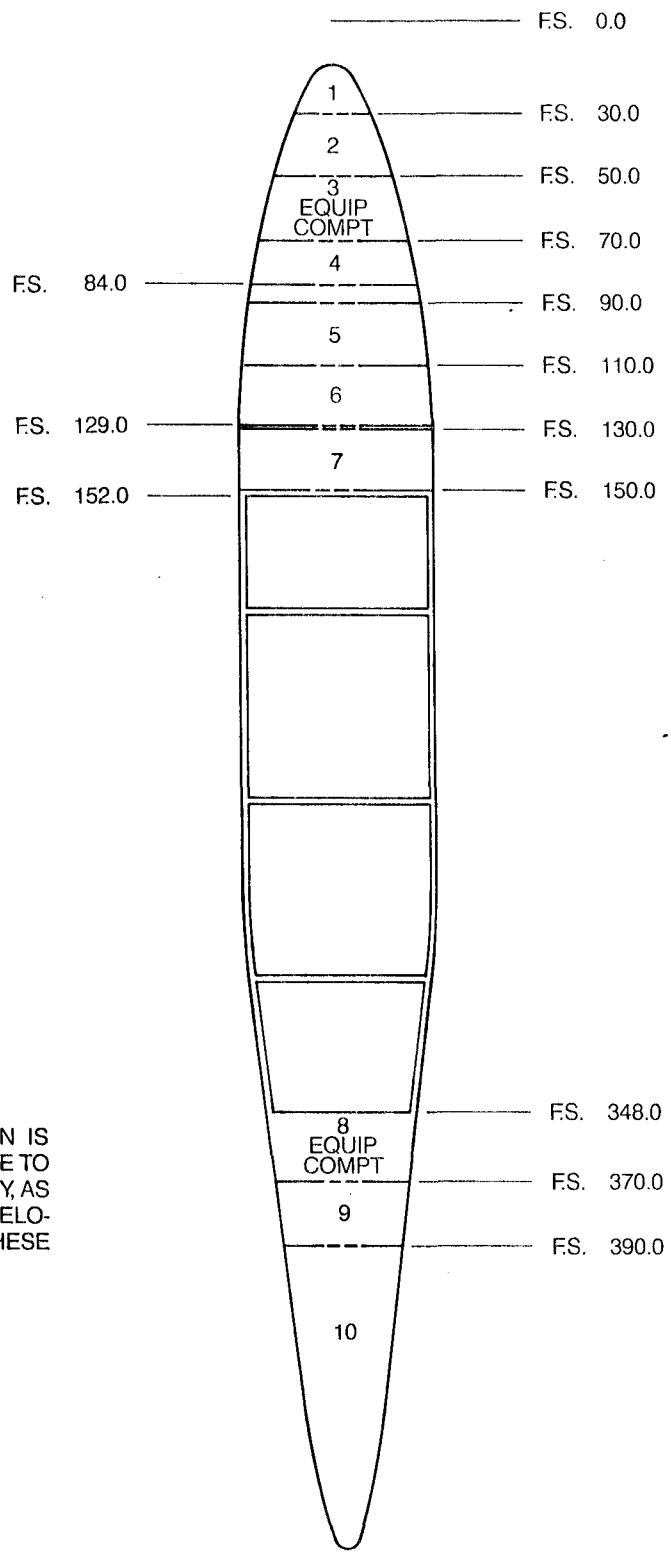
B200-603-74

LOADING LIMITS - SUPER KING AIR B200C

SECTION	MAXIMUM STRUCTURAL CAPACITY	CENTROID ARM
1	30 LB	F.S. 20
2	50 LB	F.S. 40
3	110 LB	F.S. 60
4	190 LB	F.S. 80
5	80 LB	F.S. 100
6	225 LB	F.S. 120
7	634 LB	F.S. 141
8	95 LB	F.S. 360
9	210 LB	F.S. 380
10	50 LB	F.S. 430

WARNING

WEIGHT OF THE AUXILIARY POWER UNIT INSTALLATION IS COUNTED AGAINST THE WEIGHT LIMITS LISTED ABOVE. DUE TO VARIATIONS IN ORIGINAL EQUIPMENT, IT MAY BE NECESSARY, AS PART OF THE AUXILIARY POWER UNIT INSTALLATION, TO RELOCATE OTHER COMPONENTS SO AS NOT TO EXCEED THESE LIMITS.



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SYSTEMS DESCRIPTION

Refer to Installation, Operation, Maintenance and Overhaul Instructions, Gemini Turbine-Driven Auxiliary Power Unit Category II, Class C, P/N T-20G-10C3 included in BEECHCRAFT Kit No. 101-5079.

HANDLING, SERVICING AND MAINTENANCE

COOLING SYSTEM

Refer to the SuperKing Air Model 200 Series maintenance manual for procedures relative to evacuating the air conditioner refrigerant system, system cleaning and preparation for freon charging.

Gain access to the expansion valves at each evaporator coil and adjust the valves to a full open position. This is accomplished by removing the adjustment knob cover and turning the adjustment knob clockwise until the knob is screwed into the valve body as far as it will go by hand tightening only.

Replace the adjustment knob cover and any panel assemblies or ducts that would have been removed to facilitate adjustment.

After the freon system has been evacuated, charge the system with the APU running, the compressor clutch engaged and the evaporator blower on.

Monitor the freon flow through the sight glass located in the freon line immediately aft of the pressure bulkhead. Charge the system with Refrigerant 12, approximately 1lb. 5oz., until the sight glass clears.

AUXILIARY POWER UNIT

For information relative to maintaining and servicing the APU refer to the Turbomach Installation/Maintenance Manual supplied with the APU.

Approved: Donald H Pate

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT SUPER KING AIR B200C LANDPLANE
(Serial BL-124 only)

PILOT'S OPERATING HANDBOOK

AND

FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

BEECH MODIFIED U.S. FOREST SERVICE SURVEILLANCE AIRPLANE

GENERAL

The Information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified for United States Forest Service Surveillance missions in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth below. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The U.S. Forest Service Surveillance airplane is a BEECHCRAFT Super King Air B200C with high flotation gear modified for use in fire surveillance. It is certified in the normal category for two modes of operation: the surveillance mode and the transport mode. In either mode, the airplane has the following external modifications: aft fuselage strakes, a rudder trailing edge modification, and an omni-directional downlink antenna mounted on the bottom of the fuselage.

In the surveillance mode, the following removable external modifications are utilized: an IR (infrared) scanner mounted in the bottom of the fuselage, surrounded by a fiberglass fairing. The fairing has an electrically operated scanner door that covers the IR viewing aperture, and contains an image drop tube assembly provided for the dropping of canisters from the airplane. The anti-collision light, normally mounted on the bottom of the fuselage, is replaced with a streamlined version mounted on the bottom of the fairing.

In the transport mode, the IR scanner, drop tube assembly, and the fairing described above are removed. The cover plates for the IR scanner and drop tube assembly openings are installed. A lower anti-collision light is mounted on a standoff on the bottom of the fuselage. Light dams for the anti-collision light are also installed.

LIMITATIONS

AUTOFEATHER

Must be operational and armed for all takeoffs, climbs, and approaches.

EMERGENCY PROCEDURES

ELECTRICAL SYSTEM FAILURE

GENERATOR INOPERATIVE (DC GEN annunciator ON)

NOTE

AC and DC electrical power to the mission equipment will automatically be shed with the loss of either or both generators, and cannot be reset. This will be evidenced by the extinguishing of the annunciators on the Scanner Power, Cabin DC, and RCDR PWR switches that are located on the pedestal.

1. Generator Switch - OFF, then to RESET for a minimum of 1 second, then release to ON.
2. Scanner Power, Cabin DC, and RCDR PWR Annunciator Switches - PUSH OFF, then PUSH ON to restore power to mission equipment.

If generator will not reset:

3. Generator Switch - OFF
4. Scanner Power, Cabin DC, and RCDR PWR Annunciator Switches - PUSH OFF
5. Operating Generator - DO NOT EXCEED 1.0 LOAD

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NORMAL PROCEDURES

CLIMB

— Autofeather - ARM

IN FLIGHT

OPENING OR CLOSING THE SCANNER DOOR

1. Scanner Door Annunciator Switch - PUSH (Scanner Door OPEN annunciator will flash approximately three times as scanner door opens or closes.)
2. Scanner Door Annunciator - CHECK OPEN or CLOSED - AS APPROPRIATE

NOTE

The scanner door will close automatically when the landing gear is lowered. A blinking Scanner Door OPEN annunciator indicates that the scanner door is not fully open or closed.

DROPPING CANISTERS FROM IMAGE DROP TUBE

NOTE

Successful drops have been demonstrated at the following speeds and flap configurations. The lowest speed at each flap setting provides approximately a 30% margin above the power off stall speed.

KIAS	FLAP SETTING
100 - 114	DOWN
115 - 129	APPROACH
130 - 160	UP

1. Cabin - DEPRESSURIZE
2. Access Cover - REMOVE
3. Wind Deflection Tube - INSERT IN IMAGE DROP TUBE

Approaching Drop Point:

4. Drop Tube Annunciator Switch - PUSH. READY annunciator on switch will illuminate along with READY light located by drop tube.

At Drop Point:

5. Drop Switch (Pilot's Left Grip) - PRESS. READY annunciator and READY light will extinguish, and the DROP annunciator and DROP light will illuminate.
6. Drop Tube Annunciator Switch - PUSH to extinguish DROP annunciator and DROP light.

MISSION EQUIPMENT ELECTRICAL POWER

To Provide Three-Phase AC Power to Cabin:

1. Scanner Power Annunciator Switch - PUSH ON (Aϕ, Bϕ, Cϕ, and INV DC annunciators - ILLUMINATED)

To Terminate Power to Cabin:

2. Scanner Power Annunciator Switch - PUSH OFF (Aϕ, Bϕ, Cϕ, and INV DC annunciators - EXTINGUISHED)

NOTE

When switching power OFF, the INV DC annunciator will remain illuminated for approximately 5 seconds.

To Provide DC Power to Cabin:

1. Cabin DC Annunciator Switch - PUSH ON (CABIN DC annunciator - ILLUMINATED)

To Terminate Power to Cabin:

2. Cabin DC Annunciator Switch - PUSH OFF (CABIN DC annunciator - EXTINGUISHED)

NOTE

The annunciator will not illuminate if the DC IND circuit breaker located on the right cabin circuit breaker panel is out.

To Provide Single-Phase AC Power to Cabin:

1. RCDR PWR Annunciator Switch - PUSH ON (INV DC and 115 VAC annunciators - ILLUMINATED)

To Terminate Power to Cabin:

2. RCDR PWR Annunciator Switch - PUSH OFF (INV DC and 115 VAC annunciators - EXTINGUISHED)

NOTE

When switching power OFF, the INV DC annunciator will remain illuminated for approximately 1 second.

PERFORMANCE

For performance estimation, refer to the *High Flotation Landing Gear* supplement found in the SUPPLEMENTS Section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

Notes have been provided as follows to approximate performance of the Beech Modified U.S. Forest Service Super King Air B200C for two modes of operation: the surveillance mode and the transport mode.

The performance is estimated by entering a table or graph at a higher indicated pressure altitude, or by changing the value derived from it by a specified percentage or fixed amount. The resulting performance is approximate and will vary with airspeed, temperature, and other ambient conditions.

	SURVEILLANCE MODE	TRANSPORT MODE
Take-off Weight - Flaps 0%: Before entering graph, increase indicated pressure altitude by	1000 FT	600 FT
Take-off Weight - Flaps 40%: Before entering graph, increase indicated pressure altitude by	1000 FT	800 FT
Take-off Distance - Flaps 0%: Increase total distance over 50-foot obstacle by	8%	6%
Accelerate-Stop - Flaps 0%: Increase field length by	6%	5%
Accelerate-Go - Flaps 0%: Before entering graph, increase indicated pressure altitude by	1600 FT	1400 FT
Take-off Distance - Flaps 40%: Increase total distance over 50-foot obstacle by	8%	6%
Accelerate-Stop - Flaps 40%: Increase field length by	6%	5%
Accelerate-Go - Flaps 40%: Before entering graph, increase indicated pressure altitude by	1800 FT	1600 FT
Climb - Two Engines - Flaps 0%: Reduce rate-of-climb by	100 FPM	100 FPM
Climb - Two Engines - Flaps 40%: Reduce rate-of-climb by	100 FPM	100 FPM
Climb - One-Engine-Inoperative: Reduce rate-of-climb by	80 FPM	40 FPM
All Two-Engine Cruise tables & graphs: Reduce TAS by	5%	4%
All Range Profile graphs: Reduce range values by	5%	4%
One-Engine-Inoperative Maximum Cruise Power tables: Reduce TAS by	7%	4%
Climb - Balked Landing: Reduce rate-of-climb by	125 FPM	100 FPM

The following performance information is not applicable to the U.S. Forest Service Modified airplane in either mode of operation:

- Maximum Enroute Weight (FAR 135 Operations)
- Take-off Weight - Flaps 0% - To Meet FAR 25 Take-off Climb Requirements
- Take-off Weight - Flaps 40% - To Meet FAR 25 Take-off Climb Requirements
- Take-off Flight Path
- Net Gradient of Climb - Flaps 0%
- Net Gradient of Climb - Flaps 40%
- Take-off Distance - Flaps 0%, Grass Surface
- Take-off Distance - Flaps 40%, Grass Surface
- Service Ceiling - One-Engine-Inoperative
- Time, Fuel, and Distance to Climb
- Time, Fuel, and Distance to Descend

All performance information not mentioned in this Supplement remains valid as presented in the Pilot's Operating Handbook.

WEIGHT & BALANCE

See Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

SYSTEMS DESCRIPTION

MISSION EQUIPMENT PEDESTAL PANEL

Lighting for the panel is controlled by rheostat switches located on the overhead light control panel and on the pilot's instrument panel.

The mission equipment pedestal panel contains the following annunciator switches.

Scanner Door

Activation of the Scanner Door switch, placarded OPEN (amber) and CLOSED (green), causes the scanner door to open or close. While the door is in transit, or stops before reaching the full open or closed position, the OPEN annunciator will blink.

When the LDG GEAR CONT switch is moved to the down position, the scanner door automatically closes and the Scanner Door switch becomes inoperative. Control of the Scanner Door switch can be regained by pulling the LANDING GEAR RELAY circuit breaker, located next to the LDG GEAR CONT switch.

A Ground Maintenance switch is provided on the right side of the IR scanner bay, below the cabin access cover. Raising the switch cover and lifting the switch will open the scanner door. Anytime the access cover is reinstalled, a striker plate attached to the bottom of the cover will automatically move the ground maintenance switch to the door-closed (DOWN) position if it was inadvertantly left in the up position.

NOTE

The LANDING GEAR RELAY circuit breaker must be in for operation of the Ground Maintenance switch.

Scanner Power

The Scanner Power switch provides DC power to the 3000-VA 3-phase inverter located in the aft avionics bay. Proper system operation will cause the green INV DC portion of the annunciator to illuminate, indicating that DC power is reaching the inverter. Illumination of the amber Aϕ, Bϕ, and Cϕ portions of the annunciator indicates proper output from the inverter.

Drop Tube

The Drop Tube switch is used to give visual signals to the person that manually drops canisters from the image drop tube assembly. Initial activation of the switch illuminates the green READY annunciator on the switch, and the green READY light located next to the drop tube. Pressing the lower switch on the pilot's left grip extinguishes the READY annunciator and light, and simultaneously illuminates the amber DROP annunciator and amber DROP light. Pushing the drop tube switch a second time cancels the DROP annunciator and light.

Cabin DC — The Cabin DC switch provides 28-VDC power to the mission equipment. Proper operation is indicated by illumination of the green CABIN DC annunciator.

RCDR PWR — The Recorder Power switch contains a split annunciator: a green INV DC portion and an amber 115 VAC portion. In normal operation, actuation of the switch will illuminate both annunciators. The INV DC annunciator indicates DC power is being delivered to the 1000-VA, single-phase inverter located in the aft avionics bay, and the 115 VAC annunciator indicates proper output of the inverter to operate the image recorder.

CABIN ACCESS COVERS

The IR scanner and drop tube access covers are installed in the center aisle floorboards at approximately F.S.198 and F.S.227 respectively. After depressurizing the cabin, these covers can be removed to gain access to the IR scanner head and the image drop tube. An image drop tube deflector is provided for insertion into the image drop tube prior to dropping the U.S. Forest Service canisters.

OBSERVER'S NAVIGATION PANEL

An observer's navigation panel is installed on the left cabin sidewall. The panel contains three displays: an altitude display, an airplane heading display, and a distance/groundspeed display.

ALTITUDE DISPLAY

The altitude display utilizes the copilot's pitot and static systems, an OAT probe located on the forward belly, and a computer, to provide a readout of true airspeed, pressure altitude, true OAT in °F, and time in hours and minutes.

The altitude readout indicates airplane pressure altitude within a tolerance of ±300 feet, and may be biased to a reference other than 29.92 in. Hg by utilizing the BARO SET UP or DOWN adjustment buttons located to the left of the readout. Anytime electrical power to the display (provided by the hot battery bus) is interrupted, the altitude readout will return to the 29.92 in. Hg reference.

The time readout, located below the temperature readout, can be adjusted using the three CLOCK SET adjustment buttons located below the readout.

A DIM adjustment button, located to the upper left of the display can be pressed to change the display brightness through four distinct levels.

AIRPLANE HEADING DISPLAY

The airplane heading display is a digital remote display from the copilot's compass system.

DISTANCE/GROUNDSPEED DISPLAY

The distance/groundspeed display reflects the airplane distance, groundspeed, and TTS (time-to-station) to the waypoint. Groundspeed and TTS must be alternately selected using a pushbutton on the display. Information for this display is provided only from the King KNS-660 Flight Management System and is displayed independently of the information being presented to the pilot's HSI.

OBSERVER'S COMMUNICATION PANEL

Adjacent to the observer's navigation panel is the observer's communication panel. This panel, in conjunction with the headphone jacks located below it, allows the observer to communicate inside and outside of the airplane.

The upper row of ON-OFF switches control the observer's headset. The HOT INTPH (interphone) switch must be turned ON to communicate inside the airplane. In addition, the person communicated with must also have his HOT INTPH switch ON. The MIC switch is not keyed when using HOT INTPH.

The ANNUN TEST switch produces all eights in the heading display and illuminates the Scanner Door annunciator, located on the lower row of switches.

The lower row of switches control which radio the observer transmits on, the headphone volume, and the sensitivity of the hot mic. The floating mic switch is used for communication outside of the airplane.

A lighting rheostat, placarded ANN PUSH BRIGHT, controls the brightness of the heading display, the Scanner Door annunciator, and the backlighting of the communication panel. The heading display has two levels of brightness: bright when the rheostat switch is pushed in, and dim when it is pulled out. The rheostat will variably dim the communications panel backlighting whether it is pushed in or pulled out. The rheostat will dim the Scanner Door annunciators and READY/DROP lights located by the drop tube only when pulled out.

MISSION EQUIPMENT POWER DISTRIBUTION

The following circuit breakers are installed to support the mission equipment:

On Cabin Right Circuit Breaker Panel:

Cabin Display (5-amp)	If this circuit breaker is in the out position, the true airspeed and airplane heading displays located on the observer's navigation panel will be inoperative.
Scanner (25-amp)	Self Explanatory
Downlink (25-amp)	Self Explanatory
Television (50-amp)	Self Explanatory
DC IND (1-amp)	This circuit breaker provides protection only for the Cabin DC annunciator switch. If this circuit breaker is out, the CABIN DC annunciator will extinguish.
1 Phase (10-amp)	This circuit breaker provides protection for the AC circuit from the 1000-VA inverter mounted in the aft avionics bay. If this breaker is out, the 115 VAC annunciator on the RCDR PWR annunciator switch will extinguish and the AC circuit will become inoperative.
3 Phase (10-amp)	This circuit breaker protects the AC circuit from the 3000-VA inverter. If this circuit breaker is out, the Aϕ, Bϕ, and Cϕ annunciators, located on the scanner power switch, will extinguish and the AC circuit will become inoperative.
IRFSS Control (5-amp)	This circuit breaker, located on the copilot's right side panel, controls all AC and DC power to the mission equipment. With this circuit breaker pulled out, the annunciators on the Scanner Power, Cabin DC, and RCDR PWR switches will extinguish.

Approved: Donald H. Peter

for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR B200 LANDPLANE
PILOT'S OPERATING HANDBOOK AND FAA APPROVED
AIRPLANE FLIGHT MANUAL SUPPLEMENT
for the
BEECH MODIFIED
CHINESE DUAL CAMERA WELL AERIAL SURVEILLANCE AIRPLANES
(BB-1204 & BB-1205 ONLY)
and
CHINESE FLIGHT INSPECTION AIRPLANE
(BB-1206 ONLY)

GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA-Approved Airplane Flight Manual when the airplane has been modified for Aerial Surveillance or Flight Inspection purposes as applicable in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Aerial Surveillance configuration includes dual camera wells protected by electrically actuated sliding doors, and a Litton LTN-72 PICS/ITGS System.

The Flight Inspection configuration includes a Litton LTN-76 PAFIS system with a belly-mounted pod.

These airplanes have been certified under the provisions of FAR 21.185 for Restricted Category operation.

NOTE

All information in this supplement applies to both Aerial Surveillance and Flight Inspection configurations unless otherwise noted.

MAXIMUM CERTIFIED WEIGHTS

Maximum Ramp Weight	14,090 pounds
Maximum Take-Off Weight	14,000 pounds
Maximum Landing Weight	13,500 pounds
Maximum Zero-Fuel Weight	11,000 pounds

SPECIFIC LOADINGS (14,000 POUNDS)

Wing Loading	46.2 pounds per square foot
Power Loading	8.2 pounds per shaft horsepower

LIMITATIONS

GENERAL OPERATIONS

1. The pilot must have either a type rating or a letter of authorization from the governmental office having jurisdiction over such operation.
2. The airplane shall be identified "Restricted" per FAR 45.
3. Persons required for performance of aerial surveillance or flight inspection mission are considered to be crewmembers.

AERIAL SURVEILLANCE CONFIGURATION ONLY

1. Anytime one or more internal camera well covers are removed, a camera must be installed in the camera well.
2. Takeoffs and landings shall be accomplished with the external protective doors closed.
3. The airplane shall be depressurized while removing or exchanging cameras in flight.
4. Each time the airplane is operated for aerial surveillance, a post flight inspection of the camera installation shall be made to assure camera security and that chips or cracks in the optical glass do not exist. Do not pressurize the cabin if chips or cracks in the optical glass are present.

MISCELLANEOUS INSTRUMENT MARKINGS

AERIAL SURVEILLANCE CONFIGURATION

Cabin Differential Pressure Gage:
Green Arc (Approved Operating Range) ..... 0 to 6.1 PSI
Red Arc (Unapproved Operating Range) ..... 6.1 PSI to end of scale

FLIGHT INSPECTION CONFIGURATION - No change

WEIGHT LIMITS

Maximum Ramp Weight ..... 14,090 pounds
Maximum Take-Off Weight ..... 14,000 pounds
Maximum Landing Weight ..... 13,500 pounds
Maximum Zero-Fuel Weight ..... 11,000 pounds

CENTER OF GRAVITY LIMITS

Forward Limits

188.3 inches aft of datum from 14,000 to 13,500 pounds, then straight line variation to 181.0 inches aft of datum at 11,279 pounds, 181.0 inches aft of datum at 11,279 pounds or less.

Aft Limits

196.4 Inches aft of datum at all weights.

FLIGHT LOAD FACTOR LIMITS

At the designed gross weight of 14,000 pounds:
Flaps Up ..... 3.10 positive G's; 1.24 negative G's
Flaps Down ..... 2.0 positive G's

CABIN PRESSURIZATION LIMIT

AERIAL SURVEILLANCE CONFIGURATION

Maximum Cabin Pressure Differential ..... 6.1 psi

FLIGHT INSPECTION CONFIGURATION - No change

MAXIMUM OCCUMPANCY LIMITS

Occupancy is limited to crew members only.

STRUCTURAL LIMITATIONS

Refer to chapter four of the *Super King Air 200 Series Maintenance Manual.*

MISCELLANEOUS LIMITATIONS

1. The rudder boost system must be operational and armed for all flight operations.
2. The auto-feather system must be operational and armed for all takeoffs, climbs, and approaches.
3. The Litton LTN-72 PICS/ITGS and Litton LTN-76 PAFIS systems are not approved for primary navigation.

PLACARDS

On Left Rear Portion of Cockpit Pedestal:

I.N.S. NOT TO BE USED
FOR PRIMARY NAVIGATION

EMERGENCY PROCEDURES

ELECTRICAL SYSTEM FAILURE

GENERATOR INOPERATIVE (DC GEN ANNUNCIATOR LIGHT ON)

NOTE

AC and DC power to the mission equipment will automatically be shed with the loss of either or both generator and cannot be reset until both generators are back on line.

1. Generator Switch - OFF, then to RESET for a minimum of 1 second, then release to ON.
2. Mission Power Switch - OFF, then ON to restore power.

If generator will not reset:

3. Generator Switch - OFF
4. Mission Power Switch - OFF
5. Operating Generator - DO NOT EXCEED 1.0 LOAD

INVERTER INOPERATIVE

1. Select the other inverter.

NOTE

If the PICS/ITGS or PAFIS systems are in use, the INS BLOWER annunciator will illuminate if the inverter not selected on the instrument panel fails. (PICS/ITGS and PAFIS systems require simultaneous use of both inverters.

2. PICS/ITGS or PAFIS System - SWITCH OFF

NORMAL PROCEDURES

AERIAL SURVEILLANCE CONFIGURATION

BEFORE TAKEOFF (FINAL ITEMS)

Camera Well Doors - CLOSE, using switches at the forward or aft camera operators position.

BEFORE LANDING

Camera Doors - CLOSE, using switches at the forward or aft camera operators position.

FLIGHT INSPECTION CONFIGURATION - No change

PERFORMANCE

The tables and graphs in this supplement are to be used in lieu of the corresponding tables and graphs contained in the High Flotation Landing Gear Supplement to the Beech Model B200 Pilot's Operating Handbook.

Note that some tables and graphs in this supplement are modified with additional notes to account for the decrease in performance due to external modifications. Performance is estimated by entering a table or graph at a higher indicated OAT, or by changing the value derived from it by a specified percentage or amount. The resulting performance is approximate and will vary with airspeed, temperature, and other ambient conditions.

The following graphs, found in the High Flotation Landing Gear Flight Manual Supplement, are not applicable to this airplane and must not be used:

Maximum Enroute Weight
Take-off Weight - Flaps 0%, To Meet FAR 25 Take-off Climb Requirement
Take-off Weight - Flaps 40%, To Meet FAR 25 Take-off Climb Requirement
Take-off Flight Path
Net Gradient of Climb - Flaps 0%
Net Gradient of Climb - Flaps 40%
Service Ceiling - One Engine Inoperative
Time, Fuel, and Distance to Climb
Time, Fuel, and Distance to Descend

All graphs in the High Flotation Landing Gear Flight Manual Supplement that are not modified or excluded by this supplement are valid.

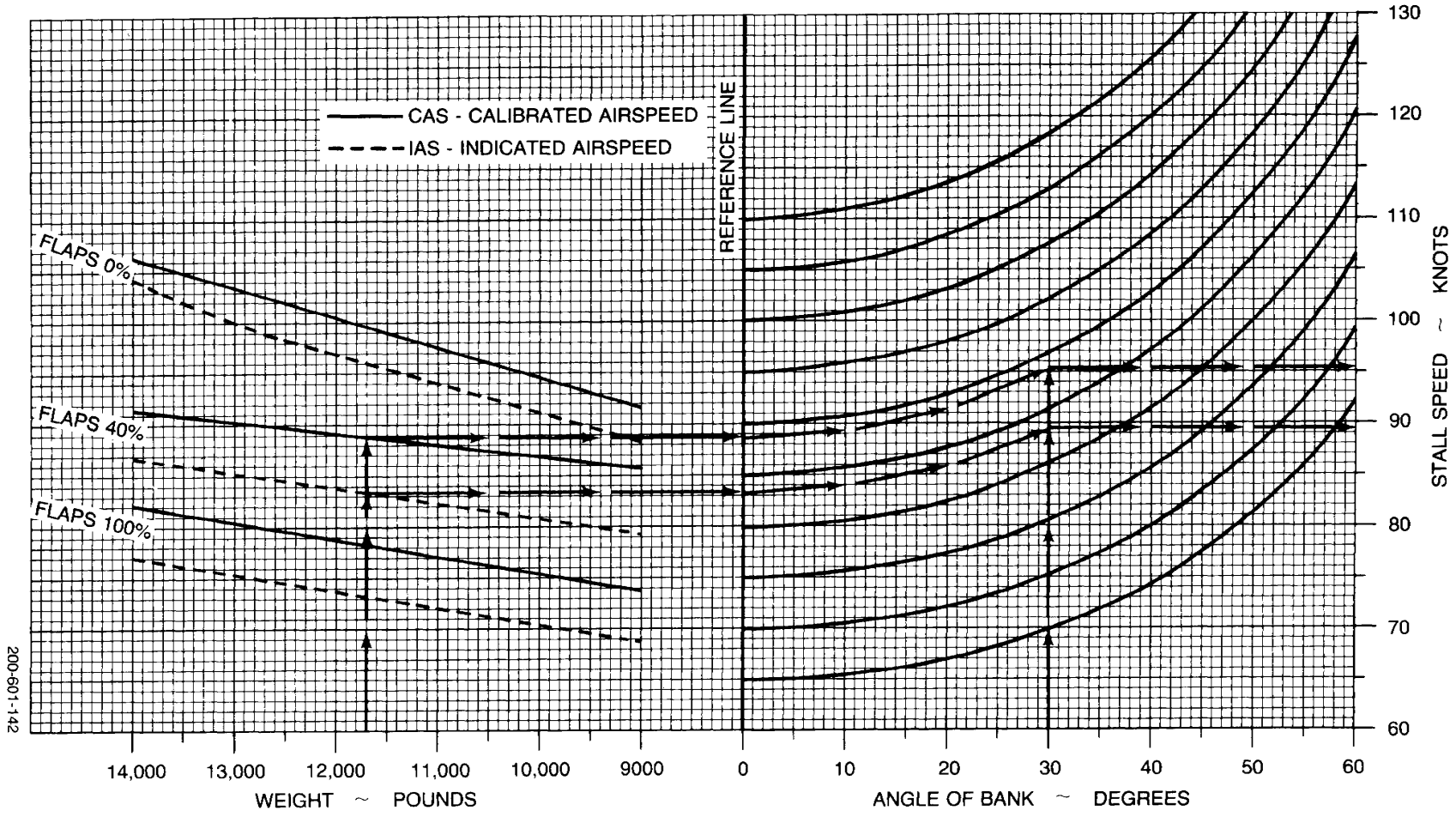
STALL SPEEDS-POWER IDLE

NOTES:

1. MAXIMUM ALTITUDE LOSS DURING A NORMAL STALL RECOVERY IS APPROXIMATELY 800 FEET
2. MAXIMUM NOSE DOWN PITCH ATTITUDE AND ALTITUDE LOSS DURING RECOVERY FROM ONE ENGINE INOPERATIVE STALLS PER FAR 23.205 ARE APPROXIMATELY 8° AND 300 FEET RESPECTIVELY.
3. A NORMAL STALL RECOVERY TECHNIQUE MAY BE USED. THE BEST PROCEDURE IS A BRISK FORWARD WHEEL MOVEMENT TO A NOSE DOWN ATTITUDE. LEVEL THE AIRPLANE AFTER AIRSPEED HAS INCREASED APPROXIMATELY 25 KNOTS ABOVE STALL.

EXAMPLE:

WEIGHT	11,700 LBS
FLAPS	40%
ANGLE OF BANK	30°
STALL SPEED	95 KTS CAS 90 KTS IAS



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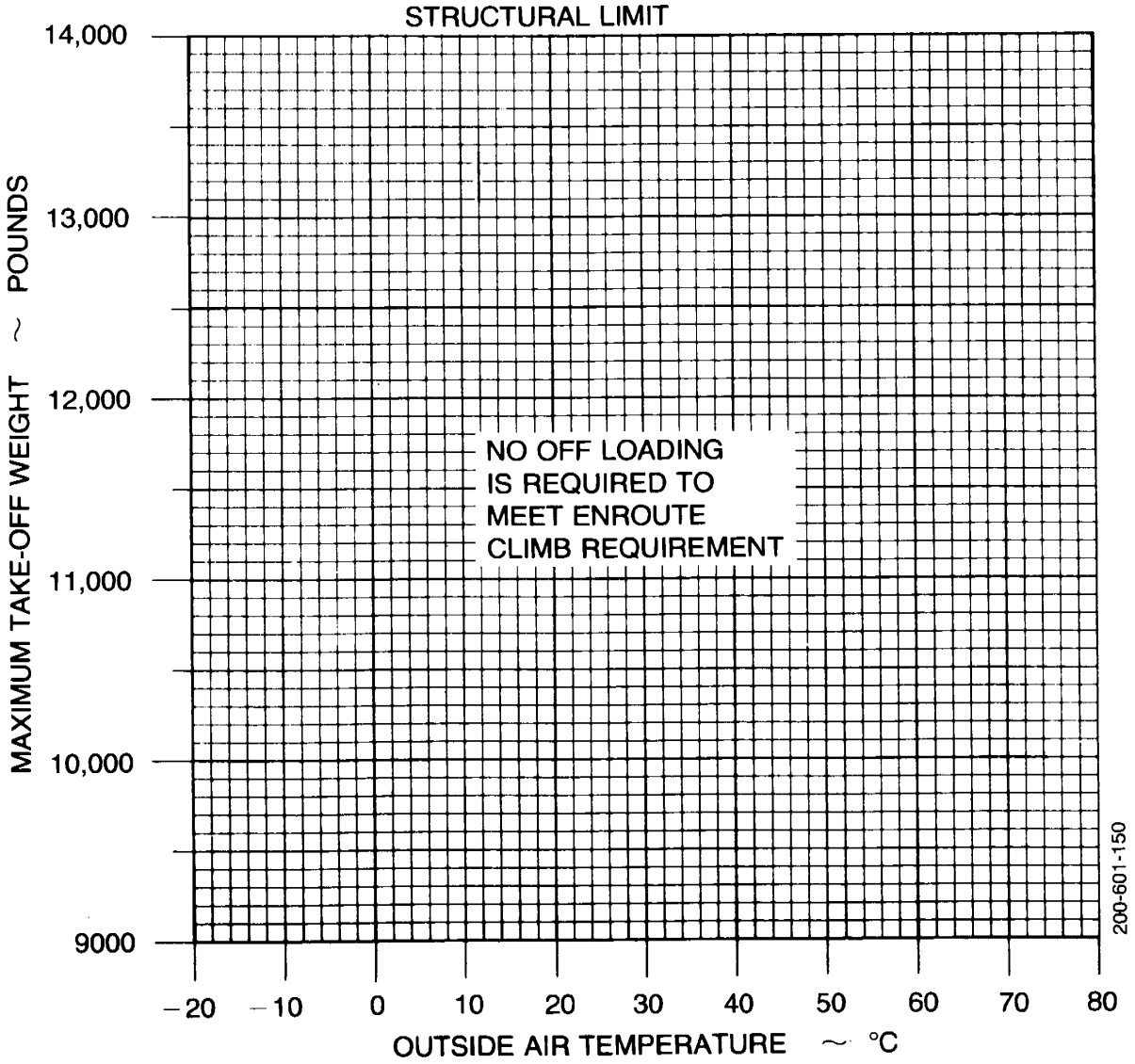
MAXIMUM TAKE-OFF WEIGHT

PERMITTED BY ENROUTE CLIMB REQUIREMENT



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TAKE-OFF WEIGHT - FLAPS 0%

TO ACHIEVE POSITIVE ONE-ENGINE INOPERATIVE CLIMB AT LIFT-OFF

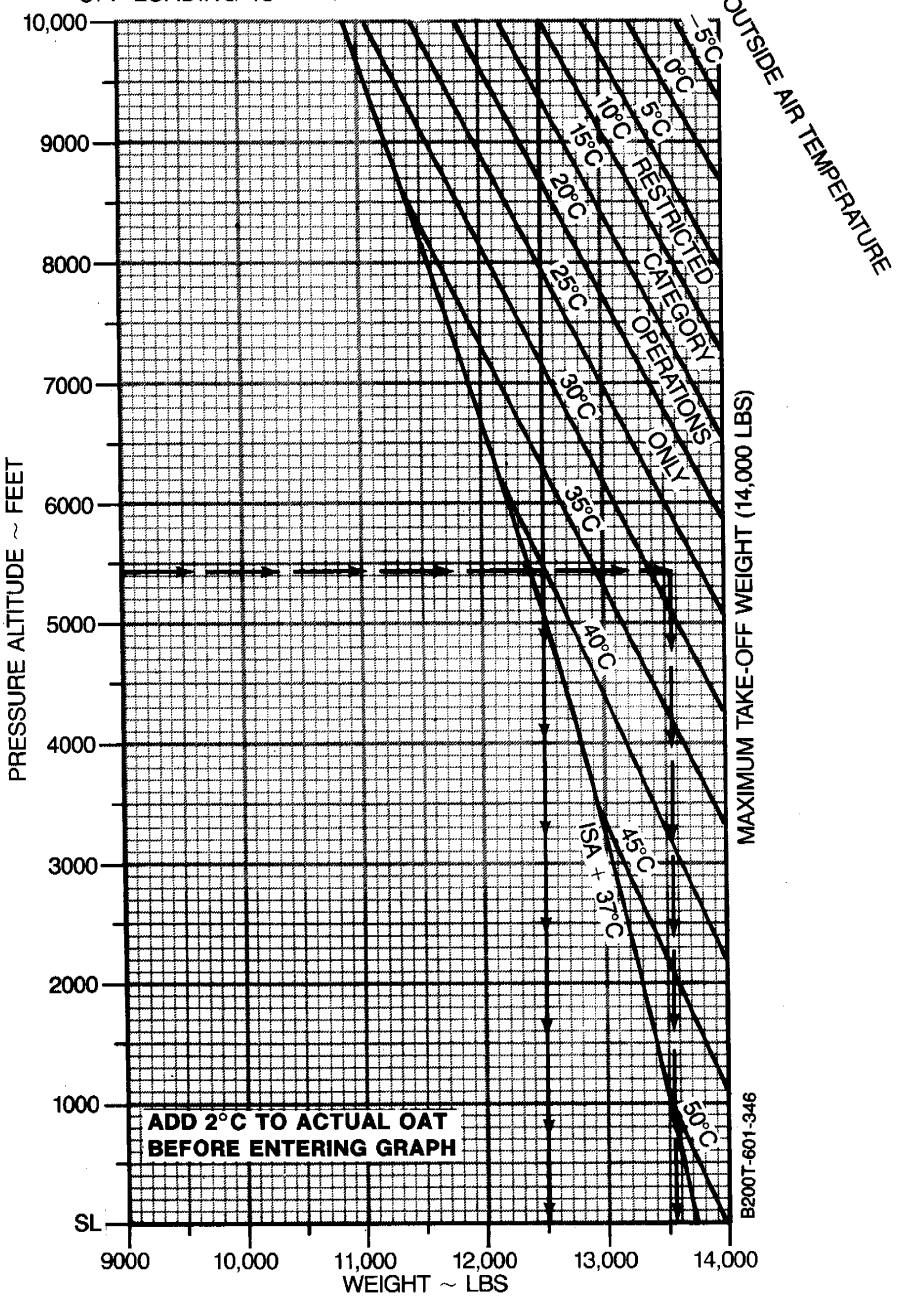
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	UP (0%)
LANDING GEAR	DOWN

EXAMPLE:

PRESSURE ALTITUDE	5433 FT
OAT	28°C
TAKE-OFF WEIGHT NORMAL CATEGORY	12,500 LBS
TAKE-OFF WEIGHT RESTRICTED CATEGORY	13,550 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, NO OFF-LOADING IS REQUIRED.



TAKE-OFF WEIGHT - FLAPS 40%

TO ACHIEVE POSITIVE ONE-ENGINE INOPERATIVE CLIMB AT LIFT-OFF

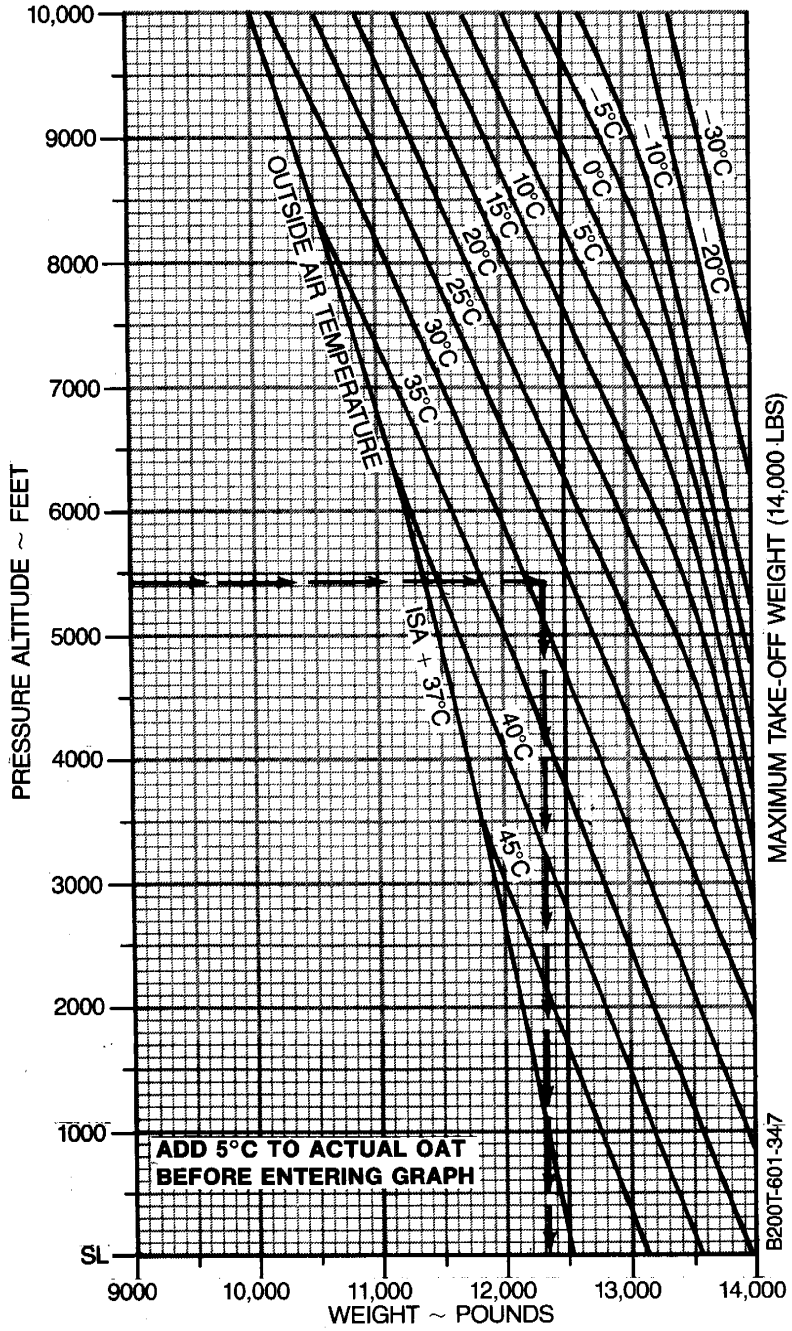
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	40%
LANDING GEAR	DOWN

EXAMPLE:

PRESSURE ALTITUDE	5433 FT
OAT	28°C
TAKE-OFF WEIGHT	12,340 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH



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TAKE-OFF DISTANCE-FLAPS 0%

ASSOCIATED CONDITIONS:

POWER . . . . . TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS 0%
LANDING GEAR . . . . . RETRACT AFTER LIFT-OFF
RUNWAY . . . . . PAVED, LEVEL, DRY SURFACE

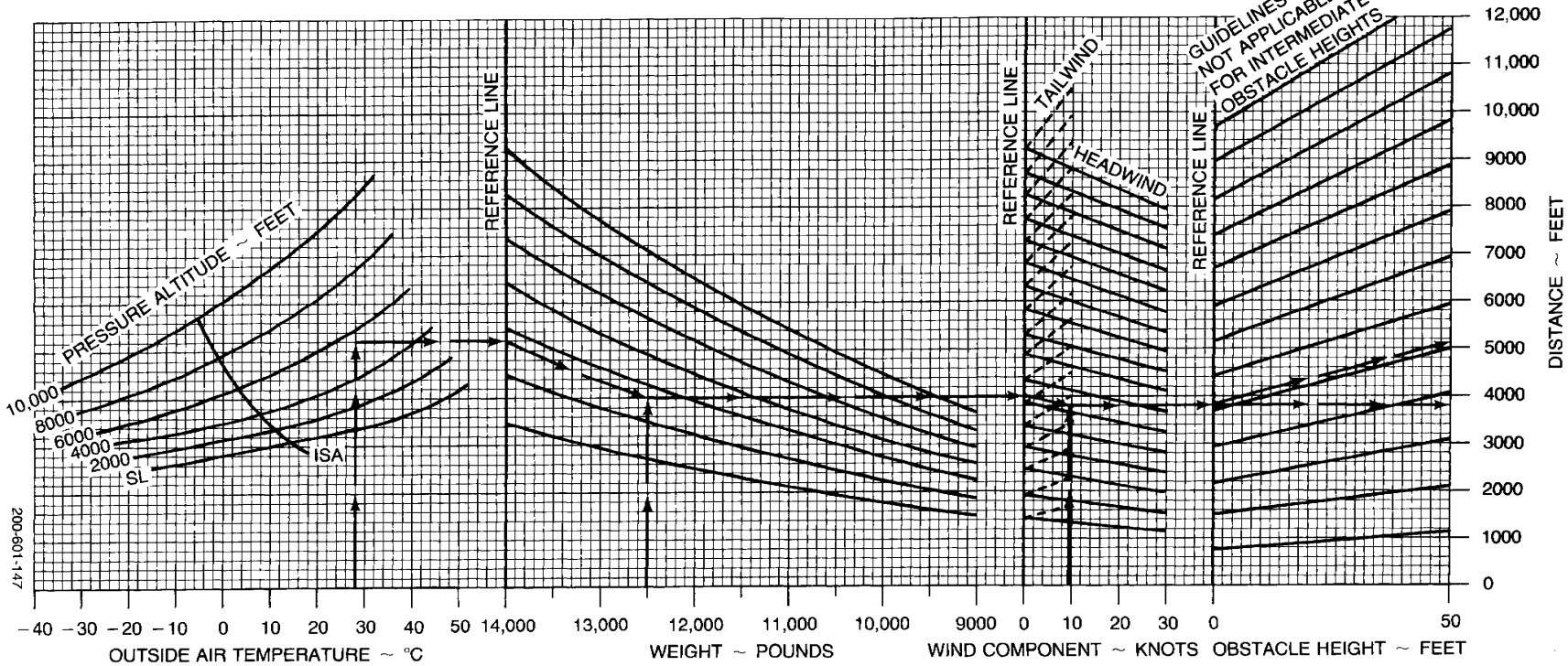
WEIGHT ~ POUNDS	TAKE-OFF SPEEDS ~ KNOTS	
	ROTATION	50 FT
14,000	117	126
13,000	113	123
12,500	111	121
12,000	109	120
11,000	105	115
10,000	101	111
9000	98	108

EXAMPLE:

OAT . . . . . 28°C
PRESSURE ALTITUDE . . . . . 5433 FT
TAKE-OFF WEIGHT . . . . . 12,500 LBS
HEADWIND COMPONENT . . . . . 9.5 KTS

GROUND ROLL . . . . . 3800 FT
TOTAL DISTANCE OVER 50 FT OBSTACLE . . . . . 5100 FT
TAKE-OFF SPEED AT ROTATION . . . . . 111 KTS
AT 50 FT . . . . . 121 KTS

ADD 4% TO TOTAL DISTANCE.



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ACCELERATE-STOP-FLAPS 0%

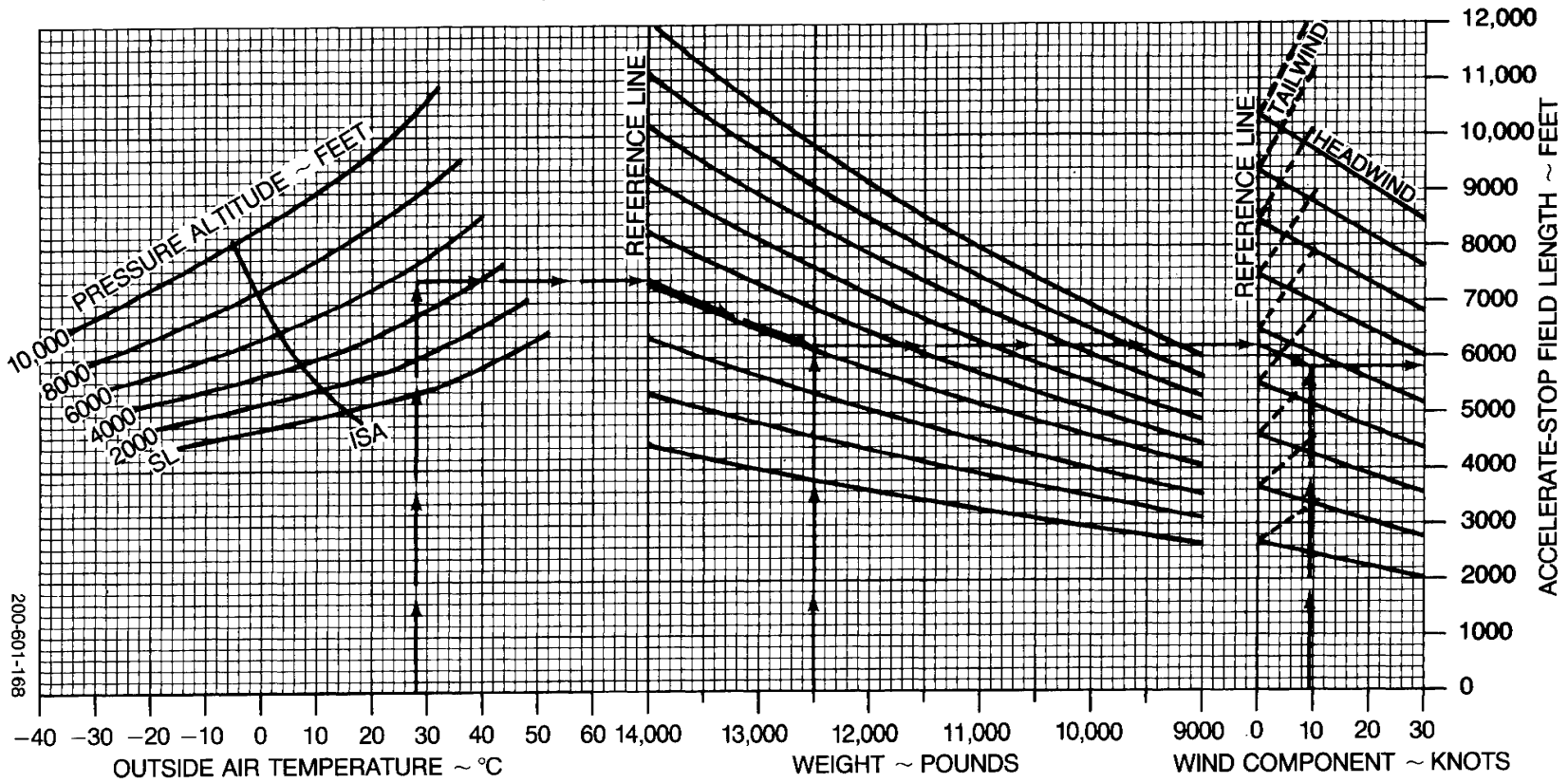
ASSOCIATED CONDITIONS:

POWER	1.	TAKE-OFF POWERSET BEFORE BRAKE RELEASE
	2.	BOTH ENGINES IDLE AT V_1 SPEED AND REVERSE OPERATING ENGINE
FLAPS	0%	
AUTOFEATHER		ARMED
BRAKING		MAXIMUM
RUNWAY		PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	V_1 ~ KNOTS
14,000	117
13,000	113
12,500	111
12,000	109
11,000	105
10,000	101
9000	98

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
WEIGHT	12500 LBS
HEADWIND COMPONENT	9.5 KTS
FIELD LENGTH	5800 FT
V_1	111 KTS



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ACCELERATE-GO - FLAPS 0%

ASSOCIATED CONDITIONS:

POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	0%
AUTOFEATHER	ARMED
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

NOTES:
1. AIR DISTANCE IS 50% OF TAKE-OFF DISTANCE OVER 35-FT OBSTACLE.
2. DISTANCES ASSUME AN ENGINE FAILURE AT ROTATION SPEED AND PROPELLER IMMEDIATELY FEATHERED.
3. USABLE CLEARWAY CANNOT EXCEED 25% OF THE RUNWAY LENGTH.
4. WEIGHTS IN ▨ AREA MAY NOT MEET FAR 25 REQUIREMENTS. REFER TO TAKE-OFF WEIGHT GRAPH TO MEET FAR 25 REQUIREMENTS.

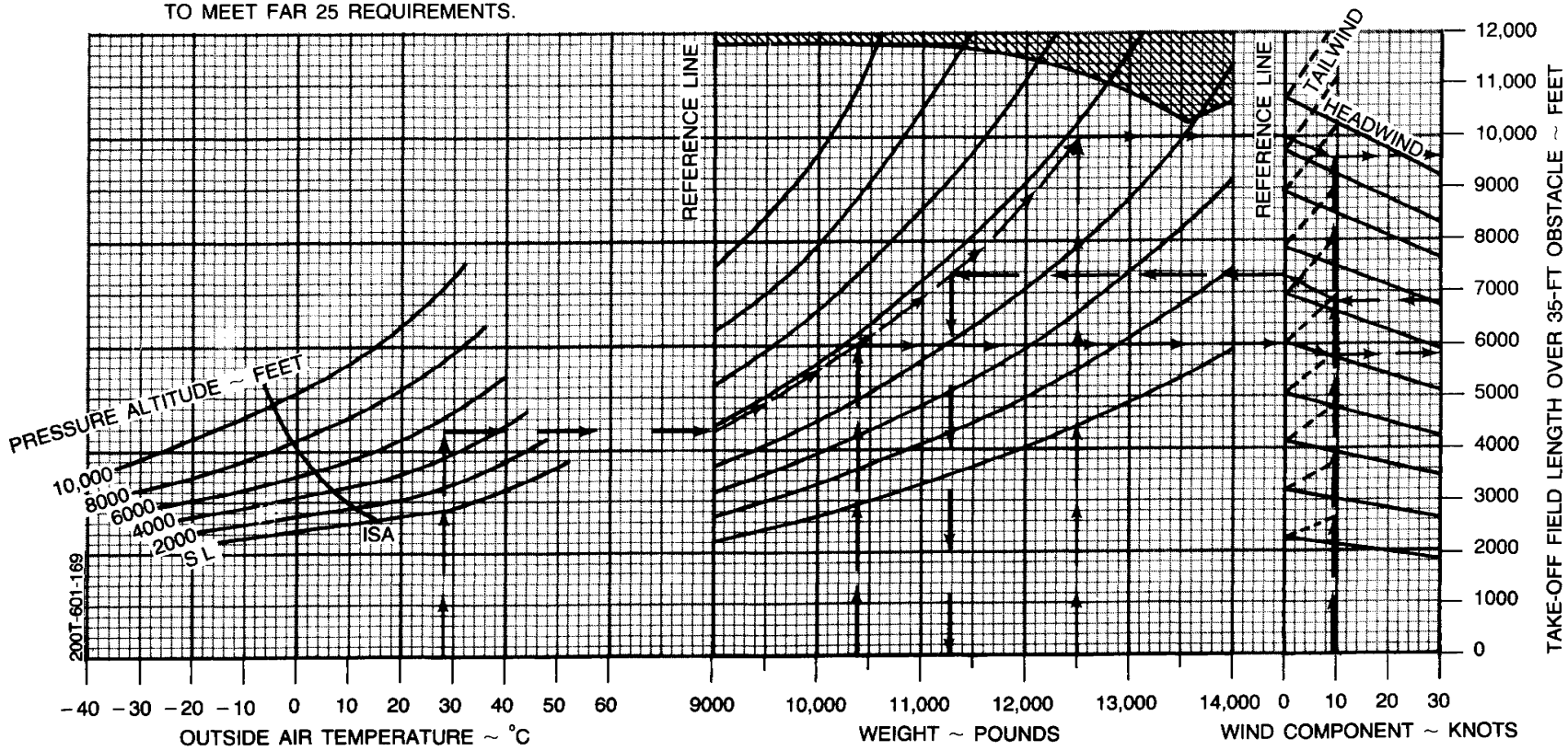
WEIGHT ~ POUNDS	SPEED ~ KNOTS V_R	SPEED ~ KNOTS V_2
14,000	117	126
13,000	113	125
12,500	111	121
12,000	109	120
11,000	105	115
10,000	101	111
9000	98	108

INCREASE INDICATED PRESSURE ALTITUDE BY 300 FT BEFORE ENTERING GRAPH

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
HEADWIND COMPONENT	9.5 KTS

TAKE-OFF WEIGHT ~ POUNDS	TAKE-OFF FIELD LENGTH ~ FEET
12,500	9600
11,280	6786
10,370	5800



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TAKE-OFF DISTANCE-FLAPS 40%

ASSOCIATED CONDITIONS:

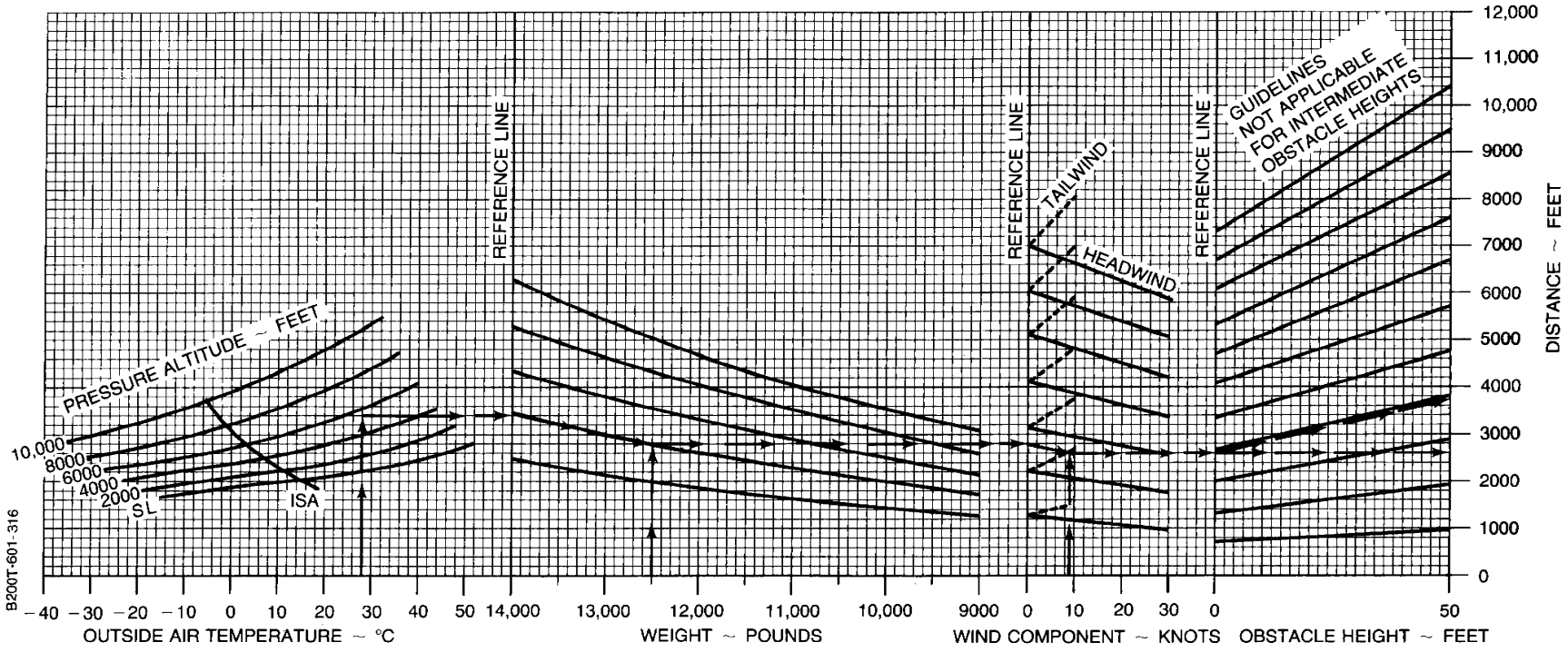
POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	40%
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	TAKE-OFF SPEED ~ KNOTS	
	ROTATION	50 FT
14,000	98	108
13,000	96	106
12,500	96	105
12,000	95	104
11,000	94	103
10,000	94	101
9000	94	99

ADD 3.5% TO TOTAL DISTANCE.

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5433 FT
TAKE-OFF WEIGHT	12,500 LBS
HEADWIND COMPONENT	9.5 KTS
GROUND ROLL	2600 FT
TOTAL DISTANCE OVER 50 FT OBSTACLE	3750 FT
TAKE-OFF SPEED AT ROTATION	96 KTS
AT 50 FT	105 KTS



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ACCELERATE-STOP - FLAPS 40%

ASSOCIATED CONDITIONS:

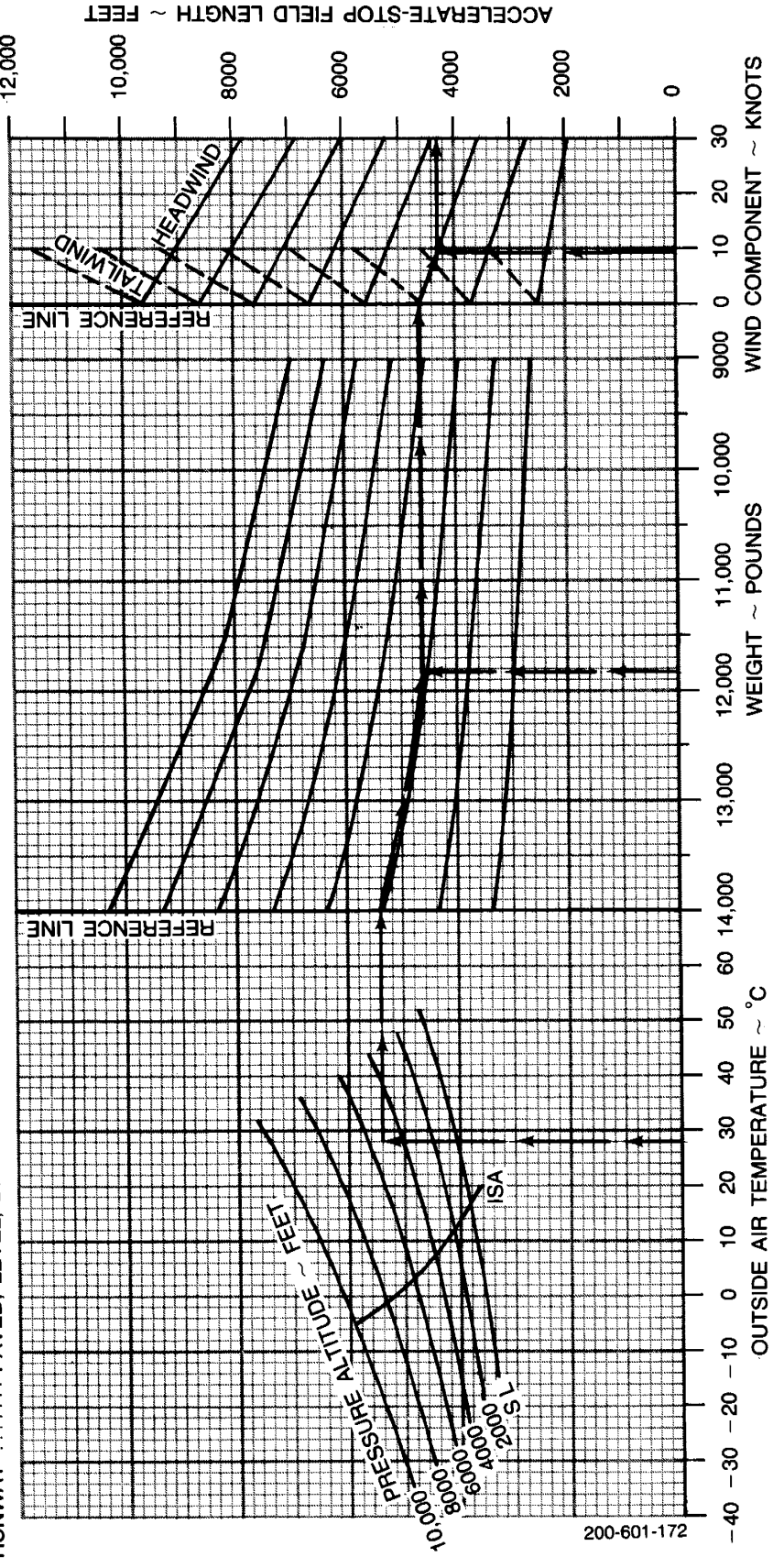
POWER ..... 1. TAKE-OFF POWER SET BEFORE BRAKE RELEASE
2. BOTH ENGINES IDLE AT V_1 SPEED AND REVERSE OPERATING ENGINE

FLAPS 40%
AUTOFEATHER ..... ARMED
BRAKING ..... MAXIMUM
RUNWAY ..... PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	V_1 ~ KNOTS
14,000	98
13,000	96
12,500	96
12,000	95
11,000	94
10,000	94
9000	94

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
WEIGHT	11,820 LBS
HEADWIND COMPONENT	9.5 KTS
FIELD LENGTH	4300 FT
V_1	95 KTS



ACCELERATE-GO - FLAPS 40%

ASSOCIATED CONDITIONS:

POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	40%
AUTOFEATHER	ARMED
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

NOTES:
1. AIR DISTANCE IS 60% OF TAKE-OFF DISTANCE OVER 35-FT OBSTACLE.
2. DISTANCES ASSUME AN ENGINE FAILURE AT ROTATION SPEED AND PROPELLER IMMEDIATELY FEATHERED.
3. USABLE CLEARWAY CANNOT EXCEED 25% OF THE RUNWAY LENGTH.
4. WEIGHTS IN ▨ AREA MAY NOT MEET FAR 25 REQUIREMENTS. REFER TO TAKE-OFF WEIGHT GRAPH TO MEET FAR 25 REQUIREMENTS.

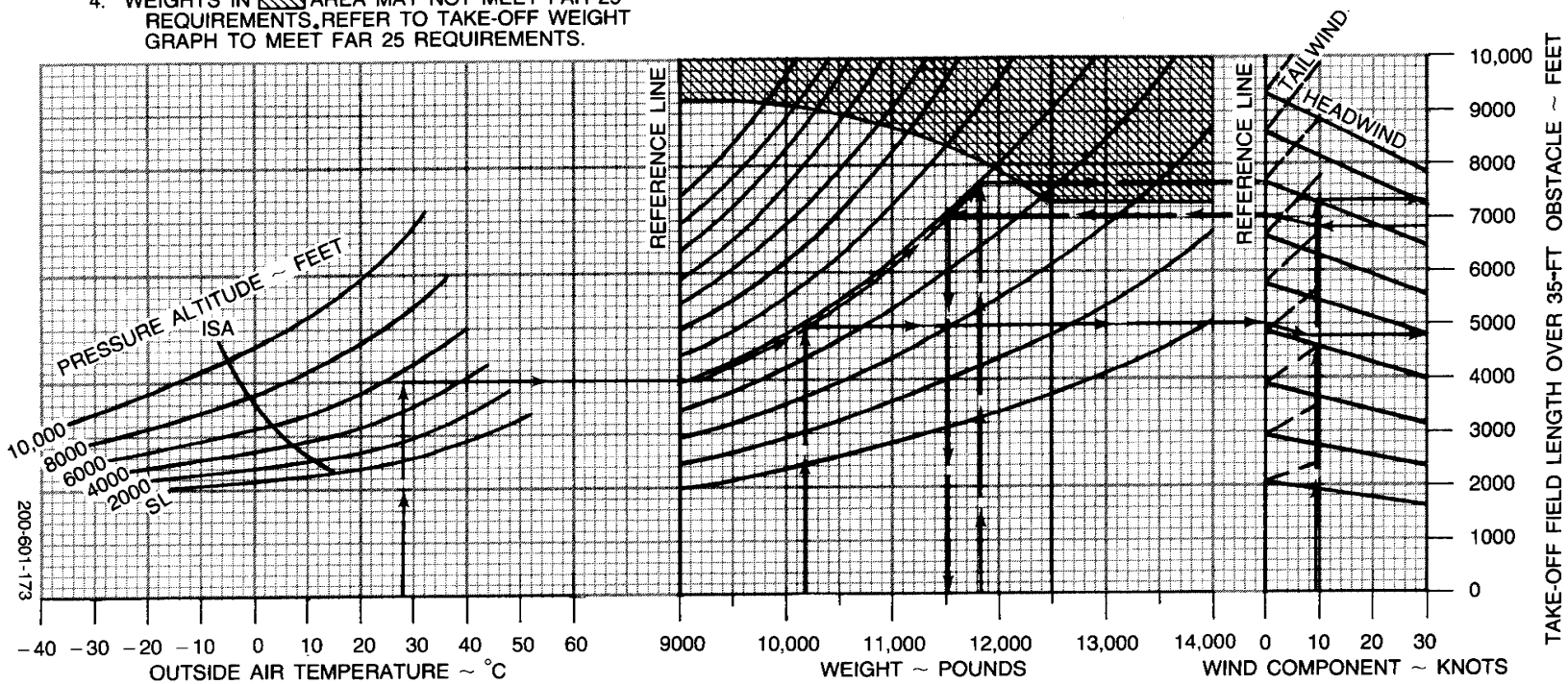
WEIGHT ~ POUNDS	SPEED ~ KNOTS V_R	SPEED ~ KNOTS V_2
14,000	98	108
13,000	96	106
12,500	96	105
12,000	95	104
11,000	94	103
10,000	94	101
9000	94	99

INCREASE INDICATED PRESSURE ALTITUDE BY 300 FT BEFORE ENTERING GRAPH

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT.
HEADWIND COMPONENT	9.5 KTS.

TAKE-OFF WEIGHT ~ POUNDS	TAKE-OFF FIELD LENGTH ~ FEET
11,820	7300
11,520	6786
10,175	4820



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CLIMB - TWO ENGINES - FLAPS 0%

ASSOCIATED CONDITIONS:

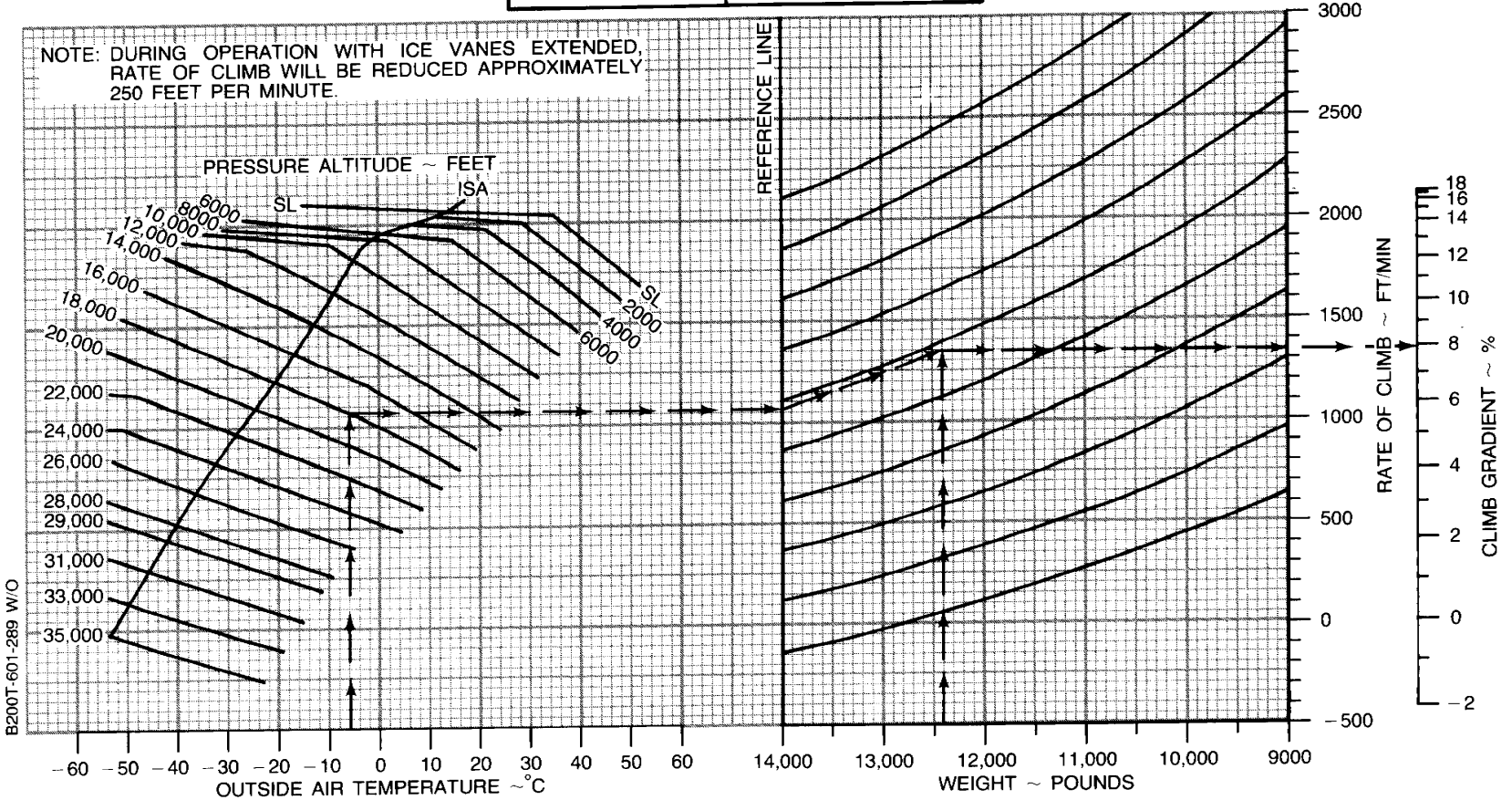
POWER	MAXIMUM CONTINUOUS
FLAPS	0%
LANDING GEAR	UP

SUBTRACT 100 FT. PER MINUTE FROM GRAPH VALUE

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	129
13,000	126
12,500	125
12,000	123
11,000	121
10,000	118
9000	115

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,391 LBS
RATE OF CLIMB	1340 FT/MIN
CLIMB GRADIENT	7.9%
CLIMB SPEED	125 KNOTS



CLIMB - TWO ENGINES - FLAPS 40%

ASSOCIATED CONDITIONS:

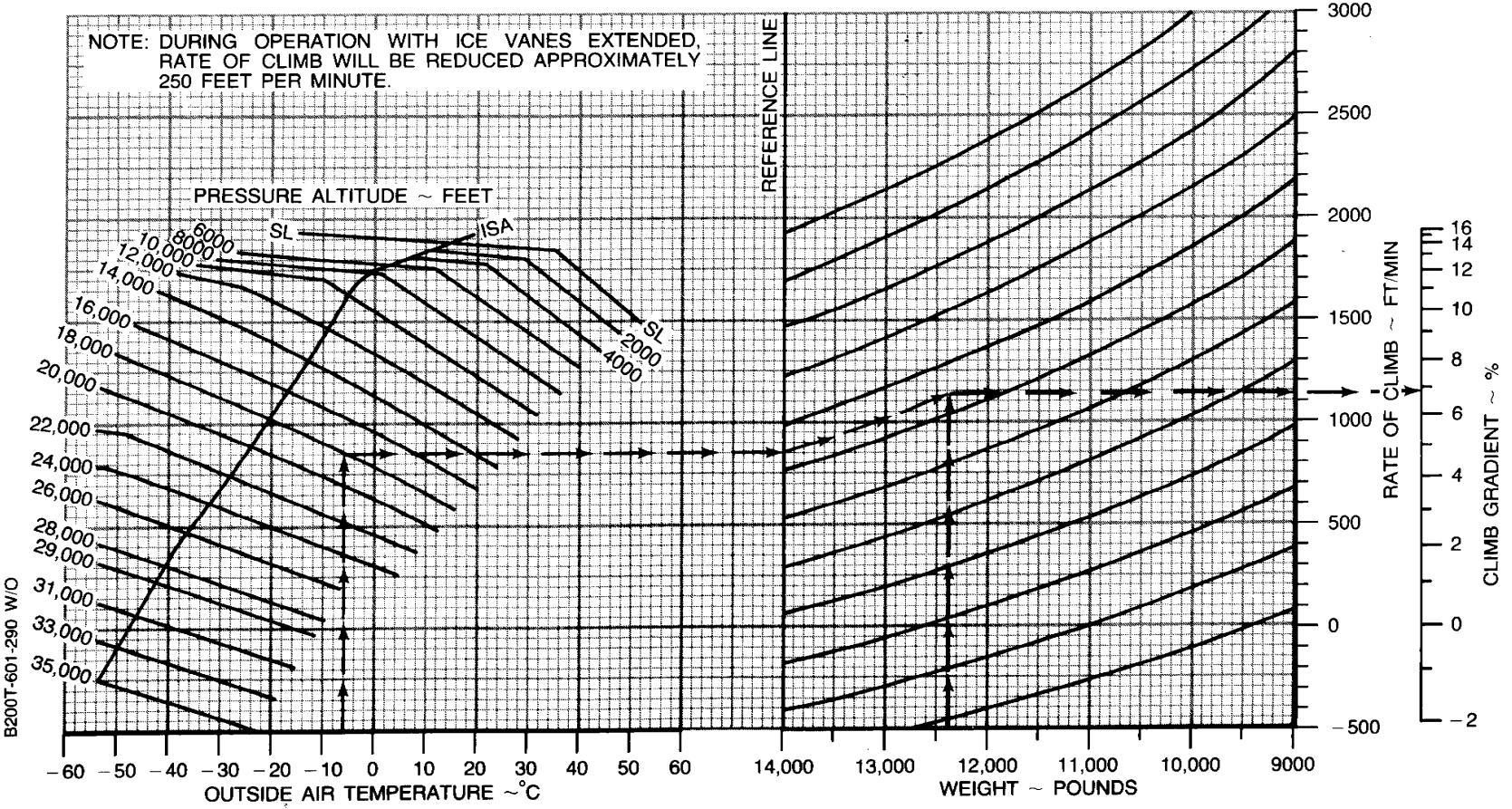
POWER	MAXIMUM CONTINUOUS
FLAPS	40%
LANDING GEAR	UP

SUBTRACT 100 FT. PER MINUTE FROM GRAPH VALUE

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	129
13,000	126
12,500	125
12,000	123
11,000	121
10,000	118
9000	115

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,391 LBS
RATE OF CLIMB	1135 FT/MIN
CLIMB GRADIENT	6.8%
CLIMB SPEED	125 KNOTS



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CLIMB - ONE ENGINE INOPERATIVE

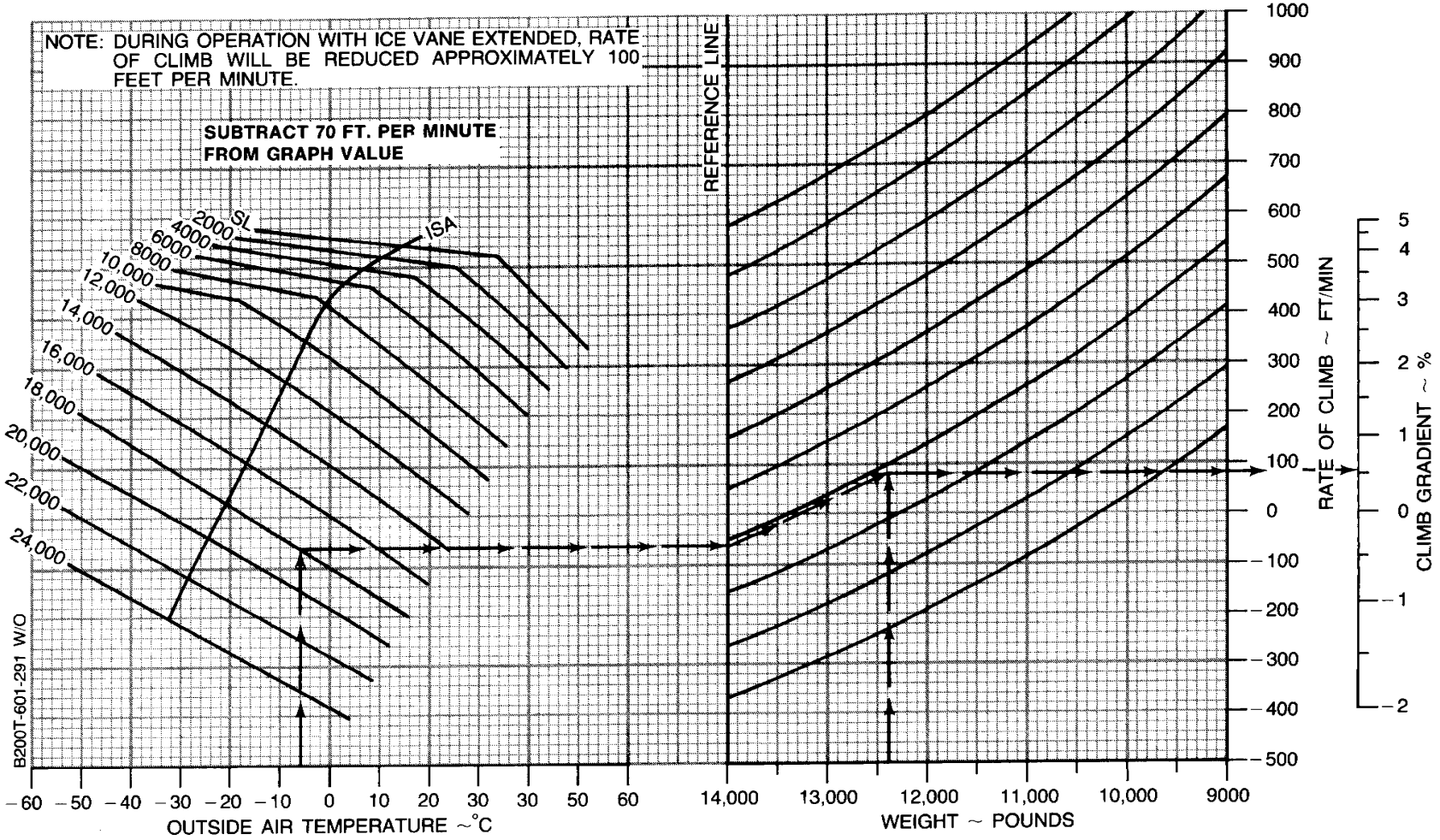
ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
FLAPS	0%
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	124
12,500	121
12,000	119
11,000	117
10,000	114
9000	111

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,391 LBS
RATE OF CLIMB	80 FT/MIN
CLIMB GRADIENT	0.5%
CLIMB SPEED	121 KNOTS



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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	465	930	240	228	241	229	242	230	242	231
2000	-14	-19	2230	453	906	237	232	238	233	239	234	240	235
4000	-18	-23	2230	441	882	235	236	236	237	237	238	238	239
6000	-22	-27	2230	429	858	233	241	234	242	235	243	236	244
8000	-25	-31	2230	417	834	231	245	232	246	233	248	234	249
10,000	-29	-35	2230	407	814	228	250	230	251	231	252	231	253
12,000	-33	-39	2230	397	794	226	255	227	256	228	257	229	258
14,000	-36	-43	2230	390	780	224	259	225	261	226	262	227	263
16,000	-40	-47	2230	384	768	221	265	222	266	224	268	225	269
18,000	-44	-51	2230	380	760	219	270	220	271	221	273	222	274
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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For Training Purposes Only

NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	467	934	238	231	239	232	240	232	241	233
2000	-4	-9	2230	454	908	236	235	237	236	238	237	239	238
4000	-8	-13	2230	440	880	233	239	234	240	235	241	236	242
6000	-11	-17	2230	429	858	231	244	232	245	233	246	234	247
8000	-15	-21	2230	419	838	229	248	230	250	231	251	232	252
10,000	-19	-25	2230	409	818	226	253	228	254	229	255	230	257
12,000	-23	-29	2230	398	796	224	258	225	259	226	260	227	262
14,000	-26	-33	2230	391	782	222	263	223	264	224	266	225	267
16,000	-30	-37	2230	386	772	219	268	221	270	222	271	223	272
18,000	-34	-41	2230	382	764	216	273	218	275	219	276	220	278
20,000	-37	-45	2230	379	758	213	278	215	280	216	281	218	283
22,000	-41	-49	2141	362	724	207	278	209	281	211	283	212	285
24,000	-45	-53	2012	340	680	199	276	201	279	203	282	205	284
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	470	940	236	233	237	234	238	235	239	236
2000	6	1	2230	457	914	234	238	235	239	236	240	237	240
4000	3	-3	2230	444	888	232	242	233	243	234	244	235	245
6000	-1	-7	2230	431	862	229	247	231	248	232	249	232	250
8000	-5	-11	2230	420	840	227	251	228	253	229	254	230	255
10,000	-9	-15	2230	409	818	225	256	226	257	227	259	228	260
12,000	-12	-19	2230	400	800	222	261	223	262	225	264	226	265
14,000	-16	-23	2230	393	786	220	266	221	268	222	269	223	270
16,000	-20	-27	2230	388	776	217	271	218	273	220	274	222	276
18,000	-24	-31	2230	384	768	214	276	216	278	217	279	218	281
20,000	-27	-35	2143	367	734	208	276	210	279	211	281	213	283
22,000	-31	-39	2043	349	698	201	276	203	279	205	281	207	284
24,000	-35	-43	1917	326	652	193	274	195	277	197	280	199	283
26,000	-40	-47	1781	303	606	184	270	187	274	189	278	191	281
28,000	-44	-51	1641	279	558	174	265	177	270	180	275	183	278
29,000	-46	-52	1573	267	534	168	261	172	267	176	272	179	277
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	473	946	235	236	236	237	237	238	238	239
2000	16	11	2230	460	920	232	240	234	241	235	242	235	243
4000	13	7	2230	446	892	230	245	231	246	232	247	233	248
6000	9	3	2230	433	866	228	249	229	251	230	252	231	253
8000	5	-1	2230	422	844	225	254	226	255	228	257	228	258
10,000	1	-5	2230	411	822	223	259	224	260	225	262	226	263
12,000	-2	-9	2230	400	800	220	264	222	266	223	267	224	268
14,000	-6	-13	2230	393	786	218	269	220	271	221	272	222	274
16,000	-10	-17	2230	388	776	215	274	216	276	218	277	219	279
18,000	-14	-21	2139	371	742	208	274	210	276	212	279	213	280
20,000	-18	-25	2045	353	706	202	274	204	277	206	279	207	281
22,000	-21	-29	1950	336	672	195	274	197	277	199	280	201	282
24,000	-26	-33	1826	314	628	187	271	189	275	192	278	194	281
26,000	-30	-37	1694	290	580	177	267	181	272	183	276	186	279
28,000	-34	-41	1559	267	534	167	261	171	267	175	272	177	276
29,000	-36	-42	1493	256	512	161	256	166	264	170	269	173	274
31,000	-40	-46	1366	236	472	149	246	156	256	160	264	164	270
33,000	-45	-50	1243	216	432	---	---	143	245	150	257	155	265
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	30	25	2230	476	952	233	238	234	239	235	240	236	241
2000	27	21	2230	462	924	231	243	232	244	233	245	234	246
4000	23	17	2230	449	898	229	247	230	249	231	250	232	251
6000	19	13	2230	437	874	226	252	227	253	228	255	229	256
8000	15	9	2230	424	848	224	257	225	258	226	259	227	261
10,000	12	5	2230	413	826	221	262	222	263	224	265	225	266
12,000	8	1	2230	402	804	219	267	220	269	221	270	222	271
14,000	4	-3	2230	395	790	216	272	217	273	219	275	220	276
16,000	0	-7	2160	378	756	210	273	212	275	213	277	215	279
18,000	-4	-11	2052	357	714	203	273	205	275	207	277	208	279
20,000	-8	-15	1964	340	680	197	273	199	276	201	278	202	281
22,000	-12	-19	1861	324	648	189	271	192	275	194	278	196	281
24,000	-16	-23	1740	302	604	181	268	184	273	186	276	189	280
26,000	-20	-27	1611	279	558	171	263	175	269	178	273	180	277
28,000	-24	-31	1480	256	512	160	256	165	263	169	269	172	274
29,000	-26	-32	1416	246	492	154	251	160	259	164	266	167	271
31,000	-31	-36	1294	226	452	139	234	149	251	154	260	158	267
33,000	-36	-40	1169	207	414	---	---	133	234	143	251	149	261
35,000	-40	-44	1069	190	380	---	---	---	---	128	233	138	251

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	40	35	2230	479	958	232	241	233	242	234	243	235	244
2000	37	31	2230	466	932	229	245	231	247	232	248	233	249
4000	33	27	2230	452	904	227	250	228	251	229	252	230	253
6000	29	23	2230	439	878	225	255	226	256	227	257	228	258
8000	25	19	2230	427	854	222	260	223	261	224	262	225	263
10,000	22	15	2230	416	832	220	265	221	266	222	268	223	269
12,000	18	11	2216	403	806	216	269	218	271	219	272	220	274
14,000	14	7	2123	381	762	210	269	212	271	213	273	215	274
16,000	10	3	2030	361	722	203	269	205	272	207	274	209	276
18,000	6	-1	1936	341	682	197	269	199	272	201	275	202	277
20,000	2	-5	1872	327	654	191	270	194	273	195	276	197	279
22,000	-2	-9	1791	312	624	184	270	187	274	189	277	191	280
24,000	-6	-13	1674	291	582	175	266	179	271	182	275	184	279
26,000	-10	-17	1540	268	536	165	260	170	266	173	271	176	275
28,000	-14	-21	1403	246	492	153	250	159	259	163	265	167	271
29,000	-17	-22	1340	236	472	145	242	153	254	158	262	162	268
31,000	-21	-26	1219	216	432	---	---	140	242	149	255	153	263
33,000	-25	-30	1125	200	400	---	---	---	---	136	243	143	256
35,000	-29	-34	1031	184	368	---	---	---	---	---	---	131	244

NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 PRM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	482	964	231	243	231	244	232	245	233	246
2000	47	41	2230	468	936	228	248	229	249	230	250	231	251
4000	43	37	2230	455	910	226	252	227	254	228	255	229	256
6000	39	33	2230	441	882	223	257	224	259	225	260	226	261
8000	36	29	2230	429	858	221	262	222	264	223	265	224	266
10,000	32	25	2174	409	818	216	264	217	266	219	268	220	269
12,000	28	21	2064	382	764	209	264	211	266	212	268	214	270
14,000	24	17	1977	362	724	203	265	205	267	206	269	208	271
16,000	20	13	1894	343	686	196	265	198	268	200	270	202	272
18,000	16	9	1817	325	650	190	265	192	268	194	271	197	273
20,000	12	5	1767	313	626	185	266	187	270	189	273	191	276
22,000	8	1	1708	301	602	179	267	182	271	184	275	186	278
24,000	4	-3	1592	280	560	169	262	173	268	176	272	179	276
26,000	0	-7	1469	258	516	159	255	164	262	168	268	171	273
28,000	-4	-11	1346	237	474	147	244	153	255	158	263	162	269
29,000	-7	-12	1285	227	454	137	232	147	250	153	259	157	266
31,000	-12	-16	1157	207	414	---	---	131	231	142	250	148	260
33,000	-16	-20	1059	191	382	---	---	---	---	126	230	137	251
35,000	-20	-24	962	175	350	---	---	---	---	---	---	121	230

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

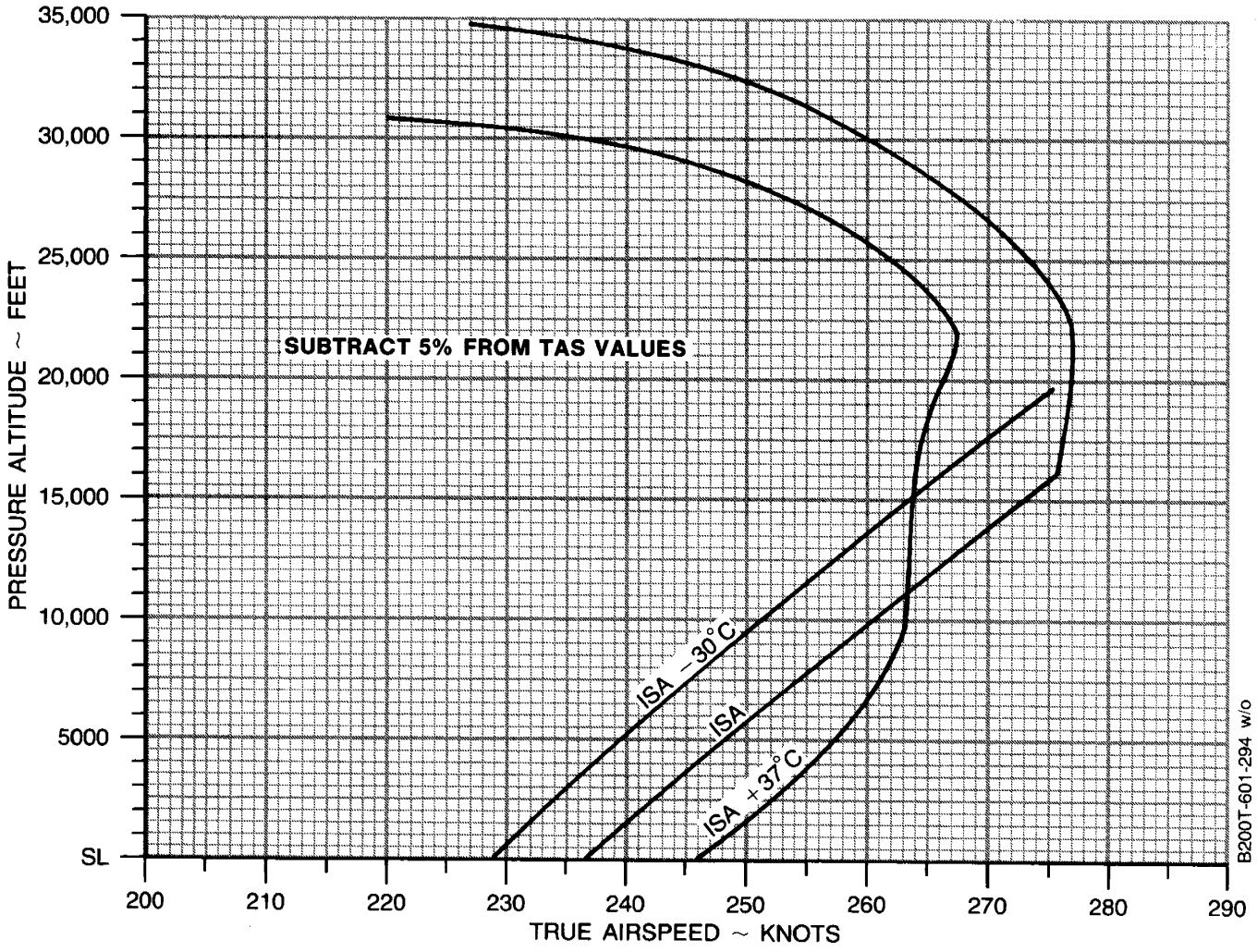
PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	484	968	229	245	231	246	232	247	232	248
2000	54	48	2230	470	940	227	249	228	251	229	252	230	253
4000	50	44	2230	456	912	225	254	226	255	227	257	228	258
6000	46	40	2229	443	886	222	259	223	260	224	262	225	263
8000	42	36	2149	419	838	216	260	218	262	219	263	220	265
10,000	39	32	2068	395	790	210	261	212	263	214	265	215	267
12,000	35	28	1963	369	738	204	261	205	263	207	265	209	267
14,000	31	24	1881	350	700	197	261	200	264	201	266	203	268
16,000	27	20	1803	332	664	191	261	194	265	196	267	198	270
18,000	23	16	1729	315	630	185	261	188	265	190	268	192	271
20,000	19	12	1685	303	606	180	263	183	267	185	270	187	273
22,000	15	8	1633	291	582	174	263	177	268	180	272	182	275
24,000	11	4	1529	271	542	165	258	169	264	172	270	175	274
26,000	7	0	1415	250	500	154	251	160	259	164	266	167	271
28,000	2	-4	1295	230	460	140	237	149	251	154	260	158	266
29,000	0	-5	1234	220	440	---	---	142	244	149	256	154	263
31,000	-4	-9	1138	203	406	---	---	---	---	137	245	144	257
33,000	-8	-13	1049	188	376	---	---	---	---	---	---	133	246
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE SPEEDS

1700 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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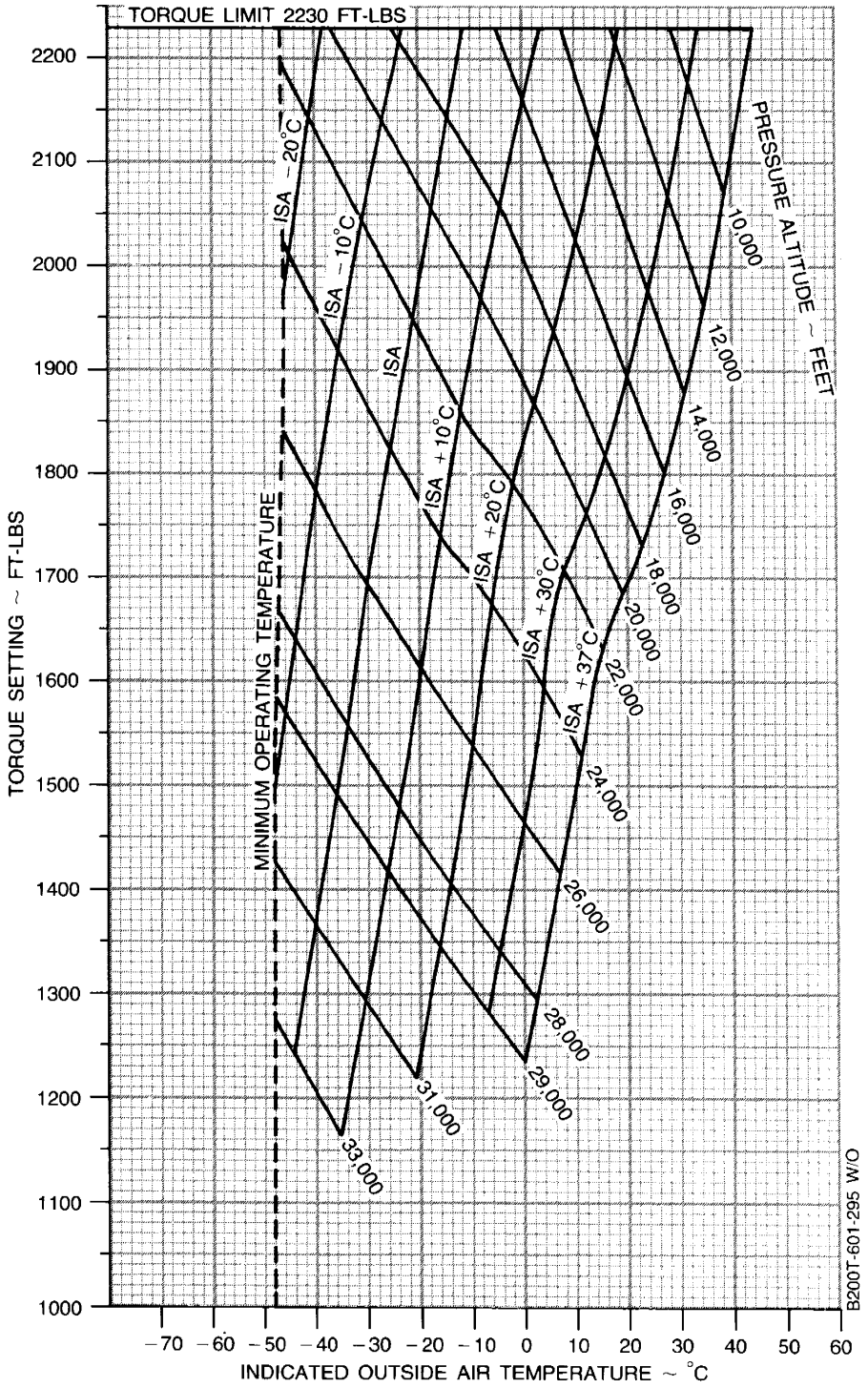
NORMAL CRUISE POWER

1700 RPM

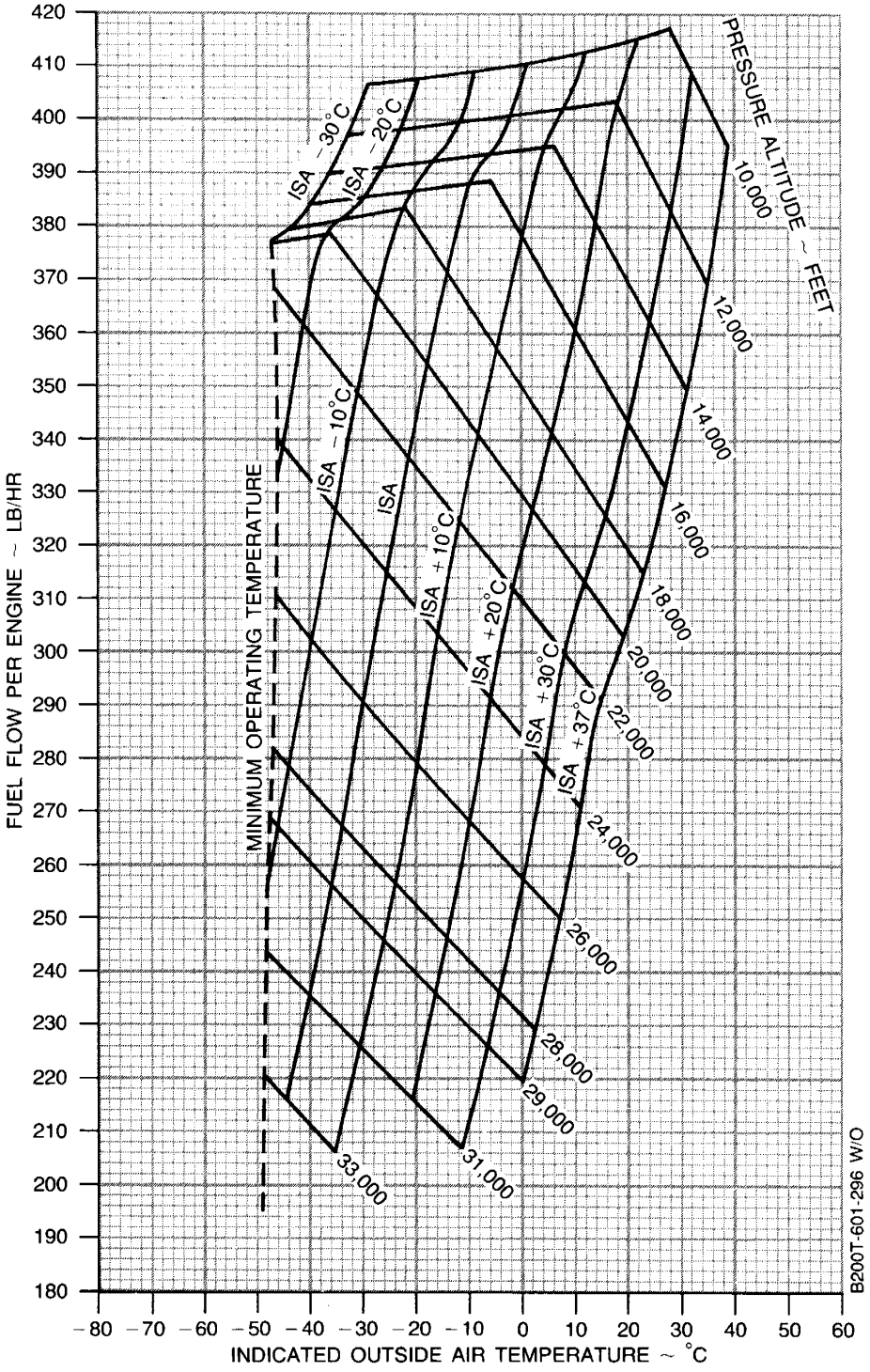
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT NORMAL CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



RANGE PROFILE - NORMAL CRUISE POWER

1700 RPM

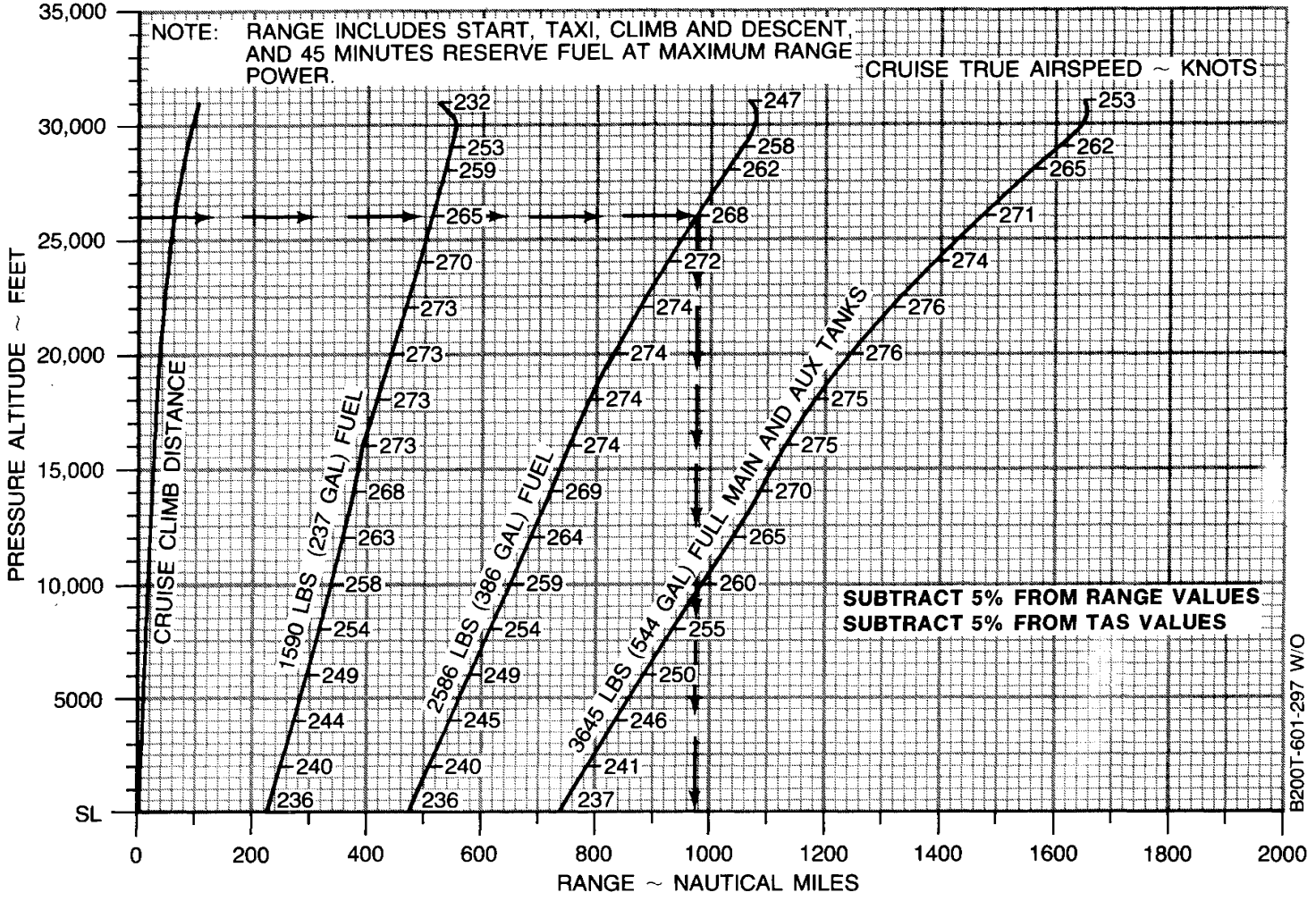
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS /GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	977 NM



For Training Purposes Only

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	465	930	240	228	241	229	242	230	242	231
2000	-14	-19	2230	453	906	237	232	238	233	239	234	240	235
4000	-18	-23	2230	441	882	235	236	236	237	237	238	238	239
6000	-22	-27	2230	429	858	233	241	234	242	235	243	236	244
8000	-25	-31	2230	417	834	231	245	232	246	233	248	234	249
10,000	-29	-35	2230	407	814	228	250	230	251	231	252	231	253
12,000	-33	-39	2230	397	794	226	255	228	256	228	257	229	258
14,000	-36	-43	2230	390	780	223	259	225	261	226	262	227	263
16,000	-40	-47	2230	384	768	221	265	222	266	224	268	225	269
18,000	-44	-51	2230	380	760	219	270	220	271	221	273	222	274
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	467	934	238	231	239	232	240	232	241	233
2000	-4	-9	2230	454	908	236	235	237	236	238	237	239	238
4000	-8	-13	2230	440	880	233	239	234	240	235	241	236	242
6000	-11	-17	2230	429	858	231	244	232	245	233	246	234	247
8000	-15	-21	2230	419	838	229	248	230	250	231	251	232	252
10,000	-19	-25	2230	409	818	226	253	228	254	229	255	230	257
12,000	-23	-29	2230	398	796	224	258	225	259	226	260	227	262
14,000	-26	-33	2230	391	782	222	263	223	264	224	266	225	267
16,000	-30	-37	2230	386	772	220	268	221	270	222	271	223	272
18,000	-34	-41	2230	382	764	216	273	218	275	219	276	220	278
20,000	-37	-45	2230	379	758	213	278	215	280	216	281	218	283
22,000	-41	-49	2230	376	752	210	283	212	285	214	287	215	288
24,000	-45	-53	2090	352	704	202	280	204	283	206	286	208	288
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	470	940	236	233	237	234	238	235	239	236
2000	6	1	2230	457	914	234	238	235	239	236	240	237	240
4000	3	-3	2230	444	888	232	242	233	243	234	244	235	245
6000	-1	-7	2230	431	862	229	247	231	248	232	249	232	250
8000	-5	-11	2230	420	840	227	251	228	253	229	254	230	255
10,000	-9	-15	2230	409	818	225	256	226	257	227	259	228	260
12,000	-12	-19	2230	400	800	222	261	223	262	225	264	226	265
14,000	-16	-23	2230	393	786	220	266	221	268	222	269	223	270
16,000	-20	-27	2230	388	776	217	271	218	273	220	274	221	276
18,000	-24	-31	2230	384	768	214	276	216	278	217	279	218	281
20,000	-27	-35	2230	381	762	211	281	213	283	214	285	215	286
22,000	-31	-39	2128	362	724	204	280	206	283	208	286	210	288
24,000	-35	-43	2002	340	680	196	279	199	282	201	285	202	287
26,000	-39	-47	1862	315	630	187	275	190	279	192	283	195	286
28,000	-43	-51	1714	290	580	177	270	181	275	184	279	186	283
29,000	-46	-52	1639	278	556	172	266	175	272	179	277	181	281
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	473	946	235	236	236	237	237	238	238	239
2000	16	11	2230	460	920	232	240	234	241	235	242	235	243
4000	13	7	2230	446	892	230	245	231	246	232	247	233	248
6000	9	3	2230	433	866	228	249	229	251	230	252	231	253
8000	5	-1	2230	422	844	225	254	226	255	228	257	228	258
10,000	1	-5	2230	411	822	223	259	224	260	225	262	226	263
12,000	-2	-9	2230	400	800	220	264	222	266	223	267	224	268
14,000	-6	-13	2230	393	786	218	269	219	271	221	272	222	274
16,000	-10	-17	2230	388	776	215	274	216	276	218	277	219	279
18,000	-13	-21	2230	386	772	212	278	213	281	215	282	217	284
20,000	-17	-25	2131	367	734	205	279	207	281	209	283	210	285
22,000	-21	-29	2031	349	698	199	278	200	281	202	284	204	286
24,000	-25	-33	1905	326	652	191	276	193	280	195	283	197	286
26,000	-29	-37	1771	302	604	181	272	184	277	187	280	189	284
28,000	-34	-41	1634	279	558	171	266	175	272	178	277	181	281
29,000	-36	-42	1567	268	536	166	263	170	269	173	275	176	279
31,000	-40	-46	1436	247	494	154	253	159	262	164	270	167	275
33,000	-45	-50	1306	226	452	137	235	148	253	154	262	158	270
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,500 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	30	25	2230	476	952	233	238	234	239	235	240	236	241
2000	27	21	2230	462	924	231	243	232	244	233	245	234	246
4000	23	17	2230	449	898	229	247	230	249	231	250	232	251
6000	19	13	2230	437	874	226	252	227	253	228	255	229	256
8000	15	9	2230	424	848	224	257	225	258	226	259	227	261
10,000	12	5	2230	413	826	221	262	223	263	224	265	225	266
12,000	8	1	2230	402	804	219	267	220	269	221	270	222	271
14,000	4	-3	2230	395	790	216	272	217	273	219	275	220	276
16,000	0	-7	2230	389	778	213	277	214	278	216	280	217	282
18,000	-3	-11	2134	370	740	206	277	209	279	210	281	211	283
20,000	-7	-15	2046	353	706	200	277	202	280	204	282	205	285
22,000	-11	-19	1937	336	672	192	276	195	279	197	282	199	285
24,000	-15	-23	1815	314	628	184	273	187	277	189	281	191	284
26,000	-20	-27	1685	290	580	175	268	178	274	181	278	184	282
28,000	-24	-31	1552	268	536	164	262	169	269	172	274	175	279
29,000	-26	-32	1487	257	514	158	257	163	265	167	271	171	277
31,000	-30	-36	1362	237	474	145	245	153	257	158	265	162	272
33,000	-35	-40	1239	217	434	---	---	140	245	148	258	153	267
35,000	-41	-44	1087	194	388	---	---	---	---	134	245	142	259

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	40	35	2230	479	958	232	241	233	242	234	243	235	244
2000	37	31	2230	466	932	229	245	231	247	232	248	233	249
4000	33	27	2230	452	904	229	240	228	251	229	252	230	253
6000	29	23	2230	439	878	225	255	226	256	227	257	228	258
8000	25	19	2230	427	854	222	260	223	261	224	262	225	263
10,000	22	15	2230	416	832	220	265	221	266	222	268	223	269
12,000	18	11	2230	405	810	217	270	218	271	220	273	221	274
14,000	14	7	2184	390	780	212	272	214	274	215	276	217	278
16,000	10	3	2104	372	744	206	273	208	275	210	278	211	279
18,000	6	-1	2023	354	708	200	274	202	276	204	279	206	281
20,000	3	-5	1944	339	678	194	274	196	277	198	280	200	283
22,000	-1	-9	1862	323	646	187	274	190	278	192	281	195	284
24,000	-6	-13	1745	302	604	179	271	182	275	185	279	187	283
26,000	-10	-17	1610	279	558	169	265	173	271	176	276	179	280
28,000	-14	-21	1472	257	514	157	257	162	264	167	271	170	276
29,000	-16	-22	1346	236	472	151	251	157	260	162	268	165	273
31,000	-21	-26	1230	217	434	133	230	146	251	152	261	156	269
33,000	-26	-30	1087	196	392	---	---	128	229	141	252	147	262
35,000	-29	-34	1034	184	368	---	---	---	---	---	---	136	252

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,500 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	482	964	230	243	231	244	232	245	233	246
2000	47	41	2230	468	936	228	248	229	249	230	250	231	251
4000	43	37	2230	455	910	226	252	227	254	228	255	229	256
6000	39	33	2230	441	882	223	257	224	259	225	260	226	261
8000	36	29	2230	429	858	221	262	222	264	223	265	224	266
10,000	32	25	2211	415	830	217	266	219	268	220	270	221	271
12,000	28	21	2095	387	774	210	266	212	268	213	270	215	271
14,000	24	17	2027	370	740	205	267	206	269	208	272	210	273
16,000	20	13	1960	353	706	199	268	201	271	203	273	204	275
18,000	16	9	1892	337	674	193	269	195	272	197	275	199	277
20,000	12	5	1844	325	650	188	271	190	274	193	277	194	288
22,000	8	1	1771	310	620	181	271	184	275	187	278	189	281
24,000	4	-3	1656	290	580	173	267	176	272	179	276	182	280
26,000	0	-7	1535	268	536	163	261	167	267	171	273	174	278
28,000	-4	-11	1413	247	494	151	252	157	261	162	268	165	274
29,000	-7	-12	1352	237	474	144	244	151	257	157	265	161	272
31,000	-11	-16	1228	217	434	---	---	138	243	146	257	151	266
33,000	-15	-20	1127	201	402	---	---	---	---	133	243	141	258
35,000	-19	-24	1027	185	370	---	---	---	---	---	---	128	243

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

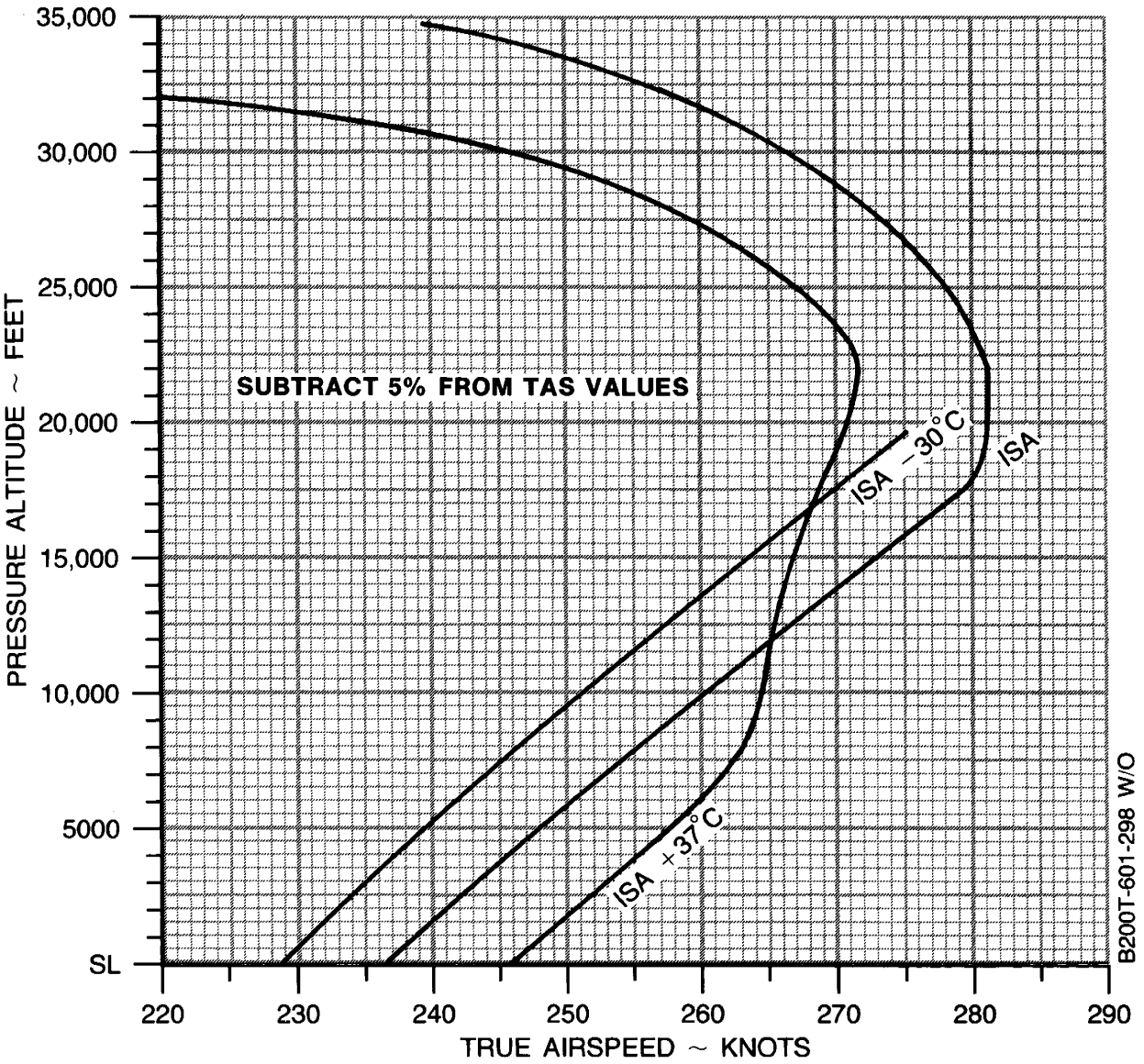
PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	484	968	230	245	231	246	232	247	232	248
2000	54	48	2230	470	940	227	249	228	251	229	252	230	253
4000	50	44	2230	456	912	225	254	226	255	227	257	228	258
6000	46	40	2230	443	886	222	259	223	260	224	262	225	263
8000	43	36	2178	423	846	217	261	219	263	220	265	221	266
10,000	39	32	2094	399	798	211	262	213	264	215	266	216	268
12,000	35	28	1985	372	744	204	262	206	264	208	266	209	268
14,000	31	24	1925	356	712	200	264	201	266	203	269	205	270
16,000	27	20	1863	340	680	194	265	196	268	198	270	200	273
18,000	23	16	1803	325	650	188	266	191	269	193	271	195	275
20,000	19	12	1757	314	628	183	267	186	271	188	274	190	277
22,000	15	8	1699	301	602	177	267	180	272	183	276	185	279
24,000	11	4	1594	281	562	168	263	172	269	175	274	178	278
26,000	7	0	1478	260	520	158	257	163	264	167	270	170	276
28,000	3	-4	1358	239	478	146	246	153	257	157	265	162	272
29,000	0	-5	1298	229	458	136	234	147	252	152	261	157	269
31,000	-5	-9	1175	210	420	---	---	134	234	142	253	148	263
33,000	-9	-13	1085	195	390	---	---	---	---	127	236	137	254
35,000	-13	-17	994	179	358	---	---	---	---	---	---	123	237

MAXIMUM CRUISE SPEEDS

1700 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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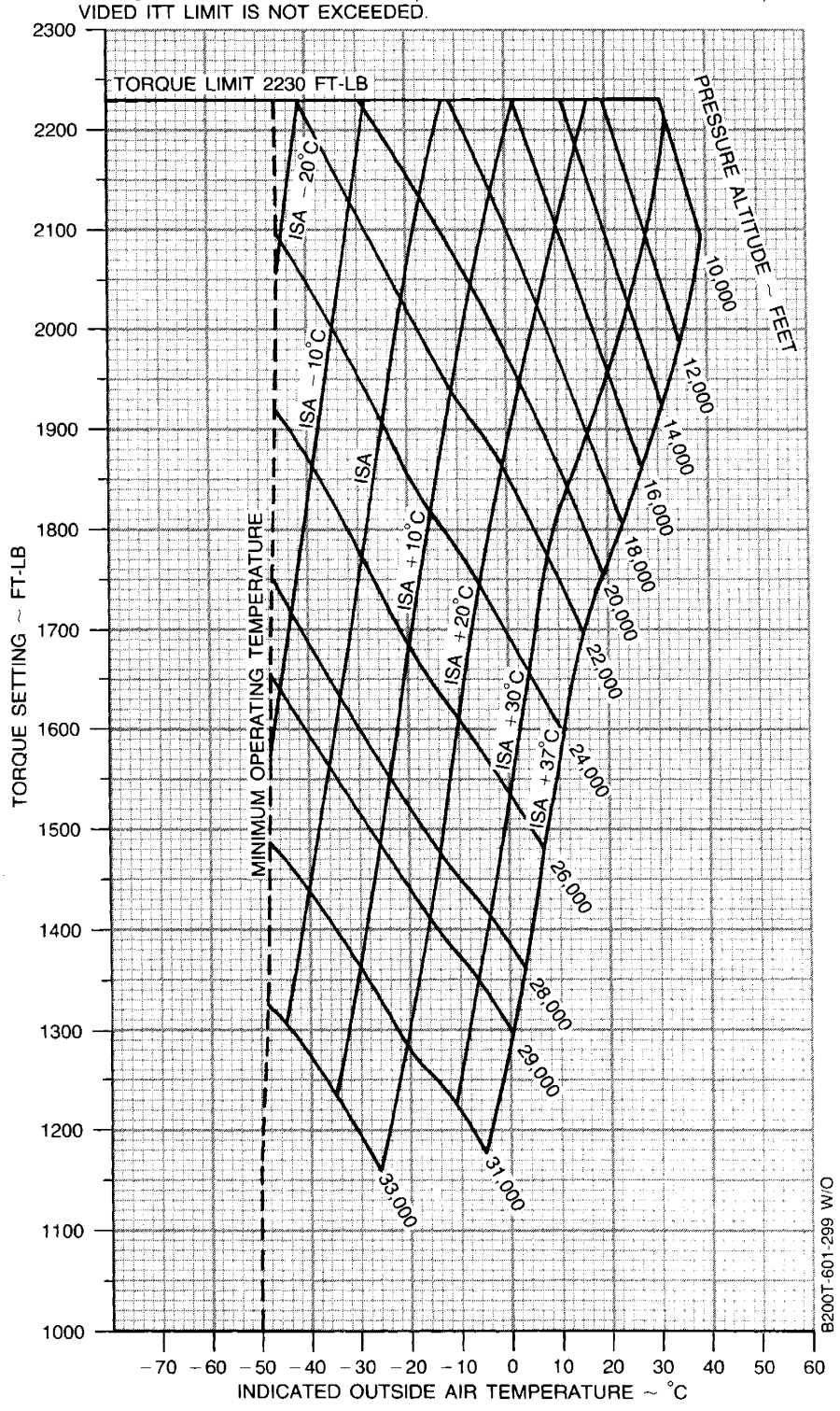
MAXIMUM CRUISE POWER

1700 RPM

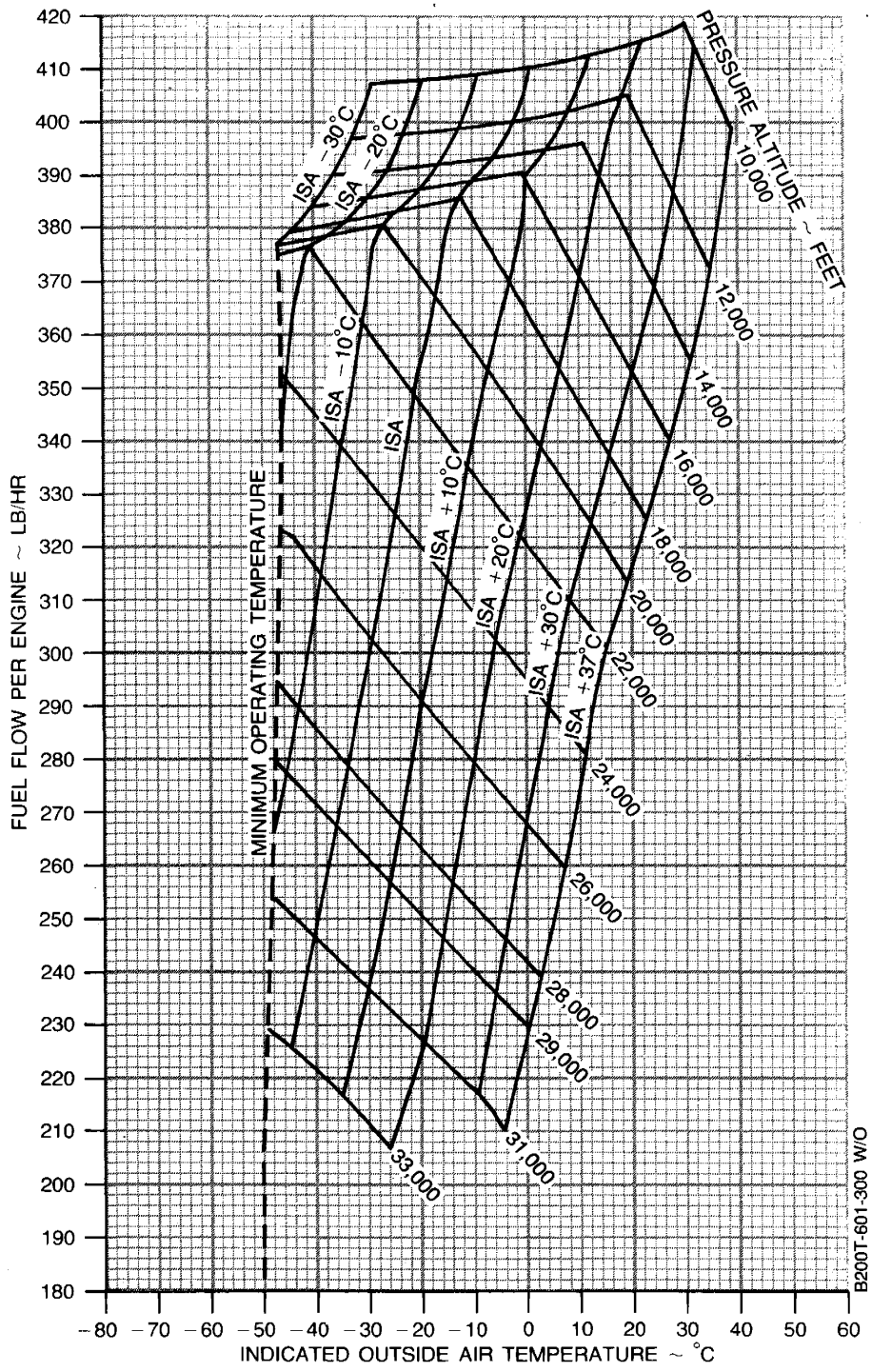
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT MAXIMUM CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - MAXIMUM CRUISE POWER

1700 RPM

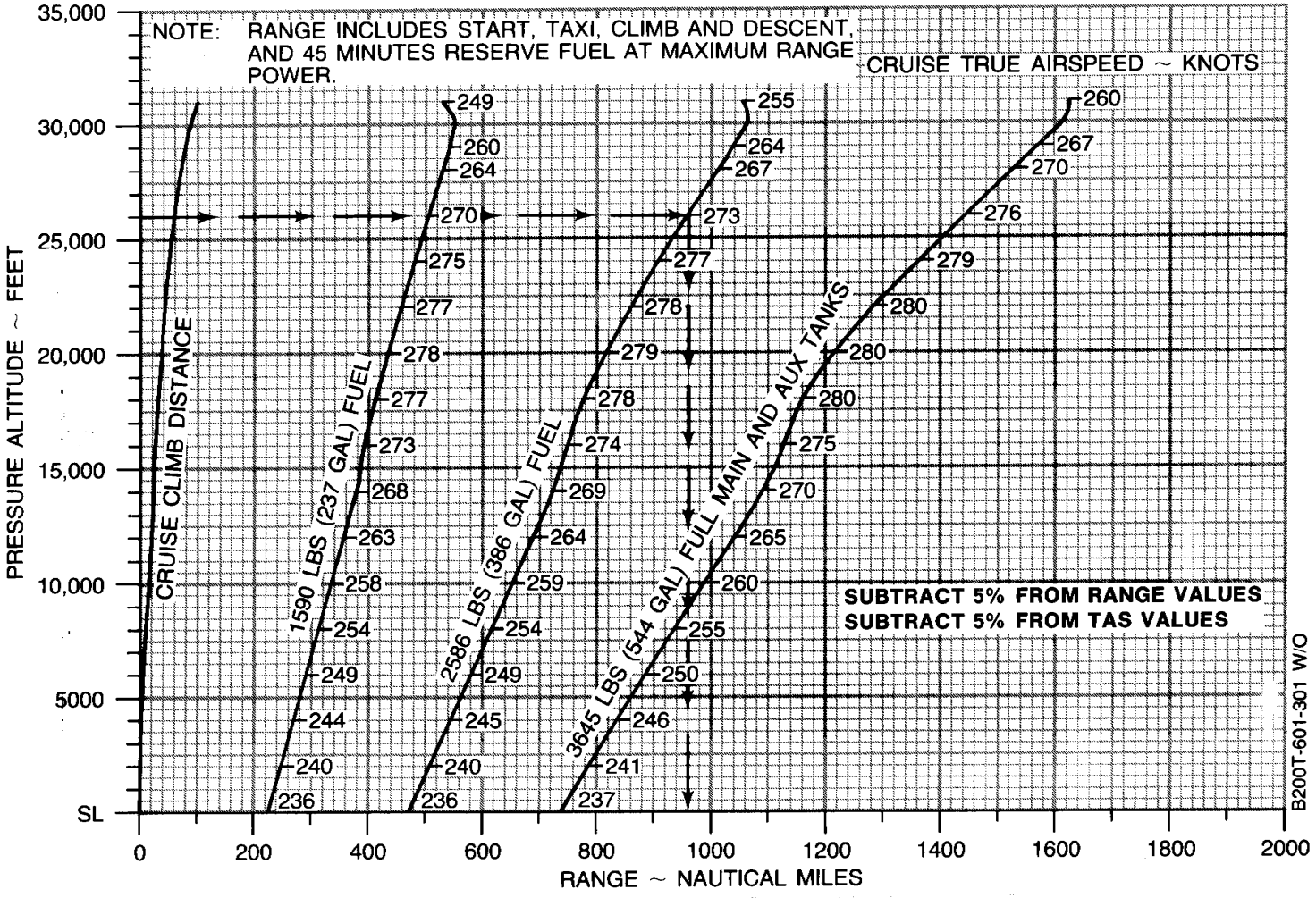
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	960 NM



B200T-601-301 W/O

For Training Purposes Only

NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2194	481	962	244	232	244	232	244	232	244	232
2000	-14	-19	2230	473	946	243	237	244	238	244	238	244	238
4000	-18	-23	2230	460	920	241	242	241	243	242	243	243	244
6000	-21	-27	2230	448	896	238	246	239	247	240	248	241	249
8000	-25	-31	2230	437	874	236	251	237	252	238	253	239	254
10,000	-29	-35	2230	427	854	233	255	235	257	236	258	236	259
12,000	-32	-39	2230	417	834	231	260	232	262	233	263	234	264
14,000	-36	-43	2230	410	820	229	266	230	267	231	268	232	269
16,000	-40	-47	2230	404	808	226	271	228	272	229	273	230	275
18,000	-44	-51	2230	399	798	224	276	225	278	226	279	228	280
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2226	486	972	243	236	244	236	244	236	244	236
2000	-4	-9	2230	474	948	241	240	242	241	243	242	244	243
4000	-7	-13	2230	461	922	239	245	240	246	241	246	241	247
6000	-11	-17	2230	449	898	236	249	237	250	238	251	239	252
8000	-15	-21	2230	438	876	234	254	235	255	236	256	237	257
10,000	-19	-25	2230	428	856	232	259	233	260	234	261	235	262
12,000	-22	-29	2230	417	834	229	264	230	265	231	266	232	267
14,000	-26	-33	2230	411	822	227	269	228	270	230	272	230	273
16,000	-30	-37	2230	405	810	224	274	226	276	227	277	228	278
18,000	-33	-41	2230	400	800	222	280	223	281	225	283	226	284
20,000	-37	-45	2142	382	764	216	281	217	283	219	285	220	286
22,000	-41	-49	2041	363	726	209	281	211	283	213	285	214	287
24,000	-45	-53	1920	340	680	201	279	203	282	205	285	207	287
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	489	978	242	238	243	239	243	240	244	241
2000	7	1	2230	475	950	239	243	240	244	241	245	242	246
4000	3	-3	2230	463	926	237	247	238	248	239	249	240	250
6000	-1	-7	2230	450	900	234	252	236	253	237	254	237	255
8000	-5	-11	2230	439	878	232	257	233	258	234	259	235	260
10,000	-8	-15	2230	428	856	230	262	231	263	232	264	233	265
12,000	-12	-19	2230	419	838	227	267	229	268	230	270	231	271
14,000	-16	-23	2230	412	824	225	272	226	274	227	275	228	276
16,000	-19	-27	2230	406	812	222	278	224	279	225	281	226	282
18,000	-23	-31	2144	388	776	216	279	218	281	219	283	221	284
20,000	-27	-35	2041	367	734	210	279	211	281	213	283	214	285
22,000	-31	-39	1947	349	698	203	278	205	282	207	284	208	286
24,000	-35	-43	1830	327	654	195	277	197	281	200	283	201	286
26,000	-39	-47	1702	304	608	186	274	189	278	191	281	194	284
28,000	-44	-51	1568	280	560	176	269	180	274	183	278	185	282
29,000	-46	-52	1503	268	536	171	265	175	271	178	276	181	280
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	492	984	240	241	241	242	242	243	243	244
2000	17	11	2230	478	956	238	246	239	247	240	248	240	248
4000	13	7	2230	464	928	235	250	236	251	237	252	238	253
6000	9	3	2230	451	902	233	255	234	256	235	257	236	258
8000	5	-1	2230	440	880	230	260	232	261	233	262	234	263
10,000	2	-5	2230	429	858	228	265	229	266	230	267	231	269
12,000	-2	-9	2230	419	838	226	270	227	272	228	273	229	274
14,000	-6	-13	2230	411	822	223	275	224	277	226	278	227	280
16,000	-9	-17	2165	396	792	218	278	219	279	221	281	222	283
18,000	-13	-21	2036	372	744	210	276	212	279	213	281	215	282
20,000	-17	-25	1948	354	708	204	277	206	279	207	282	209	284
22,000	-21	-29	1860	336	672	197	277	199	280	201	283	203	285
24,000	-25	-33	1744	315	630	189	275	192	278	194	282	196	284
26,000	-29	-37	1619	291	582	180	271	183	275	186	279	188	282
28,000	-34	-41	1490	268	536	170	265	174	271	177	275	180	279
29,000	-36	-42	1427	257	514	164	260	168	267	172	273	175	277
31,000	-40	-46	1305	237	474	151	248	158	260	163	268	166	273
33,000	-45	-50	1186	217	434	131	225	145	248	152	260	157	268
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	31	25	2230	495	990	238	244	239	245	240	246	241	246
2000	27	21	2230	481	962	236	248	237	249	238	250	239	251
4000	23	17	2230	468	936	234	253	235	254	236	255	237	256
6000	19	13	2230	455	910	231	258	232	259	233	260	234	261
8000	16	9	2230	443	886	229	263	230	264	231	265	232	266
10,000	12	5	2230	431	862	226	268	228	269	229	270	230	272
12,000	8	1	2230	420	840	224	273	225	275	226	276	227	277
14,000	4	-3	2155	401	802	218	275	220	277	221	278	222	280
16,000	0	-7	2054	378	756	212	275	213	277	215	279	216	281
18,000	-4	-11	1953	357	714	205	275	207	277	208	280	210	281
20,000	-7	-15	1871	341	682	199	275	201	278	203	281	204	283
22,000	-11	-19	1775	324	648	191	275	194	278	196	281	198	283
24,000	-15	-23	1662	303	606	183	272	186	276	188	279	191	282
26,000	-20	-27	1540	280	560	174	267	177	272	180	277	183	280
28,000	-24	-31	1415	257	514	163	260	167	267	171	272	174	277
29,000	-26	-32	1353	247	494	149	247	162	263	166	270	170	275
31,000	-31	-36	1235	227	454	133	229	150	253	156	263	161	270
33,000	-36	-40	1116	207	414	---	---	135	237	145	254	151	264
35,000	-40	-44	1021	190	380	---	---	---	---	130	237	140	254

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	497	994	237	246	238	247	239	248	240	249
2000	37	31	2230	484	968	235	251	236	252	237	253	237	254
4000	33	27	2230	470	940	232	255	233	257	234	258	235	259
6000	29	23	2230	457	914	230	261	231	262	232	263	233	264
8000	26	19	2230	445	890	227	266	228	267	230	268	230	269
10,000	22	15	2212	431	862	224	270	225	271	227	273	228	274
12,000	18	11	2102	403	806	217	270	219	271	220	273	221	275
14,000	14	7	2015	381	762	211	270	213	273	214	274	215	276
16,000	10	3	1928	361	722	205	271	207	273	208	276	210	277
18,000	6	-1	1842	341	682	198	271	200	274	202	276	204	279
20,000	2	-5	1783	328	656	193	273	195	276	197	279	199	281
22,000	-1	-9	1708	313	626	187	273	189	277	191	280	193	283
24,000	-6	-13	1599	292	584	178	270	181	274	184	278	186	282
26,000	-10	-17	1472	269	538	168	264	172	270	175	274	178	279
28,000	-14	-21	1342	247	494	155	253	161	263	165	269	169	274
29,000	-17	-22	1281	236	472	147	245	155	257	160	266	164	272
31,000	-21	-26	1163	217	434	---	---	142	245	150	258	155	266
33,000	-25	-30	1018	190	380	---	---	---	---	137	245	145	259
35,000	-29	-34	984	184	368	---	---	---	---	---	---	132	246

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	500	1000	235	249	237	250	237	251	238	251
2000	47	41	2230	486	972	233	253	234	254	235	255	236	256
4000	43	37	2230	472	944	231	258	232	259	233	261	234	262
6000	40	33	2225	458	916	228	263	229	264	230	265	231	267
8000	36	29	2142	434	868	222	264	224	266	225	267	226	268
10,000	32	25	2060	409	818	216	265	218	267	219	269	220	270
12,000	28	21	1956	383	766	209	265	211	267	213	269	214	271
14,000	24	17	1874	362	724	203	265	205	268	207	270	208	272
16,000	20	13	1798	343	686	197	266	199	269	201	271	203	273
18,000	16	9	1727	326	652	191	267	194	270	196	273	197	275
20,000	12	5	1682	313	626	186	269	189	272	191	275	193	278
22,000	8	1	1629	301	602	181	270	184	274	186	277	188	281
24,000	4	-3	1520	280	560	172	266	175	271	178	275	181	279
26,000	0	-7	1404	258	516	161	259	166	266	170	272	173	276
28,000	-4	-11	1286	237	474	148	247	156	259	160	266	164	272
29,000	-7	-12	1227	227	454	280	237	149	252	155	263	159	270
31,000	-12	-16	1107	207	414	---	---	134	236	144	253	150	263
33,000	-16	-20	1013	192	384	---	---	---	---	128	235	139	253
35,000	-20	-24	920	176	352	---	---	---	---	---	---	123	234

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NORMAL CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

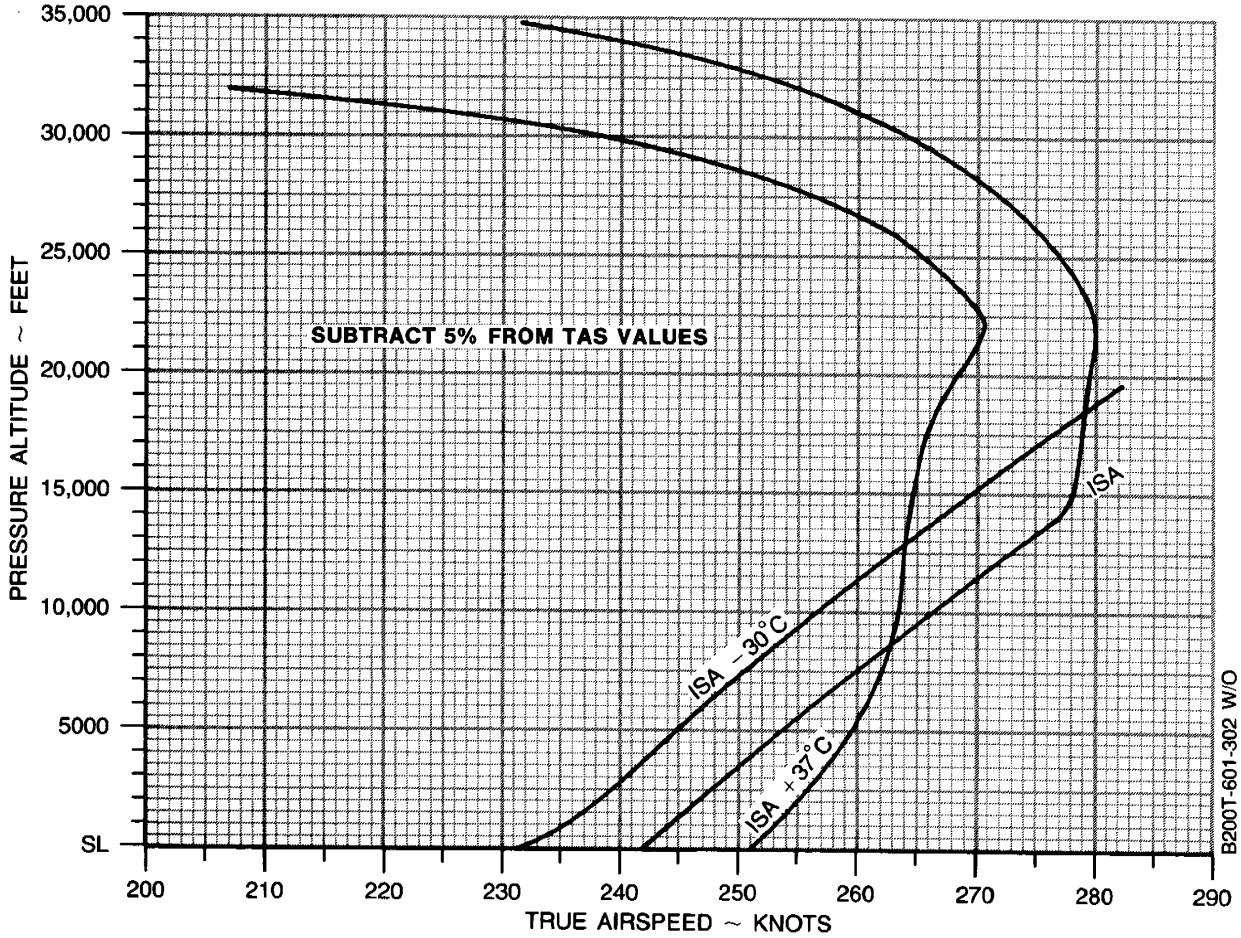
PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	501	1002	234	250	236	251	236	252	237	253
2000	54	48	2207	484	968	231	254	232	255	233	256	236	257
4000	50	44	2165	465	930	227	257	228	258	229	260	231	261
6000	46	40	2107	443	886	222	259	224	261	225	262	226	263
8000	43	36	2032	419	838	217	260	218	262	220	264	221	265
10,000	39	32	1958	396	792	211	262	213	264	214	266	215	267
12,000	35	28	1859	369	738	204	261	206	264	208	266	209	268
14,000	31	24	1783	350	700	198	262	200	265	202	267	203	269
16,000	27	20	1711	332	664	192	262	194	265	196	268	198	270
18,000	23	16	1643	315	630	186	263	189	266	191	269	193	272
20,000	19	12	1604	303	606	181	265	184	269	186	272	188	275
22,000	15	8	1558	292	584	176	266	179	271	182	274	184	278
24,000	11	4	1460	272	544	167	262	171	268	174	273	177	277
26,000	7	0	1353	251	502	156	254	162	263	166	269	169	274
28,000	3	-4	1237	230	460	142	240	151	254	156	263	160	270
29,000	1	-5	1178	220	440	130	223	143	247	151	259	156	267
31,000	-5	-9	1055	200	400	---	---	125	223	139	247	146	260
33,000	-9	-13	972	185	370	---	---	---	---	121	225	134	249
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE SPEEDS

1800 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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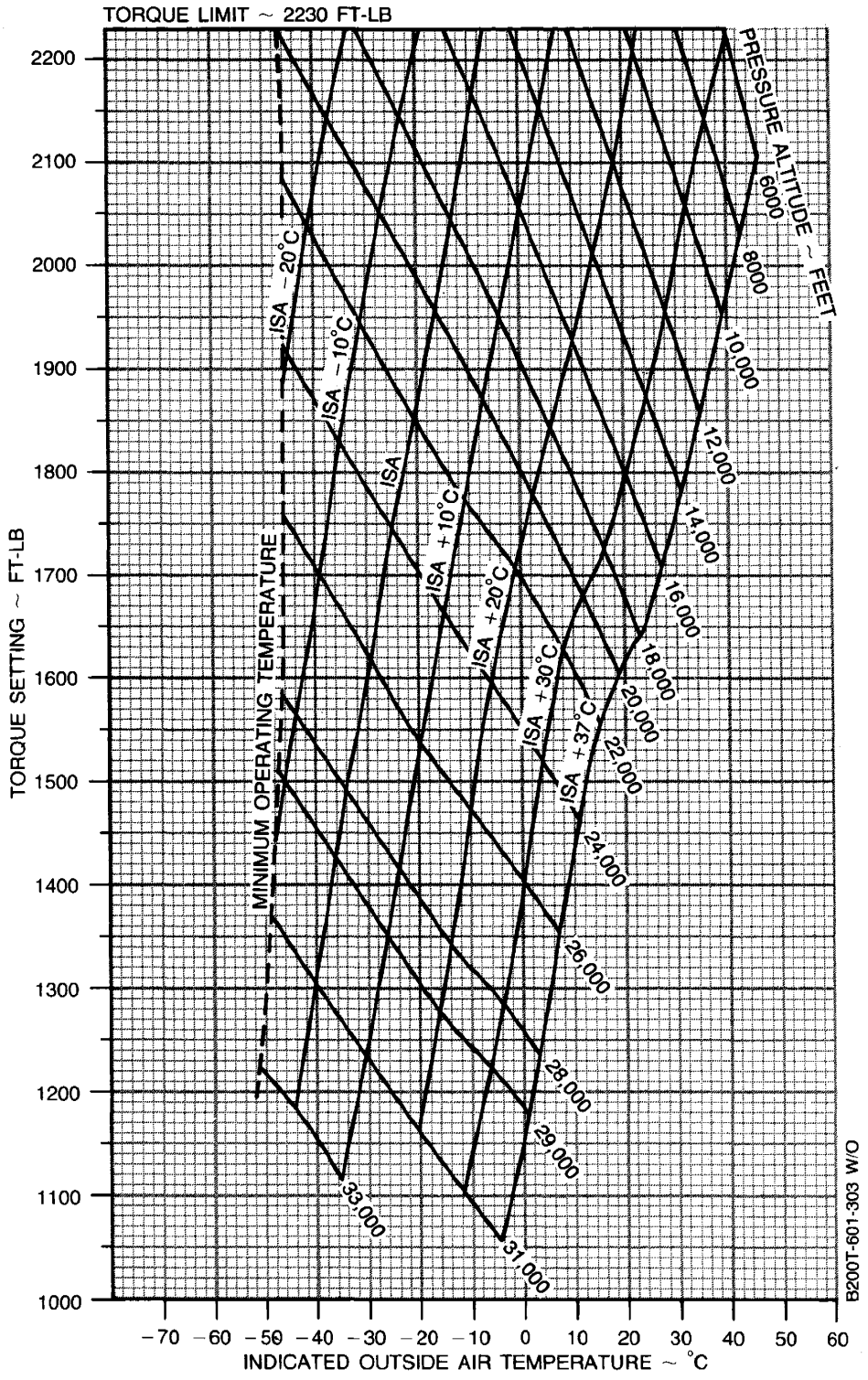
NORMAL CRUISE POWER

1800 RPM

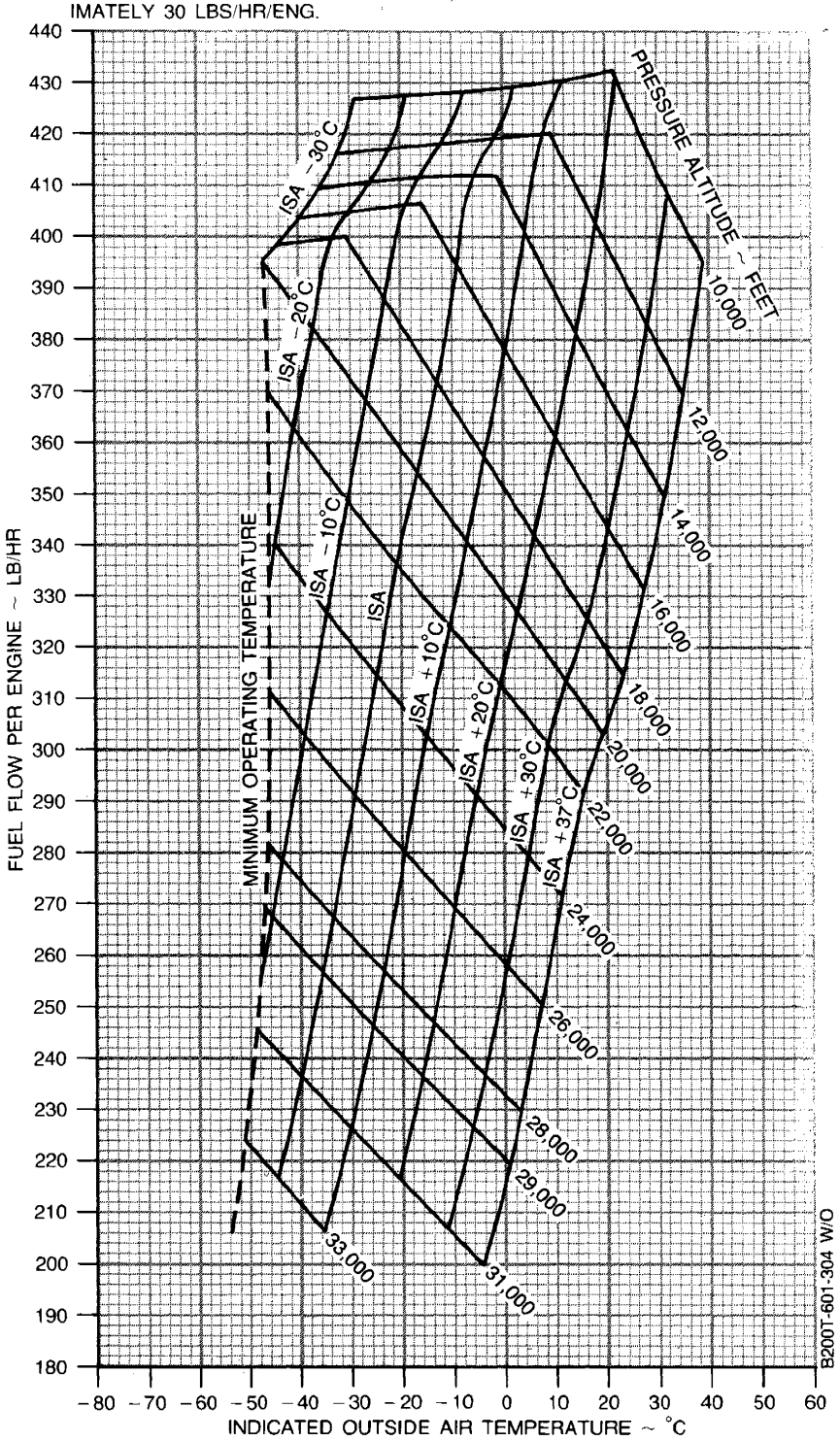
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT NORMAL CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - NORMAL CRUISE POWER

1800 RPM

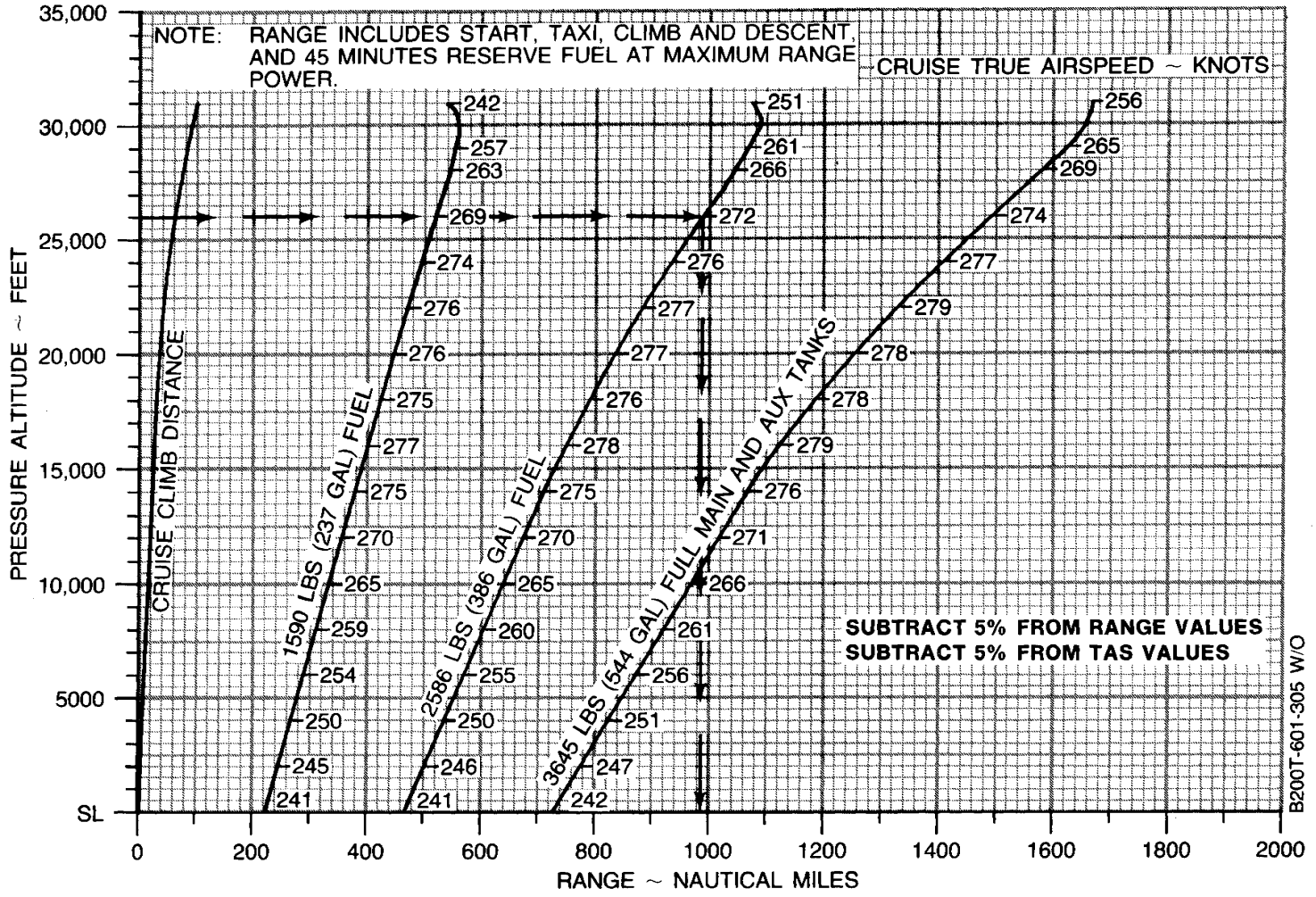
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	988 NM



For Training Purposes Only

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2194	481	962	244	232	244	232	244	232	244	232
2000	-14	-19	2230	473	946	243	237	244	238	244	238	244	238
4000	-18	-23	2230	460	920	241	242	241	243	242	243	243	244
6000	-21	-27	2230	448	896	238	246	239	247	240	248	241	249
8000	-25	-31	2230	437	874	236	251	237	238	238	253	239	254
10,000	-29	-35	2230	427	854	233	255	235	257	236	258	236	259
12,000	-32	-39	2230	417	834	231	260	232	262	233	263	234	264
14,000	-36	-43	2230	410	820	229	265	230	267	231	268	232	269
16,000	-40	-47	2230	404	808	226	271	228	272	229	273	230	275
18,000	-44	-51	2230	399	798	224	276	225	278	226	279	228	280
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2226	486	972	243	236	244	236	244	236	244	236
2000	-4	-9	2230	474	948	241	240	242	241	243	242	244	243
4000	-7	-13	2230	461	922	239	245	240	246	241	246	241	247
6000	-11	-17	2230	449	898	236	249	237	250	238	251	239	252
8000	-15	-21	2230	438	876	234	254	235	255	236	256	237	257
10,000	-19	-25	2230	428	856	232	259	233	260	234	261	235	262
12,000	-22	-29	2230	417	834	229	264	230	265	231	266	232	267
14,000	-26	-33	2230	411	822	227	269	228	270	229	272	230	273
16,000	-30	-37	2230	405	810	224	274	226	276	227	277	228	278
18,000	-33	-41	2230	400	800	222	280	223	281	225	283	226	284
20,000	-37	-45	2230	396	792	220	285	221	287	222	289	223	290
22,000	-41	-49	2133	378	756	213	286	214	288	216	290	217	292
24,000	-45	-53	1996	352	704	204	284	206	286	208	289	210	291
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	489	978	242	238	243	239	243	240	244	241
2000	7	1	2230	475	950	239	243	240	244	241	245	242	246
4000	3	-3	2230	463	926	237	247	238	248	239	249	240	250
6000	-1	-7	2230	450	900	234	252	236	253	237	254	237	255
8000	-5	-11	2230	439	878	232	257	233	258	234	259	235	260
10,000	-8	-15	2230	428	856	230	262	231	263	232	264	233	265
12,000	-12	-19	2230	419	838	227	267	229	268	230	270	231	271
14,000	-16	-23	2230	412	824	225	272	226	274	227	275	228	276
16,000	-19	-27	2230	406	812	222	278	224	279	225	281	226	282
18,000	-23	-31	2230	401	802	220	283	221	285	223	287	223	288
20,000	-27	-35	2133	382	764	213	284	215	286	217	288	218	289
22,000	-31	-39	2031	363	726	206	284	208	286	210	288	212	290
24,000	-35	-43	1911	341	682	199	282	201	285	203	288	204	290
26,000	-39	-47	1780	316	632	190	279	192	283	195	286	197	288
28,000	-43	-51	1639	291	582	181	274	183	279	186	283	188	286
29,000	-46	-52	1568	278	556	175	271	178	276	181	280	184	284
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	492	984	240	241	241	242	242	243	243	244
2000	17	11	2230	478	956	238	246	239	247	240	248	240	248
4000	13	7	2230	464	928	235	250	236	251	237	252	238	253
6000	9	3	2230	451	902	233	255	234	256	235	257	236	258
8000	5	-1	2230	440	880	230	260	232	261	233	262	234	263
10,000	2	-5	2230	429	858	228	265	229	266	230	267	231	269
12,000	-2	-9	2230	419	838	226	270	227	272	228	273	229	274
14,000	-6	-13	2230	411	822	223	275	224	277	226	278	227	280
16,000	-9	-17	2230	406	812	221	281	222	283	223	284	224	285
18,000	-13	-21	2128	386	772	213	281	216	283	217	285	218	287
20,000	-17	-25	2033	367	734	207	282	209	284	211	286	212	288
22,000	-21	-29	1938	349	698	201	282	203	284	204	287	206	289
24,000	-25	-33	1819	327	654	192	280	195	283	197	286	199	288
26,000	-29	-37	1692	303	606	184	276	186	280	189	284	191	287
28,000	-33	-41	1562	280	560	174	271	177	276	180	281	183	284
29,000	-36	-42	1499	269	538	169	268	172	273	176	278	178	283
31,000	-40	-46	1375	248	496	157	258	162	267	167	274	170	279
33,000	-45	-50	1249	227	454	140	240	150	257	157	267	161	274
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	31	25	2230	495	990	238	244	239	245	240	246	241	246
2000	27	21	2230	481	962	236	248	237	249	238	250	239	251
4000	23	17	2230	468	936	234	253	235	254	236	255	237	256
6000	19	13	2230	455	910	231	258	232	259	233	260	234	261
8000	16	9	2230	443	886	229	263	230	264	231	265	232	266
10,000	12	5	2230	431	862	226	268	228	269	229	270	230	272
12,000	8	1	2230	420	840	224	273	225	275	226	276	227	277
14,000	5	-3	2226	412	824	221	278	223	280	224	282	225	283
16,000	1	-7	2132	391	782	215	279	216	281	218	283	219	284
18,000	-3	-11	2035	371	742	208	279	210	282	212	284	213	286
20,000	-7	-15	1952	354	708	202	280	204	283	206	285	207	287
22,000	-11	-19	1848	336	672	195	279	197	282	199	285	201	288
24,000	-15	-23	1733	314	628	187	277	189	281	192	284	194	287
26,000	-19	-27	1610	291	582	177	273	181	277	183	281	186	285
28,000	-24	-31	1484	268	536	167	267	171	273	175	278	177	282
29,000	-26	-32	1422	258	516	161	262	166	270	170	275	173	280
31,000	-30	-36	1304	237	474	148	250	156	262	161	270	164	276
33,000	-35	-40	1186	218	436	---	---	143	250	150	263	155	271
35,000	-39	-44	1087	201	402	---	---	---	---	137	249	145	263

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	497	994	237	246	238	247	239	248	240	249
2000	37	31	2230	484	968	235	251	236	252	237	253	237	254
4000	33	27	2230	470	940	232	255	233	257	234	258	235	259
6000	29	23	2230	457	914	230	261	231	262	232	263	233	264
8000	26	19	2230	445	890	227	266	228	267	230	268	230	269
10,000	22	15	2230	434	868	225	271	226	272	227	273	228	275
12,000	18	11	2146	410	820	219	272	220	274	222	275	223	277
14,000	14	7	2075	390	780	213	274	215	276	216	277	218	279
16,000	10	3	2002	372	744	208	275	210	277	211	279	213	281
18,000	7	-1	1928	355	710	202	276	204	279	206	281	207	283
20,000	3	-5	1854	339	678	196	277	198	280	200	283	202	285
22,000	-1	-9	1776	324	648	190	278	192	281	194	284	196	287
24,000	-5	-13	1667	303	606	181	275	184	279	187	283	189	286
26,000	-9	-17	1539	280	560	172	270	175	275	178	279	181	283
28,000	-14	-21	1408	258	516	161	262	165	269	169	275	172	279
29,000	-16	-22	1348	247	494	154	255	160	265	164	272	168	277
31,000	-21	-26	1231	227	454	137	237	148	255	154	266	159	273
33,000	-26	-30	1112	208	416	---	---	132	237	143	256	150	267
35,000	-30	-34	1019	191	382	---	---	---	---	127	236	138	256

MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	500	1000	235	248	236	250	237	251	238	251
2000	47	41	2230	486	972	233	253	234	254	235	255	236	256
4000	43	37	2230	472	944	231	258	232	259	233	261	234	262
6000	40	33	2230	459	918	228	263	229	264	230	266	231	267
8000	36	29	2181	439	878	224	266	225	267	226	269	227	270
10,000	32	25	2095	415	830	218	267	219	269	221	270	222	272
12,000	28	21	1987	387	774	211	267	212	269	214	270	215	272
14,000	24	17	1924	370	740	205	268	207	271	209	273	210	275
16,000	20	13	1864	353	706	200	270	202	273	204	275	205	277
18,000	16	9	1806	337	674	195	272	197	275	199	277	201	279
20,000	13	5	1758	325	650	190	274	192	277	194	280	196	283
22,000	9	1	1690	311	622	184	274	187	278	189	281	191	284
24,000	5	-3	1583	290	580	175	271	179	276	181	280	184	283
26,000	0	-7	1468	269	538	166	266	170	272	173	277	176	281
28,000	-4	-11	1352	248	496	154	257	161	266	164	273	168	278
29,000	-6	-12	1293	238	476	147	248	154	261	159	269	163	276
31,000	-11	-16	1175	218	436	---	---	141	248	149	262	154	270
33,000	-15	-20	1078	202	404	---	---	---	---	135	248	144	262
35,000	-19	-24	981	185	370	---	---	---	---	---	---	130	247

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MAXIMUM CRUISE POWER

SUBTRACT 5% FROM THE TAS VALUES

1800 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

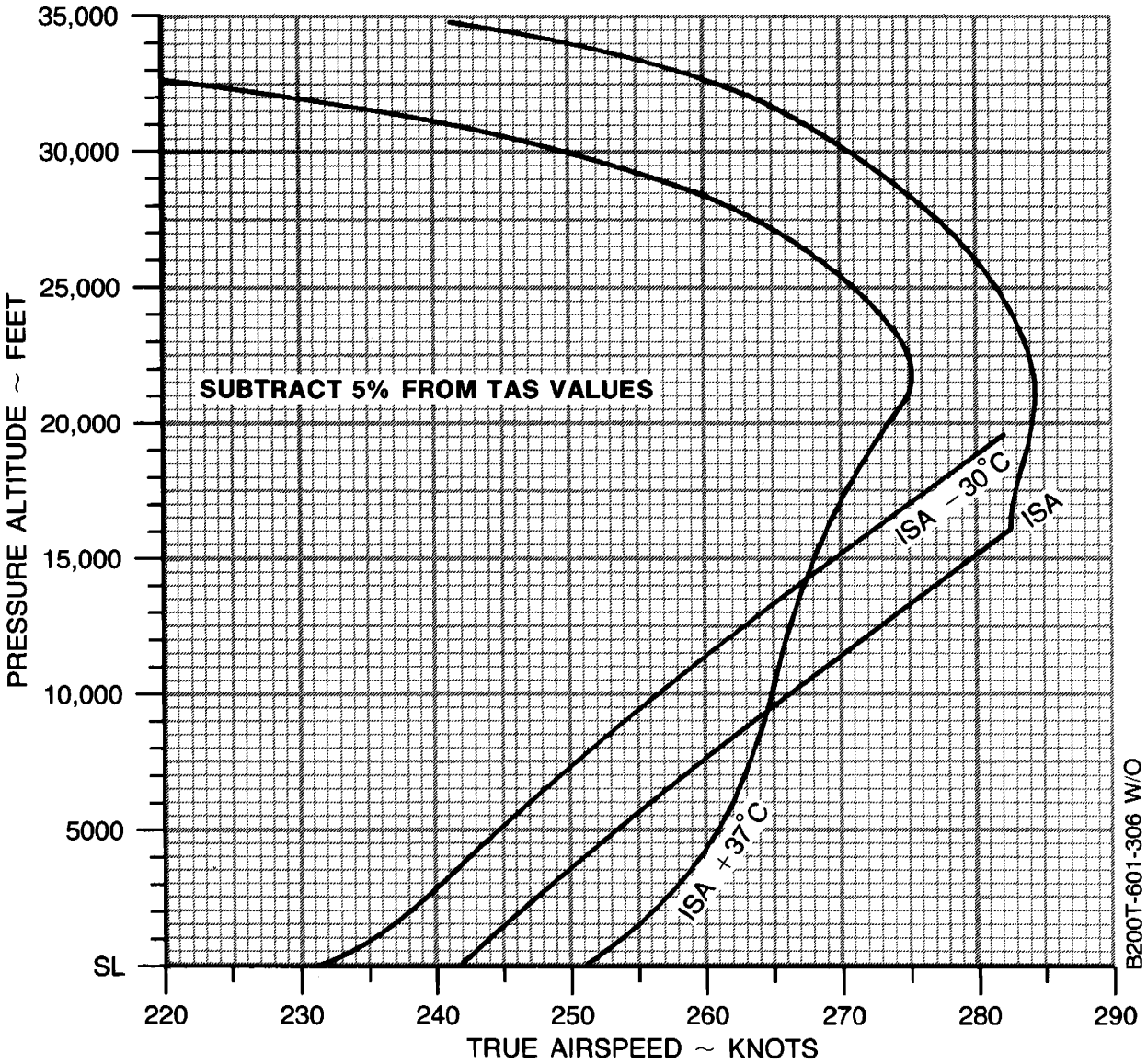
PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	501	1002	234	250	236	251	236	252	237	253
2000	54	48	2230	487	974	232	255	233	256	234	257	235	258
4000	50	44	2200	470	940	228	258	230	260	231	261	232	262
6000	46	40	2139	447	894	223	261	225	262	226	264	227	265
8000	43	36	2061	423	846	218	262	219	264	221	265	222	267
10,000	39	32	1983	399	798	212	263	214	265	215	267	216	268
12,000	35	28	1881	373	746	205	263	207	265	209	267	210	269
14,000	31	24	1826	356	712	200	264	202	267	204	269	205	271
16,000	27	20	1770	341	682	195	266	197	269	199	272	201	274
18,000	23	16	1716	325	650	190	268	192	271	194	274	196	276
20,000	19	12	1675	314	628	185	270	187	274	190	277	192	280
22,000	15	8	1621	301	602	179	271	182	275	185	279	187	282
24,000	11	4	1522	282	564	171	268	174	273	177	277	180	281
26,000	7	0	1414	261	522	161	262	166	269	169	274	172	279
28,000	3	-4	1300	240	480	148	250	156	262	160	269	164	275
29,000	0	-5	1241	230	460	139	239	149	256	155	266	159	273
31,000	-4	-9	1124	211	422	---	---	134	239	144	257	150	267
33,000	-8	-13	1038	195	390	---	---	---	---	130	241	140	258
35,000	-12	-17	951	180	360	---	---	---	---	---	---	125	242

MAXIMUM CRUISE SPEEDS

1800 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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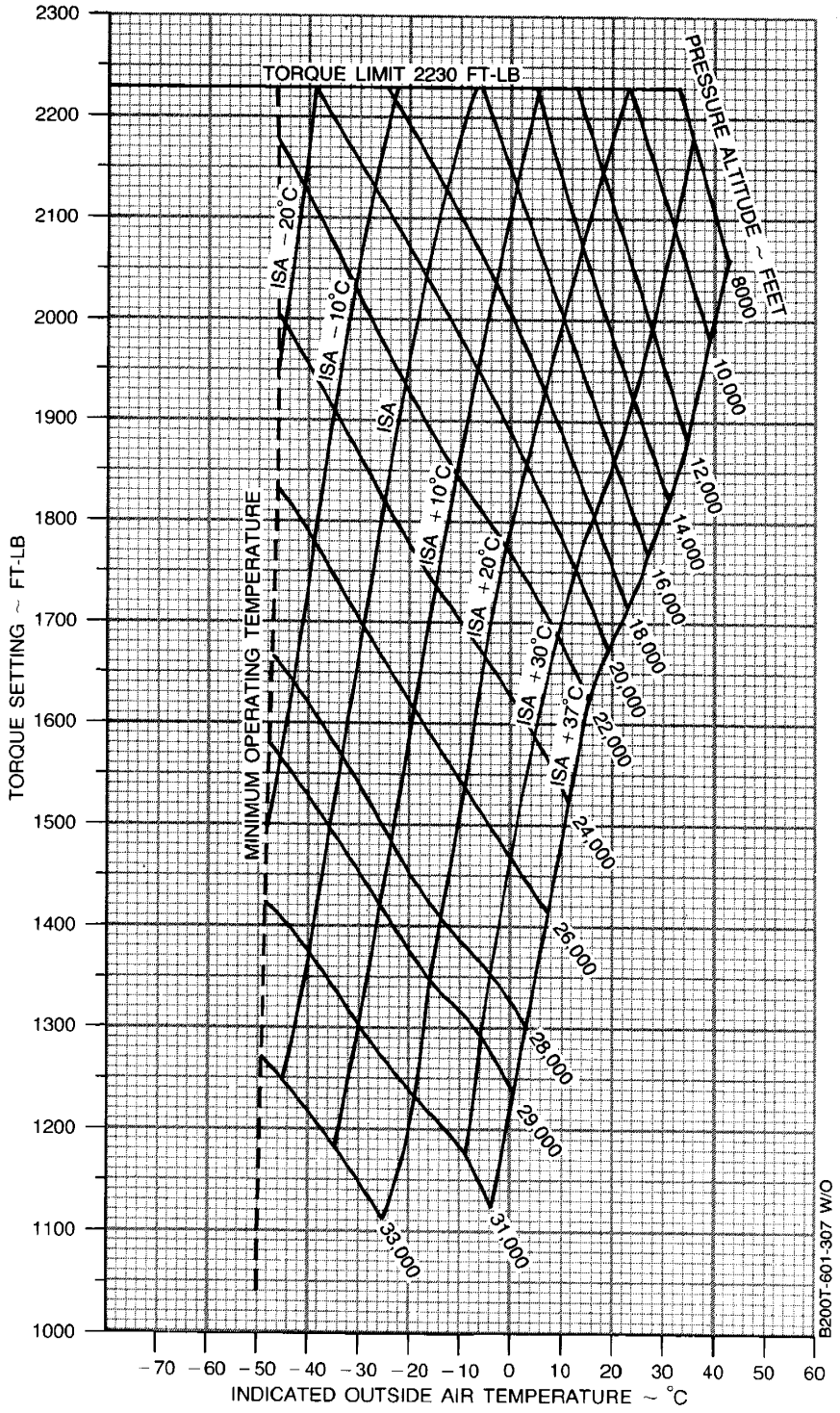
MAXIMUM CRUISE POWER

1800 RPM

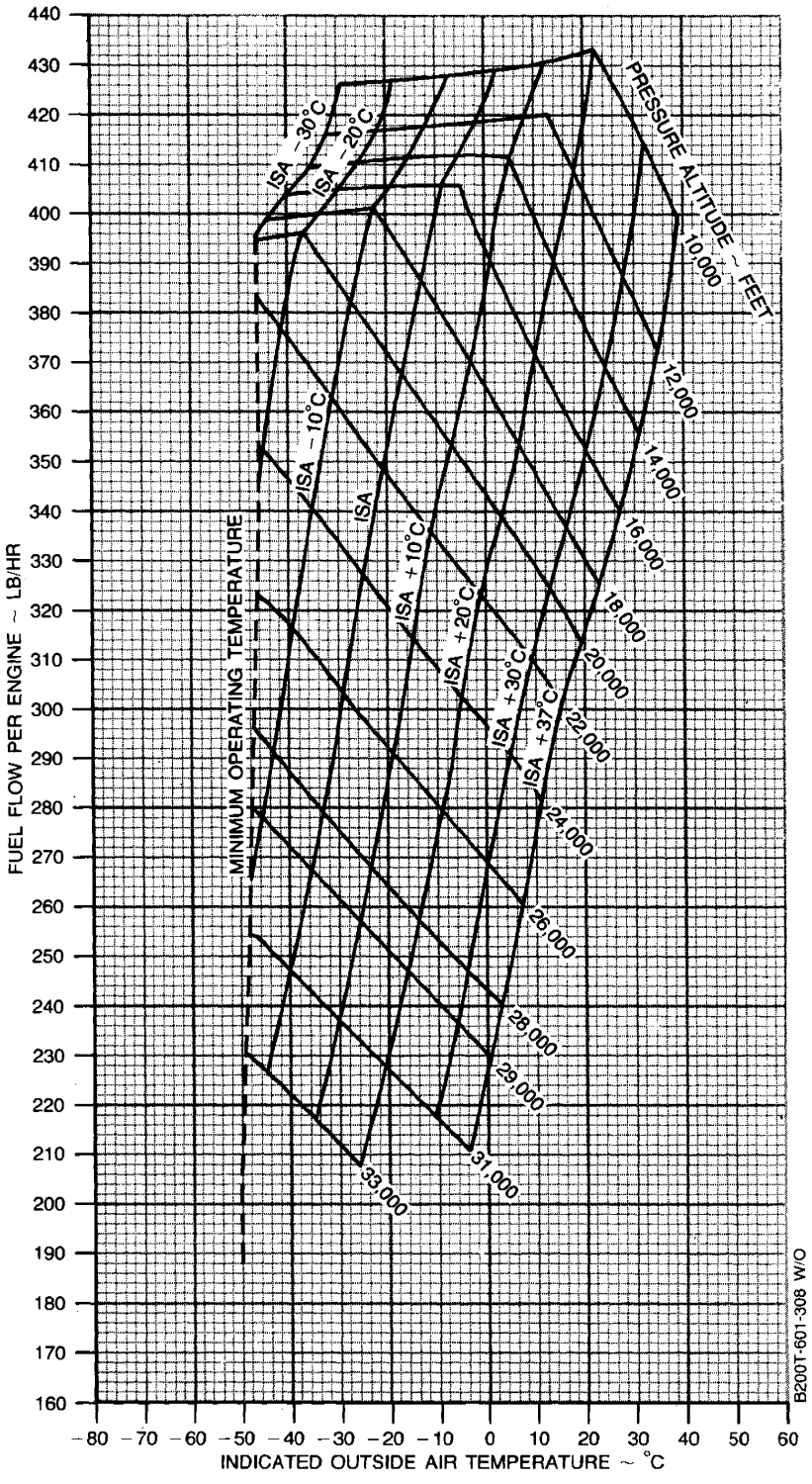
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT MAXIMUM CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - MAXIMUM CRUISE POWER

1800 RPM

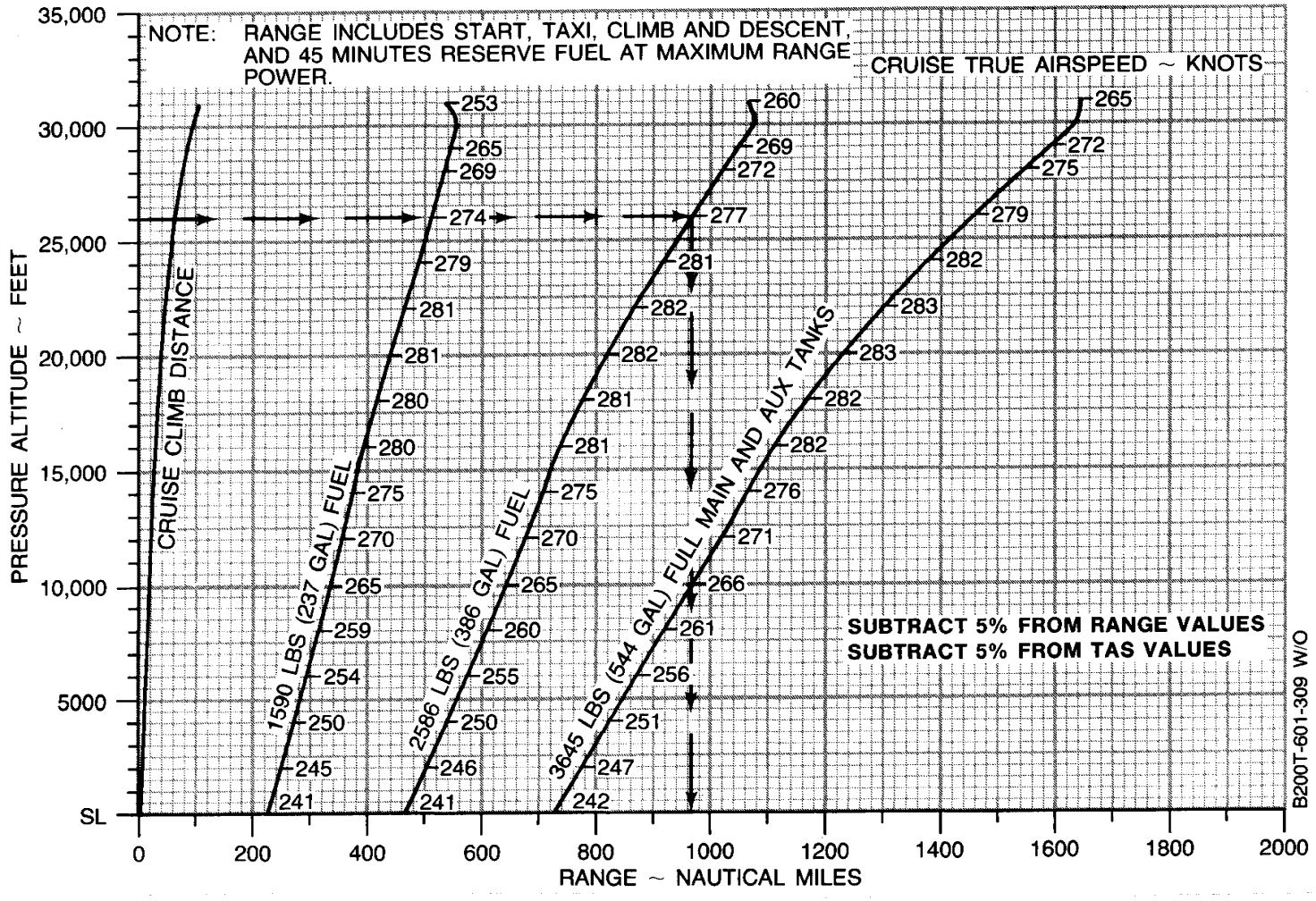
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	968 NM



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MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-11	-15	1618	394	788	210	200	1542	384	768	207	197
2000	-15	-19	1496	365	730	201	197	1422	355	710	199	195
4000	-20	-23	1395	337	674	194	195	1313	327	654	191	192
6000	-24	-27	1318	314	628	187	194	1229	302	604	184	191
8000	-28	-31	1254	292	584	181	193	1159	280	560	177	189
10,000	-32	-35	1203	273	546	175	193	1103	260	520	171	189
12,000	-35	-39	1175	258	516	171	194	1070	244	488	167	189
14,000	-39	-43	1145	245	490	167	194	1031	230	460	162	189
16,000	-43	-47	1124	234	468	163	196	1006	218	436	158	190
18,000	-47	-51	1105	223	446	158	197	978	206	412	153	190
20,000	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-11	-15	1475	376	734	205	195	1413	368	736	203	193
2000	-15	-19	1351	347	694	197	192	1280	338	676	194	190
4000	-20	-23	1230	316	632	188	189	1147	306	612	184	186
6000	-24	-27	1136	290	580	180	187	1043	278	556	176	183
8000	-28	-31	1060	267	534	173	185	961	255	510	169	180
10,000	-32	-35	1001	246	492	167	184	898	233	466	162	179
12,000	-35	-39	959	229	458	162	184	850	215	430	157	178
14,000	-39	-43	910	214	428	156	183	792	199	398	150	176
16,000	-43	-47	879	201	402	152	183	748	184	368	145	175
18,000	-47	-51	841	188	376	146	182	696	169	338	138	172
20,000	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-1	-5	1721	407	814	214	207	1643	398	796	211	205
2000	-5	-9	1602	379	758	206	205	1505	367	734	202	201
4000	-9	-13	1488	351	702	197	202	1397	339	678	194	199
6000	-13	-17	1409	327	654	191	201	1321	316	632	187	198
8000	-17	-21	1343	306	612	185	201	1254	294	588	181	197
10,000	-21	-25	1275	284	568	178	200	1185	272	544	175	196
12,000	-25	-29	1251	270	540	174	201	1154	257	514	170	197
14,000	-29	-33	1223	257	514	170	202	1118	243	486	166	198
16,000	-33	-37	1199	245	490	165	203	1085	230	460	161	198
18,000	-37	-41	1182	236	472	162	205	1061	220	440	157	199
20,000	-41	-45	1173	228	456	158	207	1039	210	420	152	200
22,000	-45	-49	1181	223	446	156	211	1042	204	408	150	203
24,000	-49	-53	1210	222	444	155	217	1068	202	404	149	208
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-1	-5	1565	388	776	208	202	1495	380	760	206	200
2000	-5	-9	1426	358	716	199	199	1357	349	698	197	196
4000	-9	-13	1320	330	660	191	196	1247	321	642	189	194
6000	-13	-17	1241	306	612	185	195	1164	296	592	182	192
8000	-17	-21	1170	284	568	178	194	1087	273	546	175	191
10,000	-21	-25	1097	261	522	171	192	1009	250	500	168	188
12,000	-25	-29	1059	245	490	167	193	963	233	466	163	188
14,000	-29	-33	1017	230	460	162	193	913	217	434	157	187
16,000	-33	-37	974	216	432	156	192	858	201	402	151	186
18,000	-37	-41	943	204	408	152	193	817	187	374	145	185
20,000	-41	-45	910	192	384	147	192	770	174	348	139	183
22,000	-45	-49	906	185	370	143	195	766	166	332	136	185
24,000	-49	-53	931	183	366	142	200	795	164	328	136	191
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	9	5	1776	415	830	216	213	1694	405	810	213	210
2000	5	1	1690	391	782	209	212	1603	380	760	206	209
4000	1	-3	1582	364	728	201	210	1488	352	704	198	207
6000	-3	-7	1496	339	678	194	209	1402	328	656	191	205
8000	-7	-11	1424	317	634	188	208	1328	305	610	184	204
10,000	-11	-15	1357	295	590	181	207	1260	283	566	178	203
12,000	-15	-19	1330	281	562	177	209	1226	268	536	173	204
14,000	-19	-23	1296	267	534	173	210	1185	253	506	168	205
16,000	-23	-27	1280	257	514	169	212	1157	241	482	164	206
18,000	-27	-31	1264	247	494	165	214	1129	230	460	159	207
20,000	-31	-35	1257	240	480	162	217	1112	221	442	156	209
22,000	-34	-39	1264	235	470	160	221	1115	215	430	153	212
24,000	-38	-43	1281	232	464	158	226	1144	213	426	152	218
26,000	-42	-47	1297	230	460	156	232	1163	211	422	151	224
28,000	-46	-51	1291	226	452	153	234	1187	210	420	150	230
29,000	-48	-52	1317	229	458	153	239	1191	210	420	149	232
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	9	5	1618	395	790	210	207	1548	387	774	207	205
2000	5	1	1519	370	740	203	206	1438	360	720	200	203
4000	1	-3	1392	341	682	194	203	1307	330	660	190	199
6000	-3	-7	1308	316	632	187	201	1225	306	612	184	198
8000	-7	-11	1234	293	586	181	200	1150	283	566	177	197
10,000	-11	-15	1169	271	542	174	199	1085	260	520	171	196
12,000	-15	-19	1128	255	510	169	200	1038	244	488	166	196
14,000	-19	-23	1082	240	480	164	200	989	228	456	160	195
16,000	-23	-27	1047	227	454	159	200	947	214	428	155	195
18,000	-27	-31	1009	214	428	154	200	897	200	400	149	194
20,000	-31	-35	985	204	408	150	201	867	189	378	145	194
22,000	-34	-39	977	197	394	147	204	851	180	360	141	196
24,000	-38	-43	1001	194	388	146	209	868	176	352	140	200
26,000	-42	-47	1024	192	384	145	215	890	173	346	138	206
28,000	-46	-51	1050	191	382	144	221	911	172	344	137	211
29,000	-48	-52	1064	192	384	144	224	925	172	344	137	214
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	19	15	1801	420	840	217	218	1730	411	822	214	215
2000	15	11	1718	395	790	210	217	1630	385	770	207	214
4000	11	7	1656	374	748	204	217	1564	362	724	201	214
6000	7	3	1560	348	696	197	216	1459	336	672	193	211
8000	3	-1	1505	328	656	191	216	1406	316	632	188	212
10,000	-1	-5	1448	307	614	186	216	1343	294	588	182	212
12,000	-5	-9	1413	291	582	181	218	1302	277	554	177	212
14,000	-9	-13	1379	277	554	177	219	1261	263	526	172	213
16,000	-12	-17	1355	266	532	173	221	1228	250	500	167	214
18,000	-16	-21	1338	257	514	169	223	1209	241	482	163	216
20,000	-20	-25	1326	250	500	165	226	1192	232	464	160	218
22,000	-24	-29	1324	244	488	162	229	1187	226	452	157	221
24,000	-28	-33	1341	241	482	161	235	1205	222	444	155	227
26,000	-31	-37	1309	232	464	155	235	1223	219	438	154	232
28,000	-35	-41	1320	231	462	153	240	1207	214	428	150	234
29,000	-37	-42	1334	233	466	152	243	1209	214	428	148	236
31,000	-41	-46	---	---	---	---	---	1225	214	428	146	242
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	19	15	1659	402	804	212	213	1597	394	788	210	211
2000	15	11	1550	375	750	204	211	1476	366	732	201	208
4000	11	7	1480	352	704	198	210	1400	342	684	195	207
6000	7	3	1370	325	650	189	208	1290	315	630	186	204
8000	3	-1	1313	304	608	184	208	1225	293	586	180	204
10,000	-1	-5	1244	282	564	178	207	1148	271	542	174	203
12,000	-5	-9	1199	265	530	173	208	1106	253	506	169	203
14,000	-9	-13	1149	249	498	167	207	1050	236	472	163	202
16,000	-12	-17	1105	235	470	162	207	999	222	444	157	202
18,000	-16	-21	1078	224	448	158	209	965	210	420	152	202
20,000	-20	-25	1052	214	428	153	210	928	198	396	147	202
22,000	-24	-29	1044	207	414	150	212	914	190	380	144	204
24,000	-28	-33	1067	204	408	149	218	925	185	370	142	208
26,000	-31	-37	1088	201	402	148	224	946	182	364	141	214
28,000	-35	-41	1107	200	400	147	230	972	181	362	141	221
29,000	-37	-42	1105	198	396	145	232	980	181	362	140	224
31,000	-41	-46	1103	195	390	142	235	997	180	360	139	230
33,000	-45	-50	1117	196	392	140	240	999	178	356	136	233
35,000	---	---	---	---	---	---	---	1008	178	356	134	238

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	OAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	29	25	1711	412	824	211	215	1671	406	812	210	215
2000	25	21	1678	393	786	207	218	1630	387	774	206	217
4000	21	17	1646	375	750	203	220	1594	368	736	202	219
6000	18	13	1605	355	710	198	221	1536	347	694	196	219
8000	14	9	1552	335	670	193	222	1468	324	648	190	219
10,000	10	5	1500	315	630	188	223	1406	303	606	184	219
12,000	6	1	1468	299	598	183	225	1381	288	576	181	221
14,000	2	-3	1432	285	570	179	226	1336	273	546	176	222
16,000	-2	-7	1404	273	546	174	228	1301	260	520	171	223
18,000	-6	-11	1384	262	524	170	230	1275	248	496	167	225
20,000	-10	-15	1365	254	508	166	232	1244	239	478	162	226
22,000	-14	-19	1356	249	498	163	235	1243	234	468	159	229
24,000	-17	-23	1320	239	478	157	235	1246	229	458	157	233
26,000	-21	-27	1339	238	476	156	240	1218	220	440	152	234
28,000	-25	-31	1340	236	472	152	243	1215	217	434	148	237
29,000	-27	-32	1354	237	474	151	246	1226	217	434	147	240
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	29	25	1623	400	800	209	214	1575	394	788	208	212
2000	25	21	1578	380	760	205	215	1511	372	744	202	213
4000	21	17	1506	357	714	199	215	1427	347	694	196	212
6000	18	13	1447	335	670	193	215	1361	325	650	190	212
8000	14	9	1374	313	626	187	215	1288	302	604	183	211
10,000	10	5	1303	291	582	180	214	1205	279	558	176	209
12,000	6	1	1280	276	552	177	216	1180	263	526	173	211
14,000	2	-3	1229	260	520	171	216	1121	247	494	167	211
16,000	-2	-7	1185	245	490	166	217	1069	231	462	161	210
18,000	-6	-11	1150	233	466	161	217	1022	217	434	155	210
20,000	-10	-15	1114	222	444	156	218	981	205	410	150	209
22,000	-14	-19	1113	217	434	153	221	976	199	398	147	212
24,000	-17	-23	1122	212	424	152	226	989	195	390	146	218
26,000	-21	-27	1132	208	416	150	231	1003	191	382	144	223
28,000	-25	-31	1121	202	404	146	234	1019	188	376	143	229
29,000	-27	-32	1109	199	398	144	234	1020	187	374	142	231
31,000	-31	-36	1116	199	398	141	239	1009	182	364	138	233
33,000	-35	-40	1144	202	404	140	246	1011	181	362	135	237
35,000	-39	-44	---	---	---	---	---	1036	184	368	134	244

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	39	35	1635	405	810	205	213	1588	399	798	204	212
2000	35	31	1596	386	772	201	215	1548	379	758	200	214
4000	31	27	1566	367	734	197	217	1531	363	726	197	217
6000	28	23	1532	349	698	193	219	1496	344	688	193	219
8000	24	19	1490	330	660	188	220	1452	325	650	188	221
10,000	20	15	1447	311	622	183	222	1404	305	610	183	222
12,000	16	11	1410	294	588	179	223	1371	289	578	179	223
14,000	12	7	1384	281	562	174	224	1341	275	550	175	225
16,000	8	3	1361	269	538	170	226	1315	263	526	171	227
18,000	4	-1	1337	258	516	166	228	1295	252	504	167	229
20,000	0	-5	1334	251	502	163	231	1285	244	488	163	232
22,000	-3	-9	1325	245	490	159	234	1267	236	472	159	234
24,000	-7	-13	1341	242	484	157	239	1231	226	452	154	234
26,000	-11	-17	1366	242	484	156	245	1230	222	444	151	238
28,000	-15	-21	1362	240	480	152	247	1254	223	446	149	244
29,000	-17	-22	---	---	---	---	---	1252	222	444	147	245
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	39	35	1544	393	786	204	212	1505	388	776	203	211
2000	35	31	1506	374	748	200	214	1459	368	736	199	213
4000	31	27	1485	356	712	196	217	1438	350	700	196	216
6000	28	23	1452	338	676	193	219	1379	329	658	190	216
8000	24	19	1405	318	636	188	220	1317	307	614	184	216
10,000	20	15	1347	297	594	182	220	1253	286	572	178	216
12,000	16	11	1314	281	562	178	222	1216	269	538	174	217
14,000	12	7	1277	267	534	173	223	1181	255	510	170	218
16,000	8	3	1242	253	506	169	224	1141	241	482	165	219
18,000	4	-1	1209	241	482	164	226	1093	227	454	159	219
20,000	0	-5	1177	230	460	159	226	1051	214	428	154	219
22,000	-3	-9	1153	221	442	155	228	1028	205	410	150	220
24,000	-7	-13	1156	216	432	153	232	1031	199	398	147	224
26,000	-11	-17	1131	208	416	148	234	1039	195	390	146	230
28,000	-15	-21	1115	203	406	144	235	1032	191	382	142	233
29,000	-17	-22	1131	204	408	144	239	1019	188	376	140	233
31,000	-21	-26	1147	204	408	142	245	1023	186	372	138	237
33,000	-25	-30	---	---	---	---	---	1037	186	372	136	243
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	49	45	1659	410	820	205	217	1570	399	798	202	214
2000	45	41	1566	384	768	198	215	1494	375	750	196	213
4000	41	37	1513	363	726	193	216	1453	355	710	192	215
6000	37	33	1480	345	690	189	218	1424	337	674	188	217
8000	34	29	1437	325	650	184	219	1383	318	636	183	218
10,000	30	25	1391	306	612	179	219	1333	298	596	178	219
12,000	26	21	1366	291	582	174	221	1306	283	566	174	220
14,000	22	17	1344	278	556	170	223	1272	269	538	169	221
16,000	18	13	1334	268	536	167	226	1255	257	514	165	224
18,000	14	9	1328	259	518	163	229	1249	248	496	162	227
20,000	10	5	1323	252	504	160	232	1241	240	480	159	230
22,000	6	1	1348	249	498	159	238	1228	232	464	155	232
24,000	3	-3	1379	249	498	158	245	1220	226	452	151	235
26,000	-1	-7	1365	243	486	153	246	1270	228	456	152	244
28,000	-5	-11	---	---	---	---	---	1259	224	448	148	246
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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For Training Purposes Only

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	49	45	1494	390	780	200	211	1438	383	766	198	209
2000	45	41	1439	368	736	195	212	1393	362	724	194	211
4000	41	37	1413	350	700	191	214	1365	344	688	190	213
6000	37	33	1382	332	664	187	216	1337	326	652	187	216
8000	34	29	1342	313	626	183	218	1300	307	614	183	218
10,000	30	25	1299	294	588	178	219	1252	287	574	178	218
12,000	26	21	1268	278	556	174	221	1224	272	544	174	220
14,000	22	17	1240	264	528	170	223	1191	258	516	169	222
16,000	18	13	1219	252	504	166	225	1167	245	490	166	224
18,000	14	9	1203	242	484	163	228	1138	233	466	161	226
20,000	10	5	1196	234	468	159	231	1102	222	444	156	226
22,000	6	1	1180	225	450	156	233	1077	212	424	152	228
24,000	3	-3	1135	214	428	150	232	1066	204	408	149	231
26,000	-1	-7	1107	205	410	144	232	1042	196	392	144	232
28,000	-5	-11	1165	209	418	146	244	1007	187	374	139	231
29,000	-7	-12	1155	207	414	144	244	1028	188	376	139	236
31,000	-11	-16	1153	205	410	140	247	1046	188	376	138	243
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	56	52	1721	420	840	207	222	1618	407	814	204	218
2000	52	48	1641	395	790	201	221	1524	380	760	197	216
4000	48	44	1555	370	740	194	220	1453	357	714	191	216
6000	44	40	1487	347	694	188	220	1396	335	670	185	216
8000	41	36	1448	328	656	183	221	1354	316	632	180	217
10,000	37	32	1410	310	620	179	222	1314	297	594	176	218
12,000	33	28	1398	296	592	175	225	1290	282	564	172	220
14,000	29	24	1381	284	568	171	227	1265	269	538	167	222
16,000	25	20	1374	274	548	168	230	1249	258	516	164	224
18,000	21	16	1380	267	534	166	235	1234	248	496	160	227
20,000	17	12	1384	261	522	163	239	1237	241	482	157	230
22,000	14	8	1386	256	512	160	243	1235	235	470	154	234
24,000	10	4	1374	249	498	156	245	1273	234	468	154	242
26,000	6	0	1394	248	496	154	250	1267	229	458	150	245
28,000	2	-4	---	---	---	---	---	1285	228	456	148	250
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

SUBTRACT 5% FROM THE TAS VALUES

1700 RPM

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	56	52	1515	394	788	200	214	1425	383	766	197	210
2000	52	48	1435	369	738	194	213	1361	360	720	191	210
4000	48	44	1383	348	696	189	214	1327	341	682	187	212
6000	44	40	1335	327	654	184	215	1294	322	644	183	214
8000	41	36	1296	309	618	179	216	1256	303	606	179	216
10,000	37	32	1257	290	580	175	217	1216	284	568	175	217
12,000	33	28	1229	274	548	171	219	1186	268	536	171	219
14,000	29	24	1195	260	520	166	220	1160	255	510	167	221
16,000	25	20	1173	248	496	162	222	1138	243	486	163	223
18,000	21	16	1164	238	476	159	225	1120	232	464	159	226
20,000	17	12	1159	231	462	156	229	1110	224	448	156	229
22,000	14	8	1150	223	446	152	231	1091	215	430	152	231
24,000	10	4	1120	213	426	147	232	1060	204	408	147	232
26,000	6	0	1150	212	424	147	239	1018	193	386	141	231
28,000	2	-4	1162	210	420	145	244	1039	190	380	139	239
29,000	0	-5	1165	209	418	143	246	1057	191	382	137	245
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

RANGE PROFILE - MAXIMUM RANGE POWER

ASSOCIATED CONDITIONS:

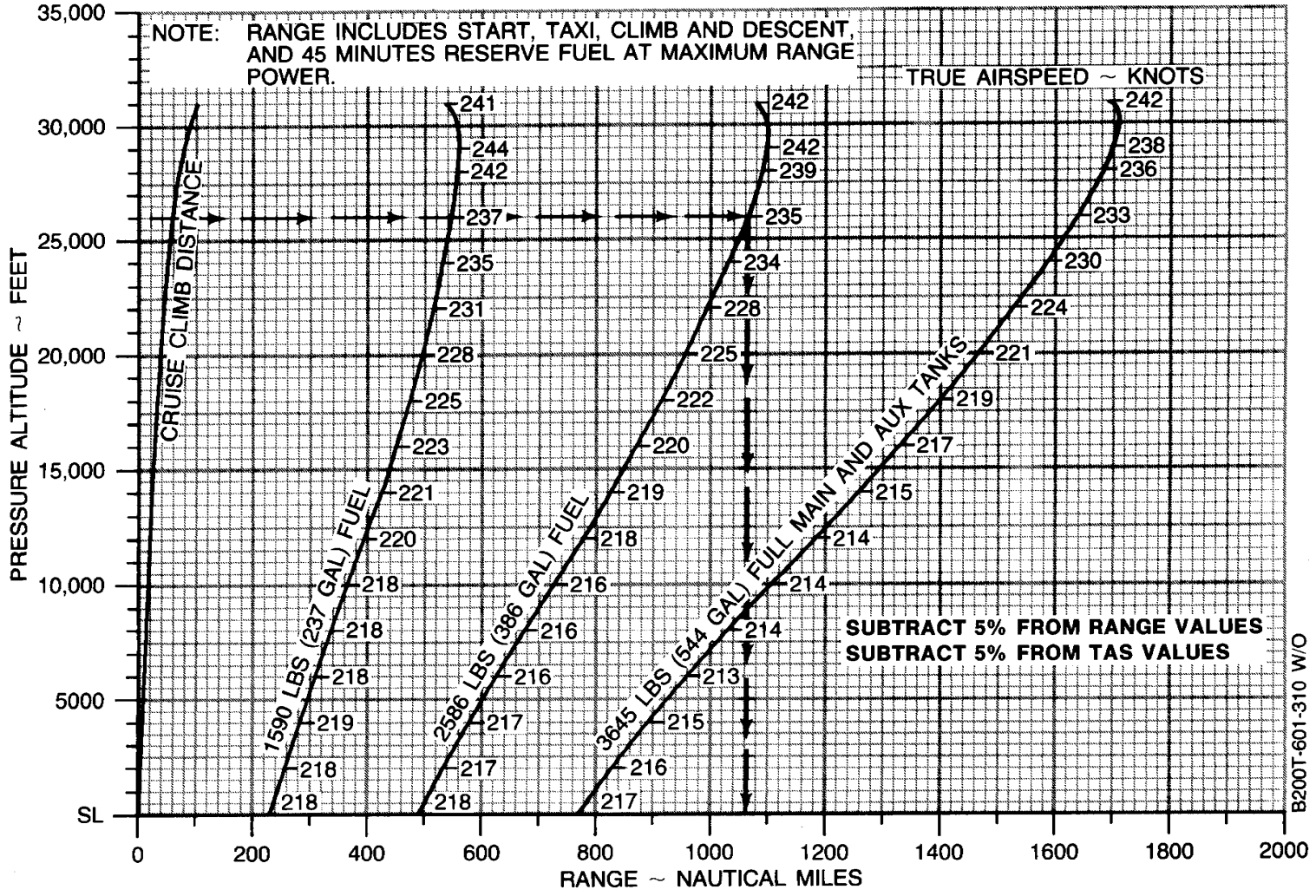
WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

1700 RPM

STANDARD DAY (ISA)
ZERO WIND

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	1065 NM



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RANGE PROFILE - FULL MAIN AND AUX TANKS

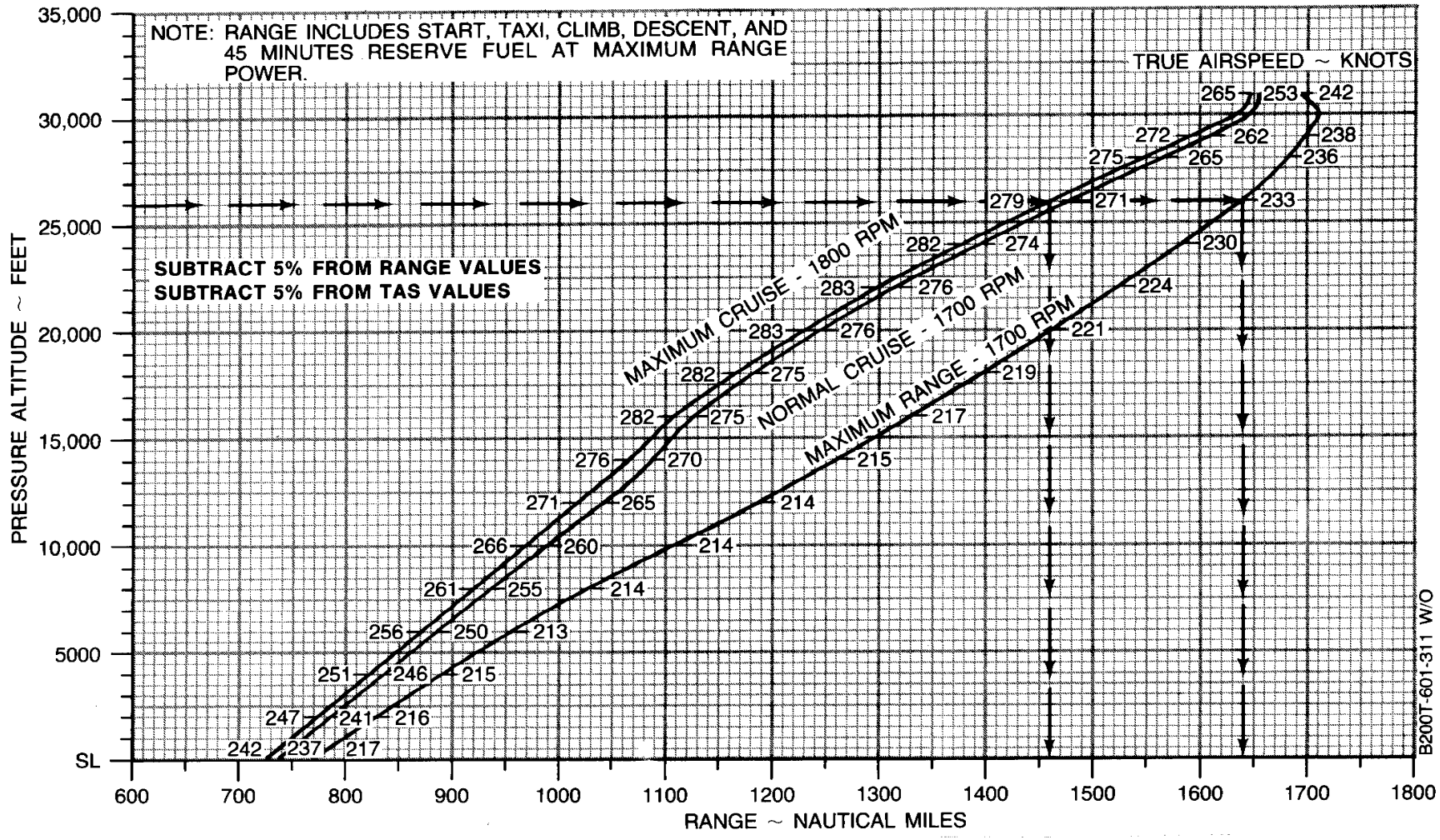
STANDARD DAY
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	1460 NM
RANGE @ MAX RANGE - 1700 RPM	1640 NM



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ENDURANCE PROFILE - FULL MAIN AND AUX TANKS

STANDARD DAY

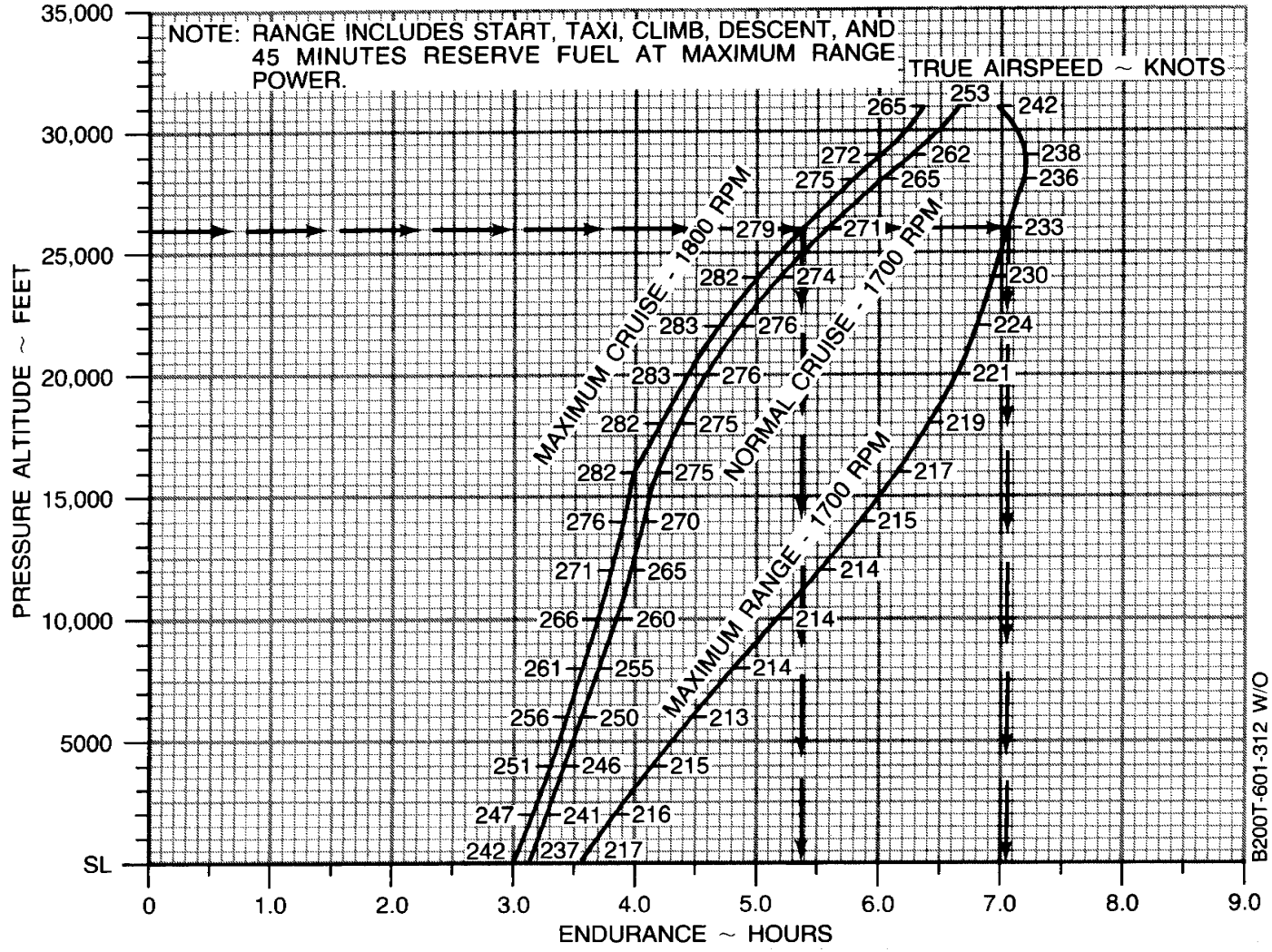
ASSOCIATED CONDITIONS:

WEIGHT ..... 14,090 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

EXAMPLE:

PRESSURE ALTITUDE ..... 26,000 FT

RANGE @ MAX CRUISE - 1800 RPM ..... 5.37 HRS
RANGE @ MAX RANGE - 1700 RPM ..... 7.05 HRS



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RANGE PROFILE - FULL MAIN TANKS

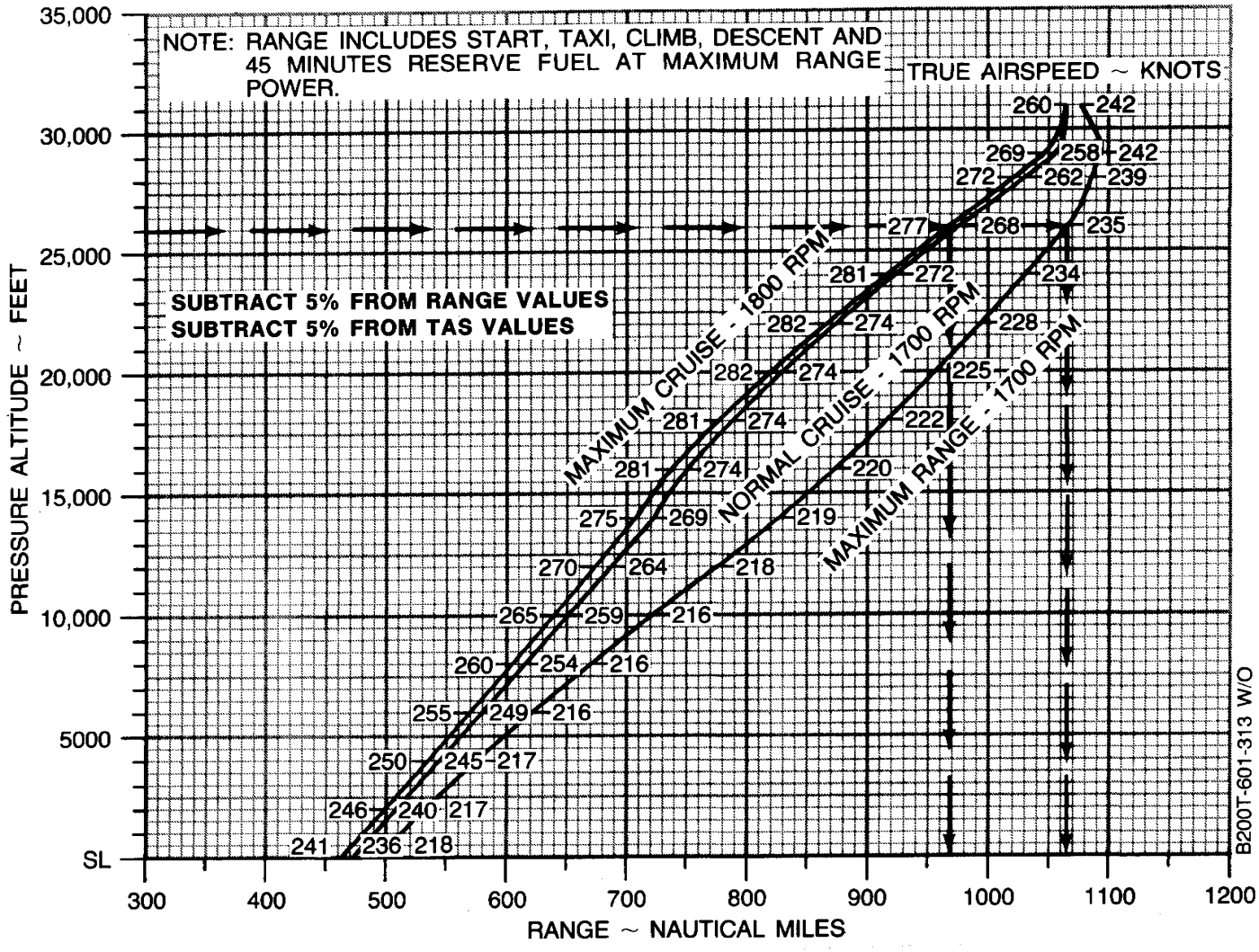
STANDARD DAY
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	968 NM
RANGE @ MAX RANGE - 1700 RPM	1065 NM



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ENDURANCE PROFILE - FULL MAIN TANKS

STANDARD DAY

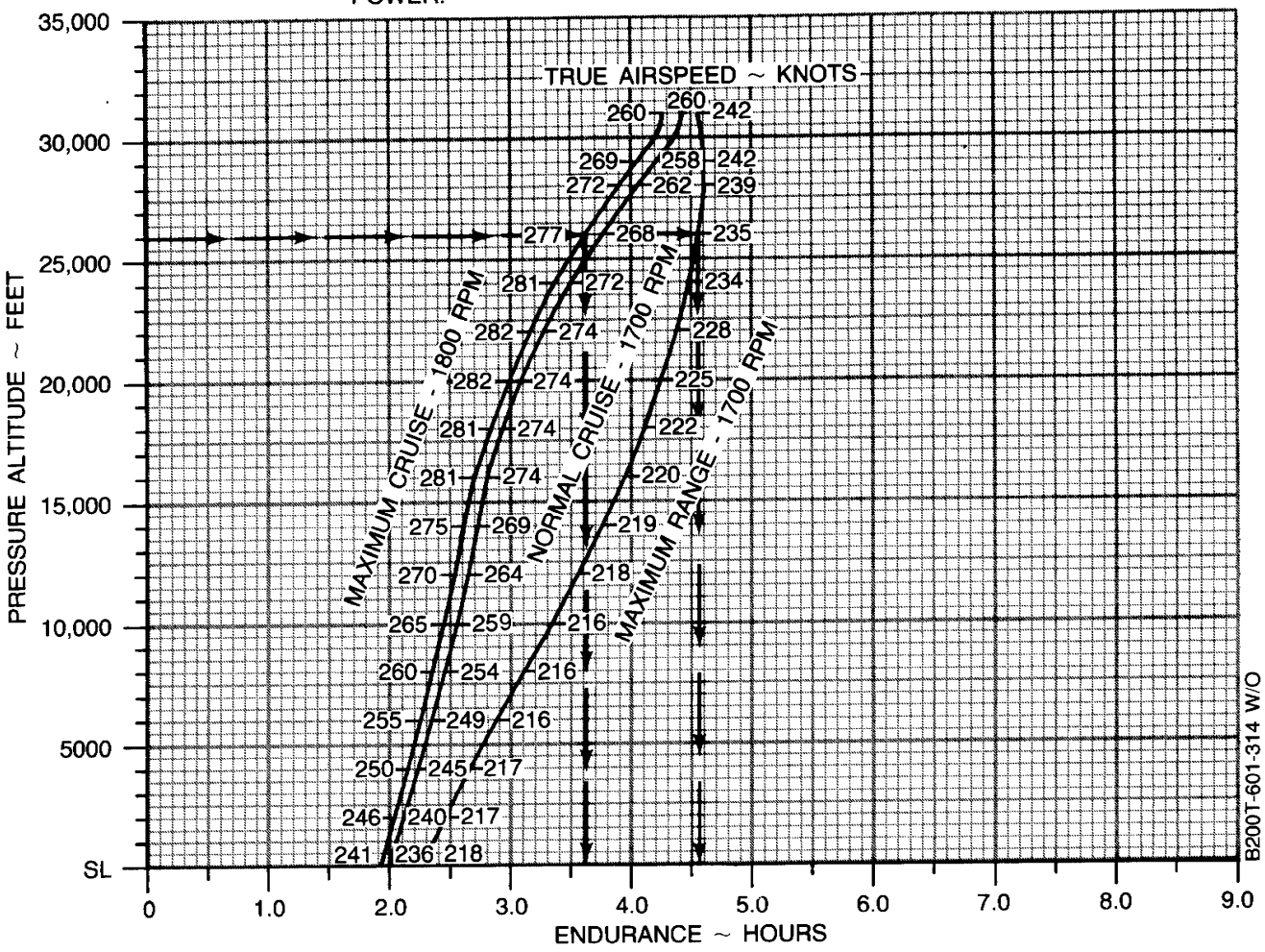
ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	3.62 HRS
RANGE @ MAX RANGE - 1700 RPM	4.57 HRS

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT MAXIMUM RANGE POWER.



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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-12	-15	2230	517	517	189	180	191	182	193	183	194	185
2000	-16	-19	2230	505	505	186	182	189	185	191	187	192	188
4000	-20	-23	2230	492	492	184	185	186	188	188	190	190	192
6000	-24	-27	2230	479	479	181	188	184	191	186	193	188	195
8000	-27	-31	2230	467	467	179	191	182	194	184	196	186	198
10,000	-31	-35	2230	455	455	176	194	179	197	182	199	184	202
12,000	-35	-39	2230	445	445	174	196	177	200	179	203	181	205
14,000	-39	-43	2230	439	439	171	199	174	203	177	206	179	208
16,000	-43	-47	2230	434	434	167	202	171	206	174	209	176	212
18,000	-47	-51	2146	415	415	160	199	165	205	169	209	172	213
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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For Training Purposes Only

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-2	-5	2230	520	520	187	181	189	184	191	185	193	187
2000	-6	-9	2230	505	505	185	184	187	187	189	189	191	190
4000	-10	-13	2230	492	492	182	187	185	190	187	192	189	194
6000	-13	-17	2230	480	480	180	190	182	193	184	195	186	197
8000	-17	-21	2230	468	468	177	193	180	196	182	198	184	201
10,000	-21	-25	2230	456	456	174	196	177	199	180	202	182	204
12,000	-25	-29	2230	446	446	171	198	175	202	177	205	179	207
14,000	-29	-33	2230	440	440	168	201	172	205	175	208	177	211
16,000	-33	-37	2159	423	423	162	199	166	204	170	209	173	212
18,000	-37	-41	2048	400	400	154	195	159	202	163	207	166	211
20,000	-41	-45	1945	378	378	144	190	151	199	156	205	160	210
22,000	-45	-49	1833	356	356	131	178	142	193	149	202	154	208
24,000	-50	-53	1675	326	326	---	---	127	179	138	194	145	203
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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For Training Purposes Only

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	8	5	2230	522	522	185	183	188	186	190	187	191	189
2000	4	1	2230	508	508	183	186	185	188	187	191	189	192
4000	0	-3	2230	494	494	180	189	183	191	185	194	187	196
6000	-3	-7	2230	481	481	178	192	181	195	183	197	185	199
8000	-7	-11	2230	468	468	175	194	178	198	181	200	183	203
10,000	-11	-15	2230	456	456	172	197	175	201	178	204	180	206
12,000	-15	-19	2230	446	446	169	200	173	204	175	207	178	209
14,000	-19	-23	2157	428	428	163	198	167	203	170	207	173	211
16,000	-23	-27	2051	406	406	155	195	160	201	165	206	167	210
18,000	-27	-31	1949	384	384	146	190	153	198	157	204	161	209
20,000	-31	-35	1857	365	365	135	182	145	194	151	202	155	208
22,000	-35	-39	1748	343	343	---	---	134	187	143	198	148	206
24,000	-39	-43	1629	318	318	---	---	---	---	132	189	140	200
26,000	-43	-47	1512	294	294	---	---	---	---	---	---	129	192
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	18	15	2230	525	525	184	185	186	187	188	189	190	191
2000	14	11	2230	511	511	181	188	184	190	186	192	188	194
4000	11	7	2230	497	497	179	190	181	193	184	196	186	198
6000	7	3	2230	483	483	176	193	179	196	181	199	183	201
8000	3	-1	2230	469	469	173	196	176	200	179	202	181	205
10,000	-1	-5	2230	457	457	170	199	174	202	176	206	179	208
12,000	-5	-9	2169	436	436	165	198	169	203	172	207	174	210
14,000	-9	-13	2065	412	412	157	195	162	201	166	206	169	209
16,000	-13	-17	1961	390	390	148	190	154	198	159	204	163	208
18,000	-17	-21	1855	369	369	138	183	146	194	152	201	156	206
20,000	-21	-25	1767	351	351	119	163	137	188	145	198	150	205
22,000	-26	-29	1658	329	329	---	---	123	175	136	193	143	202
24,000	-30	-33	1547	305	305	---	---	---	---	123	180	133	196
26,000	-33	-37	1439	283	283	---	---	---	---	---	---	121	185
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	28	25	2230	527	527	182	187	185	189	187	191	189	193
2000	24	21	2230	513	513	180	189	182	192	185	194	186	196
4000	21	17	2230	499	499	177	192	180	195	182	198	184	200
6000	17	13	2230	484	484	174	195	177	198	180	200	182	203
8000	13	9	2222	470	470	171	197	174	201	177	204	179	207
10,000	9	5	2119	441	441	164	195	168	199	171	203	174	207
12,000	5	1	2023	415	415	157	192	161	198	165	202	168	206
14,000	1	-3	1950	395	395	149	189	155	196	160	202	163	206
16,000	-3	-7	1870	376	376	141	184	148	194	153	201	158	206
18,000	-7	-11	1783	355	355	128	173	140	189	147	198	151	205
20,000	-12	-15	1700	338	338	---	---	130	182	139	195	145	203
22,000	-15	-19	1597	319	319	---	---	---	---	128	186	137	198
24,000	-19	-23	1489	296	296	---	---	---	---	---	---	126	190
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	38	35	2230	530	530	181	188	183	191	185	193	187	195
2000	35	31	2230	515	515	178	191	181	194	183	196	185	198
4000	31	27	2230	500	500	176	194	178	197	181	200	183	202
6000	27	23	2156	475	475	170	193	173	197	176	200	178	203
8000	23	19	2060	447	447	163	191	167	196	170	200	173	203
10,000	19	15	1962	418	418	155	188	160	194	164	198	167	202
12,000	15	11	1878	394	394	148	185	153	192	158	197	161	202
14,000	11	7	1818	376	376	140	180	147	190	153	197	157	202
16,000	7	3	1758	359	359	129	172	140	187	147	196	152	202
18,000	2	-1	1696	343	343	---	---	132	182	141	194	146	201
20,000	-1	-5	1631	327	327	---	---	---	---	133	189	140	199
22,000	-6	-9	1534	307	307	---	---	---	---	120	178	132	195
24,000	-9	-13	1431	285	285	---	---	---	---	---	---	120	184
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMTAELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	48	45	2154	521	521	176	186	179	189	182	192	184	194
2000	44	41	2103	499	499	172	187	175	190	178	193	180	196
4000	40	37	2062	478	478	167	188	171	192	174	195	177	198
6000	37	33	1997	453	453	162	187	166	192	169	195	172	199
8000	33	29	1908	425	425	154	184	159	190	163	194	166	198
10,000	28	25	1819	397	397	146	180	152	187	157	193	160	198
12,000	24	21	1746	375	375	137	175	145	185	151	192	155	197
14,000	20	17	1688	358	358	126	166	138	182	145	191	150	197
16,000	16	13	1635	342	342	---	---	130	177	139	189	145	197
18,000	13	9	1601	329	329	---	---	---	---	133	187	140	196
20,000	8	5	1553	316	316	---	---	---	---	125	181	134	195
22,000	5	1	1477	297	297	---	---	---	---	---	---	125	189
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

SUBTRACT 7% FROM THE TAS VALUES

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED – KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	55	52	2015	503	503	170	181	173	185	176	188	178	190
2000	51	48	1971	481	481	165	182	169	186	172	189	174	192
4000	47	44	1937	461	461	161	182	165	187	168	191	171	194
6000	43	40	1880	438	438	155	181	160	187	163	190	167	195
8000	39	36	1800	411	411	147	178	153	185	158	190	161	194
10,000	35	32	1719	384	384	138	172	146	182	151	189	156	194
12,000	31	28	1656	363	363	127	164	139	179	145	187	150	193
14,000	27	24	1603	346	346	---	---	131	175	140	186	145	193
16,000	23	20	1552	331	331	---	---	120	166	133	183	140	192
18,000	19	16	1522	318	318	---	---	---	---	126	179	135	191
20,000	16	12	1488	306	306	---	---	---	---	---	---	129	189
22,000	12	8	1414	289	289	---	---	---	---	---	---	120	182
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

CLIMB-BALKED LANDING

CLIMB SPEED 100 KNOTS (ALL WEIGHTS)

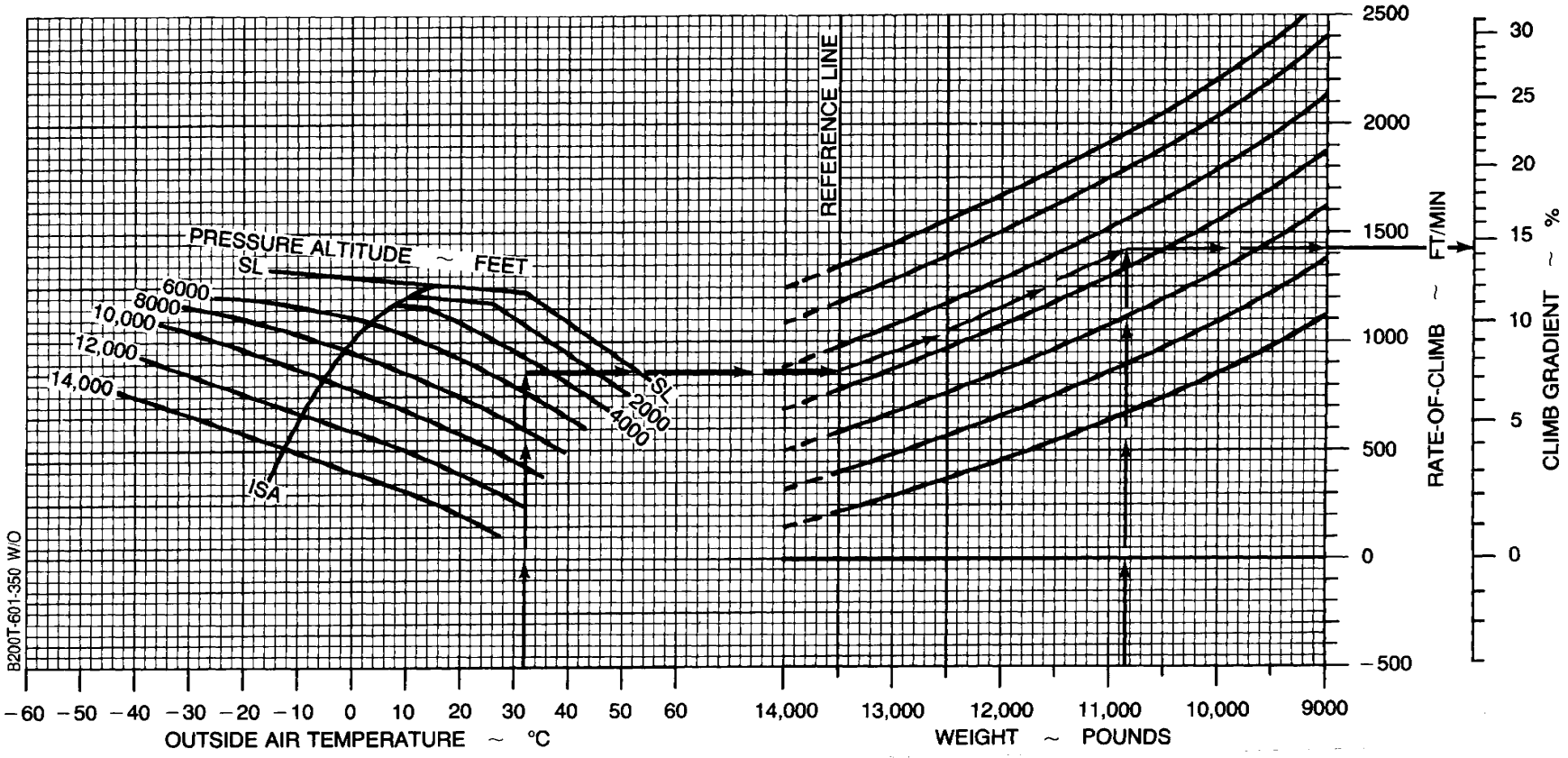
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	100%
LANDING GEAR	DOWN

EXAMPLE:

OAT	32°C
PRESSURE ALTITUDE	4732 FT
WEIGHT	10,854 LBS
RATE-OF-CLIMB	1420 FT/MIN
CLIMB GRADIENT	14.4%

SUBTRACT 250 FT. PER MINUTE FROM GRAPH VALUE.



For Training Purposes Only

NORMAL LANDING DISTANCE WITHOUT PROPELLER REVERSING - FLAPS 100%

ASSOCIATED CONDITIONS:

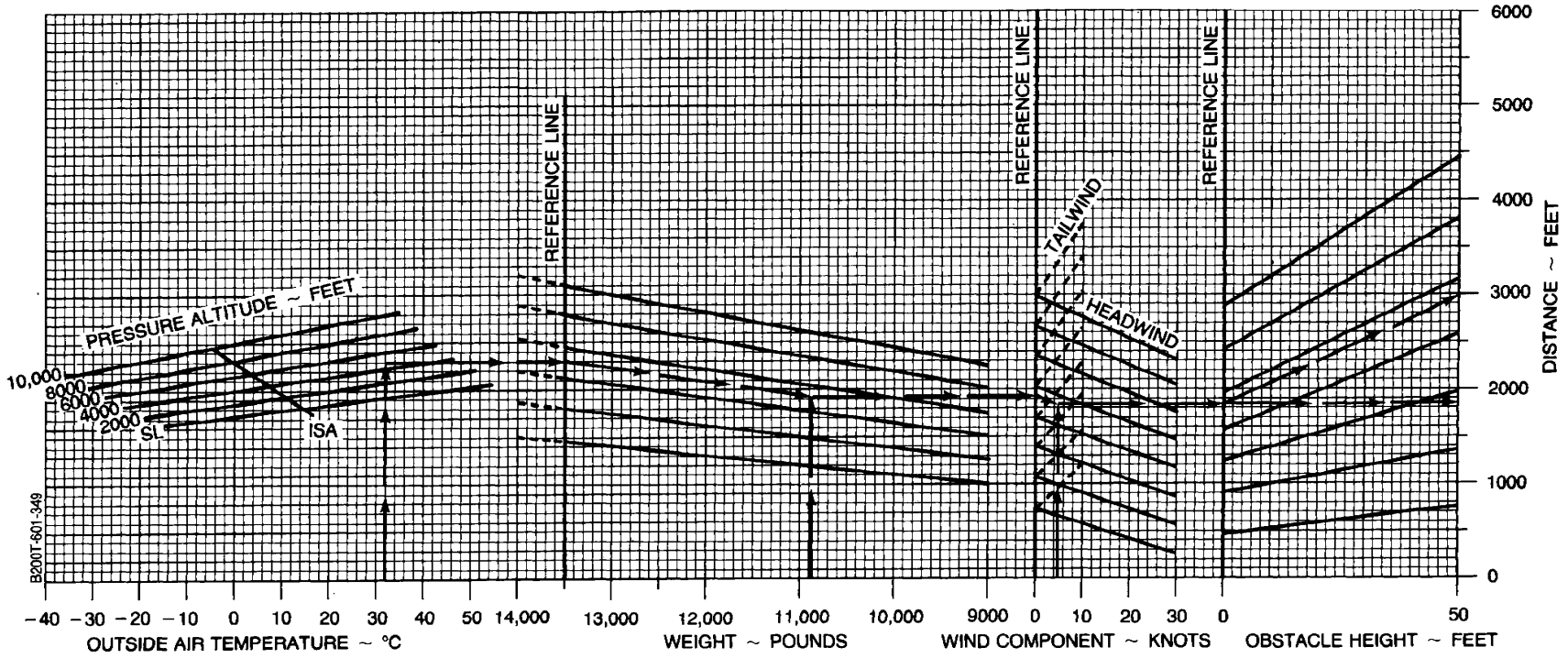
POWER.....RETARDED TO MAINTAIN 800 FT/MIN ON FINAL APPROACH
FLAPS.....100%
RUNWAY.....PAVED LEVEL DRY SURFACE
APPROACH SPEED . IAS AS TABULATED
BRAKING .....MAXIMUM

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	106
13,500	105
13,000	104
12,500	102
12,000	101
11,000	99
10,000	96
9000	94

EXAMPLE:

OAT..... 32°C
PRESSURE ALTITUDE..... 4732 FT
LANDING WEIGHT..... 10,854 LBS
HEADWIND COMPONENT..... 4.7 KTS

GROUND ROLL ..... 1850 FT
TOTAL OVER
50 FT OBSTACLE..... 3000 FT
APPROACH SPEED..... 99 KTS



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LANDING DISTANCE WITHOUT PROPELLER REVERSING-FLAPS 0%

ASSOCIATED CONDITIONS:

POWER..... RETARDED TO MAINTAIN 900 FT/MIN ON FINAL APPROACH
FLAPS..... UP
RUNWAY..... PAVED, LEVEL, DRY SURFACE
APPROACH SPEED . IAS AS TABULATED
BRAKING ..... MAXIMUM

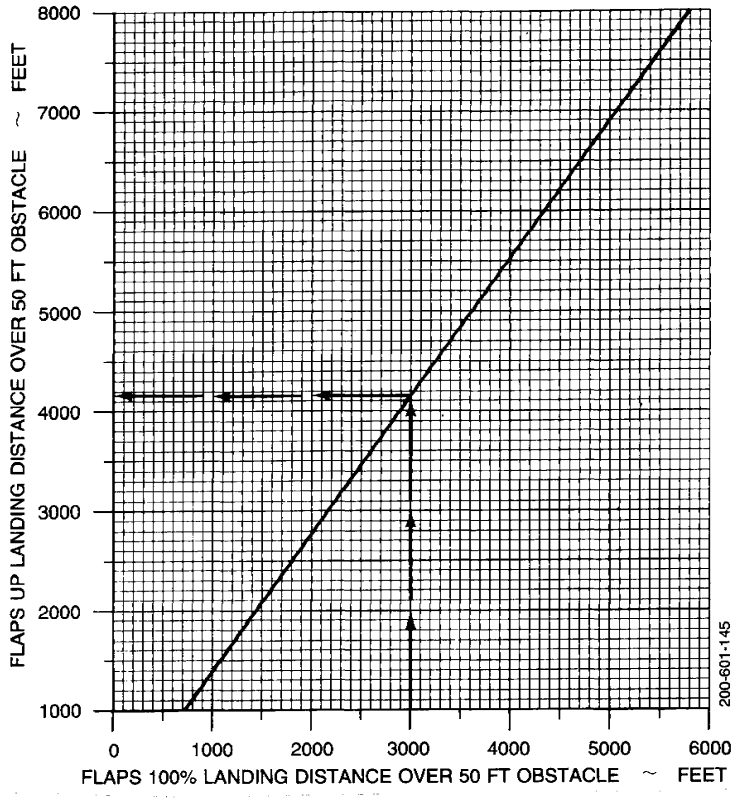
WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	137
13,500	135
13,000	133
12,500	131
12,000	129
11,000	125
10,000	121
9000	117

EXAMPLE:

FLAPS 100% LANDING DISTANCE
OVER 50 FT OBSTACLE..... 3000 FT
LANDING WEIGHT ..... 10,854 LBS

FLAPS UP LANDING DISTANCE
OVER 50 FT OBSTACLE..... 4150 FT
APPROACH SPEED ..... 124 KTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN (100%) IS NORMAL PROCEDURE. USE THIS GRAPH WHEN IT IS NECESSARY TO LAND WITH FLAPS UP (0%).
2. TO DETERMINE FLAPS-UP LANDING DISTANCE, READ FROM THE **NORMAL LANDING DISTANCE WITHOUT PROPELLER REVERSING-FLAPS 100%** GRAPH THE LANDING DISTANCE APPROPRIATE TO OAT, ALTITUDE, WEIGHT, WIND, AND 50-FT OBSTACLE. THEN ENTER THIS GRAPH WITH THE DERIVED VALUE AND READ THE FLAPS-UP LANDING DISTANCE.



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LANDING DISTANCE WITH PROPELLER REVERSING-FLAPS 100%

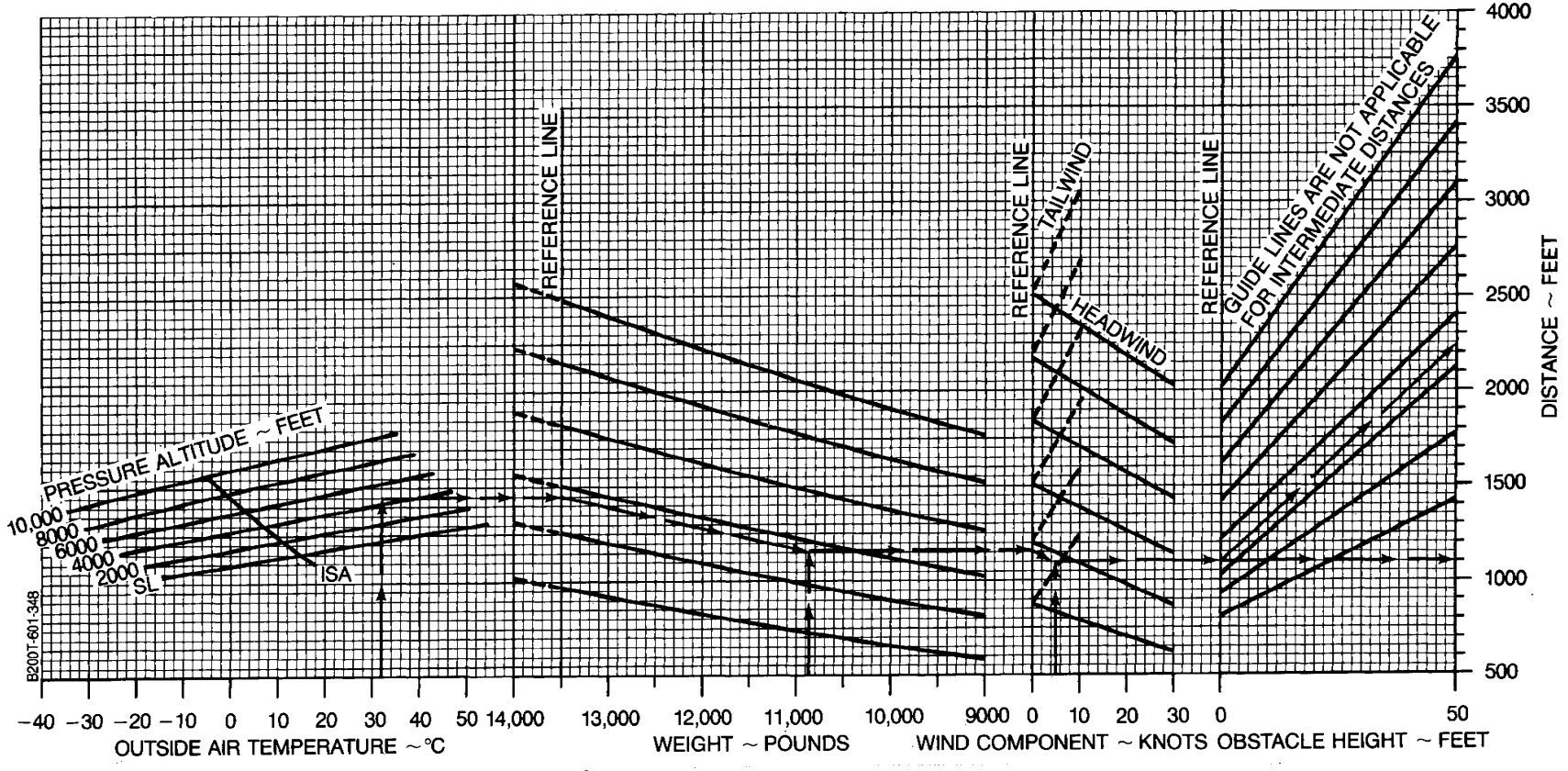
ASSOCIATED CONDITIONS:

POWER	RETARD TO MAINTAIN 1000 FT/MIN. ON FINAL APPROACH
FLAPS	100%
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS AS TABULATED
BRAKING	MAXIMUM
CONDITION LEVERS	HIGH IDLE
PROPELLER CONTROLS	FULL FORWARD
POWER LEVERS	MAXIMUM REVERSE AFTER TOUCHDOWN

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	106
13,500	105
13,000	104
12,500	102
12,000	101
11,000	99
10,000	96
9000	94

EXAMPLE:

OAT	32°C
PRESSURE ALTITUDE	4732 FT
LANDING WEIGHT	10,854 LBS
HEADWIND COMPONENT	4.7 KTS
GROUND ROLL	1100 FT
TOTAL OVER 50 FT OBSTACLE	2225 FT
APPROACH SPEED	98 KTS



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LANDING DISTANCE WITH PROPELLER REVERSING-FLAPS 0%

ASSOCIATED CONDITIONS:

POWER	RETARD TO MAINTAIN 1000 FT/MIN ON FINAL APPROACH
FLAPS	UP
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS AS TABULATED
BRAKING	MAXIMUM
CONDITION LEVERS	HIGH IDLE
PROPELLER CONTROLS	FULL FORWARD
POWER LEVERS	MAXIMUM REVERSE AFTER TOUCHDOWN

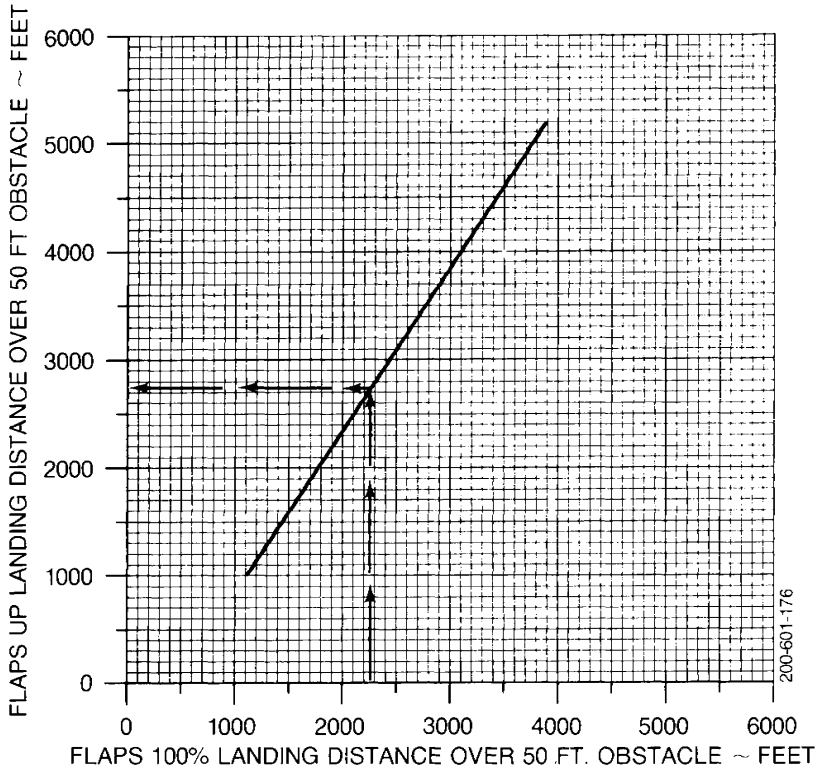
WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	137
13,500	135
13,000	133
12,500	131
12,000	129
11,000	125
10,000	121
9000	117

EXAMPLE:

FLAPS 100% LANDING DISTANCE OVER 50 FOOT OBSTACLE ..... 2260 FT
LANDING WEIGHT ..... 10,854 LBS

FLAPS UP LANDING DISTANCE OVER 50 FOOT OBSTACLE ..... 2720 FT
APPROACH SPEED ..... 124 KTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN (100%) IS NORMAL PROCEDURE. USE THIS GRAPH WHEN IT IS NECESSARY TO LAND WITH FLAPS UP (0%).
2. TO DETERMINE FLAPS-UP LANDING DISTANCE, READ FROM THE **LANDING DISTANCE WITH PROPELLER REVERSING – FLAPS 100%** GRAPH THE LANDING DISTANCE APPROPRIATE TO OAT, ALTITUDE, WEIGHT, WIND, AND 50-FT OBSTACLE. THEN ENTER THIS GRAPH WITH THE DERIVED VALUE AND READ THE FLAPS-UP LANDING DISTANCE.



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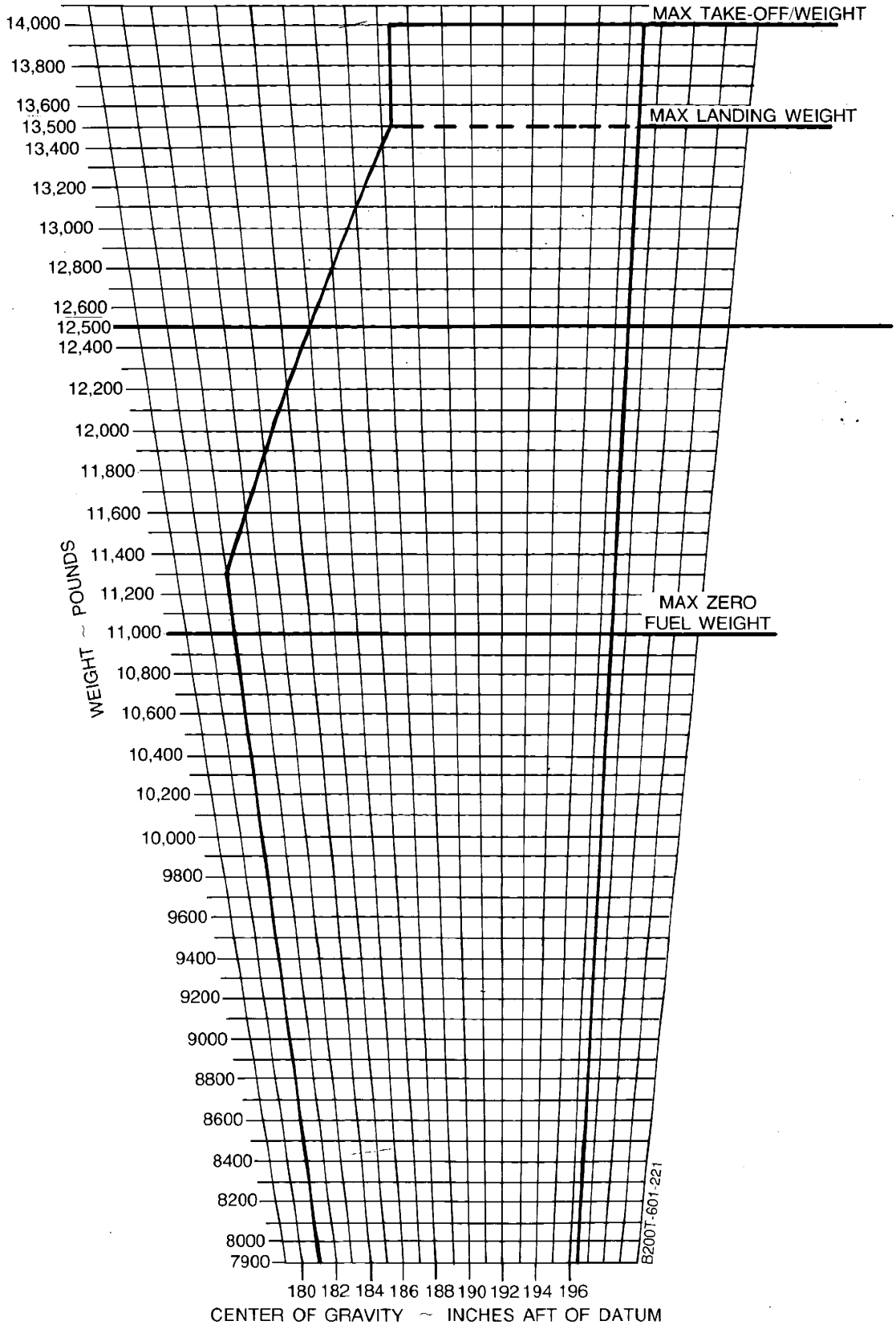
WEIGHT AND BALANCE DIAGRAM



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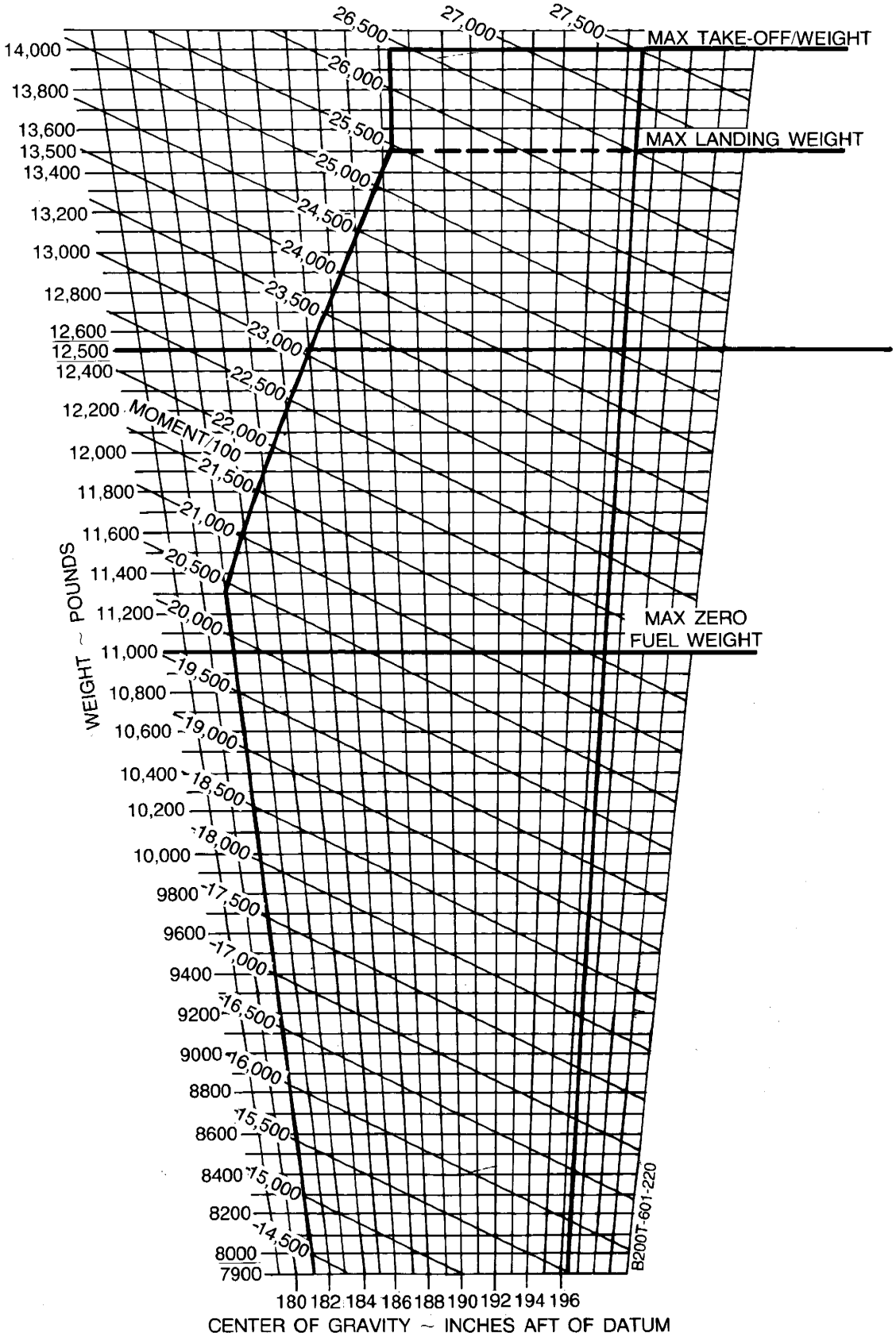
MOMENT LIMITS vs WEIGHT



MOMENT LIMITS VS. WEIGHT

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
12500	23125	24550
12550	23240	24648
12600	23355	24746
12650	23469	24845
12700	23584	24943
12750	23699	25041
12800	23814	25139
12850	23928	25237
12900	24043	25336
12950	24158	25434
13000	24273	25532
13050	24388	25630
13100	24502	25728
13150	24617	25827
13200	24732	25925
13250	24847	26023
13300	24961	26121
13350	25076	26219
13400	25191	26318
13450	25306	26416
13500	25421	26514
13550	25515	26612
13600	25609	26710
13650	25703	26809
13700	25797	26907
13750	25891	27005
13800	25985	27103
13850	26080	27201
13900	26174	27300
13950	26268	27398
14000	26362	27496

CENTER OF GRAVITY LIMITS (LANDING GEAR DOWN)

WEIGHT CONDITION	FORWARD C.G. LIMIT	AFT C.G. LIMIT
14,000 Lbs (Max. Takeoff)	188.3	196.4
13,500 Lbs or Less (Max. Landing)	188.3	196.4
12,500 Lbs	185.0	196.4
11,279 Lbs or Less	181.0	196.4

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WEIGHT AND BALANCE

Basic empty weight does not include customer furnished equipment, such as cameras and navigation sights. If such equipment is installed, a log book entry should be made reflecting the adjusted basic empty weight.

SYSTEMS DESCRIPTION - No change

HANDLING SERVICING AND MAINTENANCE - No change

FAA Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

BEECHCRAFT SUPER KING AIR

B200, B200C, B200T, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

SPERRY MLZ-910 MICROWAVE LANDING RECEIVER SYSTEM

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUALS P/N 101-590010-147 AND 101-590037-29

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Issued: February, 1988
P/N 101-590010-243

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane is flown with the Sperry MLZ-910 Microwave Landing Receiver System.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. Maximum authorized glideslope angle is 4.0 degrees.
2. Minimum altitude for an autopilot coupled MLS approach is 200 feet.

EMERGENCY PROCEDURES

No Change.

NORMAL PROCEDURES

PREFLIGHT CHECKS

1. Apply power to the system.
2. Push AUTO Pushbutton switch.
3. Push TEST switch momentarily.

    a. All segments of the displays light.
    b. Morse Code audio (— . . .) is received.
    c. AZ WARN and EL WARN indicators are illuminated as long as switch is held in TEST position.
    d. ADI and HSI vertical and lateral deviation pointer, bar, and flags sequence as follows:

        1) Centered, in view (note).
        2) Up 2 dots, left 2 dots, out of view.
        3) Centered, in view.
        4) Down 2 dots, right 2 dots, out of view.
        5) Centered, in view.

NOTE

EFIS displays remove pointers and bars if the flag is in view.

NORMAL OPERATION

The normal sequence of path selection is:

1. Select channel as published on approach plate.
2. Select AUTO.
3. Set HSI course pointer to the magnetic heading corresponding to the displayed azimuth. This keeps the NAV situation picture oriented properly.

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NOTE

The displayed azimuth will be the magnetic runway heading as transmitted from ground station.

4. Fly the approach using the normal vertical and lateral path control (flight director, coupled autopilot, raw data, etc.).

PERFORMANCE

No Change.

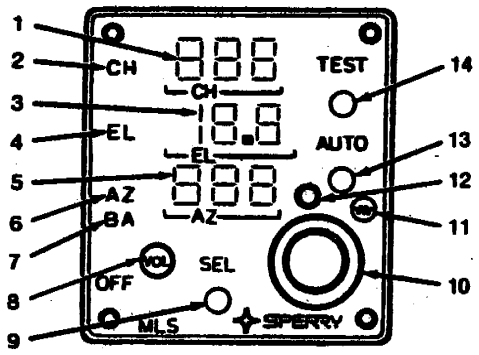
WEIGHT & BALANCE/EQUIPMENT LIST

No Change.

SYSTEMS DESCRIPTION

CONTROLS AND INDICATORS

The following diagram and table shows the Sperry MLZ-910 MLS controls and indications.



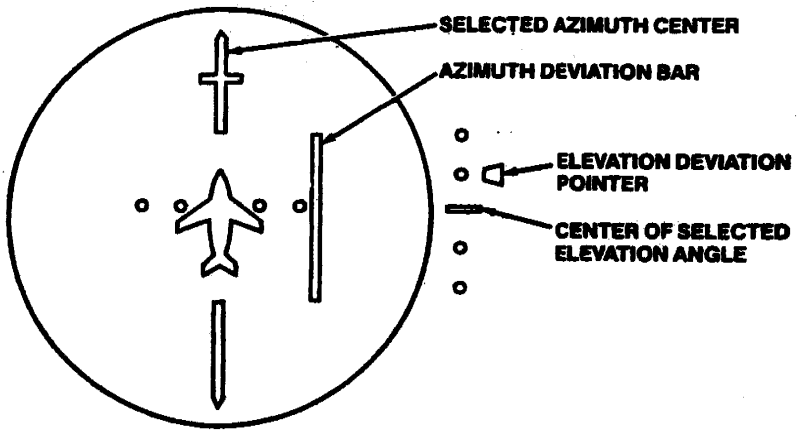
CONTROLS and INDICATORS

INDEX NO.	CONTROL/INDICATOR	FUNCTION
1	CHANNEL Display	Displays receiver channel number.
2	CHANNEL Annunciator	Indicates channel has been selected to be tuned.
3	ELEVATION Display	Displays the approach elevation angle the receiver is using to compute elevation deviation.
4	ELEVATION Annunciator	Indicates elevation angle has been selected to be changed by rotary switches.
5	AZIMUTH Display	Displays the approach magnetic heading the receiver is using to compute azimuth deviation.
6	AZIMUTH Annunciator	Indicates azimuth has been selected to be changed by rotary switches.

INDEX NO.	CONTROL/INDICATOR	FUNCTION
7	Back Azimuth Annunciator	Indicates back azimuth has been selected to be changed by rotary switches. (Presently not available)
8	ON/OFF Rotary Switch Volume Control	Applies power to Controller and adjusts volume of Audio Morse Code station identification to airplane audio system.
9	Select Pushbutton Switch	Used to select either channel, elevation or azimuth for tuning.
10	Dual Concentric Rotary Switch	Changes channel, elevation or azimuth setting. The inner knob changes the least significant digit. The outer knob changes the two higher order digits.
11	Light Sensor	Brightens displays/indicators when ambient light is increased.
12	AUTO Annunciator	Indicates Controller is in AUTO mode.
13	AUTO Pushbutton Switch	Initiates AUTO mode when pressed.
14	TEST Pushbutton Switch	Initiates receiver Built-in Test Equipment sequence and lamp test.

MLS OUTPUTS

The azimuth (lateral) and elevation (vertical) deviations are presented the same way as other flight path deviations using the standard lateral and vertical deviation displays.



LATERAL and VERTICAL DEVIATION DISPLAY

The MLS deviation signals couple to existing flight director/autopilot systems to allow fully coupled approaches using standard IFR procedures.

HANDLING, SERVICING, & MAINTENANCE

No Change.

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BEECHCRAFT SUPER KING AIR

B200, B200C, B200T, AND B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT for the KING KNS-660 FLIGHT MANAGEMENT SYSTEM with KING KLN-670 GLOBAL POSITIONING SYSTEM

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUALS P/N 101-590010-147 AND 101-590037-29

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:

Donald M Peters

For W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
Issued: January, 1988
P/N 101-590010-245

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the King KNS-660 Flight Management System with King KLN-670 Global Positioning System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The King Pilot's Guide for the KNS-660/KLN-670, P/N 006-8407-00, dated October 9, 1987 or later, is available from King Radio Corp., 400 N. Rogers Road, Olathe, Kansas 66062. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status.

LIMITATIONS

The KNS-660/KLN-670 meets the approval criteria for VOR/DME and VLF/Omega Area Navigation Systems as established by Advisory Circular 90-45A and Advisory Circular 20-101B. Operation must be in conformity with the KNS-660/KLN-670 Pilot's Guide, P/N 006-8407-00, dated October 9, 1987 or later.

The KNS-660/KLN-670 may be used as a means of VFR/IFR navigation provided the following limitations are observed:

If the data base has expired, IFR navigation is prohibited unless each waypoint and navaid is verified for accuracy by reference to current approved data.

VOR/DME OR TACAN SENSOR

1. IFR RNAV approaches are approved for published FAA certified approaches.
2. The KNS-660/KLN-670 is approved as a two-dimensional VOR/DME or TACAN RNAV system for enroute, terminal area, and approach operation in accordance with AC 90-45A and is not geographically limited.

VLF/OMEGA SENSOR

1. Only enroute operations in the conterminous United States and Alaska are approved for IFR.
2. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).
3. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.
4. IFR flight is not approved based on VLF/Omega navigation into any area in which such operations could not be approved based on the installed equipment required by paragraph 3. This includes overwater flights beyond the range of approved ground navigation facilities.
5. Auto Update function must be verified or the VLF/Omega position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:

   a. Prior to compulsory reporting points during IFR operations when not under radar surveillance or control.
   b. At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.
   c. Prior to and at hourly intervals during operation off of approved RNAV routes.

6. Navigation shall not be predicated on the use of this system during periods of dead reckoning or when a NAV flag is visible.

7. Following a period of dead reckoning, invalid operation (NAV flag visible), or operation in the Omega WARN mode, the VLF/Omega position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment. Position must also be updated if an accuracy check reveals a navigation error of greater than 2 NM.

GPS SENSOR

The GPS sensor is not approved as a means of IFR navigation.

VOR/DME, TACAN, VLF OMEGA, AND GPS BLEND SENSOR

The KNS-660/KLN-670 is approved as a means of enroute navigation in accordance with AC 90-45A.

1. IFR RNAV approaches are approved for published FAA certified approaches.
2. When "DO NOT USE FOR IFR TERM/APR" message is not present, enroute, terminal, and approach operations are permitted.
3. When "DO NOT USE FOR IFR TERM/APR" message is present, only enroute operations are approved.
4. Use of the KNS-660/KLN-670 for Jet and Victor route navigation is not approved when the message "SEL OBS MODE FOR IFR J/V ROUTES" is present and the system method of operation is other than OBS.
5. Use of the KNS-660/KLN-670 for random route navigation is not approved when the message "NOT VALID FOR IFR RANDOM RTS" is present.

MISCELLANEOUS

1. During periods when the Dead Reckoning (DR) annunciator is illuminated, no external position inputs are available. The KNS-660/KLN-670 navigation data should be used with care during these periods.
2. Navigation shall not be predicated on the KNS-660/KLN-670 when a NAV flag is visible. Following a period (5 minutes or more) of invalid operation (NAV flag visible), the KNS-660/KLN-670 position must be verified. Update only if required.

EMERGENCY PROCEDURES

1. If KNS-660/KLN-670 information is intermittent or lost, utilize the remaining operational navigation equipment as required.
2. KNS-660/KLN-670 Malfunction

   a. Press FMS NAV CTL Switch - NAV CTL illuminated.
   b. CDU ON/OFF Switch - OFF
   c. FMS Circuit Breaker - PULL
   d. VOR #1 - TUNE normally using the NAV 1 frequency control head. DME will be normally tuned by using the VOR receiver.
   e. Course Selector - MANUALLY SET.

NORMAL PROCEDURES

Normal operating procedures are outlined in the KNS-660/KLN-670 Pilot's Guide, P/N 006-8407-00, dated October 9, 1987 or later revisions.

SYSTEM ANNUNCIATORS

1. Waypoint (WPT)

   Illuminates 90 seconds prior to each waypoint when operating in the OBS method of operation, when approaching the last waypoint in the flight plan or when going direct to a waypoint not in the flight plan.

   Illuminates 15 seconds prior to the automatic course change turn when operating in the AUTO/LEG method of operation.

2. Message (MGS)

   Identical to the message light on the Control Display Unit (CDU).

3. Dead Reckoning (DR)

   Illuminates when the system is dead reckoning.

4. X-Track (XTRK)

   Illuminates whenever a left or right cross track (parallel offset) is activated.

SYSTEM SWITCHES

1. Flight Management System (FMS)

   Activation of the FMS/NAV CTL switch (FMS illuminated) allows the KNS-660/KLN-670 navigation data to be displayed on the pilot's HSI and both RMI's. Active tuning of the VOR and DME sensors is now controlled by the KNS-660/KLN-670.

2. Navigation Control (NAV CTL)

   Activation of the FMS/NAV CTL switch (NAV CTL illuminated) allows basic NAV (VOR/LOC/GS)and DME data to be displayed on the pilot's HSI and VOR bearing on both RMI's. Active tuning of the VOR and DME sensors is now controlled by the pilot's NAV 1 frequency control head.

3. HSI Reference Data (OMEGA DIST/BRG)

   Activation of the HSI REF DATA switch (OMEGA DIST/BRG illuminated) allows bearing and distance to the active KNS-660/KLN-670 VLF/OMEGA waypoint to be displayed on the pilot's HSI bearing pointer and DME readout during operation in NAV CTL. VOR #1 drives the pilot's HSI course needle and both RMI's during this operation.

PILOT'S HSI

1. FMS Selected (FMS Illuminated and "blanked" active frequency display window in NAV 1 frequency control head).

   a. The KNS-660/KLN-670 provides automatic course needle drive while using the AUTO/LEG method of operation.
   b. The KNS-660/KLN-670 does not provide course needle drive while using the OBS method of operation. The pilot must set the selected course manually either via the course knob on the HSI or through the KNS-660/KLN-670 Control Display Unit.
   c. The deviation needle provides left/right steering information from the KNS-660/KLN-670.
   d. The DME display provides distance to the waypoint in nautical miles and the selection (via a pushbutton to the right of the DME display) of either groundspeed in knots or estimated time enroute in hours and minutes.
   e. The BRG pointer provides bearing information derived from either the ADF or the KNS-660/KLN-670 depending upon whether ADF or NAV respectively are selected by the BRG PTR selector.

2. NAV CTL Selected (NAV CTL Illuminated)

   a. The pilot must set the selected course manually via the course knob on the HSI.
   b. VOR/LOC/GS tuning is via the NAV 1 frequency control head.
   c. VOR/LOC signals provide left/right steering information.
   d. The DME display provides distance to the station in nautical miles and selection of either groundspeed in knots or estimated time enroute.
   e. The BRG pointer provides bearing information derived from either the ADF or the NAV 1 receiver depending upon whether ADF or NAV respectively are selected by the BRG PTR selector.

PILOT'S AND COPILOT'S RMI

1. FMS Selected (FMS Illuminated and "blanked" active frequency display window in NAV 1 Frequency Control Head).

   The single bar (green) needle provides bearing information derived from either the ADF or KNS-660/KLN-670 depending upon whether ADF or NAV respectively are selected on the RMI.

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2. NAV CTL Selected (NAV CTL switch illuminated)

   The single bar (green) needle provides bearing information derived from either the ADF or the NAV 1 receiver depending upon whether ADF or NAV respectively are selected on the RMI.

PERFORMANCE

No Change.

WEIGHT & BALANCE/EQUIPMENT LIST

No Change.

SYSTEMS DESCRIPTION

The KNS-660/KLN-670 is an integrated long range multi-sensor flight management system. It may be used to manage the entire range of navigational functions including trip planning, long range great circle navigation, and instrument approaches. The system consists of a cockpit mounted Control Display Unit (CDU) and a remote-mounted Navigation Computer. The KNS-660/KLN-670 serves as a computer processor/display for inputs from a choice of sensors such as VOR/DME, Omega/VLF, INS, TACAN, TAS, or GPS.

The KNS-660/KLN-670 incorporates a King Data Base which must be updated every 28 days via diskette and a KDL-569 Data Loader. In addition, a self generated user data base may be loaded and maintained by the owner/operator through the CDU keyboard independent of the King Data Base. This data base consists of 175 storage locations accommodating any combination of ground stations and/or airports. If required, any number of these 175 user generated waypoints can be secured against cockpit alteration.

The KNS-660/KLN-670 will support 100 flight plans consisting of 25 alpha/numeric designated waypoints. These flight plans may be composed of any combination of waypoints taken from the data base or the user generated waypoints.

Dedicated special function keys are physically and visually separated from the alpha/numeric keys used in the generation and storage of flight plans and waypoints.

The KNS-660/KLN-670 provides for manual operation to selected waypoints or automatic operation providing uninterrupted navigation throughout a complete flight plan. Navigation sensors may be selected separately or blended within the computer.

On airplanes so equipped, the KNS-660/KLN-670 interfaces with the KRG-358 Radar Graphics Unit, providing a radar graphics presentation on the weather radar display. This includes such items as flight plan waypoints, selected course, and reference ground station.

HANDLING, SERVICING, & MAINTENANCE

No Change.

BEECHCRAFT SUPER KING AIR B200

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for

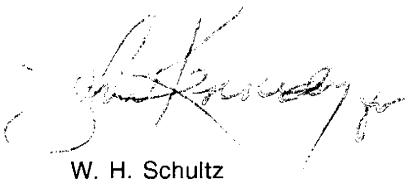
**HIGH DENSITY CONFIGURATION (HDC)
AIRPLANES EQUIPPED WITH EXTERNAL BAGGAGE
POD AND CERTIFIED IN ACCORDANCE WITH
FAR 135 APPENDIX A**

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S
OPERATING HANDBOOK AND
FAA APPROVED AIRPLANE FLIGHT
MANUAL P/N 101-590010-147

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:



W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane. Where differences are noted between the Models B200 and B200C in the basic Pilot's Operating Handbook, the information pertaining to the B200 applies unless superseded by this supplement.

This supplement applies to BEECHCRAFT Super King Air B200's in the High Density Configuration. These airplanes are certified for 10 or more passenger seats in accordance with FAR 135 Appendix A. These airplanes are configured with up to 13 passenger chairs, a second (left side) emergency escape hatch, baggage pod, nose baggage compartment, dual ventral fins and associated changes. This configuration is defined by Aircraft General Assembly Drawing 101-000015.

Complete EMERGENCY PROCEDURES, ABNORMAL PROCEDURES, and PERFORMANCE Sections for the High Density Configuration are incorporated into this supplement. The EMERGENCY PROCEDURES and PERFORMANCE data included in this supplement totally replaces all EMERGENCY PROCEDURES and PERFORMANCE data provided in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual (POH/AFM). Additionally, an ABNORMAL PROCEDURES Section has been added.

All LIMITATIONS, NORMAL PROCEDURES, WEIGHT & BALANCE, SYSTEMS, and HANDLING, SERVICING & MAINTENANCE information published in the basic POH/AFM applies, except for certain subsections that are superseded by corresponding subsections in this supplement.

DESCRIPTIVE DATA

FUEL

USABLE FUEL

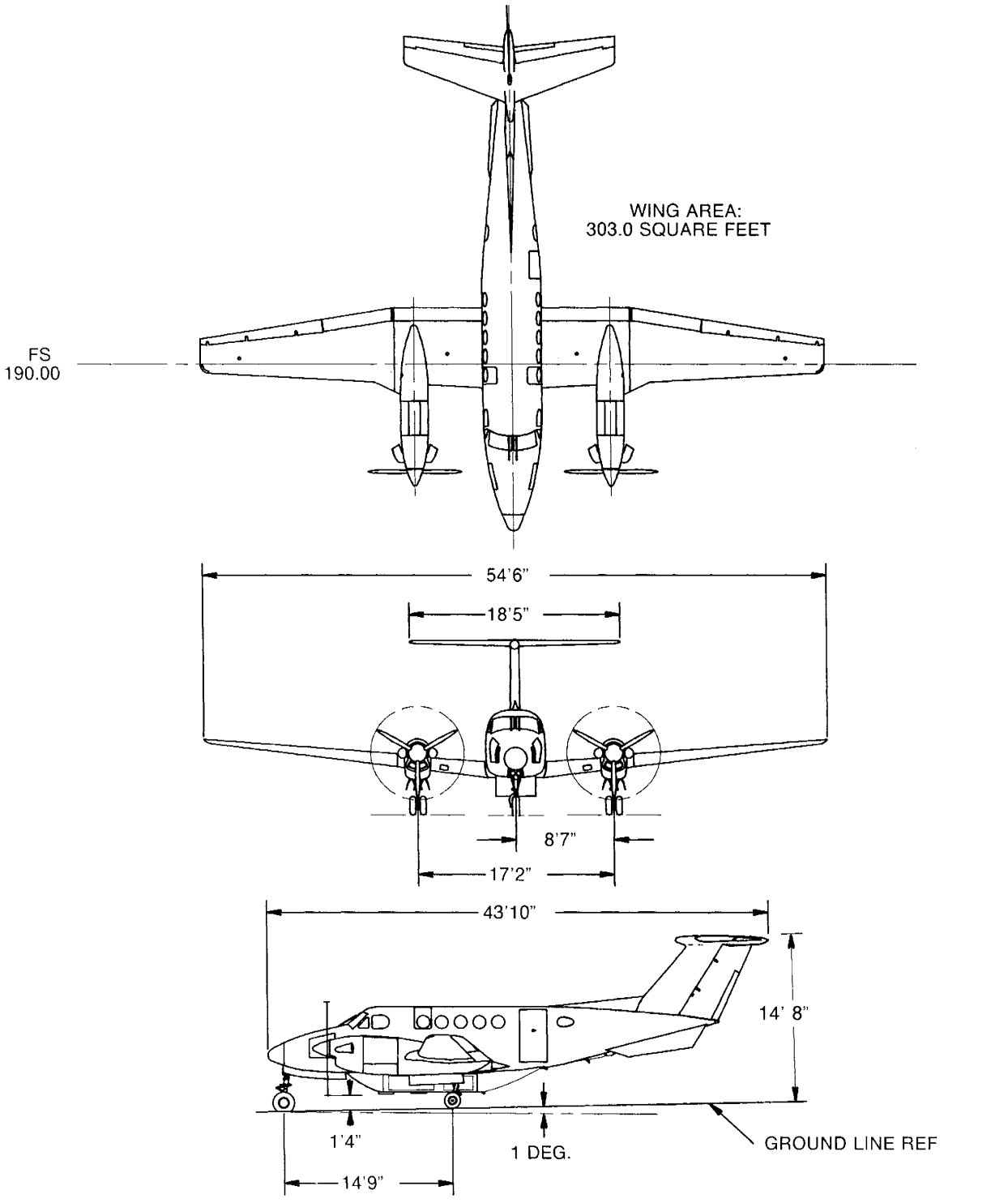
Main Fuel System
- When Filled To Outboard Filler Port . . . . . 386 gallons
- *When Filled To Mid-Span Filler Port Tab . . . . . 232 gallons

Auxiliary Fuel System . . . . . 158 gallons
Maximum Usable Fuel Quantity . . . . . 544 gallons
*Serials BB-1376 & after and prior serials with Kit 101-9058.

SYMBOLS, ABBREVIATIONS AND TERMINOLOGY

GRAPH AND TABULAR TERMINOLOGY

Take-off Distance	The all-engines take-off distance is the horizontal distance required to take off and climb to a height of 50 feet above the take-off surface under the procedures in FAR 23.51(a).



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B200 (HDC) THREE VIEW

LIMITATIONS

Except for the limitations specified herein, all limitations published in the basic Pilot's Operating Handbook apply.

The limitations included in this Section have been approved by the Federal Aviation Administration, and they must be observed in the operation of the BEECHCRAFT Super King Air B200 High Density Configuration (HDC) airplanes equipped with an external baggage pod and certificated in accordance with FAR 135, Appendix A.

AIRSPEED LIMITATIONS

SPEED	KCAS	KIAS	REMARKS
Maneuvering Speed V_A (12,500 pounds)	182	180	Do not make full or abrupt control movements above this speed.
Maximum Flap Extension/ Extended Speed V_{FE}			
Approach Position - 40%	200	199	Do not extend flaps or operate with flaps in prescribed position above these speeds.
Full Down Position - 100%	155	156	
Maximum Landing Gear Operating Speed V_{LO}			
Extension	182	180	Do not extend or retract landing gear above the speeds given.
Retraction	164	162	
Maximum Landing Gear Extended Speed V_{LE}	182	180	Do not exceed this speed with landing gear extended
Air Minimum Control Speed V_{MCA}	95	94	This is the lowest airspeed at which the airplane is directionally controllable when one engine suddenly becomes inoperative and the other engine is at take-off power. (See definition in Section I of the basic Pilot's Operating Handbook).
Maximum Operating Speed			
V_{MO}	260	260	Do not exceed this airspeed or Mach Number in any operation.
M_{MO}	.52 Mach		

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AIRSPEED INDICATOR MARKINGS*			
MARKING	KCAS VALUE OR RANGE	KIAS VALUE OR RANGE	SIGNIFICANCE
Red Line	95	94	Air Minimum Control Airspeed (V_{MCA})
White Arc	75 to 155	74 to 156	Full-flap Operating Range
Wide White Arc	75 to 98	74 to 97	Lower Limit is the Stalling Speed (V_{SO}) at maximum weight with Full Flaps (100%) and idle power.
Narrow White Arc	98 to 155	97 to 156	Lower Limit is the Stalling Speed (V_S) at maximum weight with Flaps Up (0%) and idle power. Upper limit is the maximum speed permissible with flaps extended beyond Approach (more than 40%).
White Triangle	200	199	Maximum Flaps-to/at-Approach (40%) Speed.
Blue Line	122	121	One-Engine-Inoperative Best Rate-of-Climb Speed
Red & White Hash-Marked Pointer	260 KCAS (260 KIAS) or value equal to .52 Mach, whichever is lower.		Maximum Speed for any operation.

* The airspeed indicator is marked in IAS values.

FUEL MANAGEMENT

1. Do not put any fuel into the auxiliary tanks unless the main tanks are full.
2. Do not take off if fuel quantity gages indicate in yellow arc or less than 265 pounds of fuel in each main tank.
3. Fuel imbalance.
   a. For airplanes with an aileron-rudder interconnect system (P/N 101-525183-1), the maximum allowable fuel imbalance between left and right wing fuel systems is 200 pounds.
   b. For airplanes without an aileron-rudder interconnect system, the fuel load between the left and right wing fuel systems must be balanced.
4. Crossfeeding of fuel is permitted only when required to eliminate fuel load imbalance during normal operation or when one engine is inoperative.
5. Crossfeeding of fuel is not permitted during normal (two engine) takeoff, approach or landing.
6. Crossfeed capability is required for all operations.

PROPELLER AUTOFEATHER SYSTEM

Must be armed and operational for all takeoffs, climbs, approaches and landings.

WEIGHT LIMITS

Maximum Ramp Weight is 12,590 pounds.

Maximum Take-off Weight is 12,500 pounds, or as limited by the following:

1. Maximum Take-off Weight To Achieve Take-off Climb Requirements, and;
2. Runway Length for:
   a. Accelerate-Stop Distance using Normal Take-off Procedure, or:
   b. Accelerate-Stop Distance, plus 500 feet, using Rolling Take-off Procedure.

Maximum Enroute Weight is limited by the following (as required by FAR Part 135):

1. Service Ceiling - One Engine Inoperative.

Maximum Landing Weight is 12,500 pounds, or as limited by the following:

1. Landing Field Length.

Maximum Zero Fuel Weight is 11,000 pounds.

CARRY-ON BAGGAGE

Only small items that can be stowed beneath the passenger seats may be carried on board.

MINIMUM FLIGHT CREW

FAR 135 Operations ..... One Pilot and One Co-pilot
FAR 91 Operations ..... One Pilot

KINDS OF OPERATION LIMITS

The BEECHCRAFT Super King Air B200 High Density Configuration (HDC) airplanes equipped with an external baggage pod and certificated in accordance with FAR 135, Appendix A is approved for the following types of operation when the required equipment is installed and operational as defined within the listing of Kinds Of Operations Equipment List contained in the LIMITATIONS Section of this supplement.

1. VFR Day
2. VFR Night
3. IFR Day
4. IFR Night
5. Icing Conditions

USABLE FUEL LIMITS

Main Fuel System
When Filled To Outboard Filler Port ..... 386 gallons
*When Filled To Mid-Span Filler Port Tab ..... 232 gallons
Auxiliary Fuel System ..... 158 gallons
Maximum Usable Fuel Quantity ..... 544 gallons
*Serials BB-1376 & after and prior serials with Kit 101-9058.

MAXIMUM OPERATING PRESSURE-ALTITUDE LIMITS

Normal Operation ..... 25,000 feet
Operation with Aviation Gasoline:
Both Standby Boost Pumps Operative ..... 25,000 feet
Either Standby Boost Pump Inoperative ..... Takeoff Prohibited
Climbs Without Crossfeed Capability ..... Takeoff Prohibited
Operation With Yaw Damper Inoperative ..... 25,000 feet

MAXIMUM OCCUPANCY LIMIT

FAR Part 91 Operations ..... Fifteen (15) Persons Including Crew
FAR Part 135 Operations ..... Thirteen (13) Passengers Plus Crew

STRUCTURAL LIMITATIONS

Wing and Associated Structure Interim Safelife ..... 15,000 hrs

NOTE

Wing shear fittings must be inspected and replaced per Beech Kit 101-4050 at or before 6400 hours.

Maximum Cabin Pressure Differential . . . . . 6.6 psi
Cabin Door Side Latches (or bayonets) (4) Safelife . . . . . 5000 hrs
Cabin Door Upper Latch Hooks (2) and Attaching Hardware . . . . . 12,500 hrs
Cam-Lock Actuator Cable Safelife . . . . . 10,000 hrs
Fuselage and Associated Structure First Inspection . . . . . 12,500 hrs

Refer to the *BEECHCRAFT Structural Inspection and Repair Manual* for inspection and replacement procedures.

Fuselage Side Windows:

If cracking occurs in the outer ply of any fuselage side window, pressurized operation may be continued only if all the following limitations are complied with:

1. No cracking in the inner ply is perceptible;
2. Cracking in the outer ply is in a generally circumferential direction and is approximately 2 inches from the edge of the window frame;
3. Maximum cabin differential pressure is 4.6 psi; and
4. Not more than 20 hours of operation are allowed after the outer ply cracks.

Refer to the *Super King Air 200 Series Maintenance Manual* for inspection and replacement procedures.

PLACARDS

On Center of Aft Cabin Bulkhead:

COMPARTMENT CAPACITY

TOTAL COMPARTMENT CAPACITY (INCLUDING BAGGAGE AND/OR PASSENGER) NOT TO EXCEED 510 POUNDS SEE WEIGHT AND BALANCE SECTION OF FLIGHT MANUAL FOR LOADING INSTRUCTIONS.

Inside Nose Baggage Compartment:

BAGGAGE AND/OR EQUIPMENT COMPARTMENT CAPACITY 350 LB SEE WEIGHT & BALANCE SECTION OF FLIGHT MANUAL FOR LOADING INSTRUCTIONS.

Inside Baggage Pod:

BAGGAGE POD CAPACITY

FWD SECTION:	260 LBS MAXIMUM WEIGHT
AFT SECTION:	195 LBS MAXIMUM WEIGHT
ENTIRE POD:	455 LBS: MAXIMUM WEIGHT 30 LBS/SQ.FT. MAXIMUM LOADING.

CAUTION

(1) SEE WEIGHT AND BALANCE SECTION OF PILOT'S OPERATING MANUAL.

(2) LOAD AFT POD BAGGAGE FIRST WHEN PARTITION IS RETRACTED TO ACCOMODATE LONG OBJECTS.

On Overhead Panels in Pilot's Compartment:

AIRSPEEDS (I.A.S.)

MAX GEAR EXTENSION	180 KNOTS
MAX GEAR RETRACT	162 KNOTS
MAX GEAR EXTENDED	180 KNOTS
MAX APPROACH FLAP	199 KNOTS
MAX FULL DOWN FLAP	156 KNOTS
MAX MANEUVERING	180 KNOTS

On Avionics Panel in Pilot's Compartment:

OPERATIONAL LIMITATION

THIS AIRCRAFT NOT CERTIFIED FOR OPERATIONS ABOVE 25,000 FEET

On Pilot's & Copilot's Window Sills:

WARNING

DO NOT SMOKE WHILE OXYGEN IS IN USE.

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KINDS OF OPERATIONS EQUIPMENT LIST

This airplane may be operated in day or night VFR, IFR, or icing conditions when the appropriate equipment is installed and operable.

The following equipment list identifies the systems and equipment upon which type certification for each kind of operation was predicated. The systems and items of equipment listed must be installed and operable for the partricular kind of operation indicated unless:

1. The airplane is approved to be operated in accordance with a current Minimum Equipment List (MEL) issued by the FAA.
   or:
2. An alternate procedure is provided in the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual for the inoperative state of the listed equipment and all limitations are complied with.

NOTE

The following systems and equipment list does not include all equipment required by the FAR Parts 91 and 135 Operating Requirements. It also does not include components obviously required for the airplane to be airworthy such as wings, empennage, engine, etc.

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING CONDITIONS	
ELECTRICAL POWER						
1. AC Volts/Frequency Meter	1	1	1	1	1	
2. Battery	1	1	1	1	1	
3. Battery Charge Monitor System and Annunciator	1	1	1	1	1	
4. DC Generator	2	2	2	2	2	
5. DC Generator Annunciator	2	2	2	2	2	
6. DC Load Meter	2	2	2	2	2	
7. Inverter	2	2	2	2	2	
8. Inverter Annunciator	1	1	1	1	1	
ENVIRONMENTAL						
1. Bleed Air Fail Annunciators	2	2	2	2	2	
2. Altitude Warning Annunciator (Cabin)	1	1	1	1	1	
3. Cabin Rate of Climb Indicator	1	1	1	1	1	
4. Differential Pressure/Cabin Altitude Indicator	1	1	1	1	1	
5. Duct Overtemp Annunciator	1	1	1	1	1	
6. Outflow Valve	1	1	1	1	1	
7. Pressurization Controller	1	1	1	1	1	
8. Safety Valve	1	1	1	1	1	
9. Bleed Air Shutoff Valve	2	2	2	2	2	
EQUIPMENT/FURNISHINGS						
1. Shoulder Harness - Crew	2	2	2	2	2	
2. Shoulder Harness - Aft Cabin Fold-up Seats	*	*	*	*	*	* For each occupant

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING COND-ITIONS	
FIRE PROTECTION						
1. Engine Fire Detector System and Annunciators	2	2	2	2	2	
FLIGHT CONTROLS						
1. Flap Position Indicator	1	1	1	1	1	
2. Flap System	1	1	1	1	1	
3. Stall Warning Horn	1	1	1	1	1	
4. Trim Tab Position Indicator (Rudder, Aileron and Elevator)	3	3	3	3	3	
FUEL						
1. Engine Driven Boost Pump	2	2	2	2	2	
2. Fuel Crossfeed System Including Annunciator	1	1	1	1	1	
3. Standby Fuel Boost Pump	2	2	2	2	2	
4. Fuel Pressure Annunciator	2	2	2	2	2	
5. Fuel Quantity Indicating System Including Annunciators	2	2	2	2	2	
6. Firewall Fuel Shutoff System Including Annunciators	2	2	2	2	2	
7. Jet Transfer Pump	2	2	2	2	2	
8. Motive Flow Valve	2	2	2	2	2	
9. Fuel Flow Indicator	2	2	2	2	2	
ICE AND RAIN PROTECTION						
1. Alternate Static Air Source	0	0	1	1	1	
2. Engine Auto Ignition and Annunciators	2	2	2	2	2	
3. Engine Anti-ice System and Annunciators	2	2	2	2	2	
4. Heated Fuel Vent	0	0	0	0	2	
5. Heated Windshield	0	0	0	0	2	
6. Pitot Heat	0	0	2	2	2	
7. Pneumatic Pressure Indicator	0	0	1	1	1	
8. Propeller Deicer System	0	0	0	0	1	
9. Stall Warning Heat (Lift Transducer and Mounting Plate)	0	0	0	0	1	
10. Surface Deicer System	0	0	0	0	1	
11. Wing Ice Light (Left)	0	0	0	0	1	
LANDING GEAR						
1. Landing Gear Position Indicator Lights	3	3	3	3	3	
2. Landing Gear Handle Light	1	1	1	1	1	
3. Landing Gear Aural Warning	1	1	1	1	1	
4. Alternate Landing Gear Extension System	1	1	1	1	1	
5. Hydraulic Fluid Low Annunciator	1	1	1	1	1	
6. Landing Gear Hydraulic Power Pack	1	1	1	1	1	

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING CONDITIONS	
LIGHTS						
1. Cockpit and Instrument Lighting System	0	1	0	1	0	
2. Cabin Door Annunciator	1	1	1	1	1	
3. Landing Lights	0	1	0	1	0	
4. Position Lights	0	3	0	3	0	
5. Anticollision Lights System	0	1	0	1	0	
6. Baggage Pod Door Annunciator	2	2	2	2	2	
FLIGHT/NAVIGATION INSTRUMENTS						
1. Airspeed Indicator	2	2	2	2	2	
2. Altimeter	2	2	2	2	2	
3. Magnetic Compass	1	1	1	1	1	
4. Outside Air Temperature	1	1	1	1	1	
5. Directional Gyro	2	2	2	2	2	
6. Gyroscopic Pitch & Bank Indicator System	2	2	2	2	2	
7. Gyroscopic Rate of Turn & Slip/Skid Indicators	2	2	2	2	2	
8. Weather Radar	–	–	1	1	1	
9. DME	1	1	1	1	1	
10. Marker Beacon	1	1	1	1	1	
11. Vertical Speed Indicator	–	–	2	2	2	
OXYGEN						
1. Oxygen System (115 ft^3 Bottle)	1	1	1	1	1	
VACUUM SYSTEM						
1. Instrument Air System	0	1	2	2	2	
2. Gyro Suction Indicator	1	1	1	1	1	
PROPELLER						
1. Autofeather System Including Annunciators	1	1	1	1	1	
2. Prop Reversing System Including Annunciators	2	2	2	2	2	
3. Prop Governor Test Switch	1	1	1	1	1	
4. Prop Overspeed Governor	2	2	2	2	2	
5. Propeller Primary Low-Pitch Stop	2	2	2	2	2	
ENGINE INDICATORS						
1. ITT Indicator	2	2	2	2	2	

SYSTEM and/or COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING COND-ITIONS	
2. Tachometer (Gas Generator)	2	2	2	2	2	
3. Tachometer (Propeller)	2	2	2	2	2	
4. Torque Indicator	2	2	2	2	2	
ENGINE OIL						
1. Chip Detector System Including Annunciators	2	2	2	2	2	
2. Oil Pressure Indicator	2	2	2	2	2	
3. Oil Temperature Indicator	2	2	2	2	2	
4. Oil Pressure Indicator	2	2	2	2	2	

EMERGENCY PROCEDURES

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INTRODUCTION

This EMERGENCY PROCEDURES Section totally replaces all EMERGENCY PROCEDURES data provided in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual (POH/AFM). Some of the EMERGENCY PROCEDURES in the basic POH/AFM have been reclassified and included in an ABNORMAL PROCEDURES Section which is added to this supplement.

EMERGENCY AIRSPEEDS

All airspeeds quoted in this section are indicated airspeeds (IAS) and assume zero instrument error.

One-Engine-Inoperative Best Angle-of-Climb (V_{XSE})	115 kts
One-Engine-Inoperative Best Rate-of-Climb (V_{YSE})	See Chart
Air Minimum Control Airspeed (V_{MCA})	94 kts
Emergency Descent	180 kts
Maximum Range Glide	135 kts

ENGINE FAILURE

NOTE

To obtain best performance with one engine inoperative, the airplane must be banked 3° to 5° into the operating engine while maintaining a constant heading.

EMERGENCY ENGINE SHUTDOWN

- *ENGINE TORQUE INCREASE - UNSCHEDULED (Ground or Flight) (Not Responsive to Power Lever Movement)*
- *ENGINE FIRE IN FLIGHT*
- *ENGINE FAILURE IN FLIGHT*
- *ILLUMINATION OF MAGNETIC CHIP DETECTOR ANNUNCIATOR*

Affected Engine:

1. Condition Lever - CUT-OFF
2. Propeller Lever - FEATHER
3. Fuel Firewall Valve - CLOSED
4. Fire Extinguisher - ACTUATE (if required)
5. Engine Auto Ignition - OFF
6. Generator - OFF
7. Propeller Synchrophaser - OFF
8. Electrical Load - MONITOR

ENGINE FIRE ON GROUND

Affected Engine:

1. Condition Lever - CUT-OFF
2. Fuel Firewall Valve - CLOSED
3. Starter Switch - STARTER ONLY
4. Fire Extinguisher - ACTUATE (as required)

ENGINE FAILURE DURING GROUND ROLL

1. Power Levers - IDLE
2. Brakes - AS REQUIRED
3. Operative Engine - MAXIMUM REVERSE

WARNING

Extreme care must be exercised when using single-engine reversing on surfaces with reduced traction.

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If Insufficient Runway Remains for Stopping:

4. Condition Levers - CUT-OFF
5. Fuel Firewall Valves - CLOSED
6. Master Switch - OFF (Gang bar down)

ENGINE FAILURE DURING TAKEOFF (AT OR BELOW V_1) — TAKEOFF ABORTED

1. Power Levers – IDLE
2. Brakes – MAXIMUM or as required to achieve stopping distance.

If Insufficient Runway Remains for Stopping:

3. Condition Levers – CUT-OFF
4. Fuel Firewall Valves – CLOSED
5. Master Switch – OFF (Gang bar down)

ENGINE FAILURE DURING TAKEOFF (AT OR ABOVE V_1) — TAKEOFF CONTINUED

1. Power – MAXIMUM ALLOWABLE
2. Landing Gear – UP (After lift-off)
3. Airspeed – MAINTAIN at V_2

WARNING

Do not retard the failed engine power lever until the autofeather system has completely stopped propeller rotation. To do so would deactivate the autofeather circuit and prevent automatic feathering.

4. Propeller (inoperative engine) – VERIFY FEATHERED
5. Airspeed – INCREASE TO V_{YSE} (At 400 feet AGL minimum)
6. Flaps – UP
7. Power – ADJUST AS REQUIRED
8. Clean-up (inoperative engine after reaching 1000 feet AGL)
   a. Condition Lever – CUT-OFF
   b. Fuel Firewall Valve – CLOSED
   c. Engine Auto Ignition – OFF
   d. Autofeather Switch – OFF
   e. Generator – OFF
9. Electrical Load – MONITOR

ENGINE FAILURE AFTER LIFT-OFF (If conditions preclude an immediate landing)

1. Power - MAXIMUM ALLOWABLE
2. Airspeed - MAINTAIN (take-off speed or above)
3. Landing Gear - UP

NOTE

If the autofeather system is being used, do not retard the failed engine power lever until the autofeather system has completely stopped propeller rotation. To do so will deactivate the autofeather circuit and prevent automatic feathering.

4. Propeller (inoperative engine) - FEATHER
5. Airspeed - BEST RATE-OF-CLIMB SPEED (after obstacle clearance altitude is reached)
6. Flaps - UP
7. Clean-up (inoperative engine):

a. Condition Lever - CUT-OFF
b. Fuel Firewall Valve - CLOSED
c. Engine Auto Ignition - OFF
d. Autofeather Switch - OFF
e. Generator - OFF

8. Electrical Load - MONITOR

ENGINE FAILURE IN FLIGHT BELOW AIR MINIMUM CONTROL SPEED (V_{MCA})

1. Reduce power on operative engine as required to maintain control.
2. Lower nose to accelerate above V_{MCA}.
3. Adjust power as required.
4. Secure affected engine as in EMERGENCY ENGINE SHUTDOWN.

ENGINE FLAMEOUT (2nd Engine)

1. Power Lever - IDLE
2. Propeller - DO NOT FEATHER
3. Condition Lever - CUT-OFF
4. Conduct Air Start Procedures

NOTE

The propeller will not unfeather without engine operating.

SMOKE AND FUME ELIMINATION

Attempt to identify the source of smoke or fumes. Smoke associated with electrical failures is usually gray or tan in color, and irritating to the nose and eyes. Smoke produced by environmental system failures is generally white in color, and much less irritating to the nose and eyes.

ELECTRICAL SMOKE OR FIRE

1. Oxygen –

   a. Crew – DON MASK. Insert oxygen inlet connector and twist fitting 1/4 turn clockwise. Connect audio plug.
   b. Microphone Selector Switch – OXYGEN MASK
   c. PASSENGER MANUAL O'RIDE – PULL ON
   d. Passengers – PULL LANYARD PIN, DON MASK

2. Cabin Temp Mode – OFF
3. Vent Blower – AUTO
4. Aft Blower – OFF
5. Radiant Heat – OFF
6. Avionics Master – OFF
7. Nonessential Electrical Equipment – OFF

If Fire or Smoke Ceases:

a. Individually restore avionics and equipment previously turned off.
b. Isolate defective equipment.

If Fire or Smoke Persists:

a. Cabin Pressure Switch – DUMP
b. Land as soon as practical

NOTE

Opening a storm window (after depressurizing) will facilitate smoke and fume removal.

ENVIRONMENTAL SYSTEM SMOKE OR FUMES

1. Oxygen –

   a. Crew DON MASK. Insert oxygen inlet connector and twist fitting 1/4 turn clockwise. Connect audio plug.
   b. Microphone Selector Switch – OXYGEN MASK
   c. PASSENGER MANUAL O'RIDE – PULL ON
   d. Passengers – PULL LANYARD PIN, DON MASK

2. Cabin Temp Mode – OFF
3. Vent Blower – HI
4. Left Bleed Air Valve – ENVIR OFF

If Smoke Decreases:

– Continue operation with left bleed air off.

If Smoke Does Not Decrease:

a. Left Bleed Air Valve – OPEN
b. Right Bleed Air Valve – ENVIR OFF
c. If smoke decreases, continue operation with right bleed air off.

NOTE

Each bleed air valve must remain closed long enough to allow time for smoke purging to positively identify the smoke source.

EMERGENCY DESCENT

1. Power Levers – IDLE
2. Propeller Controls – FULL FORWARD
3. Flaps – APPROACH
4. Landing Gear – EXTEND
5. Airspeed – 180 KTS

GLIDE

1. Landing Gear - UP
2. Wing Flaps - UP (0%)

WARNING

Determine that procedures for re-starting first and second failed engines are ineffective before feathering second engine propeller.

3. Propellers - FEATHERED
4. Airspeed - 135 KNOTS

NOTE

The zero wind glide ratio in this configuration is 2.0 nautical miles of glide distance for each 1000 feet of altitude. Decrease the glide ratio by 0.2 nautical miles per 1000 feet for each 10 knots of headwind.

FLIGHT CONTROLS

UNSCHEDULED ELECTRIC ELEVATOR TRIM

1. Airplane Attitude - MAINTAIN (using elevator control)
2. Control Wheel Disconnect Switch - DEPRESS FULLY (2nd level)

NOTE

Autopilot (if installed) will disengage when the disconnect switch is depressed.

3. Manually Retrim Airplane
4. ELEV TAB Control Switch (Pedestal) - OFF

CAUTION

DO NOT reactivate electric trim system until cause of malfunction has been determined.

UNSCHEDULED RUDDER BOOST ACTIVATION

Rudder boost operation without a large variation of power between the engines indicates a failure of the system.

1. Rudder Boost Switch - OFF

If condition persists:

2. Rudder Trim - ADJUST
3. Bleed Air Valves (2) - INSTR & ENVIR OFF (during approach, landing assured)
4. Conduct normal landing.

ENVIRONMENTAL SYSTEMS

PRESSURIZATION SYSTEM

Anytime the differential pressure goes into the Red Arc:

1. Cabin Altitude Controller - SELECT HIGHER SETTING

If condition persists:

2. Oxygen (crew and passengers) - AS REQUIRED
3. Bleed Air Valves - ENVIR OFF position
4. Cabin Pressure Switch - DUMP (after cabin is depressurized)
5. Descend - AS REQUIRED
6. Bleed Air Valves - OPEN

LOSS OF PRESSURIZATION

1. Oxygen (crew and passengers) - AS REQUIRED
2. Descend - AS REQUIRED

NOTE

The following table sets forth the average time of useful consciousness (time from onset of hypoxia until loss of effective performance) at various altitudes.

35,000 feet	1/2 to 1 minute
30,000 feet	1 to 2 minutes
28,000 feet	2 1/2 to 3 minutes

25,000 feet . . . . . 3 to 5 minutes
22,000 feet . . . . . .5 to 10 minutes
12-18,000 feet . . . . . .30 minutes or more

AUTO-DEPLOYMENT OXYGEN SYSTEM

1. In the event the PASS OXY ON annunciator does not illuminate following illumination of the ALT WARN annunciator, pull PASSENGER MANUAL O'RIDE valve to deploy passenger masks. First aid mask can only be deployed manually.
2. If oxygen quantity is insufficient to sustain both passengers and crew, the supply can be isolated to the crew and First Aid outlets by pulling the OXYGEN CONTROL circuit breaker located in the environmental section of the circuit breaker panel. PASSENGER MANUAL O'RIDE must be in the OFF position.

ILLUMINATION OF "DUCT OVERTEMP" ANNUNCIATOR

1. Cabin Temp Knob - SELECT LOWER TEMPERATURES
2. Vent Blower - HI position
3. Cabin Air Knob - PUSH TO INCREASE

If condition persists:

4. Cabin Temp Mode Selector - MAN HEAT position
5. Manual Temp Switch - DECR (down) position

BLEED AIR LINE FAILURE WARNING SYSTEM

Warning annunciators should be monitored during engine start procedure. Either engine will extinguish both annunciators upon starting.

Illumination of a warning annunciator in flight indicates a possible rupture of a bleed air line aft of the engine firewall.

1. Bleed Air Valve (affected engine) - INSTR & ENVIR OFF position
2. Engine Instruments - MONITOR

NOTE

The bleed air warning annunciator will not extinguish after closing the Bleed Air Valve.

EMERGENCY EXITS

— Emergency Exit Handle(s) - PULL (These are plug-type doors and open into the cabin.)

CAUTION

The outside handle may be locked from the inside with the EXIT LOCK lever. The inside EXIT-PULL handle will unlatch the door regardless of the position of the EXIT LOCK lever. Before flight, make certain the lock lever is in the unlocked position.

SPINS

If a spin is entered inadvertently:

Immediately move the control column full forward, apply full rudder opposite to the direction of the spin, and reduce power on both engines to idle. These three actions should be done as nearly simultaneously as possible; then continue to hold this control position until rotation stops and then neutralize all controls and execute a smooth pullout. Ailerons should be neutral during recovery.

NOTE

Federal Aviation Administration Regulations do not require spin demonstration of airplanes of this weight; therefore, no spin tests have been conducted. The recovery technique is based on the best available information.

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ABNORMAL PROCEDURES

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INTRODUCTION

ABNORMAL PROCEDURES for the High Density Configuration are added to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual via this Section. Some of the EMERGENCY PROCEDURES in the basic POH/AFM have been reclassified as ABNORMAL PROCEDURES and included in this Section.

AIR START

All airspeeds quoted in this section are indicated airspeeds (IAS) and assume zero instrument error.

STARTER ASSIST

CAUTION

The pilot should determine the reason for engine failure before attempting an air start.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an over-temp condition.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. Windshield Heat - OFF
5. Power Lever - IDLE
6. Condition Lever - CUT-OFF
7. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

8. Ignition and Start Switch - ON (up). Check IGNITION Annunciator - ON
9. Condition Lever - LOW IDLE
10. Ignition and Start Switch - OFF (N_1 above 50%)
11. Propeller Lever - AS REQUIRED
12. Power Lever - AS REQUIRED
13. Generator - ON
14. Engine Auto Ignition - ARM
15. Electrical Equipment - AS REQUIRED

WINDMILLING ENGINE AND PROPELLER (NO STARTER ASSIST)

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. Windshield Heat - OFF
5. Power Lever - IDLE
6. Propeller Lever - FULL FORWARD
7. Condition Lever - CUT-OFF
8. Fuel Firewall Valve - OPEN
9. Generator (inoperative engine) - OFF
10. Airspeed - 140 KNOTS MINIMUM
11. Altitude - BELOW 20,000 FEET
12. Engine Auto Ignition - ARM
13. Condition Lever - LOW IDLE
14. Power - AS REQUIRED (after ITT has peaked)
15. Generator - ON
16. Electrical Equipment - AS REQUIRED

LANDING

FLAPS-UP LANDING

Refer to Performance Section for Flaps-Up Approach Speed and Adjustment to Landing Field Length.

1. Pressurization – CHECK
2. Cabin Sign – NO SMOKE & FSB
3. Autofeather Switch – ARM
4. Flaps – UP

WARNING

Do not silence the landing gear warning horn, since the flap-actuated portion of the landing gear warning system will not be actuated during a flaps-up landing.

5. Landing Gear Control – DN
6. Lights – AS REQUIRED

NOTE

Under low visibility conditions, landing and taxi lights should be left off due to light reflections.

NOTE

If a crosswind landing is anticipated, determine Crosswind Component from PERFORMANCE Section. Immediately prior to touchdown, lower up-wind wing and align the fuselage with the runway by use of rudder. During rollout, hold aileron control into the wind and maintain directional control with rudder and brakes. Use propeller reverse as desired.

When Landing Assured:

7. Yaw Damp – OFF
8. Power Levers – IDLE (At 50 ft AGL)

After Touchdown:

9. Propeller Levers – FULL FORWARD
10. Brakes – AS REQUIRED (to achieve stopping performance)

CAUTION

To ensure constant reversing characteristics, the propeller levers must be in FULL INCREASE RPM position.

11. Power Levers – BETA RANGE or REVERSE (As required).

ONE-ENGINE-INOPERATIVE LANDING

When it is certain that the field can be reached:

1. Flaps - APPROACH
2. Landing Gear - DOWN
3. Propeller Control - FULL FORWARD
4. Airspeed - 10 KNOTS ABOVE NORMAL LANDING APPROACH SPEED

When it is certain there is no possibility of go-around:

5. Flaps - DOWN
6. Airspeed - NORMAL LANDING APPROACH SPEED
7. Execute Normal Landing

NOTE

Actual one-engine-inoperative landing distance is less than required destination or alternate landing field length.

Single-engine reverse thrust may be used with caution after touchdown on smooth, dry, paved surfaces.

ONE-ENGINE-INOPERATIVE GO-AROUND

1. Power - MAXIMUM ALLOWABLE
2. Flaps - UP
3. Landing Gear - UP
4. Airspeed - ONE-ENGINE-INOPERATIVE BEST RATE-OF-CLIMB SPEED

SYSTEMS

ENGINE OIL SYSTEM

LOW OIL PRESSURE

Oil pressure values between 60 and 85 psi are undesirable; they should be tolerated only for the completion of the flight, and then only at a reduced power setting not exceeding 1100 foot-pounds torque. Oil pressure values below 60 psi are unsafe; they require either that the engine be shut down, or that a landing be made as soon as possible, using the minimum power required to sustain flight.

FUEL SYSTEM

CROSSFEED (ONE-ENGINE-INOPERATIVE OPERATION)

1. Standby Boost Pumps - OFF
2. Crossfeed Flow Switch - LEFT or RIGHT (as required); Check FUEL CROSSFEED Annunciator - ON; Both FUEL PRESS Annunciators - OUT

CAUTION

Aux Transfer switch must be in AUTO position on side being crossfed. If the firewall valve is closed, the auxiliary fuel supply will not be available (usable), and the fuel pressure annunciator will remain illuminated on the side supplying fuel.

TO DISCONTINUE CROSSFEED:

— Crossfeed Flow Switch - OFF (Centered)

ENGINE DRIVEN BOOST PUMP FAILURE

— Standby Boost Pump (Failed Side) - ON; Check FUEL PRESS Annunciator - OFF

ELECTRICAL SYSTEM

GENERATOR INOPERATIVE (DC GEN Annunciator ON)

1. Generator Switch - OFF, then to RESET for a minimum of 1 second, then release to ON

If generator will not reset:

2. Generator Switch - OFF
3. Operating Generator - DO NOT EXCEED 1.0 LOAD

EXCESSIVE LOADMETER INDICATION (OVER 1.0)

1. Battery Switch - OFF (Monitor Loadmeter)

If loadmeter still indicates above 1.0:

2. Nonessential Electrical Equipment - OFF

If loadmeter indicates 1.0 or below:

3. Battery Switch - ON

CIRCUIT BREAKER TRIPPED

1. Nonessential Circuit - DO NOT RESET IN FLIGHT
2. Essential Circuit:
   a. Circuit Breaker - PUSH TO RESET
   b. If Circuit Breaker trips again - DO NOT RESET

BUS FEEDER CIRCUIT BREAKER TRIPPED (fuel panel bus feeders and right circuit breaker panel bus feeders)

— A short is indicated, do not reset in flight.

NOTE

The items that may be inoperative can be determined from the power distribution schematic in Section VII, SYSTEMS DESCRIPTION.

INVERTER INOPERATIVE

— Select the other inverter.

HYDRAULIC SYSTEM

If HYD FLUID LOW annunciator illuminates during flight, attempt to extend the landing gear normally upon reaching destination. If the landing gear fails to extend, follow LANDING GEAR MANUAL EXTENSION procedures.

LANDING GEAR MANUAL EXTENSION

1. Airspeed - ESTABLISH 130 KNOTS.
2. Landing Gear Relay Circuit Breaker (pilot's subpanel) - PULL
3. Landing Gear Switch Handle - DOWN
4. Extension Lever - PUMP up and down until the 3 green GEAR DOWN indicator lights are illuminated.
5. Extension Lever - STOW (Flush against floor under securing clip.)

NOTE

Ensure the pump handle is in the full down position prior to placing the pump handle in the securing clip.

WARNING

If for any reason the green GEAR DOWN indicator lights do not illuminate (e.g., in case of an electrical system failure), continue pumping until sufficient resistance is felt to ensure that the gear is down.

WARNING

After a manual (emergency) landing gear extension has been made, do not move any landing gear controls, or reset any switches or circuit breakers until the airplane is on jacks, since the failure may be in the gear-up circuit and the gear might retract on the ground.

LANDING GEAR RETRACTION AFTER PRACTICE MANUAL EXTENSION

After a practice manual extension, the gear may be retracted as follows:

1. Extension Lever - STOW (Flush against floor under securing clip.)
2. Landing Gear Relay Control Circuit Breaker (pilot's subpanel) - PUSH IN
3. Landing Gear Switch Handle - UP

ICE PROTECTION SYSTEM

ELECTROTHERMAL PROPELLER DEICE (AUTO SYSTEM)

Abnormal Readings on Deice Ammeter: (Normal operation: 14 to 18 amps)

1. Zero Amps:
   a. Propeller Deice Switch - CHECK, AUTO
   b. If OFF, reposition to AUTO after 30 seconds.
   c. If in AUTO position with zero amps reading, system is inoperative: position the switch to OFF.
   d. Use manual backup system. (No deice ammeter indication - monitor loadmeter.)
2. Zero to 14 amps:
   a. Continue operation.
   b. If propeller imbalance occurs, increase rpm briefly to aid in ice removal.
3. Over 18 amps:
   a. If circuit breaker does not trip, continue operation.
   b. If propeller imbalance occurs, increase rpm briefly to aid in ice removal.
   c. If circuit breaker trips, use manual system; monitor loadmeter for excessive current drain.
   d. If manual mode circuit breaker trips, avoid icing conditions.

ELECTROTHERMAL PROPELLER DEICE (MANUAL SYSTEM)

1. To use manual system, hold manual propeller deice switch in MANUAL position for approximately 90 seconds, or until ice is dislodged from blades.
2. Monitor manual system current requirement with the airplane's loadmeters when the manual deice switch is in the MANUAL position. A small needle deflection (approximately 5%) indicates the system is functioning.

STATIC AIR SYSTEM

PILOT'S ALTERNATE STATIC AIR SOURCE

THE PILOT'S ALTERNATE STATIC AIR SOURCE SHOULD BE USED FOR CONDITIONS WHERE THE NORMAL STATIC SOURCE HAS BEEN OBSTRUCTED. When the airplane has been exposed to moisture and/or icing conditions (especially on the ground), the possibility of obstructed static ports should be considered. Partial obstructions will result in the rate

of climb indication being sluggish during a climb or descent. Verification of suspected obstruction is possible by switching to the alternate system and noting a sudden sustained change in rate of climb. This may be accompanied by abnormal indicated airspeed and altitude changes beyond normal calibrated differences.

Whenever any obstruction exists in the Normal Static Air System, or when the Alternate Static Air System is desired for use:

1. Pilot's Static Air Source (right side panel) - ALTERNATE
2. For Airspeed Calibration and Altimeter Correction, refer to the PERFORMANCE Section.

NOTE

Be certain the static air valve is in the NORMAL position when the alternate system is not needed.

ILLUMINATION OF CABIN DOOR WARNING ANNUNCIATOR

WARNING

Do not attempt to check the security of the airstair door in flight. Remain as far from the door as possible with seatbelts securely fastened until the airplane has landed.

The airstair door locking mechanism must be in the over-centered position (indicated by positioning of safety arm around the diaphragm plunger) in order to provide complete positive locking of the airstair door.

If cabin is pressurized and door is not completely locked, any movement of the door handle toward the unlocked position may cause rapid, complete unlocking and opening of the door.

1. If the red CABIN DOOR warning annunciator illuminates, depressurize cabin (consider altitude first) by activating cabin pressurization dump switch on pedestal.
2. Do not attempt to check the airstair door for security until the cabin is depressurized and the airplane is on the ground.
3. Check security of the airstair door (on the ground) by lifting airstair door step and checking position of arm and plunger. If unlocked position of arm is indicated, turn door handle toward locked position until arm and plunger are in position.

CRACK IN ANY SIDE WINDOW OR IN WINDSHIELD

If it has been determined that a crack has developed in any side window or windshield:

— Pressurization Controller - Reset to maintain 4.0 PSI or less as required.

CAUTION

Prior to next flight, maintenance actions are required. Refer to the LIMITATIONS Section in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

NOTES (WINDSHIELD)

A. Visibility through the windshield may be significantly impaired.
B. Windshield wipers may be damaged if used on cracked surface.
C. Heating elements may be inoperative in the area of the crack.

SIMULATING ONE-ENGINE-INOPERATIVE (ZERO THRUST)

When establishing zero thrust operation, use the power setting listed below. By using this power setting to establish zero thrust, one avoids the inherent delays of restarting a shut down engine and preserves almost instant power to counter any attendant hazard.

1. Propeller - 1600 RPM
2. Power Lever - SET 102 ft-lbs torque

NOTE

This setting will approximate Zero Thrust at low altitudes using recommended One-Engine-Inoperative Climb speeds.

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NORMAL PROCEDURES

INTRODUCTION

Except for the normal procedures specified herein, all normal procedures published in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual apply.

All airspeeds quoted in this Section are Indicated Airspeeds (IAS) and assume zero instrument error.

AIRSPEEDS FOR SAFE OPERATION

Maximum Demonstrated Crosswind Component	25 kts
Takeoff (Flaps 0%):	
Rotation	See Chart
50-ft Speed	See Chart
Takeoff (Flaps 40%):	
Rotation	98 kts
50-ft Speed	100 kts
Two-Engine Best Angle-of-Climb (V_X)	100 kts
Two-Engine Best Rate-of-Climb (V_Y)	125 kts
Operational Climb:	
SL to 10,000 feet	130 kts
10,000 to 20,000 feet	125 kts
20,000 to 25,000 feet	120 kts
Turbulent Air Penetration	170 kts
Landing Approach:	
Flaps 100%	See Chart
Flaps 0%	See Chart
Balked Landing Climb	See Chart
Intentional One-Engine-Inoperative Speed (V_{SSE})	107 kts
Air Minimum Control Speed (V_{MCA})	94 kts

PROCEDURES BY FLIGHT PHASE

PREFLIGHT INSPECTION

Left Wing

1. Flaps - CHECK
2. Fuel Sump (aft of wheel well) - DRAIN
3. Aileron and Tab - CHECK
4. Flush Outboard Wing Fuel Sump - DRAIN
5. Lights - CHECK
6. Main Fuel Tank - CHECK; Cap - SECURE
7. Stall Warning - CHECK
8. Mid-Span Fuel Filler Port - CHECK; Cap - SECURE (Serials BB-1376 & after and prior serials with Kit 101-9058.)
9. Tie-down and Chocks - REMOVE
10. Outboard Deice Boot - CHECK
11. Ram Scoop Fuel Vent - CLEAR
12. Heated Fuel Vent - CLEAR
13. Wing Fuel Sump - DRAIN
14. Fire Extinguisher Pressure - CHECK
15. Landing Gear and Doors - CHECK
16. Fuel Sump (fwd of wheel well) - DRAIN
17. Engine Oil - CHECK QUANTITY; Cap - SECURE
18. Propeller - CHECK
19. Engine Air Intake - CLEAR; Ice Vane and Bypass Door - RETRACTED

20. Firewall Fuel Filter - DRAIN
21. Cowling, Doors, and Panels - CHECK
22. Auxiliary Fuel Tank - CHECK; Cap - SECURE
23. Inboard Deice Boot - CHECK
24. Heat Exchanger Inlet - CLEAR
25. Inboard Fuel Tank Sump - DRAIN
26. Lower Antennas and Beacons - CHECK

Right Wing

1. Inboard Fuel Tank Sump - DRAIN
2. Inboard Deice Boot - CHECK
3. Heat Exchanger Inlet - CLEAR
4. Battery Air Inlet - CLEAR
5. Auxiliary Fuel Tank - CHECK; Cap - SECURE
6. Engine Oil - CHECK QUANTITY; Cap - SECURE
7. Propeller - CHECK
8. Engine Air Intake - CLEAR; Ice Vane and Bypass Door - RETRACTED
9. Firewall Fuel Filter - DRAIN
10. Cowling, Doors, and Panels - CHECK
11. Fuel Sump (fwd of wheel well) - DRAIN
12. Fire Extinguisher Pressure - CHECK
13. Landing Gear and Doors - CHECK
14. Heated Fuel Vent - CLEAR
15. Ram Scoop Fuel Vent - CLEAR
16. Mid-Span Fuel Filler Port - CHECK; Cap - SECURE (Serials BB-1376 & after and prior serials with Kit 101-9058.)
17. Wing Fuel Sump - DRAIN
18. Outboard Deice Boot - CHECK
19. Tie-down and Chocks - REMOVE
20. Main Fuel Tank - CHECK; Cap - SECURE
21. Lights - CHECK
22. Aileron - CHECK
23. Flush Outboard Wing Tank Sump - DRAIN
24. Flaps - CHECK
25. Fuel Sump (aft of wheel well) - DRAIN

The following will be performed in addition to the normal PREFLIGHT INSPECTION procedures:

BAGGAGE POD

— Baggage Pod Doors – SECURE

NOSE BAGGAGE COMPARTMENT

— Nose Baggage Compartment Doors – SECURE

BEFORE ENGINE STARTING

The following will be performed in addition to the normal BEFORE ENGINE STARTING procedures:

1. Cabin Door – LOCKED

   — Verify cabin door security by attempting to turn handle to the unlocked position without depressing the release button. Check that the latch bolts align with each alignment tab (4 positions), safety arm is positioned over the diaphragm plunger, and the upper hooks are engaged.

2. Aft Exit Sign Switch – ON-TEST momentarily, check sign illumination, then OFF-RESET
3. Emergency Exits – SECURE AND UNLOCKED
4. Forward Exit Sign Switch – ON-TEST momentarily, check sign illumination, then OFF-RESET

AFTER STARTING, AND TAXIING

1. GENerator 1 and GENerator 2 Switches – OFF
2. DC Volt/Loadmeters – PRESS TO CHECK VOLTAGE (No voltage on one side indicates Current Limiter out.)
3. GENerator 1 and GENerator 2 Switches – RESET (Hold for one second) then – ON
4. Inverter – ON
5. DC Voltage and Loadmeters – CHECK
6. AC Voltage and Frequency – CHECK
7. Avionics Master – ON
8. Lights – AS REQUIRED
9. Cabin Temperature and Mode – AS REQUIRED
10. Instruments – CHECK
11. Fuel Gages – CHECK. Ensure left and right fuel loads (main plus auxiliary tank) are within imbalance limitations (see Fuel Management in LIMITATIONS Section). If they are not, the imbalance may be brought within limitations by the use of crossfeed:

    a. STANDBY PUMPS – OFF
    b. AUX TRANSFER Switches – AUTO
    c. CROSSFEED FLOW Switch – Left or right (as required); check FUEL CROSSFEED annunciator – ON; both FUEL PRESS annunciators – OUT
    d. Fuel Gages – MONITOR; CROSSFEED FLOW Switch – OFF when balance is restored.

12. Brakes – CHECK

NOTE

Propeller Beta Range may be used during taxi with minimum blade erosion up to the point where N_1 increases. Care must be exercised when taxiing on unimproved surfaces. If possible, conduct RUNUP on a hard surface free of sand and gravel, to preclude pitting of propeller blades and airplane surfaces.

BEFORE TAKEOFF (RUNUP)

1. Avionics and Radar – CHECK
2. Pressurization – SET

    a. Cabin Altitude Selector Knob –ADJUST SO THAT INNER SCALE (ACFT ALT) INDICATES THE PLANNED CRUISE ALTITUDE PLUS 1000 FEET. (If this setting does not result in an outer scale (CABIN ALT) indication of at least 500 feet above take-off field pressure altitude, adjust as required.)
    b. Rate Control Selector Knob – SET INDEX AT 12-O'CLOCK POSITION.

*3. Electric Elevator Trim Control – CHECK (If installed)

    a. Tab Control Switch – ON
    b. Pilot's and Copilot's Switches – CHECK OPERATION
    c. Trim Disconnect – DEACTIVATION OF SYSTEM
    d. Tab Control Switch – OFF, then ON

WARNING

Operation of the electric trim system should occur only by movement of pairs of switches. Any movement of the elevator trim wheel while depressing only one switch denotes a system malfunction. The elevator tab control switch must then be turned OFF and flight conducted only by manual operation of the trim wheel.

4. Trim Tabs – SET
5. Engine Control Friction Locks – SET
6. Flaps – CHECK AND SET
7. Flight Controls – CHECK PROPER DIRECTION AND FREEDOM OF MOVEMENT

*8. Overspeed Governors and Rudder Boost – TEST

a. Rudder Boost Control Switch – ON
b. Propeller Controls – FULL FORWARD (Balance of test is performed on individual engines.)
c. Prop Test Switch – HOLD TO PROP GOV TEST
d. Power Lever – INCREASE UNTIL PROP IS STABILIZED AT 1830 TO 1910 RPM. CONTINUE TO INCREASE UNTIL RUDDER MOVEMENT IS NOTED. (Observe ITT and Torque Limits.)
e. Power Lever – IDLE
f. Prop Test Switch – RELEASE. Repeat steps c, d, e, and f on the opposite engine.

*9. Primary Governors – EXERCISE AT 1800 RPM.
*10. Instrument Vacuum/Deice Pressure System — CHECK (at 1800 rpm):

a. Both Bleed Air Valves – INSTR & ENVIR OFF

(1) Pneumatic Pressure Gage – SHOULD INDICATE ZERO PRESSURE
(2) Both BL AIR FAIL Annunciators – SHOULD ILLUMINATE

b. Both Bleed Air Valves – OPEN or ENVIR OFF as desired

(1) Pneumatic Pressure Gage – SHOULD INDICATE IN GREEN ARC
(2) Gyro Suction Gage – SHOULD INDICATE IN WIDE GREEN ARC
(3) Both BL AIR FAIL Annunciators – EXTINGUISHED

*11. Engine Ice Vanes – CHECK (at 1800 rpm): EXTEND (Check torque drop): RETRACT (Regain original torque): MONITOR Ice Vane annunciators during check.
12. Autofeather – CHECK

a. Power Levers – APPROXIMATELY 500 FT-LBS TORQUE
b. Autofeather Switch – HOLD TO TEST (both autofeather annunciators illuminated)
c. Power Levers – RETARD INDIVIDUALLY:

(1) At Approximately 410 ft-lbs – OPPOSITE ANNUNCIATOR EXTINGUISHED
(2) At Approximately 260 ft-lbs – BOTH ANNUNCIATORS EXTINGUISHED (propeller starts to feather)

NOTE

Autofeather annunciator lights will cycle on and off with each fluctuation of torque as the propeller feathers.

d. Power Levers – BOTH RETARDED (both annunciators extinguished, neither propeller feathers)

13. Autofeather Switch – ARM
14. Propeller Feathering (manual) – CHECK
15. Fuel Quantity, Flight and Engine Instruments – CHECK (See LIMITATIONS section for FUEL MANAGEMENT and minimum oil temperature required for operation without anti-icing additive.)
16. CROSSFEED FLOW Switch – OFF
17. Condition Levers – LOW IDLE

*May be omitted for quick turn-around at pilot's discretion.

TAKEOFF

Refer to PERFORMANCE Section for minimum take-off power, take-off speeds (V_1 and V_2), take-off weight limits, take-off distance, and climb information.

When in take-off position:

1. Brakes – HOLD
2. Power Levers – SET TAKE-OFF POWER
3. Autofeather Annunciators – ILLUMINATED
4. Brakes – RELEASE
5. Power Levers – Adjust as required to maintain torque and ITT within limits

NOTE

Increasing airspeed will cause torque and ITT to increase.

6. V_R Speed - ROTATE
7. Landing Gear Control - UP (when safely airborne)
8. Airspeed - Maintain V_{50} until clear of obstacles
9. Flaps - UP (If extended) at V_{YSE}

ROLLING TAKEOFF

Refer to PERFORMANCE Section for minimum take-off power, take-off speeds (V_1 and V_2), take-off weight limits, take-off distance, and climb information.

Take-off, accelerate-stop, and accelerate-go distance requirements increase 500 feet when using this procedure.

When in take-off position:

1. Brakes - RELEASE
2. Power Levers - Adjust smoothly to set Take-off Torque during the first 500 ft of take-off roll.
3. Autofeather Annunciators - ILLUMINATED
4. V_R Speed - ROTATE
5. Landing Gear Control - UP (when safely airborne)
6. Airspeed - Maintain V_{50} until clear of obstacles
7. Flaps - UP (if extended) at V_{YSE}

OPERATIONAL CLIMB

1. Landing Gear - UP
2. Flaps - UP
3. Yaw Damp - AS REQUIRED
4. Climb Power - SET (Observe maximum ITT 770°C, TORQUE, and N_1 rpm limits.)
5. Propeller - 1900 RPM
6. Propeller Synchrophaser - ON
7. Engine Instruments - MONITOR
8. Cabin Sign - AS REQUIRED
9. Cabin Pressurization - CHECK
10. Aft Blower - OFF

CRUISE

WARNING

Do not lift Power Levers in flight.

1. Cruise Power - SET per CRUISE POWER TABLES or GRAPHS
2. Engine Instruments - MONITOR
3. Fuel Gages - MONITOR

    a. Ensure fuel is being transferred from auxiliary tanks whenever they contain fuel.
    b. Ensure total right and left fuel loads - main plus auxiliary - remain within fuel imbalance limitations (see LIMITATIONS Section for fuel management. If they become unbalanced, restore balance by using crossfeed as described under AFTER STARTING, AND TAXIING Procedures.)

4. Autofeather - OFF

BEFORE LANDING

CAUTION

Propeller operation in the range of 1750-1850 rpm should be avoided, as it may cause ILS interference.

1. Pressurization - CHECK
2. Cabin Sign - NO SMOKE & FSB
3. Autofeather Switch - ARM
4. CROSSFEED FLOW Switch - OFF
5. Flaps - APPROACH
6. Landing Gear - DOWN
7. Lights - AS REQUIRED

NOTE

Under low visibility conditions, landing and taxi lights should be left off due to light reflections.

8. Propeller Synchophaser (Type 1) - OFF
9. Radar - STANDBY or OFF
10. Compute Approach Speed and Balked Landing Climb Speed

NOTE

If crosswind landing is anticipated, determine Crosswind Component from PERFORMANCE Section. Immediately prior to touchdown, lower up-wind wing and align the fuselage with the runway. During rollout, hold aileron control into the wind and maintain directional control with rudder and brakes. Use propeller reverse as desired.

BALKED LANDING

1. Power - SET TAKE-OFF POWER
2. Airspeed - ESTABLISH SPEED FROM CLIMB - BALKED LANDING CHART

When Clear of Obstacles:

3. Airspeed - ESTABLISH V_Y
4. Flaps - UP
5. Landing Gear - UP

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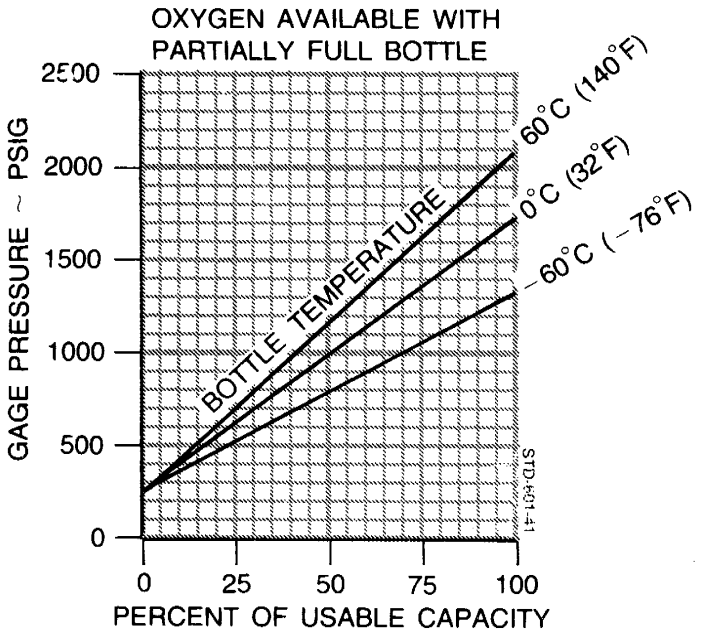
OXYGEN SYSTEM

PREFLIGHT

1. Oxygen Pressure Gage – CHECK TO ENSURE SUFFICIENT PRESSURE, AND NOTE READING.
2. Percent of Full Bottle – DETERMINE FROM "OXYGEN AVAILABLE WITH PARTIALLY FULL BOTTLE" GRAPH AT RIGHT.



3. Oxygen Duration in Minutes – Obtain from Oxygen Duration Table below using the "PERCENT OF USABLE CAPACITY" found in the "OXYGEN AVAILABLE WITH PARTIALLY FULL BOTTLE" graph above.
4. Oxygen System Control – "PULL ON SYS READY" POSITION
5. Crew Oxygen Masks – CHECK OPERATION: Plug in all masks that may be used during flight and check the flow of oxygen using the flow indicator of each mask. Unplug connections and stow masks until inflight use is required.

OXYGEN DURATION

Oxygen duration is computed for crew member masks rated at 2.7 liter per minute-NTPD (color coded: orange), passenger masks rated at 2.7 liters per minute-NTPD (color coded: green and black), and a single first aid mask rated at 3.9 liters per minute-NTPD (color coded: orange and white). All masks are constant flow. The Oxygen System is approved for altitudes up to 25,000 feet.

In order to obtain the crew oxygen duration stated in the following chart, the oxygen must be isolated to the crew outlets at the end of the passenger duration by pulling the OXYGEN CONTROL circuit breaker located in the Environmental section of the circuit breaker panel. PASSENGER MANUAL O'RIDE must be in the OFF position.

OXYGEN DURATION TABLE
115 CU. FT. BOTTLE

Number of People Using	User	Oxygen Available Percent of Usable Capacity			
		100%	75%	50%	25%
		Duration In Minutes			
1	Crew	730	548	365	182
2	Crew	430	324	216	108
3	Crew	306	230	153	101
	Passengers	306	230	153	41
4	Crew	237	178	120	92
	Passengers	237	178	118	32
5	Crew	194	145	120	86
	Passengers	194	145	83	26
6	Crew	164	122	120	82
	Passengers	164	122	64	22
7	Crew	142	120	114	79
	Passengers	142	101	54	19
8	Crew	125	120	108	77
	Passengers	125	85	48	17
9	Crew	120	120	103	75
	Passengers	109	74	43	15
10	Crew	120	120	99	73
	Passengers	96	65	39	13
11	Crew	120	118	95	72
	Passengers	86	58	35	12
12	Crew	120	114	92	71
	Passengers	78	54	32	11
13	Crew	120	110	90	70
	Passengers	71	50	30	10
14	Crew	120	106	88	69
	Passengers	65	46	28	9
15	Crew	120	103	86	69
	Passengers	60	43	26	9

INFLIGHT

The following table sets forth the average time of useful consciousness (time from onset of hypoxia until loss of effective performance) at various altitudes.

25,000 feet . . . . . 3 to 5 minutes
22,000 feet . . . . . .5 to 10 minutes
12 - 18,000 feet . . . . . .30 minutes or more

WARNING

Beards and mustaches should be carefully trimmed so that they will not interfere with the proper sealing of an oxygen mask. The fit of the oxygen mask around the beard or mustache should be checked on the ground for proper sealing. Studies conducted by the military and FAA conclude that oxygen masks do not seal over beards and mustaches.

1. Passengers
   Lanyard Pin - PULL AND DON MASK

NOTE

If system is desired for use, mask can be manually deployed by pulling PASSENGER MANUAL O'RIDE control on pilot's overhead display. The oxygen system control must be in the PULL ON SYStem READY position for the PASSENGER MANUAL O'RIDE to function.

2. Crew

   a. Don mask. Insert Oxygen Inlet Connector and twist fitting 1/4 turn clockwise. Connect Audio Plug. Adjust mask for proper fit.

NOTE

Since system efficiency is determined by the fit of the oxygen masks, make certain they fit properly and are in good condition.

   b. Oxygen Flow Indicator - CHECK that the red plunger lifts from its seat when the hose is inserted into the oxygen coupling.
   c. At the end of the stated passenger duration: PULL the OXYGEN CONTROL circuit breaker; PASSENGER MANUAL O'RIDE - OFF.

3. First Aid Oxygen

   a. Oxygen System Control - CHECK SYS READY ON
   b. Oxygen Compartment - PULL COVER OPEN
   c. ON/OFF Valve - ON position
   d. DON MASK

AFTER USING OXYGEN

1. Passengers

   a. Lanyard Pin - INSERT
   b. Masks - RETURN TO OVERHEAD CONTAINER AND SECURE DOOR

NOTE

To close overhead doors, the following conditions must be met: Cabin altitude must be below the range requiring oxygen, and the PASSENGER MANUAL O'RIDE must be in the OFF position.

2. Crew – Mask
   Unplug audio connection. Disconnect oxygen fitting by twisting 1/4 turn counterclockwise and pull from coupling. Remove and stow mask.
   Oxygen system control must stay in the PULL ON SYS READY position while in flight. Oxygen outlet for crew mask automatically seals when the mask is unplugged.
3. First Aid Oxygen

   a. ON/OFF Valve – OFF position
   b. Mask – RETURN TO COMPARTMENT AND CLOSE COVER

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GENERAL

This PERFORMANCE Section totally replaces all PERFORMANCE data provided in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

The performance data presented in this section fulfills all the requirements of FAR 135 Appendix A.

TECHNIQUE USED FOR DETERMINING TAKE-OFF AND LANDING PERFORMANCE

The take-off and accelerate-stop performance contained in this supplement was obtained using the procedures listed in the Normal and Emergency Procedures sections and the associated conditions presented on the performance charts. The take-off and accelerate-stop performance is based on the power level reflected in the appropriate Minimum Take-off Power graph. The landing performance contained in this supplement was obtained using the procedures listed in the Normal and Abnormal procedures sections and the associated conditions presented on the performance graphs.

INTRODUCTION TO PERFORMANCE AND FLIGHT PLANNING

The graphs and tables in this Section present information for takeoff, climb, landing, and flight planning at various parameters of weight, power, altitude, and temperature. All FAA-approved performance information is included within this Section. Examples have been presented on all performance graphs.

The following example presents calculations for flight time, block speed, and fuel required for a proposed flight from Denver to Reno, using the conditions listed below:

CONDITIONS

At Stapleton International (DEN):

Outside Air Temperature	28°C (82°F)
Field Elevation	5333 feet[1]
Altimeter Setting	29.82 in. Hg
Wind	340° at 10 knots
Runway 35L Length	11,500 feet[1]

[1]Source: Jeppesen Airport Diagram, AUG 26-83

Route of Trip[2]:
DEN - J116 - EKR - J173 - SLC - J154 - BAM - J32 - RNO

Route Segment Data[2]:

ROUTE SEGMENT	AVERAGE MAGNETIC COURSE	AVERAGE MAGNETIC VARIATION	DISTANCE NM	WIND AT FL 240 DIR/KNOTS	OAT AT FL 240 °C	OAT AT [3]18,000 FT °C	ALTIMETER SETTING IN. HG
DEN - EKR	263°	13°E	143[4]	350°/40	-10	-6	29.82
EKR - SLC	269°	14°E	192	350°/40	-10	-6	29.82
SLC - BVL	248°	14°E	81	340°/35	-20	0	29.75
BVL - BAM	249°	16°E	145	340°/35	-20	0	29.75
BAM - RNO	226°	17°E	146[4]	290°/45	-20	-4	29.60

[2]Source: Jeppesen High Altitude Enroute Chart US (HI) 2, NOV 18-83
[3]To illustrate the MEA requirement, the floor of the Jet Airways (18,000 ft MSL) was used in this example.
[4]Includes distance between airport and VORTAC, per Jeppesen Airport Directory, OCT 7-83.

At Cannon International (RNO):
Outside Air Temperature ..... 32°C (90°F)
Field Elevation ..... 4412 feet[5]
Altimeter Setting ..... 29.60 in. Hg
Wind ..... 290° at 13 knots
Runway 25 Length ..... 6101 feet[5]

[5]Source: Jeppesen Airport Diagram, AUG 19-83

To determine the pressure altitude at origin and destination airports, add 1000 feet to field elevation for each 1.00 in. Hg that the reported altimeter setting value is below 29.92 in. Hg, and subtract 1000 feet for each 1.00 in. Hg above 29.92 in. Hg. Always subtract the reported altimeter from 29.92 in. Hg, then multiply the answer by 1000 to find the difference in feet between field elevation and pressure altitude.

Pressure Altitude at DEN:

29.92 in. Hg - 29.82 in. Hg = 0.10
0.10 X 1000 feet = 100 feet
The pressure altitude at DEN is 100 feet above field elevation.
Pressure Altitude at DEN = 5333 + 100 = 5433 feet.

Pressure Altitude at RNO:

29.92 in. Hg - 29.60 in. Hg = 0.32
0.32 X 1000 feet = 320 feet
The pressure altitude at RNO is 320 feet above field elevation.
Pressure Altitude at RNO = 4412 + 320 = 4732 feet.

PERFORMANCE EXAMPLE

MAXIMUM TAKE-OFF WEIGHT (MAXIMUM TAKE-OFF WEIGHT LIMITATION)

Enter the graphs at 5433 feet pressure altitude, 28°C, and read:

Flaps 0% ..... 12,500 lbs
Flaps 40% ..... 11,920 lbs

MAXIMUM ENROUTE WEIGHT AT MINIMUM ENROUTE ALTITUDE (MAXIMUM ENROUTE WEIGHT LIMITATION)

To determine the maximum take-off weight to meet this requirement, the weight of the fuel used to reach the MEA is added to the maximum enroute weight obtained from the SERVICE CEILING – ONE ENGINE INOPERATIVE graph. Enter the graph at the pressure altitude of the MEA and trace right; enter again at the OAT at the MEA and trace up. The two tracings intersect at the weight limit that will exist when the subject MEA is reached. Use the OPERATIONAL CLIMB – TIME, FUEL, AND DISTANCE graph to determine the weight of the fuel used to climb. Use the Cruise Power charts to determine the weight of the fuel used to cruise to each MEA.

Enter the SERVICE CEILING – ONE ENGINE INOPERATIVE graph at 18,000 FT and –6°C.

Maximum Enroute Weight at Minimum
Enroute Altitude ..... 11,125 lbs

Add the fuel required to climb, plus any fuel used in cruise before reaching each MEA, to determine the maximum allowable take-off weight to meet the requirement for each route segment of the trip.

MINIMUM FIELD LENGTH

The following example illustrates the use of graphs which may restrict take-off weight due to field length available under existing conditions.

TAKE-OFF DISTANCE

Enter the graphs at 28°C, 5433 feet pressure altitude, 11,000 pounds, and 10 knots headwind component, and obtain the following results:

	FLAPS 0%	FLAPS 40%
Total Distance Over 50-foot Obstacle . . .	3100 ft	2870 ft
Take-off Speeds:		
Rotation	100 kts	98 kts
50 ft	108 kts	100 kts

ACCELERATE-STOP DISTANCE (MAXIMUM TAKE-OFF WEIGHT LIMITATION)

Enter the graphs at 28°C, 5433 feet pressure altitude, 11,000 pounds, and 10 knots headwind component, and obtain the following results:

	FLAPS 0%	FLAPS 40%
Field Length	4990 ft	4570 ft
V_1	100 kts	98 kts

TAKE-OFF FLIGHT PATH

The following example assumes the airplane is loaded to a take-off weight of 11,000 pounds, and illustrates the use of graphs which may restrict take-off weight due to the climb gradient necessary to clear obstacles beyond the end of the runway.

ACCELERATE-GO DISTANCE OVER 35-FOOT OBSTACLE - FLAPS 0%

Enter the graph at 28°C, 5433 feet pressure altitude, 11,000 pounds, and 10 knots headwind component:

Total Distance Over 35-foot Obstacle ..... 4700 ft
Take-off Speeds:
Rotation (V_R) ..... 100 kts
35 ft (V_2) ..... 108 kts

NET GRADIENT OF CLIMB - ONE ENGINE INOPERATIVE - FLAPS 0%

Enter the graph at 28°C, 5433 feet pressure altitude, and 11,000 pounds:

Climb Gradient 2.5%
Climb Speed ..... 108 kts

A 2.5% climb gradient is 25 feet of vertical height for each 1000 feet of horizontal distance.

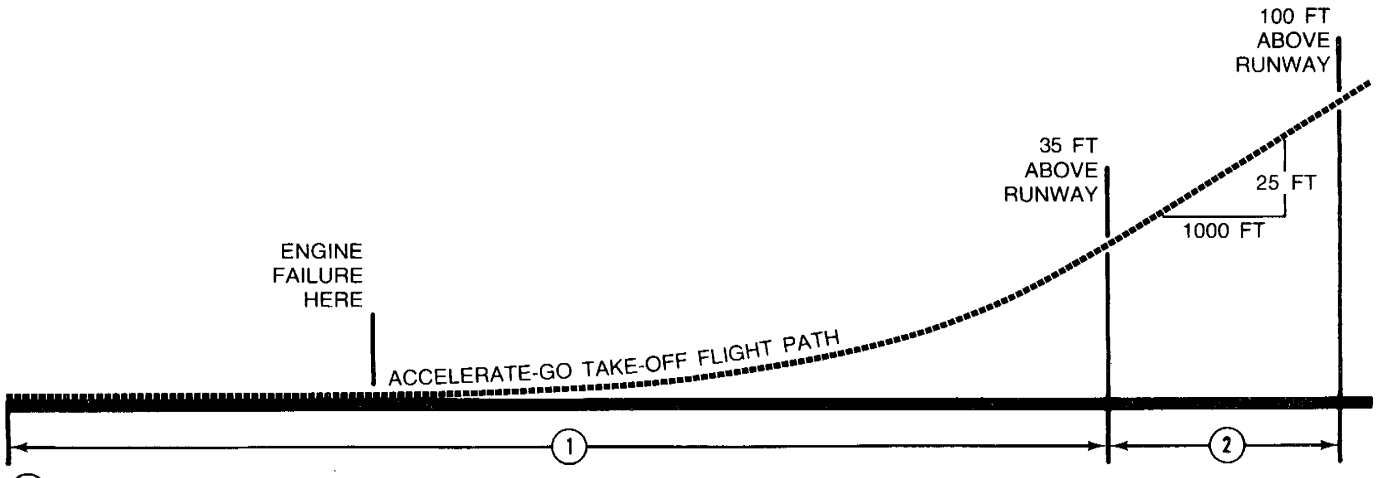
NOTE

The graphs for take-off climb gradient assume a zero-wind condition. Climbing into a headwind will result in higher angles of climb and better obstacle clearance capabilities.

Calculation of the horizontal distance to clear an obstacle 100 feet above the runway surface:

Distance from 35 feet AGL to 100 feet AGL	=65 feet
(100 - 35) (1000 ÷ 25)	=2600 feet
Total Distance = 4700 feet + 2600 feet	=7300 feet

This example is illustrated below:



(1) ACCELERATE-GO TAKE-OFF DISTANCE = 4700 FEET

(2) DISTANCE TO CLIMB FROM 35 FEET TO 100 FEET ABOVE RUNWAY = 2600 FEET

B200-601-449(HDC)

FLIGHT PLANNING EXAMPLE

The following calculations provide information for flight planning.

Calculations for flight time, block speed, and fuel requirements for the proposed flight are detailed below.

NOTE

For example purposes, the differences between MSL altitudes and pressure altitudes have been ignored in enroute calculations.

ISA CONVERSION

Enter the ISA CONVERSION Graph at the conditions indicated:

DEN	Pressure Altitude	5433 feet
	OAT	28°C
	ISA Condition	ISA + 23°C
DEN — SLC	Pressure Altitude	24,000 feet
	OAT	-10°C
	ISA Condition	ISA + 23°C
SLC — RNO	Pressure Altitude	24.000 feet
	OAT	-20°C
	ISA Condition	ISA + 13°C
RNO	Pressure Altitude	4732 feet
	OAT	32°C
	ISA Condition	ISA + 27°C

Enter the OPERATIONAL CLIMB - TIME, FUEL & DISTANCE Graph at 28°C, to 5433 feet, and to 11,000 pounds, and enter at -10°C, to 24,000 feet and to 11,000 pounds, and read:

Time to Climb ..... 14 - 2 = 12 minutes

Fuel Used to Climb ..... 196 - 40 = 156 pounds

Distance Traveled ..... 36 - 4 = 32 nautical miles

Enter the TIME, FUEL, AND DISTANCE TO DESCEND Graph at 24,000 feet, and enter again at 4732 feet and read:

Time to Descend ..... 16 - 3 = 13 minutes

Fuel Used to Descend ..... 156 - 36 = 120 pounds

Descent Distance ..... 76 - 13 = 63 nautical miles

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The estimated average cruise weight is approximately 10,100 pounds.

Enter the tables for INTERMEDIATE CRUISE POWER at 1700 RPM for ISA + 10°C, ISA + 20°C, and ISA + 30°C, and read the cruise speeds for 24,000 feet at 11,000 pounds and 10,000 pounds.

CRUISE TRUE AIRSPEEDS AT FL 240 - KTS

11,000 POUNDS			10,000 POUNDS		
ISA + 10°C	ISA + 20°C	ISA + 30°C	ISA + 10°C	ISA + 20°C	ISA + 30°C
261	260	259	263	262	261

Interpolate between these speeds for ISA + 23°C and ISA + 13°C at 10,100 pounds.

Cruise True Airspeed (ISA + 23°C) ..... 261.5 knots

Cruise True Airspeed (ISA + 13°C) ..... 262.5 knots

Enter the *INTERMEDIATE CRUISE POWER at 1700 RPM Graph at 24,000 feet, and read the recommended torque settings for ISA + 23°C (-3°C IOAT) and ISA + 13°C (-13°C IOAT):

ISA +23°C (-3°C IOAT) ..... 1635 ft-lbs torque per engine

ISA +13°C (-13°C IOAT) ..... 1700 ft-lbs torque per engine

Enter the *FUEL FLOW AT INTERMEDIATE CRUISE POWER Graph at 24,000 feet, and read the fuel flow for ISA + 23°C (-3°C IOAT) and ISA + 13°C (-13°C IOAT):

ISA + 23°C (-3°C IOAT) Fuel Flow Per Engine ..... 285.5 lbs/hr
Total Fuel Flow ..... 571 lbs/hr

ISA + 13°C (-13°C IOAT) Fuel Flow Per Engine ..... 296 lbs/hr
Total Fuel Flow ..... 592 lbs/hr

*NOTE: For flight planning, enter these graphs at the forecasted ISA condition; for enroute power settings and fuel flow, enter the graphs at the actual IOAT.

Time and Fuel Used were calculated at Intermediate Cruise Power as follows:

$$\text{Time} = \frac{\text{Distance}}{\text{Ground Speed}}$$

$$\text{Fuel Used} = (\text{Time})\,(\text{Total Fuel Flow})$$

Results are as follows:

ROUTE	DISTANCE	ESTIMATED GROUND SPEED	TIME AT CRUISE ALTITUDE	FUEL USED FOR CRUISE
	NM	KNOTS	HRS : MIN	LBS
DEN - EKR	111*	248	:27	256
EKR - SLC	192	243	:47	451
SLC - BVL	81	253	:19	190
BVL - BAM	145	251	:35	342
BAM - RNO	83*	230	:22	214

*Distance required to climb or descend has been subtracted from segment distance.

TIME — FUEL — DISTANCE

ITEM	TIME HRS : MINS	FUEL POUNDS	DISTANCE NM
Start, Runup, Taxi, and Take-off acceleration	00 : 00	90	0
Climb	00 : 12	156	32
Cruise	2 : 30	1453	612
Descent	00 : 13	120	63
Total	2 : 55	1819	707

Block Speed: 707 NM ÷ 2 hours, 55 minutes = 242 knots

RESERVE FUEL

Weight at the end of cruise is estimated to be about 9271 pounds. However, since the lowest weight listed in the power tables is 10,000 pounds, use the values for that weight in this example.

Reserve Fuel (45 minutes at Long Range Cruise Power): Enter the tables for LONG RANGE CRUISE POWER for ISA + 10°C and ISA + 20°C, and read the fuel flow for 24,000 feet at 10,000 pounds:

ISA + 10°C ..... 336 lbs/hr Total Fuel Flow

ISA + 20°C ..... 342 lbs/hr Total Fuel Flow

Interpolate to find fuel flow at ISA + 13°C:

Total fuel flow for reserve = 336 + 2 = 338 lbs/hr

Reserve Fuel = 45 minutes X 338 lbs/hr = 254 lbs

TOTAL FUEL REQUIREMENT

1819 + 254 = 2073 lbs

ZERO FUEL WEIGHT LIMITATION

For this example, the following conditions were assumed:

Ramp Weight	= 11,090 pounds
Weight of Usable Fuel Onboard	= 2073 pounds

Zero Fuel Weight = Ramp Weight - Weight of Usable Fuel Onboard

Zero Fuel Weight = 11,090 - 2073	= 9017 pounds
Maximum Zero Fuel Weight (from LIMITATIONS Section)	= 11,000 pounds

Maximum Zero Fuel Weight Limitation has not been exceeded

Anytime the Zero Fuel Weight exceeds the Maximum Zero Fuel Weight Limit, the excess weight must be off-loaded from PAYLOAD ONLY (i.e. *not* from fuel). If desired, additional fuel may then be added. However, the foregoing calculations will remain unchanged only if the fuel added is equal in weight to the payload off-loaded, since only then will the ramp weight and take-off weight remain the same as before.

LANDING INFORMATION

The estimated Landing Weight is determined by subtracting the fuel required for the trip from the Ramp Weight.

Ramp Weight	11,090 pounds
Expected fuel usage for trip	(–) 1819 pounds
Landing Weight (11,090 - 1819)	9271 pounds

LANDING FIELD LENGTH (MAXIMUM LANDING WEIGHT LIMITATION)

Enter the LANDING FIELD LENGTH - FLAPS 100% graph at 32°C, 4732 feet, 9271 pounds, and 10 knots headwind component:

Landing Field Length	4375 feet
Alternate Field Length	3750 feet
Approach Speed	95 knots

Enter the CLIMB – BALKED LANDING Graph at 32°C, 4732 feet, and 9271 pounds:

Rate of Climb ..... 1725 feet per minute

Climb Gradient 14.2%

Climb Speed ..... 95 knots

HOW TO USE GRAPHS

1. In addition to presenting the answer for a particular set of conditions, the example on the graph also presents the order in which the various scales on the graph should be used. For instance, if the first item in the example is OAT, then enter the graph at the known OAT.

2. The reference lines indicate where to begin following the guidelines. Always project to the reference line first, then follow the guidelines to the next known item by maintaining the same PROPORTIONAL DISTANCE between the guideline above and guideline below the projected line. For instance, if the projected line intersects the reference line in the ratio of 30% down/70% up between the guidelines, then maintain the same 30%/70% relationship between the guidelines all the way to the next known item or answer.

3. The associated conditions define the specific conditions from which performance parameters have been determined. They are not intended to be used as instructions; however, performance values determined from charts can only be achieved if the specified conditions exist.

4. Indicated airspeeds (IAS) were obtained by using the Airspeed Calibration – Normal System graph or the Airspeed Calibration – Normal System – Take-off Ground Roll graph.

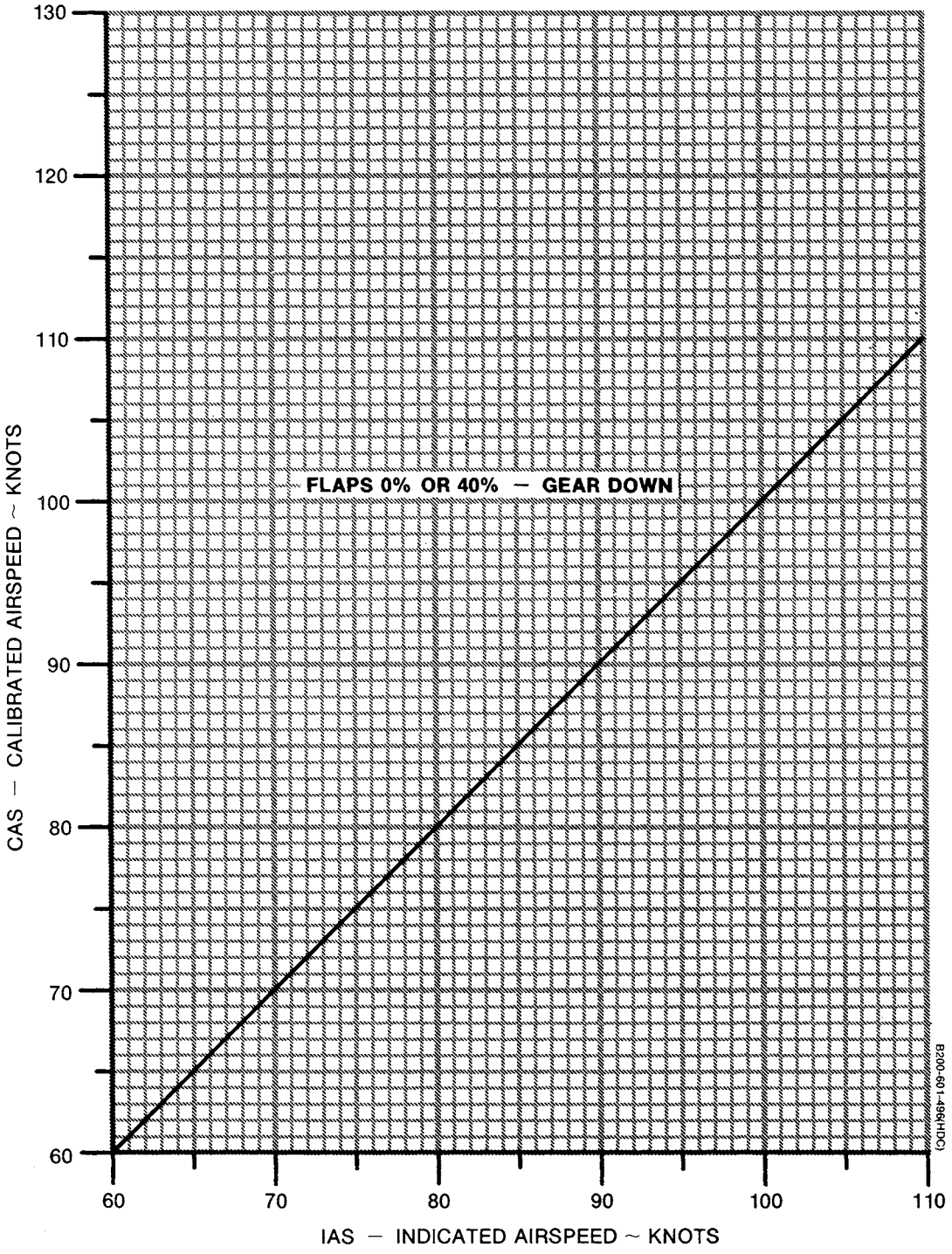
5. The full amount of usable fuel is available for all approved flight conditons.

6. Notes have been provided on various graphs and tables to approximate performance with ice vanes extended. The effect will vary, depending upon airspeed, temperature, altitude, and ambient conditions. At lower altitudes, where operation on the torque limit is possible, the effect of ice vane extension will be less, depending upon how much power can be recovered after the ice vanes have been extended.

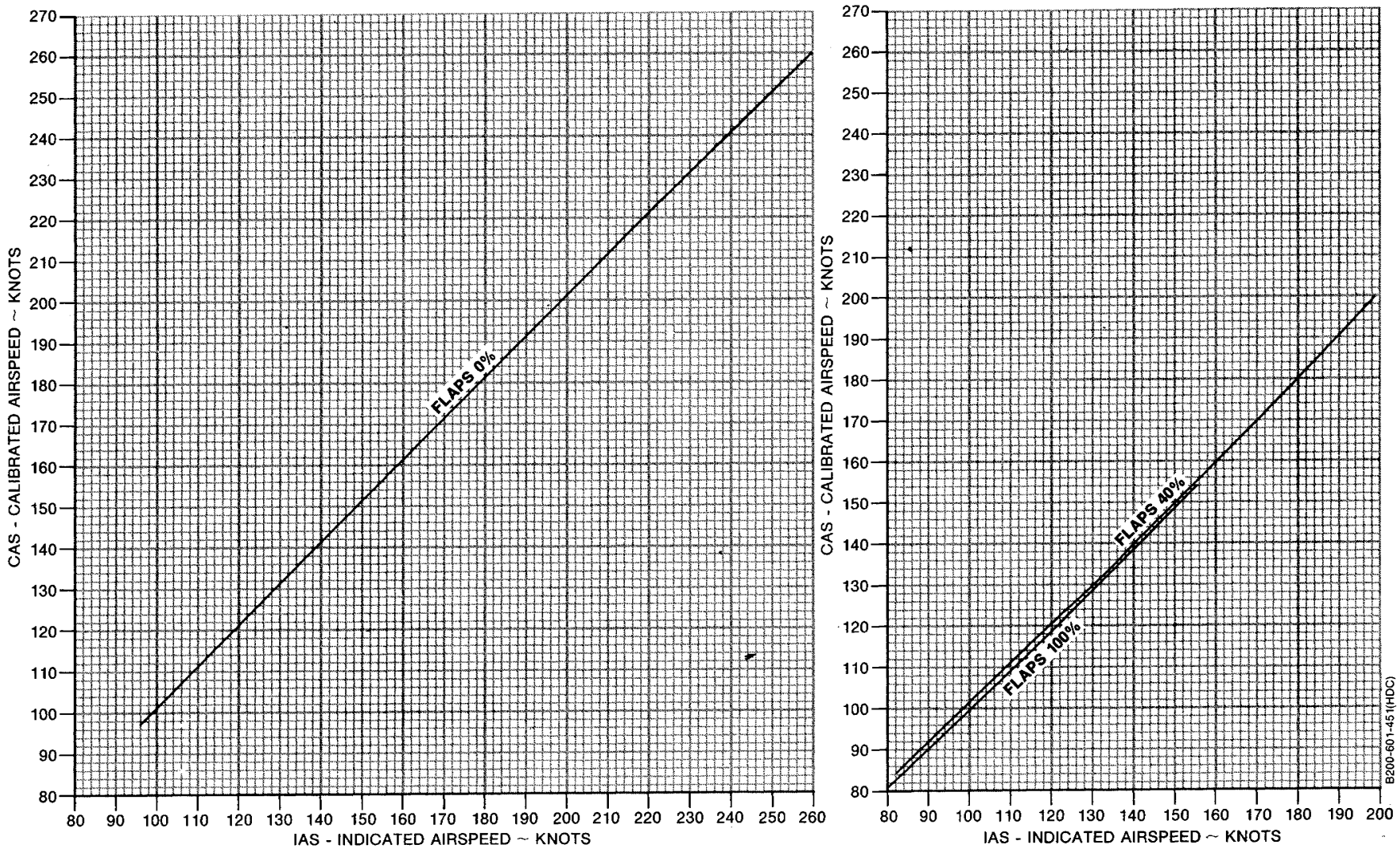
7. The Landing Field Length presented includes the required factor of 1.67 (1.43 for alternate field). This means that the field length obtained may be equated directly to the runway length available.

AIRSPEED CALIBRATION – NORMAL SYSTEM

TAKE-OFF GROUND ROLL



AIRSPEED CALIBRATION – NORMAL SYSTEM



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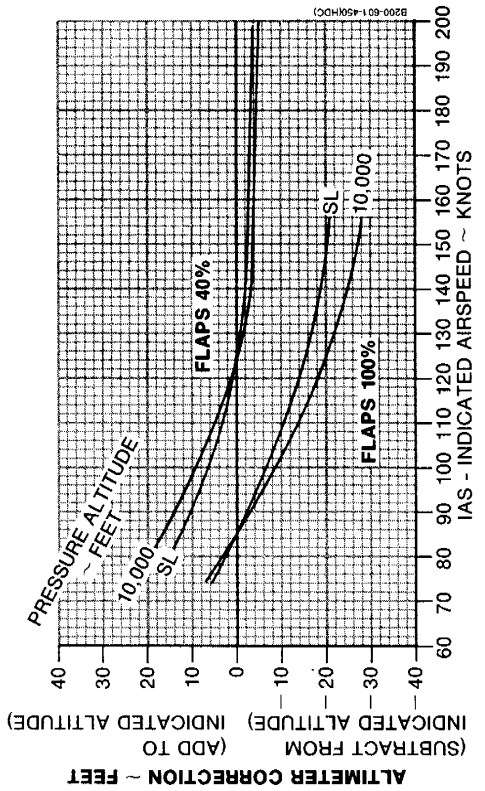
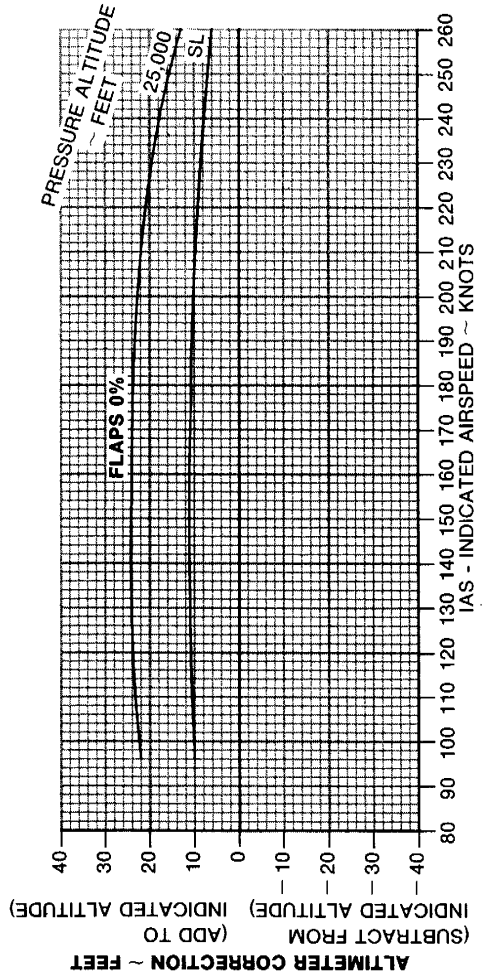
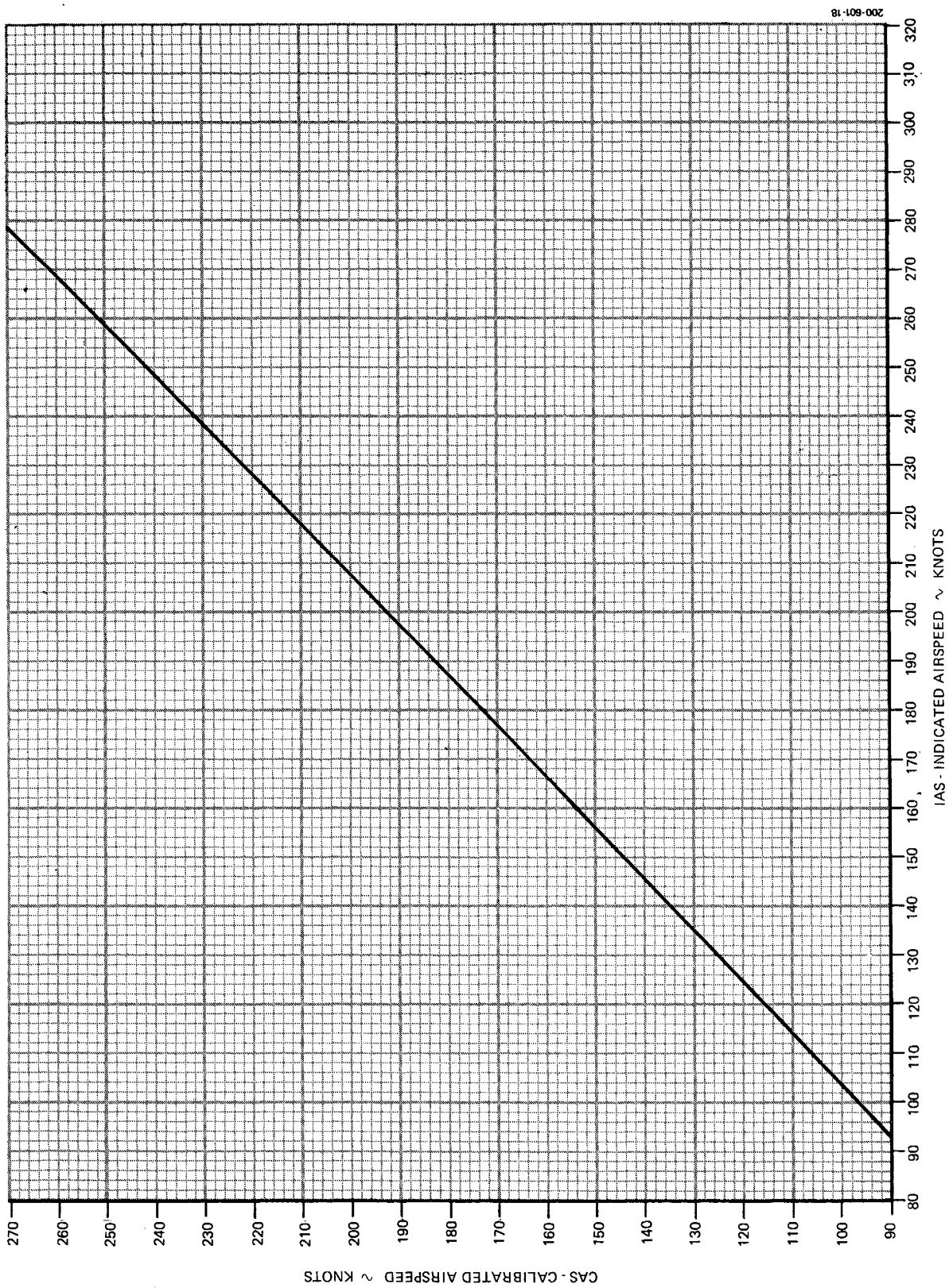
ALTIMETER CORRECTION – NORMAL SYSTEM





AIRSPEED CALIBRATION – ALTERNATE SYSTEM

APPLICABLE FOR ALL FLAP POSITIONS



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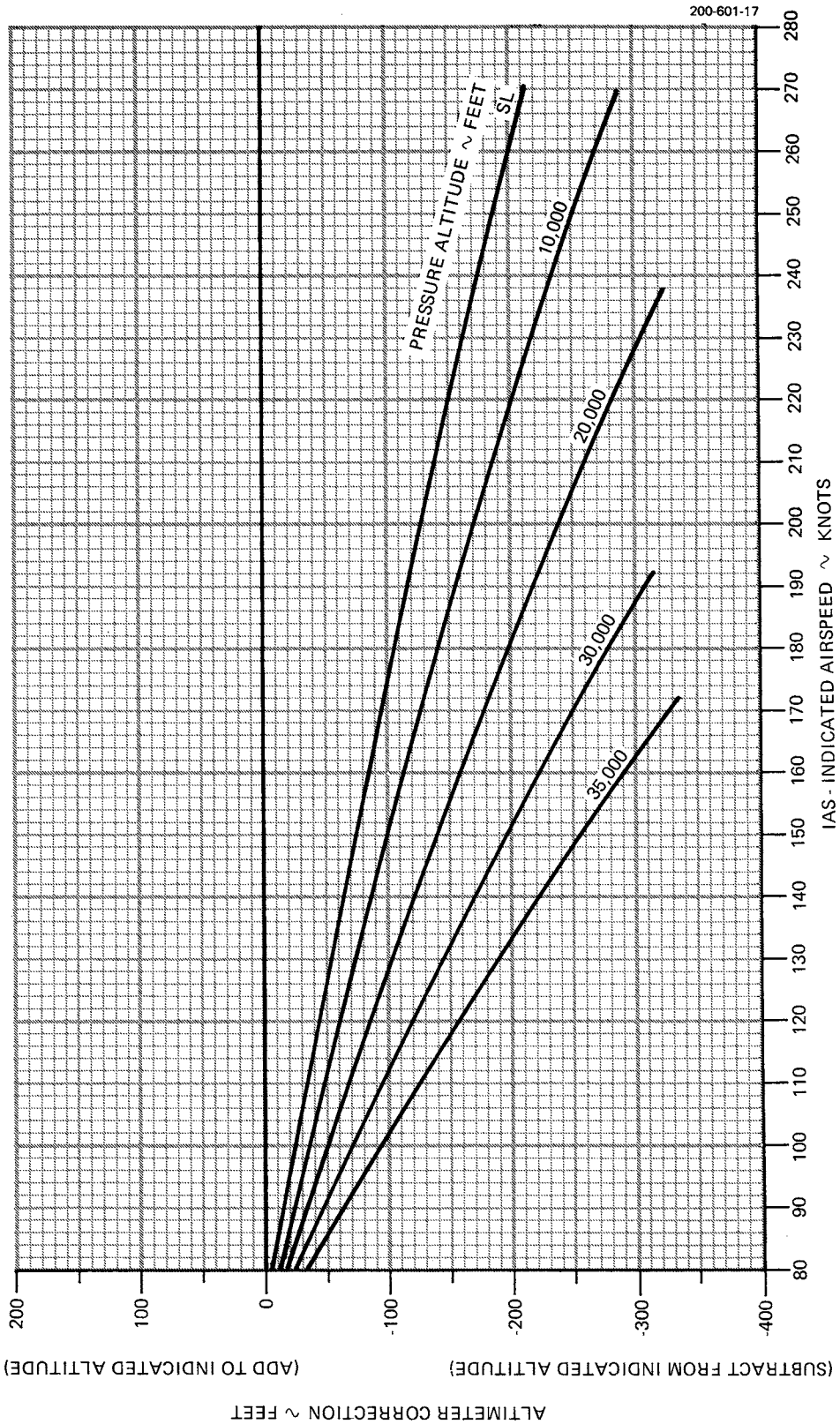
ALTIMETER CORRECTION – ALTERNATE SYSTEM

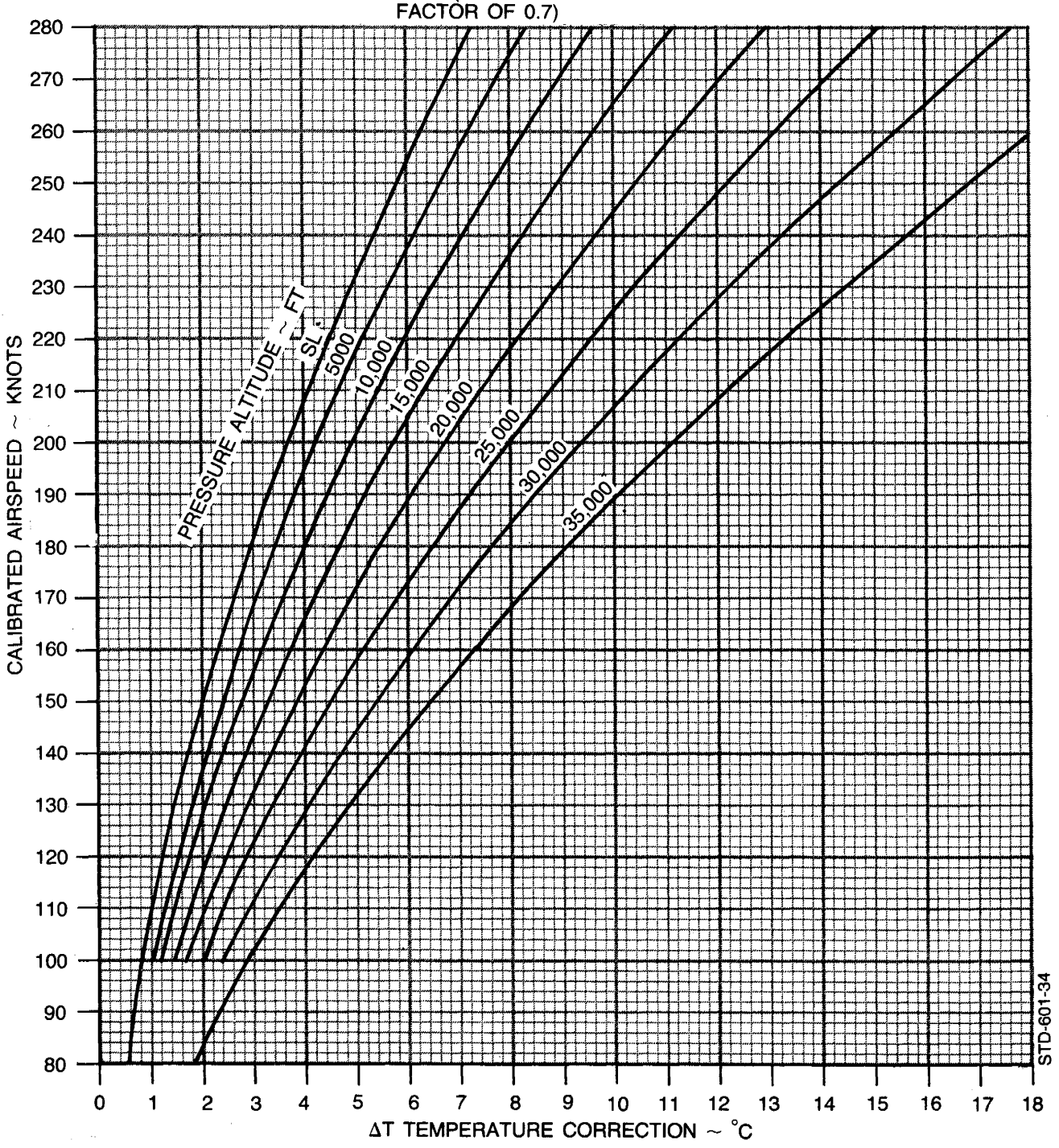
APPLICABLE FOR ALL FLAP POSITIONS



INDICATED OUTSIDE AIR TEMPERATURE CORRECTION

STANDARD DAY (ISA)

NOTE: SUBTRACT ΔT FROM INDICATED (GAGE) OAT TO OBTAIN TRUE OAT. (ΔT ASSUMES A RECOVERY FACTOR OF 0.7)



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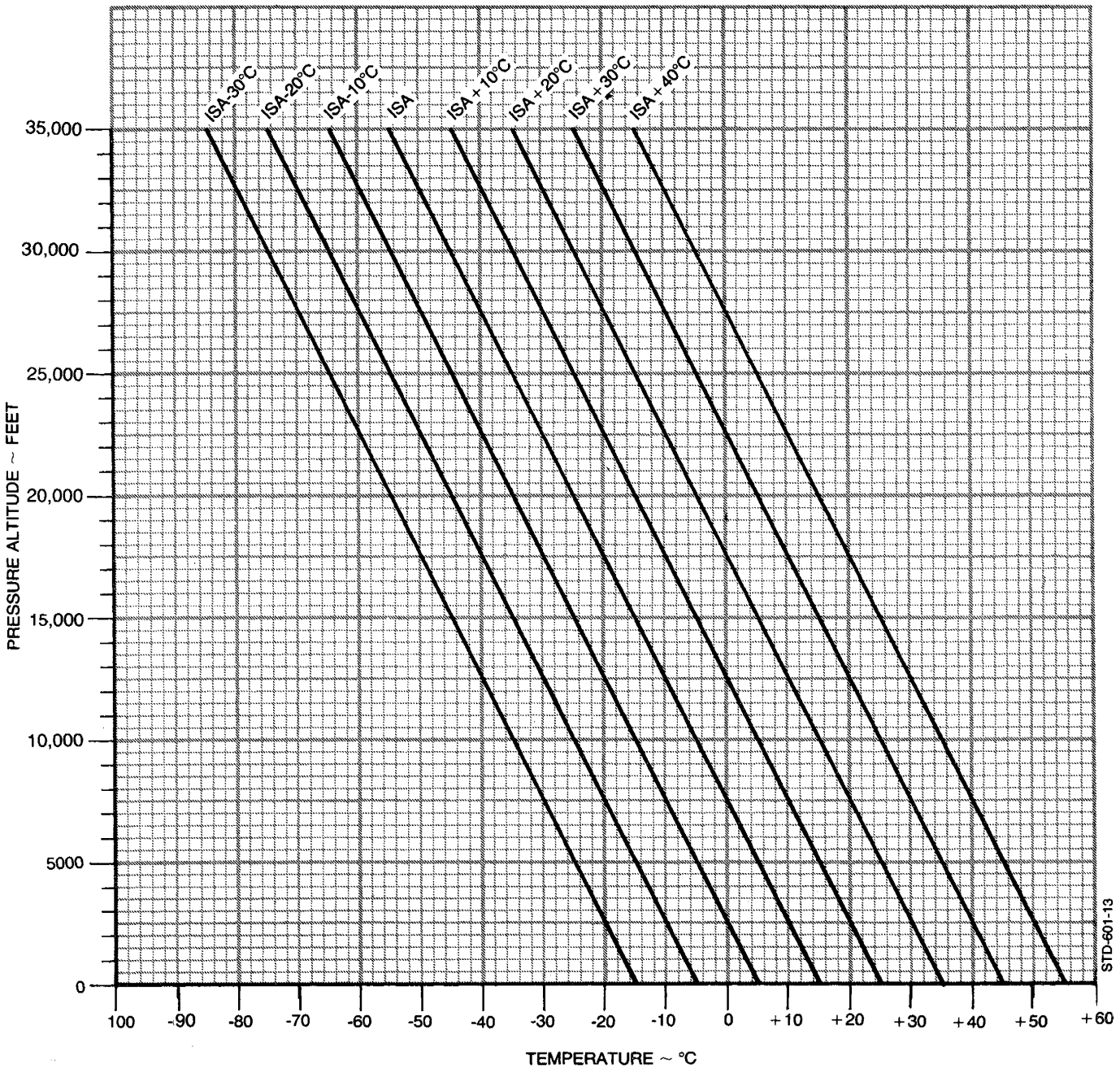
ISA CONVERSION

PRESSURE ALTITUDE vs OUTSIDE AIR TEMPERATURE



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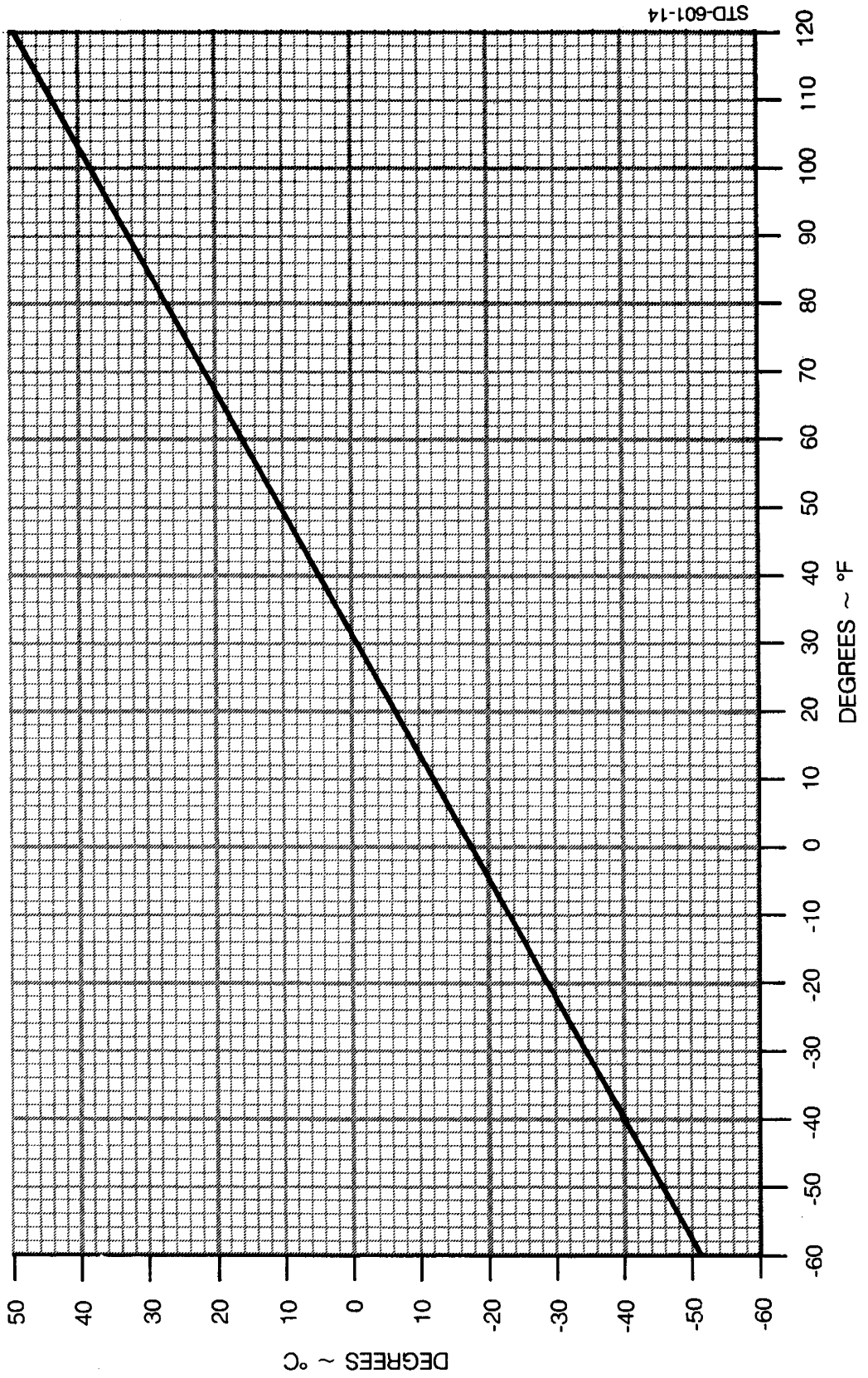
FAHRENHEIT TO CELSIUS TEMPERATURE CONVERSION



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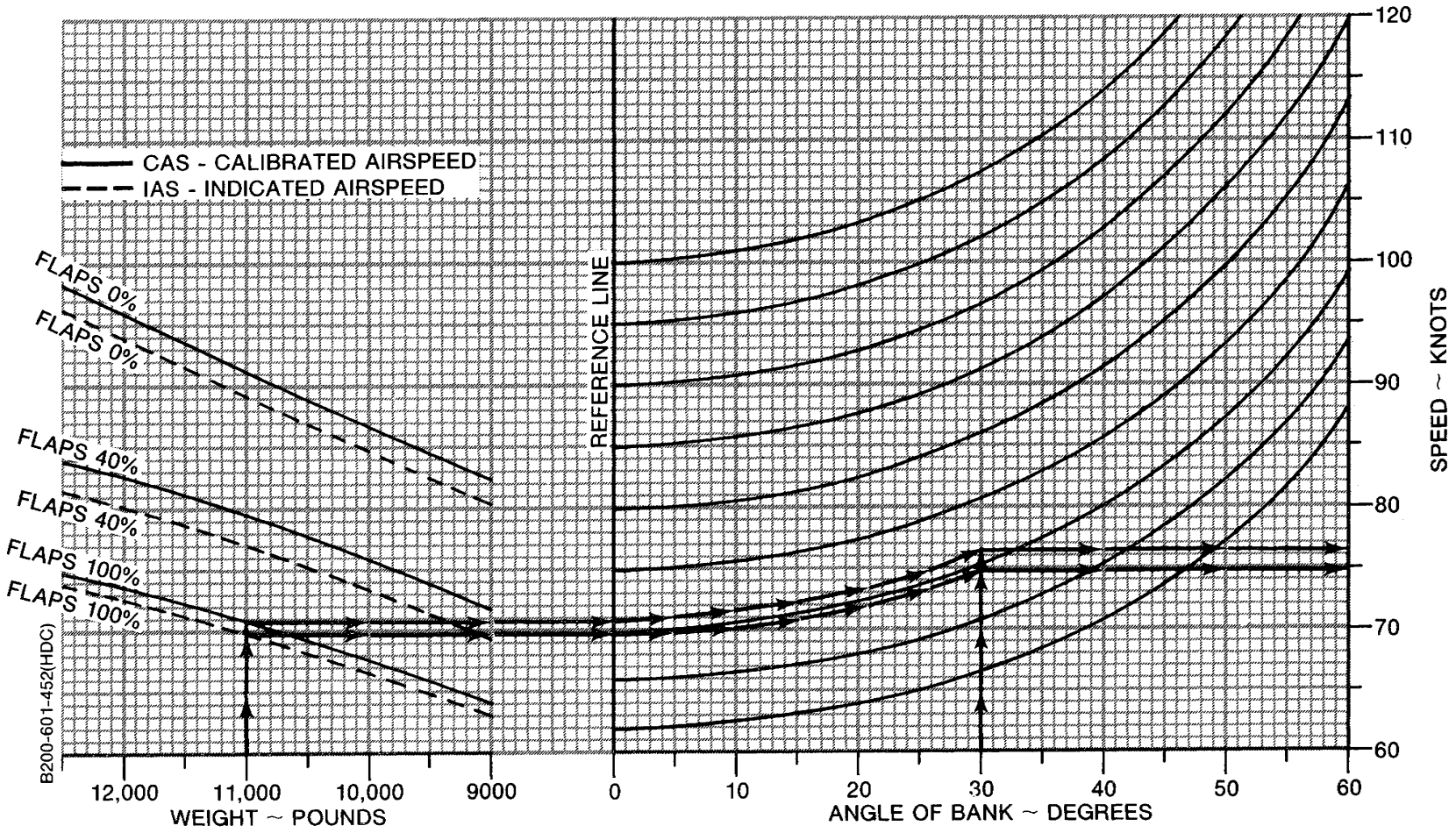
STALL SPEEDS — POWER IDLE

NOTES:
1. ALTITUDE LOSS EXPERIENCED WHILE CONDUCTING STALLS IN ACCORDANCE WITH FAR 23.201 WAS 800 FEET.
2. MAXIMUM NOSE-DOWN PITCH ATTITUDE AND ALTITUDE LOSS DURING RECOVERY FROM ONE-ENGINE-INOPERATIVE STALLS PER FAR 23.205 ARE APPROXIMATELY 13° AND 450 FEET RESPECTIVELY.
3. A NORMAL RECOVERY TECHNIQUE MAY BE USED. THE BEST PROCEDURE IS A BRISK FORWARD WHEEL MOVEMENT TO A NOSE-DOWN ATTITUDE. LEVEL THE AIRPLANE AFTER AIRSPEED HAS INCREASED APPROXIMATELY 25 KNOTS ABOVE STALL.
4. LANDING GEAR POSITION HAS NO EFFECT ON STALL SPEED.

EXAMPLE:

WEIGHT	11,000 LBS
FLAPS	100%
ANGLE OF BANK	30°
STALL SPEED	76 KCAS 75 KIAS



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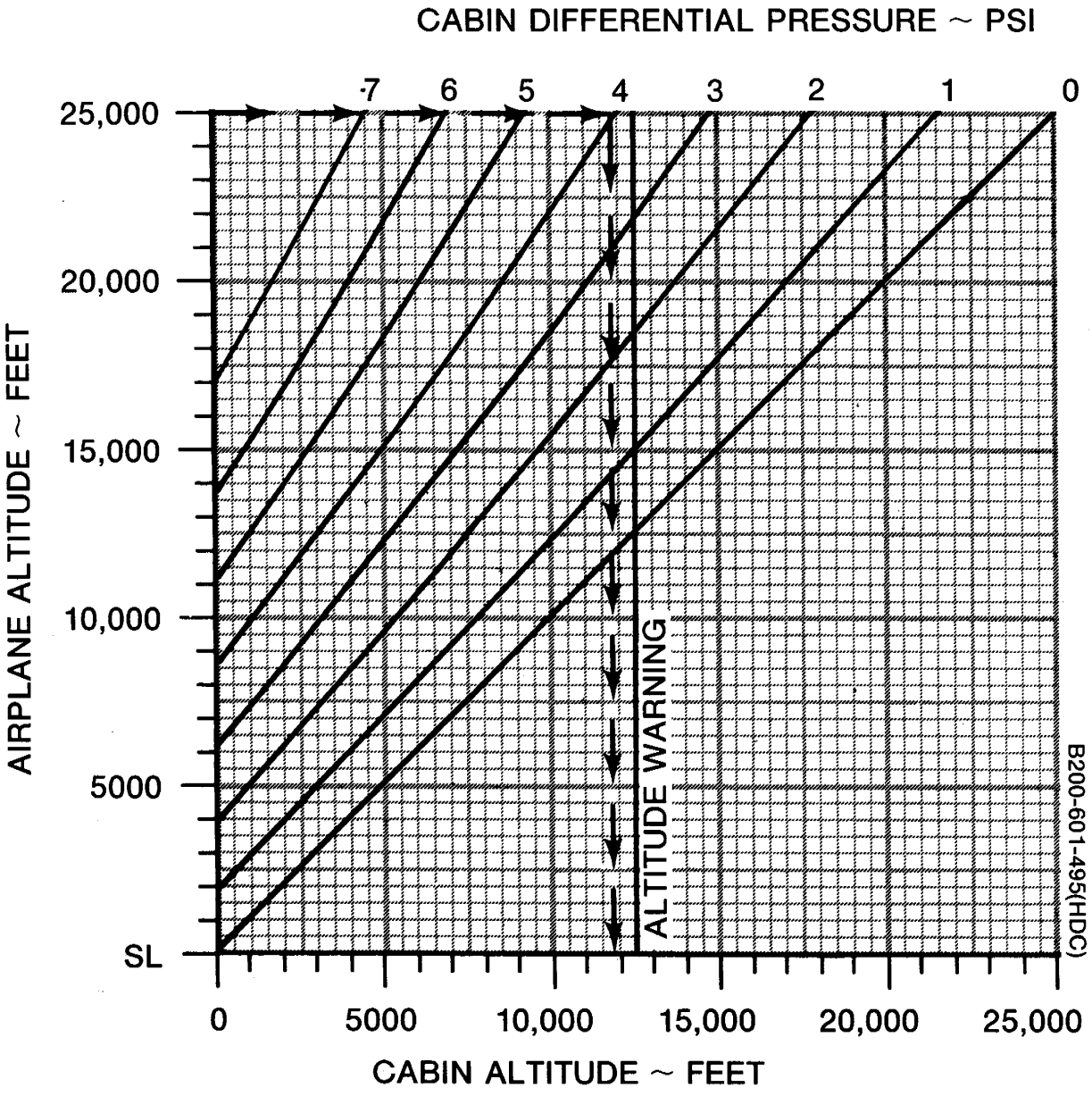
CABIN ALTITUDE FOR VARIOUS AIRPLANE ALTITUDES

EXAMPLE:

AIRPLANE ALTITUDE	25,000 FT
CABIN DIFFERENTIAL PRESSURE	4.0 PSI
CABIN ALTITUDE	11,800 FT



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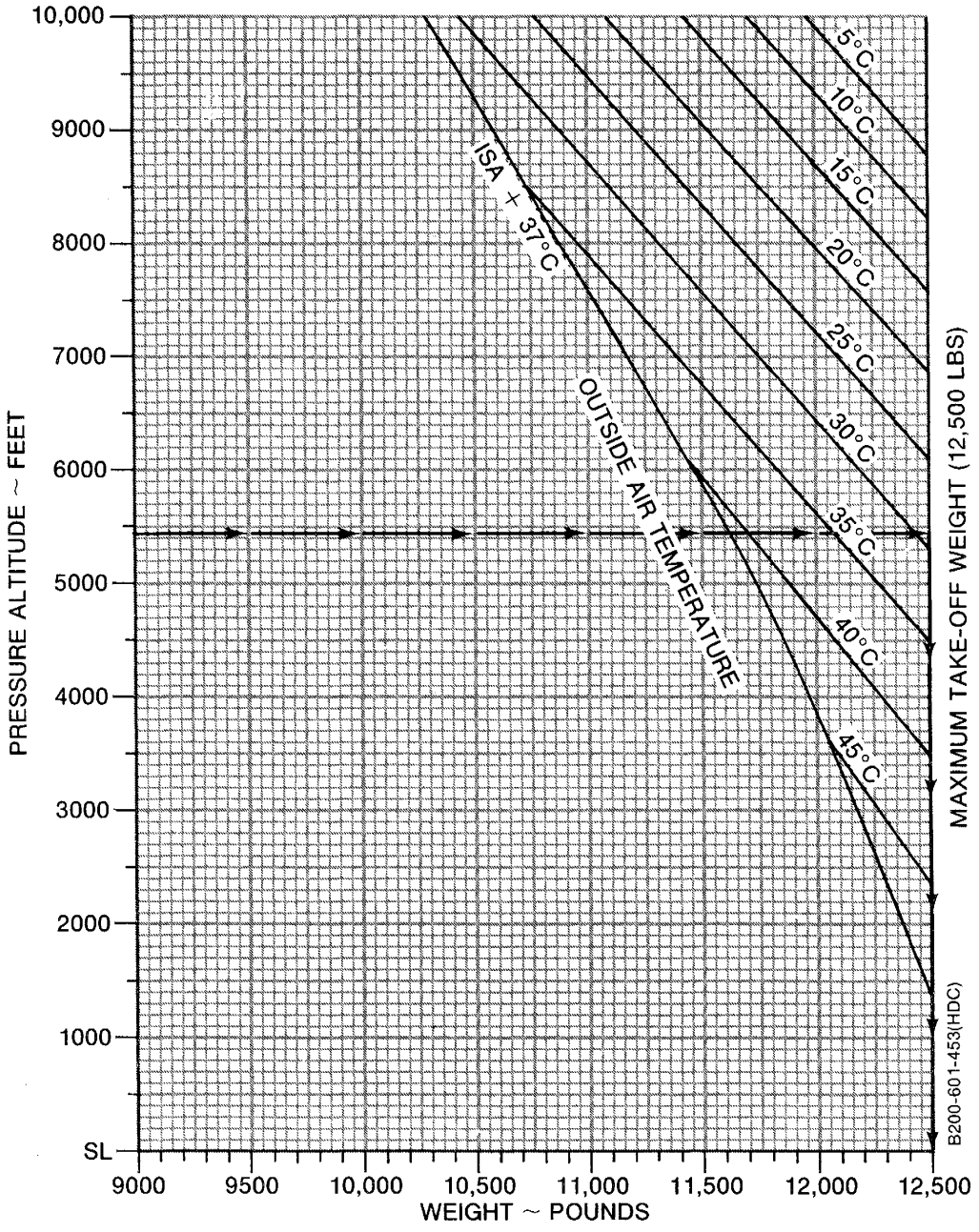
MAXIMUM TAKE-OFF WEIGHT – FLAPS 0%
TO ACHIEVE TAKE-OFF CLIMB REQUIREMENTS

NOTES:
1. WHEN OFF-LOADED TO THE WEIGHTS ON THIS GRAPH, ALL TAKE-OFF CLIMB PERFORMANCE LIMITATIONS REQUIRED FOR FLAPS SET TO THE FLAPS–0% POSITION CAN BE ACHIEVED.
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.

EXAMPLE:

PRESSURE ALTITUDE	5433 FT
OAT	28°C
MAXIMUM TAKE-OFF WEIGHT	12,500 LBS



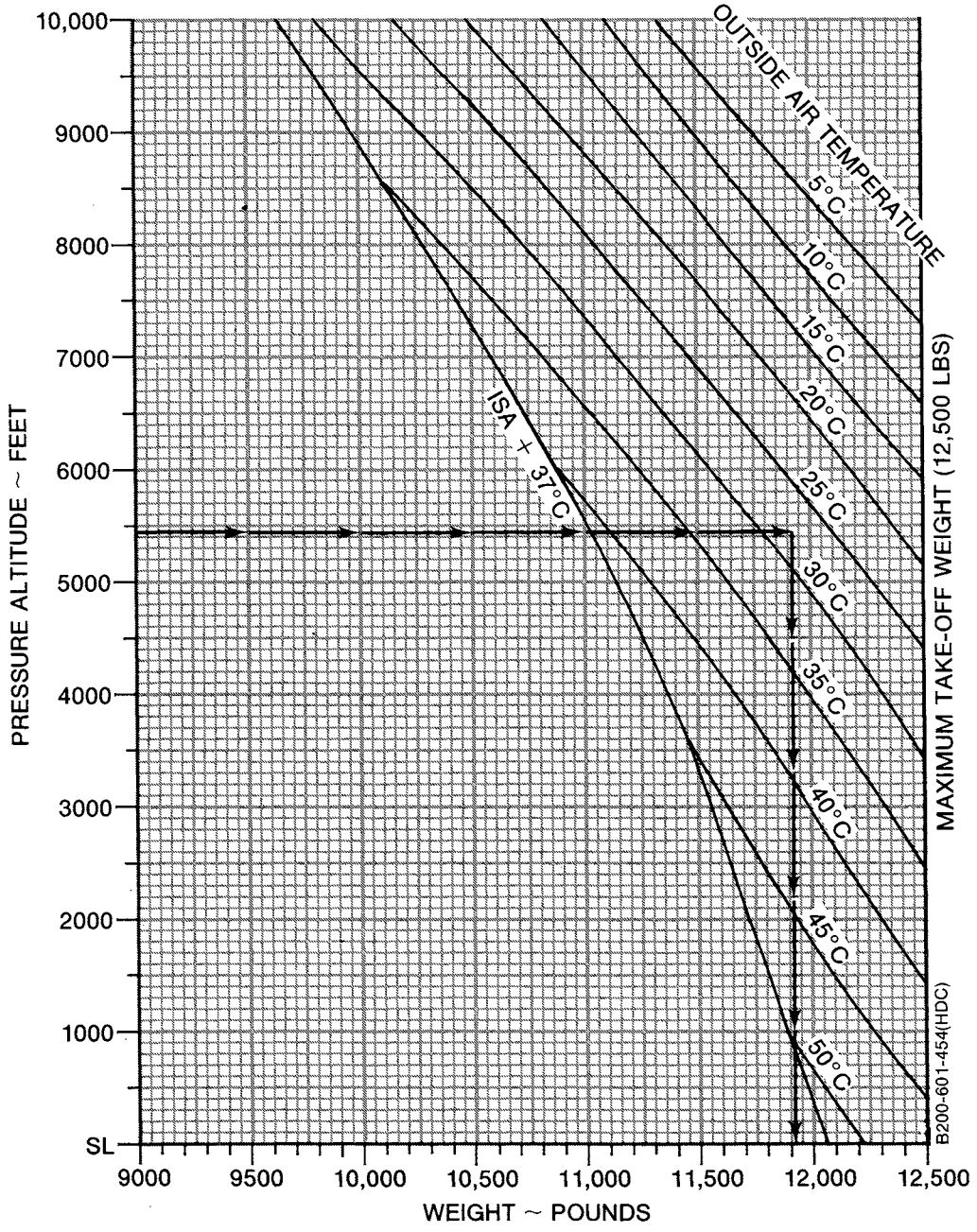
MAXIMUM TAKE-OFF WEIGHT – FLAPS 40%
TO ACHIEVE TAKE-OFF CLIMB REQUIREMENTS

NOTES:

1. WHEN OFF-LOADED TO THE WEIGHTS ON THIS CHART, ALL TAKE-OFF CLIMB PERFORMANCE LIMITATIONS REQUIRED FOR FLAPS SET TO THE FLAPS–40% POSITION CAN BE ACHIEVED.
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.

EXAMPLE:

PRESSURE ALTITUDE ..... 5433 FT
OAT ..... 28°C
MAXIMUM TAKE-OFF WEIGHT.... 11,920 LBS



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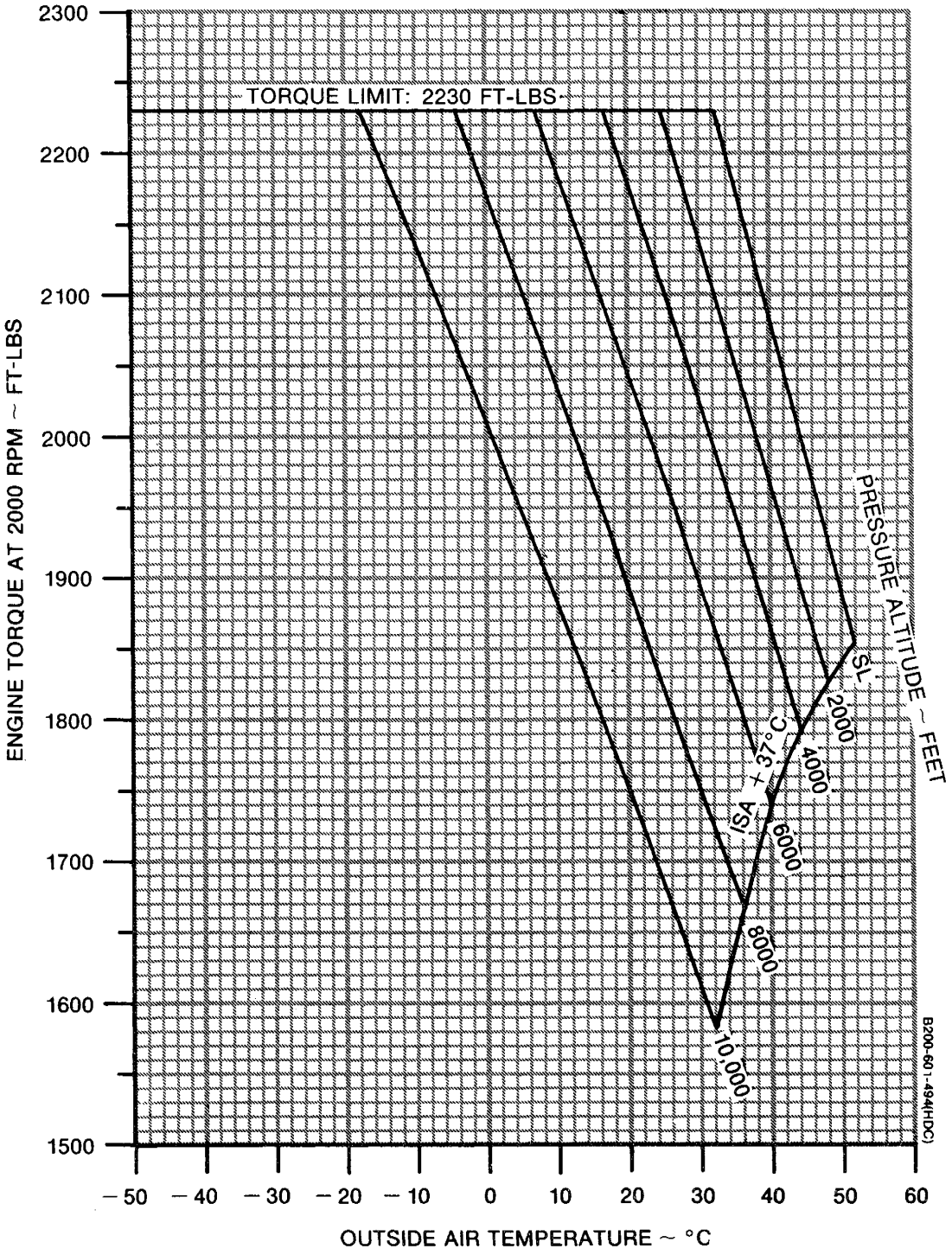
MINIMUM TAKE-OFF POWER AT 2000 RPM

WITH ICE VANES RETRACTED
(65 KNOTS)

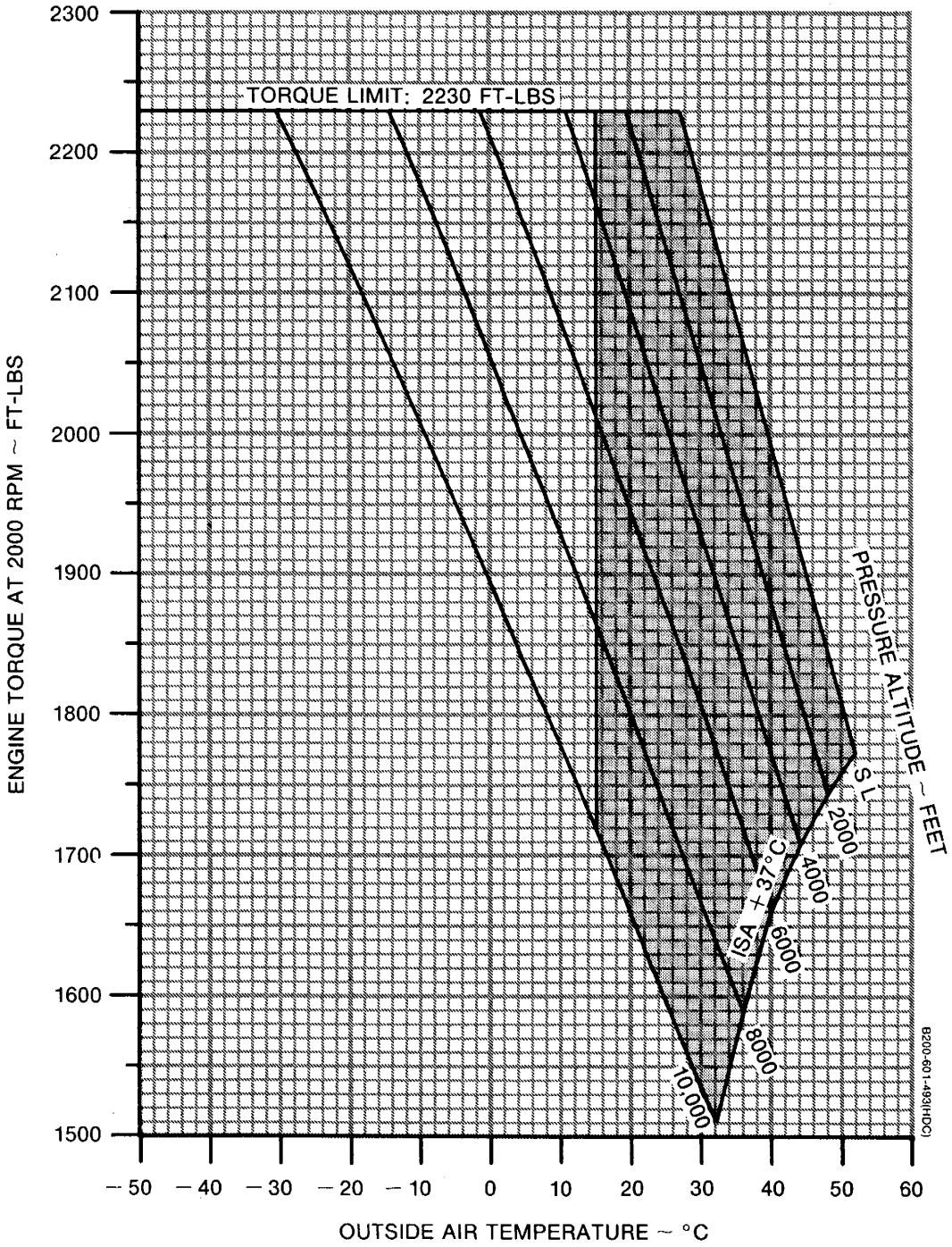
NOTES: 1. TORQUE INCREASES APPROXIMATELY 20 FT-LBS FROM ZERO TO 65 KNOTS.
2. THE POWER (TORQUE) INDICATED IS THE MINIMUM VALUE FOR WHICH TAKE-OFF PERFORMANCE IN THIS SECTION CAN BE OBTAINED.



MINIMUM TAKE-OFF POWER AT 2000 RPM
WITH ICE VANES EXTENDED
(65 KNOTS)

NOTES:

1. TORQUE INCREASES APPROXIMATELY 20 FT-LBS FROM ZERO TO 65 KNOTS.
2. THE POWER (TORQUE) INDICATED IS THE MINIMUM VALUE FOR WHICH TAKE-OFF PERFORMANCE PRESENTED IN THIS SECTION FOR ICE VANES EXTENDED CAN BE OBTAINED.
3. TAKE-OFF WITH VANES EXTENDED IS PROHIBITED IN SHADED AREA.



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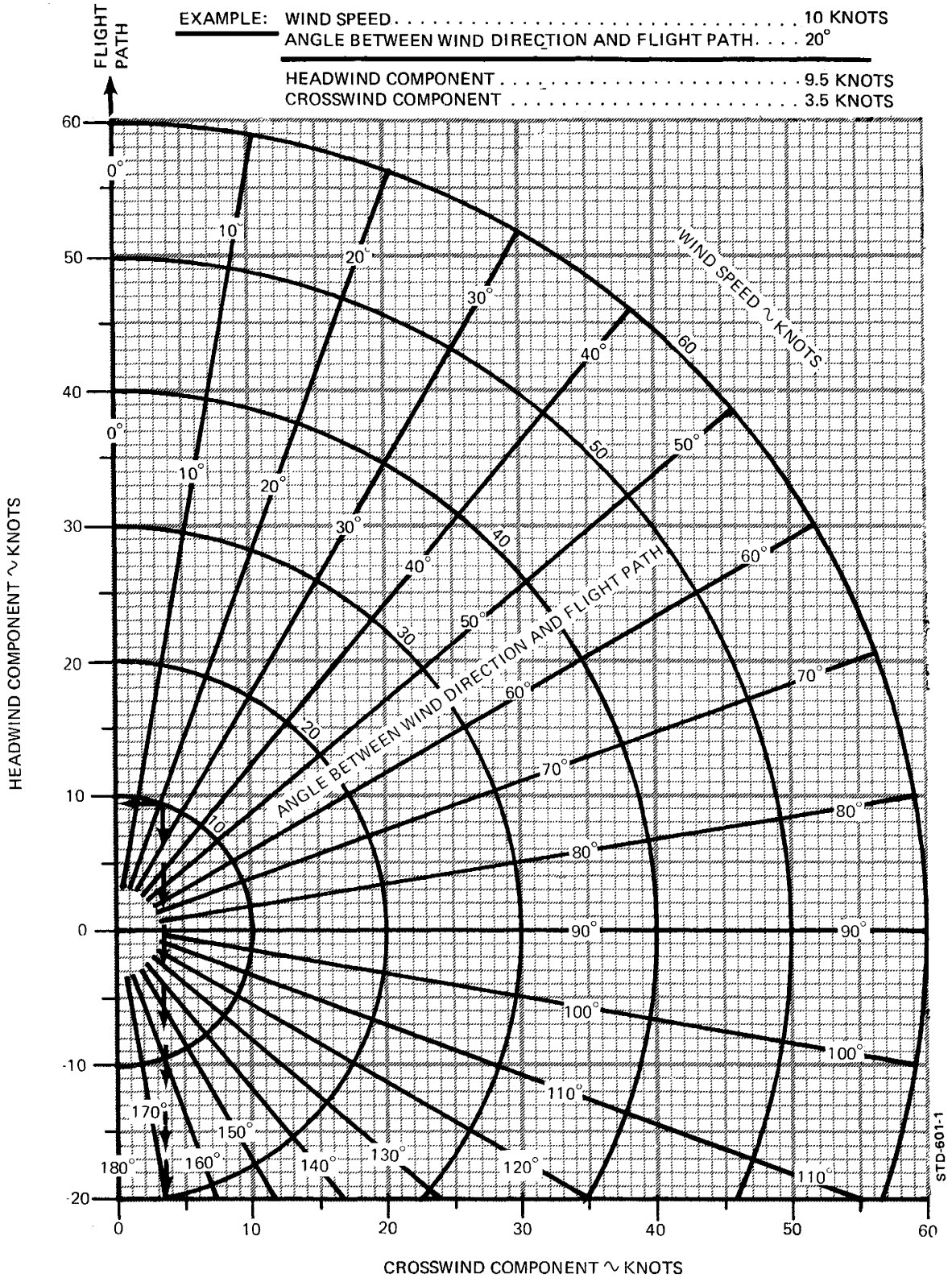
WIND COMPONENTS

EXAMPLE: WIND SPEED . . . . . 10 KNOTS
ANGLE BETWEEN WIND DIRECTION AND FLIGHT PATH . . . . 20°

HEADWIND COMPONENT . . . . . 9.5 KNOTS
CROSSWIND COMPONENT . . . . . 3.5 KNOTS



TAKE-OFF DISTANCE – FLAPS 0%
OVER 50-FT OBSTACLE

ASSOCIATED CONDITIONS:

POWER ..... TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS 0%
LANDING GEAR ..... RETRACT AFTER LIFT-OFF
RUNWAY ..... PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	TAKE-OFF SPEED ~ KNOTS	
	ROTATION	50 FT
12,500	108	117
12,000	105	114
11,000	100	108
10,000	99	103
9000	99	102

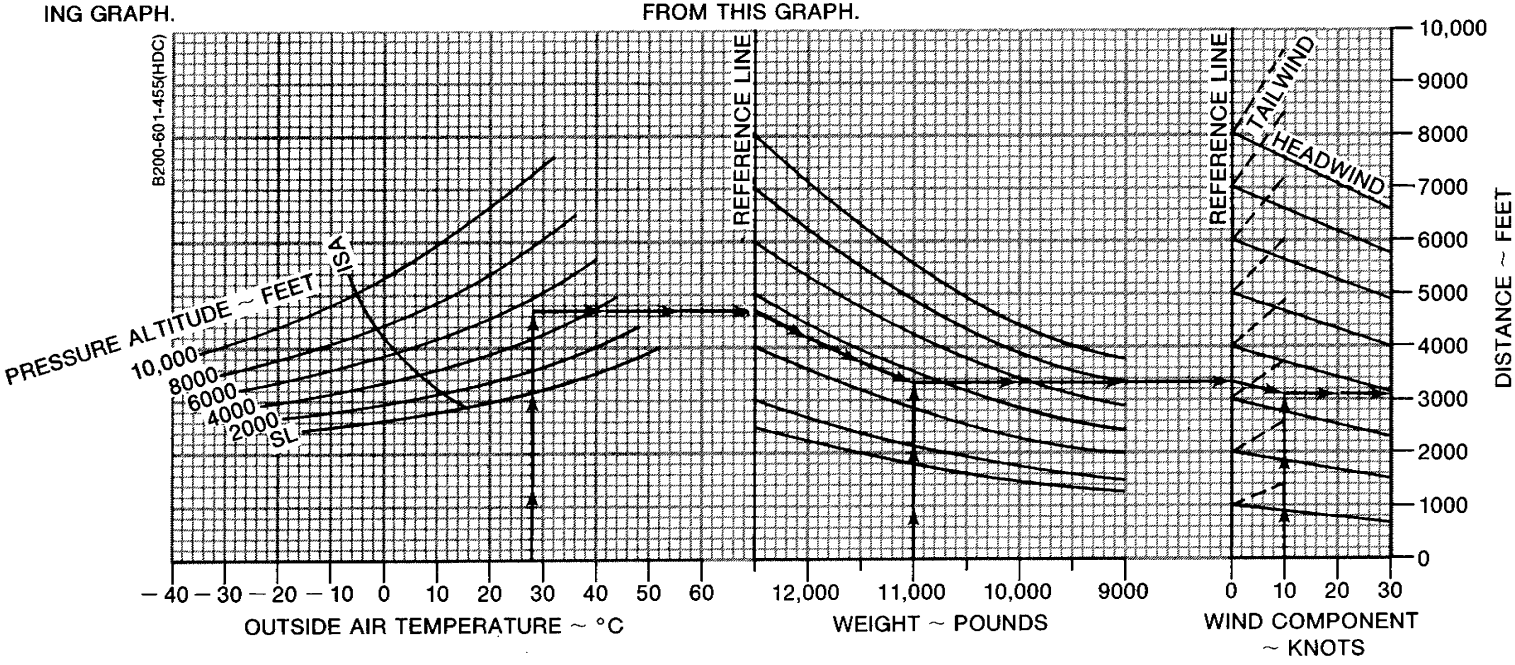
EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5433 FT
TAKE-OFF WEIGHT ..... 11,000 LBS
HEADWIND COMPONENT ..... 10 KTS

TAKE-OFF DISTANCE OVER 50-FT OBSTACLE ..... 3100 FT
TAKE-OFF SPEED AT ROTATION ..... 100 KTS
AT 50 FT ..... 108 KTS

NOTES: 1. FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.

2. WHEN USING THE ROLLING TAKE-OFF PROCEDURE, ADD 500 FEET TO THE VALUE DERIVED FROM THIS GRAPH.



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ACCELERATE-STOP — FLAPS 0%

ASSOCIATED CONDITIONS:

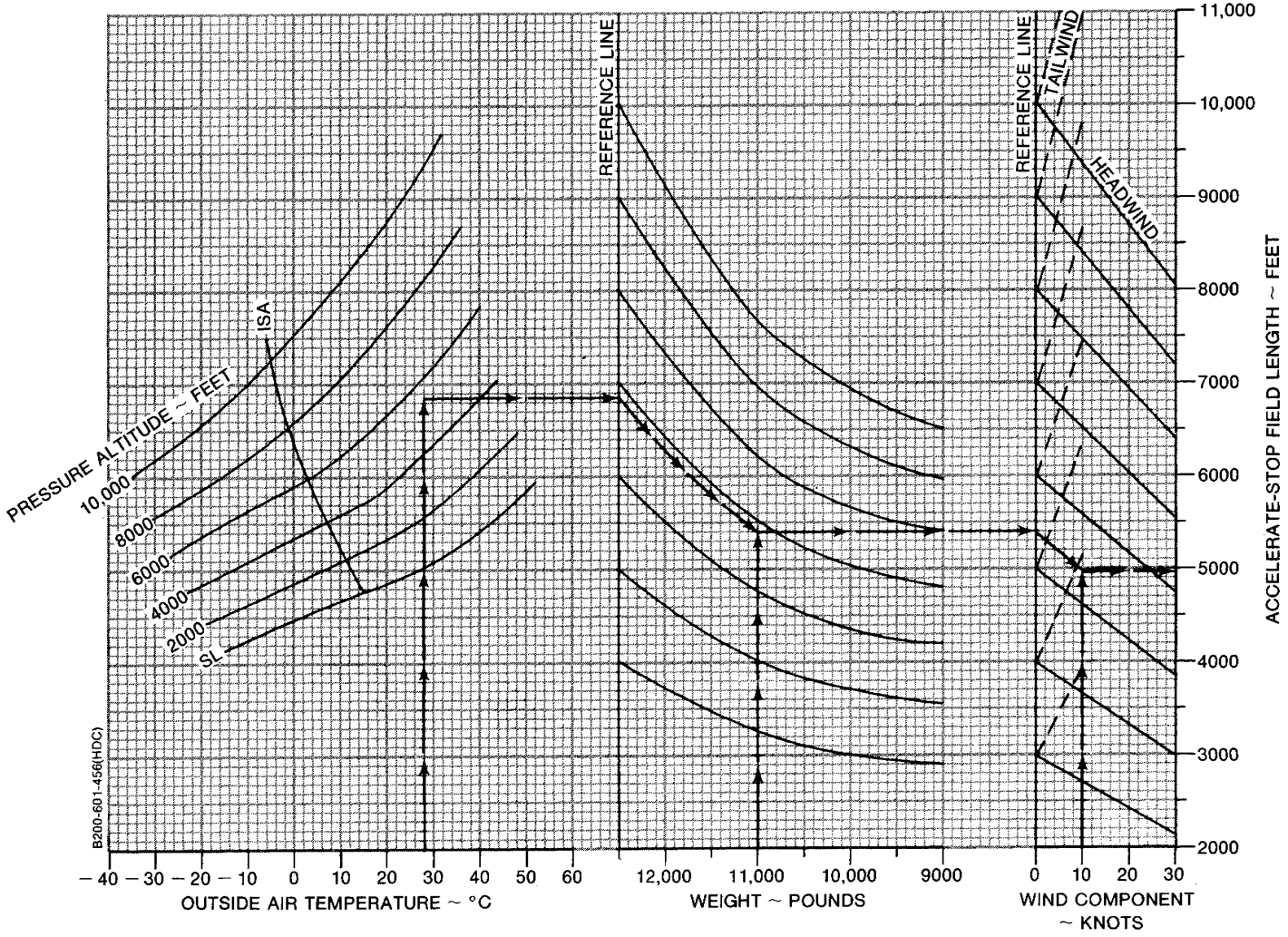
POWER ..... 1. TAKE-OFF POWER SET BEFORE BRAKE RELEASE.
2. BOTH ENGINES IDLE AT V_1 SPEED
FLAPS 0%
AUTOFEATHER ..... ARMED
BRAKING ..... MAXIMUM
RUNWAY ..... PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	V_1 ~ KNOTS
12,500	108
12,000	105
11,000	100
10,000	99
9000	99

EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5433 FT
WEIGHT ..... 11,000 LBS
HEADWIND COMPONENT ..... 10 KTS
FIELD LENGTH ..... 4990 FT
V_1 ..... 100 KTS

NOTES: 1. FOR OPERATION WITH ICE VANES EXTENDED, PERFORMANCE IS NOT AFFECTED.
2. WHEN USING THE ROLLING TAKE-OFF PROCEDURE, ADD 500 FEET TO THE VALUE DERIVED FROM THIS GRAPH.



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ACCELERATE-GO DISTANCE — FLAPS 0%
OVER 35-FT OBSTACLE

ASSOCIATED CONDITIONS:

POWER ..... TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS 0%
AUTOFEATHER ..... ARMED
LANDING GEAR ..... RETRACT AFTER LIFT-OFF
RUNWAY ..... PAVED, LEVEL, DRY SURFACE

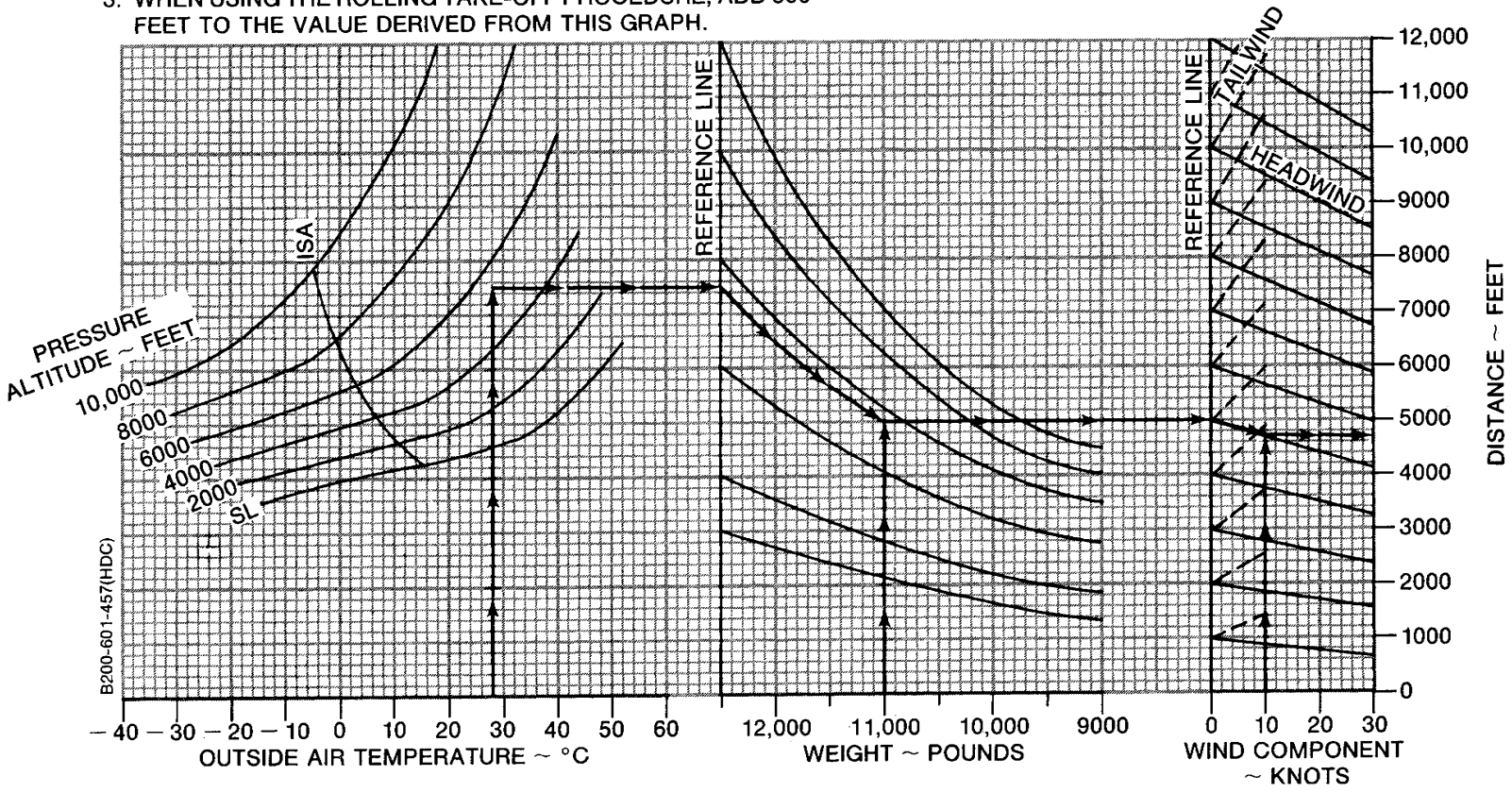
WEIGHT ~ POUNDS	SPEED ~ KNOTS	
	V_R	V_2
12,500	108	117
12,000	105	114
11,000	100	108
10,000	99	103
9000	99	102

EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5443 FT
WEIGHT ..... 11,000 LBS
HEADWIND COMPONENT ..... 10 KTS

TOTAL DISTANCE OVER 35-FT OBSTACLE ..... 4700 FT
V_R ..... 100 KTS
V_2 ..... 108 KTS

NOTES:
1. V_1 (ENGINE FAILURE SPEED) EQUALS V_R (ROTATION SPEED).
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 12°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.
3. WHEN USING THE ROLLING TAKE-OFF PROCEDURE, ADD 500 FEET TO THE VALUE DERIVED FROM THIS GRAPH.



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NET GRADIENT OF CLIMB — ONE ENGINE INOPERATIVE — FLAPS 0%

ASSOCIATED CONDITIONS:

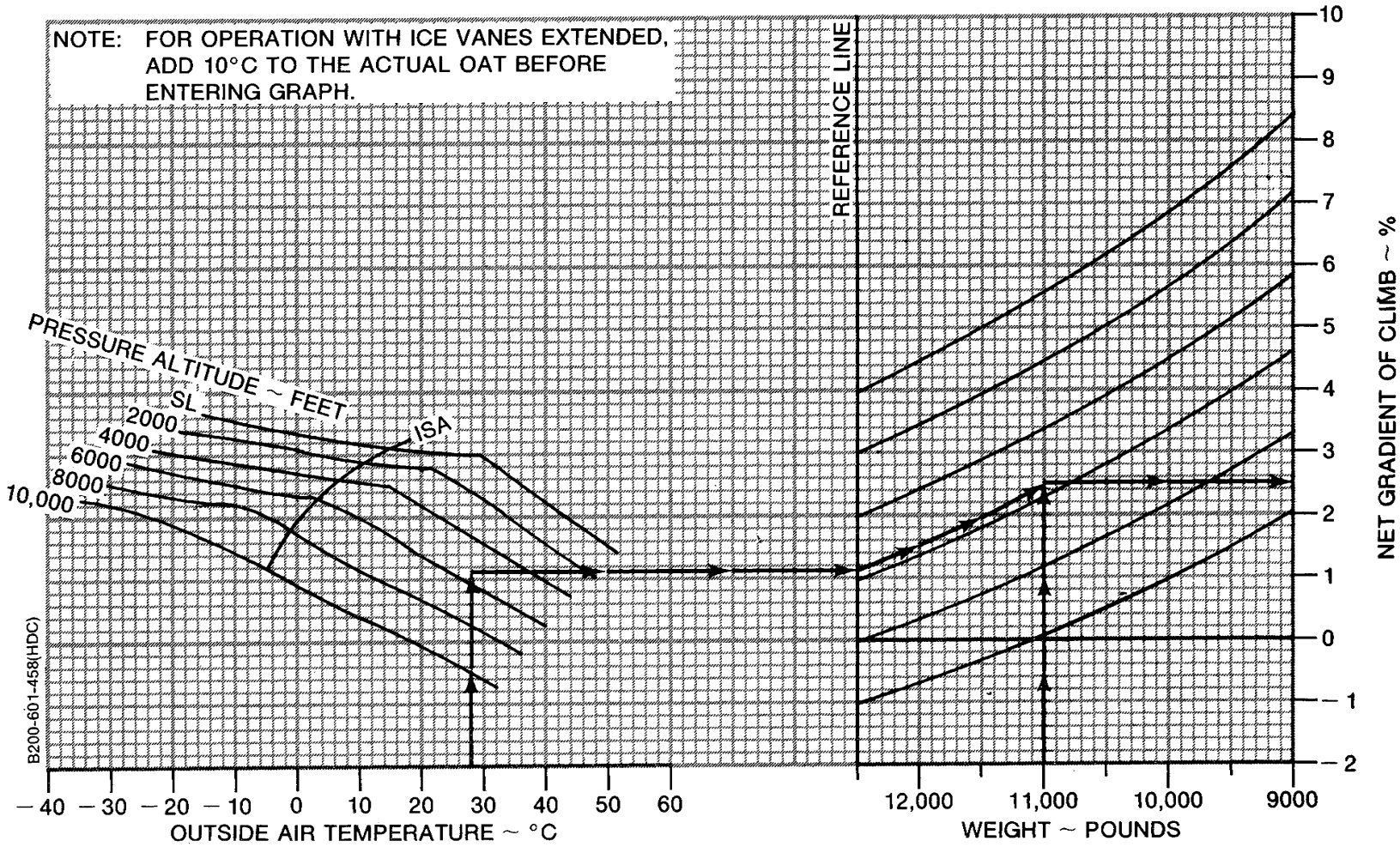
POWER ..... TAKE-OFF
FLAPS 0%
LANDING GEAR ..... UP
INOPERATIVE
PROPELLER ..... FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	117
12,000	114
11,000	108
10,000	103
9000	102

EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5433 FT
WEIGHT ..... 11,000 LBS

NET GRADIENT OF CLIMB 2.5%
CLIMB SPEED ..... 108 KNOTS



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TAKE-OFF DISTANCE — FLAPS 40%
OVER 50-FT OBSTACLE

ASSOCIATED CONDITIONS:

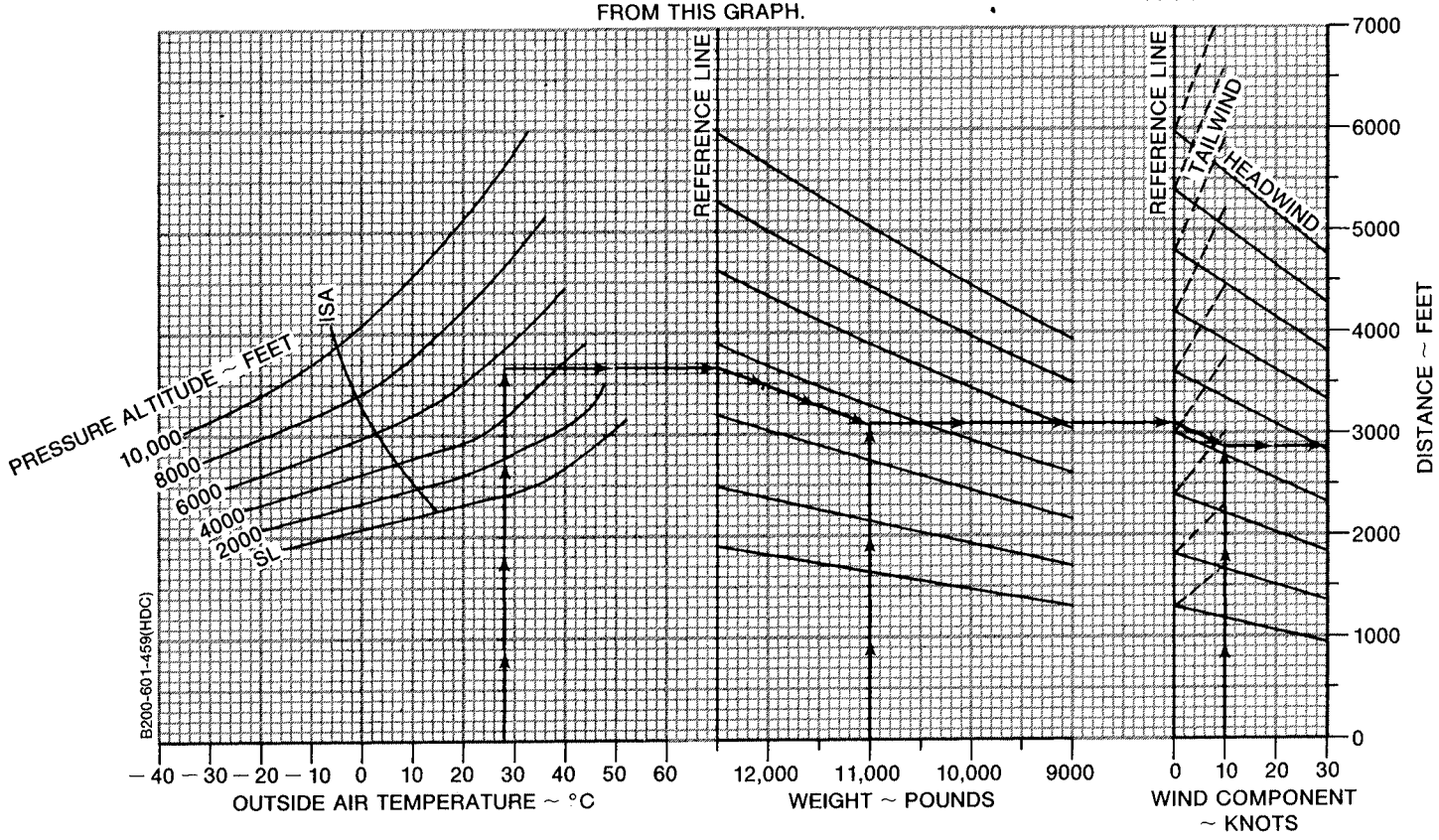
POWER ..... TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS 40%
LANDING GEAR ..... RETRACT AFTER LIFT-OFF
RUNWAY ..... PAVED, LEVEL, DRY SURFACE

TAKE-OFF SPEED AT ROTATION: 98 KNOTS (ALL WEIGHTS)
TAKE-OFF SPEED AT 50 FT: 100 KNOTS (ALL WEIGHTS)

NOTES:
1. FOR OPERATION WITH ICE VANES EXTENDED, ADD 5°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.
2. WHEN USING THE ROLLING TAKE-OFF PROCEDURE, ADD 500 FEET TO THE VALUE DERIVED FROM THIS GRAPH.

EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5433 FT
TAKE-OFF WEIGHT ..... 11,000 LBS
HEADWIND COMPONENT ..... 10 KTS
TOTAL DISTANCE OVER 50-FT OBSTACLE ..... 2870 FT



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ACCELERATE-STOP — FLAPS 40%

V_1 SPEED 98 KNOTS (ALL WEIGHTS)

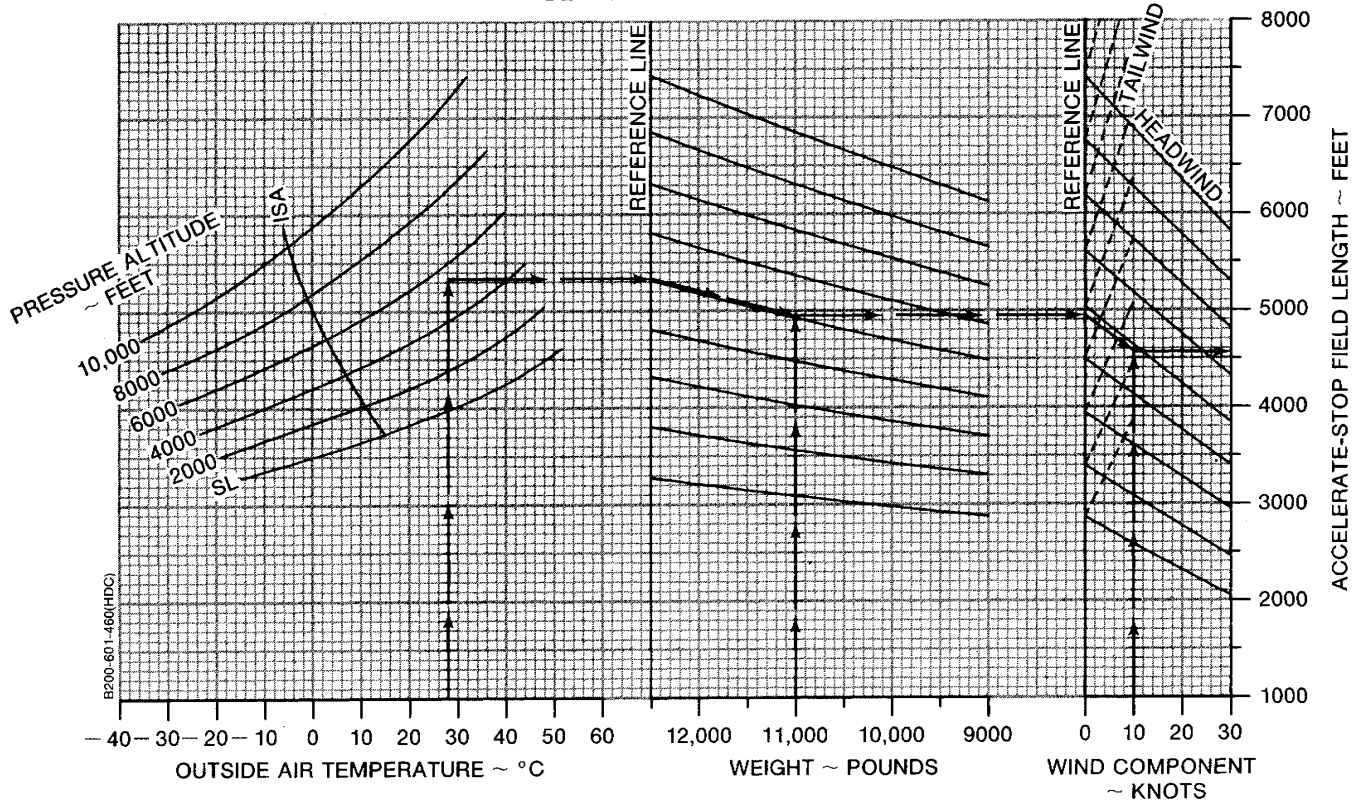
ASSOCIATED CONDITIONS:

POWER	1. TAKE-OFF POWER SET BEFORE BRAKE RELEASE 2. BOTH ENGINES IDLE AT V_1 SPEED
FLAPS	40%
AUTOFEATHER	ARMED
BRAKING	MAXIMUM
RUNWAY	PAVED, LEVEL, DRY SURFACE

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5433 FT
WEIGHT	11,000 LBS
HEADWIND COMPONENT	10 KTS
FIELD LENGTH	4570 FT

NOTES:
1. FOR OPERATION WITH ICE VANES EXTENDED, ADD 5°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.
2. WHEN USING THE ROLLING TAKE-OFF PROCEDURE, ADD 500 FEET TO THE VALUE DERIVED FROM THIS GRAPH.



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ACCELERATE-GO DISTANCE — FLAPS 40%
OVER 35-FT OBSTACLE

V_R: 98 KNOTS (ALL WEIGHTS)
V_2: 100 KNOTS (ALL WEIGHTS)

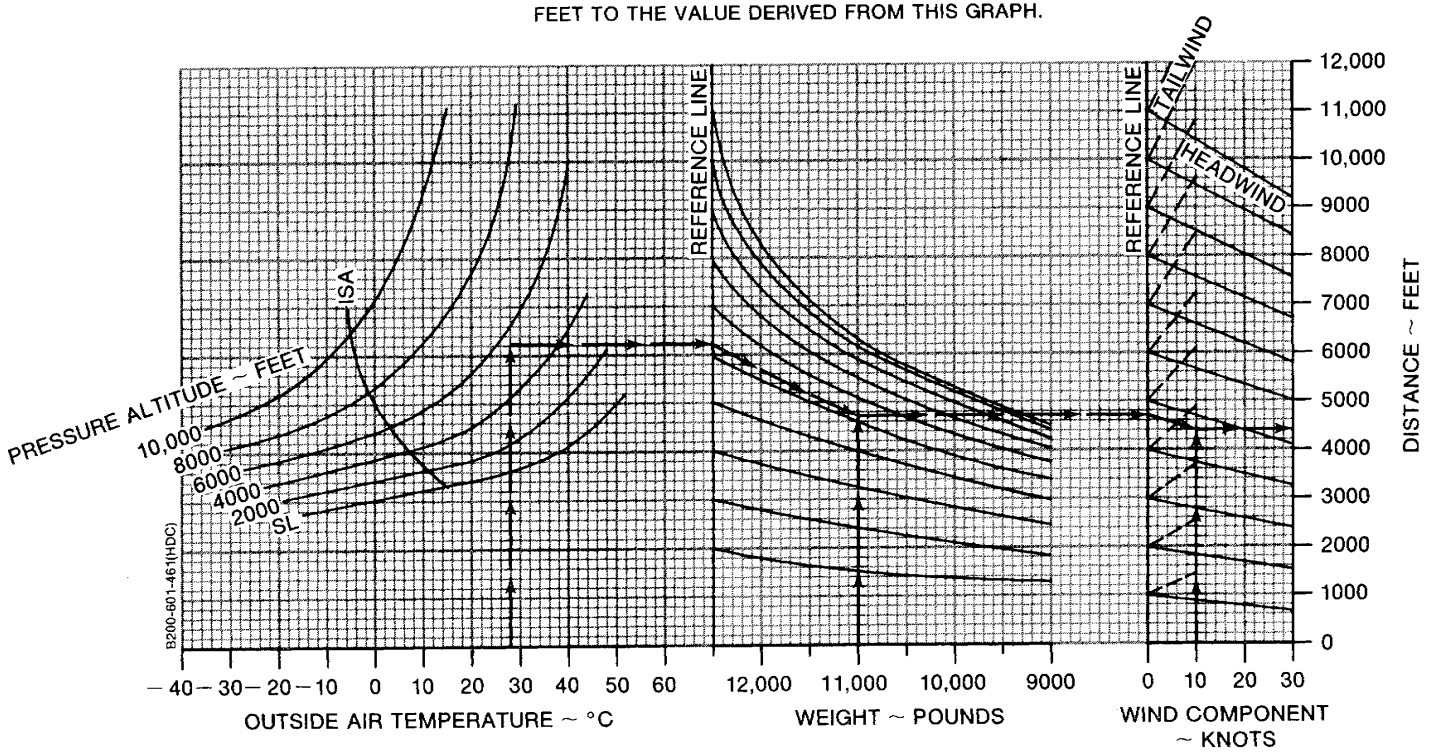
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	40%
AUTOFEATHER	ARMED
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5433 FT
WEIGHT	11,000 LBS
HEADWIND COMPONENT	10 KTS
TOTAL DISTANCE OVER 35-FT OBSTACLE	4400 FT

NOTES:
1. V_1 (ENGINE FAILURE SPEED) EQUALS V_R (ROTATION SPEED).
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.
3. WHEN USING THE ROLLING TAKE-OFF PROCEDURE, ADD 500 FEET TO THE VALUE DERIVED FROM THIS GRAPH.



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NET GRADIENT OF CLIMB — ONE ENGINE INOPERATIVE — FLAPS 40%

CLIMB SPEED: 100 KNOTS (ALL WEIGHTS)

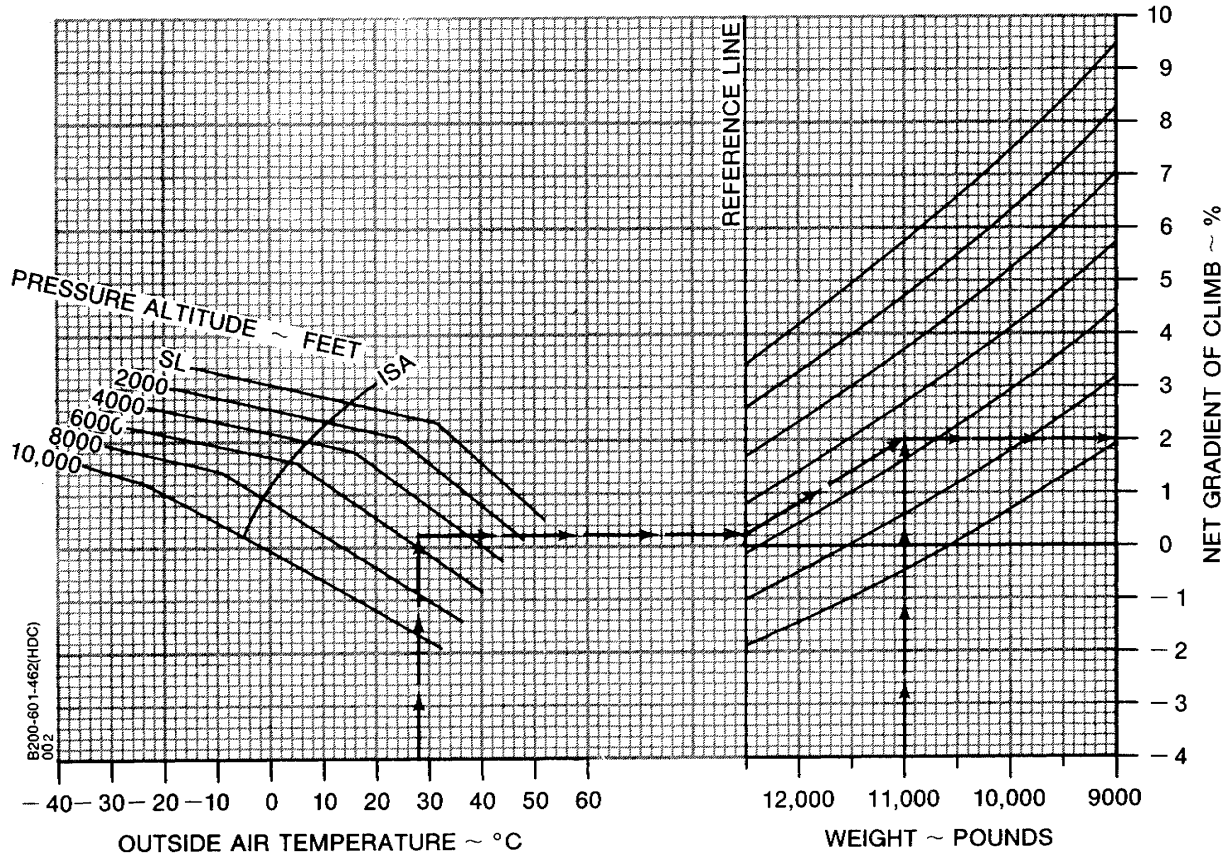
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	40%
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5433 FT
WEIGHT	11,000 LBS
NET GRADIENT	2.00%

NOTE: FOR OPERATION WITH ICE VANE EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.



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CLIMB — TWO ENGINES — MAXIMUM CONTINUOUS POWER — FLAPS 0%

ASSOCIATED CONDITIONS:

POWER ..... MAX CONTINUOUS
FLAPS 0%
LANDING GEAR ..... UP

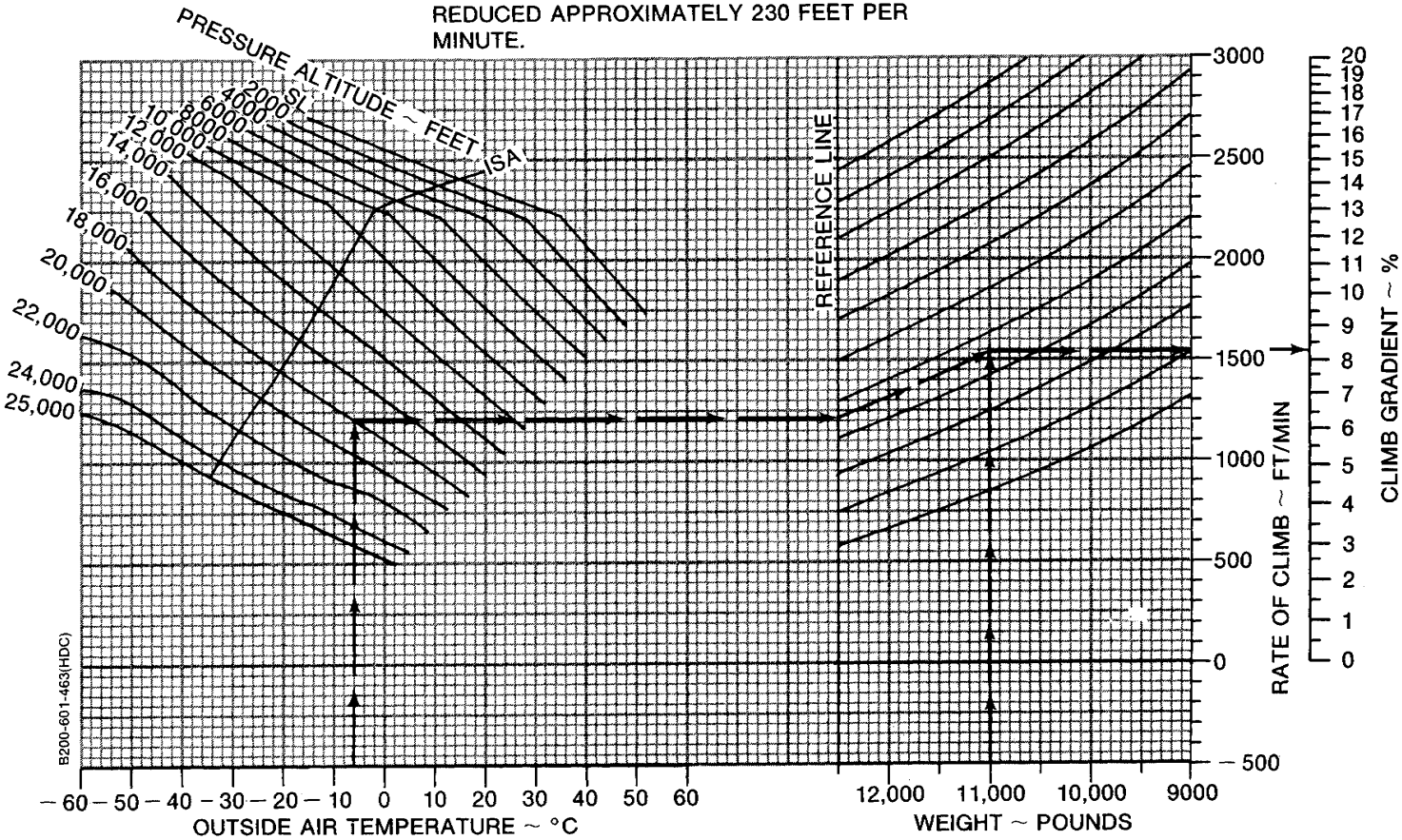
WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	125
12,000	124
11,000	121
10,000	119
9000	116

EXAMPLE:

OAT ..... − 6°C
PRESSURE ALTITUDE ..... 18,000 FT
WEIGHT ..... 11,000 LBS

RATE OF CLIMB ..... 1540 FT/MIN
CLIMB GRADIENT 8.3%
CLIMB SPEED ..... 121 KNOTS

NOTES:
1. FREQUENT OPERATION AT MAX CONTINUOUS POWER WILL SHORTEN ENGINE HOT SECTION LIFE AND INCREASE MAINTENANCE COSTS.
2. DURING OPERATION WITH ICE VANES EXTENDED, RATE OF CLIMB WILL BE REDUCED APPROXIMATELY 230 FEET PER MINUTE.



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CLIMB — TWO ENGINES — MAXIMUM CONTINUOUS POWER — FLAPS 40%

ASSOCIATED CONDITIONS:

POWER ..... MAX CONTINUOUS
FLAPS 40%
LANDING GEAR ..... UP

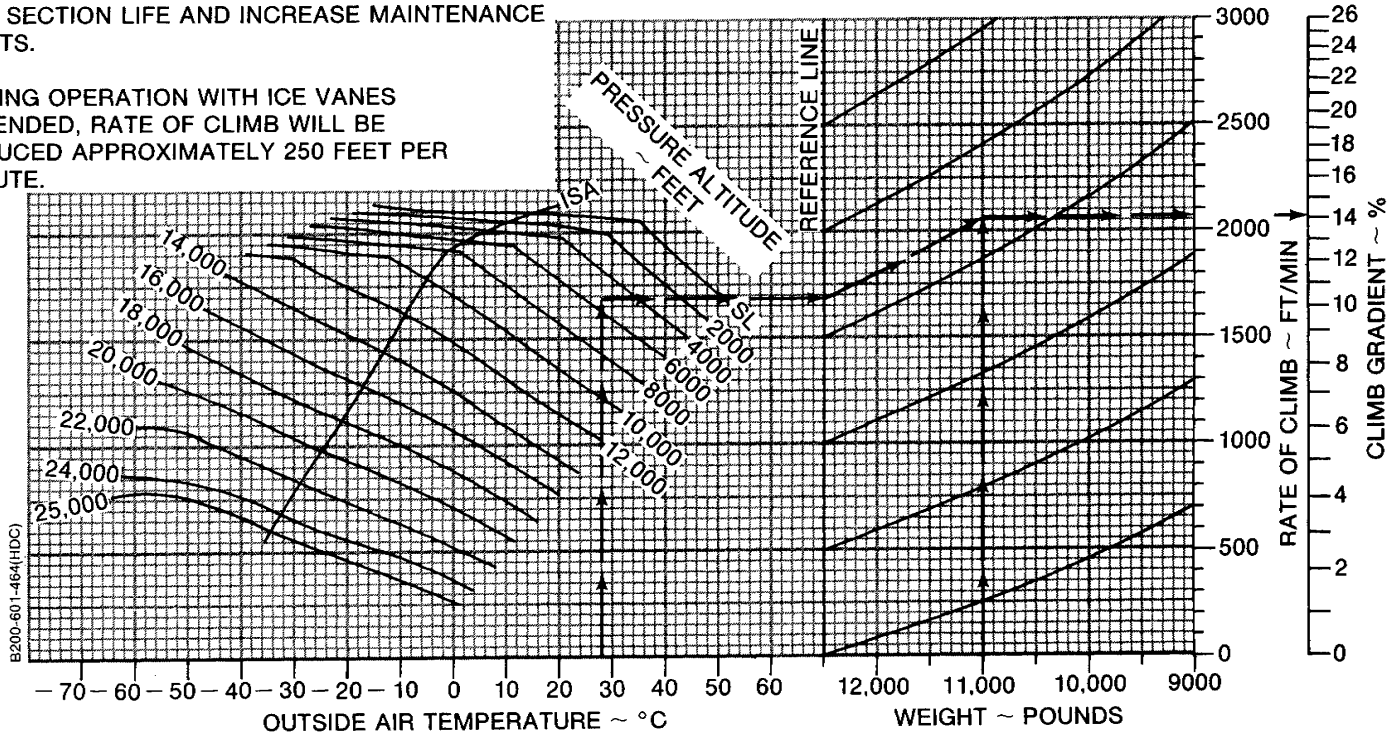
WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	125
12,000	124
11,000	121
10,000	119
9000	116

EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5433 FT
WEIGHT ..... 11,000 LBS

RATE-OF-CLIMB ..... 2060 FT/MIN
CLIMB GRADIENT 14.1%
CLIMB SPEED ..... 121 KNOTS

NOTES:
1. FREQUENT OPERATION AT MAX CONTINUOUS POWER WILL SHORTEN ENGINE HOT SECTION LIFE AND INCREASE MAINTENANCE COSTS.
2. DURING OPERATION WITH ICE VANES EXTENDED, RATE OF CLIMB WILL BE REDUCED APPROXIMATELY 250 FEET PER MINUTE.



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CLIMB — ONE ENGINE INOPERATIVE

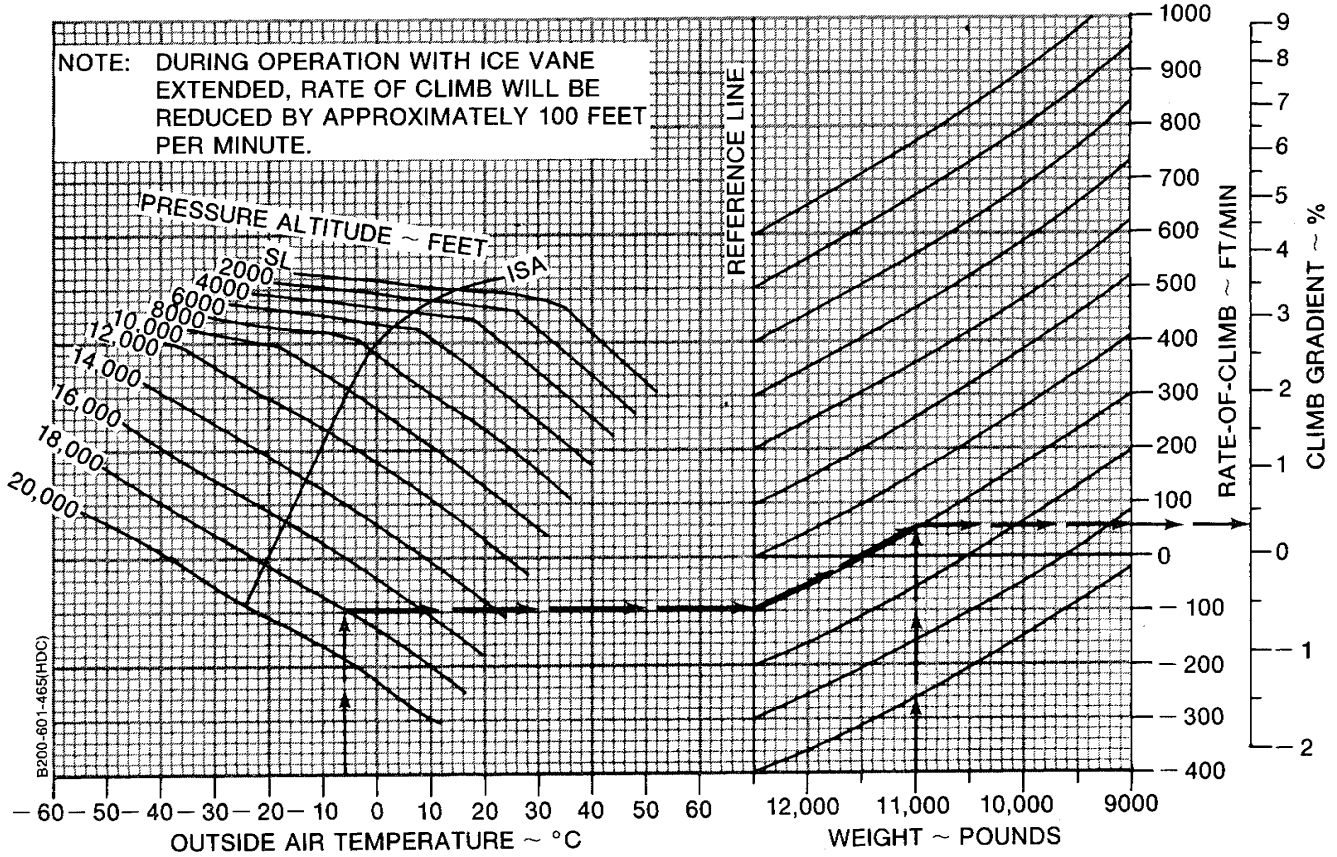
ASSOCIATED CONDITIONS:

POWER	MAX CONTINUOUS
FLAPS	0%
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	121
12,000	120
11,000	117
10,000	115
9000	112

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	11,000 LBS
RATE OF CLIMB	59 FT/MIN
CLIMB GRADIENT	0.37%
CLIMB SPEED	117 KNOTS



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SERVICE CEILING — ONE ENGINE INOPERATIVE

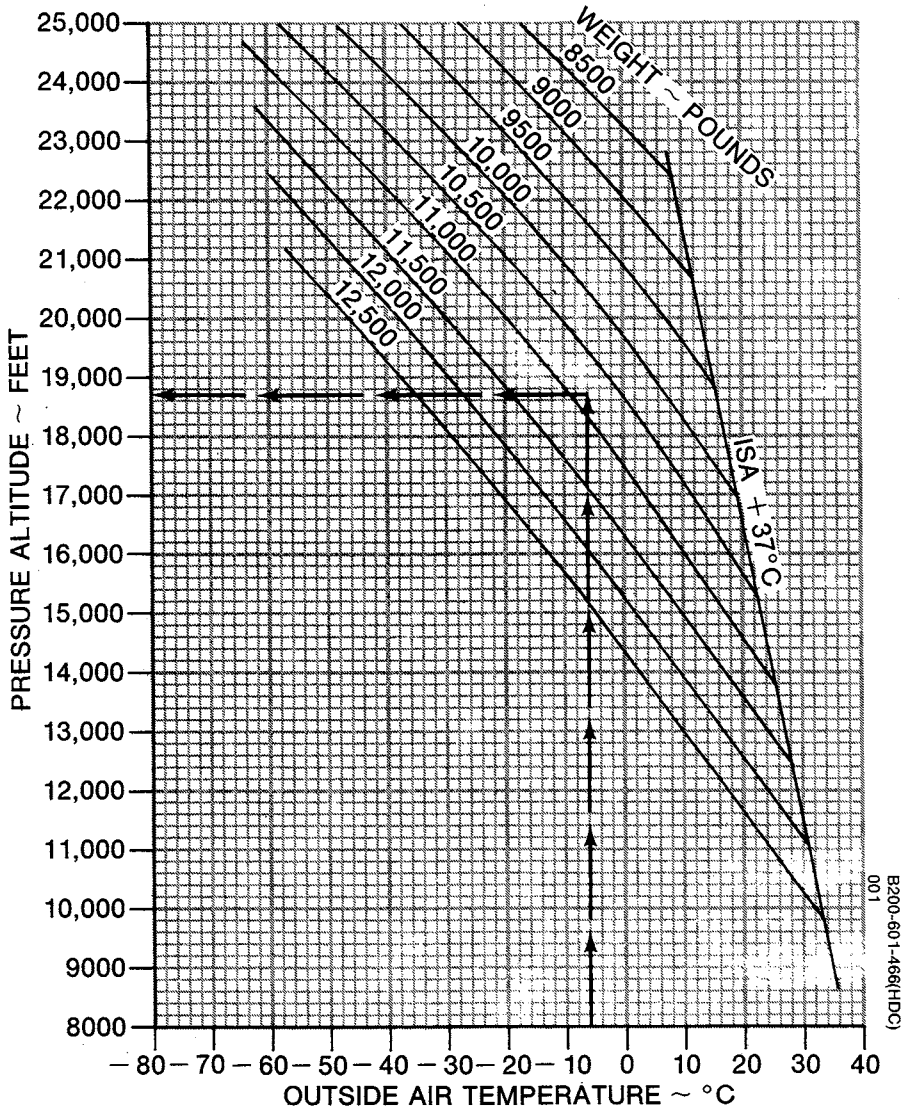
ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED
FLAPS	0%
AIRSPEED	PER CLIMB — ONE ENGINE INOPERATIVE GRAPH.

EXAMPLE:

OAT AT MEA	−6°C
WEIGHT	10,844 LBS
ROUTE SEGMENT MEA	18,000 FT
SERVICE CEILING	18,700 FT

NOTES:
1. SERVICE CEILING IS THE MAXIMUM PRESSURE ALTITUDE AT WHICH THE AIRPLANE IS CAPABLE OF CLIMBING 50 FT/MINUTE WITH ONE PROPELLER FEATHERED.
2. DURING OPERATION WITH ICE VANE EXTENDED, SERVICE CEILING WILL BE LOWERED APPROXIMATELY 1500 FEET.



OPERATIONAL CLIMB – TIME, FUEL AND DISTANCE

ASSOCIATED CONDITIONS:

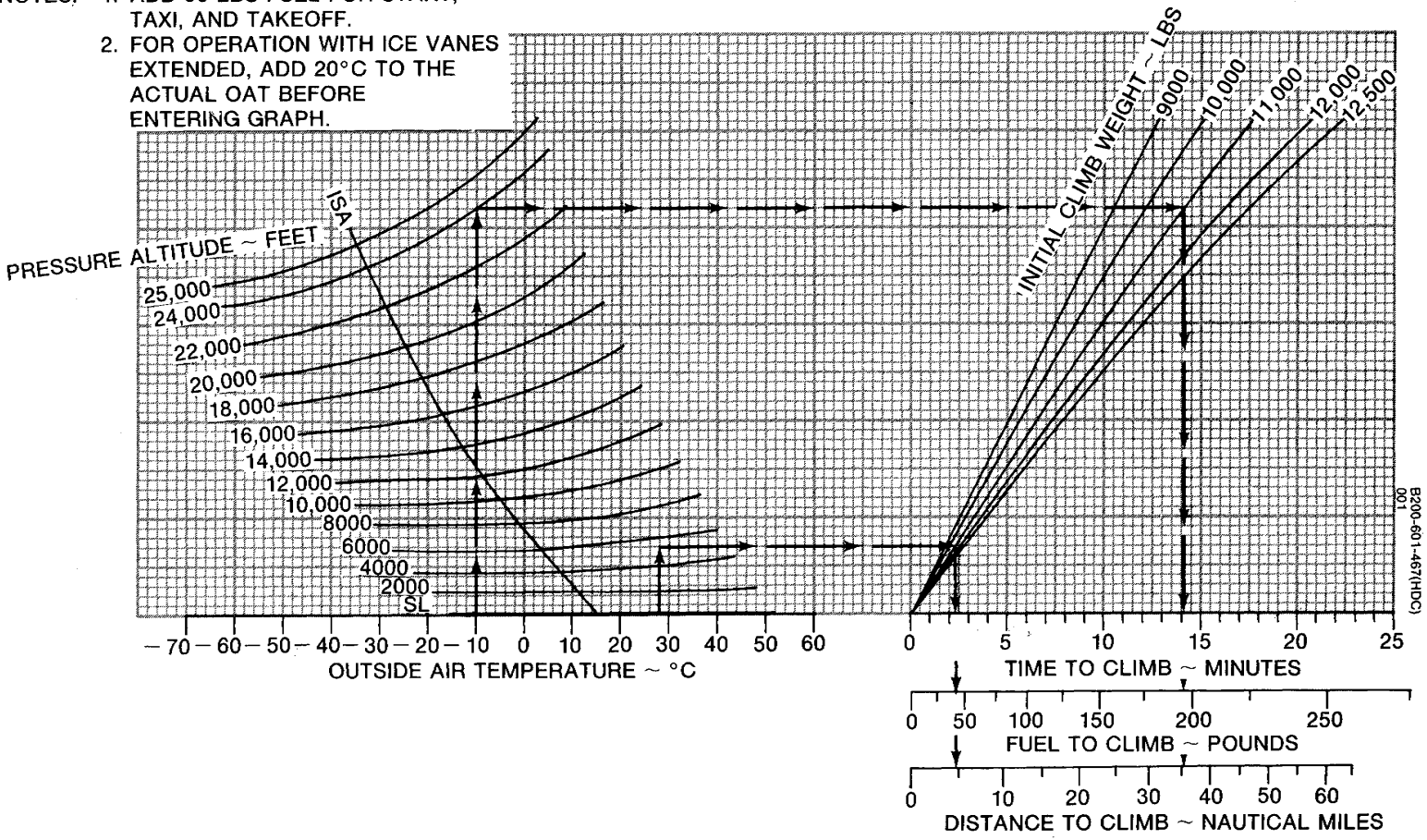
PROPELLER SPEED .... 1900 RPM
POWER:
ITT ..... 770°C
OR TORQUE ..... 2230 FT-LBS

ALTITUDE ~ FEET	CLIMB SPEED ~ KNOTS
SL TO 10,000	130
10,000 TO 20,000	125
20,000 TO 25,000	120

EXAMPLE:

OAT AT TAKEOFF ..... 28°C
OAT AT CRUISE ..... – 10°C
AIRPORT PRESSURE ALTITUDE ... 5433 FT
CRUISE ALTITUDE ..... 24,000 FT
INITIAL CLIMB WEIGHT ..... 11,000 LBS
TIME TO CLIMB (14 – 2) ..... 12 MIN
FUEL TO CLIMB (196 – 40) ..... 156 LBS
DISTANCE TO CLIMB (36 – 4) .... 32 NM

NOTES:
1. ADD 90 LBS FUEL FOR START, TAXI, AND TAKEOFF.
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 20°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH.



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HIGH SPEED CRUISE POWER

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-11	-15	2230	488	976	232	220	232	220	232	220
2000	-14	-19	2230	476	952	229	224	230	224	230	225
4000	-18	-23	2230	463	926	228	228	228	229	229	229
6000	-22	-27	2230	451	902	226	233	226	233	227	234
8000	-26	-31	2230	440	880	223	237	224	238	224	238
10,000	-29	-35	2230	429	858	221	241	222	242	222	243
12,000	-33	-39	2230	419	838	219	246	219	247	220	247
14,000	-37	-43	2230	412	824	216	251	217	252	218	252
16,000	-41	-47	2230	407	814	215	257	215	257	216	258
18,000	-44	-51	2230	402	804	213	262	214	263	214	264
20,000	-48	-55	2230	398	796	211	267	211	268	212	269
22,000	-52	-59	2169	385	770	206	270	207	271	208	272
24,000	-56	-63	2017	357	714	198	269	199	270	200	271
25,000	-58		1939	343	686	194	268	195	269	196	270

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HIGH SPEED CRUISE POWER

1800 RPM

ISA - 20°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	0	-5	2230	490	980	230	223	231	223	231	223
2000	-4	-9	2230	477	954	228	227	229	227	229	228
4000	-8	-13	2230	464	928	226	231	227	232	227	232
6000	-12	-17	2230	452	904	224	235	224	236	225	236
8000	-16	-21	2230	441	882	221	240	222	240	223	241
10,000	-19	-25	2230	430	860	219	244	220	245	220	246
12,000	-23	-29	2230	419	838	217	250	218	250	218	251
14,000	-27	-33	2230	413	826	215	255	216	255	216	256
16,000	-30	-37	2230	408	816	213	260	214	261	214	262
18,000	-34	-41	2230	403	806	211	266	212	266	212	267
20,000	-38	-45	2203	395	790	208	270	208	271	209	272
22,000	-42	-49	2097	375	750	201	270	202	272	203	273
24,000	-46	-53	1958	349	698	194	269	195	271	196	272
25,000	-48	-55	1888	336	672	190	269	192	270	192	272

HIGH SPEED CRUISE POWER

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	10	5	2230	493	986	229	225	229	226	230	226
2000	6	1	2230	479	958	227	229	227	230	228	230
4000	2	-3	2230	466	932	224	234	225	234	225	235
6000	-2	-7	2230	453	906	222	238	223	239	223	239
8000	-5	-11	2230	441	882	220	243	220	243	221	244
10,000	-9	-15	2230	430	860	218	248	218	248	219	249
12,000	-13	-19	2230	422	844	216	253	216	253	217	254
14,000	-17	-23	2230	415	830	213	258	214	259	215	259
16,000	-20	-27	2230	409	818	211	263	212	264	213	265
18,000	-24	-31	2200	399	798	208	267	209	269	209	269
20,000	-28	-35	2102	379	758	202	269	203	270	204	271
22,000	-32	-39	2001	360	720	196	269	197	271	198	272
24,000	-36	-43	1879	337	674	189	269	190	270	191	272
25,000	-38	-45	1818	326	652	186	268	187	270	188	271

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HIGH SPEED CRUISE POWER

1800 RPM

ISA

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	20	15	2230	496	992	227	228	228	228	228	228
2000	16	11	2230	482	964	225	232	225	232	226	233
4000	12	7	2230	467	934	223	236	223	237	224	237
6000	9	3	2230	454	908	221	241	221	242	222	242
8000	5	-1	2230	443	886	218	246	219	246	220	247
10,000	1	-5	2230	432	864	216	251	217	251	217	252
12,000	-3	-9	2230	421	842	214	256	215	257	215	257
14,000	-6	-13	2230	414	828	212	261	213	262	213	263
16,000	-10	-17	2209	405	810	209	266	210	267	210	267
18,000	-14	-21	2094	383	766	203	266	203	267	204	268
20,000	-18	-25	2005	365	730	197	268	198	269	199	270
22,000	-22	-29	1912	347	694	191	269	192	270	193	271
24,000	-26	-33	1792	324	648	184	268	186	269	187	271
25,000	-28	-35	1732	313	626	181	267	182	269	183	270

HIGH SPEED CRUISE POWER

1800 RPM

ISA +10°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	30	25	2230	498	996	226	230	227	231	227	231
2000	26	21	2230	485	970	224	235	224	235	225	236
4000	22	17	2230	471	942	221	239	222	240	222	240
6000	19	13	2230	458	916	219	244	220	244	220	245
8000	15	9	2230	446	892	217	248	217	249	218	250
10,000	11	5	2230	433	866	215	253	215	254	216	255
12,000	7	1	2230	422	844	212	259	213	260	214	260
14,000	4	-3	2190	408	816	209	263	210	263	210	264
16,000	0	-7	2101	388	776	204	264	204	265	205	266
18,000	-4	-11	2005	368	736	198	265	199	266	199	267
20,000	-8	-15	1927	352	704	193	267	194	268	195	269
22,000	-12	-19	1825	334	668	187	267	188	269	189	270
24,000	-16	-23	1709	312	624	180	266	181	268	182	270
25,000	-18	-25	1651	301	602	176	266	177	268	179	269

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HIGH SPEED CRUISE POWER

1800 RPM

ISA +20°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	40	35	2230	501	1002	224	233	225	233	226	234
2000	36	31	2230	487	974	222	237	223	238	223	238
4000	32	27	2230	474	948	220	241	220	242	221	243
6000	29	23	2230	460	920	218	246	218	247	219	247
8000	25	19	2230	448	896	215	251	216	252	216	252
10,000	21	15	2219	434	868	213	256	213	256	214	257
12,000	17	11	2105	406	812	206	256	207	257	208	258
14,000	14	7	2042	388	776	202	259	203	260	204	261
16,000	10	3	1975	370	740	197	261	198	262	199	263
18,000	6	-1	1905	353	706	193	263	194	265	194	266
20,000	2	-5	1833	338	676	188	265	189	267	190	268
22,000	-2	-9	1756	323	646	182	267	184	268	185	270
24,000	-6	-13	1646	301	602	176	266	177	268	178	269
25,000	-8	-15	1590	291	582	172	265	173	267	175	269

HIGH SPEED CRUISE POWER

1800 RPM

ISA +30°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	50	45	2230	504	1008	223	235	224	235	224	236
2000	46	41	2230	490	980	221	239	221	240	222	240
4000	43	37	2230	476	952	218	244	219	244	219	245
6000	39	33	2229	462	924	216	248	217	249	217	250
8000	35	29	2142	436	872	211	250	211	251	212	251
10,000	31	25	2055	411	822	205	251	206	252	207	253
12,000	27	21	1950	384	768	199	252	200	253	201	254
14,000	23	17	1895	368	736	195	254	196	255	197	257
16,000	19	13	1839	352	704	191	257	192	258	193	260
18,000	16	9	1785	336	672	186	260	188	261	189	263
20,000	12	5	1741	324	648	183	263	184	265	185	266
22,000	8	1	1673	310	620	178	265	179	266	180	268
24,000	4	-3	1565	289	578	171	263	172	266	173	267
25,000	2	-5	1511	279	558	167	263	169	265	170	267

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HIGH SPEED CRUISE POWER

1800 RPM

ISA +37°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

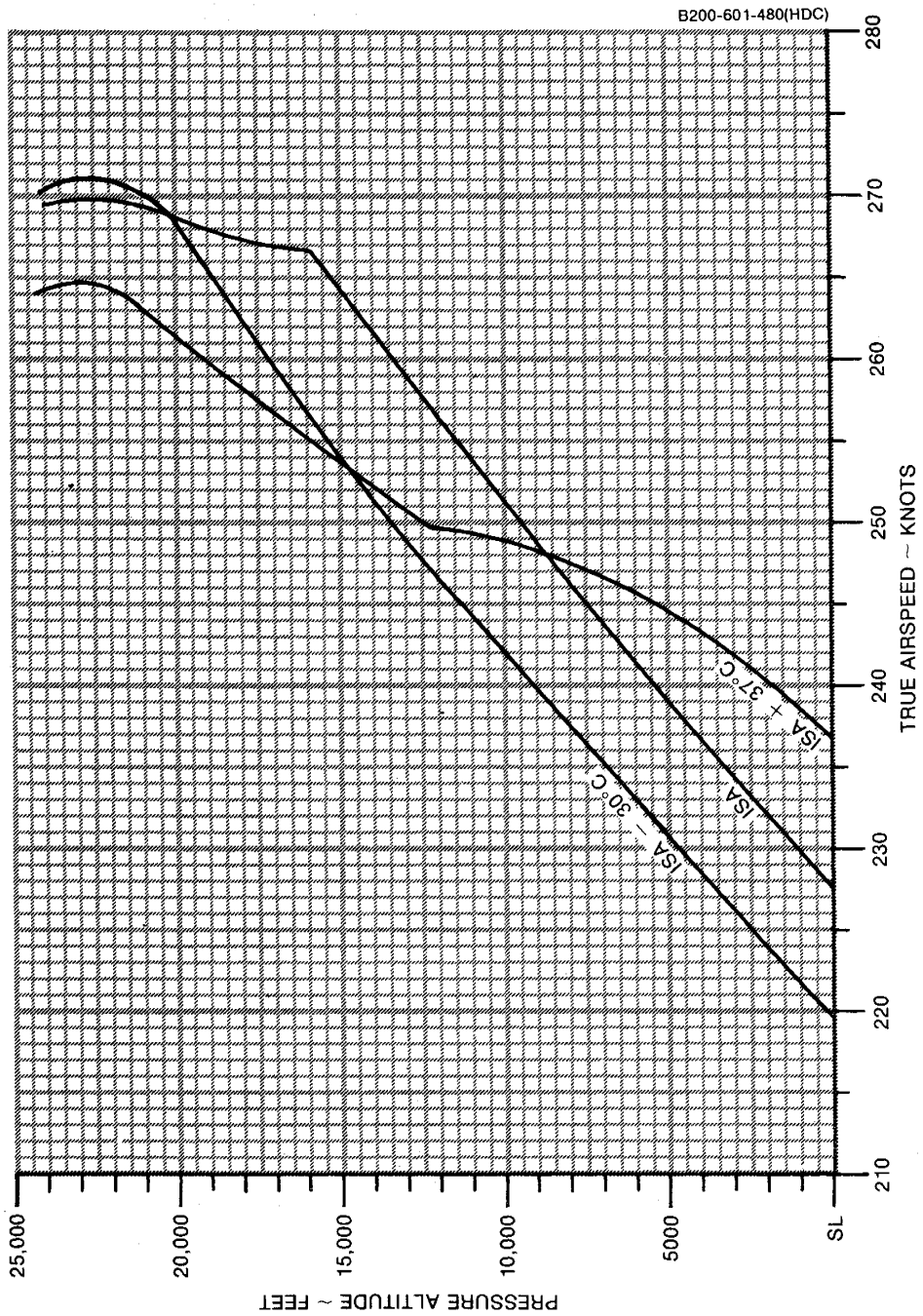
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	57	52	2230	505	1010	222	236	223	237	223	237
2000	53	48	2206	488	976	219	240	219	240	220	241
4000	50	44	2164	468	936	215	243	216	243	216	244
6000	46	40	2103	445	890	210	245	211	246	212	247
8000	42	36	2024	421	842	205	247	206	248	207	248
10,000	38	32	1946	396	792	200	248	201	249	202	250
12,000	34	28	1847	370	740	194	248	195	250	196	251
14,000	30	24	1799	354	708	190	251	191	253	192	254
16,000	26	20	1749	339	678	186	254	187	255	188	257
18,000	23	16	1697	325	650	182	256	183	258	184	260
20,000	19	12	1660	313	626	178	260	180	262	181	263
22,000	15	8	1607	301	602	174	263	175	264	176	266
24,000	11	4	1508	281	562	167	261	169	264	170	266
25,000	9	2	1458	270	540	164	261	165	263	167	266

SPEED AT HIGH SPEED CRUISE POWER

1800 RPM

WEIGHT 11,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 25 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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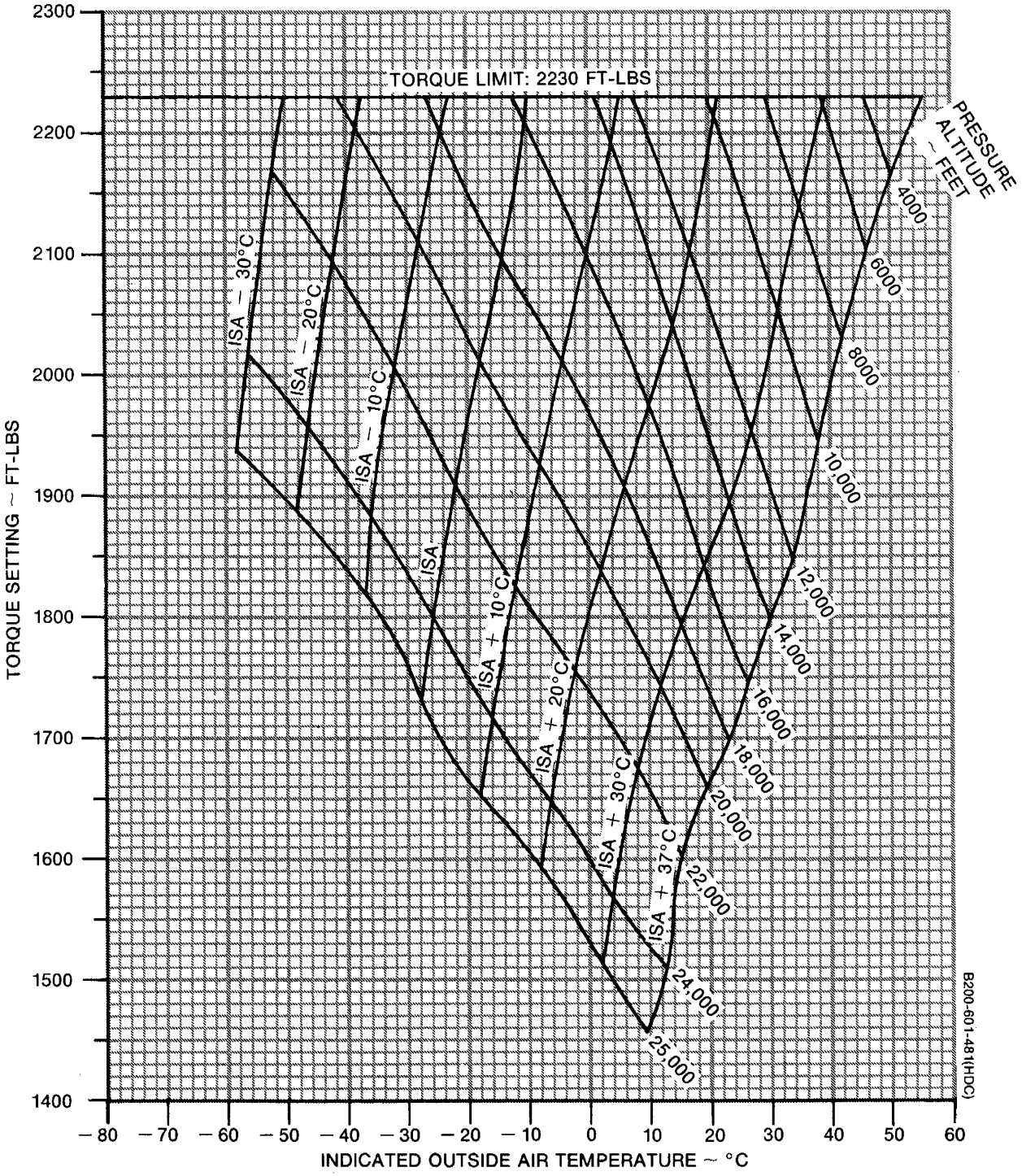
TORQUE AT HIGH SPEED CRUISE POWER

1800 RPM

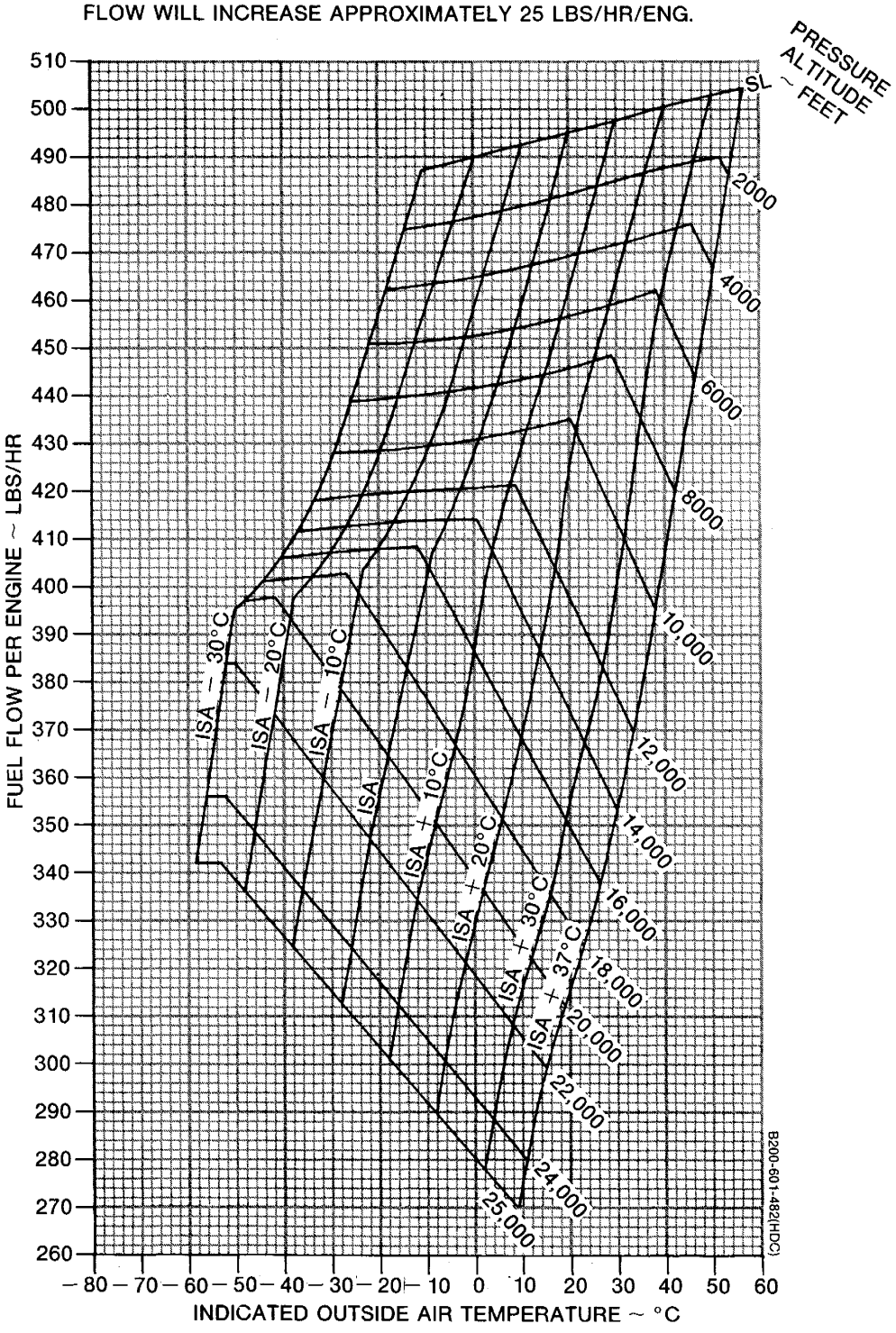
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT HIGH SPEED CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 12% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 25 LBS/HR/ENG.



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RANGE PROFILE — HIGH SPEED CRUISE POWER

1800 RPM

STANDARD DAY (ISA)
ZERO WIND

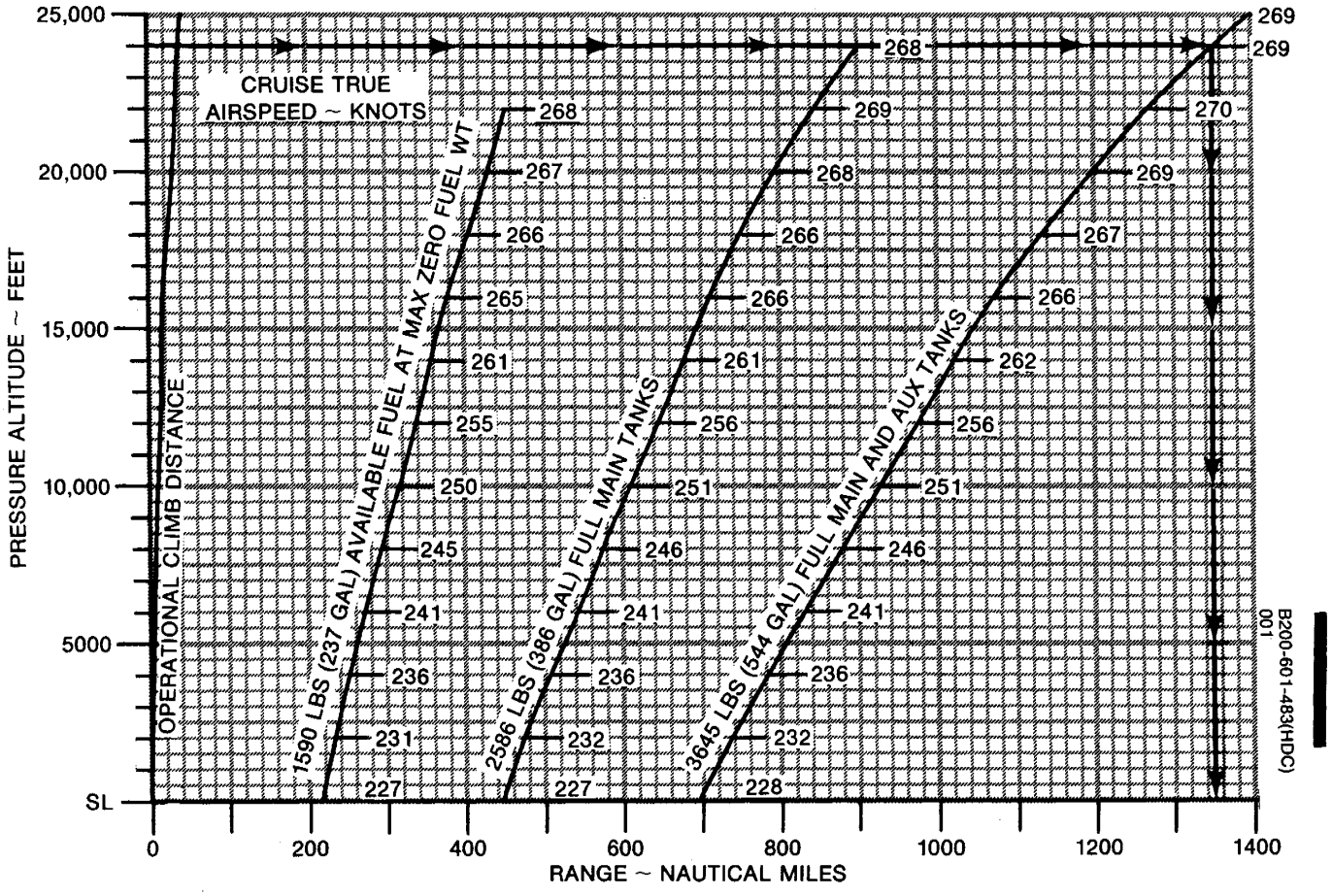
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

EXAMPLE:

PRESSURE ALTITUDE ..... 24,000 FT
FUEL LOADING ..... 3645 LBS
RANGE ..... 1350 NM

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER.



INTERMEDIATE CRUISE POWER

1700 RPM

ISA - 30°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-11	-15	2230	468	936	227	215	228	216	228	216
2000	-15	-19	2230	456	912	225	219	225	220	226	220
4000	-18	-23	2230	443	886	223	223	223	224	224	224
6000	-22	-27	2230	431	862	220	227	221	228	221	228
8000	-26	-31	2230	420	840	218	231	219	232	219	233
10,000	-30	-35	2230	409	818	216	234	217	237	218	238
12,000	-33	-39	2230	399	798	214	241	215	242	216	243
14,000	-37	-43	2230	392	784	212	246	213	247	214	248
16,000	-41	-47	2230	387	774	210	251	211	252	211	253
18,000	-45	-51	2230	382	764	208	256	209	257	209	258
20,000	-48	-55	2230	380	760	205	261	206	262	207	263
22,000	-52	-59	2195	372	744	201	264	202	265	203	266
24,000	-56	-63	2057	347	694	194	263	195	265	196	266
25,000	-58	-65	1988	335	670	191	263	192	264	193	265

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INTERMEDIATE CRUISE POWER

1700 RPM

ISA - 20°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-1	-5	2230	470	940	225	118	226	218	226	219
2000	-4	-9	2230	457	914	223	222	224	223	224	223
4000	18	-13	2230	444	888	221	226	222	227	222	227
6000	-12	-17	2230	432	864	219	230	220	231	220	232
8000	-16	-21	2230	421	842	217	235	217	235	218	236
10,000	-19	-25	2230	411	822	215	239	215	240	216	241
12,000	-23	-29	2230	400	800	213	244	213	245	214	246
14,000	-27	-33	2230	393	786	211	249	211	250	212	251
16,000	-31	-37	2230	388	776	209	255	209	256	210	256
18,000	-34	-41	2230	384	768	206	259	206	260	207	261
20,000	-38	-45	2213	379	758	202	263	203	264	204	265
22,000	-42	-49	2105	359	718	196	264	197	265	198	266
24,000	-46	-53	1973	336	672	189	263	190	264	191	266
25,000	-48	-55	1907	324	648	186	262	187	264	188	265

INTERMEDIATE CRUISE POWER

1700 RPM

ISA – 10°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	10	5	2230	474	948	224	221	225	221	225	222
2000	6	1	2230	460	920	222	225	222	225	223	226
4000	2	-3	2230	447	894	219	229	220	229	221	230
6000	-2	-7	2230	434	868	217	233	218	234	218	234
8000	-6	-11	2230	422	844	215	238	216	238	216	239
10,000	-9	-15	2230	411	822	213	242	214	243	214	244
12,000	-13	-19	2230	402	804	211	248	212	249	213	249
14,000	-17	-23	2230	396	792	209	253	210	254	210	254
16,000	-21	-27	2230	390	780	206	257	207	258	208	259
18,000	-24	-31	2213	384	768	203	261	204	262	205	263
20,000	-28	-35	2112	365	730	197	262	198	263	199	264
22,000	-32	-39	2013	346	692	191	263	192	264	193	265
24,000	-36	-43	1885	323	646	184	262	185	264	187	265
25,000	-38	-45	1820	312	624	181	261	182	263	183	264

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INTERMEDIATE CRUISE POWER

1700 RPM

ISA

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	20	15	2230	477	954	222	223	223	213	223	224
2000	16	11	2230	464	928	220	227	221	227	221	228
4000	12	7	2230	450	900	218	231	218	232	219	232
6000	8	3	2230	436	872	216	236	216	236	217	237
8000	5	-1	2230	424	848	214	241	214	241	215	242
10,000	1	-5	2230	413	826	212	246	213	247	213	247
12,000	-3	-9	2230	403	806	210	251	210	252	211	252
14,000	-7	-13	2230	395	790	207	256	208	257	209	257
16,000	-10	-17	2230	390	780	204	260	205	261	206	262
18,000	-14	-21	2104	368	736	198	260	199	261	199	262
20,000	-18	-25	2019	352	704	192	261	193	262	194	263
22,000	-22	-29	1924	334	668	186	262	188	263	188	265
24,000	-26	-33	1800	311	622	179	261	181	260	182	264
25,000	-28	-35	1737	300	600	176	260	177	262	178	269

INTERMEDIATE CRUISE POWER

1700 RPM

ISA +10°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	30	25	2230	479	958	221	225	221	226	222	226
2000	26	21	2230	466	932	219	230	219	230	220	231
4000	22	17	2230	453	906	217	234	217	235	218	235
6000	18	13	2230	440	880	214	239	215	239	216	240
8000	15	9	2230	427	854	212	244	213	244	214	245
10,000	11	5	2230	415	830	210	249	211	249	212	250
12,000	7	1	2230	404	808	208	254	209	254	209	255
14,000	3	-3	2230	397	794	205	258	206	259	207	260
16,000	-0	-7	2126	375	750	199	259	200	260	201	261
18,000	-4	-11	2023	354	708	193	259	194	261	195	262
20,000	-8	-15	1940	339	678	188	260	189	262	190	263
22,000	-12	-19	1838	322	644	182	260	183	262	184	264
24,000	-16	-23	1717	300	600	175	259	176	261	177	263
25,000	-18	-25	1656	289	578	171	258	172	260	179	262

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INTERMEDIATE CRUISE POWER

1700 RPM

ISA +20°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	40	35	2230	482	964	220	228	220	228	221	229
2000	36	31	2230	469	938	217	232	218	233	219	233
4000	32	27	2230	455	910	215	236	216	237	216	238
6000	29	23	2230	442	884	213	241	214	242	214	243
8000	25	19	2230	430	860	211	246	212	247	212	248
10,000	21	15	2230	418	836	209	251	210	252	210	253
12,000	17	11	2176	399	798	205	254	205	255	206	256
14,000	13	7	2088	378	756	199	255	200	256	201	257
16,000	9	3	2000	358	716	194	256	195	257	195	258
18,000	6	-1	1912	339	678	188	257	189	258	190	260
20,000	2	-5	1851	326	652	183	259	184	261	185	262
22,000	-2	-9	1771	311	622	178	260	179	262	180	263
24,000	-6	-13	1655	290	580	171	258	172	260	173	262
25,000	-8	-15	1595	279	558	167	257	168	260	170	262

INTERMEDIATE CRUISE POWER

1700 RPM

ISA +30°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	50	45	2230	486	972	218	230	219	230	219	231
2000	46	41	2230	472	944	216	234	217	235	217	235
4000	42	37	2230	458	916	214	239	214	240	215	240
6000	39	33	2230	444	888	212	244	212	244	213	245
8000	35	29	2226	431	862	209	249	210	249	211	250
10,000	31	25	2135	406	812	204	250	205	251	206	252
12,000	27	21	2028	379	758	198	250	199	251	200	252
14,000	23	17	1944	360	720	193	252	194	253	195	254
16,000	19	13	1869	342	684	188	253	189	254	190	256
18,000	15	9	1794	324	648	183	254	184	256	185	257
20,000	12	5	1750	313	626	178	257	179	258	180	260
22,000	8	1	1693	300	600	173	258	175	260	176	262
24,000	4	-3	1573	278	556	166	256	168	259	169	261
25,000	2	-5	1518	268	536	162	255	164	258	165	260

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INTERMEDIATE CRUISE POWER

1700 RPM

ISA +37°C

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

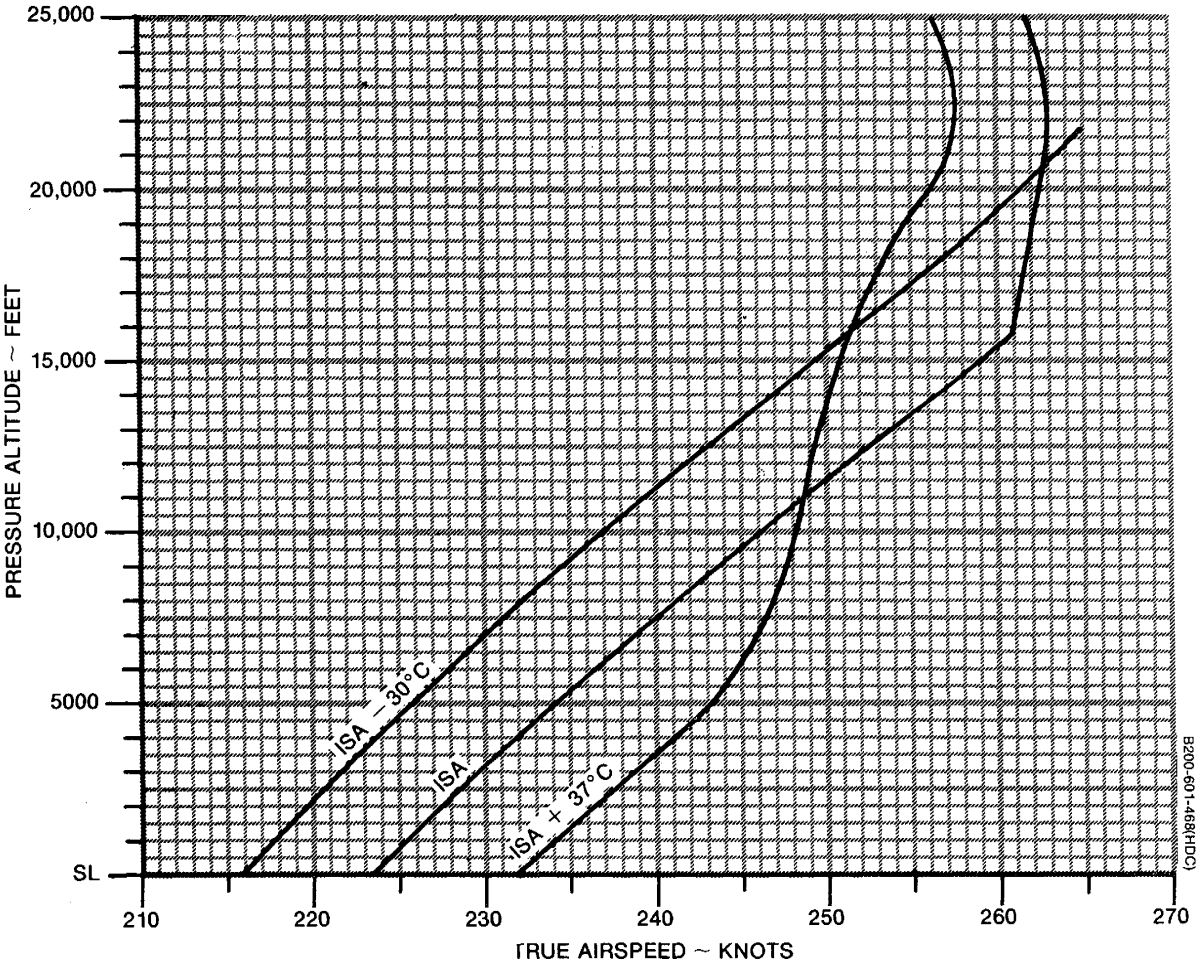
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	57	52	2230	488	976	217	231	218	232	218	232
2000	53	48	2230	474	948	215	236	216	236	216	237
4000	49	44	2230	460	920	213	240	214	241	214	242
6000	46	40	2194	441	882	209	244	210	245	211	246
8000	42	36	2114	417	834	204	246	205	247	206	247
10,000	38	32	2033	393	786	200	247	200	248	201	249
12,000	34	28	1931	367	734	194	248	195	249	195	250
14,000	30	24	1853	348	696	189	249	190	250	190	252
16,000	26	20	1780	330	660	184	250	185	252	186	253
18,000	22	16	1711	314	628	178	252	180	253	181	255
20,000	18	12	1671	303	606	174	254	175	256	177	258
22,000	15	8	1620	291	582	169	256	171	258	172	260
24,000	11	4	1517	271	542	162	254	164	257	166	259
25,000	9	2	1465	260	520	159	253	161	256	162	259

SPEED AT INTERMEDIATE CRUISE POWER

1700 RPM

WEIGHT 11,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 20 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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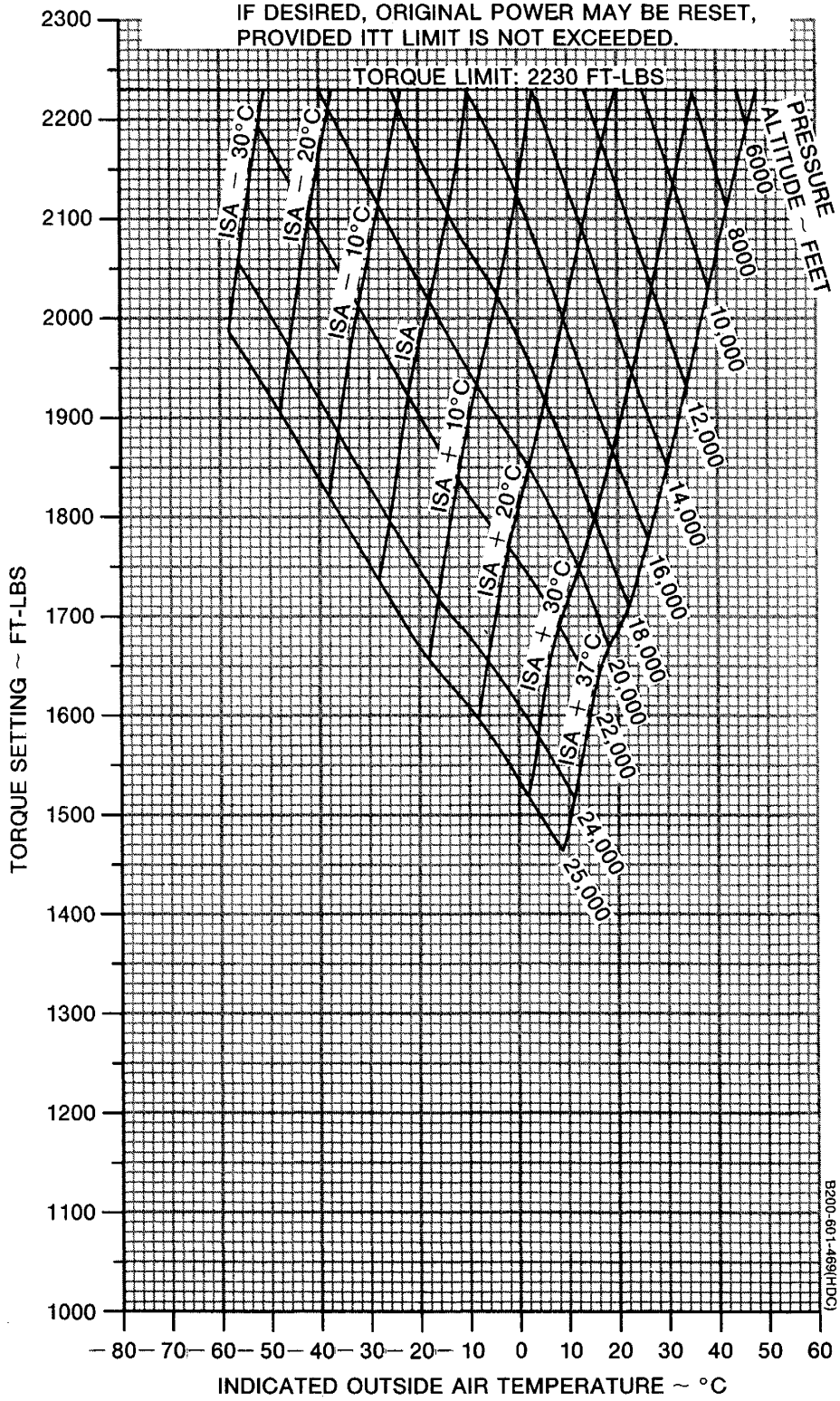
TORQUE AT INTERMEDIATE CRUISE POWER

1700 RPM

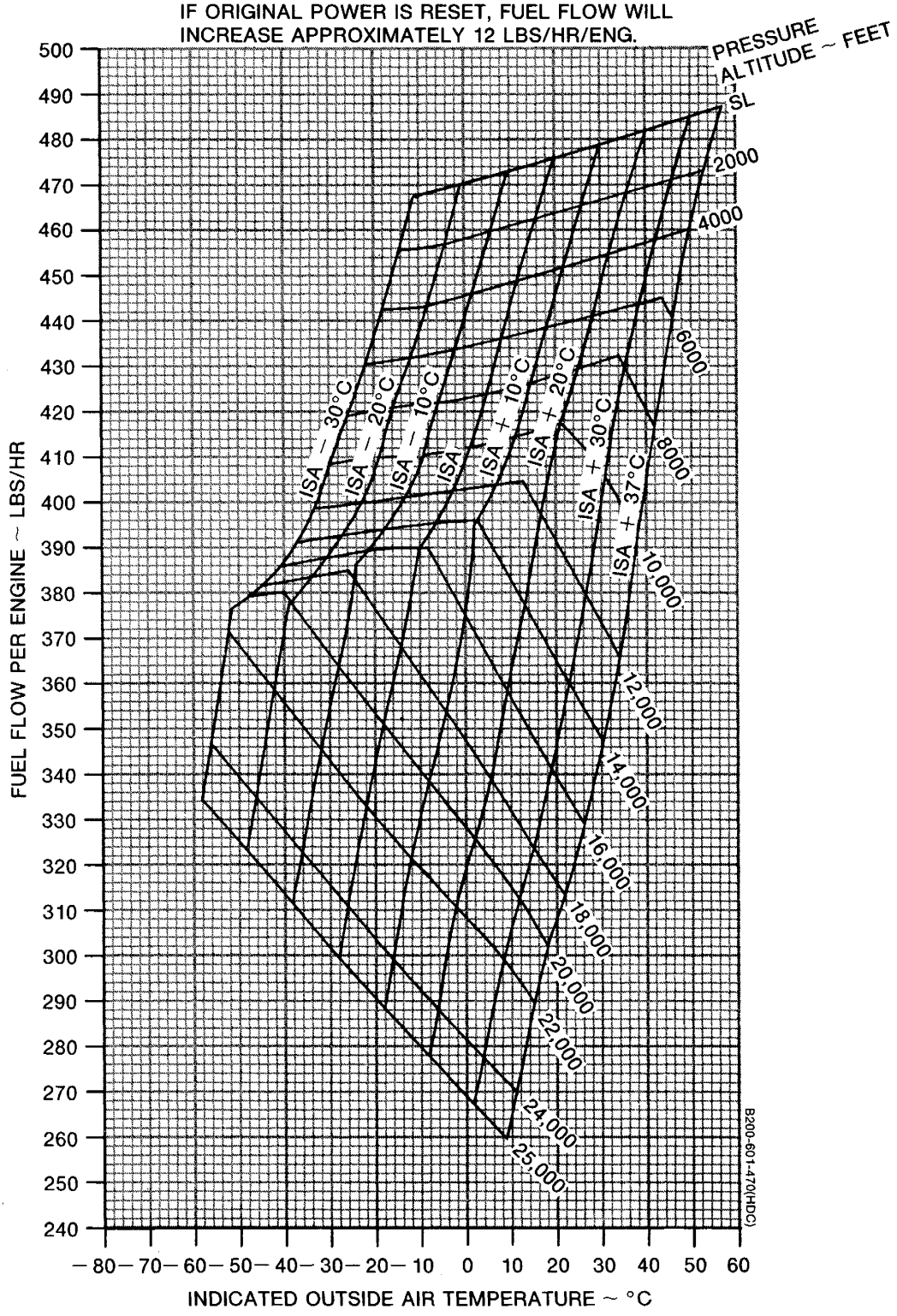
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT INTERMEDIATE CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 12% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 12 LBS/HR/ENG.



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RANGE PROFILE — INTERMEDIATE CRUISE POWER

1700 RPM

STANDARD DAY (ISA)
ZERO WIND

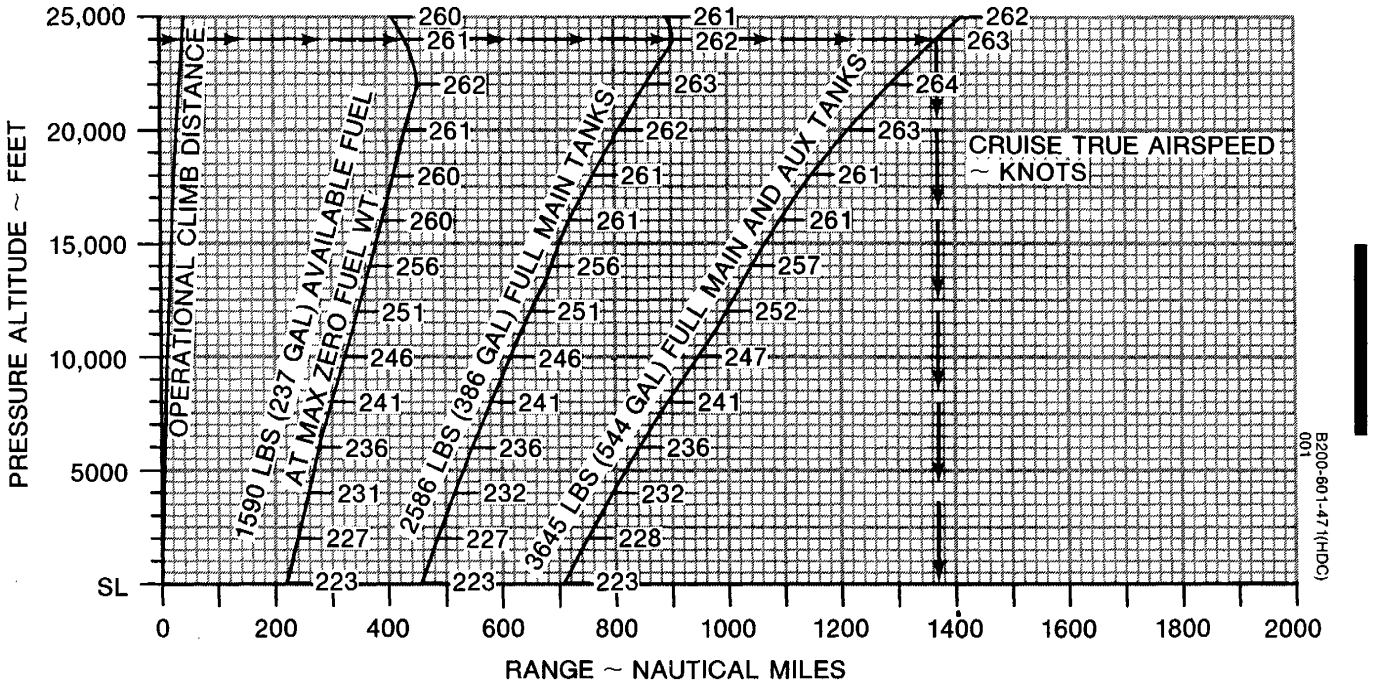
ASSOCIATED CONDITIONS:

WEIGHT	12,590 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	24,000 FT
FUEL LOADING	3645 LBS
RANGE	1370 NM

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER



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LONG RANGE CRUISE POWER

1700 RPM

ISA - 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-12	-15	1457	373	746	196	186	195	185	193	184
2000	-16	-19	1307	342	684	186	182	185	181	184	179
4000	-20	-23	1138	306	612	176	177	174	175	172	173
6000	-24	-27	1002	275	550	167	173	164	170	161	167
8000	-28	-31	927	252	504	160	171	157	168	154	165
10,000	-32	-35	857	230	460	154	170	151	166	147	161
12,000	-36	-39	817	214	428	150	170	147	166	141	160
14,000	-40	-43	767	198	396	145	170	141	165	135	158
16,000	-44	-47	808	194	388	140	169	—	—	—	—
18,000	—	—	—	—	—	—	—	—	—	—	—
20,000	—	—	—	—	—	—	—	—	—	—	—
22,000	—	—	—	—	—	—	—	—	—	—	—
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

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LONG RANGE CRUISE POWER

1700 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-2	-5	1504	380	760	197	191	196	190	194	188
2000	-6	-9	1382	351	702	189	189	188	187	186	186
4000	-10	-13	1253	322	644	181	185	179	184	178	182
6000	-14	-17	1144	295	590	173	183	171	181	169	179
8000	-18	-21	1047	269	538	166	181	163	178	161	175
10,000	-22	-25	951	244	488	159	178	156	175	152	171
12,000	-26	-29	901	226	452	154	178	150	174	147	170
14,000	-30	-33	853	211	422	149	178	145	173	141	169
16,000	-34	-37	805	196	392	144	177	140	173	136	168
18,000	-38	-41	768	183	366	140	178	135	172	131	166
20,000	—	—	—	—	—	—	—	—	—	—	—
22,000	—	—	—	—	—	—	—	—	—	—	—
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

LONG RANGE CRUISE POWER

1700 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	8	5	1442	375	750	192	189	192	189	192	189
2000	4	1	1391	354	708	188	190	187	190	187	189
4000	-0	-3	1312	330	660	183	191	181	189	180	188
6000	-4	-7	1214	304	608	176	189	174	187	172	185
8000	-8	-11	1114	279	558	168	187	166	184	164	182
10,000	-12	-15	1038	256	512	162	185	160	183	158	180
12,000	-16	-19	972	237	474	156	185	154	181	151	178
14,000	-20	-23	920	221	442	151	184	148	181	145	177
16,000	-24	-27	875	206	412	147	185	143	180	140	176
18,000	-28	-31	820	191	382	142	184	138	179	134	174
20,000	-32	-35	801	182	364	138	186	134	180	130	175
22,000	-36	-39	775	171	342	—	—	130	181	126	175
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

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LONG RANGE CRUISE POWER

1700 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	18	15	1352	368	736	186	187	186	186	185	186
2000	14	11	1303	346	692	182	188	181	187	181	187
4000	10	7	1281	328	656	179	190	179	190	178	190
6000	6	3	1233	308	616	175	192	174	191	173	190
8000	3	-1	1174	287	574	170	192	169	191	167	189
10,000	-1	-5	1102	265	530	165	192	163	190	161	187
12,000	-5	-9	1043	246	492	160	192	157	189	155	186
14,000	-10	-13	978	228	456	154	191	151	188	148	184
16,000	-13	-17	921	213	426	148	190	145	186	142	182
18,000	-17	-21	881	200	400	144	191	140	186	137	181
20,000	-21	-25	847	188	376	140	192	136	186	132	181
22,000	-25	-29	831	180	360	136	193	133	188	129	183
24,000	-29	-33	828	173	346	—	—	130	191	126	186
25,000	-31	-35	743	159	318	—	—	—	—	125	187

LONG RANGE CRUISE POWER

1700 RPM

ISA +10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	28	25	1361	371	742	187	191	185	189	184	187
2000	24	21	1268	345	690	180	189	178	188	177	187
4000	20	17	1209	322	644	174	188	173	188	173	187
6000	16	13	1171	303	606	169	189	169	189	170	189
8000	13	9	1133	284	568	165	190	165	190	165	190
10,000	9	5	1094	265	530	162	192	161	191	161	191
12,000	5	1	1069	251	502	158	194	158	193	157	193
14,000	1	-3	1018	235	470	154	195	153	193	151	191
16,000	-3	-7	972	220	440	150	196	148	193	145	190
18,000	-7	-11	919	205	410	145	196	142	192	139	187
20,000	-11	-15	879	193	386	141	196	137	191	133	186
22,000	-15	-19	869	186	372	138	199	134	194	130	188
24,000	-19	-23	867	179	358	135	203	132	197	128	192
25,000	-21	-25	867	176	352	—	—	131	199	127	193

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LONG RANGE CRUISE POWER

1700 RPM

ISA +20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	38	35	1364	373	746	185	192	184	191	183	190
2000	34	31	1296	350	700	181	193	179	191	177	189
4000	30	27	1214	325	650	175	192	173	190	171	188
6000	26	23	1135	301	602	168	190	166	189	165	188
8000	22	19	1078	280	560	162	190	161	188	160	187
10,000	19	15	1034	260	520	157	189	156	189	156	189
12,000	15	11	999	244	488	153	191	152	190	152	190
14,000	11	7	988	233	466	150	193	150	193	149	193
16,000	7	3	969	221	442	147	195	146	195	146	194
18,000	3	-1	948	210	420	144	198	143	197	141	194
20,000	-1	-5	918	199	398	141	200	139	197	135	193
22,000	-5	-9	892	189	378	137	203	135	198	131	193
24,000	-9	-13	888	181	362	133	203	132	202	129	196
25,000	-11	-15	878	177	354	—	—	130	202	127	198

LONG RANGE CRUISE POWER

1700 RPM

ISA +30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	48	45	1342	373	746	183	193	182	192	182	191
2000	44	41	1276	349	698	178	193	177	192	176	191
4000	41	37	1230	329	658	174	195	173	193	171	191
6000	37	33	1172	307	614	169	195	167	193	165	191
8000	33	29	1107	285	570	164	195	162	192	159	189
10,000	29	25	1049	264	528	158	195	156	192	154	189
12,000	25	21	995	245	490	154	195	151	191	149	189
14,000	21	17	958	231	462	149	195	146	192	144	190
16,000	17	13	933	218	436	145	196	143	194	142	193
18,000	13	9	920	208	416	141	198	140	196	140	196
20,000	9	5	914	200	400	138	200	137	199	136	197
22,000	5	1	902	191	382	134	202	134	201	132	198
24,000	1	-3	876	181	362	—	—	130	202	129	200
25,000	-1	-5	862	176	352	—	—	127	202	127	201

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LONG RANGE CRUISE POWER

1700 RPM

ISA +37°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN LONG RANGE CRUISE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	55	52	1322	372	744	181	193	180	192	179	191
2000	51	48	1256	349	698	176	193	175	192	174	191
4000	48	44	1208	328	656	171	194	170	193	169	192
6000	44	40	1173	309	618	168	196	167	195	165	193
8000	40	36	1121	288	576	163	196	162	195	160	193
10,000	36	32	1072	268	536	158	197	157	195	155	192
12,000	32	28	1034	251	502	154	198	153	196	150	193
14,000	28	24	988	236	472	150	200	148	196	145	192
16,000	24	20	946	221	442	146	200	143	196	140	192
18,000	20	16	912	208	416	142	202	138	196	137	195
20,000	16	12	893	199	398	138	203	135	198	134	197
22,000	12	8	882	190	380	—	—	132	200	131	199
24,000	8	4	804	173	346	—	—	—	—	126	199
25,000	6	2	799	169	338	—	—	—	—	125	201

RANGE PROFILE — LONG RANGE CRUISE POWER

1700 RPM

STANDARD DAY (ISA)
ZERO WIND

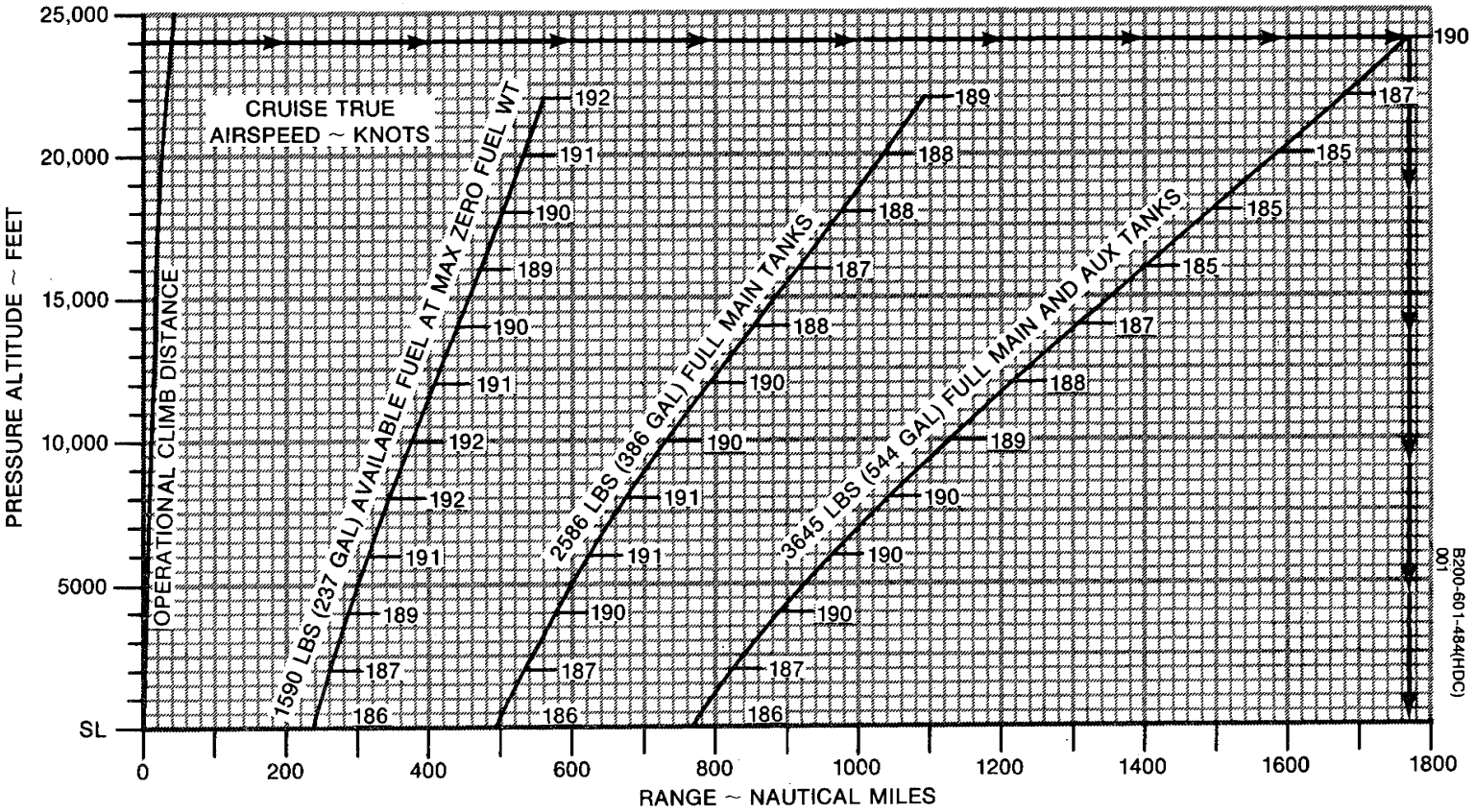
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

EXAMPLE:

PRESSURE ALTITUDE ..... 24,000 FT
FUEL LOADING ..... 3645 LBS
RANGE ..... 1770 NM

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER.



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RANGE PROFILE — FULL MAIN AND AUX TANKS

STANDARD DAY
ZERO WIND

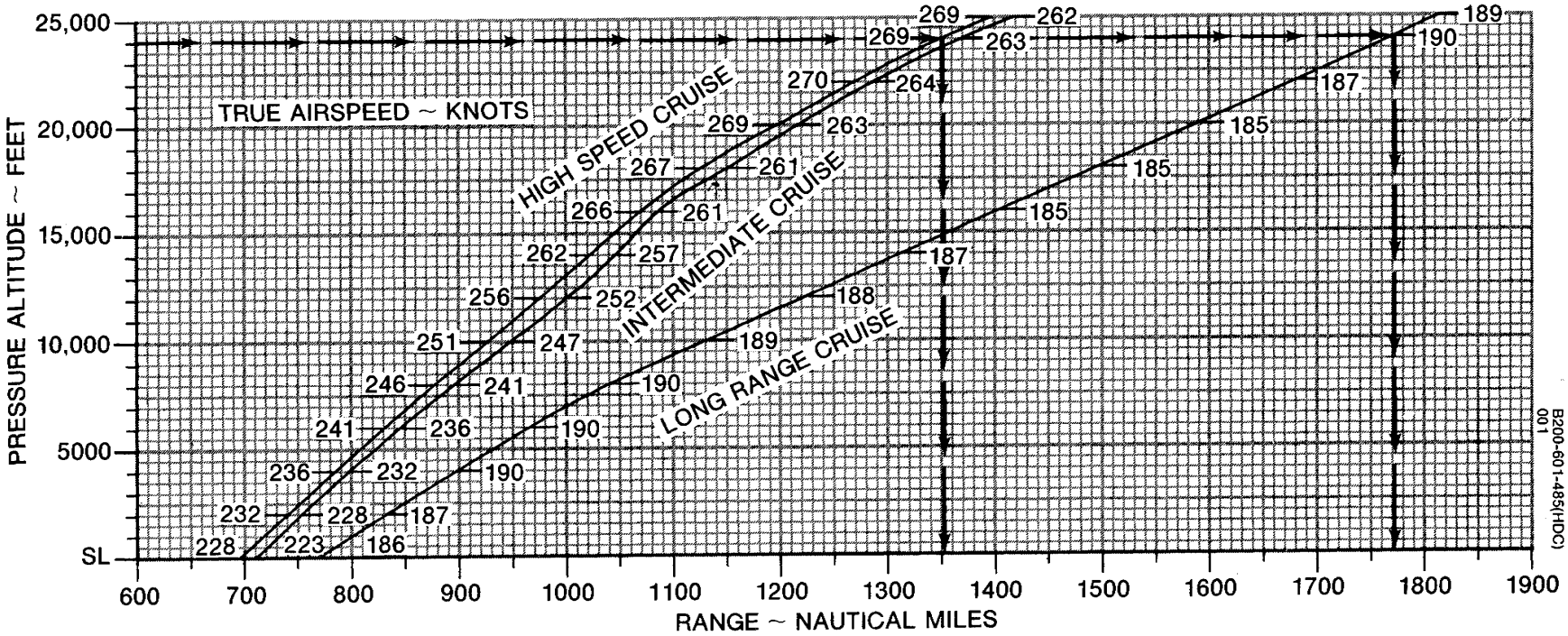
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER.

EXAMPLE:

PRESSURE ALTITUDE ..... 24,000 FT
RANGE @
HIGH SPEED CRUISE POWER ..... 1352 NM
RANGE @
LONG RANGE CRUISE POWER ..... 1772 NM



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ENDURANCE PROFILE – FULL MAIN AND AUX TANKS
STANDARD DAY

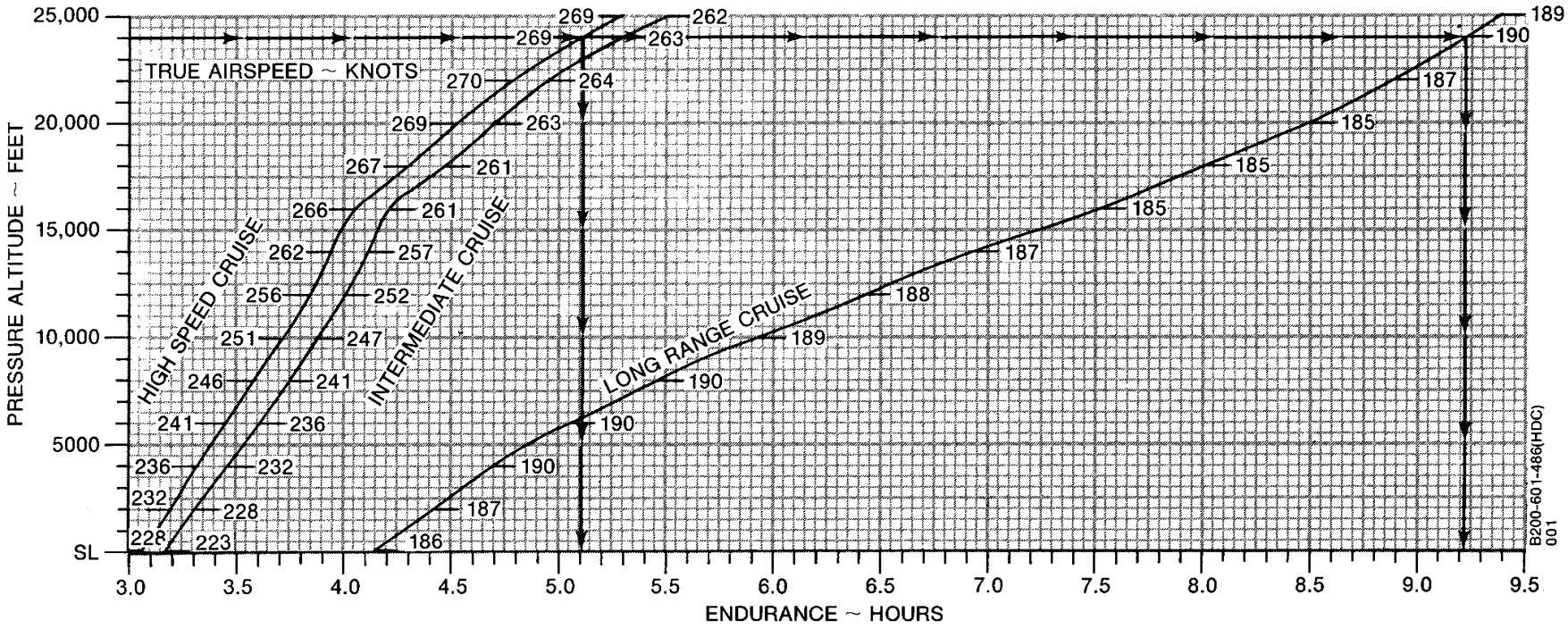
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

NOTE: ENDURANCE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER.

EXAMPLE:

PRESSURE ALTITUDE ..... 24,000 FT
ENDURANCE @
HIGH SPEED CRUISE POWER ..... 5.11 HRS
ENDURANCE @
LONG RANGE CRUISE POWER ..... 9.22 HRS



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RANGE PROFILE – FULL MAIN TANKS
STANDARD DAY
ZERO WIND

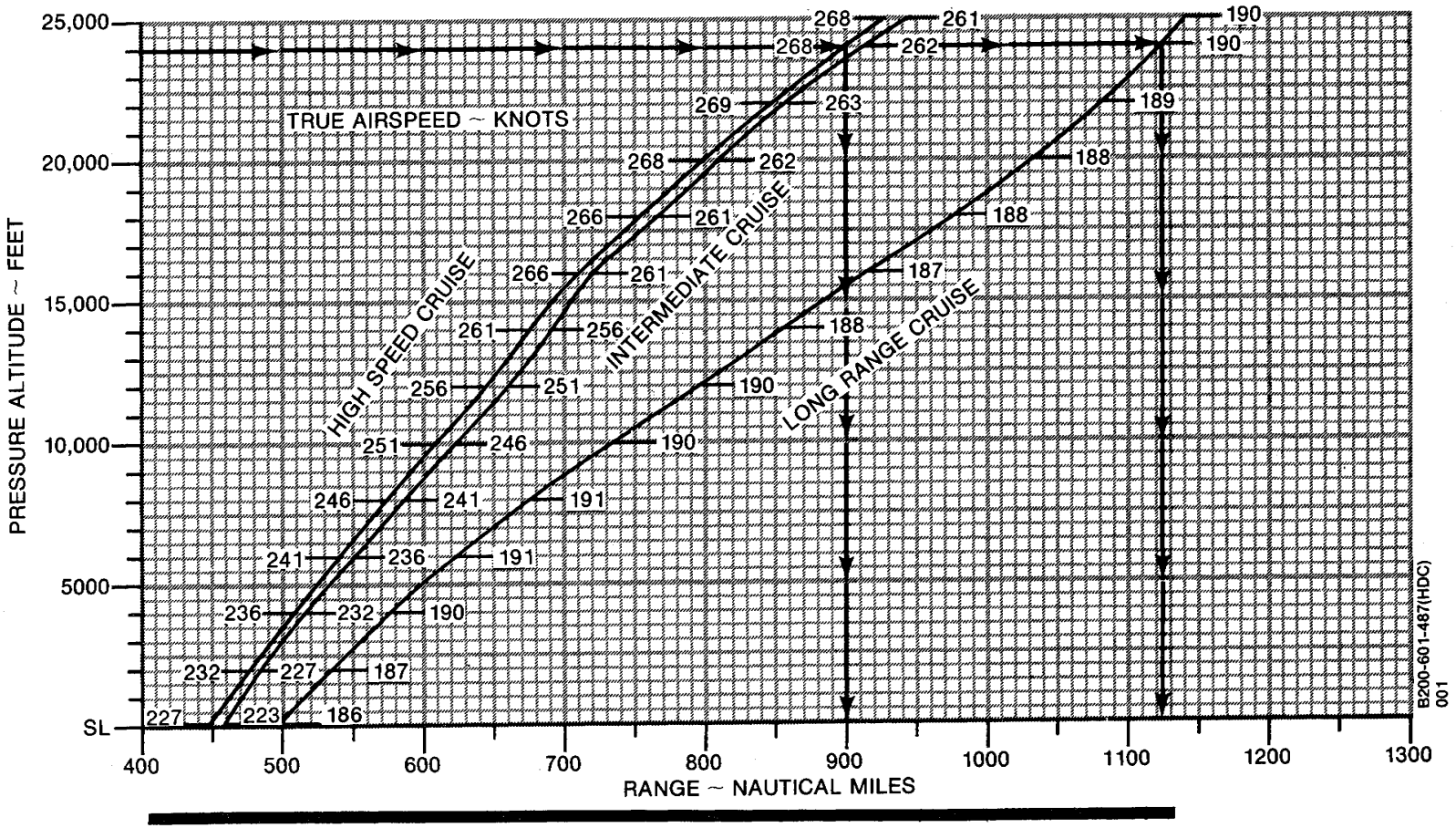
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER.

EXAMPLE:

PRESSURE ALTITUDE ..... 24,000 FT
RANGE @ HIGH SPEED CRUISE POWER ..... 900 NM
RANGE @ LONG RANGE CRUISE POWER ..... 1124 NM



ENDURANCE PROFILE — FULL MAIN TANKS
STANDARD DAY

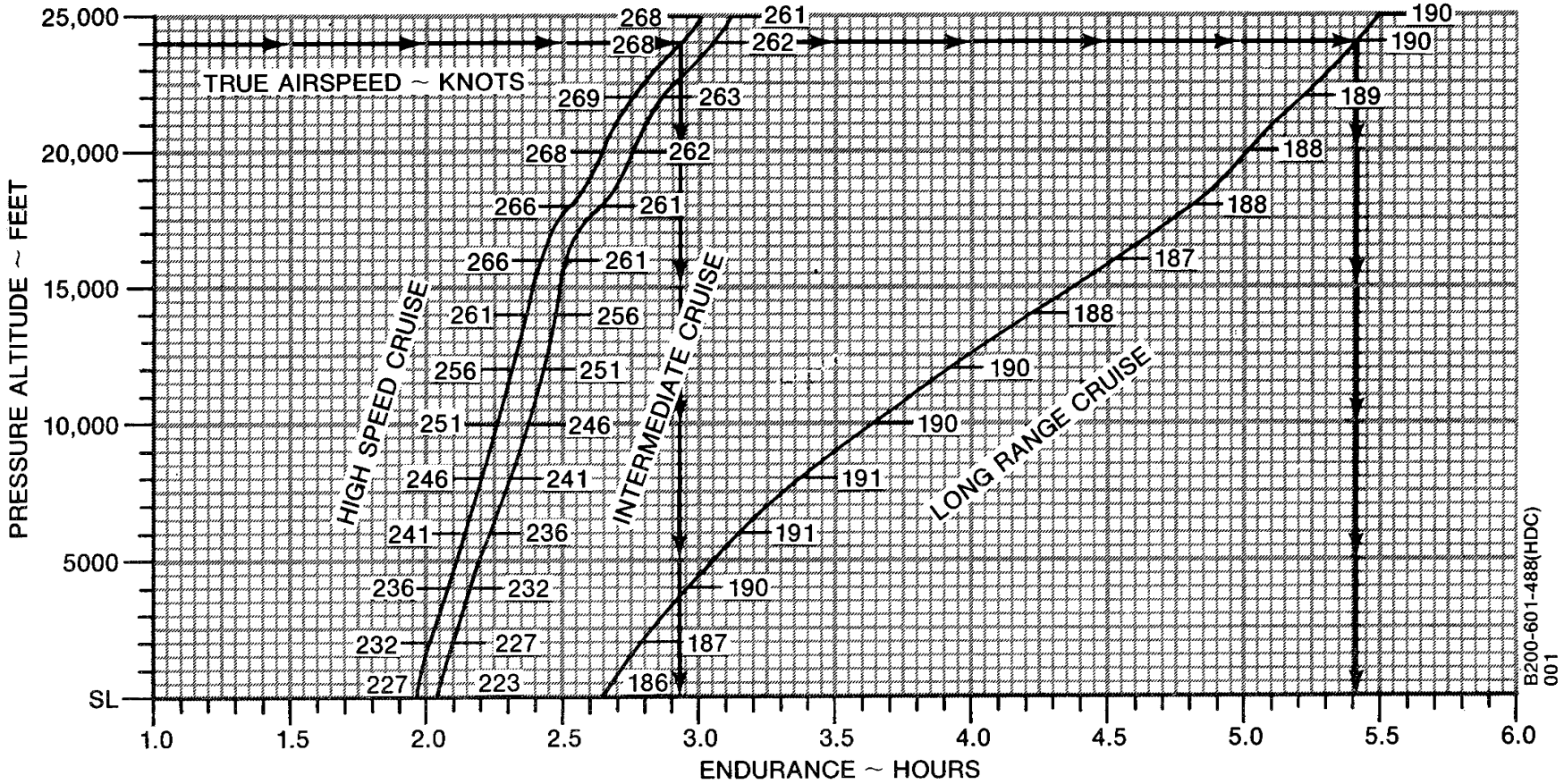
ASSOCIATED CONDITIONS:

WEIGHT ..... 12,590 LBS BEFORE ENGINE START
FUEL ..... AVIATION KEROSENE
FUEL DENSITY ..... 6.7 LBS/GAL
ICE VANES ..... RETRACTED

NOTE: ENDURANCE INCLUDES START TAXI, CLIMB DESCENT, AND 45 MINUTES RESERVE FUEL AT LONG RANGE CRUISE POWER.

EXAMPLE:

PRESSURE ALTITUDE ..... 24,000 FT
ENDURANCE @
HIGH SPEED CRUISE POWER ..... 2.93 HRS
ENDURANCE @
LONG RANGE CRUISE POWER ..... 5.41 HRS



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ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-12	-15	2230	519	519	180	171	182	173	184	175
2000	-16	-19	2230	507	507	178	174	180	176	182	178
4000	-20	-23	2230	494	494	176	177	178	179	180	181
6000	-24	-27	2230	481	481	173	179	176	182	178	184
8000	-28	-31	2230	468	468	171	182	173	185	176	187
10,000	-32	-35	2230	456	456	168	185	171	188	173	190
12,000	-35	-39	2230	447	447	166	187	169	191	171	193
14,000	-39	-43	2230	441	441	163	190	166	193	169	196
16,000	-43	-47	2230	436	436	160	192	163	196	166	200
18,000	-47	-51	2128	413	413	152	189	157	195	160	199
20,000	-51	-55	2010	390	390	144	185	150	192	154	197
22,000	-55	-59	1874	364	364	132	176	141	187	146	194
24,000	-60	-63	1677	326	326	—	—	127	174	136	186
25,000	-62	-65	1603	311	311	—	—	—	—	129	180

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ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA - 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	-2	-5	2230	522	522	179	173	181	175	183	177
2000	-6	-9	2230	507	507	176	176	179	178	181	180
4000	-10	-13	2230	494	494	174	178	176	181	178	183
6000	-14	-17	2230	481	481	171	181	174	184	176	186
8000	-18	-21	2230	469	469	169	184	172	187	174	189
10,000	-21	-25	2230	458	458	166	186	169	190	172	192
12,000	-25	-29	2230	448	448	163	189	167	192	169	195
14,000	-29	-33	2230	442	442	160	191	164	195	167	198
16,000	-33	-37	2141	422	422	154	189	158	194	161	198
18,000	-37	-41	2034	399	399	146	186	151	192	155	197
20,000	-41	-45	1932	378	378	137	180	144	189	149	196
22,000	-45	-49	1820	355	355	—	—	136	184	142	193
24,000	-49	-53	1686	328	328	—	—	—	—	132	186
25,000	-51	-55	1610	313	313	—	—	—	—	127	181

ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA - 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	8	5	2230	524	524	177	175	179	177	181	179
2000	4	1	2230	510	510	175	177	177	180	179	182
4000	0	-3	2230	496	496	172	180	175	183	177	185
6000	-4	-7	2230	482	482	170	183	172	186	175	188
8000	-8	-11	2230	470	470	167	185	170	188	172	191
10,000	-11	-15	2230	457	457	164	188	167	191	170	194
12,000	-15	-19	2230	448	448	161	190	165	194	167	197
14,000	-19	-23	2141	427	427	155	188	159	193	162	197
16,000	-23	-27	2037	405	405	147	185	152	191	156	196
18,000	-27	-31	1938	384	384	139	181	145	189	150	195
20,000	-31	-35	1847	364	364	129	173	138	186	144	193
22,000	-36	-39	1737	342	342	—	—	128	179	136	189
24,000	-39	-43	1619	317	317	—	—	—	—	126	182
25,000	—	—	—	—	—	—	—	—	—	—	—

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ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	18	15	2230	527	527	176	176	178	179	180	181
2000	14	11	2230	513	513	173	179	176	181	178	184
4000	10	7	2230	499	499	171	182	173	184	175	187
6000	6	3	2230	485	485	168	184	171	187	173	190
8000	3	-1	2230	471	471	165	187	168	190	171	193
10,000	-1	-5	2230	458	458	162	189	166	193	168	196
12,000	-5	-9	2153	435	435	157	188	160	193	163	196
14,000	-9	-13	2053	412	412	149	186	154	191	158	196
16,000	-13	-17	1950	389	389	141	181	147	189	151	194
18,000	-17	-21	1846	368	368	131	174	131	174	145	192
20,000	-22	-25	1759	350	350	—	—	131	180	138	190
22,000	-25	-29	1671	331	331	—	—	—	—	131	185
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA +10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	28	25	2230	529	529	174	178	177	180	179	182
2000	24	21	2230	515	515	172	180	174	183	176	185
4000	20	17	2230	500	500	169	183	172	186	174	188
6000	16	13	2230	486	486	166	186	169	189	172	192
8000	13	9	2208	469	469	163	187	166	191	169	194
10,000	9	5	2105	440	440	156	185	160	190	163	193
12,000	5	1	2013	414	414	149	183	154	188	157	193
14,000	1	-3	1940	395	395	142	180	148	187	152	193
16,000	-4	-7	1861	375	375	134	176	142	185	147	192
18,000	-8	-11	1775	355	355	122	166	134	181	140	190
20,000	-12	-15	1693	358	358	—	—	125	175	133	187
22,000	—	—	—	—	—	—	—	—	—	—	—
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

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ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA +20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	38	35	2230	532	532	173	179	175	182	177	184
2000	34	31	2230	517	517	170	182	173	185	175	187
4000	30	27	2223	501	501	167	184	170	187	173	190
6000	26	23	2145	475	475	162	184	165	187	168	190
8000	22	19	2048	446	446	155	182	159	186	162	190
10,000	18	15	1951	417	417	148	179	153	185	156	189
12,000	14	11	1870	394	394	141	176	147	183	151	188
14,000	10	7	1810	376	376	133	172	141	182	146	188
16,000	6	3	1751	359	359	—	—	134	179	141	187
18,000	2	-1	1690	343	343	—	—	127	175	135	186
20,000	-2	-5	1626	327	327	—	—	—	—	128	182
22,000	—	—	—	—	—	—	—	—	—	—	—
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA +30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 25 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 15 LBS/HR/ENG.

NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	48	45	2140	521	521	168	177	171	180	173	183
2000	44	41	2092	499	499	163	178	167	181	169	184
4000	40	37	2050	478	478	159	178	163	182	166	185
6000	36	33	1986	453	453	154	178	158	182	161	186
8000	32	29	1897	425	425	147	175	152	181	156	185
10,000	28	25	1808	397	397	139	171	145	179	150	184
12,000	24	21	1739	375	375	131	167	139	177	144	183
14,000	20	17	1682	358	358	—	—	133	174	139	183
16,000	16	23	1631	343	343	—	—	125	171	134	181
18,000	12	9	1597	329	329	—	—	—	—	128	180
20,000	—	—	—	—	—	—	—	—	—	—	—
22,000	—	—	—	—	—	—	—	—	—	—	—
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

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ONE-ENGINE-INOPERATIVE MAXIMUM CRUISE POWER

1900 RPM

ISA +37°C

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NOTE: IOAT, TORQUE AND FUEL FLOW BASED ON 11,000 POUNDS.

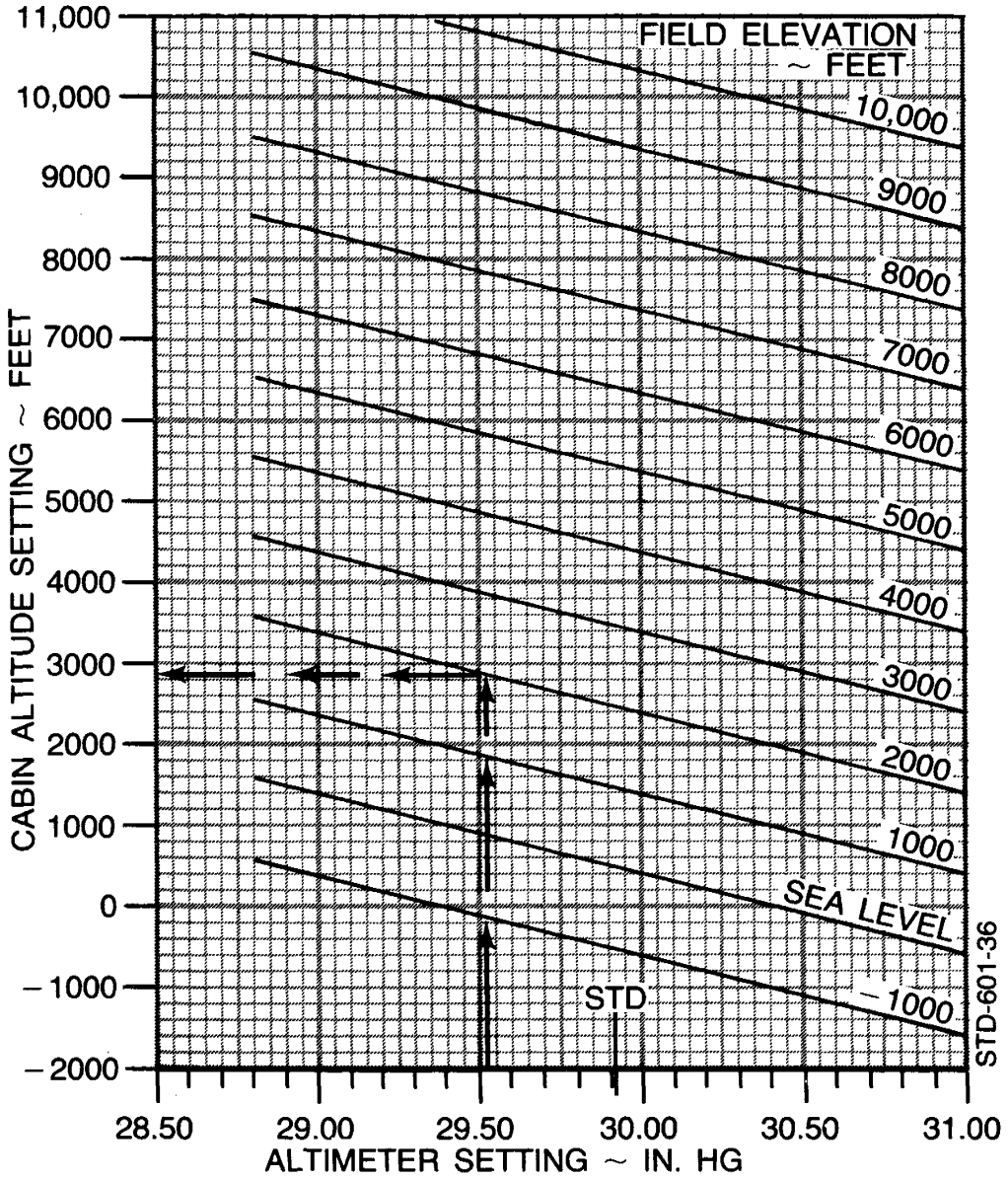
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENGINE	FUEL FLOW PER ENGINE	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS					
						@ 12,000 LBS		@ 11,000 LBS		@ 10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS
0	55	52	2002	502	502	162	173	165	176	168	179
2000	51	48	1960	481	481	157	173	161	177	164	180
4000	47	44	1927	461	461	153	174	157	178	160	182
6000	43	40	1870	438	438	148	173	152	178	156	182
8000	39	36	1790	411	411	140	170	146	176	150	182
10,000	35	32	1709	384	384	132	164	139	174	145	180
12,000	31	28	1650	363	363	—	—	133	172	139	179
14,000	27	24	1598	347	347	—	—	126	168	134	178
16,000	23	20	1560	332	332	—	—	—	—	128	177
18,000	—	—	—	—	—	—	—	—	—	—	—
20,000	—	—	—	—	—	—	—	—	—	—	—
22,000	—	—	—	—	—	—	—	—	—	—	—
24,000	—	—	—	—	—	—	—	—	—	—	—
25,000	—	—	—	—	—	—	—	—	—	—	—

PRESSURIZATION CONTROLLER SETTING FOR LANDING

EXAMPLE

ALTIMETER SETTING ..... 29.52 IN. HG
LANDING FIELD ELEVATION ... 2000 FT

CABIN ALTITUDE SETTING .... 2885 FT



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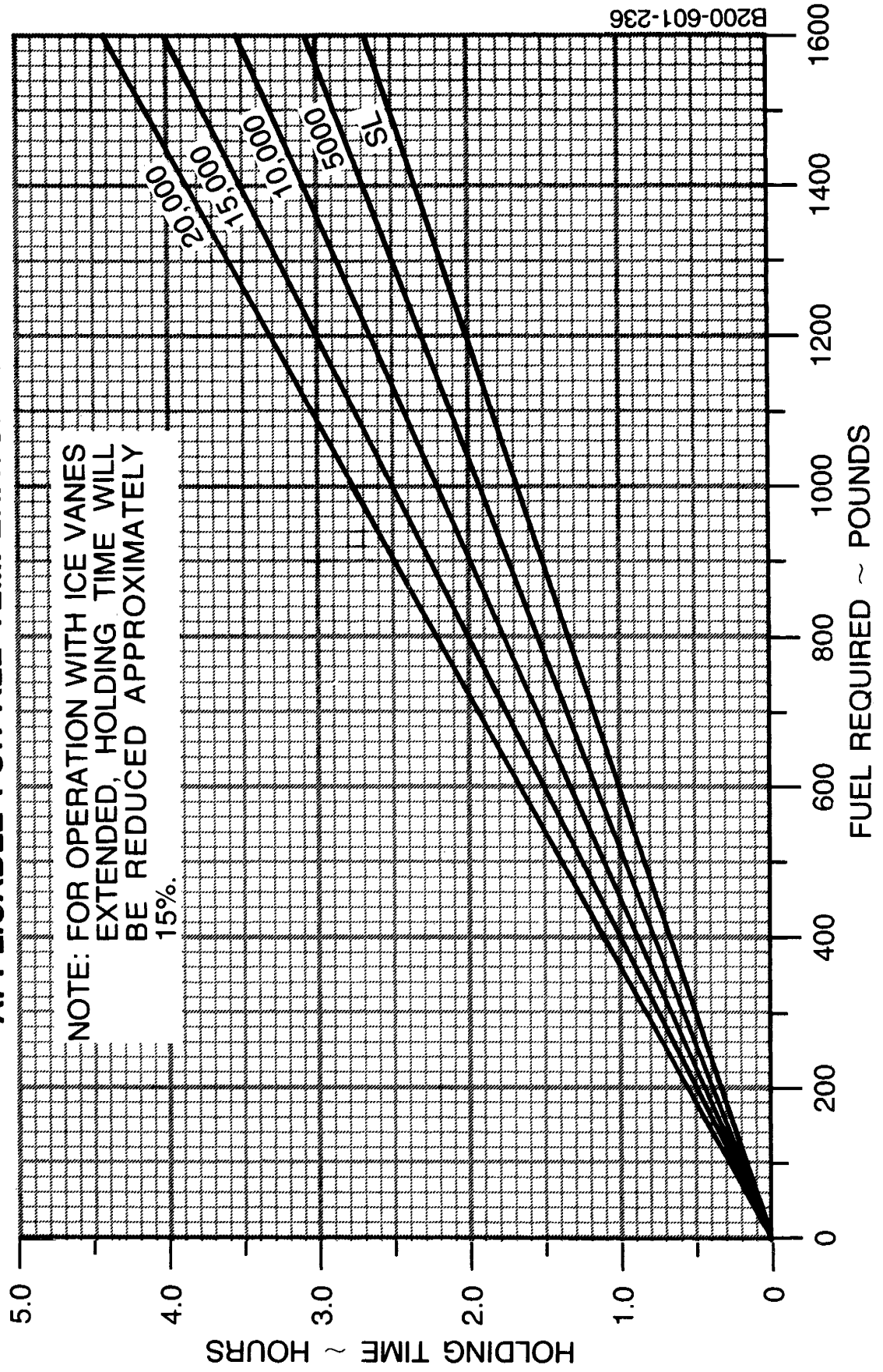
HOLDING TIME

POWER SETTING
800 FT-LBS AT 1700 RPM
APPLICABLE FOR ALL TEMPERATURES



TIME, FUEL, AND DISTANCE TO DESCEND

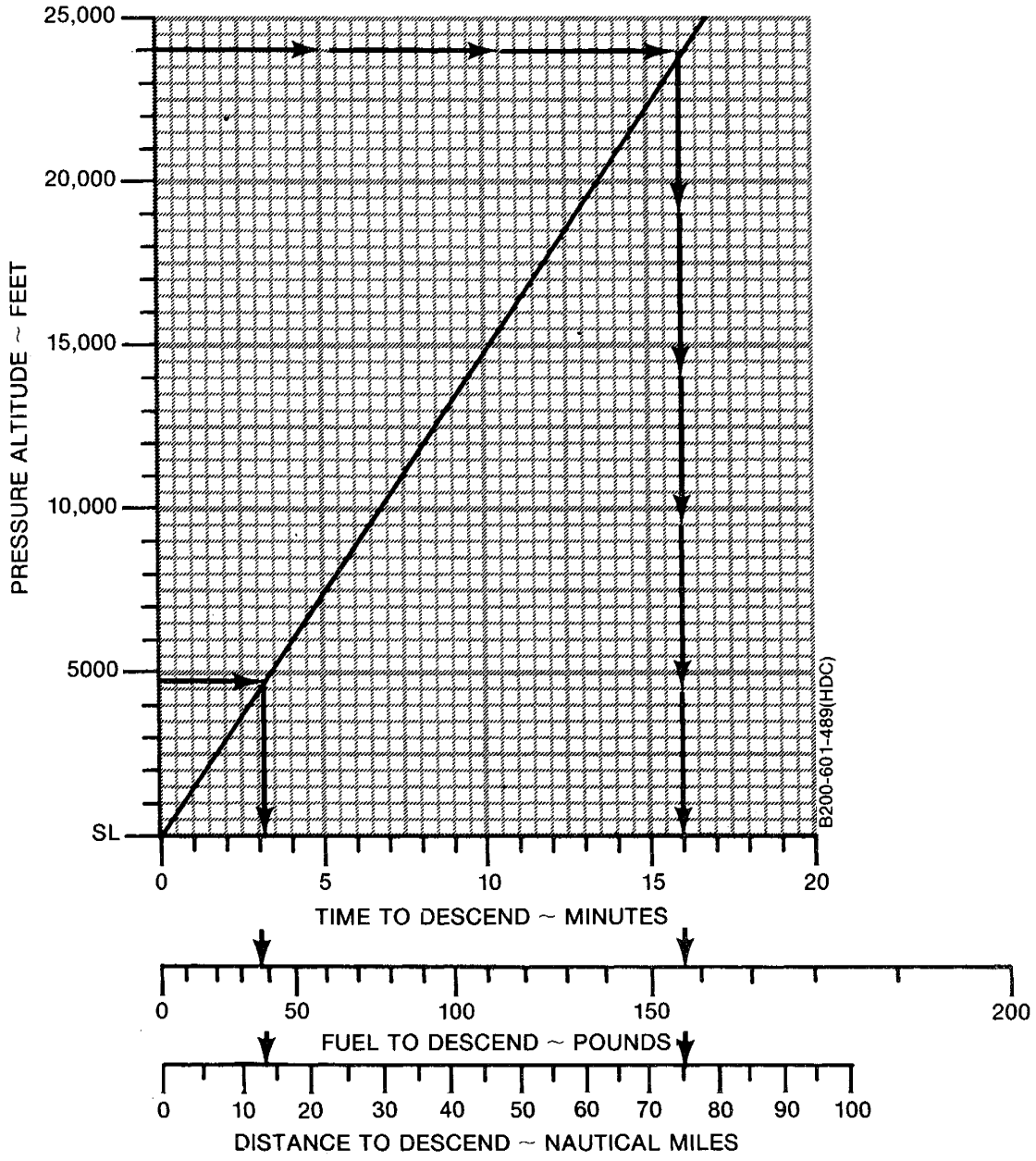
ASSOCIATED CONDITIONS:

POWER ..... AS REQUIRED TO DESCEND AT 1500 FT/MIN
GEAR ..... UP
FLAPS 0%

EXAMPLE:

INITIAL ALTITUDE	24,000 FT
FINAL ALTITUDE	4732 FT
TIME TO DESCEND (16 – 3)	13 MIN
FUEL TO DESCEND (156 – 36)	120 LBS
DISTANCE TO DESCEND (76 – 13)	63 NM

DESCENT SPEED: M_{MO} OR 250 KNOTS, WHICHEVER IS LESS



CLIMB – BALKED LANDING

ASSOCIATED CONDITIONS:

POWER ..... TAKE-OFF
FLAPS 100%
LANDING GEAR ..... DOWN

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
12,500	98
12,000	96
11,000	95
10,000	95
9000	95

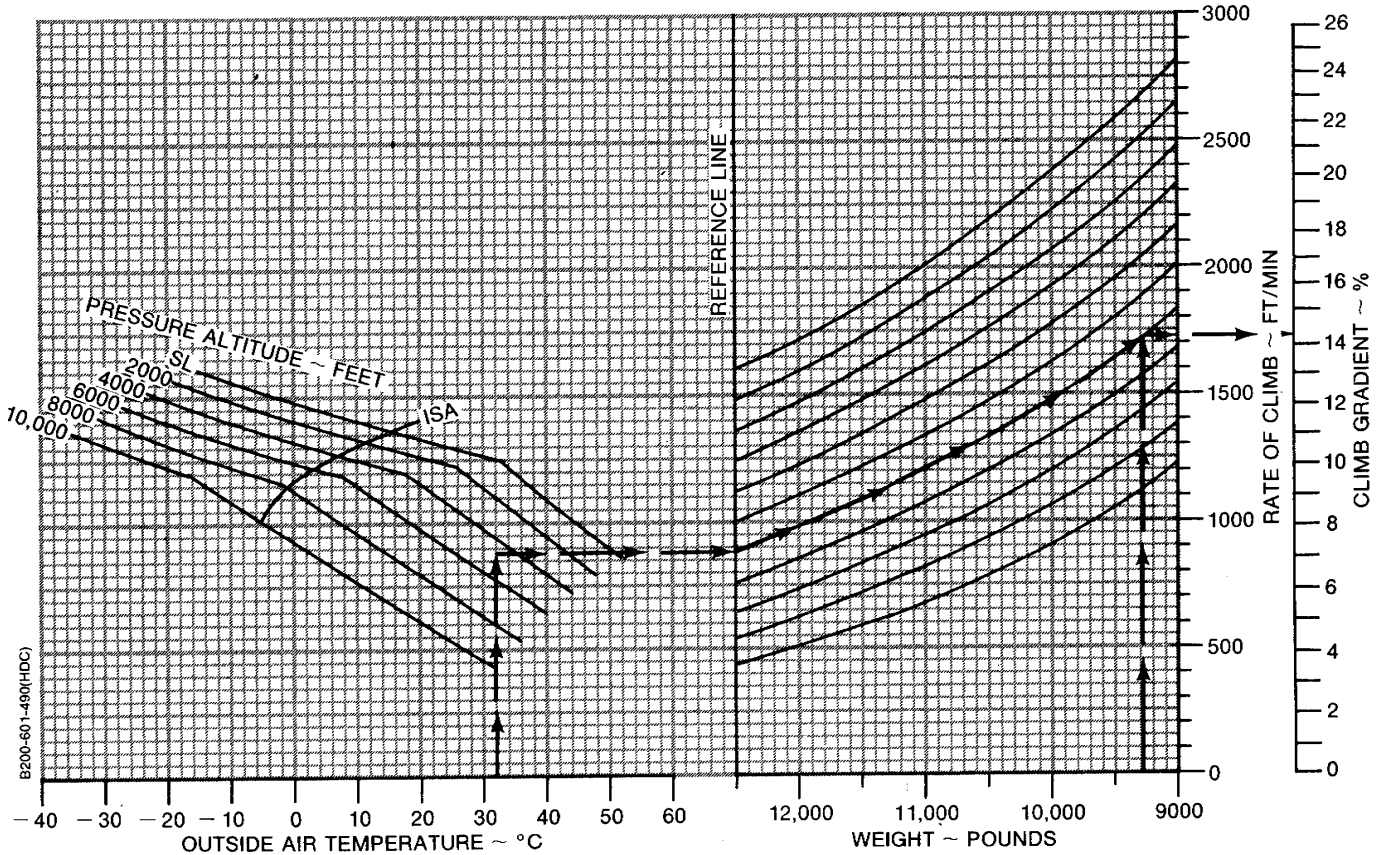
EXAMPLE:

OAT ..... 32°C
PRESSURE ALTITUDE ..... 4732 FT
WEIGHT ..... 9271 LBS

RATE-OF-CLIMB ..... 1725 FT/MIN
CLIMB GRADIENT 14.2%
CLIMB SPEED ..... 95 KTS

NOTES: 1. FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO ACTUAL OAT BEFORE ENTERING GRAPH.

2. ENTER GRAPH TO ALTITUDE FROM WHICH A BALKED LANDING WOULD BE EXECUTED.



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LANDING FIELD LENGTH — FLAPS 100%

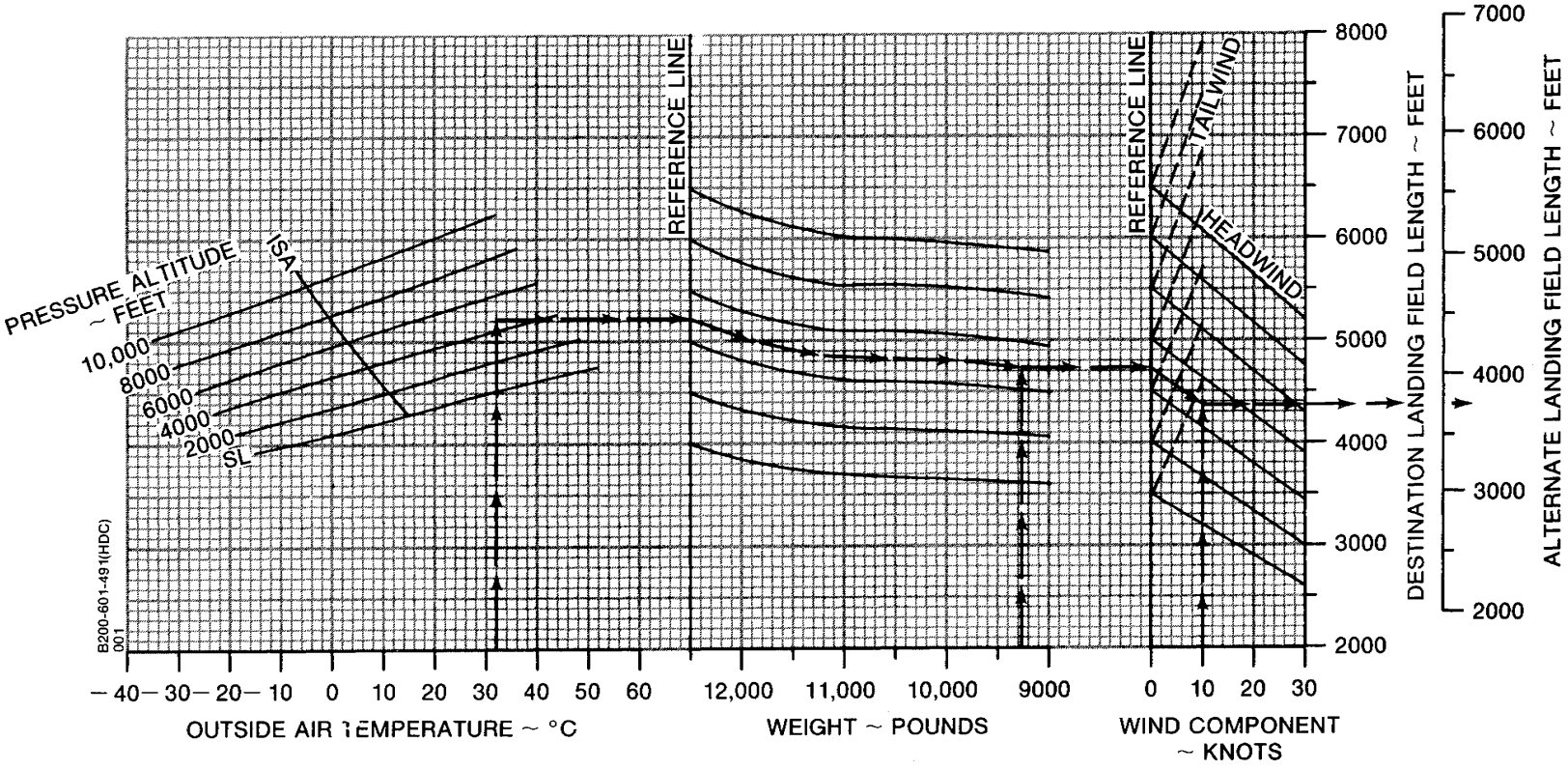
ASSOCIATED CONDITIONS:

POWER	AS REQUIRED TO MAINTAIN 800 FT/MIN ON FINAL APPROACH
FLAPS	100%
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS AS TABULATED
BRAKING	MAXIMUM
OBSTACLE HEIGHT	50 FT

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
12,500	98
12,000	96
11,000	95
10,000	95
9000	95

EXAMPLE:

OAT	32°C
PRESSURE ALTITUDE	4732 FT
LANDING WEIGHT	9271 LBS
HEADWIND COMPONENT	10 KTS
DESTINATION FIELD LENGTH	4375 FT
ALTERNATE FIELD LENGTH	3750 FT
APPROACH SPEED	95 KTS



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LANDING FIELD LENGTH – FLAPS 0%

ASSOCIATED CONDITIONS:

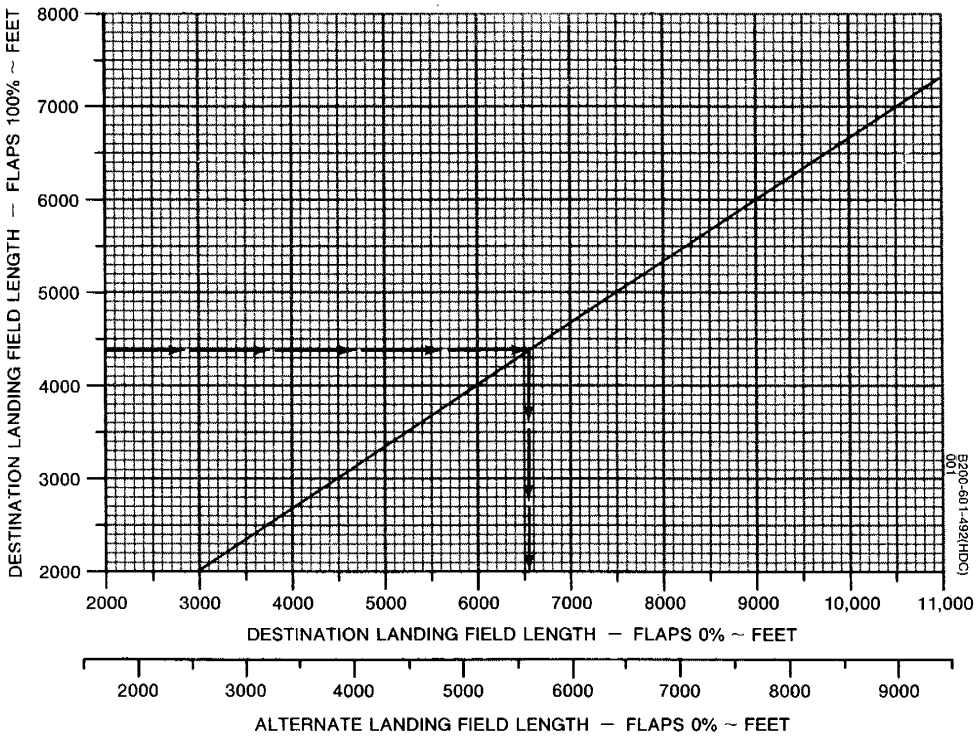
POWER	AS REQUIRED TO MAINTAIN 900 FT/MIN ON FINAL APPROACH
PROPELLER CONTROLS	FULL FORWARD
FLAPS	0%
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS TABULATED
BRAKING	MAXIMUM
OBSTACLE HEIGHT	50 FT

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
12,500	127
12,000	123
11,000	117
10,000	112
9000	106

EXAMPLE:

FLAPS-100% LANDING	
DESTINATION FIELD LENGTH	4375 FT
LANDING WEIGHT	9271 LBS
FLAPS-0% LANDING	
DESTINATION FIELD LENGTH	6550 FT
ALTERNATE FIELD LENGTH	5605 FT
APPROACH SPEED	108 KTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN (100%) IS NORMAL PROCEDURE. USE THE GRAPH BELOW WHEN IT IS NECESSARY TO LAND WITH FLAPS UP (0%).
2. TO DETERMINE FLAPS-0% LANDING FIELD LENGTH, READ FROM THE "LANDING FIELD LENGTH – FLAPS 100%" GRAPH THE DESTINATION LANDING FIELD LENGTH APPROPRIATE TO OAT, ALTITUDE, WEIGHT, AND WIND. THEN ENTER THE GRAPH BELOW WITH THE DERIVED VALUE AND READ THE FLAPS-0% LANDING DISTANCE.



WEIGHT & BALANCE/EQUIPMENT LIST

INTRODUCTION

Except for the weight & balance information specified herein, all weight & balance information published in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual applies.

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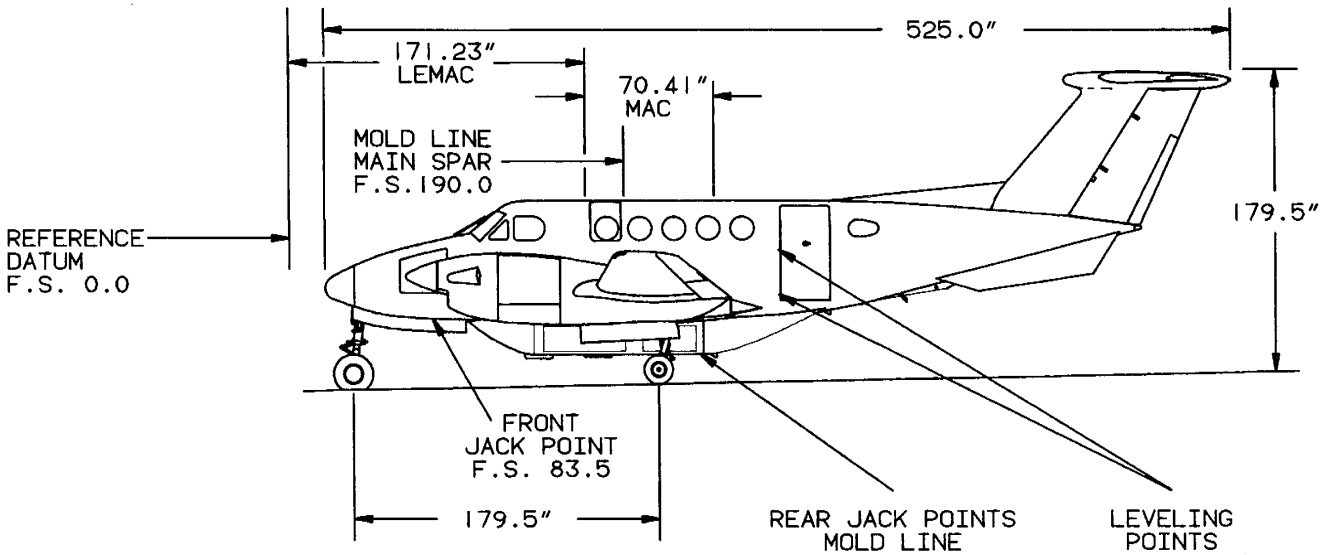
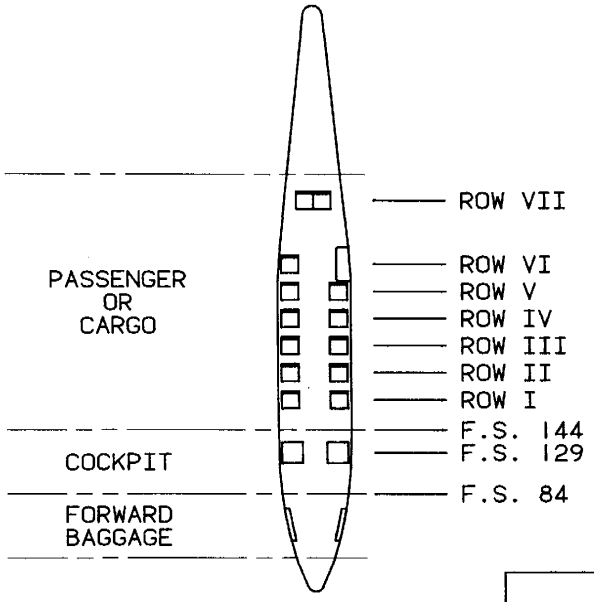
DIMENSIONAL AND LOADING DATA





PASSENGER		CENTROID	
		LEFT	RIGHT
COPILOT	F.S.	---	129
ROW I	F.S.	162	162
ROW II	F.S.	188	188
ROW III	F.S.	216	217
ROW IV	F.S.	243	245
ROW V	F.S.	270	273
ROW VI	F.S.	---	302
ROW VII	F.S.	330	330

BAGGAGE/CARGO LOCATION	CAPACITY-LBS	CENTROID (FUS STA)	
		AIRPLANES PRIOR TO BB-1376 WITHOUT KIT 101-4053 INSTALLED	BB-1376 & AFTER AND PRIOR SERIALS WITH KIT 101-4053 INSTALLED
NOSE COMPARTMENT	350	70	70
AFT CABIN	510	325	325
POD - FORWARD SECTION	260	165	156
POD - AFT SECTION	195	214	223

NOTE;

1. For compartment loadings which result in only partial utilization of total compartment volume, load items must be distributed or secured in a manner to preclude shifting under normally anticipated operating conditions.

1300-603-001 C

USEFUL LOAD WEIGHTS AND MOMENTS

OCCUPANTS

	PILOT OR	ROW I	ROW II	ROW III		ROW IV		ROW V		ROW VI	ROW VII
	COPILOT	LH & RH	LH & RH	LH	RH	LH	RH	LH	RH	RH	LH & RH
	F.S. 129	F.S. 162	F.S. 188	F.S. 216	F.S. 217	F.S. 243	F.S. 245	F.S. 270	F.S. 273	F.S. 302	F.S. 330
WEIGHT	MOMENT/100										
80	103	130	150	173	174	194	196	216	218	242	264
90	116	146	169	194	195	219	221	243	246	272	297
100	129	162	188	216	217	243	245	270	273	302	330
110	142	178	207	238	239	267	270	297	300	332	363
120	155	194	226	259	260	292	294	324	328	362	396
130	168	211	244	281	282	316	319	351	355	393	429
140	181	227	263	302	304	340	343	378	382	423	462
150	194	243	282	324	326	365	368	405	410	453	495
160	206	259	301	346	347	389	392	432	437	483	528
170	219	275	320	367	369	413	417	459	464	513	561
180	232	292	338	389	391	437	441	486	491	544	594
190	245	308	357	410	412	462	466	513	519	574	627
200	258	324	376	432	434	486	490	540	546	604	660
210	271	340	395	454	456	510	515	567	573	634	693
220	284	356	414	475	477	535	539	594	601	664	726
230	297	373	432	497	499	559	564	621	628	695	759
240	310	389	451	518	521	583	588	648	655	725	792

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USEFUL LOAD WEIGHTS AND MOMENTS

BAGGAGE

WEIGHT	NOSE BAGGAGE COMPARTMENT F.S. 70	AFT CABIN BAGGAGE COMPARTMENT F.S. 325	BAGGAGE POD			
			AIRPLANES PRIOR TO BB-1376 WITHOUT KIT 101-4053 INSTALLED		BB-1376 & AFTER AND PRIOR SERIALS WITH KIT 101-4053 INSTALLED	
			FWD SECTION F.S. 165	AFT SECTION F.S. 214	FWD SECTION F.S. 156	AFT SECTION F.S. 223
	MOMENT/100					
10	7	33	17	21	16	22
20	14	65	33	43	31	45
30	21	98	50	64	47	67
40	28	130	66	86	62	89
50	35	163	83	107	78	112
60	42	195	99	128	94	134
70	49	228	116	150	109	156
80	56	260	132	171	125	178
90	63	293	149	193	140	201
100	70	325	165	214	156	223
150	105	488	248	321	234	335
195	137	634	322	417	304	435
200	140	650	330		312	
250	175	813	413		390	
260	182	845	429		406	
300	210	975				
350	245	1138				
400		1300				
450		1463				
500		1625				
510		1658				

CAUTION

(1) Nose baggage may be required when the aft cabin compartment is utilized.
(2) Load baggage pod with partition in down position except when transporting long objects.
(3) Load aft compartment of pod first when partition is up to accomodate long objects.
(4) See WEIGHT AND BALANCE Section of Pilot's Operating Manual.

WEIGHT AND BALANCE LOADING FORM

SERIAL NO.	REGISTRATION NO.	DATE:	
FLIGHT NO.	FROM:	TO:	CAPTAIN:

PASSENGER				BAGGAGE				
LOCATION	NAME	WEIGHT	MOM/100	WEIGHT	NOSE	AFT CABIN	FWD POD	AFT POD
					MOMENT/100			
	1.							
	2.							
	3.							
	4.							
	5.							
	6.							
	7.							
	8.							
	9.							
	10.							
	11.							
	12.							
	13.							
TOTAL								

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WEIGHT AND BALANCE LOADING FORM (CONT'D)

ITEM	WEIGHT	MOM/100
BASIC EMPTY CONDITION CREW CREW'S BAGGAGE		
OPERATING CONDITION PASSENGERS		
WEIGHT - LESS FUEL & BAGGAGE NOSE COMPARTMENT BAGGAGE AFT CABIN POD - FWD SECTION POD - AFT SECTION		
ZERO FUEL CONDITION FUEL		
RAMP CONDITION LESS FUEL FOR START, WARM UP & TAXI *		
TAKEOFF CONDITION		
LANDING CONDITION		
ZERO FUEL CONDITION FUEL REMAINING AT LANDING		
LANDING CONDITION		

* FUEL FOR START, WARM UP, AND TAXI IS NORMALLY 90 LBS AT AN AVERAGE MOMENT/100 OF 177.

SYSTEMS DESCRIPTION

INTRODUCTION

Except for the systems descriptions specified herein, all systems descriptions published in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual apply.

AIRFRAME

SEATING ARRANGEMENTS

The pilot and copilot seats are mounted in a separate forward compartment. One or two fold-up seats are installed in the aft cabin area. Seating for up to 15 persons, including crew, is available. For additional information, refer to the "Cabin Arrangement Diagram" in Section VI, WEIGHT & BALANCE/EQUIPMENT LIST.

FLIGHT CONTROLS

OPERATING MECHANISMS

Some airplanes have been modified with the installation of a spring interconnect system between the aileron and rudder cables in the wing carry-through area underneath the left cabin floor. This installation provides for an increased fuel imbalance capability as noted under Fuel Management in the LIMITATIONS Section. The presence of the interconnect system can be determined on the ground by applying full right and left rudder pedal deflection. Rotation of the pilot's control wheel in the same direction as rudder deflection (eg., left rudder, left aileron) indicates that the interconnect system is installed.

CABIN BAGGAGE COMPARTMENT

A baggage compartment extends aft of the cabin door to the rear pressure bulkhead when one or both of the aft chairs are not in use. This provides up to 54 cubic feet of space which can accomodate 510 pounds of baggage. Nylon webs are provided for the restraining of loose items. See "Dimensional And Loading Data" and "Cabin Arrangement Diagrams" in Section VI, WEIGHT & BALANCE/EQUIPMENT LIST.

CAUTION

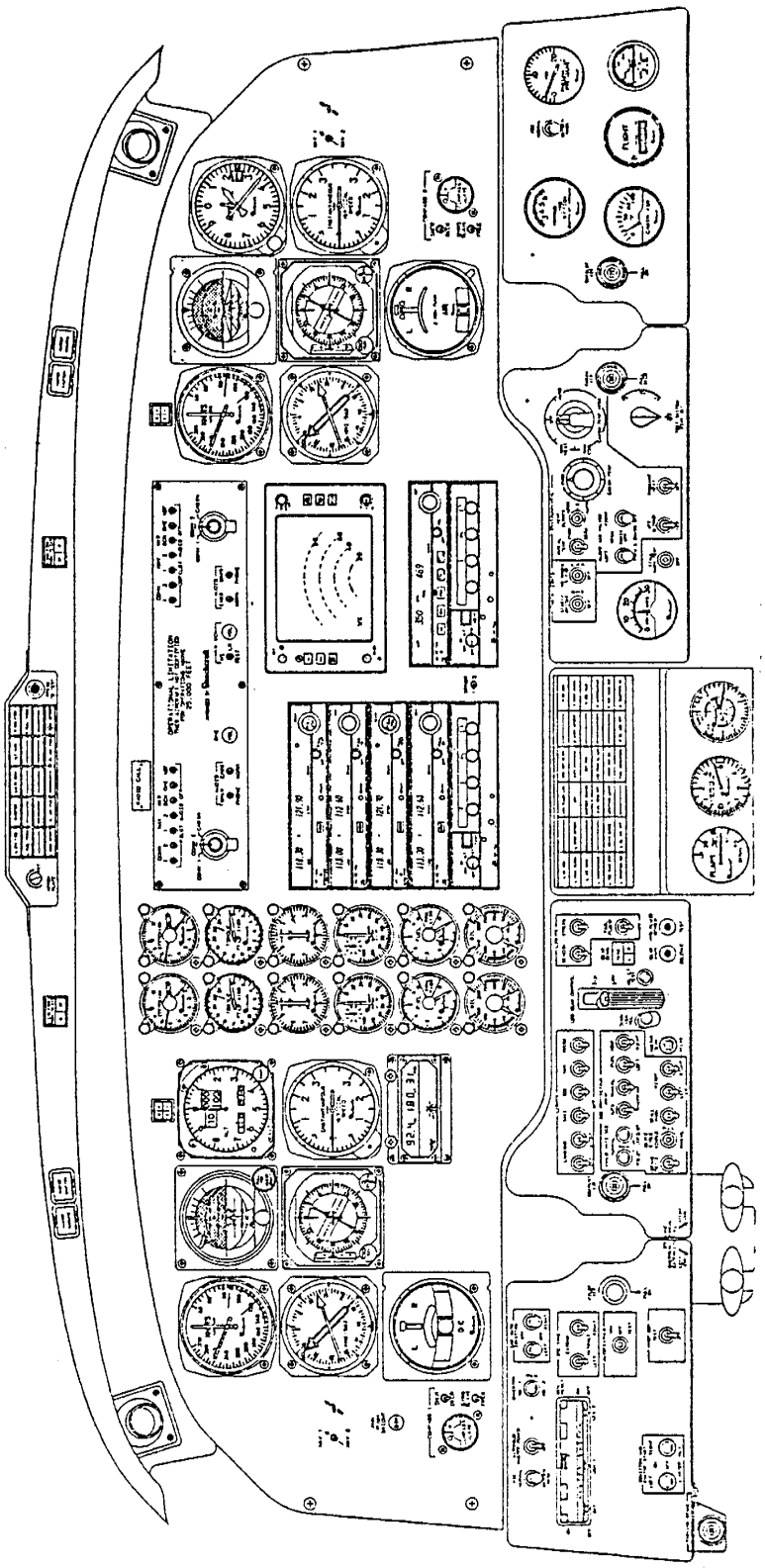
Baggage and other objects should be secured by webs, in order to prevent shifting in turbulent air.

BAGGAGE POD

The baggage pod, located on the underside of the fuselage, provides 56.5 cubic feet of space for loading up to 455 pounds of baggage or cargo. (Baggage pod volume is 44 cubic feet for serials prior to BB-1376 without Kit 101-4053 installed.) Flush-mounted latches secure each door. Interior lighting is installed to automatically illuminate the baggage compartment whenever the doors are open. The POD DOOR OPEN warning light, installed on the warning annunciator panel, informs the pilot when either baggage pod door is open or not secure.

NOSE BAGGAGE COMPARTMENT

A nose baggage compartment is provided in the nose section of the aircraft, forward of the forward pressure bulkhead. Access to this compartment is through an external door on the LH side. The access door has a safety latch to prevent inadvertent opening. The compartment provides 22 cubic feet of space that will accomodate up to 350 pounds of baggage.



TYPICAL INSTRUMENT PANEL

DOORS, WINDOWS, AND EXITS

EMERGENCY EXITS

Two emergency exit doors are provided, one located on the right side of the fuselage at the forward end of the passenger compartment, and a second exit on the left side of the fuselage, located at the forward end of the passenger compartment.

From the inside, either door is released by a pull-down handle, placarded EXIT-PULL. From the outside, either door is released by a flush-mounted, pull-out handle. The nonhinged, plug-type doors remove completely from the frame into the cabin when the latches are released.

AIRSTAIR ENTRANCE DOOR

The cabin entrance door (cabin door) is the same as described in the basic Pilot's Operating Handbook for the B200, except that it is supported by two plastic-encased cables, with the aft cable only serving as a handrail.

To close the door from inside the airplane, utilize the procedures specified in the basic Pilot's Operating Handbook. In addition, upper door hook engagement is checked by viewing through two inspection openings in the headliner just above the fore and aft upper corners of the door. The hook engagement areas are illuminated by depressing the CABIN DOOR HOOK, OBSV LT SW located between the two inspection openings in the headliner.

FUEL SYSTEM

The fuel system consists of two separate systems connected by a valve-controlled crossfeed line. The separate fuel system for each engine is further divided into a main and auxiliary fuel system. The main system consists of a nacelle tank, two wing leading edge tanks, two box section bladder tanks, and an integral (wet cell) tank, all interconnected to flow into the nacelle tank by gravity.

Each main fuel system is filled from the filler port located near the wing tip. Airplanes S/N BB-1376 & after and prior serials with Kit 101-9058 installed have an alternate fill point located near the leading edge at mid-span of the outboard wing panel. The midspan filler port incorporates an index tab positioned at the maximum payload fuel level (e.g. maximum quantity of fuel that can be carried without limiting payload). The mid-span filler cap has an integral sight gauge to indicate presence of fuel at or above the level of the port. If the sight gauge is dark, it is likely that fuel has been added at the outboard filler port.

CAUTION

Do not open mid-span fuel filler cap if sight gauge in cap is dark.

The auxiliary fuel system consists of a center section tank with its own filler opening, and an automatic fuel transfer system to transfer the fuel into the main fuel system.

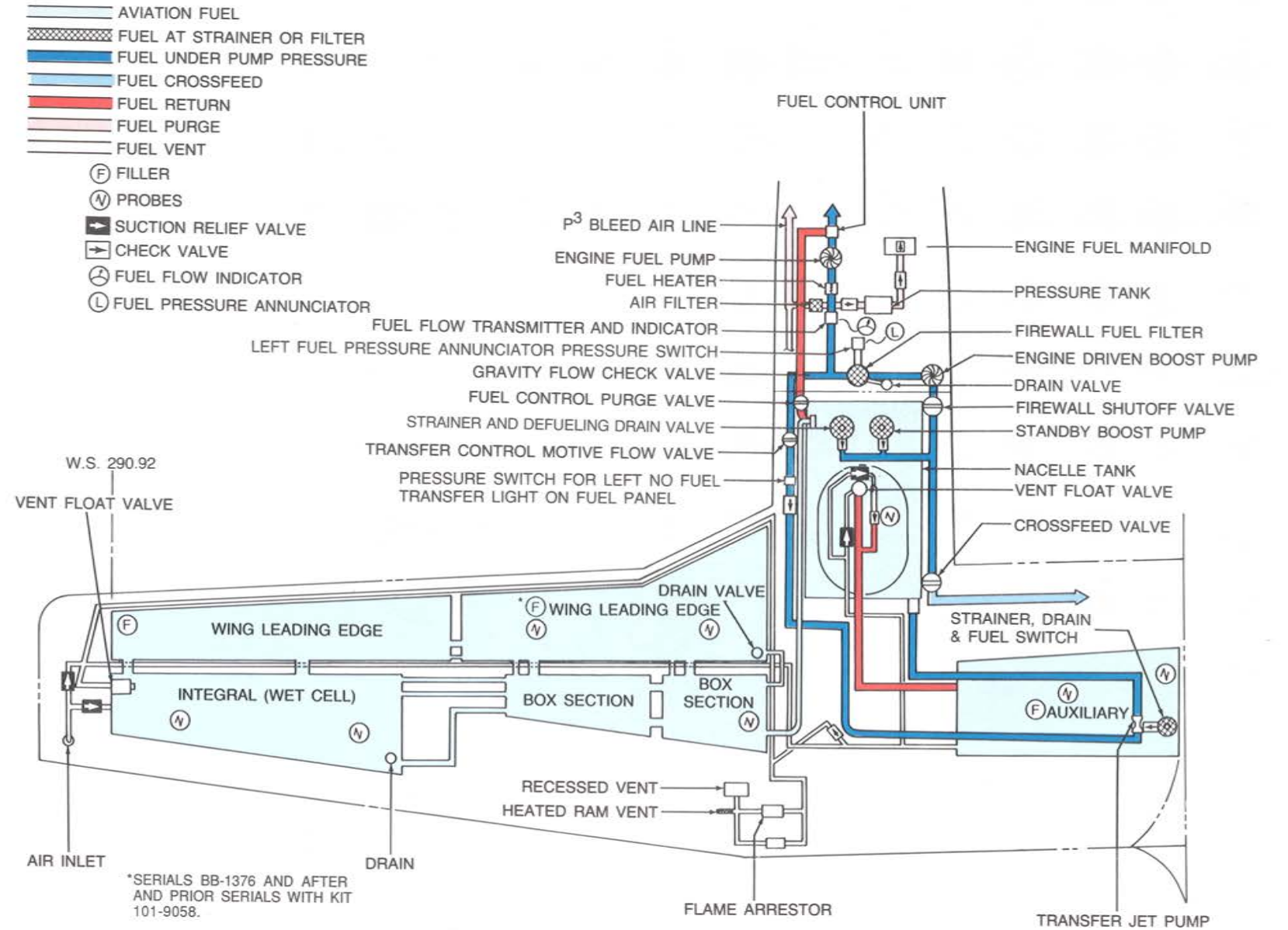
When the auxiliary tanks are filled, they will be used first. During transfer of auxiliary fuel, which is automatically controlled, the nacelle tanks are maintained full. A swing check valve in the gravity feed line from the outboard wing prevents reverse flow. Upon exhaustion of the auxiliary fuel, normal gravity transfer of the main wing fuel into the nacelle tanks will begin.

An anti-siphon valve is installed at each filler port which prevents loss of fuel or collapse of a fuel cell bladder in the event of improper securing or loss of the filler cap.

The two systems are vented through a recessed ram vent coupled to a protruding heated ram vent on the underside of the wing adjacent to the nacelle. One vent is recessed to prevent icing and the protruding vent is added as a backup and is heated to prevent icing.

All fuel is filtered with a firewall-mounted 20 micron filter. These filters incorporate an internal bypass which opens to permit uninterrupted fuel supply to the engine in the event of filter icing or blockage. In addition, a screen strainer is located at each tank outlet before the fuel reaches the boost and transfer pumps. The main engine driven fuel pump has an integral strainer to protect the pump.

A "differential pressure" fuel purge system is provided and is located in the aft compartment of each nacelle. The system purges the fuel that is left in the fuel manifolds at engine shutdown by forcing the fuel into the nozzles so that it is consumed in the combustion chamber.



FUEL SYSTEM SCHEMATIC

B200-603-41(HDC)

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LIGHTING SYSTEMS

EXIT LIGHTING SYSTEM

Two exit lights (one in the forward cabin between the emergency exits, and one in the aft cabin at the airstair door) are installed in the center of the cabin headliner. Each light is equipped with two light sources. One light source, which is illuminated during normal operations, is powered from the airplane's electrical system. The other light source is powered by internal batteries and is controlled by a three-position rocker switch (spring-loaded to the center position) labeled ON-TEST/OFF-RESET. The light illuminates when the switch is momentarily placed in the ON-TEST position. The light extinguishes when the switch is momentarily placed in the OFF-RESET position. An internal "G" switch automatically activates the light if rapid deceleration is sensed.

OXYGEN SYSTEM

The oxygen system is the same as described in the basic Pilot's Operating Handbook except for changes necessitated by commuter operating requirements. AS A RESULT OF THESE CHANGES, THE MAXIMUM OPERATING ALTITUDE HAS BEEN LIMITED TO 25,000 FEET. The oxygen duration chart in the Normal Procedures Section of this supplement reflects these changes and supersedes the chart found in the basic POH.

Specific changes to the oxygen system are as follows:

1. Crew Masks
    a. Changed to non quick don type with continuous flow at 2.7 liters per minute (LPM).
    b. Are unplugged during normal operation. They must be plugged in in order to start oxygen flow.
2. Passenger Masks
    a. Masks are unchanged except for flow rate reduction to 2.7 liters per minute (LPM).
    b. The deployment box in the aft cabin is standard equipment.

AUTOMATIC DEVICES IN THE CONTROL SYSTEM

YAW DAMP

A yaw damp system, if provided, aids in stabilizing the yaw axis of the airplane, and increases ride comfort. The system may be used at any altitude. It should be deactivated for takeoff and landing.

The components include a yaw sensor, amplifier, and control valve. Regulated air pressure from the control valve is directed to the same pneumatic servos used for the rudder boost system. The system is controlled by a YAW DAMP switch adjacent to the RUDDER BOOST switch on the pedestal. In the event the YAW DAMP switch is inadvertently left ON during takeoff or landing, the circuit for the yaw damping system will be interrupted by the left landing gear safety switch while the airplane is on the ground, rendering it inoperative.

HANDLING, SERVICING & MAINTENANCE

INTRODUCTION

Except for handling, servicing & maintenance information specified herein, all handling, servicing, and maintenance information published in the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual apply.

LAMP REPLACEMENT GUIDE

ITEM	*NUMBER*
PASSENGER COMPARTMENT	
Exit Light	425
BAGGAGE POD	
Light	5108WW

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For Training Purposes Only

Beechcraft®

Super King Air® B200, B200C, B200T & B200CT Landplanes
Pilot's Operating Handbook and
FAA Approved Airplane Flight Manual Supplement
for the
King Single or Dual KFC-400
Automatic Flight Control System
Category 1 with
Bendix EFS-10 Electronic Flight Instrument System
(3-Tube EFIS or 4-Tube EFIS) or
King Electro-Mechanical Instrument System
and optional
King KNS-660 Flight Management System

This Supplement is applicable to the following manual(s):

101-590010-147 & 101-590037-29

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane has been modified by installation of the King Single or Dual KFC-400 Automatic Flight Control System, Category 1, with Bendix EFS-10 Electronic Flight Instrument System (3-Tube EFIS or 4-Tube EFIS) or King Electro-Mechanical Instrument System and optional King KNS-660 Flight Management System in accordance with Beech-approved data.

The Bendix Pilot's Operating Manual for the EFS-10 Electronic Flight Instrument System, Control No. ACS5015, Rev. 2, dated February, 1986 or later, is available from Allied Bendix Aerospace, 2100 NW 62nd Street, Ft. Lauderdale, FL 33309. The KFC-400 Flight Control System Pilot's Guide (006-8419-00 1.5K), dated 9/87 or later is available from Bendix/King General Aviation Avionics Division, 400 North Rogers Road, Olathe, KS 66062-1212. It is the responsibility of the owner/operator to maintain the appropriate pilot's manual in current status.

The King Pilot's Guide for the KNS-660 Flight Management System, P/N 006-8407-00, dated December 2, 1985 or later, is available from Bendix/King General Aviation Avionics Division, 400 N. Rogers Road, Olathe, Kansas 66062-1212. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status. The King KNS-660 Flight Management System meets the approval criteria for VOR/DME and VLF/Omega/GPS Area Navigation Systems as established by Advisory Circular 90-45A and Advisory Circular 20-101B. Operation must be in conformity with the King KNS-660 Pilot's Guide, P/N 006-8407-00, dated December 2, 1985 or later revision. The Operational Revision Status (ORS) of the pilot's guide must match the ORS level annunciated on the SELF TEST page.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

GENERAL

The contents of the Bendix CC-2024F Checklist (if installed) is the responsibility of the owner/operator.

AUTOPILOT

1. Do not use autopilot on the Super King Air B200T or B200CT when the optional wing-tip fuel tanks are installed.
2. During autopilot operations, pilot must be seated at the controls with seat belt fastened.
3. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
4. Autopilot operation is prohibited below 200 feet AGL. This includes takeoffs and landings.
5. Do not use propeller in the range of 1750-1850 RPM during ILS approaches.
6. Operation of the KFC-400 must be in conformity with the appropriate pilot's guide.
7. The autopilot preflight check must be satisfactorily completed prior to takeoff or the AP/TRIM PWR switch must be selected OFF.
8. For proper operation in VOR NAV or VOR APR modes, the DME and NAV receivers should be channelled to the same source.

EFIS (if installed)

1. The EFIS preflight check must be satisfactorily completed prior to takeoff.
2. Instrument approaches in the MAP mode are not authorized.
3. Instrument approaches in the 120° sectored format mode are approved for inbound course guidance only.
4. The Display Down mode is approved for use only after the failure of the upper display unit.
5. The Copy On mode is approved for use only after the failure of onside symbol generator (4-tube EFIS).
6. The Pilot and Copilot Attitude and Heading Source select switches (if installed) are to be selected to the same source only after failure of one source.
7. Operation of the EFS-10 must be in conformity with the appropriate pilot's guide.

FLIGHT MANAGEMENT SYSTEM (if installed)

1. Data Base
   - If the data base has expired, IFR navigation is prohibited unless each waypoint and navaid is verified for accuracy by reference to current approved data.
2. VOR/DME Sensor
   a. IFR RNAV approaches are approved for published FAA certified approaches.
   b. The KNS-660 is approved as a two-dimensional VOR/DME RNAV system for enroute, terminal area, and approach operation in accordance with AC 90-45A and is not geographically limited.
3. VLF/Omega Sensor
   a. Only enroute operations in the conterminous United States and Alaska are approved for IFR.
   b. The system is not to be used for navigation in terminal areas or during departures from or approaches to airports or into valleys or between peaks in mountainous terrain or below Minimum Enroute Altitude (MEA).
   c. Additional equipment which would permit navigation appropriate to the available ground facilities must be installed and operating.
   d. IFR flight is not approved based on VLF/Omega/GPS navigation into any area in which such operations could not be approved based on the installed equipment required by paragraph c. This includes over-water flights beyond the range of approved ground navigation facilities.
   e. Auto Update function must be verified or the VLF/Omega/GPS position information must be checked for accuracy (reasonableness) against a visual ground fix or other approved navigation equipment under the following conditions:
      1) Prior to compulsory reporting points during IFR operations when not under radar surveillance or control.
      2) At or prior to arrival at each enroute waypoint during operation on an approved RNAV route.
      3) Prior to and at hourly intervals during operation off of approved RNAV routes.
   f. Navigation shall not be predicated on the use of this system during periods of dead reckoning or when a NAV flag is visible.
   g. Following a period of dead reckoning, invalid operation (NAV flag visible), or operation in the Omega WARN mode, the VLF/Omega/GPS position information must be verified and updated as required by a visual ground fix or by using other approved navigation equipment. Position must also be updated if an accuracy check reveals a navigation error of greater than 2 NM.
4. VOR/DME and VLF/Omega/GPS Blend Sensor
   - The KNS-660 is approved as a means of enroute navigation in accordance with AC 90-45A.
5. VNAV
   a. VNAV RNAV enroute operation is prohibited below 800 feet AGL.
   b. VNAV operation in the terminal area is restricted to the VOR/DME sensor.
   c. VNAV RNAV approach operation is prohibited below the published MDA.
   d. Published approach procedures take precedence over computed VNAV signals.
   e. Approach profile cannot be set up prior to establishment of initial approach altitude.

CAUTION

Before attempting an actual VNAV approach, complete familiarity with set up procedure is necessary.

PLACARDS

Below Pilot's and Copilot's HSI's when separate DME Control Heads are installed:

DME NOT CHANNELLED

BY NAV CONTROL HEAD

EMERGENCY PROCEDURES

ENGINE FAILURE

1. Follow the EMERGENCY ENGINE SHUTDOWN Procedure in the airplane POH/AFM.
2. Disengage the autopilot and yaw damper.
3. Retrim and re-engage the autopilot if desired.
4. Disengage the yaw damper and manually trim the rudder if maximum airplane performance is desired.
5. See ONE-ENGINE-INOPERATIVE LANDING OR GO-AROUND.

CAUTION

If the yaw damper is engaged, rudder trim will operate to center the ball. Do not attempt to help the yaw damper. Rudder trim will run to oppose pilot effort. Disengage the yaw damper and manually trim the rudder if maximum airplane performance is desired.

AIR START

STARTER ASSIST

The procedure found in the POH/AFM should be modified as follows:

NOTE

EFIS power (if installed) should be turned off prior to a STARTER ASSIST air start if conditions permit.

WARNING

Standby Attitude and Air Data displays must be monitored during STARTER ASSIST air starts.

LANDING

ONE-ENGINE-INOPERATIVE LANDING OR GO-AROUND

1. One-Engine-Inoperative procedures remain unchanged from those found in the airplane POH/AFM.
2. Disengage the autopilot and retrim for large variations in speed or power.
3. Disengage the yaw damper and manually trim the rudder if maximum airplane performance is desired.

CAUTION

If the yaw damper is engaged, rudder trim will operate to center the ball. Do not attempt to help the yaw damper. Rudder trim will run to oppose pilot effort. Disengage the yaw damper and manually trim the rudder if maximum airplane performance is desired.

FLIGHT CONTROLS

AUTOPILOT OR TRIM MALFUNCTION (AP DISC OR AP TRIM FAIL ANNUNCIATOR)

An uncommanded deviation in airplane attitude, flight path, or control movement, or the presence of an AP DISC or AP TRIM FAIL annunciator, possibly accompanied by an alert tone, indicates an autopilot or trim malfunction.

1. Maintain airplane control - Manually override the autopilot and/or mistrim as required.

NOTE

Pitch and yaw trim may run to oppose pilot effort.

2. Simultaneously:
3. Disengage the autopilot. DEPRESS and HOLD AP DISC & TRIM INTRPT button on the pilot's or copilot's control wheel.
4. After recovery - Release the AP DISC & TRIM INTRPT button while observing the pitch trim wheel.
5. If the pitch trim wheel is in motion:
   a. AP DISC & TRIM INTRPT Button - DEPRESS and HOLD
   b. AP/TRIM PWR Switch - OFF
   c. AP PWR Circuit Breaker - PULL

CAUTION

When disconnecting the autopilot after a trim malfunction, hold the control wheel and rudder pedals firmly; be prepared to accept out-of-trim control forces.

THE FOLLOWING ARE ALTERNATE MEANS OF DISCONNECTING THE AUTOPILOT:

- Turn off the AP/TRIM PWR switch.
- Push the AP and YD push button on the autopilot controller.
- Operate the manual electric trim on the control wheel. (Yaw damper remains engaged.)
- Pull the AP PWR or FLT DIR circuit breaker on the copilot's right sidepanel.
- Press the GA button. (Yaw damper remains engaged.)
- Press the AP Transfer Switch (Dual FCS only).

MAXIMUM ALTITUDE LOSSES DURING MALFUNCTION TESTS WERE:

1. Cruise ..... 150 ft
2. Maneuvering ..... 120 ft
3. Approach - ILS Coupled ..... 80 ft
4. One-Engine-Inoperative Approach - ILS Coupled ..... 100 ft

AUTOMATIC DISENGAGEMENT OF AUTOPILOT (AP DISC ANNUNCIATOR)

Illumination of the AP DISC annunciator (accompanied by the Master Warning flasher and an aural alert tone) indicates that the monitor has disengaged the autopilot. The pilot should manually control the airplane. Refer to AUTOPILOT ENGAGEMENT FOLLOWING AUTOMATIC DISENGAGEMENT - before attempting to re-engage the autopilot.

AUTOMATIC DISENGAGEMENT OF ELECTRIC TRIM (AP TRIM FAIL ANNUNCIATOR)

Illumination of the AP TRIM FAIL annunciator (accompanied by the Master Warning flasher and an aural alert tone) indicates that the monitor has turned off electric trim (pitch and yaw). The pilot should acknowledge the warning by assuming manual control of the airplane and depressing the AP DISC & TRIM INTRPT switch on the control wheel.

CAUTION

When disconnecting the autopilot after a trim malfunction, hold the control wheel and rudder pedals firmly; be prepared to accept out-of-trim control forces.

SYSTEMS

AUTOPILOT ENGAGEMENT FOLLOWING AUTOMATIC DISENGAGEMENT

The autopilot can be re-engaged following automatic disengagement for certain monitor-detected failures. Prior to reengaging, observe the AP DISC and AP TRIM FAIL annunciators on the glareshield and the R (roll), P (pitch), and Y (yaw) LED annunciators on the Autopilot Control Panel to determine the nature of the failure and appropriate pilot action following re-engagement.

AP DISC, R - ROLL AXIS FAILURE

- Pilot must manually control the roll attitude of the airplane.

AP DISC, P - PITCH AXIS FAILURE

- Pilot must manually control the pitch attitude of the airplane.

AP DISC, Y - YAW DAMPER FAILURE

- Descend to 17,000 ft or below. Manually operate the rudder as required.

AP DISC, AP TRIM FAIL, P - PITCH TRIM FAILURE

- Temporarily disengage the autopilot and retrim for variations in airspeed or power. Be prepared to accept pitch axis trim forces upon disengagement.

AP DISC, AP TRIM FAIL, Y - YAW TRIM FAILURE

- Manually trim the rudder as required.

CAUTION

If the pitch or yaw axis and respective trim are engaged, do not help the autopilot in that axis. Trim will run to oppose pilot effort.

EFIS FAILURE

If the EFIS displays are missing or abnormal, refer to alternate instruments for usable data. The offside instruments are approved as backup to onside instruments. (A standby attitude indicator may also be installed.) Refer to EFIS WARNING/ CAUTION FLAGS.

EFIS WARNING/CAUTION FLAGS

CAUTION

Following failure of a red gun in any display tube, red warning flags will not be visible. If EFIS displays are missing or abnormal, refer to alternate instruments for usable data.

RED SG OR LOSS OF EADI AND EHSI - SYMBOL GENERATOR FAILURE

1. (3-tube EFIS)

- Select STBY SG. The EADI, EHSI, and MFD will be driven by the standby symbol generator.

2. (4-tube EFIS)
   - Select COPY/ON. The EADI and EHSI will be driven by the offside symbol generator. COPY appears in yellow on all four affected displays and onside autopilot engagement will be inhibited.

CAUTION

Failure of attitude, heading or navigation sources will affect both pilot's and copilot's displays.

YELLOW SG - SYMBOL GENERATOR COOLING FAILURE

1. (3-tube EFIS)
   - Select STBY SG. The EADI, EHSI, and MFD will be driven by the standby symbol generator.
2. (4-tube EFIS)
   - Select COPY/ON. The EADI and EHSI will be driven by the offside symbol generator. COPY appears in yellow on all four affected displays and onside autopilot engagement will be inhibited.

CAUTION

Failure of attitude, heading or navigation sources will affect both pilot's and copilot's displays.

YELLOW DU-1/DU-2 - EADI/EHSI COOLING FAN FAILURE

If a fan failure is indicated on the ground, do not take off until appropriate repairs are accomplished.

If a fan failure is indicated in flight:

1. Monitor the presentation for an abnormal appearance which will indicate an impending failure.
2. Although an EADI or EHSI failure is very unlikely, consideration should be given to switching to a reversion mode or shutting off the system and flying the offside instruments.
3. System heating can be reduced by lowering the lighting intensity of the presentation.
4. Accomplish repairs as necessary after landing.

LOSS OF EADI - SELECT DISPLAY/DOWN

1. (3-tube EFIS)
   - EADI presentation will be presented on the EHSI. EHSI data will be presented on the MFD regardless of the position of the MFD/FMS switch.
2. (4-tube EFIS)
   - EADI presentation will be presented on the EHSI.

CAUTION

Lateral modes on onside FLT DIR will disengage and cannot be re-engaged until Display/Down is deselected.

RED CP - EFIS CONTROL PANEL FAILURE

1. (3-tube EFIS)
   a. Continue using EADI. Decision height control cannot be set. Use EHSI with caution. Verify validity of displayed data. If control panel functions are inoperable or abnormal, select OFF on the system function select switch:
      1) The red CP annunciation will erase.
      2) "CP OFF" appears in yellow on upper left section of EHSI.

3) The EHSI will revert to Full Compass HSI mode.
4) The active NAV system will revert to VOR-1 or LOC-1.
5) The course pointer will function as an RMI pointer and continuously point to the VOR station selected on VOR-1.
6) If a localizer frequency is selected on NAV-1 receiver, the course pointer will slew to the lubber line on the EHSI and remain there regardless of airplane heading, to allow correct orientation of the left/right deviation indicator for an ILS or localizer approach.
7) For a localizer back course approach, left/right deviation indicator sensing will be reversed because ILS inbound course cannot be selected.
8) The heading bug will initially slew to the existing airplane heading and remain at that setting on the compass card.
9) RMI-1 pointer will revert to ADF-1.
10) RMI-2 pointer will revert to VOR-1.
11) Decision height setting on the EADI will revert to 200 feet.

2. (4-tube System)
   a. If onside control panel fails, set function select switch on failed control panel to OFF. This transfers control to opposite side control panel.
   b. If both control panels are OFF, both go into the default mode. This is the same as that described for the three-tube system except that copilot's pointers are dedicated as follows:
      1) The active NAV system will revert to VOR-2 or LOC-2.
      2) RMI-1 pointer will revert to VOR-1.
      3) RMI-2 pointer will revert to ADF-2 (or TCN).

RED HDG - HEADING FAILURE

Discontinue using EHSI.

NOTE

On airplanes equipped with compass switching, the compass select switch can be used to select a different source and allow continued use of EHSI. Also, the annunciation "HDG" in a yellow box will be displayed on the left side of EHSI when both pilot's and copilot's EFIS systems are single source (i.e., both sides on compass 2 system).

CAUTION

Failures of the remaining source affect all stabilized headings.

NOTE

On airplanes equipped with MAG/TRUE switching, the MAG/TRUE heading source select switch can be used to select a different heading source and allow continued use of the EHSI, if the NAV source selected uses either MAG or True Heading. If True Heading is selected as the source, a white "T" will be displayed by the lubber line on the EHSI. Also, the annunciation "HDG" in a yellow box will be displayed on the left side of EHSI when both pilot's and copilot's EFIS systems are single source (i.e., both sides on True Heading).

CAUTION

Failures of the remaining source affect all stabilized headings.

RED ATTITUDE FAIL - ATTITUDE FAILURE

Discontinue using EADI.

NOTE

On airplanes equipped with attitude switching, the attitude select switch can be used to select a different source and allow continued use of EADI. Also, the annunciation "ATT" in a yellow box will be displayed on the left side of EADI when both pilot's and copilot's are single source (i.e., both sides on Vertical Gyro No. 2).

CAUTION

Failures of the remaining attitude source affect all primary attitude displays.

COURSE/HEADING FLAGGED

Failure of the course pointer, the heading bug, or its external source, is annunciated by a red "X" drawn through the head of the pointer or heading bug.

RMI AND DEVIATION FLAGGED

If the Nav Source receiver is OFF or indicating an internal fault, a double red line will be drawn through the affected deviation scale. The affected RMI pointer will be visible and flagged by a red "X" drawn through the head and tail of the pointer. The affected pointer assignment annunciation will be flagged.

DISTANCE REMOVED

In the event that the VORTAC or DME station is out of range or not in operation, or if for any reason the DME receiver is operational but not providing computed data to the EFS-10 system, no distance or ground speed will be displayed on the EHSI and no flags will be displayed.

DISTANCE FLAGGED

If the DME receiver is indicating an internal fault or turned off, distance and ground speed readout will be flagged.

ALPHANUMERICS FLAGGED

Failure of an external source for an alphanumeric readout, i.e., course, DME, DH, etc., is annunciated by a double red line drawn through the readout.

WEATHER RADAR FLAGGED

In the event of a failure of the weather radar while operating the EHSI in the WX mode, a white annunciation indicating the nature of the failure will be displayed in the lower left section of the EHSI. A WX FAIL message on the EHSI will be accompanied by a clearing of the weather picture. If this occurs in rapid succession it will appear as a flickering of the EHSI. If flickering persists, deselect weather on the EHSI by pressing the WX button on the EFS-10 Control Panel.

CROSS COMPARITOR ANNUNCIATIONS (4-TUBE EFIS)

The EFS-10 System provides cross comparison between the pilot's and copilot's systems and annunciates discrepancies. Discrepancy annunciations are shown as double-ended yellow arrows drawn either through, on top of, or on the side of the specific function. These annunciations include HDG, ATT and GS (both displays), RA (EADI only), and rising runway on the EADI that corresponds to localizer deviation on the EHSI. It should be noted that the GS and LOC comparitors are only active when both pilot's and copilot's sides are tuned to the same ILS frequency.

The following table lists the magnitude of discrepancy and conditions for each cross comparitor annunciation:

FUNCTION	PARAMETER SPECIFICATION	CONDITIONS
HDG	>6° up to a 6° bank, >20° difference	Pilot and copilot both using either magnetic or true north reference.

FUNCTION	PARAMETER SPECIFICATION	CONDITIONS
RA	8 ft at 0 ft altitude	Increasing by 5.6 ft per 100 ft of altitude.
	13.6 ft at 100 ft altitude	
	64 ft at 1000 ft altitude	
	148 ft at 2500 ft altitude	
ATT	>6° roll, 5° pitch	
LOC DEV	Approximately one-half dot on EHSI.	Pilot and copilot select ILS frequency and both select NAV.
GS	Approximately one-half dot on EHSI.	Same as LOC DEV, above.

FLIGHT MANAGEMENT SYSTEM FAILURE

1. If KNS-660 System information is intermittent or lost, utilize the remaining operational navigation equipment as required.
2. KNS-660 Malfunction (when equipped with Bendix EFS-10):
   a. EFIS Controller System Function Select Switch - SELECT NAV
   b. VOR - Information displayed on EHSI
   c. Course Selector - MANUALLY SET
   d. KNS-660 Circuit Breaker - PULL
3. KNS-660 Malfunction (when equipped with King mechanical instruments):
   a. FMS/NAV CTL Switch - PRESS to select NAV CTL (NAV CTL is illuminated).
   b. VOR-1 - Information displayed on HSI.
   c. Course Selector - MANUALLY SET
   d. KNS-660 Circuit Breaker - PULL

NORMAL PROCEDURES

BEFORE ENGINE STARTING

1. AP/TRIM PWR Switch(es) - OFF
2. EFIS ON/OFF Switch(es) (if installed) - OFF

BEFORE TAXI

1. AP/TRIM PWR Switch(es) - ON
2. EFIS ON/OFF Switch(es) (if installed) - ON
3. Autopilot Self-Test
   a. Allow 3-4 minutes for gyros to erect, HDG and ATTITUDE flags clear.
   b. AP DISC and AP TRIM FAIL - Annunciators illuminate upon initial application of AP/TRIM PWR and then extinguish (followed by an audio test tone) after successful completion of the self-test. Allow 60 seconds after gyros are valid.

CAUTION

Taxi with caution. The autopilot temporarily engages the servos during the automatic self-test. Be prepared to overpower the autopilot as required.

NOTE

Illumination of the AP DISC annunciator other than during initial power-up indicates a failure. This failure annunciation will result in power being removed from the roll, pitch, yaw, pitch trim, and rudder trim servos. The flight director may remain functional depending upon the nature of the failure. The continual self-test feature may also inhibit flight director, autopilot and electric trim use without illumination of the AP DISC annunciator.

c. Repeat self-test for offside autopilot using AP Transfer Switch on AP control (if dual FCS is installed).

BEFORE TAKEOFF (RUNUP)

Items marked by an "" may be omitted for a fast turnaround at the discretion of the pilot.*

1. Autopilot - CHECK
   a. AP Transfer Switch - SELECT pilot's side (dual FCS)
   * b. GLARESHIELD ANNUN LAMP TEST or TEST BUTTON on KCI-310 (if installed) - PRESS, observe autopilot control panel mode annunciation.
   c. AP Mode Selector Button - PRESS TO ENGAGE AUTOPILOT.
   d. Flight Controls - OVERPOWER AUTOPILOT in pitch, roll and yaw axes.

WARNING

If unable to overpower the autopilot in any axis, do not use.

   * e. Auto Trim - CHECK
      1) Apply nose up force on control wheel - note nose down trim motion after approximately 3 seconds.
      2) Apply nose down force on control wheel - note nose up trim motion after approximately 3 seconds.
      3) Press right rudder and note left rudder trim motion after approximately 3 seconds.
      4) Press left rudder and note right rudder trim motion after approximately 3 seconds.
   * f. HDG Mode - SELECT HDG MODE and observe FD commands and control wheel motion correspond to movement of the heading selector knob.
   g. AP DISC & TRIM INTRPT - PRESS and RELEASE. Note autopilot disconnection, flashing AP annunciation, and aural disconnect tone.
   * h. Manual Electric Trim - CHECK
      1) Pilot and Copilot Control Wheel Trim Switches - CHECK OPERATION

WARNING

Operation of the electric trim switch system should occur only by movement of pairs of switches. Any movement of the elevator trim wheel while depressing only one switch denotes a system malfunction. The AP/TRIM PWR switch(es) must be turned OFF and flight conducted only by manual operation of the trim wheel.

      2) AP DISC & TRIM INTRPT - DEPRESS and HOLD while running pitch trim. Observe that trim motion is interrupted.
   i. AP TRANSFER - Select copilot's side and repeat items b thru h (dual FCS).
   j. AP TRANSFER SWITCH - SELECT desired side for operation in flight (dual FCS).
   k. Trim Tabs - SET
2. KAV-485 Altimeter - CHECK

NOTE

Airplanes equipped with dual FCS systems have dual Air Data Systems installed with a single KAV-485 and an ADC 1/2 select switch. The ADC 1/2 switch selects desired Air Data source for KAV-485. Source selected is also the same source used by other airplane systems for obtaining altitude information (optional) (i.e., transponder altitude reporting).

a. ADC 1/2 Switch - SELECT 1 (dual FCS)

b. BARO Set Knob - momentarily press to commence Air Data System automatic test.

c. Observe the following:

1) All LED, flags and annunciation momentarily come on.

2) Altitude preselect indicates 28,000 feet. Altitude increases at approximately 6000 feet/min and altitude alert activates at 27,000 feet. Altitude decreases at approximately 6000 feet/min and altitude alert activates at 27,700 feet.

3) ALT, VS, and FAIL momentarily flag.

4) Density altitude momentarily displays in altitude select window.

d. ADC 1/2 Switch - SELECT 2, repeat item b (dual FCS).

e. ADC 1/2 Switch - SELECT desired Air Data source for operation in flight (dual FCS).

3. KCI-310 Flight Command Indicator - CHECK

- Test Button - PRESS and HOLD. Observe the following:

All KFC-400 System mode annunciators should illuminate, including the marker lights on the KA-419 Mode Annunciator. In addition, the artificial horizon should indicate 10° pitch up and 10° right roll.

4. EFS-10 - CHECK

a. Brightness - ADJUST BRIGHTNESS of the EADI(s), EHSI(s), and MFD (if installed) to desired level.

b. Function Select Switch - SELECT TST (on pilot's control panel if 4-tube EFIS installed), observe:

1) TEST IN PROGRESS will briefly display on EADI, EHSI, and MFD (if installed).

2) EADI will then display the full ADI format including glideslope, expanded localizer, turn rate pointer, rising runway, all types of failure and warning annunciations, AUTO TEST VALID, and (on 4-tube EFIS) PUSH SYS TO TEST ATT MON.

3) The EHSI and MFD (if installed), will display standard full compass format with glideslope, RMI pointers, all types of failure and warning annunciation, AUTO TEST VALID, and TST 1.

4) Press SYS button on onside control panel (4-tube EFIS system), observe:

a) Attitude comparitors on offside EADI.

b) TEST COMPLETE on EADI and EHSI.

c) TST 2 on EHSI.

NOTE

Failure annunciations consist of a displayed message surrounded by a red box, double parallel red lines drawn through alphanumeric readouts, and red crosses through pointers. Cooling fan failure annunciation is displayed as a yellow message surrounded by a yellow box. CP and DU 2 occupy the same display area as DU 1 and are not shown in the test mode. Any of the following conditions indicate a malfunction of the EFS-10 system:

- EADI or EHSI blank
- Data or message missing, abnormal, or unsymetrical on EADI or EHSI.
- Absence of a failure or warning annunciation during test.
- TEST IN PROGRESS message remains on EADI or EHSI. This message should only be dispayed for a few seconds at the beginning of the test.
- AUTO TEST FAILED message on EADI or EHSI.
- AUTO TEST VALID message does not appear on both the EADI and EHSI.

c. Repeat item b for copilot's (4-tube EFIS), except TST 2 will be displayed in step 3), TST 1 in step 4).

d. Switching - CHECK OPERATION of the DISPLAY/DOWN, SG 1/STBY SG, MFD/FMS, COPY/ON, ADC 1/2, HDG source, and ATT source switches (if installed).

e. EHSI(s) - SELECT NAV Source, display format, and weather radar as required.

f. MFD (if installed) - SELECT the MFD/FMS switch as desired. In the MFD position, MFD data is controlled by the position of the Function Select Switch. In the FMS postion, the MFD display is dedicated to the KNS-660.

IN FLIGHT

1. Autopilot Operation

a. AP TRANSFER - SELECT as desired (dual FCS).

b. AP Mode Selector Button - PRESS. Note AP, YD, and FD annunciators on. If no other modes are selected, the autopilot will operate in pitch and roll attitude hold.

NOTE

Airplanes with 4-tube EFIS installed will annunciate AP L/R and YD as green on onside EADI and in white on the offside EADI. A Disconnect of the AP (left or right) or YD will be displayed in a flashing Red on both sides.

CAUTION

Do not help the autopilot or yaw damper as the pitch and yaw trim will run to oppose the help.

If the flight director has already been engaged, avoid engaging the autopilot into large unsatisfied flight director commands or large airplane mistrim conditions in any axis.

c. SOFT RIDE Mode Selector Button - PRESS if desired. Note annunciation.

d. HALF BANK Mode Selector Button - PRESS if desired. Note annunciation.

e. Vertical Trim (DN - UP) Control - ACTIVATE and HOLD as desired to continuously trim pitch attitude hold (PAH), ALT, IAS, and VS. ACTIVATE momentarily and RELEASE as desired to discretely trim the above mentioned vertical modes.
Vertical trim will cancel GS, VNAV, CLB and GA and revert to pitch attitude hold.
Vertical trim will cancel DES and revert to VS.
Vertical trim is disabled on initial application while in GS or VNAV ARM modes (to avoid unintentional interruption). Upon reapplication, GS or VNAV will be cancelled.

f. Roll Trim (L - R) Control - ROTATE and POSITION as desired to progressively command left or right roll rate. RELEASE to hold the acquired bank angle.
Roll trim will cancel all lateral modes as well as GS.
Roll trim is disabled on initial application while in NAV or APR ARM mode to avoid unintentional interruption. Upon reapplication, NAV or APR will be cancelled.

g. CWS Button - PRESS and HOLD to allow Control Wheel Steering, (Disengages the pitch, roll and trim servo clutches to allow hand flying and manual trim adjustment.)
Control wheel steering will sync pitch attitude hold (PAH), VS, IAS, ALT and roll attitude hold (RAH).
Control wheel steering will cancel GA and revert to PAH and RAH.

CAUTION

With dual flight control systems, onside control wheel steering affects only the onside flight control system. Depressing the onside CWS button will not disengage the servo clutches if the offside autopilot is engaged. Trim may run to oppose pilot effort.

h. Pitch Attitude Hold (PAH)

1) Vertical Trim Control (DN - UP) - ACTIVATE continuously to slew pitch attitude at a rate corresponding to .13 g's or discretely to program a pitch attitude change of .5 degrees per activation.

2) Control Wheel Steering - ACTIVATE and hand fly to the desired pitch attitude. See CAUTION, item g.

NOTE

In accordance with FAA recommendation (AC00-24A), use of basic PITCH ATTITUDE HOLD is recommended during operation in severe turbulence.

i. CLB Mode Selector Button - PRESS to achieve an indicated airspeed profile. Note CLB annunciation.

- When climb mode is activated, the autopilot smoothly establishes a preselected airspeed. The airspeed command is scheduled as a function of altitude, 160 knots from sea level to 10,000 feet, decreasing at 2 knots per 1000 feet to an airspeed of 130 knots at 25,000 feet, and then decreasing at 1 knot per 1000 feet to an airspeed of 120 knots at 35,000 feet.

j. DES Mode Selector Button - PRESS to initiate a 2000 fpm descent in VS mode. Note DES annunciation followed shortly by VS annunciation and DES extinguishing.
An automatic ALT ARM (Pilot only) is possible if the selected altitude is set to an altitude lower than the present altitude prior to DES mode selection.

k. Vertical Speed Hold (VS)

1) To engage the present vertical speed:

a) If the vertical speed bug is out of view: VS Mode Selector button - PRESS, or VS ENG Button on the KAV-485 (Pilot only) - PRESS. Note annunciation and vertical speed bug (Pilot only) in view.

b) If the vertical speed bug is in view: SYNC Button - PRESS and RELEASE to sync the vertical speed bug to the present vertical speed. VS Mode Selector Button, or VS ENG Button on the KAV-485 - PRESS. Note annunciation (Pilot only).

2) To engage a preselected vertical speed (Pilot only):

a) Vertical Speed/Altitude Select Knob - PULL inner knob out to bring the vertical speed bug into view, and ROTATE in combination with the outer knob to select the desired vertical speed.

b) VS Mode Selector Button or VS ENG Button on the KAV-485 - PRESS. Note annunciation.

3) To adjust the vertical speed after engagement:

a) Vertical Speed Bug - SET using the Vertical Speed/Altitude Select Knob (Pilot only).

b) Vertical Trim (DN - UP) Control - ACTIVATE continuously to slew vertical speed at a rate of approximately 100 fpm per second; or discretely to program a vertical speed change of 100 fpm per activation.

c) Control Wheel Steering - ACTIVATE and hand fly to the desired vertical speed. See CAUTION, item g.

l. Indicated Airspeed Hold (IAS)

1) IAS Mode Selector Button - PRESS to engage. Note annunciation.

2) Vertical Trim (DN - UP) Control - ACTIVATE continuously to slew indicated airspeed at a rate of approximately 1 knot per second; or discretely to program an indicated airspeed change of 2 knots per activation.

m. Vertical Navigation (VNAV) (Optional) - VNAV mode operates in conjunction with the King KNS-660. Prior to vertical path capture, the VNAV scale displayed on the flight instrument is ±10° full scale. After vertical path capture, the scale is ±1000 feet in the enroute mode and ±500 feet in approach. Refer to the KNS-660 Pilot's Guide for complete details regarding operation of the VNAV mode.

n. Altitude Hold (ALT)

1) ALT Mode Selector Button - PRESS to engage. Note annunciation.

2) Vertical Trim (DN - UP) Control - ACTIVATE continuously to slew altitude at a rate of approximately 500 fpm; or discretely to program an altitude change of 20 ft per activation.

3) Control Wheel Steering - ACTIVATE and hand fly to the desired altitude. See CAUTION, item g.

o. Altitude Select (ALT SEL) - Pilot's side only.

1) Vertical Speed/Altitude Select Knob - PUSH inner knob in and ROTATE in combination with the outer knob to select the desired altitude in the Altitude Select Display.

2) Altitude Select (ALT SEL) Mode Select Button (on the KAV-485 or the KMS-446) - PRESS to arm the Altitude Select mode. Note ALT ARM annunciation.

3) Select desired vertical mode to intercept the selected altitude.

p. Heading Hold (HDG)

1) Heading Bug - SET to desired heading using the Heading Selector Knob.

2) HDG Mode Selector Button - PRESS to engage. Note annunciation.

3) To change heading, Heading Bug - RESET.

q. NAV Coupling

1) Course Bearing Pointer - SET to desired course.

2) Airplane - ESTABLISH HEADING necessary to intercept the selected course.

3) NAV Mode Selector Button - PRESS to engage.

a) If the Course Deviation Bar is outside the computed capture point, the airplane will continue in HDG (or wings level if HDG not selected) with HDG and NAV ARM annunciated. When the computed capture point is reached, HDG will disengage, NAV will annunciate and the selected course will be automatically captured and tracked.

b) If the D-Bar is inside the computed capture point, HDG will disengage, the NAV annunciator will illuminate, and the capture/track sequence will automatically begin.

r. Approach (APR) Coupling

1) Course Bearing Pointer - SET to desired course.

a) When using VOR or RNAV navigation sources, the course pointer must be set to the desired course (outbound or inbound).

b) For an ILS, the course pointer is always set to the front course inbound course whether tracking inbound or outbound.

2) Airplane - ESTABLISH HEADING necessary to intercept the selected course.
When a localizer frequency is selected, the intercept heading must be within 105° of the course bearing pointer setting to enable a front course approach.

3) APR Mode Selector Button - PRESS to engage.
The automatic capture/track sequence is similar to that described under NAV coupling.

NOTE

Fly coupled approaches at or above 120 knots, with 40% flaps, gear down.

s. Back Course (BC) Approach Coupling

1) Course Bearing Pointer - SET to the ILS front course inbound heading.

2) Airplane - ESTABLISH HEADING within 75° of the tail of the Course Bearing Pointer to automatically enable the BC logic.

3) APR Mode Selector Button - PRESS to engage.
The automatic capture/track sequence is similar to that described under NAV coupling.

t. Glideslope Coupling

1) APR Mode - ENGAGED.

2) At glideslope centering - NOTE GS annunciator ON.

NOTE

Autopilot can capture glideslope from above or below the beam while operating in either pitch attitude hold, IAS Hold, VS Hold or ALT Hold Modes. If, after glideslope coupling, the glideslope signal becomes inadequate (GS flag in view), the GS annunciator will flash continuously and the system will transfer to pitch attitude hold. If a valid glideslope signal returns, the APR mode must be recycled and the airplane must once again pass through the glideslope beam to achieve glideslope coupling.

Glideslope coupling is inhibited when operating in NAV or APR BC Modes. Glideslope coupling occurs automatically in the APR Mode.

u. Missed Approach

1) Power Lever GA Switch (or GA switch on copilot's control wheel, dual FCS) - PRESS to disengage the autopilot (if engaged) and engage the flight director (if not engaged) in a wings level, pitch up command. Note GA mode annunciator ON.

2) Missed Approach - EXECUTE.

3) Airplane Trim - ESTABLISH.

4) Lateral Guidance (Select one)

a) Heading Mode - SET BUG and PRESS HDG.

b) NAV Mode - PRESS NAV.

c) APR Mode - PRESS APR. Glideslope coupling will be inhibited so that the LOC can be tracked outbound and the autopilot will not couple to false glideslope signals.

v. AP DISC & TRIM INTRPT Switch - PRESS to disengage AP and YD.

2. Flight Director Operation

The flight director modes of operation are the same as those used for autopilot operations except the autopilot is not engaged and the pilot must maneuver the airplane to satisfy the flight director commands.

3. EFIS Operation

Operation of the EFS-10 should be in conformity with the Bendix Pilot's Operating Manual, Control No. ACS5015, Rev. 2, dated February, 1986 or later issue.

NOTE

The autopilot will not allow NAV coupled operation when ADF is selected by the function select switch.

4. Flight Management System Operation

Normal operating procedures are outlined in the KNS-660 Pilot's Guide, P/N 006-8407-00, dated December 2, 1985 or later revisions.

SHUTDOWN AND SECURING

1. EFIS ON/OFF Switch(es) (if installed) - OFF (before FUEL CUTOFF)
2. AP/TRIM PWR Switch(es) - OFF (before FUEL CUTOFF)

PERFORMANCE

No Change

WEIGHT AND BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

FLIGHT CONTROL SYSTEM

The King KFC-400 Digital AFCS is certified in this airplane with 3-axis control; pitch, roll, and yaw. The various instruments and the controls for the operation of the KFC-400 System are described as follows:

The KFC-400 Digital AFCS has an electric pitch trim system which provides pitch autotrim during autopilot operation and manual electric trim for the pilot and copilot when the autopilot is not engaged. The yaw damper system employs an automatic rudder trim feature whenever the yaw damper is engaged.

A lockout device prevents autopilot engagement unless the system successfully completes a self test.

The KFC-400 AFCS is compatible with the Bendix EFS-10. Refer to the Bendix EFS-10 Pilot's Operating Manual and KFC-400 Pilot's Guide for a description of the Electronic Flight Instrument System and Flight Control System.

The following conditions will cause the autopilot and/or flight director to automatically disengage (note that the flight director may be used independently of the autopilot; however, the autopilot may not be used without the flight director):

1. Power Failure.
2. Internal Flight Control System Failures - Internal flight control system failures may be annunciated in one of several ways.

Instrument flags, the AP DISC, and AP TRIM FAIL annunciators on the glareshield, and the individual roll, pitch and yaw servo loop monitor lights on the KMC-440 autopilot mode controller will annunciate monitored failures or exceeded performance limits and may disconnect the autopilot. The illumination of an annunciator should be investigated to determine the extent of the disability prior to re-engagement. The single illumination of a roll, pitch or yaw servo loop monitor light will disable that axis alone; however, the combined failures of the roll and pitch axes will inhibit further engagement. Failure of pitch trim will inhibit autopilot engagement. Failure of yaw trim will inhibit yaw damper engagement. Servo loop failures will not inhibit continued flight director operation.

3. A loss of compass valid (HDG flagged) defaults the autopilot and flight director to roll attitude hold (wings level).
4. The following sensor inputs will cause the autopilot and/or yaw damper to disengage (unless the CWS switch is depressed) or will inhibit engagement.
   a. Roll rates in excess of 15° per second.
   b. Pitch rates in excess of 5° per second.
   c. Bank angles in excess of 40°.
   d. Pitch attitudes in excess of ±20°.
   e. Vertical accelerations outside of a 0.3g to 1.7g envelope (1.0g being normal straight and level flight).
   f. Yaw rates in excess of 9° per second will cause the yaw damper to disengage.
   g. Lateral accelerations in excess of 0.5g's will cause the yaw damper to disengage.

Flight director modes may be automatically cancelled in favor of a default mode if the conditions required for continued safe operation in that mode are no longer met. For instance, the vertical modes of GS and VNAV will default to pitch attitude hold after six seconds of invalid data. Lateral modes such as NAV and APR will default to roll attitude hold (in a wings level attitude) after fifteen seconds of invalid data. All modes automatically disengaged will be annunciated by the continued flashing of the cancelled mode annunciation until the pilot (1) presses the subject mode selector button to cancel the flashing annunciation (it requires a second press of the button to attempt re-engagement of the mode) or (2) chooses an alternate mode.

Any mode that results in excessive airspeed will default to IAS hold at Vmo/Mmo plus 5 knots and reduce the airspeed to V_{MO} M_{MO}

Autopilot disengagement, whether manual or automatic, will always be accompanied by a flashing AP annunciation (approximately six seconds) and an aural tone (approximately two seconds). The Master Warning flasher and AP DISC or AP TRIM FAIL warning may be present on automatic disconnects.

EFIS

The Bendix EFS-10 Electronic Flight Instrument System consists of identical and interchangable electronic Display Units (DU's), two Symbol Generators (SG's), Main EFIS Control Panel(s), one Auxiliary Course/Heading Control Panel (Electro-Mechanical System), and a Radar Control Panel (if installed).

Each Main Control Panel contains the NAV System Selector Switch, the EHSI Format Select Buttons, and concentric Bright/Off knobs. The Auxiliary Control Panel contains the pilot's heading (HDG) knobs, Course (CRS) knob, and the radar altimeter Decision Height (DH) set knobs.

Two electronic Display Units are mounted one above the other in the pilot's (or copilot's, 4-tube EFIS) instrument panel. The top DU serves as an Electronic Attitude Director Indicator (EADI). The lower DU is the Electronic Horizontal Situation Indicator (EHSI). The third tube (3-tube EFIS) is mounted in the center panel and serves as a Multifunction Display (MFD) and Radar Indicator.

The two remotely mounted Symbol Generators accept all pertinent signals from associated systems and sensors. This information is processed and presented on the EADI, EHSI, and MFD Display Units. The Symbol Generators also provide navigational data to the Autopilot/Flight Director.

Refer to the Bendix EFS-10 Pilot's Operating Manual for a more complete system description.

FLIGHT MANAGEMENT SYSTEM

The King KNS-660 is an integrated long range multi-sensor flight management system. It may be used to manage the entire range of navigational functions including trip planning, long range great circle navigation, and instrument approaches. The system consists of a cockpit mounted Control Display Unit (CDU) and a remote-mounted Navigation Computer. The KNS-660 serves as a computer processor/display for inputs from a choice of sensors such as VOR/DME, VLF/Omega/GPS, INS, TACAN, or TAS.

The KNS-660 incorporates a King Data Base which must be updated every 28 days via diskette and a KDL-589 Data Loader. In addition, a self generated user data base may be loaded and maintained by the owner/operator through the CDU keyboard independent of the King Data Base. This user generated data base consists of 175 storage locations accommodating any combination of ground stations and/or airports. If required, any number of these 175 user generated waypoints can be secured against cockpit alteration.

The KNS-660 will support 100 flight plans consisting of 25 alpha/numeric designated waypoints. These flight plans may be composed of any combination of waypoints taken from the data base or the user generated waypoints.

Dedicated special function keys are physically and visually separated from the alphanumeric keys used in the generation and storage of flight plans and waypoints.

The KNS-660 provides for manual operation to selected waypoints or automatic operation providing uninterrupted navigation throughout a complete flight plan. Navigation sensors may be selected separately or blended within the computer.

On airplanes so equipped, the KNS-660 interfaces with the KRG-358 Radar Graphics Unit or EFS-10 MFD, providing a radar graphics presentation on the weather radar display. This includes such items as flight plan waypoints, selected course, and reference ground station.

SWITCHES/ANNUNCIATORS

CONTROL WHEEL SWITCHES

1. PITCH SYNC AND CWS BUTTON - When pressed with the autopilot engaged, decouples the pitch, roll, and trim servo clutches to allow hand flying and manual trim adjustment. The effects on existing flight director modes are as follows:
   a. Will sync IAS, VS, pitch attitude hold, ALT and roll attitude hold.
   b. Will cancel GA and revert to pitch attitude hold.
   c. When pressed prior to flight director engagement, will engage the flight director in pitch and roll attitude hold.

NOTE

The Pitch Synch and CWS button affects only the onside AP/FD when dual FCS Systems are installed. If the offside autopilot is engaged, pressing the onside CWS button does not disengage the autopilot servo clutches. The trim may run to oppose pilot effort.

2. PITCH TRIM (NOSE DN - NOSE UP) - A split-switch unit designed to insure malfunction protection. Both segments of the split trim switch must be depressed to engage pitch trim in the desired direction. With the autopilot engaged, operation of the left segment will disconnect the pitch, roll, and pitch trim servos. The flight director, yaw damper, and rudder trim will remain engaged. Operation of any one segment of the switch, in either direction, for approximately 3 seconds, will illuminate the AP TRIM FAIL annunciator in the glareshield.
3. AP DISC & TRIM INTRPT SWITCH - When depressed will disengage the autopilot and yaw damper and cancel all operating flight director modes. When depressed and held will interrupt all electric trim power (stop trim motion), disengage the autopilot, cancel all operating flight director modes and suppress the autopilot disconnect tone.

POWER LEVER GO AROUND (GA) BUTTON (or Copilot's Control Wheel with Dual FCS)

Disengages the autopilot, NAV, and APR modes, if engaged, while commanding a fixed 7° pitch up, wings level attitude. The autopilot and any lateral mode may subsequently be re-engaged. Initiation of any other vertical mode cancels GO AROUND.

MC-103 OR C-14A COMPASS SYSTEM

1. GYRO SLAVE/FREE Switch - Selects either the manual or automatic slaving mode for the compass system.
2. INCR/DECR Switch - With the GYRO SLAVE/FREE Switch in the FREE position, allows manual compass card slaving in either the clockwise (INCR) or counterclockwise (DECR) direction. The switch is spring-loaded to the center position.
3. Slaving Meter - Indicates the difference between the displayed heading and the magnetic heading.

KCS-305 COMPASS SYSTEM

1. FREE/SLAVE COMPASS SLAVE SWITCH - Selects either the manual or automatic slaving mode for the compass system.
2. CW/CCW COMPASS MANUAL SLAVE SWITCH - With the FREE/SLAVE compass slave switch in the FREE position, allows manual compass card slaving in either the clockwise or counterclockwise direction. The switch is spring-loaded to the center position.

3. SLAVING METER - Indicates the difference between the displayed heading and the magnetic heading. Up deflection indicates a clockwise error of the compass card. Down deflection indicates a counterclockwise error of the compass card.

DISPLAY/DOWN SWITCH (3-tube EFIS)

This switch is a reversionary mode for the EADI presentation when the top Display Unit fails. Depressing the switch causes the power to be removed from the top electronic Display Unit, moves the EADI presentation to the bottom Display Unit, and moves the EHSI presentation to the MFD. The DISPLAY/DOWN switch cannot be selected if the SG/STBY switch has been selected first.

DISPLAY/DOWN SWITCH (4-tube EFIS)

This switch is a reversionary mode for the EADI presentation when the top Display Unit fails. Depressing the switch causes the power to be removed from the top electronic Display Unit, moves the EADI presentation to the bottom Display Unit, and inhibits lateral modes on the onside FLT DIR. Lateral modes cannot be re-engaged on onside FLT DIR until DISPLAY/DOWN Switch is deselected. The DISPLAY/DOWN Switch cannot be selected if the onside COPY/ON Switch has been selected first. Also, both pilot and copilot can select DISPLAY/DOWN on both sides at the same time.

SG/STBY SWITCH (3-tube EFIS)

This switch is a reversionary mode for an SG failure. Depressing the switch causes the standby symbol generator to control the top, bottom, and MFD Display Units, presentation of EADI information to be displayed on the top Display Unit, presentation of EHSI information to be displayed on bottom and MFD Display Units, and annunciates STBY in white on upper left sections of EADI and EHSI presentations. The SG/STBY Switch cannot be selected if the DISPLAY/DOWN Switch has been selected first.

COPY/ON SWITCH (4-tube EFIS)

This switch is a reversionary mode for an onside SG failure. Depressing the switch causes a copy of the offside EADI and EHSI presentations to be displayed on the onside top and bottom Display Units respectively, COPY is annunciated in yellow on the upper left section of the Display Units, and onside autopilot engagement is inhibited. The COPY/ON switch cannot be selected if the DISPLAY/DOWN switch has been selected first or if COPY/ON switch has been selected on the offside system.

AP LT/AP RT ANNUNCIATOR

On airplanes with dual Flight Control Systems, AP LT and AP RT are annunciated to display the position of the AP transfer relay at all times.

ADC 1/2 SWITCH (Dual FCS)

This switch is used to select either Air Data System 1 or 2 for display on the KAV-485 and other systems using Air Data information.

CMPS 1/CMPS 2 SWITCH (Optional)

This switch is used to select the heading source for use on the onside EFIS and FCS and offside RMI. If both the onside and offside heading source select switches are selected to the same source, "HDG" appears in a yellow box on the upper left section of the EHSI.

MAG HDG/TRUE HDG SWITCH (Optional)

This switch is used to select either MAG or TRUE heading for use on the onside EFIS and FCS, and will place a white "T" next to the lubber line on the EHSI. The switch only functions if the selected NAV source uses either MAG or TRUE heading. If both the onside and offside heading source select switches have TRUE selected, "HDG" appears in a yellow box on the upper left section of the EHSI.

ISS/VERT GYRO SWITCH OR VERT GYRO 1/VERT GYRO 2 SWITCH (Optional)

This switch is used to select the attitude source for use on the onside EFIS and FCS. If both the onside and offside attitude source select switches are selected to the same attitude source, "ATT" appears in a yellow box on the upper left section of the EADI.

ATTITUDE GYRO FAST ERECT SWITCH(ES)

A momentary ON switch used to increase the KVG-350 or VG 14 Vertical Gyro erection rate from 2°/minute (normal) to 11°/minute (fast). Fast erection should only be used in level flight.

AP DISC/AP TRIM FAIL ANNUNCIATORS

Annunciates monitored flight control system failures. Will illuminate for a period of time on initial power application as a part of the preflight test sequence.

EFIS ON/OFF SWITCH(ES)

Controls power to the EFIS symbol generators and display units. The switch(es) should be selected OFF during engine start to protect the EFIS system from low/transient voltages.

AP/TRIM PWR SWITCH(ES)

Provides 28 vdc power to the KCP-420 computer to power all autopilot modes, electric trim, and each axis servo. Cycling between OFF and ON will restart the autopilot preflight test sequence. Placing the switch(es) in the OFF position will inhibit operation of all AP modes and all control servos. The switch(es) should be selected OFF during engine start to protect the Flight Control System from low/transient voltages.

FMS ANNUNCIATORS

1. Waypoint (WPT)
   Illuminates 90 seconds prior to each waypoint when operating in the OBS method of operation, when approaching the last waypoint in the flight plan or when going direct to a waypoint not in the flight plan.
   Illuminates 15 seconds prior to the automatic course change turn when operating in the AUTO/LEG method of operation.
2. Message (MGS)
   Identical to the message light on the Control Display Unit (CDU).
3. Dead Reckoning (DR)
   Illuminates when the system is dead reckoning.
4. X-Track (XTRK)
   Illuminates whenever a left or right cross track (parallel offset) is activated.
5. VNAV Approve (when VNAV option is installed)
   Illuminates 90 seconds prior to a VNAV angle capture or before a 3-D flight path angle change. Must be pressed to approve the flight path angle change.

FMS SWITCHES (BENDIX EFS-10 EQUIPPED)

A function select switch on the EFS controller selects the OFF, TEST, NAV, FMS, or ADF modes. The NAV and FMS positions are the only ones pertinent to this discussion.

1. The NAV position selects the VOR navigation receiver and the DME receiver as the primary navigation source for navigation instruments.
2. The FMS position selects the KNS-660 as the navigation source for the navigation instruments.
   a. The KNS-660 provides automatic course pointer drive while using the AUTO/LEG method of operation.
   b. The KNS-660 does not provide automatic course pointer drive while using the OBS method of operation. The pilot must set the selected course manually.
   c. The deviation needle provides left/right steering information from the KNS-660.
   d. The DME display (upper right) provides distance to the waypoint in nautical miles.
   e. The cyan RMI pointer provides bearing and DME information derived from the KNS-660.
   f. The white RMI pointer provides bearing to the VOR station currently tuned by the KNS-660. Although the information is valid, the station selection is randomly selected by the KNS-660. Therefore, the white RMI pointer is always flagged by RED X's. DME information to the VOR is not displayed.
   g. With VNAV installed, VNAV flight path angle is presented in degrees prior to VNAV engagement (VNAV ANG annunciated). Deviation from the VNAV flight path is presented in feet after VNAV engagement (VNAV annunciated).
   h. ILS frequencies cannot be tuned via the KNS-660.
   i. If VOR sensor and NAV mode are selected on the KNS-660, the EHSI navigation information will be flagged. Select NAV on the function select switch.

FMS SWITCHES (WHEN EQUIPPED WITH KING MECHANICAL INSTRUMENTS)

A push on/push off switch on the instrument panel selects the FMS or NAV CTL positions. The active selection is annunciated on the switch.

1. The NAV CTL position selects the VOR navigation receiver and the DME receiver as the primary navigation source for the navigation instruments.
2. The FMS position selects the KNS-660 as the navigation source for the navigation instruments.
   a. The KNS-660 provides automatic course pointer drive while using the AUTO/LEG method of operation.
   b. The KNS-660 does not provide automatic course pointer drive while using the OBS method of operation. The pilot must set the selected course manually.
   c. The deviation needle provides left/right steering information from the KNS-660.
   d. The DME display provides distance to the waypoint in nautical miles and the selection (via a pushbutton to the right of the DME display) of either groundspeed in knots or estimated time enroute in hours and minutes.
   e. The green RMI pointer provides bearing information derived from either the ADF or the KNS-660 depending upon whether ADF or NAV respectively are selected by the RMI selector.
   f. With VNAV installed, when the ANG annunciator is illuminated the VNAV scale is presenting the VNAV flight path angle in degrees. When the ANG annunciator is not illuminated, the VNAV scale is presenting the deviation from the VNAV flight path in feet.
3. HSI Reference Data (OMEGA DIST/BRG)
   Activation of the HSI REF DATA Switch (OMEGA DIST/BRG illuminated) allows bearing and distance to the active KNS-660 VLF/OMEGA waypoint to be displayed on the pilot's HSI bearing pointer and DME readout during operation in NAV CTL with an ILS frequency tuned. VOR-1 drives the pilot's HSI course needle and both RMI's during this operation.

PILOT'S AND COPILOT'S RMI (FMS)

1. FMS selected (FMS illuminated and "blanked" active frequency display window in NAV-1 Frequency Control Head).
   The single bar (green) needle provides bearing information derived from either the ADF or KNS-660 depending upon whether ADF or NAV respectively are selected on the RMI.
2. NAV selected on EFS-10 or NAV CTL for mechanical instruments
   The single bar (green) needle provides bearing information derived from either the ADF or the NAV 1 receiver depending upon whether ADF or NAV respectively are selected on the RMI.

MFD/FMS SWITCH (Bendix EFS-10 Equipped)

A push on/off annunciator switch located on the instrument panel controls the data source for the multifunction display. The active selection is annunciated on the switch.

1. The MFD position will repeat navigation selected on the EHSI on the multifunction display.
2. The FMS position will dedicate the multifunction display to the KNS-660. Selected course must be inserted via the KNS-660 OBS window when the EHSI primary navigation source is other than FMS.

HANDLING, SERVICING AND MAINTENANCE

No Change

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BEECHCRAFT

KING AIR C90A (LJ-1138 and after, except LJ-1146)

and

SUPER KING AIR B200, B200C, B200T, & B200CT

LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

FOSTER LNS616B LONG RANGE NAV SYSTEM

THIS SUPPLEMENT IS APPLICABLE TO
PILOT'S OPERATING HANDBOOK AND
FAA APPROVED AIRPLANE FLIGHT MANUALS
90-590024-23, 90-590024-35,
101-590010-147, AND 101-590037-29

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
P/N 101-590010-253
Revised: November, 1991

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Foster LNS616B Long Range NAV System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Foster Pilot's Guide for the LNS616B Long Range NAV System, P/N 002A0559, dated 1987 or later, is available from Foster Airdata Systems, Inc., 7020 Huntley Road, Columbus, Ohio 43229. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status.

LIMITATIONS

1. The Foster LNS616B Pilot's Guide, P/N 002A0559, dated 1987 or later, must be immediately available to the flight crew whenever navigation is predicated on the use of the LNS616B.
2. If the LNS616B data base has expired, IFR navigation is prohibited unless each REFerence is verified for accuracy against current approved data.
3. No flight operation shall be predicated on the LNS616B whenever a NAVigation flag is displayed in the pilot's HSI or whenever the system annunciates an unreliable condition.
4. The vertical guidance system (VNAV) is not approved for IFR.
5. The LNS616B must be checked for accuracy (reasonableness) prior to use as a means of navigation and following a period of operation with a NAV flag or aquisition of a new VORTAC or LORAN-C Group Repetition Interval (GRI).

VOR/DME ONLY (VO) MODE

1. The LNS616B is approved as a two-dimensional VOR/DME RNAV system in accordance with AC 90-45A for enroute terminal area and approach operation for IFR or VFR and is not geographically limited.
2. RNAV functions may be used only with co-located VOR/DME facilities.
3. IFR approaches must be conducted in the APProach mode in accordance with approved instrument approach procedures.
4. APProach mode is limited to approach operation with ground speeds of under 180 knots.

LORAN ONLY (LO) OR LORAN/VOR/DME (LV) MODES (IF INSTALLED)

1. The LNS616B is approved as a two-dimensional LORAN-C navigation system in accordance with AC 20-121 for enroute and terminal area operations for IFR or VFR and is limited to the following LORAN-C coverage areas:

LORAN-C COVERAGE AREA	GRI
SOUTHEAST U.S. CHAIN	7980
GREAT LAKES CHAIN	8970
NORTHEAST U.S. CHAIN	9960
U.S. WEST COAST CHAIN	9940

Charts of the approved coverage areas are included in this supplement.

2. During LORAN-C operation, other navigation equipment required for the specific type of operation must be installed and operable.
3. The "LO" mode is not approved for IFR approaches.

EMERGENCY PROCEDURES

1. If the NAV flag appears enroute, verify correct frequency and decoded station ident or utilize other navigation equipment as required.
2. If the NAV flag appears during approach, execute missed approach and utilize another approved navigation facility.

NORMAL PROCEDURES

Normal operating procedures are outlined in the Foster Pilot's Guide for the LNS616B, P/N 002A0559, dated 1987, or later.

SYSTEM ANNUNCIATORS

Waypoint (WPT)

- Illuminates approximately three miles prior to waypoint passage.

SYSTEM INTEGRATION

1. The LNS616B utilizes the #1 NAV control head to tune VOR/DME. LORAN-C is tuned by the LNS616B control head.
2. VOR/DME and RNAV steering information from the LNS616B is supplied to the pilot's HSI only. RMI indicators provide raw VOR information only.
3. The HSI will display RNAV information when the control head is in the "NAV", "TME/WND", "GRI", or "AUX" position. RNAV information will also be supplied in the "PRG" and "REF" mode when these modes are entered from the "NAV" mode. When "REF" and "PRG" are entered from the "VOR/LOC" mode, VOR or Localizer steering information will be displayed.
4. The HSI will display the following LNS616B system derived information:

   a. OBS Course Arrow

      - VOR mode selects course to VOR
      - NAV mode selects course to Waypoint

   b. CDI - Left or Right course information with full scale sensitivity as follows:

      "VOR" mode - 10 degrees
      "NAV" mode - a) APPROACH - 1.25 NM
                   b) ENROUTE - 5.0 NM

   c. To/From Symbol

      - The selected course will take the airplane To/From the VOR or waypoint.

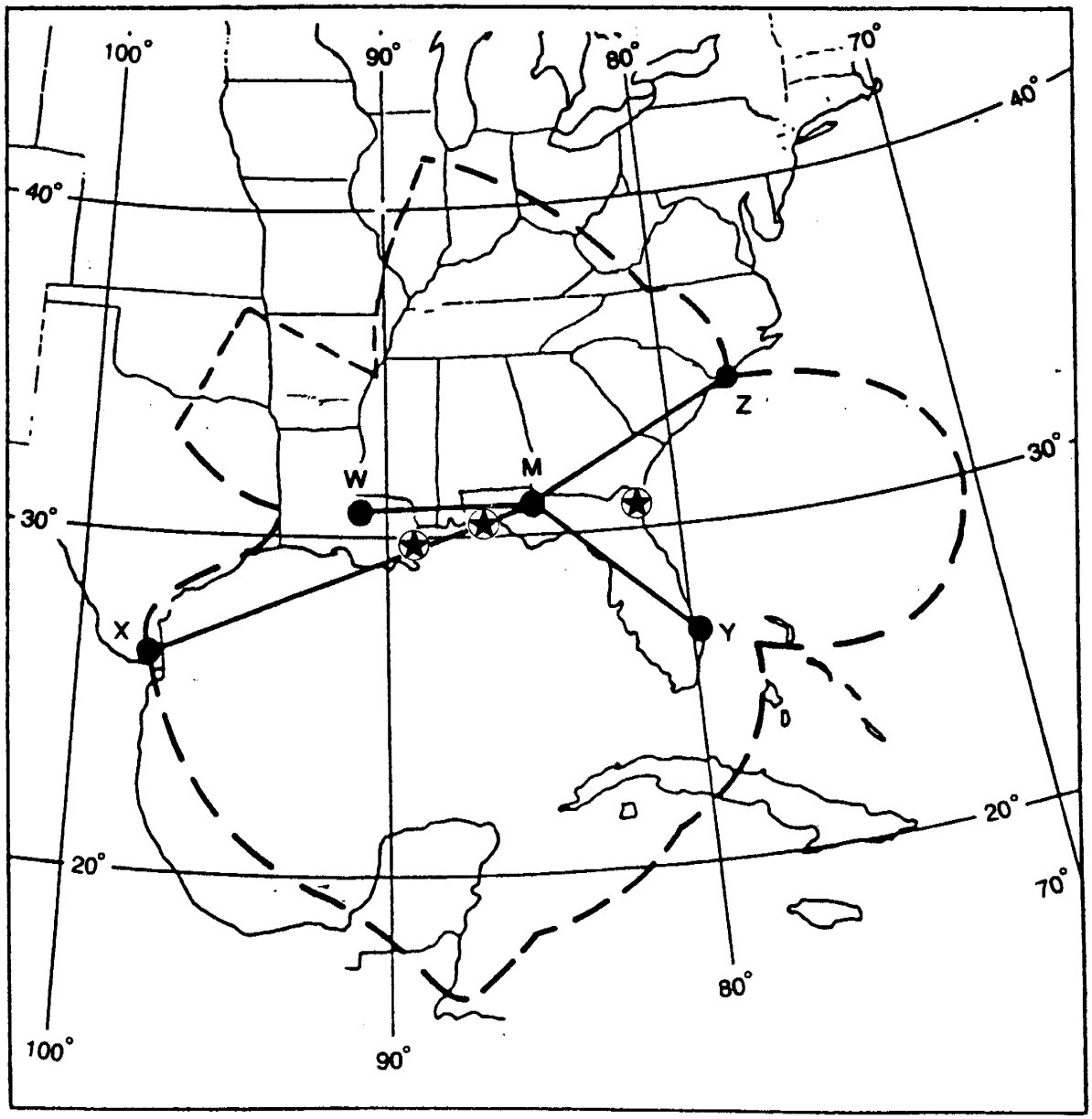
   d. NAV Flag

      - LNS616B Navigation is unreliable.

5. When operating in "LEG" mode, the pilot must manually set the course indicator to agree with desired track (DTK) appearing on the NAV page of the LNS616B control head unless the installation includes an automatic slewing capability.

LORAN-C

SOUTHEAST U.S. CHAIN ——— GRI 7980



Charts reproduced courtesy of the Department of Transportation

LEGEND:

- ● TRANSMITTING
- ⍟ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	MALONE
W	GRANGEVILLE
X	RAYMONDVILLE
Y	JUPITER
Z	CAROLINA BEACH

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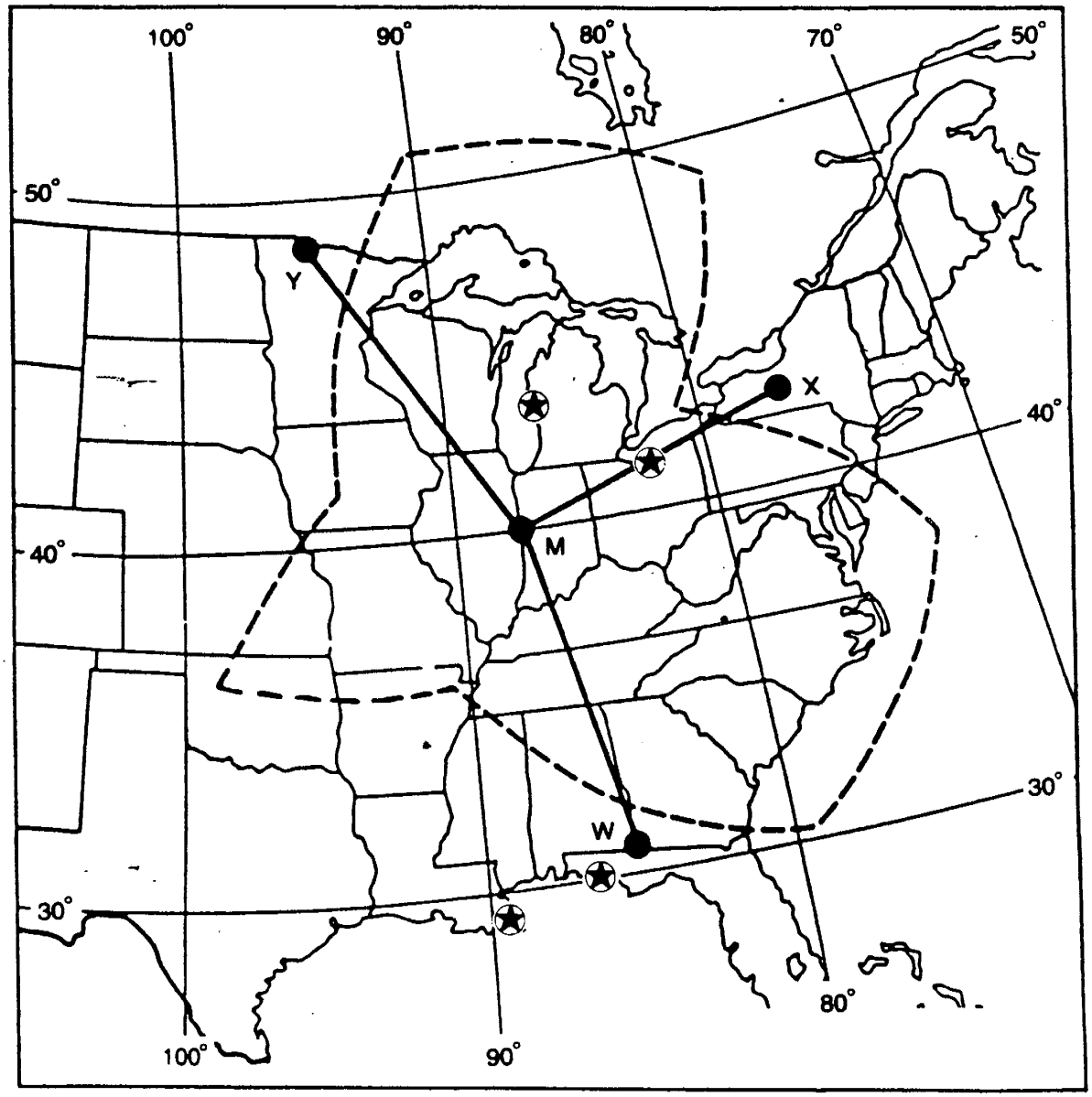
LORAN-C

GREAT LAKES CHAIN ——————— GRI 8970



Charts reproduced courtesy of the Department of Transportation

LEGEND:
- ● TRANSMITTING
- ⍟ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	DANA
W	MALONE
X	SENECA
Y	BAUDETTE

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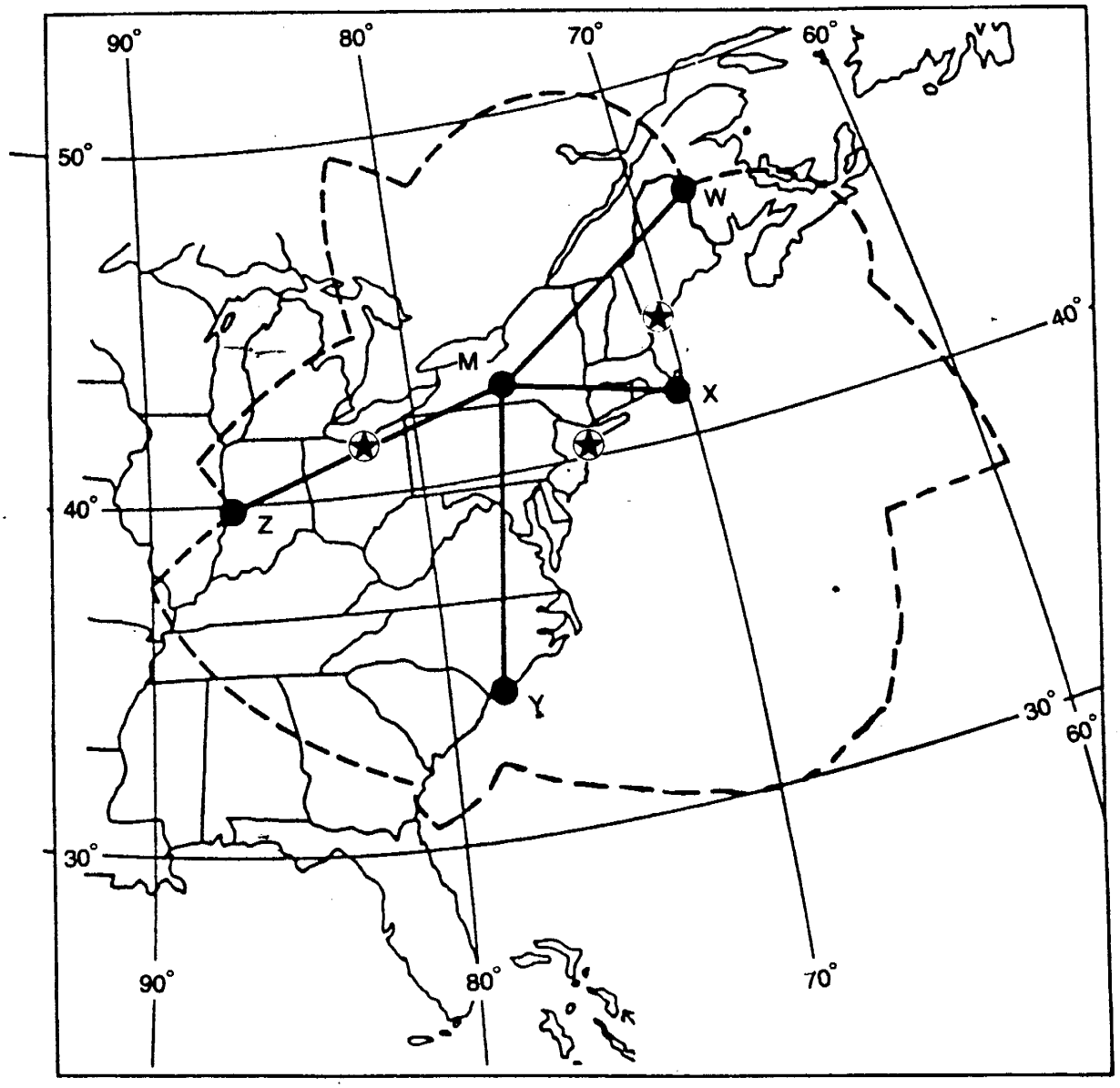
LORAN-C

NORTHEAST U.S. CHAIN ——— GRI 9960



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LEGEND:
- ● TRANSMITTING
- ⍟ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	SENECA
W	CARIBOU
X	NANTUCKET
Y	CAROLINA BEACH
Z	DANA

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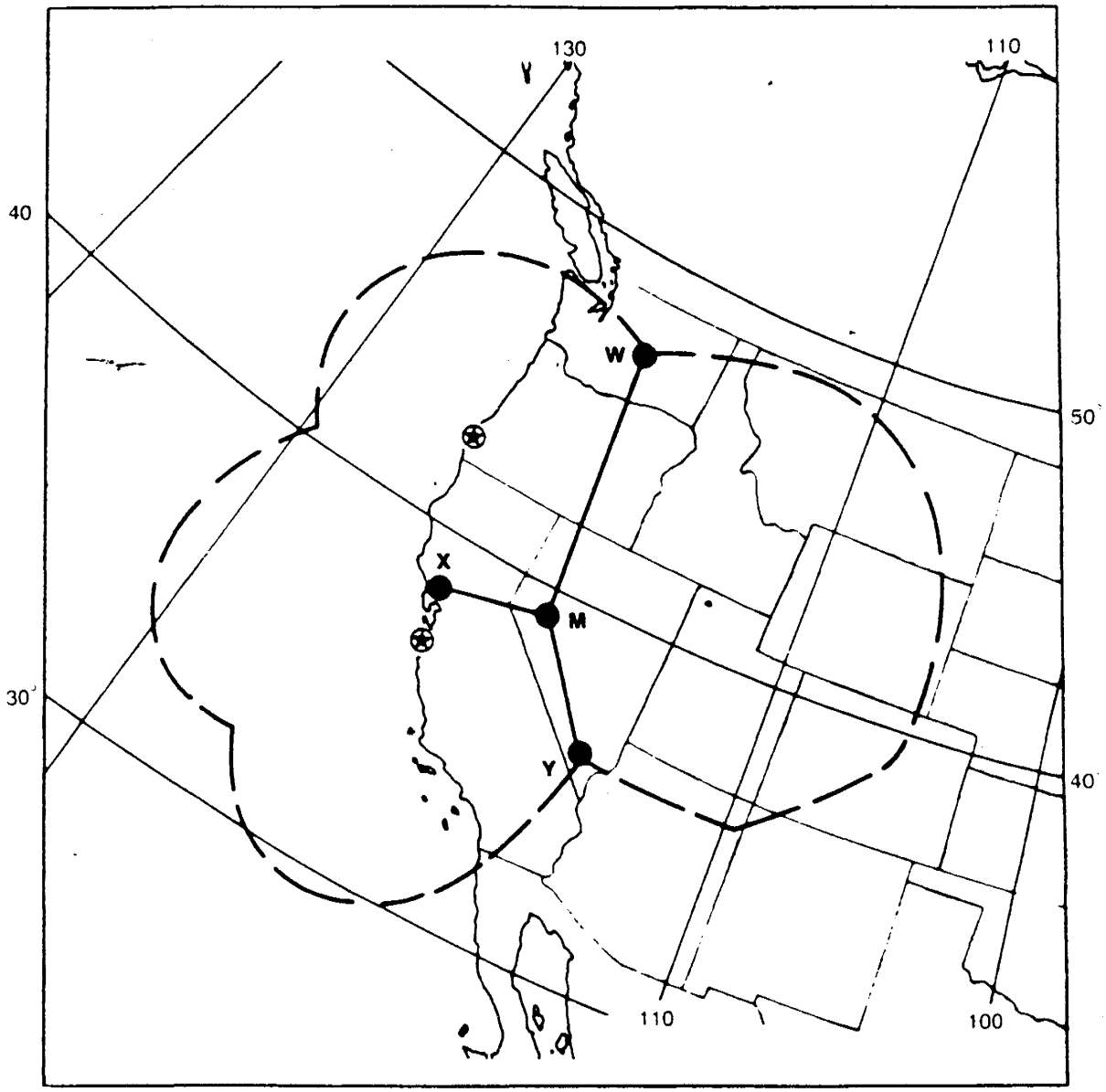
LORAN-C

U.S. WEST COAST CHAIN ——— GRI 9940



Charts reproduced courtesy of the Department of Transportation

LEGEND:

● TRANSMITTING

⊛ MONITOR (AUTOMATED)

STATION NAME DESIGNATION

M	FALLON
W	GEORGE
X	MIDDLETOWN
Y	SEARCHLIGHT

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FAILURES AND WARNINGS

The LNS616B Long Range Navigation System annunciates failures or conditions in which LORAN-C navigation is unreliable in the following manner:

1. The LNS616B constantly monitors signal-to-noise ratio, and displays signal quality (MWXYZ) on a scale from 0 to 9 (best case is 9, worst case is 0) on the control head. Use only stations which display signal-to-noise ratings of 3 or greater.
2. Geometry quality (Q) is constantly monitored and displayed in "GRI" mode on the control head on a scale from 0 to 9 (best case is 9, worst case is 0). Low "Q" indicates marginal geometry when station ratings are acceptable.
3. Excessive range is indicated by a low signal-to-noise number and a low "Q" number. Use only stations which display signal-to-noise ratios of 3 or greater.
4. A blink condition is indicated by a signal-to-noise ratio of "0" for the appropriate station that the computer has detected a blink condition.
5. A steering failure will be indicated by a NAV flag on the HSI.

SYSTEM OPERATION WITHOUT LORAN-C INSTALLED

1. When LNS616B's LORAN-C receiver is not installed, "LO" mode causes the HSI to flag and navigation computations of LNS616B control head to cease functioning. The initial display after turn-on will also delete the LORAN-C software revision levels, "N 0.0 C 0.0" (bottom display line), when the receiver is not installed.
2. Without LNS616B LORAN-C receiver installed, "POS" pushbutton will only display the airplane's position relative to received VORTAC stations.
3. GRI mode is ineffective in the use of stored GRI data (PAGE ONE and PAGE TWO) unless the LNS616B remote LORAN-C receiver is installed. When the LNS616B LORAN-C receiver is not installed, GRI mode is operationally useful only to display the airplane's position in Latitude/Longitude when "POS" pushbutton is pressed.
4. LNS616B installations without LORAN-C receivers may use the "POS" pushbutton in all modes except "GRI" and "AUX" to display the decoded Morse code IDENT in the upper left of the display, and the airplane's radial and distance from the received VORTAC station. In "GRI" and "AUX" mode, pressing "POS" displays the airplane's Latitude/Longitude and during "REF" mode, pressing "POS" pushbutton programs the airplane's Latitude/Longitude into REF "PPP". If IDENT, VOR, or DME signals are not available, pressing "POS" displays bars (---) in the associated display elements.
5. Messages relating to LORAN-C operation do not apply when LORAN-C receiver component of the LNS616B system is not installed.
6. DATA control of GRI pages may be implemented as described, but will not result in operational use without the LORAN-C receiver installed or operational in the LNS616B system.
7. When the LORAN-C receiver portion of LNS616B is not installed or is inoperative, the LAT/LNG page of "AUX" mode shows bars (---).

REFERENCE DATA BASE

It is the pilot's responsibility to ensure that current and accurate VORTAC reference data is being used for navigation and that the current data base cartridge is installed in the D1681A Database/Interface Unit. Effective date for the preprogrammed navigation references may be determined in two ways.

1. Upon turning on the LNS616B system, Software Revision Level and Database Effective Date will be displayed. The bottom line of the third display indicates chart effective date expressed in month, day, and year (i.e., 05/23/84). Database is valid for 56 days following effective date.
2. Database Chart Effective date is also printed on the Database Cartridge which is installed in the D1681A Database/Interface Unit located in the nose avionics compartment.

AUTOPILOT/FLIGHT DIRECTOR COUPLING

No change.

PERFORMANCE

No change.

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WEIGHT AND BALANCE/EQUIPMENT LIST

No change.

SYSTEMS DESCRIPTION

No change.

HANDLING, SERVICE, AND MAINTENANCE

No change.

BEECHCRAFT® SUPER KING AIR®

B200 LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

THAILAND FLIGHT INSPECTION AIRPLANE
(S/N BB-1315 Only)

and

THAILAND AERIAL SURVEILLANCE AIRPLANES
(S/N BB-1436, BB-1441 & BB-1443 Only)

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL P/N 101-590010-147

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved by:

A. C. Jackson
Beech Aircraft Corporation
DOA CE-2

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Revised: April, 1992
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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Super King Air B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane is operated in dual categories in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Super King Air B200/B200C Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to the operation of the airplane.

This airplane has been certified under the provisions of FAR 21.187 for Multiple Airworthiness Certification in both Normal and Restricted Categories.

All information contained in this supplement which applies to Restricted Category only is indicated by the use of shading.

MAXIMUM CERTIFICATED WEIGHTS (NORMAL CATEGORY)

Maximum Ramp Weight . . . . . 12,590 pounds
Maximum Take-off Weight . . . . . 12,500 pounds
Maximum Landing Weight . . . . . 12,500 pounds
Maximum Zero-Fuel Weight . . . . . 11,000 pounds

MAXIMUM CERTIFICATED WEIGHTS (RESTRICTED CATEGORY)

Maximum Ramp Weight . . . . . 14,090 pounds
Maximum Take-off Weight . . . . . 14,000 pounds
Maximum Landing Weight . . . . . 13,500 pounds
Maximum Zero-Fuel Weight . . . . . 11,000 pounds
Maximum Weight in Baggage Compartment:
When Equipped With Fold-up Seats . . . . . 510 pounds
When Not Equipped With Fold-up Seats . . . . . 550 pounds

SPECIFIC LOADINGS (12,500 POUNDS)

Wing Loading . . . . . 41.3 pounds per square foot
Power Loading . . . . . 7.4 pounds per shaft horsepower

SPECIFIC LOADINGS (14,000 POUNDS)

Wing Loading . . . . . 46.2 pounds per square foot
Power Loading . . . . . 8.2 pounds per shaft horsepower

LIMITATIONS

RESTRICTED CATEGORY OPERATIONS

1. For operation above 12,500 pounds maximum take-off gross weight, the pilot must have either a type rating or a letter of authorization from the governmental office having jurisdiction over such operation.
2. The airplane shall be identified "Restricted" per FAR 45.
3. Persons required for performance of flight inspection/aerial surveillance missions are considered to be crew members.
4. Each time the airplane is converted to the Restricted Category, a record of inspection shall be accomplished.
5. Maximum take-off weight is 14,000 pounds. (See shaded area of the Performance section of this supplement.)

WEIGHT LIMITS (NORMAL CATEGORY)

Maximum Ramp Weight	12,590 pounds
Maximum Take-off Weight	12,500 pounds
Maximum Landing Weight	12,500 pounds
Maximum Zero-Fuel Weight	11,000 pounds
Maximum Weight in Baggage Compartment:	
When Equipped with Fold-up Seats	510 pounds
When Not Equipped with Fold-up Seats	550 pounds

WEIGHT LIMITS (RESTRICTED CATEGORY)

Maximum Ramp Weight	14,090 pounds
Maximum Take-off Weight	14,000 pounds
Maximum Landing Weight	13,500 pounds
Maximum Zero-Fuel Weight	11,000 pounds
Maximum Weight in Baggage Compartment:	
When Equipped with Fold-up Seats	510 pounds
When Not Equipped with Fold-up Seats	550 pounds

CENTER OF GRAVITY LIMITS

FORWARD LIMITS

NORMAL CATEGORY

185.0 inches aft of datum at 12,500 pounds, with straight line variation to 181.0 inches aft of datum at 11,279 pounds. 181.0 inches aft of datum at 11,279 pounds or less.

RESTRICTED CATEGORY

188.3 inches aft of datum from 14,000 to 13,500 pounds, then straight line variation to 181.0 inches aft of datum at 11,279 pounds. 181.0 inches aft of datum at 11,279 pounds or less.

AFT LIMITS

196.4 inches aft of datum at all weights.

FLIGHT LOAD FACTOR LIMITS (NORMAL CATEGORY)

At the designed gross weight of 12,500 pounds:

Flaps Up	3.17 positive g's; 1.27 negative g's
Flaps Down	2.0 positive g's; 1.27 negative g's

FLIGHT LOAD FACTOR LIMITS (RESTRICTED CATEGORY)

At the designed gross weight of 14,000 pounds:

Flaps Up . . . . . 3.10 positive g's; 1.24 negative g's
Flaps Down . . . . . 2.0 positive g's

MAXIMUM OCCUPANCY LIMITS (RESTRICTED CATEGORY)

Occupancy is limited to crew members only.

STRUCTURAL LIMITATIONS

Refer to Chapter 4 of the Super King Air 200 Series Maintenance Manual for structural limitations.

MISCELLANEOUS LIMITATIONS

1. The auto-feather system must be operational and armed for all takeoffs, climbs, and approaches.
2. Maximum allowable fuel imbalance between wing fuel systems is 300 pounds.

EMERGENCY PROCEDURES

No change.

NORMAL PROCEDURES

No change.

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INTRODUCTION

The PERFORMANCE data included in this supplement totally replaces all PERFORMANCE data provided in the PERFORMANCE section of the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

HOW TO USE GRAPHS

1. In addition to presenting the answer for a particular set of conditions, the example on the graph also presents the order in which the various scales on the graph should be used. For instance, if the first item in the example is OAT, then enter the graph at the existing OAT.

2. The reference lines indicate where to begin following the guidelines. Always project to the reference line first, then follow the guidelines to the next known item by maintaining the same PROPORTIONAL DISTANCE between the guideline above and guideline below the projected line. For instance, if the projected line intersects the reference line in the ratio of 30% down/70% up between the guidelines, then maintain this same 30%/70% relationship between the guidelines all the way to the next known item or answer.

3. The associated conditions define the specific conditions from which performance parameters have been determined. They are not intended to be used as instructions; however, performance values determined from charts can only be achieved if the specified conditions exist.

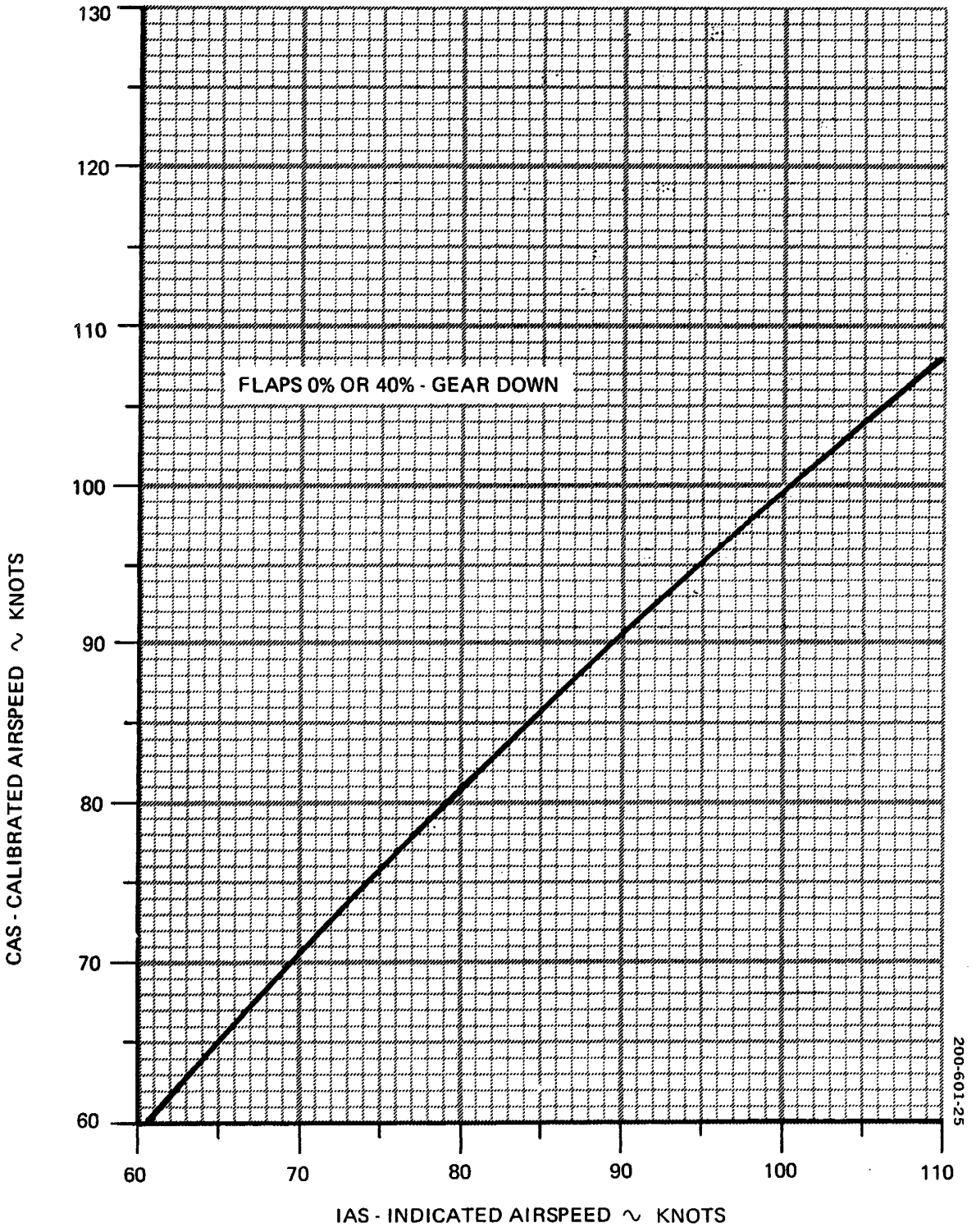
4. Indicated airspeeds (IAS) were obtained using the Airspeed Calibration - Normal System graph or the Airspeed Calibration - Normal System - Take-off Ground Roll graph.

5. The full amount of usable fuel is available for all approved flight conditions.

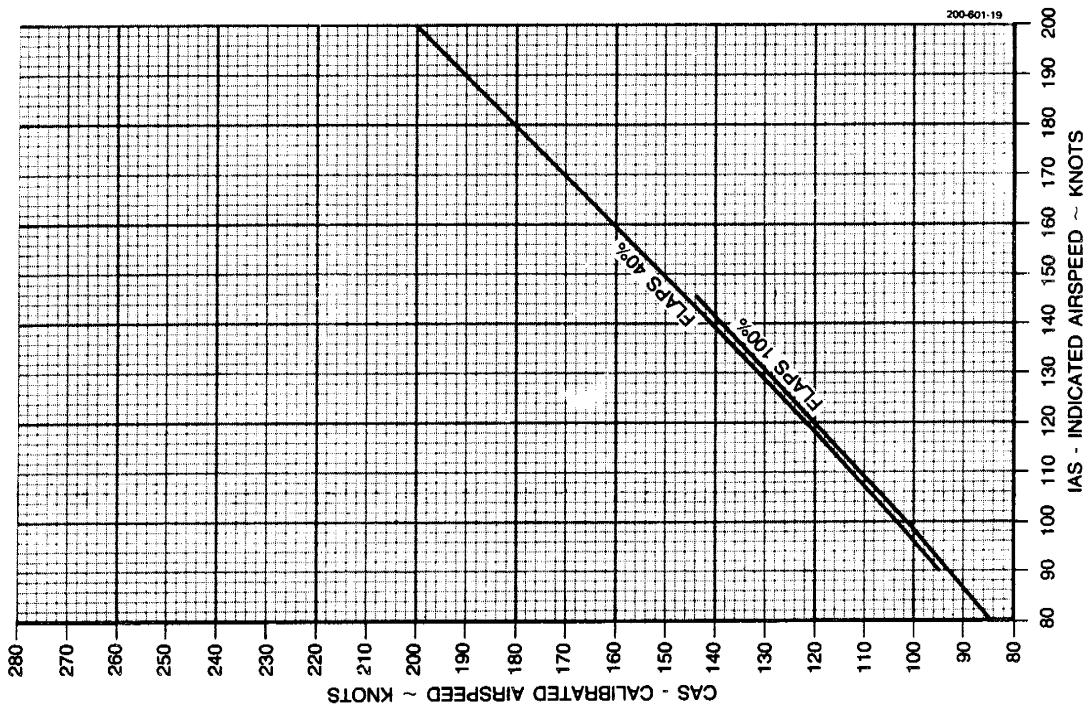
6. Notes have been provided on various graphs and tables to approximate performance with ice vanes extended. The effect will vary, depending upon airspeed, temperature, altitude, and ambient conditions. At lower altitudes, where operation on the torque limit is possible, the effect of ice vanes extended will be less, depending upon how much power can be recovered after the ice vanes have been extended.

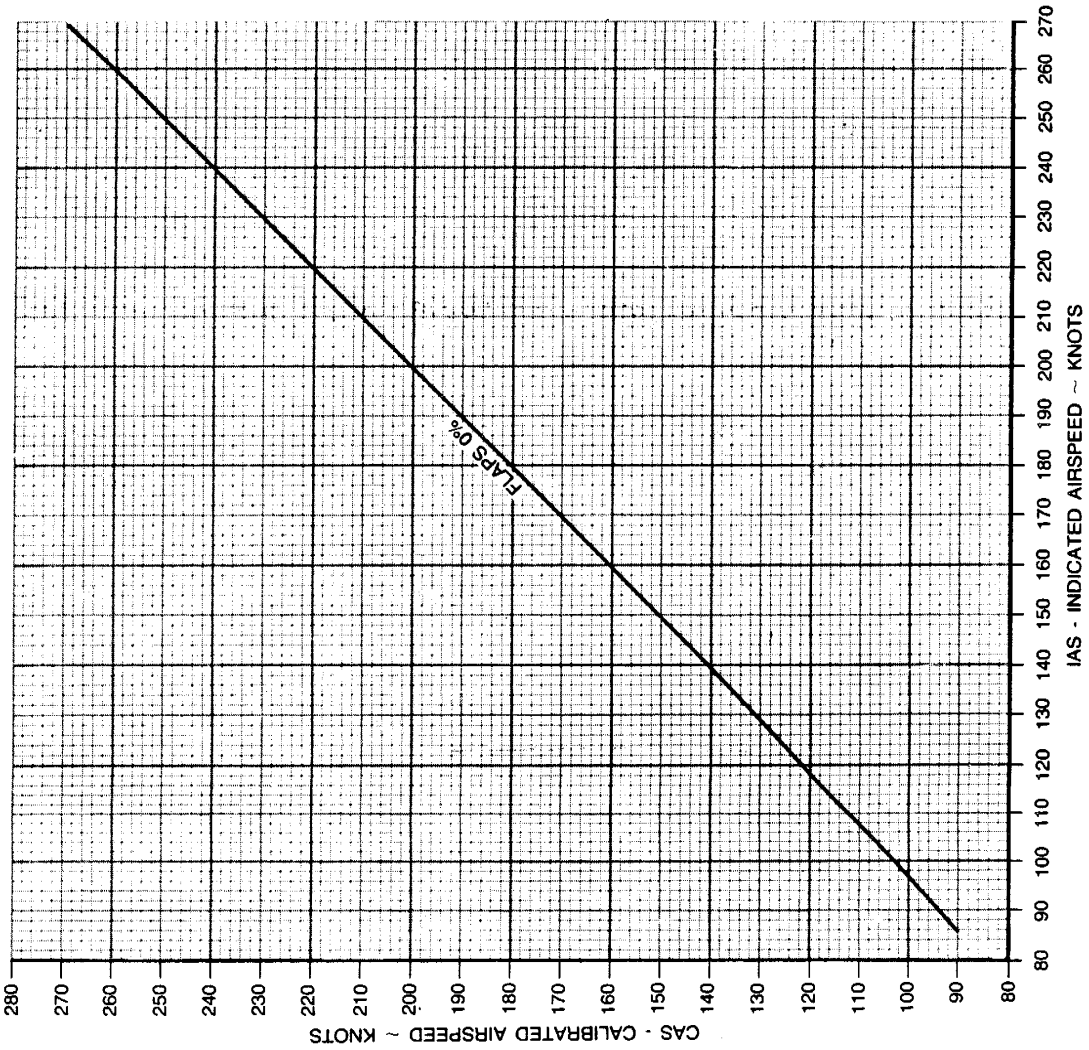
AIRSPEED CALIBRATION – NORMAL SYSTEM

TAKE-OFF GROUND ROLL



AIRSPEED CALIBRATION – NORMAL SYSTEM





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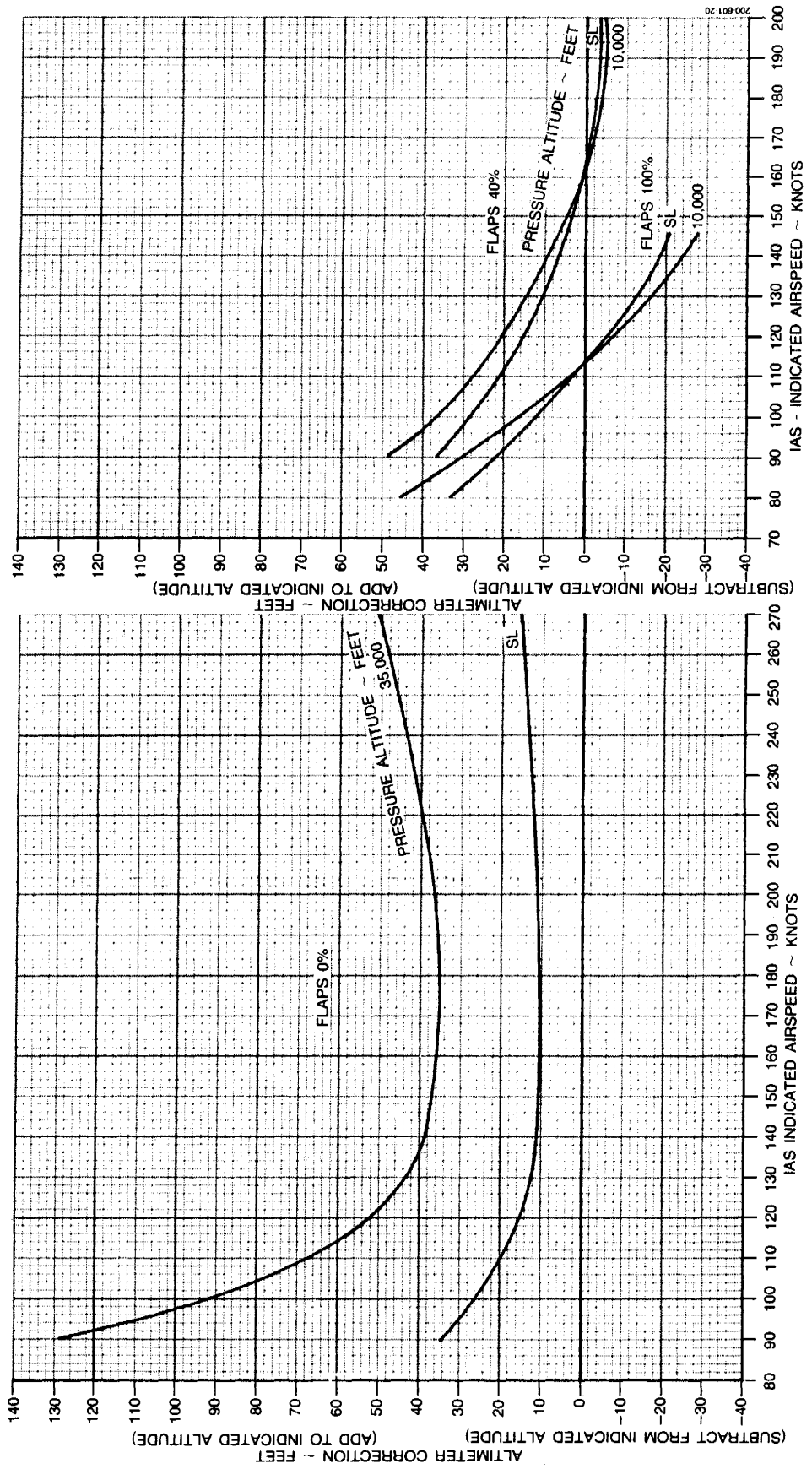
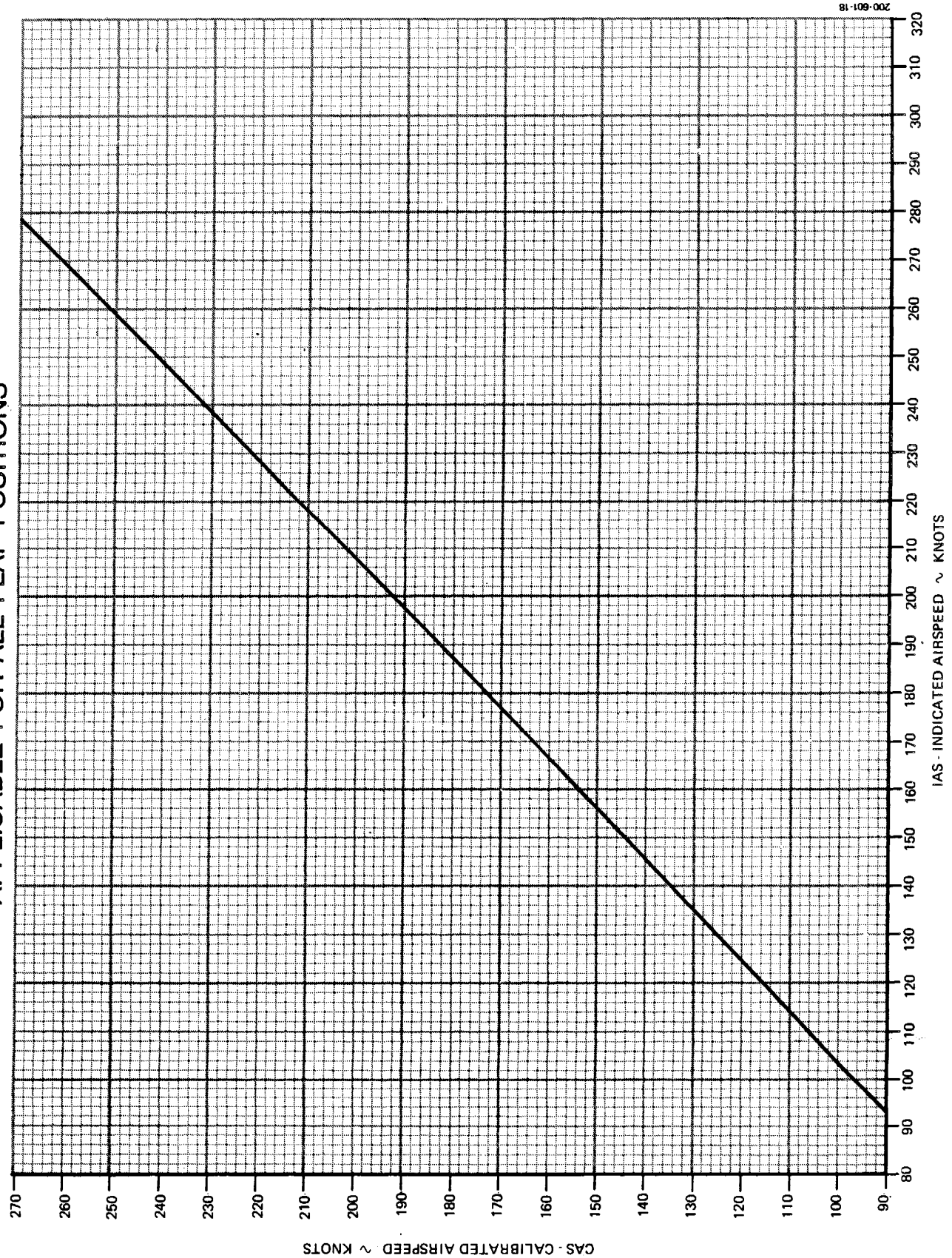
ALTIMETER CORRECTION – NORMAL SYSTEM



AIRSPEED CALIBRATION – ALTERNATE SYSTEM

APPLICABLE FOR ALL FLAP POSITIONS



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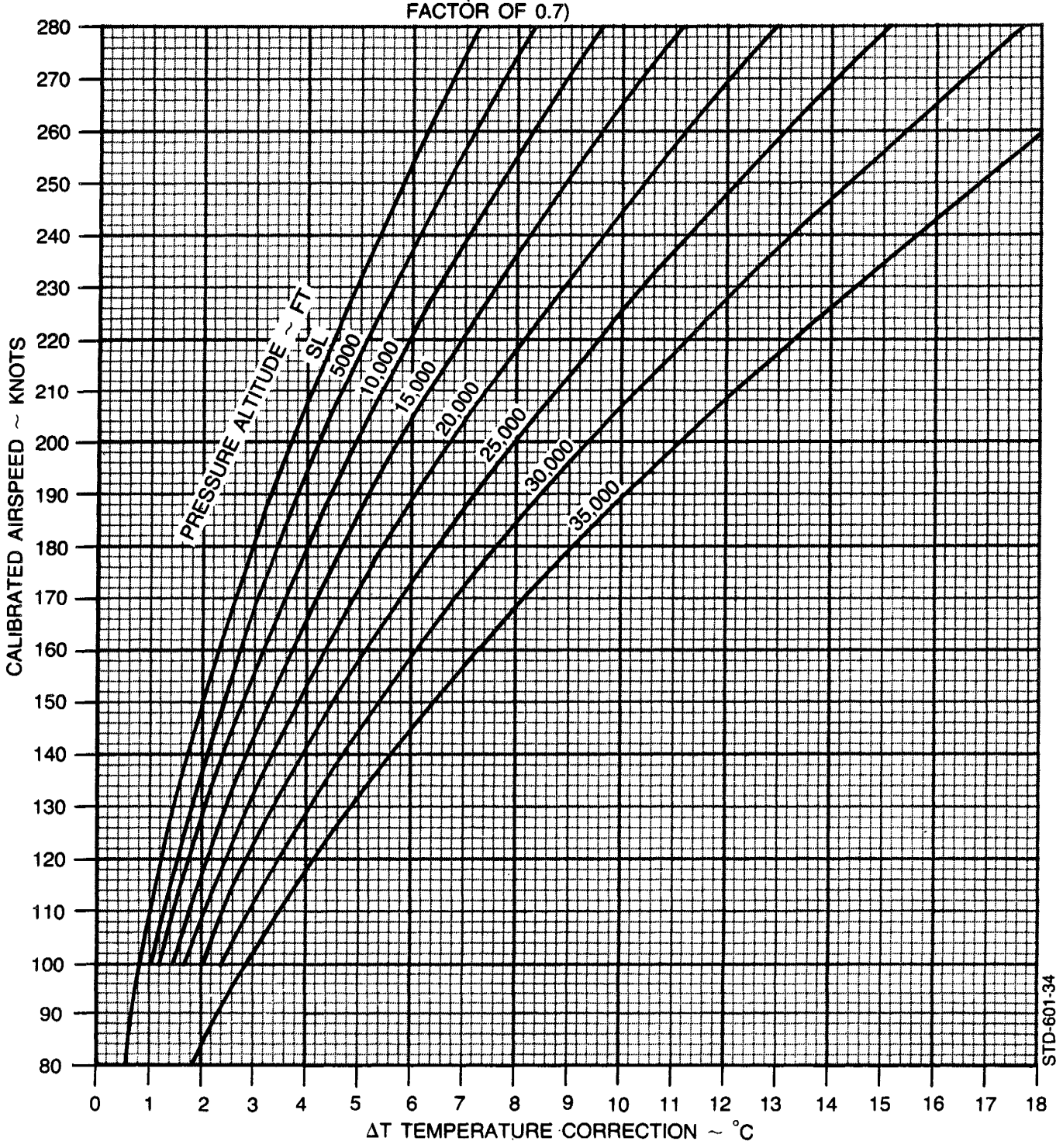
ALTIMETER CORRECTION – ALTERNATE SYSTEM

APPLICABLE FOR ALL FLAP POSITIONS

INDICATED OUTSIDE AIR TEMPERATURE CORRECTION

STANDARD DAY (ISA)

NOTE: SUBTRACT ΔT FROM INDICATED (GAGE) OAT TO OBTAIN TRUE OAT. (ΔT ASSUMES A RECOVERY FACTOR OF 0.7)



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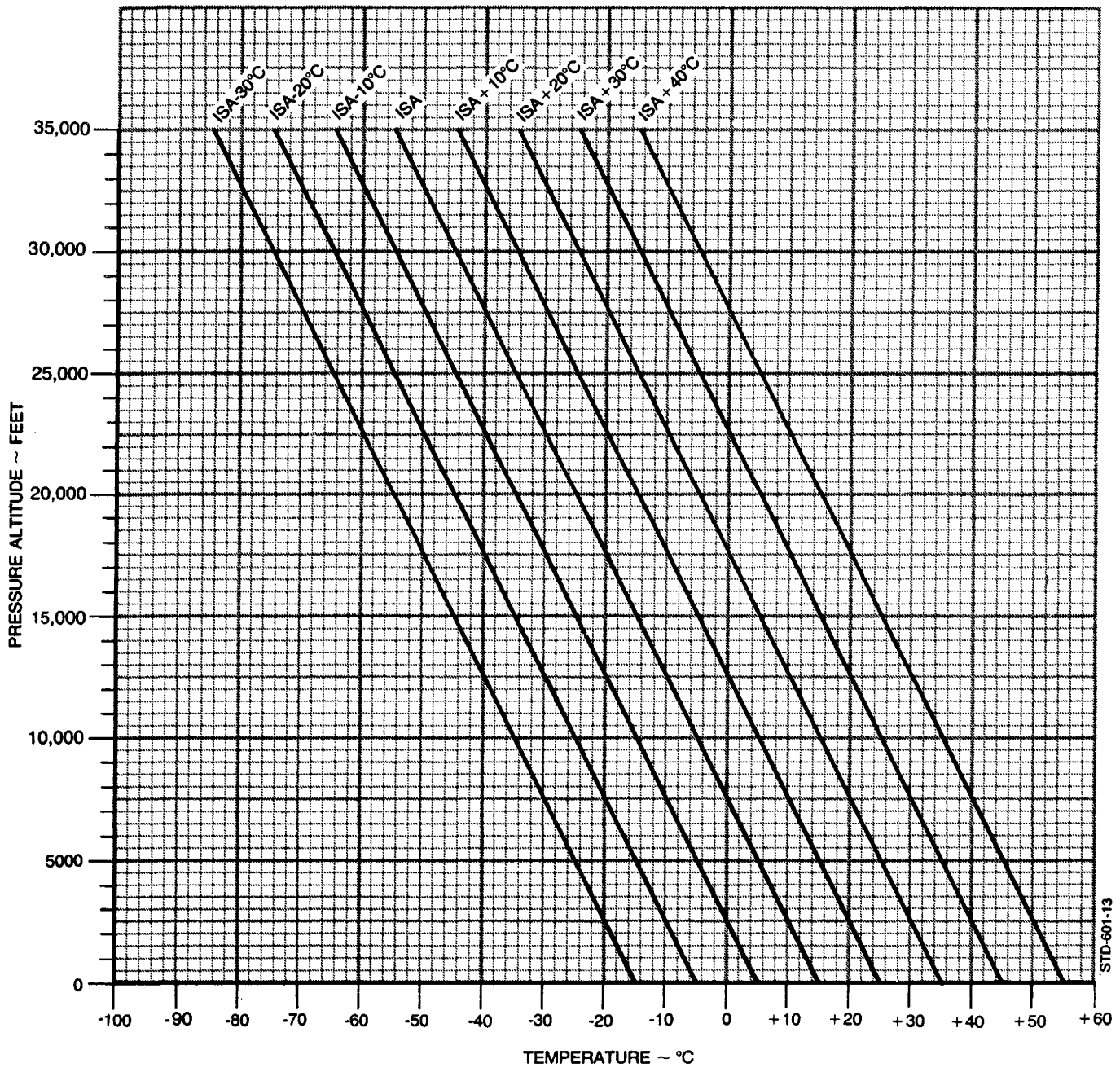
ISA CONVERSION

PRESSURE ALTITUDE vs OUTSIDE AIR TEMPERATURE



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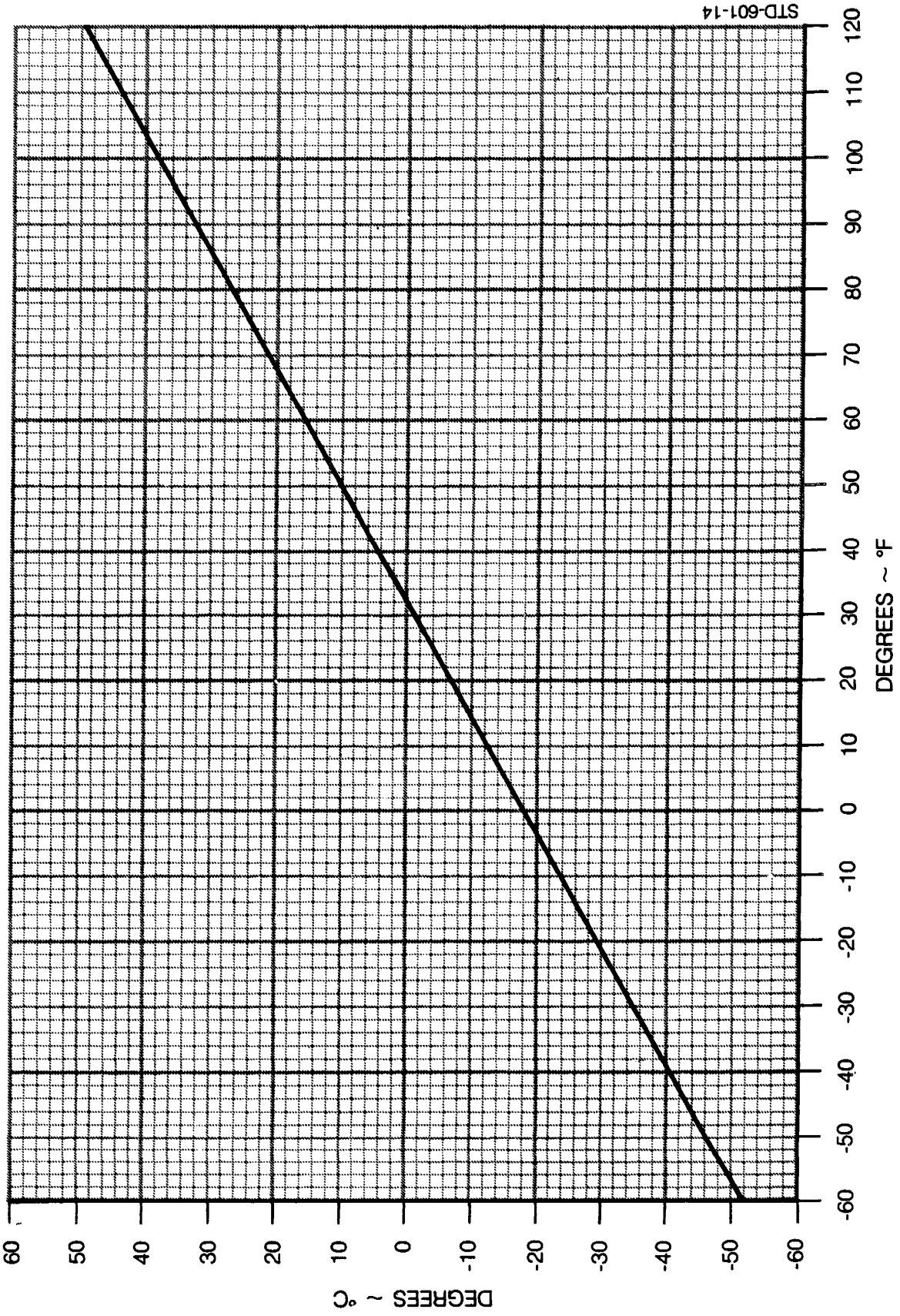
FAHRENHEIT TO CELSIUS TEMPERATURE CONVERSION



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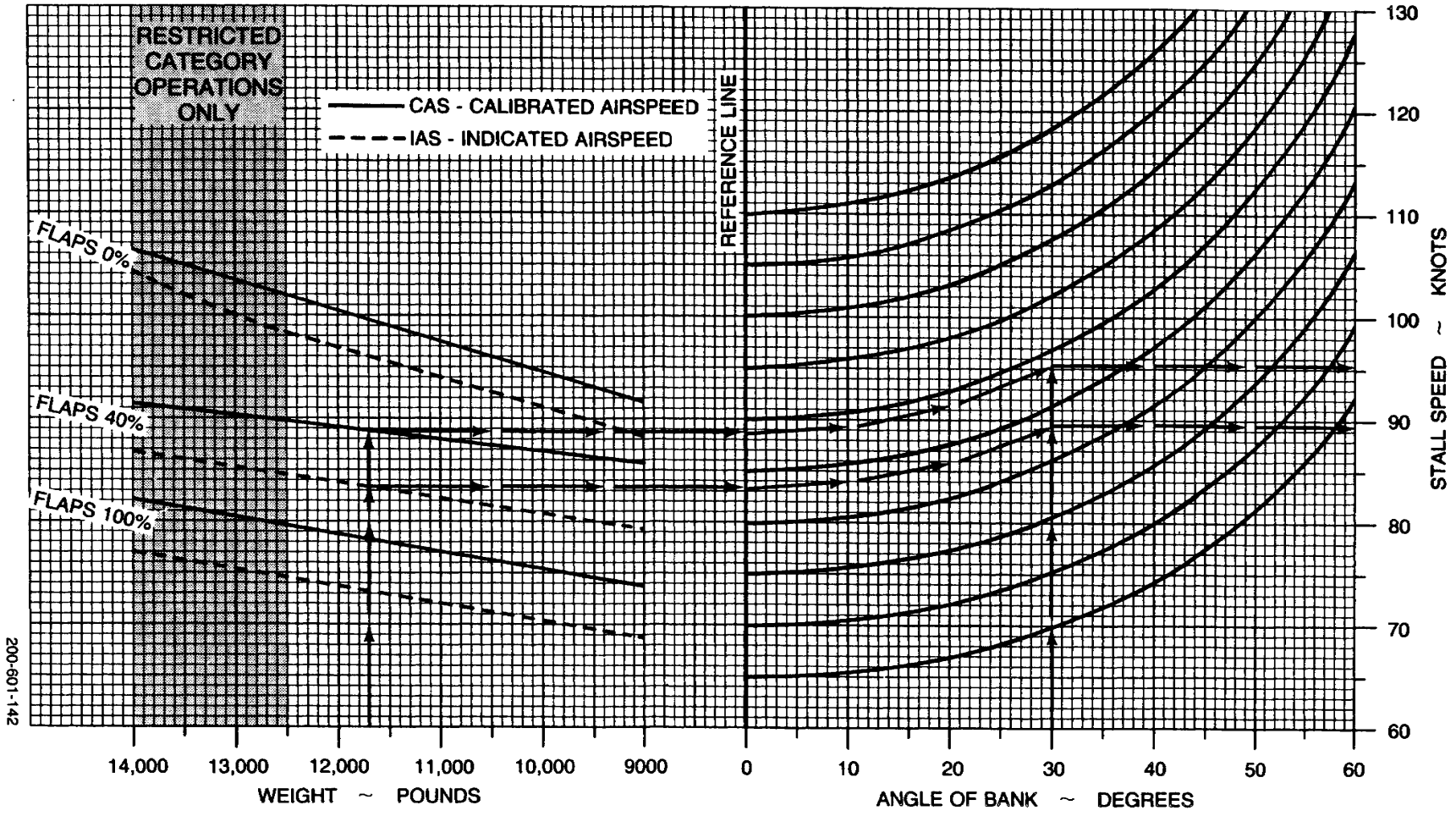
STALL SPEEDS-POWER IDLE

NOTES:

1. MAXIMUM ALTITUDE LOSS DURING A NORMAL STALL RECOVERY IS APPROXIMATELY 800 FEET
2. MAXIMUM NOSE DOWN PITCH ATTITUDE AND ALTITUDE LOSS DURING RECOVERY FROM ONE ENGINE INOPERATIVE STALLS PER FAR 23.205 ARE APPROXIMATELY 8° AND 300 FEET RESPECTIVELY.
3. A NORMAL STALL RECOVERY TECHNIQUE MAY BE USED. THE BEST PROCEDURE IS A BRISK FORWARD WHEEL MOVEMENT TO A NOSE DOWN ATTITUDE. LEVEL THE AIRPLANE AFTER AIRSPEED HAS INCREASED APPROXIMATELY 25 KNOTS ABOVE STALL.

EXAMPLE:

WEIGHT	11,700 LBS
FLAPS	40%
ANGLE OF BANK	30°
STALL SPEED	95 KTS CAS 90 KTS IAS



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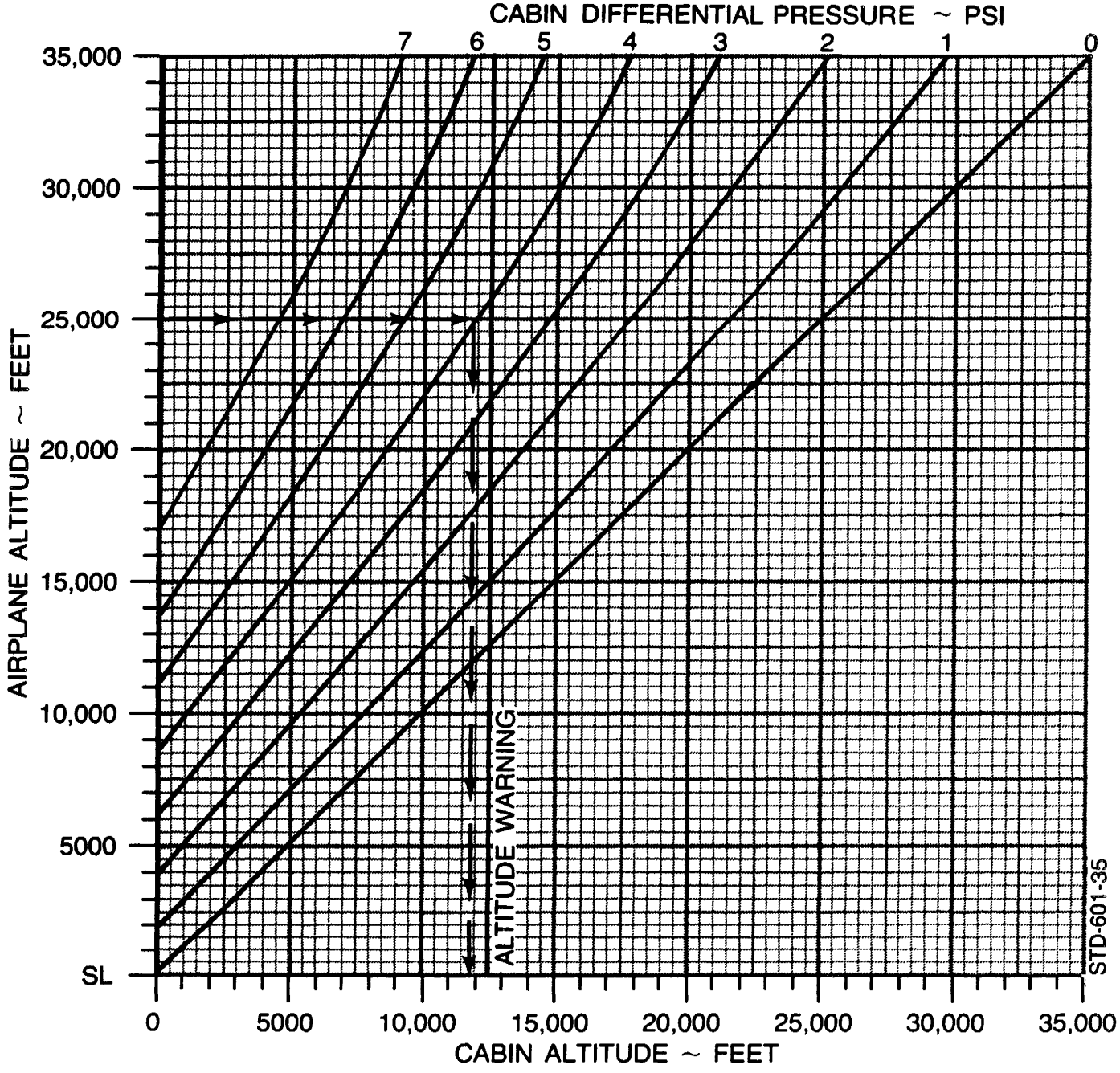
CABIN ALTITUDE FOR VARIOUS AIRPLANE ALTITUDES

EXAMPLE:

AIRPLANE ALTITUDE	25,000 FT
CABIN DIFFERENTIAL PRESSURE	4.0 PSI
CABIN ALTITUDE	11,800 FT



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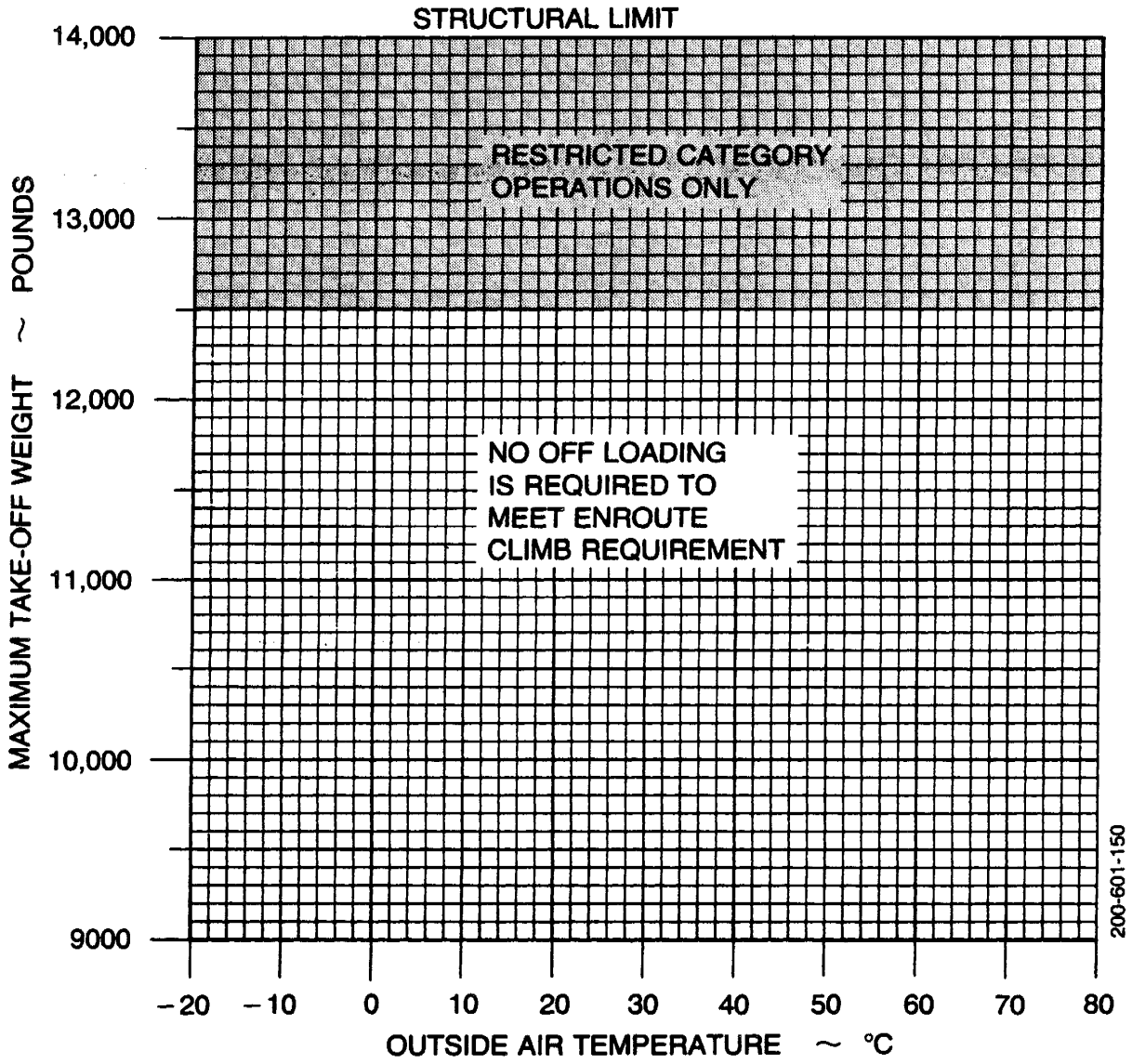
MAXIMUM TAKE-OFF WEIGHT

PERMITTED BY ENROUTE CLIMB REQUIREMENT



TAKE-OFF WEIGHT - FLAPS 0%

TO ACHIEVE POSITIVE ONE-ENGINE-INOPERATIVE CLIMB AT LIFT-OFF

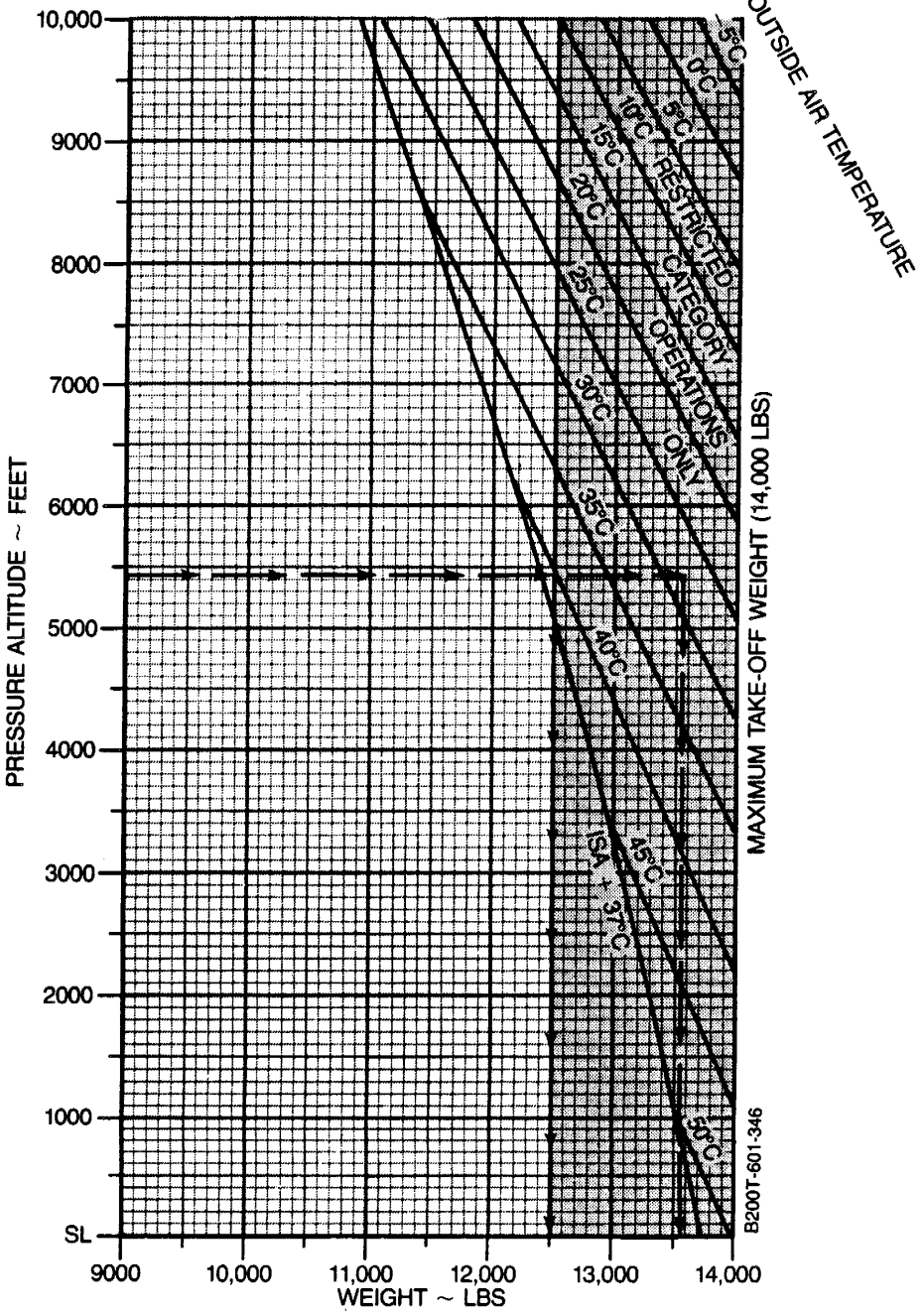
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	UP (0%)
LANDING GEAR	DOWN

EXAMPLE:

PRESSURE ALTITUDE	5433 FT
OAT	28°C
TAKE-OFF WEIGHT NORMAL CATEGORY	12,500 LBS
TAKE-OFF WEIGHT RESTRICTED CATEGORY	13,550 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, NO OFF-LOADING IS REQUIRED.



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TAKE-OFF WEIGHT - FLAPS 40%

TO ACHIEVE POSITIVE ONE-ENGINE-INOPERATIVE CLIMB AT LIFT-OFF

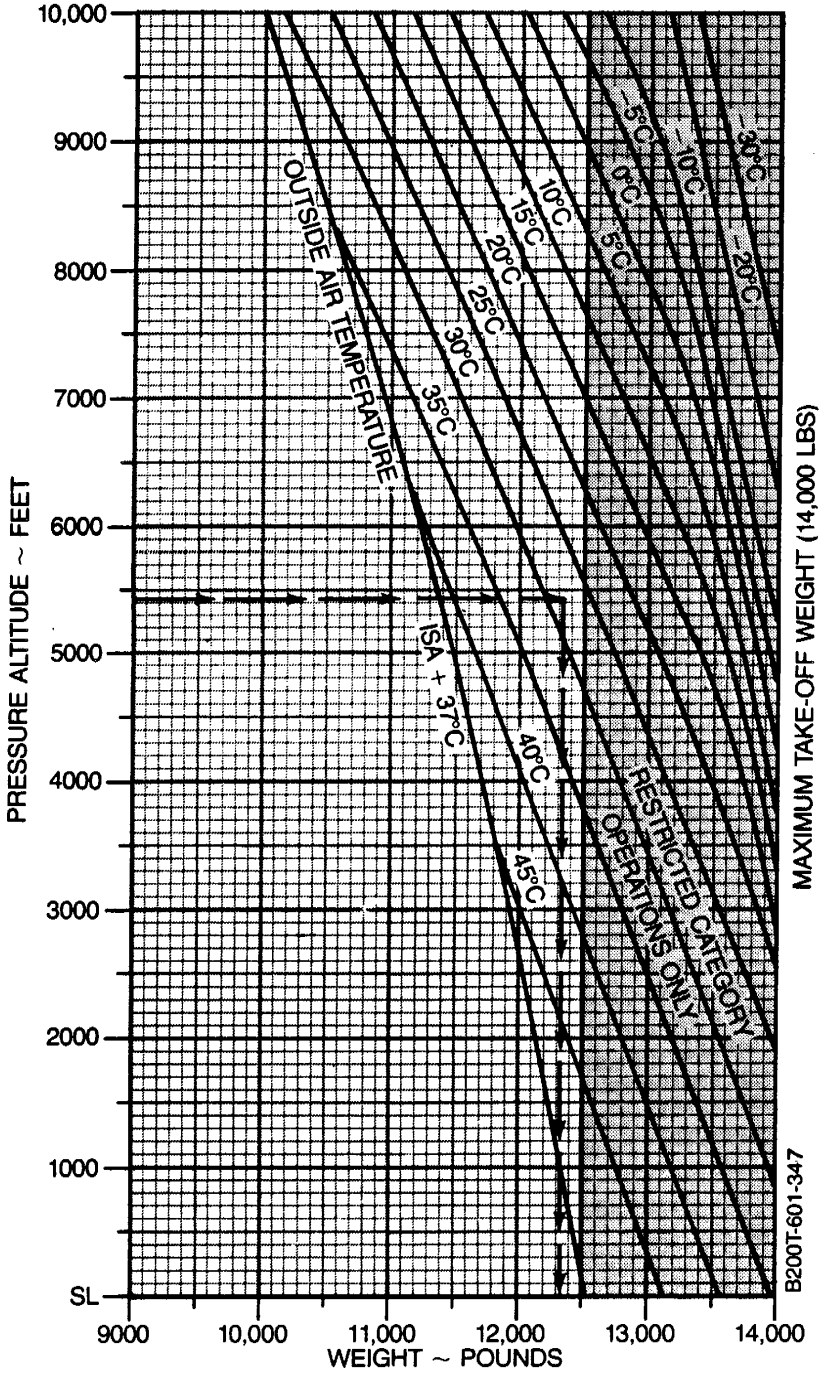
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	40%
LANDING GEAR	DOWN

EXAMPLE:

PRESSURE ALTITUDE	5433 FT
OAT	28°C
TAKE-OFF WEIGHT	12,340 LBS

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, ADD 10°C TO THE ACTUAL OAT BEFORE ENTERING GRAPH



MAXIMUM ENROUTE WEIGHT

(FAR PART 135 OPERATIONS)

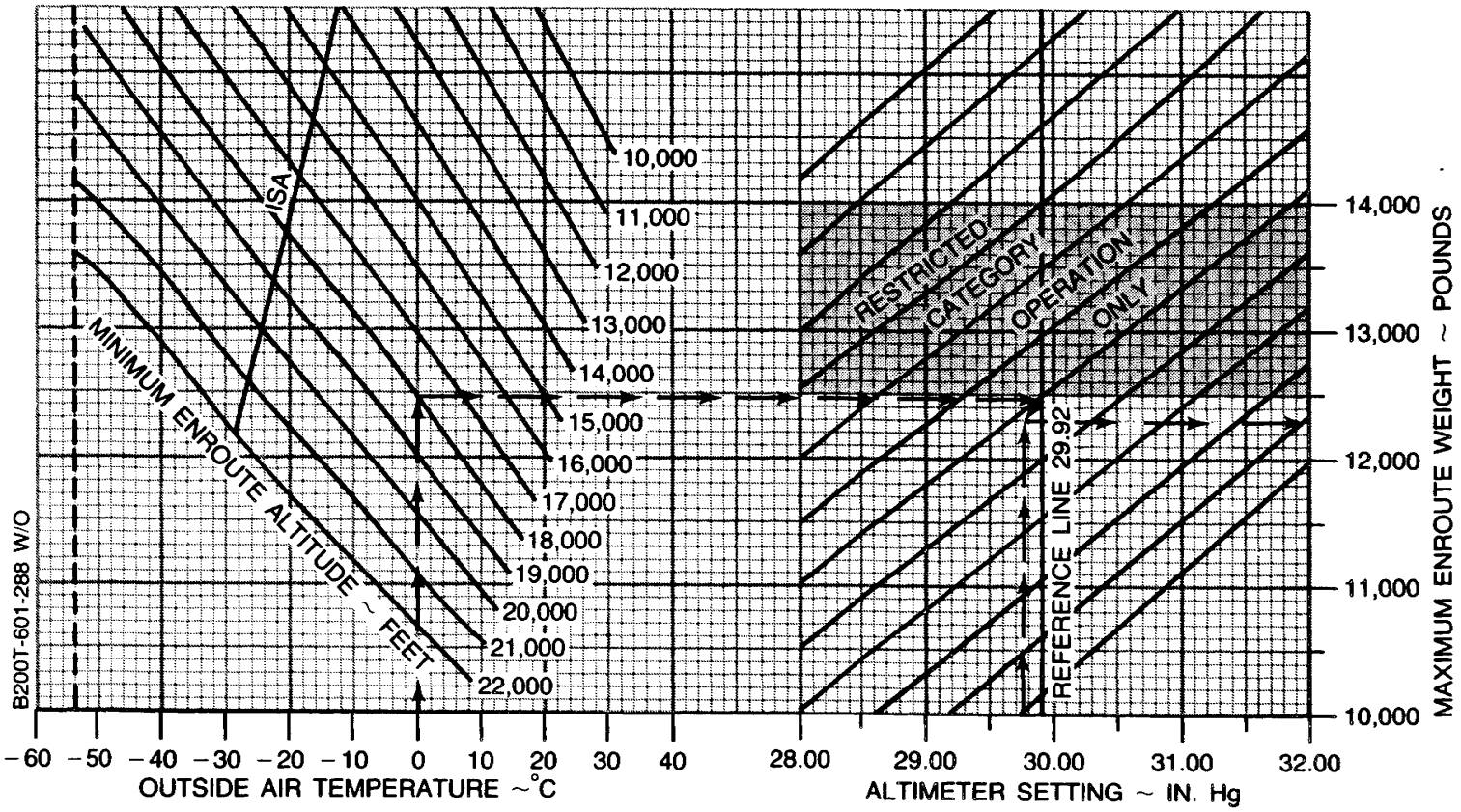
ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
FLAPS	0%
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

EXAMPLE:

OAT AT MEA (WORST LEG)	0°C
MINIMUM ENROUTE ALTITUDE	18,000 FT
ALTIMETER SETTING	29.75 IN. Hg
MAXIMUM ALLOWABLE WEIGHT	12,300 LBS

NOTE: PER FAR 135, OPERATIONS OVER THE TOP OR IN IFR CONDITIONS REQUIRE THAT THE AIRPLANE BE CAPABLE OF CLIMBING 50 FT/MIN AT THE MEA'S OF THE PROPOSED ROUTE OR 5000 FEET MSL, WHICHEVER IS HIGHER.



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TAKE-OFF WEIGHT-FLAPS 0%

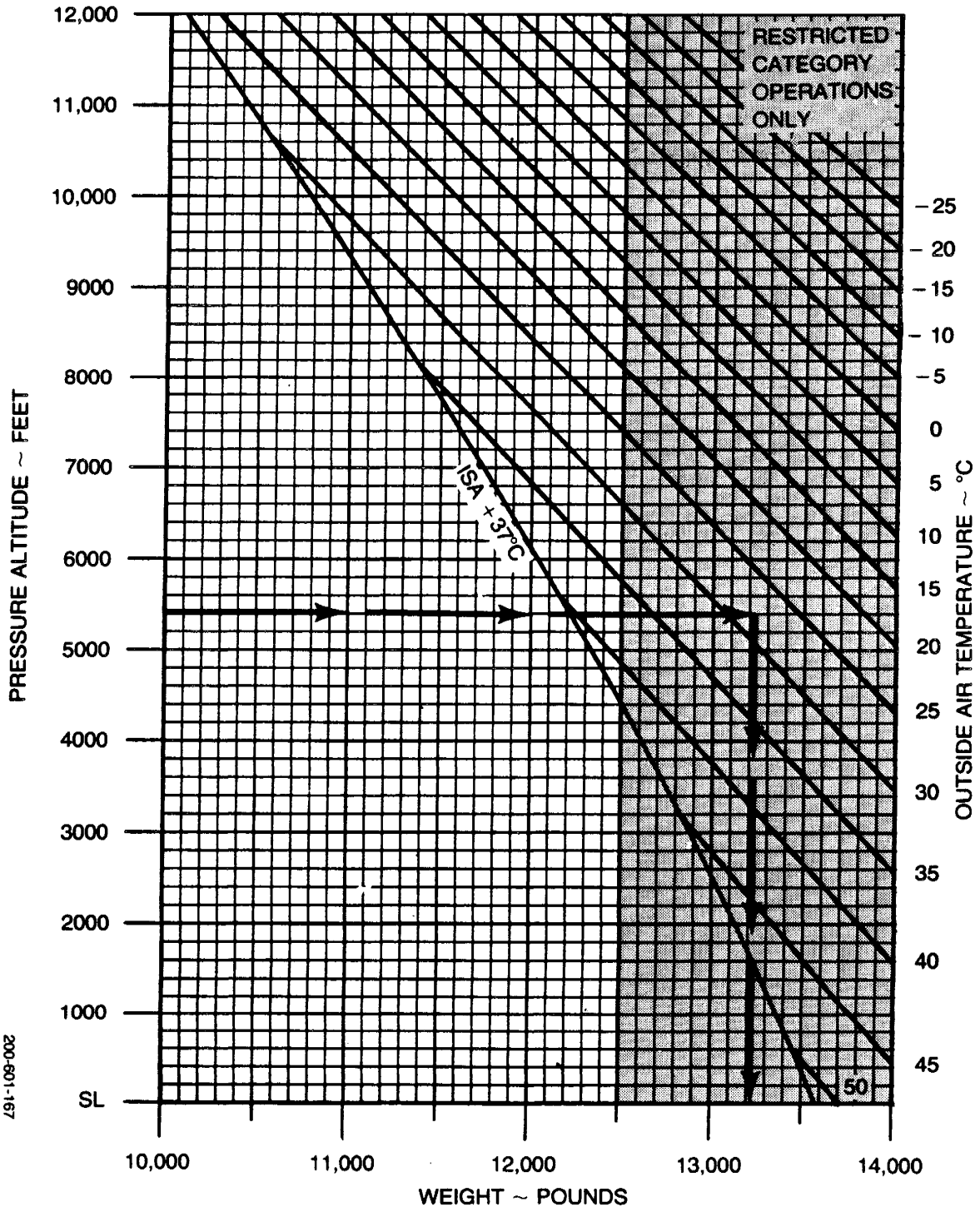
TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS

ASSOCIATED CONDITIONS:

AIRPLANE	AIRBORNE
POWER	TAKE-OFF
FLAPS	0%
INOPERATIVE PROPELLER	FEATHERED

EXAMPLE:

PRESSURE ALTITUDE	5430 FT
OAT	28°C
TAKE-OFF WEIGHT	13,225 LBS



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TAKE-OFF WEIGHT - FLAPS 40%

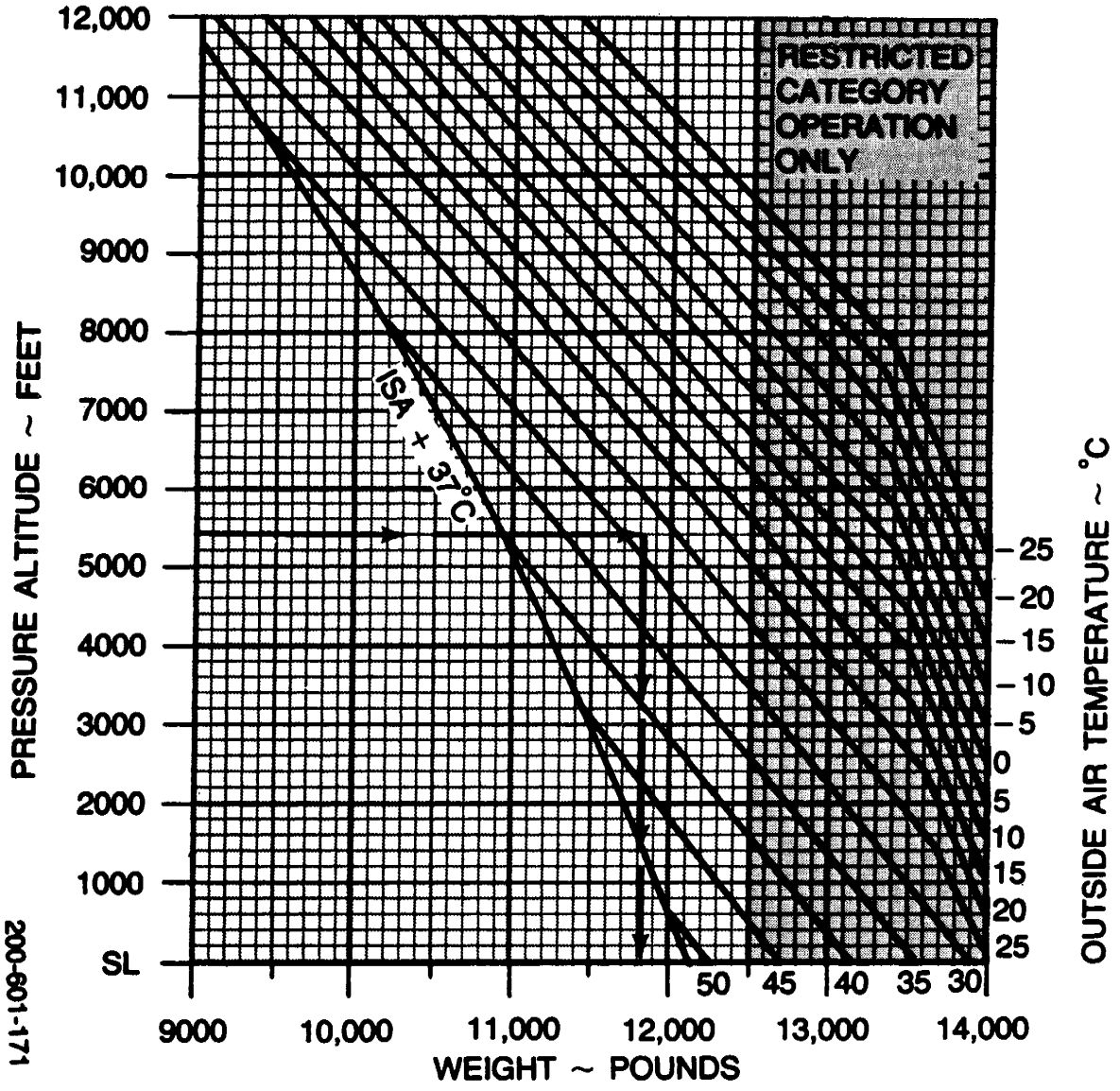
TO MEET FAR 25 TAKE-OFF CLIMB REQUIREMENTS

ASSOCIATED CONDITIONS:

AIRPLANE	AIRBORNE
POWER	TAKE-OFF
FLAPS	40%
INOPERATIVE PROPELLER	FEATHERED

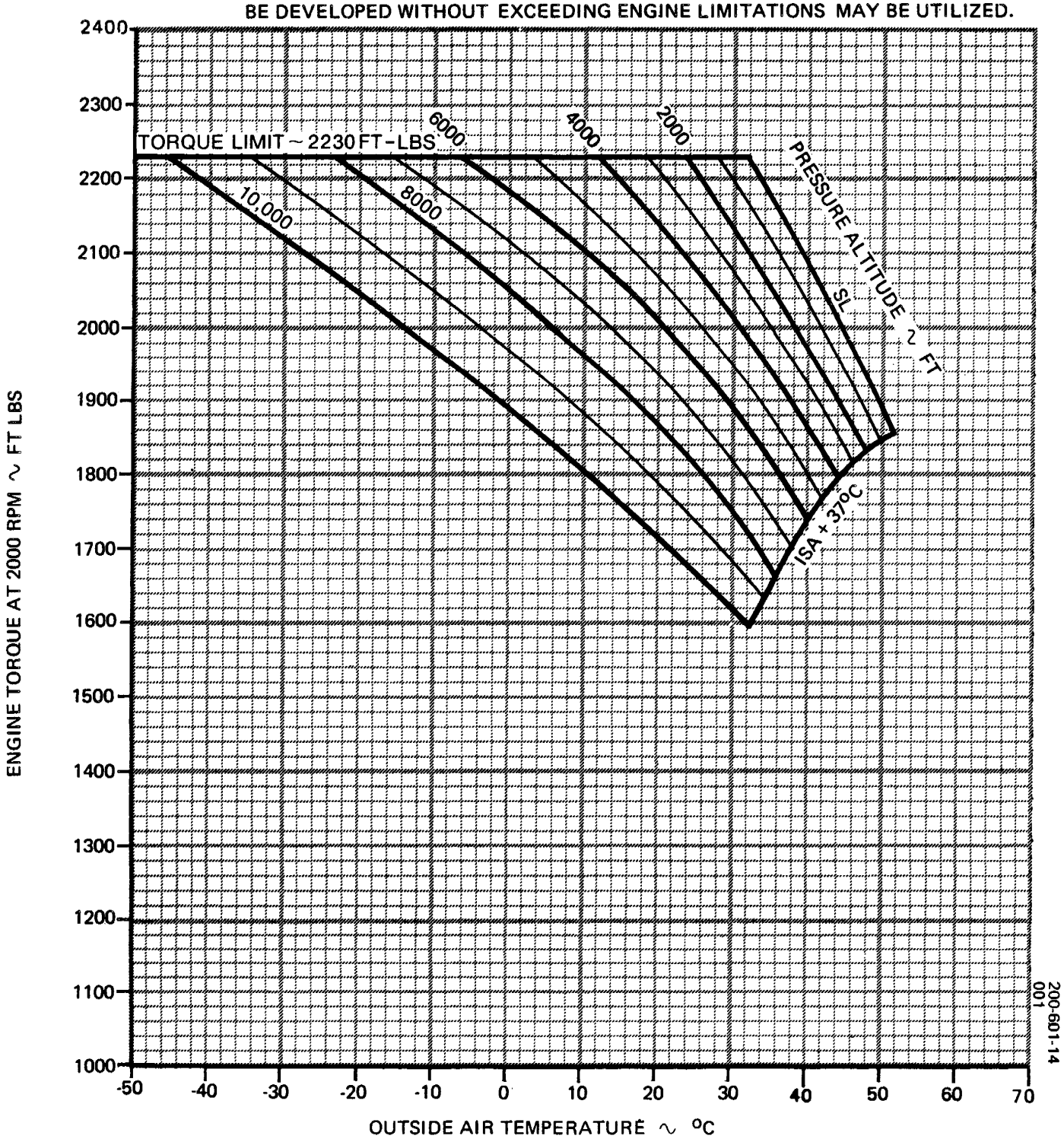
EXAMPLE:

PRESSURE ALTITUDE	5430 FT
OAT	28°C
TAKE-OFF WEIGHT	11,820 LB



MINIMUM TAKE-OFF POWER AT 2000 RPM
WITH ICE VANES RETRACTED
(65 KNOTS)

NOTES: 1. TORQUE INCREASES APPROXIMATELY 20 FT-LBS FROM ZERO TO 65 KNOTS.
2. THE POWER (TORQUE) INDICATED IS THE MINIMUM VALUE FOR WHICH TAKE-OFF PERFORMANCE IN THIS SECTION CAN BE OBTAINED. EXCESS POWER WHICH CAN BE DEVELOPED WITHOUT EXCEEDING ENGINE LIMITATIONS MAY BE UTILIZED.



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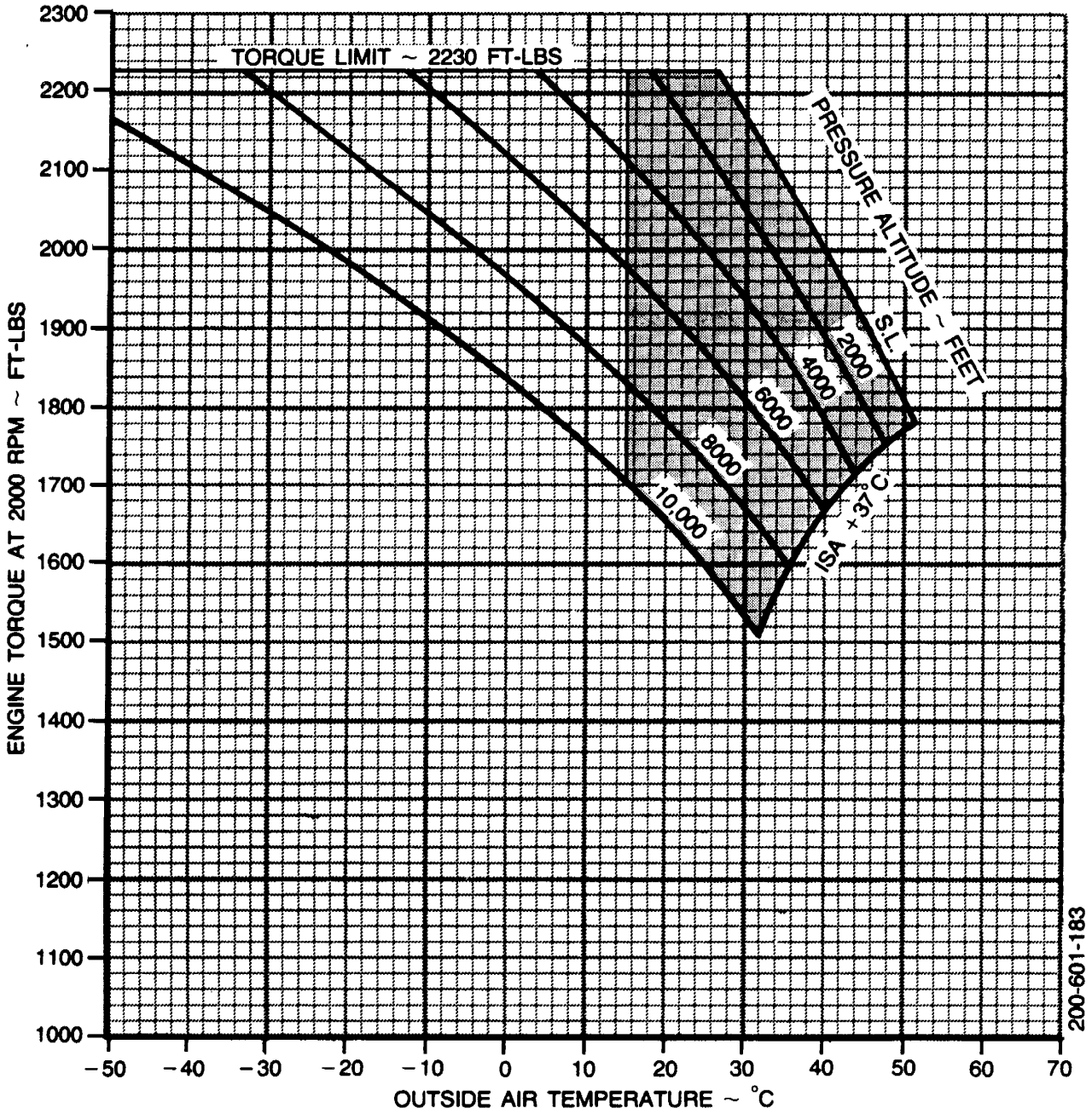
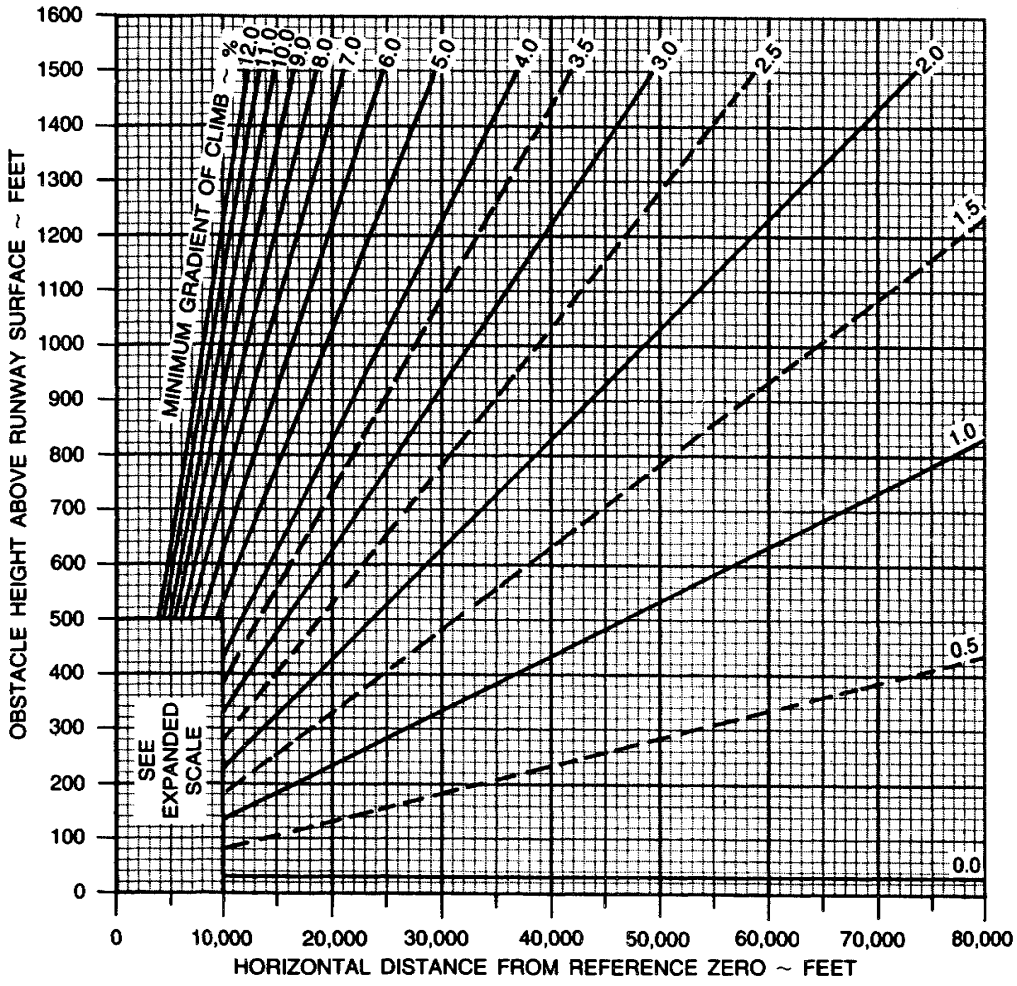
MINIMUM TAKE-OFF POWER
WITH ICE VANES EXTENDED
(65 KNOTS)

NOTES:
1. TORQUE INCREASES APPROXIMATELY 20 FT-LBS FROM ZERO TO 65 KNOTS
2. THE POWER (TORQUE) INDICATED IS THE MINIMUM VALUE FOR WHICH TAKE-OFF PERFORMANCE WITH EXTENDED ICE VANES CAN BE OBTAINED EXCESS POWER WHICH CAN BE DEVELOPED WITHOUT EXCEEDING ENGINE LIMITATIONS MAY BE UTILIZED
3. TAKE-OFF WITH VANES EXTENDED IS PROHIBITED IN SHADED AREA.



TAKE-OFF FLIGHT PATH

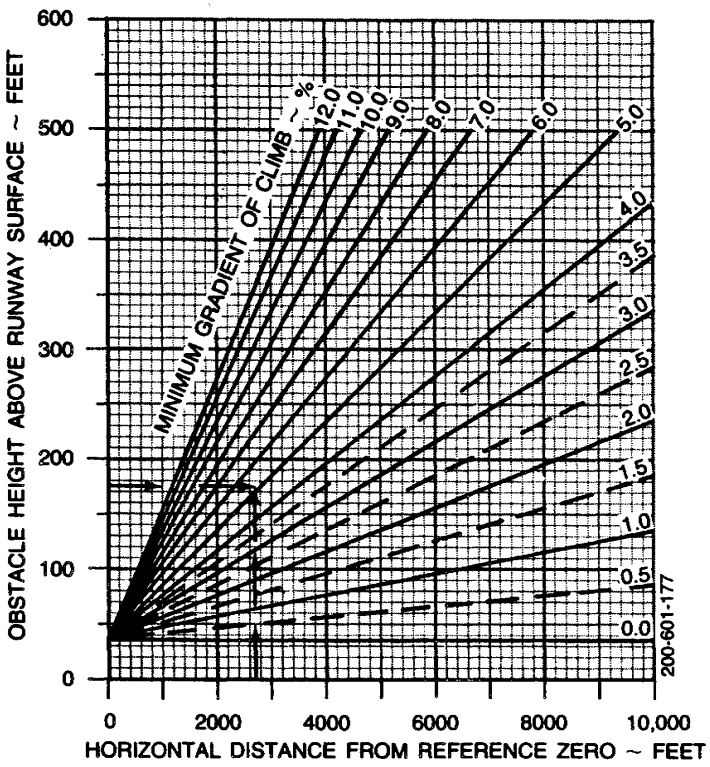
REFERENCE ZERO: THE POINT AT THE END OF THE TAKE-OFF RUN AT WHICH THE AIRPLANE IS 35 FEET ABOVE THE RUNWAY SURFACE.



EXAMPLE:

OBSTACLE HEIGHT ..... 175 FEET

HORIZONTAL DISTANCE FROM REFERENCE ZERO ~ FEET	MINIMUM GRADIENT OF CLIMB ~ %
2700	5.2



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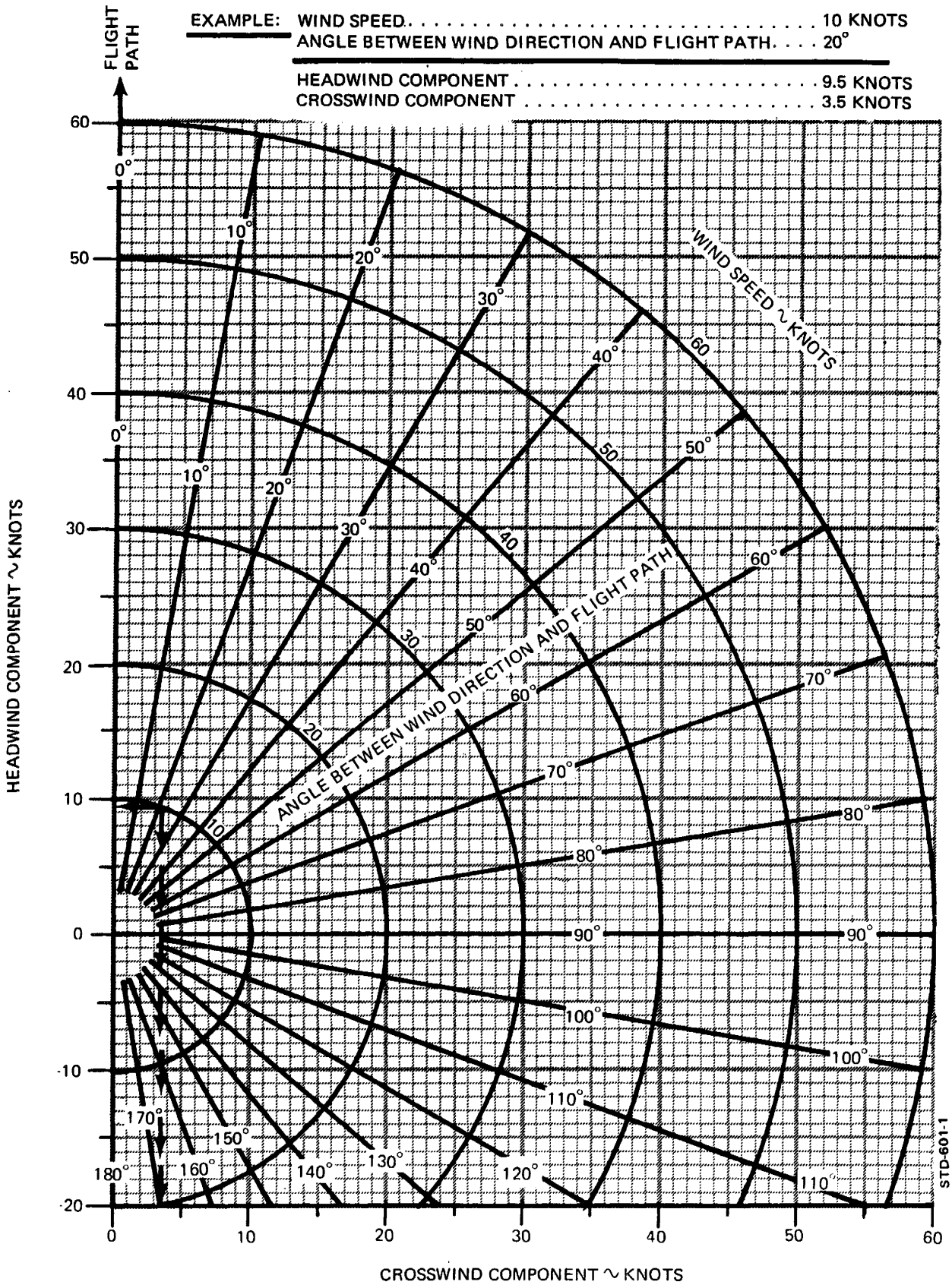
WIND COMPONENTS



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TAKE-OFF DISTANCE-FLAPS 0%

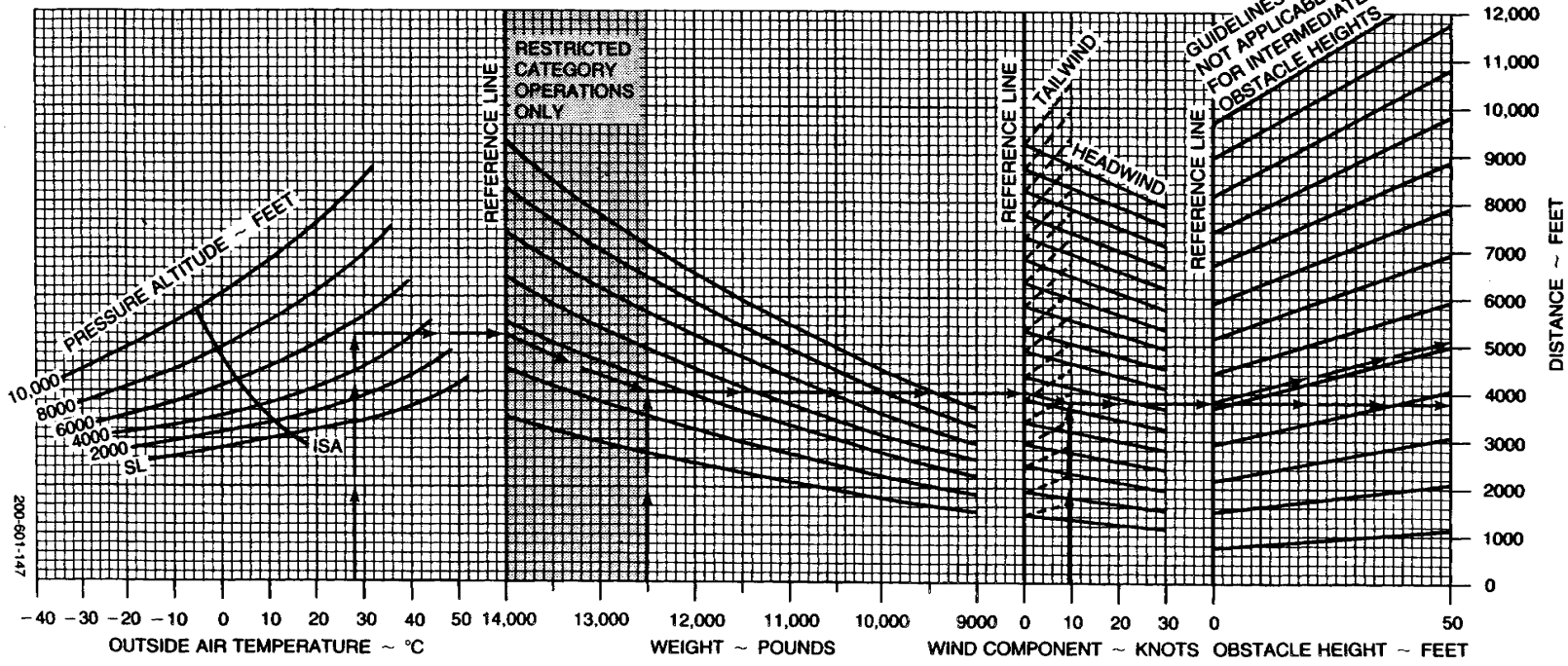
ASSOCIATED CONDITIONS:

POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	0%
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	TAKE-OFF SPEEDS ~ KNOTS	
	ROTATION	50 FT
14,000	117	126
13,000	113	123
12,500	111	121
12,000	109	120
11,000	105	115
10,000	101	111
9000	98	108

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5433 FT
TAKE-OFF WEIGHT	12,500 LBS
HEADWIND COMPONENT	9.5 KTS
GROUND ROLL	3800 FT
TOTAL DISTANCE OVER 50 FT OBSTACLE	5100 FT
TAKE-OFF SPEED AT ROTATION	111 KTS
AT 50 FT	121 KTS



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ACCELERATE-STOP-FLAPS 0%

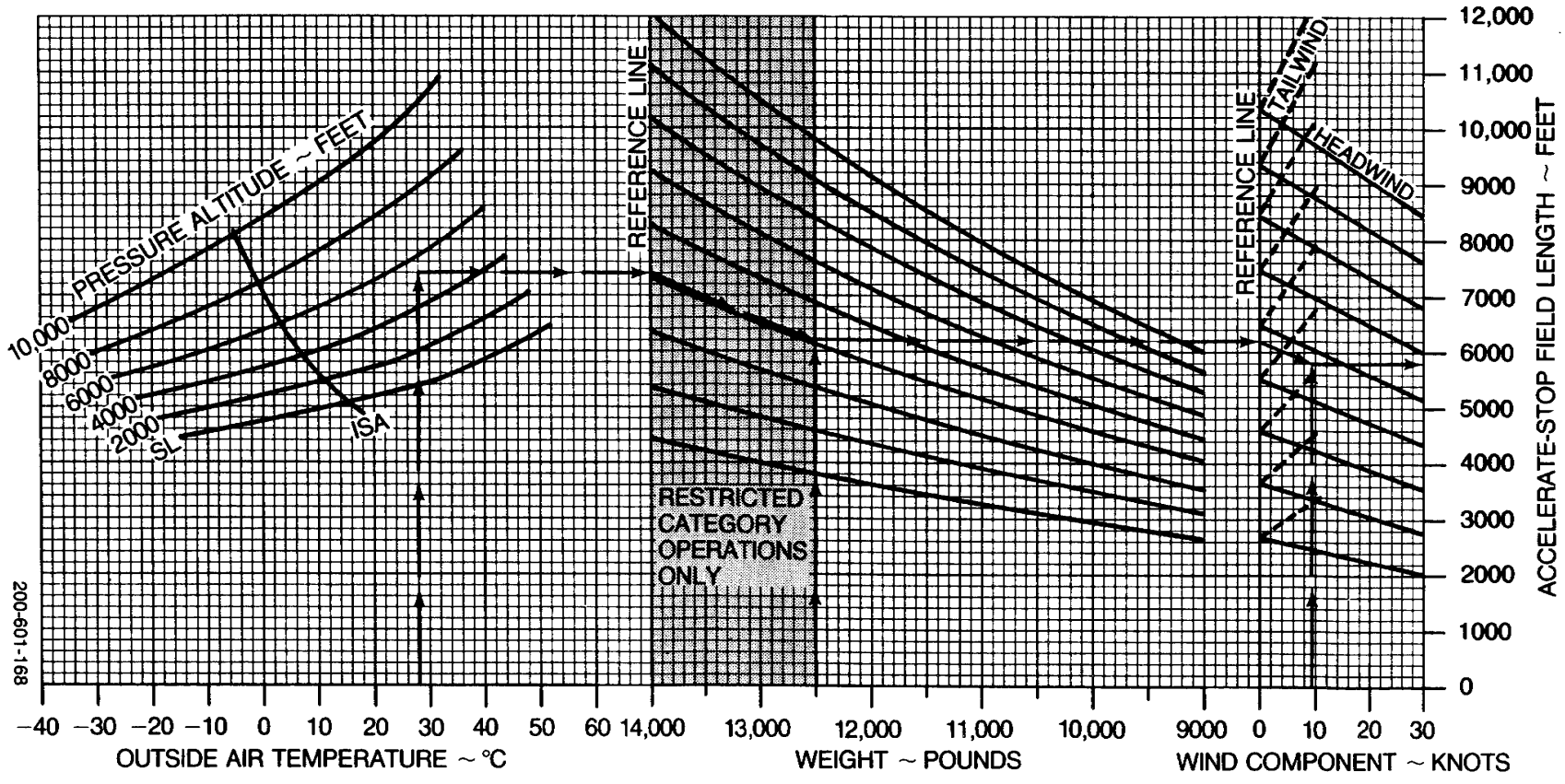
ASSOCIATED CONDITIONS:

POWER	1.	TAKE-OFF POWERSET BEFORE BRAKE RELEASE
	2.	BOTH ENGINES IDLE AT V_1 SPEED AND REVERSE OPERATING ENGINE
FLAPS		0%
AUTOFEATHER		ARMED
BRAKING		MAXIMUM
RUNWAY		PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	V_1 ~ KNOTS
14,000	117
13,000	113
12,500	111
12,000	109
11,000	105
10,000	101
9000	98

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
WEIGHT	12,500 LBS
HEADWIND COMPONENT	9.5 KTS
FIELD LENGTH	5800 FT
V_1	111 KTS



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ACCELERATE-GO - FLAPS 0%

ASSOCIATED CONDITIONS:

POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	0%
AUTOFEATHER	ARMED
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

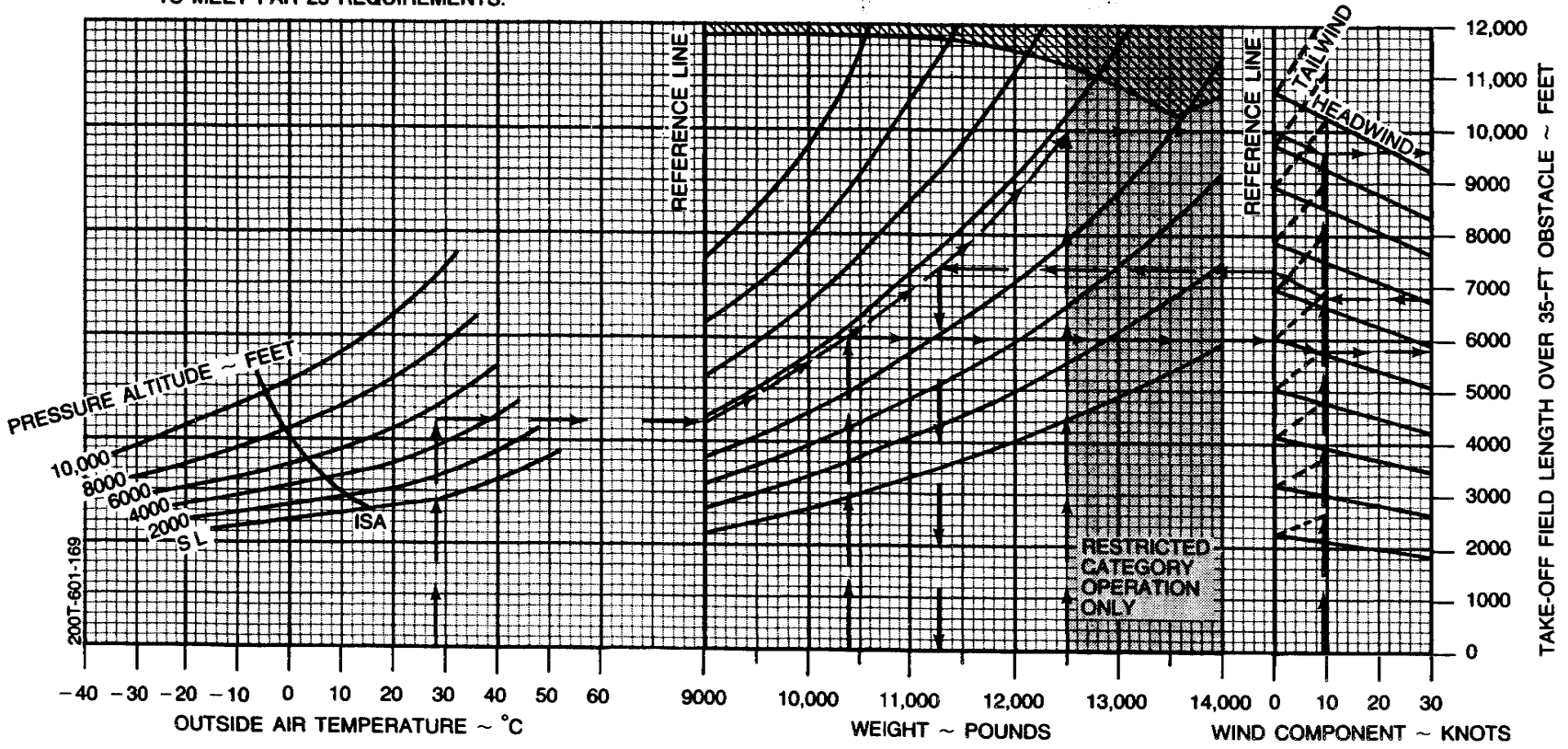
NOTES:
1. AIR DISTANCE IS 50% OF TAKE-OFF DISTANCE OVER 35-FT OBSTACLE.
2. DISTANCES ASSUME AN ENGINE FAILURE AT ROTATION SPEED AND PROPELLER IMMEDIATELY FEATHERED.
3. USABLE CLEARWAY CANNOT EXCEED 25% OF THE RUNWAY LENGTH.
4. WEIGHTS IN ▨ AREA MAY NOT MEET FAR 25 REQUIREMENTS. REFER TO TAKE-OFF WEIGHT GRAPH TO MEET FAR 25 REQUIREMENTS.

WEIGHT ~ POUNDS	SPEED ~ KNOTS V_R	SPEED ~ KNOTS V_2
14,000	117	128
13,000	113	125
12,500	111	121
12,000	109	120
11,000	105	115
10,000	101	111
9000	98	108

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT
HEADWIND COMPONENT	9.5 KTS

TAKE-OFF WEIGHT ~ POUNDS	TAKE-OFF FIELD LENGTH ~ FEET
12,500	9600
11,280	6786
10,370	5800



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NET GRADIENT OF CLIMB-FLAPS 0%

ASSOCIATED CONDITIONS:

POWER ..... TAKEOFF
FLAPS 0%
GEAR ..... UP
INOPERATIVE PROPELLER ..... FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	126
13,000	123
12,500	121
12,000	120
11,000	115
10,000	111
9000	108

EXAMPLE:

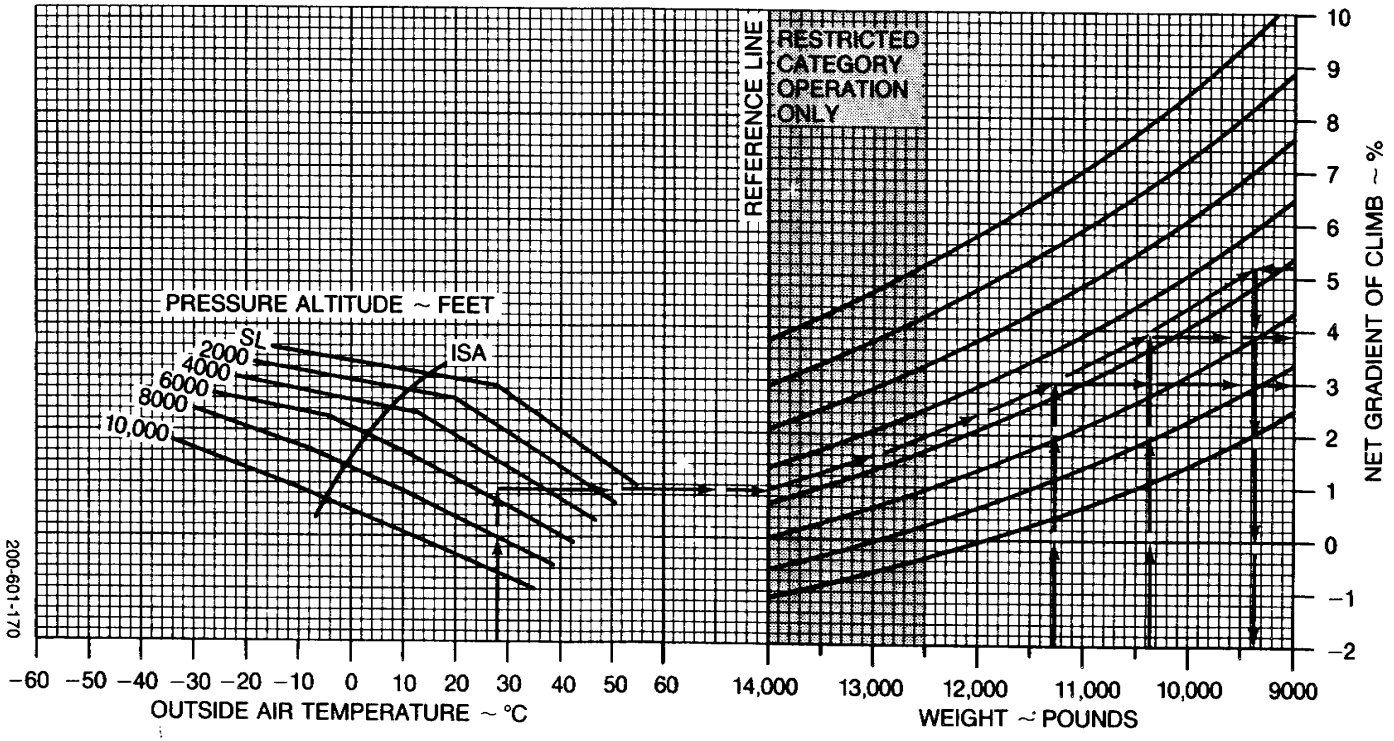
OAT ..... 28°C
PRESSURE ALTITUDE ..... 5430 FT

TAKE-OFF WEIGHT ~ POUNDS	NET GRADIENT ~ %
11,280	3.0
9380	5.2
10,370	3.9

CLIMB SPEED @ 10,370 LBS ..... 112 KTS

NOTE:

NET GRADIENT OF CLIMB
IS EQUAL TO ACTUAL GRADIENT OF CLIMB IN PERCENT MINUS 0.8 PERCENT



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TAKE-OFF DISTANCE-FLAPS 40%

ASSOCIATED CONDITIONS:

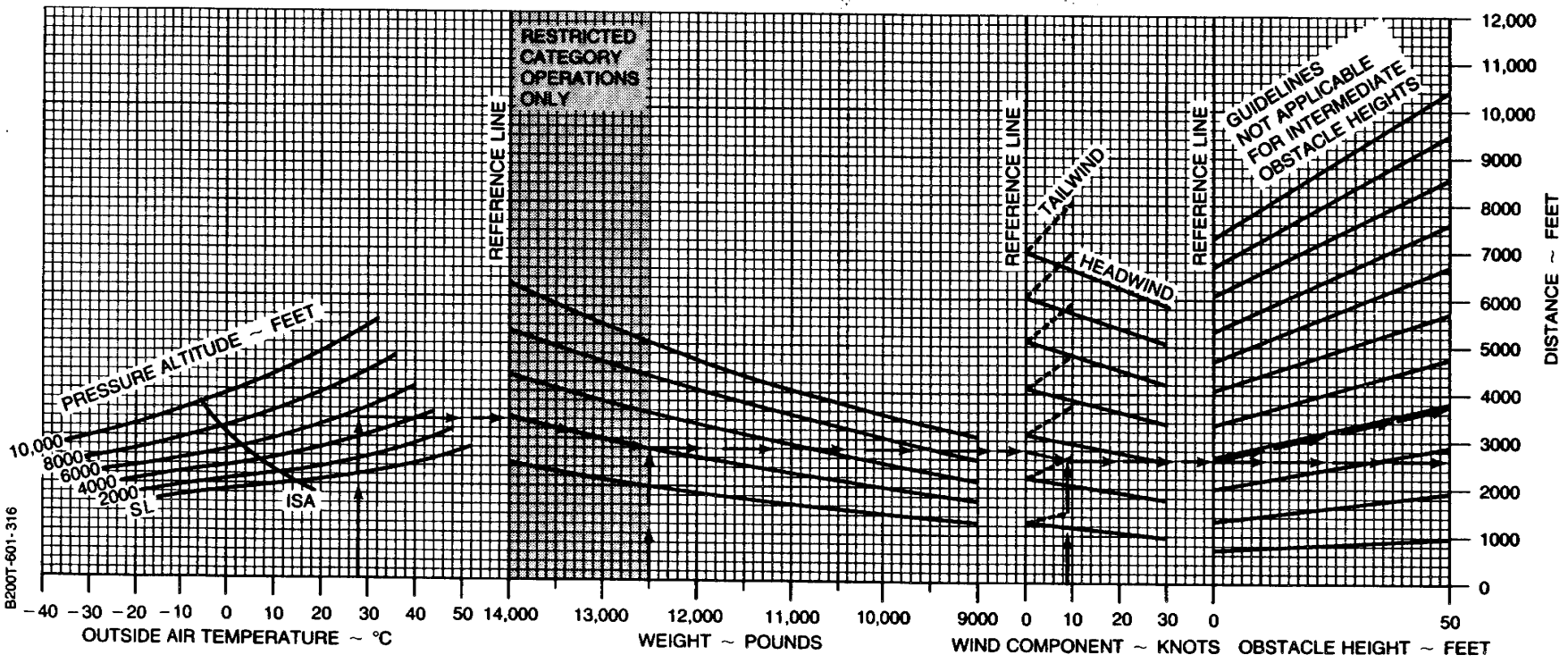
POWER.....TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS.....40%
LANDING GEAR.....RETRACT AFTER LIFT-OFF
RUNWAY.....PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	TAKE-OFF SPEED ~ KNOTS	
	ROTATION	50 FT
14,000	98	108
13,000	96	106
12,500	96	105
12,000	95	104
11,000	94	103
10,000	94	101
9000	94	99

EXAMPLE:

OAT.....28°C
PRESSURE ALTITUDE.....5433 FT
TAKE-OFF WEIGHT.....12,500 LBS
HEADWIND COMPONENT.....9.5 KTS

GROUND ROLL.....2600 FT
TOTAL DISTANCE OVER 50 FT OBSTACLE.....3750 FT
TAKE-OFF SPEED AT ROTATION.....96 KTS
AT 50 FT.....105 KTS



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ACCELERATE-STOP - FLAPS 40%

ASSOCIATED CONDITIONS:

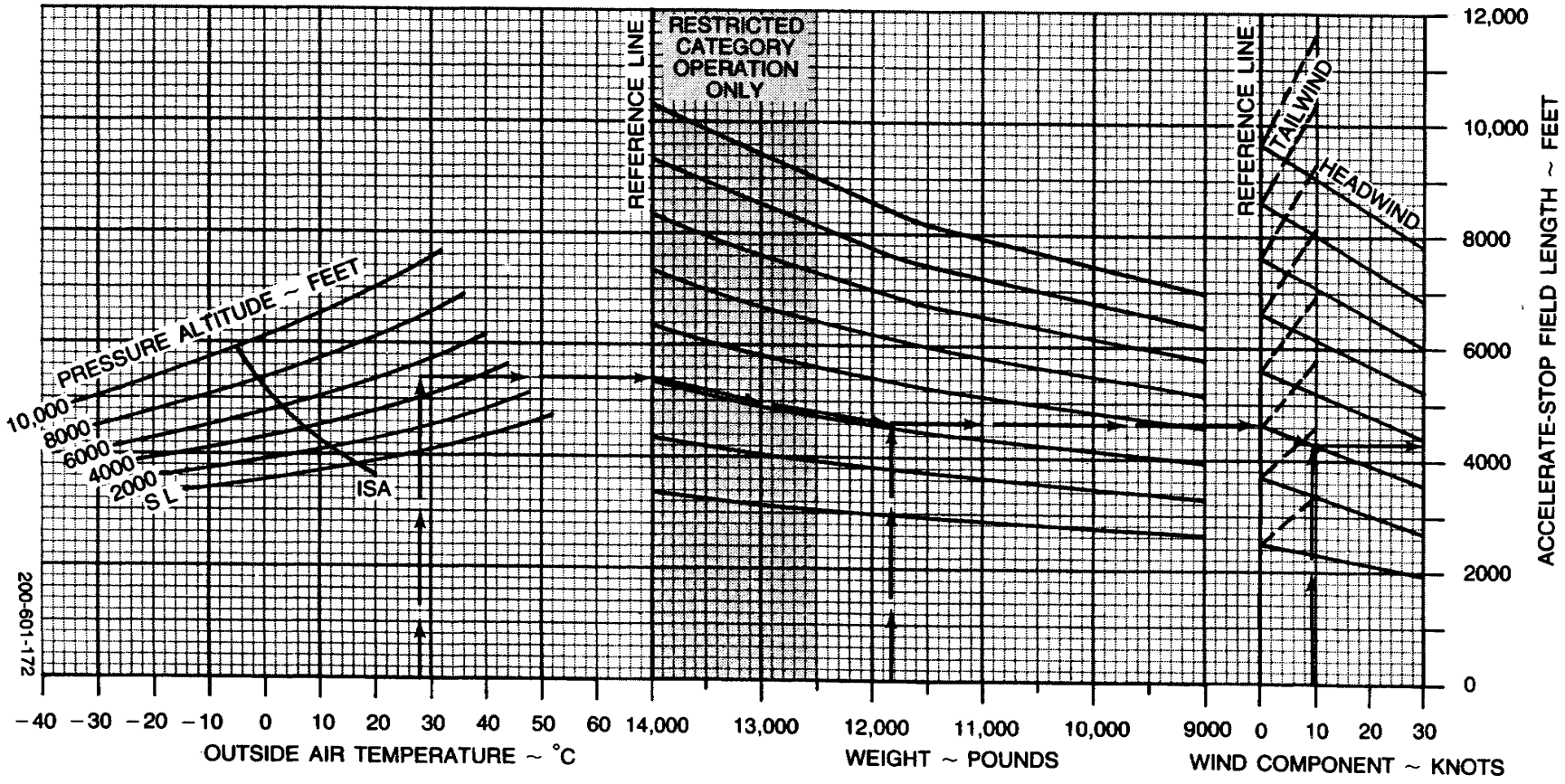
POWER ..... 1. TAKE-OFF POWER SET BEFORE BRAKE RELEASE
2. BOTH ENGINES IDLE AT V_1 SPEED AND REVERSE OPERATING ENGINE
FLAPS 40%
AUTOFEATHER . ARMED
BRAKING ..... MAXIMUM
RUNWAY ..... PAVED, LEVEL, DRY SURFACE

WEIGHT ~ POUNDS	V_1 ~ KNOTS
14,000	98
13,000	96
12,500	96
12,000	95
11,000	94
10,000	94
9000	94

EXAMPLE:

OAT ..... 28°C
PRESSURE ALTITUDE ..... 5430 FT
WEIGHT ..... 11,820 LBS
HEADWIND COMPONENT ..... 9.5 KTS

FIELD LENGTH ..... 4300 FT
V_1 ..... 95 KTS



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ACCELERATE-GO - FLAPS 40%

ASSOCIATED CONDITIONS:

POWER	TAKE-OFF POWER SET BEFORE BRAKE RELEASE
FLAPS	40%
AUTOFEATHER	ARMED
LANDING GEAR	RETRACT AFTER LIFT-OFF
RUNWAY	PAVED, LEVEL, DRY SURFACE

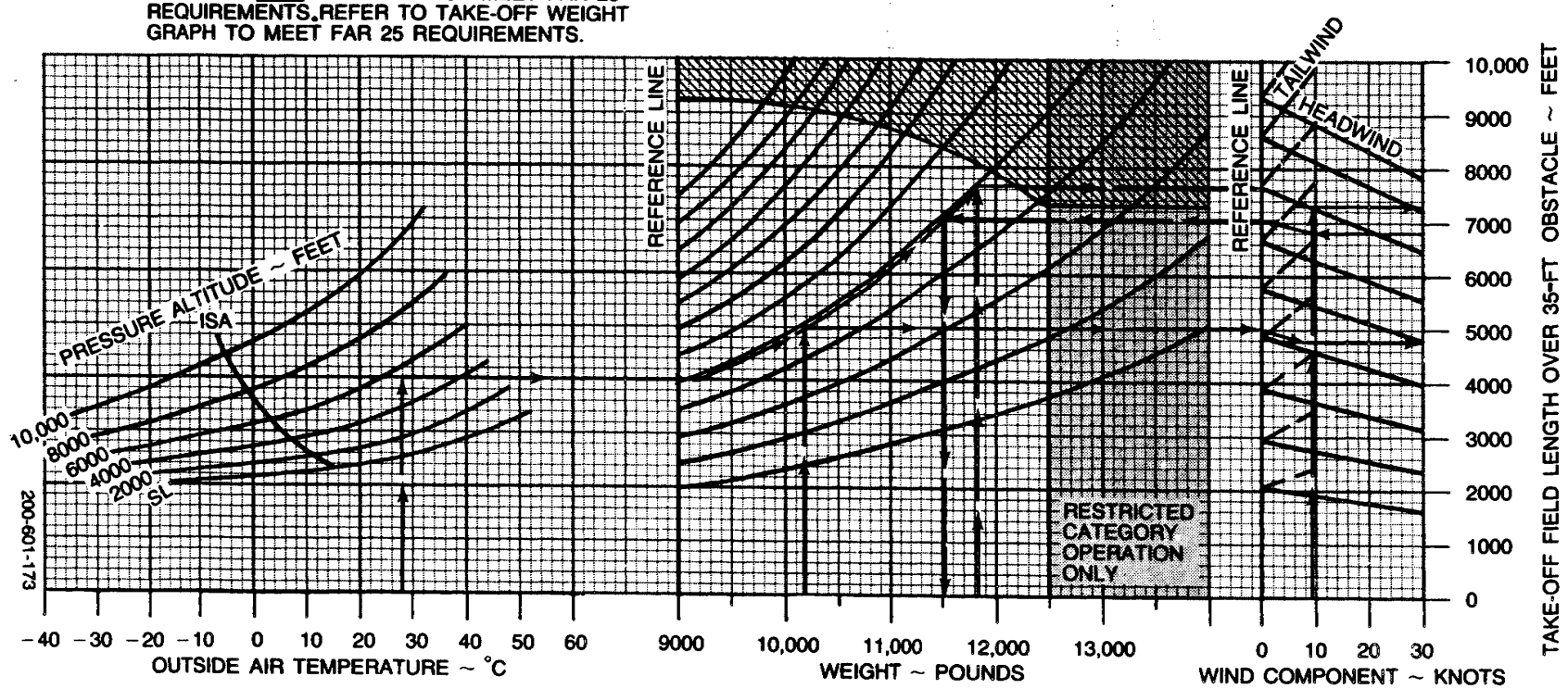
NOTES:
1. AIR DISTANCE IS 60% OF TAKE-OFF DISTANCE OVER 35-FT OBSTACLE.
2. DISTANCES ASSUME AN ENGINE FAILURE AT ROTATION SPEED AND PROPELLER IMMEDIATELY FEATHERED.
3. USABLE CLEARWAY CANNOT EXCEED 25% OF THE RUNWAY LENGTH.
4. WEIGHTS IN ▨ AREA MAY NOT MEET FAR 25 REQUIREMENTS. REFER TO TAKE-OFF WEIGHT GRAPH TO MEET FAR 25 REQUIREMENTS.

WEIGHT ~ POUNDS	SPEED ~ KNOTS V_R	SPEED ~ KNOTS V_2
14,000	98	108
13,000	96	106
12,500	96	105
12,000	95	104
11,000	94	103
10,000	94	101
9000	94	99

EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT.
HEADWIND COMPONENT	9.5 KTS.

TAKE-OFF WEIGHT ~ POUNDS	TAKE-OFF FIELD LENGTH ~ FEET
11,820	7300
11,520	6786
10,175	4820



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NET GRADIENT OF CLIMB - FLAPS 40%

ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	40%
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	108
13,000	106
12,500	105
12,000	104
11,000	103
10,000	101
9000	99

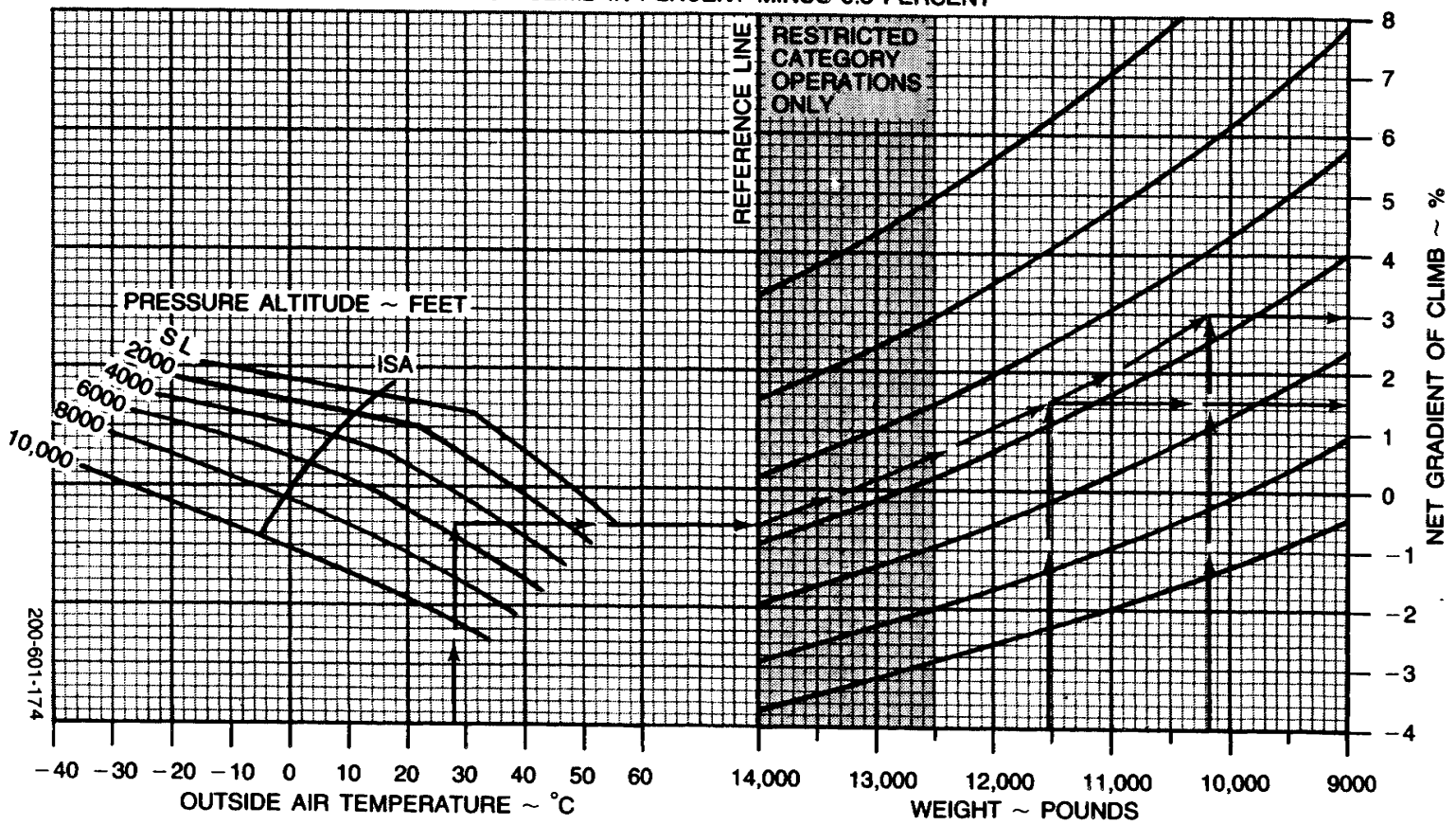
EXAMPLE:

OAT	28°C
PRESSURE ALTITUDE	5430 FT.

TAKE-OFF WEIGHT ~ POUNDS	NET GRADIENT ~ %
11,520	1.5
LESS THAN 9000	5.2
10,175	3.0

CLIMB SPEED @ 10,175 LBS. ..... 101 KTS.

NOTE: NET GRADIENT OF CLIMB IS EQUAL TO ACTUAL GRADIENT OF CLIMB IN PERCENT MINUS 0.8 PERCENT



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CLIMB - TWO ENGINES - FLAPS 0%

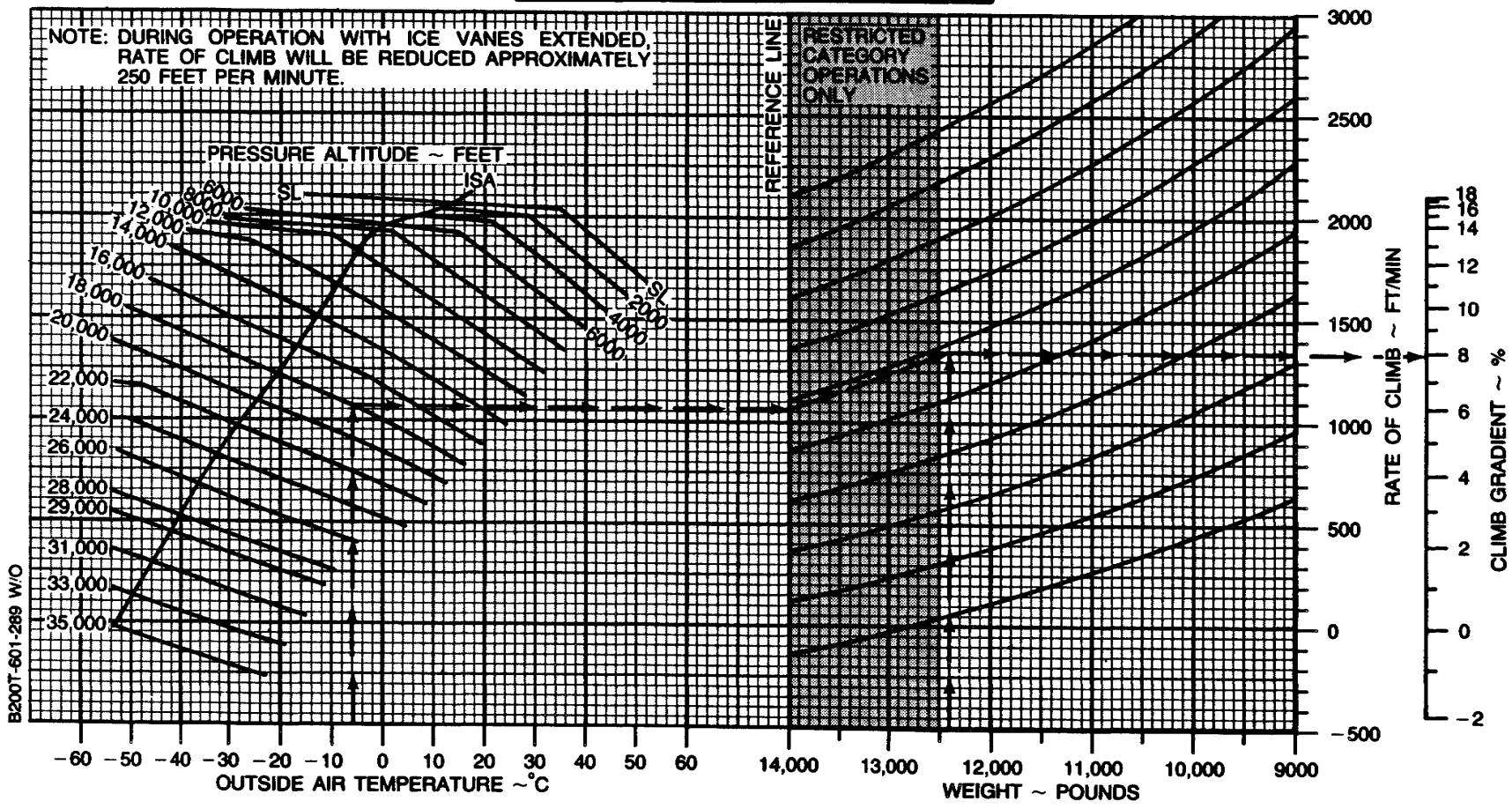
ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
FLAPS	0%
LANDING GEAR	UP

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	129
13,000	126
12,500	125
12,000	123
11,000	121
10,000	118
9000	115

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,391 LBS
RATE OF CLIMB	1340 FT/MIN
CLIMB GRADIENT	7.9%
CLIMB SPEED	125 KNOTS



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CLIMB - TWO ENGINES - FLAPS 40%

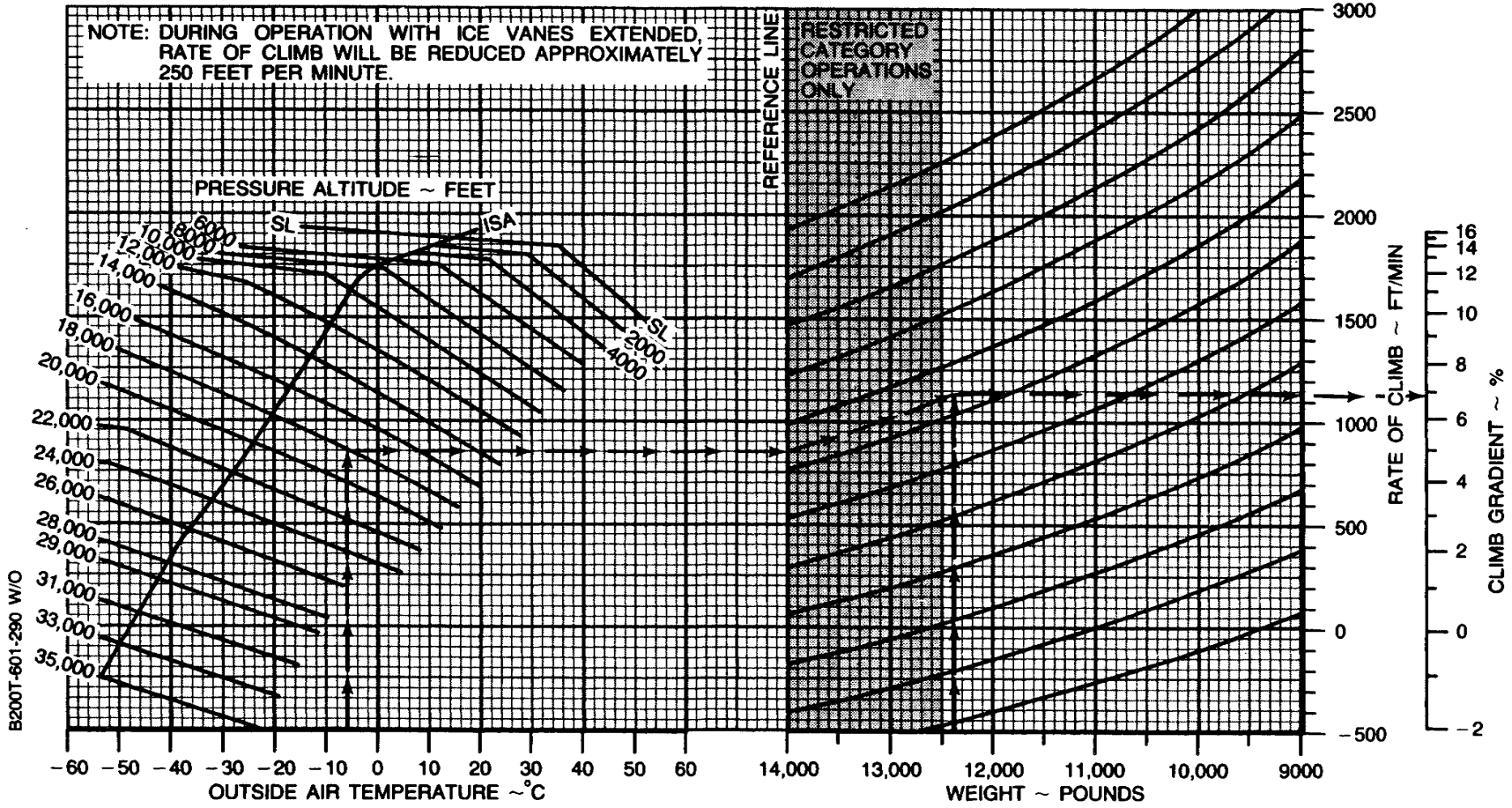
ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
FLAPS	40%
LANDING GEAR	UP

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	129
13,000	126
12,500	125
12,000	123
11,000	121
10,000	118
9000	115

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,391 LBS
RATE OF CLIMB	1135 FT/MIN
CLIMB GRADIENT	6.8%
CLIMB SPEED	125 KNOTS



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CLIMB - ONE ENGINE INOPERATIVE

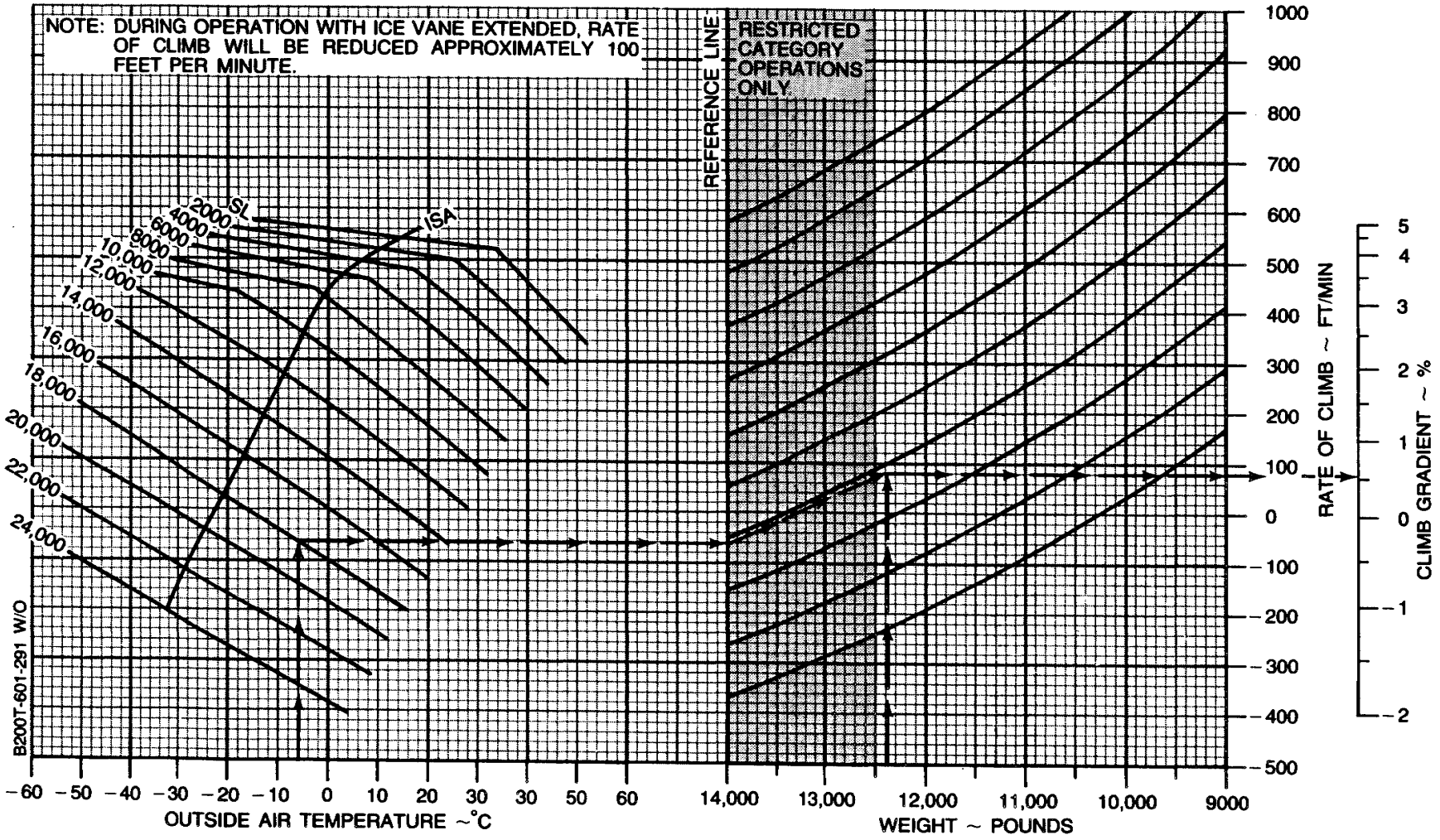
ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
FLAPS	0%
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED

WEIGHT ~ POUNDS	CLIMB SPEED ~ KNOTS
14,000	124
12,500	121
12,000	119
11,000	117
10,000	114
9000	111

EXAMPLE:

OAT	−6°C
PRESSURE ALTITUDE	18,000 FT
WEIGHT	12,391 LBS
RATE OF CLIMB	80 FT/MIN
CLIMB GRADIENT	0.5%
CLIMB SPEED	121 KNOTS



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SERVICE CEILING - ONE ENGINE INOPERATIVE

ASSOCIATED CONDITIONS:

POWER	MAXIMUM CONTINUOUS
LANDING GEAR	UP
INOPERATIVE PROPELLER	FEATHERED
FLAPS	0%

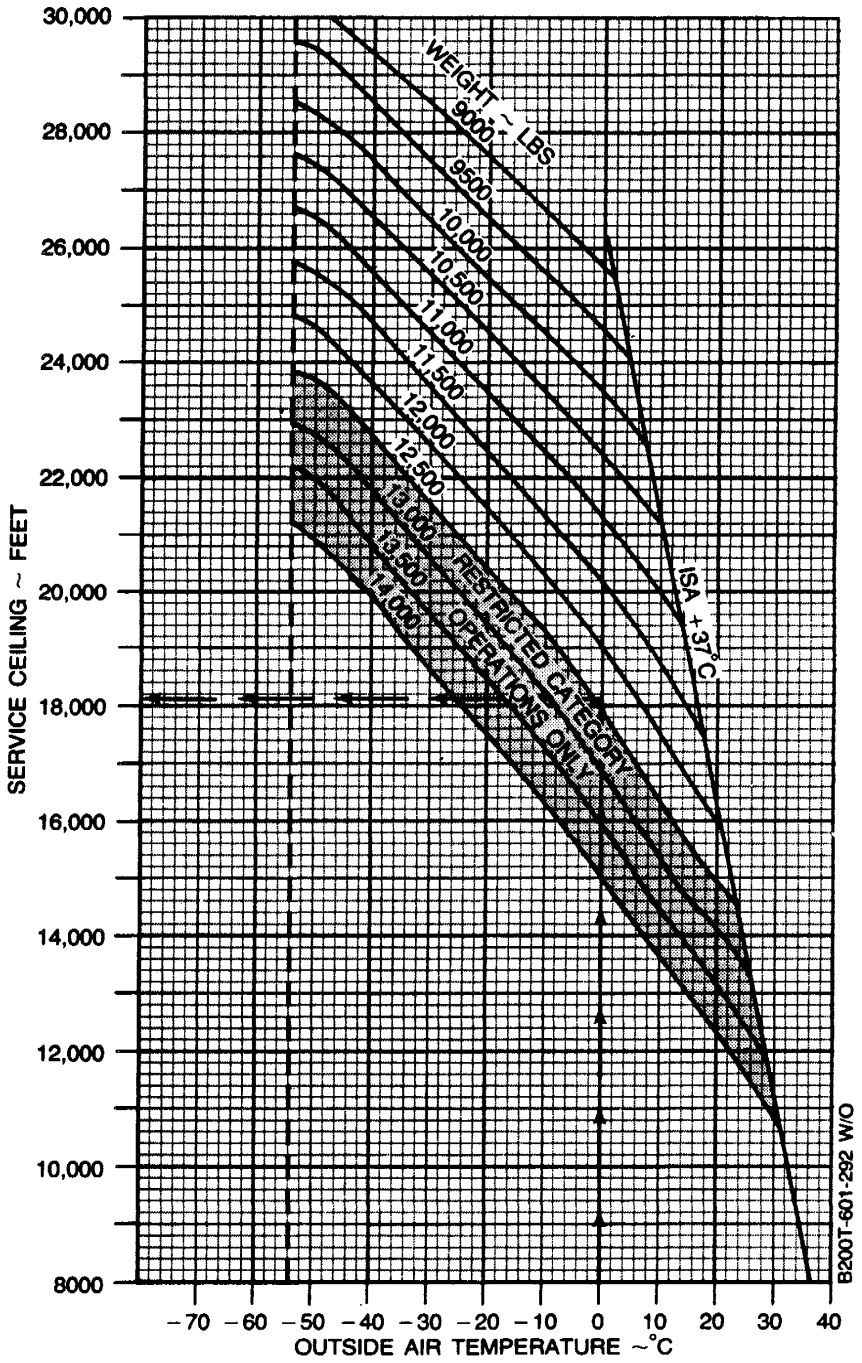
NOTE: SERVICE CEILING IS THE MAXIMUM PRESSURE ALTITUDE AT WHICH THE AIRPLANE IS CAPABLE OF CLIMBING 50 FT/MINUTE WITH ONE PROPELLER FEATHERED.

EXAMPLE:

OAT AT MEA (WORST LEG)	0°C
WEIGHT	12,391 LBS
ROUTE SEGMENT MEA	18,000 FT
SERVICE CEILING	18,100 FT

NOTE: SERVICE CEILING IS ABOVE ENROUTE MEA.

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, SERVICE CEILING WILL BE LOWERED APPROXIMATELY 1500 FEET.



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TIME, FUEL, AND DISTANCE TO CLIMB

ASSOCIATED CONDITIONS:

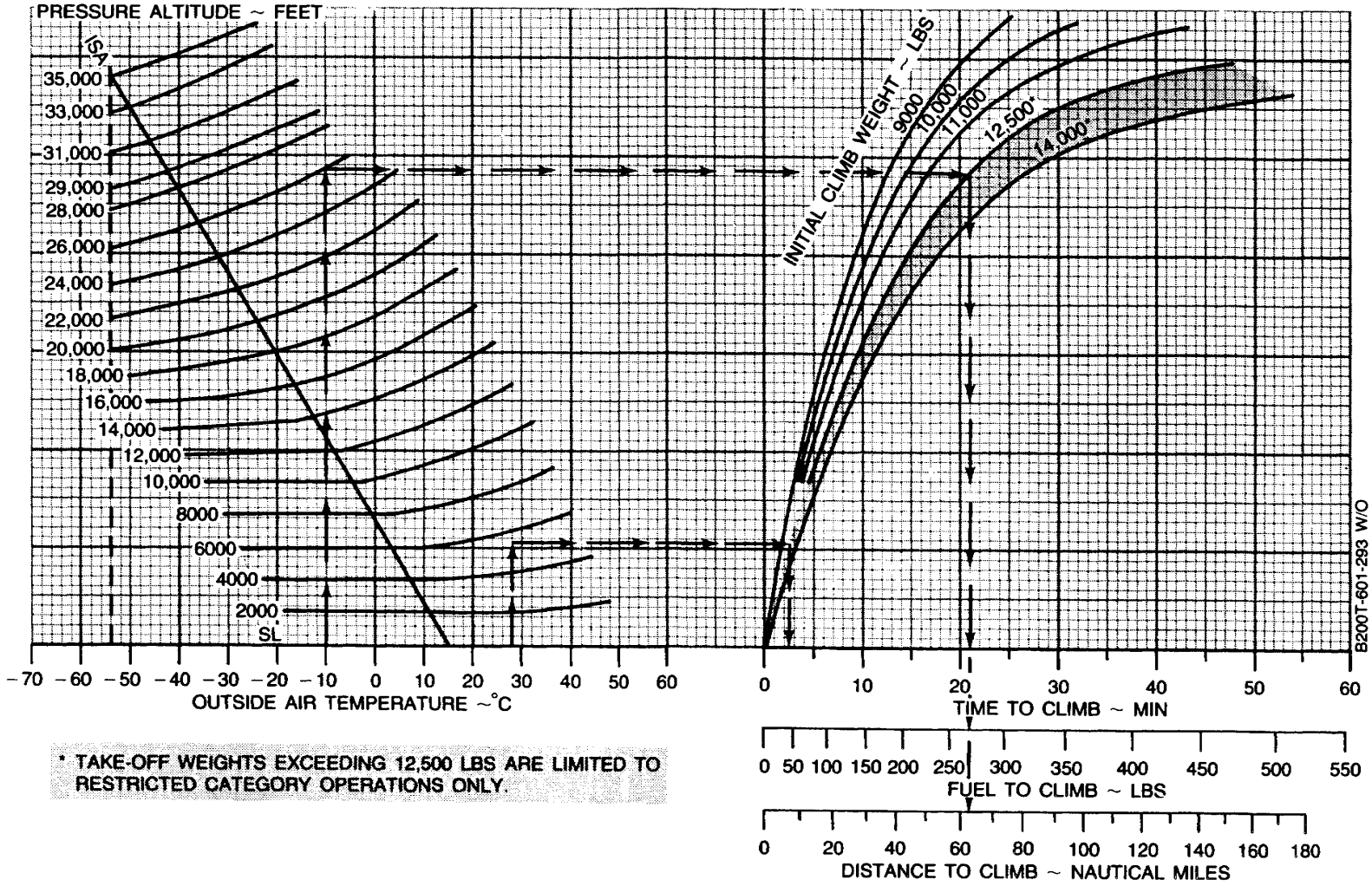
PROPELLER SPEED	1900 RPM
ITT	770°C
OR TORQUE	2230 FT-LBS

ALTITUDE ~ FEET	CLIMB SPEED ~ KNOTS
SL TO 10,000	160
10,000 TO 20,000	140
20,000 TO 25,000	130
25,000 TO 35,000	120

NOTE:
1. ADD 90 LBS FUEL FOR START, TAXI, AND TAKEOFF.
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 20°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.

EXAMPLE:

OAT AT TAKEOFF	28°C
OAT AT CRUISE	−10°C
AIRPORT PRESSURE ALTITUDE	5433 FT
CRUISE ALTITUDE	26,000 FT
INITIAL CLIMB WEIGHT	12,500 LBS
TIME TO CLIMB (21-3)	18 MIN
FUEL TO CLIMB (265-43)	222 LBS
DISTANCE TO CLIMB (63-8)	55 NM



* TAKE-OFF WEIGHTS EXCEEDING 12,500 LBS ARE LIMITED TO RESTRICTED CATEGORY OPERATIONS ONLY.

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NORMAL CRUISE POWER

1700 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	465	930	240	228	241	229	242	230	242	231
2000	-14	-19	2230	453	906	237	232	238	233	239	234	240	235
4000	-18	-23	2230	441	882	235	236	236	237	237	238	238	239
6000	-22	-27	2230	429	858	233	241	234	242	235	243	236	244
8000	-25	-31	2230	417	834	231	245	232	246	233	248	234	249
10,000	-29	-35	2230	407	814	228	250	230	251	231	252	231	253
12,000	-33	-39	2230	397	794	226	255	227	256	228	257	229	258
14,000	-36	-43	2230	390	780	224	259	225	261	226	262	227	263
16,000	-40	-47	2230	384	768	221	265	222	266	224	268	225	269
18,000	-44	-51	2230	380	760	219	270	220	271	221	273	222	274
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

1700 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	467	934	238	231	239	232	240	232	241	233
2000	-4	-9	2230	454	908	236	235	237	236	238	237	239	238
4000	-8	-13	2230	440	880	233	239	234	240	235	241	236	242
6000	-11	-17	2230	429	858	231	244	232	245	233	246	234	247
8000	-15	-21	2230	419	838	229	248	230	250	231	251	232	252
10,000	-19	-25	2230	409	818	226	253	228	254	229	255	230	257
12,000	-23	-29	2230	398	796	224	258	225	259	226	260	227	262
14,000	-26	-33	2230	391	782	222	263	223	264	224	266	225	267
16,000	-30	-37	2230	386	772	219	268	221	270	222	271	223	272
18,000	-34	-41	2230	382	764	216	273	218	275	219	276	220	278
20,000	-37	-45	2230	379	758	213	278	215	280	216	281	218	283
22,000	-41	-49	2141	362	724	207	278	209	281	211	283	212	285
24,000	-45	-53	2012	340	680	199	276	201	279	203	282	205	284
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

1700 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	470	940	236	233	237	234	238	235	239	236
2000	6	1	2230	457	914	234	238	235	239	236	240	237	240
4000	3	-3	2230	444	888	232	242	233	243	234	244	235	245
6000	-1	-7	2230	431	862	229	247	231	248	232	249	232	250
8000	-5	-11	2230	420	840	227	251	228	253	229	254	230	255
10,000	-9	-15	2230	409	818	225	256	226	257	227	259	228	260
12,000	-12	-19	2230	400	800	222	261	223	262	225	264	226	265
14,000	-16	-23	2230	393	786	220	266	221	268	222	269	223	270
16,000	-20	-27	2230	388	776	217	271	218	273	220	274	222	276
18,000	-24	-31	2230	384	768	214	276	216	278	217	279	218	281
20,000	-27	-35	2143	367	734	208	276	210	279	211	281	213	283
22,000	-31	-39	2043	349	698	201	276	203	279	205	281	207	284
24,000	-35	-43	1917	326	652	193	274	195	277	197	280	199	283
26,000	-40	-47	1781	303	606	184	270	187	274	189	278	191	281
28,000	-44	-51	1641	279	558	174	265	177	270	180	275	183	278
29,000	-46	-52	1573	267	534	168	261	172	267	176	272	179	277
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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For Training Purposes Only

NORMAL CRUISE POWER

1700 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	473	946	235	236	236	237	237	238	238	239
2000	16	11	2230	460	920	232	240	234	241	235	242	235	243
4000	13	7	2230	446	892	230	245	231	246	232	247	233	248
6000	9	3	2230	433	866	228	249	229	251	230	252	231	253
8000	5	-1	2230	422	844	225	254	226	255	228	257	228	258
10,000	1	-5	2230	411	822	223	259	224	260	225	262	226	263
12,000	-2	-9	2230	400	800	220	264	222	266	223	267	224	268
14,000	-6	-13	2230	393	786	218	269	220	271	221	272	222	274
16,000	-10	-17	2230	388	776	215	274	216	276	218	277	219	279
18,000	-14	-21	2139	371	742	208	274	210	276	212	279	213	280
20,000	-18	-25	2045	353	706	202	274	204	277	206	279	207	281
22,000	-21	-29	1950	336	672	195	274	197	277	199	280	201	282
24,000	-26	-33	1826	314	628	187	271	189	275	192	278	194	281
26,000	-30	-37	1694	290	580	177	267	181	272	183	276	186	279
28,000	-34	-41	1559	267	534	167	261	171	267	175	272	177	276
29,000	-36	-42	1493	256	512	161	256	166	264	170	269	173	274
31,000	-40	-46	1366	236	472	149	246	156	256	160	264	164	270
33,000	-45	-50	1243	216	432	---	---	143	245	150	257	155	265
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE POWER

1700 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	30	25	2230	476	952	233	238	234	239	235	240	236	241
2000	27	21	2230	462	924	231	243	232	244	233	245	234	246
4000	23	17	2230	449	898	229	247	230	249	231	250	232	251
6000	19	13	2230	437	874	226	252	227	253	228	255	229	256
8000	15	9	2230	424	848	224	257	225	258	226	259	227	261
10,000	12	5	2230	413	826	221	262	222	263	224	265	225	266
12,000	8	1	2230	402	804	219	267	220	269	221	270	222	271
14,000	4	-3	2230	395	790	216	272	217	273	219	275	220	276
16,000	0	-7	2160	378	756	210	273	212	275	213	277	215	279
18,000	-4	-11	2052	357	714	203	273	205	275	207	277	208	279
20,000	-8	-15	1964	340	680	197	273	199	276	201	278	202	281
22,000	-12	-19	1861	324	648	189	271	192	275	194	278	196	281
24,000	-16	-23	1740	302	604	181	268	184	273	186	276	189	280
26,000	-20	-27	1611	279	558	171	263	175	269	178	273	180	277
28,000	-24	-31	1480	256	512	160	256	165	263	169	269	172	274
29,000	-26	-32	1416	246	492	154	251	160	259	164	266	167	271
31,000	-31	-36	1294	226	452	139	234	149	251	154	260	158	267
33,000	-36	-40	1169	207	414	---	---	133	234	143	251	149	261
35,000	-40	-44	1069	190	380	---	---	---	---	128	233	138	251

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NORMAL CRUISE POWER

1700 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	40	35	2230	479	958	232	241	233	242	234	243	235	244
2000	37	31	2230	466	932	229	245	231	247	232	248	233	249
4000	33	27	2230	452	904	227	250	228	251	229	252	230	253
6000	29	23	2230	439	878	225	255	226	256	227	257	228	258
8000	25	19	2230	427	854	222	260	223	261	224	262	225	263
10,000	22	15	2230	416	832	220	265	221	266	222	268	223	269
12,000	18	11	2216	403	806	216	269	218	271	219	272	220	274
14,000	14	7	2123	381	762	210	269	212	271	213	273	215	274
16,000	10	3	2030	361	722	203	269	205	272	207	274	209	276
18,000	6	-1	1936	341	682	197	269	199	272	201	275	202	277
20,000	2	-5	1872	327	654	191	270	194	273	195	276	197	279
22,000	-2	-9	1791	312	624	184	270	187	274	189	277	191	280
24,000	-6	-13	1674	291	582	175	266	179	271	182	275	184	279
26,000	-10	-17	1540	268	536	165	260	170	266	173	271	176	275
28,000	-14	-21	1403	246	492	153	250	159	259	163	265	167	271
29,000	-17	-22	1340	236	472	145	242	153	254	158	262	162	268
31,000	-21	-26	1219	216	432	---	---	140	242	149	255	153	263
33,000	-25	-30	1125	200	400	---	---	---	---	136	243	143	256
35,000	-29	-34	1031	184	368	---	---	---	---	---	---	131	244

NORMAL CRUISE POWER

1700 PRM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	482	964	231	243	231	244	232	245	233	246
2000	47	41	2230	468	936	228	248	229	249	230	250	231	251
4000	43	37	2230	455	910	226	252	227	254	228	255	229	256
6000	39	33	2230	441	882	223	257	224	259	225	260	226	261
8000	36	29	2230	429	858	221	262	222	264	223	265	224	266
10,000	32	25	2174	409	818	216	264	217	266	219	268	220	269
12,000	28	21	2064	382	764	209	264	211	266	212	268	214	270
14,000	24	17	1977	362	724	203	265	205	267	206	269	208	271
16,000	20	13	1894	343	686	196	265	198	268	200	270	202	272
18,000	16	9	1817	325	650	190	265	192	268	194	271	197	273
20,000	12	5	1767	313	626	185	266	187	270	189	273	191	276
22,000	8	1	1708	301	602	179	267	182	271	184	275	186	278
24,000	4	-3	1592	280	560	169	262	173	268	176	272	179	276
26,000	0	-7	1469	258	516	159	255	164	262	168	268	171	273
28,000	-4	-11	1346	237	474	147	244	153	255	158	263	162	269
29,000	-7	-12	1285	227	454	137	232	147	250	153	259	157	266
31,000	-12	-16	1157	207	414	---	---	131	231	142	250	148	260
33,000	-16	-20	1059	191	382	---	---	---	---	126	230	137	251
35,000	-20	-24	962	175	350	---	---	---	---	---	---	121	230

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NORMAL CRUISE POWER

1700 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

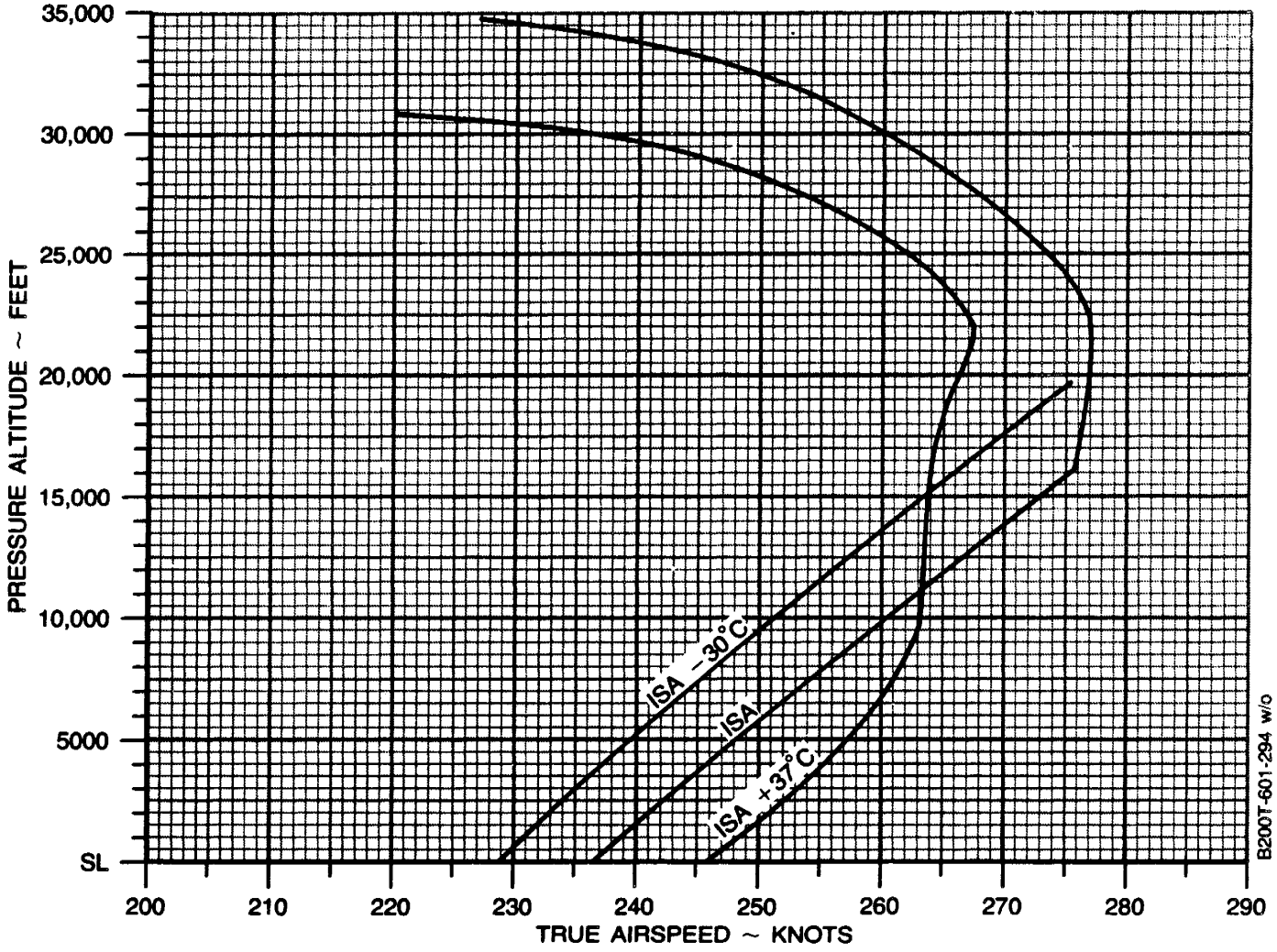
PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	484	968	229	245	231	246	232	247	232	248
2000	54	48	2230	470	940	227	249	228	251	229	252	230	253
4000	50	44	2230	456	912	225	254	226	255	227	257	228	258
6000	46	40	2229	443	886	222	259	223	260	224	262	225	263
8000	42	36	2149	419	838	216	260	218	262	219	263	220	265
10,000	39	32	2068	395	790	210	261	212	263	214	265	215	267
12,000	35	28	1963	369	738	204	261	205	263	207	265	209	267
14,000	31	24	1881	350	700	197	261	200	264	201	266	203	268
16,000	27	20	1803	332	664	191	261	194	265	196	267	198	270
18,000	23	16	1729	315	630	185	261	188	265	190	268	192	271
20,000	19	12	1685	303	606	180	263	183	267	185	270	187	273
22,000	15	8	1633	291	582	174	263	177	268	180	272	182	275
24,000	11	4	1529	271	542	165	258	169	264	172	270	175	274
26,000	7	0	1415	250	500	154	251	160	259	164	266	167	271
28,000	2	-4	1295	230	460	140	237	149	251	154	260	158	266
29,000	0	-5	1234	220	440	---	---	142	244	149	256	154	263
31,000	-4	-9	1138	203	406	---	---	---	---	137	245	144	257
33,000	-8	-13	1049	188	376	---	---	---	---	---	---	133	246
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

NORMAL CRUISE SPEEDS

1700 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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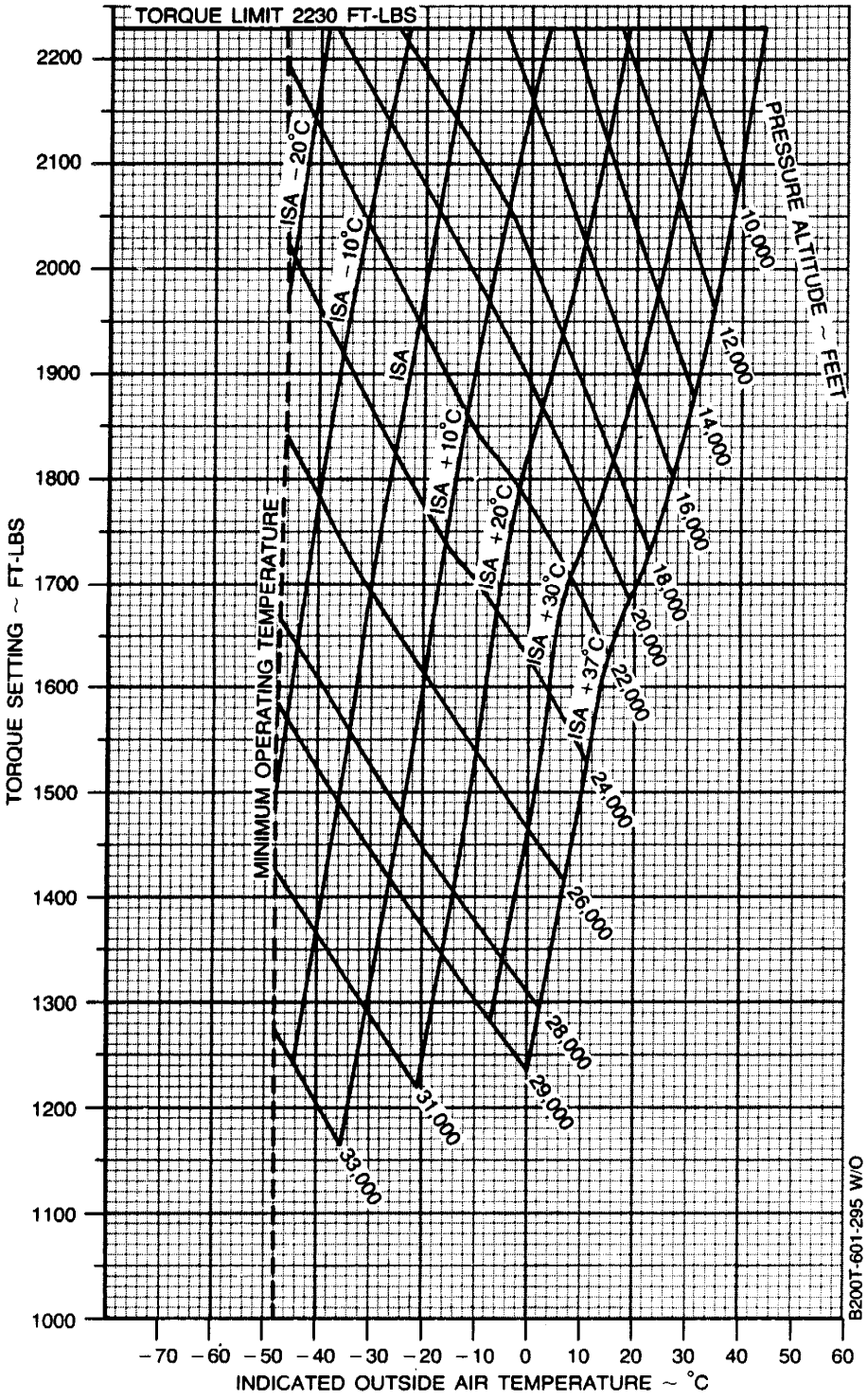
NORMAL CRUISE POWER

1700 RPM

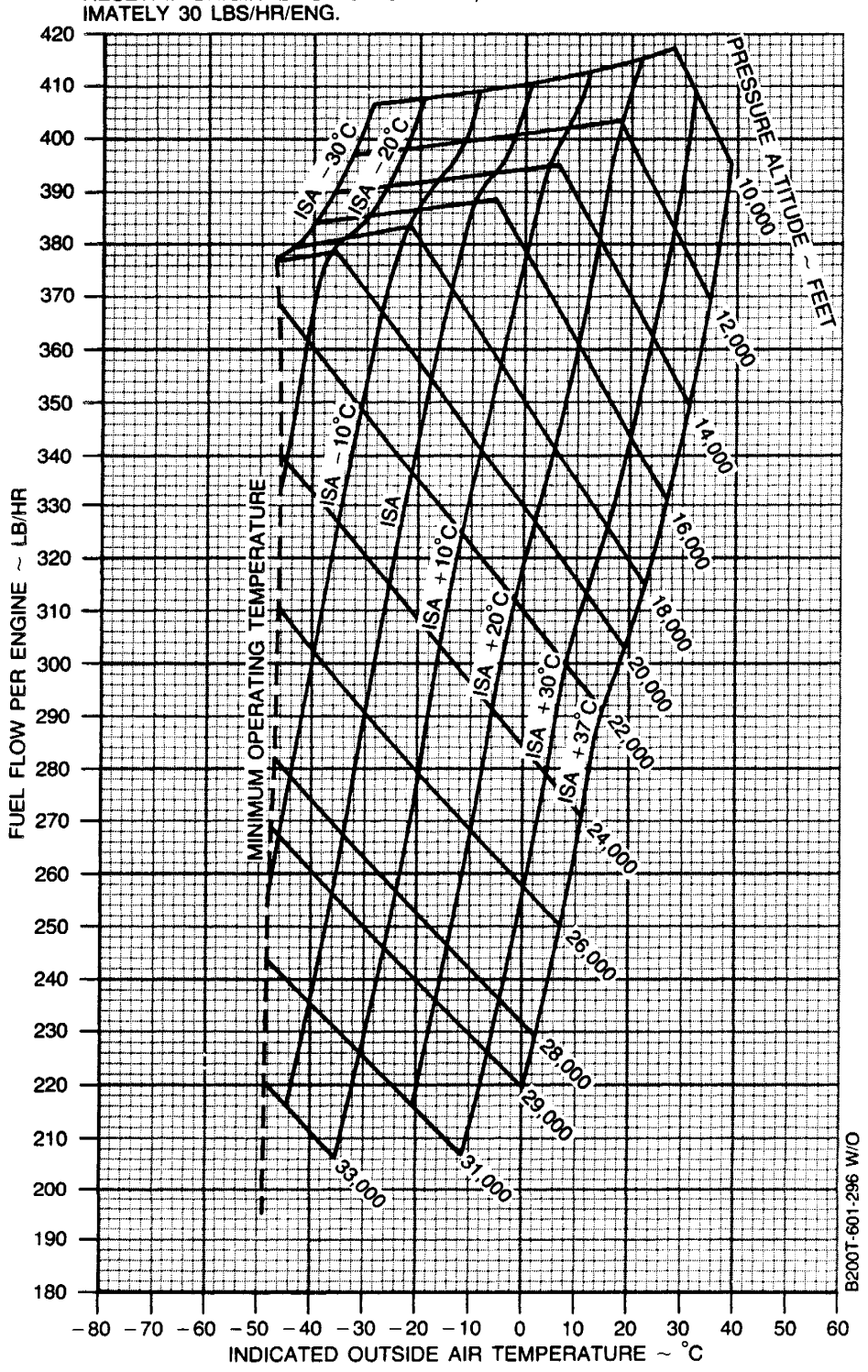
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT NORMAL CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - NORMAL CRUISE POWER

1700 RPM

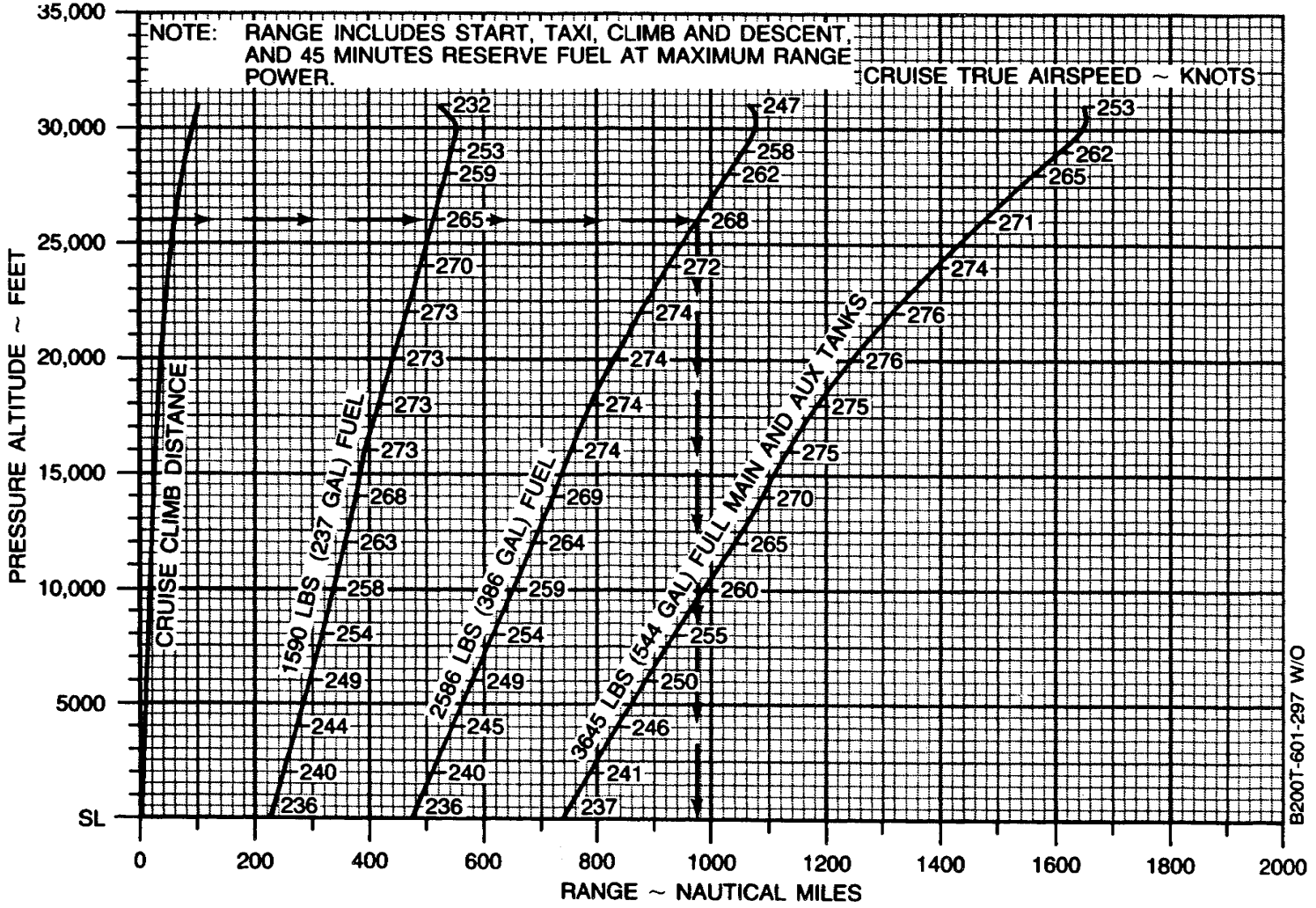
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS /GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	977 NM



For Training Purposes Only

MAXIMUM CRUISE POWER

1700 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2230	465	930	240	228	241	229	242	230	242	231
2000	-14	-19	2230	453	906	237	232	238	233	239	234	240	235
4000	-18	-23	2230	441	882	235	236	236	237	237	238	238	239
6000	-22	-27	2230	429	858	233	241	234	242	235	243	236	244
8000	-25	-31	2230	417	834	231	245	232	246	233	248	234	249
10,000	-29	-35	2230	407	814	228	250	230	251	231	252	231	253
12,000	-33	-39	2230	397	794	226	255	228	256	228	257	229	258
14,000	-36	-43	2230	390	780	223	259	225	261	226	262	227	263
16,000	-40	-47	2230	384	768	221	265	222	266	224	268	225	269
18,000	-44	-51	2230	380	760	219	270	220	271	221	273	222	274
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

1700 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2230	467	934	238	231	239	232	240	232	241	233
2000	-4	-9	2230	454	908	236	235	237	236	238	237	239	238
4000	-8	-13	2230	440	880	233	239	234	240	235	241	236	242
6000	-11	-17	2230	429	858	231	244	232	245	233	246	234	247
8000	-15	-21	2230	419	838	229	248	230	250	231	251	232	252
10,000	-19	-25	2230	409	818	226	253	228	254	229	255	230	257
12,000	-23	-29	2230	398	796	224	258	225	259	226	260	227	262
14,000	-26	-33	2230	391	782	222	263	223	264	224	266	225	267
16,000	-30	-37	2230	386	772	220	268	221	270	222	271	223	272
18,000	-34	-41	2230	382	764	216	273	218	275	219	276	220	278
20,000	-37	-45	2230	379	758	213	278	215	280	216	281	218	283
22,000	-41	-49	2230	376	752	210	283	212	285	214	287	215	288
24,000	-45	-53	2090	352	704	202	280	204	283	206	286	208	288
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	470	940	236	233	237	234	238	235	239	236
2000	6	1	2230	457	914	234	238	235	239	236	240	237	240
4000	3	-3	2230	444	888	232	242	233	243	234	244	235	245
6000	-1	-7	2230	431	862	229	247	231	248	232	249	232	250
8000	-5	-11	2230	420	840	227	251	228	253	229	254	230	255
10,000	-9	-15	2230	409	818	225	256	226	257	227	259	228	260
12,000	-12	-19	2230	400	800	222	261	223	262	225	264	226	265
14,000	-16	-23	2230	393	786	220	266	221	268	222	269	223	270
16,000	-20	-27	2230	388	776	217	271	218	273	220	274	221	276
18,000	-24	-31	2230	384	768	214	276	216	278	217	279	218	281
20,000	-27	-35	2230	381	762	211	281	213	283	214	285	215	286
22,000	-31	-39	2128	362	724	204	280	206	283	208	286	210	288
24,000	-35	-43	2002	340	680	196	279	199	282	201	285	202	287
26,000	-39	-47	1862	315	630	187	275	190	279	192	283	195	286
28,000	-43	-51	1714	290	580	177	270	181	275	184	279	186	283
29,000	-46	-52	1639	278	556	172	266	175	272	179	277	181	281
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

1700 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	473	946	235	236	236	237	237	238	238	239
2000	16	11	2230	460	920	232	240	234	241	235	242	235	243
4000	13	7	2230	446	892	230	245	231	246	232	247	233	248
6000	9	3	2230	433	866	228	249	229	251	230	252	231	253
8000	5	-1	2230	422	844	225	254	226	255	228	257	228	258
10,000	1	-5	2230	411	822	223	259	224	260	225	262	226	263
12,000	-2	-9	2230	400	800	220	264	222	266	223	267	224	268
14,000	-6	-13	2230	393	786	218	269	219	271	221	272	222	274
16,000	-10	-17	2230	388	776	215	274	216	276	218	277	219	279
18,000	-13	-21	2230	386	772	212	278	213	281	215	282	217	284
20,000	-17	-25	2131	367	734	205	279	207	281	209	283	210	285
22,000	-21	-29	2031	349	698	199	278	200	281	202	284	204	286
24,000	-25	-33	1905	326	652	191	276	193	280	195	283	197	286
26,000	-29	-37	1771	302	604	181	272	184	277	187	280	189	284
28,000	-34	-41	1634	279	558	171	266	175	272	178	277	181	281
29,000	-36	-42	1567	268	536	166	263	170	269	173	275	176	279
31,000	-40	-46	1436	247	494	154	253	159	262	164	270	167	275
33,000	-45	-50	1306	226	452	137	235	148	253	154	262	158	270
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

1700 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,500 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	30	25	2230	476	952	233	238	234	239	235	240	236	241
2000	27	21	2230	462	924	231	243	232	244	233	245	234	246
4000	23	17	2230	449	898	229	247	230	249	231	250	232	251
6000	19	13	2230	437	874	226	252	227	253	228	255	229	256
8000	15	9	2230	424	848	224	257	225	258	226	259	227	261
10,000	12	5	2230	413	826	221	262	223	263	224	265	225	266
12,000	8	1	2230	402	804	219	267	220	269	221	270	222	271
14,000	4	-3	2230	395	790	216	272	217	273	219	275	220	276
16,000	0	-7	2230	389	778	213	277	214	278	216	280	217	282
18,000	-3	-11	2134	370	740	206	277	209	279	210	281	211	283
20,000	-7	-15	2046	353	706	200	277	202	280	204	282	205	285
22,000	-11	-19	1937	336	672	192	276	195	279	197	282	199	285
24,000	-15	-23	1815	314	628	184	273	187	277	189	281	191	284
26,000	-20	-27	1685	290	580	175	268	178	274	181	278	184	282
28,000	-24	-31	1552	268	536	164	262	169	269	172	274	175	279
29,000	-26	-32	1487	257	514	158	257	163	265	167	271	171	277
31,000	-30	-36	1362	237	474	145	245	153	257	158	265	162	272
33,000	-35	-40	1239	217	434	---	---	140	245	148	258	153	267
35,000	-41	-44	1087	194	388	---	---	---	---	134	245	142	259

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MAXIMUM CRUISE POWER

1700 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	40	35	2230	479	958	232	241	233	242	234	243	235	244
2000	37	31	2230	466	932	229	245	231	247	232	248	233	249
4000	33	27	2230	452	904	229	240	228	251	229	252	230	253
6000	29	23	2230	439	878	225	255	226	256	227	257	228	258
8000	25	19	2230	427	854	222	260	223	261	224	262	225	263
10,000	22	15	2230	416	832	220	265	221	266	222	268	223	269
12,000	18	11	2230	405	810	217	270	218	271	220	273	221	274
14,000	14	7	2184	390	780	212	272	214	274	215	276	217	278
16,000	10	3	2104	372	744	206	273	208	275	210	278	211	279
18,000	6	-1	2023	354	708	200	274	202	276	204	279	206	281
20,000	3	-5	1944	339	678	194	274	196	277	198	280	200	283
22,000	-1	-9	1862	323	646	187	274	190	278	192	281	195	284
24,000	-6	-13	1745	302	604	179	271	182	275	185	279	187	283
26,000	-10	-17	1610	279	558	169	265	173	271	176	276	179	280
28,000	-14	-21	1472	257	514	157	257	162	264	167	271	170	276
29,000	-16	-22	1346	236	472	151	251	157	260	162	268	165	273
31,000	-21	-26	1230	217	434	133	230	146	251	152	261	156	269
33,000	-26	-30	1087	196	392	---	---	128	229	141	252	147	262
35,000	-29	-34	1034	184	368	---	---	---	---	---	---	136	252

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MAXIMUM CRUISE POWER

1700 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,500 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	482	964	230	243	231	244	232	245	233	246
2000	47	41	2230	468	936	228	248	229	249	230	250	231	251
4000	43	37	2230	455	910	226	252	227	254	228	255	229	256
6000	39	33	2230	441	882	223	257	224	259	225	260	226	261
8000	36	29	2230	429	858	221	262	222	264	223	265	224	266
10,000	32	25	2211	415	830	217	266	219	268	220	270	221	271
12,000	28	21	2095	387	774	210	266	212	268	213	270	215	271
14,000	24	17	2027	370	740	205	267	206	269	208	272	210	273
16,000	20	13	1960	353	706	199	268	201	271	203	273	204	275
18,000	16	9	1892	337	674	193	269	195	272	197	275	199	277
20,000	12	5	1844	325	650	188	271	190	274	193	277	194	288
22,000	8	1	1771	310	620	181	271	184	275	187	278	189	281
24,000	4	-3	1656	290	580	173	267	176	272	179	276	182	280
26,000	0	-7	1535	268	536	163	261	167	267	171	273	174	278
28,000	-4	-11	1413	247	494	151	252	157	261	162	268	165	274
29,000	-7	-12	1352	237	474	144	244	151	257	157	265	161	272
31,000	-11	-16	1228	217	434	---	---	138	243	146	257	151	266
33,000	-15	-20	1127	201	402	---	---	---	---	133	243	141	258
35,000	-19	-24	1027	185	370	---	---	---	---	---	---	128	243

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MAXIMUM CRUISE POWER

1700 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	484	968	230	245	231	246	232	247	232	248
2000	54	48	2230	470	940	227	249	228	251	229	252	230	253
4000	50	44	2230	456	912	225	254	226	255	227	257	228	258
6000	46	40	2230	443	886	222	259	223	260	224	262	225	263
8000	43	36	2178	423	846	217	261	219	263	220	265	221	266
10,000	39	32	2094	399	798	211	262	213	264	215	266	216	268
12,000	35	28	1985	372	744	204	262	206	264	208	266	209	268
14,000	31	24	1925	356	712	200	264	201	266	203	269	205	270
16,000	27	20	1863	340	680	194	265	196	268	198	270	200	273
18,000	23	16	1803	325	650	188	266	191	269	193	271	195	275
20,000	19	12	1757	314	628	183	267	186	271	188	274	190	277
22,000	15	8	1699	301	602	177	267	180	272	183	276	185	279
24,000	11	4	1594	281	562	168	263	172	269	175	274	178	278
26,000	7	0	1478	260	520	158	257	163	264	167	270	170	276
28,000	3	-4	1358	239	478	146	246	153	257	157	265	162	272
29,000	0	-5	1298	229	458	136	234	147	252	152	261	157	269
31,000	-5	-9	1175	210	420	---	---	134	234	142	253	148	263
33,000	-9	-13	1085	195	390	---	---	---	---	127	236	137	254
35,000	-13	-17	994	179	358	---	---	---	---	---	---	123	237

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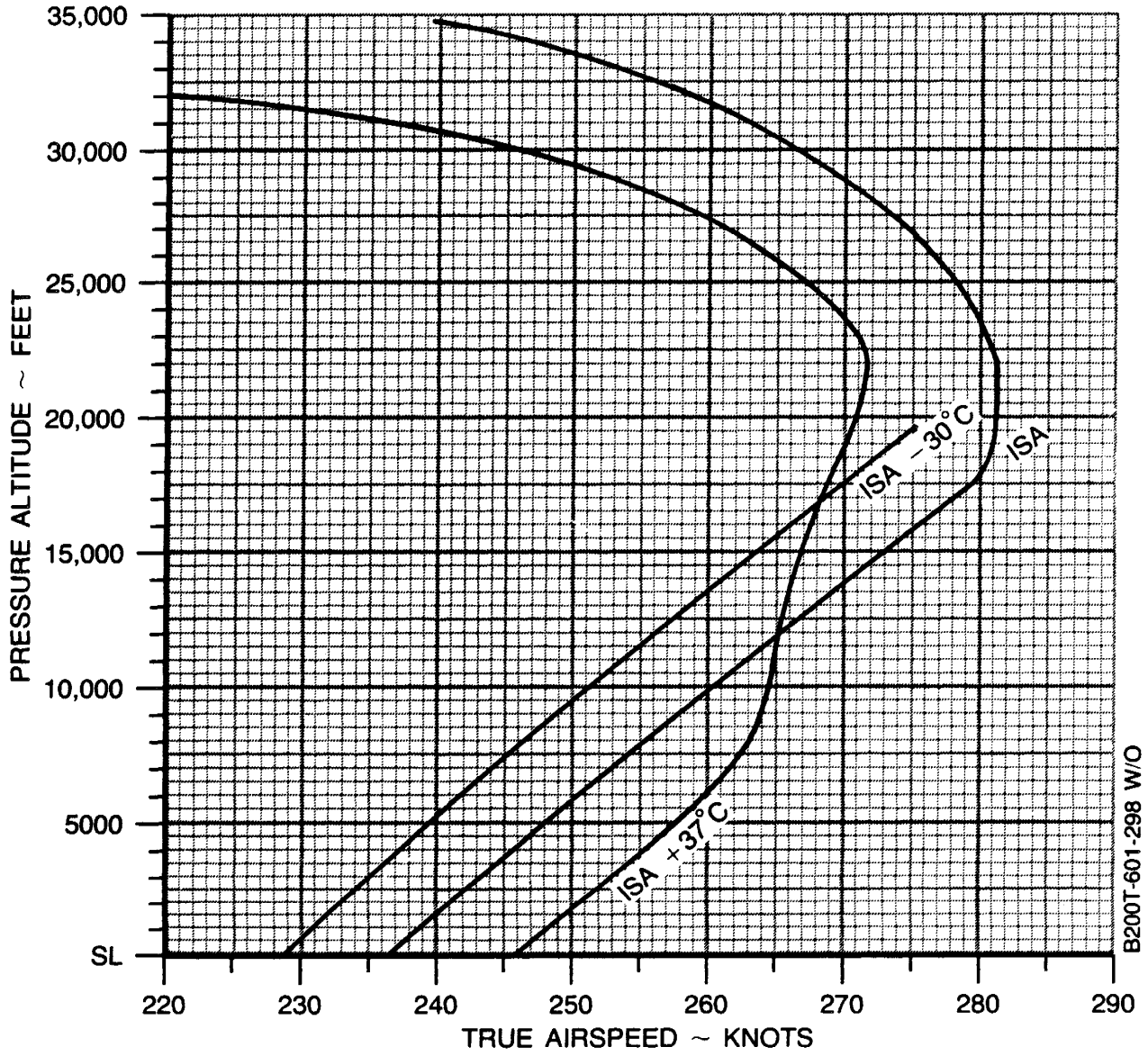
MAXIMUM CRUISE SPEEDS

1700 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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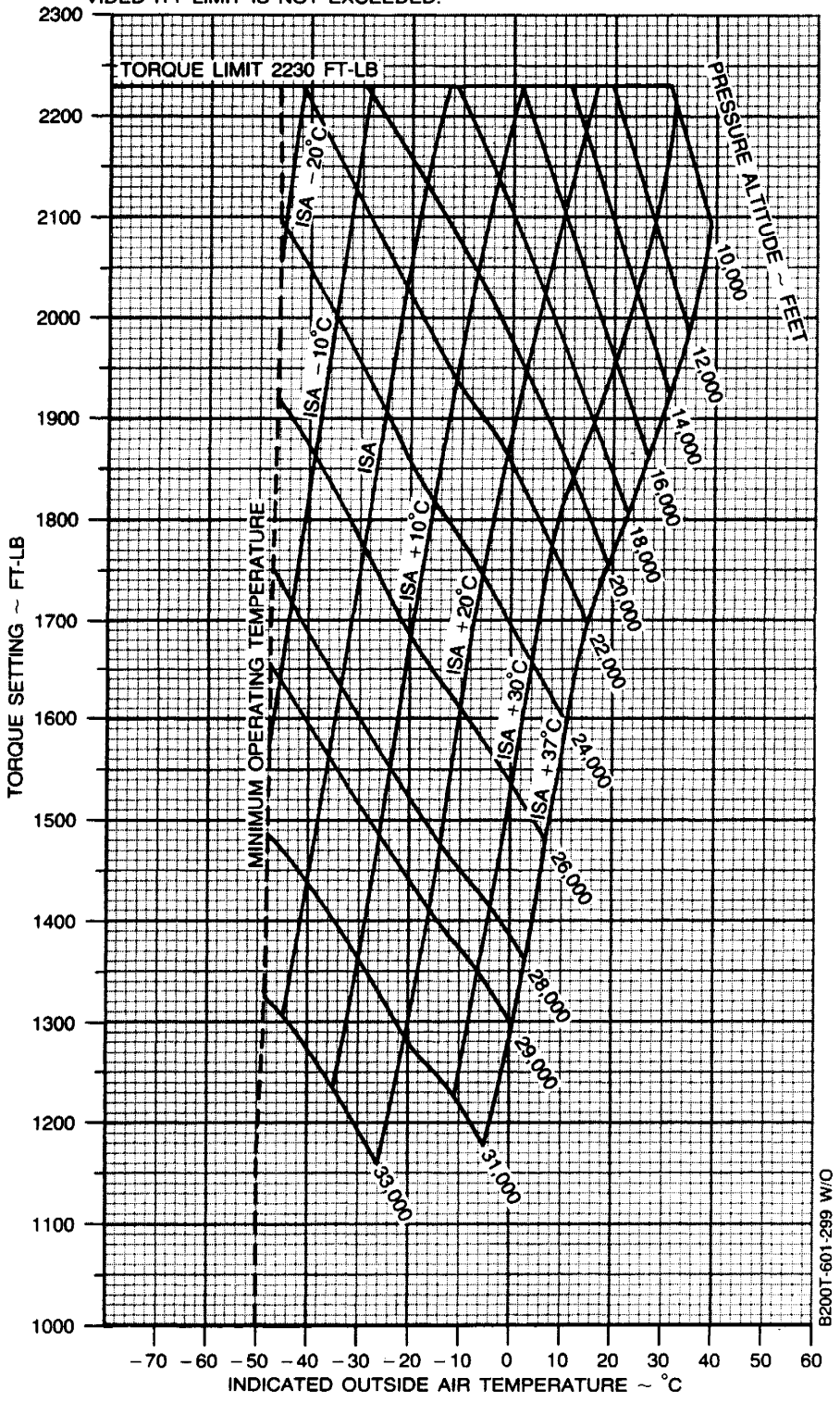
MAXIMUM CRUISE POWER

1700 RPM

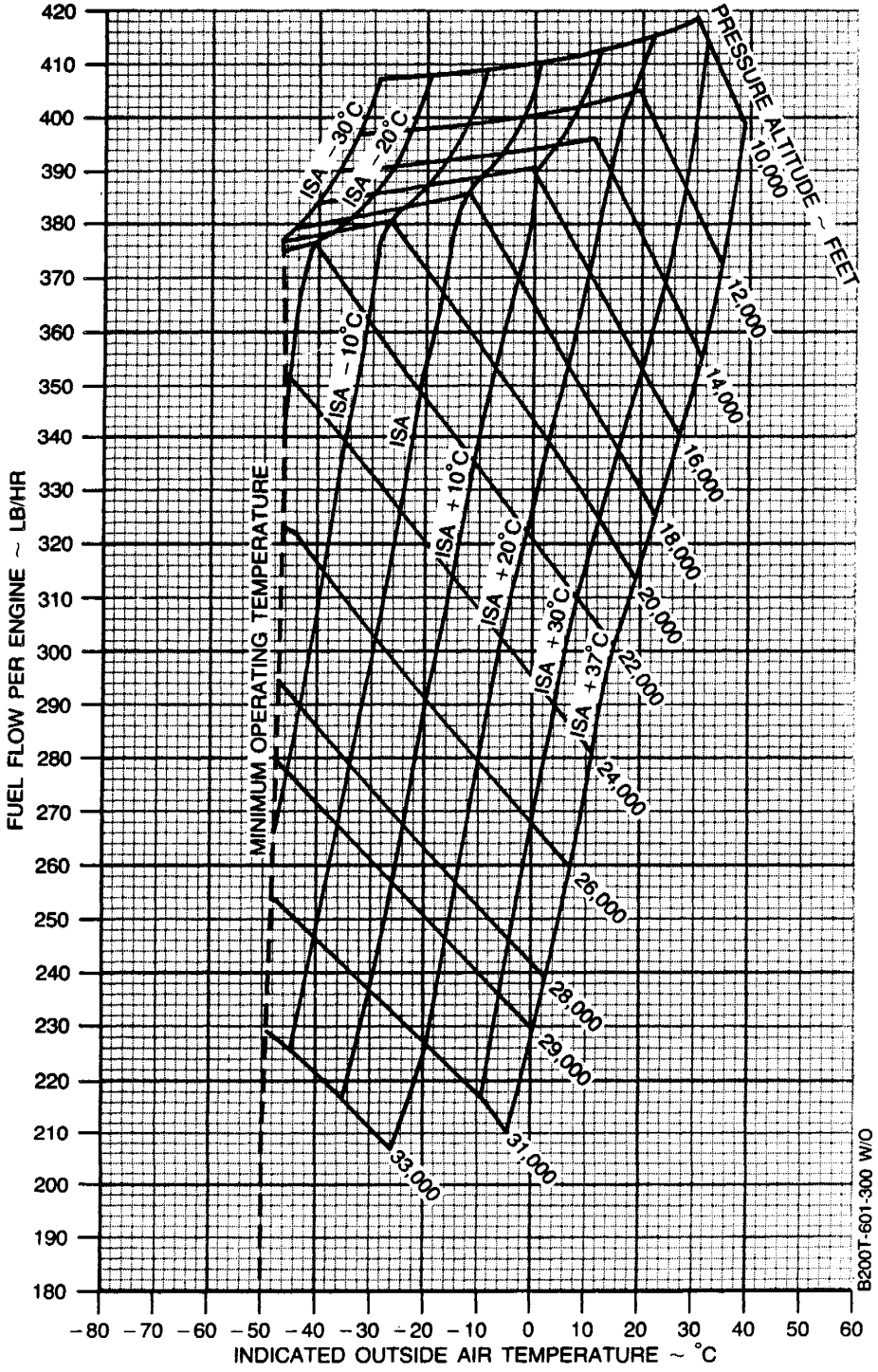
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT MAXIMUM CRUISE POWER

1700 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - MAXIMUM CRUISE POWER

1700 RPM

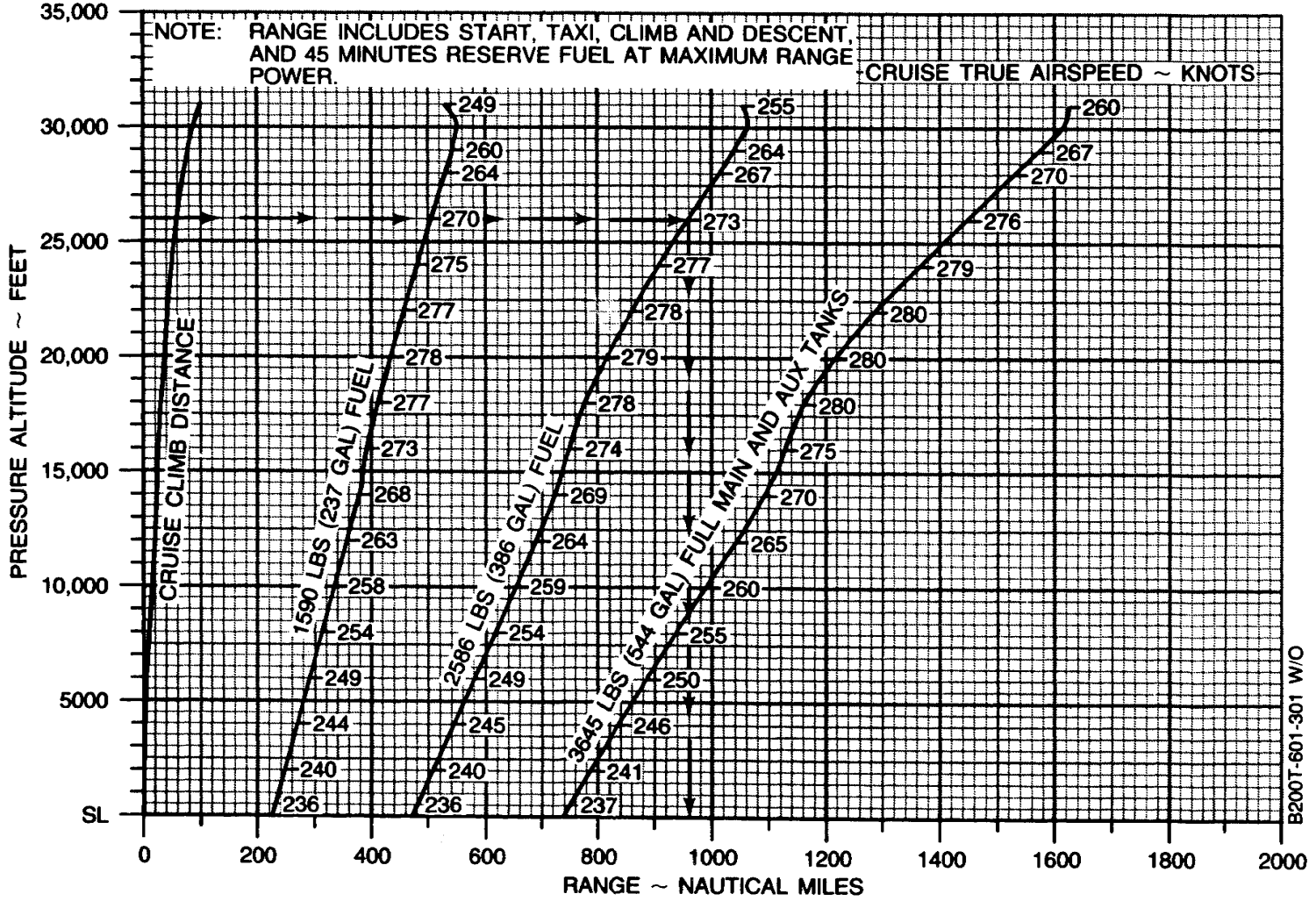
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	960 NM



For Training Purposes Only

NORMAL CRUISE POWER

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2194	481	962	244	232	244	232	244	232	244	232
2000	-14	-19	2230	473	946	243	237	244	238	244	238	244	238
4000	-18	-23	2230	460	920	241	242	241	243	242	243	243	244
6000	-21	-27	2230	448	896	238	246	239	247	240	248	241	249
8000	-25	-31	2230	437	874	236	251	237	252	238	253	239	254
10,000	-29	-35	2230	427	854	233	255	235	257	236	258	236	259
12,000	-32	-39	2230	417	834	231	260	232	262	233	263	234	264
14,000	-36	-43	2230	410	820	229	266	230	267	231	268	232	269
16,000	-40	-47	2230	404	808	226	271	228	272	229	273	230	275
18,000	-44	-51	2230	399	798	224	276	225	278	226	279	228	280
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

1800 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2226	486	972	243	236	244	236	244	236	244	236
2000	-4	-9	2230	474	948	241	240	242	241	243	242	244	243
4000	-7	-13	2230	461	922	239	245	240	246	241	246	241	247
6000	-11	-17	2230	449	898	236	249	237	250	238	251	239	252
8000	-15	-21	2230	438	876	234	254	235	255	236	256	237	257
10,000	-19	-25	2230	428	856	232	259	233	260	234	261	235	262
12,000	-22	-29	2230	417	834	229	264	230	265	231	266	232	267
14,000	-26	-33	2230	411	822	227	269	228	270	230	272	230	273
16,000	-30	-37	2230	405	810	224	274	226	276	227	277	228	278
18,000	-33	-41	2230	400	800	222	280	223	281	225	283	226	284
20,000	-37	-45	2142	382	764	216	281	217	283	219	285	220	286
22,000	-41	-49	2041	363	726	209	281	211	283	213	285	214	287
24,000	-45	-53	1920	340	680	201	279	203	282	205	285	207	287
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	489	978	242	238	243	239	243	240	244	241
2000	7	1	2230	475	950	239	243	240	244	241	245	242	246
4000	3	-3	2230	463	926	237	247	238	248	239	249	240	250
6000	-1	-7	2230	450	900	234	252	236	253	237	254	237	255
8000	-5	-11	2230	439	878	232	257	233	258	234	259	235	260
10,000	-8	-15	2230	428	856	230	262	231	263	232	264	233	265
12,000	-12	-19	2230	419	838	227	267	229	268	230	270	231	271
14,000	-16	-23	2230	412	824	225	272	226	274	227	275	228	276
16,000	-19	-27	2230	406	812	222	278	224	279	225	281	226	282
18,000	-23	-31	2144	388	776	216	279	218	281	219	283	221	284
20,000	-27	-35	2041	367	734	210	279	211	281	213	283	214	285
22,000	-31	-39	1947	349	698	203	278	205	282	207	284	208	286
24,000	-35	-43	1830	327	654	195	277	197	281	200	283	201	286
26,000	-39	-47	1702	304	608	186	274	189	278	191	281	194	284
28,000	-44	-51	1568	280	560	176	269	180	274	183	278	185	282
29,000	-46	-52	1503	268	536	171	265	175	271	178	276	181	280
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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NORMAL CRUISE POWER

1800 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	492	984	240	241	241	242	242	243	243	244
2000	17	11	2230	478	956	238	246	239	247	240	248	240	248
4000	13	7	2230	464	928	235	250	236	251	237	252	238	253
6000	9	3	2230	451	902	233	255	234	256	235	257	236	258
8000	5	-1	2230	440	880	230	260	232	261	233	262	234	263
10,000	2	-5	2230	429	858	228	265	229	266	230	267	231	269
12,000	-2	-9	2230	419	838	226	270	227	272	228	273	229	274
14,000	-6	-13	2230	411	822	223	275	224	277	226	278	227	280
16,000	-9	-17	2165	396	792	218	278	219	279	221	281	222	283
18,000	-13	-21	2036	372	744	210	276	212	279	213	281	215	282
20,000	-17	-25	1948	354	708	204	277	206	279	207	282	209	284
22,000	-21	-29	1860	336	672	197	277	199	280	201	283	203	285
24,000	-25	-33	1744	315	630	189	275	192	278	194	282	196	284
26,000	-29	-37	1619	291	582	180	271	183	275	186	279	188	282
28,000	-34	-41	1490	268	536	170	265	174	271	177	275	180	279
29,000	-36	-42	1427	257	514	164	260	168	267	172	273	175	277
31,000	-40	-46	1305	237	474	151	248	158	260	163	268	166	273
33,000	-45	-50	1186	217	434	131	225	145	248	152	260	157	268
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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P/N 101-590010-257

For Training Purposes Only

NORMAL CRUISE POWER

1800 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	31	25	2230	495	990	238	244	239	245	240	246	241	246
2000	27	21	2230	481	962	236	248	237	249	238	250	239	251
4000	23	17	2230	468	936	234	253	235	254	236	255	237	256
6000	19	13	2230	455	910	231	258	232	259	233	260	234	261
8000	16	9	2230	443	886	229	263	230	264	231	265	232	266
10,000	12	5	2230	431	862	226	268	228	269	229	270	230	272
12,000	8	1	2230	420	840	224	273	225	275	226	276	227	277
14,000	4	-3	2155	401	802	218	275	220	277	221	278	222	280
16,000	0	-7	2054	378	756	212	275	213	277	215	279	216	281
18,000	-4	-11	1953	357	714	205	275	207	277	208	280	210	281
20,000	-7	-15	1871	341	682	199	275	201	278	203	281	204	283
22,000	-11	-19	1775	324	648	191	275	194	278	196	281	198	283
24,000	-15	-23	1662	303	606	183	272	186	276	188	279	191	282
26,000	-20	-27	1540	280	560	174	267	177	272	180	277	183	280
28,000	-24	-31	1415	257	514	163	260	167	267	171	272	174	277
29,000	-26	-32	1353	247	494	149	247	162	263	166	270	170	275
31,000	-31	-36	1235	227	454	133	229	150	253	156	263	161	270
33,000	-36	-40	1116	207	414	---	---	135	237	145	254	151	264
35,000	-40	-44	1021	190	380	---	---	---	---	130	237	140	254

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NORMAL CRUISE POWER

1800 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	497	994	237	246	238	247	239	248	240	249
2000	37	31	2230	484	968	235	251	236	252	237	253	237	254
4000	33	27	2230	470	940	232	255	233	257	234	258	235	259
6000	29	23	2230	457	914	230	261	231	262	232	263	233	264
8000	26	19	2230	445	890	227	266	228	267	230	268	230	269
10,000	22	15	2212	431	862	224	270	225	271	227	273	228	274
12,000	18	11	2102	403	806	217	270	219	271	220	273	221	275
14,000	14	7	2015	381	762	211	270	213	273	214	274	215	276
16,000	10	3	1928	361	722	205	271	207	273	208	276	210	277
18,000	6	-1	1842	341	682	198	271	200	274	202	276	204	279
20,000	2	-5	1783	328	656	193	273	195	276	197	279	199	281
22,000	-1	-9	1708	313	626	187	273	189	277	191	280	193	283
24,000	-6	-13	1599	292	584	178	270	181	274	184	278	186	282
26,000	-10	-17	1472	269	538	168	264	172	270	175	274	178	279
28,000	-14	-21	1342	247	494	155	253	161	263	165	269	169	274
29,000	-17	-22	1281	236	472	147	245	155	257	160	266	164	272
31,000	-21	-26	1163	217	434	---	---	142	245	150	258	155	266
33,000	-25	-30	1018	190	380	---	---	---	---	137	245	145	259
35,000	-29	-34	984	184	368	---	---	---	---	---	---	132	246

NORMAL CRUISE POWER

1800 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	500	1000	235	249	237	250	237	251	238	251
2000	47	41	2230	486	972	233	253	234	254	235	255	236	256
4000	43	37	2230	472	944	231	258	232	259	233	261	234	262
6000	40	33	2225	458	916	228	263	229	264	230	265	231	267
8000	36	29	2142	434	868	222	264	224	266	225	267	226	268
10,000	32	25	2060	409	818	216	265	218	267	219	269	220	270
12,000	28	21	1956	383	766	209	265	211	267	213	269	214	271
14,000	24	17	1874	362	724	203	265	205	268	207	270	208	272
16,000	20	13	1798	343	686	197	266	199	269	201	271	203	273
18,000	16	9	1727	326	652	191	267	194	270	196	273	197	275
20,000	12	5	1682	313	626	186	269	189	272	191	275	193	278
22,000	8	1	1629	301	602	181	270	184	274	186	277	188	281
24,000	4	-3	1520	280	560	172	268	175	271	178	275	181	279
26,000	0	-7	1404	258	516	161	259	166	266	170	272	173	276
28,000	-4	-11	1286	237	474	148	247	156	259	160	266	164	272
29,000	-7	-12	1227	227	454	280	237	149	252	155	263	159	270
31,000	-12	-16	1107	207	414	---	---	134	236	144	253	150	263
33,000	-16	-20	1013	192	384	---	---	---	---	128	235	139	253
35,000	-20	-24	920	176	352	---	---	---	---	---	---	123	234

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NORMAL CRUISE POWER

1800 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	501	1002	234	250	236	251	236	252	237	253
2000	54	48	2207	484	968	231	254	232	255	233	256	236	257
4000	50	44	2165	465	930	227	257	228	258	229	260	231	261
6000	46	40	2107	443	886	222	259	224	261	225	262	226	263
8000	43	36	2032	419	838	217	260	218	262	220	264	221	265
10,000	39	32	1958	396	792	211	262	213	264	214	266	215	267
12,000	35	28	1859	369	738	204	261	206	264	208	266	209	268
14,000	31	24	1783	350	700	196	262	200	265	202	267	203	269
16,000	27	20	1711	332	664	192	262	194	265	196	268	198	270
18,000	23	16	1643	315	630	186	263	189	266	191	269	193	272
20,000	19	12	1604	303	606	181	265	184	269	186	272	188	275
22,000	15	8	1558	292	584	176	266	179	271	182	274	184	278
24,000	11	4	1460	272	544	167	262	171	268	174	273	177	277
26,000	7	0	1353	251	502	156	254	162	263	166	269	169	274
28,000	3	-4	1237	230	460	142	240	151	254	156	263	160	270
29,000	1	-5	1178	220	440	130	223	143	247	151	259	156	267
31,000	-5	-9	1055	200	400	---	---	125	223	139	247	146	260
33,000	-9	-13	972	185	370	---	---	---	---	121	225	134	249
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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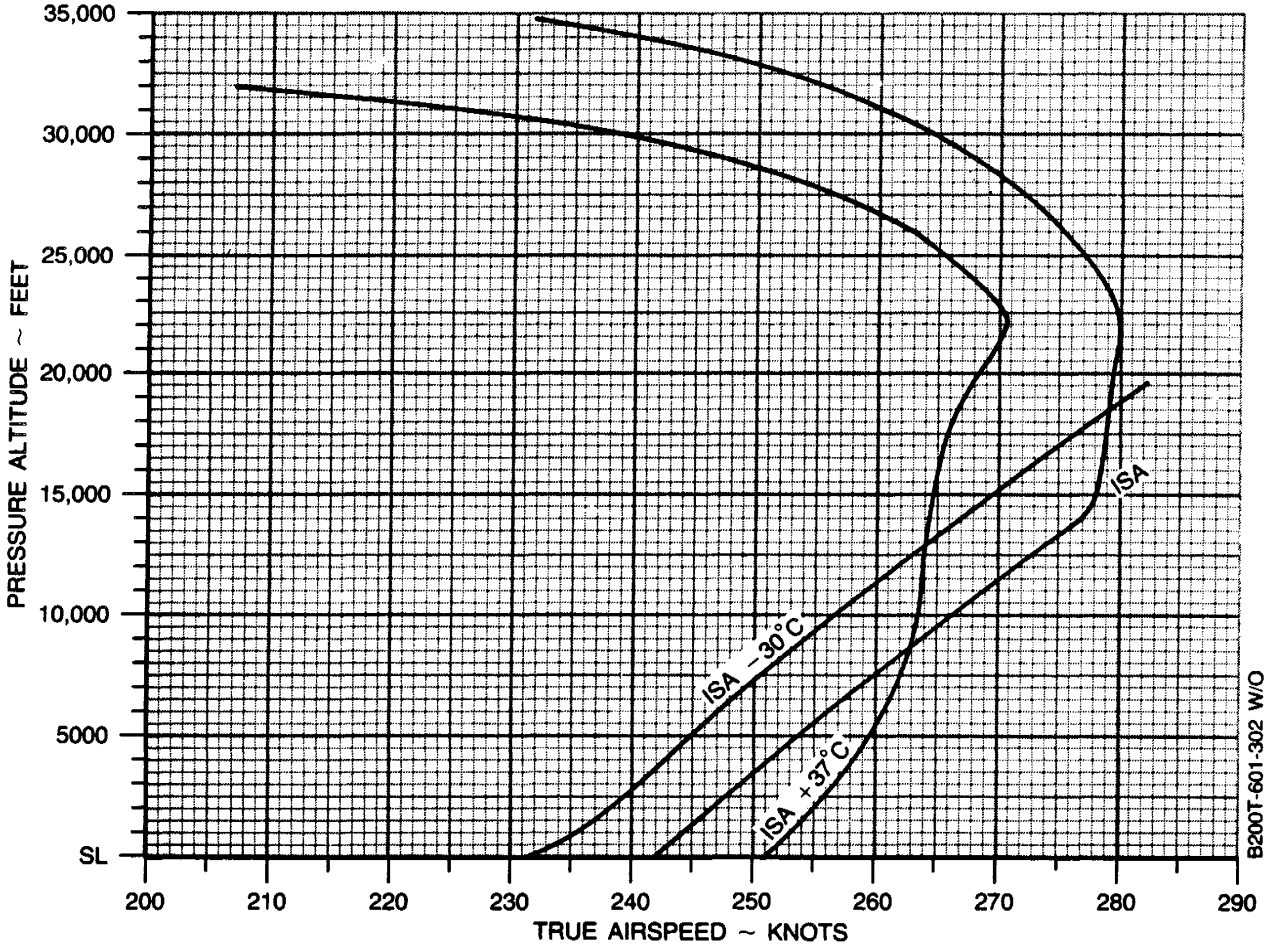
NORMAL CRUISE SPEEDS

1800 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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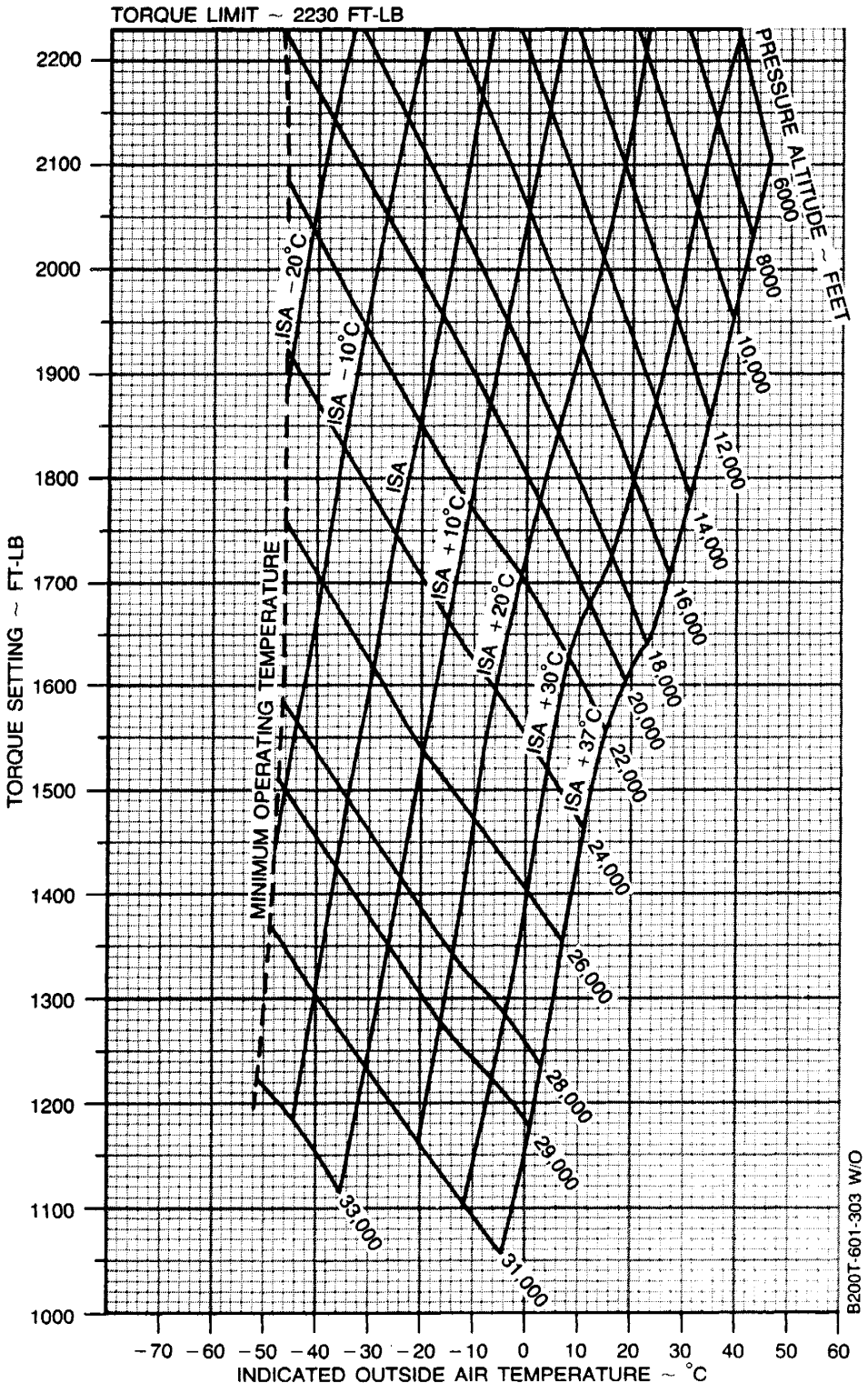
NORMAL CRUISE POWER

1800 RPM

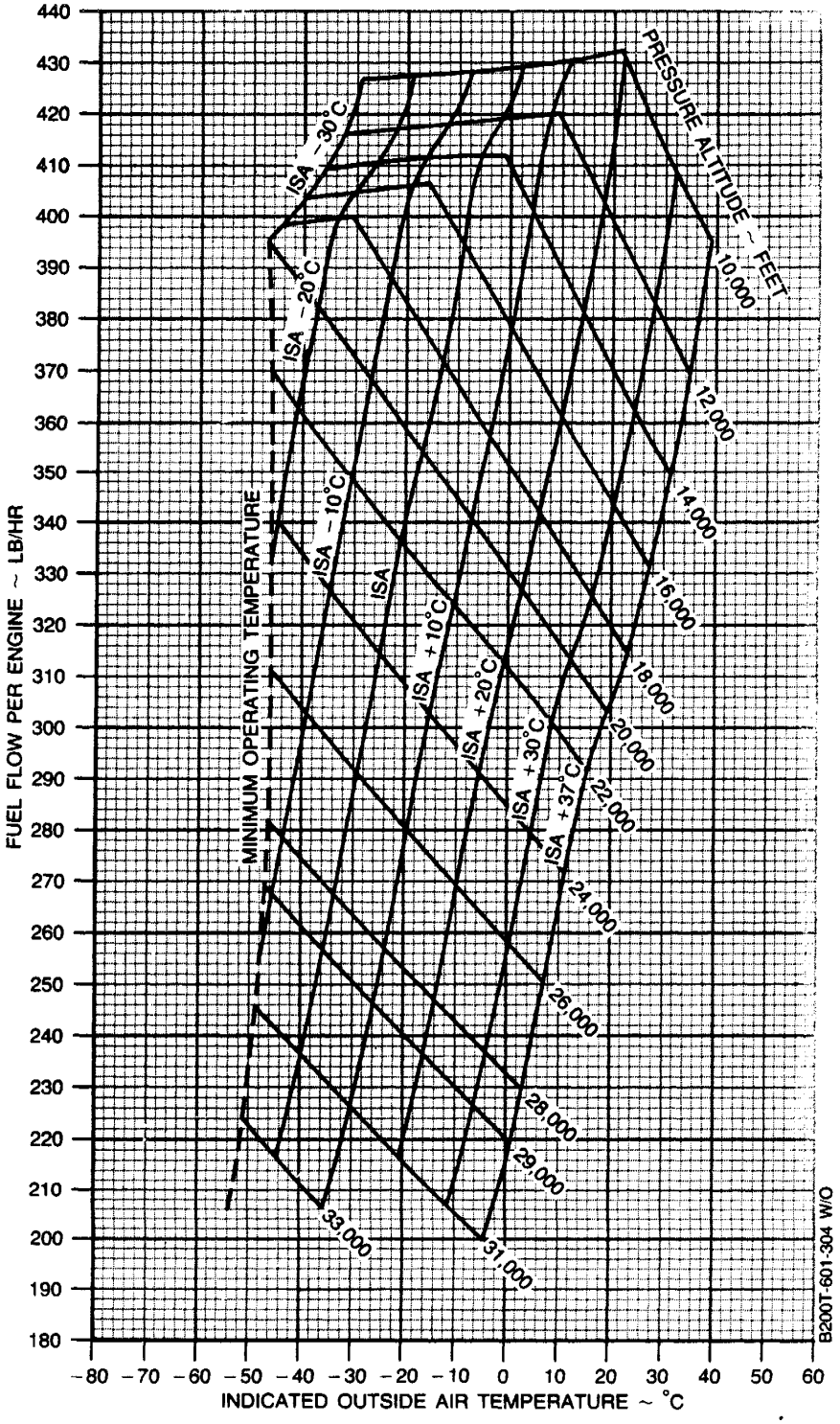
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED, ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT NORMAL CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - NORMAL CRUISE POWER

1800 RPM

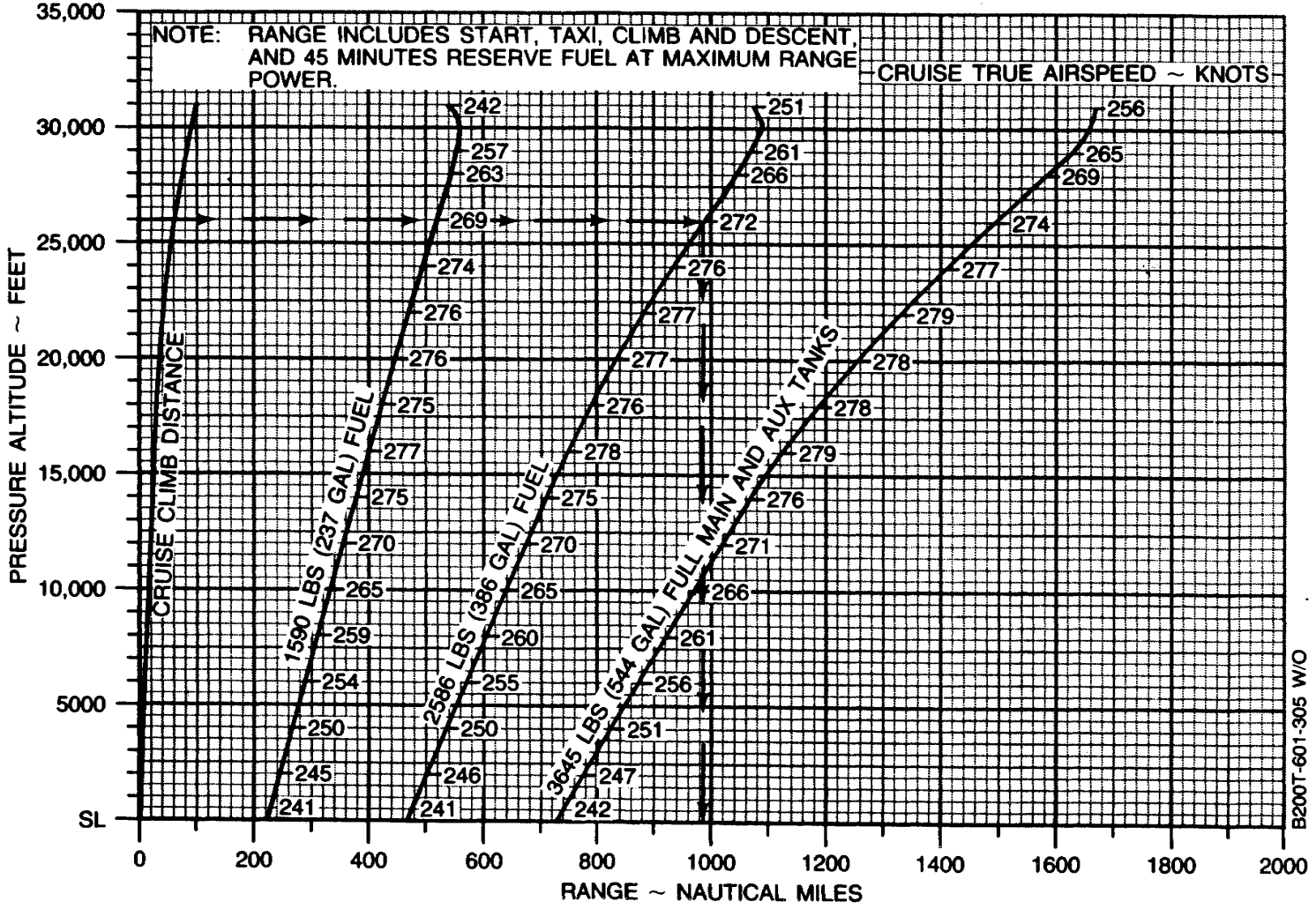
STANDARD DAY (ISA)
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	988 NM



For Training Purposes Only

MAXIMUM CRUISE POWER

1800 RPM

ISA – 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-10	-15	2194	481	962	244	232	244	232	244	232	244	232
2000	-14	-19	2230	473	946	243	237	244	238	244	238	244	238
4000	-18	-23	2230	460	920	241	242	241	243	242	243	243	244
6000	-21	-27	2230	448	896	238	246	239	247	240	248	241	249
8000	-25	-31	2230	437	874	236	251	237	238	238	253	239	254
10,000	-29	-35	2230	427	854	233	255	235	257	236	258	236	259
12,000	-32	-39	2230	417	834	231	260	232	262	233	263	234	264
14,000	-36	-43	2230	410	820	229	265	230	267	231	268	232	269
16,000	-40	-47	2230	404	808	226	271	228	272	229	273	230	275
18,000	-44	-51	2230	399	798	224	276	225	278	226	279	228	280
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

1800 RPM

ISA – 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	0	-5	2226	486	972	243	236	244	236	244	236	244	236
2000	-4	-9	2230	474	948	241	240	242	241	243	242	244	243
4000	-7	-13	2230	461	922	239	245	240	246	241	246	241	247
6000	-11	-17	2230	449	898	236	249	237	250	238	251	239	252
8000	-15	-21	2230	438	876	234	254	235	255	236	256	237	257
10,000	-19	-25	2230	428	856	232	259	233	260	234	261	235	262
12,000	-22	-29	2230	417	834	229	264	230	265	231	266	232	267
14,000	-26	-33	2230	411	822	227	269	228	270	229	272	230	273
16,000	-30	-37	2230	405	810	224	274	226	276	227	277	228	278
18,000	-33	-41	2230	400	800	222	280	223	281	225	283	226	284
20,000	-37	-45	2230	396	792	220	285	221	287	222	289	223	290
22,000	-41	-49	2133	378	756	213	286	214	288	216	290	217	292
24,000	-45	-53	1996	352	704	204	284	206	286	208	289	210	291
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM CRUISE POWER

1800 RPM

ISA – 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	10	5	2230	489	978	242	238	243	239	243	240	244	241
2000	7	1	2230	475	950	239	243	240	244	241	245	242	246
4000	3	-3	2230	463	926	237	247	238	248	239	249	240	250
6000	-1	-7	2230	450	900	234	252	236	253	237	254	237	255
8000	-5	-11	2230	439	878	232	257	233	258	234	259	235	260
10,000	-8	-15	2230	428	856	230	262	231	263	232	264	233	265
12,000	-12	-19	2230	419	838	227	267	229	268	230	270	231	271
14,000	-16	-23	2230	412	824	225	272	226	274	227	275	228	276
16,000	-19	-27	2230	406	812	222	278	224	279	225	281	226	282
18,000	-23	-31	2230	401	802	220	283	221	285	223	287	223	288
20,000	-27	-35	2133	382	764	213	284	215	286	217	288	218	289
22,000	-31	-39	2031	363	726	206	284	208	286	210	288	212	290
24,000	-35	-43	1911	341	682	199	282	201	285	203	288	204	290
26,000	-39	-47	1780	316	632	190	279	192	283	195	286	197	288
28,000	-43	-51	1639	291	582	181	274	183	279	186	283	188	286
29,000	-46	-52	1568	278	556	175	271	178	276	181	280	184	284
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

1800 RPM

ISA

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	20	15	2230	492	984	240	241	241	242	242	243	243	244
2000	17	11	2230	478	956	238	246	239	247	240	248	240	248
4000	13	7	2230	464	928	235	250	236	251	237	252	238	253
6000	9	3	2230	451	902	233	255	234	256	235	257	236	258
8000	5	-1	2230	440	880	230	260	232	261	233	262	234	263
10,000	2	-5	2230	429	858	228	265	229	266	230	267	231	269
12,000	-2	-9	2230	419	838	226	270	227	272	228	273	229	274
14,000	-6	-13	2230	411	822	223	275	224	277	226	278	227	280
16,000	-9	-17	2230	406	812	221	281	222	283	223	284	224	285
18,000	-13	-21	2128	386	772	213	281	216	283	217	285	218	287
20,000	-17	-25	2033	367	734	207	282	209	284	211	286	212	288
22,000	-21	-29	1938	349	698	201	282	203	284	204	287	206	289
24,000	-25	-33	1819	327	654	192	280	195	283	197	286	199	288
26,000	-29	-37	1692	303	606	184	276	186	280	189	284	191	287
28,000	-33	-41	1562	280	560	174	271	177	276	180	281	183	284
29,000	-36	-42	1499	269	538	169	268	172	273	176	278	178	283
31,000	-40	-46	1375	248	496	157	258	162	267	167	274	170	279
33,000	-45	-50	1249	227	454	140	240	150	257	157	267	161	274
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM CRUISE POWER

1800 RPM

ISA + 10°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	31	25	2230	495	990	238	244	239	245	240	246	241	246
2000	27	21	2230	481	962	236	248	237	249	238	250	239	251
4000	23	17	2230	468	936	234	253	235	254	236	255	237	256
6000	19	13	2230	455	910	231	258	232	259	233	260	234	261
8000	16	9	2230	443	886	229	263	230	264	231	265	232	266
10,000	12	5	2230	431	862	226	268	228	269	229	270	230	272
12,000	8	1	2230	420	840	224	273	225	275	226	276	227	277
14,000	5	-3	2226	412	824	221	278	223	280	224	282	225	283
16,000	1	-7	2132	391	782	215	279	216	281	218	283	219	284
18,000	-3	-11	2035	371	742	208	279	210	282	212	284	213	286
20,000	-7	-15	1952	354	708	202	280	204	283	206	285	207	287
22,000	-11	-19	1848	336	672	195	279	197	282	199	285	201	288
24,000	-15	-23	1733	314	628	187	277	189	281	192	284	194	287
26,000	-19	-27	1610	291	582	177	273	181	277	183	281	186	285
28,000	-24	-31	1484	268	536	167	267	171	273	175	278	177	282
29,000	-26	-32	1422	258	516	161	262	166	270	170	275	173	280
31,000	-30	-36	1304	237	474	148	250	156	262	161	270	164	276
33,000	-35	-40	1186	218	436	---	---	143	250	150	263	155	271
35,000	-39	-44	1087	201	402	---	---	---	---	137	249	145	263

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MAXIMUM CRUISE POWER

1800 RPM

ISA + 20°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	41	35	2230	497	994	237	246	238	247	239	248	240	249
2000	37	31	2230	484	968	235	251	236	252	237	253	237	254
4000	33	27	2230	470	940	232	255	233	257	234	258	235	259
6000	29	23	2230	457	914	230	261	231	262	232	263	233	264
8000	26	19	2230	445	890	227	266	228	267	230	268	230	269
10,000	22	15	2230	434	868	225	271	226	272	227	273	228	275
12,000	18	11	2146	410	820	219	272	220	274	222	275	223	277
14,000	14	7	2075	390	780	213	274	215	276	216	277	218	279
16,000	10	3	2002	372	744	208	275	210	277	211	279	213	281
18,000	7	-1	1928	355	710	202	276	204	279	206	281	207	283
20,000	3	-5	1854	339	678	196	277	198	280	200	283	202	285
22,000	-1	-9	1776	324	648	190	278	192	281	194	284	196	287
24,000	-5	-13	1667	303	606	181	275	184	279	187	283	189	286
26,000	-9	-17	1539	280	560	172	270	175	275	178	279	181	283
28,000	-14	-21	1408	258	516	161	262	165	269	169	275	172	279
29,000	-16	-22	1348	247	494	154	255	160	265	164	272	168	277
31,000	-21	-26	1231	227	454	137	237	148	255	154	266	159	273
33,000	-26	-30	1112	208	416	---	---	132	237	143	256	150	267
35,000	-30	-34	1019	191	382	---	---	---	---	127	236	138	256

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MAXIMUM CRUISE POWER

1800 RPM

ISA + 30°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	51	45	2230	500	1000	235	248	236	250	237	251	238	251
2000	47	41	2230	486	972	233	253	234	254	235	255	236	256
4000	43	37	2230	472	944	231	258	232	259	233	261	234	262
6000	40	33	2230	459	918	228	263	229	264	230	266	231	267
8000	36	29	2181	439	878	224	266	225	267	226	269	227	270
10,000	32	25	2095	415	830	218	267	219	269	221	270	222	272
12,000	28	21	1987	387	774	211	267	212	269	214	270	215	272
14,000	24	17	1924	370	740	205	268	207	271	209	273	210	275
16,000	20	13	1864	353	706	200	270	202	273	204	275	205	277
18,000	16	9	1806	337	674	195	272	197	275	199	277	201	279
20,000	13	5	1758	325	650	190	274	192	277	194	280	196	283
22,000	9	1	1690	311	622	184	274	187	278	189	281	191	284
24,000	5	-3	1583	290	580	175	271	179	276	181	280	184	283
26,000	0	-7	1468	269	538	166	266	170	272	173	277	176	281
28,000	-4	-11	1352	248	496	154	257	161	266	164	273	168	278
29,000	-6	-12	1293	238	476	147	248	154	261	159	269	163	276
31,000	-11	-16	1175	218	436	---	---	141	248	149	262	154	270
33,000	-15	-20	1078	202	404	---	---	---	---	135	248	144	262
35,000	-19	-24	981	185	370	---	---	---	---	---	---	130	247

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MAXIMUM CRUISE POWER

1800 RPM

ISA + 37°C

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	58	52	2230	501	1002	234	250	236	251	236	252	237	253
2000	54	48	2230	487	974	232	255	233	256	234	257	235	258
4000	50	44	2200	470	940	228	258	230	260	231	261	232	262
6000	46	40	2139	447	894	223	261	225	262	226	264	227	265
8000	43	36	2061	423	846	218	262	219	264	221	265	222	267
10,000	39	32	1983	399	798	212	263	214	265	215	267	216	268
12,000	35	28	1881	373	746	205	263	207	265	209	267	210	269
14,000	31	24	1826	356	712	200	264	202	267	204	269	205	271
16,000	27	20	1770	341	682	195	266	197	269	199	272	201	274
18,000	23	16	1716	325	650	190	268	192	271	194	274	196	276
20,000	19	12	1675	314	628	185	270	187	274	190	277	192	280
22,000	15	8	1621	301	602	179	271	182	275	185	279	187	282
24,000	11	4	1522	282	564	171	268	174	273	177	277	180	281
26,000	7	0	1414	261	522	161	262	166	269	169	274	172	279
28,000	3	-4	1300	240	480	148	250	156	262	160	269	164	275
29,000	0	-5	1241	230	460	139	239	149	256	155	266	159	273
31,000	-4	-9	1124	211	422	---	---	134	239	144	257	150	267
33,000	-8	-13	1038	195	390	---	---	---	---	130	241	140	258
35,000	-12	-17	951	180	360	---	---	---	---	---	---	125	242

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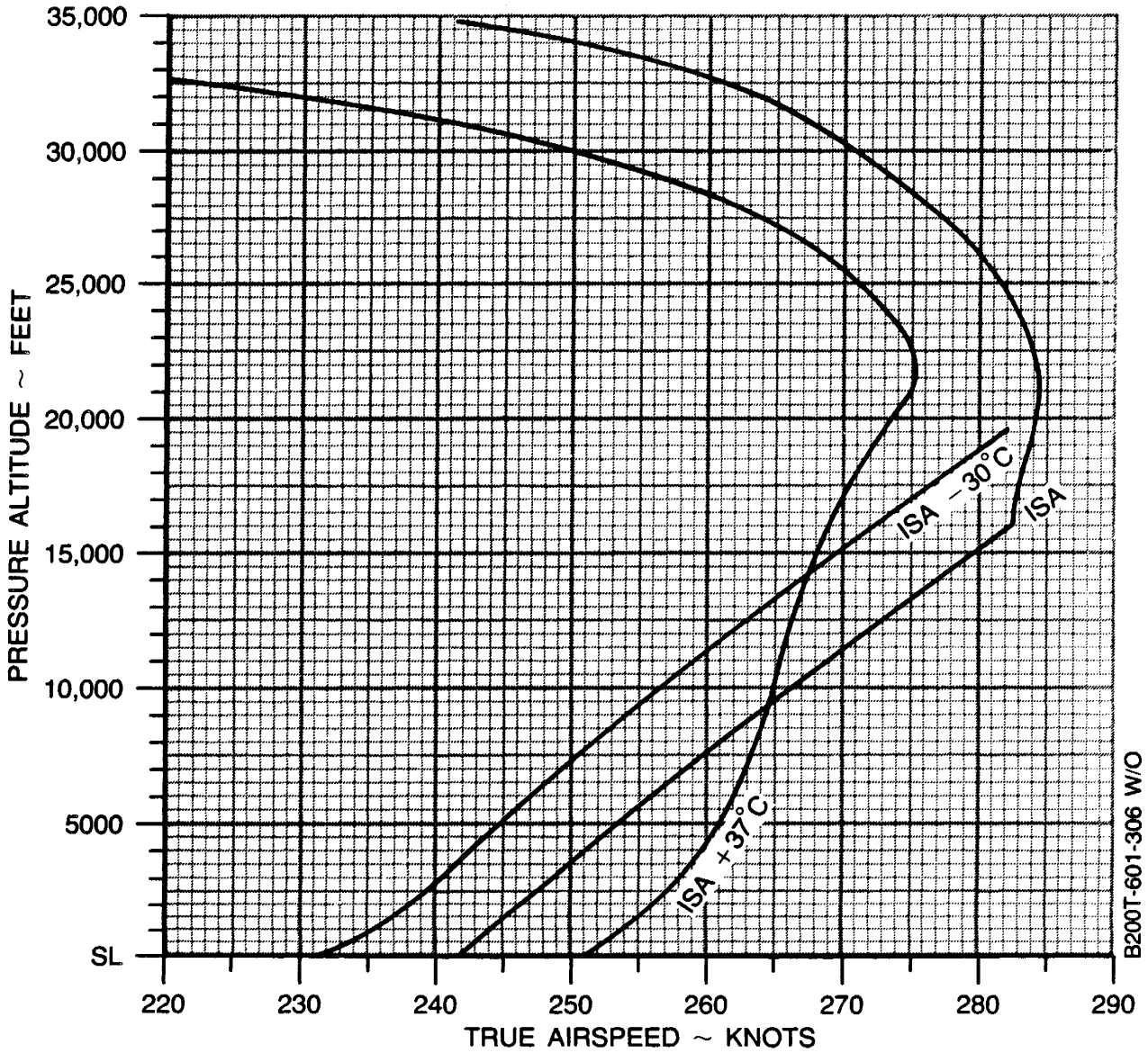
MAXIMUM CRUISE SPEEDS

1800 RPM

WEIGHT 12,000 LBS

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 30 KNOTS IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, BUT WILL BE UNCHANGED IF THE ORIGINAL POWER IS RESET.



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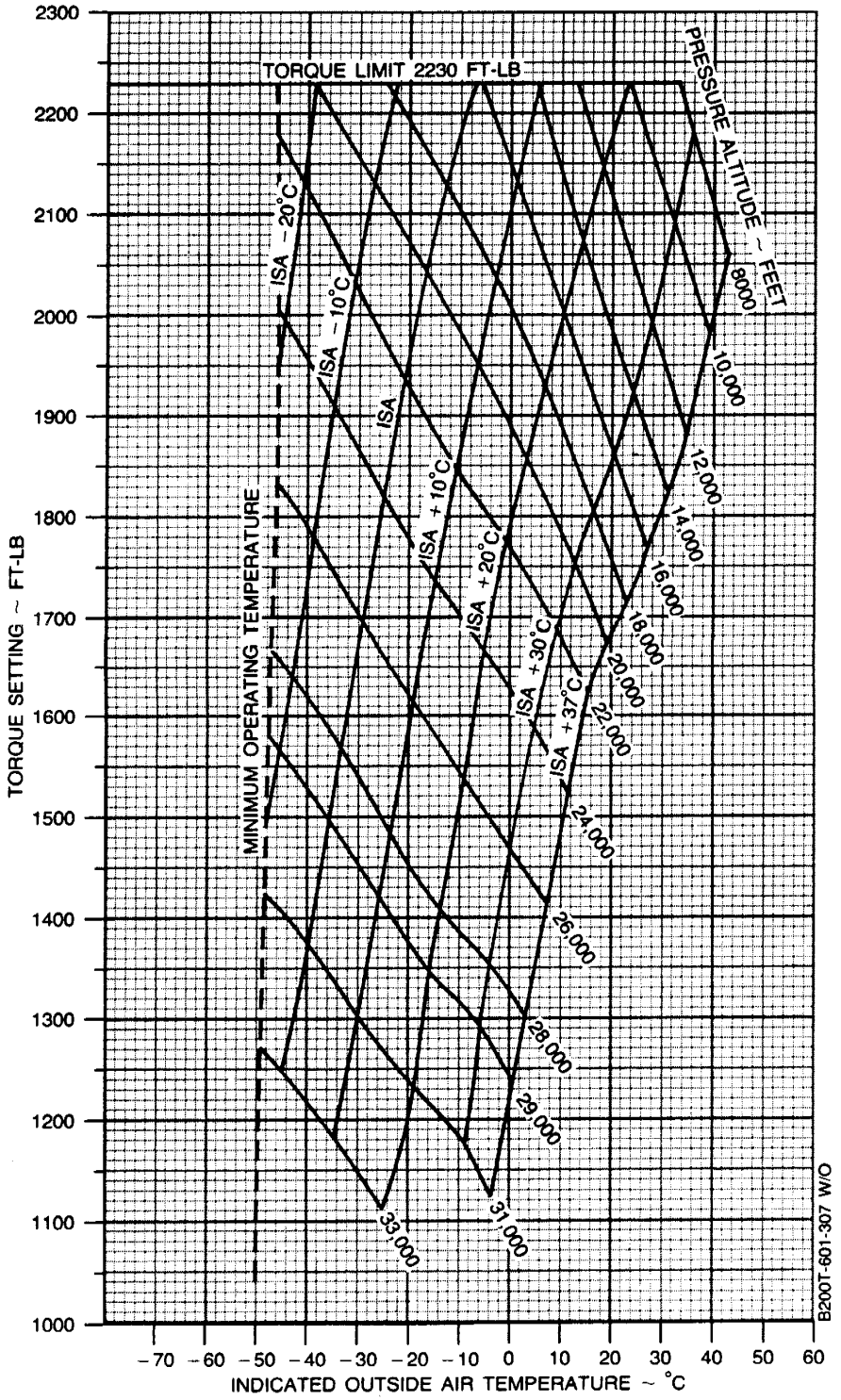
MAXIMUM CRUISE POWER

1800 RPM

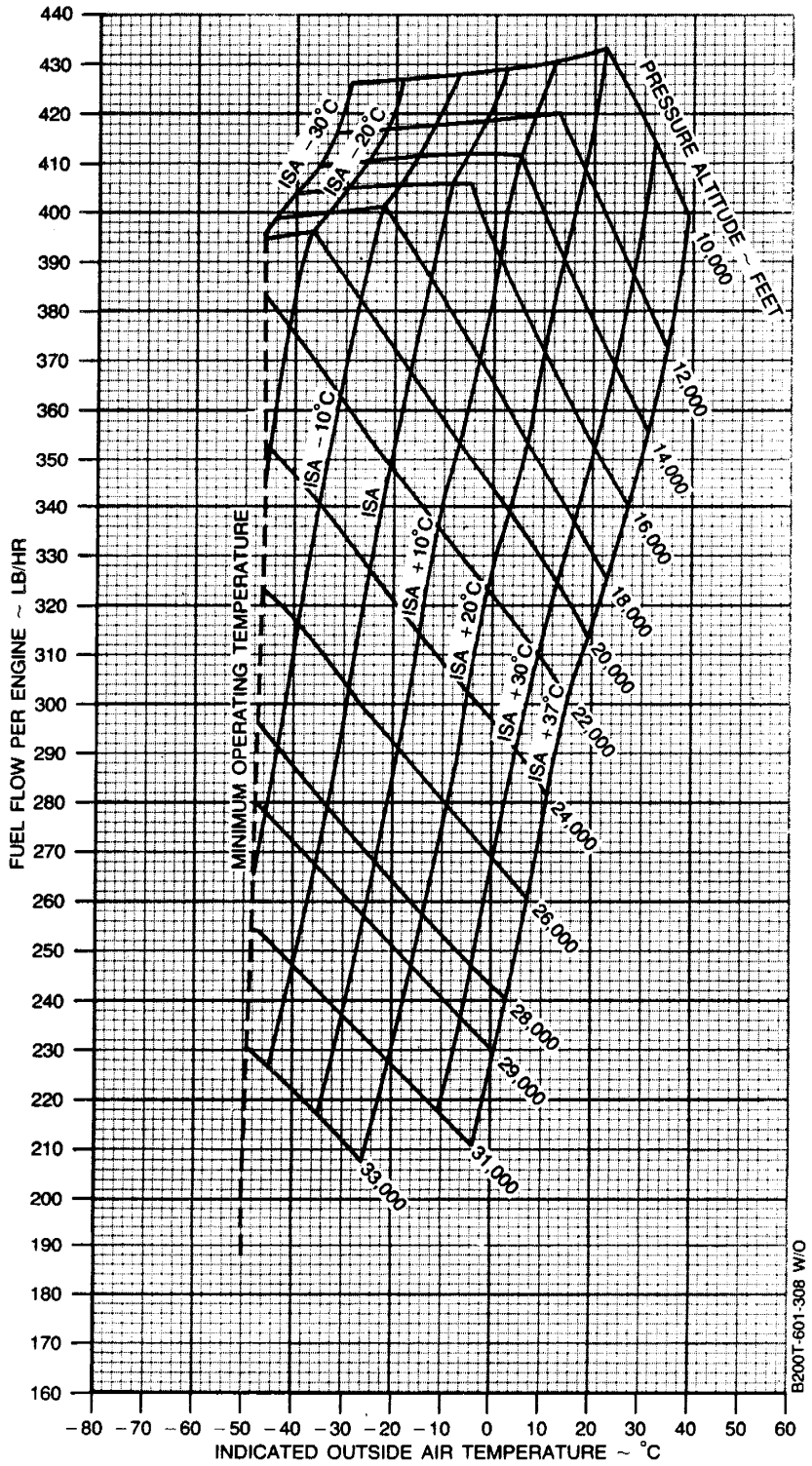
NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE APPROXIMATELY 20%. IF DESIRED ORIGINAL POWER MAY BE RESET, PROVIDED ITT LIMIT IS NOT EXCEEDED.



FUEL FLOW AT MAXIMUM CRUISE POWER

1800 RPM

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, FUEL FLOW WILL DECREASE APPROXIMATELY 10% IF ORIGINAL POWER IS NOT OR CANNOT BE RESET. IF ORIGINAL POWER IS RESET, FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.



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RANGE PROFILE - MAXIMUM CRUISE POWER

1800 RPM

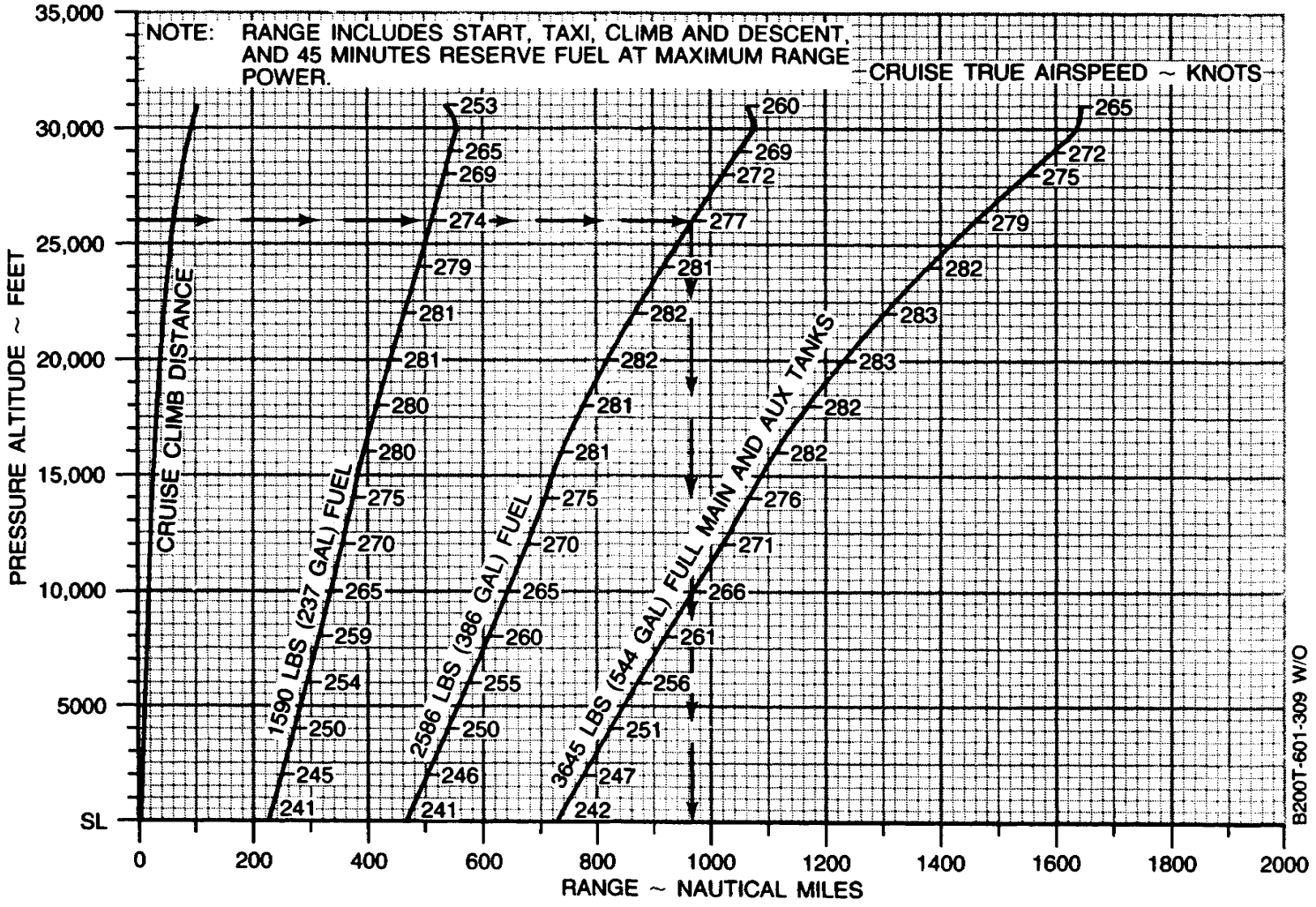
ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

STANDARD DAY (ISA)
ZERO WIND

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	968 NM



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MAXIMUM RANGE POWER

1700 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-11	-15	1618	394	788	210	200	1542	384	768	207	197
2000	-15	-19	1496	365	730	201	197	1422	355	710	199	195
4000	-20	-23	1395	337	674	194	195	1313	327	654	191	192
6000	-24	-27	1318	314	628	187	194	1229	302	604	184	191
8000	-28	-31	1254	292	584	181	193	1159	280	560	177	189
10,000	-32	-35	1203	273	546	175	193	1103	260	520	171	189
12,000	-35	-39	1175	258	516	171	194	1070	244	488	167	189
14,000	-39	-43	1145	245	490	167	194	1031	230	460	162	189
16,000	-43	-47	1124	234	468	163	196	1006	218	436	158	190
18,000	-47	-51	1105	223	446	158	197	978	206	412	153	190
20,000	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-11	-15	1475	376	734	205	195	1413	368	736	203	193
2000	-15	-19	1351	347	694	197	192	1280	338	676	194	190
4000	-20	-23	1230	316	632	188	189	1147	306	612	184	186
6000	-24	-27	1136	290	580	180	187	1043	278	556	176	183
8000	-28	-31	1060	267	534	173	185	961	255	510	169	180
10,000	-32	-35	1001	246	492	167	184	898	233	466	162	179
12,000	-35	-39	959	229	458	162	184	850	215	430	157	178
14,000	-39	-43	910	214	428	156	183	792	199	398	150	176
16,000	-43	-47	879	201	402	152	183	748	184	368	145	175
18,000	-47	-51	841	188	376	146	182	696	169	338	138	172
20,000	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-1	-5	1721	407	814	214	207	1643	398	796	211	205
2000	-5	-9	1602	379	758	206	205	1505	367	734	202	201
4000	-9	-13	1488	351	702	197	202	1397	339	678	194	199
6000	-13	-17	1409	327	654	191	201	1321	316	632	187	198
8000	-17	-21	1343	306	612	185	201	1254	294	588	181	197
10,000	-21	-25	1275	284	568	178	200	1185	272	544	175	196
12,000	-25	-29	1251	270	540	174	201	1154	257	514	170	197
14,000	-29	-33	1223	257	514	170	202	1118	243	486	166	198
16,000	-33	-37	1199	245	490	165	203	1085	230	460	161	198
18,000	-37	-41	1182	236	472	162	205	1061	220	440	157	199
20,000	-41	-45	1173	228	456	158	207	1039	210	420	152	200
22,000	-45	-49	1181	223	446	156	211	1042	204	408	150	203
24,000	-49	-53	1210	222	444	155	217	1068	202	404	149	208
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	OAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	-1	-5	1565	388	776	208	202	1495	380	760	206	200
2000	-5	-9	1426	358	716	199	199	1357	349	698	197	196
4000	-9	-13	1320	330	660	191	196	1247	321	642	189	194
6000	-13	-17	1241	306	612	185	195	1164	296	592	182	192
8000	-17	-21	1170	284	568	178	194	1087	273	546	175	191
10,000	-21	-25	1097	261	522	171	192	1009	250	500	168	188
12,000	-25	-29	1059	245	490	167	193	963	233	466	163	188
14,000	-29	-33	1017	230	460	162	193	913	217	434	157	187
16,000	-33	-37	974	216	432	156	192	858	201	402	151	186
18,000	-37	-41	943	204	408	152	193	817	187	374	145	185
20,000	-41	-45	910	192	384	147	192	770	174	348	139	183
22,000	-45	-49	906	185	370	143	195	766	166	332	136	185
24,000	-49	-53	931	183	366	142	200	795	164	328	136	191
26,000	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	9	5	1776	415	830	216	213	1694	405	810	213	210
2000	5	1	1690	391	782	209	212	1603	380	760	206	209
4000	1	-3	1582	364	728	201	210	1488	352	704	198	207
6000	-3	-7	1496	339	678	194	209	1402	328	656	191	205
8000	-7	-11	1424	317	634	188	208	1328	305	610	184	204
10,000	-11	-15	1357	295	590	181	207	1260	283	566	178	203
12,000	-15	-19	1330	281	562	177	209	1226	268	536	173	204
14,000	-19	-23	1296	267	534	173	210	1185	253	506	168	205
16,000	-23	-27	1280	257	514	169	212	1157	241	482	164	206
18,000	-27	-31	1264	247	494	165	214	1129	230	460	159	207
20,000	-31	-35	1257	240	480	162	217	1112	221	442	156	209
22,000	-34	-39	1264	235	470	160	221	1115	215	430	153	212
24,000	-38	-43	1281	232	464	158	226	1144	213	426	152	218
26,000	-42	-47	1297	230	460	156	232	1163	211	422	151	224
28,000	-46	-51	1291	226	452	153	234	1187	210	420	150	230
29,000	-48	-52	1317	229	458	153	239	1191	210	420	149	232
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	9	5	1618	395	790	210	207	1548	387	774	207	205
2000	5	1	1519	370	740	203	206	1438	360	720	200	203
4000	1	-3	1392	341	682	194	203	1307	330	660	190	199
6000	-3	-7	1308	316	632	187	201	1225	306	612	184	198
8000	-7	-11	1234	293	586	181	200	1150	283	566	177	197
10,000	-11	-15	1169	271	542	174	199	1085	260	520	171	196
12,000	-15	-19	1128	255	510	169	200	1038	244	488	166	196
14,000	-19	-23	1082	240	480	164	200	989	228	456	160	195
16,000	-23	-27	1047	227	454	159	200	947	214	428	155	195
18,000	-27	-31	1009	214	428	154	200	897	200	400	149	194
20,000	-31	-35	985	204	408	150	201	867	189	378	145	194
22,000	-34	-39	977	197	394	147	204	851	180	360	141	196
24,000	-38	-43	1001	194	388	146	209	868	176	352	140	200
26,000	-42	-47	1024	192	384	145	215	890	173	346	138	206
28,000	-46	-51	1050	191	382	144	221	911	172	344	137	211
29,000	-48	-52	1064	192	384	144	224	925	172	344	137	214
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	19	15	1601	420	840	217	218	1730	411	822	214	215
2000	15	11	1718	395	790	210	217	1630	385	770	207	214
4000	11	7	1656	374	748	204	217	1564	362	724	201	214
6000	7	3	1560	348	696	197	216	1459	336	672	193	211
8000	3	-1	1505	328	656	191	216	1406	316	632	188	212
10,000	-1	-5	1448	307	614	186	216	1343	294	588	182	212
12,000	-5	-9	1413	291	582	181	218	1302	277	554	177	212
14,000	-9	-13	1379	277	554	177	219	1261	263	526	172	213
16,000	-12	-17	1355	266	532	173	221	1228	250	500	167	214
18,000	-16	-21	1338	257	514	169	223	1209	241	482	163	216
20,000	-20	-25	1326	250	500	165	226	1192	232	464	160	218
22,000	-24	-29	1324	244	488	162	229	1187	226	452	157	221
24,000	-28	-33	1341	241	482	161	235	1205	222	444	155	227
26,000	-31	-37	1309	232	464	155	235	1223	219	438	154	232
28,000	-35	-41	1320	231	462	153	240	1207	214	428	150	234
29,000	-37	-42	1334	233	466	152	243	1209	214	428	148	236
31,000	-41	-46	---	---	---	---	---	1225	214	428	146	242
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	19	15	1659	402	804	212	213	1597	394	788	210	211
2000	15	11	1550	375	750	204	211	1476	366	732	201	208
4000	11	7	1480	352	704	198	210	1400	342	684	195	207
6000	7	3	1370	325	650	189	208	1290	315	630	186	204
8000	3	-1	1313	304	608	184	208	1225	293	586	180	204
10,000	-1	-5	1244	282	564	178	207	1148	271	542	174	203
12,000	-5	-9	1199	265	530	173	208	1106	253	506	169	203
14,000	-9	-13	1149	249	498	167	207	1050	236	472	163	202
16,000	-12	-17	1105	235	470	162	207	999	222	444	157	202
18,000	-16	-21	1078	224	448	158	209	965	210	420	152	202
20,000	-20	-25	1052	214	428	153	210	928	198	396	147	202
22,000	-24	-29	1044	207	414	150	212	914	190	380	144	204
24,000	-28	-33	1067	204	408	149	218	925	185	370	142	208
26,000	-31	-37	1088	201	402	148	224	946	182	364	141	214
28,000	-35	-41	1107	200	400	147	230	972	181	362	141	221
29,000	-37	-42	1105	198	396	145	232	980	181	362	140	224
31,000	-41	-46	1103	195	390	142	235	997	180	360	139	230
33,000	-45	-50	1117	196	392	140	240	999	178	356	136	233
35,000	---	---	---	---	---	---	---	1008	178	356	134	238

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MAXIMUM RANGE POWER

1700 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	29	25	1711	412	824	211	215	1671	406	812	210	215
2000	25	21	1678	393	786	207	218	1630	387	774	206	217
4000	21	17	1646	375	750	203	220	1594	368	736	202	219
6000	18	13	1605	355	710	198	221	1536	347	694	196	219
8000	14	9	1552	335	670	193	222	1468	324	648	190	219
10,000	10	5	1500	315	630	188	223	1406	303	606	184	219
12,000	6	1	1468	299	598	183	225	1381	288	576	181	221
14,000	2	-3	1432	285	570	179	226	1336	273	546	176	222
16,000	-2	-7	1404	273	546	174	228	1301	260	520	171	223
18,000	-6	-11	1384	262	524	170	230	1275	248	496	167	225
20,000	-10	-15	1365	254	508	166	232	1244	239	478	162	226
22,000	-14	-19	1356	249	498	163	235	1243	234	468	159	229
24,000	-17	-23	1320	239	478	157	235	1246	229	458	157	233
26,000	-21	-27	1339	238	476	156	240	1218	220	440	152	234
28,000	-25	-31	1340	236	472	152	243	1215	217	434	148	237
29,000	-27	-32	1354	237	474	151	246	1226	217	434	147	240
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	29	25	1623	400	800	209	214	1575	394	788	208	212
2000	25	21	1578	380	760	205	215	1511	372	744	202	213
4000	21	17	1506	357	714	199	215	1427	347	694	196	212
6000	18	13	1447	335	670	193	215	1361	325	650	190	212
8000	14	9	1374	313	626	187	215	1288	302	604	183	211
10,000	10	5	1303	291	582	180	214	1205	279	558	176	209
12,000	6	1	1280	276	552	177	216	1180	263	526	173	211
14,000	2	-3	1229	260	520	171	216	1121	247	494	167	211
16,000	-2	-7	1185	245	490	166	217	1069	231	462	161	210
18,000	-6	-11	1150	233	466	161	217	1022	217	434	155	210
20,000	-10	-15	1114	222	444	156	218	981	205	410	150	209
22,000	-14	-19	1113	217	434	153	221	976	199	398	147	212
24,000	-17	-23	1122	212	424	152	226	989	195	390	146	218
26,000	-21	-27	1132	208	416	150	231	1003	191	382	144	223
28,000	-25	-31	1121	202	404	146	234	1019	188	376	143	229
29,000	-27	-32	1109	199	398	144	234	1020	187	374	142	231
31,000	-31	-36	1116	199	398	141	239	1009	182	364	138	233
33,000	-35	-40	1144	202	404	140	246	1011	181	362	135	237
35,000	-39	-44	---	---	---	---	---	1036	184	368	134	244

MAXIMUM RANGE POWER

1700 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	39	35	1635	405	810	205	213	1588	399	798	204	212
2000	35	31	1596	386	772	201	215	1548	379	758	200	214
4000	31	27	1566	367	734	197	217	1531	363	726	197	217
6000	28	23	1532	349	698	193	219	1496	344	688	193	219
8000	24	19	1490	330	660	188	220	1452	325	650	188	221
10,000	20	15	1447	311	622	183	222	1404	305	610	183	222
12,000	16	11	1410	294	588	179	223	1371	289	578	179	223
14,000	12	7	1384	281	562	174	224	1341	275	550	175	225
16,000	8	3	1361	269	538	170	226	1315	263	526	171	227
18,000	4	-1	1337	258	516	166	228	1295	252	504	167	229
20,000	0	-5	1334	251	502	163	231	1285	244	488	163	232
22,000	-3	-9	1325	245	490	159	234	1267	236	472	159	234
24,000	-7	-13	1341	242	484	157	239	1231	226	452	154	234
26,000	-11	-17	1366	242	484	156	245	1230	222	444	151	238
28,000	-15	-21	1362	240	480	152	247	1254	223	446	149	244
29,000	-17	-22	---	---	---	---	---	1252	222	444	147	245
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	39	35	1544	393	786	204	212	1505	388	776	203	211
2000	35	31	1506	374	748	200	214	1459	368	736	199	213
4000	31	27	1485	356	712	196	217	1438	350	700	196	216
6000	28	23	1452	338	676	193	219	1379	329	658	190	216
8000	24	19	1405	318	636	188	220	1317	307	614	184	216
10,000	20	15	1347	297	594	182	220	1253	286	572	178	216
12,000	16	11	1314	281	562	178	222	1216	269	538	174	217
14,000	12	7	1277	267	534	173	223	1181	255	510	170	218
16,000	8	3	1242	253	506	169	224	1141	241	482	165	219
18,000	4	-1	1209	241	482	164	226	1093	227	454	159	219
20,000	0	-5	1177	230	460	159	226	1051	214	428	154	219
22,000	-3	-9	1153	221	442	155	228	1028	205	410	150	220
24,000	-7	-13	1156	216	432	153	232	1031	199	398	147	224
26,000	-11	-17	1131	208	416	148	234	1039	195	390	146	230
28,000	-15	-21	1115	203	406	144	235	1032	191	382	142	233
29,000	-17	-22	1131	204	408	144	239	1019	188	376	140	233
31,000	-21	-26	1147	204	408	142	245	1023	186	372	138	237
33,000	-25	-30	---	---	---	---	---	1037	186	372	136	243
35,000	---	---	---	---	---	---	---	---	---	---	---	---

MAXIMUM RANGE POWER

1700 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	49	45	1659	410	820	205	217	1570	399	798	202	214
2000	45	41	1566	384	768	198	215	1494	375	750	196	213
4000	41	37	1513	363	726	193	216	1453	355	710	192	215
6000	37	33	1480	345	690	189	218	1424	337	674	188	217
8000	34	29	1437	325	650	184	219	1383	318	636	183	218
10,000	30	25	1391	306	612	179	219	1333	298	596	178	219
12,000	26	21	1366	291	582	174	221	1306	283	566	174	220
14,000	22	17	1344	278	556	170	223	1272	269	538	169	221
16,000	18	13	1334	268	536	167	226	1255	257	514	165	224
18,000	14	9	1328	259	518	163	229	1249	248	496	162	227
20,000	10	5	1323	252	504	160	232	1241	240	480	159	230
22,000	6	1	1348	249	498	159	236	1228	232	464	155	232
24,000	3	-3	1379	249	498	158	245	1220	226	452	151	235
26,000	-1	-7	1365	243	486	153	246	1270	228	456	152	244
28,000	-5	-11	---	---	---	---	---	1259	224	448	148	246
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	49	45	1494	390	780	200	211	1438	383	766	198	209
2000	45	41	1439	368	736	195	212	1393	362	724	194	211
4000	41	37	1413	350	700	191	214	1365	344	688	190	213
6000	37	33	1382	332	664	187	216	1337	326	652	187	216
8000	34	29	1342	313	626	183	218	1300	307	614	183	218
10,000	30	25	1299	294	588	178	219	1252	287	574	178	218
12,000	26	21	1268	278	556	174	221	1224	272	544	174	220
14,000	22	17	1240	264	528	170	223	1191	258	516	169	222
16,000	18	13	1219	252	504	166	225	1167	245	490	166	224
18,000	14	9	1203	242	484	163	228	1138	233	466	161	226
20,000	10	5	1196	234	468	159	231	1102	222	444	156	226
22,000	6	1	1180	225	450	156	233	1077	212	424	152	228
24,000	3	-3	1135	214	428	150	232	1066	204	408	149	231
26,000	-1	-7	1107	205	410	144	232	1042	196	392	144	232
28,000	-5	-11	1165	209	418	146	244	1007	187	374	139	231
29,000	-7	-12	1155	207	414	144	244	1028	188	376	139	236
31,000	-11	-16	1153	205	410	140	247	1046	188	376	138	243
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			13,000 POUNDS					12,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	56	52	1721	420	840	207	222	1618	407	814	204	218
2000	52	48	1641	395	790	201	221	1524	380	760	197	216
4000	48	44	1555	370	740	194	220	1453	357	714	191	216
6000	44	40	1487	347	694	188	220	1396	335	670	185	216
8000	41	36	1448	328	656	183	221	1354	316	632	180	217
10,000	37	32	1410	310	620	179	222	1314	297	594	176	218
12,000	33	28	1398	296	592	175	225	1290	282	564	172	220
14,000	29	24	1381	284	568	171	227	1265	269	538	167	222
16,000	25	20	1374	274	548	168	230	1249	258	516	164	224
18,000	21	16	1380	267	534	166	235	1234	248	496	160	227
20,000	17	12	1384	261	522	163	239	1237	241	482	157	230
22,000	14	8	1386	256	512	160	243	1235	235	470	154	234
24,000	10	4	1374	249	498	156	245	1273	234	468	154	242
26,000	6	0	1394	248	496	154	250	1267	229	458	150	245
28,000	2	-4	---	---	---	---	---	1285	228	456	148	250
29,000	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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MAXIMUM RANGE POWER

1700 RPM

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANES EXTENDED, TORQUE WILL DECREASE. IN ORDER TO MAINTAIN MAXIMUM RANGE CONFIGURATION, DO NOT RESET POWER TO ORIGINAL SETTING. FUEL FLOW WILL REMAIN ABOUT THE SAME, BUT TRUE AIRSPEED WILL BE REDUCED APPROXIMATELY 10 KNOTS.

WEIGHT →			11,000 POUNDS					10,000 POUNDS				
PRESSURE ALTITUDE	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	IAS	TAS
FEET	°C	°C	FT-LB	LB/HR	LB/HR	KTS	KTS	FT-LB	LB/HR	LB/HR	KTS	KTS
SL	56	52	1515	394	788	200	214	1425	383	766	197	210
2000	52	48	1435	369	738	194	213	1361	360	720	191	210
4000	48	44	1383	348	696	189	214	1327	341	682	187	212
6000	44	40	1335	327	654	184	215	1294	322	644	183	214
8000	41	36	1296	309	618	179	216	1256	303	606	179	216
10,000	37	32	1257	290	580	175	217	1216	284	568	175	217
12,000	33	28	1229	274	548	171	219	1186	268	536	171	219
14,000	29	24	1195	260	520	166	220	1160	255	510	167	221
16,000	25	20	1173	248	496	162	222	1138	243	486	163	223
18,000	21	16	1164	238	476	159	225	1120	232	464	159	226
20,000	17	12	1159	231	462	156	229	1110	224	448	156	229
22,000	14	8	1150	223	446	152	231	1091	215	430	152	231
24,000	10	4	1120	213	426	147	232	1060	204	408	147	232
26,000	6	0	1150	212	424	147	239	1018	193	386	141	231
28,000	2	-4	1162	210	420	145	244	1039	190	380	139	239
29,000	0	-5	1165	209	418	143	246	1057	191	382	137	245
31,000	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---

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RANGE PROFILE - MAXIMUM RANGE POWER

ASSOCIATED CONDITIONS:

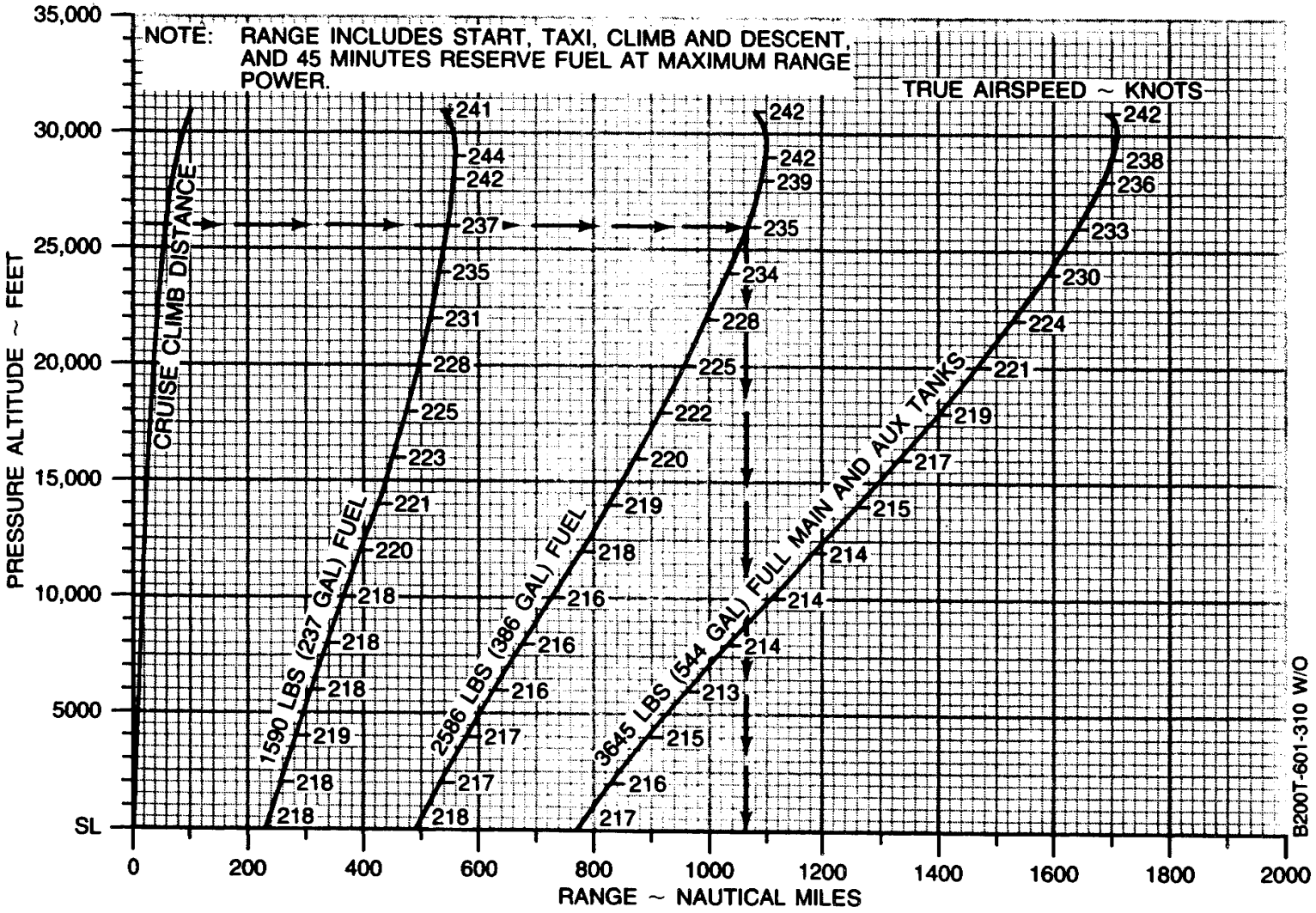
WEIGHT	14,000 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

1700 RPM

STANDARD DAY (ISA)
ZERO WIND

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
FUEL USED	2586 LBS
RANGE	1065 NM



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RANGE PROFILE - FULL MAIN AND AUX TANKS

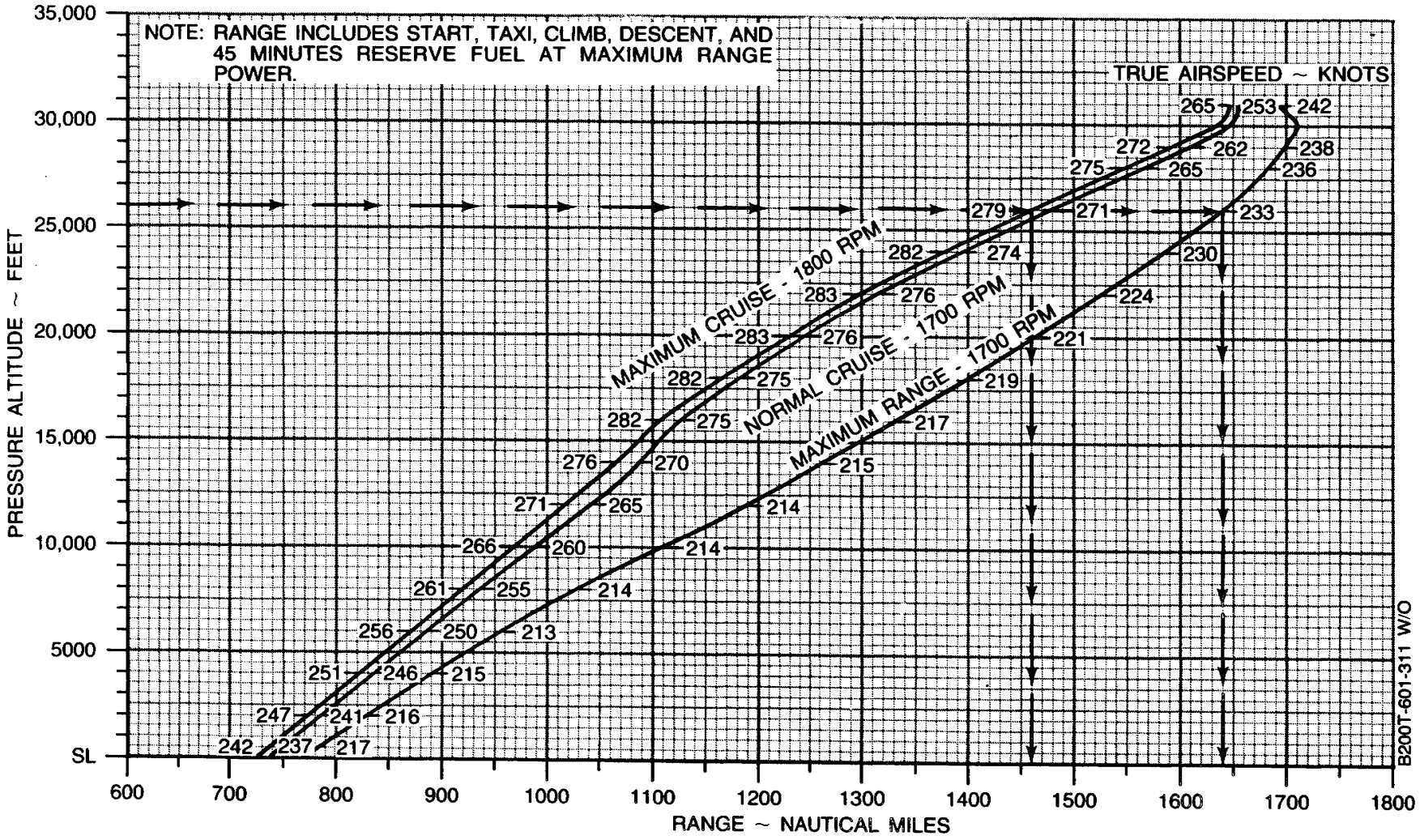
STANDARD DAY
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	1460 NM
RANGE @ MAX RANGE - 1700 RPM	1640 NM



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ENDURANCE PROFILE - FULL MAIN AND AUX TANKS

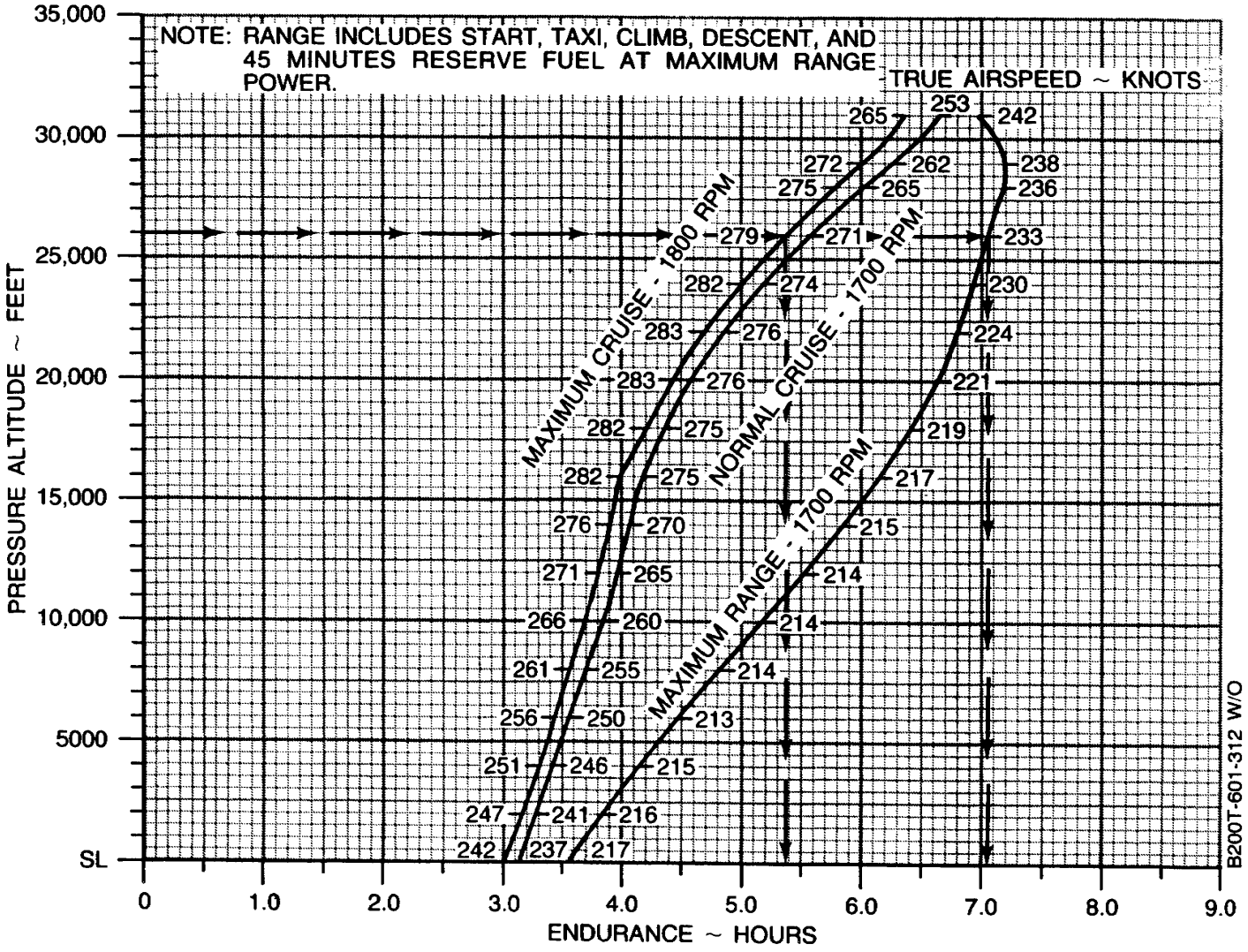
STANDARD DAY

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	5.37 HRS
RANGE @ MAX RANGE - 1700 RPM	7.05 HRS



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RANGE PROFILE - FULL MAIN TANKS

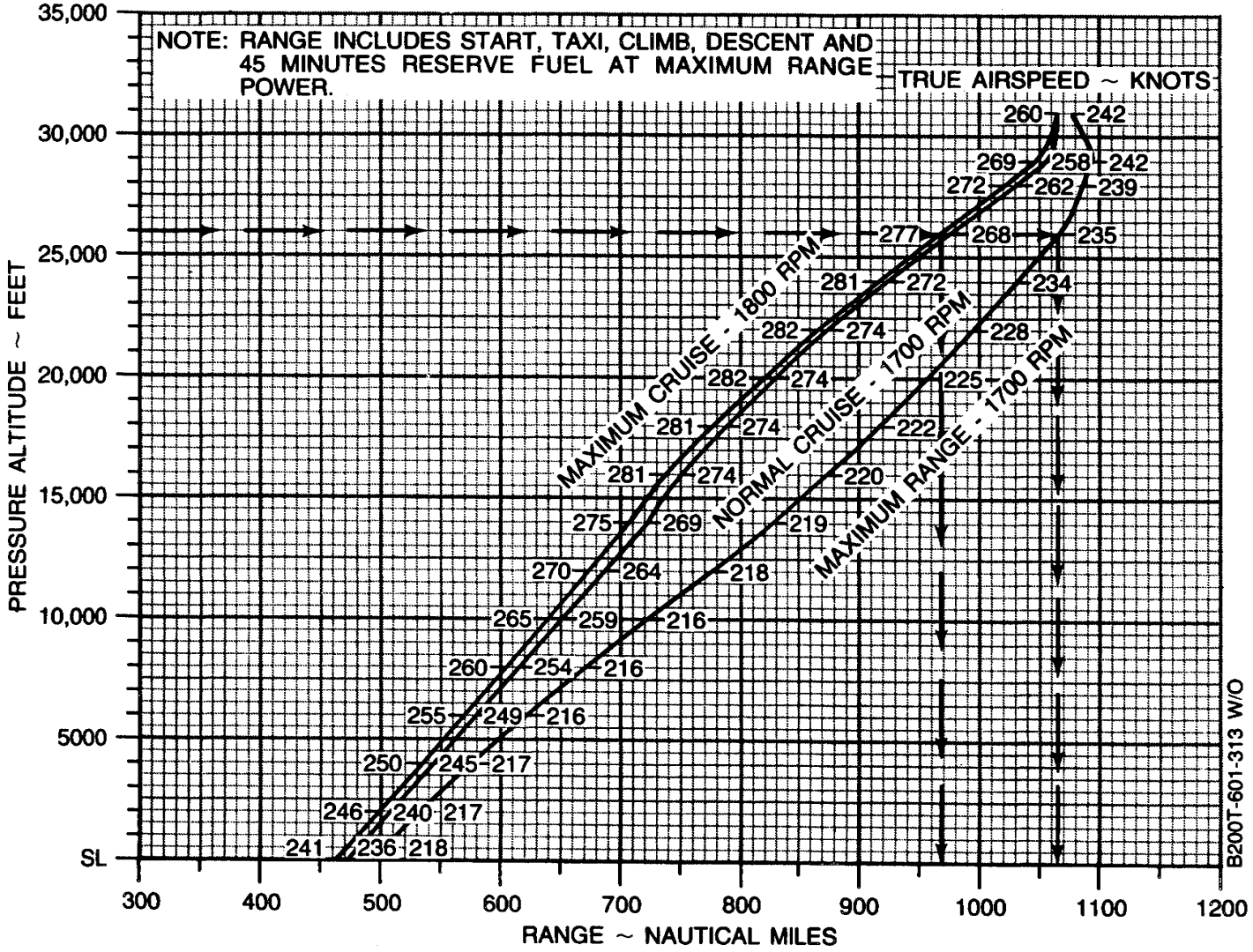
STANDARD DAY
ZERO WIND

ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	968 NM
RANGE @ MAX RANGE - 1700 RPM	1065 NM



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ENDURANCE PROFILE - FULL MAIN TANKS

STANDARD DAY

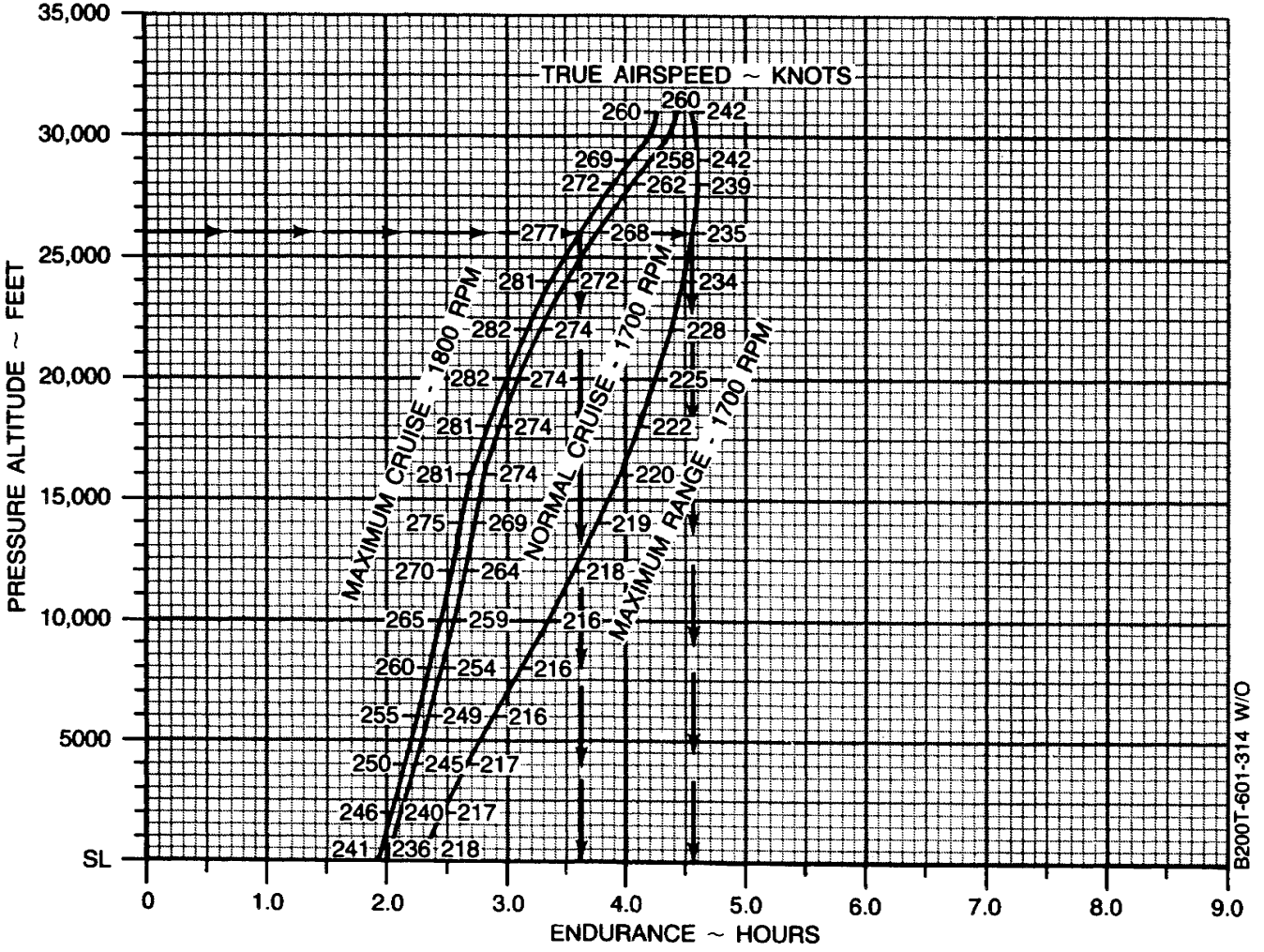
ASSOCIATED CONDITIONS:

WEIGHT	14,090 LBS BEFORE ENGINE START
FUEL	AVIATION KEROSENE
FUEL DENSITY	6.7 LBS/GAL
ICE VANES	RETRACTED

EXAMPLE:

PRESSURE ALTITUDE	26,000 FT
RANGE @ MAX CRUISE - 1800 RPM	3.62 HRS
RANGE @ MAX RANGE - 1700 RPM	4.57 HRS

NOTE: RANGE INCLUDES START, TAXI, CLIMB, DESCENT, AND 45 MINUTES RESERVE FUEL AT MAXIMUM RANGE POWER.



For Training Purposes Only

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA – 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-12	-15	2230	517	517	189	180	191	182	193	183	194	185
2000	-16	-19	2230	505	505	186	182	189	185	191	187	192	188
4000	-20	-23	2230	492	492	184	185	186	188	188	190	190	192
6000	-24	-27	2230	479	479	181	188	184	191	186	193	188	195
8000	-27	-31	2230	467	467	179	191	182	194	184	196	186	198
10,000	-31	-35	2230	455	455	176	194	179	197	182	199	184	202
12,000	-35	-39	2230	445	445	174	196	177	200	179	203	181	205
14,000	-39	-43	2230	439	439	171	199	174	203	177	206	179	208
16,000	-43	-47	2230	434	434	167	202	171	206	174	209	176	212
18,000	-47	-51	2146	415	415	160	199	165	205	169	209	172	213
20,000	---	---	---	---	---	---	---	---	---	---	---	---	---
22,000	---	---	---	---	---	---	---	---	---	---	---	---	---
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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For Training Purposes Only

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA – 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	-2	-5	2230	520	520	187	181	189	184	191	185	193	187
2000	-6	-9	2230	505	505	185	184	187	187	189	189	191	190
4000	-10	-13	2230	492	492	182	187	185	190	187	192	189	194
6000	-13	-17	2230	480	480	180	190	182	193	184	195	186	197
8000	-17	-21	2230	468	468	177	193	180	196	182	198	184	201
10,000	-21	-25	2230	456	456	174	196	177	199	180	202	182	204
12,000	-25	-29	2230	446	446	171	198	175	202	177	205	179	207
14,000	-29	-33	2230	440	440	168	201	172	205	175	208	177	211
16,000	-33	-37	2159	423	423	162	199	166	204	170	209	173	212
18,000	-37	-41	2048	400	400	154	195	159	202	163	207	166	211
20,000	-41	-45	1945	378	378	144	190	151	199	156	205	160	210
22,000	-45	-49	1833	356	356	131	178	142	193	149	202	154	208
24,000	-50	-53	1675	326	326	---	---	127	179	138	194	145	203
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA – 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	8	5	2230	522	522	185	183	188	186	190	187	191	189
2000	4	1	2230	508	508	183	186	185	188	187	191	189	192
4000	0	-3	2230	494	494	180	189	183	191	185	194	187	196
6000	-3	-7	2230	481	481	178	192	181	195	183	197	185	199
8000	-7	-11	2230	468	468	175	194	178	198	181	200	183	203
10,000	-11	-15	2230	456	456	172	197	175	201	178	204	180	206
12,000	-15	-19	2230	446	446	169	200	173	204	175	207	178	209
14,000	-19	-23	2157	428	428	163	198	167	203	170	207	173	211
16,000	-23	-27	2051	406	406	155	195	160	201	165	206	167	210
18,000	-27	-31	1949	384	384	146	190	153	198	157	204	161	209
20,000	-31	-35	1857	365	365	135	182	145	194	151	202	155	208
22,000	-35	-39	1748	343	343	---	---	134	187	143	198	148	206
24,000	-39	-43	1629	318	318	---	---	---	---	132	189	140	200
26,000	-43	-47	1512	294	294	---	---	---	---	---	---	129	192
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	18	15	2230	525	525	184	185	186	187	188	189	190	191
2000	14	11	2230	511	511	181	188	184	190	186	192	188	194
4000	11	7	2230	497	497	179	190	181	193	184	196	186	198
6000	7	3	2230	483	483	176	193	179	196	181	199	183	201
8000	3	-1	2230	469	469	173	196	176	200	179	202	181	205
10,000	-1	-5	2230	457	457	170	199	174	202	176	206	179	208
12,000	-5	-9	2169	436	436	165	198	169	203	172	207	174	210
14,000	-9	-13	2065	412	412	157	195	162	201	166	206	169	209
16,000	-13	-17	1961	390	390	148	190	154	198	159	204	163	208
18,000	-17	-21	1855	369	369	138	183	146	194	152	201	156	206
20,000	-21	-25	1767	351	351	119	163	137	188	145	198	150	205
22,000	-26	-29	1658	329	329	---	---	123	175	136	193	143	202
24,000	-30	-33	1547	305	305	---	---	---	---	123	180	133	196
26,000	-33	-37	1439	283	283	---	---	---	---	---	---	121	185
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 10°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	28	25	2230	527	527	182	187	185	189	187	191	189	193
2000	24	21	2230	513	513	180	189	182	192	185	194	186	196
4000	21	17	2230	499	499	177	192	180	195	182	198	184	200
6000	17	13	2230	484	484	174	195	177	198	180	200	182	203
8000	13	9	2222	470	470	171	197	174	201	177	204	179	207
10,000	9	5	2119	441	441	164	195	168	199	171	203	174	207
12,000	5	1	2023	415	415	157	192	161	198	165	202	168	206
14,000	1	-3	1950	395	395	149	189	155	196	160	202	163	206
16,000	-3	-7	1870	376	376	141	184	148	194	153	201	158	206
18,000	-7	-11	1783	355	355	128	173	140	189	147	198	151	205
20,000	-12	-15	1700	338	338	---	---	130	182	139	195	145	203
22,000	-15	-19	1597	319	319	---	---	---	---	128	186	137	198
24,000	-19	-23	1489	296	296	---	---	---	---	---	---	126	190
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 20°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	38	35	2230	530	530	181	188	183	191	185	193	187	195
2000	35	31	2230	515	515	178	191	181	194	183	196	185	198
4000	31	27	2230	500	500	176	194	178	197	181	200	183	202
6000	27	23	2156	475	475	170	193	173	197	176	200	178	203
8000	23	19	2060	447	447	163	191	167	196	170	200	173	203
10,000	19	15	1962	418	418	155	188	160	194	164	198	167	202
12,000	15	11	1878	394	394	148	185	153	192	158	197	161	202
14,000	11	7	1818	376	376	140	180	147	190	153	197	157	202
16,000	7	3	1758	359	359	129	172	140	187	147	196	152	202
18,000	2	-1	1696	343	343	---	---	132	182	141	194	146	201
20,000	-1	-5	1631	327	327	---	---	---	---	133	189	140	199
22,000	-6	-9	1534	307	307	---	---	---	---	120	178	132	195
24,000	-9	-13	1431	285	285	---	---	---	---	---	---	120	184
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 30°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMTAELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	48	45	2154	521	521	176	186	179	189	182	192	184	194
2000	44	41	2103	499	499	172	187	175	190	178	193	180	196
4000	40	37	2062	478	478	167	188	171	192	174	195	177	198
6000	37	33	1997	453	453	162	187	166	192	169	195	172	199
8000	33	29	1908	425	425	154	184	159	190	163	194	166	198
10,000	28	25	1819	397	397	146	180	152	187	157	193	160	198
12,000	24	21	1746	375	375	137	175	145	185	151	192	155	197
14,000	20	17	1688	358	358	126	166	138	182	145	191	150	197
16,000	16	13	1635	342	342	---	---	130	177	139	189	145	197
18,000	13	9	1601	329	329	---	---	---	---	133	187	140	196
20,000	8	5	1553	316	316	---	---	---	---	125	181	134	195
22,000	5	1	1477	297	297	---	---	---	---	---	---	125	189
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

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ONE ENGINE INOPERATIVE
MAXIMUM CRUISE POWER

1900 RPM

ISA + 37°C

NOTE: DURING OPERATION WITH ICE VANE EXTENDED, TORQUE WILL DECREASE. IF ORIGINAL POWER IS NOT OR CANNOT BE RESET, TRUE AIRSPEED WILL DECREASE APPROXIMATELY 20 KNOTS AND FUEL FLOW WILL DECREASE APPROXIMATELY 7%. IF ORIGINAL POWER IS RESET, TRUE AIRSPEED WILL BE UNCHANGED AND FUEL FLOW WILL INCREASE APPROXIMATELY 30 LBS/HR/ENG.

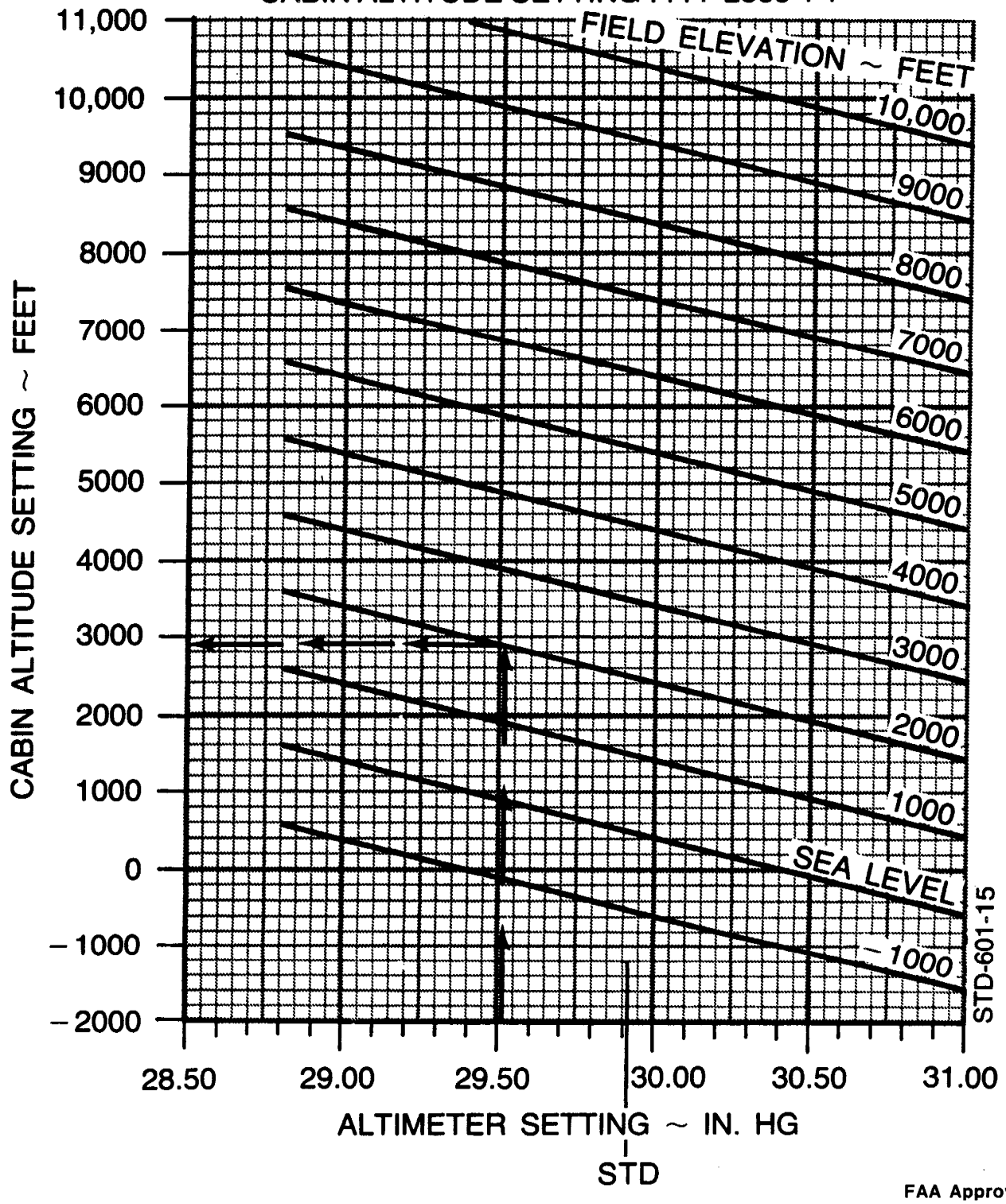
NOTE: IOAT, TORQUE, AND FUEL FLOW BASED ON 12,000 POUNDS.

PRESS. ALT	IOAT	OAT	TORQUE PER ENG	FUEL FLOW PER ENG	TOTAL FUEL FLOW	AIRSPEED ~ KNOTS							
						13,000 LBS		12,000 LBS		11,000 LBS		10,000 LBS	
FEET	°C	°C	FT-LBS	LBS/HR	LBS/HR	IAS	TAS	IAS	TAS	IAS	TAS	IAS	TAS
SL	55	52	2015	503	503	170	181	173	185	176	188	178	190
2000	51	48	1971	481	481	165	182	169	186	172	189	174	192
4000	47	44	1937	461	461	161	182	165	187	168	191	171	194
6000	43	40	1880	438	438	155	181	160	187	163	190	167	195
8000	39	36	1800	411	411	147	178	153	185	158	190	161	194
10,000	35	32	1719	384	384	138	172	146	182	151	189	156	194
12,000	31	28	1656	363	363	127	164	139	179	145	187	150	193
14,000	27	24	1603	346	346	---	---	131	175	140	186	145	193
16,000	23	20	1552	331	331	---	---	120	166	133	183	140	192
18,000	19	16	1522	318	318	---	---	---	---	126	179	135	191
20,000	16	12	1488	306	306	---	---	---	---	---	---	129	189
22,000	12	8	1414	289	289	---	---	---	---	---	---	120	182
24,000	---	---	---	---	---	---	---	---	---	---	---	---	---
26,000	---	---	---	---	---	---	---	---	---	---	---	---	---
28,000	---	---	---	---	---	---	---	---	---	---	---	---	---
29,000	---	---	---	---	---	---	---	---	---	---	---	---	---
31,000	---	---	---	---	---	---	---	---	---	---	---	---	---
33,000	---	---	---	---	---	---	---	---	---	---	---	---	---
35,000	---	---	---	---	---	---	---	---	---	---	---	---	---

PRESSURIZATION CONTROLLER SETTING FOR LANDING

EXAMPLE

ALTIMETER SETTING. . . . . 29.52 IN. HG
LANDING FIELD ELEVATION . . . 2000 FT
CABIN ALTITUDE SETTING . . . . 2885 FT



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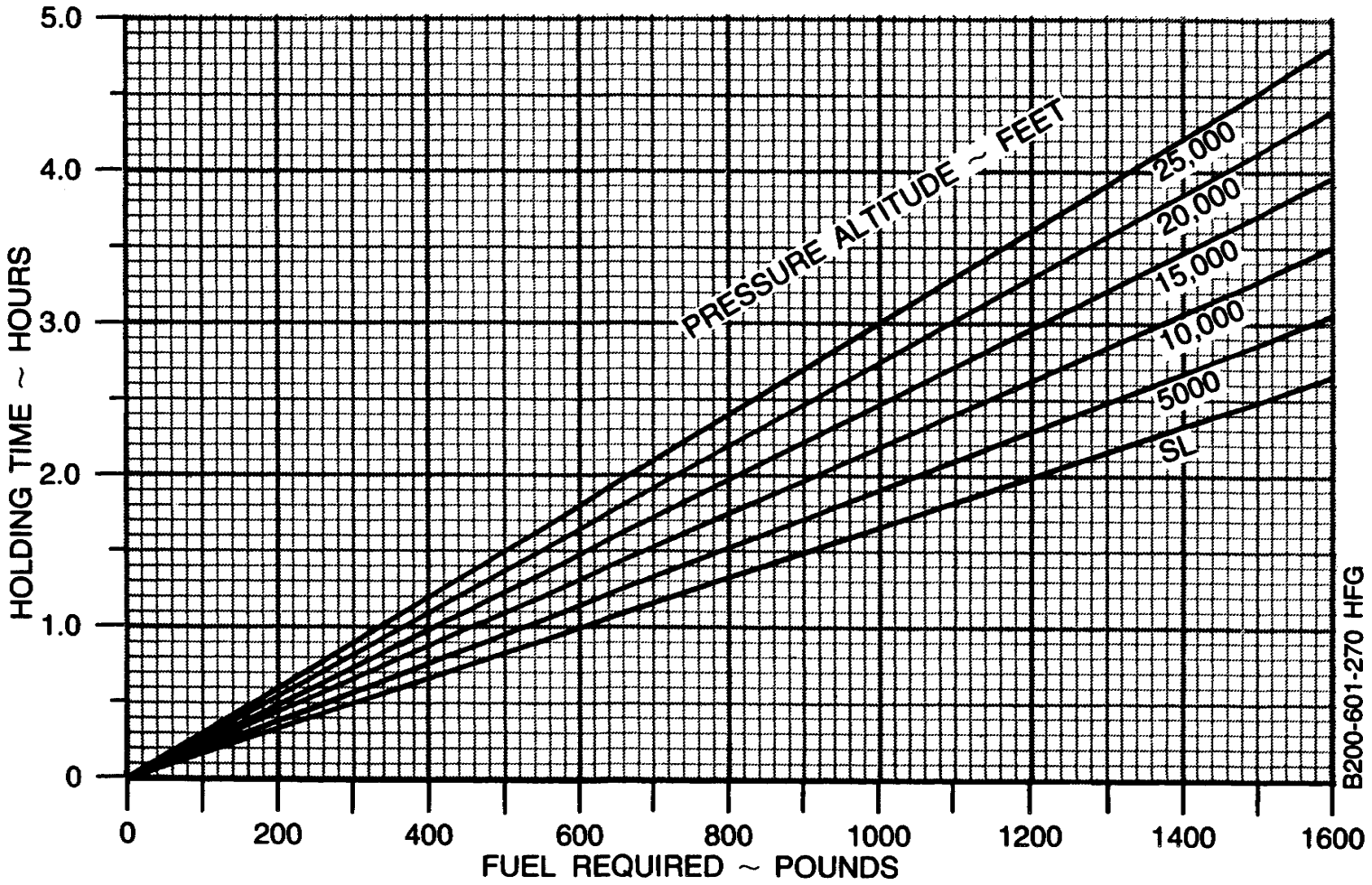
HOLDING TIME

POWER SETTING
800 FT-LBS AT 1700 RPM

APPLICABLE FOR ALL TEMPERATURES

NOTE: FOR OPERATION WITH ICE VANES EXTENDED, HOLDING TIME WILL BE REDUCED APPROXIMATELY 15%.



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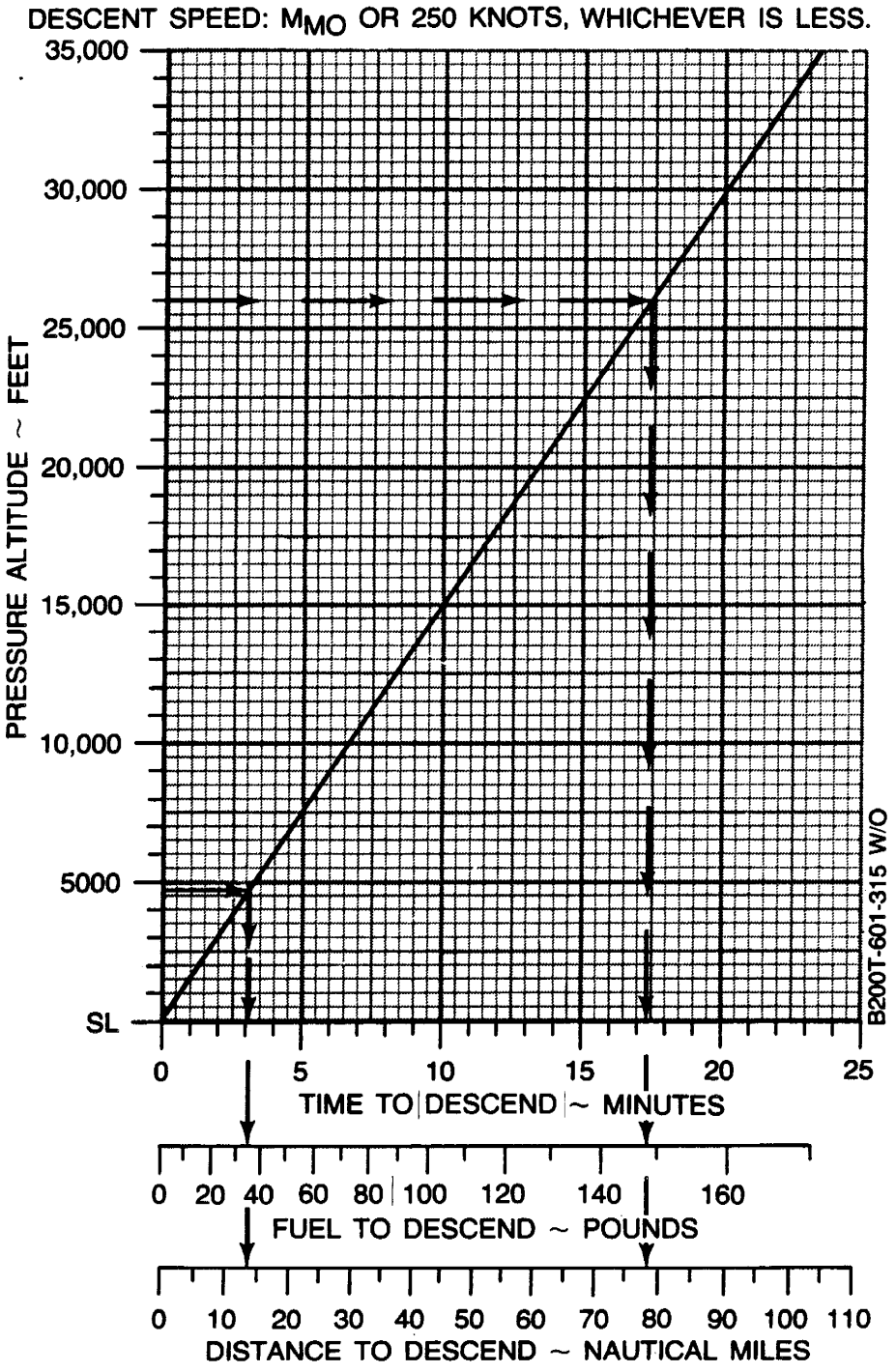
TIME, FUEL, AND DISTANCE TO DESCEND

ASSOCIATED CONDITIONS:

POWER	AS REQUIRED TO DESCEND AT 1500 FT/MIN
GEAR	UP
FLAPS	0%

EXAMPLE:

INITIAL ALTITUDE	26,000 FT
FINAL ALTITUDE	4732 FT
TIME TO DESCEND (17-3)	14 MIN
FUEL TO DESCEND (148-34)	114 LBS
DISTANCE TO DESCEND (78-13)	65 NM



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CLIMB-BALKED LANDING

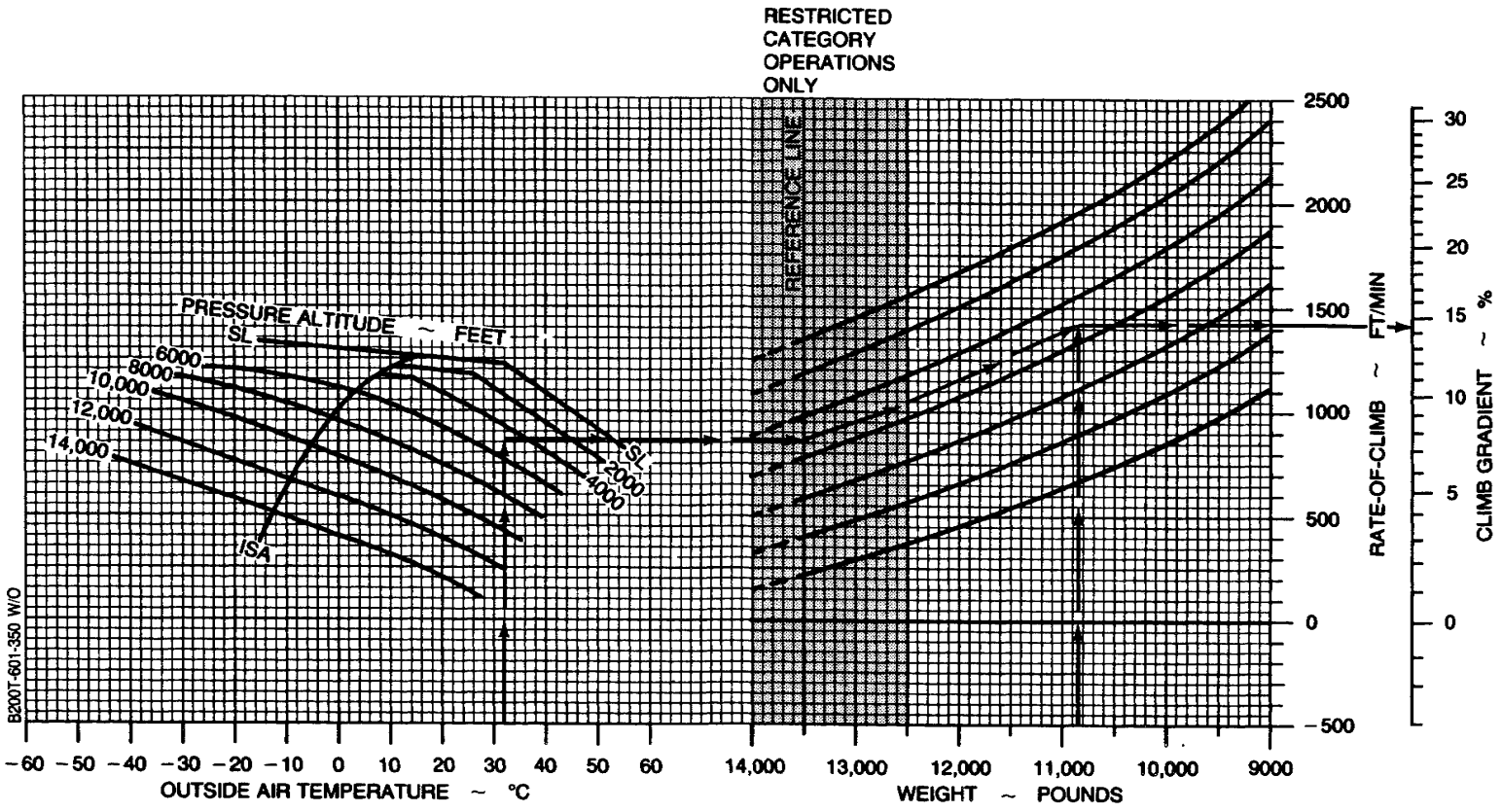
CLIMB SPEED 100 KNOTS (ALL WEIGHTS)

ASSOCIATED CONDITIONS:

POWER	TAKE-OFF
FLAPS	100%
LANDING GEAR	DOWN

EXAMPLE:

OAT	32°C
PRESSURE ALTITUDE	4732 FT
WEIGHT	10,854 LBS
RATE-OF-CLIMB	1420 FT/MIN
CLIMB GRADIENT	14.4%



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NORMAL LANDING DISTANCE WITHOUT PROPELLER REVERSING - FLAPS 100%

ASSOCIATED CONDITIONS:

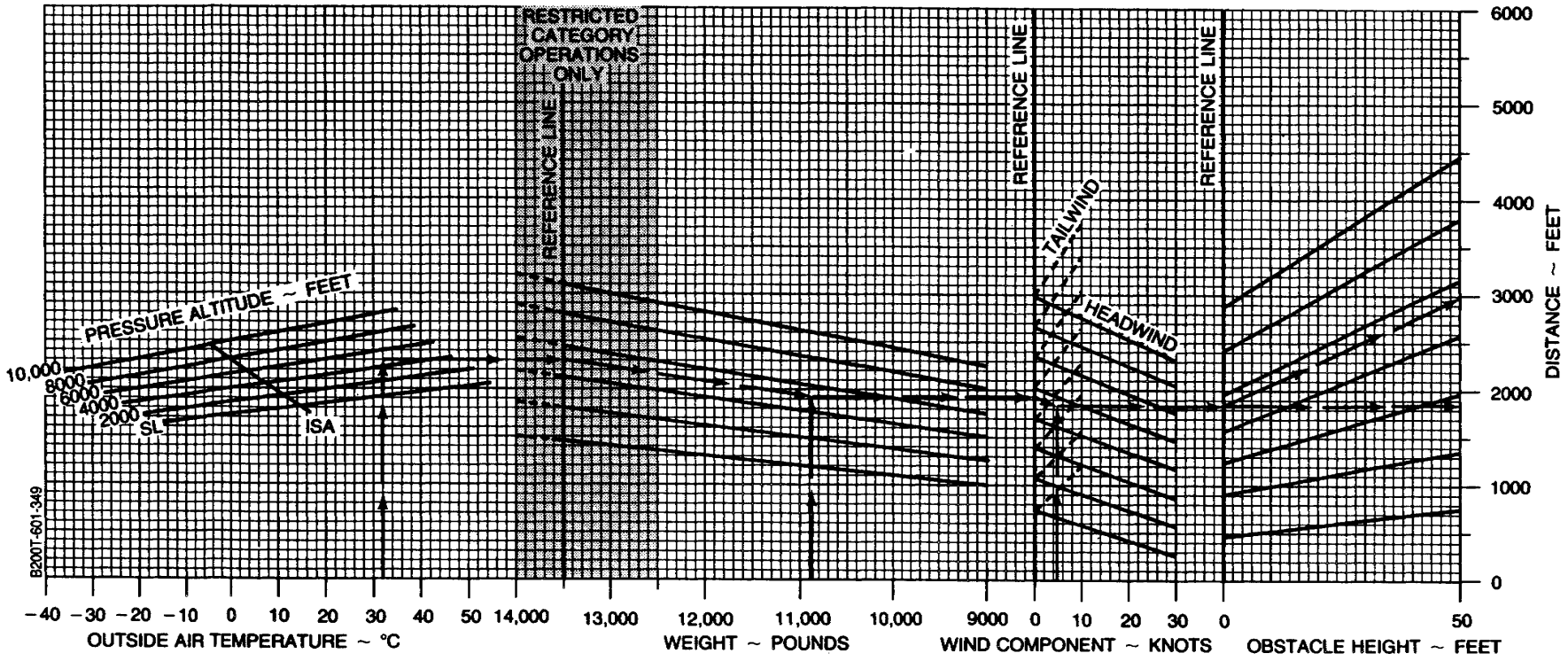
POWER.....RETARDED TO MAINTAIN 800 FT/MIN ON FINAL APPROACH
FLAPS.....100%
RUNWAY.....PAVED LEVEL DRY SURFACE
APPROACH SPEED.IAS AS TABULATED
BRAKING .....MAXIMUM

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	106
13,500	105
13,000	104
12,500	102
12,000	101
11,000	99
10,000	96
9000	94

EXAMPLE:

OAT..... 32°C
PRESSURE ALTITUDE..... 4732 FT
LANDING WEIGHT..... 10,854 LBS
HEADWIND COMPONENT..... 4.7 KTS

GROUND ROLL ..... 1850 FT
TOTAL OVER
50 FT OBSTACLE..... 3000 FT
APPROACH SPEED..... 99 KTS



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LANDING DISTANCE WITHOUT PROPELLER REVERSING-FLAPS 0%

ASSOCIATED CONDITIONS:

POWER..... RETARDED TO MAINTAIN 900 FT/MIN ON FINAL APPROACH
FLAPS..... UP
RUNWAY..... PAVED, LEVEL, DRY SURFACE
APPROACH SPEED . IAS AS TABULATED
BRAKING ..... MAXIMUM

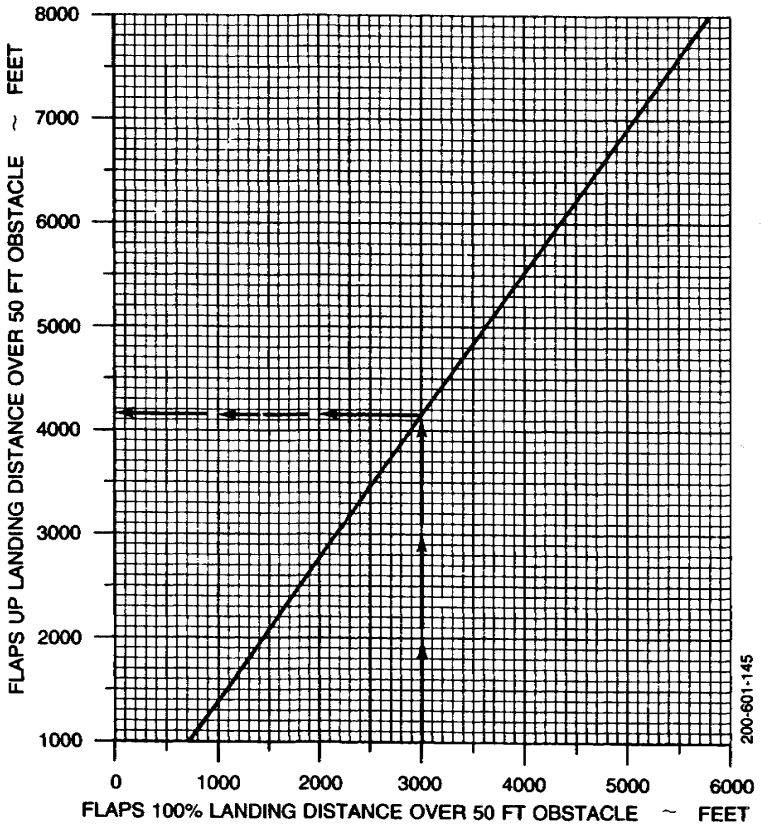
WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	137
13,500	135
13,000	133
12,500	131
12,000	129
11,000	125
10,000	121
9000	117

EXAMPLE:

FLAPS 100% LANDING DISTANCE
OVER 50 FT OBSTACLE..... 3000 FT
LANDING WEIGHT ..... 10,854 LBS

FLAPS UP LANDING DISTANCE
OVER 50 FT OBSTACLE..... 4150 FT
APPROACH SPEED ..... 124 KTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN (100%) IS NORMAL PROCEDURE. USE THIS GRAPH WHEN IT IS NECESSARY TO LAND WITH FLAPS UP (0%).
2. TO DETERMINE FLAPS-UP LANDING DISTANCE, READ FROM THE **NORMAL LANDING DISTANCE WITHOUT PROPELLER REVERSING-FLAPS 100%** GRAPH THE LANDING DISTANCE APPROPRIATE TO OAT, ALTITUDE, WEIGHT, WIND, AND 50-FT OBSTACLE. THEN ENTER THIS GRAPH WITH THE DERIVED VALUE AND READ THE FLAPS-UP LANDING DISTANCE.



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LANDING DISTANCE WITH PROPELLER REVERSING-FLAPS 100%

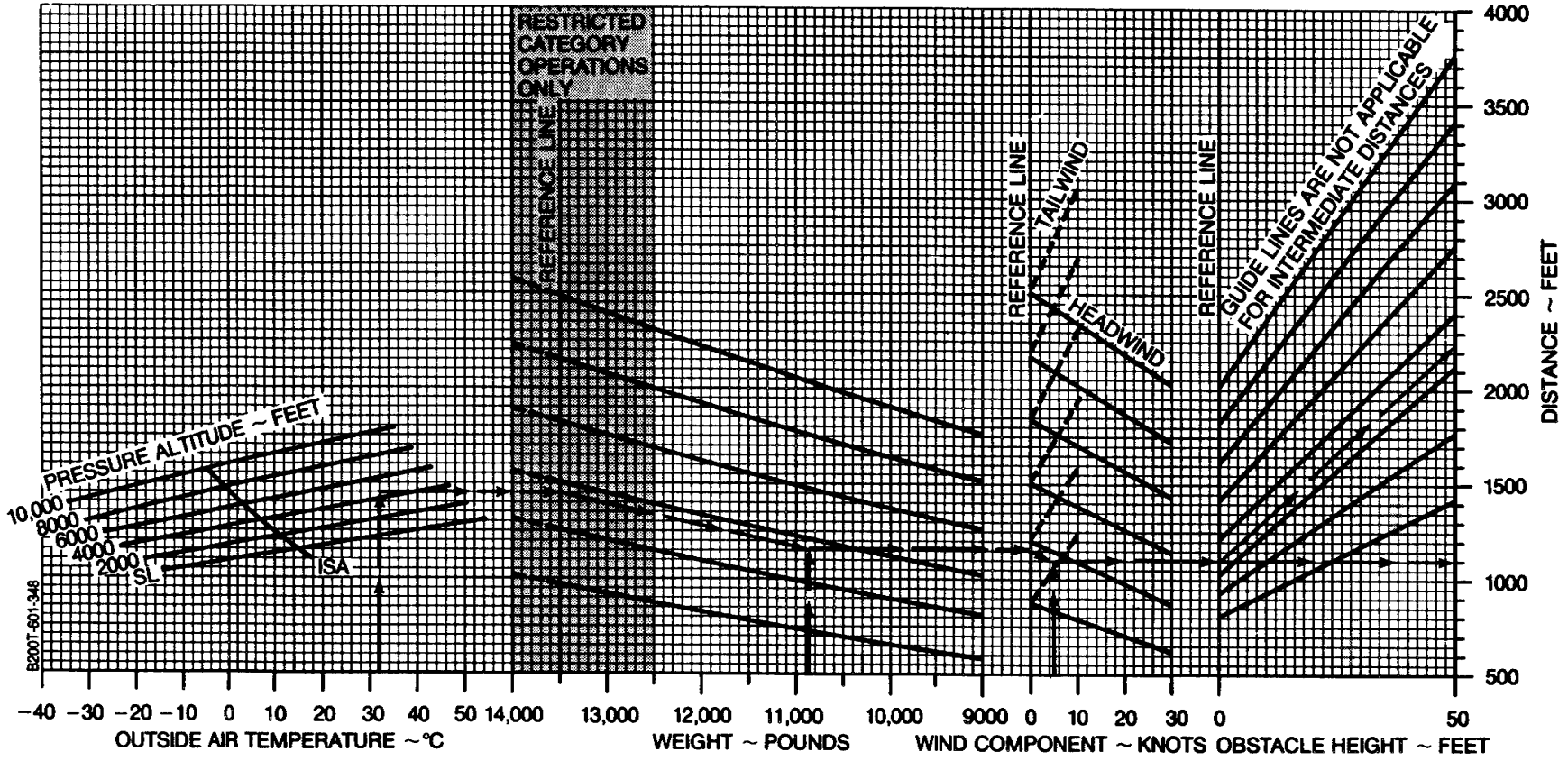
ASSOCIATED CONDITIONS:

POWER	RETARD TO MAINTAIN 1000 FT/MIN. ON FINAL APPROACH
FLAPS	100%
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS AS TABULATED
BRAKING	MAXIMUM
CONDITION LEVERS	HIGH IDLE
PROPELLER CONTROLS	FULL FORWARD
POWER LEVERS	MAXIMUM REVERSE AFTER TOUCHDOWN

WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	108
13,500	106
13,000	104
12,500	102
12,000	101
11,000	99
10,000	96
9000	94

EXAMPLE:

OAT	32°C
PRESSURE ALTITUDE	4732 FT
LANDING WEIGHT	10,854 LBS
HEADWIND COMPONENT	4.7 KTS
GROUND ROLL	1100 FT
TOTAL OVER 50 FT OBSTACLE	2225 FT
APPROACH SPEED	98 KTS



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LANDING DISTANCE WITH PROPELLER REVERSING-FLAPS 0%

ASSOCIATED CONDITIONS:

POWER	RETARD TO MAINTAIN 1000 FT/MIN ON FINAL APPROACH
FLAPS	UP
RUNWAY	PAVED, LEVEL, DRY SURFACE
APPROACH SPEED	IAS AS TABULATED
BRAKING	MAXIMUM
CONDITION LEVERS	HIGH IDLE
PROPELLER CONTROLS	FULL FORWARD
POWER LEVERS	MAXIMUM REVERSE AFTER TOUCHDOWN

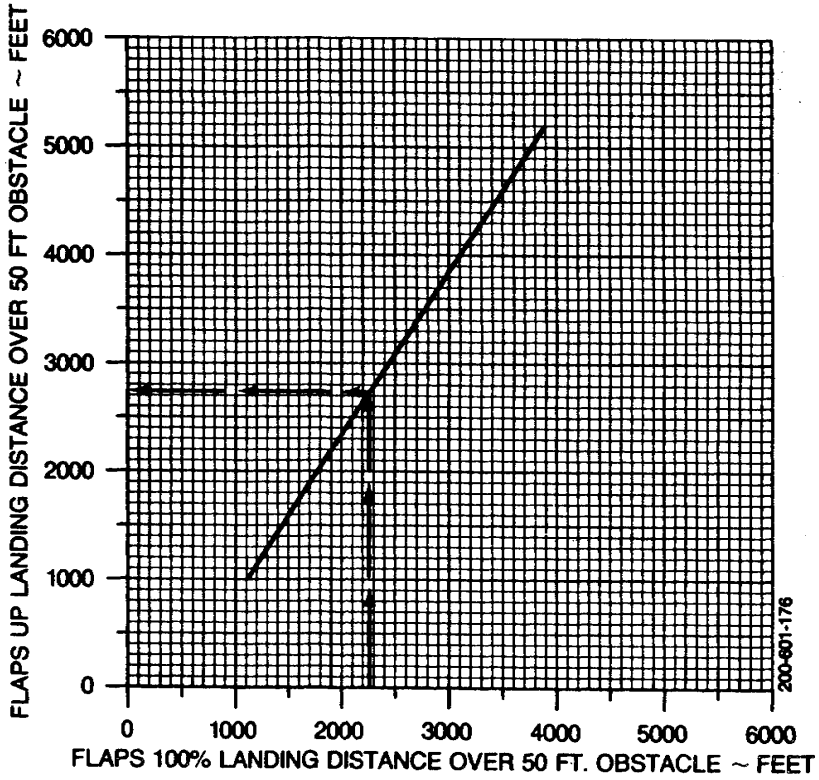
WEIGHT ~ POUNDS	APPROACH SPEED ~ KNOTS
14,000	137
13,500	135
13,000	133
12,500	131
12,000	129
11,000	125
10,000	121
9000	117

EXAMPLE:

FLAPS 100% LANDING DISTANCE OVER 50 FOOT OBSTACLE ..... 2260 FT
LANDING WEIGHT ..... 10,854 LBS

FLAPS UP LANDING DISTANCE OVER 50 FOOT OBSTACLE ..... 2720 FT
APPROACH SPEED ..... 124 KTS

NOTES:
1. LANDING WITH FLAPS FULL DOWN (100%) IS NORMAL PROCEDURE. USE THIS GRAPH WHEN IT IS NECESSARY TO LAND WITH FLAPS UP (0%).
2. TO DETERMINE FLAPS-UP LANDING DISTANCE, READ FROM THE **LANDING DISTANCE WITH PROPELLER REVERSING – FLAPS 100%** GRAPH THE LANDING DISTANCE APPROPRIATE TO OAT, ALTITUDE, WEIGHT, WIND, AND 50-FT OBSTACLE. THEN ENTER THIS GRAPH WITH THE DERIVED VALUE AND READ THE FLAPS-UP LANDING DISTANCE.



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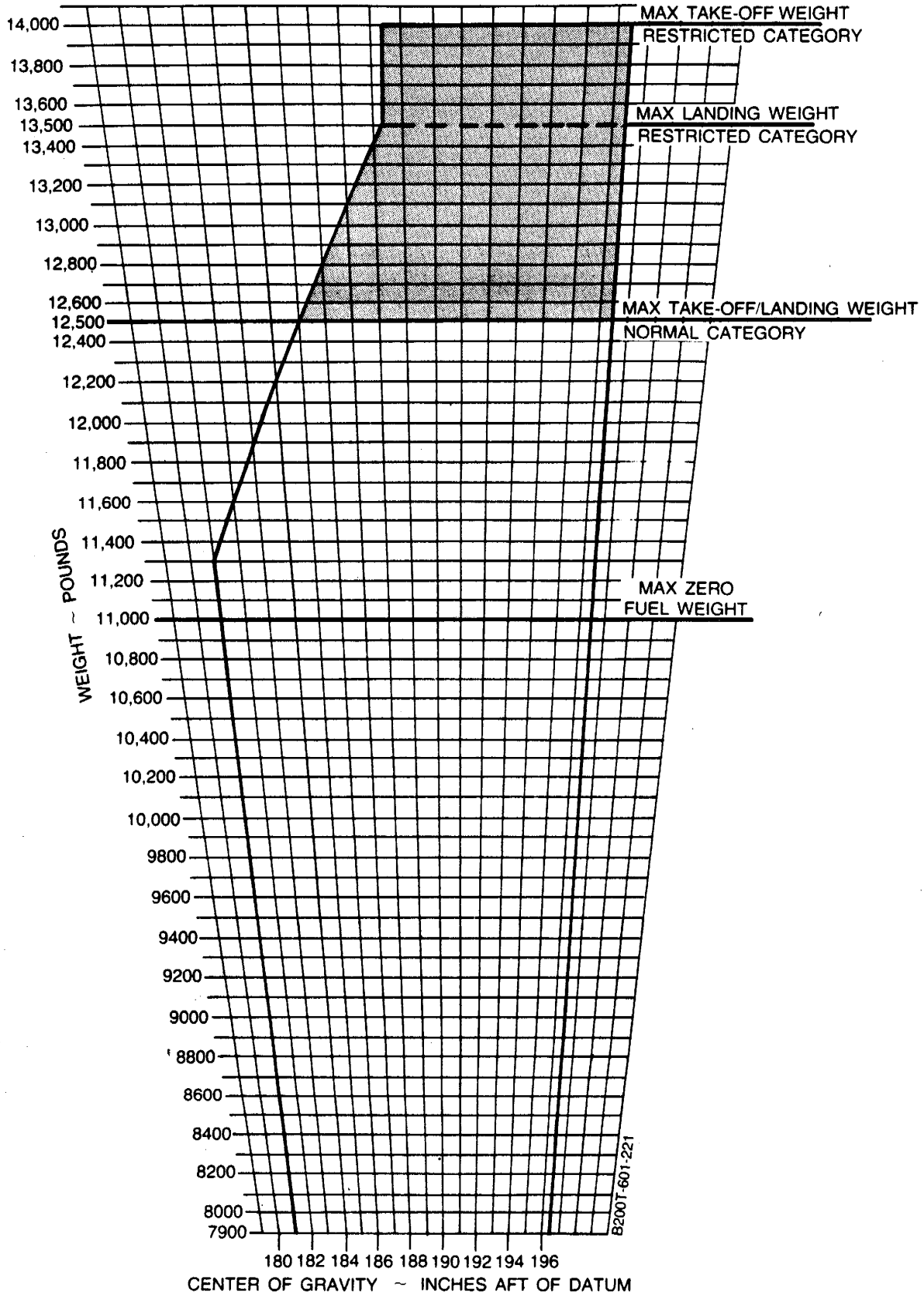
WEIGHT AND BALANCE DIAGRAM



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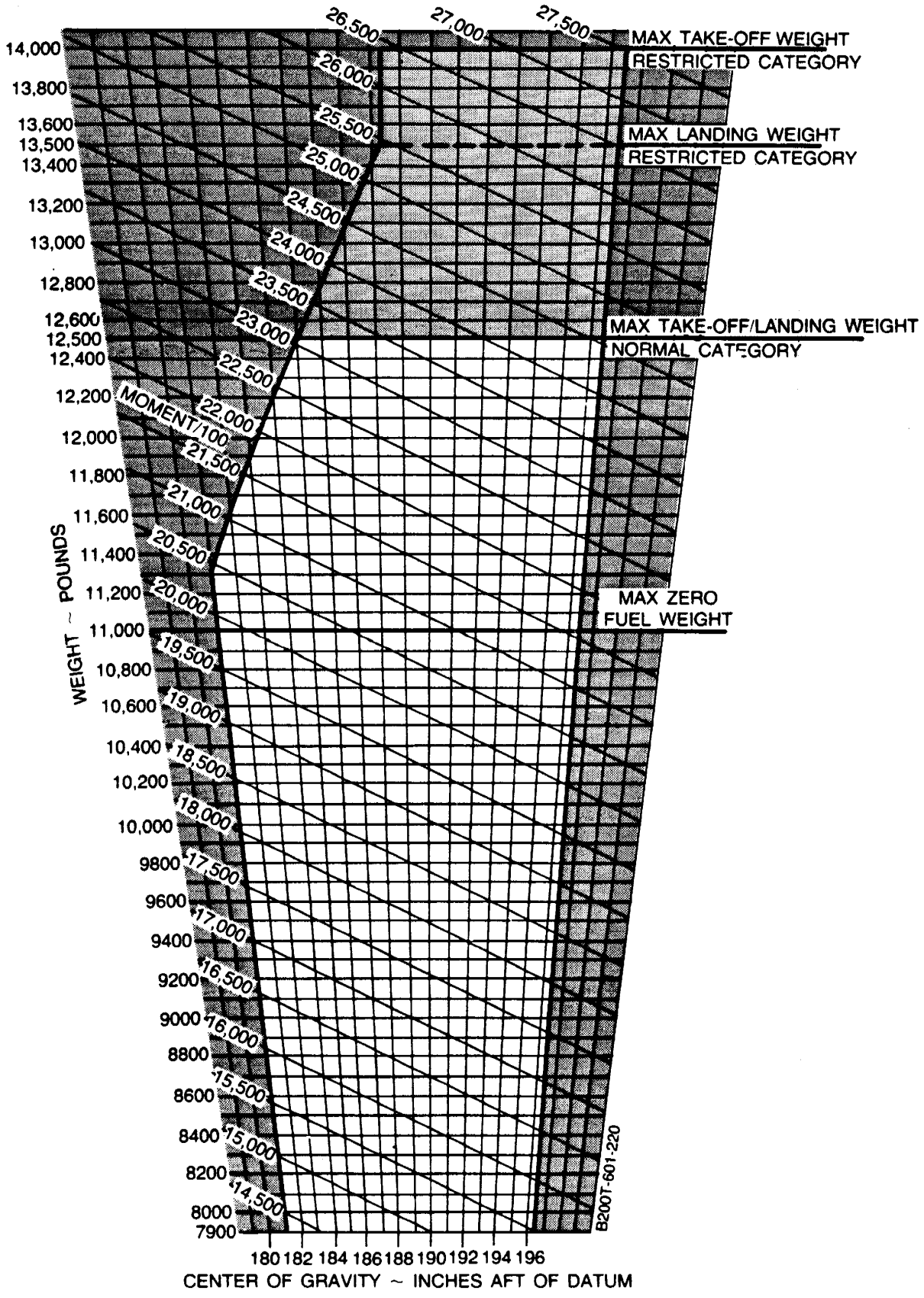
MOMENT LIMITS vs WEIGHT



MOMENT LIMITS VS. WEIGHT
NORMAL CATEGORY

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
7200	13032	14141
7250	13122	14239
7300	13213	14337
7350	13304	14435
7400	13394	14534
7450	13484	14632
7500	13575	14730
7550	13666	14828
7600	13756	14926
7650	13846	15025
7700	13937	15123
7750	14028	15221
7800	14118	15319
7850	14208	15417
7900	14299	15516
7950	14390	15614
8000	14480	15712
8050	14570	15810
8100	14661	15908
8150	14752	16007
8200	14842	16105
8250	14932	16203
8300	15023	16301
8350	15114	16399
8400	15204	16498
8450	15294	16596
8500	15385	16694
8550	15476	16792
8600	15566	16890
8650	15656	16989
8700	15747	17087
8750	15838	17185
8800	15928	17283
8850	16018	17381
8900	16109	17480
8950	16200	17578
9000	16290	17676
9050	16380	17774
9100	16471	17872
9150	16562	17971
9200	16652	18069
9250	16742	18167
9300	16833	18265
9350	16924	18363
9400	17014	18462
9450	17104	18560
9500	17195	18658
9550	17286	18756
9600	17376	18854
9650	17466	18953
9700	17557	19051
9750	17648	19149
9800	17738	19247
9850	17828	19345

MAX ZERO FUEL WEIGHT

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
9900	17919	19444
9950	18010	19542
10000	18100	19640
10050	18190	19738
10100	18281	19836
10150	18372	19935
10200	18462	20033
10250	18552	20131
10300	18643	20229
10350	18734	20327
10400	18824	20426
10450	18914	20524
10500	19005	20622
10550	19096	20720
10600	19186	20818
10650	19276	20917
10700	19367	21015
10750	19458	21113
10800	19548	21211
10850	19638	21309
10900	19729	21408
19050	19820	21506
11000	19910	21604
11050	20000	21702
11100	20091	21800
11150	20182	21899
11200	20272	21997
11250	20362	22095
11300	20461	22193
11350	20570	22291
11400	20679	22390
11450	20789	22488
11500	20898	22586
11550	21008	22684
11600	21118	22782
11650	21228	22881
11700	21338	22979
11750	21449	23077
11800	21559	23175
11850	21670	23273
11900	21781	23372
11950	21892	23470
12000	22003	23568
12050	22115	23666
12100	22226	23764
12150	22338	23863
12200	22450	23961
12250	22562	24059
12300	22674	24157
12350	22787	24255
12400	22899	24354
12450	23012	24452
12500	23125	24550

CENTER OF GRAVITY LIMITS (LANDING GEAR DOWN)

WEIGHT CONDITION	FORWARD CG LIMIT	AFT CG LIMIT
12,500 LBS (MAX. TAKE-OFF OR LANDING)	185.0	196.4
11,279 LBS OR LESS	181.0	196.4

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MOMENT LIMITS VS. WEIGHT

RESTRICTED CATEGORY

WEIGHT	MINIMUM MOMENT/100	MAXIMUM MOMENT/100
12500	23125	24550
12550	23240	24648
12600	23355	24746
12650	23469	24845
12700	23584	24943
12750	23699	25041
12800	23814	25139
12850	23928	25237
12900	24043	25336
12950	24158	25434
13000	24273	25532
13050	24388	25630
13100	24502	25728
13150	24617	25827
13200	24732	25925
13250	24847	26023
13300	24961	26121
13350	25076	26219
13400	25191	26318
13450	25306	26416
13500	25421	26514
13550	25515	26612
13600	25609	26710
13650	25703	26809
13700	25797	26907
13750	25891	27005
13800	25985	27103
13850	26080	27201
13900	26174	27300
13950	26268	27398
14000	26362	27496

CENTER OF GRAVITY LIMITS (LANDING GEAR DOWN)

WEIGHT CONDITION	FORWARD C.G. LIMIT	AFT C.G. LIMIT
14,000 Lbs (Max. Takeoff - Restricted Category)	188.3	196.4
13,500 Lbs or Less (Max. Landing - Restricted Category)	188.3	196.4
12,500 Lbs (Max. Takeoff or Landing - Normal Category)	185.0	196.4
11,279 Lbs or Less	181.0	196.4

BASIC EMPTY WEIGHT

Basic empty weight does not include customer furnished equipment, such as cameras and navigation sights. If such equipment is installed, a log book entry should be made reflecting the adjusted empty weight.

SYSTEMS DESCRIPTION

No change.

HANDLING, SERVICING, & MAINTENANCE

Each time the airplane is converted from one category to the other, an inspection will be entered in the airplane logbook along with the signature of the person making the inspection.

BEECHCRAFT SUPER KING AIR

B200, S/N BB-1322 LANDPLANE

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

SPERRY SPZ-4000 AUTOMATIC FLIGHT CONTROL SYSTEM CATEGORY I

and

SPERRY EDZ-605 ELECTRONIC FLIGHT INSTRUMENT SYSTEM (EFIS)

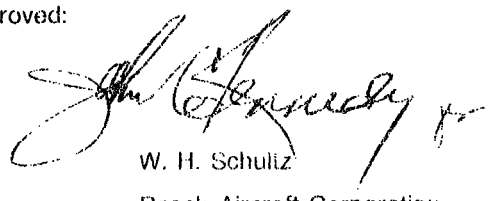
THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL

P/N 101-590010-147

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:



W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Sperry SPZ-4000 Automatic Flight Control System, alone or in conjunction with the Sperry Electronic Flight Instrument System, in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

Sperry Pilot's Operating Handbooks for the Electronic Flight Instrument System are available from Sperry Flight Systems, P.O. Box 29000, Phoenix, Arizona 85038. It is the responsibility of the owner/operator to maintain the appropriate Pilot's Operating Handbook in current status.

LIMITATIONS

1. During autopilot operations, pilot must be seated at the controls with seat belt fastened.
2. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
3. Do not use autopilot under 200 feet above the terrain. This includes takeoffs and landings.
4. Do not use autopilot or yaw damper during takeoff or landing.
5. Do not use propeller in the range of 1750-1850 rpm during coupled ILS approach.
6. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
7. Set desired heading and select HDG on the Flight Director Mode Selector prior to changing the VOR/ILS receiver frequency if that receiver is the navigation source and the NAV, APR, or BC mode is selected.
8. Prior to dispatch, the EDZ-605 system must be verified to be operational by a satisfactory preflight test as contained in the NORMAL PROCEDURES.
9. A copy of the Honeywell Pilot's Operating Handbook for the EDZ-605 Electronic Flight Instrument System (EFIS) with Multifunction Display (MFD), 3-Tube System Pub. No. A62-0088-00-00, Sept, 1987 or later revision, must be available to the pilot while operating this system.
10. The EADI and EHSI must be operational in the normal mode for takeoff.
11. Dispatch is prohibited with the EFIS FAN FAIL or MFD FAN FAIL annunciator lights illuminated.
12. The composite mode is to be used in IFR operations only after a failure of an EADI or EHSI, the Flight Director is to be used for instrument approaches.
13. At least one symbol generator must be installed and operational for dispatch.

PLACARDS

Above VN 800 VNAV Programmer:

VNAV DISABLED

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EMERGENCY PROCEDURES

AIR START

STARTER ASSIST

CAUTION

The pilot should determine the reason for engine failure before attempting an air start.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an over-temp condition.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. EFIS ON/OFF Switches - OFF (if conditions permit)
5. Windshield Heat - OFF
6. Power Lever - IDLE
7. Condition Lever - CUT-OFF
8. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

9. Ignition and Start Switch - ON (up), Check IGNITION Annunciator - ON
10. Condition Lever - LOW IDLE
11. Ignition and Start Switch - OFF (N_1 above 50%)
12. Propeller Lever - AS REQUIRED
13. Power Lever - AS REQUIRED
14. Generator - RESET, then ON
15. Engine Auto Ignition - ARM
16. EFIS ON/OFF Switches - ON
17. Electrical Equipment - AS REQUIRED

AUTOPILOT MALFUNCTION

If the AP TRIM FAIL and MASTER WARNING annunciators illuminate while the autopilot is engaged, immediately disconnect while restraining the control wheel for a possible out-of-trim force. Disengagement can be accomplished by any one of the following:

1. Press (momentarily) to the first level the pilot's or copilot's AP/YD & TRIM DISC button on the control wheel.
2. Open the autopilot (AP CONTROL) circuit breaker.
3. If necessary, the autopilot may be overpowered by either pilot.

CAUTION

Trim will run to oppose pilot effort.

MAXIMUM ALTITUDE LOSSES DURING MALFUNCTION TESTS WERE:

CONFIGURATION	ALTITUDE LOSS
Climb	130 ft
Cruise	310 ft
Maneuvering	140 ft
Descent	510 ft
Approach/ILS Coupled	90 ft
One-engine-inoperative Approach/ILS Coupled	80 ft

ENGINE FAILURE

— Disengage the autopilot, retrim the airplane and reengage the autopilot. Maintain at least 120 knots for one-engine-inoperative approach.

EFIS FAILURE

If the EFIS displays are missing or abnormal, refer to alternate instruments for usable data. The copilot instruments are approved as backup to the pilot instruments. EFIS reversionary modes may be selected as follows:

1. In the event of an Electronic Horizontal Situation Indicator (EHSI) failure:

   — Turn off the HSI/DIM control on the Display Controller. The EADI will provide a composite attitude/navigation display.

   or;

   — Select HSI on the Mode Controller. The MFD will display HSI information.

2. In the event of an Electronic Attitude-Director Indicator (EADI) failure, turn off the ADI/DIM control and adjust the WX/DIM control on the Display Controller. The EHSI will provide a composite attitude/navigation display.

CAUTION

A suitable composite attitude/navigation display on the EHSI requires that the WX/DIM control on the EFIS Display Controller be adjusted to provide an appropriate earth/sky background color for the attitude indicator.

3. In the event of primary symbol generator failure (both EADI and EHSI go blank or annunciate SG FAIL):

   — Select the SG position on the MFD controller. The Multifunction generator will drive the EFIS and the MFD will blank.

4. Use flight director for instrument approaches in the composite mode.

NOTE

The composite format utilizes a course deviation indicator with automatic reverse course steering. Always fly toward the needle to intercept course centerline.

CAUTION

Flight Director mode annunciation is not present in composite format. Refer to the Flight Director Mode Selector for mode engagement annunciation.

5. EFIS FAN FAIL or MFD FAN FAIL Annunciator Illuminated.

Illumination of the EFIS FAN FAIL or MFD FAN FAIL annunciator (located above the EADI) indicates that the respective display indicator cooling system has failed. If this occurs on the ground, do not take off. If this occurs while in flight, the following actions are recommended:

a. Monitor the Display Indicator for possible failures.
b. Consider using one of the Composite or Reversionary modes, or turn the indicator off.
c. Lower the lighting intensity on the display.
d. Repair on the ground prior to the next flight.

NORMAL PROCEDURES

The Automatic Flight Control System (AFCS) is an integrated Flight Director, Autopilot, and Yaw Damper System. Flight Director modes are selected using push-button switches on the Flight Director Mode Selector. The Autopilot and Yaw Damper are engaged using push-button switches on the Autopilot Controller. Whenever the autopilot is engaged, it will automatically couple to the selected flight director mode. The autopilot may also be controlled manually using the Pitch Wheel and Turn Knob on the controller.

BEFORE ENGINE STARTING

EFIS

— EFIS ON/OFF Switches - OFF

AFTER STARTING, AND TAXIING

EFIS

— EFIS ON/OFF Switches - ON

Altitude Alert

1. Altitude Alert Controller (or VNAV if installed) — Set to 1500 feet above altitude indicated on pilot's altimeter.
2. Pilot's baro set — ADJUST to increase altitude toward alerter setting.
3. VERIFY — alert light on pilot's altimeter illuminates and Aural Alert sounds at 1000 feet from alerter setting.
4. VERIFY — alert light is extinguished at 250 feet from alerter setting.
5. Pilot's baro set — ADJUST toward original setting.
6. VERIFY — alert light illuminates and horn sounds at 250 feet from alerter setting.
7. Altitude Alert Controller (or VNAV if installed) — SET to initial clearance altitude.
8. VERIFY — alert light extinguished.

Autopilot

Perform prior to each flight to verify proper operation of torque limiters, torque monitors, autopilot electric trim system, and failure annunciators.

1. Autopilot Controller TRIM UP, TRIM DN Annunciators — CHECK not illuminated. A steady illumination of TRIM UP or TRIM DN annunciator indicates that automatic synchronization is not functioning and autopilot should not be engaged.
2. Turn Knob — IN CENTER DETENT POSITION
3. Elevator Trim Control Switch — ON
4. Control Wheel — TO MID TRAVEL
5. Autopilot Controller AP Button — PRESS. AP ENGAGE and YD ENGAGE annunciators on autopilot controller will flash. Servo clutches will engage. FD flag on ADI/EADI in view
6. Control Movement — CHECK when pushed ON.
    a. Rudder Pedals — OVERPOWER SLOWLY. YD ENGAGE annunciator stops flashing.
    b. Control Wheel — OVERPOWER SLOWLY in both pitch and roll axes. AP ENGAGE annunciator stops flashing. FD flag retracts.

WARNING

If autopilot or yaw damper disengages during overpower test, DO NOT USE.
If AP ENGAGE or YD ENGAGE annunciator continues to flash, DO NOT USE.

7. Elevator Trim Follow-up — CHECK
   a. Control Wheel — HOLD AFT OF MID TRAVEL. Trim wheel should run nose down after approximately 3 seconds. Trim down annunciator should illuminate after approximately 8 seconds.
   b. Control Wheel — HOLD FORWARD OF MID TRAVEL. Trim wheel should run nose up after approximately 3 seconds, TRIM UP annunciator should illuminate after approximatley 8 seconds, and AP TRIM FAIL annunciator and MASTER WARNING flasher should illuminate after approximately 15 seconds.
8. AP/YD & TRIM DISC Button — DEPRESS THROUGH SECOND LEVEL. Autopilot and yaw damper should disengage and ELECT TRIM OFF annunciator should illuminate. AP ENGAGE and YD ENGAGE annunciators on instrument panel should flash 5 times and autopilot off aural alert should sound for one second (or Voice Advisory System, if installed, should announce "AUTOPILOT").
9. MASTER WARNING Flasher — PRESS TO RESET.
10. Elevator Trim Control Switch — OFF, then ON (resets electric trim; ELECT TRIM OFF annunciator should extinguish)
11. Elevator Trim — SET AS REQUIRED.

NOTE

Autopilot Preflight Check is disabled during inflight operations.

EFIS

1. SELF TEST

Press the "TEST" button on the Display Controller., Verify the following:

- Radar altimeter reads test value.
- Course Select, Heading Select, DH Set and ET are replaced with dashes.
- All flags in view.
- Command cue out of view.
- If Flight Director Lamp test is good, "TEST" will appear in upper right corner of EADI. "FD FAIL" indicates a problem in the EFIS components or interconnection of the flight director computer.

NOTE

Self-test is inhibited during glideslope capture.

2. SWITCHING TEST
   — Check each switch function on the EFIS Display Controller. Switching should operate as described under Item 18 of GENERAL OPERATION.

GENERAL OPERATION

All flight director modes (except Go-Around) are selected by push-button switches on the Flight Director Mode Selector. Go-Around is selected by pressing the Go-Around button on the left engine power lever. Annunciation for the mode selected is provided on the mode selector and pilot's instrument panel.

The Electronic Flight Instrument System replaces the conventional electro-mechanical Attitude-Director (ADI) and Horizontal Situation (HSI) indicators. The CRT displays (EADI & EHSI) provide the necessary information for the pilot to respond to flight director and navigation commands in the same manner as conventional instruments. The EDZ-605 utilizes standard (4.6" x 5") displays. The system includes additional features which allow the pilot to select alternate formats as navigation aids or in the event of specific failures. The operator is cautioned that the primary displays (attitude and directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate formats is described in the appropriate Sperry Pilot's Operating Handbooks. The owner/operator of the airplane is responsible for obtaining and maintaining the appropriate handbook.

1. Standby Lamp Test (STBY) Mode selector button — PRESS. All flight director modes are reset and all flight control system annunciators are illuminated. Flight director command cue is biased out of view and FD FAIL on the EADI is in view as long as STBY is pressed. If STBY is pressed for more than 5 seconds:

   a. AP TRIM FAIL and MASTER WARNING will illuminate.
   b. Autopilot off aural alert will sound (or Voice Advisory System, if installed, will announce "AUTOPILOT").

2. Heading Select Mode (HDG)

   a. Heading Cursor — SET desired heading on EHSI.
   b. Mode Selector HDG button — PRESS. Flight director commands a turn to capture and hold selected heading.

3. VOR Enroute Mode (NAV), and VOR approaches when more than 20 DME from the station:

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired VOR frequency
   d. Course Selector — SET desired course on EHSI.
   e. Mode Selector NAV button — PRESS. Mode selector annunciates (NAV) ARM (white NAV on EADI)
   f. At the computed capture point — HDG mode is cancelled, mode selector annunciates (NAV) CAP (green NAV on EADI). Flight director commands a turn to capture and track selected course.

NOTE

At transition from armed to captured flight director modes, a white box appears for five seconds around the green capture mode annunciators on the EADI.

CAUTION

Do not select a VOR frequency when F/D is in NAV CAPTURE or ARM. NAV CAPTURE without mode annunciation may result.

4. VOR Approach Mode (NAV/APR) for optimized NAV mode tracking, for VOR approaches:

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired VOR frequency.
   d. Course Selector — SET desired course on EHSI.
   e. Mode Selector APR button — PRESS. Mode selector annunciates (NAV) ARM and (APR) ARM. VOR APR is annunciated on the instrument panel (white VAPP on the EADI).
   f. At computed capture point — HDG mode is cancelled and mode selector annunciates (NAV) CAP and (APR) CAP (green VAPP on EADI). Flight director commands a turn to capture and track selected course.

5. Localizer Approach (NAV)

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired localizer frequency.
   d. Course Selector — SET inbound localizer course on EHSI.
   e. Mode Selector NAV Button — PRESS. Mode selector annunciates (NAV) ARM (white LOC on EADI).
   f. At computed capture point — HDG mode is cancelled. Mode selector annunciates (NAV) CAP (green LOC on EADI). Flight director commands a turn to capture and track localizer beam.
   g. Mode Selector — SELECT vertical flight director mode, as required.

6. Back Course Approach (BC)

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired localizer frequency.
   d. Course Selector — SET inbound front course on EHSI.
   e. Mode Selector BC Button — PRESS. Mode selector annunciates (BC) ARM (white BC on EADI).

f. At computed capture point HDG mode is cancelled. Mode Selector annunciates (BC) CAP (green BC on EADI). Flight director commands a turn to capture and track localizer back course.
g. Mode Selector -- SELECT vertical flight director mode, as required.

7. ILS Approach (APR)

a. Heading Cursor — SET desired intercept heading on EHSI.
b. Mode Selector — Select HDG
c. Navigation Receiver — TUNE to desired ILS frequency.
d. Course Selector — SET inbound localizer course on EHSI.
e. Mode Selector APR Button — PRESS. Mode selector annunciates (NAV) ARM and (APR) ARM (white LOC and GS on EADI).
f. Mode Selector — SELECT vertical flight director mode, as required, to intercept glideslope.
g. At computed localizer capture point — HDG mode is cancelled. Mode selector annunciates (NAV) CAP and (APR) ARM (green LOC on EADI). Flight director commands a turn to capture and track localizer beam.
h. At computed glideslope capture point — Flight Director Vertical modes are cancelled (except ALT SEL ARM). Mode selector annunciates (NAV) CAP and (APR) CAP (green LOC and GS on EADI). Flight director commands a pitch change to capture and track glideslope.

NOTE

Glideslope capture will not occur until after localizer has been captured. Pitch wheel is disabled while in glideslope (APR CAP) mode.

8. KNS-660 RNAV Enroute or Approach

a. KNS-660 - Program for desired operation per Pilot's Guide
b. Mode Selector NAV Button - PRESS. Mode Selector annunciates NAV, EADI annunciates LNAV and EHSI annunciates selected source as FMS 1 or 2. The KNS-660 provides steering commands for course intercept and tracking.

CAUTION

Do not use the map format on the EHSI for KNS-660 RNAV approaches.

9. Pitch Hold/Sync Mode
With no vertical flight director modes selected:

a. Heading Cursor — SET to desired heading.
b. Mode Selector — SELECT lateral flight director mode as required. Flight director pitch command is to hold attitude existing when lateral mode was selected.

To set/change command pitch attitude:

a. TCS Switch — PRESS and HOLD
b. Maneuver airplane to desired pitch attitude. Trim elevator as required.
c. TCS Switch — RELEASE. Flight director command is to hold new pitch attitude.

10. Altitude Hold Mode (ALT)
To hold altitude:

a. Maneuver airplane to desired altitude.
b. Mode Selector ALT Button — PRESS.

To change altitude hold reference:

a. TCS Switch — PRESS and HOLD
b. Maneuver airplane to new altitude.
c. TCS Switch — RELEASE. Flight director command is to hold new altitude.

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NOTE

Use ALT SEL mode for large altitude changes.

11. Altitude Alerting
The desired altitude is selected on the Altitude Alert controller (or the ALT PRE position on the VNAV controller, if installed) using the SET knob. Then altitude can be attained by using the altitude preselect mode of the flight director.

12. Altitude Preselect Mode (ALT SEL)

a. Altitude Preselect (or VNAV) Controller — SET desired altitude (ALT PRE position on VNAV).
b. Mode Selector ALT SEL Button — PRESS. Mode Selector annunciates (ALT SEL) ARM (white ASEL on EADI).
c. Maneuver airplane to intercept selected altitude. (Use Pitch Hold, IAS, VS or GA mode, as required.)
d. Altitude Capture Sequence — VERIFY:

1) 1000 feet prior to selected altitude - alert light on pilot's altimeter illuminates and aural alert sounds for one second.
2) 250 feet prior to selected altitude - alert light extinguishes.
3) At computed altitude capture point — mode selector annunciates (ALT SEL) CAP (green ASEL on EADI). Other vertical flight director modes are cancelled. Flight director commands a flare to selected altitude.
4) At selected altitude — Altitude Select Mode is cancelled, and Altitude Hold Mode is automatically selected (green ALT is annunciated on EADI).

NOTE

Changing selected altitude on Altitude Preselect (or VNAV) Controller during altitude capture (ALT SEL CAP) will automatically cancel ALT SEL CAP, and ALT SEL ARM will be selected.

ALT SEL mode is referenced to baro corrected altitude displayed on pilot's altimeter.

13. Airspeed Hold Mode (IAS)
To hold airspeed:

a. Maneuver airplane to desired indicated airspeed
b. Mode Selector IAS Button — PRESS.

To change airspeed hold reference:

a. TCS Switch — PRESS and HOLD
b. Maneuver airplane to new airspeed.
c. TCS Switch — RELEASE. Flight director command is to hold new indicated airspeed.

14. Vertical Speed Hold Mode (VS)
Operation of VS mode is identical to IAS mode except that flight director command is to maintain vertical speed.

15. Go-Around Mode (GA)
Go-Around Button (on left power lever) — PRESS. All flight director modes are cancelled. Flight director commands a 7-degree-pitch-up, wings-level attitude, then Heading Hold is commanded when roll attitude becomes less than 6 degrees (green GA is annunciated on EADI).
To cancel heading hold:

a. Heading Cursor — SET desired heading on HSI.
b. Mode Selector — SELECT desired lateral mode.

To cancel go-around mode:

a. Mode Selector — SELECT vertical flight director mode (ALT, IAS, VS)
   or;
b. TCS Switch — PRESS
   or;
c. Autopilot — ENGAGE

16. Vertical Navigation Mode (VNAV) (Disabled)

17. Air Data Command Display (Optional)
To Command Airspeed Hold:

a. Mode Selector IAS Button — PRESS. Command display will indicate existing IAS in knots.
b. Autopilot Controller Pitch Wheel — ROTATE to increase or decrease airspeed shown on Command Display. Flight director will command pitch up, or down, to capture and track displayed airspeed.

To Command Vertical Speed Hold:

a. Mode Selector VS Button — PRESS. Command Display will indicate existing vertical speed in feet per minute.
b. Autopilot Controller Pitch Wheel — ROTATE to increase or decrease vertical speed shown on Command Display. Flight director will command pitch up, or down, to capture and track displayed vertical speed.

18. DC-811 EFIS Display Controller Button Selections:

a. FULL/MAP — Toggles the EHSI display from Full Compass to Partial Compass format to MAP format.
b. SC/CP — Toggles the Flight Director command from Single Cue to Cross-Pointer configuration.
c. GSPD/TTG — Toggles display on EHSI from Ground Speed to Time-To-Go.
d. ET — Toggles Elapsed Time, when displayed, between Initialize, Start, and Stop.
e. HDG REV — Toggles display from HDG-1 to HDG-2.
f. V/L — Toggles SRN Nav source between NAV-1 and NAV-2, when selected.
g. FMS — Toggles LRN Nav source between FMS-1 and FMS-2, when selected.

19. MC-800 MFD Controller Button Selections:

a. MAP/PLAN — Alternately selects the Heading-Up MAP display or the North-Up PLAN mode for display.
b. SRC — Alternately selects the FMS-1 and FMS-2 source of LRN data for mapping.
c. WX — Used to call up weather radar returns on the MFD Map display or Weather Radar Only display.
d. VOR — Used to add VOR/DME symbols to the Map and Plan displays.
e. DAT — Used to add LRN information to the Map and Plan displays.
f. APT — Used to add airport designators to the MFD Map and Plan displays.
g. SKP, RCL, PAG, ENT, and Joystick — These buttons and Joystick are used to control the checklist and operation of the designator on the map.

20. EFIS Source Controller Initiated Flight Director Mode resets:
Pilot input to the EFIS through the DC-811 Source Controller initiates Flight Director and Autopilot mode changes as follows:

PILOT ACTION	FD AP MODE CHANGE
Select a NAV sourse	All lateral modes reset except HDG
Select a HDG source	All lateral modes reset

AUTOPILOT OPERATION

1. In-Flight Engagement of Autopilot

a. All AFCS and Autopilot Circuit Breakers — IN
b. Autopilot Controller TRIM UP and TRIM DN annunciators — CHECK. If either annunciator is illuminated, do not engage autopilot.
c. Turn Knob — IN CENTER DETENT POSITION.
d. Autopilot Controller AP Button — PRESS. Controller will annunciate AP Engage and YD ENGAGE.

2. Disengaging the Autopilot
The autopilot may be disengaged by:

a. Pressing the AP/YD & TRIM DISC button on either control wheel to the first level.

NOTE

The AP/YD & TRIM DISC button is a two-level switch. The first level disengages the autopilot and yaw damper, the second level disengages the electric trim.

b. Pressing the Go-Around button on the left power lever (Yaw Damper remains engaged).
c. Actuating the Trim Switches on either control wheel (YAW Damper remains engaged).
d. Pressing the AP button on the Autopilot Controller (Yaw Damper remains engaged).

NOTE

Upon autopilot disengagement, AP ENGAGE annunciator on the instrument panel will flash 5 times, then extinguish, and the autopilot-off aural alert will sound for 1 second (or the Voice Advisory System, if installed, will announce "AUTOPILOT").

In addition, if the autopilot disenages automatically due to a detected failure, the AP DISC annunciator and the MASTER WARNING flasher will illuminate. The AP DISC annunciator is extinguished by pressing the AP/YD & TRIM DISC button on either control wheel. The MASTER WARNING flasher is reset by pressing its face.

3. Engaging the Yaw Damper
The Yaw Damper is automatically engaged whenever the Autopilot is engaged. The Yaw Damper may be engaged without engaging the Autopilot by pressing the YD button on the Autopilot Controller.

4. Disengaging the Yaw Damper
The Yaw Damper may be disengaged by:

a. Pressing the AP/YD & TRIM DISC button on either control wheel to the first level.
b. Pressing the YD button on the Autopilot Controller.

NOTE

Upon Yaw Damper disengagement, the YD ENGAGE annunciator on the instrument panel will flash 5 times, then extinguish.

5. Flight Director Coupling
When engaged, the Autopilot automatically couples to any flight director mode selected (except Go-Around).

6. Attitude Hold
When the Autopilot is engaged with no flight director modes selected, pitch attitude and heading are held. (If roll attitude is greater than 6 degrees, the Autopilot will command wings level, then heading hold).
To change Attitude

a. TCS Switch — PRESS and HOLD
b. Manually maneuver airplane to desired attitude
c. TCS Switch — RELEASE. New pitch attitude and heading will be held. (If roll attitude is greater than 6 degrees, new roll attitude will be held.)

7. Turn Knob
To command a turn — ROTATE Turn Knob on Autopilot Controller out of detent in desired direction. Autopilot will command roll attitude proportional to amount of Turn Knob rotation.

NOTE

Rotating Turn Knob out of detent position with Autopilot engaged will cancel all lateral flight director modes.

8. Pitch Wheel
To change pitch attitude — ROTATE Pitch Wheel on Autopilot Controller in desired direction. Autopilot will command pitch attitude change proportional to amount of Pitch Wheel movement.

NOTE

Rotating the Pitch Wheel with the Autopilot engaged will cancel ALT, ALT SEL/CAP, IAS, and VS. (If optional Air Data Command Display is installed, IAS or VS will not be cancelled, but will be changed proportional to amount of wheel movement.)

9. Touch Control Steering (TCS)
TCS allows the pilot to override autopilot modes (by disengaging pitch and roll servos) in two ways:

   a. In Attitude Hold, ALT, IAS, or VS mode, TCS synchronizes the hold references. When TCS is released, the new reference is held.
   b. In HDG, VOR, VOR APR, LOC, BC, or ILS mode, TCS allows corrective or supervisory override without mode disengagement. When TCS is released, the autopilot will recouple to the selected mode.

10. Soft Ride
To soften autopilot response for turbulence penetration — PRESS Soft Ride Button on Autopilot Controller.

NOTE

Soft Ride is automatically cancelled if the autopilot is disengaged.

11. Bank Limit
To limit bank angle in HDG mode — PRESS Bank Limit Button on Autopilot Controller. Autopilot and Flight Director commands are limited to 17 degrees.

NOTE

Low Bank Limit will be cancelled if HDG mode is cancelled, but will be reselected automatically if HDG is reselected. To cancel Low Bank Limit PRESS Bank Limit Button when LOW is annunciated

SHUTDOWN AND SECURING

— EFIS Switches - OFF before FUEL CUTOFF

PERFORMANCE

No change.

WEIGHT & BALANCE

No change.

SYSTEMS DESCRIPTION

No change.

HANDLING, SERVICING AND MAINTENANCE

No change.

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For Training Purposes Only

Beechcraft®

Super King Air®
B200, B200C, B200T & B200CT Landplanes

Pilot's Operating Handbook and
FAA Approved Airplane Flight Manual Supplement
for the
Collins FCS-65 Automatic Flight Control System
Category I
and
Collins EFIS-85B(14)
Electronic Flight Instrument System

This Supplement is applicable to the following manual(s):

101-590010-147,
101-590037-29 and 101-590037-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane has been modified by installation of the Collins FCS-65 Automatic Flight Control System with Collins EFIS-85B(14) Electronic Flight Instrument System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Collins Pilot's Guide for the Electronic Flight Instrument System (EFIS-85B(14)) No. 523-0775919 dated August 15, 1989 or later issue, is available from Collins General Aviation Division, 400 Collins Road NE, Cedar Rapids, Iowa 52498. It is the responsibility of the owner/operator to maintain the appropriate pilot's guides in current status.

LIMITATIONS

1. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
2. Do not use autopilot below the following altitudes:
   a. Approach - 200 feet above ground level.
   b. Climb - 500 feet above ground level.
   c. All other operations - 1000 feet above ground level.
3. Do not use autopilot or yaw damper during takeoff or landing.
4. Pilot must be seated at the controls with the seat belt fastened during autopilot operations.
5. Do not use propeller speeds in the range of 1750-1850 rpm during ILS approaches.
6. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
7. Operation of the EFIS-85B(14) system must be in conformity with the Collins Pilot's Guide, No. 523-0775919, dated August 15, 1989 or later issue.
8. Navigation and checklist data on the Multifunction Display (MFD-85) is provided only as an aid and is not approved as primary information.
9. The pilot's EADI and EHSI must be operational in the normal mode for take off. Aux Drive may be used if the Primary Drive has failed.
10. The composite mode is to be used after failure of an EADI or EHSI.
11. The Airspeed readout is not to be used as the primary reference.

CAUTION

FMS course information is unreliable when switched from "CRS PRE" to the "CRS ACT" mode. Course information must be programmed into the FMS when the FMS is in the ACT mode.

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For Training Purposes Only

AIR START

WARNING

Air start using the STARTER ASSIST procedures may momentarily cause the loss of attitude display on Electronic Flight Instrument System (EFIS) equipped airplanes, and lead to premature system failures. IF FLIGHT CONDITIONS DO NOT PERMIT THE TEMPORARY LOSS OF ATTITUDE REFERENCE, CONDUCT AIR START USING THE NO STARTER ASSIST PROCEDURES.

CAUTION

The pilot should determine the reason for engine failure before attempting an air start. Do not attempt an air start if N_1 indicates zero.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an overtemperature condition.

WINDMILLING ENGINE AND PROPELLER (NO STARTER ASSIST)

1. Power Lever - IDLE
2. Propeller Lever - FULL FORWARD
3. Condition Lever - CUT-OFF
4. Fuel Firewall Valve - OPEN
5. Generator (inoperative engine) - OFF
6. Airspeed - 140 KNOTS MINIMUM
7. Altitude - BELOW 20,000 FEET
8. Engine Auto Ignition - ARM
9. Condition Lever - LOW IDLE
10. Power - AS REQUIRED (after ITT has peaked)
11. Generator - ON
12. Electrical Equipment - AS REQUIRED

STARTER ASSIST

NOTE

EFIS power should be turned OFF prior to a STARTER ASSIST air start if conditions permit.

WARNING

Standby attitude and air data displays must be monitored during STARTER ASSIST air starts.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. Windshield Heat - OFF
5. Power Lever - IDLE

6. Condition Lever - CUT-OFF
7. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

8. Ignition and Start Switch - ON (up). Check IGNITION Annunciator - ON
9. Condition Lever - LOW IDLE
10. Ignition and Start Switch - OFF (N_1 above 50%)
11. Propeller Lever - AS REQUIRED
12. Power Lever - AS REQUIRED
13. Generator - ON
14. Engine Auto Ignition - ARM
15. Electrical Equipment - AS REQUIRED

AUTOPILOT/ELEVATOR TRIM MALFUNCTIONS

1. Hold control wheel firmly. Be prepared to accept control wheel loads.
2. Fully depress red DISC TRIM/AP YD switch on pilot's or copilot's control wheel.
3. Recover - The maximum altitude losses observed during malfunction tests were:

Climb	450 ft
Cruise	440 ft
Maneuvering	150 ft
Descent	450 ft
Approach	100 ft

NOTE

An elevator trim malfunction will activate the master warning flasher and the red trim annunciator. The master warning flasher may be delayed for small trim requirements.

If necessary, the autopilot can also be disengaged by:

4. Pressing the AP ENG switch on the autopilot control head. (Yaw damper remains engaged.)
5. Activating the PITCH TRIM switch on the pilot's or copilot's control wheel. (Yaw damper remains engaged.)
6. Pressing the go around (GA) switch. (Yaw damper remains engaged.)
7. Pulling the autopilot circuit breaker.
8. Moving the avionics master switch to the OFF position.
9. Moving the gang bar to the OFF position.

CAUTION

If the autopilot is overpowered in pitch, the electric elevator trim will operate in the direction opposing the overpower force.

AUTOMATIC AUTOPILOT DISENGAGEMENT

The following conditions will cause the autopilot to disengage or servo torque to go to approximately zero torque:

DISENGAGE

1. Major degradation, interruption, or failure of electric power.
2. Autopilot internal monitors.
   a. System integrity checks.

b. Loss of vertical gyro.

c. Pitch attitudes in excess of 30°.

d. Roll attitudes in excess of 45°.

ZERO TORQUE

The autopilot also monitors pitch and roll rate and normal acceleration. Pitch rates in excess of 4°/sec, roll rates in excess of 10°/sec, or accelerations in excess of 1.0g cause autopilot servo torque to decrease to approximately zero torque until the rate or acceleration is below the above limits.

ENGINE FAILURE

1. Disengage autopilot.
2. Follow POH/AFM engine failure procedures.
3. Retrim.
4. Re-engage autopilot.
5. Maintain at least 120 KIAS until landing is assured.
6. Disengage autopilot for large power or airspeed variations.

FOR THE EFIS-85 SYSTEM:

1. In the event of a Display Processor failure: Activate the AUX DRIVE switch.
2. In the event of an EADI or EHSI display failure: Activation of the CMPST switch will provide a composite display on the remaining indicator.
3. In the event of a number 1 dual fed electrical bus failure: Activation of the AUX DRIVE and CMPST switches will provide attitude information (minimum) on the EHSI in the composite format (inverter #2 is required for AC power).

NORMAL PROCEDURES

PREFLIGHT CHECKS

1. Position pitch trim to the take-off position, turn ELEV TRIM switch on and engage the autopilot. Check that YAW and AP annunciators are on.
2. With control wheel in the forward position and the autopilot engaged, operate pilot then copilot pitch trim switches in both directions to ensure that the autopilot disengages. The yaw damper remains engaged.
3. Center the control wheel and engage the autopilot. Apply forward pressure on control wheel. Note that pitch trim travels nose up and that the amber TRIM light comes on. Apply rearward pressure on control wheel. Note that pitch trim travels nose down and that the amber TRIM light comes on.
4. Hold the control wheel and disengage the autopilot by depressing the DISC TRIM/AP YD button on the pilot's control column to the first level. Note that the YAW DIS and AP DIS annunciators illuminate. Further depress the DISC TRIM/AP YD switch. Note that the ELEC TRIM OFF annunciator illuminates. Cycle the electric trim switch on the console, engage the autopilot, and repeat the check using the copilot's DISC TRIM/AP YD switch.
5. Engage yaw damper. Note YAW annunciator is on. Check for additional resistance to movement of rudder pedals. Disengage yaw damper.
6. Push the TEST button on the mode annunciator panel to test the annunciators.
7. Reset pitch trim to the take-off position.
8. Move all primary flight controls through their full travel in both directions. Verify that controls move in proper direction and no restrictions to free movement are present.

ELECTRIC ELEVATOR TRIM

1. Verify the ELEV TRIM is ON.
2. Check operation of dual-element thumb switches as detailed in the POH/AFM.

WARNING

Operation of the electric trim system should occur only by movement of pairs of switches. Any movement of the elevator trim wheel while actuating only one switch denotes a system malfunction. If a malfunction of the electric trim system is indicated, electric trim must be disengaged and trim changes made with manual trim ONLY.

DIGITAL AIR DATA SYSTEM (if installed)

1. Press and hold the cockpit-mounted ADC TEST switch for the Air Data Computer. Check for the following:
   a. Air data instrument flags come into view, and the instruments go into the loss-of-data mode.
   b. Altitude pointer goes to the 250-foot mark.
   c. Vertical speed pointer goes to 6000-ft/min rate-of-descent mark.
   d. Airspeed Indicator IAS and V_{MO} pointers go to zero.
2. Release the ADC TEST switch and check for the following:
   a. Data transmissions to the air data instruments are restored.
   b. All air data instrument flags go out of view.
3. Press and hold the TEST switch on the Vertical Speed Indicator, and check for the following:
   a. The warning flag comes into view.
   b. The pointer slews to the 6000-ft/min rate of climb mark.
4. Release the TEST switch, and check for the following:
   a. The flag goes out of view.
   b. The pointer returns to zero.
5. Press and hold the PUSH TEST switch on the Airspeed Indicator, and check for the following:
   a. The warning flag comes into view, and both pointers (IAS and V_{MO}) slew to 160 knots.
   b. After approximately 1 second, both pointers slew to zero.
6. Release the PUSH TEST switch and check for the following:
   a. The flag goes out of view.
   b. The barber pole (V_{MO} indicator) returns to 260 knots.
7. Press and hold the TEST switch on the Barometric Altimeter, and check for the following:
   a. The warning flag comes into view in approximately 1/2 second.
   b. The altitude pointer goes to the 750-foot mark.

NOTE

The output from the altimeter to the transponder is inhibited during the self test.

8. Release the altimeter TEST switch, and check for the following:
   a. The flag goes out of view.
   b. The altitude pointer returns to the actual altitude.
9. Turn the BARO knob on the altimeter and observe that the baroset digits and altitude pointer respond accordingly.
10. Push BARO knob in to obtain baroset values in in. Hg. Set the barometric display to the correct local pressure value. (Pull BARO knob out if display of baroset values in millibars is desired.)

11. Press and hold the ALT ALERT annunciator/switch on the Altitude Preselector/Alerter, and check for the following:
    a. The ALT ALERT annunciator illuminates.
    b. The warning flag comes into view.
    c. Remote ALT ALERT annunciators illuminate.
12. Release the ALT ALERT annunciator/switch, and check for the following:
    a. The ALT ALERT annunciator extinguishes.
    b. The warning flag goes out of view.
    c. Remote ALT ALERT annunciators extinguish.

BEFORE STARTING

- EFIS ON/OFF Switches - OFF

BEFORE TAXI

- EFIS ON/OFF Switches - ON

IN FLIGHT OPERATION

AUTOPILOT OPERATION

To engage the autopilot, press the AP ENG switch on the autopilot control panel. If a red fault annunciator is on (AP, TRIM), it requires one operation of the AP ENG switch to clear the fault and another operation to engage the autopilot. If the fault does not clear, the autopilot will not engage.

An automatic pre-engagement test is performed each time the autopilot is engaged. This test requires approximately 2 seconds and is identified during the test by annunciation of red TRIM and AP.

The autopilot may be engaged in any reasonable attitude. When engaged with no lateral mode selected beyond 30° bank, 15° pitch up or 10° pitch down, the autopilot will return the airplane to these limits. No steering information will be presented on the flight director until a lateral mode is selected.

To disengage the autopilot, press the AP ENG switch or operate the DISC TRIM/AP YD DISC switch on the pilot's or copilot's control wheel to first detent. Refer to EMERGENCY PROCEDURES for other means of disconnecting.

CONTROL WHEEL SYNCHRONIZATION (CWS)

Autopilot sync mode is controlled by the PITCH SYNC & CWS (control wheel steering) button on the control wheel. With the autopilot engaged and in either manual or guidance mode, depressing the PITCH SYNC & CWS button cancels vertical modes and allows the pilot to maneuver the airplane without disengaging the autopilot. When the PITCH SYNC & CWS button is released, the autopilot maintains the attitude existing at the time the button is released if in manual mode, or maintains the new pitch attitude and follows commands received from the autopilot computer if in a lateral mode. CWS mode does not cancel GS when in APPR mode.

VERTICAL TRIM SWITCH

This control switch is used for vertical control of the flight control system. Activating the switch for more than one second with a vertical hold mode selected causes the vertical mode to be cancelled (except GA, GS, and ALT SEL ARM) and the system to revert to basic pitch operation. Refer to Special Note 1.

1/2 BANK (1/2Φ)

In 1/2ϕ mode, the bank limit in HDG or NAV mode is reduced to approximately 1/2 the normal value. Capture of NAV or APPR mode clears the 1/2ϕ mode, if selected.

SOFT RIDE (SR)

Soft Ride mode is used to reduce the autopilot response to turbulent conditions. The SR mode may be used at the pilot's discretion. SR is cancelled by APPR mode and when engaging the autopilot.

YAW DAMPER OPERATION

The yaw damper is automatically engaged any time the autopilot is engaged.

The rudder channel of the autopilot may be selected separately for yaw damping by the YAW ENG switch on the autopilot panel.

To disengage the yaw damper, operate the DISC TRIM/AP YD disc switch, or press the YAW ENG switch. The YAW ENG switch is disabled if the autopilot is engaged.

ELECTRIC PITCH TRIM OPERATION

The autopilot computer provides manual electric trim operated by the split pitch trim switches located on the pilot's and copilot's control wheels. Electric trim operation is annunciated by the amber TRIM annunciator (when the trim speed is above the present annunciation threshold).

MODE SELECTION

All modes with the exception of GA are selected by pushing the desired PUSH-ON/PUSH-OFF button on the mode control panel. Annunciator lights on the mode control panel and mode annunciator panel indicate the selected mode of operation.

The flight director system supplies steering commands for the pilot and the autopilot. When the autopilot is engaged, the pilot monitors autopilot performance on the attitude director indicator. When the autopilot is not engaged, the pilot flies the airplane manually in response to the attitude director indicator commands. A lateral mode must be selected to bring the command bars into view for manual flight director operation.

The EFIS-85B(14) Electronic Flight Instrument System replaces the conventional electro-mechanical Attitude Director (ADI) and Horizontal Situation (HSI) Indicators. The CRT displays (EADI & EHSI) provide the necessary information for the pilot to respond to flight director and navigation commands in the same manner as with conventional instruments. The system also includes additional features which allow the pilot to select alternate formats as a navigation aid or in the event of specific failures. The operator is cautioned that the primary displays (attitude and directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate modes is described in the Collins EFIS-85B(14) Pilot's Guide. *The Owner/Operator of the airplane is responsible for obtaining and maintaining this manual.*

HEADING MODE (HDG)

Set desired heading and select HDG mode. The autopilot will turn the airplane to and maintain the selected heading. For proper operation, the HDG marker should not be displaced from the airplane heading by more than 150 degrees when the HDG mode is selected.

NAV MODE (NAV)

Set desired heading and course prior to selecting NAV mode. When NAV mode is selected, the HDG and NAV ARM annunciators will illuminate. The airplane will follow the selected heading until the centerline of the selected radio course is approached. HDG mode will then clear and the airplane will turn to track the beam centerline. The ARM annunciator will extinguish to indicate course capture and NAV will remain on. Crosswind correction, up to 30 degrees, is automatically computed after course capture.

CAUTION

Premature Glideslope and localizer capture may occur in the APPR ARM mode if selected when the airplane is far from the localizer centerline. Therefore, use the HDG mode to position the airplane near the localizer intercept before selecting the APPR ARM mode.

NOTE

1. A vertical mode, if selected, will automatically clear at glideslope capture.

2. All approaches must be made in APPR or B/C mode.

BACK COURSE MODE (B/C)

When the B/C mode is selected, localizer capture follows the same sequence as a front course approach. Correct lateral steering information is displayed on the flight director, but vertical information is displayed as a function of the vertical mode selected.

NOTE

The inbound (front) course must be selected on the course indicator.

ALTITUDE HOLD MODE (ALT)

ALT can be selected during all modes of operation except after glideslope capture in APPR mode. Deviations from the selected altitude are displayed on the flight director.

NOTE

The autopilot maintains the airplane at the selected altitude by changing the pitch attitude of the airplane. The pilot must maintain proper power settings to assure a safe airspeed.

INDICATED AIRSPEED MODE (IAS)

The IAS mode may be selected during all modes of operation except after glideslope capture in APPR mode. Deviations from the airspeed at the time the mode was selected are displayed on the flight director as pitch commands.

VERTICAL SPEED MODE (VS)

The VS mode may be selected during all modes of operation except after glideslope capture in APPR mode. Deviation in vertical speed provides commands to maintain the vertical speed present at the time of mode selection. Rate of ascent/descent should be established for 10 seconds before selecting VS mode.

ALTITUDE PRESELECT MODE (ALT SEL) (if installed)

Dial in the preselected altitude on the Altitude Preselector/Alerter and press the ALT SEL mode on the mode selector. The ALT SEL annunciator will illuminate. Any vertical mode except altitude hold may be used in conjunction with ALT SEL.

An aural alert tone sounds and annunciator lights on the altitude panel and flight director illuminate when the airplane reaches 1000 ft from the preselected altitude. At 300 feet from the preselected altitude the altitude annunciator extinguishes. Thereafter, the alert tone and altitude annunciators will warn when the airplane deviates more than 300 feet from the preselected altitude. As the airplane reaches the preselected altitude the ALT SEL annunciator extinguishes and ALT illuminates.

The use of vertical trim switch or altitude preselect dial, when the airplane is within 1000 ft of the preselected altitude and before altitude capture, will cancel the ALT SEL mode.

DESCENT MODE (DSC)

When the DSC mode is selected, the autopilot begins a gradual descent, stabilizing at a rate-of-descent of 950 to 1350 feet per minute. The stabilized rate-of-descent will vary depending on altitude, with the lower rates-of-descent occurring at high altitude and the higher rates-of-descent occurring at lower altitudes. After the vertical speed has stabilized (DSC annunciator extinguished and VS annunciator illuminated), the pilot may vary the rate-of-descent by operation of the vertical trim switch.

On those airplanes equipped with the digital air data system, the altitude preselect mode is automatically activated when the DSC mode is selected.

NOTE

Desired altitude must be selected on the altitude alerter prior to selection of the DSC mode.

GO-AROUND MODE

The GA mode is activated by the GA switch. When the GA switch is pushed, while in any lateral mode, the autopilot will disengage and a 7° pitch up, wings level attitude will be displayed on the flight director. GA annunciator will illuminate and remain on until a lateral mode is reselected.

Operation of the PITCH SYNC & CWS switch will clear the go-around mode and synchronize the vertical command to the airplane attitude. Re-engagement of the autopilot while in go-around clears the GA mode and synchronizes the autopilot commands to the airplane pitch attitude at the time of engagement and continues to maintain wings level.

Selection of a lateral mode will also clear the go-around mode and synchronize the vertical commands to airplane pitch attitude at the time of mode selection.

SPECIAL NOTES

1. Vertical Trim Switch Operation
    a. In basic pitch mode, with the autopilot or a lateral mode selected, activating the vertical trim switch gives a 0.5° command followed by a continuous slew of 1.0°/second.

b. In VS hold mode the synchronized vertical speed is changed in 200 ft/min increments with each momentary activation of the vertical trim switch.

c. In IAS hold mode the synchronized airspeed is changed in one-knot increments with each momentary activation of the switch.

d. In the altitude hold mode, the synchronized altitude is changed by 25 ft with each momentary activation of the switch.

2. Basic Attitude Mode - With no lateral mode selected, the steering display is biased out of view and the EADI is used as a standard artificial horizon. No computed steering information is displayed on the EADI.
3. Bank command limit is ±30° with turn knob, ±25° in lateral modes, ±15° in APPR (after GS capture), and ±10° in NAV track.
4. Vertical mode information is displayed by pushing the desired vertical mode on the mode select panel. If a vertical mode other than GS is selected during manual flight director operation, it will automatically clear when the autopilot is engaged.
5. To synchronize the vertical commands in flight director function to the airplane attitude, actuate the PITCH SYNC & CWS switch on the control wheel after any lateral mode has been selected.
6. When making any coupled approach, primary flight instruments must be monitored by the pilot. Presence of a VOR/LOC/GS flag must be considered as a warning of system or signal failure.
7. When flying inbound toward a VOR, do not set up a VOR capture problem within approximately 5 miles of the VOR. Depending on the speed of the airplane, the system may not have time to solve the problem and stabilize on course before passage over the station.
8. Course may be changed over a VOR station when operating in NAV mode, as long as the course change is not more than 30°. If the course change is more than 30°, the HDG mode should be selected to establish a new intercept and then NAV mode should be reselected to set up a new capture.
9. For optimum NAV mode operation, select intercept angle so the system maintains straight and level flight in the NAV mode a minimum of 30 seconds prior to capture.
10. When making a station passage in NAV mode, the computer may automatically switch into heading memory sub-mode to allow a smooth station passage. During this time, the dead reckoning (DR) annunciator will illuminate.
11. Maximum recommended capture angles are 90° or less in NAV mode and 60° or less in APPR mode.
12. Fly all coupled approaches at or above 120 KIAS, flaps 40% and landing gear extended.
13. Always check the mode select panel annunciator display to verify that proper switching has taken place.
14. Do not select any vertical modes until 3 minutes after applying power.
15. Optional Dual Flight Director Systems:
    a. Disengage the autopilot prior to transferring the autopilot to either flight director.
    b. Deselect 1/2ϕ mode prior to transferring the autopilot to either flight director.
16. The vertical gyro connected to the autopilot system is not transferred if optional VG switching feature is installed.

SHUTDOWN AND SECURING

EFIS ON/OFF Switches - OFF (before FUEL CUTOFF)

PERFORMANCE

No Change

WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

The Collins FCS-65 is an integrated flight control system which can be functionally divided into two general subsystems.

1. Flight Director System (FDS)
   The FDS consists of an Electronic Attitude Director Indicator (EADI), an Electronic Horizontal Situation Indicator (EHSI) and associated components. The EADI presents attitude data, raw radio information and steering commands. The EHSI

presents the navigation situation.
EFIS ON/OFF Switches - Control power to the EFIS symbol generators, EHSI, and EADI. The switches should be selected OFF during engine start to protect the EFIS system from low/transient voltages.

2. Autopilot System (APS)
The APS consists of an autopilot computer, mode select panel, air data sensor, three primary servos, a vertical reference and a trim servo. The system also requires compass and navigation system inputs from the avionics system.

ANNUNCIATORS

Annunciation is displayed on the mode select panel and mode annunciator panel. Annunciation is color-keyed for better recognition of mode status with green for active condition, amber for arm condition, and red for warning condition. The following table shows the annunciations and their association with each other:

ANNUNCIATOR	CONDITION TO ACTIVATE	COLOR ACTIVE	COLOR ARM	COLOR WARNING
HDG	Shows heading mode selected. Also, will light automatically if NAV or APPR is selected, but capture has not occurred.	Green		
NAV	Selection of navigation mode.	Green		
APPR	Selection of approach mode.	Green		
GS	Indicates glideslope mode.	Green		
AP	Indicates autopilot is engaged.	Green		
ALT	Selection of altitude hold mode.	Green		
IAS	Selection of IAS hold mode.	Green		
VS	Selection of vertical speed hold mode.	Green		
GA	Selection of go-around mode.	Green		
YAW	Indicates yaw damper is engaged.	Green		
DR	Computation is in Heading Memory.	Green		
ARM	Indicates an armed condition prior to capture.		Amber	
DSC	Indicates the selection of descent mode.	Green		
1/2ϕ	Indicates selection of half bank mode.	Green		
SR	Indicates selection of soft ride mode.	Green		
B/C	Indicates selection of back course mode.	Green		
DIS	Disengage annunciation for YAW and AP. Flashes for 5 seconds, then steady indication when system is disengaged. Lights steady when PITCH SYNC button is depressed.		Amber	
ALT SEL	Shows that altitude preselect is armed for automatic capture.		Amber	
AP FAIL	Indicates first failure in dual safety functions - Pilot should take over and disengage.			Red
TRIM	Indicates Trim Servo failure in autopilot or manual electric operation.			Red
TRIM	Indicates Elevator Trim-in-Motion, both manual electric and autopilot operation.	Amber		*Amber

**Sustained Annunciation*

ANNUNCIATOR NOTES:

1. During annunciator test, CLM annunciator will illuminate even though the function is not installed.

2. When a mode annunciator is flashing it indicates that signals used for that mode are not usable.

HANDLING, SERVICING & MAINTENANCE

If a Collins altimeter is installed, one additional static-line drain is installed. It is located inside the nose avionics compartment in the pilot's static airline that provides static pressure to the digital air data computer. This drain should be opened to release any trapped moisture at each 150-hour inspection and after exposure to visible moisture on the ground. It must be closed after draining.

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For Training Purposes Only

BEECHCRAFT SUPER KING AIR

B200 AND B200C LANDPLANES
SERIALS BL-65 AND BB-1114 ONLY

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

AUXILIARY POWER UNIT
KIT NO. 101-5120

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL:
101-590010-147

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved:

W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

FAA Approved
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For Training Purposes Only

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Auxiliary Power Unit, BEECHCRAFT Kit No. 101-5120, in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

This supplement replaces supplement 101-590010-231 in its entirety for serials BL-65 and BB-1114 only.

LIMITATIONS

1. The Auxiliary Power Unit is not approved for in-flight operation.
2. Minimum allowable Auxiliary Power Unit engine RPM is 90%.
3. Maximum allowable Auxiliary Power Unit engine RPM is 106%.
4. Maximum rated Auxiliary Power Unit EGT is 704°C.
5. Maximum Auxiliary Power Unit electrical load is 300 amps.
6. Aviation gasoline grades are not approved.
7. Auxiliary Air Conditioning System is not to be operated above 20,000 feet.

EMERGENCY PROCEDURES

ILLUMINATION OF "APU FIRE-PUSH" ANNUNCIATOR ON GROUND

1. **APU FIRE-PUSH Annunciator - PUSH**
2. **Electrical Equipment - GANG BAR DOWN**

If engines operating:

   a. **Condition Levers - CUTOFF**
   b. **Propellers - FEATHER**

3. **Evacuate the airplane.**

ILLUMINATION OF "APU FIRE-PUSH" ANNUNCIATOR IN FLIGHT

1. **APU FIRE-PUSH Annunciator - PUSH**
2. **Land as soon as practicable.**

NORMAL PROCEDURES

ENGINE STARTING WITH APU GENERATOR OPERATING (BATTERY)

NOTE

This procedure protects electrical equipment operating at the time of start from possible transient effects due to engine starting.

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NOTE

Starting with cabin bleed air OFF will provide cooler engine starts.

1. BUS ISO Switch - ON
2. Right Ignition and Engine Start Switch - ON (R FUEL PRESS annunciator - EXTINGUISHED)
3. Right Condition Lever - LOW IDLE (after N_1 indicates 12% minimum)
4. ITT and N_1 - MONITOR (1000°C maximum)

CAUTION

If no ITT rise is observed within 10 seconds after moving the Condition Lever to LOW IDLE, move the Condition Lever to CUTOFF and release the Ignition and Engine Start Switch to OFF. Allow 5 minutes for fuel to drain and starter to cool, then follow ENGINE CLEARING procedures in the NORMAL PROCEDURES section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

5. Right Oil Pressure - CHECK
6. Right Ignition and Engine Start Switch - OFF (at 50% N_1 or above)
7. Right Generator - RESET, then ON
8. BUS ISO Switch - OFF, CHARGE BATTERY
9. BUS ISO Switch - ON
10. Left Ignition and Engine Start - ON (L FUEL PRESS annunciator - EXTINGUISHED)
11. Left Condition Lever - LOW IDLE (after N_1 indicates 12% minimum)
12. ITT and N_1 - MONITOR (1000°C maximum)
13. Left Oil Pressure - CHECK
14. Left Ignition and Engine Start Switch - OFF (at 50% N_1 or above)
15. Left Generator - RESET, then ON
16. BUS ISO Switch - OFF
17. Battery Charge Annunciator - ILLUMINATED

NOTE

The BATTERY CHARGE annunciator will illuminate approximately 6 seconds after selecting BUS ISO - OFF. If the annunciator does not extinguish within 5 minutes, refer to NICKEL-CADMIUM BATTERY CHECK procedure in the NORMAL PROCEDURES section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

ENGINE STARTING WITH APU GENERATOR OPERATING (EXTERNAL POWER)

CAUTION

NEVER CONNECT AN EXTERNAL POWER SOURCE TO THE AIRPLANE UNLESS A BATTERY INDICATING A CHARGE OF AT LEAST 20 VOLTS IS IN THE AIRPLANE. If the battery voltage is less than 20 volts, the battery must be recharged, or replaced with a battery indicating at least 20 volts, before connecting external power. Only use an external power source fitted with an AN-type plug.

When an external power source is used, ascertain that the External Power Unit has a capacity of 600 amperes (minimum intermittent) output.

1. Avionics Master Power Switch - AS REQUIRED
2. GEN 1 and GEN 2 Switches - OFF
3. BAT Switch - ON (The battery will tend to absorb transients present in the external power source.)
4. BUS ISO Switch - ON

5. External Power Source - TURN OFF, then CONNECT TO AIRPLANE
6. External Power Source - TURN ON
7. Right Propeller Control - FEATHER
8. Right Ignition and Engine Start Switch - ON (R FUEL PRESS annunciator - EXTINGUISHED)
9. Right Condition Lever - LOW IDLE (after N_1 indicates 12% minimum)
10. ITT and N_1 - MONITOR (1000°C maximum)

CAUTION

If no ITT rise is observed within 10 seconds after moving the Condition Lever to LOW IDLE, move the Condition Lever to CUTOFF and release the Ignition and Engine Start Switch to OFF. Allow 5 minutes for fuel to drain and starter to cool, then follow ENGINE CLEARING procedures in the NORMAL PROCEDURES section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

11. Right Oil Pressure - CHECK
12. Right Ignition and Engine Start Switch - OFF (at 50% N_1 or above)
13. Left Ignition and Engine Start Switch - ON (L FUEL PRESS annunciator - EXTINGUISHED)
14. Left Condition Lever - LOW IDLE (after N_1 indicates 12% minimum)
15. ITT and N_1 - MONITOR (1000°C maximum)
16. Left Oil Pressure - CHECK
17. Left Ignition and Engine Start Switch - OFF (at 50% N_1 or above)
18. External Power Source - TURN OFF; DISCONNECT; Door - SECURE
19. BUS ISO Switch - OFF
20. GEN 1 and GEN 2 Switches - RESET, then ON

NOTE

If the battery is partially discharged, the BATTERY CHARGE annunciator will illuminate approximately 6 seconds after the external power is on-line. If the annunciator does not extinguish within 5 minutes, refer to the NICKEL-CADMIUM BATTERY CHECK procedure in the NORMAL PROCEDURES section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

21. Right Propeller Control - FULL FORWARD

ENGINE STARTING WITH APU GENERATOR OPERATING (APU ASSISTED)

NOTE

Battery must be on-line for APU assisted starts.

1. BUS ISO Switch - OFF
2. Right Ignition and Engine Start Switch - ON (R FUEL PRESS annunciator - EXTINGUISHED)
3. Right Condition Lever - LOW IDLE (after N_1 indicates 12% minimum)
4. ITT and N_1 - MONITOR (1000°C maximum)

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CAUTION

If no ITT rise is observed within 10 seconds after moving the Condition Lever to LOW IDLE, move the Condition Lever to CUTOFF and release the Ignition and Engine Start Switch to OFF. Allow 5 minutes for fuel to drain and starter to cool, then follow ENGINE CLEARING procedures in the NORMAL PROCEDURES section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

5. Right Oil Pressure - CHECK
6. Right Ignition and Engine Start Switch - OFF (at 50% N_1 or above)
7. CHARGE BATTERY
8. Left Ignition and Engine Start - ON (L FUEL PRESS annunciator - EXTINGUISHED)
9. Left Condition Lever - LOW IDLE (after N_1 indicates 12% minimum)
10. ITT and N_1 - MONITOR (1000°C maximum)
11. Left Oil Pressure - CHECK
12. Left Ignition and Engine Start Switch - OFF (at 50% N_1 or above)
13. GEN 1 and GEN 2 Switches - RESET, then ON
14. Battery Charge Annunciator - ON

NOTE

The BATTERY CHARGE annunciator will ILLUMINATE approximately 6 seconds after generator is on-line. If the annunciator does not extinguish within 5 minutes, refer to NICKEL-CADMIUM BATTERY CHECK procedure in the NORMAL PROCEDURES section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

OTHER NORMAL PROCEDURES

AUXILIARY POWER UNIT

NOTE

Prior to turning battery switch ON, ensure that all APU switches are OFF.

AUXILIARY POWER UNIT STARTING (BATTERY OR AIRPLANE GENERATOR POWERED)

1. APU Inlets and Outlets - CLEAR
2. Perform BEFORE ENGINE STARTING procedures.
3. APU FIRE TEST - TEST (APU FIRE-PUSH annunciator will flash.)
4. START SELECT Switch - AS REQUIRED
5. APU Switch - ON (ON annunciator will ILLUMINATE and start sequence will begin.)
6. APU EGT - MONITOR

CAUTION

If no EGT rise is observed within 10 seconds after start initiation, turn the APU switch OFF. Allow 30 seconds before the next start attempt.

7. When APU RPM is stable, GEN switch - ON (FAIL annunciator may ILLUMINATE along with the ON annunciator until the generator is on-line - approximately 20-30 seconds.)
8. APU generator load - MONITOR

NOTE

Allow 30 seconds between APU starts or between shutdown and next start.

The APU generator or the battery must be on-line during APU operation.

AUXILIARY POWER UNIT STARTING (EXTERNAL POWER)

CAUTION

NEVER CONNECT AN EXTERNAL POWER SOURCE TO THE AIRPLANE UNLESS A BATTERY INDICATING A CHARGE OF AT LEAST 20 VOLTS IS IN THE AIRPLANE. If the battery voltage is less than 20 volts, the battery must be recharged, or replaced with a battery indicating at least 20 volts, before connecting external power. Only use an external power source fitted with an AN-type plug.

When an external power source is used, ascertain that the Auxiliary Power Unit has a capacity of 600 amperes (minimum intermittent) output.

1. APU Inlets and Outlets - CLEAR
2. Perform BEFORE ENGINE STARTING procedures.
3. External Power Source - TURN OFF, then CONNECT TO AIRPLANE
4. External Power Source - TURN ON
5. APU FIRE TEST - TEST (APU FIRE-PUSH annunciator will FLASH.)
6. START SELECT Switch - AS REQUIRED
7. APU Switch - ON (ON annunciator will ILLUMINATE and start sequence will begin.)
8. APU EGT - MONITOR

CAUTION

If no EGT rise is observed within 10 seconds after start initiation, turn the APU switch OFF. Allow at least 30 seconds before next start attempt.

9. When APU RPM is stable, External Power Source - TURN OFF; DISCONNECT; Door - SECURE (AS REQUIRED)
10. GEN Switch - ON (FAIL annunciator may ILLUMINATE along with the ON annunciator until the generator is on-line - approximately 20-30 seconds.)
11. APU Generator Load - MONITOR

NOTE

Allow at least 30 seconds between APU starts or between shutdown and next start.

AUXILIARY POWER UNIT SHUTDOWN

1. APU GEN Switch - OFF. The segment placarded OFF will ILLUMINATE and display will indicate no generator output.
2. O/S TEST Switch - PUSH. The segment placarded APU FAIL will illuminate and the APU will be deactivated.
3. APU Switch - OFF. The segment placarded OFF will ILLUMINATE.

CAUTION

If the APU is deactivated by other than normal procedures, ensure that all the APU Switches are OFF.

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AUXILIARY AIR CONDITIONER OPERATION

CAUTION

Do not operate the Auxiliary Air Conditioning System above 20,000 feet.

NOTE

Erratic operation of the magnetic compass may occur when the auxiliary air conditioner is being used.

— AIR COND Switch - ON

PERFORMANCE

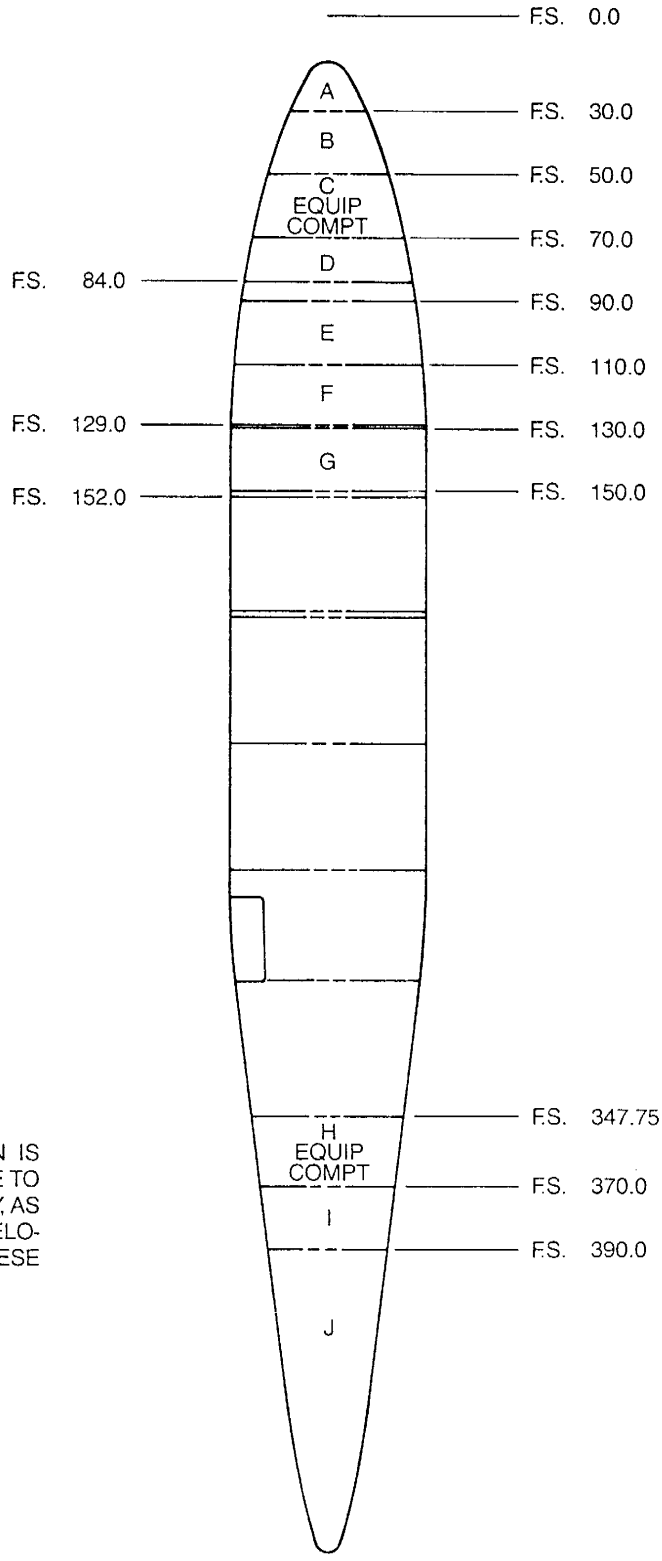
No Change

LOADING LIMITS - SUPER KING AIR B200

SECTION	MAXIMUM STRUCTURAL CAPACITY	CENTROID ARM
A	30 LB	F.S. 20
B	50 LB	F.S. 40
C	110 LB	F.S. 60
D	190 LB	F.S. 80
E	80 LB	F.S. 100
F	225 LB	F.S. 120
G	545 LB	F.S. 140
H	95 LB	F.S. 360
I	210 LB	F.S. 380
J	50 LB	F.S. 430

WARNING

WEIGHT OF THE AUXILIARY POWER UNIT INSTALLATION IS COUNTED AGAINST THE WEIGHT LIMITS LISTED ABOVE. DUE TO VARIATIONS IN ORIGINAL EQUIPMENT, IT MAY BE NECESSARY, AS PART OF THE AUXILIARY POWER UNIT INSTALLATION, TO RELOCATE OTHER COMPONENTS SO AS NOT TO EXCEED THESE LIMITS.



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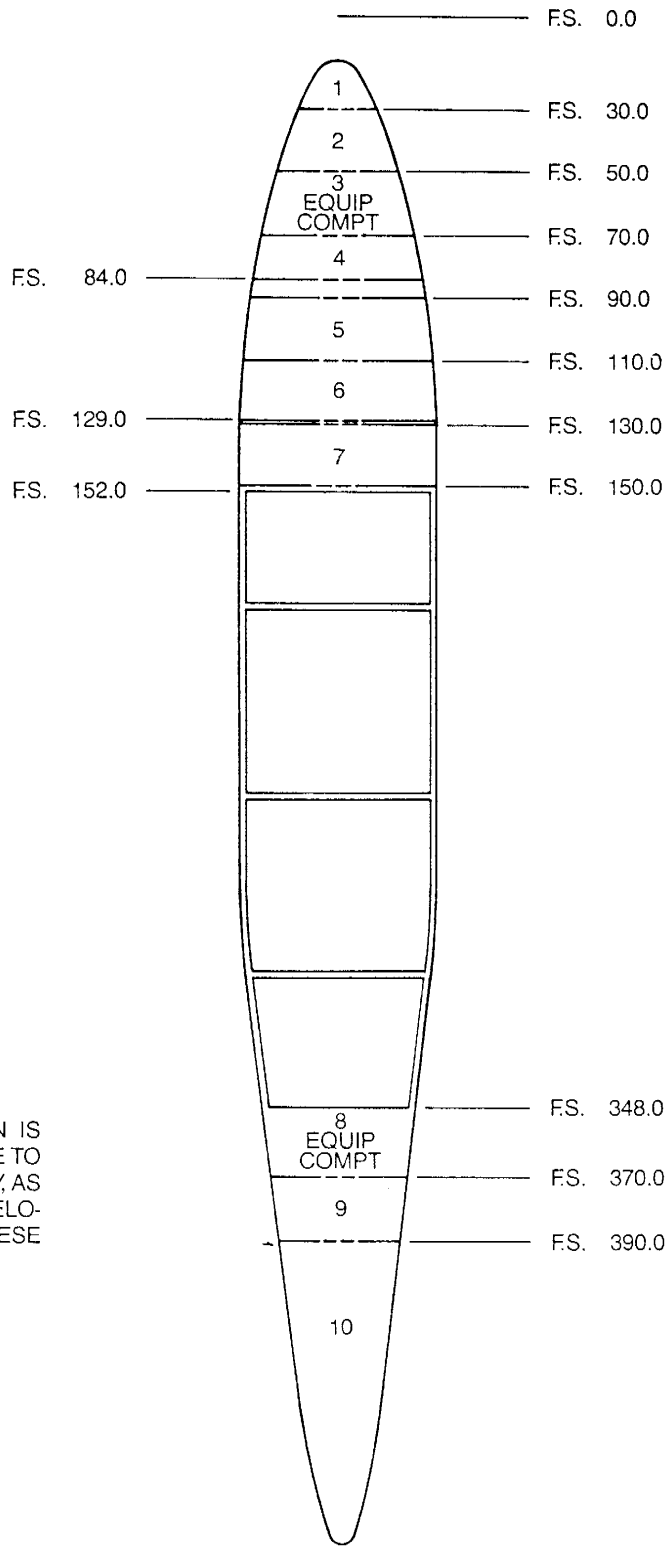
LOADING LIMITS - SUPER KING AIR B200C

SECTION	MAXIMUM STRUCTURAL CAPACITY	CENTROID ARM
1	30 LB	F.S. 20
2	50 LB	F.S. 40
3	110 LB	F.S. 60
4	190 LB	F.S. 80
5	80 LB	F.S. 100
6	225 LB	F.S. 120
7	634 LB	F.S. 141
8	95 LB	F.S. 360
9	210 LB	F.S. 380
10	50 LB	F.S. 430

WARNING

WEIGHT OF THE AUXILIARY POWER UNIT INSTALLATION IS COUNTED AGAINST THE WEIGHT LIMITS LISTED ABOVE. DUE TO VARIATIONS IN ORIGINAL EQUIPMENT, IT MAY BE NECESSARY, AS PART OF THE AUXILIARY POWER UNIT INSTALLATION, TO RELOCATE OTHER COMPONENTS SO AS NOT TO EXCEED THESE LIMITS.



B200-603-75

SYSTEMS DESCRIPTION

AUXILIARY POWER UNIT GENERATOR

The Auxiliary Power Unit starter-generator is rated at 300 amps. The APU starter-generator is capable of powering the electrical system during ground operations only. The generator is controlled by a lighted pushbutton switch in the APU control head. When the generator switch is in the OFF position, the legend GEN OFF is illuminated. When the generator switch is in the ON position the legend GEN ON is illuminated. The generator may be selected at anytime. But if it is selected before the APU engine is ready to load, it will not come on-line until the ready to load signal is received which occurs approximately 20 seconds after engine speed reaches 98%. Anytime the generator switch is in the ON position and the generator is not on-line, the FAIL legend in the generator control switch will illuminate.

The APU generator control unit provides voltage regulation, differential voltage, reverse current protection, paralleling, current limiting, and overvoltage protection. The voltage regulation circuit controls the generator to maintain a constant 28-volt output. The differential voltage circuit senses the generator output voltage and the airplane bus voltage then controls the APU line contactor. When the generator voltage is within the acceptable limits of the airplane bus voltage and the APU is ready to load, the APU line contactor will close and connect the APU generator to the airplane bus.

Reverse current protection circuitry opens the APU line contactor and disconnects the APU generator from the airplane bus if a reverse current condition occurs. If the condition corrects itself, the APU line contactor will re-close automatically. The paralleling circuit provides load sharing with the main engine generators. The current limiting circuit limits the maximum output of the generator so excessive loads such as engine starting will not overload the APU engine.

The overvoltage protection circuit senses APU generator output voltage. This circuit de-excites the generator and opens the APU line contactor if an overvoltage occurs. If the APU generator is disconnected for overvoltage, it will be necessary to select the generator switch OFF, then ON, to reset the generator.

The digital meter in the APU control panel provides a readout of generator load in amps. Generator voltage can be monitored by depressing the switch of either airplane volt-loadmeter.

The BUS ISOLATION relay provides the ability to isolate the battery, main engine starters, and external power from the airplane's electrical system. It is only functional when the APU generator is on-line and it is manually selected with the BUS ISO control switch.

BUS ISO is used during main engine start so that constant APU voltage is maintained at the airplane bus while the engines are started with the battery or an external power unit. Also, BUS ISO should be selected anytime the APU generator is on-line and an external power unit is connected to the airplane to prevent paralleling the two power sources. During bus isolation the battery cannot be charged from the APU and main engine generators.

AUXILIARY POWER UNIT

The Auxiliary Power Unit supplies electrical power to airplane and mission equipment systems. The APU may be operated, on the ground only, by itself or in parallel with the airplane's generators. The APU system is located aft of the aft pressure bulkhead and consists of a gas turbine engine which is controlled by an electronic sequence unit (ESU), a starter generator, and a regulator controller. A control head in the cockpit provides start and stop commands as well as monitoring generator output and APU engine parameters. The APU obtains fuel from the main gravity feed line of the left main fuel tank.

NOTE

If APU FAIL annunciator is illuminated, the display reads a code which indicates the cause of shutdown and should be recorded for maintenance purposes.

The ESU provides circuit protection and directs the APU main functions. The ESU also monitors APU engine RPM, EGT, and other engine parameters and automatically shuts down the APU if operating limitations are exceeded. The APU regulator controller provides voltage regulation and allows parallel operations between the APU and airplane generators when both systems are on-line.

APU operation is controlled from the cockpit with a START SELECT switch and the APU control head. The START SELECT switch, labeled START SELECT-NORM-COLD, selects APU starter source resistance. In the NORM position, the starter is powered through a series start resistor. In the COLD position, the starter is powered directly from the airplane bus without the start resistor. The START SELECT switch should be in the NORM position to start the APU when it is

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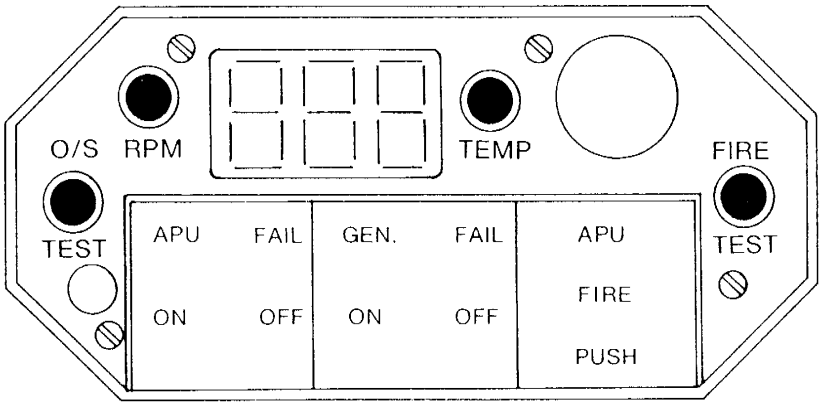
above approximately 10°F. If the APU is cold soaked in an ambient temperature below approximately 10°F, the START SELECT switch should be placed in the COLD position before starting the APU.

An APU start is initiated by pressing the APU switch after which the ESU controls the start sequence. RPM or EGT may be monitored by pressing the momentary switches labeled RPM and TEMP. When neither of these switches is pressed, the display indicates starter generator current. The APU generator is selected on-line or off-line by pressing the GEN switch. Normal APU shutdowns are initiated by pressing the O/S TEST switch which electrically simulates an APU engine overspeed causing the ESU to command a shutdown. This use of the O/S TEST switch provides the only test of the O/S protection circuits.



APU Control Head

FIRE DETECTION SYSTEM

The fire detection system provides immediate warning in the event of a fire or an overheat condition. The system utilizes a temperature sensing tube mounted inside the APU upper shroud. Fire or overheat conditions are indicated by an APU FIRE - PUSH lighted switch in the APU Control Head. The APU is automatically shut-down when the APU FIRE - PUSH annunciator is illuminated except when FIRE TEST switch is pushed. Pressing the APU FIRE - PUSH switch also commands an APU shutdown.

The system can be tested by depressing the FIRE TEST switch prior to APU operation. The FIRE TEST switch is located on the right side of the APU Control Head. If the circuit is operating properly, the APU FIRE - PUSH switch/annunciator will flash.

AUXILIARY AIR CONDITIONER

CAUTION

Do not operate the Auxiliary Air Conditioning System above 20,000 feet.

An electric auxiliary air conditioner located in the aft fuselage provides additional cooling air. Cooling air from the auxiliary air conditioner is introduced directly into the cabin where it is circulated by blowers. The auxiliary air conditioner is manually controlled by a switch in the cockpit labeled AIR COND-OFF-ON.

HANDLING, SERVICING, & MAINTENANCE

COOLING SYSTEM

Refer to the Super King Air Model 200 Series Maintenance Manual for procedures relative to evacuating the air conditioner refrigerant system, system cleaning, and preparation for freon charging.

The Auxiliary Air Conditioning System contains a total refrigerant charge between 1lb 8oz to 2lb 8oz.

The following pressures are measured with the system operating:

Suction (low) pressures must not exceed 38 PSIG.
The discharge (high) pressure should not exceed the following:

OAT (F°)	MAXIMUM PRESSURE (PSIG)
70°	180
80°	190
90°	205
100°	230
120°	320

Charging of the system using the sight gage may result in an overcharged system, and thus, may result in exceeding the above maximum pressures.

AUXILIARY POWER UNIT

For information relative to maintaining and servicing the APU refer to the Turbomach Installation/Maintenance Manual supplied with the APU.

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Beechcraft®

SUPER KING AIR®
B200/B200C LANDPLANES
(BB-1401, BB-1406 and BB-1463)

Pilot's Operating Handbook and
FAA Approved Airplane Flight Manual Supplement

for the

JAPAN AIR SYSTEM TRAINER

This Supplement is applicable to the following manual(s):

101-590010-147
101-590010-327

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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Revised: November, 1993
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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified for service as a Japan Air System training airplane in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

PLACARDS

Adjacent to Observer's Chair:

SEAT MUST BE LOCATED IN
OUTBOARD POSITION FOR
TAKEOFF AND LANDING

On Overhead Panel:

CAUTION

STALL WARNING IS
INOPERATIVE WHEN
MASTER SWITCH
IS OFF.
STANDBY COMPASS
IS ERRATIC WHEN
WINDSHIELD ANTI-ICE
AND/OR
AIR CONDITIONING
AND/OR
EFIS IS ON.

EMERGENCY PROCEDURES

No change.

NORMAL PROCEDURES

NOISE CHARACTERISTICS

Approach to and departure from an airport should be made so as to avoid prolonged flight at low altitude near noise-sensitive areas. Avoidance of noise-sensitive areas, if practical, is preferable to overflight at relatively low altitudes.

For VFR operations over outdoor assemblies of persons, recreational and park areas, and other noise-sensitive areas, pilots should make every effort to fly not less than 2000 feet above the surface, weather permitting, even though flight at a lower level may be consistent with the provisions of government regulations.

NOTE

The preceding recommended procedures do not apply where they would conflict with Air Traffic Control clearances or instructions, or where, in the pilot's judgement, an altitude of less than 2000 feet is necessary to adequately exercise his duty to see and avoid other airplanes.

The flyover noise level established in compliance with FAR 36 is: 75.4 dB(A)

No determination has been made by the Federal Aviation Administration that the noise level of this airplane is or should be acceptable or unacceptable for operation at, into, or out of any airport.

TIME, FUEL, AND DISTANCE TO CLIMB

CLIMB SPEED: 160 KNOTS

EXAMPLE:
OAT AT TAKEOFF ..... 28°C
OAT AT CRUISE ..... -10°C
AIRPORT PRESSURE ALTITUDE ..... 5433 FT
CRUISE ALTITUDE ..... 26,000 FT
INITIAL CLIMB WEIGHT ..... 12,500 LBS
TIME TO CLIMB (25-3) ..... 22 MIN
FUEL TO CLIMB (297-47) ..... 230 LBS
DISTANCE TO CLIMB (92-8) ..... 84 NM

NOTES: 1. ADD 90 LBS FUEL FOR START, TAXI, AND TAKEOFF.
2. FOR OPERATION WITH ICE VANES EXTENDED, ADD 20°C TO THE ACTUAL OAT BEFORE ENTERING THE GRAPH.

ASSOCIATED CONDITIONS:
PROPELLER SPEED .... 1900 RPM
POWER:
ITT ..... 770°C
OR TORQUE ..... 2230 FT-LBS

PRESSURE ALTITUDE ~ FEET

OUTSIDE AIR TEMPERATURE ~ °C

INITIAL CLIMB WEIGHT ~ LBS

TIME TO CLIMB ~ MINUTES

FUEL TO CLIMB ~ POUNDS

DISTANCE TO CLIMB ~ NAUTICAL MILES

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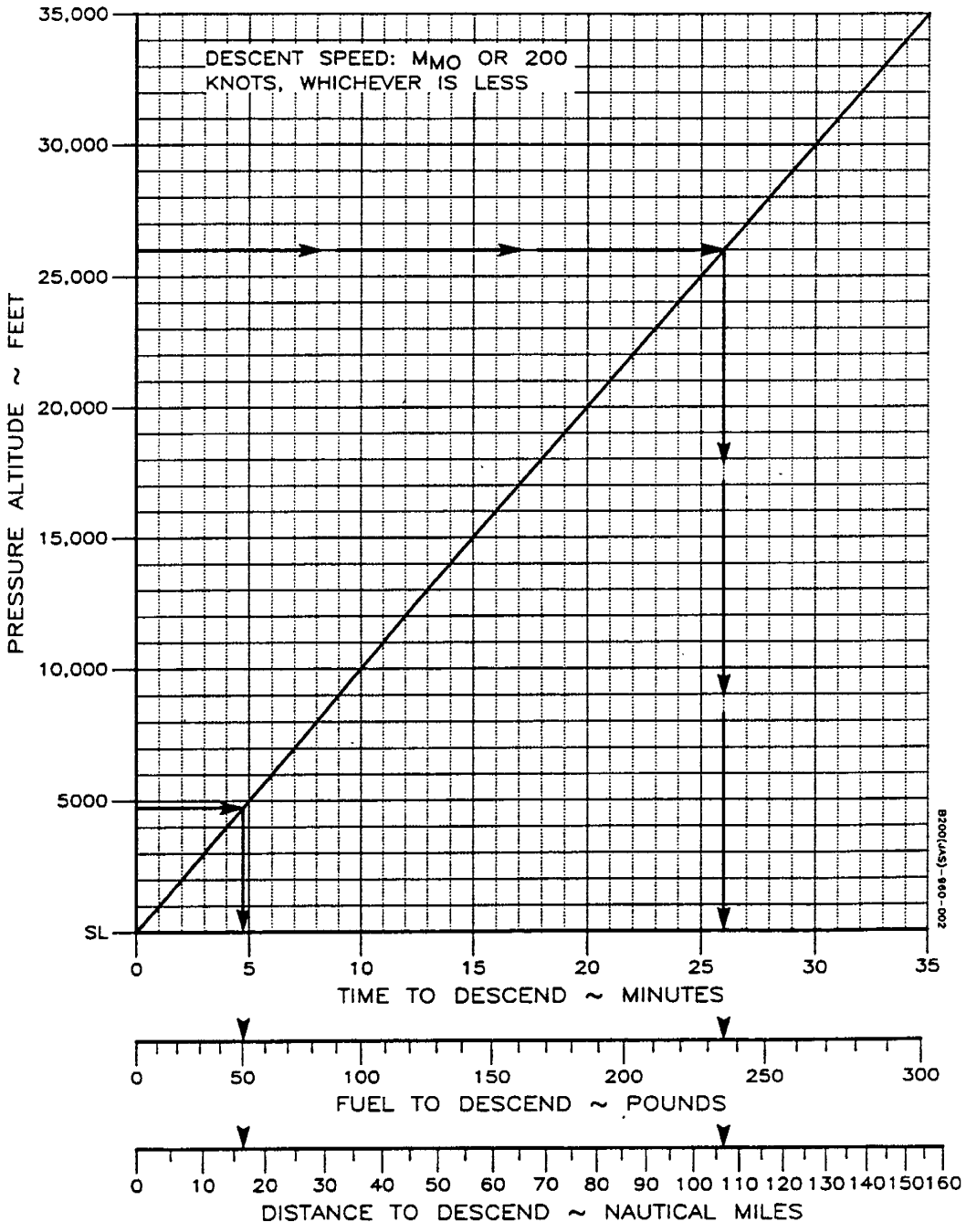
TIME, FUEL, AND DISTANCE TO DESCEND

ASSOCIATED CONDITIONS:

POWER.....AS REQUIRED TO DESCEND AT 1000 FT/MIN
LANDING GEAR....UP
FLAPS.....0%

EXAMPLE:

INITIAL ALTITUDE.....26,000 FT
FINAL ALTITUDE.....4732 FT
TIME TO DESCEND (26-5).....21 MIN
FUEL TO DESCEND (235-50).....185 LBS
DISTANCE TO DESCEND (107-17).....90 NM



WEIGHT & BALANCE/EQUIPMENT LIST

No change.

SYSTEMS DESCRIPTION

FLAPS

Two flaps are installed on each wing. Power is delivered from an electric motor to a gearbox mounted on the forward side of the rear spar. Flap travel is measured from the 0% (full up position), to 100% (full down position). The flaps are of the infinite selection type which allows for travel to any selected position from any pre-selected position. A side detent allows for quick selection of the approach flap position (40%).

The flaps are operated by a sliding switch handle on the pedestal just below the condition levers. Flap travel is registered on an electric indicator located on the top of the pedestal.

Flap motor power is provided by a 20-ampere circuit breaker while the control circuitry is powered by a 5-ampere circuit breaker. Protective circuitry is provided to detect either an open or a short in the command circuit. Should a malfunction occur, the protective device will open the control circuit preventing a runaway flap.

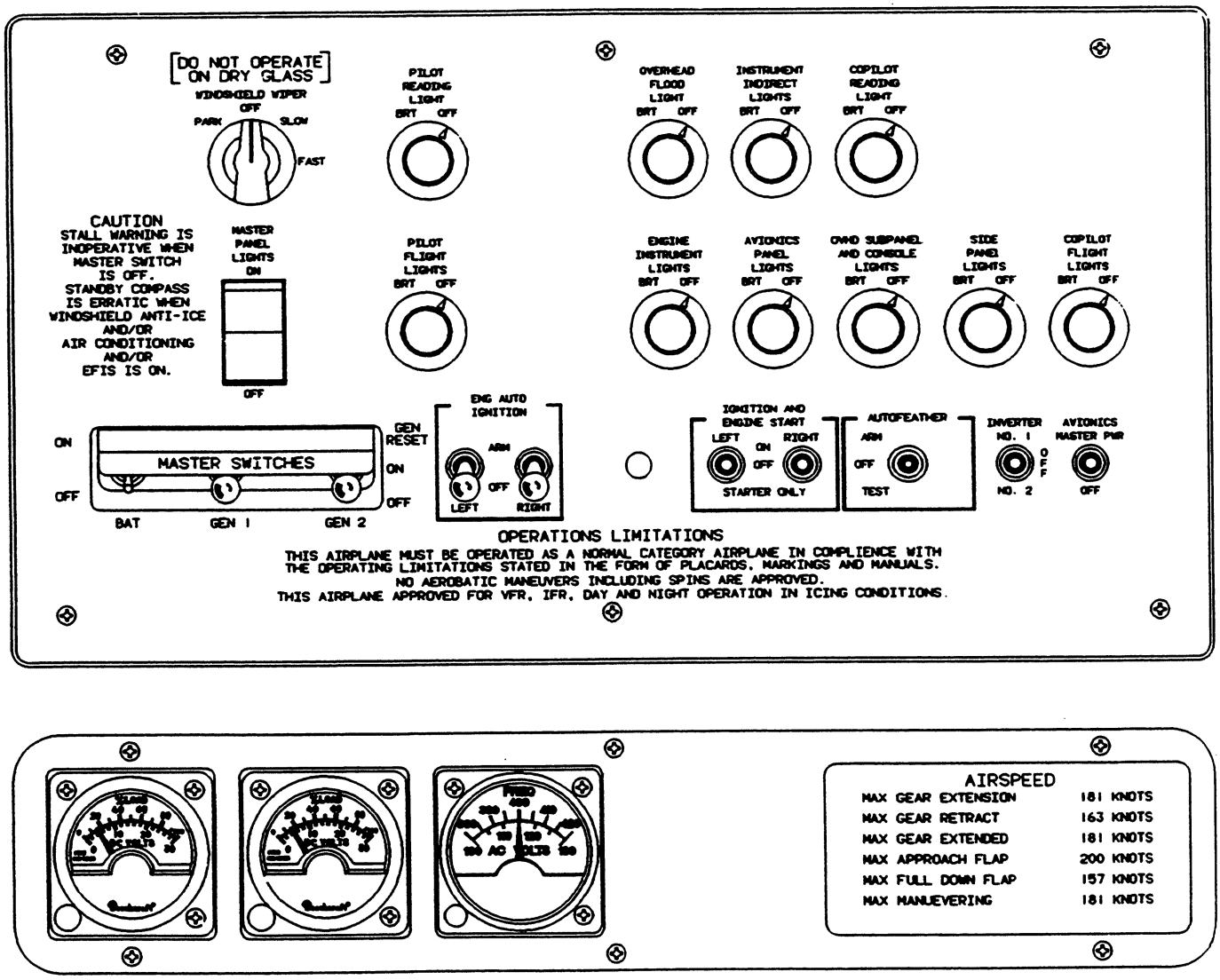
Lowering the flaps will produce these results:

ATTITUDE - Nose Up

AIRSPEED - Reduced

STALL SPEED - Lowered

TRIM - Nose-down Adjustment Required to Maintain Attitude



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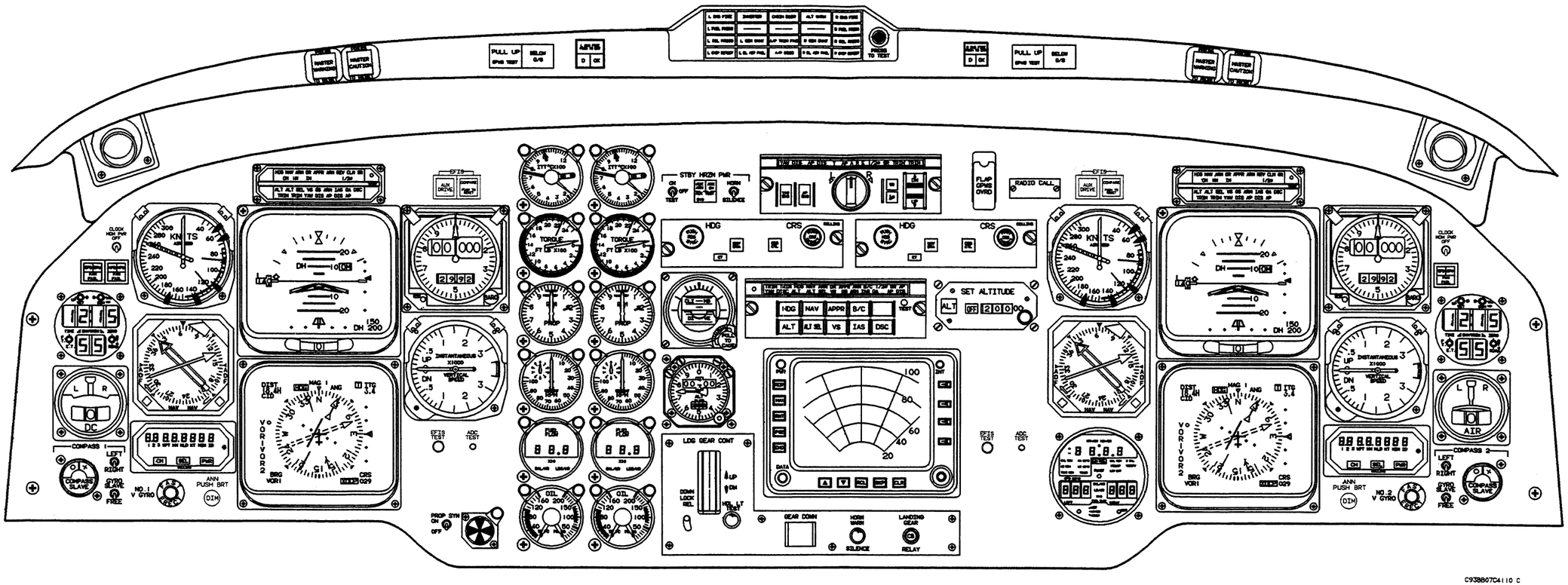
OVERHEAD LIGHT CONTROL PANEL



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INSTRUMENT PANEL
(BB-1401, BB-1406)

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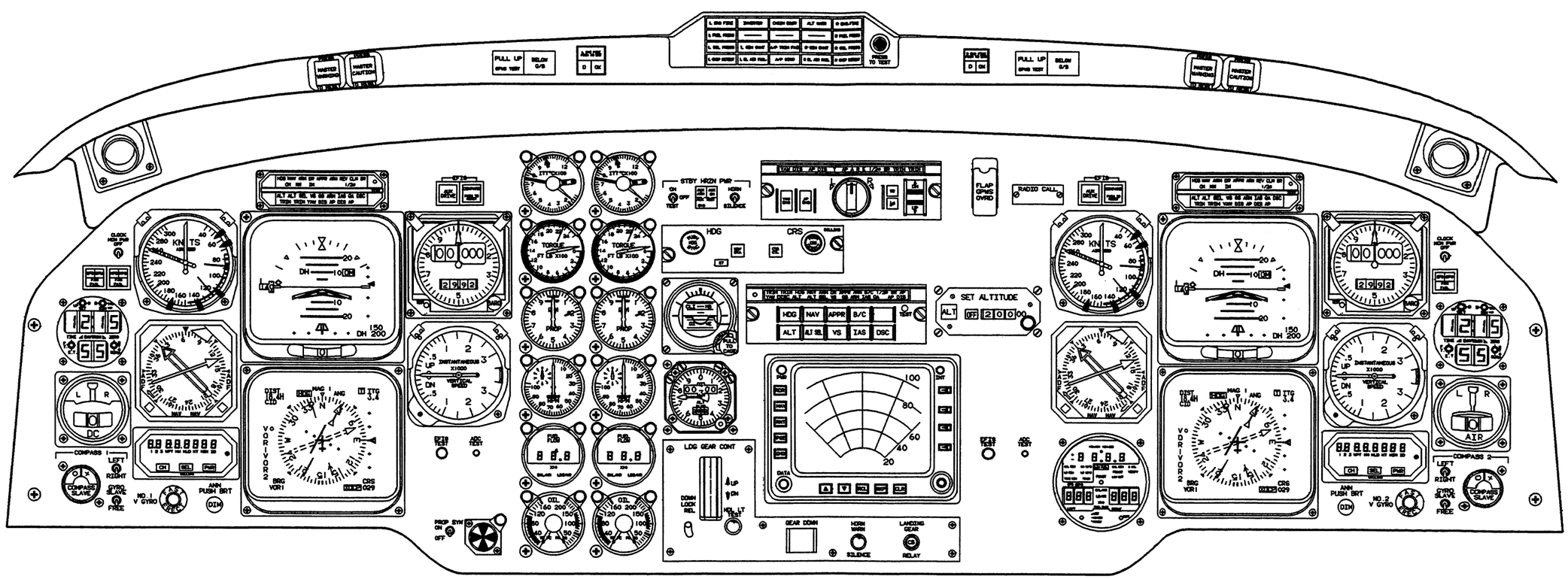
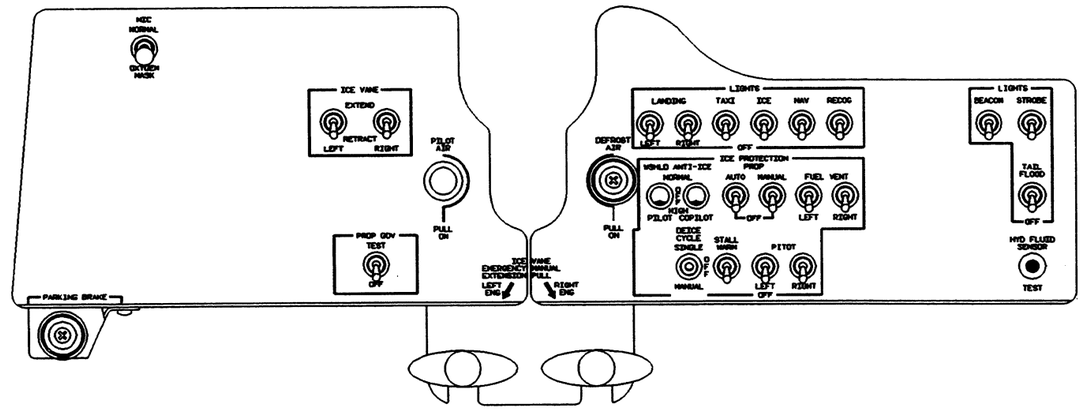


INSTRUMENT PANEL
(BB-1463)



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PILOT'S SUBPANEL

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SWITCHING MODIFICATIONS

The IGNITION and ENGINE START Switches, MASTER SWITCHES, ENG AUTO IGNITION, AUTOFEATHER, INVERTER, and AVIONIC Switches are located in the overhead light control panel. The following items are located on the avionic panel; GEAR WARN SILENCE Switch, GEAR DOWN Indicator Lights, LDG GEAR CONT, DOWNLOCK REL, HDL LT TEST Switch, and LANDING GEAR RELAY Circuit Breaker.

HANDLING, SERVICING & MAINTENANCE

No change.

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Beechcraft®

Super King Air®
B200, B200C, B200T & B200CT Landplanes

Pilot's Operating Handbook and
FAA Approved Airplane Flight Manual Supplement
for the
Collins FCS-65 Automatic Flight Control System
Category I
and
Collins EFIS-85B(14) (5-Tube)
Electronic Flight Instrument System (EFIS)

This Supplement is applicable to the following manual(s):

101-590010-147,
101-590010-327 and 101-590037-29

Airplane Serial Number: _____

Airplane Registration Number: _____

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Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane has been modified by installation of the Collins FCS-65 Automatic Flight Control System with the Collins EFIS-85B(14) 5-Tube Electronic Flight Instrument System in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth below. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

The Collins Pilot's Guide for the EFIS-85B(14)/86B(14) Electronic Flight Instrument System (5-Tube) No. 523-0775345, dated June 15, 1989 or later revision, is available from Collins General Aviation Division, 400 Collins Road NE, Cedar Rapids, Iowa 52498. It is the responsibility of the owner/operator to maintain the appropriate pilot's guide in current status.

LIMITATIONS

1. The Collins Pilot's Guide for the EFIS-85B(14)/86B(14) Electronic Flight Instrument System (5-Tube) No. 523-0775345, dated June 15, 1989 or later revision, must be immediately available to the flight crew.
2. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
3. Do not use autopilot below the following altitudes:
   a. Approach - 200 feet above ground level.
   b. Climb - 500 feet above ground level.
   c. All other operations - 1000 feet above ground level.
4. Do not use autopilot or yaw damper during takeoff or landing.
5. Pilot must be seated at the controls with the seat belt fastened during autopilot operations.
6. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
7. The standby attitude must be operational for takeoff.
8. The pilot's and copilot's EADI and EHSI must be operational in normal mode for takeoff. If a DPU is inoperative, DR XFR (Drive Transfer) must be selected for use on the failed side.
9. The attitude, heading, and DCP switches must be selected to normal mode for takeoff.
10. Do not use propeller speeds in the range of 1750-1850 RPM during ILS approaches.
11. The T-bar airspeed is not to be used as a primary source of airspeed information.

EMERGENCY PROCEDURES

AIR START

WARNING

Air start using the STARTER ASSIST procedures may momentarily cause the loss of attitude display on Electronic Flight Instrument System (EFIS) equipped airplanes, and lead to premature system failures. IF FLIGHT CONDITIONS DO NOT PERMIT THE TEMPORARY LOSS OF ATTITUDE REFERENCE, CONDUCT AIR START USING THE NO STARTER ASSIST PROCEDURES.

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For Training Purposes Only

CAUTION

The pilot should determine the reason for engine failure before attempting an air start. Do not attempt an air start if N_1 indicates zero.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an overtemperature condition.

WINDMILLING ENGINE AND PROPELLER (NO STARTER ASSIST)

1. Power Lever - IDLE
2. Propeller Lever - FULL FORWARD
3. Condition Lever - CUT-OFF
4. Fuel Firewall Valve - OPEN
5. Generator (inoperative engine) - OFF
6. Airspeed - 140 KNOTS MINIMUM
7. Altitude - BELOW 20,000 FEET
8. Engine Auto Ignition - ARM
9. Condition Lever - LOW IDLE
10. Power - AS REQUIRED (after ITT has peaked)
11. Generator - ON
12. Electrical Equipment - AS REQUIRED

STARTER ASSIST

NOTE

EFIS power should be turned OFF prior to a STARTER ASSIST air start if conditions permit.

WARNING

Standby attitude and air data displays must be monitored during STARTER ASSIST air starts.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. Windshield Heat - OFF
5. Power Lever - IDLE
6. Condition Lever - CUT-OFF
7. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

8. Ignition and Start Switch - ON (up). Check IGNITION Annunciator - ON
9. Condition Lever - LOW IDLE
10. Ignition and Start Switch - OFF (N_1 above 50%)
11. Propeller Lever - AS REQUIRED
12. Power Lever - AS REQUIRED

13. Generator - ON
14. Engine Auto Ignition - ARM
15. Electrical Equipment - AS REQUIRED

AUTOPILOT/ELEVATOR TRIM MALFUNCTIONS

1. Hold control wheel firmly. Be prepared to accept control wheel loads.
2. Fully depress red DISC TRIM/AP YD switch on pilot's or copilot's control wheel.
3. Recover - The maximum altitude losses observed during malfunction tests were:

Climb	450 ft
Cruise	440 ft
Maneuvering	150 ft
Descent	450 ft
Approach	100 ft

NOTE

An elevator trim malfunction will activate the master warning flasher and the red trim annunciator. The master warning flasher may be delayed for small trim requirements.

If necessary, the autopilot can also be disengaged by:

1. Pressing the AP ENG switch on the autopilot control head. (Yaw damper remains engaged.)
2. Activating the PITCH TRIM switch on the pilot's or copilot's control wheel. (Yaw damper remains engaged.)
3. Pressing the go around (GA) switch. (Yaw damper remains engaged.)
4. Pulling the autopilot circuit breaker.
5. Moving the avionics master switch to the OFF position.
6. Moving the gang bar to the OFF position.

CAUTION

If the autopilot is overpowered in pitch, the electric elevator trim will operate in the direction opposing the overpower force.

AUTOMATIC AUTOPILOT DISENGAGEMENT

The following conditions will cause the autopilot to disengage or servo torque to go to approximately zero torque:

DISENGAGE

1. Major degradation, interruption, or failure of electric power.
2. Autopilot internal monitors;
   a. System integrity checks.
   b. Loss of vertical gyro.
   c. Pitch attitudes in excess of 30°.
   d. Roll attitudes in excess of 45°.

ZERO TORQUE

The autopilot also monitors pitch and roll rate and normal acceleration. Pitch rates in excess of 4°/sec, roll rates in excess of 10°/sec, or accelerations in excess of 1.0g cause autopilot servo torque to decrease to approximately zero torque until the rate or acceleration is below the above limits.

ENGINE FAILURE

1. Disengage autopilot.

2. Follow POH/AFM engine failure procedures.
3. Retrim.
4. Re-engage autopilot.
5. Maintain at least 120 KIAS until landing is assured.
6. Disengage autopilot for large power or airspeed variations.

EFIS MALFUNCTION

1. In the unlikely event of a number 1 dual fed electrical bus failure: Select the #2 inverter to provide AC power for attitude and heading information on the copilot's instruments. Use the copilot's instruments as back-up to the pilot's instruments.
2. In the unlikely event of a number 2 dual fed electrical bus failure: Select the #1 inverter to provide AC power for attitude and heading information on the pilot's instruments.
3. If the EFIS displays are missing or abnormal, refer to alternate instruments for usable data. The copilot's instruments are approved as back-up to the pilot's instruments. Additionally, a standby attitude indicator is installed. Refer to EFIS MALFUNCTIONS in the OTHER NORMAL PROCEDURES section of this supplement.
4. If failure of both Display Processor Units occurs: Select DR XFR to PLT or COPLT as desired. The selected EADI and EHSI will be driven by the Multi-function Processor Unit.

NORMAL PROCEDURES

BEFORE STARTING

EFIS

- EFIS POWER Switches - OFF

BEFORE TAXI

EFIS

1. EFIS POWER Switches PUSH ON - Illuminated PILOT, COPILOT, and MFD.
2. STBY HRZN PWR - ON
3. UNCAGE Standby Attitude Indicator (after sufficient time for spin-up).
4. SET Standby Altimeter.

BEFORE TAKEOFF (RUNUP)

EFIS TEST

EFIS TEST switch will select test mode on the EFIS system. (See Collins Pilot's Guide for TEST mode description.)

AUTOPILOT

1. Momentarily push and release TEST button on the flight control panel to initiate annunciator test.

NOTE

If TEST button is held too long, the Flight Control System will stay in Test mode. The TEST button must be momentarily pushed and released again to clear the Test mode.

2. Position pitch trim to the take-off position, turn the ELEV TRIM switch ON and engage the autopilot. Check that YAW and AP annunciators are illuminated.
3. With control wheel in the forward position and the autopilot engaged, operate pilot then copilot pitch trim switches in both directions to ensure that the autopilot disengages. The yaw damper remains engaged.
4. Center the control wheel and engage the autopilot. Apply forward pressure on control wheel. Note that pitch trim travels nose up. Apply rearward pressure on control wheel. Note that pitch travels nose down, and that the amber trim light comes ON.
5. Hold the control wheel and disengage the autopilot by depressing the DISC TRIM/AP YD button on the pilot's control

wheel to the first level. Note that the YAW DIS and AP DIS annunciators illuminate. Further depress the DISC TRIM/AP YD switch. Note that the ELEC TRIM OFF annunciator illuminates. Cycle the electric trim switch on the console, engage the autopilot, and repeat the check using the copilot's DISC TRIM/AP YD switch.

6. Engage yaw damper. Note YAW annunciator is illuminated. Check for additional resistance to movement of rudder pedals. Disengage yaw damper.
7. Reset pitch trim to the take-off position.
8. Move all primary flight controls through their full travel in both directions. Verify that controls move in proper direction and no restrictions to free movement are present.

ELECTRIC ELEVATOR TRIM

1. Verify the ELEV TRIM is ON.
2. Check operation of dual-element thumb switches as detailed in the POH/AFM.

WARNING

Operation of the electric trim system should occur only by movement of pairs of switches. Any movement of the elevator trim wheel while actuating only one switch denotes a system malfunction. If a malfunction of the electric trim system is indicated, electric trim must be disengaged and trim changes made with manual trim ONLY.

DIGITAL AIR DATA SYSTEM

1. Press and hold the pilot's cockpit-mounted ADC TEST switch for the air data computer. Check for the following:
   a. Air data instrument flags come into view, and the instruments go into the loss-of-data mode.
   b. Altitude pointer goes to the 250-foot mark.
2. Release the pilot's ADC TEST switch and check for the following:
   a. Data transmissions to the air data instruments are restored.
   b. All air data instrument flags go out of view.
3. Press and hold the copilot's cockpit mounted ADC TEST switch for the air data computer. Check the following:
   a. Air data instrument flags come into view, and the instruments go into loss-of-data mode.
   b. Altitude pointer goes to the 250-foot mark.
4. Release the copilot's ADC TEST switch and check the following:
   a. Data transmissions to the air data instruments are restored.
   b. All air data instrument flags go out of view.
5. Press and hold the TEST switch on the pilot's barometric altimeter, and check for the following:
   a. The warning flag comes into view in approximately 1/2 second.
   b. The altitude pointer goes to the 750-foot mark.

NOTE

The output from the altimeter to the transponder is inhibited during the self test.

6. Release the pilot's altimeter TEST switch, and check for the following:
   a. The flag goes out of view.
   b. The altitude pointer returns to the actual altitude.
7. Press and hold the TEST switch on the copilot's barometric altimeter, and check for the following:
   a. The warning flag comes into view in approximately 1/2 second.
   b. The altitude pointer goes to the 750-foot mark.

NOTE

The output from the altimeter to the transponder is inhibited during the self test.

8. Release the copilot's altimeter TEST switch, and check for the following:
   a. The flag goes out of view.
   b. The altitude pointer returns to the actual altitude.
9. Turn the BARO knob on the pilot's and copilot's altimeter and observe that the baroset digits and altitude pointer respond accordingly.
10. Push in BARO knob to obtain baroset values in in. Hg. Set the barometric display to the correct local pressure value. (Pull out BARO knob if display of baroset values in millibars is desired.)
11. Press and hold the ALT ALERT annunciator/switch on the altitude preselector/alerter, and check for the following:
   a. The ALT ALERT annunciator illuminates.
   b. The warning flag comes into view.
   c. Remote ALT annunciators on pilot's and copilot's EADI's illuminate.
12. Release the ALT ALERT annunciator/switch, and check for the following:
   a. The ALT ALERT annunciator extinguishes.
   b. The warning flag goes out of view.
   c. Remote ALT annunciators extinguish.

IN FLIGHT OPERATION

AUTOPILOT OPERATION

To engage the autopilot, press the AP ENG switch on the autopilot control panel. If a red fault annunciator is illuminated (AP, TRIM), it requires one operation of the AP ENG switch to clear the fault and another operation to engage the autopilot. If the fault does not clear, the autopilot will not engage.

An automatic pre-engagement test is performed each time the autopilot is engaged. This test requires approximately 2 seconds and is identified during the test by annunciation of red TRIM and AP.

The autopilot may be engaged in any reasonable attitude. When engaged with no lateral mode selected beyond 30 degrees bank, 17.5 degrees pitch up, or 10 degrees pitch down, the autopilot will return the airplane to these limits. No steering information will be presented on the flight director until a lateral mode is selected.

To disengage the autopilot, press the AP ENG switch or operate the DISC TRIM/AP YD DISC switch on the pilot's or copilot's control wheel to first detent. Refer to EMERGENCY PROCEDURES in this supplement for other means of disconnecting.

CONTROL WHEEL SYNCHRONIZATION (CWS)

Autopilot sync mode is controlled by the PITCH SYNC & CWS (control wheel synchronization) button on the control wheel. With the autopilot engaged, and in either manual or guidance mode, depressing the PITCH SYNC & CWS button cancels vertical modes and allows the pilot to maneuver the airplane without disengaging the autopilot. When the PITCH SYNC & CWS button is released, the autopilot maintains the new pitch attitude and follows commands received from the autopilot computer if in a lateral mode. CWS mode does not cancel GS when in APPR mode.

VERTICAL TRIM SWITCH

This control switch is used for vertical control of the flight control system. Activating the switch for more than one second with a vertical hold mode selected causes the vertical mode to be cancelled (except GA, GS, and ALT SEL) and the system to revert to basic pitch operation. Refer to special note 1.

1/2 BANK (1/2Φ)

In 1/2ϕ mode, the bank limit in HDG or NAV mode is reduced to approximately 1/2 the normal value. Capture of NAV or APPR mode clears the 1/2ϕ mode, if selected.

SOFT RIDE (SR)

SR mode is used to reduce the autopilot response to turbulent conditions. The pilot may use SR mode at his discretion. SR is cancelled by APPR mode and when engaging the autopilot.

YAW DAMPER OPERATION

The yaw damper is automatically engaged anytime the autopilot is engaged.

The rudder channel of the autopilot may be selected separately for yaw damping by the YAW ENG switch on the autopilot panel.

To disengage the yaw damper, operate the DISC TRIM/AP YD disc switch, or press the YAW ENG switch. The YAW ENG switch is disabled if the autopilot is engaged.

ELECTRIC PITCH TRIM OPERATION

The autopilot computer provides manual electric trim operated by the split pitch trim switches located on the pilot's and copilot's control wheels. Electric trim operation is annunciated by the amber TRIM annunciator when the trim speed is above the preset annunciation threshold for more than 5 seconds.

MODE SELECTION

All modes with the exception of GA are selected by depressing the PUSH ON/PUSH OFF switch on the mode control panel. Annunciators on the mode control panel and mode annunciator panel indicate the selected mode of operation.

The flight director system supplies steering commands for the pilot and the autopilot. When the autopilot is engaged, the pilot monitors autopilot performance on the attitude director indicator. When the autopilot is not engaged, the pilot flies the airplane manually in response to the attitude director indicator commands. A lateral mode must be selected to bring the command bars into view for manual flight director operation.

The EFIS-85B(14) Electronic Flight Instrument System replaces the conventional electro-mechanical attitude director (ADI) and horizontal situation indicators (HSI). The CRT displays (EADI & EHSI) provide the necessary information for the pilot or copilot to respond to flight director and navigation commands in the same manner as with conventional instruments. The system also includes additional features which allow the pilot or copilot to select alternate formats as a navigation aid or in the event of specific failures. The operator is cautioned that the primary displays (attitude and directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate modes is described in the Collins Pilot's Guide.

HEADING MODE (HDG)

Set desired heading and select HDG mode. The autopilot will turn the airplane to and maintain the selected heading. For proper operation, the HDG marker should not be displaced from the airplane heading by more than 150 degrees when the HDG mode is selected.

NAV MODE (NAV)

Set desired heading and course prior to selecting NAV mode. When NAV mode is selected, the HDG and NAV ARM annunciators will illuminate. The airplane will follow the selected heading until the centerline of the selected radio course is approached. HDG mode will then clear and the airplane will turn to track the beam centerline. The amber ARM annunciator will extinguish to indicate course capture and NAV will remain on. Crosswind correction, up to 30 degrees, is automatically computed after course capture.

APPROACH MODE (APPR)

When the APPR mode is selected, localizer capture is the same as described in NAV mode. The APPR mode also provides glideslope arm and capture. After localizer beam capture is achieved, the system will automatically arm to capture the glideslope. The green GS and amber ARM annunciators will illuminate at this time. When the glideslope is intercepted, either from above or below, the amber ARM annunciator will extinguish indicating GS capture. Crosswind correction, up to 30 degrees, is automatically computed after course capture.

CAUTION

Premature glideslope and localizer capture may occur in the APPR ARM mode if selected when the airplane is far from the localizer centerline. Therefore, use the HDG mode to position the airplane near the localizer intercept before selecting the APPR ARM mode.

NOTE

A vertical mode, if selected, will automatically clear at glideslope capture.

All approaches must be made in APPR or B/C mode.

EFIS-85B(14) equipped airplanes display the linear deviation with a full-scale offset at 10 NM in NAV mode, and 2 NM in APPR mode.

BACK COURSE MODE (B/C)

When the B/C mode is selected, localizer capture follows the same sequence as a front course approach. Correct lateral steering information is displayed on the flight director, but vertical information is displayed as a function of the vertical mode selected.

NOTE

The inbound (front) course must be selected on the course indicator.

ALTITUDE HOLD MODE (ALT)

ALT can be selected during all modes of operation except after glideslope capture in APPR mode. Deviations from the selected altitude are displayed on the flight director.

NOTE

The autopilot maintains the airplane at the selected altitude by changing the pitch attitude of the airplane. The pilot must maintain proper power settings to assure a safe airspeed.

INDICATED AIRSPEED MODE (IAS)

The IAS mode may be selected during all modes of operation except after glideslope capture in APPR mode. Deviations from the airspeed at the time the mode was selected are displayed on the flight director as pitch commands. Selecting the IAS mode synchronizes the airspeed bug to the existing airspeed. Changes in IAS may be made utilizing the vertical trim switch.

VERTICAL SPEED MODE (VS)

The VS mode may be selected during all modes of operation except after glideslope capture in APPR mode. Deviation in vertical speed provides commands to maintain the vertical speed present at the time of mode selection. Rate of ascent/descent should be established for 10 seconds before selecting VS mode.

ALTITUDE PRESELECT MODE (ALT SEL)

Dial in the preselected altitude on the altitude preselector/alerter and press the ALT SEL mode on the mode selector. The ALT SEL annunciator will illuminate. Any vertical mode except altitude hold may be used in conjunction with ALT SEL.

An aural alert tone sounds and annunciator lights on the altitude panel and flight director illuminate when the airplane reaches 1000 ft from the preselected altitude. At 300 feet from the preselected altitude, the altitude annunciator extinguishes. Thereafter, the alert tone and altitude annunciators will warn when the airplane deviates more than 300 feet from the preselected altitude. As the airplane reaches the preselected altitude, the ALT SEL annunciator extinguishes and ALT illuminates.

The use of vertical trim switch or altitude preselect dial when the airplane is within 1000 ft of the preselected altitude and before altitude capture will cancel the ALT SEL mode.

DESCENT MODE (DSC)

When the DSC mode is selected, the autopilot begins a gradual descent, stabilizing at a rate-of-descent of 950 to 1350 feet per minute. The stabilized rate-of-descent will vary depending on altitude, with the lower rates-of-descent occurring at high altitude and the higher rates-of-descent occurring at lower altitudes. After the vertical speed has stabilized (DSC annunciator extinguished and VS annunciator illuminated), the pilot may vary the rate-of-descent by operation of the vertical trim switch.

On those airplanes equipped with the digital air data system, the altitude preselect mode is automatically activated when the DSC mode is selected.

NOTE

Desired altitude must be selected on the altitude alerter prior to selection of the DSC mode.

GO-AROUND MODE

The GA mode is activated by the GA switch. When the GA switch is pushed, while in any lateral mode, the autopilot will disengage and a 7° pitch up, wings level attitude will be displayed on the flight director. GA annunciator will illuminate and remain on until a lateral mode is re-selected.

Operation of the PITCH SYNC & CWS switch will clear the GA mode and synchronize the vertical command to the airplane attitude. Re-engagement of the autopilot while in go-around clears the GA mode and synchronizes the autopilot commands to the airplane pitch attitude at the time of engagement and continues to maintain wings level.

Selection of a lateral mode will also clear the go-around mode and synchronize the vertical commands to airplane pitch attitude at the time of mode selection.

SPECIAL NOTES

1. Vertical Trim Switch Operation
   a. In basic pitch mode, with the autopilot or a lateral mode selected, activating the vertical trim switch gives a 0.5° command followed by a continuous slew of 1.0°/second.
   b. In VS or DSC mode, the synchronized vertical speed is changed in 200 ft/min increments with each momentary activation of the vertical trim switch.
   c. In IAS or Climb mode, the synchronized airspeed is changed in one-knot increments with each momentary activation of the switch.
   d. In the ALT mode, the synchronized altitude is changed by 25 ft with each momentary activation of the switch.
2. Basic Attitude Mode - With no lateral mode selected, the steering display is biased out of view and the ADI is used as a standard artificial horizon. No computed steering information is displayed on the ADI.
3. Bank command limit is ±30° with turn knob, ±25° in lateral modes, ±15° in APPR (after GS capture), and ±10° in NAV track.
4. Vertical mode information is displayed by pushing the desired vertical mode on the mode select panel. If a mode other than GS is selected during manual flight director operation, it will automatically clear when the autopilot is engaged.
5. To synchronize the vertical commands in flight director function to the airplane attitude, actuate the PITCH SYNC & CWS switch on the control wheel after any lateral mode has been selected.
6. When making any coupled approach, primary flight instruments must be monitored by the pilot. Presence of a VOR/LOC/GS flag must be considered as a warning of system or signal failure.
7. When flying inbound toward a VOR, do not set up a VOR capture problem within approximately 5 miles of the VOR. Depending on the speed of the airplane, the system may not have time to solve the problem and stabilize on course before passage over the station.
8. Course may be changed over a VOR station when operating in NAV mode, as long as the course change is not more than 30 degrees. If the course change is more than 30 degrees, the HDG mode should be selected to establish a new intercept and then NAV mode should be re-selected to set up a new capture.
9. For optimum NAV mode operation, select intercept angle so the system maintains straight and level flight in the NAV mode a minimum of 30 seconds prior to capture.
10. When making a station passage in NAV mode, the computer may automatically switch into heading memory sub-mode to allow a smooth station passage. During this time, the dead reckoning (DR) annunciator will illuminate.
11. Maximum recommended capture angles are 90 degrees or less in NAV mode and 60 degrees or less in APPR mode.
12. Fly all coupled approaches at or above 120 KIAS, flaps 40%, and landing gear extended.
13. Always check the mode select panel annunciator display to verify that proper switching has taken place.
14. Do not select any vertical modes until 3 minutes after applying power.
15. The vertical gyro connected to the autopilot system is not transferred if optional VG switching feature is installed.

SHUTDOWN AND SECURING

EFIS

1. EFIS ON/OFF Switches - OFF (before FUEL CUTOFF)
2. CAGE Standby Attitude Indicator.
3. STBY HRZN PWR - OFF

OTHER NORMAL PROCEDURES

EFIS MALFUNCTIONS

COMPARATOR WARNING

Illumination of the amber COMPARE annunciator on the instrument panel along with a yellow HDG annunciation on the EHSI, and/or a yellow PIT (pitch) or ROL (roll) annunciation on the EADI indicates a mismatch of pilot's and copilot's heading or attitude data.

If mismatch is EHSI HDG:

1. Establish airplane in straight and level, unaccelerated flight.
2. Compare magnetic compass and cross-side heading indications to determine actual heading.
   - If single system is in error, associated HEADING SLAVE-FREE Switch - FREE, then SLAVE.
   - If heading cannot be determined, both HEADING SLAVE-FREE Switches - FREE, then SLAVE.
   - If EHSI HDG is still annunciated - HDG Switch - ALL ON, NO. 1 or NO. 2, as required.

If mismatch is EADI PIT or ROL:

3. Cross-check attitude indications with cross-side indications and standby attitude indicator to determine faulty system.
4. ATT Switch - ALL ON, NO. 1 or NO. 2, as required.

EADI FAILURE

1. Select pilot or copilot Composite switch to the CMPST position.
2. The ALTN HSI on MFD switch may then be selected to PLT or COPLT position to provide an HSI display on the MFD, if desired.
3. Pull the appropriate circuit breaker for the affected indicator.

EHSI FAILURE

1. Select pilot or copilot Composite switch to the CMPST position.
   or;
2. Select ALTN HSI on MFD switch to PLT or COPLT position to provide an HSI display on the MFD.
3. Pull the appropriate circuit breaker for the affected indicator.

EADI AND EHSI FAILURE

Display Processor Unit failure is indicated by the DPU FAIL flag on EADI and EHSI displays.

- Select the DR XFR switch to the appropriate PLT or COPLT position.

NOTE

The EADI and EHSI will be driven by the Multi-function Processor Unit and the amber AUX DRIVE annunciator will illuminate on the selected side and the MFD will display the selected EHSI display.

MFD FAILURE

There is no back-up for MFD failure. Weather modes are available on the pilot and copilot EHSI.

DCP FAIL

Display Control Panel failure is indicated by the DCP flag on the EADI.

- Select DCP switch to PLT or COPLT position.

NOTE

This will cause the DCP on the selected side to control the cross-side DPU navigation data.

ATTITUDE FAILURE

Attitude failure is indicated by ATT flag on the appropriate EADI. The autopilot will disengage if it is selected to source that fails.

- ATT Switch - ALL ON, NO. 1 or NO. 2, as required.

HEADING FAILURE

Heading failure is indicated by HDG flag on the appropriate EHSI.

- HDG Switch - ALL ON, NO. 1 or NO. 2, as required.

EFIS FAN FAILURE

The pilot and copilot electronic flight display (EFD) fan will illuminate EFD FAN FAIL. The multi-function display fan will illuminate MFD FAN FAIL. The pilot and copilot display processor unit fan will illuminate DPU FAN FAIL. The multi-function processor unit fan will illuminate MPU FAN FAIL.

- Monitor the affected unit for further performance degradation. If a system component failure occurs, pull the appropriate circuit breaker.

PERFORMANCE

No Change

WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

The Collins FCS-65 is an integrated flight control system which can be functionally divided into two general subsystems.

1. Flight Instrument System (FIS)
   The FIS consists of an attitude director indicator (ADI), a horizontal situation indicator (HSI), and associated components. The ADI presents attitude data, raw radio information, and steering commands. The HSI presents the navigation situation. EFIS ON/OFF SWITCHES - Control power to the EFIS symbol generators. The switches should be selected OFF during engine start to protect the EFIS system from low/transient voltages.
2. Flight Control System (FCS)
   The FCS consists of an autopilot computer, mode select panel, air data sensor, three primary servos, a vertical reference and a trim servo. The system also requires compass and navigation system inputs from the avionics system.

EFIS SWITCHES

EFIS POWER

- With airplane power on, the PILOT, COPILOT, and MFD EFIS switches apply power to the respective EFIS equipment.

SOURCE SELECTION

- PLT CMPST (pilot composite) will select composite mode of operation on the pilot EADI and EHSI. A composite format replaces the normal EADI and EHSI display.
- COPLT CMPST (copilot composite) will select composite mode of operation on the copilot EADI and EHSI. A composite format replaces the normal EADI and EHSI display.
- DR XFR (drive transfer) will select the MPU to drive the EADI and EHSI on the pilot and copilot side as selected. The amber AUX DRIVE annunciator will illuminate on the selected side.
- DCP (display control panel) will cause the DCP on the selected side to control the cross-side DPU navigation data.
- CMD DSPL (command display) will select airplane symbol as VBAR or XPT, as selected.
- ALTN HSI ON MFD will transfer the pilot or copilot EHSI display to the multi-function display (MFD).
- ATT ALL ON will select NO. 1 pitch and roll information or NO. 2 pitch and roll information to be displayed on pilot and copilot EADI. ATT 1 or ATT 2 will be displayed (yellow) on both EADI displays.
- HDG ALL ON will select NO. 1 heading information or NO. 2 heading information to be displayed on pilot and copilot EHSI. MAG 1 or MAG 2 will be displayed (yellow) on both EHSI displays.
- AP FLT DIR (if two autopilot flight directors are installed) will select autopilot flight director NO. 1 or NO. 2. AP/L or AP/R will be displayed on both pilot and copilot EADI displays.

NOTE

See OTHER NORMAL PROCEDURES for further description of source selection switches.

ANNUNCIATORS

Annunciation is displayed on the mode select panel and mode annunciator panel. Annunciation is color-keyed for better recognition of mode status with green for active condition, amber for arm condition, and red for warning condition. The following table shows the annunciations and their association with each other:

ANNUNCIATOR	CONDITION TO ACTIVATE	COLOR ACTIVE	COLOR ARM	COLOR WARNING
HDG	Shows heading mode selected. Also, will light automatically if NAV or APPR is selected, but capture has not occurred.	Green		
NAV	Selection of navigation mode.	Green		
APPR	Selection of approach mode.	Green		
GS	Indicates glideslope mode.	Green		
AP	Indicates autopilot is engaged.	Green		
ALT	Selection of altitude hold mode.	Green		
IAS	Selection of IAS hold mode.	Green		
VS	Selection of vertical speed hold mode.	Green		
GA	Selection of go-around mode.	Green		
YAW	Indicates yaw damper is engaged.	Green		
DR	Computation is in Heading Memory.	Green		
ARM	Indicates an armed condition prior to capture.		Amber	
DSC	Indicates the selection of descent mode.	Green		
1/2ϕ	Indicates selection of half bank mode.	Green		
SR	Indicates selection of soft ride mode.	Green		
B/C	Indicates selection of back course mode.	Green		
DIS	Disengage annunciation for YAW and AP. Flashes for 5 seconds, then steady indication when system is disengaged. Lights steady when PITCH SYNC button is depressed.		Amber	
ALT SEL	Shows that altitude preselect is armed for automatic capture.	Green		
AP	Indicates first failure in dual safety functions - Pilot should take over and disengage.			Red
TRIM	Indicates Trim Servo failure in autopilot or manual electric operation.			Red
TRIM	Indicates Autopilot Elevator Trim-in-Motion, when trim speed is above the preset threshold for more than approximately 3 seconds.		Amber	

ANNUNCIATOR NOTES:

1. During annunciator test, CLM annunciator will illuminate even though the function is not installed.
2. When a mode annunciator is flashing, it indicates that signals used for that mode are not usable.
3. EFIS modes flash briefly when selected.

HANDLING, SERVICING & MAINTENANCE

1. If a Collins altimeter is installed, one additional static-line drain is installed. It is located inside the nose avionics compartment in the pilot's static air line that provides static pressure to the digital air data computer. This drain should be opened to release any trapped moisture at each 150-hour inspection, and after exposure to visible moisture on the ground. It must be closed after draining.

2. If a Collins EFIS-85B(14) is installed, fans are installed to aid in cooling of the Display Processor Unit (DPU-85), Multi-function Processor Unit (MPU-85), and the pilot CRT displays. At each 150-hour inspection and at any sign of abnormal equipment operation, verify these fans are operating properly.

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BEECHCRAFT® SUPER KING AIR®

B200 & B200C LANDPLANES

(S/N BB-1425 and after), (S/N BL-138 and after)

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

HONEYWELL SPZ-4000 AUTOMATIC FLIGHT CONTROL SYSTEM

CATEGORY I

and

HONEYWELL EDZ-605 ELECTRONIC FLIGHT INSTRUMENT SYSTEM (EFIS)

THIS SUPPLEMENT IS APPLICABLE TO PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL:
P/N 101-590010-147

Airplane Serial Number:_____

Airplane Registration Number:_____

FAA Approved by:

A. C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of the Honeywell SPZ-4000 Automatic Flight Control System, alone or in conjunction with the Honeywell Electronic Flight Instrument System, in accordance with Beech-approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

Honeywell Pilot's Operating Handbooks for the Electronic Flight Instrument System are available from Honeywell Flight Systems, P.O. Box 29000, Phoenix, Arizona 85038. It is the responsibility of the owner/operator to maintain the appropriate Pilot's Operating Handbook in current status.

LIMITATIONS

1. During autopilot operations, pilot must be seated at the controls with seat belt fastened.
2. Maximum speed limit for autopilot operation is unchanged from the airplane maximum airspeed limit (V_{MO}/M_{MO}).
3. Do not use autopilot under 200 feet above the terrain. This includes takeoffs and landings.
4. Do not use autopilot or yaw damper during takeoff or landing.
5. Do not use propeller in the range of 1750-1850 rpm during coupled ILS approach.
6. Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the autopilot is to be used.
7. Set desired heading and select HDG on the Flight Director Mode Selector prior to changing the VOR/ILS receiver frequency if that receiver is the navigation source and the NAV, APR, or BC mode is selected.
8. Prior to dispatch, the EDZ-605 system must be verified to be operational by a satisfactory preflight test as contained in the NORMAL PROCEDURES.
9. A copy of the Honeywell Pilot's Operating Handbook for the EDZ-605 Electronic Flight Instrument System (EFIS) with Multifunction Display (MFD), 3-Tube System Pub. No. A62-0088-00-00, Sept, 1987 or later revision, must be available to the pilot while operating this system.
10. The EADI and EHSI must be operational in the normal mode for takeoff.
11. Dispatch is prohibited with the EFIS FAN FAIL or MFD FAN FAIL annunciator lights illuminated.
12. The composite mode is to be used in IFR operations only after a failure of an EADI or EHSI, the Flight Director is to be used for instrument approaches.
13. At least one symbol generator must be installed and operational for dispatch.
14. The accuracy of the VNAV system has not been shown to meet the requirements of Advisory Circular 90-45A and is therefore not to be used as the primary source of vertical guidance information. VNAV is to be used only when proceeding directly toward or away from a VORTAC or RNAV waypoint.

EMERGENCY PROCEDURES

AIR START

STARTER ASSIST

CAUTION

The pilot should determine the reason for engine failure before attempting an air start.

Above 20,000 feet, starts tend to be hotter. During engine acceleration to idle speed, it may become necessary to move the condition lever periodically into CUT-OFF in order to avoid an over-temp condition.

1. Cabin Temp Mode - OFF; Blower - AUTO; Aft Blower - OFF
2. Radiant Heat - OFF
3. Radar - STANDBY or OFF
4. EFIS ON/OFF Switches - OFF (if conditions permit)
5. Windshield Heat - OFF
6. Power Lever - IDLE
7. Condition Lever - CUT-OFF
8. Fuel Firewall Valve - OPEN

NOTE

If conditions permit, retard operative engine ITT to 700°C or less to reduce the possibility of exceeding ITT limit. Reduce electrical load to minimum consistent with flight conditions.

9. Ignition and Start Switch - ON (up), Check IGNITION Annunciator - ON
10. Condition Lever - LOW IDLE
11. Ignition and Start Switch - OFF (N_1 above 50%)
12. Propeller Lever - AS REQUIRED
13. Power Lever - AS REQUIRED
14. Generator - RESET, then ON
15. Engine Auto Ignition - ARM
16. EFIS ON/OFF Switches - ON
17. Electrical Equipment - AS REQUIRED

AUTOPILOT MALFUNCTION

If the AP TRIM FAIL and MASTER WARNING annunciators illuminate while the autopilot is engaged, immediately disconnect while restraining the control wheel for a possible out-of-trim force. Disengagement can be accomplished by any one of the following:

1. Press (momentarily) to the first level the pilot's or copilot's AP/YD & TRIM DISC button on the control wheel.
2. Open the autopilot (AP CONTROL) circuit breaker.
3. If necessary, the autopilot may be overpowered by either pilot.

CAUTION

Trim will run to oppose pilot effort.

MAXIMUM ALTITUDE LOSSES DURING MALFUNCTION TESTS WERE:

CONFIGURATION	ALTITUDE LOSS
Climb	130 ft
Cruise	310 ft
Maneuvering	140 ft
Descent	510 ft
Approach/ILS Coupled	90 ft
One-engine-inoperative Approach/ILS Coupled	80 ft

ENGINE FAILURE

— Disengage the autopilot, retrim the airplane and reengage the autopilot. Maintain at least 120 knots for one-engine-inoperative approach.

EFIS FAILURE

If the EFIS displays are missing or abnormal, refer to alternate instruments for usable data. The copilot instruments are approved as backup to the pilot instruments. EFIS reversionary modes may be selected as follows:

1. In the event of an Electronic Horizontal Situation Indicator (EHSI) failure:

   — Turn off the HSI/DIM control on the Display Controller. The EADI will provide a composite attitude/navigation display.

   or;

   — Select HSI on the Mode Controller. The MFD will display HSI information.

2. In the event of an Electronic Attitude-Director Indicator (EADI) failure, turn off the ADI/DIM control and adjust the WX/DIM control on the Display Controller. The EHSI will provide a composite attitude/navigation display.

CAUTION

A suitable composite attitude/navigation display on the EHSI requires that the WX/DIM control on the EFIS Display Controller be adjusted to provide an appropriate earth/sky background color for the attitude indicator.

3. In the event of primary symbol generator failure (both EADI and EHSI go blank or annunciate SG FAIL):

   — Select the SG position on the MFD controller. The Multifunction generator will drive the EFIS and the MFD will blank.

4. Use flight director for instrument approaches in the composite mode.

NOTE

The composite format utilizes a course deviation indicator with automatic reverse course steering. Always fly toward the needle to intercept course centerline.

CAUTION

Flight Director mode annunciation is not present in composite format. Refer to the Flight Director Mode Selector for mode engagement annunciation.

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5. Interpretation of the Course Deviation Indicator (CDI) in the composite mode.

    a. If a localizer is the selected navigation source, set the desired course to the normal front course. The CDI uses reverse course sensing to indicate the location of course centerline relative to the nose of the airplane.
    b. If other than a localizer is the selected navigation source, the CDI does not use reverse course sensing. When the airplane heading is more than 90° from the selected course, the CDI displacement is opposite the location of course centerline relative to the nose of the airplane.

6. EFIS FAN FAIL or MFD FAN FAIL Annunciator Illuminated.

Illumination of the EFIS FAN FAIL or MFD FAN FAIL annunciator (located above the EADI) indicates that the respective display indicator cooling system has failed. If this occurs on the ground, do not take off. If this occurs while in flight, the following actions are recommended:

a. Monitor the Display Indicator for possible failures.
b. Consider using one of the Composite or Reversionary modes, or turn the indicator off.
c. Lower the lighting intensity on the display.
d. Repair on the ground prior to the next flight.

NORMAL PROCEDURES

The Automatic Flight Control System (AFCS) is an integrated Flight Director, Autopilot, and Yaw Damper System. Flight Director modes are selected using push-button switches on the Flight Director Mode Selector. The Autopilot and Yaw Damper are engaged using push-button switches on the Autopilot Controller. Whenever the autopilot is engaged, it will automatically couple to the selected flight director mode. The autopilot may also be controlled manually using the Pitch Wheel and Turn Knob on the controller.

BEFORE ENGINE STARTING

EFIS

— EFIS ON/OFF Switches - OFF

AFTER STARTING, AND TAXIING

EFIS

— EFIS ON/OFF Switches - ON

Altitude Alert

1. Altitude Alert Controller (or VNAV if installed) — Set to 1500 feet above altitude indicated on pilot's altimeter.
2. Pilot's baro set — ADJUST to increase altitude toward alerter setting.
3. VERIFY — alert light on pilot's altimeter illuminates and Aural Alert sounds at 1000 feet from alerter setting.
4. VERIFY — alert light is extinguished at 250 feet from alerter setting.
5. Pilot's baro set — ADJUST toward original setting.
6. VERIFY — alert light illuminates and horn sounds at 250 feet from alerter setting.
7. Altitude Alert Controller (or VNAV if installed) — SET to initial clearance altitude.
8. VERIFY — alert light extinguished.

Autopilot

Perform prior to each flight to verify proper operation of torque limiters, torque monitors, autopilot electric trim system, and failure annunciators.

1. Autopilot Controller TRIM UP, TRIM DN Annunciators — CHECK not illuminated. A steady illumination of TRIM UP or TRIM DN annunciator indicates that automatic synchronization is not functioning and autopilot should not be engaged.
2. Turn Knob — IN CENTER DETENT POSITION
3. Elevator Trim Control Switch — ON
4. Control Wheel — TO MID TRAVEL

5. Autopilot Controller AP Button — PRESS. AP ENGAGE and YD ENGAGE annunciators on autopilot controller will flash. Servo clutches will engage. FD flag on ADI/EADI in view.
6. Control Movement — CHECK when pushed ON.

   a. Rudder Pedals — OVERPOWER SLOWLY. YD ENGAGE annunciator stops flashing.
   b. Control Wheel — OVERPOWER SLOWLY in both pitch and roll axes. AP ENGAGE annunciator stops flashing. FD flag retracts.

WARNING

If autopilot or yaw damper disengages during overpower test, DO NOT USE.
If AP ENGAGE or YD ENGAGE annunciator continues to flash, DO NOT USE.

7. Elevator Trim Follow-up — CHECK

   a. Control Wheel — HOLD AFT OF MID TRAVEL. Trim wheel should run nose down after approximately 3 seconds. Trim down annunciator should illuminate after approximately 8 seconds.
   b. Control Wheel — HOLD FORWARD OF MID TRAVEL. Trim wheel should run nose up after approximately 3 seconds, TRIM UP annunciator should illuminate after approximatley 8 seconds, and AP TRIM FAIL annunciator and MASTER WARNING flasher should illuminate after approximately 15 seconds.

8. AP/YD & TRIM DISC Button — DEPRESS THROUGH SECOND LEVEL. Autopilot and yaw damper should disengage and ELECT TRIM OFF annunciator should illuminate. AP ENGAGE and YD ENGAGE annunciators on instrument panel should flash 5 times and autopilot off aural alert should sound for one second (or Voice Advisory System, if installed, should announce "AUTOPILOT").
9. MASTER WARNING Flasher — PRESS TO RESET.
10. Elevator Trim Control Switch — OFF, then ON (resets electric trim; ELECT TRIM OFF annunciator should extinguish)
11. Elevator Trim — SET AS REQUIRED.

NOTE

Autopilot Preflight Check is disabled during inflight operations.

EFIS

1. SELF TEST

Press the "TEST" button on the Display Controller. Verify the following:

- Radar altimeter reads test value.
- Course Select, Heading Select, DH Set and ET are replaced with dashes.
- All flags in view.
- Command cue out of view.
- If Flight Director Lamp test is good, "TEST" will appear in upper right corner of EADI. "FD FAIL" indicates a problem in the EFIS components or interconnection of the flight director computer.

NOTE

Self-test is inhibited during glideslope capture.

2. SWITCHING TEST
   — Check each switch function on the EFIS Display Controller. Switching should operate as described under Item 18 of GENERAL OPERATION.

GENERAL OPERATION

All flight director modes (except Go-Around) are selected by push-button switches on the Flight Director Mode Selector. Go-Around is selected by pressing the Go-Around button on the left engine power lever. Annunciation for the mode selected is provided on the mode selector and pilot's instrument panel.

The Electronic Flight Instrument System replaces the conventional electro-mechanical Attitude-Director (ADI) and Horizontal Situation (HSI) indicators. The CRT displays (EADI & EHSI) provide the necessary information for the pilot to respond to flight director and navigation commands in the same manner as conventional instruments. The EDZ-605 utilizes standard (4.6" x 5") displays. The system includes additional features which allow the pilot to select alternate formats as navigation aids or in the event of specific failures. The operator is cautioned that the primary displays (attitude and directional information) are required to be operational prior to initiating flight in conditions for which these displays are required. Operation of the alternate formats is described in the appropriate Honeywell Pilot's Operating Handbooks. The owner/operator of the airplane is responsible for obtaining and maintaining the appropriate handbook.

1. Standby Lamp Test (STBY) Mode selector button — PRESS. All flight director modes are reset and all flight control system annunciators are illuminated. Flight director command cue is biased out of view and FD FAIL on the EADI is in view as long as STBY is pressed. If STBY is pressed for more than 5 seconds:

   a. AP TRIM FAIL and MASTER WARNING will illuminate.
   b. Autopilot off aural alert will sound (or Voice Advisory System, if installed, will announce "AUTOPILOT").

2. Heading Select Mode (HDG)

   a. Heading Cursor — SET desired heading on EHSI.
   b. Mode Selector HDG button — PRESS. Flight director commands a turn to capture and hold selected heading.

3. VOR Enroute Mode (NAV), and VOR approaches when more than 20 DME from the station:

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired VOR frequency.
   d. Course Selector — SET desired course on EHSI.
   e. Mode Selector NAV button — PRESS. Mode selector annunciates (NAV) ARM (white NAV on EADI).
   f. At the computed capture point — HDG mode is cancelled, mode selector annunciates (NAV) CAP (green NAV on EADI). Flight director commands a turn to capture and track selected course.

NOTE

At transition from armed to captured flight director modes, a white box appears for five seconds around the green capture mode annunciators on the EADI.

CAUTION

Do not select a VOR frequency when F/D is in NAV CAPTURE or ARM. NAV CAPTURE without mode annunciation may result.

4. VOR Approach Mode (NAV/APR) for optimized NAV mode tracking, for VOR approaches:

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired VOR frequency.
   d. Course Selector — SET desired course on EHSI.
   e. Mode Selector APR button — PRESS. Mode selector annunciates (NAV) ARM and (APR) ARM. VOR APR is annunciated on the instrument panel (white VAPP on the EADI).
   f. At computed capture point — HDG mode is cancelled and mode selector annunciates (NAV) CAP and (APR) CAP (green VAPP on EADI). Flight director commands a turn to capture and track selected course.

5. Localizer Approach (NAV)

   a. Heading Cursor — SET desired intercept heading on EHSI.
   b. Mode Selector — Select HDG
   c. Navigation Receiver — TUNE to desired localizer frequency.
   d. Course Selector — SET inbound localizer course on EHSI.
   e. Mode Selector NAV Button — PRESS. Mode selector annunciates (NAV) ARM (white LOC on EADI).
   f. At computed capture point — HDG mode is cancelled. Mode selector annunciates (NAV) CAP (green LOC on EADI). Flight director commands a turn to capture and track localizer beam.

g. Mode Selector — SELECT vertical flight director mode, as required.

6. Back Course Approach (BC)

a. Heading Cursor — SET desired intercept heading on EHSI.
b. Mode Selector — Select HDG
c. Navigation Receiver — TUNE to desired localizer frequency.
d. Course Selector — SET inbound front course on EHSI.
e. Mode Selector BC Button — PRESS. Mode selector annunciates (BC) ARM (white BC on EADI).
f. At computed capture point HDG mode is cancelled. Mode Selector annunciates (BC) CAP (green BC on EADI). Flight director commands a turn to capture and track localizer back course.
g. Mode Selector — SELECT vertical flight director mode, as required.

7. ILS Approach (APR)

a. Heading Cursor — SET desired intercept heading on EHSI.
b. Mode Selector — Select HDG
c. Navigation Receiver — TUNE to desired ILS frequency.
d. Course Selector — SET inbound localizer course on EHSI.
e. Mode Selector APR Button — PRESS. Mode selector annunciates (NAV) ARM and (APR) ARM (white LOC and GS on EADI).
f. Mode Selector — SELECT vertical flight director mode, as required, to intercept glideslope.
g. At computed localizer capture point — HDG mode is cancelled. Mode selector annunciates (NAV) CAP and (APR) ARM (green LOC on EADI). Flight director commands a turn to capture and track localizer beam.
h. At computed glideslope capture point — Flight Director Vertical modes are cancelled (except ALT SEL ARM). Mode selector annunciates (NAV) CAP and (APR) CAP (green LOC and GS on EADI). Flight director commands a pitch change to capture and track glideslope.

NOTE

Glideslope capture will not occur until after localizer has been captured. Pitch wheel is disabled while in glideslope (APR CAP) mode.

8. KNS-660 RNAV Enroute or Approach

a. KNS-660 - Program for desired operation per Pilot's Guide
b. Mode Selector NAV Button - PRESS. Mode Selector annunciates NAV, EADI annunciates LNAV and EHSI annunciates selected source as FMS 1 or 2. The KNS-660 provides steering commands for course intercept and tracking.

CAUTION

Do not use the map format on the EHSI for KNS-660 RNAV approaches.

9. Pitch Hold/Sync Mode
With no vertical flight director modes selected:

a. Heading Cursor — SET to desired heading.
b. Mode Selector — SELECT lateral flight director mode as required. Flight director pitch command is to hold attitude existing when lateral mode was selected.

To set/change command pitch attitude:

a. TCS Switch — PRESS and HOLD
b. Maneuver airplane to desired pitch attitude. Trim elevator as required.
c. TCS Switch — RELEASE. Flight director command is to hold new pitch attitude.

10. Altitude Hold Mode (ALT)
To hold altitude:

a. Maneuver airplane to desired altitude.
b. Mode Selector ALT Button — PRESS

To change altitude hold reference:

a. TCS Switch — PRESS and HOLD
b. Maneuver airplane to new altitude.
c. TCS Switch — RELEASE. Flight director command is to hold new altitude.

NOTE

Use ALT SEL mode for large altitude changes.

11. Altitude Alerting
The desired altitude is selected on the Altitude Alert controller (or the ALT PRE position on the VNAV controller, if installed) using the SET knob. Then altitude can be attained by using the altitude preselect mode of the flight director.

12. Altitude Preselect Mode (ALT SEL)

a. Altitude Preselect (or VNAV) Controller — SET desired altitude (ALT PRE position on VNAV).
b. Mode Selector ALT SEL Button — PRESS. Mode Selector annunciates (ALT SEL) ARM (white ASEL on EADI).
c. Maneuver airplane to intercept selected altitude. (Use Pitch Hold, IAS, VS or GA mode, as required.)
d. Altitude Capture Sequence — VERIFY:

1) 1000 feet prior to selected altitude — alert light on pilot's altimeter illuminates and aural alert sounds for one second.
2) 250 feet prior to selected altitude — alert light extinguishes.
3) At computed altitude capture point — mode selector annunciates (ALT SEL) CAP (green ASEL on EADI). Other vertical flight director modes are cancelled. Flight director commands a flare to selected altitude.
4) At selected altitude — Altitude Select Mode is cancelled, and Altitude Hold Mode is automatically selected (green ALT is annunciated on EADI).

NOTE

Changing selected altitude on Altitude Preselect (or VNAV) Controller during altitude capture (ALT SEL CAP) will automatically cancel ALT SEL CAP, and ALT SEL ARM will be selected.

ALT SEL mode is referenced to baro corrected altitude displayed on pilot's altimeter.

13. Airspeed Hold Mode (IAS)
To hold airspeed:

a. Maneuver airplane to desired indicated airspeed.
b. Mode Selector IAS Button — PRESS

To change airspeed hold reference:

a. TCS Switch — PRESS and HOLD
b. Maneuver airplane to new airspeed.
c. TCS Switch — RELEASE. Flight director command is to hold new indicated airspeed.

14. Vertical Speed Hold Mode (VS)
Operation of VS mode is identical to IAS mode except that flight director command is to maintain vertical speed.

15. Go-Around Mode (GA)
Go-Around Button (on left power lever) — PRESS. All flight director modes are cancelled. Flight director commands a 7-degree-pitch-up, wings-level attitude, then Heading Hold is commanded when roll attitude becomes less than 6 degrees (green GA is annunciated on EADI).
To cancel heading hold:

a. Heading Cursor — SET desired heading on HSI.

b. Mode Selector — SELECT desired lateral mode.

To cancel go-around mode:

a. Mode Selector — SELECT vertical flight director mode (ALT, IAS, VS)
   or;
b. TCS Switch — PRESS
   or;
c. Autopilot — ENGAGE

16. Vertical Navigation Mode (VNAV) (Optional)
The VNAV mode operates in conjunction with an optional dual waypoint VNAV controller.
Normal VNAV (Two independent waypoints) (See illustration).

a. Navigation Receiver - TUNE to desired VORTAC frequency.
b. DME Receiver - CHECK, not in HOLD.
c. WPT Select Button - PRESS to select WPT 1 or WPT 2 as desired.
d. Course Selector - SET DESIRED COURSE.
e. VNAV Display Selector Switch - SET to WPT ALT. SET desired waypoint altitude using SET knob.
f. VNAV Display Selector Switch - SET to STA EL. SET VORTAC station elevation using SET knob.
g. VNAV Display Selector Switch - SET to TO or FROM, as required. SET distance (alongtrack offset) that the waypoint is ahead of (TO) or beyond (FROM) the VORTAC.
h. VNAV Display Selector Switch - SET to VANG. SELECT desired vertical angle for climb or descent by:

   1) OBSERVE angle displayed on VNAV Controller. Angle will increase as airplane approaches the waypoint.
   2) Mode Selector VNAV Button - PRESS when desired angle is displayed. Mode selector annunciates (VNAV) ARM (white VNAV on EADI) momentarily, then (VNAV) CAP (green VNAV on EADI), and aural alert will sound in conjunction with (VNAV) CAP. Displayed angle will stop increasing. HSI/EHSI glideslope pointer will display deviation from vertical angle. Flight director commands a flare to selected vertical angle.

   OR;

   1) SELECT desired vertical angle using SET knob.
   2) Mode Selector VNAV Button - PRESS. Mode selector annunciates (VNAV) ARM (white VNAV on EHSI). HSI/EHSI glideslope pointer displays deviation from vertical angle.
   3) When airplane reaches selected angle - Mode selector annunciates (VNAV) CAP (green VNAV on EHSI), and the aural alert sounds. Flight director commands a flare to selected vertical angle.

NOTE

VNAV angles are limited to 6 degrees and angle displayed in VANG position will flash for angles between 5 and 6 degrees.

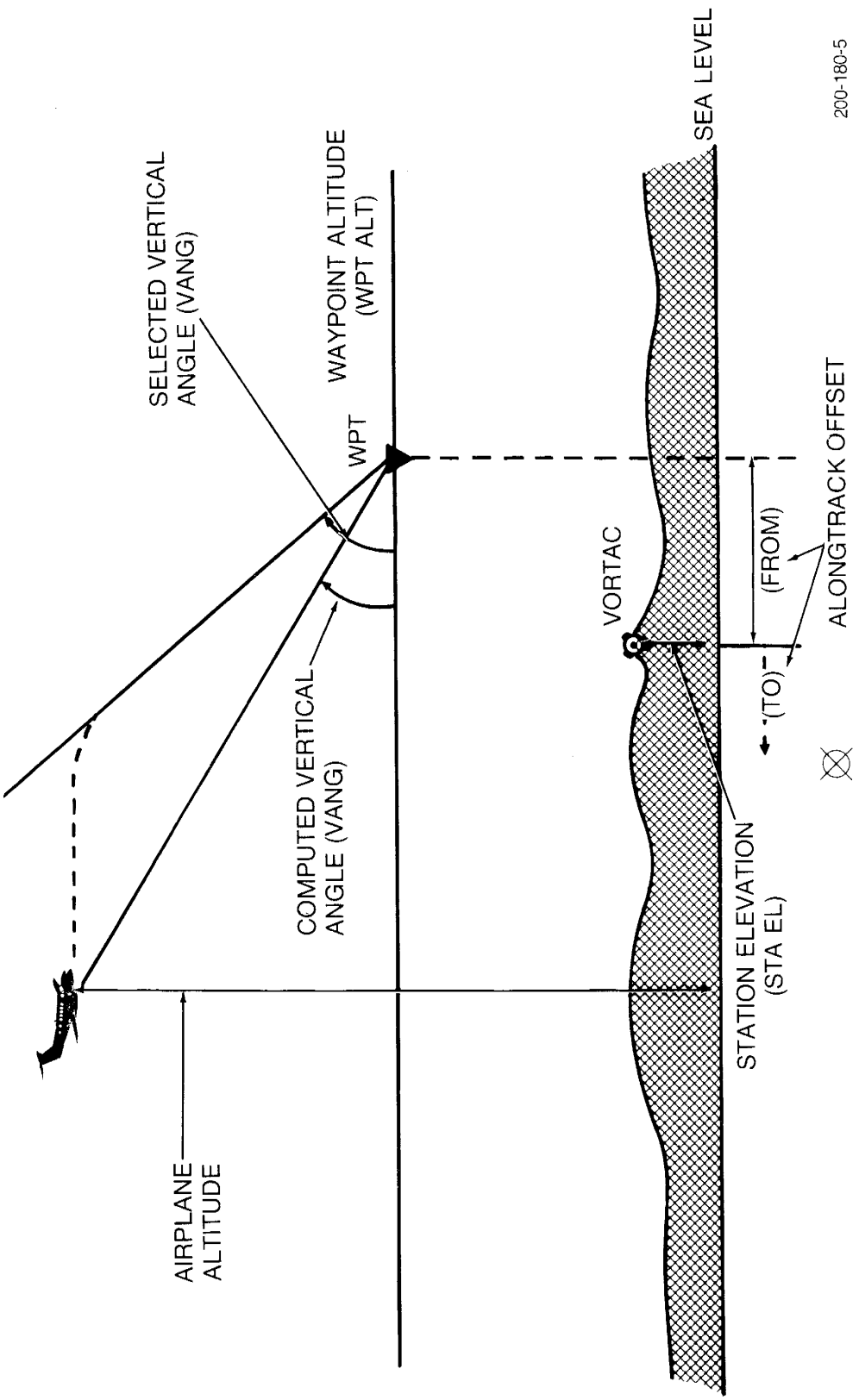
Preselect altitude (ALT PRE) will be set equal to WPT ALT when VNAV mode is selected. Changes to ALT PRE are not allowed when VNAV mode is selected.

i. Approaching WPT ALT - Flight director commands a flare to WPT.
j. At WPT ALT - VNAV mode is cancelled, (WPT ALT, VANG set to TO, FROM set to zero) and Altitude Hold mode is automatically selected.

NOTE

Once the VNAV mode has been selected, no changes to the active WPT data are allowed. Data for the other (alternate) WPT may be entered by pressing the WPT select button on the VNAV controller. The number of the active WPT will remain illuminated (steady) and the number of the alternate WPT will flash.

If changes to the active waypoint are required, disengage the VNAV mode. The autopilot will maintain pitch hold. Reset VNAV controller and reengage the VNAV.


SEA LEVEL
200-180-5
SELECTED VERTICAL ANGLE (VANG)
WAYPOINT ALTITUDE (WPT ALT)
WPT
VORTAC
(FROM)
ALONGTRACK OFFSET
(TO)
COMPUTED VERTICAL ANGLE (VANG)
STATION ELEVATION (STA EL)
AIRPLANE ALTITUDE

NORMAL VNAV

VNAV Profile (Two waypoints linked together) (See illustration)

a. SET desired WPT data for both WPT 1 and WPT 2.
b. VNAV Controller PROFILE Button - PRESS, PROFILE button will illuminate.

NOTE

Profile mode is inhibited if:

1) Station elevation (STA EL) for WPT 1 and WPT 2 is not the same.
2) VANG for second WPT has been slewed ahead.
3) Airplane has passed computed flare point from first VANG to second VANG.

No changes will be allowed to WPT 1 or WPT 2 data while in the PROFILE mode.

Before PROFILE is selected, VANG for second WPT is computed angle between airplane and second WPT ALT. After PROFILE is selected, VANG for second WPT is computed angle between first WPT ALT and second WPT ALT.

To cancel PROFILE mode, press illuminated PROFILE button.

c. Mode Selector VNAV Button - PRESS. Mode selector annunciates (VNAV) ARM. HSI/EHSI glideslope pointer displays deviation from first vertical angle.
d. When airplane reaches VANG for first WPT - Mode Selector annunciates (VNAV) CAP. Flight director commands a flare to first VANG.
e. At computed flare point to second VANG - PROFILE button flashes then extinguishes; WPT data for first WPT is cancelled; mode selector VNAV annunciators extinguish momentarily, then annunciate (VNAV) ARM momentarily, then annunciate (VNAV) CAP. VNAV controller WPT button annunciates number of second WPT. Flight director commands a flare to second VANG. ALT PRE is set to WPT ALT for second WPT.
f. Approaching second WPT ALT - Flight director commands a flare to second WPT ALT.
g. At second WPT ALT - VNAV mode is cancelled and Altitude Hold is automatically selected.

17. Air Data Command Display (Optional)
To Command Airspeed Hold:

a. Mode Selector IAS Button - PRESS. Command display will indicate existing IAS in knots.
b. Autopilot Controller Pitch Wheel - ROTATE to increase or decrease airspeed shown on command display. Flight director will command pitch up, or down, to capture and track displayed airspeed (autopilot engaged).

To Command Vertical Speed Hold:

a. Mode Selector VS Button - PRESS. Command display will indicate existing vertical speed in feet per minute.
b. Autopilot Controller Pitch Wheel - ROTATE to increase or decrease vertical speed shown on Command Display. Flight director will command pitch up, or down, to capture and track displayed vertical speed (autopilot engaged).

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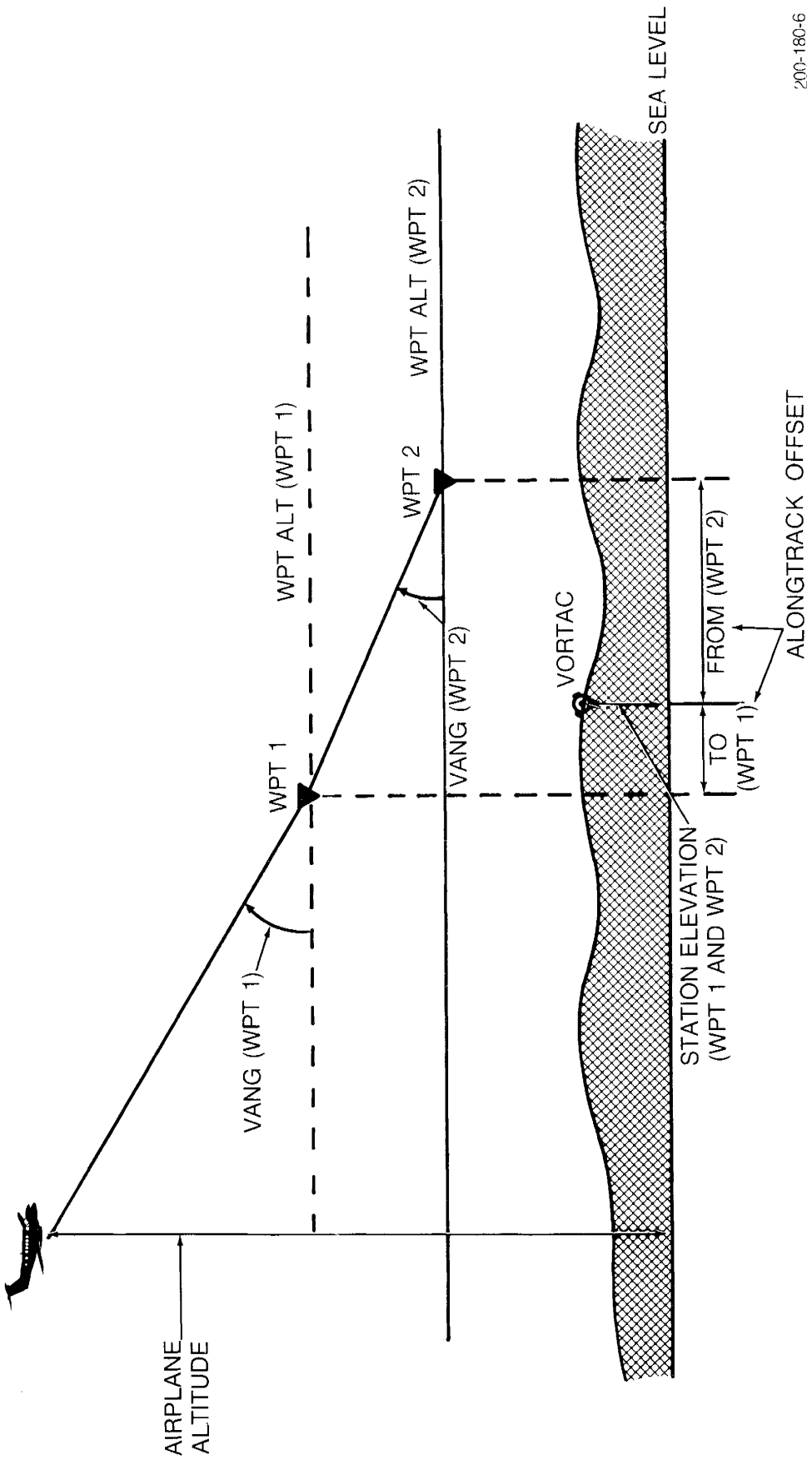

VNAV - TWO WAYPOINTS LINKED TOGETHER

18. DC-811 EFIS Display Controller Button Selections:

    a. FULL/MAP — Toggles the EHSI display from Full Compass to Partial Compass to MAP format.
    b. SC/CP — Toggles the Flight Director command from Single Cue to Cross-Pointer configuration.
    c. GSPD/TTG — Toggles display on EHSI from Ground Speed to Time-To-Go.
    d. ET — Toggles Elapsed Time, when displayed, between Initialize, Start, and Stop.
    e. HDG REV — Toggles display from HDG-1 to HDG-2.
    f. V/L — Toggles SRN Nav source between NAV-1 and NAV-2, when selected.
    g. FMS — Toggles LRN Nav source between FMS-1 and FMS-2, when selected.

19. MC-800 MFD Controller Button Selections:

    a. MAP/PLAN — Alternately selects the Heading-Up MAP display or the North-Up PLAN mode for display.
    b. SRC — Alternately selects the FMS-1 and FMS-2 source of LRN data for mapping.
    c. WX — Used to call up weather radar returns on the MFD Map display or Weather Radar Only display.
    d. VOR — Used to add VOR/DME symbols to the Map and Plan displays.
    e. DAT — Used to add LRN information to the Map and Plan displays.
    f. APT — Used to add airport designators to the MFD Map and Plan displays.
    g. SKP, RCL, PAG, ENT, and Joystick — These buttons and Joystick are used to control the checklist and operation of the designator on the map.

20. EFIS Source Controller Initiated Flight Director Mode resets:
    Pilot input to the EFIS through the DC-811 Source Controller initiates Flight Director and Autopilot mode changes as follows:

PILOT ACTION	FD/AP MODE CHANGE
Select a NAV sourse	All lateral modes reset except HDG
Select a HDG source	All lateral modes reset

AUTOPILOT OPERATION

1. In-Flight Engagement of Autopilot

    a. All AFCS and Autopilot Circuit Breakers — IN
    b. Autopilot Controller TRIM UP and TRIM DN annunciators — CHECK. If either annunciator is illuminated, do not engage autopilot.
    c. Turn Knob — IN CENTER DETENT POSITION.
    d. Autopilot Controller AP Button — PRESS. Controller will annunciate AP Engage and YD ENGAGE.

2. Disengaging the Autopilot
    The autopilot may be disengaged by:

    a. Pressing the AP/YD & TRIM DISC button on either control wheel to the first level.

NOTE

The AP/YD & TRIM DISC button is a two-level switch. The first level disengages the autopilot and yaw damper, the second level disengages the electric trim.

    b. Pressing the Go-Around button on the left power lever (Yaw Damper remains engaged).
    c. Actuating the Trim Switches on either control wheel (YAW Damper remains engaged).
    d. Pressing the AP button on the Autopilot Controller (Yaw Damper remains engaged).

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NOTE

Upon autopilot disengagement, AP ENGAGE annunciator on the instrument panel will flash 5 times, then extinguish, and the autopilot-off aural alert will sound for 1 second (or the Voice Advisory System, if installed, will announce "AUTOPILOT").

In addition, if the autopilot disengages automatically due to a detected failure, the AP DISC annunciator and the MASTER WARNING flasher will illuminate. The AP DISC annunciator is extinguished by pressing the AP/YD & TRIM DISC button on either control wheel. The MASTER WARNING flasher is reset by pressing its face.

3. Engaging the Yaw Damper
The Yaw Damper is automatically engaged whenever the Autopilot is engaged. The Yaw Damper may be engaged without engaging the Autopilot by pressing the YD button on the Autopilot Controller.

4. Disengaging the Yaw Damper
The Yaw Damper may be disengaged by:

   a. Pressing the AP/YD & TRIM DISC button on either control wheel to the first level.
   b. Pressing the YD button on the Autopilot Controller.

NOTE

Upon Yaw Damper disengagement, the YD ENGAGE annunciator on the instrument panel will flash 5 times, then extinguish.

5. Flight Director Coupling
When engaged, the Autopilot automatically couples to any flight director mode selected (except Go-Around).

6. Attitude Hold
When the Autopilot is engaged with no flight director modes selected, pitch attitude and heading are held. (If roll attitude is greater than 6 degrees, the Autopilot will command wings level, then heading hold).
To change Attitude

   a. TCS Switch — PRESS and HOLD
   b. Manually maneuver airplane to desired attitude
   c. TCS Switch — RELEASE. New pitch attitude and heading will be held. (If roll attitude is greater than 6 degrees, new roll attitude will be held.)

7. Turn Knob
To command a turn — ROTATE Turn Knob on Autopilot Controller out of detent in desired direction. Autopilot will command roll attitude proportional to amount of Turn Knob rotation.

NOTE

Rotating Turn Knob out of detent position with Autopilot engaged will cancel all lateral flight director modes.

8. Pitch Wheel
To change pitch attitude — ROTATE Pitch Wheel on Autopilot Controller in desired direction. Autopilot will command pitch attitude change proportional to amount of Pitch Wheel movement.

NOTE

Rotating the Pitch Wheel with the Autopilot engaged will cancel ALT, ALT SEL/CAP, IAS, and VS. (If optional Air Data Command Display is installed, IAS or VS will not be cancelled, but will be changed proportional to amount of wheel movement.)

9. Touch Control Steering (TCS)
   TCS allows the pilot to override autopilot modes (by disengaging pitch and roll servos) in two ways:

   a. In Attitude Hold, ALT, IAS, or VS mode, TCS synchronizes the hold references. When TCS is released, the new reference is held.
   b. In HDG, VOR, VOR APR, LOC, BC, ILS, or VNAV mode, TCS allows corrective or supervisory override without mode disengagement. When TCS is released, the autopilot will recouple to the selected mode.

10. Soft Ride
    To soften autopilot response for turbulence penetration — PRESS Soft Ride Button on Autopilot Controller.

NOTE

Soft Ride is automatically cancelled if the autopilot is disengaged.

11. Bank Limit
    To limit bank angle in HDG mode — PRESS Bank Limit Button on Autopilot Controller. Autopilot and Flight Director commands are limited to 17 degrees.

NOTE

Low Bank Limit will be cancelled if HDG mode is cancelled, but will be reselected automatically if HDG is reselected. To cancel Low Bank Limit — PRESS Bank Limit Button when LOW is annunciated.

SHUTDOWN AND SECURING

— EFIS Switches - OFF before FUEL CUTOFF

PERFORMANCE

No change.

WEIGHT & BALANCE

No change.

SYSTEMS DESCRIPTION

No change.

HANDLING, SERVICING AND MAINTENANCE

No change.

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P/N 101-590010-313

For Training Purposes Only

Beechcraft®

Super King Air® B200 Landplanes

Pilot's Operating Handbook and

FAA Approved Airplane Flight Manual Supplement

for the

Turkish Ministry of Defense

Aerial Surveillance Airplane

(BB-1434 Only)

This Supplement is applicable to the following manual(s):

101-590010-147

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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September, 1992
P/N 101-590010-317

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* (POH/AFM) when the airplane has been modified for aerial surveillance purposes by installation of a forward camera well in accordance with BEECH-approved data.

The information in this supplement supersedes or adds to the basic *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual* only as set forth within this document. Users of the Handbook are advised always to refer to the Supplement for possibly superseding information and placarding applicable to operation of the airplane.

The photographic configuration consists of a mid-cabin camera well, protected by electrically actuated sliding doors.

The airplane is eligible for certification in the Restricted Category for special photographic purposes when it is equipped with a BEECH-installed camera well installation and when an appropriate *Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement* is attached to the Handbook.

The airplane has been issued a Multiple Airworthiness Certificate permitting operation in the Restricted Category under FAR 21.187.

LIMITATIONS

OPERATION UNDER RESTRICTED CATEGORY AIRWORTHINESS CERTIFICATION

Operation shall be in accordance with the provisions and limitations of FAR 91.39 (Restricted Category Civil Aircraft; Operation Limitations).

Occupancy shall be restricted to those persons specified in FAR 91.39. Persons required for performance of aerial surveillance are considered to be crew members.

The airplane shall be identified "RESTRICTED", per FAR 45.23.

Anytime an internal camera well cover is removed, a camera shall be installed in the camera well for pressurized flight. The cabin shall be unpressurized while removing or exchanging a camera in flight.

Each time the airplane is operated in the Restricted Category, a post-flight inspection of the camera installation shall be made to assure camera security, and to assure that no chips or cracks are present in the optical glass, and a record of the inspection shall be made. Do not pressurize the cabin with chips or cracks in the camera glass.

Maximum cabin pressurization shall be 6.1 PSI.

The external protective doors must be closed for takeoff and landing.

OPERATION UNDER NORMAL CATEGORY AIRWORTHINESS CERTIFICATION

The internal camera well cover must be installed.

The external protective doors must be closed for all flight operations.

Maximum cabin pressurization shall be 6.1 PSI.

MISCELLANEOUS INSTRUMENT MARKINGS

Cabin Differential Pressure Gage:

Green Arc (Approved Operating Range) ..... 0 to 6.1 PSI.

Red Arc (Unapproved Operating Range) ..... 6.1 PSI to end of scale.

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EMERGENCY PROCEDURES

No Change.

NORMAL PROCEDURES

PREFLIGHT INSPECTION

To protect the optical glass from flying debris, the camera well protective doors should be closed before starting engines.

TAKEOFF, CLIMB, CRUISE, DESCENT, APPROACH, AND LANDING (RESTRICTED CATEGORY ONLY)

To protect the optical quality of the glass against unnecessary exposure to possible degradation, the external protective doors should remain closed except during the actual photographic session.

PERFORMANCE FOR NORMAL AND RESTRICTED CATEGORIES

Takeoff and Landing - No Change.

Climb Performance - Subtract 20 feet per minute in all configurations.

Cruise Performance and Range Calculation - 2% reduction in true airspeed. The existing torque and fuel flow information will remain the same.

RECORD OF INSPECTION

Each time the airplane is converted from one category to the other, an inspection will be conducted by a certified pilot, or an appropriately certified mechanic, for chips and cracks in the optical glass, camera security, and camera well protective covers, as appropriate. A record of this inspection will be entered in the aircraft logbook with the signature of the person making the inspection.

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P/N 101-590010-317

For Training Purposes Only

Beechcraft®

Super King Air®
200, 200C, B200, B200C, 200T, 200CT, B200T and B200CT Landplanes
BB-2, BB-6 thru BB-1438, BB-1440 thru BB-1443; BL-1 thru BL-72, BL-124 thru BL-138; BT-1 thru BT-34; BN-1 thru BN-4
Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement
for the
Continuous Loop Fire and Overheat Detection System

This Supplement is applicable to the following manual(s):

101-590010-127, 101-590010-147, 101-590037-3, and 101-590037-29

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved:

A.C. Jackson
Beech Aircraft Corporation
DOA CE-2

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GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual (POH/AFM) when the airplane is equipped with the Continuous Loop Fire and Overheat Detection System installed in accordance with Beechcraft Kit No. 101-9077-1 or 101-9077-3

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

LIMITATIONS

1. Flight will not be initiated with any malfunction of either the right or left fire detection systems.

EMERGENCY PROCEDURES

No Change

ABNORMAL PROCEDURES

No Change

NORMAL PROCEDURES

No Change

PERFORMANCE

No Change

WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SYSTEMS DESCRIPTION

The fire detection system is designed to provide an immediate warning in the event of fire in either engine compartment. The system consists of a temperature sensing cable for each engine; two red warning annunciators, FIRE L ENG and FIRE R ENG (200 series) and L ENG FIRE and R ENG FIRE (B200 series) located on the warning annunciator panel; a test switch on the copilot's left subpanel and one circuit breaker, labeled FIRE DET, on the right side panel. When the fire has been extinguished (if the integrity of the system has not been destroyed), the system will reset itself.

The test switch, on the copilot's left subpanel, is placarded TEST SWITCH FIRE DETECT, LH and RH. When the switch is placed in the LH or RH position, the corresponding FIRE L ENG/FIRE R ENG (200 series) or L ENG FIRE/R ENG FIRE (B200 series) annunciator, along with the flashing MASTER WARNING annunciator, should illuminate. The system may be tested anytime, either on the ground or in flight.

This system replaces the photoconductive system originally installed in these airplanes.

HANDLING, SERVICING AND MAINTENANCE

No Change

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P/N 101-590010-353

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Raytheon Aircraft

Beech King Air® 65-90, 65-A90, B90, C90, C90A (Serials LJ-1 thru LJ-1402)
E90, F90, 100 (Serials B-2 thru B-89, B-93), A100
(Serials B-1, B-90 thru B-92, B-94, B-100 thru B-247), B100 Landplanes
Beech Super King Air® 200/200C, B200/B200C Landplanes
(Serials BB-2, BB-6 thru BB-1438, BB-1440 thru BB-1443 and BB-1463; BL-1 thru BL-72, BL-113 thru BL-117, BL-124 thru BL-138), 200T/200CT, B200T/B200CT Landplanes (Serials BT-1 thru BT-38; BN-1 thru BN-4)

PILOT'S OPERATING HANDBOOK
AND
FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

ARTEX ELT 110-4-002 With Remote Cockpit Switch
(Raytheon Aircraft Kit P/N 101-3210 or 101-3214)

This Supplement is applicable to the following Manual(s):

130695B, 130766B, 65-001123-7, 65-001123-13, 65-001123-25, 65-001123-31, 90-590010-5, 90-590010-53, 90-590010-61, 90-590010-87, 90-590012-3, 90-590024-5, 90-590024-23, 90-590024-35, 90-590024-43, 90-590024-61, 90-590024-69, 100-590026-1, 100-590032-1, 100-590038-1, 101-590010-127, 101-590010-147, 101-590010-327, 101-590037-3, 101-590037-29, 101-590037-57, 109-590010-3, 109-590010-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: _____
A.C. Jackson
Raytheon Aircraft Company
DOA CE-2

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Issued: December, 1998
P/N 101-590010-373

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SECTION I - GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane is operated with an ARTEX ELT 110-4-002 With Remote Cockpit Switch in accordance with Raytheon Aircraft Kit P/N 101-3210 or 101-3214.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the handbook are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION II - LIMITATIONS

No Change

SECTION III - EMERGENCY PROCEDURES

Add the following procedure.

IN FLIGHT EMERGENCIES

If needed, the ELT may be activated in flight as follows:

- ELT Switch - ON, Yellow XMT Annunciator - BLINKING

SECTION IIIA - ABNORMAL PROCEDURES

No Change

SECTION IV - NORMAL PROCEDURES

PREFLIGHT INSPECTION

Delete reference to arming the ELT.

BEFORE ENGINE STARTING

Add the following step:

- ELT Switch - ARM, XMT Annunciator - EXTINGUISHED

SECTION V - PERFORMANCE

No Change

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Issued: December, 1998
P/N 101-590010-373

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SECTION VI - WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SECTION VII - SYSTEMS DESCRIPTION

The ARTEX 110-4-002 Emergency Locator Transmitter (ELT) System is designed to meet the requirements of TSO C91a. The system consists of the ELT transmitter, located in the aft fuselage area, an antenna mounted on the aft fuselage, and a remote switch with a yellow transmit light, located on the left cockpit sidewall next to the OAT gage. The switch is lever-locked in the ARM and the ON positions. Neither this switch, nor the switch on the ELT transmitter, can be positioned to prevent the automatic activation of the ELT transmitter. The system is independent from other airplane systems except for the transmit light, which is hot-wired to the airplane battery, and the edge lit panel which is controlled by the side panel lights rheostat located on the overhead panel.

The ELT will automatically activate during a crash and transmit a sweeping tone on 121.5 and 243.0 MHz. This activation is independent of the remote switch setting or availability of aircraft power. The remote switch is installed to perform the following functions:

- Test the ELT.
- Deactivate the ELT if it has been inadvertently activated by the "G" switch.
- Activate the ELT in an in-flight emergency if an off-airport landing is anticipated.
- Activate the ELT after an off-airport landing, if the impact did not automatically activate it.

The ELT should be tested every three months. The test consists of turning the unit on and then resetting it using the following procedures:

- Tests should be conducted between the times of on-the-hour until 5 minutes after the hour.
- Notify any nearby control towers.
- Provide power to an aircraft radio and tune it to 121.5 MHz.
- Place the ELT remote switch on ON. Wait for at least 3 sweeping tones on the aircraft radio, which will take about 1 second, then return the switch to ARM.
- The test is successful if the sweeping tones are heard and the transmit light next to the switch illuminates immediately. If there is a delay in the illumination of the transmit light, the system is not working properly.

If the ELT should be inadvertently activated by the "G" switch, the transmit light next to the switch will blink. The ELT can be deactivated by momentarily placing the remote switch ON and then back to ARM.

SECTION VIII - HANDLING, SERVICING & MAINTENANCE

No Change

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Raytheon Aircraft

Beech Super King Air® B200/B200C Landplanes
(Serials BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-990, BB-992 thru BB-1438, BB-1440 thru BB-1443, BB-1463; BL-37 thru BL-138)

Beech Super King Air® B200T/B200CT Landplanes
(Serials BT-23 and After, except BT-28; BN-2 and After)

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for

Pitot Heat Annunciators

This Supplement is applicable to the following Manual(s):

101-590010-147, 101-590010-327, 101-590037-29, 101-590037-57

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: _____
for A.C. Jackson
Raytheon Aircraft Company
DOA CE-2

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SECTION I - GENERAL

The information in this supplement is FAA approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when the airplane has been modified by installation of Pitot Heat Annunciators in accordance with Raytheon Aircraft approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION II - LIMITATIONS

No Change

SECTION III - EMERGENCY PROCEDURES

No Change

SECTION IIIA - ABNORMAL PROCEDURES

SYSTEMS

ICE PROTECTION SYSTEMS

PITOT-STATIC MAST HEAT FAILURE (L OR R PITOT HEAT ANNUNCIATOR)

Illumination of the L or R PITOT HEAT annunciator indicates that pitot mast heat is inoperative. The annunciator will also illuminate anytime the PITOT switch is in the OFF position.

SECTION IV - NORMAL PROCEDURES

BEFORE TAKEOFF (FINAL ITEMS)

Change Pitot Heat step in POH/AFM to read as follows:

- Left and Right Pitot Heat ..... ON
  (L & R PITOT HEAT annunciators - EXTINGUISHED)

SECTION V - PERFORMANCE

No Change

SECTION VI - WEIGHT AND BALANCE/EQUIPMENT LIST

No Change

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P/N 101-590010-391

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ANNUNCIATOR SYSTEM

Delivered by Raytheon Aircraft or equipped with Raytheon Aircraft Kit P/N 101-3238-1

Serials BB-734, BB-793, BB-829, BB-854 thru BB-870, BB-874 thru BB-891, BB-894, BB-896 thru BB-911, BB-913 thru BB-1438, BB-1440 thru BB-1443, BB-1463; BL-37 thru BL-138; BT-23 thru BT-34; BN-2 thru BN-4:

CAUTION/ADVISORY ANNUNCIATOR PANEL ILLUSTRATION

L DC GEN	- - -	HYD FLUID LOW	RVS NOT READY	- - -	R DC GEN
- - -	- - -	- - -	DUCT OVERTEMP	- - -	- - -
L ICE VANE	L PITOT HEAT	BATTERY CHARGE	EXT PWR	R PITOT HEAT	R ICE VANE
* L AUTO FEATHER	- - -	* ELEC TRIM OFF	AIR CND N_1 LOW	- - -	* R AUTO FEATHER
L ICE VANE EXT	* BRAKE DEICE ON	LDG/TAXI LIGHT	PASS OXY ON	ELEC HEAT ON	R ICE VANE EXT
L IGNITION ON	L BL AIR OFF	- - -	FUEL CROSSFEED	R BL AIR OFF	R IGNITION ON

*Optional/If Installed

PMG0074

Serials BT-35 and After:

CAUTION/ADVISORY ANNUNCIATOR PANEL ILLUSTRATION

L DC GEN	- - -	HYD FLUID LOW	RVS NOT READY	- - -	R DC GEN
L CHIP DETECT	- - -	- - -	DUCT OVERTEMP	- - -	R CHIP DETECT
L ENG ICE FAIL	L PITOT HEAT	** BATTERY CHARGE	EXT PWR	R PITOT HEAT	R ENG ICE FAIL
* L AUTO FEATHER	- - -	* ELEC TRIM OFF	AIR CND N_1 LOW	- - -	* R AUTO FEATHER
L ENG ANTI-ICE	* BRAKE DEICE ON	LDG/TAXI LIGHT	PASS OXY ON	* ELEC HEAT ON	R ENG ANTI-ICE
L IGNITION ON	L BL AIR OFF	- - -	FUEL CROSSFEED	R BL AIR OFF	R IGNITION ON

*Optional/If Installed
**Airplanes Prior To BT-39

PMG0073

CAUTION/ADVISORY ANNUNCIATOR PANEL DESCRIPTION

NOMENCLATURE	COLOR	CAUSE FOR ILLUMINATION
L PITOT HEAT	Amber	Left Pitot Heat Inoperative or switch is in the OFF position.
R PITOT HEAT	Amber	Right Pitot Heat Inoperative or switch is in the OFF position.

ICE PROTECTION SYSTEMS

PITOT MAST HEAT

Heating elements are installed in the pitot masts located on the nose. Each heating element is controlled by an individual circuit breaker switch placarded PITOT - LEFT - RIGHT, located on the pilot's right subpanel. A failure is indicated by the illumination of the L PITOT HEAT or R PITOT HEAT annunciator in the caution/advisory annunciator panel. Illumination of these annunciators indicates that pitot mast heat is inoperative. The annunciators will also illuminate anytime the PITOT switches are in the OFF position. It is not advisable to operate the pitot heat system on the ground except for testing or for short intervals of time to remove ice or snow from the masts.

SECTION VIII - HANDLING, SERVICING & MAINTENANCE

No Change

Raytheon Aircraft

Beech Super King Air® B200/B200C

Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement

for Wing Fatigue Limitations (BB-944 & BB-1048 Only)

This Supplement is Applicable to the Following Manual(s):
101-590010-147

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: _____
A.C. Jackson
Raytheon Aircraft Company
DOA-230339-CE

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FAA Approved
Revised: March, 2001
P/N 101-590010-397

For Training Purposes Only

Raytheon Aircraft

LOG OF REVISIONS

Beech Super King Air® B200/B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-944 & BB-1048)

REV NO.	PAGE NO(S).	DESCRIPTION	DATE OF REV	FAA APPROVED
0	1 thru 3	Original Issue	October, 1999	John Kennedy
1	1 thru 4	Added Log of Revisions, Revised "Structural Limitations" and Shifted Data	March, 2001	

P/N 101-590010-397

For Training Purposes Only

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SECTION VII - SYSTEMS DESCRIPTION ..... Page 4
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SECTION I - GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when BB-944 and BB-1048 are operated in accordance with Raytheon Aircraft approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION II - LIMITATIONS

STRUCTURAL LIMITATIONS

LOCATION	A/C SN	SAFE LIFE / INITIAL INSPECTION	SUBSEQUENT INSPECTION
Wing and Associated Structure	BB-944	23,000 HRS	N/A
	BB-1048	24,000 HRS	N/A
Lower Main Spar Cap Bathtub Fitting	BB-944	20,000 HRS	1200 HRS
	BB-1048	20,000 HRS	1200 HRS

Refer to the *Beech Structural Inspection and Repair Manual* and the *Beech Super King Air 200 Series Maintenance Manual* for inspection and replacement procedures.

The wing and associated structural items are analyzed based on the following mission profiles.

FLIGHT PROFILES

Profile 1: Transit (Short) - 22.5% - Flight Time 1.5 Hrs; Take-off Weight 14,500 lbs; Take-off Fuel 3485 lbs; Cruise 20,000 FT AGL @ 200 KIAS

Profile 2: Transit (Long) - 22.5% - Flight Time 4.0 Hrs; Take-off Weight 14,500 lbs; Take-off Fuel 3485 lbs; Cruise 30,000 FT AGL @ 165 KIAS

Profile 3: PAR - 11.6% - Flight Time 2.0 Hrs; 8 Circuits; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 3000 FT AGL @ 200 KIAS

Profile 4: ILS - 12.6% - Flight Time 3.0 Hrs; 7 Circuits; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 5000 FT AGL @ 200 KIAS

Profile 5: VOR - 7.7% - Flight Time 3.5 Hrs; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 3000 FT AGL @ 195 KIAS, 4000 FT AGL @ 195 KIAS, 5000 FT AGL @ 190 KIAS, 7000 FT AGL @ 190 KIAS, 10,000 FT AGL @ 155 KIAS

Profile 6: TACAN - 4.4% - Flight Time 2.0 Hrs; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 5000 FT AGL @ 195 KIAS

Profile 7: MSSR - 5.5% - Flight Time 4.0 Hrs; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 5000 FT AGL @ 190 KIAS, 8000 FT AGL @ 190 KIAS, 18,000 FT AGL @ 160 KIAS, 24,000 FT AGL @ 150 KIAS, 33,000 FT AGL @ 140 KIAS

Profile 8: WATCHMAN - 11.0% - Flight Time 5.0 Hrs; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 18,000 FT AGL @ 165 KIAS, 24,000 FT AGL @ 150 KIAS, 33,000 FT AGL @ 140 KIAS

Profile 9: PLUTO - 2.2% - Flight Time 6.0 Hrs; Take-off Weight 12,500 lbs; Take-off Fuel 3485 lbs; Cruise 35,000 FT AGL @ 135 KIAS

SECTION III - EMERGENCY PROCEDURES

No Change

SECTION IIIA - ABNORMAL PROCEDURES

No Change

SECTION IV - NORMAL PROCEDURES

No Change

SECTION V - PERFORMANCE

No Change

SECTION VI - WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SECTION VII - SYSTEMS DESCRIPTION

No Change

SECTION VIII - HANDLING, SERVICING & MAINTENANCE

No Change

FAA Approved
Revised: March, 2001
P/N 101-590010-397

For Training Purposes Only

Raytheon Aircraft

Beech Super King Air® B200 & B200C
(N^os de Sèrie BB-734, BB-793, BB-829, BB-854 à BB-870, BB-874 à BB-891, BB-894, BB-896 à BB-911, BB-913 à BB-1438, BB-1440 à BB-1443; BL-37 à BL-138)

SUPPLEMENT AU MANUEL DE VOL AVION
pour les avions
immatriculés en France

Ce supplément est applicable au manuel de vol P/N 101-590010-147

Ce supplément doit être uniquement utilisé pour les avions immatriculés en France et il est défini en accord avec les instructions pour approbation par la DGAC. Ce supplément ne doit pas être utilisé à l'intérieur des USA et il n'est ni approuvé, ni signé par la FAA.

Ce manuel inclut les informations que les conditions de certification exigent de fournir au pilote.

Numéro de Série de l'Avion : _____

Immatriculation de l'Avion : _____

Approuvé par la D.G.A.C. Le: 13 JUIN 2002

DIRECTION GÉNÉRALE DE L'AVIATION SERVICE DE LA FORMATION AÉRONAUTIQUE ET DU CONTRÔLE

Hugues LE CARDINAL
Bureau Navigabilité
Aviation Générale

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PARU: Juin, 2002
P/N 101-590010-411

For Training Purposes Only

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LISTE DES PERSONNES AUTORISEES A UTILISER LE MANUEL DE VOL EN LANGUE ANGLAISE ..... Page 7
LISTE DES SUPPLEMENTS ..... Page 9

SECTION I - GENERAL

Ce supplément au manuel de vol doit être attaché au manuel de vol du Beech Super King Air B200 & B200C approuvé FAA, P/N 101-590010-147, quand l'avion est immatriculé en France.

Les informations remplacent ou complètent celles du Manuel de Vol agréé par la FAA, tel que défini ci-après. Il est recommandé aux utilisateurs de se reporter systématiquement au Supplément afin de vérifier la présence éventuelle de mise à jour ainsi que le marquage des indications applicables à l'utilisation de l'avion.

MISE A JOUR DES PAGES

Ajouter les pages 1 et 2 de ce supplément aussitot après la page 9 de ce supplément.

Ajouter la page 3 de ce supplément devant la page de garde du manuel de vol approuvé FAA.

Ajouter la page 5 « LISTE DES REVISIONS S» de ce supplément, devant la page A « LOG OF REVISIONS » existante pour la révision C5 du manuel P/N 101-590010-147 daté de Mai 2000.

Ajouter la page 7 « LISTE DES PERSONNES AUTORISEES A UTILISER LE MANUEL DE VOL EN LANGUE ANGLAISE » page LP-1 de ce supplément, à la suite de la page a « Introduction » datée de Decembre 1994, ou après.

Ajouter la page 9 «LISTE DES SUPPLEMENT S» de ce supplément, à la suite de la page 9-2 de la section Suppléments du manuel de vol.

Raytheon Aircraft

Beech Super King Air®
B200 & B200C

(N^{os} de Série BB-734, BB-793, BB-829, BB-854 à BB-870, BB-874 à BB-891, BB-894, BB-896 à BB-911, BB-913 à BB-1438, BB-1440 à BB-1443; BL-37 à BL-138)

MANUEL DE VOL

Le Manuel de Vol de type de l'avion Beech Super King Air B200 & B200C est constitué du "Pilot's Operating Handbook and FAA Approved Airplane Flight Manual," référence 101-590010-147C5 de Mai 2000, Ou Toute Suivante Revision C.

Dans ce Manuel de Vol en langue anglaise, toutes les références á la FAA doivent s'étendre pour la DGAC.

Ce Manuel de Vol, approuvé par la DGAC, est conforme á la note DGAC/SFACT/N No. 952734 du 3 Juillet 1995.

Ce Manuel de Vol en langue anglaise ne peut être utilisé qu'en Transport Public ou par les pilotes mentionnés á la page suivante.

Numéro de Série de l'Avion : _____

Immatriculation de l'Avion : _____

Approuvé par la D.G.A.C.
Le : 13 JUIN 2002

DIRECTION GÉNÉRALE DE L'AVIATION CIVILE
DIRECTION DE LA NAVIGATION AÉRONAUTIQUE ET DU TRANSPORT AÉRIEN

Hugues LE CARDINAL
Bureau Navigabilité
Aviation Générale

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Raytheon Aircraft

Beech Super King Air® B200 & B200C

(N^{os} de Sèrie BB-734, BB-793, BB-829, BB-854 à BB-870, BB-874 à BB-891, BB-894, BB-896 à BB-911, BB-913 à BB-1438, BB-1440 à BB-1443; BL-37 à BL-138)

MANUEL DE VOL

P/N 101-590010-147

(Revision C5 et suivante)

LISTE DE REVISIONS

Edition originale. . . . . .Juin, 2002

Page	Description
Page de Garde	Nouvelle - Approbation DGAC (France).
Page LP-1	Nouvelle - Liste des Personnes Autorisées à Utiliser le Manuel de Vol en langue anglaise en dehors d'une exploitation en Transport Public (France).
Page AA	Nouvelle - Liste des révisions du Manuel de Vol en langue anglaise, approuvè par DGAC (France).
Page A	"Log of Revisions" conforme à l'édition de la FAA appropriée.
1-1 à la jusqu'à la fin	Section I "General," conforme à l'édition de la FAA appropriée.
2-1 à la jusqu'à la fin	Section II "Limitations," conforme à l'édition de la FAA appropriée.
3-1 à la jusqu'à la fin	Section III "Emergency Procedures," conforme à l'édition de la FAA appropriée.
3A-1 à la jusqu'à la fin	Section IIIA "Abnormal Procedures," conforme à l'édition de la FAA appropriée.
4-1 à la jusqu'à la fin	Section IV "Normal Procedures," conforme à l'édition de la FAA appropriée.
5-1 à la jusqu'à la fin	Section V "Performance," conforme à l'édition de la FAA appropriée.
6-1 à la jusqu'à la fin	Section VI "Weight & Balance/Equip List," conforme à l'édition de la FAA appropriée.
7-1 à la jusqu'à la fin	Section VII "Systems Description," conforme à l'édition de la FAA appropriée.
8-1 à la jusqu'à la fin	Section VIII "Handling, Serv & Maint," conforme à l'édition de la FAA appropriée.
9-1 à la jusqu'à la fin	Section IX "Supplements," conforme à l'édition de la FAA appropriée.
1 sur 1 (Suppléments)	Nouvelle - Liste des Suppléments approuvés par DGAC incorporés au Manuel de Vol (France).

C5 FR

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Raytheon Aircraft

Beech Super King Air® B200 & B200C

(N^OS de Sèrie BB-734, BB-793, BB-829, BB-854 à BB-870, BB-874 à BB-891, BB-894, BB-896 à BB-911, BB-913 à BB-1438, BB-1440 à BB-1443; BL-37 à BL-138)

MANUEL DE VOL

P/N 101-590010-147

(Revision C5 et suivante)

LISTE DE PERSONNES AUTORISEES A UTILISER LE MANUEL DE VOL EN LANGUE ANGLAISE DE L'AERONEF IMMATRICULE EN DEHORS D'UNE EXPLOITATION EN TRANSPORT PUBLIC.

Pilote	Qualification	Approbation DGAC

CETTE PAGE LAISSÉE BLANCE INTENTIONNELLENENT

RAYTHEON AIRCRAFT

BEECH SUPER KING AIR® B200 & B200C LANDPLANES

(N^{os} de Série BB-734, BB-793, BB-829, BB-854 à BB-870, BB-874 à BB-891, BB-894, BB-896 à BB-911, BB-913 à BB-1438, BB-1440 à BB-1443; BL-37 à BL-138)

MANUEL DE VOL

P/N 101-590010-147

(Revision C5 et suivante)

LISTE DES SUPPLEMENTS

Le Supplément doit être dans l'avion pour utilisation quand l'équipement est installé			
Référence	Sujet	Rév	Date

CETTE PAGE LAISSÉE BLANCE INTENTIONNELLENENT

Raytheon Aircraft

Beech Super King Air® B200/B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-950 Only)

This Supplement is Applicable to the Following Manual(s):
101-590010-147

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: Dawn M Wingfield
for John Tigue
Raytheon Aircraft Company
DOA-230339-CE

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Issued: July, 2001
P/N 101-590010-417

For Training Purposes Only

Raytheon Aircraft

LOG OF REVISIONS

Beech Super King Air® B200/B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-950 Only)

REV NO.	PAGE NO(S).	DESCRIPTION	DATE OF REV	FAA APPROVED
0	1 thru 4	Original Issue	July, 2001	Dawn M. Winfield

P/N 101-590010-417

For Training Purposes Only

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SECTION I - GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION II - LIMITATIONS

STRUCTURAL LIMITATIONS

LOCATION	A/C SN	SAFE LIFE / INITIAL INSPECTION	SUBSEQUENT INSPECTION
Wing and Associated Structure	BB-950	20,377 HRS	N/A
Lower Main Spar Cap Bathtub Fitting	BB-950	12,000 HRS	600 HRS

Refer to the *Beech Structural Inspection and Repair Manual* and the *Beech Super King Air 200 Series Maintenance Manual* for inspection and replacement procedures.

The wing and associated structural items are analyzed based on the following flight profiles.

FLIGHT PROFILES

Profile 1: Cruise - 18% - Flight Time 25 Minutes; Take-off Weight 12,230 lbs; Take-off Fuel 2730 lbs; Cruise 4000 FT AGL @ 233 KIAS.

Profile 2: Cruise - 32% - Flight Time 45 Minutes; Take-off Weight 11,646 lbs; Take-off Fuel 2146 lbs; Cruise 8000 FT AGL @ 248 KIAS.

Profile 3: Cruise - 32% - Flight Time 45 Minutes; Take-off Weight 10,929 lbs; Take-off Fuel 1429 lbs; Cruise 8000 FT AGL @ 248 KIAS.

Profile 4: Cruise - 18% - Flight Time 25 Minutes; Take-off Weight 10,343 lbs; Take-off Fuel 843 lbs; Cruise 5000 FT AGL @ 237 KIAS.

SECTION III - EMERGENCY PROCEDURES

No Change

SECTION IIIA - ABNORMAL PROCEDURES

No Change

SECTION IV - NORMAL PROCEDURES

No Change

SECTION V - PERFORMANCE

No Change

SECTION VI - WEIGHT & BALANCE/EQUIPMENT LIST

No Change

SECTION VII - SYSTEMS DESCRIPTION

No Change

SECTION VIII - HANDLING, SERVICING & MAINTENANCE

No Change

FAA Approved
Issued: July, 2001
P/N 101-590010-417

For Training Purposes Only

Raytheon Aircraft Company

Beechcraft

Super King Air® B200 & B200C
Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-1125 and BB-1189)

This Supplement is Applicable to the Following Manual(s):
101-590010-147

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: Bob Redus for
John Tigue
Raytheon Aircraft Company
DOA-230339-CE

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Issued: September, 2003
P/N 101-590010-441

For Training Purposes Only

Raytheon Aircraft Company

LOG OF REVISIONS

Super King Air® B200 & B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-1125 and BB-1189)

REV NO.	PAGE NO(S).	DESCRIPTION	DATE OF REV
0	1 thru 4	Original Issue	September, 2003

P/N 101-590010-441

For Training Purposes Only

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SECTION I - GENERAL

The information in this supplement is FAA-approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual (POH/AFM) when BB-1125 and BB-1189 are operated in accordance with Raytheon Aircraft Company-approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth within this document. Users of the manual are advised always to refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION II - LIMITATIONS

STRUCTURAL LIMITATIONS

LOCATION	INITIAL INSPECTION INTERVAL/LIFE LIMIT	RECURRING INSPECTION INTERVAL	COMMENTS
Wing and Associated Structure	23,000 HRS		Life Limited
Tension Fitting	20,000 HRS	1,200 HRS	

Refer to the Raytheon Aircraft Company Structural and Inspection Repair Manual and the Super King Air 200 Series Maintenance Manual for inspection and replacement procedures.

The wing and associated structural items are analyzed based on the following mission profiles.

MISSION PROFILES

Mission Profile	Mission Mix Percentages	
	BB-1125	BB-1189
Basic General Flying (BAS GF)	9.52%	7.34%
Basic Navigation (BAS NAV)	35.69%	33.82%
Basic Low-Level (BAS LL)	5.60%	5.61%
Basic Low-Level Over Water (BAS LW)	1.73%	0.89%
Maritime (MAR)	9.68%	9.16%
Tactical (TAC A)	6.22%	6.59%
Tactical (TAC B)	3.88%	4.11%
Navigation Continuation Training (NAVCONT)	12.79%	12.93%
Pilot Continuation Training (PCONT)	14.91%	19.56%
Mission Profile Total Hours	**4039.2**	**3429.5**

NOTE

After mission profile total hours are completed, these airplanes will return to flying the typical executive profiles. Mission profiles differing from those defined above can affect the structural limitations. The operator should contact Raytheon Aircraft Company for a fatigue evaluation if the mission profiles change.

SECTION III - EMERGENCY PROCEDURES

No Change

SECTION IIIA - ABNORMAL PROCEDURES

No Change

SECTION IV - NORMAL PROCEDURES

No Change

SECTION V - PERFORMANCE

No Change

SECTION VI - WEIGHT AND BALANCE/EQUIPMENT LIST

No Change

SECTION VII - SYSTEMS DESCRIPTION

No Change

SECTION VIII - HANDLING, SERVICING & MAINTENANCE

No Change

Raytheon Aircraft Company

Beechcraft

Super King Air® B200 & B200C

Pilot's Operating Handbook and FAA Approved Airplane Flight Manual Supplement

for Wing Fatigue Limitations (BB-968, BB-1008, BB-1054, BB-1087, and BB-1178)

This Supplement is Applicable to the Following Manual(s):
101-590010-147

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved by: _____
for John Tigue
Raytheon Aircraft Company
DOA-230339-CE

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Issued: September, 2005
P/N 101-590010-465

For Training Purposes Only

Raytheon Aircraft Company

LOG OF REVISIONS

Super King Air® B200 & B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-968, BB-1008, BB-1054, BB-1087, and BB-1178)

REV NO.	PAGE NO(S).	DESCRIPTION	DATE OF REV
0	1 thru 18	Original Issue	Sep, 2005

For Training Purposes Only

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SECTION 1 - GENERAL

The information in this supplement is FAA approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual (POH/AFM) when BB-968, BB-1008, BB-1054, BB-1087, and BB-1178 are operated in accordance with Raytheon Aircraft approved data.

The information in this supplement supersedes or adds to the basic POH/AFM only as set forth below. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION 2 - LIMITATIONS

STRUCTURAL LIMITATIONS

LOCATION	INITIAL INSPECTION INTERVAL/LIFE LIMIT	COMMENTS
Wing and Associated Structure	30,000 HRS	Provided special inspections defined in SIRM supplement, P/N 101-590010-467, are adhered to.

NOTE

The Structural Limitations are based on the following mission profiles.

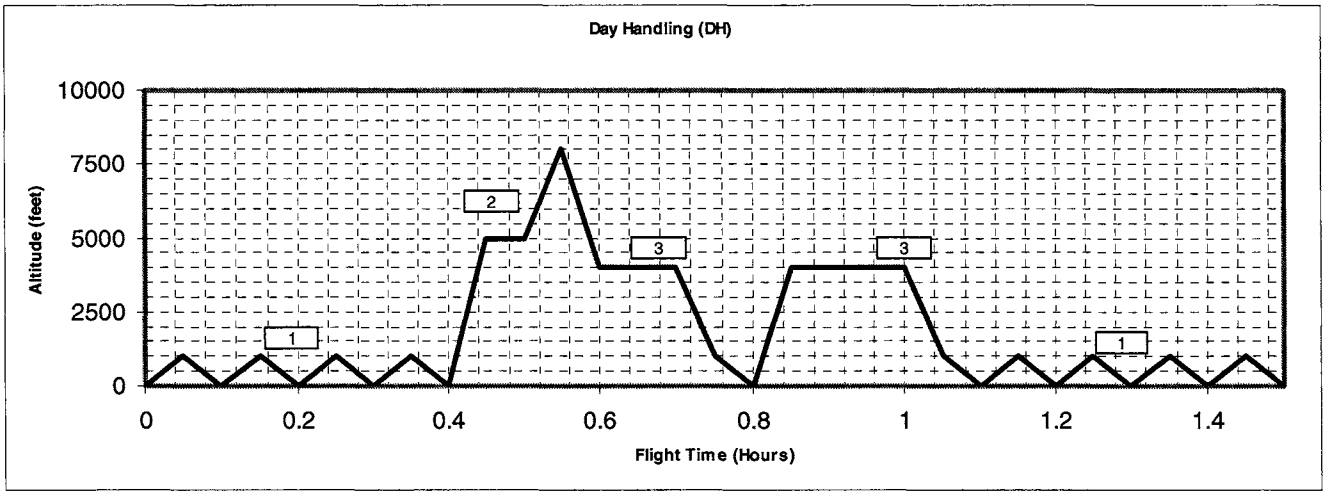
MISSION PROFILES

DAY HANDLING (DH) (34.4% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	1.00	Engine Restarts	0.10	
Fuel Weight	2500 lbs	Touch & Goes	8.00	Additional L/G Cycles	2.00	
Flight Time	1.50 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	8.00	4.00

NOTE

1. Circuits in accordance with Pilots Operating Handbook.
2. General Handling up to 210 KIAS including Stalling and Steep Turns at 60 AOG.
3. Instrument Approaches in accordance with Pilots Operating Handbook.

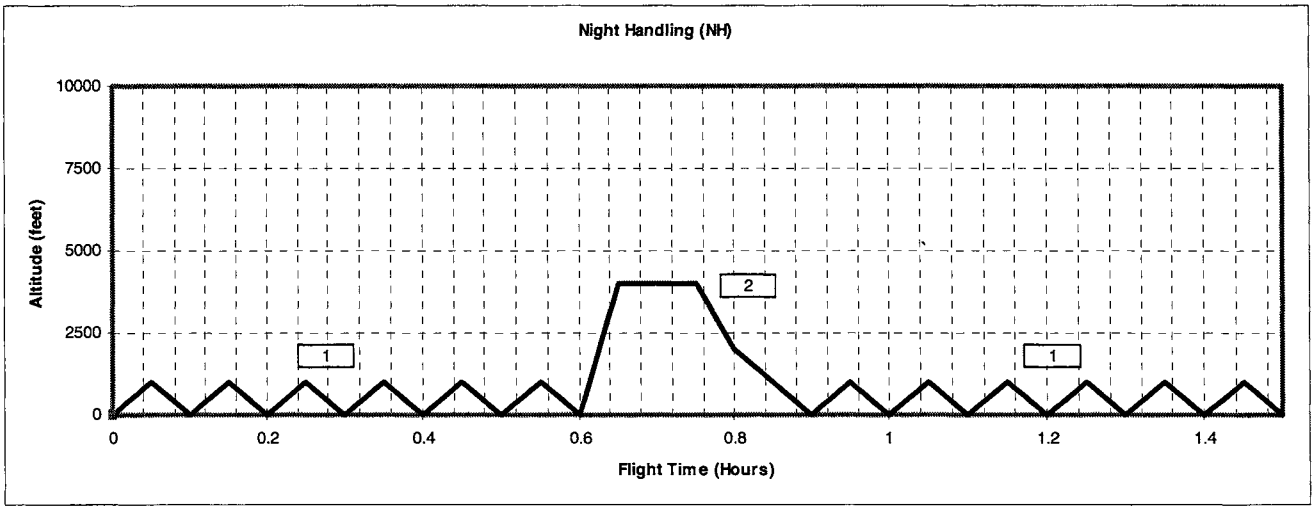


NIGHT HANDLING (NH) (2.5% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	0.00	Engine Restarts	0.00	
Fuel Weight	2500 lbs	Touch & Goes	7.00	Additional L/G Cycles	0.00	
Flight Time	1.00 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	0.00	0.00

NOTE

1. Circuits in accordance with Pilots Operating Handbook.
2. Instrument Approaches in accordance with Pilots Operating Handbook.

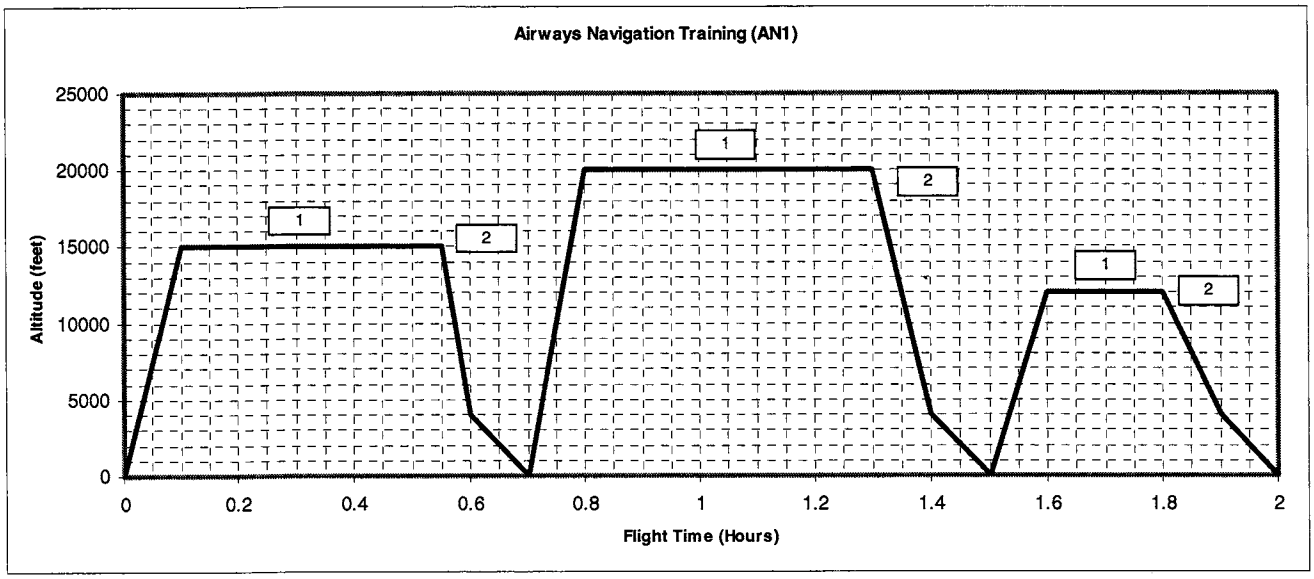


AIRWAYS NAVIGATION TRAINING (AN1) (15% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	3.00	Engine Restarts	0.00	
Fuel Weight	2500 lbs	Touch & Goes	2.00	Additional L/G Cycles	0.00	
Flight Time	1.75 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	1.00	0.00

NOTE

1. Airways Flying 200 KIAS.
2. Instrument Approaches in accordance with Pilots Operating Handbook.

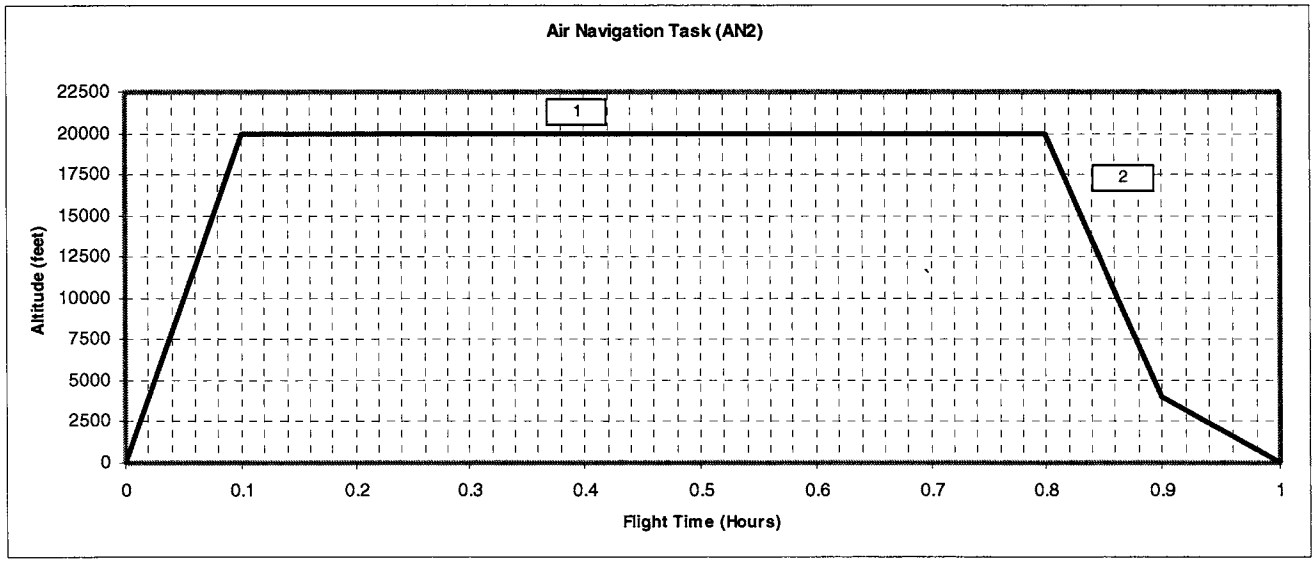


AIR NAVIGATION TASK (AN2) (27% OF FLIGHT TIME)

Gross Weight	12,000 lbs	Press	1.00	Engine Restarts	0.00	
Fuel Weight	2000 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	1.25 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	1.00	0.00

NOTE

1. Airways Flying 180 KIAS.
2. Instrument or Visual Approaches in accordance with Pilots Operating Handbook.

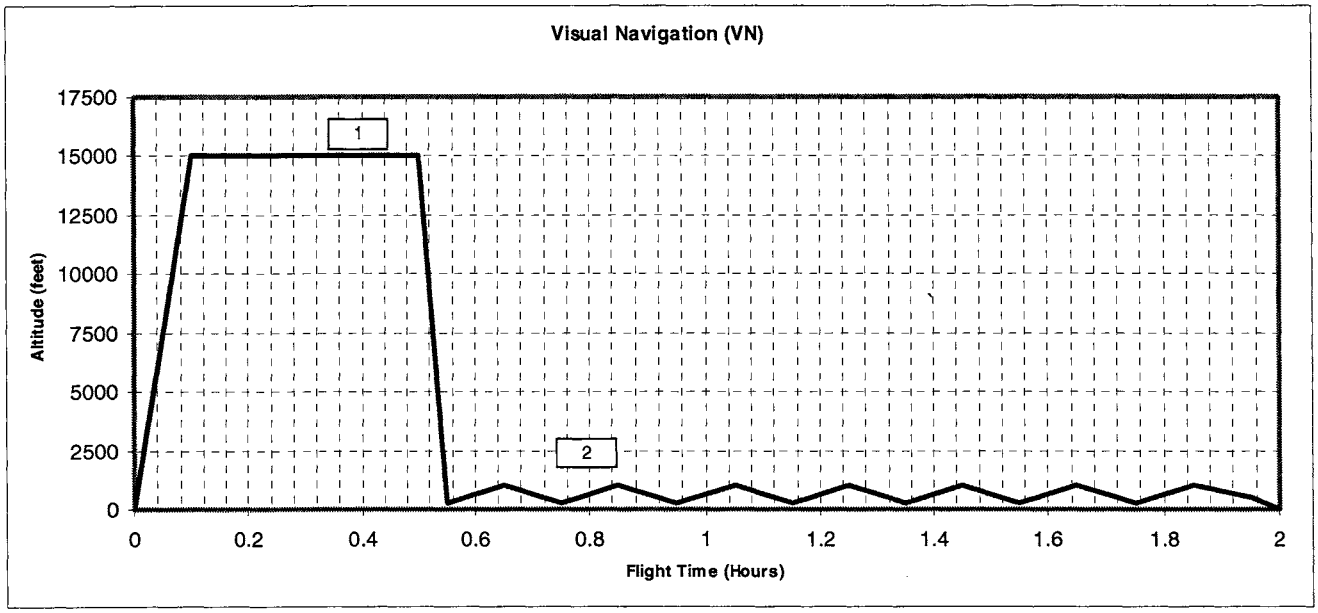


VISUAL NAVIGATION (VN) (5% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	1.00	Engine Restarts	0.00	
Fuel Weight	2500 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	1.75 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	20.00	2.00

NOTE

1. Airways Flying 200 KIAS.
2. Low Level Navigation 210 KIAS, 250 ft min, Steep Turns to 60 deg AOB.

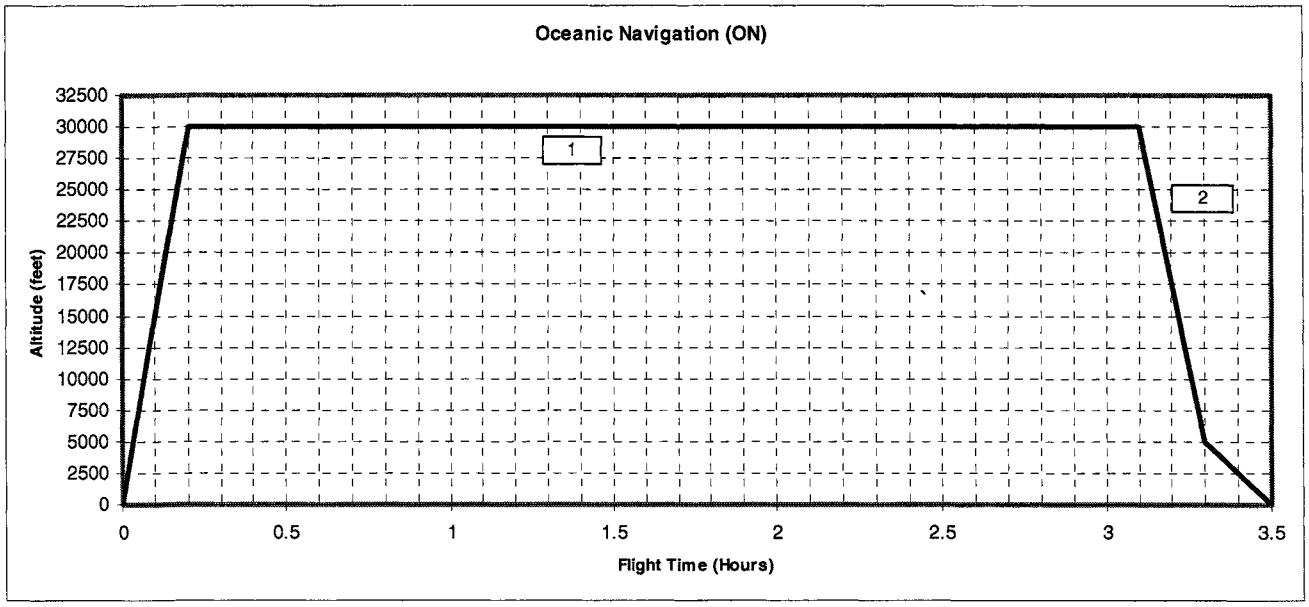


OCEANIC NAVIGATION (ON) (5% OF FLIGHT TIME)

Gross Weight	12,500 lbs	Press	1.00	Engine Restarts	0.00	
Fuel Weight	3000 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	2.5 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	1.00	0.00

NOTE

1. Airways Flying 180 KIAS.
2. Radar Let Down, Instrument or Visual Approach in accordance with Pilots Operating Handbook.

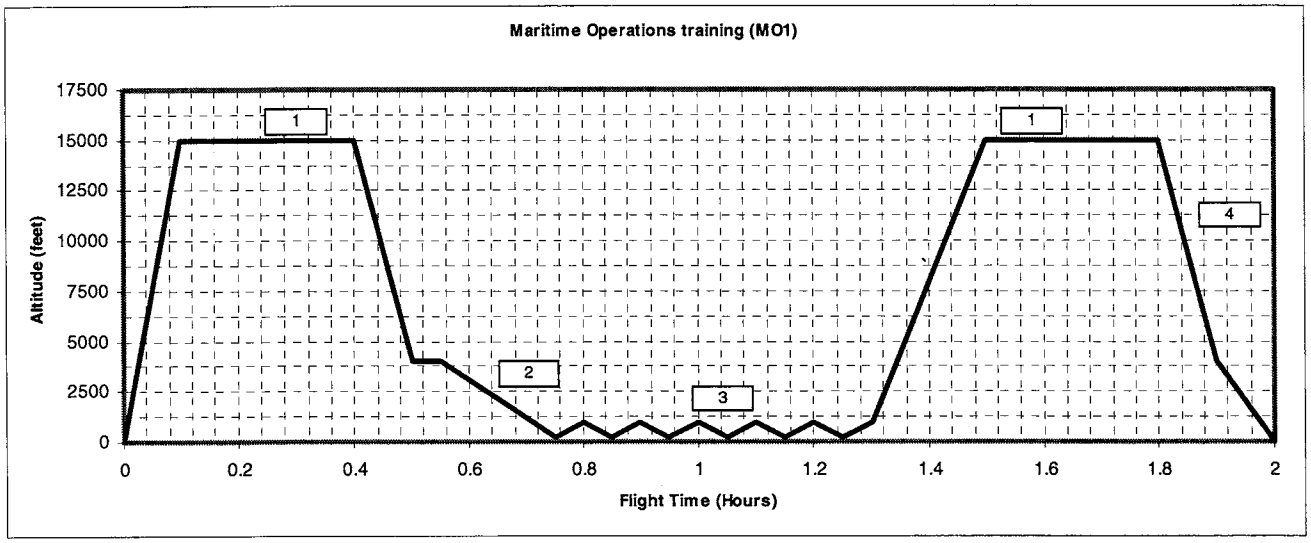


MARITIME OPERATIONS TRAINING (MO1) (1% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	2.00	Engine Restarts	0.00	
Fuel Weight	2500 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	2.75 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	20.00	10.00

NOTE

1. Airways Flying 210 KIAS.
2. Radar Let Down at 150 KIAS.
3. Maritime Handling 210 KIAS, Steep Turns to 60 deg AOB down to 200 ft.
4. Instrument or Visual Approach in accordance with Pilots Operating Handbook.

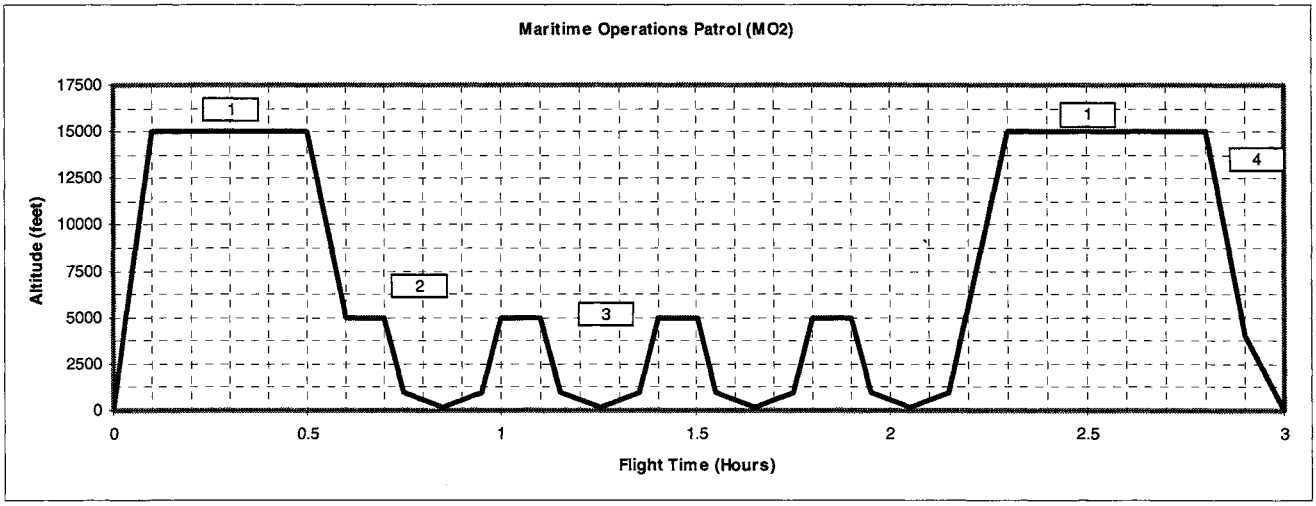


MARITIME OPERATIONS PATROL (MO2) (1% OF FLIGHT TIME)

Gross Weight	12,500 lbs	Press	2.00	Engine Restarts	0.00	
Fuel Weight	3500 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	2.75 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	5.00	0.00

NOTE

1. Airways Flying 200 KIAS.
2. Radar Let Down at 150 KIAS.
3. Maritime Handling 210 KIAS, Steep Turns to 60 deg AOB down to 200 ft.
4. Instrument or Visual Approach in accordance with Pilots Operating Handbook.

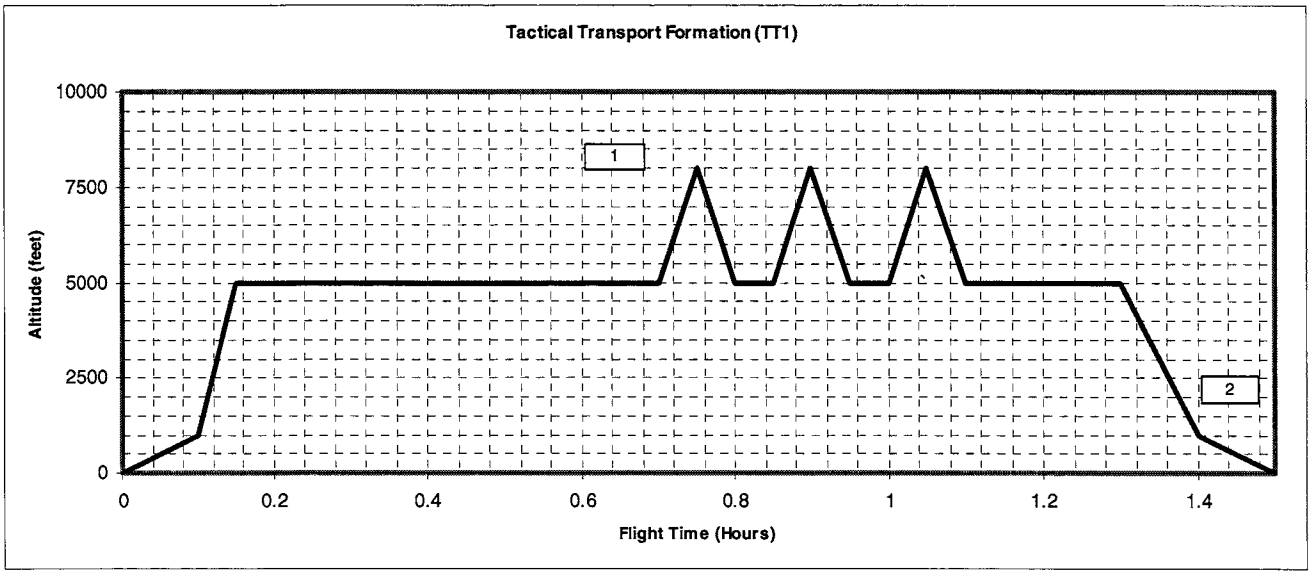


TACTICAL TRANSPORT FORMATION (TT1) (2% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	1.00	Engine Restarts	0.00	
Fuel Weight	2500 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	1.75 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	5.00	0.00

NOTE

1. Formation Handling 210 KIAS, Turns to 45 deg AOB.
2. Formation Buzz and Break 60 deg AOB followed by Visual Approach.

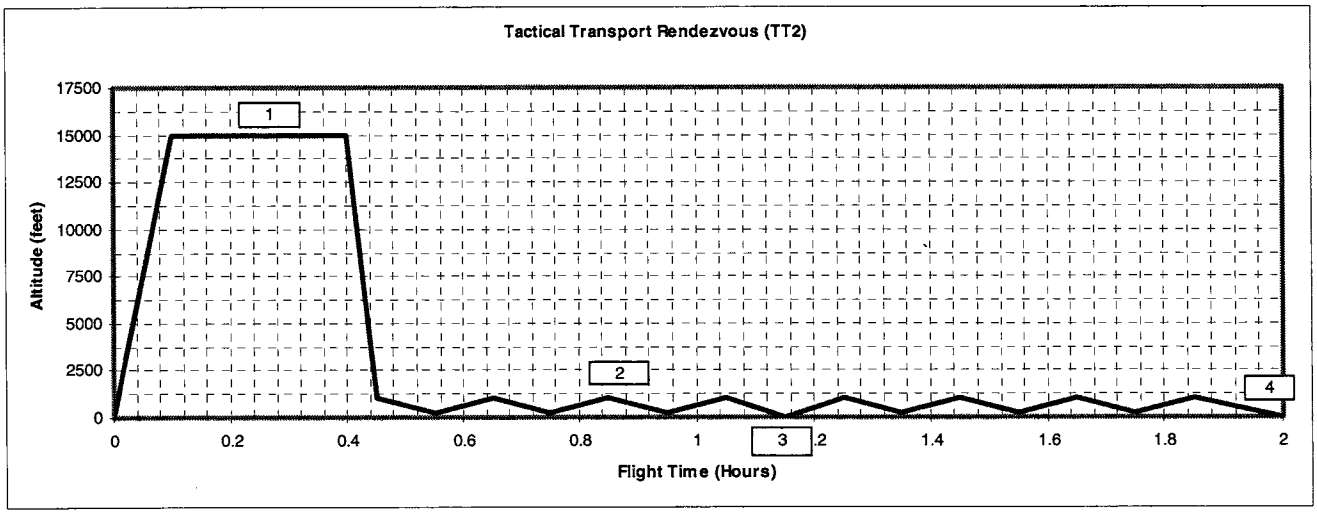


TACTICAL TRANSPORT RENDEZVOUS (TT2) (3.5% OF FLIGHT TIME)

Gross Weight	11,500 lbs	Press	1.00	Engine Restarts	0.00	
Fuel Weight	2500 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	1.75 hrs	Full Stop	2.00	G Maneuvers (1.5, 2)	20.00	2.00

NOTE

1. Airways Flying 200 KIAS.
2. Low Level Navigation 210 KIAS.
3. Grass Airfield Full Stop Landing (no refuel).
4. Buzz and Break 60 deg AOB followed by Visual Approach

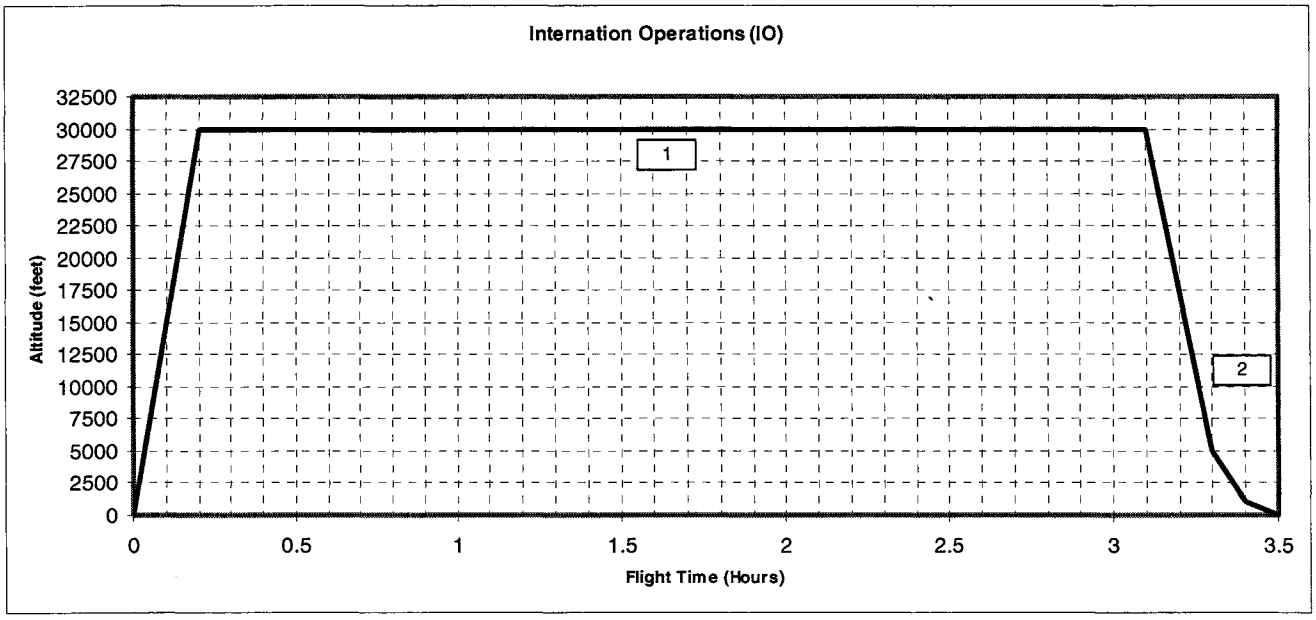


INTERNATIONAL OPERATIONS (IO) (2.5% OF FLIGHT TIME)

Gross Weight	13,000 lbs	Press	1.00	Engine Restarts	0.00	
Fuel Weight	3000 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	3.5 hrs	Full Stop	1.00	G Maneuvers (1.5, 2)	1.00	0.00

NOTE

1. Airways Flying 180 KIAS.
2. Instrument or Visual Approach in accordance with Pilots Operating Handbook.



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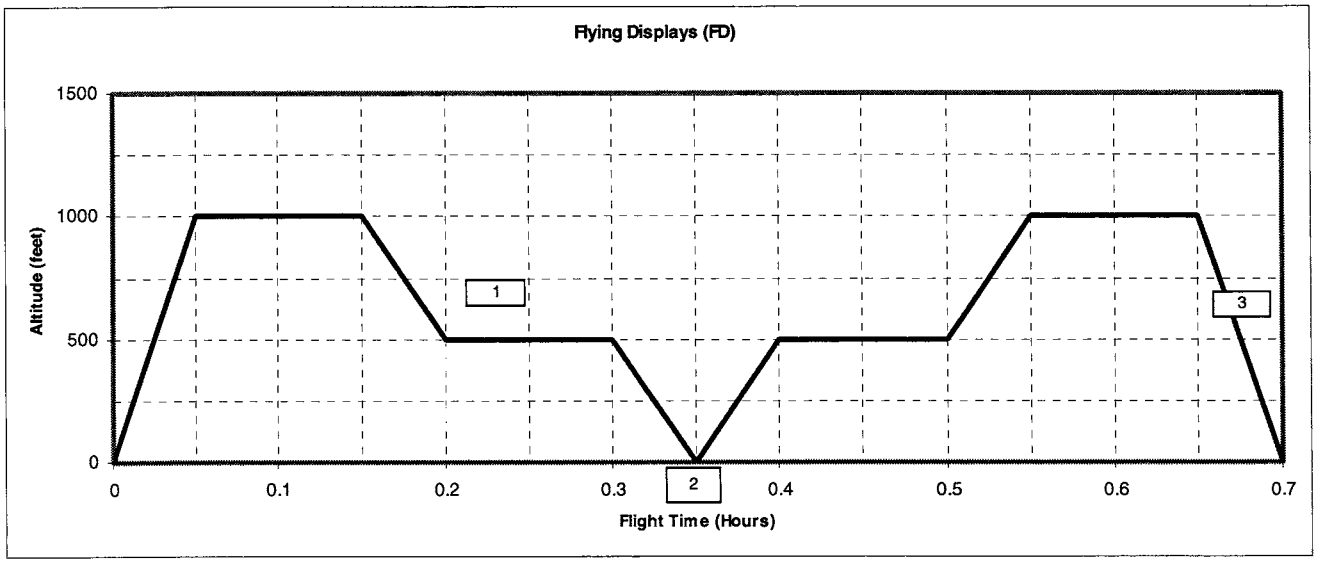
FLYING DISPLAYS (FD) (0.1% OF FLIGHT TIME)

Gross Weight	11,000 lbs	Press	0.00	Engine Restarts	0.00	
Fuel Weight	2000 lbs	Touch & Goes	0.00	Additional L/G Cycles	0.00	
Flight Time	0.75 hrs	Full Stop	2.00	G Maneuvers (1.5, 2)	5.00	2.00

NOTE

1. Handling Display up to 240 KIAS Steep Turns to 60 deg AOB.
2. Full Stop Landing, Back Track and Takeoff (no refuel).
3. Visual Approach in accordance with Pilots Operating Handbook.

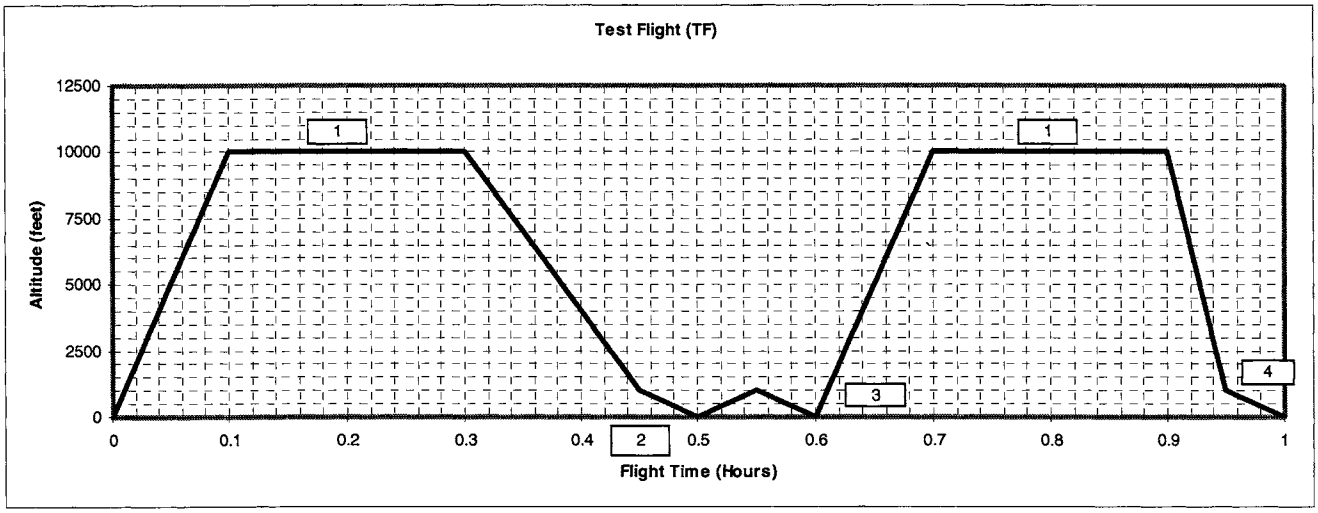


TEST FLIGHT (TF) (1% OF FLIGHT TIME)

Gross Weight	11,000 lbs	Press	2.00	Engine Restarts	0.00	
Fuel Weight	2000 lbs	Touch & Goes	1.00	Additional L/G Cycles	0.00	
Flight Time	0.75 hrs	Full Stop	2.00	G Maneuvers (1.5, 2)	1.00	0.00

NOTE

1. General Handling including Stalling and Steep Turns at 60 deg AOB.
2. Touch and Go in accordance with Pilots Operating Handbook.
3. Full Stop Landing in accordance with Pilots Operating Handbook.
4. Instrument Approach in accordance with Pilots Operating Handbook.



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SECTION 3 - EMERGENCY PROCEDURES

No Change

SECTION 3A - ABNORMAL PROCEDURES

No Change

SECTION 4 - NORMAL PROCEDURES

No Change

SECTION 5 - PERFORMANCE

No Change

SECTION 6 - WEIGHT AND BALANCE\EQUIPMENT LIST

No Change

SECTION 7 - SYSTEMS DESCRIPTION

No Change

SECTION 8 - HANDLING, SERVICING & MAINTENANCE

No Change

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Hawker Beechcraft Corporation

Beech Super King Air®
B200/B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-944 & BB-1048 Only)

This Supplement is Applicable to the Following Manual(s):
101-590010-147

Airplane Serial Number: _____

Airplane Registration Number: _____

FAA Approved R.C. Shields
Randolph Shields, ODA Lead administrator
Hawker Beechcraft Corporation
ODA-230339-CE

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P/N 101-590010-505

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Hawker Beechcraft Corporation

LOG OF REVISIONS

Beech Super King Air® B200/B200C

Pilot's Operating Handbook
and
FAA Approved Airplane Flight Manual Supplement

for
Wing Fatigue Limitations
(BB-944 & BB-1048 Only)

REV NO.	PAGE NO(S).	DESCRIPTION	DATE OF REV
0	1 thru 12	Original Issue	June, 2010

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SECTION I - GENERAL

The information in this supplement is FAA approved material and must be attached to the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual when BB-944 and BB-1048 are operated in accordance with Hawker Beechcraft Corporation approved data.

The information in this supplement supersedes or adds to the basic Pilot's Operating Handbook and FAA Approved Airplane Flight Manual only as set forth within this document. Users of the manual are advised to always refer to the supplement for possibly superseding information and placarding applicable to operation of the airplane.

SECTION II - LIMITATIONS

STRUCTURAL LIMITATIONS

Area	Safelife
Wing and Associated Structure Fatigue Safelife	22,700 Hours

Refer to the *Structural Inspection and Repair Manual Supplement (101-590010-501)* and the *Super King Air® 200 Series Maintenance Manual* for inspection and replacement procedures.

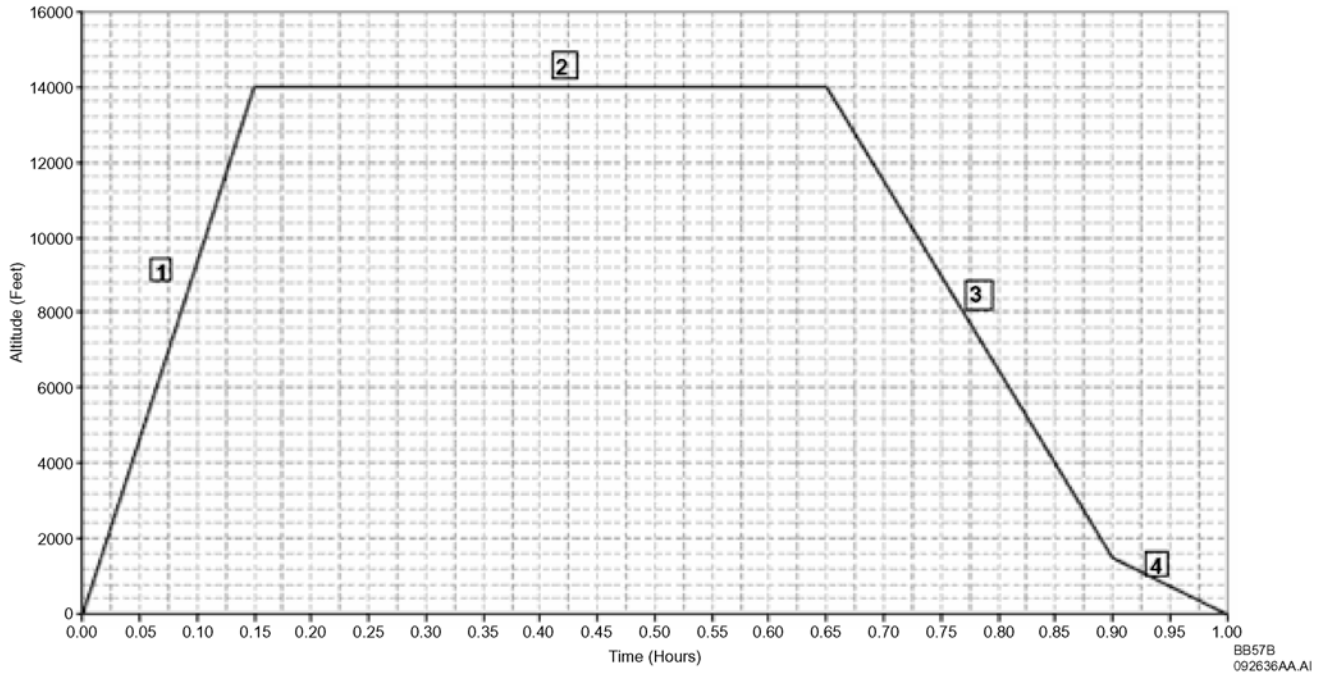
The wing and associated structural items are analyzed based on the following mission profiles:

MISSION PROFILES

SHORT TRANSIT - 20%

Gross Weight:	12,500 lbs
Fuel Weight:	2,500 lbs
Total Flight Time:	1.0 hrs
Cabin Diff:	4.5 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Climb	0.13	7,000	150
2. Cruise	0.52	14,000	200
3. Descent	0.25	8,000	210
4. Approach	0.1	1,000	140

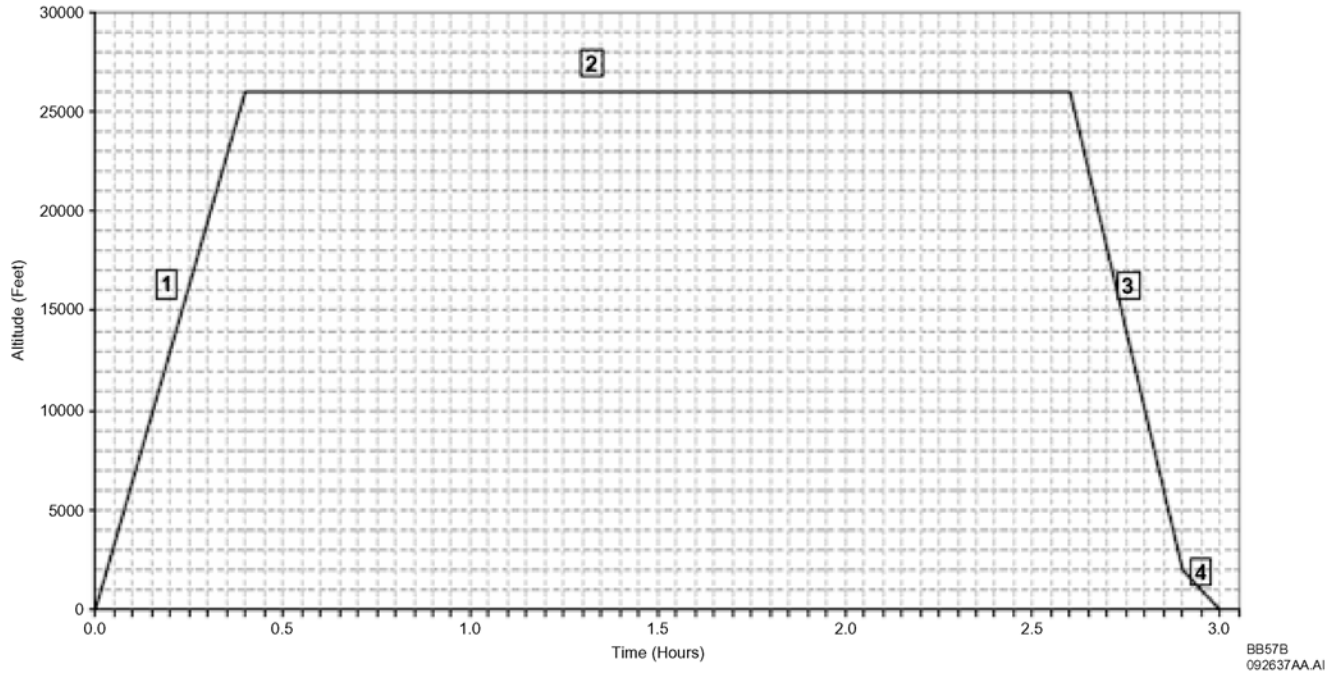


SHORT TRANSIT - 20%

LONG TRANSIT - 13%

Gross Weight:	12,500 lbs
Fuel Weight:	2,500 lbs
Total Flight Time:	3.0 hrs
Cabin Diff:	6.4 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Climb	0.3	13,000	140
2. Cruise	2.2	26,000	190
3. Descent	0.4	13,000	220
4. Approach	0.1	1,000	140

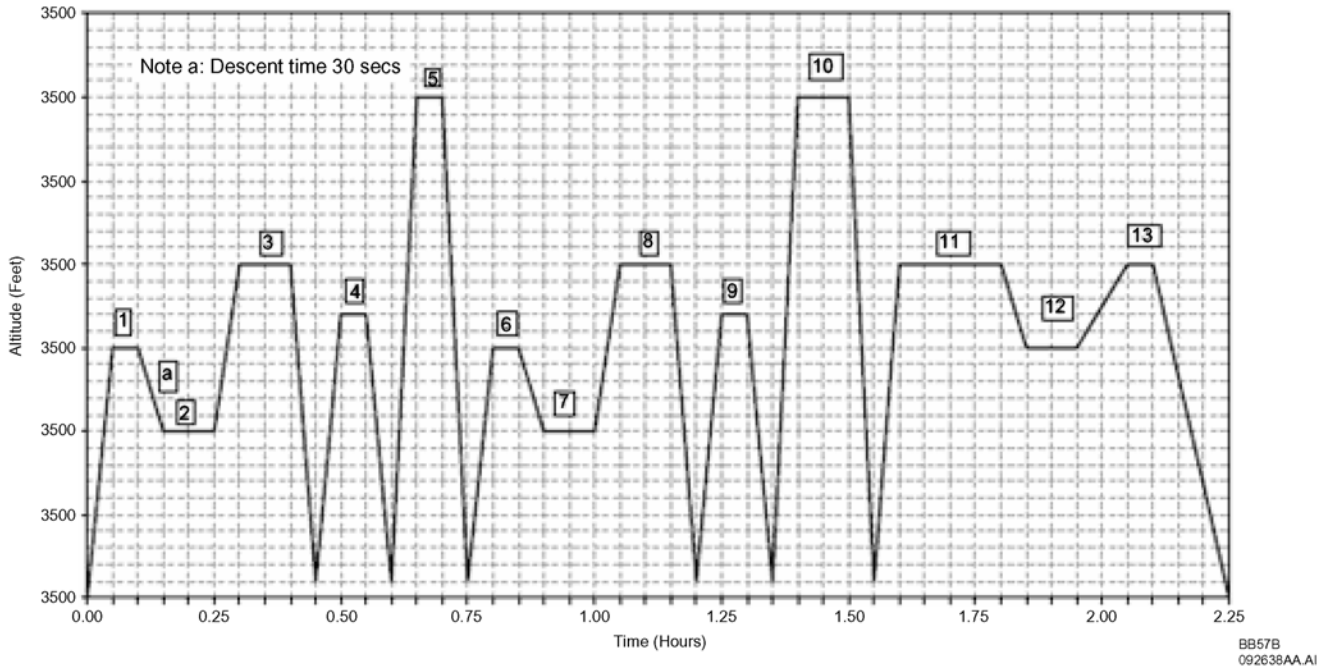


LONG TRANSIT - 20%

ILS - 20%

Gross Weight:	12,500 lbs
Fuel Weight:	3,000 lbs
Total Flight Time:	2.25 hrs
Cabin Diff:	1.5 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Part Orbit	0.10	1,500	170
2. Slice	0.17	1,000	170
3. Approach - Above GP	0.17	1,000	170
4. Approach - Below GP	0.17	850	170
5. Approach	0.17	1,500	170
6. Part Orbit	0.10	1,500	170
7. Slice	0.17	1,000	170
8. Approach - Above GP	0.17	1,000	170
9. Approach - Below GP	0.17	850	170
10. Approach	0.17	1,500	170
11. Range Run	0.27	2,000	170
12. 8 Degree Runs	0.17	1,500	170
13. Approach	0.25	500	140



ILS - 20%

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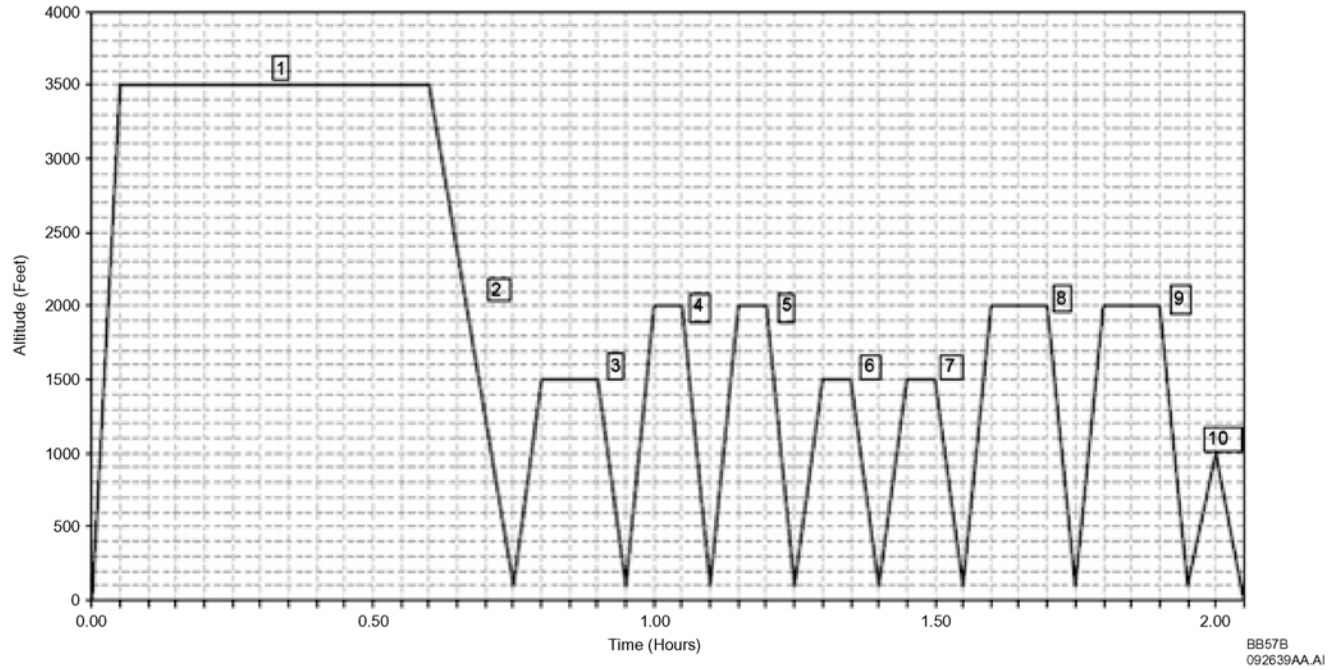
RPAR - 6%

Gross Weight:	12,500 lbs
Fuel Weight:	3,000 lbs
Total Flight Time:	2.06 hrs
Cabin Diff:	1.5 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Performance Checks	0.6	3,500	170
2. Approach	0.17	750	170
3. Approach	0.17	750	170
4. Approach	0.17	1,000	170
5. Approach	0.17	1,000	170
6. Approach	0.17	750	170
7. Approach	0.17	750	170
8. Approach	0.17	1,000	170
9. Approach	0.17	1,000	170
10. Approach	0.1	500	140

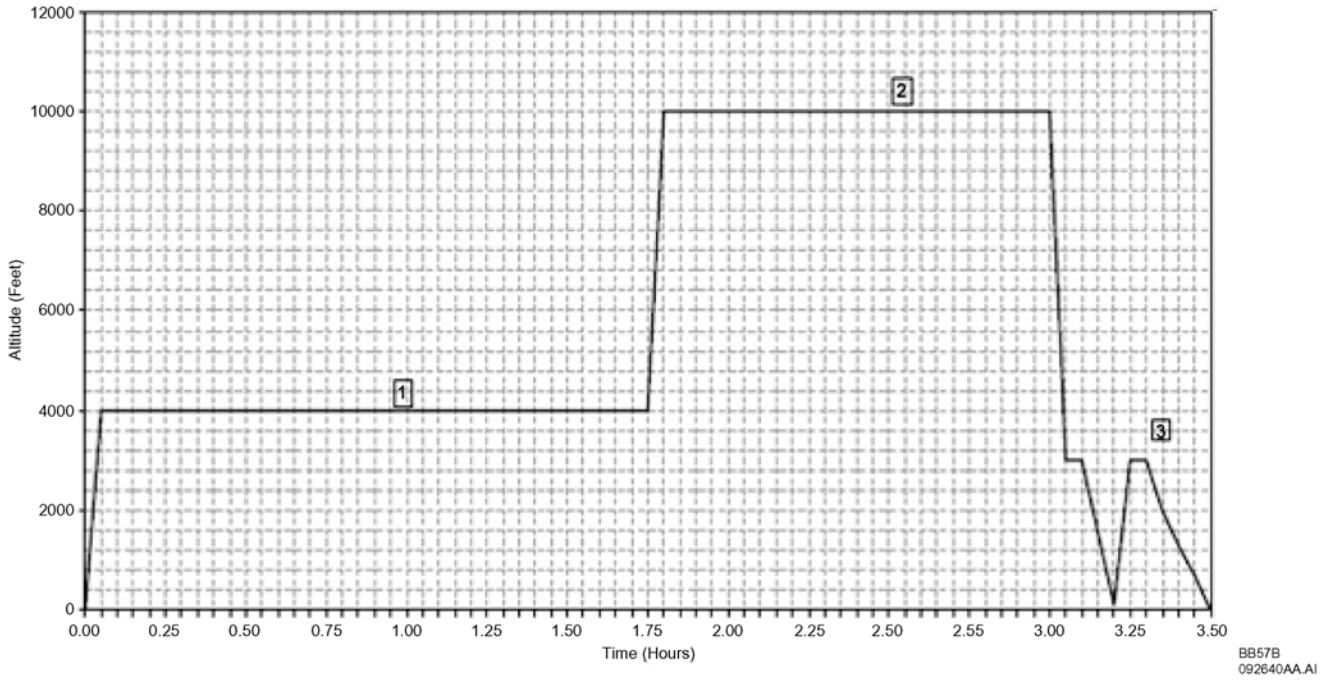


RPAR - 6%

VOR/TACAN/NDB/DME - 22%

Gross Weight:	12,500 lbs
Fuel Weight:	3,000 lbs
Total Flight Time:	3.5 hrs
Cabin Diff:	4.0 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Two Stab Checks and Two Orbits	1.75	4,000	190
2. Radial Flying	1.25	10,000	200
3. Approaches	0.5	1,000	170



VOR/TACAN/NDB/DME - 22%

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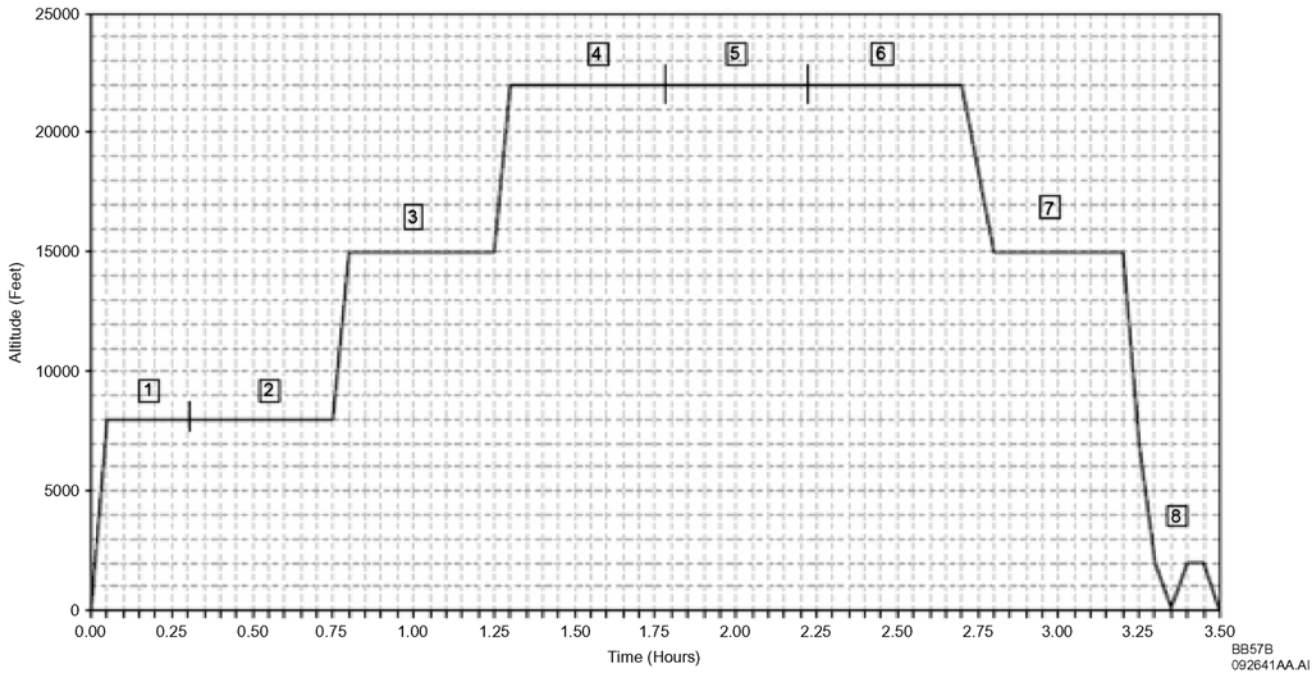
SURVEILLANCE RADARS/MSSR - 17%

Gross Weight:	12,500 lbs
Fuel Weight:	3,200 lbs
Total Flight Time:	3.5 hrs
Cabin Diff:	6.2 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Transit to Start	0.25	4,000	200
2. 8,000 ft Runs	0.5	8,000	200
3. 15,000 ft Runs	0.5	15,000	190
4. 22,000 ft Runs	0.5	22,000	170
5. Overhead Run	0.45	22,000	170
6. 22,000 ft Runs	0.5	22,000	170
7. 15,000 ft Runs	0.5	15,000	190
8. Approaches	0.3	1,000	170

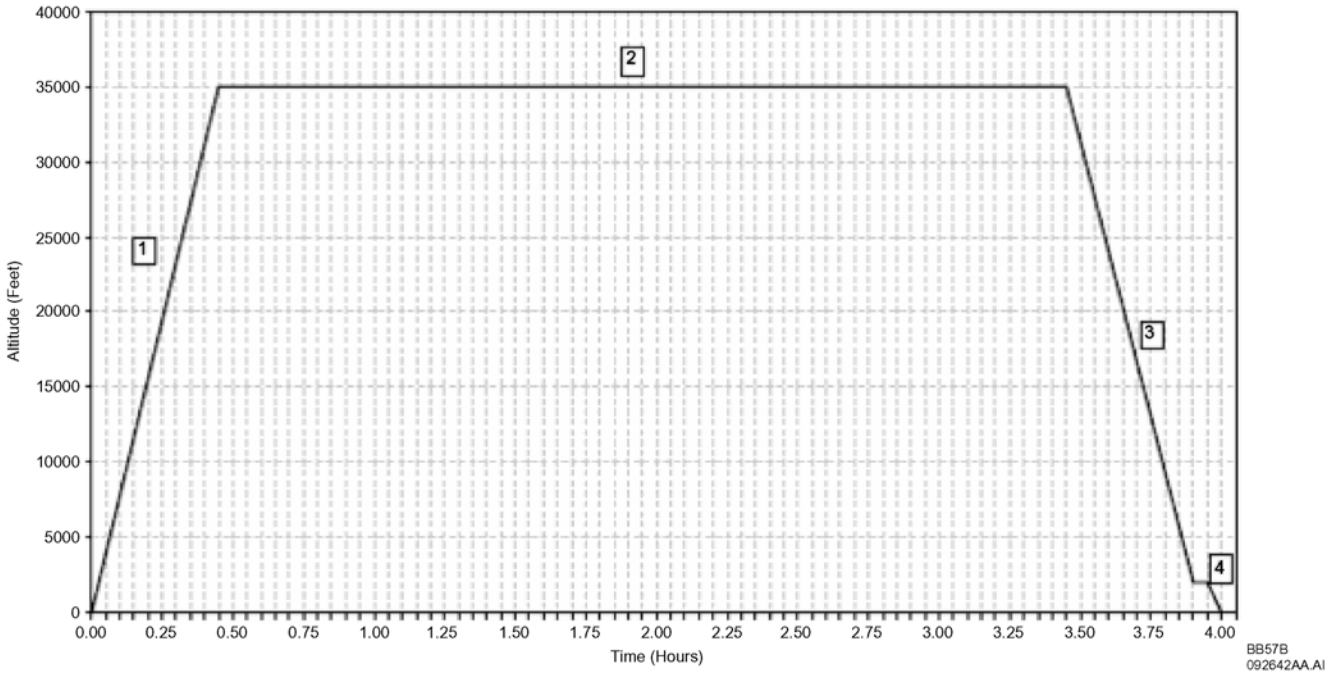


SSR/MSSR - 17%

HIGH LEVEL RADARS - 2%

Gross Weight:	12,500 lbs
Fuel Weight:	3,000 lbs
Total Flight Time:	4.0 hrs
Cabin Diff:	6.6 psi
Landing Gear Cycles:	1

Profile	Time (hrs)	Mean Altitude (Ft)	Mean Speed (KIAS)
1. Climb	0.45	17,500	140
2. Cruise	3.12	35,000	150
3. Descent	0.33	35,000	210
4. Approach	0.1	1,000	140



HIGH LEVEL RADAR - 2%

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SECTION III - EMERGENCY PROCEDURES

No change.

SECTION IIIA - ABNORMAL PROCEDURES

No change.

SECTION IV - NORMAL PROCEDURES

No change.

SECTION V - PERFORMANCE

No change.

SECTION VI - WEIGHT & BALANCE/EQUIPMENT LIST

No change.

SECTION VII - SYSTEMS DESCRIPTION

No change.

SECTION VIII - HANDLING, SERVICING & MAINTENANCE

No change.

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BEECHCRAFT SUPER KING AIR 200, 200C, 200T, 200CT, B200, B200C, B200T, B200CT LANDPLANES

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL SUPPLEMENT

for the

DECCA DOPPLER 72 NAVIGATION SYSTEM (TANS COMPUTER, TYPE 9447C)

GENERAL

The information in this supplement is FAA Approved material which along with the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual is applicable to the operation of the airplane when modified by the installation of the Decca Doppler 72 Navigation System installed in accordance with BEECHCRAFT FAA Approved data. The information in this supplement supersedes or adds to that of the basic airplane flight manual. Users of this manual are advised always to refer to the supplement for possible superseding information and placarding applicable to the operation of this airplane.

The operation of the Decca Doppler 72 Navigation System is to be accomplished in conformity with the procedures included in the Decca Doppler 70 Series Aircrew Operating Instructions (October 1973 or later issue). This publication is available from the Decca Navigator Company Ltd., Aviation Electronics Division, Decca House, 9 Albert Embankment, London SE 1 7SW, England. The Aircrew Operating Instructions must be in the airplane and available to the pilot during all flight operations using the Decca Doppler 72 Navigation System.

This navigation system does not adversely affect any other airplane system. However, the functional and performance characteristics have not been fully evaluated.

LIMITATIONS

Flight operations must not be predicated on its use as the primary source of navigation.

EMERGENCY PROCEDURES

The Doppler system drives the normal VOR/LOC warning flag when Doppler information is displayed on the horizontal situation indicator (HSI). The warning flag will be in view anytime the Doppler information is invalid (e.g. Doppler failure or computer malfunction).

Refer to the Decca Doppler 70 Series Aircrew Operating Instructions (October 1973 or later issue).

NORMAL PROCEDURES

Refer to the Decca Doppler 70 Series Aircrew Operating Instructions (October 1973 or later issue).

DOPPLER CONTROL

A rotary switch is provided to activate the Doppler system power, conduct the appropriate system tests, and select the Doppler for presentation on the HSI.

1. Test A - Allows manual selection of the "Memory" condition to permit simulated Doppler speeds to be fed into the TANS computer for ground checking purposes.

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2. Test B - Provides an artificial spectrum for self-testing the Doppler system.

3. OPR - Activates the Doppler ground speed and drift meter.

4. FD CPLD - Provided the TANS computer is properly programmed, Doppler steering information is presented on the flight director.

NAVIGATION DISPLAYS

The course deviation indicator on the HSI is activated whenever a waypoint number is shown on the TANS display and the Doppler control switch is selected to the FD CPLD position. The NAV 1 frequency selector must be placed in the VOR range. Frequency selection in the ILS range will automatically switch the flight director to the ILS mode. A green FLT DIR DOPPLER light will illuminate on the flight director annunciator panel when Doppler information is being supplied to the HSI.

The heading error indication as shown by the course deviation indicator is appropriate to the waypoint number shown, and will continue to indicate so long as the waypoint number is displayed even through the remainder of the TANS display may be showing present position, wind velocity, etc. The HSI DME readout will display raw DME distance to a tuned VORTAC, not distance to a Doppler waypoint.

The course deviation indicator displacement on the HSI is dependent on the airplane heading relative to the bearing to the waypoint. Therefore, to get relative bearing as well as displacement information, the current selected waypoint bearing as shown in the "Bearing-Distance" mode of the TANS display must be set on the HSI course selector.

AUTOPILOT OPERATION

Once a waypoint has been selected and the Doppler control switch set to the FD CPLD position, the autopilot can be coupled to the Doppler.

1. Set the waypoint bearing on the HSI course selector.

2. Depress the NAV mode on the flight director controller to "arm" the navigation function.

3. Adjust the heading bug to agree with the intercept heading as shown in the "intercept" mode of the TANS display. Small additional adjustments to the heading bug may be required until the flight director transitions to the "capture" mode.

NOTE

If it is required to intercept a predetermined track to a waypoint, set up an appropriate intercept angle to the desired track. Monitor the displayed bearing in the "Bearing-Distance" mode. When the displayed bearing indicates the desired track angle, proceed as in Steps 1 through 3 above.

PERFORMANCE - No Change

Approved: Donald H Pete
for W. H. Schultz
Beech Aircraft Corporation
DOA CE-2

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INTRODUCTION

Beech Aircraft Corporation has developed this special summary publication of Safety Information to refresh pilots' and owners' knowledge of safety related subjects. Topics in this publication are dealt with in more detail in FAA Advisory Circulars and other publications pertaining to the subject of safe flying.

The skilled pilot recognizes that safety consciousness is an integral - and never ending - part of his or her job. Be thoroughly familiar with your airplane. Know its limitations and your own. Maintain your currency, or fly with a qualified instructor until you are current and proficient. Practice emergency procedures at safe altitudes and airspeeds, preferably with a qualified instructor pilot, until the required action is instinctive. Periodically review this Safety Information as part of your recurrency training regimen.

BEECHCRAFT airplanes are designed and built to provide you with many years of safe and efficient transportation. By maintaining your BEECHCRAFT properly and flying it prudently you will realize its full potential.

. . . . . Beech Aircraft Corporation

WARNING

Because your airplane is a high performance, high speed transportation vehicle, designed for operation in a three-dimensional environment, special safety precautions must be observed to reduce the risk of fatal or serious injuries to the pilot(s) and occupant(s).

It is mandatory that you fully understand the contents of this manual and the other manuals which accompany the airplane; that FAA requirements for ratings, certifications and review be scrupulously complied with; and that you allow only persons who are properly licensed and rated, and thoroughly familiar with the contents of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual to operate the airplane.

IMPROPER OPERATION OR MAINTENANCE OF AN AIRPLANE, NO MATTER HOW WELL BUILT INITIALLY, CAN RESULT IN CONSIDERABLE DAMAGE OR TOTAL DESTRUCTION OF THE AIRPLANE, ALONG WITH SERIOUS OR FATAL INJURIES TO ALL OCCUPANTS.

GENERAL

As a pilot, you are responsible to yourself and to those who fly with you, to other pilots and their passengers and to people on the ground, to fly wisely and safely.

The following material in this Safety Information Section covers several subjects in limited detail.

SOURCES OF INFORMATION

There is a wealth of information available to the pilot, created for the sole purpose of making flying safer, easier and more efficient. Take advantage of this knowledge and be prepared for an emergency in the remote event that one should occur.

PILOT'S OPERATING HANDBOOK AND FAA APPROVED AIRPLANE FLIGHT MANUAL

You must be thoroughly familiar with the contents of your operating manuals, placards, and check lists to ensure safe utilization of your airplane. When the airplane was manufactured, it was equipped with the following: placards, Pilot's Operating Handbook and FAA Approved Airplane Flight Manual, and Pilot's Checklist. Beech has revised and reissued many of the early manuals for certain models of airplanes in GAMA Standard Format as Pilot's Operating Handbooks and FAA Approved Airplane Flight Manuals. For simplicity and convenience, all official manuals for various models are referred to in this publication as the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual. If the airplane has changed ownership, the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual may have been misplaced or may not be current. If missing or out of date, a replacement handbook must be obtained from any BEECHCRAFT Aviation Center.

BEECHCRAFT SERVICE PUBLICATIONS

Beech Aircraft Corporation publishes a wide variety of manuals, service letters, service instructions, service bulletins, safety communiques and other publications for the various models of BEECHCRAFT airplanes. Information on how to obtain publications relating to your airplane is contained in BEECHCRAFT Service Bulletin Number 2001, entitled "General - BEECHCRAFT Service Publications - What Is Available and How to Obtain It."

Beech Aircraft Corporation automatically mails original issues and revisions of BEECHCRAFT Service Bulletins (Mandatory, Recommended and Optional), FAA Approved Airplane Flight Manual Supplements, reissues and revisions of FAA Approved Airplane Flight Manuals, Flight Handbooks, Owner's Manuals, Pilot's Operating Manuals and Pilot's Operating Handbooks, and original issues and revisions of BEECHCRAFT Safety Communiques to BEECHCRAFT owner addresses as listed by the FAA Aircraft Registration Branch List and the BEECHCRAFT International Owner Notification Service List. While this information is distributed by Beech Aircraft Corporation, we can not make changes in the name or address furnished by the FAA. The owner must contact the FAA regarding any changes to name or address. Their address is: FAA Aircraft Registration Branch (AAC250) P.O. Box 25082, Oklahoma City, OK 73125, Phone (405) 680-2131.

It is the responsibility of the FAA owner of record to ensure that any mailings from Beech are forwarded to the proper persons. Often the FAA registered owner is a bank, financing company or an individual not in possession of the airplane. Also, when an airplane is sold, there is a lag in processing the change in registration with the FAA. If you are a new owner, contact your BEECHCRAFT dealer and ensure that your manuals are up to date.

Beech Aircraft Corporation provides a subscription service which provides for direct factory mailing of BEECHCRAFT publications applicable to a specific serial number airplane. Details concerning the fees and ordering information for this owner subscription service are contained in Service Bulletin Number 2001.

For owners who choose not to apply for a publications revision subscription service, Beech provides a free Owner Notification Service by which owners are notified by post card of BEECHCRAFT manual reissues, revisions and supplements which are being issued applicable to the airplane owned. On receipt of such notification, the owner may obtain the publication through a BEECHCRAFT Aviation Center, Aero Center or International Distributor. This notification service is available when requested by the owner. This request may be made by using the owner notification request card furnished with the loose equipment of each airplane at the time of delivery, or by a letter requesting this service, referencing the specific airplane serial number owned. Write To:

Supervisor, Special Services
Dept. 52
Beech Aircraft Corporation
P.O. Box 85
Wichita, Kansas 67201-0085

From time to time Beech Aircraft Corporation issues BEECHCRAFT Safety Communiques dealing with the safe operation of a specific series of airplanes, or airplanes in general. It is recommended that each owner/operator maintain a current file of these publications. Back issues of BEECHCRAFT Safety Communiques may be obtained without charge by sending a request including airplane model and serial number to the Supervisor, Special Services, at the address listed above.

FEDERAL AVIATION REGULATIONS

FAR Part 91, General Operating and Flight Rules, is a document of law governing operation of airplanes and the owner's and pilot's responsibilities. Some of the subjects covered are:

Responsibilities and authority of the pilot-in-command

Certificates required

Liquor and drugs

Flight plans

Preflight action
Fuel requirements
Flight rules
Maintenance, preventive maintenance, alterations, inspection and maintenance records

You, as a pilot, have responsibilities under government regulations. The regulations are designed for your protection, the protection of your passengers and the public. Compliance is mandatory.

AIRWORTHINESS DIRECTIVES

FAR Part 39 specifies that no person may operate a product to which an Airworthiness Directive issued by the FAA applies, except in accordance with the requirements of that Airworthiness Directive.

Airworthiness Directives (AD's) are not issued by the manufacturer. They are issued and available from the FAA.

AIRMAN'S INFORMATION MANUAL

The Airman's Information Manual (AIM) is designed to provide airmen with basic flight information and ATC procedures for use in the national airspace system of the United States. It also contains items of interest to pilots concerning health and medical facts, factors affecting flight safety, a pilot/controller glossary of terms in the Air Traffic Control system, information on safety, and accident/hazard reporting. It is revised at six-month intervals and can be purchased from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

This document contains a wealth of pilot information. Among the subjects are:

Controlled Airspace
Emergency Procedures
Service Available to Pilots
Weather and Icing
Radio Phraseology and Technique
Mountain Flying
Airport Operations
Wake Turbulence - Vortices
Clearances and Separations
Medical Facts for Pilots
Preflight
Bird Hazards
Departures - IFR
Good Operating Practices
En route - IFR
Airport Location Director
Arrival - IFR

All pilots must be thoroughly familiar with and use the information in the AIM.

ADVISORY INFORMATION

Notams (Notices to Airmen) are documents that have information of a time-critical nature that would affect a pilot's decision to make a flight; for example, an airport closed, terminal radar out of service, or enroute navigational aids out of service.

FAA ADVISORY CIRCULARS

The FAA issues Advisory Circulars to inform the aviation public in a systematic way of nonregulatory material of interest. Advisory Circulars contain a wealth of information with which the prudent pilot should be familiar. A complete list of current FAA Advisory Circulars is published in AC OO-2, which lists Advisory Circulars that are for sale, as well as those distributed free of charge by the FAA, and provides ordering information. Many Advisory Circulars which are for sale can be purchased locally in aviation bookstores or at FBO's. These documents are subject to periodic revision. Be certain the Advisory Circular you are using is the latest revision available. Some of the Advisory Circulars of interest to pilots are:

***00-6**	Aviation Weather
00-24	Thunderstorms
00-30	Rules of Thumb For Avoiding or Minimizing Encounters with Clear Air Turbulence
***00-45**	Aviation Weather Services
00-46	Aviation Safety Reporting Program
20-32	Carbon Monoxide (CO) Contamination in Aircraft - Detection and Prevention
20-35	Tie-Down Sense
20-43	Aircraft Fuel Control Control Surfaces
20-105	Engine Power-Loss Accident Prevention
20-125	Water in Aviation Fuels
21-4	Special Flight Permits for Operation of Overweight Aircraft
43-9	Maintenance Records: General Aviation Aircraft
43-12	Preventive Maintenance
60-4	Pilot's Spatial Disorientation
60-6	Airplane Flight Manuals (AFM), Approved Manual Materials, Markings and Placards - Airplanes
60-12	Availability of Industry-Developed Guidelines for the Conduct of the Biennial Flight Review
60-13	The Accident Prevention Counselor Program
***61-21**	Flight Training Handbook

*61-23	Pilot's Handbook of Aeronautical Knowledge
*61-27	Instrument Flying Handbook
61-67	Hazards Associated with Spins in Airplanes Prohibited from Intentional Spinning.
61-84	Role of Preflight Preparation
*67-2	Medical Handbook for Pilots
90-23	Aircraft Wake Turbulence
90-42	Traffic Advisory Practices at Nontower Airports
90-48	Pilot's Role in Collision Avoidance
90-66	Recommended Standard Traffic Patterns for Airplane Operations at Uncontrolled Airports
90-85	Severe Weather Avoidance Plan (SWAP)
91-6	Water, Slush and Snow On the Runway
91-8	Use of Oxygen by General Aviation Pilots/Passengers
91-13	Cold Weather Operation of Aircraft
*91-23	Pilot's Weight and Balance Handbook
91-26	Maintenance and Handling of Air Driven Gyroscopic Instruments
91-35	Noise, Hearing Damage, and Fatigue in General Aviation Pilots
91-43	Unreliable Airspeed Indications
91-44	Operational and Maintenance Practices for Emergency Locator Transmitters and Receivers
91-46	Gyroscopic Instruments - Good Operating Practices
91-50	Importance of Transponder Operations and Altitude Reporting
91-51	Airplane Deice and Anti-Ice Systems
91-65	Use of Shoulder Harness in Passenger Seats
103-4	Hazards Associated with Sublimation of Solid Carbon Dioxide (Dry Ice) Aboard Aircraft
210-5A	Military Flying Activities

* For Sale

FAA GENERAL AVIATION NEWS

FAA General Aviation News is published by the FAA in the interest of flight safety. The magazine is designed to promote safety in the air by calling the attention of general aviation airmen to current technical, regulatory and procedural matters affecting the safe operation of airplanes. FAA General Aviation News is sold on subscription by the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.

FAA ACCIDENT PREVENTION PROGRAM

The FAA assigns accident prevention specialists to each Flight Standards and General Aviation District Office to organize accident prevention program activities. In addition, there are over 3,000 volunteer airmen serving as accident prevention counselors, sharing their technical expertise and professional knowledge with the general aviation community. The FAA conducts seminars and workshops, and distributes invaluable safety information under this program.

Usually the airport manager, the FAA Flight Service Station (FSS), or Fixed Base Operator (FBO) will have a list of accident prevention counselors and their phone numbers available. All Flight Standards and General Aviation District Offices have a list of the counselors serving the district.

Before flying over unfamiliar territory, such as mountainous terrain or desert areas, it is advisable for transient pilots to consult with local counselors. They will be familiar with the more desirable routes, the wind and weather conditions, and the service and emergency landing areas that are available along the way. They can also offer advice on the type of emergency equipment you should be carrying.

ADDITIONAL INFORMATION

The National Transportation Safety Board and the Federal Aviation Administration periodically issue, in greater detail, general aviation pamphlets concerning aviation safety. FAA Regional Offices also publish material under the FAA General Aviation Accident Prevention Program. These can be obtained at FAA Offices, Weather Stations, Flight Service Stations or Airport Facilities. Some of these are titled:

12 Golden Rules for Pilots
Weather or Not
Disorientation
Plane Sense
Weather Info Guide for Pilots
Wake Turbulence

Don't Trust to Luck, Trust to Safety
Rain, Fog, Snow
Thunderstorm - TRW
Icing
Pilot's Weather Briefing Guide
Thunderstorms Don't Flirt . . . Skirt 'em
IFR-VFR - Either Way Disorientation Can Be Fatal
IFR Pilot Exam-O-Grams
VFR Pilot Exam-O-Grams
Impossible Turn
Wind Shear
Estimating Inflight Visibility
Is the Aircraft Ready for Flight
Tips on Mountain Flying
Tips on Desert Flying
Always Leave Yourself An Out
Tips on the Use of Ailerons and Rudder
Some Hard Facts About Soft Landings
Propeller Operation and Care
Torque "What it Means to the Pilot"
Weight and Balance - An Important Safety Consideration for Pilots

GENERAL INFORMATION ON SPECIFIC TOPICS

MAINTENANCE

Safety of flight begins with a well maintained airplane. Make it a habit to keep your airplane and all of its equipment in first-class, airworthy condition. Keep a "Squawk List" on board, and see that all discrepancies, however minor, are noted and promptly repaired.

Schedule your maintenance regularly, and have your airplane serviced by a reputable organization. Be suspicious of bargain prices for maintenance, repairs and inspections.

If repairs or modifications are made to the flight control system, make sure the control surfaces are properly balanced and the controls can be moved freely from the cockpit in the proper direction and through their designed range of travel.

It is the responsibility of the owner and the operator to assure that the airplane is maintained in an airworthy condition and that proper maintenance records are kept.

Use only genuine BEECHCRAFT or BEECHCRAFT approved parts obtained from BEECHCRAFT approved sources, in connection with the maintenance and repair of Beech airplanes.

Genuine BEECHCRAFT Parts are produced and inspected under rigorous procedures to ensure airworthiness and suitability for use in Beech airplane applications. Parts purchased from sources other than BEECHCRAFT, even though outwardly identical in appearance, may not have had the required tests and inspections performed, may be different in fabrication techniques and materials, and may be dangerous when installed in an airplane.

Salvaged airplane parts, reworked parts obtained from non-BEECHCRAFT approved sources, or parts, components, or structural assemblies, the service history of which is unknown or cannot be authenticated, may have been subjected to unacceptable stresses or temperatures, or have other hidden damage not discernible through routine visual or nondestructive testing techniques. This may render the part, component or structural assembly, even though originally manufactured by BEECHCRAFT, unsuitable and unsafe for airplane use.

BEECHCRAFT expressly disclaims any responsibility for malfunctions, failures, damage or injury caused by use of non-BEECHCRAFT parts.

Airplanes operated for Air Taxi or other than normal operation, and airplanes operated in humid tropics, or cold and damp climates, etc., may need more frequent inspections for wear, corrosion and/or lack of lubrication. In these areas, periodic inspections should be performed until the operator can set his own decreased inspection periods based on experience.

NOTE

The required periods do not constitute a guarantee that the item will reach the period without malfunction, as the aforementioned factors cannot be controlled by the manufacturer.

Corrosion and its effects must be treated at the earliest possible opportunity. A clean, dry surface is virtually immune to corrosion. Make sure that all drain holes remain unobstructed. Protective films and sealants help to keep corrosive agents from contacting metallic surfaces. Corrosion inspections should be made most frequently under high-corrosion-risk operating conditions, such as in areas of excessive airborne salt concentrations (e.g., near the sea) and in high-humidity areas (e.g., tropical regions).

If you have purchased a used airplane, have your mechanic inspect the airplane registration records, logbooks and maintenance records carefully. An unexplained period of time for which the airplane has been out of service, or unexplained significant repairs, may well indicate the airplane has been seriously damaged in a prior accident. Have your mechanics inspect a used airplane carefully. Take the time to ensure that you really know what you are buying when you buy a used airplane.

HAZARDS OF UNAPPROVED MODIFICATIONS

Many airplane modifications are approved under Supplemental Type Certificates (STC's). Before installing an STC on your airplane, check to make sure that the STC does not conflict with other STC's that have already been installed. Because approval of an STC is obtained by the individual STC holder, based upon modification of the original type design, it is possible for two STC's to interfere with each other when both are installed. Never install an unapproved modification of any type, however innocent the apparent modification may seem. Always obtain proper FAA approval.

Airplane owners and maintenance personnel are particularly cautioned not to make attachments to, or otherwise modify, seats from original certification without approval from the FAA Engineering and Manufacturing District Office having original certification responsibility for that make and model.

Any unapproved attachment or modification to seat structure may increase load factors and metal stress which could cause failure of seat structure at a lesser "G" force than exhibited for original certification. Examples of unauthorized attachments are drilling holes in seat tubing to attach fire extinguishers and drilling holes to attach approach plate book bins to seats.

FLIGHT PLANNING

FAR Part 91 requires that each pilot in command, before beginning a flight, familiarize himself with all available information concerning that flight.

Obtain a current and complete preflight briefing. This should consist of local, enroute and destination weather and enroute navaid information. Enroute terrain and obstructions, alternate airports, airport runways active, length of runways, and take-off and landing distances for the airplane for conditions expected should be known.

The prudent pilot will review his planned enroute track and stations and make a list for quick reference. It is strongly recommended a flight plan be filed with Flight Service Stations, even though the flight may be VFR. Also, advise Flight Service Stations of changes or delays of one hour or more and remember to close the flight plan at destination.

The pilot must be completely familiar with the performance of the airplane and performance data in the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual. The effects of temperature and pressure altitude must be taken into account in performance. An applicable FAA Approved Airplane Flight Manual must be aboard the airplane at all times, including the weight and balance forms and equipment list.

PASSENGER INFORMATION CARDS

Beech has available, for your Beech airplane, passenger information cards which contain important information on the proper use of restraint systems, oxygen masks, emergency exits and emergency bracing procedures. Passenger information cards may be obtained at any BEECHCRAFT Aviation Center. A pilot should not only be familiar with the information contained in the cards, but should, prior to flight, always inform the passengers of the information contained in the information cards. The pilot should orally brief the passengers on the proper use of restraint systems, doors and emergency exits, and other emergency procedures, as required by Part 91 of the FAR's.

STOWAGE OF ARTICLES

Airplane seats are designed to absorb energy in a downward direction. In order to accomplish this action, the space between the seat pan and the floor is utilized to provide space for seat displacement. If hard, solid objects are stored beneath seats, the energy absorbing feature is lost and severe spinal injuries can occur to occupants.

Prior to flight, pilots should assure that articles are not stowed beneath seats that would restrict seat pan energy absorption or penetrate the seat in event of a high vertical velocity accident.

Ensure that cargo and baggage is stowed and properly secured with tie-down straps and cargo nets.

FLIGHT OPERATIONS

GENERAL

The pilot must be thoroughly familiar with all information published by the manufacturer concerning the airplane, and is required by law to operate the airplane in accordance with the FAA Approved Airplane Flight Manual and placards installed.

PREFLIGHT INSPECTION

In addition to maintenance inspections and preflight information required by FAR Part 91, a complete, careful preflight inspection is imperative.

Each airplane has a checklist for the preflight inspection which must be followed. USE THE CHECKLIST.

WEIGHT AND BALANCE

Maintaining center of gravity within the approved envelope throughout the planned flight is an important safety consideration.

The airplane must be loaded so as not to exceed the weight and center of gravity (C.G.) limitations. Airplanes that are loaded above the maximum take-off or landing weight limitations will have an overall lower level of performance compared to that shown in the Performance section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual. If loaded above maximum takeoff weight, the take-off distance and the landing distance will be longer than that shown in the Performance section; the stalling speed will be higher; rate of climb, cruising speed, and range of the airplane will all be lower than shown in the Performance section.

If an airplane is loaded so that the C.G. is forward of the forward limit, it will require additional control movements for maneuvering the airplane with correspondingly higher control forces. The pilot may experience difficulty during takeoff and landing because of the elevator control limits.

If an airplane is loaded aft of the aft C.G. limit, the pilot will experience a lower level of stability. Airplane characteristics that indicate a lower stability level are; lower control forces, difficulty in trimming the airplane, lower control forces for maneuvering with attendant danger of structural overload, decayed stall characteristics, and a lower level of lateral-directional damping.

Ensure that all cargo and baggage is properly secured before takeoff. A sudden shift in balance at rotation can cause controllability problems.

AUTOPILOTS AND ELECTRIC TRIM SYSTEMS

Because there are several different models of autopilots and electric trim systems installed in Beech airplanes and different installations and switch positions are possible from airplane to airplane, it is essential that every owner/operator review his Airplane Flight Manual (AFM) Supplements and ensure that the supplements properly describe the autopilot and trim installations on his specific airplane. Each pilot, prior to flight, must be fully aware of the proper procedures for operation, and particularly disengagement, for the system as installed.

In addition to ensuring compliance with the autopilot manufacturer's maintenance requirements, all owners/operators should thoroughly familiarize themselves with the operation, function and procedures described in the Airplane Flight Manual Supplements. Ensure a full understanding of the methods of engagement and disengagement of the autopilot and trim systems.

Compare the descriptions and procedures contained in the supplements to the actual installation in the airplane to ensure that the supplement accurately describes your installation. Test that all buttons, switches and circuit breakers function as described in the supplements. If they do not function as described, have the system repaired by a qualified service agency. If field service advice or assistance is necessary, contact Beech Aircraft Corporation, Customer Support Department.

As stated in all AFM Supplements for autopilot systems and trim systems installed on Beech airplanes, the preflight check must be conducted before every flight. The preflight check assures not only that the systems and all of their features are operating properly, but also that the pilot, before flight, is familiar with the proper means of engagement and disengagement of the autopilot and trim system.

Autopilot AFM Supplements caution against trying to override the autopilot system during flight without disengaging the autopilot because the autopilot will continue to trim the airplane and oppose the pilot's actions. This could result in a severely out-of-trim condition. This is a basic feature of all autopilots with electric trim followup.

Do not try to manually override the autopilot during flight

IN CASE OF EMERGENCY, YOU CAN OVERPOWER THE AUTOPILOT TO CORRECT THE ATTITUDE, BUT THE AUTOPILOT AND ELECTRIC TRIM MUST THEN IMMEDIATELY BE DISENGAGED.

It is often difficult to distinguish an autopilot malfunction from an electric trim system malfunction. The safest course is to deactivate both. Do not re-engage either system until after you have safely landed. Then have the systems checked by a qualified service facility prior to further flight.

Depending upon the installation on your airplane, the following additional methods may be available to disengage the autopilot or electric trim in the event that the autopilot or electric trim does not disengage utilizing the disengage methods specified in the supplements.

CAUTION

Transient control forces may occur when the autopilot is disengaged.

1. Turn off the autopilot master switch, if installed.
2. Pull the autopilot and trim circuit breaker(s) or turn off the autopilot switch breaker, if installed.
3. Turn off the RADIO MASTER SWITCH, if installed (if the autopilot system and the trim system are wired through this switch).

CAUTION

Radios, including VHF COMM, are also disconnected when the radio master switch is off.

4. Turn off the ELECTRIC MASTER SWITCH.

WARNING

Almost all electrically powered systems will be inoperative. Therefore, the cabin will depressurize. Consult the AFM for further information.

5. Push the GA switch on throttle grip, if installed (depending upon the autopilot system).
6. Push TEST EACH FLT switch on the autopilot controller, if installed.
7. Position inverter switch(es) (INV 1/INV 2) to OFF momentarily, then return to original position.

CAUTION

While the switch(es) are placed OFF, the AC power will also be removed from AC-driven equipment.

NOTE

After the autopilot is positively disengaged, it may be necessary to restore other electrical functions. Be sure when the master switches are turned on that the autopilot does not re-engage.

It is essential that you read your airplane's Pilot's Operating Handbook and FAA Approved Airplane Flight Manual and applicable supplements for your autopilot system and check the function and operation of your system.

The engagement of the autopilot must be done in accordance with the instructions and procedures contained in the AFM Supplement.

Particular attention must be paid to the autopilot settings prior to engagement. If you attempt to engage the autopilot when the airplane is out of trim, a large attitude change may occur.

IT IS ESSENTIAL THAT THE PROCEDURES SET FORTH IN THE APPROVED AFM SUPPLEMENTS FOR YOUR SPECIFIC INSTALLATION BE FOLLOWED BEFORE ENGAGING THE AUTOPILOT.

FLUTTER

Flutter is a phenomenon that can occur when an aerodynamic surface begins vibrating. The energy to sustain the vibration is derived from airflow over the surface. The amplitude of the vibration can (1) decrease, if airspeed is reduced; (2) remain constant, if airspeed is held constant and no failures occur; or (3) increase to the point of self-destruction, especially if airspeed is high and/or is allowed to increase. Flutter can lead to an in-flight break up of the airplane. Airplanes are designed so that flutter will not occur in the normal operating envelope of the airplane as long as the airplane is properly maintained. In the case of any airplane, decreasing the damping and stiffness of the structure or increasing the trailing edge weight of control surfaces will tend to cause flutter. If a combination of those factors is sufficient, flutter can occur within the normal operating envelope.

Owners and operators of airplanes have the primary responsibility for maintaining their airplanes. To fulfill that responsibility, it is imperative that all airplanes receive a thorough preflight inspection. Improper tension on the control cables or any other loose condition in the flight control system can also cause or contribute to flutter. Pilots should pay particular attention to control surface attachment hardware including tab pushrod attachment during preflight inspection. Looseness of fixed surfaces or movement of control surfaces other than in the normal direction of travel should be rectified before flight. Further, owners should take their airplanes to mechanics who have access to current technical publications and prior experience in properly maintaining that make and model of airplane. The owner should make certain that control cable tension inspections are performed as outlined in the applicable Beech Inspection Guide. Worn control surface attachment hardware must be replaced. Any repainting or repair of a moveable control surface will require a verification of the control surface balance before the airplane is returned to service. Control surface drain holes must be open to prevent freezing of accumulated moisture, which could create an increased trailing-edge-heavy control surface and flutter.

If an excessive vibration, particularly in the control column and rudder pedals, is encountered in flight, this may be the onset of flutter and the procedure to follow is:

1. IMMEDIATELLY REDUCE AIRSPEED (lower the landing gear, if necessary).
2. RESTRAIN THE CONTROLS OF THE AIRPLANE UNTIL THE VIBRATION CEASES.
3. FLY AT THE REDUCED AIRSPEED AND LAND AT THE NEAREST SUITABLE AIRPORT.
4. HAVE THE AIRPLANE INSPECTED FOR AIRFRAME DAMAGE, CONTROL SURFACE ATTACHING HARDWARE CONDITION/SECURITY, TRIM TAB FREE PLAY, PROPER CONTROL CABLE TENSION, AND CONTROL SURFACE BALANCE BY ANOTHER MECHANIC WHO IS FULLY QUALIFIED.

TURBULENT WEATHER

A complete and current weather briefing is a requirement for a safe trip.

Updating of weather information enroute is also essential. The wise pilot knows that weather conditions can change quickly, and treats weather forecasting as professional advice, rather than an absolute fact. He obtains all the advice he can, but stays alert to any sign or report of changing conditions.

Plan the flight to avoid areas of reported severe turbulence. It is not always possible to detect individual storm areas or find the in-between clear areas.

The National Weather Service classifies turbulence as follows:

Class of Turbulence	Effect
Extreme	Airplane is violently tossed about and is practically impossible to control. May cause structural damage.
Severe	Airplane may be momentarily out of control. Occupants are thrown violently against the belts and back into the seat. Unsecured objects are tossed about.
Moderate	Occupants require seat belts and occasionally are thrown against the belt. Unsecured objects move about.

Light	Occupants may be required to use seat belts, but objects in the airplane remain at rest.

Thunderstorms, squall lines and violent turbulence should be regarded as extremely dangerous and must be avoided. Hail and tornadic wind velocities can be encountered in thunderstorms that can destroy any airplane, just as tornadoes destroy nearly everything in their path on the ground.

Thunderstorms also pose the possibility of a lightning strike on an airplane. Any structure or equipment which shows evidence of a lightning strike, or being subjected to a high current flow due to a strike, or is a suspected part of a lightning strike path through the airplane, should be thoroughly inspected and any damage repaired prior to additional flight.

A roll cloud ahead of a squall line or thunderstorm is visible evidence of violent turbulence; however, the absence of a roll cloud should not be interpreted as denoting that severe turbulence is not present.

Even though flight in severe turbulence must be avoided, flight in turbulent air may be encountered unexpectedly under certain conditions.

The following recommendations should be observed for airplane operation in turbulent air:

Flying through turbulent air presents two basic problems, the answer to both is proper airspeed. On one hand, if you maintain an excessive airspeed, you run the risk of structural damage or failure; on the other hand, if your airspeed is too low, you may stall.

If turbulence is encountered, reduce speed to the turbulent air penetration speed, if given, or to the maneuvering speed, which is listed in the Limitations Section of the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual. These speeds give the best assurance of avoiding excessive stress loads, and at the same time providing the proper margin against inadvertent stalls due to gusts.

Beware of overcontrolling in an attempt to correct for changes in attitude; applying control pressure abruptly will build up G-forces rapidly and could cause structural damage or even failure. You should watch particularly your angle of bank, making turns as wide and shallow as possible. Be equally cautious in applying forward or back pressure to keep the airplane level. Maintain straight and level attitude in either up or down drafts. Use trim sparingly to avoid being grossly out of trim as the vertical air columns change velocity and direction. If necessary to avoid excessive airspeeds, lower the landing gear.

WIND SHEAR

Wind shears are rapid, localized changes in wind direction, which can occur vertically as well as horizontally. Wind shear can be very dangerous to all airplanes, large and small, particularly on approach to landing when airspeeds are slow.

A horizontal wind shear is a sudden change in wind direction or speed that can, for example, transform a headwind into a tailwind, producing a sudden decrease in airspeed because of the inertia of the airplane. A vertical wind shear is a sudden updraft or downdraft. Microbursts are intense, highly localized severe downdrafts.

The prediction of wind shears is far from an exact science. Monitor your airspeed carefully when flying in storms, particularly on approach. Be mentally prepared to add power and go around at the first indication that a wind shear is being encountered.

FLIGHT IN ICING CONDITIONS

Every pilot should be intimately acquainted with the FAA Approved National Weather Service definitions for ice intensity and accumulation which we have reprinted below:

Intensity	Ice Accumulation
Trace	Ice becomes perceptible. Rate of accumulation slightly greater than rate of sublimation. It is not hazardous even though deicing/anti-icing equipment is not utilized, unless encountered for an extended period of time (over 1 hour).
Light	The rate of accumulation may create a problem if flight is prolonged in this environment (over 1 hour). Occasional use of deicing/anti-icing equipment will prevent or remove accumulation. It does not present a problem if the deicing/anti-icing equipment is used.
Moderate	The rate of accumulation is such that even short encounters become potentially hazardous and use of deicing/anti-icing equipment, or diversion, is necessary.
Severe	The rate of accumulation is such that deicing/anti-icing equipment fails to reduce or control the hazard. Immediate diversion is necessary.

It is no longer unusual to find deicing and anti-icing equipment on a wide range of airplane sizes and types. Since the

capability of this equipment varies, it becomes the pilot's primary responsibility to understand limitations which restrict the use of the airplane in icing conditions and the conditions which may exceed the systems capacity.

Pilots and airplane owners must carefully review the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual in order to ascertain the required operable equipment needed for flight in icing conditions. In addition, they must ascertain from the same sources the limits of approval or certification of their airplane for flight in icing conditions, and plan the flight accordingly if icing conditions are known or forecast along the route.

Remember that regardless of its combination of deicing/anti-icing equipment, any airplane not fully equipped and functional for IFR flight is not properly equipped for flight in icing conditions. An airplane which does not have all critical areas protected in the required manner by fully operational equipment must not be exposed to icing encounters of any intensity. When icing is detected, the pilot of such an airplane must make immediate diversion by flying out of the area of visible moisture or going to an altitude where icing is not encountered.

Even airplanes fully equipped and certified for flight in the icing conditions described in Appendix C to FAR Part 25 must avoid flights into those conditions defined by the National Weather Service as "Severe". No airplane equipped with any combination of deicing/anti-icing equipment can be expected to cope with such conditions. As competent pilots know, there appears to be no predictable limits for the severest weather conditions. For essentially the same reasons that airplanes, however designed or equipped for IFR flight, cannot be flown safely into conditions such as thunderstorms, tornadoes, hurricanes or other phenomena likely to produce severe turbulence, airplanes equipped for flight in icing conditions cannot be expected to cope with "Severe" icing conditions as defined by the National Weather Service. The prudent pilot must remain alert to the possibility that icing conditions may become "Severe" and that his equipment will not cope with them. At the first indication that such condition may have been encountered or may lie ahead, he should immediately react by selecting the most expeditious and safe course for diversion.

Every pilot of a properly fully-equipped Beech airplane who ventures into icing conditions must maintain the minimum speed (KIAS) for operation in icing conditions, which is set forth in the Normal Procedures section, and in the Limitations section, of his Pilot's Operating Handbook and FAA Approved Airplane Flight Manual. The pilot must remain aware of the fact that if he allows his airspeed to deteriorate below this minimum speed, he will increase the angle of attack of his airplane to the point where ice may build up on the under side of the wings aft of the area protected by the deice/anti-icing equipment.

Ice build-up, and its extent in unprotected areas may not be directly observable from the cockpit. Due to distortion of the wing airfoil, increased drag and reduced lift, stalling speeds will increase as ice accumulates on the airplane. For the same reasons, stall warning devices are not accurate and cannot be relied upon in icing conditions.

Even though the pilot maintains the prescribed minimum speeds for operating in icing conditions, ice is still likely to build up on the unprotected areas. Under some atmospheric conditions, it may even build up aft of the de-iced areas despite the maintenance of the prescribed minimum speed. The effect of ice accumulation on any unprotected surface is aggravated by length of exposure to the icing conditions. Ice buildup on unprotected surfaces will increase drag, add weight, reduce lift, and generally, adversely affect the aerodynamic characteristics and performance of the airplane. It can progress to the point where the airplane is no longer capable of flying. Therefore, the pilot operating even a fully-equipped airplane in sustained icing conditions must remain sensitive to any indication, such as observed ice accumulation, loss of airspeed, the need for increased power, reduced rate of climb, or sluggish response, that ice is accumulating on unprotected surfaces and that continued flight in these conditions is extremely hazardous, regardless of the performance of the deicing/anti-icing equipment.

Since flight in icing conditions is not an everyday occurrence, it is important that you maintain a proper proficiency and awareness of the operating procedures necessary for safe operation of the airplane and that the airplane is in a condition for safe operation.

Ensure moisture drains in the airplane structure are maintained open as specified in the Aircraft Maintenance Manual, so that moisture will not collect and cause freezing in the control cable area. Also, control surface tab hinges should be maintained and lubricated as specified in the Aircraft Maintenance Manual.

In icing conditions the autopilot should be disengaged at an altitude sufficient to permit the pilot to gain the feel of the airplane prior to landing. In no case should this be less than the minimum altitude specified in the Autopilot Airplane Flight Manual Supplement.

Observe the procedures set forth in your Pilot's Operating Handbook and FAA Approved Airplane Flight Manual during operation in icing conditions.

Activate your deice and anti-icing systems before entering an area of moisture where you are likely to go through a freezing level.

For any owner or pilot whose use pattern for an airplane exposes it to icing encounters, the following references are required reading for safe flying:

- The airplane's Pilot's Operating Handbook and FAA Approved Airplane Flight Manual, especially the sections on Normal Procedures, Emergency Procedures, Abnormal Procedures, Systems, and Safety Information.
- FAA Advisory Circular 91-51 - Airplane Deice and Anti-ice Systems.
- Weather Flying by Robert N. Buck.

Finally, the most important ingredients to safe flight in icing conditions - regardless of the airplane or the combination of deicing/anti-icing equipment - are a complete and current weather briefing, sound pilot judgment, close attention to the rate and type of ice accumulations, and the knowledge that "severe icing" as defined by the National Weather Service is beyond the capability of modern airplanes and an immediate diversion must be made. It is the inexperienced or uneducated pilot who presses on "regardless" hoping that steadily worsening conditions will improve, only to find himself flying an airplane which has become so loaded with ice that he can no longer maintain altitude. At this point he has lost most, if not all, of his safety options, including perhaps a 180 degree turn to return along the course already traveled.

The responsible and well-informed pilot recognizes the limitations of weather conditions, his airplane and its systems, and reacts promptly.

WEATHER RADAR

Airborne weather avoidance radar is, as its name implies, for avoiding severe weather - not for penetrating it. Whether to fly into an area of radar echoes depends on echo intensity, spacing between the echoes, and the capabilities of you and your airplane. Remember that weather radar detects precipitation drops. Except for the most advanced radar units, it does not detect turbulence. Therefore, the radar scope provides no assurance of avoiding turbulence. The radar scope also does not provide assurance of avoiding instrument weather from clouds and fog. Your scope may be clear between intense echoes; this clear area does not necessarily mean you can fly between the storms and maintain visual sighting of them.

Thunderstorms build and dissipate rapidly. Therefore, do not attempt to plan a course between echoes. The best use of ground radar information is to isolate general areas and coverage of echoes. You must avoid individual storms by in-flight observations either by visual sighting or airborne radar. It is better to avoid the whole thunderstorm area than to detour around individual storms, unless they are scattered.

Remember that while hail always gives a radar echo, it may fall several miles from the nearest visible cloud and hazardous turbulence may extend to as much as 20 miles from the echo edge. Avoid intense or extreme level echoes by at least 20 miles; that is, such echoes should be separated by at least 40 miles before you fly between them. With weaker echos you can reduce the distance by which you avoid them.

Above all, remember this; never regard any thunderstorm lightly. Even when radar observers report the echoes are of light intensity, avoiding thunderstorms is the best policy. The following are some do's and don'ts of thunderstorm avoidance:

1. Don't land or take off in the face of an approaching thunderstorm. Sudden gust-front low level turbulence could cause loss of control.
2. Don't attempt to fly under a thunderstorm even if you can see through to the other side. Turbulence and wind shear under the storm could be disastrous.
3. Don't fly without airborne radar into a cloud mass containing scattered embedded thunderstorms. Scattered thunderstorms not embedded usually can be visually circumnavigated.
4. Don't trust the visual appearance to be a reliable indicator of the turbulence inside a thunderstorm.
5. Do avoid by at least 20 miles any thunderstorm identified as severe or giving an intense radar echo. This is especially true under the anvil of a large cumulonimbus.
6. Do circumnavigate the entire area if the area has 6/10 or more thunderstorm coverage.
7. Do remember that vivid and frequent lightning indicates the probability of a severe thunderstorm.
8. Do regard as extremely hazardous any thunderstorm with tops 35,000 feet or higher whether the top is visually sighted or determined by radar.

If you cannot avoid penetrating a thunderstorm, the following are some do's BEFORE entering the storm:

9. Tighten your safety belt, put on your shoulder harness, and secure all loose objects.
10. Plan and hold your course to take you through the storm in minimum time.
11. To avoid the most critical icing, establish a penetration altitude below the freezing level or an altitude where the OAT is -15°C or colder.
12. Verify that pitot heat is on, and activate anti-ice systems. Icing can be rapid at any altitude and can cause almost instantaneous power failure and/or loss of airspeed indication.

MOUNTAIN FLYING

Pilots flying in mountainous areas should inform themselves of all aspects of mountain flying, including the effects of topographic features on weather conditions. Many good articles have been published, and a synopsis of mountain flying operations is included in the FAA Airman's Information Manual, Part 1.

Avoid flight at low altitudes over mountainous terrain, particularly near the lee slopes. If the wind velocity near the level of the ridge is in excess of 25 knots and approximately perpendicular to the ridge, mountain wave conditions are likely over and near the lee slopes. If the wind velocity at the level of the ridge exceeds 50 knots, a strong mountain wave is probable with extreme up and down drafts and severe turbulence. The worst turbulence will be encountered in and below the rotor zone, which is usually 8 to 10 miles downwind from the ridge. This zone is sometimes characterized by the presence of "roll clouds" if sufficient moisture is present. Altocumulus standing lenticular clouds are also visible signs that a mountain wave exists, but their presence is likewise dependent upon moisture. Mountain wave turbulence can, of course, occur in dry air and the absence of such clouds should not be taken as assurance that mountain wave turbulence will not be encountered. A mountain

wave downdraft may exceed the climb capability of your airplane. Avoid mountain wave downdrafts.

VFR AT NIGHT

When flying VFR at night, in addition to the altitude appropriate for the direction of flight, pilots should maintain a safe minimum altitude as dictated by terrain, obstacles such as TV towers, or communities in the area. This is especially true in mountainous terrain, where there is usually very little ground reference. Minimum clearance is 2,000 feet above the highest obstacle enroute. Do not depend on your ability to see obstacles in time to miss them. Flight on dark nights over sparsely populated country can be the same as IFR.

VERTIGO - DISORIENTATION

Disorientation can occur in a variety of ways. During flight, inner-ear balancing mechanisms are subjected to varied forces not normally experienced on the ground. This, combined with loss of outside visual reference, can cause vertigo. False interpretations (illusions) result, and may confuse the pilot's conception of the attitude and position of his airplane.

Under VFR conditions, the visual sense, using the horizon as a reference, can override the illusions. Under low visibility conditions (night, fog, clouds, haze, etc.) the illusions predominate. Only through awareness of these illusions, and proficiency in instrument flight procedures, can an airplane be operated safely in a low visibility environment.

Flying in fog, dense haze or dust, cloud banks, or very low visibility, with strobe lights or rotating beacons turned on can contribute to vertigo. They should be turned off in these conditions, particularly at night.

Motion sickness often precedes or accompanies disorientation and may further jeopardize the flight.

Disorientation in low visibility conditions is not limited to VFR pilots. Although IFR pilots are trained to look at their instruments to gain an artificial visual reference as a replacement for the loss of a visual horizon, they do not always do so. This can happen when: the pilot's physical condition will not permit him to concentrate on his instruments, when the pilot is not proficient in flying instrument conditions in the airplane he is flying, or when the pilot's work load of flying by reference to his instruments is compounded by such factors as turbulence. Even an instrument rated pilot encountering instrument conditions, intentional or unintentional, should ask himself whether or not he is sufficiently alert and proficient in the airplane he is flying to fly under low visibility conditions and the turbulence anticipated or encountered.

All pilots should check the weather and use good judgement in planning flights. If any doubt exists, the flight should not be made or it should be discontinued as soon as possible.

The result of vertigo is loss of control of the airplane. If the loss of control is sustained, it will result in an excessive speed accident. Excessive speed accidents occur in one of two manners - either as an inflight airframe separation or as a high speed ground inpact, and they are fatal accidents in either case. All airplanes are subject to this form of accident.

Excessive speed accidents occur at airspeeds greatly in excess of two operating limitations which are specified in the manuals (Maximum maneuvering speed and the "red line" or maximum operating speed). Such speed limits are set to protect the structure of an airplane. For example, control surfaces are designed to be used to their fullest extent only below a the airplane's maximum maneuvering speed. As a result, the control surfaces should never be suddenly or fully deflected above maximum maneuvering speed. Turbulence penetration should not be performed above that speed. The accidents we are discussing here occur at airspeeds greatly in excess of these limitations. No airplane should ever be flown beyond its FAA approved operating limitations.

FLIGHT WITH ONE ENGINE INOPERATIVE

Safe flight with one engine out requires an understanding of the basic aerodynamics involved, as well as proficiency in engine-out procedures.

Loss of power from one engine affects both climb performance and controllability of twin-engine airplanes. Climb performance depends on an excess of power over that required for level flight. Loss of power from one engine obviously represents a 50% loss of power but, in virtually all twin-engine airplanes, climb performance is reduced by at least 80%. A study of the charts in your Pilot's Operating Handbook and FAA Approved Airplane Flight Manual will confirm this fact. Single-engine climb performance depends on four factors:

Airspeed	Too little, or too much, will decrease climb performance
Drag	Gear, flaps, and windmilling prop
Power	Amount available in excess of that needed for level flight
Weight	Passengers, baggage, and fuel load greatly affect climb performance

Loss of power on one engine creates yaw due to asymetric thrust. Yaw forces must be balanced with the rudder. Loss of power on one engine also reduces airflow over the wing. In addition, yaw affects the lift distribution over the wing causing a roll toward the "dead" engine. These roll forces may be balanced by banking slightly (up to 5°) into the operating engine.

Airspeed is the key to safe single-engine operations. For most twin-engine airplanes there is:

Symbol	Description
V_{MCA}	Airspeed below which directional control cannot be maintained
V_{SSE}	Airspeed below which an intentional engine cut should never be made

For Training Purposes Only

V_{YSE}	Airspeed that will give the best single engine rate-of-climb (or the slowest loss of altitude)
V_{XSE}	Airspeed that will give the steepest angle-of-climb with one engine out

AIR MINIMUM CONTROL SPEED (V_{MCA})

V_{MCA} is designated by the red radial on the airspeed indicator and indicates the minimum control speed, airborne at sea level. V_{MCA} is determined by FAA regulations as the minimum airspeed at which it is possible to recover directional control of the airplane within 20 degrees heading change, and therefore maintain straight flight, with not more than 5 degrees of bank if one engine fails suddenly with:

- Takeoff power on the operative engine
- Rearmost allowable center of gravity
- Flaps in takeoff position
- Propeller on failed engine windmilling (feathered if Auto-Feather system is required)

However, sudden engine failures rarely occur with all factors listed above, and therefore, the actual V_{MCA} under any particular situation may be a little slower than the red radial on the airspeed indicator. Most airplanes will not maintain level flight at speeds at or near V_{MCA}. Consequently, it is not advisable to fly at speeds approaching V_{MCA}, except in training situations or during flight tests. Adhering to the practice of never flying at or below the published V_{MCA} speed for your airplane will virtually eliminate loss of directional control as a problem in the event of an engine failure.

INTENTIONAL ONE-ENGINE-INOPERATIVE SPEED (V_{SSE})

V_{SSE} is specified by the airplane manufacturer and is the minimum speed to perform intentional engine cuts. Use of V_{SSE} is intended to reduce the accident potential from loss of control after engine cuts at or near minimum control speed. V_{MCA} demonstrations are necessary in training but should only be made at safe altitude above the terrain and with power reduction on one engine made at or above V_{SSE}.

ONE-ENGINE-INOPERATIVE BEST RATE-OF-CLIMB SPEED (V_{YSE})

V_{YSE} is designated by the blue radial on the airspeed indicator. V_{YSE} delivers the greatest gain in altitude in the shortest possible time, and is based on the following criteria:

- Critical engine inoperative, and its propeller in the minimum drag position.
- Operating engine set at not more than the maximum continuous power.
- Landing gear retracted.
- Wing flaps in the most favorable (i.e., best lift/drag ratio) position.
- Airplane flown at recommended bank angle.

Drag caused by a windmilling propeller, extended landing gear, or flaps in the landing position, will severely degrade or destroy single-engine climb performance. Since climb performance varies widely with weight, temperature, altitude, and airplane configuration, the climb gradient (altitude gain or loss per mile) may be marginal - or even negative - under some conditions. Study the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual for your specific airplane and know what performance to expect with one engine out.

ONE-ENGINE-INOPERATIVE BEST ANGLE-OF-CLIMB SPEED (V_{XSE})

V_{XSE} is used only to clear obstructions during initial climb-out as it gives the greatest altitude gain per unit of horizontal distance. It requires more rudder control input than V_{YSE}.

SINGLE ENGINE SERVICE CEILING

The single engine service ceiling is the maximum altitude at which an airplane will climb at a rate of at least 50 feet per minute in smooth air, with one engine inoperative.

The single-engine service ceiling graph should be used during flight planning to determine whether the airplane, as loaded, can maintain the Minimum Enroute Altitude (MEA) if IFR, or terrain clearance if VFR, following an engine failure.

BASIC SINGLE ENGINE PROCEDURES

Know and follow, to the letter, the single-engine emergency procedures specified in your Pilot's Operating Handbook and FAA Approved Airplane Flight Manual for your airplane. However, the basic fundamentals of all the procedures are as follows:

1. Maintain airplane control and airspeed at all times. **THIS IS CARDINAL RULE NUMBER ONE.**
2. Usually, apply 100% torque to the operating engine. However, if the engine failure occurs at a speed below V_{MCA}, or during cruise or in a steep turn, you may elect to use only enough power to maintain a safe speed and altitude. If the failure occurs on final approach, use power only as necessary to complete the landing.
3. Reduce drag to an absolute minimum.
4. Secure the failed engine and related sub-systems.

The first three steps should be done promptly and from memory. The check list should then be consulted to be sure that the inoperative engine is secured properly and that the appropriate switches are placed in the correct position. The airplane must be banked about 5° into the live engine, with the "slip/skid" ball slightly out of center toward the live engine, to achieve rated performance.

ANOTHER NOTE OF CAUTION. Be sure to identify the dead engine positively, before securing it. Remember: First identify the suspected engine (i.e., "Dead foot means dead engine"), second, verify with cautious throttle movement, then secure.

ENGINE FAILURE ON TAKEOFF

If an engine fails before attaining V_1, the only proper action is to discontinue the takeoff. If the engine fails after V_1, the takeoff may be continued using the procedures specified in the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.

Your Pilot's Operating Handbook and FAA Approved Airplane Flight Manual contains charts that are used in calculating the runway length required to stop if the engine fails at V_1 speed and also has charts showing the single-engine performance after takeoff.

Study your charts carefully. No airplane is capable of climbing out on one engine under all weight, pressure altitude, and temperature conditions. The maximum take-off weight must be limited to achieve the required performance as specified in the LIMITATIONS section.

WHEN YOU FLY V_X, V_Y, V_{XSE} AND V_{YSE}

During normal two-engine operations always fly the published take-off speeds on initial climb out. Then, accelerate to your cruise climb airspeed after you have obtained a safe altitude. Use of cruise climb airspeed will give you increased inflight visibility and better fuel economy. However, at first indication of an engine failure during climb out, or while on approach, establish V_{YSE} or V_{XSE}, whichever is appropriate. (Consult your Pilot's Operating Handbook and FAA Approved Airplane Flight Manual for specifics.)

STALLS, SLOW FLIGHT AND TRAINING

The stall warning system must be kept operational at all times and must not be deactivated by interruption of circuits or circuit breakers. Compliance with this requirement is especially important in all high performance multi-engine airplanes during engine-out practice or stall demonstrations, because the stall speed is critical in operation of high-performance airplanes.

The single-engine stall speed of a twin-engine airplane is generally slightly below the power off (engines idle) stall speed for a given weight condition. Single-engine stalls in multi-engine airplanes are not recommended. Single-engine stalls should not be conducted in high performance airplanes by other than qualified engineering test pilots.

V_{MCA} demonstrations should not be attempted when the altitude and temperature are such that the engine-out minimum control speed is known, or discovered to be, close to the stalling speed. Loss of directional or lateral control, just as a stall occurs, is potentially hazardous.

V_{SSE}, the airspeed below which an engine should not be intentionally rendered inoperative for practice purposes, was established because of the apparent practice of some pilots, instructors, and examiners, of intentionally rendering an engine inoperative at a time when the airplane is being operated at a speed close to, or below, the flight idle stall speed. Unless the pilot takes immediate and proper corrective action under such circumstances, it is possible to enter an inadvertent spin.

It is recognized that flight below V_{SSE} with one engine inoperative, or simulated inoperative, may be required for conditions such as practice demonstration of V_{MCA} for multi-engine pilot certification. Refer to the procedure set forth in the Pilot's Operating Handbook and FAA Approved Airplane Flight Manual for your airplane. This procedure calls for simulating one engine inoperative by reducing the power lever on one engine to zero thrust while operating at an airspeed above V_{SSE}. Power on the other engine is set at maximum, then airspeed is reduced at approximately one knot per second until either V_{MCA} or stall warning is obtained. During this transition, rudder should be used to maintain directional control, and ailerons should be used to maintain a 5° bank toward the operative engine. At the first sign of either V_{MCA} or stall warning (which may be evidenced by inability to maintain longitudinal, lateral or directional control, aerodynamic stall buffet, or stall warning horn sound), recovery must be initiated immediately by reducing power to zero thrust on the operative engine and lowering the nose to regain V_{SSE}. Resume normal flight. This entire procedure should be used at a safe altitude of at least 5,000 feet above the ground in clear air only.

If stall warning is detected prior to the first sign of V_{MCA}, an engine-out minimum control speed demonstration cannot be accomplished under the existing gross weight conditions and should not be attempted.

SPINS

A major cause of fatal accidents in general aviation airplanes is a spin. Stall demonstrations and practice are a means for a pilot to acquire the skills to recognize when a stall is about to occur and to recover as soon as the first signs of a stall are evident. IF A STALL DOES NOT OCCUR - A SPIN CANNOT OCCUR. It is important to remember, however, that a stall can occur in any flight attitude, at any airspeed, if controls are misused.

Your airplane has not been tested for spin recovery characteristics, and is placarded against intentional spins.

The pilot of an airplane placarded against intentional spins should assume that the airplane may become uncontrollable in a spin, since its performance characteristics beyond certain limits specified in the FAA regulations have not been tested and are unknown. This is why airplanes are placarded against intentional spins, and this is why stall avoidance is your protection against an inadvertent spin.

Pilots are taught that intentional spins are entered by deliberately inducing a yawing moment with the controls as the airplane is stalled. Inadvertent spins result from the same combination - stall plus yaw. That is why it is important to use coordinated controls and to recover at the first indication of a stall when practicing stalls.

In any twin engine airplane, fundamental aerodynamics dictate that if the airplane is allowed to become fully stalled while one engine is providing lift-producing thrust, the yawing moment which can induce a spin will be present. Consequently, it is important to immediately reduce power on the operating engine, lower the nose to reduce the angle of attack, and increase the airspeed to recover from the stall.

In any twin engine airplane, if application of stall recovery controls is delayed, a rapid rolling and yawing motion may develop, even against full aileron and rudder, resulting in the airplane becoming inverted during the onset of a spinning motion. Once the airplane has been permitted to progress beyond the stall and is allowed to reach the rapid rolling and yawing condition, the pilot must then immediately initiate the generally accepted spin recovery procedure for multi-engine airplanes, which is as follows:

Immediately move the control column full forward, apply full rudder opposite to the direction of the spin and reduce power on both engines to idle. These three actions should be done as near simultaneously as possible; then continue to hold this control position until rotation stops and then neutralize all controls and execute a smooth pullout. Ailerons should be neutral during recovery. THE LONGER THE PILOT DELAYS BEFORE TAKING CORRECTIVE ACTION, THE MORE DIFFICULT RECOVERY WILL BECOME.

Always remember that extra alertness and good pilot techniques are required for slow flight maneuvers, including the practice or demonstration of stalls or V_{MCA}. In addition to the foregoing mandatory procedure, always:

- Be certain that the center of gravity of the airplane is as far forward as possible. Forward C.G. aids stall recovery, spin avoidance and spin recovery. An aft C.G. can create a tendency for a spin to flatten out, which delays recovery.
- Conduct any maneuvers which could possibly result in a spin at altitudes in excess of five thousand (5,000) feet above ground level in clear air only.
- Remember that an airplane, at or near traffic pattern and approach altitudes, cannot recover from a spin, or perhaps even a stall, before impact with the ground. When descending to traffic altitude and during pattern entry and all other flight operations, maintain speed no lower than V_{SSE}. On final approach maintain at least the airspeed shown in the flight manual. Recognize that under some conditions of weight, density altitude, and airplane configuration, a twin engine airplane cannot climb or accelerate on a single engine; hence a single engine go-around is impossible and the airplane is committed to a landing. Plan your approach accordingly.
- Remember that, if a stall or spin occurs under instrument conditions, the pilot, without reference to the horizon, is certain to become disoriented. He may be unable to recognize a stall, spin entry, or the spin condition and he may be unable to determine even the direction of the rotation.
- Finally, never forget that stall avoidance is your best protection against an inadvertent spin. MAINTAIN YOUR AIRSPEED.

VORTICES - WAKE TURBULENCE

Every airplane generates wakes of turbulence while in flight. Part of this is from the propeller or jet engine, and part from the wing tip vortices. The larger and heavier the airplane, the more pronounced and turbulent the wakes will be. Wing tip vortices from large, heavy airplanes are very severe at close range, degenerating with time, wind and distance. These are rolling in nature, from each wing tip. In tests, vortex velocities of 133 knots have been recorded. Encountering the rolling effect of wing tip vortices within two minutes after passage of large airplanes is extremely hazardous to small airplanes. This roll effect can exceed the maximum counter roll obtainable in a small airplane. The turbulent areas may remain for three minutes or more, depending on wind conditions, and may extend several miles beyond the airplane. Plan to fly slightly above and to the windward side of other airplanes. Because of the wide variety of conditions that can be encountered, there is no set rule to follow to avoid wake turbulence in all situations. However, the Airman's Information Manual, and to a greater extent Advisory Circular 90-23, Aircraft Wake Turbulence, provides a thorough discussion of the factors you should be aware of when wake turbulence may be encountered.

TAKEOFF AND LANDING CONDITIONS

When taking off on runways covered with water or freezing slush, the landing gear should remain extended for approximately ten seconds longer than normal, allowing the wheels to spin and dissipate the freezing moisture. The landing gear should then be cycled up, then down, wait approximately five seconds and then retracted again. Caution must be exercised to ensure that the entire operation is performed below Maximum Landing Gear Operating Airspeed.

Use caution when landing on runways that are covered by water or slush which cause hydroplaning (aquaplaning), a phenomenon that renders braking and steering ineffective because of the lack of sufficient surface friction. Snow and ice covered runways are also hazardous. The pilot should also be alert to the possibility of the brakes freezing.

Use caution when taking off or landing during gusty wind conditions. Also be aware of the special wind conditions caused by buildings or other obstructions located near the runway in a crosswind pattern.

MEDICAL FACTS FOR PILOTS

GENERAL

When the pilot enters the airplane, he becomes an integral part of the man-machine system. He is just as essential to a successful flight as the control surfaces. To ignore the pilot in preflight planning would be as senseless as failing to inspect the integrity of the control surfaces or any other vital part of the machine. The pilot has the responsibility for determining his reliability prior to entering the airplane for flight. When piloting an airplane, an individual should be free of conditions which are harmful to alertness, ability to make correct decisions, and rapid reaction time.

FATIGUE

Fatigue generally slows reaction time and causes errors due to inattention. In addition to the most common cause of fatigue, insufficient rest and loss of sleep, the pressures of business, financial worries, and family problems can be important contributing factors. If you are tired, don't fly.

For Training Purposes Only

HYPOXIA

Hypoxia, in simple terms, is a lack of sufficient oxygen to keep the brain and other body tissues functioning properly. There is a wide individual variation in susceptibility to hypoxia. In addition to progressively insufficient oxygen at higher altitudes, anything interfering with the blood's ability to carry oxygen can contribute to hypoxia (anemias, carbon monoxide, and certain drugs). Also, alcohol and various drugs decrease the brain's tolerance to hypoxia.

Your body has no built-in alarm system to let you know when you are not getting enough oxygen. It is impossible to predict when or where hypoxia will occur during a given flight, or how it will manifest itself. Some of the common symptoms of hypoxia are increased breathing rate, a lightheaded or dizzy sensation, tingling or warm sensation, sweating, reduced visual field, sleepiness, blue coloring of skin, fingernails and lips, and behavior changes. A particularly dangerous feature of hypoxia is an increased sense of well-being, called euphoria. It obscures a person's ability and desire to be critical, slows reaction time, and impairs thinking ability. Consequently, an hypoxic individual commonly believes things are getting progressively better while he nears total collapse.

The symptoms are slow but progressive, insidious in onset, and are most marked at altitudes above ten thousand feet. Night vision, however, can be impaired starting at an altitude of 5,000 feet. Persons who have recently overindulged in alcohol, who are moderate to heavy smokers, or who take certain drugs, may be more susceptible to hypoxia. Susceptibility may also vary in the same individual from day to day or even morning to evening.

Depending upon altitude, a hypoxic individual requires more time to make decisions and perform useful acts, even though he may remain conscious for a longer period. If pressurization equipment fails, the pilot and passengers have only a certain amount of time to get an oxygen mask on before they exceed their time of useful consciousness. The time of useful consciousness is approximately 3-5 minutes at 25,000 feet of altitude for the average individual and diminishes markedly as altitude increases. At 30,000 feet, altitude, for example, the time of useful consciousness is approximately 1-2 minutes. Therefore, in the event of depressurization, oxygen masks should be used immediately.

Should symptoms occur that cannot definitely be identified as either hypoxia or hyperventilation, try three or four deep breaths of oxygen. The symptoms should improve markedly if the condition was hypoxia (recovery from hypoxia is rapid).

Pilots who fly to altitudes that require or may require the use of supplemental oxygen should be thoroughly familiar with the operation of the airplane oxygen systems. A preflight inspection of the system should be performed, including proper fit of the mask.

The passengers should be briefed on the proper use of their oxygen system before flight.

Pilots who wear beards should be careful to ensure that their beard is carefully trimmed so that it will not interfere with proper sealing of the oxygen masks. If you wear a beard or mustache, test the fit of your oxygen mask on the ground for proper sealing. Studies conducted by the military and oxygen equipment manufacturers conclude that oxygen masks do not seal over beards or heavy facial hair.

Federal Aviation Regulations related to the use of supplemental oxygen by flight crew and passengers must be adhered to if flight to higher altitudes is to be accomplished safely. Passengers with significant circulatory or lung disease may need to use supplemental oxygen at lower altitudes than specified by these regulations.

Pilots of pressurized airplanes should receive physiological training with emphasis on hypoxia and the use of oxygen and oxygen systems. Pilots of airplanes with pressure demand oxygen systems should undergo training, experience altitude chamber decompression, and be familiar with pressure breathing before flying at high altitude. This training is available throughout the United States at nominal cost. Information regarding this training may be obtained by request from the Chief, Civil Aeromedical Institute, Attention: Aeromedical Education Branch, AAC-140, Mike Monroney Aeronautical Center, P.O. Box 25082, Oklahoma City, Oklahoma 73125

HYPERVENTILATION

Hyperventilation, or overbreathing, is a disturbance of respiration that may occur in individuals as a result of emotional tension or anxiety. Under conditions of emotional stress, fright, or pain, breathing rate may increase, causing increased lung ventilation, although the carbon dioxide output of the body cells does not increase. As as result, carbon dioxide is "washed out" of the blood. The most common symptoms of hyperventilation are: dizziness, nausea, sleepiness, and finally, unconsciousness. If the symptoms persist, discontinue use of oxygen and consciously slow your breathing rate until symptoms clear, and then resume normal breathing rate. Normal breathing can be aided by talking aloud.

ALCOHOL

Common sense and scientific evidence dictate that you must not fly as a crew member while under the influence of alcohol. Alcohol, even in small amounts, produces (among other things):

- A dulling of critical judgement.
- A decreased sense of responsibility.
- Diminished skill reactions and coordination.
- Decreased speed and strength of muscular reflexes (even after one ounce of alcohol).
- Decreases in efficiency of eye movements during reading (after one ounce of alcohol).
- Increased frequency of errors (after one ounce of alcohol).
- Constriction of visual fields.
- Decreased ability to see under dim illuminations.

- Loss of efficiency of sense of touch.
- Decrease of memory and reasoning ability.
- Increased susceptibility to fatigue and decreased attention span.
- Decreased relevance of response.
- Increased self confidence with decreased insight into immediate capabilities.

Tests have shown that pilots commit major errors of judgment and procedure at blood alcohol levels substantially less than the minimum legal levels of intoxication for most states. These tests further show a continuation of impairment from alcohol up to as many as 14 hours after consumption, with no appreciable diminution of impairment. The body metabolizes ingested alcohol at a rate of about one-third of an ounce per hour. Even after the body completely destroys a moderate amount of alcohol, a pilot can still be severely impaired for many hours by hangover. The effects of alcohol on the body are magnified at altitudes; 2 oz. of alcohol at 18,000 feet produce the same adverse effects as 6 oz. at sea level.

Federal Aviation Regulations have been amended to reflect the FAA's growing concern with the effects of alcohol impairment. FAR 91 states:

"Alcohol or drugs.

(a) No person may act or attempt to act as a crewmember of a civil aircraft -

(1) Within 8 hours after the consumption of any alcoholic beverage;

(2) While under the influence of alcohol;

(3) While using any drug that affects the person's faculties in any way contrary to safety; or

(4) While having .04 percent by weight or more alcohol in the blood.

(b) Except in an emergency, no pilot of a civil aircraft may allow a person who appears to be intoxicated or who demonstrates by manner or physical indications that the individual is under the influence of drugs (except a medical patient under proper care) to be carried in that aircraft."

Because of the slow destruction of alcohol by the body, a pilot may still be under influence eight hours after drinking a moderate amount of alcohol. Therefore, an excellent rule is to allow at least 12 to 24 hours between "bottle and throttle," depending on the amount of alcoholic beverage consumed.

DRUGS

Self-medication or taking medicine in any form when you are flying can be extremely hazardous. Even simple home or over-the-counter remedies and drugs such as aspirin, antihistamines, cold tablets, cough mixtures, laxatives, tranquilizers, and appetite suppressors may seriously impair the judgment and coordination needed while flying. The safest rule is to take no medicine before or while flying, except after consultation with your Aviation Medical Examiner.

SCUBA DIVING

Flying shortly after any prolonged scuba diving could be dangerous. Under the increased pressure of the water, excess nitrogen is absorbed into your system. If sufficient time has not elapsed prior to takeoff for your system to rid itself of this excess gas, you may experience the bends at cabin altitudes even under 10,000 feet.

CARBON MONOXIDE AND NIGHT VISION

The presence of carbon monoxide results in hypoxia which will affect night vision in the same manner and extent as hypoxia from high altitudes. Even small levels of carbon monoxide have the same effect as an altitude increase of 8,000 to 10,000 feet. Smoking several cigarettes can result in a carbon monoxide saturation sufficient to affect visual sensitivity equal to an increase of 8,000 feet altitude.

DECOMPRESSION SICKNESS

Pilots flying unpressurized airplanes at altitudes in excess of 10,000 feet should be alert for the symptoms of 'decompresson sickness'. This phenomenon, while rare, can impair the pilot's ability to perform and in extreme cases, can result in the victim being rendered unconscious. Decompression sickness, also known as dysbarism and aviator's "bends", is caused by nitrogen bubble formation in body tissue as the ambient air pressure is reduced by climbing to higher altitudes. The symptoms are pain in the joints, abdominal cramps, burning sensations in the skin, visual impairment and numbness. Some of these symptoms are similar to hypoxia. The only known remedy for decompression sickness is recompression, which can only be accomplished in an unpressurized airplane by descending. The pilot should immediately descend if it is suspected that this condition existed, since the effect will only worsen with continued exposure to the reduced pressure environment at altitude and could result if uncorrected, in complete incapacitation. The possibility of decompression sickness can be greatly reduced by pre-breathing oxygen prior to flight and by commencing oxygen breathing well below the altitudes where it is legally mandatory.

A FINAL WORD

Airplanes are truly remarkable machines. They enable us to shrink distance and time, and to expand our business and personal horizons in ways that, not too many years ago, were virtually inconceivable. For many businesses, the general aviation airplane has become the indispensable tool of efficiency.

Advances in the mechanical reliability of the airplane we fly have been equally impressive, as attested by the steadily declining statistics of accidents attributed to mechanical causes, at a time when the airframe, systems and powerplants have grown infinitely more complex. The explosion in capability of avionics systems is even more remarkable. Radar, RNAV, LORAN, sophisticated autopilots, EFIS and other devices which, just a few years ago, were too large and prohibitively expensive for general aviation size airplanes, are becoming increasingly commonplace in even the smallest airplanes.

Thus, this SAFETY INFORMATION is directed to the pilot, for it is in the area of the skill and proficiency of you, the pilot, that the greatest gains in safe flying are to be made over the years to come. Intimate knowledge of your airplane, its capabilities and its limitations, and disciplined adherence to the procedures for your airplane's operation, will enable you to transform potential tragedy into an interesting hangar story when - as it inevitably will - the abnormal situation is presented.

Know your airplane's limitations, and your own. Never exceed either.

Safe Flying,

BEECH AIRCRAFT CORPORATION